

Topics in Mathematics with Applications in Finance

Lecture notes

based on MIT 18.642 (Fall 2024) and MIT 18.S096 (Fall 2013)

Personal study notes of Alessio Martini

These notes are a personal, non-commercial study aid derived from the MIT OpenCourseWare courses *18.642 Topics in Mathematics with Applications in Finance* (Fall 2024; instructors Peter Kempthorne, Vasily Strela, Jake Xia) and *18.S096 Topics in Mathematics with Applications in Finance* (Fall 2013; instructors Peter Kempthorne, Choongbum Lee, Vasily Strela, Jake Xia), and on the guest lectures given in those courses. The original material is licensed under Creative Commons BY-NC-SA 4.0 (<https://ocw.mit.edu/terms>); these notes are shared under the same licence. Lecture content is paraphrased and re-derived, not transcribed; any mistakes are mine.

How to use these notes

Each chapter covers one *topic* of the course. Chapters are ordered as the topics first appear in the 2024 course (18.642); lectures from 2013 (18.S096) on the same topic are merged into the same chapter, while 2013 lectures with no 2024 counterpart form Part II. Every chapter opens with a **Sources** box listing the lectures (with their videos), slides, notes, case studies and problem sets it is built on.

Boxes.

- **Definitions** (blue), **theorems** (red), **examples** (green), **exercises** and **remarks** are marked with a coloured bar on the left.
- **Key concepts** boxes close each chapter: questions to answer without looking, then the concepts to remember with their formulas (older chapters still have **Key formulas**).
- **Try first** exercises are computations to attempt before reading the solution, printed in a *Solutions* section at the end of the chapter (with links both ways). Short exercises sit right after the paragraph they practise; long, structured ones are at the end of the chapter.
- **Intuition** boxes give the “physicist’s reading” of a result.
- **In finance** boxes explain where the mathematics is used in practice.
- **Lab** boxes turn the R case studies of the course into problems to redo in Python (data in `data/`, a reference script in `labs/`); the course’s results are their solution. Older chapters still have **Case study** boxes, which only summarise them.
- **Beyond the core** blocks (grey bar on the left, smaller type) hold material that can be skipped on a first reading: auxiliary derivations, numerical checks, 2013-vs-2024 details.

The right column. The page is laid out for a screen: the text is always on the left and a column on the right holds **proofs**, **examples**, **course notes** (errata in the slides, differences between the 2013 and 2024 versions, material that was not recorded; numbered with your notes, a grey-blue number in the text marks what they are about), **history** boxes and **asides** (curiosities), your **notes** with their **answers**, and **feedback** on your solutions. Each item starts level with the paragraph it belongs to; when the right column is longer, the text leaves a gap.

Video timestamps. A tag such as `[► 2024 L12 @ 23:10]` means: lecture 12 of the 2024 course, at minute 23:10 of the video, is where the point is explained in class. The 2024 lecture 1 is split into three videos, written `[► 2024 L1-II @ 12:36]`.

¹ Why is the duration of a zero-coupon bond $T/(1+y)$ and not simply T ? (Example: delete it from `main.tex`.)

Your annotations. While reading, write directly in the chapter files. The macro `\note{...}` prints a comment in the margin next to the text where it is written: a doubt, an idea, a reminder. For example `\note{Why is duration divided by  $1+y$ ??}` produces the box in the margin here. When a doubt is answered, the note and its answer become one box in the right column. Requests about the notes themselves are collected in the chapter *Requests about these notes*, which also lists how read and unread notes are marked.¹

Requests about these notes

Your margin notes are of two kinds: notes on the *content* (a doubt, an idea about a formula), which stay next to the text they refer to, and requests about the *notes themselves* (layout, boxes, conventions), which do not depend on where they were written. The second kind is collected here, with what was done about it, so that the chapters read more smoothly. Write new ones anywhere with `\note{...}` and they will be moved here, or add them directly at the end of this list.

How the margin notes are marked.

- `\note{...}`: a *new* note, not yet read. “Read my notes” means these, and only these.
- A note that has been answered becomes one box with the question on top and a dark yellow **ANSWER** bar below it (`\begin{answer}{question}[topic] ... \end{answer}`). A note with several questions gets one ANSWER bar per question. A question asked a second time gets a short answer that links to the first one.
- `\note*{...}`: a request that has been carried out (green DONE tag).
- `\note+{...}`: a comment that has been read and needs no answer (blue SEEN tag).
- `\feedback{...}`: a comment on one of your own solutions (indigo).

The list

DONE

Chapter 3, moments. quando scrivi i valori di aspettazione usa sempre le parentesi quadre

WHAT WAS DONE

Done in the whole book: about 460 expectations written as $\mathbb{E}X$, $\mathbb{E}|X|$, $\mathbb{E}(X)$ or $\mathbb{E} X$ became $\mathbb{E}[X]$, $\mathbb{E}[|X|]$, ... (with a script, checked by hand where the argument was ambiguous). A few long ones such as $\mathbb{E} \int_0^T f^2 \, ds$ are left as they are, where brackets would not make the reading clearer.

DONE

Chapter 3, normal distribution. in general, whenever it can be useful plot graphs and functions.

WHAT WAS DONE

Rule adopted for all chapters. In chapter 3: daily-return distributions with the course data, the volatility smile, the normal density, call and put payoffs, all the laws of the table.

DONE

Chapter 2, exercises. allinea i feedback con la soluzione corrispondente. colora di rosso i feedback se la soluzione ha un errore

WHAT WAS DONE

A feedback now starts level with the top of your solution (it is written just before the solution box), and it is red, with the tag “FEEDBACK: ERROR”, when the solution contains a mistake; indigo when it is right or only incomplete. All the feedback of chapter 2 converted.

NOTED

Chapter 3, start. voglio che le note appaiano a lato di dove le scrivo, non alla fine del paragrafo

WHAT WAS DONE

Not possible with the two-column layout: the right column can only be filled between paragraphs, so a note appears level with the end of its paragraph. (Real margin notes, the old layout, can sit at their line, but they overflowed and overlapped as soon as there were many.) What helps: the number in the text; and a long paragraph can be split where you write a note.

NOTED

Chapter 3, independence. avrebbe senso che nella fase di studio (quando leggo, faccio gli esercizio, scrivo annotazioni) il pdf non sia diviso in pagine ma sia una unica pagina lunghissima? e' possibile? me lo consigli?

WHAT WAS DONE

Not for a whole chapter: a page cannot be taller than about 5.7 m in $\text{\LaTeX}$ and 5 m in a PDF, and a chapter is 8–10 m. Also the right column needs page breaks to restart. I do not recommend it: set the viewer to continuous vertical scrolling with no gap between pages, which reads almost the same.

NOTED

Chapter 3, expected call payoff. stai attento che teoremi e dimostrazioni non sono perfettamente allineati

WHAT WAS DONE

A proof starts level with its theorem when the right column is free there; if a note or answer from the paragraph above is still in that column, the proof starts just below it (e.g. [theorem 3.9](#)). The proof of that theorem also had a formula wider than the column: split on two lines.

DONE

Chapter 2, one-period model. the professor here introduced 2 new concepts: contingent claim and replicating portfolios, whenever you introduce new concept, reserve few line or maybe a dedicated box for new concepts to introduce it, remarking that it is a new concept. maybe you should use the definition box

WHAT WAS DONE

Contingent claim and replicating portfolio now have their own definition box, titled “New concepts” ([definition 2.4](#)), announced by a sentence in the text. From now on every new concept gets a blue definition box when it first appears; older chapters get them during the core/secondary pass.

DONE

Chapter 2, one-period model and Markov-chain exercise. voglio che le note siano numerate, in stile footnote, in modo tale che possa vedere velocemente qual'e' il punto nel testo a cui fanno riferimento | vorrei che le note appaiano a fianco di dove le scrivo. inoltre voglio che siano numerate in stile footnotes. [in realta' forse l'idea migliore e' che vengano messe in lista sulla destra una dopo l'altra senza spazio in mezzo, con la numerazione]

WHAT WAS DONE

Notes and answered notes are numbered per chapter: a small dark-yellow number is printed in the text where you wrote the note, and the same number starts its box in the right column. The boxes are stacked one after the other in the right column, at the height of the paragraph that contains them (they cannot be exactly level with the line: the column is filled paragraph by paragraph).

DONE

Chapter 2, n assets. voglio che queste indicazioni di video e minuto siano link effettivi al video e al minuto-secondo giusto

WHAT WAS DONE

Every video tag [[▶ 2024 L2 @ 1:06:57](#)] is now a link to the lecture on MIT OpenCourseWare's YouTube channel at that second (youtu.be/<id>?t=<seconds>); the video IDs come from the course data downloaded with the material. Lectures without a video (2013 L4, 2024 L3) print the tag without link.

DONE

Chapter 2, Markov chains. vorrei che le answers fossero strutturate in modo diverso: cioè, che nel box della nota a lato venga attaccato sotto un altro box con la risposta [magari lo stesso box diviso dalla barra gialla più scura "ANSWER", sopra domanda e sotto risposta]

WHAT WAS DONE

Question and answer are now one box in the right column: your note on top, then a dark yellow bar "ANSWER n : topic" and the answer, in `\tiny`. All the answers of chapters 1–2 were converted.

DONE

Chapter 2, Markov chains. penso potrebbe essere una buona idea avere il pdf in formato draft, e non impaginato come se fosse la versione finale, in modo tale che la colonna di testo principale stia sulla sinistra (non come ora che le pagine pari stanno da un lato e le dispari dall'altro) e che le note, commenti, answers, historical notex, curiosities stiano tutte dall'altro a destra. in modo tale che ottimizzo meglio l'uso dello spazio sul mio schermo del laptop. forse metterli a lato anche gli esempi

WHAT WAS DONE

Screen layout: the page is 25 cm wide instead of 21, the text column is always on the left and keeps its old width (15 cm, so no formula or figure changes), and a 6.5 cm column on the right holds notes, answers, feedback, history boxes and asides. A column that overflows continues on the next page. Short examples will move to the right column with the `sideexample` box during the core/secondary pass through the chapters; long ones stay in the text.

DONE

Chapter 2, one-period model. problem: links to other chapters don't work. furthermore, i want the note's text smaller

WHAT WAS DONE

Notes are now in `\tiny`. Links to other chapters: see the last request below.

DONE

Chapter 2, one-period model. qualche volta scrivo più di una domanda nella stessa nota, e qualche volta scrivo la stessa domanda in più note diverse. scrivi una risposta per ogni domanda, separandole, e metti un link alla risposta che hai già a una domanda dato se è la seconda volta che ti chiedo la stessa cosa

WHAT WAS DONE

One ANSWER bar per question, also in the old answers of chapter 1. A repeated question gets a one-line answer with a link, e.g. "non capisco cos'è la claim" points to [answer 2.14](#).

DONE

Chapter 2, one-period model. trova il modo di differenziare le note che hai già letto e a cui hai già dato una risposta o hai già risolto il problema che chiedevo di risolvere, da quelle nuove, in modo tale che quando ti chiedo di leggere le note vai a guardare quelle nuove e eviti di modificare cose che andavano già bene

WHAT WAS DONE

Every note that has been read is now marked: answered (ANSWER box), DONE or SEEN, as listed above. A plain `\note{...}` is always new, and only those are read when you ask.

DONE

Chapter 2, one-period model. direi che ci sono due tipi di note che scrivo, quelle riferite al contenuto delle letture notes e quelle riferite alla struttura delle lecture notes. siccome quelle della struttura non penso siano strettamente legate a dove le scrivo, pensavo che sarebbe una buona idea raggrupparle in un punto specifico delle lecture notes senza lasciarle lungo il testo, così la seconda lettura dovrebbe risultare più facile e scorrevole

WHAT WAS DONE

This chapter.

NOTED

Chapter 2, one-period model. non so se ha senso e se è fattibile, ma ho pensato che potrebbe essere utile avere sempre una copia delle build di ogni capitolo e/o una dell'intero book. e quando sono in un capitolo e clicco il link a un altro semplicemente mi apre un altro pdf

WHAT WAS DONE

Tried and removed (29 Sep 2026): in the fast build that runs on save, links into other chapters were sent to a copy of the whole book, but the PDF viewer inside VS Code (LaTeX Workshop) cannot open a second PDF from a link. What remains: every full build (recipe *Whole book (latexmk)*) keeps a copy of the whole book in `build/book.pdf`, which the fast builds never overwrite, so it is always there to consult the other chapters.

NOTED

Chapter 1, the course at a glance. Nowadays everyone uses Python, so I will use it to illustrate the maths.

WHAT WAS DONE

The numerical checks in these notes are done in Python; the course's own code is R and is quoted as it is. Python snippets next to the formulas can be added on request.

DONE

Chapter 1, what is traded. Leave some space between lines in the table.

WHAT WAS DONE

Every table in the notes has its rows spaced (`\arraystretch 1.35`).

DONE

Chapter 1, what is traded. Here I am sure there were picture from the presentation. We should include them to make the presentation clearer and more interesting.

WHAT WAS DONE

The figures of the chapter-1 slides are redrawn in TikZ. Doing the same for all chapters is planned ("more figures like the slides").

DONE

Chapter 1, where mathematics enters. the titles of the boxes should be given more importance. here id call it something like: INTUITION: KT prospect theory, reflection effect and utility (this is just an idea)

WHAT WAS DONE

Every box title reads "TYPE: topic" in bold capitals, and every content box has a topic.

DONE

Chapter 1, compounding. I love history facts! can we implement a special box for them? like something that looks like old, antique. Furthermore, factcheck them and add references.

WHAT WAS DONE

History box framed as a classical temple front, fact-checked, with a References line; now in the right column.

DONE

Chapter 1, duration and convexity. in general, i'd like to have graphs and conceptual maps when possible

WHAT WAS DONE

More figures and a conceptual map at the end of chapter 1; continued chapter by chapter.

DONE

Chapter 1, key formulas. forse aiuterebbe aggiungere le unita' di misura, tipo T^{-1} per r

WHAT WAS DONE

Units added to the key formulas of chapter 1 (e.g. r in year^{-1}).

DONE

Chapter 1, exercises (twice). My solution moved into the box below.

WHAT WAS DONE

Your own solutions go in a **My solution** box (snippet `sol`) right below the exercise; comments on them are **Feedback** boxes in the right column.

DONE

All chapters, exercises. Gli esercizi brevi vanno lungo il testo, subito dopo la sezione che esercitano; alla fine del capitolo restano solo quelli complessi (che richiedono tutto il capitolo).

WHAT WAS DONE

Done in all chapters (1–25), with your refinement: standalone exercises (self-contained, practising one section) moved as whole blocks, with your solutions, notes and feedback, to the end of the section they practise; exercises that form a connected series (consecutive problems referring to each other, such as “the matrix of the previous exercise”) stay together, in their original order, at the end of the chapter, with the integrative ones. In total 212 of the 236 exercises are now in the text; 24 stay in a final Exercises section (chapters 1, 2, 5, 7, 9, 10, 13, 25). Chapters whose exercises were all standalone no longer have an Exercises section. Chapter 2 was corrected: PS1 (2024) Problem 3 is back next to Problem 4, and PS1 (2013) A-2 to A-4 follow the eigenvalue section.

DONE

Chapter 3, the CAPM. non ha senso mettere due link a due citazioni consecutive, in questi casi tieni solo il link che punta all'inizio del discorso

WHAT WAS DONE

Done in the whole book: wherever two or more video tags of the *same* lecture followed each other, only the first (the start of the explanation) is kept; 611 tags removed in 16 chapters. Consecutive tags of *different* lectures (a 2024 and a 2013 lecture on the same point) are kept: they lead to two different videos.

DONE

Chapter 3, put–call parity exercise. voglio che il colore degli esercizi sia piu diverso dal colore delle definizioni (non cambiare il colore definizioni)

WHAT WAS DONE

Exercises, your solutions and feedback are now lime (7CB342) instead of indigo. Lime is too light for text on white, so titles and tags use a darker shade of it (**like this**); the bars, frames and backgrounds are lime. Same colour in the study HTML page.

DONE

Chapter 3, heavy tails. questa proof non e' allineata

WHAT WAS DONE

Cause: the figure just before the proposition was a *float*. $\text{\LaTeX}$ moved it to the top of the next page after the right column had already been lined up there, so the proof was printed beside the figure. Figures and tables now stay exactly where they are written (in the whole book), so they can no longer slide above the right column; the price is some white space at the bottom of a page when a figure does not fit.

DONE

Chapter 3, principal components. spesso le course notes si riferiscono a qualcosa di preciso nel testo. direi di numerarle in stile footnotes come abbiamo fatto con le note, con l'indicazione nel testo. poi le posizionerei non bloccate al testo, ma di lato a destra in alto.

WHAT WAS DONE

Course notes are numbered with the notes (one sequence per chapter): a grey-blue number in the text marks the words the course note is about, and the same number starts its box in the right column. In chapter 3 every course note was moved right after the words it refers to (the one on the scope of the PCA section is about the whole section and has no number in the text). As for the notes, the box starts level with the end of the paragraph that contains the number, not with the line: the right column is filled paragraph by paragraph. Other chapters: their course notes already have a number in the box; the number in the text is added when each chapter is revised.

DONE

Chapter 3, lognormal example. dalla ricerca su efficacia studio, mi sembra di aver capito che al mio livello e' meglio non avere esercizi svolti, ma provare a farli da zero. quindi pensavo di trasformare gli esempi in esercizi, magari con la soluzione nascosta, e una volta che ho provato a risolverli, mostri/compari con la soluzione.

WHAT WAS DONE

Right, with one qualification from the research. For beginners, studying worked examples beats solving problems (the *worked-example effect*; meta-analysis in mathematics: Barbieri, Clerjuste and Chawla 2023, $g \approx 0.5$), but the advantage reverses as prior knowledge grows (the *expertise reversal effect*, Kalyuga 2007), and trying a problem before being shown the solution helps conceptual understanding (*productive failure*, meta-analysis Sinha and Kapur 2021, $d \approx 0.36$). So: examples of computations on material you already master become **Try first** exercises; examples that introduce a new idea stay worked examples. The solution is printed in a *Solutions* section at the end of the chapter, with a link from the exercise and one back (in the PDF it cannot be truly hidden; on the study HTML page it will be folded). Chapter 3: the lognormal(0.2, 0.4²) example and "How many observations?" converted ([Exercises 3.1](#) and [3.10](#)); "Estimating a mean" stays an example. Other chapters next, after you have tried these two.

DONE

Chapter 3, exercises on MGFs. qua hai messo una lista di molti esercizi. penso che sarebbe piu efficace avere esercizi brevi, molto selezionati, e mirati lungo il testo, e solo alla fine esercizi piu lunghi e struttirati, comunque mirati a verificare la comprensione.

WHAT WAS DONE

Agreed (it is the evidence on adjunct questions and interpolated testing quoted for the previous exercise request). Chapter 3 now has short exercises right after the *paragraph* they practise (e.g. the Poisson MGF after the normal MGF, exponential families right after their definition), and a final *Exercises* section ([section 3.11.5](#)) with the five long structured problems: 2024 PS2 Problem 2, 2013 PS2 B-4, A-1 and B-1 (B-1 uses A-1), and 2024 PS3 Problem 2. No exercise was dropped. Other chapters next.

DONE

Introduction, boxes. non so se la cosa delle key formulas e' utile. non mi piace particolarmente. forse apprezzerei di piu key concepts, magari presentati lungo il testo in risalto o riportati alla fine o entrambi

WHAT WAS DONE

The Key formulas box becomes **Key concepts**: first a few questions to answer without looking (retrieval practice is one of the best-supported study techniques: Dunlosky et al. 2013; meta-analysis Adesope, Trevisan and Sundararajan 2017, $g \approx 0.6$), then each concept in one or two sentences with its formula and a link to where it is stated. Along the text the concepts are already highlighted by the blue *New concepts* definition boxes (highlighting by the author helps a little: signalling meta-analysis, Schneider et al. 2018). Every formula of the old box is kept. Chapter 3 done; the other chapters next.

DONE

Introduction, boxes. questi box non so quanto abbiano senso. dobbiamo ragionarci.

WHAT WAS DONE

Agreed with you: read passively, a summary of someone else's R analysis teaches little; redoing it does more (learning by doing beats reading: the ICAP framework, Chi and Wylie 2014; active learning in STEM, meta-analysis Freeman et al. 2014). So each case study becomes a **Lab**: a question, the data (official source, in data/), the steps to redo in Python, and the course's results as its solution at the end of the chapter, with a reference script in labs/ that you can open after trying. Pilot: the high-yield spread of chapter 5 ([section 5.1](#), data from FRED, script labs/ch05_hyspread.py, which reproduces the $R^2 = 0.995$ of the autoregression shown in class). About 30 case studies remain; they are converted one chapter at a time.

NOTED

Chapter 3, kinds of random variables. questa nota non appare affianco di dove e' stata scritta

WHAT WAS DONE

It was written inside a bulleted list, and a whole list counts as one paragraph for the right column: the items queued in it are printed when the list ends, so the box starts level with the end of the list, not with the item. Splitting the list would break its layout; the number in the text still marks the exact point.

NOTED

Chapter 3, estimating a mean; the CAPM. qua c'e' un problema di impaginazione, e spazi vuoti. inoltre, questo box non lo chiamarei esercizio/esempio, e' qualcos'altro | qua c'e' un problema con l'impaginazione

WHAT WAS DONE

Fixed. "Estimating a mean" is now a paragraph of the text, not an example: it was an example box broken across two pages with a course note inside it. In the CAPM section a paragraph broke across pages with a long course note anchored in it: the note was pushed to the next page and the text waited, leaving a page almost empty; the paragraph is now split, and the note sits beside it.

OPEN

Chapter 3, solutions. vorrei che tutti gli esercizi avessero la soluzione cosi'

WHAT WAS DONE

Chapter 3: every exercise now has a solution in the *Solutions* section at the end of the chapter, with a "Solution" link under the exercise and one back (numbers checked numerically). Try each exercise first. Other chapters next, one at a time (about 220 exercises). A remark: in 2013 PS2 B-4 the author's numbers do not follow from the stated parameters; the solution shows the correct ones.

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Part I

The 2024 course (18.642)

Chapter 1

Financial Markets, Instruments and Bond Mathematics

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L1-I	V. Strela, P. Kempthorne	Organisation of the class, guest lectures, R/RStudio and financial data.
2024 L1-II	J. Xia	Markets, products, participants, trader types, investment strategies, investment game.
2024 L1-III	V. Strela	Compounding, discounting, zero-coupon and coupon bonds, yield, yield curve, duration.
2013 L1	J. Xia, V. Strela	Same introduction with more detail on hedging, market making and prop trading; risk-aversion quiz; two intern projects (noisy delta, Kalman filter).

Files. Slides `mit18_642_f24_lec01_1.pdf` (intro, markets, bond math); `mit18_642_f24_lec01_2.pdf` (R/RStudio) with the notebook `FM_Intro1.Rmd` in `mit18_642_f24_rstudio.zip`; 2013 slides `MIT18_S096F13_lecnote1.pdf`; the 2024 *Investment Game* page. No problem set of either year covers this chapter directly (the 2024 problem sets start with linear algebra, [chapter 2](#)).

Motivation

The first lecture has two jobs: to give a vocabulary of markets, instruments and players that every later application lecture takes for granted, and to introduce the single most important idea in quantitative finance, *pricing by no-arbitrage*, on the simplest possible instruments: bonds. Everything in this chapter is elementary mathematically; what matters is the way of reasoning. The discounting argument of [section 1.7.2](#) is the ancestor of the replication argument behind Black–Scholes ([chapter 15](#)) and the forward-rate formula of [section 1.8.5](#) is the seed of the HJM framework ([chapter 24](#)).¹

1.1 The course at a glance

The course alternates *mathematics lectures* (linear algebra, probability, stochastic processes, regression, time series, volatility, stochastic calculus; P. Kempthorne in 2024, P. Kempthorne and C. Lee in 2013) with *application lectures* by practitioners (banks, hedge funds, exchanges, academia) [► 2024 L1-I @ 4:05]. The mathematics is chosen because it is used on trading floors, not for its own sake [► 2024 L1-I @ 8:17]. Illustrations and case studies are written in R; organisational details (problem sets, group project, final paper, investment game, grading)

¹ What are the discounting argument? and the replication argument? and the forward-rate formula? and the HJM framework?

ANSWER 1.1: Discounting argument

The sentence is a roadmap; each item is developed later. [Section 1.7.2](#): the price today of \$1 paid in one year is fixed by comparison with riskless borrowing and lending, $x = 1/(1+r)$; any other price gives a riskless profit.

ANSWER 1.2: Replication argument

If a portfolio of traded assets pays exactly what a contract pays in every state of the world, the contract must cost what the portfolio costs. Discounting is the simplest case (the claim to \$1 is replicated by lending $1/(1+r)$); Black–Scholes ([chapter 15](#)) replicates an option with a continuously rebalanced position of Δ shares plus cash.

ANSWER 1.3: Forward-rate formula

[Proposition 1.13](#): the interest fixed today for a loan from t to T is read off today's discount factors, $F_{tT} = \Lambda_t / \Lambda_T - 1$.

ANSWER 1.4: HJM framework

Heath–Jarrow–Morton, [chapter 24](#): a model for the random evolution of the whole forward curve $f(t, T)$; no-arbitrage fixes its drift in terms of its volatility.

are collected in [chapter C](#).

1.2 Markets and products

1.2.1 Where trading happens

- **Exchanges:** centralised venues where standardised products (*listed* securities, futures) trade under common rules [► 2024 L1-II @ 8:23].
- **Over-the-counter (OTC):** bilateral contracts between two counterparties, not necessarily standardised and not subject to exchange rules; most swaps and many options trade OTC [► 2013 L1 @ 12:30]. OTC trading creates *counterparty risk* ([chapter 8](#)).
- **Electronic communication networks (ECN):** electronic platforms acting as distributed exchanges [► 2024 L1-II @ 9:25].

1.2.2 What is traded

[Table 1.1](#) lists the main product classes with the chapters in which their mathematics appears.

Table 1.1: Main product classes.

Class	What it is	Where
Money, FX	Currencies issued by different states; also gold and, arguably, crypto.	9, 23
Equity	Ownership of a company (stocks); baskets (indices such as the S&P 500, ETFs).	5, 10
Fixed income	Loans, securitised as bonds (government, corporate, municipal); interest-rate curves.	this ch., 6
Credit	Instruments whose value depends on default of an issuer.	8, 23
Real estate	Mortgages, mortgage-backed (MBS) and asset-backed securities (ABS).	–
Commodities	Metals, energy, agricultural products; mostly traded as futures.	21
Derivatives	Contracts whose payoff depends on another asset: forwards, futures, swaps, options, structured products.	15, 6

A few points from the lectures deserve emphasis.

HISTORY: The course

The class grew out of recruiting trips of J. Xia and V. Strela (then at Morgan Stanley) to MIT: students with strong quantitative backgrounds asked more questions about finance than an interview slot could answer, so a seminar was set up, later extended to a full term with mathematics lectures by P. Kempthorne [► 2024 L1-II @ 4:14]. The 2013 edition (18.S096) was the first to be recorded; the 2024 edition (18.642) keeps the same structure with new guest speakers.

Source. J. Xia's own account in the 2024 lecture (timestamp above); both editions are on MIT OpenCourseWare (<https://ocw.mit.edu>).

2 Difference between stock, equity, security? Define future, forward, option and swap. What are the exchange rules? Intuition behind counterparty risk?

ANSWER 1.5: Stock, equity, security

Security: any tradable financial claim (share, bond, note, ...). **Equity:** ownership of a company, as opposed to debt. A **stock** (share) is the security that represents equity; in market jargon "equities" simply means stocks.

ANSWER 1.6: Future, forward, option, swap

Forward: private (OTC) agreement to buy or sell an asset at a date T at a price K fixed today; nothing is paid today, everything is settled at T . **Future:** the same, but standardised (size, dates) and exchange-traded, with gains and losses settled daily through margin accounts. **Option:** the right, not the obligation, to buy (call, payoff $(S_T - K)^+$) or sell (put, payoff $(K - S_T)^+$); the buyer pays a premium today. **Swap:** exchange of two streams of cash flows on a notional, e.g. fixed vs floating interest; it is a portfolio of forwards.

ANSWER 1.7: Exchange rules

Standardised contracts; only members trade directly (others go through brokers); orders are matched in an order book by price, then time; tick sizes, trading hours, circuit breakers and reporting duties. Trades are cleared by a central counterparty that collects margin from both sides, which is what removes bilateral counterparty risk.

ANSWER 1.8: Counterparty risk

In an OTC contract your gain is only a promise of the other side. If they default when the contract is worth something to you, you lose that value (the cost of replacing the contract). It is like having lent money to them: the exposure is what they would owe you if they disappeared now, and it matters only when the contract has positive value to you. Collateral and clearing reduce it ([chapter 8](#)).

3 Is the primary market an actual market? what is a market?

ANSWER 1.9: Primary market

The primary market is not a place but a process: the first sale of newly issued securities by the issuer to investors (an IPO for shares, auctions or bank syndicates for bonds). The money goes to the issuer, and investment banks organise the sale (underwriting, book-building). In the secondary market investors trade existing securities among themselves and the issuer receives nothing.

ANSWER 1.10: What a market is

In general a *market* is any mechanism that brings buyers and sellers together and lets a price form: an exchange with an order book, a network of dealers quoting prices (OTC), or an auction.

- **Primary vs secondary market.** A company becomes listed through an *initial public offering* (IPO), the primary market; afterwards its shares are traded among investors, the secondary market [► 2024 L1-II @ 11:33].
- **A bond is a securitised loan.** The issuer pays periodic interest (*coupons*) and repays the *principal* or *notional* at maturity. Debt is “more generic” than equity: finance is fundamentally about someone who has money and someone who needs it [► 2013 L1 @ 14:36]. Since the issuer may fail to pay, a bond carries *credit risk* [► 2024 L1-II @ 12:36]; the 2008 crisis was a credit crisis that started in mortgages and spread through the derivatives built on them [► 2024 L1-II @ 13:36].
- **Asset-backed securities:** debt issued against a pool of assets (e.g. mortgages), whose risk depends on the income stream of the pool [► 2013 L1 @ 16:44].
- **Derivatives:** an *option* is the right but not the obligation to buy (call) or sell (put) at a fixed price; a *forward* is an agreement today to trade at a future date; a *future* is a standardised, exchange-traded forward; a *swap* exchanges two streams of cash flows (e.g. fixed vs floating interest) [► 2024 L1-II @ 15:41].

IN FINANCE: Impossible events happen

In April 2020 the front-month WTI crude-oil futures contract traded at a significantly *negative* price: holders of long contracts had to pay to get rid of them because nobody could store the oil. Some brokers’ systems had not been programmed for negative prices [► 2024 L1-I @ 14:30]. Negative interest rates, once considered impossible, were the norm in the eurozone and Japan for years. Lesson: a model’s hard constraints (“prices are positive”) are assumptions, not facts.

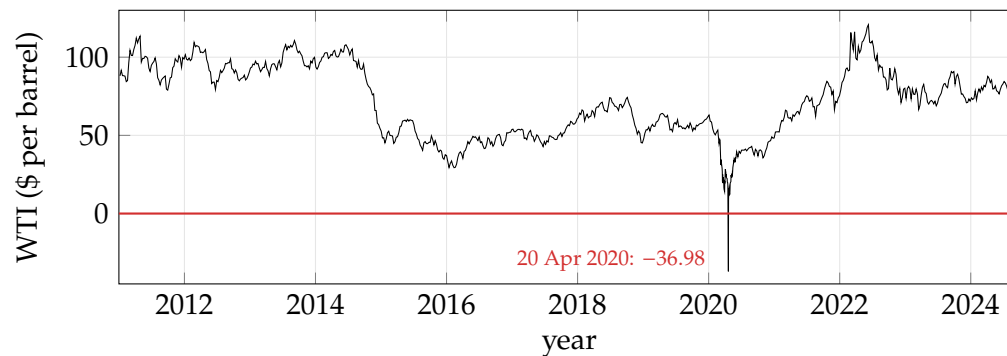


Figure 1.1: WTI crude oil (FRED series DCOILWTIC0, weekly closes, daily in spring 2020), 2011–2024, with the zero line in red: re-drawn from the data of the course notebook FM_Intro1.Rmd (section 1.9), shown in class [► 2024 L1-I @ 14:54].

1.3 Market participants

- **Commercial banks** take deposits and lend. **Investment banks** raise capital and trade. The two were separated in the US by the Glass–Steagall Act (1933); its repeal in 1999 is often blamed as a cause of the 2008 crisis [► 2013 L1 @ 17:44]. An investment bank typically has three client divisions [► 2024 L1-II @ 17:45]: *Equity* (stocks and equity derivatives), *Fixed Income* (bonds, rates, FX, commodities and their derivatives) and the *Investment Banking Division* (IBD: corporate finance, IPOs, bond issuance, M&A

advisory); plus asset and wealth management.

- **Dealers vs brokers.** A dealer (market maker) quotes a price and takes the other side of the trade, i.e. takes *principal risk*; a broker only matches buyers and sellers and earns a commission [► 2013 L1 @ 23:00].
- **Asset owners and managers:** mutual funds (usually long-only), insurance companies and pension funds (assets invested to meet liabilities), sovereign wealth funds, endowments, family offices [► 2013 L1 @ 24:04].
- **Hedge funds** trade their own and their investors' capital to generate returns; **private equity** buys stakes in companies to improve and later sell them.
- **Retail investors, central banks** (policy-driven: set short rates, provide liquidity, intervene in FX) and **corporates** (mostly hedgers).

The driving force. Every participant tries to maximise gains while limiting losses under uncertainty. The fundamental trade is between *lenders* (have money, need return) and *borrowers* (need money, bear risk) [► 2024 L1-II @ 20:49]. The same pairing appears as *limited partners* (LP, the investors in a fund) vs *general partners* (GP, the fund manager). Whether markets are a zero-sum game depends on the product: many trades net to zero, but markets exist because they transfer capital from those who have it to those who can use it productively [► 2013 L1 @ 20:53]. ⁷

1.4 Three types of traders

1.4.1 Hedgers

Hedgers already have an exposure and want to *reduce* it [► 2013 L1 @ 26:13]: a homeowner with a floating-rate mortgage who locks in a fixed rate; an exporter with future euro revenues who sells euros forward; a company in an M&A deal hedging currency and rate exposure. ⁸

INTUITION: Covered interest parity

The yen/AUD trade is an example of a general principle you will see formalised in [chapter 15](#): once the exchange-rate risk is hedged with a forward, the forward price itself incorporates the interest-rate differential (*covered interest parity*). A risk-free “free lunch” from the rate differential would be an arbitrage.

1.4.2 Market makers

4 repeal? liabilities?

ANSWER 1.11: Repeal

To *repeal* a law is to abolish it with a later law: the Gramm–Leach–Bliley Act (1999) removed the Glass–Steagall separation, so commercial and investment banks could merge again (e.g. Citigroup).

ANSWER 1.12: Liabilities

Liabilities are what an entity owes: for a bank, its deposits; for a pension fund, the pensions it has promised; for an insurer, future claims. “Invested to meet liabilities” means choosing assets whose cash flows match those future payments (asset–liability management), e.g. long bonds against long-dated pensions, with matching durations ([section 1.8.4](#)).

DONE

5 Central bank decides interest rate. Right now in 2024 the interest curve is inverted! The central bank of US is Federal Reserve Bank.

6 what is a typical shape of interest curve?

ANSWER 1.13: Central banks and the yield curve

The central bank sets only the *short-term* policy rate (in the US the federal funds target range, decided by the FOMC; in the eurozone the ECB deposit rate). Longer yields are set by the market: expectations of future policy rates plus a term premium. The US central bank is the *Federal Reserve System*: the Board of Governors in Washington plus twelve regional Federal Reserve Banks. On the inversion: the 2-year vs 10-year Treasury spread was negative from July 2022 until early September 2024, the longest inversion on record; the September 2024 curve of [fig. 1.3](#) is still inverted at the short end.

Typical shape: upward sloping and concave, steep over the first few years and flattening beyond about ten years (1992 and 2021 in [fig. 1.3](#)). The other shapes are flat, inverted (short above long, when the market expects the central bank to cut) and humped (a maximum at medium maturities).

7 Nice point: what is the point of a (financial) market?

ANSWER 1.14: What financial markets are for

- *Capital allocation*: savings flow to those who can use them productively;
- *risk transfer*: hedgers pass risks to those willing to bear them for a premium;
- *price discovery*: prices aggregate dispersed information into a public signal;
- *liquidity*: assets can be turned into cash quickly, which makes holding them cheaper;
- *moving consumption in time*: saving now to spend later, or the reverse.

Even when payoffs net to zero (derivatives), a trade can leave both sides better off, because they start from different exposures and preferences.

8 Exposure = risk? what is M&A?

ANSWER 1.15: Exposure

Exposure is the amount of value that moves when a risk factor moves: an exporter expecting EUR 10 million of revenues has a EUR 10 million EUR/USD exposure. *Risk* combines the exposure with how much the factor moves: roughly, the size of the typical loss is exposure $\times$ volatility of the factor. Exposure is thus a sensitivity (for derivatives, the delta).

ANSWER 1.16: M&A

M&A = mergers and acquisitions: one company buying another, or two merging. In a cross-border deal the buyer has to pay in a foreign currency at a future closing date, hence the FX and rate hedges.

Example 1.1 (Samurai bonds and the carry trade

[► 2013 L1 @ 30:29]). Australian rates have historically been well above Japanese rates; the 2013 slide quotes a 10-year swap-rate difference of $4.5\% - 1\% = 3.5\%$. An Australian company can therefore issue a bond in Japanese yen (JPY) (a *Samurai bond*), paying $\approx 1\%$, and use the proceeds in Australia where money earns $\approx 4.5\%$. Why doesn't everybody do this forever? Because the Australian dollar (AUD) may weaken against the yen, wiping out the rate differential; and

A market maker quotes a *bid* (price at which it buys) and an *offer* or *ask* (price at which it sells), with $\text{bid} < \text{offer}$, and aims to earn the spread rather than to take a view [► 2024 L1-II @ 21:56]. For transparent products (a liquid stock) the spread is tiny; for opaque products (say a two-month call on Apple struck at 500) the market maker provides liquidity and must then manage the residual inventory of risk [► 2013 L1 @ 33:41]. This is done by balancing the *Greeks* of the book [► 2013 L1 @ 34:46], the partial derivatives of the value V of the option (or of the whole book) with respect to the price S of the underlying, time t and the volatility σ of the underlying: ¹⁰

Greek	Definition	Meaning
Delta Δ	$\partial V / \partial S$	sensitivity to the underlying price
Gamma Γ	$\partial^2 V / \partial S^2$	curvature, change of delta
Theta Θ	$\partial V / \partial t$	“carry” or time decay when nothing moves
Vega ν (or kappa)	$\partial V / \partial \sigma$	sensitivity to volatility

Beyond the Greeks, a dealer book is managed through tail-risk measures such as *Value at Risk* (chapter 19), capital usage, balance sheet and leverage. Before 2008 many banks were levered about 40 times: with \$1 of capital and \$40 of exposure, a 2.5% move wipes out the capital [► 2013 L1 @ 35:47]. ¹¹

1.4.3 Proprietary traders and portfolio managers

Risk takers deliberately take positions to earn a return [► 2013 L1 @ 36:50]. Styles include:

- **Directional:** long or short on a view (including “gut” trading). ¹³
- **Arbitrage:** exploit violations of *deterministic* price relationships, e.g. the same gold priced differently in two markets, or a spot price inconsistent with the forward price [► 2013 L1 @ 37:50].
- **Value and relative value:** estimate the “fair value” of an instrument relative to others and trade on the difference over a longer horizon.
- **Systematic:** computer models such as trend following, momentum, statistical arbitrage across many stocks.
- **Fundamental / global macro:** trade balances, earnings, policy changes [► 2013 L1 @ 40:00]; **special situations:** distressed companies.

if everybody holds the same position, the exit moves the market against them. To hedge, the company must *receive* JPY and *pay* AUD in the future, which in practice means FX forwards, interest-rate swaps and cross-currency (basis) swaps.

DONE

9 JPY = japanese yen, AUD = australian dollar.

DONE

10 def: V is the value of the option, S is the underlying price, t is time, σ is volatility.

11 def of liquidity, residual inventory of risk. Dealer book = order book? tail-risk measures?

ANSWER 1.17: Liquidity

The ability to trade a large size quickly without moving the price much; measured by the bid–ask spread, the depth of the market and traded volume.

ANSWER 1.18: Residual inventory of risk

After buying an option from a client the market maker holds a position it did not choose, and cannot sell the same option on at once (it is illiquid). It hedges what it can with liquid instruments (shares for delta, other options for vega); what is left is the residual risk it carries in inventory.

ANSWER 1.19: Dealer book ≠ order book

An order book is the exchange’s list of pending buy and sell orders at each price. A dealer’s *book* is its portfolio of positions, managed as a whole.

ANSWER 1.20: Tail-risk measures

They describe the extreme losses of the profit-and-loss distribution: *Value at Risk* (the loss exceeded with probability 1%, say, over one day), *Expected Shortfall* (the average loss beyond the VaR), stress tests (chapter 19).

12 COURSE NOTE: Vega or kappa?

J. Xia’s anecdote [► 2013 L1 @ 6:15]: on his first day on the Morgan Stanley options desk he asked for the “vega report”; the training manual explained that the house term was *kappa*, since vega is not a Greek letter. Both names are in use; this reflects how young quantitative finance is as a discipline (the Black–Scholes model dates to 1973) [► 2013 L1 @ 9:19].

13 what is gut trading?

ANSWER 1.21: Gut trading

Trading on instinct (“gut feeling”) and experience rather than on a model or a fixed rule: the extreme form of *discretionary* trading, as opposed to *systematic* trading.

Definition 1.2 (Alpha and beta). Let R_b be the return of a benchmark (e.g. an S&P 500 index fund) and R_a the return of a portfolio. The linear regression

$$R_a = \alpha + \beta R_b + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Cov}(\varepsilon, R_b) = 0,$$

decomposes the portfolio return into a part βR_b explained by co-movement with the benchmark (*beta*, market exposure, cheap to obtain by buying the index) and an excess return α (*alpha*, the manager's skill) [► 2013 L1 @ 28:21]. The least-squares estimate is $\hat{\beta} = \widehat{\text{Cov}}(R_a, R_b) / \widehat{\text{Var}}(R_b)$; regression is developed in [chapter 5](#) and the CAPM interpretation in [chapter 10](#).

1.5 Where mathematics enters

Mathematics is used in three ways [► 2024 L1-II @ 23:01]:

- **Pricing models:** relative value and *arbitrage-free* prices of complex instruments (options need PDEs; [chapter 15](#)); a pricing model also delivers the risk parameters (Greeks) needed to hedge.
- **Risk management:** position sizing, diversification, when to take profit or cut losses, leverage, liquidity, counterparty risk; it counteracts greed and fear [► 2024 L1-II @ 24:10].
- **Trading strategies:** the “holy grail” or perpetual money machine. Such strategies, when they exist, are kept secret, and markets adapt: no model can be built and left alone forever [► 2013 L1 @ 42:03]. Firms protect them with confidentiality obligations on trade secrets that do not expire when an employee leaves, and with non-compete clauses (often a paid “garden leave” of some months) that keep departing traders and quants away from competitors.

Efficient markets vs behavioural finance. Can one learn from historical patterns and extrapolate? The *efficient market hypothesis* says no: a \$100 bill lying on the ground would have been picked up already. *Behavioural finance* replies that people are not always rational, which creates opportunities [► 2024 L1-II @ 26:15]. Deterministic (physics-like) relationships are nowadays almost always arbitrated away; what remains are *statistical* relationships, which hold only with some probability [► 2024 L1-II @ 29:25]. Market cycles are long and memories short: people over-weight recent experience [► 2013 L1 @ 51:23].

14 def of what is a return. what is CAPM interpretation?

ANSWER 1.22: Return

Over one period, with price P_t and dividend D_t , the *simple return* is $R_t = (P_t + D_t - P_{t-1})/P_{t-1}$ and the *log return* is $r_t = \ln(1 + R_t)$; log returns add up over time and $r_t \approx R_t$ for small moves.

ANSWER 1.23: CAPM interpretation

The CAPM ([chapter 10](#)) states that in equilibrium expected *excess* returns are proportional to beta with respect to the market portfolio M : $\mathbb{E}[R_a] - r_f = \beta_a (\mathbb{E}[R_M] - r_f)$. Written for excess returns with the market as benchmark, the regression above then has $\alpha = 0$: a non-zero alpha is return that market risk does not explain, i.e. skill or an unmodelled risk factor.

15 COURSE NOTE: Slide glitch

On the 2013 slide the formula reads “ $R(a) = a + b \cdot R(b)$ ”: the Greek letters α, β were lost in the conversion, as J. Xia remarks in class. In the same lecture a quote is given as “\$0.99 bid, \$0.95 offer”; the bid must be below the offer, and indeed later it becomes “\$0.90 bid, \$0.95 offer”.

DONE

16 Life-long no-disclosure contracts for trading strategies.

Example 1.3 (Risk-aversion quiz [► 2013 L1 @ 44:04]). *Loss framing.*

- (A) lose \$500 with probability 0.8, win \$500 with probability 0.2;
- (B) lose \$280 for sure.

Gain framing.

- (A) win \$500 with probability 0.8, lose \$500 with probability 0.2;
- (B) win \$280 for sure. ¹⁸

In the loss framing $\mathbb{E}[A] = 0.8(-500) + 0.2(500) = -300 < -280 = \mathbb{E}[B]$, so B has the higher expectation; yet many people choose A, to avoid “locking in” a loss. In the gain framing $\mathbb{E}[A] = +300 > 280 = \mathbb{E}[B]$, but most people choose B. A student's answer for B in class was “higher Sharpe”: indeed $\text{sd}(A) = 1000\sqrt{0.8 \cdot 0.2} = 400$

Sharpe ratio and other performance metrics. For a strategy with return R (per period), risk-free rate r_f and volatility $\sigma = \text{sd}(R)$, the *Sharpe ratio* is

$$\text{SR} = \frac{\mathbb{E}[R] - r_f}{\sigma},$$

the excess return per unit of risk. It is usually annualised, $\text{SR}_{\text{year}} \approx \sqrt{252} \text{SR}_{\text{day}}$ for daily returns. It does not change under leverage, which is why it is the standard basis for comparing strategies ([proposition 10.10](#)). A riskless payoff has $\sigma = 0$, hence the student's "higher Sharpe" for B. Other common metrics:

- **Sortino ratio:** like Sharpe, but divides by the downside deviation, so upside volatility is not penalised;
- **maximum drawdown:** largest peak-to-trough loss of the cumulative P&L; the **Calmar ratio** is annual return divided by maximum drawdown;
- **information ratio:** active return over a benchmark divided by the tracking error;
- **alpha and beta** with respect to a benchmark ([definition 1.2](#));
- **Value at Risk** and **Expected Shortfall:** tail losses ([chapter 19](#));
- **hit rate** (fraction of winning trades) and **profit factor** (gross gains over gross losses), close to the ratio $(G - L)/(G + L)$ of the Investment Game ([Exercise 1.1](#));
- **turnover** and **capacity:** how much trading the strategy needs and how much capital it can absorb before costs eat the returns.

INTUITION: Prospect theory: reflection effect and utility

The pattern "risk-seeking over losses, risk-averse over gains" is the *reflection effect* of Kahneman and Tversky's prospect theory (not named in the lecture). In trading it appears as the tendency to sell winners too early and hold losers too long, which is why discipline and stop-loss rules are part of risk management [[▶ 2013 L1 @ 46:10](#)]. With a concave utility u (risk aversion), B is preferred in the gain framing whenever $u(280) > 0.8u(500) + 0.2u(-500)$; the answer depends on wealth: a student with \$800 in the bank and an investor with \$100,000 should not answer the same way [[▶ 2013 L1 @ 50:20](#)].

while B is riskless.

DONE

17 Brief def of Sharpe ratio, and mention the other metrics to assess an investment strategy.

18 I don't understand what's the question. What's the best strategy?

ANSWER 1.24: What the quiz asks

Each framing asks: *which of the two would you choose?* It is a question about your preferences, and there is no single right answer. Xia says so explicitly: he is not telling the class which choice is right [[▶ 2013 L1 @ 47:49](#)].

ANSWER 1.25: The "best" strategy

A pure expected-value maximiser picks B in the loss framing ($-280 > -300$) and A in the gain framing ($300 > 280$). Most people do the opposite in both. The point is that the *framing* (losing vs winning the same kind of bet) flips the choice, while a rational choice should depend only on the payoffs, one's utility and one's wealth (see the intuition box below).

1.6 Choosing an investment strategy

J. Xia's list of choices every investor faces [[▶ 2024 L1-II @ 27:19](#)]:

Internal (direct investing)	vs	external (fund of funds)
Public assets (equity, bonds, FX, commodities)	vs	private assets (PE, VC, real estate, natural resources)
Passive (beta, benchmark)	vs	active (alpha)
Systematic (quantitative rules)	vs	discretionary (fundamental judgement)
Deterministic relationships	vs	statistical relationships
Trend following (buy strength)	vs	mean reversion (buy the dip)
Short term (fast, liquid)	vs	long term (slow, illiquid)
Value: $V(0) \gg P$	vs	growth: $V(T) \gg V(0)$
History (repeating patterns)	vs	future (extrapolation)

What Xia said about each choice:

- **Internal vs external:** do it yourself, or pay other people to do it for you (invest in their funds, or in a fund of funds) [► 2024 L1-II @ 27:44].
- **Public vs private:** venture capital, private equity, buyouts, real-estate and natural-resource funds are not transparent, “but they seem to make a lot of returns, too”. The two are very different markets and you need to pick one [► 2024 L1-II @ 28:04].
- **Passive vs active:** stock picking, or just buying the index. Buying the S&P 500 over the last ten years beat most fund managers, “so why shouldn’t you just do the passive?” [► 2024 L1-II @ 28:23].
- **Systematic vs discretionary:** decisions from quantitative rules or models, or case by case with your own judgement [► 2024 L1-II @ 28:52].
- **Deterministic vs statistical:** a deterministic relationship is like physics (A plus B always gives C); a statistical one gives C only with some probability. Deterministic arbitrage is nowadays very hard to find (the \$100 bills have been picked up), “but sometimes, they still show up” [► 2024 L1-II @ 29:18].
- **Trend following vs mean reversion:** when the market goes up, buy more, or sell (play the contrarian, buy the dip) [► 2024 L1-II @ 29:50].
- **Short vs long term:** short-term trading is fast, in liquid markets, and you can change your mind easily. Long-term decisions are slow: a start-up in a venture-capital portfolio may take ten years to prove itself [► 2024 L1-II @ 30:10].
- **Value vs growth:** Warren Buffett-style value investing wants the price well below today’s value. Growth investing projects, say, NVIDIA’s income over the next ten years and buys if the future value exceeds the current one. In recent years more and more people focus on growth, “you need to figure out why” [► 2024 L1-II @ 30:37].
- **History vs future:** believe that history repeats and follow the patterns, or refuse to simply extrapolate [► 2024 L1-II @ 31:09].

Here P is the current price and $V(t)$ the (estimated) value of the company at time t : a value investor buys what is cheap relative to today’s value, a growth investor buys expected growth. ²⁰ The key message [► 2024 L1-II @ 31:28]: understand your own objectives and constraints (what the money is for, how much you can afford to lose, your horizon), pick the area where you have an edge over the rest of the market, and “if math is your strength, make math your edge”. ²¹

DONE

19 I am pretty sure he said something more about each choice

20 This is just a different time-scale approach: connection to renormalization.

ANSWER 1.26: Value vs growth as time scales

Partly right. Both styles bet that price converges to value; they differ in *which* value and over what horizon. Value bets that the gap $V(0) - P$ closes (a mean-reversion time, typically years). Growth bets on the drift of $V(t)$ itself up to a distant T . So the choice is indeed about which time scale carries your signal. The analogy with the renormalization group is loose, though: there is no coarse-graining map and no fixed point here, only two different observables. The solid physics connections are elsewhere: *separation of time scales* (fast noise vs slow fundamental drift, as in the Ornstein–Uhlenbeck model of chapter 18) and the empirical *scaling laws* of returns across time scales (Mandelbrot’s fat tails and multifractals; R. N. Mantegna and H. E. Stanley, *An Introduction to Econophysics*, 2000), where renormalization-like ideas do appear.

21 what is an edge?

ANSWER 1.27: Edge

An *edge* is a persistent advantage that makes your expected return, after costs, positive, and that you can explain: “you need to understand why you should make money. Other people should give you the money” [► 2024 L1-II @ 32:10]. Typical sources:

- better *information* or better processing of public information;
- better *models* (the quant’s edge);
- *speed* or lower *costs*;
- *risk capacity* or *patience*: holding what others must sell, or for longer than they can;
- being paid for providing a service (liquidity, insurance).

If you cannot say who is on the other side and why they lose, you probably have no edge.

Exercise 1.1 (2024 Investment Game (ungraded)). With a hypothetical \$10,000, buy a single publicly traded stock, ETF, currency or bond (to be short, buy an inverse ETF) at the close of Friday 13 September 2024 and sell at the close of Friday 15 November 2024, with one optional switch of the whole balance at the close of 11 October [► 2024 L1-II @ 32:28]. Track in a spreadsheet the number of shares, the daily P&L in \$ and %, the total P&L, the sum G of the daily gains and the sum L of the daily losses (in absolute value), and the ratio $(G - L)/(G + L)$. Repeat the game on any two-month window of your choice.

- Show that $G - L$ is the total P&L and that $(G - L)/(G + L) \in [-1, 1]$.
- Interpret the ratio: which values correspond to a “clean” trend, which to a noisy path with the same final P&L?

MY SOLUTION

- Easy to see: imposta le due disuguaglianze, sono rispettate trivialmente.
- Il rateo è il profitto totale diviso per il valore assoluto di tutte le variazioni, quindi grandi variazioni implicano piccolo rateo e piccole variazioni grande rateo. Inoltre, grande profitto implica grande rateo, piccolo profitto piccolo rateo. Quindi questa è una metrica di quanto profittevole e con bassa volatilità l’investimento è stato.

1.7 Time value of money

1.7.1 Compounding

Invest \$1 at a constant annual interest rate r [► 2024 L1-III @ 1:00].²² With annual compounding (interest earns interest) it grows to $1 + r$ after one year and to $(1 + r)^n$ after n years. If interest is paid and reinvested m times a year, each period earns r/m and

$$1 \mapsto \left(1 + \frac{r}{m}\right)^{mn} \text{ after } n \text{ years.} \quad (1.1)$$

Letting $m \rightarrow \infty$ (continuous compounding) [► 2024 L1-III @ 3:02],

$$\lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^{mt} = e^{rt}, \quad (1.2)$$

since $\ln(1 + r/m)^{mt} = mt \ln(1 + r/m) = mt(r/m + O(m^{-2})) \rightarrow rt$.

Remark 1.4 (Equivalent rates). The same growth can be quoted with different conventions: a rate r_m compounded m times a year and a continuously compounded rate r_c are equivalent iff $(1 + r_m/m)^m = e^{r_c}$, i.e. $r_c = m \ln(1 + r_m/m)$. A rate is meaningless without its compounding convention.

²² what are typical values of r ?

ANSWER 1.28: Typical values of r

A few percent per year, but the range is wide:

- US policy rate (federal funds): 0–0.25% in 2008–2015 and 2020–2022, then 5.25–5.50% from July 2023 to September 2024;
- 10-year US Treasury yield: about 0.5% at its 2020 low, about 5% in October 2023, above 15% in 1981 (when the policy rate was near 20% to fight inflation);
- negative rates: the ECB deposit rate was –0.5% in 2019–2022, and the Bank of Japan’s policy rate was –0.1% in 2016–2024;
- borrowers pay more than governments: mortgages, corporate bonds and credit cards carry a credit spread on top (from 1% to more than 20% a year for credit cards).

HISTORY: Compound interest and the number e

V. Strela credits Jacob Bernoulli with “maybe the first mathematical-finance paper in history” [► 2024 L1-III @ 4:02]. The facts:

- in 1683 Jacob Bernoulli (1655–1705) studied continuously compounded interest and asked for the limit of $(1 + 1/n)^n$ as $n \rightarrow \infty$, showing that it lies between 2 and 3;
- he published it in May 1690 in *Acta Eruditorum*, in the paper “Quaestiones nonnullae de usuris, cum solutione problematis de sorte alearum” (“Some questions on interest, with the solution of a problem on games of chance”). Finance and probability were born together;
- the letter e is Euler’s: it first appears in a letter to Goldbach (1731) and in print in his *Mechanica* (1736).

“First” is generous: Fibonacci’s *Liber Abaci* (1202) already compares cash flows at different dates with present-value calculations (W. N. Goetzmann).

References. J. Bernoulli, *Acta Eruditorum*, May 1690, pp. 219–223; J. J. O’Connor and E. F. Robertson, “The number e ”, MacTutor History of Mathematics (<https://mathshistory.st-andrews.ac.uk/HistTopics/e/>); W. N. Goetzmann, “Fibonacci and the Financial Revolution”, NBER Working Paper 10352 (2004).

23 is there any situation where is not allowed mathematically to use continuous compound? or where we get totally different results and we should be careful which one to use?

ANSWER 1.29: When the compounding convention matters

Mathematically nothing forbids continuous compounding. Every positive growth factor G over a year can be written as e^{r_c} with $r_c = \ln G$. Once rates are converted with the formula above, all conventions give the same prices. Trouble comes from forgetting to convert:

- **the size of the gap grows with r and t :** 5% annual is $r_c = 4.88\%$, a small difference; 50% annual is only $r_c = 40.5\%$. Over 30 years at 5%, $1.05^{30} = 4.32$ but $e^{1.5} = 4.48$, so quoting one rate in the other convention misprices long bonds by several percent;
- **very negative rates:** $1 + r_m/m$ must be positive, so a simple rate $r_m \leq -m$ is meaningless (you would lose more than everything), while $e^{r_c t}$ is always positive;
- **contracts fix the convention:** coupons, money-market deposits and swaps specify compounding and day-count rules (chapter 6), and a price must follow the contract;
- **continuous time:** in stochastic models (Black–Scholes, short-rate models) continuous compounding is the natural choice, because $e^{r t}$ is what $dB = rB dt$ produces.

24 what does it mean buy the claim?

ANSWER 1.30: Claim

A *claim* is a contract that entitles its holder to a payment: here, the right to receive \$1 in one year (a zero-coupon bond, section 1.8). *Buying the claim* means paying its price x today to become the one who will be paid.

1.7.2 Discounting by no-arbitrage

How much should one pay today for the right to receive \$1 in one year [► 2024 L1-III @ 5:03]? Call the price x and consider the strategy: *borrow x today and buy the claim.*²⁴ Today the net cash flow is zero.²⁵ In one year we receive \$1 and repay $x(1 + r)$, so the final profit is

$$\Pi = 1 - x(1 + r) \quad \text{with certainty.}$$

If $\Pi > 0$ we would do this on a huge scale (riskless profit from nothing); if $\Pi < 0$ the counterparty would do the opposite. Absence of arbitrage forces $\Pi = 0$, i.e. $x = 1/(1+r)$ [► 2024 L1-III @ 6:08].

Definition 1.5 (Discount factor). The present value of \$1 received at time t is the *discount factor* Λ_t . With a constant rate,

$$\Lambda_n = \frac{1}{(1+r)^n} \text{ (annual compounding),} \quad \Lambda_t = e^{-rt} \text{ (continuous compounding),}$$

and the present value of a cash flow N at time t is $N\Lambda_t$ [► 2024 L1-III @ 8:09].

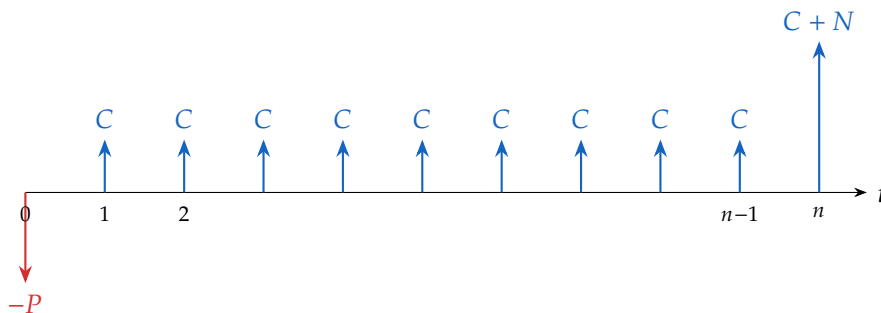
INTUITION: No probabilities needed

The argument never uses probabilities or preferences: only the existence of a riskless strategy whose profit must vanish. The same logic, applied to a portfolio that is riskless only *instantaneously* (option + Δ shares), yields the Black-Scholes equation.

1.8 Bonds

1.8.1 Zero-coupon and coupon bonds

A *zero-coupon* (discount) bond pays the notional N at maturity T and nothing else: it is exactly “the right to receive N at T ” [► 2024 L1-III @ 9:09]. ²⁷ A *coupon bond* pays periodic coupons C and the notional at maturity; US Treasuries pay coupons semi-annually [► 2024 L1-III @ 10:10]. ²⁸ The cash flows from the buyer’s point of view (the buyer lends P) are:



Proposition 1.6 (Bond prices at a flat rate). With annual coupons and a constant annual rate r ,

$$P_{\text{zero}} = \frac{N}{(1+r)^n}, \quad (1.3)$$

$$P_{\text{coupon}} = \sum_{k=1}^n \frac{C}{(1+r)^k} + \frac{N}{(1+r)^n} = C \frac{(1+r)^n - 1}{r(1+r)^n} + \frac{N}{(1+r)^n}. \quad (1.4)$$

Corollary 1.7 (Par bonds). A bond trades at par ($P = N$) iff its coupon rate equals the rate: $C = rN$.

25 what is net cash flow?

ANSWER 1.31: Net cash flow

The *net cash flow* at a date is money in minus money out at that date. Today we receive $+x$ from the loan and pay $-x$ for the claim: net 0, so the strategy needs no capital of our own. In one year: $+1$ from the claim, $-x(1+r)$ to repay the loan.

26 am i right if i say: given X money now, its value in the future is computed via the interest rate, while its value in the past is computed via the discount factor [we consider time as having an intrinsic value - different if we compound or not]. I AM RIGHT! look at:
 $X_{\text{now}} = X_{\text{future}} \cdot e^{-rt}$ and notice you can easily invert it to get
 $X_{\text{future}} = X_{\text{now}} \cdot e^{rt}$. [it is even simpler if you allow time to be negative, then you can use the same formula for both past and future]

ANSWER 1.32: Moving money along the time axis

Yes. With a constant rate, $\Lambda_t = e^{-rt}$ moves a value *back* from time t to today and its inverse $1/\Lambda_t = e^{rt}$ (the growth or compounding factor) moves it *forward*; with signed time the single formula $X_{t_2} = X_{t_1} e^{r(t_2-t_1)}$ does both. Two refinements. First, the value is not intrinsic to time: it is set by the rate the market pays for lending over that period. Second, when the rate depends on maturity (a yield curve, [section 1.8](#)) the factor between two dates is $\Lambda_{t_2}/\Lambda_{t_1}$, not a single exponential, and this ratio is exactly the forward rate of [proposition 1.13](#). Reading every pricing formula as “price now = $\Lambda \times$ payoff later” is the habit to keep: Λ carries values back in time.

27 id say that the contract is an obligation, for the other party to pay you in the future: a future/forward contract?

ANSWER 1.33: Bond vs forward

A bond is a loan, not a forward. The buyer pays the price P *today* and from then on only the issuer has obligations (to pay coupons and notional). In a forward or future nothing is paid today and *both* parties are obliged at T , one to buy and one to sell.

28 what about other countries and eu?

ANSWER 1.34: Coupon frequency elsewhere

Coupon frequency by issuer:

- annual: Germany (Bund), France (OAT), Spain (Bonos), Netherlands (DSL);
- semi-annual: Italy (BTP), United Kingdom (gilts), Japan (JGB), United States (Treasury notes and bonds).

The pricing formulas are the same with n periods and the rate per period ($r/2$ for semi-annual coupons).

1.8.2 Yield

Prices of bonds with different coupons and maturities are not directly comparable. The market quotes instead the *yield* [► 2024 L1-III @ 11:10].

Definition 1.8 (Yield to maturity). The yield y of a bond with market price P is the constant discount rate that reproduces the price: $P = \sum_k CF_k(1+y)^{-t_k}$. For an n -year zero-coupon bond with $N = 1$,

$$y = P^{-1/n} - 1 = \frac{1}{\sqrt[n]{P}} - 1.$$

For coupon bonds the equation is a polynomial in $(1+y)^{-1}$ and is inverted numerically [► 2024 L1-III @ 12:14].

Since every cash flow is positive, $P(y)$ is strictly decreasing and strictly convex in y , so the yield is unique. Price decreases with yield and increases with coupon (fig. 1.2); by corollary 1.7 the yield of a par bond equals its coupon.³³

29 credo che il discount rate sulla parte del coupon debba essere divisa per n . no non e' vero, n e' il numero di anni, e noi calcoliamo il valore al tempo zero se riceviamo soldi al tempo k , dato un certo tasso di interesse.

ANSWER 1.35: Why the coupon is not divided by n

Your second thought is right. Each coupon C is a separate payment at time k and is discounted by its own factor $(1+r)^{-k}$; n only fixes how many coupons there are and when the notional comes back. Nothing is shared out over the n years.

Proof. Each cash flow is discounted with its discount factor. With $v = 1/(1+r)$ the coupon part is the geometric sum $C \sum_{k=1}^n v^k = C v \frac{1-v^n}{1-v} = C \frac{1-(1+r)^{-n}}{r}$, which equals the stated expression. $\square$

30 does par stand for parity?

ANSWER 1.36: Par

Not parity: *par* is the face (nominal) value of the bond, from Latin *par*, "equal". "At par" means $P = N$; above par is a *premium*, below par a *discount*. Prices are quoted per 100 of nominal, so a par bond is quoted at 100.

31 good! in general, when a new term is introduced, explain why it has been given the name it has

ANSWER 1.37: Where the names of this chapter come from

- *Coupon*: French for "piece cut off" (*couper*). Paper bonds carried detachable coupons, one per payment date, which the holder cut off and handed in to be paid; hence "coupon clipping".
- *Principal*: the "chief" sum (Latin *principalis*), as opposed to the interest. *Notional*: the amount on which payments are computed. In a swap it is only a notion and is never exchanged.
- *Maturity*: the date on which the bond "ripens" (Latin *maturus*, "ripe") and the principal is repaid.
- *Yield*: what an investment produces, as a field yields a crop (Old English *gielðan*, "to pay"). The *yield curve* plots it against maturity.
- *Spread*: the gap by which two rates or prices are spread apart (e.g. BTP–Bund, bid–ask).
- *Duration*: literally "how long". It is the present-value-weighted average waiting time for the cash flows, introduced under this name by F. Macaulay (1938). *Convexity*: the price–yield curve is convex, and C measures its curvature.
- *Arbitrage*: from French *arbitrer*, "to judge": the arbitrageur judges the price of the same thing in two markets (historically, bills of exchange in different cities) and trades the difference.
- *Hedge*: like a hedge around a field, a fence that limits exposure. *Delta* and the other *Greeks* are named after the letters used for the sensitivities (Δ for a change in the price).

From now on a new term gets a short note on its name where it first appears.

Proof. Set $C = cN$ in (1.4): $P/N = \frac{c}{r}(1 - (1+r)^{-n}) + (1+r)^{-n} = 1 + (\frac{c}{r} - 1)(1 - (1+r)^{-n})$, which equals 1 iff $c = r$ (for $n \geq 1$). The same identity shows $P > N$ (premium) iff $c > r$ and $P < N$ (discount) iff $c < r$. $\square$

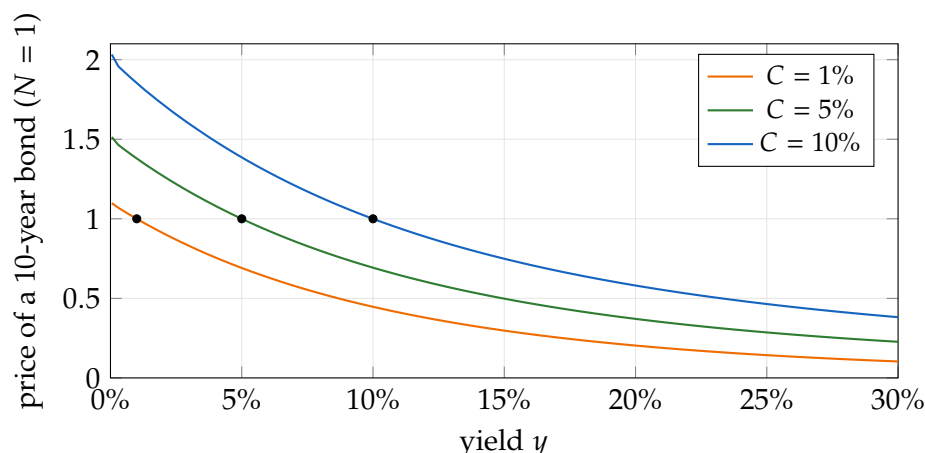


Figure 1.2: Price–yield curves of 10-year annual-coupon bonds, from (1.4) (re-drawn from slide 22 of lec01_1). At $y = 0$ the price is $1 + 10C$; the dots mark the par points $y = C$.

1.8.3 The yield curve

In reality bonds of different maturities have different yields: the *term structure of interest rates*, or *yield curve* [► 2024 L1-III @ 14:19]. Normally it slopes upward (lenders want more yield for longer commitments); when short yields exceed long yields the curve is *inverted*. Inversions preceded the 2008 and 2020 recessions, which is why an inverted curve is widely (though not infallibly) read as a recession signal [► 2024 L1-III @ 16:24].³⁴ The short end is anchored by the central bank’s policy rate [► 2024 L1-II @ 14:41]; the long end reflects expectations, risk premia, supply and demand.

32 C is the coupon, what is F_k ? e’ giusta la formula di P? perche’ non usiamo continuous framework, sembra molto piu semplice-pulito? coupon non funzionerebbe! aggiungi valori tipici di yield to maturity (e sopraa anche di coupon, prezzo di un bond, discount rate)

ANSWER 1.38: What is CF_k

CF_k (one symbol, “cash flow”) is the k -th payment: C at the coupon dates and $C + N$ at maturity.

ANSWER 1.39: Is the formula for P right?

The formula is right: it is (1.4) with a rate y that is *solved for* instead of given.

ANSWER 1.40: Why not the continuous framework? Would coupons break it?

The continuous framework works equally well with coupons. The payments stay discrete, only the discounting changes: $P = \sum_k CF_k e^{-y_c t_k}$ with $y_c = \ln(1 + y)$ (for $y = 5\%$, $y_c = 4.88\%$). Both describe the same price. Markets quote the annual (or semi-annual) compounded yield by convention; theory (chapters 18 and 24) prefers y_c because exponentials compose and differentiate cleanly.

ANSWER 1.41: Typical values

Official data, September 2026:

- **yield to maturity:** US 4.2% (3 months) to 5.5% (30 years), 25 Sep 2026; Germany 10 y: 3.4%, Italy 10 y: 4.0% (August 2026 averages);
- **coupon rate:** close to the yield at issue: 3–5% for new bonds, 0–1.5% for 2020–21 issues;
- **price:** quoted per 100 of nominal, near 100 at issue; e.g. 5 years left, yield 4.9%: coupon 1.5% $\rightarrow$ 85.2, coupon 4% $\rightarrow$ 96.1, coupon 5.5% $\rightarrow$ 102.6;
- **discount factor:** $\Lambda_{10} \approx 1.0517^{-10} = 0.60$ at the 10-year US yield.

Sources: US Treasury Daily Par Yield Curve Rates; Deutsche Bundesbank (BBSIS); Banca d’Italia (BTP benchmark gross yields, BMK0100).

33 l’idea e’ che un bond verra’ pagato una certa quantita X il giorno Y, e magari restituisce coupon C. Il problema e’ trovare quale sia il prezzo adesso. [per me yield=interest rate=discount rate]

ANSWER 1.42: Yield, interest rate, discount rate

Your reading of the problem is exactly right: known future payments, unknown price today. The three words coincide only when the curve is flat. In general each date has its own rate (the yield curve below). The *discount rate* is the rate used for one specific date, and the *yield* is the single constant rate that gives the right total price, a cash-flow-weighted average of the curve’s rates. Two bonds of the same issuer with the same maturity but different coupons can therefore have slightly different yields.

34 what is the definition of recession?

ANSWER 1.43: Recession

A common rule of thumb is two consecutive quarters of falling real GDP. In the US the official dating is done by the NBER’s Business Cycle Dating Committee: a significant decline in economic activity that is spread across the economy and lasts more than a few months (it looks at employment, income, spending and production, not GDP alone). The 2020 recession, for example, lasted only two months (February–April 2020) but was dated as one because it was so deep and widespread.

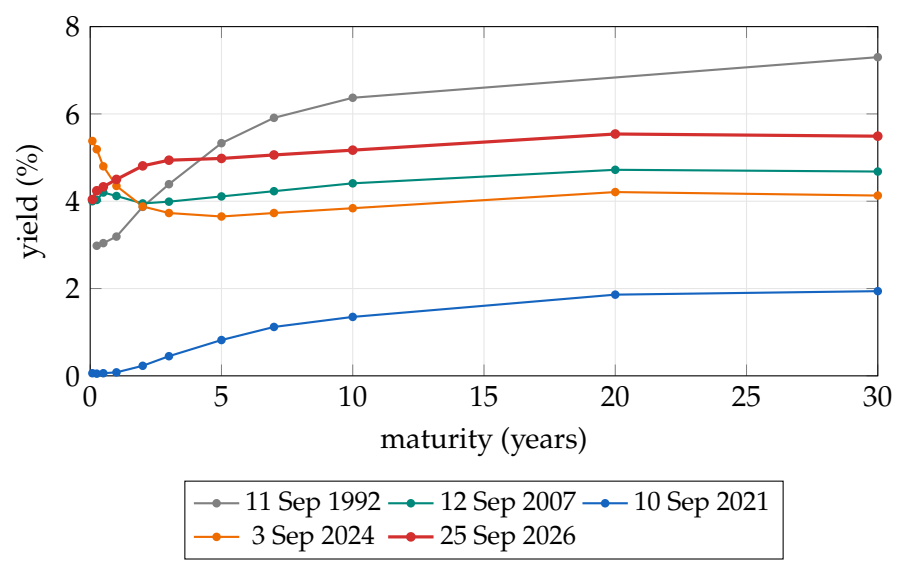
35 non capisco, se la maturity e’ zero allora yield dovrebbe essere zero, no? non e’ veramente a zero! perche’ un prestito per un anno dovrebbe avere tasso di interesse piu alto che di due? cosi’ incentivi a comprare bond a scadenza breve, perche’?

ANSWER 1.44: Yield at maturity zero

Yields are *annualised*. A loan for one day earns almost nothing in euros or dollars, but its rate per year does not go to zero: as maturity $\rightarrow 0$ the yield tends to the overnight rate, which the central bank steers with its policy rate (in fig. 1.5 the 2-year yield follows the Fed funds rate closely). So the curve starts at the policy rate, not at zero.

ANSWER 1.45: Why would 1 year pay more than 2? Does it push people to short bonds?

An inverted curve does not reward short bonds in a way anyone can exploit. A long yield is roughly the average of the short rates expected over its life, plus a premium for tying money up. If the market expects the central bank to *cut* rates (typically because a slowdown is coming), rolling over short bonds is expected to earn the high short rate only for a while and then less. The 2-year yield therefore sits below today’s 1-year yield, and investors are indifferent. That is also why inversions are read as recession signals: they encode expected rate cuts.



DONE
36 vorrei aggiungere yield curves of italian bonds

Figure 1.3: US Treasury par yield curves (official Daily Treasury Par Yield Curve Rates, treasury.gov; the first four dates are those of slide 23 of lec01_1). 1992, 2021 and 2026: normal upward slope, at very different levels; 2007 and 2024: inverted short end.

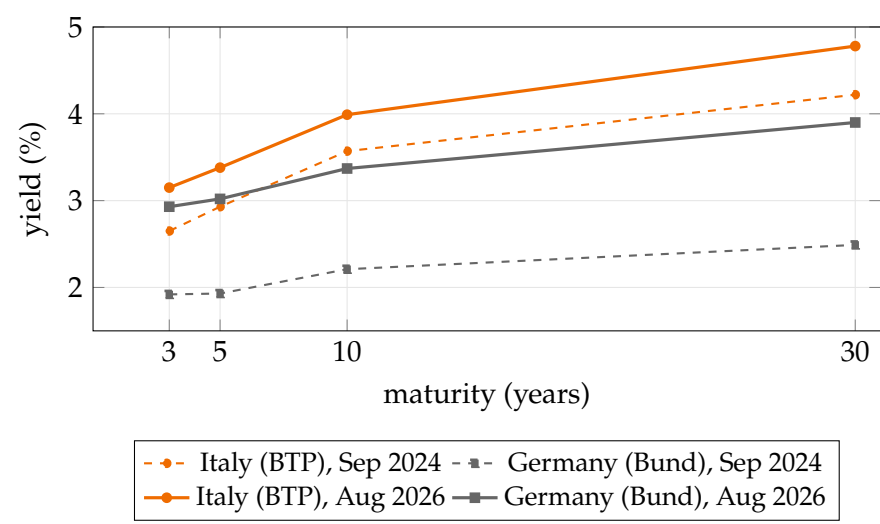


Figure 1.4: Italian and German government yield curves, monthly averages at 3, 5, 10 and 30 years. Sources: Banca d'Italia, gross yields of benchmark BTPs (table BMK0100); Deutsche Bundesbank, yield curve of listed Federal securities (BBSIS). Both curves slope upward. The gap between them is the BTP-Bund spread, the extra yield investors ask to hold Italian rather than German debt: about 135 bp at 10 years in September 2024 and 60 bp in August 2026.

DONE
37 fai il check con dati ufficiali di queste yield curves e aggiungi quella piu aggiornata di oggi

DONE
38 i found this nice picture to see correlation with fed interest rate (?)
<https://www.gwkinvest.com/wp-content/uploads/Figure1-GP-June26.svg>

39 what happens to economy after rate cut? people are not incentivised to invest, right?

ANSWER 1.46: What a rate cut does
The other way round: a cut *encourages* investment. Lower rates make borrowing cheaper, so firms invest in projects that did not pay at the old rate, households take mortgages and consumer loans, and existing debt costs less. What is rewarded less is *saving* in deposits and short bonds. Money therefore moves towards spending and riskier assets: stock and bond prices rise, because future cash flows are discounted at a lower rate. The currency tends to weaken. This is why central banks cut in slowdowns and raise rates against inflation. The effect comes with a lag of many months, and if the slowdown is severe, cheap credit may not be enough to restart spending.

40 la figura fig:fedrates e molto bella, non esisiste qualcosa di simile per italia germania? mi sembra abbia lo stesso contenuto se non di piu, mi sbaglio?

ANSWER 1.47: An Italian-German version of fig. 1.5
The two figures contain different things. Figure 1.4 is a *snapshot* of the curve on two dates: yield against maturity. Figure 1.5 is a *time series*: a few maturities over 36 years, next to the central-bank rate. The euro analogue is fig. 1.6. It uses the ECB deposit rate, the rate that anchors overnight euro rates since the banking system holds ample reserves, and the 10-year yields of BTPs and Bunds. Three things stand out:

- the 2011–12 sovereign-debt crisis, when the BTP-Bund spread peaked at about 520 bp (November 2011);
- negative deposit rates from June 2014 to July 2022, with the Bund yield below zero for long stretches (2016, 2019–21);
- the 2022–23 hiking cycle: from 2023 to early 2025 the deposit rate stayed above the 10-year Bund yield, an inverted curve.

41 it doesn't have to, right? it is just a choice $P=N$ imposed to 1

ANSWER 1.48: Par price 1
Yes, this is a normalisation: at par $P = N$, and we measure money in units of the nominal, so $N = 1$ and $P = 1$. With $N = 100$ everything is multiplied by 100 and cancels.

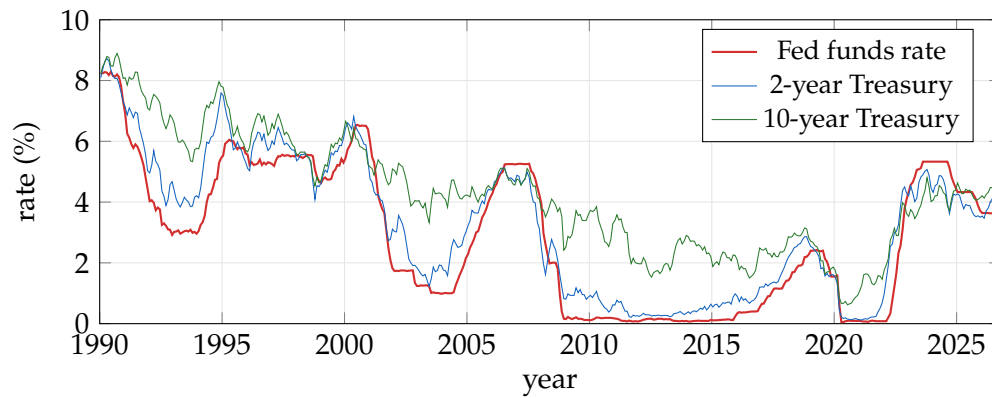


Figure 1.5: Fed funds rate and 2- and 10-year Treasury yields, monthly averages 1990–September 2026 (FRED series DFF, DGS2, DGS10; re-drawn after the chart linked in the margin). The 2-year yield tracks the *expected* path of the policy rate and moves before it. The 10-year yield moves much less. When the Fed funds rate is above the 10-year yield (e.g. 2000, 2006–07, 2019, 2023–24) the curve is inverted.

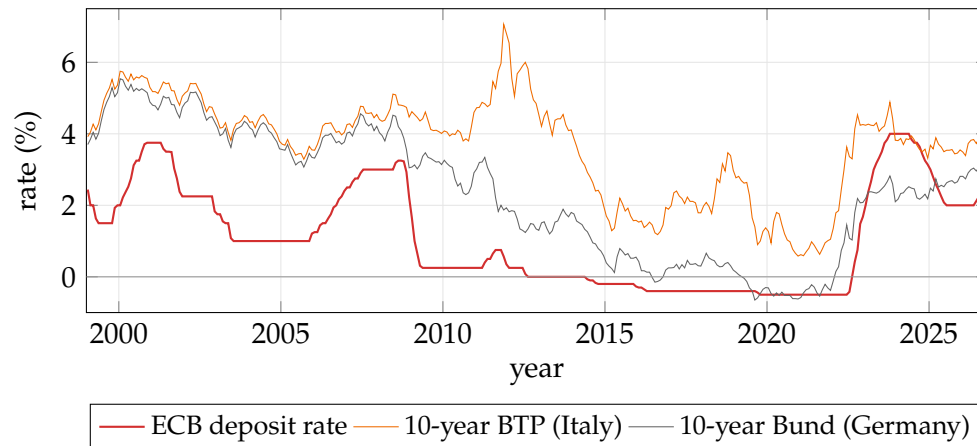


Figure 1.6: ECB deposit facility rate and 10-year Italian and German government yields, monthly averages January 1999–August 2026. Source: ECB Data Portal, series FM.D.U2.EUR.4F.KR.DFR.LEV and IRS.M.{IT,DE}.L.L40.CI.0000.EUR.N.Z (long-term interest rates for convergence purposes). The gap between the two yields is the BTP–Bund spread.

Bootstrapping. “Discount factors for all times can be bootstrapped from the yield curve” (slide 23). Concretely, suppose we observe par yields $c_1, c_2, \dots$ of annual-coupon bonds with maturities 1, 2, $\dots$ years. A par bond with maturity n has price 1,⁴¹ so

$$1 = c_n \sum_{k=1}^n \Lambda_k + \Lambda_n \quad \Rightarrow \quad \Lambda_n = \frac{1 - c_n \sum_{k=1}^{n-1} \Lambda_k}{1 + c_n},$$

⁴² which determines $\Lambda_1, \Lambda_2, \dots$ recursively. Real curve construction (many instrument types, interpolation, day-count conventions) is the subject of [chapter 6](#).⁴³

1.8.4 Duration and convexity

⁴² non capisco da dove viene questa formula con i c_k , al loro posto dovrebbe esserci C il coupon, giusto?

ANSWER 1.49: Where the c_n come from

There is not one bond but a *family*, one for each maturity $n = 1, 2, \dots$, and each has its own coupon: c_n is the coupon rate of the n -year bond, i.e. $C = c_n N = c_n$ for that bond. Since the bond is at par, its coupon equals its yield ([corollary 1.7](#)), hence “par yields”. The discount factors, on the other hand, are the *same* for every bond: \$1 paid in year k is worth Λ_k whoever pays it. Pricing the n -year bond with them,

$$1 = \underbrace{c_n \Lambda_1 + \dots + c_n \Lambda_n}_{\text{coupons}} + \underbrace{1 \cdot \Lambda_n}_{\text{nominal}} = c_n \sum_{k=1}^{n-1} \Lambda_k + (1 + c_n) \Lambda_n,$$

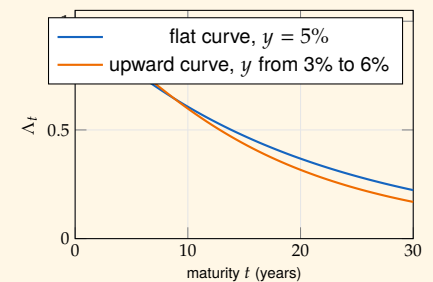
and solving for Λ_n gives the formula. The 1-year bond gives $\Lambda_1 = 1/(1 + c_1)$. The 2-year bond contains Λ_1 (known) and Λ_2 (unknown), and so on: each new maturity adds one unknown.

⁴³ i need some more intuition on what is the discount factor and what is the yield, yield curve

ANSWER 1.50: Discount factor, yield, yield curve: one picture

- The *discount factor* Λ_t is a price: what \$1 delivered at time t costs today. It starts at $\Lambda_0 = 1$ and decreases, because waiting costs money.
- The *zero yield* $y(t)$ is the same information written as a rate. It is defined by $\Lambda_t = e^{-y(t)t}$, i.e. $y(t) = -\ln \Lambda_t / t$, the average speed at which Λ decays up to t .
- The *yield curve* is $t \mapsto y(t)$. A flat curve means Λ_t is a pure exponential; an upward curve means Λ_t falls faster and faster at long horizons.

Prices (Λ) are what you use to compute; rates (y) are what you use to compare different maturities, the way a physicist compares decay constants rather than surviving fractions.



A bondholder is exposed to yield changes, and the dependence is non-linear [► 2024 L1-III @ 19:29]. The sensitivities are derivatives, conventionally scaled by the price.⁴⁴

Definition 1.9 (Duration and convexity).

$$D = -\frac{1}{P} \frac{dP}{dy}, \quad C = \frac{1}{P} \frac{d^2P}{dy^2},$$

so that, by Taylor expansion,

$$\frac{\Delta P}{P} \approx -D \Delta y + \frac{1}{2} C (\Delta y)^2. \quad (1.5)$$

Proposition 1.10.

(i) For cash flows CF_k at times t_k discounted with annual compounding,

$$D = \frac{1}{1+y} \underbrace{\sum_k t_k \frac{CF_k(1+y)^{-t_k}}{P}}_{D_{\text{Mac}}},$$

where the Macaulay duration D_{Mac} is the present-value-weighted average time of the cash flows; D is also called modified duration.

(ii) Zero-coupon bond: $D_Z = \frac{T}{1+y}$, $C_Z = \frac{T(T+1)}{(1+y)^2}$.

(iii) “Continuous” par bond (coupon paid continuously at rate $c = y$, continuous discounting, $N = 1$): $D_p = \frac{1 - e^{-yT}}{y}$.

INTUITION: Duration as a centre of mass

Macaulay duration is the “centre of mass” of the cash flows in time, with present values as masses. A zero-coupon bond has all its mass at T ; coupons pull the centre of mass earlier, so a coupon bond has $D_{\text{Mac}} < T$. For the continuous par bond $D_p = (1 - e^{-yT})/y < T$ and $D_p \rightarrow 1/y$ as $T \rightarrow \infty$: even a perpetual bond has finite duration because distant cash flows weigh almost nothing.

IN FINANCE: Why convexity is valuable

Because $C > 0$, the price rises more when yields fall than it drops when yields rise by the same amount. Two portfolios with the same price and duration but different convexity are therefore not equivalent: the more convex one gains from large rate moves in either direction. Trading desks hedge duration (first order, like the delta $\Delta = \partial V / \partial S$ of an option) and monitor convexity (second order, like the gamma $\Gamma = \partial^2 V / \partial S^2$). For a bond the “underlying” is the yield: $\partial P / \partial y = -DP$ plays the role of Δ and $\partial^2 P / \partial y^2 = CP$ that of Γ .

44 so they are dimensionless?

ANSWER 1.51: Units of duration and convexity

No. y is a rate per year, so $D = -P'/P$ has units of years and $C = P''/P$ of years² (the Macaulay duration is literally an average time). What is dimensionless is the product: $D \Delta y$ and $C (\Delta y)^2$, and hence $\Delta P/P$. Example: $D = 4.3$ years and $\Delta y = +1 \text{ bp} = 0.0001$ per year give $\Delta P/P \approx -0.043\%$.

45 the convention for the signs is because we expect D and C to be positive. look at fig 1.2

ANSWER 1.52: Signs

Correct. In fig. 1.2 every curve decreases ($P' < 0$) and bends upward ($P'' > 0$), so the minus sign in D and the plus sign in C make both numbers positive for ordinary bonds.

46 COURSE NOTE: Erratum in the slides

On slide 24 of lec01_1 convexity is written with the same letter and sign as duration, “ $D = -\frac{1}{P} \frac{d^2P}{dy^2}$ ”. The standard definition is $C = +\frac{1}{P} \frac{d^2P}{dy^2}$, which is positive for bonds with positive cash flows.

Proof. (i) Differentiate $P = \sum_k CF_k(1+y)^{-t_k}$: $dP/dy = -\sum_k t_k CF_k(1+y)^{-t_k-1}$. (ii) For $P = (1+y)^{-T}$: $P' = -T(1+y)^{-T-1}$ and $P'' = T(T+1)(1+y)^{-T-2}$; divide by P . (iii) With coupon rate c fixed, $P(y) = \int_0^T c e^{-ys} ds + e^{-yT} = c \frac{1-e^{-yT}}{y} + e^{-yT}$. Then

$$P'(y) = c \frac{yTe^{-yT} - (1 - e^{-yT})}{y^2} - Te^{-yT},$$

and at $c = y$ (where $P = 1$) the terms $\pm Te^{-yT}$ cancel, leaving $P'(y) = -(1 - e^{-yT})/y$. □

Example 1.11 (How good is the quadratic approximation?). A 10-year zero at $y = 4\%$: $P = 1.04^{-10} = 0.6756$, $D = 10/1.04 = 9.615$, $C = 110/1.04^2 = 101.7$. If the yield rises by $\Delta y = 1\%$: first order -9.62% ; adding convexity $+\frac{1}{2} \cdot 101.7 \cdot 0.0001 = +0.01\%$ gives -9.11% ; the exact change is $(1.04/1.05)^{10} - 1 = -9.13\%$.

47 we are still considering the yield to be fixed along the duration of the bond maturity, right?

ANSWER 1.53: Flat yield, parallel shifts

Right. One yield y discounts all the cash flows (a flat curve), and Δy moves it by the same amount at every maturity (a parallel shift). Real curves also steepen and twist. Desks therefore also use *key-rate durations*, the sensitivity to the yield at each single maturity, and the principal components of curve moves (level, slope, curvature, chapter 7).

Exercise 1.2 (not from the course). A 5-year bond pays annual coupons of 6% on $N = 100$.

- Price it at $y = 5\%$ and $y = 7\%$.
- Compute its Macaulay and modified duration at $y = 5\%$ and check them against the price change for $\Delta y = \pm 1$ bp.

MY SOLUTION

(a)

$$P_{y=0.05} = 6 \frac{1 - (1 + 0.05)^{-5}}{0.05} + \frac{100}{(1 + 0.05)^5} = 104.33,$$

$$P_{y=0.07} = 6 \frac{1 - (1 + 0.07)^{-5}}{0.07} + \frac{100}{(1 + 0.07)^5} = 95.90.$$

(b)

$$D_{\text{Mac}} = \frac{1}{104.33} (1 \cdot 6 \cdot 1.05^{-1} + 2 \cdot 6 \cdot 1.05^{-2} + 3 \cdot 6 \cdot 1.05^{-3} + 4 \cdot 6 \cdot 1.05^{-4} + 5 \cdot (6 + 100) \cdot 1.05^{-5}) = 4.48 \text{ years},$$

$$D_{\text{mod}} = 4.48/1.05 = 4.27 \text{ years},$$

$$P_{\text{Taylor}} = P - P D \Delta y = P \cdot (1 - D \Delta y),$$

$$P^{\pm 0.01\%} = 104.33 (1 - 4.27 \cdot (\pm 0.0001)) = 104.285 (+1 \text{ bp}) \text{ \& } 104.375 (-1 \text{ bp}),$$

$$P_{\text{exact}} \approx 104.285 (+1 \text{ bp}) \text{ \& } 104.375 (-1 \text{ bp}).$$

Exercise 1.3 (not from the course). Compute the convexity of the continuous par bond of [proposition 1.10\(iii\)](#) and its limit as $T \rightarrow \infty$.

MY SOLUTION

$$C = \frac{1}{P} \frac{d^2 P}{dy^2} = -\frac{dD}{dy} = \frac{e^{-yT}(Ty + 1) - 1}{y^2} \rightarrow_{T \rightarrow \infty} -\frac{1}{y^2}$$

48 first appearance of the Greeks, right? use the symbols Δ , Γ , etc.

ANSWER 1.54: The Greeks

Yes, this is their first appearance, as an analogy. The Greeks (Δ , Γ , $\Theta = \partial V / \partial t$, Ψ "vega" $= \partial V / \partial \sigma$, $\rho = \partial V / \partial r$) are defined properly with the Black-Scholes formula in [chapter 15](#). A conceptual map of this chapter is at the end, before the key formulas.

SEEN

49 bp stands for basis point and it is 0.01%

FEEDBACK

All correct. Exact -1 bp price: 104.374; at 1 bp convexity is negligible ($\sim 10^{-7} P$).

FEEDBACK

$C < 0$ is impossible here. $C = D^2 - dD/dy$, not $-dD/dy$; and differentiate $P = c(1 - e^{-yT})/y + e^{-yT}$ at fixed c , then set $c = y$. Result: $C = 2[1 - e^{-yT}(1 + yT)]/y^2 \rightarrow 2/y^2$.

1.8.5 Forward rates

Definition 1.12 (Forward rate agreement). A *forward rate agreement* (FRA) fixes today the interest F_{tT} to be paid at time T on a loan of N that starts at a future date $t < T$ (slide 25): the lender pays N at t and receives $N(1 + F_{tT})$ at T .

Proposition 1.13. In the absence of arbitrage, $F_{tT} = \frac{\Lambda_t}{\Lambda_T} - 1$.

Remark 1.14. F_{tT} is the interest over the whole period $[t, T]$; the annualised simple rate is $F_{tT}/(T - t)$. In continuous time one defines the *instantaneous forward rate*

DONE

50 reminder: price now = Lambda payoff at maturity. Lambda serve a trasportare il valore lungo il tempo - nel passato

Proof. Entering an FRA costs nothing today, so the present value of its cash flows must be zero: $-N\Lambda_t + N(1 + F_{tT})\Lambda_T = 0$. Equivalently (the form on the slide), $1 = \frac{1}{\Lambda_t}(1 + F_{tT})\Lambda_T$. If F_{tT}

$f(0, T) = -\partial_T \ln \Lambda_T$, so that $\Lambda_T = \exp(-\int_0^T f(0, s) ds)$: the continuously compounded zero yield $-\frac{1}{T} \ln \Lambda_T$ is the *average* of the forward rates up to T . Modelling the dynamics of the whole curve $f(t, T)$ is the HJM framework ([chapter 24](#)).

Exercise 1.4 (not from the course). Three bonds with maturities 1, 2, 3 years all trade at par (price 1 per unit of nominal). The bond of maturity n pays the *same* annual coupon c_n every year (not a coupon that changes over time), with $c_1 = 4\%$, $c_2 = 4.5\%$, $c_3 = 5\%$; these c_n are the par yields. At par $P/N = 1$ for each bond, and the discount factors Λ_k are common to all three bonds. Bootstrap $\Lambda_1, \Lambda_2, \Lambda_3$, the zero yields and the one-year forward rates $F_{1,2}, F_{2,3}$. Why is the zero curve above the par curve when the curve slopes upward?

MY SOLUTION

Recursively,

$$1. \Lambda_{c_1} = \frac{1}{1 + c_1} = 0.9615$$

$$2. \Lambda_{c_2}^1 = \frac{1}{1 + c_2} = 0.9569; \Lambda_{c_2}^2 = \frac{1 - c_2 \cdot \Lambda_{c_2}^1}{1 + c_2} = 0.9157$$

$$3. \Lambda_{c_3}^1 = \frac{1}{1 + c_3} = 0.9524; \Lambda_{c_3}^2 = \frac{1 - c_3 \cdot \Lambda_{c_3}^1}{1 + c_3} = 0.9070; \Lambda_{c_3}^3 = \frac{1 - c_3 \cdot (\Lambda_{c_3}^1 + \Lambda_{c_3}^2)}{1 + c_3} = 0.8638$$

Forward rate:

$$F_{12}^{c_2} = \frac{\Lambda_{c_2}^1}{\Lambda_{c_2}^2} - 1 = \dots;$$

Exercise 1.5 (not from the course). Complete the proof of [proposition 1.13](#): if the market quotes an FRA rate $F > \Lambda_t / \Lambda_T - 1$, write down an explicit zero-cost strategy with zero-coupon bonds and the FRA that has a positive payoff at T .

MY SOLUTION

I borrow N from a commercial bank, given to me at a r interest rate, I lend N with the F bigger that it should, when I am payed back I repay N plus interests to the bank and I am left with money.

were larger, one could lock in a riskless profit by borrowing $N\Lambda_T(1+F_{1T})$ until T ... ([Exercise 1.5](#)). $\square$

51 it feels that f plays the role of r , the interest rate. and Λ is the corresponding of the compound rate, right?

ANSWER 1.55: f vs r , Λ vs compounding

Yes. $f(0, s)$ is the interest rate agreed today for the infinitesimal period $[s, s + ds]$. With a constant rate $f(0, s) = r$ for every s and $\Lambda_T = e^{-rT}$, so f is "allowed to depend on the future date". Λ_T is the *inverse* of the compounding factor: $1/\Lambda_T = \exp(\int_0^T f(0, s) ds)$ is what \$1 grows to by T if you lock in all the forward rates today (continuous compounding at a time-varying rate).

DONE

52 non capisco se il coupon c è variabile nel tempo, nel senso che il primo anno riceviamo 4% di N il nominale, il secondo 4.5, il terzo 5.

53 come faccio a calcolare lo zero yield per i tre bonds senza il rapporto P/N ? se la premessa è che sono par bonds non capisco. L'esercizio è scritto male, non è chiaro.

ANSWER 1.56: P/N is known

You are right that the statement was unclear; it now says so explicitly. "At par" means $P = N$, so $P/N = 1$ for all three bonds. The only other data are the coupons c_n , and the discount factors Λ_k are common to the three bonds. The zero yield of maturity n is then $y_n = \Lambda_n^{-1/n} - 1$ (see the feedback below).

FEEDBACK

Recursion right, but Λ_k is shared by all bonds: reuse $\Lambda_1 = 0.9615$, don't recompute it per bond. $\Lambda_2 = 0.9155$, $\Lambda_3 = 0.8630$; zero yields 4.00, 4.51, 5.03%; $F_{12} = 5.03\%$, $F_{23} = 6.09\%$.

FEEDBACK

Right idea, but a future bank rate is unknown: lock funding today. Buy 1 zero to t , sell Λ_t / Λ_T zeros to T (cost 0); lend the \$1 at t via the FRA. Profit at T : $1 + F - \Lambda_t / \Lambda_T > 0$.

1.9 Working with data: R and RStudio

CASE STUDY: FM_Intro1.Rmd — collecting and plotting financial time series

Goal. Show how easily financial data can be downloaded and displayed [[► 2024 L1-I @ 10:22](#)]. **Tools.** R packages tidyquant (downloads), tidyverse (data handling), fpp2 (forecasting); later lectures use quantmod, xts/zoo (time series), car (regression), rugarch (GARCH), fxregime (regimes). **Data (2011–2024).** From Yahoo Finance: S&P 500 and VIX indices, stocks NVDA, GE, AAPL, GOOG, AMZN, XOM, GME, AMC, the ARKK and XLF ETFs, BTC-USD. From FRED (St. Louis Fed, about

80,000 series): Treasury constant-maturity yields 3m, 1y, 5y, 10y (DGS3M0...DGS10), Moody's Aaa/Baa corporate yields (DAAA, DBAA), WTI crude oil (DCOILWTIC0), Coinbase Bitcoin. **Steps.** Download, keep (symbol, date, price), drop missing values, merge, plot each series, save to data_fm_intro1.Rdata for later scripts.

What to look at [► 2024 L1-I @ 12:25]: the VIX ("fear gauge", the market's implied volatility of the S&P 500) spikes when the index drops, a relationship formalised by option pricing (chapters 13 and 15); NVIDIA and Bitcoin raise the question of detecting bubbles; WTI goes negative in April 2020; the Baa–Treasury spread is the credit spread modelled in chapter 5.

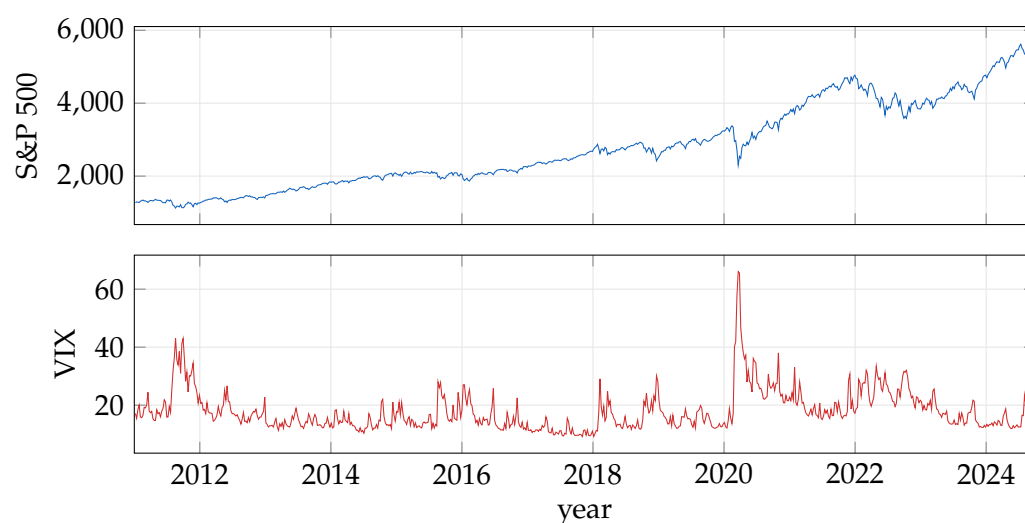


Figure 1.7: S&P 500 index and VIX, weekly closes 2011–2024, re-drawn from the data of FM_Intro1.Rmd. Every VIX spike (2011, 2015, 2018, 2020, 2022) coincides with a drop of the index; the record daily close, 82.7, was on 16 March 2020.

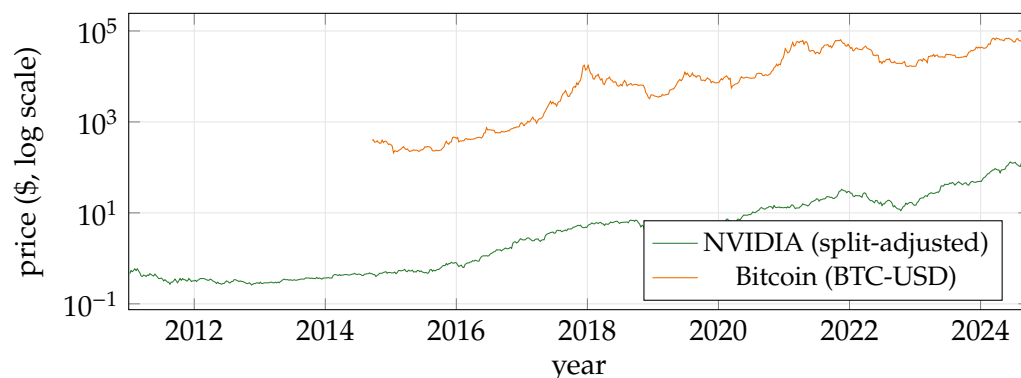


Figure 1.8: NVIDIA and Bitcoin, weekly closes, from the data of FM_Intro1.Rmd. On a log scale a straight line is constant exponential growth: NVIDIA rose about 500-fold in 2012–2024, the kind of run that raises the bubble question [► 2024 L1-I @ 14:36].

The notebook runs on RStudio Cloud (<https://posit.cloud>, free account, nothing to install) or on desktop RStudio (faster, works offline); lec01_2 gives step-by-step instructions, summarised in chapter B.

1.10 Two examples of quant work (2013)

V. Strela showed two projects done by a student intern at Morgan Stanley [► 2013 L1 @ 54:30].

Noisy delta. Prices of complex derivatives are often computed by Monte Carlo, so the pricing function $\hat{f}(x)$ is only known up to statistical noise of standard deviation $\text{sd}(x)$. Delta is computed by a centred difference $\hat{\Delta}_\varepsilon = [\hat{f}(x + \varepsilon) - \hat{f}(x - \varepsilon)]/(2\varepsilon)$. A small shift ε amplifies the noise, a large one introduces discretisation bias. The project chose ε to minimise the mean squared error $\mathbb{E}[|f'(x) - \hat{\Delta}_\varepsilon|^2]$ and found a rule of the form $\varepsilon_{\text{opt}} \propto (\text{sd}(x)/(d + |\Gamma(x)|))^{1/3}$, with Γ the option's gamma and c, d tuning constants (slide 19 of 1ecnote1); the resulting delta is visibly smoother than with a constant shift.

INTUITION: Where the cube root comes from

Taylor-expanding, the centred difference has bias $\frac{\varepsilon^2}{6} f'''(x)$; with independent noise at the two points its variance is $2 \text{sd}^2/(2\varepsilon)^2 = \text{sd}^2/(2\varepsilon^2)$. Hence

$$\text{MSE}(\varepsilon) \approx \frac{\varepsilon^4 f'''(x)^2}{36} + \frac{\text{sd}^2}{2\varepsilon^2},$$

$$\frac{d \text{MSE}}{d\varepsilon} = 0 \iff \varepsilon^6 = \frac{9 \text{sd}^2}{f'''(x)^2} \iff \varepsilon_{\text{opt}} = \left(\frac{3 \text{sd}}{|f'''(x)|} \right)^{1/3}.$$

The same bias–variance trade-off appears in numerical differentiation of noisy experimental data. The course's formula uses gamma as a proxy for the local curvature scale and regularises it with d ; the exact derivation is not in the material.

Kalman filter for FX quotes. Five brokers quoted forward points for USD/RUB at random, non-uniform times; the aim was to combine these noisy observations into a better estimate of the true price with a Kalman filter [► 2013 L1 @ 57:41]. State-space models and the Kalman filter are part of time-series analysis (chapter 9).

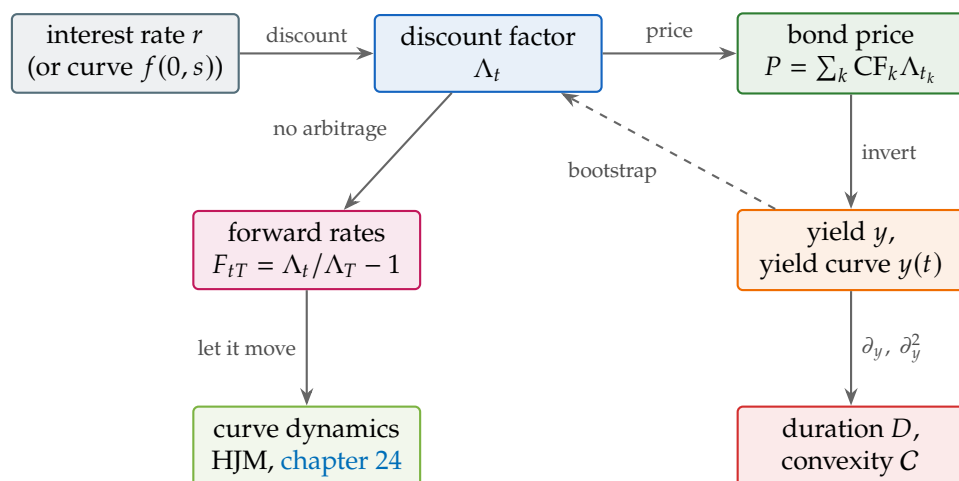


Figure 1.9: Conceptual map of the chapter: one object, the discount factor, and what is derived from it.

KEY FORMULAS

Compounding:	$(1 + r)^n, \quad (1 + r/m)^{mn}, \quad e^{rt}$
Discount factor:	$\Lambda_n = (1 + r)^{-n}, \quad \Lambda_t = e^{-rt}$
Coupon bond:	$P = C \frac{1-(1+r)^{-n}}{r} + N(1 + r)^{-n}; \quad P = N \iff C = rN$
Zero yield:	$y = P^{-1/n} - 1$
Bootstrapping:	$\Lambda_n = (1 - c_n \sum_{k < n} \Lambda_k)/(1 + c_n)$
Duration, convexity:	$D = -P'/P, \quad C = P''/P, \quad \Delta P/P \approx -D\Delta y + \frac{1}{2}C\Delta y^2$
Zero-coupon:	$D_Z = T/(1 + y), \quad C_Z = T(T + 1)/(1 + y)^2$
Continuous par bond:	$D_p = (1 - e^{-yT})/y$
Forward rate:	$F_{tT} = \Lambda_t/\Lambda_T - 1$
Alpha, beta:	$R_a = \alpha + \beta R_b + \varepsilon$
Units:	r, y, c, f [year ⁻¹]; $t, T, n, D, D_{\text{Mac}}$ [years]; C [years ²] $\Lambda, F_{tT}, P/N, \Delta P/P$ [dimensionless] $1 \text{ bp} = 0.01\% = 10^{-4} \text{ year}^{-1}$

Exercises

No problem set covers this chapter; [Exercises 1.1](#) and [1.6](#) are the course’s own activities, the others are added for practice (marked *not from the course*). Standalone exercises follow the section they practise; the exercise below is the integrative one.

Exercise 1.6 (2013, optional homework). Build your own financial glossary: list every term of this chapter you could not define precisely and look it up [► 2013 L1 @ 53:27](#).

Chapter 2

Linear Algebra and One-Period Financial Models

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L2	P. Kempthorne	Vectors as portfolios, P&L, arbitrage; matrices; stochastic matrices and Markov chains; single-period market model, completeness, pricing measure; linear systems; eigenvalues and diagonalisation.
2024 L4 (first 28 min)	P. Kempthorne	Diagonalisation and state equations, symmetric matrices, SVD, Perron–Frobenius; equal-weighted portfolio case study and rebalancing.
2013 L2	C. Lee	Eigenvalues, symmetric matrices (proofs), SVD with a worked example, Perron–Frobenius and its link to the Ross recovery theorem.

Files. Slides `mit18_642_f24_lec02_1.pdf`; lecture note *One-Period Financial Models* `mit18_642_f24_lec02_2.pdf`; `mit18_642_f24_lec02_3.pdf` with `mit18_642_f24_equal_weighted_portfolios.zip` (`Script_SP500.Rmd`, `Equal_Weighted_Portfolios.Rmd`, `SP500_data_subset.RData`); 2013 notes `MIT18_S096F13_lecnote2.pdf`. **Problem sets.** 2024 PS1 (all problems); 2013 PS1 (all problems). **Not recorded.** 2024 L3 and 2013 L4, see [section 2.10](#).

Motivation

Linear algebra is the language in which portfolios, scenarios and covariance structures are written. Three ideas of this chapter recur for the rest of the course:

- a portfolio is a vector and its value is a dot product, so P&L is linear in holdings;
- in a market with finitely many scenarios, *no arbitrage* is equivalent to the existence of positive *state prices*, and derivative prices are discounted expectations under a “risk-neutral” measure ([section 2.4](#)): this is the discrete skeleton of Black–Scholes;
- spectral decompositions (eigen-decomposition, SVD, Perron–Frobenius) describe long-run behaviour of Markov chains and the dominant directions of co-movement in data (PCA, [chapter 7](#)).

The basic material (vector spaces, rank, determinants) is only recalled; a physicist can skim [section 2.2](#) and focus on [sections 2.1, 2.3, 2.4](#) and [2.8](#).

2.1 Vectors as portfolios

Let $p(t) \in \mathbb{R}_+^{501}$ collect the closing prices on day t of the 500 stocks of the S&P 500 plus *cash* as asset 0 with $p_0(t) \equiv 1$ [► 2024 L2 @ 3:21].¹ A portfolio is a vector of holdings $q(t)$ (shares, and dollars of cash), and its value is the dot product

$$V_t = q^T p(t) = \sum_{j=0}^{500} q_j p_j(t).$$

Timing and rebalancing. The holdings $q(t+1)$ carried through day $t+1$ are chosen at the end of day t , using only information available then [► 2024 L2 @ 8:43]. Rebalancing by trading $\delta_j(t)$ shares, $q_j(t+1) = q_j(t) + \delta_j(t)$, is *self-financing* (no money added or withdrawn) if $\sum_j \delta_j(t) p_j(t) = 0$ [► 2024 L2 @ 4:26]. Then $V_t = q(t+1)^T p(t)$ and the daily P&L is

$$\text{PnL}(t+1) = V_{t+1} - V_t = q(t+1)^T [p(t+1) - p(t)], \quad (2.1)$$

where the cash term drops out because p_0 does not change.

INTUITION: Left-point sums

Summing (2.1) over days, the total gain is $\sum_t q(t+1)^T \Delta p(t)$: holdings decided *before* the price move multiply the price increment. This “evaluate the integrand at the left end point” rule is exactly the discrete version of the Itô integral $\int \theta_t dS_t$ of chapter 17; using q at the right end point would mean trading on tomorrow’s prices.

Long/short and zero-cost portfolios. Holdings may be negative. To *sell short* means to borrow shares through the broker (who must “locate” them), sell them now and buy them back later [► 2024 L2 @ 11:44]; the sale proceeds are credited to the (margin) account, which allows long positions worth more than 100% of one’s capital.² For two portfolios q, w the difference $d = q - w$ is a long/short portfolio with $\text{PnL}(d) = \text{PnL}(q) - \text{PnL}(w)$. A *zero-cost* portfolio has $d^T p(t-1) = 0$; an *arbitrage* would be a zero-cost portfolio whose P&L is positive [► 2024 L2 @ 13:45]. The catch: at the time of the decision $p(t)$ is random, so the P&L is random. A pure arbitrage requires a model of what $p(t)$ can be (section 2.4); strategies whose random P&L merely tends to be positive are called *statistical arbitrage* [► 2024 L2 @ 16:49].

Geometry. $\|v\| = \sqrt{v^T v}$, $v^T w = \|v\| \|w\| \cos \theta$; $v^T w / \|w\|$ is the length of the projection of v on the direction of w [► 2024 L2 @ 17:50]. Orthogonal projections are the geometry of least squares (chapter 5).³

2.2 Matrices: a quick recap

For $A = [a_1 \cdots a_n] \in \mathbb{R}^{m \times n}$ the two readings of Av are

$$\begin{aligned} Av &= \sum_{j=1}^n v_j a_j && \text{(combination of columns),} \\ (Av)_i &= \text{row}_i(A) v && \text{(dot products with rows);} \end{aligned}$$

the first is the more useful one [► 2024 L2 @ 23:03]. Similarly $AB = [Ab_1 \cdots Ab_p]$. Remember $(AB)^T = B^T A^T$ and $AB \neq BA$ in general. Notation used later: $I_n = [e_1 \cdots e_n]$, $\mathbf{1}$ the vector of ones, $J = \mathbf{1}\mathbf{1}^T$ the matrix of ones, $\text{diag}(d_1, \dots, d_n)$.

¹ prices at the beginning vs end of the day, is the end of the day price equal to the beginning price of the day after?

ANSWER 2.1: Close of one day, open of the next

In general they differ. News that arrives overnight (earnings are usually released after the close) and trading in after-hours sessions and on foreign exchanges move the price, so the next day opens at a different level: the *overnight gap*, often a sizeable part of the daily move. Here only closing prices are used, so the daily return $p(t+1)/p(t) - 1$ is close-to-close and the gap is simply part of it.

SEEN
2 cool!

³ I am a bit concerned by the dot product $d \cdot p$ because it is not an inner product, d and p don't live in the same space. I'd rather have a function value that eats them and gives the value, am I wrong?

ANSWER 2.2: Holdings and prices are dual objects

You are right. Holdings (units: shares) and prices (units: dollars per share) are different kinds of vectors: a holding vector q is best read as a *linear functional* on price vectors, $p \mapsto V(p) = \langle q, p \rangle$, which returns dollars. In the standard basis, where each coordinate is one asset, this pairing is computed as the dot product $q^T p$. This is the same as a covector α_i acting on a vector v^i in physics, where the metric is not needed. So the value is basis-free. Lengths and angles between q and p have no financial meaning, because they would add quantities with different units. The geometry above is meaningful between vectors of the same kind: two return vectors, two portfolios, a data vector and its fit in least squares.

Linear systems [► 2024 L2 @ 1:14:25]. For $Ax = b$ with $A \in \mathbb{R}^{m \times n}$: if $m = n = \text{rank } A$ then $\det A \neq 0$ and $x = A^{-1}b$; otherwise the system is under-determined ($m < n$), over-determined ($m > n$, solved in the least-squares sense in [chapter 5](#)) or rank-deficient. Solvability: $b \in \text{col}(A)$; ⁴ uniqueness: $\ker A = \{0\}$. These two facts are exactly what decides replicability of derivatives in [section 2.4](#).

4 what is $\text{col}(A)$?

ANSWER 2.3: Column space

$\text{col}(A) = \{Ax : x \in \mathbb{R}^n\} = \text{span}(a_1, \dots, a_n) \subseteq \mathbb{R}^m$, the *column space* (image, range) of A . By the first reading of Av above, $Ax = b$ asks for a combination of the columns equal to b , so it has a solution iff b lies in their span. Its dimension is $\text{rank } A$.

2.3 Stochastic matrices and Markov chains

Definition 2.1 (Stochastic matrix, Markov chain). A square matrix A with $a_{ij} \geq 0$ whose *columns* sum to 1 is (column-)stochastic [► 2024 L2 @ 29:29]. A Markov chain on states $\{1, \dots, m\}$ with stationary transition probabilities $a_{ij} = \mathbb{P}(S_{t+1} = i \mid S_t = j)$ propagates the distribution $\pi(t) = (\mathbb{P}(S_t = 1), \dots, \mathbb{P}(S_t = m))^T$ as

$$\pi(t+1) = A\pi(t), \quad \pi(t) = A^t\pi(0).$$

A *stationary distribution* is a probability vector with $\pi^* = A\pi^*$, i.e. an eigenvector of eigenvalue 1 [► 2024 L2 @ 35:53].

5 cool! i've never seen the definition before. but the distribution can be non stationary, right? when you have cycles, periodic

ANSWER 2.4: Stationary distribution vs distribution at time t

Yes. $\pi(t)$ is in general *not* stationary: it changes with t until it (possibly) settles. A finite chain always *has* at least one stationary π^* ([proposition 2.2](#) and Perron–Frobenius), but $\pi(t)$ need not converge to it. The periodic chain just below the proposition is the standard counterexample: it has $\pi^* = (\frac{1}{2}, \frac{1}{2})^T$, yet starting from state 1 the distribution flips between $(1, 0)^T$ and $(0, 1)^T$ forever. Convergence needs the chain to be irreducible *and* aperiodic; this is the condition “some $A^k > 0$ ” below.

DONE

6 it would be nice to have a picture of a markov chain, the usual one: oriented graph with weights

7 COURSE NOTE: Conventions

The course uses column-stochastic matrices acting on column vectors. Many probability texts use the transpose (row-stochastic P , row vectors, $\pi^T(t+1) = \pi^T(t)P$). Nothing changes except transposes.

Proof. $\mathbf{1}^T A = \mathbf{1}^T$, so 1 is an eigenvalue of A^T , hence of A (same characteristic polynomial).⁸

If $Av = \lambda v$ then $|\lambda| \sum_i |v_i| = \sum_i |\sum_j a_{ij} v_j| \leq \sum_j |v_j| \sum_i a_{ij} = \sum_j |v_j|$. $\square$

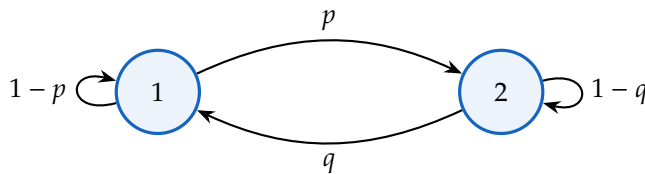


Figure 2.1: The two-state chain of [example 2.3](#). Each arrow $j \rightarrow i$ carries the transition probability $a_{ij} = \mathbb{P}(S_{t+1} = i \mid S_t = j)$, and the arrows leaving a state sum to 1 (the columns of A sum to 1). In PS1 state 1 is “buy order”, state 2 “sell order”.

Proposition 2.2. If A is column-stochastic then 1 is an eigenvalue and every eigenvalue satisfies $|\lambda| \leq 1$.

8 remind me what is the characteristic polynomial

ANSWER 2.5: Characteristic polynomial

$\chi_A(\lambda) = \det(A - \lambda I)$, a polynomial of degree n in λ . Its roots are the eigenvalues, because $Av = \lambda v$ has a solution $v \neq 0$ iff $A - \lambda I$ is singular. For $n = 2$: $\chi_A(\lambda) = \lambda^2 - (\text{tr } A)\lambda + \det A$. Since $\det M^T = \det M$, $\chi_{A^T}(\lambda) = \det((A - \lambda I)^T) = \chi_A(\lambda)$. So A and A^T have the same eigenvalues, although in general not the same eigenvectors: here $\mathbf{1}$ is an eigenvector of A^T , while the eigenvector of A is π^* .

A stationary distribution need not be the limit of $\pi(t)$: for $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ the chain alternates deterministically and $\pi(t)$ oscillates (a periodic chain, eigenvalue -1) [► 2024 L2 @ 36:53]. If some power A^k has all entries positive, the chain “mixes”: by the Perron–Frobenius theorem ([section 2.8](#)) all other eigenvalues have $|\lambda| < 1$ and $\pi(t) \rightarrow \pi^*$ from any start [► 2024 L2 @ 37:54].

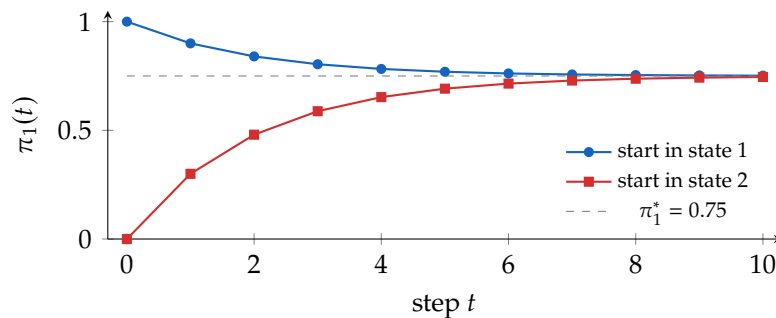


Figure 2.2: The two-state chain with $p = 0.1$, $q = 0.3$: from either starting state the probability of being in state 1 approaches $\pi_1^* = q/(p + q) = 0.75$, and the distance shrinks by the factor $\lambda_2 = 0.6$ at each step.

IN FINANCE: Order-flow as a Markov chain

In PS1 (2024) the side (buy/sell) of successive market orders is modelled as a two-state chain: buys tend to follow buys and sells follow sells. The stationary distribution gives the long-run fraction of buy orders; λ_2 measures the persistence of order flow [► 2024 L2 @ 38:56].

Example 2.3 (Two-state chain). Let $p = \mathbb{P}(1 \rightarrow 2)$, $q = \mathbb{P}(2 \rightarrow 1)$, $0 < p, q < 1$: $A = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}$. From $\text{tr } A = 2 - p - q$ and $\det A = 1 - p - q$ the eigenvalues are 1 and $\lambda_2 = 1 - p - q$, with eigenvectors $\pi^* = \frac{1}{p+q}(q, p)^T$ and $(1, -1)^T$.¹⁰ Since $\pi(0) - \pi^*$ has zero sum it is a multiple of $(1, -1)^T$, hence

$$\pi(t) = \pi^* + (1 - p - q)^t (\pi(0) - \pi^*).$$

Convergence is geometric with relaxation time $\tau = -1/\ln|1 - p - q|$: the spectral gap $1 - |\lambda_2|$ controls how fast the chain forgets its initial state.

9 non ho capito molto bene questo esempio

ANSWER 2.6: The idea

Split the starting distribution into the stationary part and a deviation, $\pi(0) = \pi^* + d$. Both $\pi(0)$ and π^* sum to 1, so d sums to 0: it is a multiple of $(1, -1)^T$, an eigenvector of λ_2 . Each step leaves π^* unchanged and multiplies d by λ_2 , hence $\pi(t) = \pi^* + \lambda_2^t d$. Because $|\lambda_2| < 1$, the deviation dies out geometrically, like the decay of a normal mode.

ANSWER 2.7: Numbers

$p = 0.1$, $q = 0.3$: $\lambda_2 = 0.6$ and $\pi^* = (0.75, 0.25)^T$. Starting in state 1, $\pi(0) = (1, 0)^T$ and $d = (0.25, -0.25)^T$, so $\pi_1(t) = 0.75 + 0.25 \cdot 0.6^t$: 1, 0.90, 0.84, 0.804, ... $\rightarrow 0.75$ (fig. 2.2). The first step can be checked directly: $A(1, 0)^T = (1 - p, p)^T = (0.9, 0.1)^T$. The relaxation time is $\tau = -1/\ln 0.6 \approx 2$ steps.

10 how do you get eigenvalues and vectors from tr and det?

ANSWER 2.8: Eigenvalues from tr and det

For a 2×2 matrix, $\chi_A(\lambda) = \lambda^2 - (\text{tr } A)\lambda + \det A$, so $\lambda_1 + \lambda_2 = \text{tr } A$ and $\lambda_1 \lambda_2 = \det A$. We already know $\lambda_1 = 1$ (proposition 2.2), hence $\lambda_2 = \det A = 1 - p - q$; check: $1 + (1 - p - q) = 2 - p - q = \text{tr } A$.

ANSWER 2.9: Eigenvectors

They solve $(A - \lambda I)v = 0$:

- $\lambda = 1$: $\begin{pmatrix} -p & q \\ p & -q \end{pmatrix} v = 0$ gives $pv_1 = qv_2$, so $v \propto (q, p)^T$. Normalising to sum 1 gives $\pi^* = \frac{1}{p+q}(q, p)^T$.
- $\lambda = 1 - p - q$: $\begin{pmatrix} q & q \\ p & p \end{pmatrix} v = 0$ gives $v_1 + v_2 = 0$, so $v \propto (1, -1)^T$.

11 cos'e' ps1?

ANSWER 2.10: PS1

PS1 is *Problem Set 1*, the first homework of the 2024 course (18.642). Its problems are spread through this chapter: each standalone one follows the section it practises, while the connected ones (PS1 2024 Problems 3–4 and 2013 A-1) are at the end; the order-flow chain is *Exercise 2.11*. In these notes “PS*n* (2024)” and “PS*n* (2013)” always mean the problem sets of the two course editions, and the Sources box of each chapter lists which ones it covers.

2.4 The one-period market model

This section follows the lecture note *One-Period Financial Models* (lec02_2) and [► 2024 L2 @ 41:03]–[► 2024 L2 @ 1:13:22]. It is the most important part of the chapter.

2.4.1 Two assets, two states

One period $[0, T]$ and two assets [► 2024 L2 @ 43:17]:

- a *bond* (risk-free asset) with $B_T = B_0(1 + r_f T)$ known in advance;¹²
- a *stock* with known S_0 and random $S_T \in \{S_T^d, S_T^u\}$, $S_T^d < S_T^u$, depending on the state $\omega \in \{d, u\}$ of the economy at T .

No probabilities of u and d are specified: they will turn out to be irrelevant for pricing.¹³ A portfolio $\theta = (\theta_B, \theta_S)^T$ costs $V_0 = \theta_B B_0 + \theta_S S_0$ and pays $V_T(\omega) = \theta_B B_T + \theta_S S_T(\omega)$.

12 the professor made a parenthesis about risk-free rate, which doesn't exist, but...

ANSWER 2.11: “Risk-free”

The professor's parenthesis [► 2024 L2 @ 46:45]: nothing in the world is risk-free, but a government that can print money to pay its debts is essentially risk-free, and one worries less about countries with stronger economies. In practice “the” risk-free rate is a short-term government-bill rate or an overnight interbank rate (SOFR in dollars, ESTR in euros). Such an asset has no default risk in nominal terms. It still has inflation risk, and a country borrowing in a currency it cannot print (e.g. euro-area members) can default.

Two new concepts enter here: the contract to be priced and the portfolio that copies it.

Definition 2.4 (New concepts: contingent claim, replicating portfolio).

- A *contingent claim* C is a contract that pays, at time T , an amount depending on the state: C_T^u in state u , C_T^d in state d .¹⁴
- A portfolio θ *replicates* C (Latin *replicare*, “to repeat”) if it pays exactly what C pays, in every state:

$$\begin{pmatrix} B_T & S_T^d \\ B_T & S_T^u \end{pmatrix} \begin{pmatrix} \theta_B \\ \theta_S \end{pmatrix} = \begin{pmatrix} C_T^d \\ C_T^u \end{pmatrix}. \quad (2.2)$$

The determinant is $B_T(S_T^u - S_T^d) \neq 0$, so every claim is replicable, uniquely:

$$\theta_S = \frac{C_T^u - C_T^d}{S_T^u - S_T^d}, \quad \theta_B = \frac{C_T^d S_T^u - C_T^u S_T^d}{B_T (S_T^u - S_T^d)}. \quad (2.3)$$

θ_S is a finite-difference *delta*, the hedge ratio of [section 1.4](#). If the claim trades at a price different from the cost of its replicating portfolio, buying the cheap one and selling the expensive one is an arbitrage; so $C_0 = \theta_B B_0 + \theta_S S_0$.¹⁵ Substituting (2.3) and rearranging:

$$C_0 = \frac{1}{1 + r_f T} [q_u C_T^u + q_d C_T^d], \quad q_u = \frac{(1 + r_f T)S_0 - S_T^d}{S_T^u - S_T^d}, \quad q_d = 1 - q_u. \quad (2.4)$$

16

Proposition 2.5 (No arbitrage in the binomial model). *The two-asset model is arbitrage-free iff $S_T^d < S_0(1 + r_f T) < S_T^u$, i.e. iff $0 < q_u < 1$.*

The “only if” part is PS1 Problem 5 ([Exercise 2.1](#)); both directions follow from the general theorem below.¹⁷

13 crazy! is this true? maybe for risk it would be important!

ANSWER 2.12: Is it true?

Yes, and it is the central surprise of derivative pricing. The claim can be *replicated* exactly with the bond and the stock, in both states. Its price must then equal the cost of the replicating portfolio, whatever the chances of u and d ; otherwise there is an arbitrage. The probabilities are not lost: they are already in S_0 , since the market set the stock price knowing them.

ANSWER 2.13: And for risk?

You are right that they matter for *risk*: the expected return, the variance and the probability of losing money all depend on them (real-world measure $\mathbb{P}$). The q_u, q_d that appear below look like probabilities but are only pricing weights (the *risk-neutral* measure, [chapter 15](#)).

14 what is a contingent claim? richiedere i soldi al tempo T? l'ho cercato e dovrei aver capito. ma l'obiettivo ora e' capire quale sia il prezzo di acquisto a tempo 0 se c'e' la contingent claim C, giusto?

ANSWER 2.14: Contingent claim

A *claim* is a right to receive a payment. It is *contingent* because the amount depends on which state occurs at T (from Latin *contingere*, “to happen”). It is just a contract that pays C_T^u if the stock goes up and C_T^d if it goes down. Examples: a call pays $(S_T - K)^+$, a forward pays $S_T - K$, a digital pays 1 if $S_T > K$.

ANSWER 2.15: Asking for the money at T ?

You do not “claim” the money: you hold the contract, and it pays automatically at T .

ANSWER 2.16: The goal is its price at time 0?

Exactly as you say: find the fair price C_0 to pay at time 0 for this contract. The answer is the cost of the portfolio of bond and stock that pays the same in both states.

15 non capisco cos'e' la claim

ANSWER 2.17: Same question as before

You asked this above: see [answer 2.14](#). In one line: the claim is the contract C , which pays C_T^u or C_T^d at T , and C_0 is its price today.

16 quindi, abbiamo creato un portafoglio binario con l'outcome desiderato tra due valori a scelta al tempo T, giusto?

ANSWER 2.18: Any pair of payoffs

Yes. With the bond and the stock we can build a portfolio that pays any pair (C_T^d, C_T^u) we choose, because the 2×2 matrix in (2.2) is invertible: the payoffs of the two assets span all of $\mathbb{R}^2$. A market where every claim can be replicated is called *complete*. The portfolio is not “binary” in itself: it takes two values only because the model has two states. With three states and two assets the matrix is 3×2 , and not every claim can be replicated ([section 2.2](#): b must lie in the column space).

17 ovvio: se sono tutti e due piu grandi allora ti fai prestare soldi a risk-free interest e compri stock, se sono tutti e due piu bassi, vai in short, e compri il bond

ANSWER 2.19: Right: that is the proof

Correct, and this is exactly the “only if” proof. If $S_T^d \geq S_0(1 + r_f T)$, borrow S_0 and buy the stock: the cost is 0, and at T you hold $S_T - S_0(1 + r_f T) \geq 0$, which is > 0 at least in state u . If $S_T^u \leq S_0(1 + r_f T)$, do the reverse: short the stock and lend the proceeds. One detail matters: an arbitrage needs *no loss in any state and a gain in at least one*. This is why the inequalities in [proposition 2.5](#) are strict.

Example 2.6 (Pricing a call). Take the numbers of the lecture note: $B_0 = 100$, $B_T = 103$ ($r_f T = 3\%$), $S_0 = 100$, $S_T^u = 130$, $S_T^d = 90$, and a call with strike $K = 100$ ¹⁹: $C_T^u = 30$, $C_T^d = 0$. Then $\theta_S = 30/40 = 0.75$ shares, $\theta_B = -0.75 \cdot 90/103 = -0.6553$ bonds, and $C_0 = -65.53 + 75 = 9.466$. Equivalently $q_u = (103 - 90)/40 = 0.325$ and $C_0 = 0.325 \cdot 30/1.03 = 9.466$. Note that q_u has nothing to do with how likely the up move really

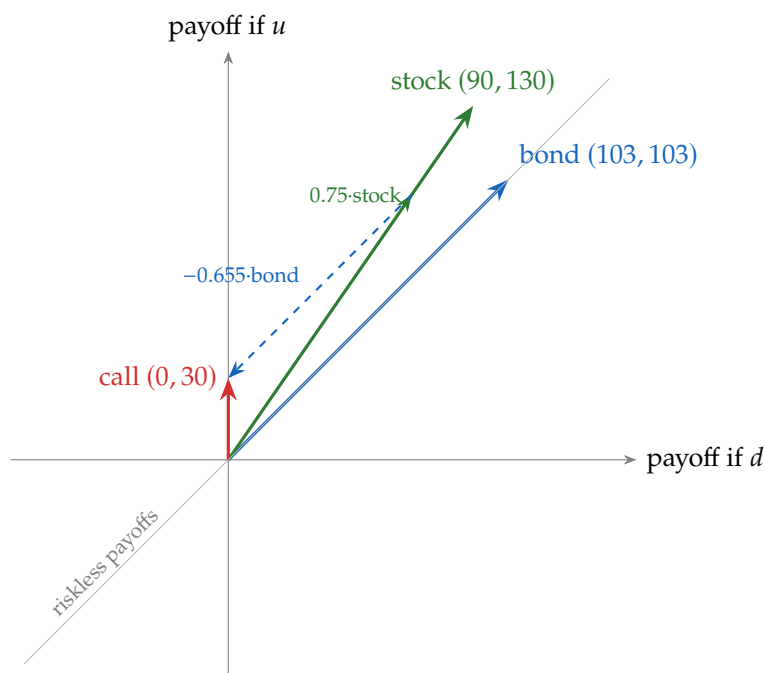


Figure 2.3: The two-state model as a plane of payoff vectors (x: state d , y: state u), as in Figure 1 of lec02_2. Bond and stock span $\mathbb{R}^2$, so every claim, e.g. the call of example 2.6, is a combination of them [► 2024 L2 @ 55:18].

INTUITION: Why “risk-neutral”

From (2.4) applied to the stock itself, $q_u S_T^u + q_d S_T^d = (1 + r_f T) S_0$: under $Q = (q_u, q_d)$ every asset earns the risk-free rate on average, as it would in a world of risk-neutral investors. The real-world probabilities and investors’ risk preferences are already encoded in the current price S_0 ; ¹⁸ given S_0 , replication makes them irrelevant for the claim. This is why option prices do not depend on the stock’s expected return, a fact that will reappear as the absence of the drift μ from the Black–Scholes formula.

Exercise 2.1 (PS1 (2024), Problem 5; lecture note lec02_2, Exercise 1). In the two-asset one-period model prove that no arbitrage requires $S_T^d < S_0(1 + r_f T) < S_T^u$. *Hint*: if an inequality fails, build an explicit portfolio whose cost is lower than its payoff in every state.

MY SOLUTION

first fails: $S_0(1 + r_f T) \leq S_T^d$ then buy the stock and short the bond, the cost is negative and the payoff is positive in both states. this means $\theta = (\theta_B, \theta_S)$ where $V_0 = \theta_B B_0 + \theta_S S_0 = 0$ but $\theta_S > 0$.

second fails: then the zero cost portfolio in which you have positive quantity of bond and negative of stock will have positive payoff in both states.

IDEA: if you invest in the stock with uncertainty, you could have invested in the risk-free bond and have a guaranteed return. if the stock is guaranteed to be lower than the bond, then you are better off investing in the bond. vice versa, if the stock is guaranteed to be higher than

is.

¹⁸ is this the concept of martingale?

ANSWER 2.20: Almost: the discounted price is a Q -martingale

Very close, but the martingale is the formula just before, not the sentence about S_0 . Divide $q_u S_T^u + q_d S_T^d = (1 + r_f T) S_0$ by $B_T/B_0 = 1 + r_f T$:

$$\mathbb{E}_Q \left[\frac{S_T}{B_T} \right] = \frac{S_0}{B_0}.$$

The price measured in units of the bond has, under Q , expected future value equal to its value today. Over one period this is the defining property of a *martingale*: a fair game with no drift (section 4.4). In general: in an arbitrage-free market there is a measure Q under which all discounted prices are martingales. This is the *fundamental theorem of asset pricing*: theorem 2.9 for n assets, and chapter 15 for continuous time. Under the real probabilities $\mathbb{P}$ the discounted stock is usually *not* a martingale: it drifts up by the risk premium.

¹⁹ cos’e’ questa call? K lho visto usare per options, futures e forwards, e’ lo stesso K?

ANSWER 2.21: What a call is

A (European) *call* is the right, not the obligation, to *buy* the stock at time T at the price K fixed today. You exercise it only if $S_T > K$, so it pays $(S_T - K)^+ = \max(S_T - K, 0)$: here $130 - 100 = 30$ in state u and 0 in state d . The *put* is the right to sell, paying $(K - S_T)^+$. See also answer 1.6.

ANSWER 2.22: Is it the same K ?

Same role, different contract. K is always the price fixed today at which the exchange happens at T . For an option it is the *strike* (exercise price), and only the holder decides whether to trade. For a forward or a future it is the *delivery price*, and both sides must trade. A forward is usually struck at the K that makes it cost 0 today, the forward price.

FEEDBACK: ERROR

Right idea, and the IDEA paragraph is the intuition. Fix the details. You write “the cost is negative” and then $V_0 = 0$: it is zero. Give the portfolio explicitly: $\theta_S = 1$, $\theta_B = -S_0/B_0$. Then $V_0 = 0$ and $V_T = S_T - S_0(1 + r_f T)$, which is ≥ 0 in state d (it can be = 0 if equality holds) and > 0 in state u . So the payoff is not “positive in both states”, but it does not need to be: an arbitrage needs ≥ 0 everywhere and > 0 somewhere.

Second case: $\theta_S = -1$, $\theta_B = S_0/B_0$, same check.

the bond, you better sell/short the bond and buy the stock.

Exercise 2.2 (lec02_2, Exercise 2). With $B_0 = 100$, $B_T = 103$, $S_0 = 100$, $S_T^u = 130$, $S_T^d = 90$, consider the put (option) with strike $K = 100$,²⁰ payoff $\max(0, K - S_T)$.

- Find its replicating portfolio.²¹
- Find its cost at $t = 0$.
- Is this price arbitrage-free? Check put-call parity against [example 2.6](#).

MY SOLUTION

a. we want $C_d^T = \min(0, K - S_T)$ and $C_u^T = \max(0, K - S_T)$, using the formulas to find the portfolio we want:

$q_u = \frac{(1+r_f T)S_0 - S^d}{S^u - S^d} = \frac{(1+0.03)100 - 90}{130 - 90} = \frac{103 - 90}{40} = \frac{13}{40}$, $q_d = 1 - q_u = \frac{27}{40}$ [q was not needed, but measures how much higher from the lowest outcome would the risk free bring the stock value]

$\theta_S = \frac{C_u - C^d}{S^u - S^d} = \frac{\max(0, K - S_T)}{130 - 90} = \frac{\max(0, K - S_T)}{40}$, $\theta_B = \frac{C^d S^u - C_u S^d}{B_T(S^u - S^d)} = \frac{\min(0, K - S_T) \cdot 130 - \max(0, K - S_T) \cdot 90}{103 \cdot 40}$

b. $q_d = 1 - q_u = 1 - \frac{13}{40} = \frac{27}{40}$

$C_0 = \theta_B B_0 + \theta_S S_0 = \frac{\min(0, K - S_T) \cdot 130 - \max(0, K - S_T) \cdot 90}{103 \cdot 40} \cdot 100 + \frac{\max(0, K - S_T)}{40} \cdot 100$

or equivalently $C_0 = \frac{q_u C_u + q_d C^d}{1 + r_f T} = \frac{\frac{13}{40} \max(0, K - S_T) + \frac{27}{40} \min(0, K - S_T)}{1.03}$

c. I DON'T KNOW HOW TO CHECK, we have min and max functions and not values.

Exercise 2.3 (lec02_2, Exercise 3). Starting from $C_0 = \theta_B B_0 + \theta_S S_0$, derive (2.4) and interpret $(1 + r_f T)^{-1} S_T^u$, $(1 + r_f T)^{-1} S_T^d$ and $S_0(1 + r_f T)$ (the forward value of the bond position obtained by shorting one share). Show that $q_u + q_d = 1$ and that $q_u, q_d > 0$ under no arbitrage.

MY SOLUTION

l'interpretazione dei due termini e' il valore attuale dei due possibili putcomes del prezzo del titolo, mentre il terzo e' il valore del titolo alla fine del periodo se fosse risk-free. quello che dici nelle parentesi e': se prendiamo in prestito il titolo ora, lo vendiamo, riinvestiamo il ricavato in un titolo risk-free, poi lo venderemo alla fine del periodo al valore che hai scritto. [ci aspettiamo che il valore del titolo alla fine del periodo oscilli attorno al valore del titolo risk-free]

SECONDA PARTE NON HO VOGLIA ORA

²⁰ we are betting stock goes down and hedging if goes up

ANSWER 2.23: Half right

Buying a put is indeed a bet that the stock goes down: it pays $K - S_T$ when $S_T < K$. But there is no hedge if it goes up: then the put pays 0 and you have lost the premium you paid (the most you can lose). The *hedge* is the other way round: someone who owns the stock buys a put to be protected if it falls (stock + put = "protective put", an insurance policy on the stock).

²¹ replicating portfolio = portfolio with the desired outcome possibilities?

ANSWER 2.24: Yes

Yes: the portfolio of bond and stock whose payoff equals the put's payoff in every state ([definition 2.4](#)); see also [answer 2.18](#).

FEEDBACK: ERROR

One misunderstanding blocks everything: C^u and C^d are *numbers*, the payoff $\max(0, K - S_T)$ evaluated in each state. It is not min in one state and max in the other.

- State u : $S_T^u = 130$, so $C^u = \max(0, 100 - 130) = 0$.

- State d : $S_T^d = 90$, so $C^d = \max(0, 100 - 90) = 10$.

(a) $\theta_S = (0 - 10)/40 = -0.25$ (short a quarter share),

$\theta_B = (10 \cdot 130 - 0 \cdot 90)/(103 \cdot 40) = 0.3155$ bonds.

(b) $C_0 = 0.3155 \cdot 100 - 0.25 \cdot 100 = 6.553$; check:

$q_d C^d / 1.03 = 0.675 \cdot 10 / 1.03 = 6.553$.

(c) Put-call parity: call - put = $S_0 - K / (1 + r_f T)$, i.e.

$9.466 - 6.553 = 2.913 = 100 - 100 / 1.03$. It holds, and

$0 < q_u < 1$, so the price is arbitrage-free. Your formulas in (a), (b)

are right: only the inputs were wrong. Your reading of q_u is nice: it is the position of the forward value $S_0(1 + r_f T)$ inside $[S^d, S^u]$, as a fraction.

FEEDBACK

Interpretations correct: $S_T^{u,d} / (1 + r_f T)$ are present values of the two outcomes, and $S_0(1 + r_f T)$ is what you hold at T after shorting one share and lending S_0 . Careful with "we expect the stock to oscillate around it": true under Q ($\mathbb{E}_Q[S_T] = S_0(1 + r_f T)$), not under the real probabilities, where the stock earns more on average (risk premium). Still to do: the derivation (substitute (2.3) into $C_0 = \theta_B B_0 + \theta_S S_0$,

use $B_0/B_T = 1/(1 + r_f T)$, collect C^u and C^d), and the second part:

$q_u + q_d = 1$ holds by definition, $q_u > 0 \iff S_0(1 + r_f T) > S^d$,

$q_d = \frac{S^u - S_0(1 + r_f T)}{S^u - S^d} > 0 \iff S_0(1 + r_f T) < S^u$: together,

[proposition 2.5](#).

2.4.2 n assets, m states

Generalise to n assets and m possible states $\omega_1, \dots, \omega_m$ at time T [► 2024 L2 @ 1:06:57]:

- $A_0 = (P_0^1, \dots, P_0^n) \in \mathbb{R}^{1 \times n}$: known prices at $t = 0$;
- $A \in \mathbb{R}^{m \times n}$ with $A_{ij} = P_T^j(\omega_i)$: price of asset j in state i . Column j is the payoff vector of asset j , row i the prices in scenario i ;
- a portfolio $\theta \in \mathbb{R}^n$ costs $V_0 = A_0 \theta$ and pays $V_T = A \theta \in \mathbb{R}^m$.

Implicit assumptions: prices do not depend on quantities traded (unlimited liquidity), holdings can be fractional, short selling is unrestricted, no transaction costs [► 2024 L2 @ 1:05:55].

Definition 2.7 (Replication, completeness, arbitrage, state prices). In the model (A_0, A) :

- A claim with payoff $C_T \in \mathbb{R}^m$ is *replicable* if $A\theta = C_T$ for some θ . The market is *complete* if every claim is replicable.
- An *arbitrage* is a portfolio with $V_0 \leq 0$, $V_T \geq 0$ (componentwise) and $(V_0, V_T) \neq (0, 0)$: it never loses and either costs nothing and may gain, or pays you today and never costs anything later [► 2024 L2 @ 1:09:06].
- A vector $\psi \in \mathbb{R}^m$ with all $\psi_i > 0$ is a vector of *state prices* if $A_0 = \psi^\top A$, i.e. $P_0^j = \sum_i \psi_i P_T^j(\omega_i)$ for every asset. Writing $\beta = \sum_i \psi_i$ and $q_i = \psi_i/\beta$, this reads

$$P_0^j = \beta \mathbb{E}_Q[P_T^j], \quad j = 1, \dots, n,$$

and $Q = (q_1, \dots, q_m)$ is a *pricing measure* with discount factor β [► 2024 L2 @ 1:11:16].

ψ_i is the price of the *Arrow–Debreu security* e_i paying \$1 in state i and nothing otherwise (when it is replicable). If a riskless bond exists, $B_0 = \psi^\top \mathbf{1} B_T$ gives $\beta = 1/(1 + r_f T)$, the ordinary discount factor.

Lemma 2.8 (Stiemke). Let $L \subset \mathbb{R}^k$ be a linear subspace [not affine]. Then $L \cap \mathbb{R}_+^k = \{0\}$ iff there exists $y \in \mathbb{R}^k$ with all $y_i > 0$ and $y \perp L$.

Theorem 2.9 (Fundamental theorem of asset pricing, one period). In the one-period model (A_0, A) :

- (Part 1) The model is arbitrage-free iff a state-price vector $\psi > 0$ exists.
- (Part 2) An arbitrage-free model is complete iff the state-price vector is unique.
- If C_T is replicable, its only arbitrage-free price is $C_0 = \psi^\top C_T = \beta \mathbb{E}_Q[C_T]$, for any state-price vector ψ .

Remark 2.10. In the binomial model $A = \begin{pmatrix} B_T & S_T^d \\ B_T & S_T^u \end{pmatrix}$ is invertible: $\psi^\top = A_0 A^{-1}$ is unique and positive exactly under the condition of [proposition 2.5](#), and (2.4) is (iii). With more states than independent assets ($\text{rank } A < m$) the market is incomplete: state prices are not unique and a non-replicable claim has an *interval* of arbitrage-free prices $(\inf_\psi \psi^\top C_T, \sup_\psi \psi^\top C_T)$.

Finding an arbitrage. In a complete market, solve $A_0 = \psi^\top A$. If some $\psi_i \leq 0$, the portfolio replicating the Arrow–Debreu security e_i ($A\theta = e_i$) costs $A_0\theta = \psi^\top e_i = \psi_i \leq 0$ and pays \$1 in state i : an arbitrage (lecture note, section 2.4).

22 COURSE NOTE: Notation

The lecture note writes the portfolio as $\tilde{\pi}$; here it is θ , to keep π for probability vectors.

23 COURSE NOTE: Definition of arbitrage

The lecture note requires $V_0 \leq 0$ and $V_T \geq 0$ with $V_T(\omega_i) > 0$ for at least one state, which omits the case “negative cost, zero payoff”. When a riskless asset exists the two definitions are equivalent (invest the negative cost in the bond); the version above makes the theorem below true without that assumption.

Proof. ($\Leftarrow$) If $x \in L$, $x \geq 0$, $x \neq 0$, then $y^\top x > 0$, contradicting $y \perp L$. ($\Rightarrow$) The simplex $\Delta = \{x \geq 0 : \sum_i x_i = 1\}$ is compact, convex and disjoint from the closed convex set L . By the separating hyperplane theorem there are y and c with $y^\top l < c < y^\top x$ for all $l \in L$, $x \in \Delta$. Since L is a subspace, $y^\top l$ bounded above on L forces $y^\top l = 0$; then $c > 0$ and $y_i = y^\top e_i > 0$ for every i . $\square$

Proof. (i) Let $L = \{(-A_0\theta, A\theta) : \theta \in \mathbb{R}^n\} \subset \mathbb{R}^{1+m}$. By definition an arbitrage is a non-zero element of $L \cap \mathbb{R}_+^{1+m}$. By [lemma 2.8](#), no arbitrage iff there is $y = (y_0, y_1, \dots, y_m) > 0$ with $-y_0 A_0\theta + \sum_{i \geq 1} y_i (A\theta)_i = 0$ for all θ , i.e. $A_0 = \psi^\top A$ with $\psi_i = y_i/y_0 > 0$. (ii) State prices solve $A^\top \psi = A_0^\top$. The solution is unique iff $\ker A^\top = \{0\}$ iff $\text{rank } A = m$ iff $\text{col}(A) = \mathbb{R}^m$, i.e. iff the market is complete. (iii) If $A\theta = C_T$ then C_0 must equal the cost of θ (otherwise buy the cheaper, sell the dearer), and $A_0\theta = \psi^\top A\theta = \psi^\top C_T$. $\square$

INTUITION: Pricing is a positive linear functional

Replication makes the price a *linear* function of the payoff vector, and no arbitrage makes it *positive* (non-negative payoffs have non-negative prices). A positive linear functional on $\mathbb{R}^m$ is represented by a positive vector, ψ , which normalised becomes a probability measure. This is the finite-dimensional Riesz representation theorem; in continuous time it becomes “price = discounted expectation under the risk-neutral measure” (chapter 15).

CASE STUDY: Visualising portfolios (lec02_2, section 3)

For the binomial example of example 2.6, the R code (ggplot2) plots the payoff vectors (V_T^d, V_T^u) of many portfolios [► 2024 L2 @ 1:01:39]:

- long-only portfolios with weights summing to 1 lie on the segment between the bond and the stock;
- weights in $[0, 1]^2$ fill a parallelogram, coloured by the cost V_0 ;
- allowing shorting (weights in $[-2, 2]^2$) fills a large parallelogram around the origin. In the last plot the zero-cost portfolios (white) lie on a line through the origin that avoids the positive quadrant: no zero-cost portfolio pays off in both states, i.e. no arbitrage [► 2024 L2 @ 1:04:48].

The three plots are redrawn in fig. 2.4.

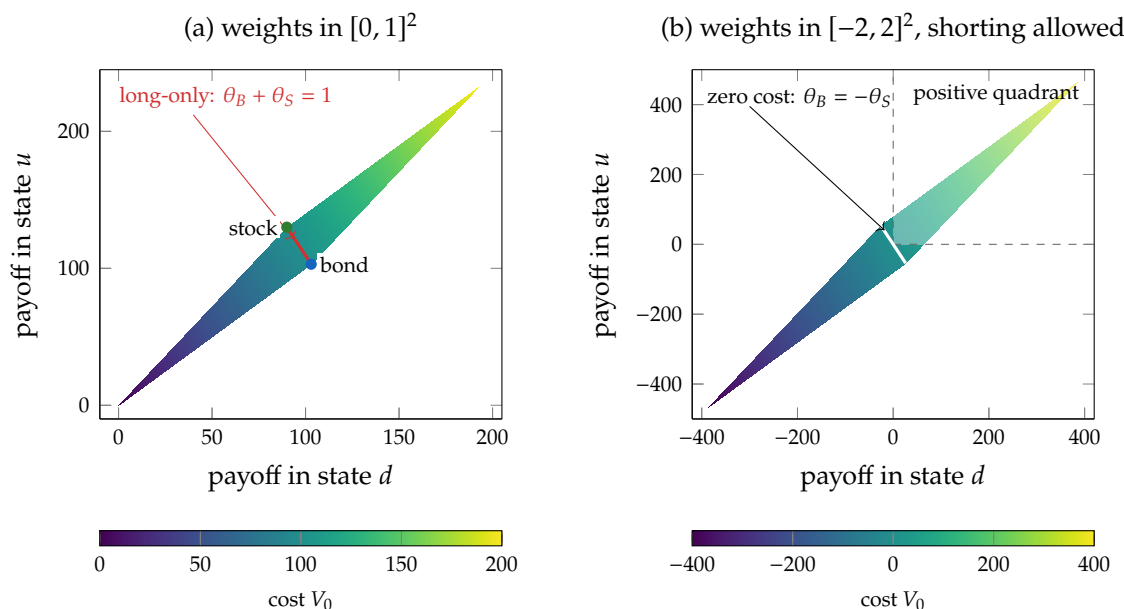


Figure 2.4: Payoff vectors (V_T^d, V_T^u) of portfolios of the bond (103, 103) and the stock (90, 130) of example 2.6, coloured by their cost $V_0 = 100\theta_B + 100\theta_S$ (redrawn from the R plots of lec02_2, section 3). (a) Long positions only: a parallelogram; the portfolios whose weights sum to 1 lie on the red segment between bond and stock. (b) With shorting the payoffs cover a large region around the origin. The zero-cost portfolios $\theta_S(-13, 27)$ (white line) never enter the positive quadrant except at the origin: no zero-cost portfolio pays off in both states, i.e. no arbitrage.

DONE

24 aggiungi i grafici di cui stai parlando

25 what is a tree in this context?

ANSWER 2.25: Binomial tree

A diagram of all possible price paths. Start from S_0 (the root); after one period the price is uS_0 or dS_0 (two branches); from each of these it moves up or down again, and so on. Because up-then-down and down-then-up give the same price udS_0 , branches recombine: after n steps there are only $n + 1$ nodes, $u^k d^{n-k} S_0$. Each little fork is the one-period model of this section. One prices a claim backwards, from its payoffs at the leaves (the last nodes), applying (2.4) at every fork (chapter 15). Name: drawn sideways it branches like a tree (in maths, a graph without loops). Cox, Ross and Rubinstein introduced it in 1979.

IN FINANCE: From the binomial tree to Black–Scholes

The binomial model iterated over many periods is the Cox–Ross–Rubinstein tree;²⁵ letting the step go to zero gives Black–Scholes ([chapter 15](#)). Part 2 of the theorem explains why stochastic-volatility or jump models, which have more sources of randomness than traded assets, are incomplete: prices there depend on a choice of pricing measure calibrated to market option prices.

2.5 Eigenvalues and diagonalisation

$\lambda \in \mathbb{C}$ is an eigenvalue of $A \in \mathbb{R}^{n \times n}$ with eigenvector $v \neq 0$ if $Av = \lambda v$, i.e. iff $\det(A - \lambda I) = 0$, a polynomial equation of degree n : there is always at least one (complex) eigenvalue [[2013 L2 @ 6:17](#)]. Eigenvectors are the directions in which A acts as a pure rescaling.²⁷

If A has n linearly independent eigenvectors, $S = [v_1 \cdots v_n]$ is invertible and

$$AS = S\Lambda, \quad A = S\Lambda S^{-1}, \quad A^k = S\Lambda^k S^{-1}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n).$$

State equations [[2024 L4 @ 4:14](#)]. For the linear dynamics $u_t = Au_{t-1}$ (the state equation of a Kalman filter, [chapter 9](#)), expand the initial state in the eigenbasis, $u_0 = \sum_i c_i v_i$ with $c = S^{-1}u_0$; then

$$u_t = A^t u_0 = \sum_{i=1}^n c_i \lambda_i^t v_i.$$

²⁸ Modes with $|\lambda_i| < 1$ die out, those with $|\lambda_i| > 1$ explode, and if the largest eigenvalue is 1 and simple, u_t converges to a multiple of its eigenvector [[2024 L4 @ 6:21](#)]. (For a physicist: normal-mode decomposition of a linear discrete-time system.)²⁹

Exercise 2.4 (PS1 (2024), Problem 1: Fibonacci matrix). Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $F_0 = 0$, $F_1 = 1$, $F_{k+2} = F_{k+1} + F_k$ and $u_k = (F_{k+1}, F_k)^T$.

- Show $u_{k+1} = Au_k$ and $u_k = A^k u_0$.
- Find the eigenvalues λ_1, λ_2 of A and show that $s = (\lambda, 1)^T$ is an eigenvector for λ .
- With $S = [s_1 \ s_2]$ compute S^{-1} , verify $A = S\Lambda S^{-1}$, and show $c = S^{-1}u_0 = a(1, -1)^T$; find a .
- Deduce $F_k = (\lambda_1^k - \lambda_2^k)/\sqrt{5}$ and the limit of F_{k+1}/F_k (the golden ratio, also used in “technical analysis” of prices, e.g. Fibonacci retracements).

²⁶ i need to get familiar with the definitions before being able to understand this and the theorem

ANSWER 2.26: A reading guide

Read in this order, each with the binomial case in mind:

- *claim and replication*: [definition 2.4](#);
- *arbitrage*: a free lunch, never loses and sometimes wins;
- *state price* ψ_i : today's price of \$1 paid only in state i ;
- *complete*: every claim is replicable.

These are the items of [definition 2.7](#). Then the theorem says, in words: (i) no free lunch iff every state has a positive price; (ii) every claim replicable iff these prices are unique; (iii) price of a claim = sum over states of (state price $\times$ payoff). This box says: real models have more randomness (volatility, jumps) than traded assets, so (ii) fails and the price is not unique. [Exercise 2.1](#) and the put exercise ([Exercise 2.2](#)) are good practice.

SEEN

²⁷ nice insight!

²⁸ quello che hai fatto e' stato diagonalizzare l'hamiltoniana/l'evoluzione temporale. l'equazione e' uno step/infinitesima (NO NON E' VERO! e' un tempo finito: all'esponenziale)

ANSWER 2.27: Yes: A is the one-step propagator

Your correction is right. A is not the generator (the Hamiltonian) but the evolution over one *finite* step, like $U(\Delta t) = e^{-iH\Delta t/\hbar}$; then A^t plays the role of $U(t) = U(\Delta t)^t$. Diagonalising A is the same as diagonalising H : same eigenvectors, and $\lambda_i = e^{-iE_i\Delta t/\hbar}$, so $\lambda_i^t = e^{-iE_i t/\hbar}$. One difference: U is unitary ($|\lambda_i| = 1$, nothing decays), while a general A has $|\lambda_i| \neq 1$, so modes grow or die. The continuous-time analogue is $\dot{u} = Gu$, $u(t) = e^{Gt}u_0$, with $A = e^{G\Delta t}$.

SEEN

²⁹ bello, mi piace!

30 COURSE NOTE: Erratum in the slides

Slide “Eigenvalues and Eigenvectors” of [lec02_1](#) states $\max_{\|v\|=1} \|Av\| = \max |\lambda(A)|$ for every square A (and it is repeated in class [[2024 L2 @ 1:17:34](#)]). This holds for *symmetric* (more generally normal) matrices, as in the 2013 notes (Theorem 4 of [lecnote2](#)). In general $\max_{\|v\|=1} \|Av\| = \sigma_1(A)$, the largest singular value ([section 2.7](#)), and $\sigma_1 \geq \max |\lambda|$: for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ all eigenvalues are 0 but $\|Ae_2\| = 1$.

MY SOLUTION

- a. $\begin{pmatrix} F_{k+2} \\ F_{k+1} \end{pmatrix} = u_{k+1} = Au_k = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \begin{pmatrix} F_{k+1} + F_k \\ F_{k+1} \end{pmatrix}$ and the second is trivial by induction.
- b. $0 = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - \lambda - 1 \implies \lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}$
- $$\begin{pmatrix} \lambda+1 \\ \lambda \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ 1 \end{pmatrix} = Av = \lambda v = \begin{pmatrix} \lambda^2 \\ \lambda \end{pmatrix} \iff \frac{3 \pm \sqrt{5}}{2} = \frac{1 \pm 5 \pm 2\sqrt{5}}{4} = \lambda^2 = \lambda + 1 = \frac{3 \pm \sqrt{5}}{2}$$
- c. $S = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \implies S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix}$
- $$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = A = S \Lambda S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 & -\lambda_1 \lambda_2 \\ -\lambda_2 & \lambda_1 \lambda_2 \end{pmatrix} =$$
- $$\frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1^2 - \lambda_2^2 & \lambda_1 \lambda_2 (-\lambda_1 + \lambda_2) \\ \lambda_1 - \lambda_2 & 0 \end{pmatrix} = \begin{pmatrix} \lambda_1 + \lambda_2 & -\lambda_1 \lambda_2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$
- $u_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ then $c = S^{-1}u_0 = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = a \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ with $a = 1/(\lambda_1 - \lambda_2) = 1/\sqrt{5} \implies a = \frac{1}{\lambda_1 - \lambda_2} = \frac{1}{\sqrt{5}}$
- d. $\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = u_k = A^k u_0 = S \Lambda^k S^{-1} u_0 = S \Lambda^k c = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^k a \begin{pmatrix} 1 \\ -1 \end{pmatrix} = a \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^k \\ -\lambda_2^k \end{pmatrix} =$
- $$a \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^k \\ -\lambda_2^k \end{pmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{pmatrix} \implies F_k = \frac{\lambda_1^k - \lambda_2^k}{\sqrt{5}}$$
- Golden Ratio: $\lim_{k \rightarrow \infty} \frac{F_{k+1}}{F_k} = \lim_{k \rightarrow \infty} \frac{\lambda_1^{k+1} - \lambda_2^{k+1}}{\lambda_1^k - \lambda_2^k} = \lim_{k \rightarrow \infty} \frac{\lambda_1^{k+1}(1 - (\lambda_2/\lambda_1)^{k+1})}{\lambda_1^k(1 - (\lambda_2/\lambda_1)^k)} =$
- $$\lim_{k \rightarrow \infty} \lambda_1 \frac{(1 - (\lambda_2/\lambda_1)^{k+1})}{(1 - (\lambda_2/\lambda_1)^k)} = \lambda_1 \text{ since } |\lambda_2/\lambda_1| < 1.$$

FEEDBACK

All correct. Two small points.

- (b) No need to plug in the numbers: the second component of λ_1 is λ , and the first is $\lambda + 1 = \lambda^2$ by the characteristic equation itself.
- (c) Say what you use in the last step: $\lambda_1 + \lambda_2 = 1$ (trace) and $\lambda_1 \lambda_2 = -1$ (determinant), so $\lambda_1^2 - \lambda_2^2 = \lambda_1 - \lambda_2$. Bonus: since $|\lambda_2| < 1$, F_k is the integer nearest to $\lambda_1^k / \sqrt{5}$.

Exercise 2.5 (PS1 (2024), Problem 2). Let $A = SAS^{-1}$ be diagonalisable. Under which conditions on the eigenvalues does $\lim_k A^k$

- (a) not exist,
 (b) exist and equal the zero matrix,
 (c) exist and be non-zero?

MY SOLUTION

- a. if an eigenvalue is bigger than 1 in abs value.
 b. if all the eigenvalues are strictly smaller than 1 in abs value.
 c. at least one eigen value is equal to 1, and the others are smaller than 1 in abs value.

Exercise 2.6 (PS1 (2013), A-2 to A-4). (A-2) Row and column rank of $\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 4 & 0 & -2 \end{pmatrix}$ and

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 1 & -1 & 0 \\ 0 & 0 & 4 & 0 & -2 & 0 \end{pmatrix}.$$

(A-3) Determinant and inverse of $\begin{pmatrix} 1 & 2 \\ 4 & -1 \end{pmatrix}$ and $\begin{pmatrix} -1 & -2 & 3 \\ 1 & 2 & 0 \\ 4 & 6 & 3 \end{pmatrix}$.

(A-4) Characteristic polynomial, eigenvalues and eigenvectors of $\begin{pmatrix} -3 & 3 & 2 \\ 1 & -1 & -2 \\ -1 & -3 & 0 \end{pmatrix}$.

MY SOLUTION

A-2. first matrix: $columnrank = 3; rowrank = 3$; second matrix: $columnrank = 3; rowrank = 3$.

A-3. first matrix: $\det = -1 - 8 = -9$; $inv = \frac{1}{-9} \begin{pmatrix} -1 & -2 \\ 4 & -1 \end{pmatrix}$; second matrix: $\det = -1 * 6 - (-2 * 3) + 3 * (6 - 8) = -6 + 6 - 6 = -6$; $inv = \frac{1}{-6} \begin{pmatrix} 4 & 24 & -6 \\ -3 & -15 & 3 \\ -2 & -2 & 0 \end{pmatrix}$

A-4. $\det(A - \lambda I) = \det \begin{pmatrix} -3-\lambda & 3 & 2 \\ 1 & -1-\lambda & -2 \\ -1 & -3 & 0-\lambda \end{pmatrix} = (-3 - \lambda)[(-1 - \lambda)(-\lambda) - 6] - 3[(-\lambda) - 2] + 2[(-3) +$

31 matrix inverse, adjoint method or gauss-jordan elimination.

$$A^{-1} = \frac{1}{\det(A)} \text{Cofactor}(A)^T \text{ - i think this works only for } 3 \times 3 \text{ matrices.}$$

ANSWER 2.28: It works for every n

$A^{-1} = \text{adj}(A)/\det A$, with $\text{adj}(A) = \text{Cof}(A)^T$ (the *adjugate*), holds for every invertible $n \times n$ matrix. Reason: expanding $\det A$ along row i gives $\sum_j a_{ij} C_{ij} = \det A$, while $\sum_j a_{kj} C_{ij} = 0$ for $k \neq i$ (it is the determinant of a matrix with two equal rows); so $A \text{Cof}(A)^T = (\det A)I$. For $n = 2$ it is the rule "swap the diagonal, change the sign of the off-diagonal". In practice it is used only for $n \leq 3$: it needs n^2 determinants of size $n - 1$, while Gauss-Jordan costs $O(n^3)$ operations.

FEEDBACK: ERROR

- (b) correct.
 (a) incomplete: the limit also fails to exist when $|\lambda| = 1$ but $\lambda \neq 1$, e.g. $\lambda = -1$ ($(-1)^k$ oscillates) or $\lambda = e^{i\theta}$ (rotates).
 (c) the other eigenvalues may also be equal to 1. Complete answer, since $A^k = S \Lambda^k S^{-1}$ and $\lim \lambda^k$ exists iff $|\lambda| < 1$ or $\lambda = 1$:
 (a) some $|\lambda_i| > 1$, or $|\lambda_i| = 1$ with $\lambda_i \neq 1$; (b) all $|\lambda_i| < 1$; (c) every λ_i is 1 or has $|\lambda_i| < 1$, and at least one is 1. Then $\lim A^k = S \text{diag}(1 \text{ or } 0) S^{-1}$ is the projection onto the eigenvectors of 1.

FEEDBACK: ERROR

- A-2: both ranks 3 correct (for the second, row 4 $-2 \cdot$ row 3 $\propto$ row 2 $-$ row 1).
 A-3: determinants correct, first inverse correct. Second inverse: one entry wrong, the $(1, 1)$ cofactor is $\det \begin{pmatrix} 2 & 0 \\ 6 & 3 \end{pmatrix} = 6$, not 4:
 $A^{-1} = -\frac{1}{6} \begin{pmatrix} 6 & 24 & -6 \\ -3 & -15 & 3 \\ -2 & -2 & 0 \end{pmatrix}$. Check by computing AA^{-1} .
 A-4: polynomial and eigenvalues correct. The eigenvectors are not: $A(1, -1, 0)^T = (-6, 2, 2)^T$ is not a multiple of $(1, -1, 0)^T$. Correct ones: $\lambda = 2$: $(0, -2, 3)^T$; $\lambda = -2$: $(2, 0, 1)^T$; $\lambda = -4$: $(-3, 1, 0)^T$. Always check $Av = \lambda v$, it takes ten seconds.

$(-1 - \lambda)] = -\lambda^3 - 4\lambda^2 + 4\lambda + 16 = -(\lambda - 2)(\lambda + 2)(\lambda + 4)$; this means that the eigenvalues are 2, -2, -4.

Solving $(A - \lambda I)x = 0$, we find the eigenvectors $v_1 = (1, -1, 0)^T$, $v_2 = (1, 1, 2)^T$, $v_3 = (1, 3, 4)^T$. I DIDN'T DO THIS LAST EIGENVECTOR CALCULATION.

2.6 Symmetric matrices

Most matrices in this course (covariance and correlation matrices, $A^T A$, Hessians) are symmetric, and for them everything works [► 2013 L2 @ 17:01].

Proof. (i) If $Av = \lambda v$, then $\bar{v}^T Av = \lambda \|v\|^2$; conjugating and using $A = \bar{A} = A^T$, the same quantity equals $\bar{\lambda} \|v\|^2$, so $\lambda = \bar{\lambda}$ [► 2013 L2 @ 19:12].

(ii) $\lambda_1 v_2^T v_1 = v_2^T Av_1 = (Av_2)^T v_1 = \lambda_2 v_2^T v_1$, so $v_2^T v_1 = 0$ if $\lambda_1 \neq \lambda_2$.

(iii) Induction: take a unit eigenvector v_1 ; its orthogonal complement W is invariant, because $w \perp v_1 \Rightarrow (Aw)^T v_1 = w^T Av_1 = \lambda_1 w^T v_1 = 0$; restrict A to W (still symmetric) and repeat (2013 notes, Proposition 3). □

Theorem 2.11 (Spectral theorem). Let $A = A^T \in \mathbb{R}^{n \times n}$. Then

- (i) all eigenvalues are real;
- (ii) eigenvectors of distinct eigenvalues are orthogonal;
- (iii) $A = U \Lambda U^T$ with U orthogonal ($U^T U = I$) and Λ real diagonal.

Consequences used later:

- *Rayleigh quotient*: the top eigenvector attains^{32,33}

$$\max_{\|v\|=1} v^T A v = \lambda_{\max}, \quad \text{and} \quad \max_{\|v\|=1} \|A v\| = \max_i |\lambda_i|.$$

This variational characterisation is the basis of PCA.³⁴

- A is *positive semi-definite* ($v^T A v \geq 0$ for all v) iff all $\lambda_i \geq 0$ (2013 PS1 B-3). Every covariance matrix Σ is PSD, because $w^T \Sigma w = \text{Var}(w^T X) \geq 0$ is the variance of the portfolio with weights w (2024 PS2 Problem 3, chapter 7).

Exercise 2.7 (PS1 (2013), B-3). Prove that a symmetric PSD matrix can be written $D = U^T A U$ with U orthogonal and D diagonal with non-negative entries.

MY SOLUTION

NON LO FACCIO ESPPLICITAMENTE PERCHE NON SO FARLO.

Ma, se A e' sym allora si puo diagonalizzare usando U sia a destra che a sinistra, e se e' PSD allora gli autovalori sono non-negativi.

32 what attain means?

ANSWER 2.29: "Attains"

"Reaches": the maximum is not only an upper bound, some vector actually achieves it. Here $v^T A v \leq \lambda_{\max}$ for every unit v , with equality for v = the top eigenvector.

33 i dont get if v is a generic unit vector or it is the eigenvector corresponding to the top eigenvalue, normalized

ANSWER 2.30: Both, in two steps

In the formula v is a *generic* unit vector: the max runs over all of them. The vector that achieves the max is the normalised top eigenvector. Proof in one line: write $v = \sum_i c_i u_i$ in the orthonormal eigenbasis, $\sum_i c_i^2 = 1$; then $v^T A v = \sum_i \lambda_i c_i^2 \leq \lambda_{\max}$, with equality when all the weight is on the top eigenvector. The name: Lord Rayleigh used $v^T A v / v^T v$ to estimate vibration frequencies (*The Theory of Sound*, 1877).

34 what is PCA?

ANSWER 2.31: Principal component analysis

Principal component analysis (chapter 7). Given many correlated series (e.g. returns of 500 stocks) with covariance matrix Σ , find the portfolio direction w , $\|w\| = 1$, with the largest variance $w^T \Sigma w$: by the Rayleigh quotient it is the top eigenvector of Σ (the first *principal component*). The next one maximises the variance among directions orthogonal to the first, and so on. A few components often explain most of the variance, so the data are effectively low-dimensional. Physics analogue: the principal axes of the inertia tensor (hence "principal").

FEEDBACK

Your two sentences are the proof; add one line for each. By the spectral theorem (theorem 2.11) $A = U \Lambda U^T$ with U orthogonal, so $U^T A U = \Lambda =: D$. For a unit eigenvector u_i , $\lambda_i = \lambda_i u_i^T u_i = u_i^T A u_i \geq 0$ by PSD.

2.7 Singular value decomposition

Theorem 2.12 (SVD). Every $A \in \mathbb{R}^{m \times n}$ of rank r can be written as $A = U\Sigma V^T$ with $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ orthogonal and $\Sigma \in \mathbb{R}^{m \times n}$ diagonal with entries $\sigma_1 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots$. Equivalently

$$A = \sum_{k=1}^r \sigma_k \mathbf{u}_k \mathbf{v}_k^T, \quad (2.5)$$

a sum of r rank-one matrices. The σ_k (singular values) are the square roots of the non-zero eigenvalues of $A^T A$ (equivalently of AA^T).

INTUITION: Rotate, stretch, rotate

Read $Ax = U\Sigma V^T x$ from the right: V^T rotates the input frame onto the axes $\mathbf{v}_1, \dots, \mathbf{v}_n$, Σ stretches axis k by σ_k (and kills the axes with $\sigma_k = 0$), U rotates onto the output frame $\mathbf{u}_1, \dots, \mathbf{u}_m$. An eigen-decomposition uses *one* frame in which A scales; the SVD uses *two* frames, $A\mathbf{v}_k = \sigma_k \mathbf{u}_k$, but exists for every matrix, even rectangular [► 2013 L2 @ 27:52].

Why it matters for data. Let A be a data matrix with one row per company and one column per date (e.g. 5×365). The reduced SVD stores 5×5 , 5×5 and 5×365 matrices instead of a 365×365 one, and if the data effectively live in a low-dimensional subspace, a few singular values dominate [► 2013 L2 @ 56:35]. AA^T (indexed by companies) measures how the companies' series move together, so the leading singular vectors describe groups of co-moving stocks [► 2013 L2 @ 58:47]; with centred data this is exactly principal component analysis (chapter 7). The best rank- k approximation of A in the Frobenius or operator norm is the truncation of (2.5) to its first k terms (Eckart–Young theorem, not proved in the course).

Exercise 2.8 (PS1 (2013), A-5 and A-6). (A-5) Gram–Schmidt on $\mathbf{v}_1 = (1, 0, 1, 0, 1)$, $\mathbf{v}_2 = (1, 1, 1, 0, 0)$, $\mathbf{v}_3 = (0, 0, 1, 1, 1)$; complete to an orthonormal basis of $\mathbb{R}^5$ and find the matrix U mapping it to the standard basis. (A-6) SVD of $\begin{pmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

MY SOLUTION

A-5. $\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{3}}(1, 0, 1, 0, 1)$; $\mathbf{u}_2 = \frac{\mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{u}_1)\mathbf{u}_1}{\|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{u}_1)\mathbf{u}_1\|} = \frac{1}{\sqrt{6}}(1, 2, -1, 0, -1)$; $\mathbf{u}_3 = \frac{\mathbf{v}_3 - (\mathbf{v}_3 \cdot \mathbf{u}_1)\mathbf{u}_1 - (\mathbf{v}_3 \cdot \mathbf{u}_2)\mathbf{u}_2}{\|\mathbf{v}_3 - (\mathbf{v}_3 \cdot \mathbf{u}_1)\mathbf{u}_1 - (\mathbf{v}_3 \cdot \mathbf{u}_2)\mathbf{u}_2\|} = \frac{1}{\sqrt{6}}(-2, 0, 1, 2, 1)$; complete to an orthonormal basis with $\mathbf{u}_4 = \frac{1}{\sqrt{2}}(0, 0, 1, 0, -1)$ and $\mathbf{u}_5 = (0, 0, 0, 1, 0)$ [U5 E' SBAGLIATO]. The matrix U mapping this basis to the standard basis is $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \ \mathbf{u}_4 \ \mathbf{u}_5]$ [CREDO SIA L'INVERSA DI U].

Exercise 2.9 (PS1 (2013), B-1 and B-2). (B-1) For an SVD $A = U\Sigma V^T$, true or false with explanation:

- (a) the number of non-zero diagonal entries of Σ equals rank A ;
 - (b) Σ is unique up to permutations;
 - (c) for square A one can choose $U = V$;
 - (d) for symmetric A one can choose $U = V$.
- (B-2) Prove that AA^T and $A^T A$ have the same non-zero eigenvalues.

Construction. $A^T A$ is symmetric PSD with rank r : by theorem 2.11 it has orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0 = \dots$. Put $\mathbf{u}_k = A\mathbf{v}_k / \sigma_k$ for $k \leq r$. These are orthonormal: $\mathbf{u}_i^T \mathbf{u}_k = \mathbf{v}_i^T A^T A \mathbf{v}_k / (\sigma_i \sigma_k) = \sigma_i^2 \mathbf{v}_i^T \mathbf{v}_k / (\sigma_i \sigma_k) = \delta_{ik}$. Complete $\{\mathbf{u}_k\}$ and $\{\mathbf{v}_k\}$ to orthonormal bases; then $U^T A V = \Sigma$ because $A\mathbf{v}_k = \sigma_k \mathbf{u}_k$ ($k \leq r$) and $A\mathbf{v}_k = 0$ ($k > r$) [► 2013 L2 @ 33:05]. □

Example 2.13 (SVD by hand [► 2013 L2 @ 47:05]). $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$. Then $AA^T = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$ has eigenvalues 25 and 9, so $\sigma_1 = 5$, $\sigma_2 = 3$ ($A^T A$ is 3×3 with eigenvalues 25, 9, 0). Eigenvectors of $A^T A$: $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T$, $\mathbf{v}_2 = \frac{1}{\sqrt{18}}(1, -1, 4)^T$; then $\mathbf{u}_1 = A\mathbf{v}_1/5 = \frac{1}{\sqrt{2}}(1, 1)^T$, $\mathbf{u}_2 = A\mathbf{v}_2/3 = \frac{1}{\sqrt{2}}(1, -1)^T$, and

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{18}} & -\frac{1}{\sqrt{18}} & \frac{4}{\sqrt{18}} \end{pmatrix}.$$

The eigenvector of eigenvalue 0 is not needed: this is the *reduced* SVD, keeping only the r non-zero singular values [► 2013 L2 @ 54:33].

FEEDBACK: ERROR

$\mathbf{u}_1$ correct; $\mathbf{u}_2$ is not. $\mathbf{v}_2 \cdot \mathbf{u}_1 = 2/\sqrt{3}$, so $\mathbf{v}_2 - \frac{2}{3}(1, 0, 1, 0, 1) = \frac{1}{3}(1, 3, 1, 0, -2)$ and $\mathbf{u}_2 = \frac{1}{\sqrt{15}}(1, 3, 1, 0, -2)$; then $\mathbf{u}_3 = \frac{1}{2\sqrt{10}}(-3, 1, 2, 5, 1)$. Your $\mathbf{u}_4, \mathbf{u}_5$ are not orthogonal to $\mathbf{u}_2, \mathbf{u}_3$. To complete, solve $\mathbf{v}_1 \cdot \mathbf{x} = \mathbf{v}_2 \cdot \mathbf{x} = \mathbf{v}_3 \cdot \mathbf{x} = 0$: the solutions are spanned by $(-1, 0, 1, -1, 0)$ and $(-1, 1, 0, -1, 1)$; Gram–Schmidt on them gives $\mathbf{u}_4 = \frac{1}{\sqrt{3}}(-1, 0, 1, -1, 0)$, $\mathbf{u}_5 = \frac{1}{2\sqrt{6}}(-1, 3, -2, -1, 3)$. Your doubt is right: $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_5]$ maps $\mathbf{e}_i \mapsto \mathbf{u}_i$; the map $\mathbf{u}_i \mapsto \mathbf{e}_i$ is $U^{-1} = U^T$. A-6 still to do (singular values: $2\sqrt{3}, \sqrt{10}, \sqrt{10}; 2, 1, 0$).

MY SOLUTION

B-1. a. true; b. true; c. i believe false, but not sure. i think it is true only if A is symmetric; d. true.

B-2. assume $AA^T v = \lambda v$, then we can construct a vector $v = (A^T)^{-1}u$ such that $A^T u = \lambda(A^T)^{-1}u$; multiply from the left by A^T to get what we want. Notice we assumed A^T is invertible.

Exercise 2.10 (PS1 (2013), B-4: least squares via SVD). Minimise $L = \|Ax - w\|$ for $A \in \mathbb{R}^{m \times n}$.

- (a) Solve it when A is diagonal.
- (b) Using $Ax - w = U(\Sigma V^T x - U^T w)$, reduce the general case to (a).
- (c) Show that fitting a line $y = ax + b$ to points (x_i, y_i) by least squares is a special case;
- (d) same for a parabola $y = ax^2 + bx + c$. (Continued in [chapter 5](#).)

MY SOLUTION

a. if A is diagonal, then the problem reduces to minimizing each component separately. For each diagonal entry a_{ii} , we have $L_i = |a_{ii}x_i - w_i|$. [Taking the derivative and setting it to zero gives the optimal solution for each component. WRONG! IT IS NOT DIFFERENTIABLE!]. it is always positive, and in a point it is equal to zero: $x_i^* = \frac{w_i}{a_{ii}}$

b. $V_{ij}^T x_j^* = \frac{U_{ij}^T w_j}{\sigma_i}$, then $x_k^* = \frac{(V^T)^{-1}_{ki} U_{ij}^T w_j}{\sigma_i}$.

c. NON HO CAPITO COSA FARE. ok ho chiesto a gemini una spiegazione. Dobbiamo scrivere la somma delle distanze assolute tra i punti e la retta, e minimizzare questa somma. $L = \sum_i |y_i - (a \cdot x_i + b)|$. NON SO COME ANDARE AVANTI.

d. same as c, but with a parabola instead of a line. BUT IN C I WAS STUCK.

FEEDBACK: ERROR

B-1: (a), (b) correct.

(c) False, and the condition is stronger than symmetric: $U = V$ gives $A = U\Sigma U^T$, symmetric and PSD.

(d) Hence false: $A = -I$ is symmetric with $\Sigma = I$, and $U = V$ would give $A = I$. For symmetric $A = Q\Lambda Q^T$ take $U = Q$, $V = Q \text{diag}(\text{sign } \lambda_i)$: they differ by the signs of the columns with $\lambda_i < 0$.

B-2: A^T is usually not invertible (not even square), so the proof must avoid it. Take $AA^T v = \lambda v$, $\lambda \neq 0$, and set $w = A^T v$. Then $A^T A w = A^T (AA^T v) = \lambda w$, and $w \neq 0$ because $w = 0$ would give $\lambda v = Aw = 0$. Same argument in the other direction.

FEEDBACK: ERROR

Key point: $\|\cdot\|$ is the Euclidean norm, so minimising L = minimising $L^2 = \sum_i ((Ax)_i - w_i)^2$, a sum of *squares*. Gemini's sum of absolute values is a different problem (least absolute deviations), with no closed form: that is why you got stuck.

(a) $L^2 = \sum_{i \leq n} (a_{ii}x_i - w_i)^2 + \sum_{i > n} w_i^2$: take $x_i = w_i/a_{ii}$ when $a_{ii} \neq 0$, anything (say 0) when $a_{ii} = 0$; the components w_i with no matching a_{ii} ($i > n$, or $a_{ii} = 0$) stay as residual.

(b) Missing step: U is orthogonal, so $\|Uy\| = \|y\|$ and $L = \|\Sigma z - U^T w\|$ with $z = V^T x$. This is (a): $z_i = (U^T w)_i / \sigma_i$ for $i \leq r$, and $x = Vz$, i.e. $x = V\Sigma^+ U^T w$ (pseudo-inverse). Your index formula is this, if the sum runs only over $i \leq r$.

(c) Least squares minimises $\sum_i (y_i - ax_i - b)^2 = \|Ax - w\|^2$ with rows $(x_i, 1)$ of A , $x = (a, b)^T$, $w = (y_1, \dots, y_N)^T$.

(d) Same with rows $(x_i^2, x_i, 1)$ and $x = (a, b, c)^T$: still linear in the unknowns.

2.8 The Perron–Frobenius theorem

Proof. Write $x \geq 0$ (resp. > 0) for componentwise inequalities. Following the slides, let $T = \{t \geq 0 : Ax \geq tx \text{ for some } x \geq 0, x \neq 0\}$ and $\lambda_0 = \sup T$. T contains $\min_i a_{ii} > 0$ and is bounded (if $Ax \geq tx$ with $\sum x_i = 1$, look at the largest component $x_k \geq 1/n$: $t \leq (Ax)_k / x_k \leq n \max a_{ij}$); by compactness of the simplex the sup is attained at some x^* .

Step 1: λ_0 is an eigenvalue with a positive eigenvector. Suppose $y = Ax^* - \lambda_0 x^* \geq 0$ were non-zero. Since all $a_{ij} > 0$, $Ay > 0$, i.e. $Az > \lambda_0 z$ for $z = Ax^* > 0$, and then $Az \geq (\lambda_0 + \varepsilon)z$ for small $\varepsilon > 0$, contradicting the maximality of λ_0 . So $Ax^* = \lambda_0 x^*$, and $x^* = Ax^* / \lambda_0 > 0$. The same argument shows: any $x \geq 0, x \neq 0$ with $Ax \geq \lambda_0 x$ is an eigenvector of λ_0 .

Step 2: $|\lambda| \leq \lambda_0$. If $Az = \lambda z$ (z possibly complex), then $|\lambda||z| = |Az| \leq A|z|$ componentwise, where $|z| = (|z_1|, \dots, |z_n|)$; hence $|\lambda| \in T$.

Step 3: strict inequality. If $|\lambda| = \lambda_0$, Step 2 gives $A|z| \geq \lambda_0 |z|$, so by Step 1 $A|z| = \lambda_0 |z|$ and $|z| > 0$. Then $|\sum_j a_{ij} z_j| = \sum_j a_{ij} |z_j|$ for each i : equality in the triangle inequality with positive weights forces all z_j to have the same phase, $z = e^{i\phi} |z|$. Substituting, $\lambda|z| = A|z| = \lambda_0 |z|$, so $\lambda = \lambda_0$.

Step 4: simplicity. By Step 3 every eigenvector of λ_0 is a multiple of a positive vector. If $z > 0$ were an eigenvector not proportional to x^* , let

Theorem 2.14 (Perron–Frobenius). Let $A \in \mathbb{R}^{n \times n}$ have all entries $a_{ij} > 0$. Then:

- (i) there is a real eigenvalue $\lambda_0 > 0$ with $|\lambda| < \lambda_0$ for every other eigenvalue λ ;
- (ii) λ_0 has an eigenvector with all entries positive;
- (iii) λ_0 is simple: its eigenvectors form a one-dimensional space.

$c = \min_i x_i^*/z_i$; then $w = x^* - cz \geq 0$ has a zero component and $w \neq 0$, yet $w = Aw/\lambda_0 > 0$: contradiction. $\square$

Applications in this course.

- **Markov chains:** a column-stochastic A with some $A^k > 0$ has $\lambda_0 = 1$ (proposition 2.2), a unique positive stationary distribution, and $\pi(t) \rightarrow \pi^*$ geometrically (2024 PS1 Problem 4).³⁶
- **Covariance matrices** with all positive entries: the first principal component has all weights positive, a “market factor” (2024 PS2 Problem 3, chapter 7).³⁷
- **Ross recovery theorem:** recovering real-world probabilities from option prices relies on Perron–Frobenius applied to a matrix of state prices [► 2013 L2 @ 1:01:54](chapter 25).³⁸

2.9 Case study: equal-weighted portfolios

CASE STUDY: Script_SP500.Rmd and Equal_Weighted_Portfolios.Rmd

Data. Script_SP500.Rmd downloads (tidyquant) daily prices of the stocks currently in the S&P 500 from 2019-01-01 to 2024-08-31 and keeps those with no missing data: 1426 days $\times$ 487 stocks (R prints 1426 $\times$ 488: the first column is the date), saved in SP500_data_subset.RData.

Portfolio as a matrix–vector product. Invest \$1000 equally at the first date: $q_j = (1000/n)/p_j(0)$ shares, then *never rebalance* (buy and hold). With P the 1426 $\times$ n matrix of prices, the value path is simply $V = Pq$ [► 2024 L4 @ 23:31]. The same is done for four “hot” stocks (AMZN, AAPL, NFLX, GOOG), \$250 each.

Results (from the knitted output). The \$1000 in the four stocks grows to \$3541.76 by 2024-08-30; the four \$250 stakes end at AAPL \$1516.55, GOOG \$790.26, NFLX \$655.08, AMZN \$579.87. The concentrated portfolio beats the equal-weighted S&P 500 portfolio (which ends at \$2631.59) but is riskier, being less diversified: it “decreases rather dramatically at some points” [► 2024 L4 @ 24:33]. From the data: annualised volatility of daily returns 28% against 22%, and it fell 41% (from \$3012 at the end of 2021 to \$1765 at the end of 2022) and regained its peak only in January 2024, while the S&P 500 portfolio’s worst fall was the Covid crash of early 2020 (–38%), recovered within about five months. See fig. 2.5.

35 COURSE NOTE: On the proofs in the course material

The slides prove Steps 1–2 as above and argue simplicity from “ $x - y$ has entries of both signs”; this needs the fact, proved in Step 3, that every eigenvector of λ_0 has entries of one sign. The 2013 notes only sketch the symmetric case. The theorem also holds (with $|\lambda| \leq \lambda_0$) for non-negative *irreducible* matrices, the form needed for Markov chains [► 2013 L2 @ 1:10:23]. The eigenvalue is in fact algebraically simple as well (not shown here).

36 non ho capito

ANSWER 2.32: Markov chains and Perron–Frobenius

Unpacked (details in section 2.3):

- $\pi(t)$ = probabilities of the states at time t , and $\pi(t) = A^t \pi(0)$.
- *Column-stochastic*: each column of A sums to 1 (from state j you go somewhere). Then $\mathbf{1}^T A = \mathbf{1}^T$, so 1 is an eigenvalue.
- “Some $A^k > 0$ ”: every state can be reached from every state in k steps; then Perron–Frobenius applies to A^k .
- P–F says: the eigenvalue 1 is the largest, simple, with a positive eigenvector π^* , which normalised to sum 1 is the stationary distribution ($A\pi^* = \pi^*$).
- All other $|\lambda_i| < 1$, so in $A^t \pi(0) = \sum_i c_i \lambda_i^t v_i$ every other mode dies like $|\lambda_2|^t$ (“geometrically” = like a geometric sequence) and $\pi(t) \rightarrow \pi^*$.

Exercise 2.11 and the next exercise are exactly this, with $\lambda_2 = 0.5$.

37 significa che non c’è nessuna posizione short? la first principal component è quella che domina lon-time behaviour?

ANSWER 2.33: No shorts

Yes: all weights have the same sign (an eigenvector is defined up to sign, so take them positive), so the first principal component is a long-only portfolio, roughly “buy the whole market”. It describes the common movement: when the market rises, almost all stocks rise together.

ANSWER 2.34: Long-time behaviour?

No, here nothing evolves in time: Σ is not applied repeatedly. The first component *dominates the variance*: it is the direction along which the returns spread the most. P–F gives the same mathematical object in both cases (the top eigenvector), but in a Markov chain it is the long-run limit, in PCA the largest-variance direction. (It *would* dominate $\Sigma^t x$ for large t , which is how the power method computes it.)

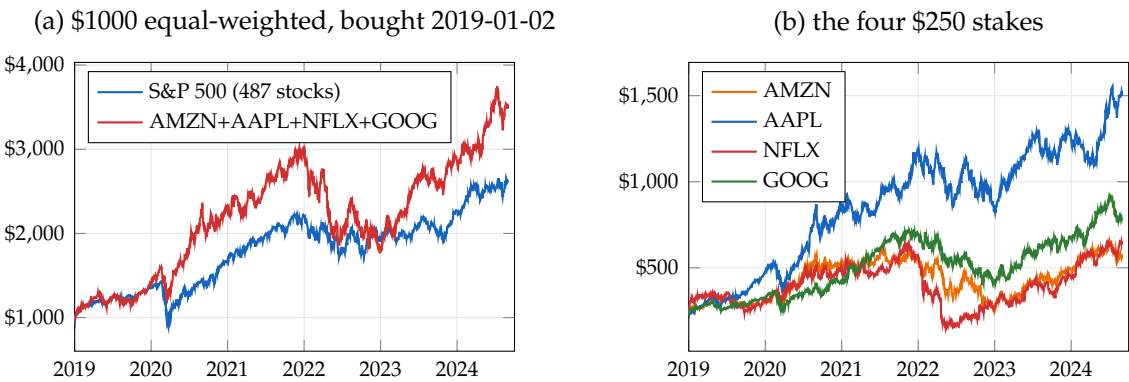
38 perche’ si fa sempre riferimento a options e non a derivatives in generale?

ANSWER 2.35: Why options

Because options are the derivatives that reveal *probabilities*. A forward or future on the stock has a linear payoff $S_T - K$: its price gives only one number, the (risk-neutral) mean of S_T . Calls with many strikes K have kinked payoffs $(S_T - K)^+$, and from their prices one recovers the whole distribution: the price of \$1 paid if S_T is near K is $\partial^2 C / \partial K^2$ (Breeden–Litzenberger, 1978, chapter 22). Every other payoff of S_T is a combination of calls (they are a “basis”), and index options are among the most liquid derivatives, with dozens of strikes listed. So “options” is not a restriction: they are the building blocks.

DONE

39 aggiungi grafici



SEEN
40 makes sense

Figure 2.5: The buy-and-hold portfolios of `Equal_Weighted_Portfolios.Rmd`, recomputed from `SP500_data_subset.RData` (daily, 2019-01-02 to 2024-08-30). (a) The four “hot” stocks end at \$3541.76, the equal-weighted S&P 500 at \$2631.59, but with larger swings. (b) Apple alone ends at \$1516.55, 43% of the four-stock portfolio: the weight drift discussed below.

Weight drift and rebalancing. Without rebalancing, the weight of asset j at time t is

$$w_j(t) = \frac{q_j p_j(t)}{\sum_k q_k p_k(t)} = \frac{w_j(0) G_j(t)}{\sum_k w_k(0) G_k(t)}, \quad G_j(t) = \frac{p_j(t)}{p_j(0)},$$

so winners take over: Apple’s weight goes from 25% to $1516.55/3541.76 \approx 43\%$, and the portfolio becomes less diversified and riskier [► 2024 L4 @ 25:37]. Rebalancing every day back to equal weights is not obviously better: it systematically sells winners and buys losers, which hurts when there are trends [► 2024 L4 @ 26:40]. One principled alternative is to rebalance so as to keep the *risk* of the portfolio at a target level (chapter 10).⁴⁰

IN FINANCE: Survivorship bias (not discussed in class)

The script uses the stocks that are in the S&P 500 *today* and have complete histories since 2019. Companies that were in the index in 2019 but later dropped out (often because they performed badly) are excluded, so the back-test of the equal-weighted portfolio is biased upward. Any historical study built on current index membership has this problem.

2.10 Lectures without material

SEEN
41 importance of awareness of your data!

42 COURSE NOTE: 2024 L3 and 2013 L4

2024 L3, *Quantitative Equity Investing* (Jeff Shen, BlackRock): not recorded and no slides published (the guest lecturer did not grant IP permission).⁴³

2013 L4, *Matrix Primer* (Morgan Stanley Matrix team): no notes and no video; OCW only points to the Morgan Stanley Matrix microsite (the bank’s electronic trading and research platform for clients).⁴⁴

KEY FORMULAS

Portfolio value, P&L:	$V_t = \mathbf{q}^\top \mathbf{p}(t), \quad \text{PnL}(t+1) = \mathbf{q}(t+1)^\top \Delta \mathbf{p}(t)$
Markov chain:	$\boldsymbol{\pi}(t) = A^t \boldsymbol{\pi}(0), \quad \boldsymbol{\pi}^* = A \boldsymbol{\pi}^*$
Two-state chain:	$\lambda_2 = 1 - p - q, \quad \boldsymbol{\pi}^* = (q, p)/(p + q)$
Binomial replication:	$\theta_S = \frac{C^u - C^d}{S^u - S^d},$ $C_0 = \frac{q_u C^u + q_d C^d}{1 + r_f T},$ $q_u = \frac{(1 + r_f T) S_0 - S^d}{S^u - S^d}$
State prices:	$A_0 = \boldsymbol{\psi}^\top A, \quad \boldsymbol{\psi} > 0 \iff \text{no arbitrage};$ $C_0 = \boldsymbol{\psi}^\top C_T = \beta \mathbb{E}_Q[C_T]$
Completeness:	$\text{rank } A = m \iff \boldsymbol{\psi} \text{ unique}$
Diagonalisation:	$A = S \Lambda S^{-1}, \quad A^t \mathbf{u}_0 = \sum_i c_i \lambda_i^t \mathbf{v}_i$
Spectral theorem:	$A = A^\top \Rightarrow A = U \Lambda U^\top, \quad U^\top U = I$
SVD:	$A = U \Sigma V^\top = \sum_{k \leq r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top,$ $\sigma_k^2 = \lambda_k(A^\top A)$

Exercises

From 2024 Problem Set 1

Exercise 2.11 (PS1 (2024), Problem 3: order-side Markov chain). The side of successive market orders is a two-state chain (1 = buy, 2 = sell) with $A = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$, $a_{ij} = \mathbb{P}(X_t = i \mid X_{t-1} = j)$.⁴⁵

- Prove $\mathbf{u}_t = A \mathbf{u}_{t-1}$ and $\mathbf{u}_t = A^t \mathbf{u}_0$ for the distribution $\mathbf{u}_t$ of X_t .
- Starting from $X_0 = 1$, compute $\mathbf{u}_1, \dots, \mathbf{u}_5$ in R (A %% ut in a loop) and increase t until convergence; repeat from $X_0 = 2$. Is the limit the same?
- What is the stationary distribution of the order side? Relate it to the eigenvector of eigenvalue 1 returned by `eigen(A)` (which is normalised to unit Euclidean length, not to sum 1).

MY SOLUTION

- mi sembra ovvio, non capisco cosa intendi con "prove".
- python code: [Google Colab](#) the first initial condition converges faster.
- $\boldsymbol{\pi} = A \boldsymbol{\pi} \implies \boldsymbol{\pi} = (0.6, 0.4)^T$, look at the link

Exercise 2.12 (PS1 (2024), Problem 4). For the matrix of the previous exercise:

- find its eigenvalues explicitly;
- explain which case of Problem 2 applies;

43 what is IP?

ANSWER 2.36: IP

Intellectual property: the legal rights (copyright, trademarks, patents, trade secrets) on creations of the mind. MIT can publish a lecture only if the speaker grants permission; a BlackRock speaker's slides contain the firm's proprietary methods, so they were not published.

44 what is OCW?

ANSWER 2.37: OCW

MIT *OpenCourseWare* (<https://ocw.mit.edu>): MIT's free online publication of course material (notes, slides, problem sets, videos), started in 2002. Both courses of these notes come from there.

SEEN

45 imagine every market order is an event, and each one can be buy or sell

FEEDBACK: ERROR

(a) "Prove" means: say which rule gives it. By the law of total probability, $u_t(i) = \mathbb{P}(X_t = i) = \sum_j \mathbb{P}(X_t = i \mid X_{t-1} = j) \mathbb{P}(X_{t-1} = j) = \sum_j a_{ij} u_{t-1}(j)$, using that the transition probabilities depend only on the last state and not on t (Markov, time-homogeneous). Then $\mathbf{u}_t = A^t \mathbf{u}_0$ by induction.
 (b) You did not state the answer to "is the limit the same?": yes, $(0.6, 0.4)$ from both starts. "Converges faster" is not quite right: $\mathbf{u}_t - \boldsymbol{\pi}^* = 0.5^t (\mathbf{u}_0 - \boldsymbol{\pi}^*)$ and $\mathbf{u}_0 - \boldsymbol{\pi}^*$ is $(0.4, -0.4)$ from state 1 and $(-0.6, 0.6)$ from state 2. Same rate 0.5^t ; state 1 just starts closer.
 (c) $\boldsymbol{\pi}^*$ correct. Write the relation in the notes too: `eigen(A)` returns $(0.832, 0.555)$, a multiple of $(3, 2)$, and dividing by the sum of its entries gives $(0.6, 0.4)$.

- (c) explain the link between the stationary distribution and the Perron–Frobenius theorem.

MY SOLUTION

- a. done in the google colab link.
 b. the eigenvalues are 1 and 0.5, so the limit exists and is non-zero, case C.
 c. (i) the eigen value 1 is the biggest, (ii) its eigenvector has all positive entries, then it can be normalized and interpreted as prob distribution, (iii) it is simple, i.e. the probability distribution is unique.

FEEDBACK

(b), (c) correct.
 (a) By hand it takes one line: $\text{trace} = 1.5$, $\det = 0.56 - 0.06 = 0.5$, so $\lambda^2 - 1.5\lambda + 0.5 = 0$, $\lambda \in \{1, 0.5\}$.
 (c) One missing link: why is the P–F eigenvalue equal to 1? Because the columns sum to 1, $\mathbf{1}^T A = \mathbf{1}^T$, so 1 is an eigenvalue with a positive (left) eigenvector; P–F's λ_0 is the only eigenvalue with a positive eigenvector, so $\lambda_0 = 1$. And add the consequence: the other $|\lambda| < 1$, so $\mathbf{u}_t \rightarrow \pi^*$ from any start (that is your exercise 3(b)).

From 2013 Problem Set 1

Exercise 2.13 (PS1 (2013), A-1). True or false:

- (a) row rank equals column rank;
 (b) $\text{rank}(AB) \leq \text{rank}(A)$;
 (c) there are matrices with $BA = I$, $AC = I$ and $B \neq C$;
 (d) full rank iff invertible;
 (e) symmetric matrices are invertible;
 (f) distinct eigenvalues imply invertible;
 (g) diagonalisable iff invertible.

MY SOLUTION

- a. vero, ma non so perche'
 b. vero, se A proietta un vettore su uno spazio n-dim, allora applicare B non puo enlarge la dimensione dello spazio immagine.
 c. falso, moltiplica per C da destra la prima equazione e ottieni $B = C$
 d. vero, ho solo intuizione su questa domanda, per me rank e' la dimensione dello spazio immagine
 e. falso, la matrice zero e' simemtrica ma non invertibile.
 f falso, non-zero eigenvalues imply invertible, anche se due sono uguali, significa soltanto che in due direzioni (un piano) [o piu direzioni] la matrice scala della stessa quantita'.
 g. credo sia vera, ma non so perche'

FEEDBACK: ERROR

(a) True. Why: if the column space has dimension r , write $A = CR$ with C ($m \times r$) a basis of it; every row of A is then a combination of the r rows of R , so $\text{row rank} \leq r$; the same on A^T gives the reverse.
 (b) True, but the order is inverted: AB applies B first. Reason: $ABx = A(Bx)$, so $\text{col}(AB) \subseteq \text{col}(A)$.
 (c) Correct, and the proof is just $B = B(AC) = (BA)C = C$.
 (d) Only for square matrices: a 2×3 matrix of rank 2 has full rank but no inverse.
 (e) Correct.
 (f) False, but your reason does not prove it: distinct eigenvalues can include 0, e.g. $\text{diag}(0, 1)$. What matters for invertibility is only $0 \notin \{\lambda_i\}$.
 (g) **False**, in both directions: $\text{diag}(0, 1)$ is diagonalisable and singular; $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is invertible but not diagonalisable (one eigenvector only). The two properties are unrelated.

Chapter 3

Probability Theory

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L4 (from 27:40)	P. Kempthorne	Discrete, continuous and mixed variables; CDF and probability integral transform; moments, skewness, kurtosis; normal and lognormal laws, change of variables; expected call payoff; moment-generating functions; independence, covariance, random vectors and portfolio variance; start of PCA.
2024 L5 (to 1:12:25)	P. Kempthorne	PCA of a random vector; χ^2 , t and F laws; weak law of large numbers, central limit theorem; <i>Probability Theory for Asset Pricing</i> : Stein's lemma, expected-utility pricing, CARA utility, CAPM. (The rest of L5 is in chapter 4 .)
2013 L3	C. Lee	pmf/pdf, mutual vs pairwise independence; why prices are lognormal; change of variables; exponential families; MGFs and the moment problem; weak LLN (casino and poker); CLT with proof.

Files. Slides *Probability Theory* [mit18_642_f24_lec04_1.pdf](#); slides *Probability Theory for Asset Pricing* [mit18_642_f24_lec04_2.pdf](#); 2013 notes [MIT18_S096F13_lecnote3.pdf](#). Suggested readings on the 2024 Week 3 page: J. Lintner (1965), *The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets*, and *Security prices, risk, and maximal gains from diversification*. **Problem sets.** 2024 PS2 Problem 2; 2024 PS3 Problems 1–2; 2013 PS2 A-1, A-2, B-1 to B-4. The other problems of these sets (random walks, Markov chains, martingales, PCA of yields) are in [chapters 4](#) and [7](#). **Not recorded.** Nothing: all three lectures are on video.

Motivation

Every model in this course is a probability model. This chapter collects the probability that the rest of the notes take for granted, organised around four questions that a quant asks about a position:

- *What is the distribution of a price or a return?* Normal for log-returns, lognormal for prices, and in reality heavier tails than either ([sections 3.3](#) and [3.9.1](#)). The distribution fixes the expected payoff of an option ([section 3.4](#)).
- *How do assets move together?* Covariance matrices, portfolio variance $a^\top \Sigma a$, diversification, the multivariate normal law and its principal components ([sections 3.6](#) to [3.8](#)).
- *What survives aggregation?* Laws of large numbers and the central limit theorem explain why a small edge repeated many times makes money and why log-returns

over long horizons look Gaussian (section 3.10).

- *What should an asset cost?* With nothing more than expectations, covariances and one Gaussian identity (Stein's lemma), an expected-utility argument yields the capital asset pricing model (section 3.11).

A physicist can skim sections 3.1 and 3.2 and the definitions of section 3.9; the less standard material is the option-payoff formula, the pitfalls of moment-generating functions, conditional expectation, Stein's lemma and the CAPM derivation.

3.1 Random variables and distribution functions

Financial random variables come in three kinds [► 2024 L4 @ 28:47]:

- *discrete*: default (1) or no default (0) of a counterparty, the FOMC decision on the Fed funds rate, the side (buy/sell) and the share size of the next market order in AAPL;¹
- *continuous*: the value of an asset, the waiting time until the next market order [► 2024 L4 @ 29:49];
- *mixed*: the value of a stock in six months, which is continuous except for an atom at 0 (bankruptcy) or at the buyout price (the company is taken private) [► 2024 L4 @ 30:55].

Definition 3.1 (Distribution of a random variable). Let $\mathcal{X}$ be the set of possible values (the *sample space*) of X .

- If X is discrete, its *probability mass function* (pmf) is $f_X(x) = \mathbb{P}(X = x) \geq 0$ with $\sum_{x \in \mathcal{X}} f_X(x) = 1$.
- If X is continuous, its *probability density function* (pdf) $f_X \geq 0$ satisfies $\mathbb{P}(X \in [x, x + dx]) = f_X(x) dx$ and $\int_{\mathcal{X}} f_X(x) dx = 1$.
- The *cumulative distribution function* (CDF) is $F_X(x) = \mathbb{P}(X \leq x)$ [► 2024 L4 @ 31:56]. It is non-decreasing, right-continuous, with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$, and it determines the distribution.

Then $\mathbb{P}(X \in A) = \sum_{x \in A} f_X(x)$ or $\int_A f_X(x) dx$, and $F_X(x) = \int_{-\infty}^x f_X(u) du$ in the continuous case, so $f_X = F'_X$.

Remark 3.2. Both courses identify the sample space with the set of values of X (2013 notes, section 1). The measure-theoretic set-up (a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and X a measurable map $\Omega \rightarrow \mathbb{R}$) is not needed here; it becomes useful for martingales and filtrations in chapter 4.

Independence. X and Y are *independent* if $\mathbb{P}(\{X \in A\} \cap \{Y \in B\}) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$ for all sets A, B ; equivalently, the joint pmf/pdf factorises, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ [► 2013 L3 @ 6:16]. For n variables one must distinguish [► 2013 L3 @ 7:17]: $X_1, \dots, X_n$ are *mutually independent* if $\mathbb{P}(\cap_i \{X_i \in A_i\}) = \prod_i \mathbb{P}(X_i \in A_i)$ for all $A_1, \dots, A_n$, and *pairwise independent* if every pair is independent. Mutual implies pairwise, not conversely (Exercise 3.7).² "Independent" always means mutually independent.

1 what is FOMC?

ANSWER 3.1: FOMC

The *Federal Open Market Committee*: the body of the US Federal Reserve that sets the target for the Fed funds rate (the overnight rate between banks). It meets eight times a year; each decision (cut, hold, hike, usually by 0.25%) is a discrete random variable for markets. "Open market" because the Fed implements the decision by buying and selling bonds in the open market.

SEEN

2 interesting new subtlety

Proof. If F is strictly increasing, $\mathbb{P}(F(X) \leq u) = \mathbb{P}(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$ for $u \in (0, 1)$. In general, with $q = F^{-1}(u)$ continuity gives $F(q) = u$, and $\{F(X) \geq u\} = \{X \geq q\}$ (if $X \geq q$

Proposition 3.3 (Probability integral transform). Let X have a continuous CDF F . Then $U = F(X)$ is uniform on $(0, 1)$ [► 2024 L4 @ 32:57]. Conversely, if $U \sim \text{Uniform}(0, 1)$ and $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ is the quantile function of an arbitrary CDF F , then $F^{-1}(U)$ has CDF F .

then $F(X) \geq F(q) = u$; if $X < q$ then $F(X) < u$ by definition of the infimum, so $\mathbb{P}(F(X) \geq u) = 1 - F(q) = 1 - u$. For the converse, $F^{-1}(u) \leq x \iff u \leq F(x)$ by right-continuity, so $\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$. $\square$

IN FINANCE: Uses of the probability integral transform

- *Model checking* [► 2024 L4 @ 33:57]: if a risk model predicts that tomorrow's return has CDF F_t , the numbers $u_t = F_t(x_t)$ computed from realised returns should look like an i.i.d. uniform sample. A histogram of the u_t piling up near 0 and 1 reveals tails that are too thin: this is how value-at-risk and density forecasts are back-tested (chapter 19).³
- *Simulation*: $F^{-1}(U)$ turns uniform random numbers into draws from any distribution (inverse-transform sampling).
- *Copulas* (not discussed in class): applying F_i to each coordinate of a random vector separates the marginal laws from the dependence structure.⁴

SEEN

3 cool way to spot the heavy tails

4 non credo di aver capito quest'ultima

ANSWER 3.2: Copulas

Take two returns (X, Y) and set $U = F_X(X)$, $V = F_Y(Y)$. By proposition 3.3 each of U, V is uniform: all the information about the marginal laws (fat tails, skew) has been removed. What is left, the joint law of (U, V) on the unit square, is the pure dependence: the copula (Latin, "link"). Sklar's theorem: $F_{X,Y}(x, y) = C(F_X(x), F_Y(y))$. So one can model each asset's tails and their co-movement separately, e.g. fat-tailed marginals glued with a Gaussian copula (the model blamed for CDO losses in 2008).

3.2 Expectation, moments and the shape of a distribution

Definition 3.4 (Moments, skewness, kurtosis). For a random variable X (sums replace integrals in the discrete case):

- mean $\mu = \mathbb{E}[X] = \int x f_X(x) dx$; k -th moment $m_k = \mathbb{E}[X^k]$;
- variance $\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = m_2 - m_1^2$ and standard deviation $\sigma = \sqrt{\text{Var}(X)}$, which has the units of X [► 2024 L4 @ 35:03];
- with the standardised variable $Z = (X - \mu)/\sigma$ [► 2024 L4 @ 36:07], the skewness $\gamma = \mathbb{E}[Z^3]$ and the kurtosis $\kappa = \mathbb{E}[Z^4]$, both dimensionless [► 2024 L4 @ 37:12].

$\gamma > 0$ indicates a long right tail, $\gamma < 0$ a long left tail. The normal law has $\gamma = 0$ and $\kappa = 3$; X is leptokurtic (fat-tailed relative to the Gaussian) if $\kappa > 3$ [► 2024 L4 @ 38:18]. The excess kurtosis is $\kappa - 3$.

Proposition 3.5 (Affine transformations). If $Y = a + bX$ with $b \neq 0$, then $\mathbb{E}[Y] = a + b\mu$, $\text{Var}(Y) = b^2\sigma^2$, $\gamma_Y = \text{sign}(b)\gamma_X$ and $\kappa_Y = \kappa_X$.⁵

Proof. $(Y - \mathbb{E}[Y])/\sigma_Y = b(X - \mu)/(|b|\sigma) = \text{sign}(b)Z_X$; take third and fourth powers. $\square$

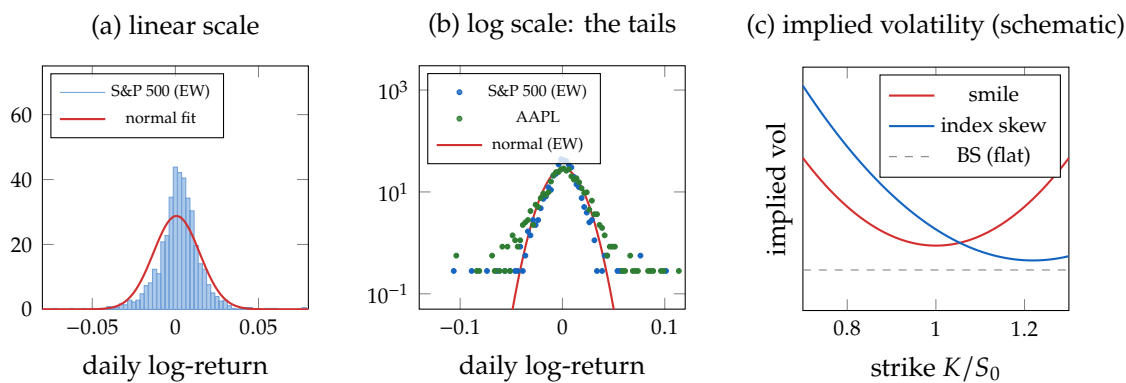
5 COURSE NOTE: Erratum (in class) and a caveat on the slides

At [► 2024 L4 @ 1:02:42] it is said that linear transformations do not change the skewness but might change the kurtosis. It is the other way round (proposition 3.5): kurtosis is invariant, skewness changes sign when $b < 0$ (a short position has the opposite skew of the long one). Also, slide 6 of lec04_1 lists " $\gamma = 0$: X is symmetric about μ ". Symmetry implies $\gamma = 0$, but not conversely: $X \in \{-3, -1, 2\}$ with probabilities 0.1, 0.5, 0.4 has $\mathbb{E}[X] = \mathbb{E}[X^3] = 0$ and is not symmetric. Software differs on

whether “kurtosis” means κ or $\kappa - 3$: check before comparing numbers.

IN FINANCE: What the moments of returns mean

For a return, the mean is the drift and the standard deviation the *volatility* (quoted annualised). Equity-index returns are typically negatively skewed (crashes are sudden, rallies are gradual) and daily returns of almost all assets are strongly leptokurtic. Both facts matter for risk: a model matched only to mean and variance underestimates the probability of large losses (section 3.9.1), and they are visible in option markets as the volatility smile (chapter 13).



6 can you include some graphs of: a typical probability distribution of an equity index, of a daily return of an asset. what is the volatility smile?

ANSWER 3.3: Graphs

Done in fig. 3.1, with the course’s own data (daily closes of the S&P 500 stocks, 2019–2024). The equal-weighted portfolio of all 487 stocks stands in for the index.

ANSWER 3.4: Volatility smile

Take the Black–Scholes formula, which has one unknown, σ . From the market price of each option solve for the σ that reproduces it: the *implied volatility*. If returns were normal (BS true), it would be the same number for every strike. It is not: plotted against the strike it is a curve, U-shaped (a “smile”) or, for equity indices, falling from left to right (a “skew”, or “smirk”), sketched in panel (c). Options far from the money are dearer than BS says, because the market knows the tails are fat (and, for indices, that crashes go down). Full story in chapter 13.

Figure 3.1: Daily log-returns, 2019-01-02 to 2024-08-30, from SP500_data_subset.RData (the data of the case study in section 2.9). (a) The equal-weighted S&P 500 portfolio against the normal with the same mean and standard deviation (1.39% a day): a sharper peak and fatter tails; skewness -0.91 , kurtosis 17 . (b) On a log scale the normal is a parabola and the data are not: both the index and Apple (kurtosis 8.5) have 11 days beyond 4σ , where the normal would expect 0.09 . The extreme left points are March 2020. (c) A sketch, not data, of the volatility smile and of the skew typical of index options (chapter 13).

3.3 Normal and lognormal distributions

3.3.1 The normal distribution

Definition 3.6 (Normal law). $X \sim N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$, if

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

ϕ and Φ denote the density and CDF of the standard normal $N(0, 1)$; $\Phi(-x) = 1 - \Phi(x)$.⁷

The density is symmetric about μ , maximal at μ , with inflection points at $\mu \pm \sigma$ [► 2024 L4 @ 39:29]; the probabilities of lying within 1, 2, 3 standard deviations are 0.6827, 0.9545, 0.9973 [► 2024 L4 @ 40:31]. That f integrates to 1 is the Gaussian integral (2013 PS2 B-2, Exercise 3.2); see fig. 3.2.⁸

7 perche’ non restiamo con una notazione unica? $\phi = f_{\text{normal}}$ e $\Phi = F_{\text{normal}}$

ANSWER 3.5: Why ϕ, Φ

Because ϕ and Φ are universal: every paper and the Black–Scholes formula use them, so you must recognise them. They are also a special case, not a different notation: for $Z \sim N(0, 1)$, $f_Z = \phi$ and $F_Z = \Phi$; for $X \sim N(\mu, \sigma^2)$, $F_X(x) = \Phi((x - \mu)/\sigma)$.

DONE

8 insert a graph of the distribution. in general, whenever it can be useful plot graphs and functions.

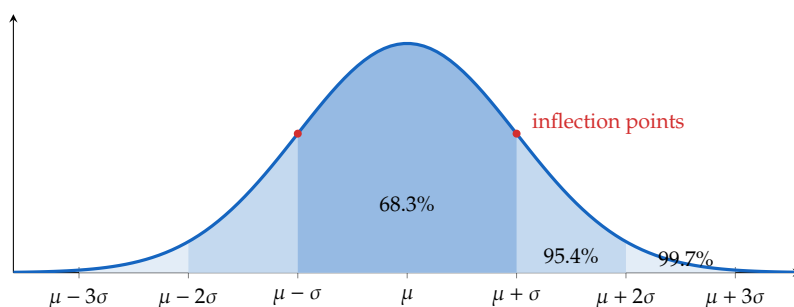


Figure 3.2: The normal density $N(\mu, \sigma^2)$ with the probability of lying within 1, 2, 3 standard deviations of the mean. The inflection points are at $\mu \pm \sigma$.

3.3.2 Why prices are not normal

Suppose daily price changes $P_n - P_{n-1}$ were i.i.d. normal [► 2013 L3 @ 10:28]. Two things go wrong [► 2013 L3 @ 11:30]:

- the size of a move would not depend on the price level, whereas a \$50 stock typically moves about five times as many dollars as a \$10 stock: it is the *percentage* change that is roughly stationary [► 2013 L3 @ 12:32];
- P_n would be normal with variance growing like n , so it eventually becomes negative with substantial probability [► 2013 L3 @ 19:05]. Locally (short horizons, small moves) the additive model is harmless.

The remedy is to model *log-returns* $r_n = \ln(P_n/P_{n-1})$ as i.i.d. normal. Then $\ln P_n = \ln P_0 + \sum_{i \leq n} r_i$ is normal (sums of independent normals are normal, [section 3.5](#)), and P_n is *lognormal*; positivity is automatic [► 2024 L4 @ 41:34]. In continuous time the same construction with Brownian increments gives geometric Brownian motion [► 2024 L4 @ 42:35]([chapters 17 and 18](#)).

3.3.3 Change of variables

Theorem 3.7 (Change of variables). Let X have density f_X , let g be differentiable and strictly monotone with inverse $h = g^{-1}$, and $Y = g(X)$. If g is increasing, $F_Y(y) = F_X(h(y))$ [► 2024 L4 @ 43:35]; in all cases

$$f_Y(y) = f_X(h(y)) |h'(y)|.$$

Proof. If g is increasing, $\{Y \leq y\} = \{X \leq h(y)\}$, so $F_Y = F_X \circ h$; differentiating with the chain rule and $F'_X = f_X$ gives $f_Y = f_X(h) h'$ [► 2024 L4 @ 44:43]. If g is decreasing, $F_Y(y) = \mathbb{P}(X \geq h(y)) = 1 - F_X(h(y))$ and $f_Y = -f_X(h)h' = f_X(h)|h'|$. $\square$

The 2013 notes (Theorem 1.3) call h' the *Jacobian* [► 2013 L3 @ 16:47]; in $\mathbb{R}^n$ it becomes $f_Y(\mathbf{y}) = f_X(\mathbf{h}(\mathbf{y})) |\det D\mathbf{h}(\mathbf{y})|$.

INTUITION: Conservation of probability

Probability is a conserved “mass”: $f_Y(y) |dy| = f_X(x) |dx|$. Where g stretches the axis the density thins out, where it compresses the axis the density piles up. This is the same bookkeeping as for a particle density under a change of coordinates.

3.3.4 The lognormal distribution

Definition 3.8 (Lognormal law). Y is lognormal(μ, σ^2) if $\ln Y \sim N(\mu, \sigma^2)$, i.e. $Y = e^X$ with $X \sim N(\mu, \sigma^2)$ [► 2024 L4 @ 41:34]. By [theorem 3.7](#) with $h(y) = \ln y$, $h'(y) = 1/y$ [► 2024 L4 @ 45:45]:

$$f_Y(y) = \frac{1}{y \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right), \quad y > 0$$

[► 2024 L4 @ 46:56][► 2013 L3 @ 21:12].

Properties (the first three from monotonicity of $\ln$ and a derivative):

- quantiles map to quantiles: $q_p(Y) = e^{\mu + \sigma z_p}$ with $z_p = \Phi^{-1}(p)$; the median is e^μ ;
- the mode is $e^{\mu - \sigma^2}$;
- $\mathbb{E}[Y^k] = e^{k\mu + k^2\sigma^2/2}$ for every real k ([section 3.5](#)); in particular $\mathbb{E}[Y] = e^{\mu + \sigma^2/2}$, so mode < median < mean: the law is skewed to the right;
- products and powers: if Y_1, Y_2 are independent lognormal, $Y_1 Y_2$ and Y_1^c are lognormal.

The parameters μ, σ are the mean and standard deviation of $\ln Y$, *not* of Y [► 2013 L3 @ 23:15]; they are kept because the law is defined through the logarithm [► 2013 L3 @ 24:18].

Exercise 3.1 (Try first: lognormal(0.2, 0.4²), slides 15–16 of [lec04_1](#)). Let $X_0 = 1$ and X_1 lognormal with $\mu = 0.2$, $\sigma = 0.4$: a one-year log-return with mean 0.2 and standard deviation 0.4 [► 2024 L4 @ 47:56]. With $z_{0.95} = 1.645$ and $z_{0.75} = 0.674$:

- Find the 5%, 25%, 50%, 75% and 95% quantiles of X_1 .
- Find its mode, mean and standard deviation, and sketch the density.
- In what sense are the quantiles symmetric about the median?

Solution →

IN FINANCE: Log-returns and the square-root-of-time rule

Log-returns add over time: $\ln(S_T/S_0) = \sum_{\text{days}} r_i$. If daily log-returns are i.i.d. with mean μ_d and variance σ_d^2 , the T -day log-return has mean $T\mu_d$ and standard deviation $\sigma_d\sqrt{T}$. This is why volatilities are quoted annualised and scaled by $\sqrt{t}$ (as in 2024 PS2 Problem 2(c), where $\sigma = \sqrt{t} \ln 1.3$), and why the Black–Scholes model takes S_T lognormal ([chapter 15](#)). For small moves the simple return $e^r - 1 \approx r$.

Exercise 3.2 (PS2 (2013), B-2).

- Verify that $\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$ is a probability density (compute $(\int \phi)^2$ in polar coordinates).
- Compute the mean and variance of the lognormal law (it suffices to treat $\mu = 0$; use (a) to simplify). (This is also Exercise 1.5(i) of [lecnote3](#); Exercise 1.2 there asks to verify that the mean of $N(\mu, \sigma^2)$ is μ .)

Solution →

9 vorrei che la notazione fosse consistente per tutto il libro. S e' il prezzo della stock, giusto? cos'e' P allora? B e' il prezzo del bond?

ANSWER 3.6: S, P, B

Yes: S = stock price, B = bond (risk-free asset) price. P is a *generic* asset price, any asset; above it comes from the 2013 lecture, which uses P_n for the price on day n of an unspecified asset (as P^j in [definition 2.7](#)). Here P_n and S_n are the same thing.

Exercise 3.3 (lec04_1, slide 21).

- (a) Find the skewness of a lognormal(μ, σ^2) variable (use the lognormal moments).
 (b) Show directly from the density that the kurtosis of $N(\mu, \sigma^2)$ is 3.

Solution →

3.4 Expected option payoffs

A European call gives the right to buy the asset at time T at the strike K ; if X is the price at T , the payoff is the “hockey stick” $C = (X - K)^+ = \max(0, X - K)$ [► 2024 L4 @ 50:01](fig. 3.3).¹⁰

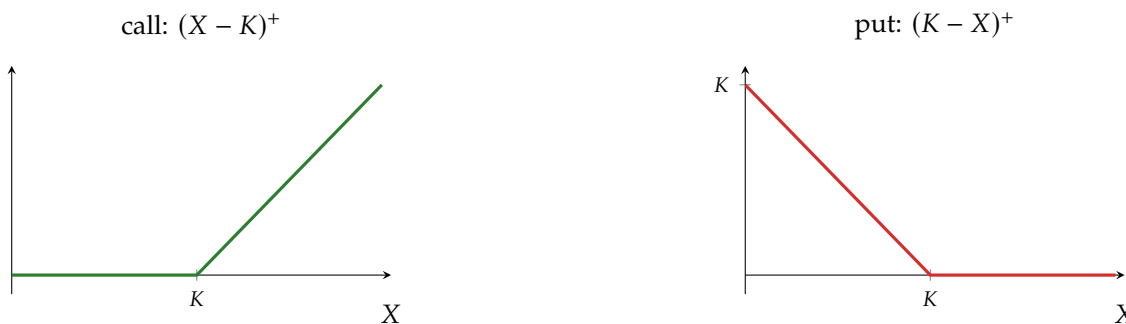


Figure 3.3: Payoffs at expiry as functions of the price X of the underlying: the call (the “hockey stick”: flat blade, then slope 1) and the put, which pays at most K .

Theorem 3.9 (Expected call payoff). Let X have density f , CDF F and $\mathbb{E}[|X|] < \infty$. For every constant K ,

$$\mathbb{E}[(X - K)^+] = \int_K^\infty \int_x^\infty f(t) dt dx = \int_K^\infty [1 - F(x)] dx.$$

This is the familiar fact that for a non-negative variable $\mathbb{E}[W] = \int_0^\infty \mathbb{P}(W > s) ds$ [► 2024 L4 @ 52:11], applied to $W = (X - K)^+$. The slides assume finite variance; a finite mean is enough.

Corollary 3.10 (Strike derivatives; not discussed in class). $C(K) = \mathbb{E}[(X - K)^+]$ satisfies $C'(K) = -\mathbb{P}(X > K)$ and $C''(K) = f(K)$.

So call prices across strikes determine the distribution of X (under the pricing measure): this is the Breeden–Litzenberger formula behind chapters 22 and 25.¹¹

Corollary 3.11 (Normal and lognormal payoffs, slide 18 of lec04_1).

- (i) If $X \sim N(\mu, \sigma^2)$ and $d = (\mu - K)/\sigma$, then

$$\mathbb{E}[(X - K)^+] = (\mu - K) \Phi(d) + \sigma \phi(d) = (\mu - K) \Phi(d) + \frac{\sigma}{\sqrt{2\pi}} e^{-(K-\mu)^2/(2\sigma^2)}.$$

- (ii) If X is lognormal(μ, σ^2) and $d = (\mu - \ln K)/\sigma$, then

$$\mathbb{E}[(X - K)^+] = e^{\mu + \sigma^2/2} \Phi(d + \sigma) - K \Phi(d)$$

[► 2024 L4 @ 53:14].

DONE

10 plot it

Proof. Following the slides [► 2024 L4 @ 51:05], for $M > K$ integrate by parts with $u = x - K$, $v = -[1 - F(x)]$:

$$\begin{aligned} \int_K^M (x - K) f(x) dx &= -(M - K)[1 - F(M)] \\ &\quad + \int_K^M [1 - F(x)] dx. \end{aligned}$$

The boundary term vanishes as $M \rightarrow \infty$ because $0 \leq (M - K)[1 - F(M)] \leq \int_M^\infty (x - K) f(x) dx$, the tail of a convergent integral. (Alternatively: $(x - K)^+ = \int_K^\infty \mathbf{1}\{x > s\} ds$; take expectations and swap the integrals.) $\square$

11 questa parte non l'ho capita

ANSWER 3.7: Breeden–Litzenberger

Read $C(K)$ as the (undiscounted) price of a call, for every strike K . Differentiate twice in K : the corollary gives $C''(K) = f(K)$. So if the market quotes calls at many strikes, the curvature of the price curve is the density of S_T (under Q , answer 3.9). Intuition: a “butterfly” (buy calls at $K - h$ and $K + h$, sell two at K) pays a small triangle around K , i.e. roughly h dollars if $S_T \approx K$; its price $\approx h^2 C''(K)$ is the price of “ S_T near K ”.

IN FINANCE: From expected payoffs to option prices

A price is a *discounted* expectation under a *pricing* measure (theorem 2.9):¹³ $C_0 = e^{-rT} \mathbb{E}_Q[(S_T - K)^+]$. In the Black–Scholes model, under Q , $\ln S_T \sim N(\ln S_0 + (r - \sigma^2/2)T, \sigma^2 T)$; inserting these parameters in corollary 3.11(ii) gives $e^{\mu + \sigma^2 T/2} = S_0 e^{rT}$, $d = d_2$, $d + \sigma\sqrt{T} = d_1$ and hence $C_0 = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)$ (chapter 15). Part (1) is the *Bachelier* model (1900), still used for interest rates and spreads, which can be negative. At the money ($K = \mu$) it gives $\sigma/\sqrt{2\pi} \approx 0.4\sigma$: the traders’ rule of thumb “an at-the-money option is worth about 0.4 standard deviations of the underlying at expiry” (not discussed in class).¹⁴

Exercise 3.4 (not from the course: puts and parity). With the notation of theorem 3.9, show that $\mathbb{E}[(K - X)^+] = \int_{-\infty}^K F(x) dx$ and that $\mathbb{E}[(X - K)^+] - \mathbb{E}[(K - X)^+] = \mathbb{E}[X] - K$. Interpret the second identity as put–call parity: a long call plus a short put is a forward contract (section 1.8.5).

Solution →

3.5 Moment-generating functions

Definition 3.12 (MGF, characteristic function). The *moment-generating function* of X is $M_X(t) = \mathbb{E}[e^{tX}]$, $t \in \mathbb{R}$ [► 2024 L4 @ 53:14][► 2013 L3 @ 36:02]. It always satisfies $M_X(0) = 1$, and the set where it is finite is an interval containing 0 (by convexity of $t \mapsto e^{tx}$). We say that X has an MGF if M_X is finite on some $(-\delta, \delta)$. The *characteristic function* $\varphi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos tX] + i \mathbb{E}[\sin tX]$ is finite for every X and t , since $|e^{itX}| = 1$ [► 2024 L4 @ 56:28].

Proposition 3.13 (Moments from the MGF). If M_X is finite on $(-\delta, \delta)$, all moments are finite, $\mathbb{E}[X^k] = M_X^{(k)}(0)$ and $M_X(t) = \sum_{k \geq 0} \frac{t^k}{k!} m_k$ for $|t| < \delta$ [► 2013 L3 @ 38:14][► 2024 L4 @ 54:18].

12 non c’è un errore nell’ultima uguaglianza di (i)?

ANSWER 3.8: No error

It is correct: $\phi(d) = e^{-d^2/2}/\sqrt{2\pi}$ with $d^2 = (\mu - K)^2/\sigma^2$, and $(\mu - K)^2 = (K - \mu)^2$.

Proof. (i) Write $X = \mu + \sigma Z$: $(X - K)^+ = (\mu - K + \sigma Z)\mathbf{1}\{Z > -d\}$. Then $\mathbb{P}(Z > -d) = \Phi(d)$ and $\mathbb{E}[Z\mathbf{1}\{Z > a\}] = \int_a^\infty z\phi(z) dz = \phi(a)$, with $\phi(-d) = \phi(d)$.
(ii) is 2024 PS2 Problem 2(d) (Exercise 3.13). □

13 cos’è una pricing measure Q ?

ANSWER 3.9: Pricing measure

The weights $q_i = \psi_i/\sum \psi$ of definition 2.7: positive numbers summing to 1, so they look like probabilities, chosen so that every price is a discounted average of payoffs, $P_0 = \beta \mathbb{E}_Q[P_T]$. They are not the real chances: they also include risk aversion (bad states get more weight). Also called the *risk-neutral* measure, see chapter 15.

14 can you give a brief example?

ANSWER 3.10: Rule of thumb

A stock at \$100 with volatility 20%/year, a 3-month at-the-money option. The standard deviation of S_T is about $100 \cdot 0.2 \cdot \sqrt{0.25} = \10 , so the option is worth about $0.4 \cdot 10 = \$4$. Black–Scholes ($r = 0$) gives \$3.99.

15 they are related by analytical continuation. posso dire che una è la laplace transform e l’altra la fourier transform?

ANSWER 3.11: Yes

Yes: $M_X(t)$ is the (two-sided) Laplace transform of the density at $-t$, $\varphi_X(t)$ its Fourier transform (up to the sign convention), and $\varphi_X(t) = M_X(it)$ when M_X is finite near 0. As in physics, the Fourier version always exists, the Laplace one only with enough decay.

Proof. For $|s| < \delta$, $e^{s|x|} \leq e^{sx} + e^{-sx}$, so $\mathbb{E}[e^{s|x|}] < \infty$; this dominates $\sum_k |tX|^k/k!$ for $|t| \leq s$, so expectation and series (and derivatives) can be exchanged. □

The MGF need not exist ▶ 2024 L4 @ 55:21]. For the Cauchy density $f(x) = 1/(\pi(1 + x^2))$ (fig. 3.4),¹⁶ $\mathbb{E}[|X|] = \infty$ and $M_X(t) = \infty$ for every $t \neq 0$; its characteristic function is $e^{-|t|}$. The lognormal law has all moments but no MGF ▶ 2013 L3 @ 37:09]¹⁷: for $t > 0$ and every k , $e^{tY} \geq (tY)^k/k!$, hence

$$M_Y(t) \geq \frac{t^k \mathbb{E}[Y^k]}{k!} = \frac{t^k e^{k\mu + k^2\sigma^2/2}}{k!} \geq \exp\left(\frac{k^2\sigma^2}{2} + k(\mu + \ln t) - k \ln k\right) \xrightarrow{k \rightarrow \infty} \infty,$$

using $k! \leq k^k$. For $t \leq 0$, instead, $0 < e^{tY} \leq 1$ and $M_Y(t)$ is finite.

Theorem 3.14 (Uniqueness and continuity).

- (i) If $M_X(t) = M_Y(t) < \infty$ for all t in a neighbourhood of 0, then X and Y have the same distribution¹⁸ ▶ 2024 L4 @ 56:28] ▶ 2013 L3 @ 39:18].
- (ii) If $M_{X_n}(t) \rightarrow M_X(t) < \infty$ for all t in a neighbourhood of 0, then X_n converges to X in distribution: $F_{X_n}(x) \rightarrow F_X(x)$ at every point where F_X is continuous ▶ 2024 L4 @ 57:31] ▶ 2013 L3 @ 43:39].

The same statements hold for characteristic functions, with “for all $t \in \mathbb{R}$ ” and no existence assumption. (Not proved in class; see Durrett, Probability: Theory and Examples.)

Remark 3.15 (Moments do not determine a distribution). If two variables have the same moments of all orders, they need not have the same law: the hole in the argument “same moments $\Rightarrow$ same MGF $\Rightarrow$ same law” is that the MGF may not exist ▶ 2013 L3 @ 41:26]. The standard example (Durrett, pp. 106–107, cited by lecnote3) is the lognormal. Let f be the lognormal(0, 1) density and, for $|a| \leq 1$, $f_a(y) = f(y) [1 + a \sin(2\pi \ln y)] \geq 0$. With $y = e^u$, $\phi(u)e^{ku} = e^{k^2/2}\phi(u - k)$ and $v = u - k$,

$$\int_0^\infty y^k f(y) \sin(2\pi \ln y) dy = e^{k^2/2} \int_{\mathbb{R}} \phi(v) \sin(2\pi v + 2\pi k) dv = 0 \quad (k = 0, 1, 2, \dots),$$

since $\sin$ is odd. So every f_a is a density with exactly the lognormal moments. (Not discussed in class.)

The normal MGF. For $Z \sim N(0, 1)$, complete the square, $tx - \frac{1}{2}x^2 = -\frac{1}{2}(x - t)^2 + \frac{1}{2}t^2$ ▶ 2024 L4 @ 58:34]:

$$M_Z(t) = \int_{\mathbb{R}} \frac{e^{-\frac{1}{2}(x-t)^2}}{\sqrt{2\pi}} dx e^{t^2/2} = e^{t^2/2}.$$

Since $M_{a+bX}(t) = e^{at} M_X(bt)$ ▶ 2024 L4 @ 1:01:42], for $X = \mu + \sigma Z \sim N(\mu, \sigma^2)$:

$$M_X(t) = e^{\mu t + \sigma^2 t^2/2}. \quad (3.1)$$

Consequences:

- **Lognormal moments** ▶ 2024 L4 @ 1:02:42]: $\mathbb{E}[e^{kX}] = M_X(k)$, so $\mathbb{E}[Y^k] = e^{k\mu + k^2\sigma^2/2}$ for $Y = e^X$ (2024 PS2 Problem 2(a,b)).
- **Sums:** if $X_1, \dots, X_n$ are independent, $M_{\sum X_i}(t) = \prod_i M_{X_i}(t)$, because the expectation of a product of independent variables is the product of expectations. Hence a sum of independent normals is $N(\sum \mu_i, \sum \sigma_i^2)$, by theorem 3.14(i).
- **Normal moments:** $e^{t^2/2} = \sum_k t^{2k}/(2^k k!)$ gives $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!)$, e.g. $\mathbb{E}[Z^4] = 3$: the normal kurtosis is 3.

DONE
16 plot it

17 COURSE NOTE: Erratum in the 2013 notes

lecnote3 (and the lecture, ▶ 2013 L3 @ 46:47]) state that for the lognormal law $\mathbb{E}[e^{tX}]$ does not exist for all $t \neq 0$. It is infinite for all $t > 0$ but finite for all $t < 0$; what fails is finiteness on a neighbourhood of 0, which is what the theorems below need. In the same paragraph, $\frac{d^k M_X}{dx^k}(0)$ should read $\frac{d^k M_X}{dt^k}(0)$.

18 COURSE NOTE: Statement in the slides and notes

Slide 19 of lec04_1 and Theorem 2.2 of lecnote3 assume equality or convergence “for all t ”. The neighbourhood version above is the standard one and is the one needed in practice: the MGF of a χ^2 variable, for instance, is finite only for $t < 1/2$ (section 3.9).

19 per ovviare al fatto che alcune distribuzioni non hanno i momenti ben definiti, perche' non aggiungiamo un weight esponenziale all'integrale cosi' abbiamo quantita' finite per caratterizzare le nostre distribuzioni heavy-tailed?

ANSWER 3.12: It is done

Good idea, and it is used. A damping $e^{-a|x|}$ makes every “moment” finite, but the numbers then depend on a and lose their meaning (mean, variance). The bounded weight e^{itx} is the standard answer: the characteristic function always exists and determines the law. Practitioners also use quantile-based measures (median, interquartile range) and the tail index α of $\mathbb{P}(|X| > x) \sim x^{-\alpha}$. In option pricing, damped transforms $e^{-\theta x}$ are the Carr–Madan (1999) method of pricing options by FFT.

20 pero' non c'e' il problema che la pdf non e' definita positiva? inoltre, i momenti non si possono ricavare dalla characteristic function che esiste sempre?

ANSWER 3.13: Positivity

No: $|a \sin(\cdot)| \leq 1$, so $1 + a \sin(2\pi \ln y) \geq 0$ and $f_a \geq 0$; it integrates to 1 (the $k = 0$ case).

ANSWER 3.14: Moments from φ ?

Yes, $\mathbb{E}[X^k] = i^{-k} \varphi^{(k)}(0)$ when the moments exist. But the converse fails: the moments give only the Taylor coefficients of φ at 0, and for the lognormal that Taylor series has radius 0, so it does not reconstruct φ . Different laws, different φ , same derivatives at 0.

Exercise 3.5 (PS2 (2013), A-2: Poisson distribution). X has pmf $f_X(k) = \lambda^k e^{-\lambda} / k!$, $k = 0, 1, 2, \dots$

- Compute its MGF.
- Deduce mean and variance.
- Prove, with MGFs, that a sum of independent Poisson variables is Poisson.

Solution →

INTUITION: Partition function and cumulants (not discussed in class)

$M_X(t)$ is a Laplace transform, like a partition function $Z(\beta)$, and $K_X(t) = \ln M_X(t)$ plays the role of a free energy. Its Taylor coefficients are the *cumulants*: $K_X(t) = \mu t + \sigma^2 \frac{t^2}{2} + \kappa_3 \frac{t^3}{6} + \kappa_4 \frac{t^4}{24} + \dots$ with $\kappa_3 = \gamma \sigma^3$ and $\kappa_4 = (\kappa - 3)\sigma^4$. For independent variables cumulants *add*. The Gaussian is the “free theory”: K is exactly quadratic. For the standardised sum $S_n/\sqrt{n}$ of i.i.d. variables the k -th cumulant is $n\kappa_k/n^{k/2} \rightarrow 0$ for $k \geq 3$: all non-Gaussian features wash out, which is the central limit theorem in one line (theorem 3.34).

Exponential families (2013). A family of pdfs or pmfs indexed by a parameter vector θ is an *exponential family* if ▶ 2013 L3 @ 26:24]

$$f(x | \theta) = h(x) c(\theta) \exp\left(\sum_{i=1}^k w_i(\theta) t_i(x)\right),$$

with h, t_i depending only on x and c, w_i only on θ . The lognormal is one ▶ 2013 L3 @ 29:37]: expanding the square,

$$f(x) = \frac{1}{x} \cdot \frac{e^{-\mu^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{(\ln x)^2}{2\sigma^2} + \frac{\mu \ln x}{\sigma^2}\right),$$

so $h(x) = 1/x^{22}$, $c(\theta) = e^{-\mu^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$, $w_1 = -1/(2\sigma^2)$, $t_1 = (\ln x)^2$, $w_2 = \mu/\sigma^2$, $t_2 = \ln x$. The point, stressed by P. Kempthorne from the audience ▶ 2013 L3 @ 28:35]: the joint density of an i.i.d. sample $\prod_j f(x_j | \theta) = \prod_j h(x_j) c(\theta)^n \exp(\sum_i w_i(\theta) \sum_j t_i(x_j))$ has the same form and depends on the data only through the *sufficient statistics* $\sum_j t_i(x_j)$. Normal, lognormal, exponential, Poisson and gamma laws are exponential families (2013 PS2 B-3); Cauchy and Student t are not.

Exercise 3.6 (PS2 (2013), B-3: exponential families). Prove that the lognormal, Poisson and exponential distributions are exponential families, i.e. that their densities or pmfs can be written as $h(x)c(\theta)\exp(\sum_i w_i(\theta)t_i(x))$ with $c \geq 0$, $h \geq 0$.

Solution →

21 beautiful box. i want to know more.

ANSWER 3.15: More on cumulants

Three directions:

- *Large deviations* (Cramér): $\mathbb{P}(S_n/n \approx x) \approx e^{-nI(x)}$ with I the Legendre transform of K , exactly free energy $\leftrightarrow$ entropy.
- *Edgeworth expansion*: corrections to the CLT in powers of $\kappa_3/\sqrt{n}$, κ_4/n .
- *Exponential tilting* $f_\theta(x) = e^{\theta x - K(\theta)} f(x)$ (the canonical ensemble): the Esscher transform, a way to build a pricing measure.

Reference: Feller, vol. 2, ch. XVI; Dembo–Zeitouni, *Large Deviations*.

22 COURSE NOTE: Erratum in the 2013 notes

In the lognormal example of **lecnote3** (p. 3) the factor is printed $h(x) = \frac{1}{2}$; it should be $h(x) = \frac{1}{x}$, as in the displayed formula just above it.

Law	pmf / pdf	Where it appears
Bernoulli(p)	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	default indicators, side of an order, event contracts (chapter 14)
Poisson(λ)	$\lambda^k e^{-\lambda} / k!, k \in \mathbb{N}$	number of events (orders, jumps, defaults) in a period; 2013 PS2 A-2, B-1
Exponential(λ)	$\lambda e^{-\lambda x}, x \geq 0$	waiting times between events (memoryless); 2013 PS2 A-1
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	log-returns, Brownian increments, regression errors
Lognormal(μ, σ^2)	$\frac{1}{x\sigma\sqrt{2\pi}} e^{-(\ln x - \mu)^2/(2\sigma^2)}, x > 0$	prices (Black–Scholes)
Gamma(α, λ)	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, x > 0$	contains $\chi_n^2 = \text{Gamma}(n/2, 1/2)$
$\chi_n^2, t_r, F_{m,n}$	section 3.9	inference in regression (chapter 5); t : heavy-tailed returns
Cauchy	$1/(\pi(1+x^2))$	a law without mean, variance or MGF

Table 3.1: Distributions used in the course.

DONE
23 plot all the mentioned distributions

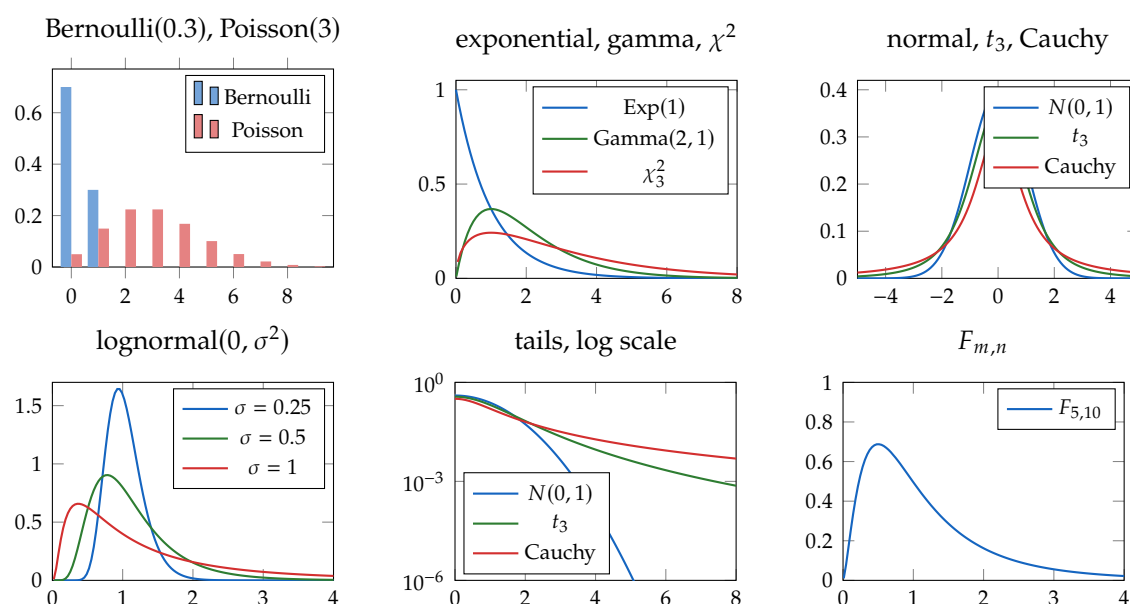


Figure 3.4: The laws of [table 3.1](#). The t_3 and Cauchy densities look like the normal near 0 but their tails decay like powers (x^{-4} , x^{-2}), not like $e^{-x^2/2}$: on the log scale the gap at $x = 5$ is a factor 10^3 – 10^4 .

3.6 Several random variables

Independence of X and Y means that knowing one gives no information about the other; for densities it is the factorisation $f_{X,Y} = f_X f_Y$ [► 2024 L4 @ 1:03:47].

Definition 3.16 (Covariance, correlation). $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] =$

$\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ (so $\text{Cov}(X, X) = \text{Var } X$) and

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var } X \text{ Var } Y}} = \mathbb{E}\left[\frac{X - \mu_X}{\sigma_X} \cdot \frac{Y - \mu_Y}{\sigma_Y}\right]$$

[► 2024 L4 @ 1:04:52]. X, Y are *uncorrelated* if $\text{Cov}(X, Y) = 0$.

Covariance is bilinear; by Cauchy–Schwarz $|\text{Corr}(X, Y)| \leq 1$, with equality iff $Y = a + bX$ almost surely. Independent variables are uncorrelated, but not conversely: correlation detects only *linear* dependence [► 2024 L4 @ 1:06:58]. Example: $X \sim N(0, 1)$ and $Y = X^2$ have $\text{Cov}(X, Y) = \mathbb{E}[X^3] = 0$, although Y is a function of X .

Proposition 3.17 (Mean and variance of a linear combination). Let $\mathbf{X} = (X_1, \dots, X_n)^\top$ have mean vector $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ and covariance matrix $\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$, $\sigma_{ij} = \text{Cov}(X_i, X_j)$ [► 2024 L4 @ 1:07:59]. For a constant $\mathbf{a} \in \mathbb{R}^n$ and $Y = \mathbf{a}^\top \mathbf{X}$ [► 2024 L4 @ 1:09:07],

$$\mathbb{E}[Y] = \mathbf{a}^\top \boldsymbol{\mu}, \quad \text{Var } Y = \mathbf{a}^\top \Sigma \mathbf{a} = \sum_i a_i^2 \text{Var } X_i + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

More generally $\mathbb{E}[B\mathbf{X} + \mathbf{b}] = B\boldsymbol{\mu} + \mathbf{b}$ and $\text{Cov}(B\mathbf{X} + \mathbf{b}) = B\Sigma B^\top$.

Proof. $\text{Var } Y = \mathbb{E}[(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}))^2] = \mathbb{E}[\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{a}^\top \Sigma \mathbf{a}$: constants come out of the expectation, which acts entry by entry [► 2024 L4 @ 1:10:10]. The matrix version is identical. $\square$

In particular Σ is positive semi-definite, since $\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0$ (section 2.6). If $\mathbf{X}$ are asset returns and $\mathbf{a}$ portfolio weights, Y is the portfolio return: its variance is a quadratic form in the weights.

Diversification [► 2024 L4 @ 1:11:13]. If the X_i are uncorrelated with common variance σ^2 and $a_i = 1/n$, then $\sum a_i^2 = 1/n$ and $\text{Var } Y = \sigma^2/n$ [► 2024 L4 @ 1:13:27]. With a common correlation ρ instead (not discussed in class),

$$\text{Var}\left(\frac{1}{n} \sum_i X_i\right) = \frac{\sigma^2}{n} + \frac{n-1}{n} \rho \sigma^2 \xrightarrow{n \rightarrow \infty} \rho \sigma^2.$$

IN FINANCE: Systematic and idiosyncratic risk

Diversification removes the part of the risk that is specific to each asset, but not the part common to all of them: with $\rho = 0.3$ and $\sigma = 30\%$, even an infinitely diversified equal-weighted portfolio keeps a volatility of $\sqrt{0.3} \cdot 30\% \approx 16\%$. Only this common, *systematic* risk should command a risk premium in equilibrium: this is the content of the CAPM (section 3.11) and the starting point of factor models (chapter 7).

3.6.1 Conditional distributions and conditional expectation

This material is only used implicitly in the lectures; it is needed for martingales (chapter 4) and in 2024 PS3 Problem 2.

Definition 3.18 (Conditional density and expectation). If (X, Y) has joint density $f_{X,Y}$, the conditional density of Y given $X = x$ (for $f_X(x) > 0$) is $f_{Y|X}(y | x) = f_{X,Y}(x, y) / f_X(x)$, and $m(x) = \mathbb{E}[Y | X = x] = \int y f_{Y|X}(y | x) dy$. The *conditional expectation* $\mathbb{E}[Y | X]$ is the random variable $m(X)$. Similarly $\text{Var}(Y | X) = \mathbb{E}[(Y - m(X))^2 | X]$. (Discrete case: replace densities by pmfs.)

Proposition 3.19 (Properties). For integrable Y and functions g :

- (i) tower property (law of iterated expectations): $\mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y]$;
- (ii) taking out what is known: $\mathbb{E}[g(X)Y | X] = g(X)\mathbb{E}[Y | X]$; if X, Y are independent, $\mathbb{E}[Y | X] = \mathbb{E}[Y]$;
- (iii) best prediction: $\mathbb{E}[Y | X]$ minimises $\mathbb{E}[(Y - h(X))^2]$ over all functions h ;
- (iv) law of total variance: $\text{Var } Y = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])$.

Proof. (i) $\int m(x)f_X(x) dx = \iint y f_{X,Y}(x, y) dy dx = \mathbb{E}[Y]$.
(ii) Given $X = x$, $g(X) = g(x)$ is a constant; under independence $f_{Y|X} = f_Y$.
(iii) $\mathbb{E}[(Y - h)^2] = \mathbb{E}[(Y - m)^2] + \mathbb{E}[(m - h)^2] + 2\mathbb{E}[(Y - m)(m - h)(X)]$, and by (i)–(ii) the cross term is $2\mathbb{E}[(m - h)(X) \mathbb{E}[Y - m(X) | X]] = 0$.
(iv) Apply the same decomposition to $Y - \mathbb{E}[Y] = (Y - m) + (m - \mathbb{E}[Y])$: the cross term vanishes, $\mathbb{E}[(Y - m)^2] = \mathbb{E}[\text{Var}(Y | X)]$ by (i), and $\mathbb{E}[m(X)] = \mathbb{E}[Y]$. $\square$

INTUITION: Conditional expectation is a projection

In the Hilbert space of square-integrable variables with $\langle U, V \rangle = \mathbb{E}[UV]$, $\mathbb{E}[Y | X]$ is the orthogonal projection of Y on the subspace of functions of X : (iii) is Pythagoras. Linear regression projects on the smaller subspace of affine functions $a + bX$ (chapter 5); for jointly normal variables the two projections coincide (section 3.7). For a general information set $\mathcal{G}$ (e.g. the price history up to time t) $\mathbb{E}[Y | \mathcal{G}]$ is defined by the same orthogonality, $\mathbb{E}[(Y - \mathbb{E}[Y | \mathcal{G}])W] = 0$ for every bounded $\mathcal{G}$ -measurable W : this is the object in the definition of a martingale.

Exercise 3.7 (not from the course). Let X_1, X_2 be independent with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$, and $X_3 = X_1X_2$. Show that X_1, X_2, X_3 are pairwise independent but not mutually independent.

Solution $\rightarrow$

3.7 The multivariate normal distribution

Definition 3.20 (Multivariate normal). $X \in \mathbb{R}^n$ is multivariate normal, $X \sim N_n(\mu, \Sigma)$, if every linear combination $a^\top X$ is normal (a constant counting as $N(c, 0)$); then $\mu = \mathbb{E}[X]$ and $\Sigma = \text{Cov } X$. Components of such a vector are called jointly normal.

Proposition 3.21.

- (i) $X \sim N_n(\mu, \Sigma)$ iff $\mathbb{E}[e^{t^\top X}] = \exp(t^\top \mu + \frac{1}{2}t^\top \Sigma t)$ for all $t \in \mathbb{R}^n$.
- (ii) $X \stackrel{d}{=} \mu + AZ$ with Z a vector of i.i.d. $N(0, 1)$ and any A with $AA^\top = \Sigma$ (e.g. Cholesky, or $A = U\Lambda^{1/2}U^\top$ from theorem 2.11).
- (iii) If Σ is invertible, X has density $f(x) = (2\pi)^{-n/2}(\det \Sigma)^{-1/2} \exp(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu))$.

Proof. (i) $t^\top X \sim N(t^\top \mu, t^\top \Sigma t)$; evaluate (3.1) at 1. Conversely this MGF restricted to $t = sa$ is the MGF of $N(a^\top \mu, a^\top \Sigma a)$.
(ii) $\mathbb{E}[e^{t^\top(\mu + AZ)}] = e^{t^\top \mu} \prod_i \mathbb{E}[e^{(A^\top t)_i Z_i}] = e^{t^\top \mu + \frac{1}{2}|A^\top t|^2}$, and $|A^\top t|^2 = t^\top \Sigma t$; conclude by the n -dimensional version of theorem 3.14.
(iii) Change variables $x = \mu + Az$ in the density $(2\pi)^{-n/2}e^{-|z|^2/2}$ of Z : $|\det A| = (\det \Sigma)^{1/2}$ and $|z|^2 = (x - \mu)^\top \Sigma^{-1}(x - \mu)$. $\square$

Theorem 3.22 (Properties of the multivariate normal). Let $X \sim N_n(\mu, \Sigma)$.

- (i) $BX + b \sim N(B\mu + b, B\Sigma B^\top)$; in particular all marginals are normal.
- (ii) Jointly normal variables are independent iff they are uncorrelated.
- (iii) Partition $X = (X_1, X_2)$ with Σ_{22} invertible. Then

$$X_1 | X_2 = x_2 \sim N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

Proof. (i) Every linear combination of $BX + b$ is a linear combination of X plus a constant; the moments follow from proposition 3.17.
(ii) If the cross-covariances vanish, the MGF of proposition 3.21(i) factorises into the MGFs of the blocks.
(iii) Let $\epsilon = X_1 - \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}(X_2 - \mu_2)$. By (i) (ϵ, X_2) is jointly normal, and $\text{Cov}(\epsilon, X_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0$, so ϵ is independent of X_2 by (ii), with mean 0 and covariance $\Sigma_{11} - 2\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$. Given $X_2 = x_2$, X_1 is a constant plus ϵ . $\square$

Bivariate case. If (X, Y) is jointly normal with means μ_X, μ_Y , standard deviations σ, τ and

correlation ρ , then

$$Y = a + bX + \varepsilon, \quad b = \frac{\rho\tau}{\sigma}, \quad a = \mu_Y - b\mu_X, \quad \varepsilon \sim N(0, (1-\rho^2)\tau^2) \text{ independent of } X. \quad (3.2)$$

So $\mathbb{E}[Y | X] = \mu_Y + \rho \frac{\tau}{\sigma}(X - \mu_X)$ is exactly the regression line, and the conditional variance does not depend on X . This decomposition is the hint of 2024 PS3 Problem 1 ([Exercise 3.12](#)).

Geometry. The level sets of the density are the ellipsoids $(x - \mu)^\top \Sigma^{-1} (x - \mu) = c$, centred at μ , with principal axes along the eigenvectors of Σ and semi-axes $\sqrt{c\lambda_i}$; in two dimensions they are ellipses tilted upward for $\rho > 0$ and downward for $\rho < 0$ [► 2024 L5 @ 3:12]. Since $(X - \mu)^\top \Sigma^{-1} (X - \mu) = |Z|^2$, this quadratic form has the χ_n^2 law of [section 3.9](#).

Remark 3.23 (Normal marginals are not enough; not discussed in class). Let $Z \sim N(0, 1)$ and $S = \pm 1$ with probability $\frac{1}{2}$ each, independent of Z , and $W = SZ$. Then $W \sim N(0, 1)$ and $\text{Cov}(Z, W) = \mathbb{E}[S] \mathbb{E}[Z^2] = 0$, yet Z and W are dependent ($|W| = |Z|$) and $Z + W = 0$ with probability $\frac{1}{2}$, so (Z, W) is not jointly normal. Joint normality is a strong assumption about *dependence*, and in particular it makes simultaneous extreme moves of several assets very unlikely.

3.8 Principal components of a random vector

Let $x \in \mathbb{R}^m$ have $\mathbb{E}[x] = \alpha$ and $\text{Cov } x = \Sigma$ [► 2024 L4 @ 1:14:27]. Σ is symmetric and positive semi-definite [► 2024 L4 @ 1:15:32], so by [theorem 2.11](#)

$$\Sigma = \Gamma \Lambda \Gamma^\top = \sum_{i=1}^m \lambda_i \gamma_i \gamma_i^\top, \\ \Lambda = \text{diag}(\lambda_1, \dots, \lambda_m), \quad \lambda_1 \geq \dots \geq \lambda_m \geq 0, \quad \Gamma = [\gamma_1 \cdots \gamma_m], \quad \Gamma^\top \Gamma = I_m.$$

Definition 3.24 (Principal component variables). $p = \Gamma^\top(x - \alpha)$, i.e. $p_i = \gamma_i^\top(x - \alpha)$ [► 2024 L4 @ 1:17:39].

Proposition 3.25.

- (i) $\mathbb{E}[p] = 0$ and $\text{Cov } p = \Gamma^\top \Sigma \Gamma = \Lambda$: the PC variables are uncorrelated, with variances $\lambda_1 \geq \dots \geq \lambda_m$ [► 2024 L4 @ 1:18:39]; and $x = \alpha + \Gamma p$.
- (ii) Total variance: $\sum_i \text{Var}(x_i) = \text{tr } \Sigma = \text{tr}(\Lambda \Gamma^\top \Gamma) = \sum_k \lambda_k = \sum_k \text{Var}(p_k)$ [► 2024 L5 @ 21:04].
- (iii) Variational characterisation [► 2024 L5 @ 17:53]: γ_1 maximises $\text{Var}(w^\top x) = w^\top \Sigma w$ over unit vectors w ; γ_k maximises it over unit vectors orthogonal to $\gamma_1, \dots, \gamma_{k-1}$ [► 2024 L5 @ 18:55]; the maxima are $\lambda_1, \dots, \lambda_k$. The last component has the smallest variance, and $\lambda_m = 0$ iff $\gamma_m^\top(x - \alpha) = 0$ almost surely, an exact linear relation among the x_i .

INTUITION: Shift and rotate

$x \mapsto p$ moves the origin to the mean and rotates the axes onto the principal axes of the density ellipsoids; no distance is changed [► 2024 L5 @ 8:39]. Σ is the “inertia tensor” of

24 COURSE NOTE: Scope

This section is the probabilistic part of PCA (slides 26–32 of [lec04_1](#), [► 2024 L4 @ 1:13:27] to [► 2024 L5 @ 21:04]). Applications to equity indices and to the yield curve [► 2024 L5 @ 15:51], announced in L5 for S. Andreev’s lecture, are in [chapter 7](#).

Proof. (i) and (ii) are [proposition 3.17](#) and $\text{tr}(AB) = \text{tr}(BA)$. (iii) Write $w = \Gamma c$, $|c| = |w| = 1$: $w^\top \Sigma w = \sum_i \lambda_i c_i^2 \leq \lambda_1$, with equality at $c = e_1$. Orthogonality to $\gamma_1, \dots, \gamma_{k-1}$ means $c_1 = \dots = c_{k-1} = 0$, and then the bound is λ_k . A variable with zero variance is a.s. equal to its mean. $\square$

the cloud of outcomes and PCA diagonalises it. A zero eigenvalue is a flat direction: before the euro the Deutsche Mark and the French franc moved so closely that the smallest PC of currency returns was almost degenerate [► 2024 L5 @ 19:59].

Factor form [► 2024 L5 @ 9:43]. Split $\Gamma = [\Gamma_1 \Gamma_2]$ and $p = (p_1, p_2)$ with the $K < m$ largest eigenvalues in the first block. Then

$$x = \alpha + \Gamma_1 p_1 + \Gamma_2 p_2 = \alpha + Bf + \varepsilon, \quad B = \Gamma_1, \quad f = p_1, \quad \varepsilon = \Gamma_2 p_2,$$

like a factor model with K factors, except that $\text{Cov } \varepsilon = \Gamma_2 \Lambda_2 \Gamma_2^\top$ is not diagonal [► 2024 L5 @ 11:48]. If $\sum_{k \leq K} \lambda_k / \sum_k \lambda_k$ is close to 1, the first K components carry almost all the variability.

Empirical PCA [► 2024 L5 @ 12:48]. For data $X = [x_1 \cdots x_T] \in \mathbb{R}^{m \times T}$ (one column per date): row means $\bar{x} = \frac{1}{T} X \mathbf{1}_T$; de-meaned matrix $X^* = X - \bar{x} \mathbf{1}_T^\top$; sample covariance $\hat{\Sigma} = \frac{1}{T} X^* X^{*\top} = \hat{\Gamma} \hat{\Lambda} \hat{\Gamma}^\top$; sample PCs $P = \hat{\Gamma}^\top X^* (m \times T)$. Equivalently, from the (reduced) SVD $X^* = V D U^\top$ one gets $\hat{\Lambda} = D^2/T$, $\hat{\Gamma} = V$, $P = D U^\top$ [► 2024 L5 @ 14:49] (Exercise 3.8).

Exercise 3.8 (lec04_1, slide 30: PCA through the SVD). Let X^* be the de-meaned $m \times T$ data matrix and $X^* = V D U^\top$ its reduced SVD (V orthogonal $m \times m$, D diagonal with $d_1 \geq \cdots \geq d_m \geq 0$, U of size $T \times m$ with $U^\top U = I_m$). Show that $\hat{\Lambda} = \frac{1}{T} D^2$, $\hat{\Gamma} = V$ and $P = \hat{\Gamma}^\top X^* = D U^\top$.

Solution →

3.9 Sampling distributions: χ^2 , t and F

These laws arise from sums of squares of Gaussians and reappear in every test of chapter 5 [► 2024 L5 @ 21:04].

Definition 3.26 (χ^2 distribution). If $Z \sim N(0, 1)$, $U = Z^2$ has the *chi-squared law with 1 degree of freedom*, χ_1^2 . If $Z_1, \dots, Z_n$ are i.i.d. $N(0, 1)$, $V = Z_1^2 + \cdots + Z_n^2 \sim \chi_n^2$ [► 2024 L5 @ 22:06].

Proposition 3.27. $\chi_1^2 = \text{Gamma}(\frac{1}{2}, \frac{1}{2})$ ²⁶ and $\chi_n^2 = \text{Gamma}(\frac{n}{2}, \frac{1}{2})$, with

$$f_U(u) = \frac{u^{-1/2} e^{-u/2}}{\sqrt{2\pi}}, \quad f_V(v) = \frac{v^{n/2-1} e^{-v/2}}{2^{n/2} \Gamma(n/2)}, \quad M_V(t) = (1 - 2t)^{-n/2} \quad (t < \frac{1}{2}),$$

$$\mathbb{E}[V] = n, \quad \text{Var } V = 2n.$$

25 COURSE NOTE: Erratum in the slides

Slide 30 of lec04_1 describes U as an “ $(m \times T)$ row-orthonormal matrix, $UV' = I_m$ ”. For $X^* = V D U'$ to be conformable, U must be $T \times m$ with orthonormal columns, $U^\top U = I_m$ (equivalently $U^\top$ is $m \times T$ with orthonormal rows); UV' is not even defined unless $T = m$. The formulas $\hat{\Lambda} = D^2/T$, $\hat{\Gamma} = V$, $P = D U'$ are correct (for $T \geq m$).

Proof. $F_U(u) = \mathbb{P}(-\sqrt{u} \leq Z \leq \sqrt{u}) = 2\Phi(\sqrt{u}) - 1$, so $f_U(u) = \phi(\sqrt{u})/\sqrt{u}$ (the map $z \mapsto z^2$ is two-to-one, so theorem 3.7 is applied on each branch). For the $\text{Gamma}(\alpha, \lambda)$ density, $\int_0^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-(\lambda-t)x} dx = (1-t/\lambda)^{-\alpha}$ for $t < \lambda$, which is $(1-2t)^{-1/2}$ for $\alpha = \lambda = \frac{1}{2}$. The MGF of a sum of n independent copies is the n -th power, $(1-2t)^{-n/2}$, the MGF of $\text{Gamma}(\frac{n}{2}, \frac{1}{2})$; conclude by theorem 3.14. Mean $\alpha/\lambda = n$, variance $\alpha/\lambda^2 = 2n$. □

26 COURSE NOTE: Erratum in the slides

Slide 34 of lec04_1 calls λ in $\text{Gamma}(\alpha, \lambda)$ “the scale parameter”. With the density $\lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha)$ used on slide 33, λ is the *rate*; the scale is $1/\lambda$ ($= 2$ for χ^2). R uses the same convention: `dgamma(x, shape = a, rate = 1)`.

Definition 3.28 (Student t and Fisher F). Let $Z \sim N(0, 1)$, $U \sim \chi_r^2$, $V \sim \chi_n^2$, $W \sim \chi_m^2$, independent. Then $T = Z/\sqrt{U/r} \sim t_r$ (t distribution with r degrees of freedom) [► 2024 L5 @ 23:11] and $F = \frac{W/m}{V/n} \sim F_{m,n}$ [► 2024 L5 @ 24:13].

Properties:

- t_r has density $\frac{\Gamma((r+1)/2)}{\sqrt{r\pi}\Gamma(r/2)}(1 + x^2/r)^{-(r+1)/2}$: a power-law tail $|x|^{-(r+1)}$, so moments of order $\geq r$ are infinite; $\text{Var } T = r/(r-2)$ for $r > 2$ and $\kappa = 3 + 6/(r-4)$ for $r > 4$. t_1 is the Cauchy law, and $t_r \rightarrow N(0, 1)$ as $r \rightarrow \infty$ because $U/r \rightarrow 1$.
- t_r has heavier tails than the normal: Z is divided by a random factor that is only around 1 [► 2024 L5 @ 24:13].
- $\mathbb{E}[F] = \mathbb{E}[W/m] \mathbb{E}[n/V] = n/(n-2)$ for $n > 2$ (using $\mathbb{E}[1/V] = 1/(n-2)$), and $T^2 \sim F_{1,r}$ if $T \sim t_r$.
- F statistics compare two independent variance estimates (analysis of variance, regression F tests) [► 2024 L5 @ 25:16]. When the null hypothesis is false, the numerator is a non-central χ^2 and the statistic has a non-central F law, which governs the power of the test [► 2024 L5 @ 26:16].

Where they come from (preview of [chapter 5](#)): for an i.i.d. $N(\mu, \sigma^2)$ sample, $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$ independently of $\bar{X}$, hence $\sqrt{n}(\bar{X} - \mu)/s \sim t_{n-1}$ (Student, 1908).

3.9.1 Heavy tails

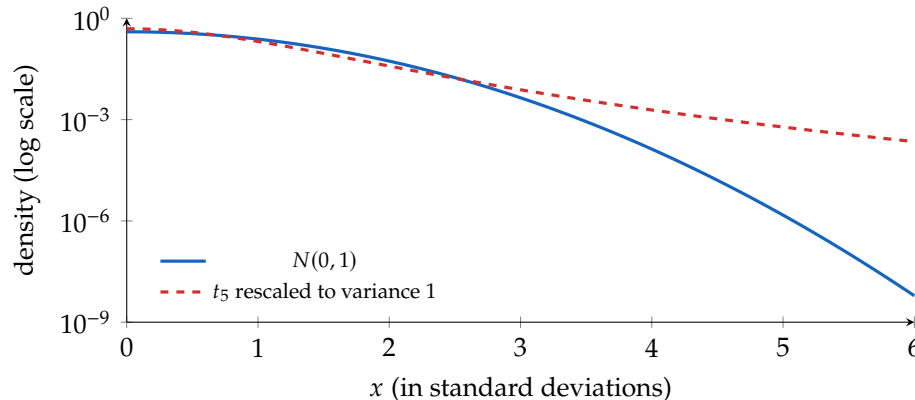


Figure 3.5: Right tails of the standard normal and of a Student t_5 scaled to unit variance (density $0.490(1 + x^2/3)^{-3}$, kurtosis 9). On a log scale the Gaussian tail is a parabola, the t tail a logarithm.

With equal variances the two laws of [fig. 3.5](#) look alike near the centre, but $\mathbb{P}(|X| > 4)$ is 6.3×10^{-5} for the normal and 3.6×10^{-3} for the rescaled t_5 , about 56 times more. For daily returns with 252 trading days a year, a 4-standard-deviation day happens on average once every 63 years under the normal model and about once a year under the t_5 model.

Proposition 3.29 (Normal variance mixtures; not discussed in class). Let $X | V \sim$

Proof. Condition on V : $\mathbb{E}[(X - \mu)^2 | V] = \sigma^2 V$ and $\mathbb{E}[(X - \mu)^4 | V] = 3\sigma^4 V^2$; take expectations (tower property). $\square$

$N(\mu, \sigma^2 V)$ with $V > 0$ random and $\mathbb{E}[V^2] < \infty$. Then $\mathbb{E}[X] = \mu$, $\text{Var } X = \sigma^2 \mathbb{E}[V]$ and

$$\kappa_X = 3 \frac{\mathbb{E}[V^2]}{(\mathbb{E}[V])^2} = 3 \left(1 + \frac{\text{Var } V}{(\mathbb{E}[V])^2} \right) \geq 3,$$

with equality iff V is constant.

The t_r law is such a mixture: $T = Z\sqrt{r/U}$, so $T | U \sim N(0, r/U)$. Stochastic volatility models, in which the variance of returns is itself random, are of this type, and always produce kurtosis above 3. 2024 PS3 Problem 2 extends Stein's lemma (section 3.11) to them (Exercise 3.17).

INTUITION: Superstatistics

A Gaussian whose variance fluctuates is like a Brownian particle in a medium whose temperature fluctuates slowly: locally Gaussian, globally fat-tailed. Physicists call this superposition of statistics *superstatistics*; in finance the fluctuating temperature is the volatility, which clusters in time (chapters 9 and 13).

IN FINANCE: Heavy tails in practice

Daily returns of stocks and indices have sample kurtosis far above 3, and many studies find tails decaying roughly like a power law, with exponent near 3 for individual stocks (the “inverse cubic law”). Consequences: normal-based value-at-risk understates extreme losses (chapter 19); sample moments of high order are unstable or meaningless; averages converge more slowly than Gaussian intuition suggests, and not at all for a Cauchy-like law (Exercise 3.11).

Exercise 3.9 (not from the course: a heavy-tailed mixture). Let $X | V \sim N(0, V)$ with V exponential with rate 1. Show that $M_X(t) = 1/(1 - t^2/2)$ for $|t| < \sqrt{2}$, identify the law of X (Laplace, density $\frac{1}{\sqrt{2}}e^{-\sqrt{2}|x|}$), and check proposition 3.29: $\text{Var } X = 1$, $\kappa = 6$.

Solution →

3.10 Limit theorems

Definition 3.30 (Modes of convergence). $X_n \rightarrow X$ in probability if $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$; in distribution if $F_{X_n}(x) \rightarrow F_X(x)$ at every continuity point of F_X ; almost surely if $\mathbb{P}(X_n \rightarrow X) = 1$. Almost sure convergence implies convergence in probability, which implies convergence in distribution.

Lemma 3.31 (Markov and Chebyshev inequalities). For $a > 0$: $\mathbb{P}(|X| \geq a) \leq \mathbb{E}[|X|]/a$. If $\text{Var } X = \sigma^2 < \infty$ and $\varepsilon > 0$: $\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \sigma^2/\varepsilon^2$.

Proof. $a \mathbf{1}\{|X| \geq a\} \leq |X|$ pointwise; take expectations. For Chebyshev apply Markov to $(X - \mu)^2$ and $a = \varepsilon^2$: the parabola $(x - \mu)^2$ lies above the step function $\varepsilon^2 \mathbf{1}\{|x - \mu| \geq \varepsilon\}$ (fig. 3.6), and averaging preserves the inequality [► 2024 L5 @ 29:29]. □

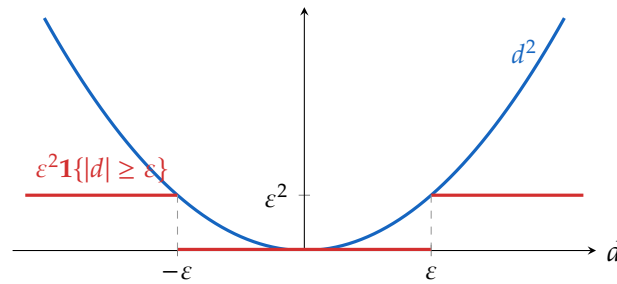


Figure 3.6: The picture behind Chebyshev's inequality, as drawn in class [► 2024 L5 @ 29:29]: with $d = \bar{X}_n - \mu$, $d^2 \geq \varepsilon^2 \mathbf{1}\{|d| \geq \varepsilon\}$, so $\text{Var } \bar{X}_n = \mathbb{E}[d^2] \geq \varepsilon^2 \mathbb{P}(|d| \geq \varepsilon)$.

Theorem 3.32 (Weak law of large numbers). Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var } X_i = \sigma^2 < \infty$, and $\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$. Then $\bar{X}_n \rightarrow \mu$ in probability [► 2024 L5 @ 27:20] [► 2013 L3 @ 49:04]: $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0$.

Proof. $\mathbb{E}[\bar{X}_n] = \mu$ by linearity. $\text{Var } \bar{X}_n = \frac{1}{n^2} \sum_{i,j} \text{Cov}(X_i, X_j) = \frac{\sigma^2}{n}$, since the cross terms vanish [► 2013 L3 @ 58:43]. Apply lemma 3.31 [► 2024 L5 @ 28:28]. $\square$

Remark 3.33.

- The proof only uses that the X_i are pairwise *uncorrelated* with equal means and variances.
- The conclusion holds with only $\mathbb{E}[|X_i|] < \infty$ (Khinchin's WLLN) [► 2024 L5 @ 31:38], and the *strong* law gives $\bar{X}_n \rightarrow \mu$ almost surely under the same assumption [► 2013 L3 @ 1:02:53].
- Without a finite mean it fails: the mean of n i.i.d. Cauchy variables is again Cauchy (Exercise 3.11).

Exercise 3.10 (Try first: how many observations?). You want to be 99% sure that $|\bar{X}_n - \mu| < \varepsilon = 0.1$ [► 2013 L3 @ 1:00:50], for i.i.d. observations with $\sigma = 1$.

- How large must n be according to Chebyshev's inequality?
- And according to the normal approximation of the CLT (theorem 3.34; $z_{0.995} = 2.576$)?
- Why are the two answers so different, and which one would you trust?

Solution →

IN FINANCE: The casino, the poker room and the trader (2013)

Casino games against the house are designed so that the house has an edge of about 1–5% (2013 notes) while the variance of each game is large compared with the edge [► 2013 L3 @ 50:22]. If a player wins an even-money bet with probability 0.48 (the indicative figure quoted in class for blackjack [► 2013 L3 @ 51:22]), the expected P&L per \$1 bet is $0.48 - 0.52 = -\$0.04$ with a standard deviation of about \$1: over an evening the edge is invisible, but the casino plays an enormous n and the LLN makes its profit almost certain [► 2013 L3 @ 52:22]. In poker the players play against each other and the house simply charges a fee per hand (the *rake*) [► 2013 L3 @ 54:32]; a player whose edge over the others exceeds the rake can let the LLN work for him. The same logic drives a hedge fund or a high-frequency trader: a small edge per trade, many (nearly) independent trades [► 2013 L3 @ 55:32]. The practical difficulty is the

variance: drawdowns make it hard to tell a real edge from bad luck [► 2013 L3 @ 56:34], and correlated trades reduce the effective n .

Theorem 3.34 (Central limit theorem). Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$. Then

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) = \sqrt{n} (\bar{X}_n - \mu) \longrightarrow N(0, \sigma^2) \text{ in distribution}$$

[► 2024 L5 @ 31:38][► 2013 L3 @ 1:07:29].

Remark 3.35.

- The statement “ $\bar{X}_n \approx N(\mu, \sigma^2/n)$ ” is written with quotes in class, because a limit law cannot depend on n ; it is the practical approximation [► 2024 L5 @ 32:47].
- The MGF is only a convenience: with characteristic functions the same proof works whenever $\sigma^2 < \infty$.
- Identical distribution is not needed (Lindeberg–Lyapunov CLT) [► 2024 L5 @ 38:04], and weak dependence is allowed (mixing sequences, martingale differences).

To a student’s question whether the CLT is of any use when returns are neither independent nor identically distributed [► 2024 L5 @ 38:04], the answer in class: what matters is that a quantity be a sum of many small, roughly independent contributions. Projections of a high-dimensional vector on a random direction, and the residuals of a regression, are such sums and are often close to Gaussian [► 2024 L5 @ 39:09].

Estimating a mean (2013 notes, Example 4.2). To estimate an unknown mean from i.i.d. observations one uses $\bar{X}_n$ [► 2013 L3 @ 1:16:15]²⁷: by the LLN it converges to μ , and by the CLT its error is approximately $N(0, \sigma^2/n)$ [► 2013 L3 @ 1:17:21], which gives the usual interval $\bar{X}_n \pm 1.96 s/\sqrt{n}$.

IN FINANCE: Aggregation of returns

A T -day log-return is a sum of T daily log-returns, so by the CLT it is approximately normal for long horizons even when daily returns are fat-tailed (as long as their variance is finite): return distributions look more Gaussian at monthly than at daily frequency. Volatility clustering slows this convergence down. The same theorem makes the binomial tree converge to the lognormal model (chapter 15) and the random walk to Brownian motion (chapter 4).

Exercise 3.11 (not from the course: no LLN for Cauchy). Assume that the standard Cauchy law has characteristic function $e^{-|t|}$. Show that if $X_1, \dots, X_n$ are i.i.d. standard Cauchy, $\bar{X}_n$ is again standard Cauchy. Why does this not contradict theorem 3.32? What does it suggest about averaging returns with very heavy tails?

Solution →

Proof when the MGF exists. Both courses prove it this way [► 2024 L5 @ 33:48] [► 2013 L3 @ 1:09:46]. Take $\mu = 0$ and let M be the MGF of X_i , finite near 0. By independence the expectation of the product is the product of expectations [► 2024 L5 @ 34:51]:

$$M_{Z_n}(t) = \mathbb{E} \left[\prod_{i=1}^n e^{tX_i/\sqrt{n}} \right] = M \left(\frac{t}{\sqrt{n}} \right)^n.$$

M is smooth near 0 with $M(0) = 1$, $M'(0) = \mathbb{E}[X] = 0$, $M''(0) = \mathbb{E}[X^2] = \sigma^2$, so by Taylor $M(s) = 1 + \frac{1}{2}\sigma^2 s^2 + o(s^2)$ as $s \rightarrow 0$ [► 2024 L5 @ 35:54]. For fixed t ,

$$\ln M_{Z_n}(t) = n \ln \left(1 + \frac{\sigma^2 t^2}{2n} + o\left(\frac{1}{n}\right) \right) \longrightarrow \frac{\sigma^2 t^2}{2},$$

as in $(1 + r/n)^n \rightarrow e^r$, the limit behind continuous compounding [► 2024 L5 @ 37:01]. So $M_{Z_n}(t) \rightarrow e^{\sigma^2 t^2/2}$, the MGF of $N(0, \sigma^2)$, and theorem 3.14(ii) concludes. □

27 COURSE NOTE: Is the sample mean the best estimator?

In the 2013 lecture the sample mean is called the maximum likelihood estimator [► 2013 L3 @ 1:18:23]. It is the MLE for normal (and Poisson, exponential) data but not in general, and the notes rightly add that for some distributions it is not the best choice: for Cauchy data it does not converge at all,²⁸ while the sample median does; for heavy-tailed data robust estimators are preferable (chapter 5).

SEEN
28 insane!

3.11 Probability theory for asset pricing: from expected utility to the CAPM

The slides lec04_2 derive the capital asset pricing model with nothing more than this chapter: normal payoffs, covariances and a Gaussian identity [► 2024 L5 @ 40:17]. The development follows F. de Jong and B. Rindi, *The Microstructure of Financial Markets* (2009) [► 2024 L5 @ 1:11:22].

3.11.1 One risky asset

One period, from t_0 (trading) to t_1 (end of period) [► 2024 L5 @ 41:21]. Before t_0 an agent is endowed with Q_a shares of a risky asset and Q_f units of a risk-free asset. At t_0 the risk-free asset costs \$1 per unit and the risky asset p_a per share [► 2024 L5 @ 42:26], so the initial wealth is $w_0 = Q_a p_a + Q_f$. At t_1 the risk-free asset pays $1 + r_f$ per unit and the risky one pays $\tilde{F} \sim N(\mu, \sigma^2)$ per share [► 2024 L5 @ 43:28]. (A normal payoff can be negative; the model is useful if this has negligible probability [► 2024 L5 @ 44:28].) The agent buys X shares at t_0 ($X < 0$: sells), financed with the risk-free asset, so the final wealth is [► 2024 L5 @ 45:31]

$$\tilde{w} = (Q_a + X)\tilde{F} + (Q_f - Xp_a)(1 + r_f), \quad \frac{\partial \tilde{w}}{\partial X} = \tilde{F} - p_a(1 + r_f). \quad (3.3)$$

Expected utility. The agent chooses X to maximise $\mathbb{E}[U(\tilde{w})]$ for a utility function with $U' > 0$ [► 2024 L5 @ 46:37] (more wealth is better) and $U'' < 0$ [► 2024 L5 @ 47:37] (decreasing marginal utility). Why not simply $U(w) = w$ [► 2024 L5 @ 48:39]? Because then there is no penalty for risk [► 2024 L5 @ 49:47]²⁹: with U concave, Jensen's inequality gives $\mathbb{E}[U(\tilde{w})] \leq U(\mathbb{E}[\tilde{w}])$, so the agent prefers a sure amount to a gamble with the same mean. The ratio $A(w) = -U''(w)/U'(w)$ is the (Arrow–Pratt) coefficient of absolute risk aversion.

3.11.2 Stein's lemma

Lemma 3.36 (Stein). Let Y and Y^* be jointly normal with $Y \sim N(\mu, \sigma^2)$, and let g be differentiable with $\mathbb{E}[|g(Y)(Y - \mu)|] < \infty$ and $\mathbb{E}[|g'(Y)|] < \infty$. Then [► 2024 L5 @ 50:58]

$$\mathbb{E}[g(Y)(Y - \mu)] = \sigma^2 \mathbb{E}[g'(Y)] \quad \text{and} \quad \text{Cov}(g(Y), Y^*) = \mathbb{E}[g'(Y)] \text{Cov}(Y, Y^*).$$

The proof is 2024 PS3 Problem 1 (Exercise 3.12). The first identity is a Gaussian integration by parts, the second reduces to the first through the decomposition (3.2).

INTUITION: Gaussian integration by parts

For a centred Gaussian, $\mathbb{E}[Y g(Y)] = \sigma^2 \mathbb{E}[g'(Y)]$ is the identity behind Wick's theorem and the Schwinger–Dyson equations: with $g(y) = y^{k-1}$ it gives $\mathbb{E}[Y^k] = (k-1)\sigma^2 \mathbb{E}[Y^{k-2}]$, hence $\mathbb{E}[Y^4] = 3\sigma^4$. The second form says that the covariance of a nonlinear function $g(Y)$ with any jointly Gaussian Y^* is that of its linearisation, $g' \cdot \text{Cov}(Y, Y^*)$, with the slope g' averaged over the law of Y . In the pricing argument $g = U'$: it converts “covariance of marginal utility with the payoff” into “covariance of wealth with the payoff”.

29 COURSE NOTE: Shannon, Kelly and the log-utility debate

The lecturer recalls a controversy between Claude Shannon and (he thinks) Milton Friedman on maximising the expected logarithm of wealth [► 2024 L5 @ 49:47]. The well-documented debate is between Paul Samuelson, who argued against maximising expected log-wealth, and the proponents of the *Kelly criterion* (J. Kelly, C. Shannon, E. Thorp); see W. Poundstone, *Fortune's Formula* (2005). Log utility has $A(w) = 1/w$.

Exercise 3.12 (PS3 (2024), Problem 1: Stein's lemma). Prove lemma 3.36 (with X in place of Y and Y in place of Y^*): if X, Y are jointly normal with $X \sim N(\mu, \sigma^2)$ and g is differentiable with $\mathbb{E}[g(X)(X-\mu)]$ and $\mathbb{E}[g'(X)]$ finite, then $\mathbb{E}[g(X)(X-\mu)] = \sigma^2 \mathbb{E}[g'(X)]$ and $\text{Cov}(g(X), Y) = \mathbb{E}[g'(X)] \text{Cov}(X, Y)$. *Hint*: integrate by parts; you may use that for jointly normal (X, Y) with standard deviations σ, τ and correlation ρ , $Y = a + bX + \varepsilon$ with $a = \mu_Y - b\mu_X$, $b = \rho\tau/\sigma$ and $\varepsilon \sim N(0, (1 - \rho^2)\tau^2)$ independent of X .

Solution →

3.11.3 First-order condition and the equilibrium price

Assume one may differentiate under the expectation, which holds by dominated convergence if, e.g., U' is bounded, or for the exponential utility below since normal variables have exponential moments [► 2024 L5 @ 53:03]. By (3.3) the first-order condition is

$$\begin{aligned} 0 &= \frac{\partial}{\partial X} \mathbb{E}[U(\tilde{w})] = \mathbb{E}[U'(\tilde{w})(\tilde{F} - p_a(1 + r_f))] \\ &= \text{Cov}(U'(\tilde{w}), \tilde{F}) + \mathbb{E}[U'(\tilde{w})](\mu - p_a(1 + r_f)), \end{aligned}$$

using $\mathbb{E}[AB] = \text{Cov}(A, B) + \mathbb{E}[A] \mathbb{E}[B]$ and that a constant shift does not change a covariance [► 2024 L5 @ 55:19]. Now $\tilde{w}$ is normal (affine in $\tilde{F}$), so lemma 3.36 with $g = U'$, $Y = \tilde{w}$, $Y^* = \tilde{F}$ gives

$$\begin{aligned} \text{Cov}(U'(\tilde{w}), \tilde{F}) &= \mathbb{E}[U''(\tilde{w})] \text{Cov}(\tilde{w}, \tilde{F}) \\ &= \mathbb{E}[U''(\tilde{w})](Q_a + X)\sigma^2 \end{aligned}$$

[► 2024 L5 @ 56:22]. Solving for p_a :

$$p_a = \frac{1}{1 + r_f} \left[\mathbb{E}[\tilde{F}] + \frac{\mathbb{E}[U''(\tilde{w})]}{\mathbb{E}[U'(\tilde{w})]} (Q_a + X) \text{Var}(\tilde{F}) \right]. \quad (3.4)$$

Since $U'' < 0 < U'$ the second term is negative: the price is the discounted expected payoff minus a risk adjustment [► 2024 L5 @ 58:29].

CARA utility [► 2024 L5 @ 59:30]. For $U(w) = c - e^{-Aw}$ with $A > 0$: $U'(w) = Ae^{-Aw} > 0$ and $U''(w) = -A^2 e^{-Aw} = -A U'(w)$ ³⁰, so the ratio in (3.4) is the constant $-A$ and

$$p_a = \frac{1}{1 + r_f} \left[\mathbb{E}[\tilde{F}] - A (Q_a + X) \text{Var}(\tilde{F}) \right]. \quad (3.5)$$

The risk adjustment is proportional to the risk aversion A , to the position held $Q_a + X$ and to the variance of the payoff [► 2024 L5 @ 1:00:36].

Remark 3.37 (CARA and normal = mean–variance; not discussed in class). Since $-A\tilde{w}$ is normal, (3.1) at $t = 1$ gives $\mathbb{E}[U(\tilde{w})] = c - \exp(-A \mathbb{E}[\tilde{w}] + \frac{A^2}{2} \text{Var} \tilde{w})$. Maximising expected utility is therefore equivalent to maximising the *certainty equivalent* $\mathbb{E}[\tilde{w}] - \frac{A}{2} \text{Var} \tilde{w}$, a mean–variance criterion (chapter 10). Its maximiser is the demand $Q_a + X = (\mathbb{E}[\tilde{F}] - (1 + r_f)p_a)/(A\sigma^2)$, and (3.5) is this demand curve solved for the price: the price at which the agent is content to hold $Q_a + X$ shares.

30 COURSE NOTE: Erratum in the slides

Slide 6 of lec04_2 writes $U'(\tilde{w}) = AU(\tilde{w})$ and $U''(\tilde{w}) = -A^2 U(\tilde{w})$. For $U = c - e^{-Aw}$ one has $U' = Ae^{-Aw} = A(c - U)$ and $U'' = -A^2 e^{-Aw}$; even with $c = 0$ the signs are wrong ($U' = -AU$, $U'' = A^2 U$). What the derivation uses, $U''/U' = -A$, and hence (3.5), are correct.

Expected return. With $\tilde{r} = (\tilde{F} - p_a)/p_a$, (3.4) gives

$$\mathbb{E}[\tilde{r}] = r_f - \frac{\mathbb{E}[U''(\tilde{w})]}{\mathbb{E}[U'(\tilde{w})]} \frac{(Q_a + X) \text{Var}(\tilde{F})}{p_a} \stackrel{\text{CARA}}{=} r_f + A \frac{(Q_a + X) \text{Var}(\tilde{F})}{p_a} :$$

the risk-free rate plus a risk premium that grows with risk aversion, position and variance.

3.11.4 Many risky assets

Now the agent holds $Q \in \mathbb{R}^n$ shares of n risky assets priced p at t_0 , whose payoffs $\tilde{F} \sim N_n(\mu_F, \Sigma_F)$ are jointly normal [► 2024 L5 @ 1:01:39], and buys X : $\tilde{w} = (Q + X)^\top \tilde{F} + (Q_f - X^\top p)(1 + r_f)$, again normal. The n first-order conditions are handled exactly as before, with Stein's lemma applied to the jointly normal pair $(\tilde{w}, \tilde{F}_i)$:

$$p_i = \frac{1}{1 + r_f} \left[\mathbb{E}[\tilde{F}_i] + \frac{\mathbb{E}[U''(\tilde{w})]}{\mathbb{E}[U'(\tilde{w})]} \text{Cov}(\tilde{w}, \tilde{F}_i) \right], \quad \text{Cov}(\tilde{w}, \tilde{F}_i) = [\Sigma_F(Q + X)]_i.$$

For CARA, $p = [\mu_F - A \Sigma_F(Q + X)]/(1 + r_f)$. With $\tilde{r}_i = (\tilde{F}_i - p_i)/p_i$, so that $\text{Cov}(\tilde{w}, \tilde{r}_i) = \text{Cov}(\tilde{w}, \tilde{F}_i)/p_i$,

$$\mathbb{E}[\tilde{r}_i] = r_f - \frac{\mathbb{E}[U''(\tilde{w})]}{\mathbb{E}[U'(\tilde{w})]} \text{Cov}(\tilde{w}, \tilde{r}_i) \stackrel{\text{CARA}}{=} r_f + A \text{Cov}(\tilde{w}, \tilde{r}_i)$$

[► 2024 L5 @ 1:03:48]. The premium of asset i is set by its covariance with the agent's *total* wealth, not by its own variance: an asset that hedges wealth ($\text{Cov} < 0$) is expected to earn less than r_f .

3.11.5 The capital asset pricing model

Market portfolio. In equilibrium prices must be such that the assets are willingly held. With identical agents (or a representative agent) holding no initial risky endowment, $Q = 0$, the agent's equilibrium purchase X is the whole supply of risky assets: the *market portfolio* [► 2024 L5 @ 1:03:48]. Its payoff, price and return are

$$\tilde{F}_m = \sum_i X_i \tilde{F}_i, \quad p_m = \sum_i X_i p_i, \quad \tilde{r}_m = \frac{\tilde{F}_m - p_m}{p_m} = \sum_i \omega_i \tilde{r}_i, \quad \omega_i = \frac{X_i p_i}{p_m},$$

with ω_i the market-value weights [► 2024 L5 @ 1:05:56]³¹.

The key identity is $\tilde{w} = \tilde{F}_m + (Q_f - p_m)(1 + r_f) = p_m \tilde{r}_m + \text{const}$, hence $\text{Cov}(\tilde{w}, \cdot) = p_m \text{Cov}(\tilde{r}_m, \cdot)$.

Theorem 3.38 (CAPM, security market line). In this equilibrium, with $\beta_i = \text{Cov}(\tilde{r}_i, \tilde{r}_m)/\text{Var}(\tilde{r}_m)$,

$$\mathbb{E}[\tilde{r}_i] = r_f + \beta_i (\mathbb{E}[\tilde{r}_m] - r_f), \quad p_i = \frac{\mathbb{E}[\tilde{F}_i]}{1 + r_f + \beta_i (\mathbb{E}[\tilde{r}_m] - r_f)}.$$

31 COURSE NOTE: Notation and assumptions in lec04_2

The slides denote the market weights by μ_i , which clashes with the means; here they are ω_i . The step “the agent's portfolio is the market portfolio” is the market-clearing condition with identical agents, stated on slide 12 as an assumption. The CAPM was formulated by W. Sharpe (1964) and J. Lintner (1965) from mean-variance portfolio choice [► 2024 L5 @ 1:11:22]; by [remark 3.37](#) the CARA-normal route taken here is equivalent. A Lintner paper was distributed on Canvas; the OCW Week 3 page lists both Lintner (1965) papers.

Proof. Let $\theta = -\mathbb{E}[U''(\tilde{w})]/\mathbb{E}[U'(\tilde{w})] > 0$. Averaging the expected returns with weights ω_i [► 2024 L5 @ 1:07:04]: $\mathbb{E}[\tilde{r}_m] = r_f + \theta \text{Cov}(\tilde{w}, \tilde{r}_m) = r_f + \theta p_m \text{Var}(\tilde{r}_m)$, so $\theta = (\mathbb{E}[\tilde{r}_m] - r_f)/(p_m \text{Var}(\tilde{r}_m))$. Then for each asset $\mathbb{E}[\tilde{r}_i] - r_f = \theta \text{Cov}(\tilde{w}, \tilde{r}_i) = \theta p_m \text{Cov}(\tilde{r}_m, \tilde{r}_i) = \beta_i (\mathbb{E}[\tilde{r}_m] - r_f)$ [► 2024 L5 @ 1:08:10]. Finally $\mathbb{E}[\tilde{r}_i] = \mathbb{E}[\tilde{F}_i]/p_i - 1$ gives the price [► 2024 L5 @ 1:10:18]. □

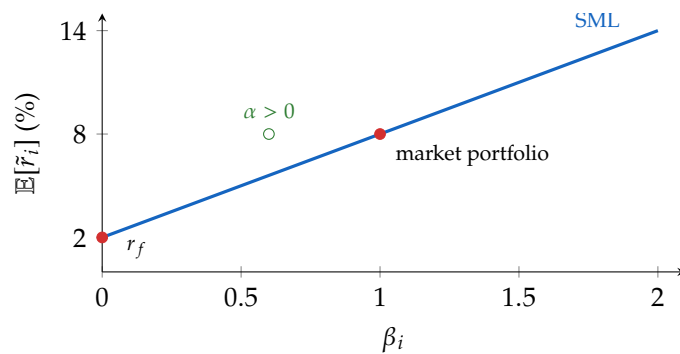


Figure 3.7: Security market line with illustrative numbers $r_f = 2\%$, $\mathbb{E}[\tilde{r}_m] = 8\%$. In the CAPM every asset lies on the line; an asset above it (open dot) would have a positive “alpha”.

IN FINANCE: Reading and using the CAPM

- β_i is the slope of the regression of $\tilde{r}_i$ on $\tilde{r}_m$ [► 2024 L5 @ 1:09:14]([definition 1.2, chapter 5](#)). High-beta stocks must offer higher expected returns; $\beta < 1$ is “defensive”.
- Only systematic risk is priced: the idiosyncratic part of $\text{Var } \tilde{r}_i$ can be diversified away ([section 3.6](#)) and earns nothing.
- Practice (not discussed in class): the cost of equity in corporate valuation is often set to $r_f + \beta \times (\text{equity risk premium})$; fund performance is measured by the *alpha* over the SML. Empirically the SML is flatter than predicted (low-beta stocks earn more than the CAPM says), which motivated multi-factor models ([chapters 7 and 10](#)).

Exercises

The short exercises are in the text, right after the paragraph they practise. These are longer, structured problems that use several sections: the 2024 lognormal problem set (moments and options), time diversification, the exponential–Poisson pair of 2013 PS2 (A-1 then B-1, which uses it), and Stein’s lemma for stochastic volatility.

Exercise 3.13 (PS2 (2024), Problem 2: lognormal distribution). Let X be lognormal(μ, σ^2), i.e. $Y = \ln X \sim N(\mu, \sigma^2)$.

- Prove that $\mu_* = \mathbb{E}[X] = e^{\mu + \sigma^2/2}$.
- Prove that $\sigma_*^2 = \text{Var } X = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = \mu_*^2 (e^{\sigma^2} - 1)$. For (a) and (b) use the MGF of the normal law; no integrals are needed.
- A stock has price S_0 at $t = 0$ and $S_0 X$ at time t (years), where X is lognormal with $\mu = t \ln 1.15$ and $\sigma = \sqrt{t} \ln 1.3$. Find the mean and standard deviation of the total return X for $t = 1/12, 3/12, 1$.
- Show that for a call with strike K on a lognormal(μ, σ^2) price, $\mathbb{E}[(X - K)^+] = e^{\mu + \sigma^2/2} \Phi\left(\frac{\mu - \ln K}{\sigma} + \sigma\right) - K \Phi\left(\frac{\mu - \ln K}{\sigma}\right)$.
- For the stock of (c) with $S_0 = \$100$, compute the expected payoff of the call for $T = 1/12, 3/12, 1$ and strikes $K = \$100$ (at the money) and $K = \$120$ (out of the

money). Comment on the effect of a longer time to expiration.

Solution →

Exercise 3.14 (PS2 (2013), B-4: does time diversify risk?). (From M. Kritzman, *Puzzles of Finance*, ch. 3; see also Kritzman and Rich, *Beware of dogma: the truth about time diversification*.) A stock has independent annual returns, each lognormal with $\mu = \ln 1.1$ and $\sigma = \ln 1.2$. According to the author, the fraction of wealth lost with 0.1% probability over T years is 40.5%, 58.43%, 61.48% and 73.92% for $T = 1, 5, 10, 20$, and he concludes that if risk is measured by the magnitude of potential loss, it *increases* with the horizon.

- Compute the fraction $h(T)$ of wealth lost with probability 0.1% over T years (in terms of Φ).
- Find the T that maximises $h(T)$. Is h increasing in T ?
- Criticise the argument using (a) and (b).

Solution →

Exercise 3.15 (PS2 (2013), A-1: exponential distribution). X has density $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$.

- Compute its MGF.
- Deduce mean and variance from the MGF.
- Compute $\mathbb{P}(X > t)$ for $t > 0$.
- Prove the memoryless property $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$ for all $s, t > 0$.
- If $X_1, \dots, X_n$ are i.i.d. $\text{exponential}(\lambda)$, prove that $\min\{X_1, \dots, X_n\}$ is exponential.
- Entering a bank you find all three tellers busy and nobody waiting. Service times of the three customers and yours are i.i.d. exponential. What is the probability that you are the last of the four to leave?

Solution →

Exercise 3.16 (PS2 (2013), B-1: from exponential waiting times to Poisson counts). Let $X_1, X_2, \dots$ be i.i.d. $\text{exponential}(\lambda)$ and $\tau = \max\{t \geq 0 : X_1 + \dots + X_t \leq 1\}$, the number of events with exponential waiting times in an interval of length 1.

- Compute $\mathbb{P}(\tau = 0)$.
- Compute $\mathbb{P}(\tau = n)$ using that $S_n = X_1 + \dots + X_n$ has CDF $F_{S_n}(x) = 1 - \sum_{k=0}^{n-1} \frac{1}{k!} e^{-\lambda x} (\lambda x)^k$ ($x > 0$), and conclude that τ is Poisson.

Solution →

Exercise 3.17 (PS3 (2024), Problem 2: Stein's lemma for stochastic volatility). (From A. Gron, B. Jørgensen, N. Polson, *Optimal portfolio choice and stochastic volatility*, Applied Stochastic Models in Business and Industry, 2012; cited as a 2011 working paper in the problem set.) X has *stochastic volatility* if $X \mid V \sim N(\mu, \sigma^2 V)$ with $V > 0$ random with density $p(V)$; the law of X is $p(X) = \int_0^\infty p(X \mid V) p(V) dV$ and has heavy tails. Assume $0 < \mathbb{E}[V] < \infty$ and define the *size-biased* density $q(V) = V p(V) / \mathbb{E}[V]$ and the induced $q(X) = \int_0^\infty p(X \mid V) q(V) dV$. The authors' theorem states

$$\text{Cov}(g(X), X) = \mathbb{E}^Q[g'(X)] \text{Var } X, \quad (**)$$

with $\mathbb{E}^Q$ the expectation under the measure induced by $q(V)$. Their proof reads

$$\begin{aligned}\text{Cov}(g(X), X) &\stackrel{(1)}{=} \mathbb{E}[g(X)(X - \mathbb{E}[X])] \\ &\stackrel{(2)}{=} \mathbb{E}_V[\mathbb{E}_{X|V}[g(X)(X - \mathbb{E}[X])]] \\ &\stackrel{(3)}{=} \mathbb{E}_V[\mathbb{E}_{X|V}[g(X)(X - \mathbb{E}[X | V])]].\end{aligned}$$

- Show that q is a probability density on $(0, \infty)$.
- Prove (1).
- Does (2) follow from the law of iterated expectations?
- Prove (3) (compare $\mathbb{E}[X | V]$ and $\mathbb{E}[X]$).
- Applying Stein's lemma conditionally on V they obtain $\text{Cov}(g(X), X) = \mathbb{E}_V\{\mathbb{E}_{X|V}[g'(X)] \sigma^2 V\} = \mathbb{E}[g'(X) \frac{V}{\mathbb{E}[V]}] \sigma^2 \mathbb{E}[V]$ (*). Does (*) imply (**)? Explain.

Solution →

KEY CONCEPTS

Can you answer these without looking?

- Why are prices modelled as lognormal rather than normal? What are the parameters μ, σ of a lognormal law?
- For a lognormal variable, put mode, median and mean in order, and give $\mathbb{E}[Y^k]$.
- Write the expected payoff of a call in terms of the distribution function, and explain why it holds.
- What does an MGF determine, and why is it convenient for sums? Name a law with all moments and no MGF.
- What are the mean and variance of a portfolio $a^\top X$? Why does diversification stop at a floor?
- For jointly normal (X, Y) , what are $\mathbb{E}[Y | X]$ and $\text{Var}(Y | X)$?
- What are the principal components of a random vector, and what do their variances add up to?
- Why does a random variance produce a kurtosis above 3?
- What do the law of large numbers and the central limit theorem say, and how fast does $\bar{X}_n - \mu$ shrink?
- State Stein's lemma. Where is it used in the chapter?
- In the CARA-normal model, the price is the discounted expected payoff minus what?
- State the CAPM. Which risk earns a premium, and why?

The concepts

Distribution function and PIT. F_X determines the law; $F_X(X) \sim U(0, 1)$ if F_X is continuous and $F^{-1}(U) \sim F$, which is how random numbers with any law are simulated (proposition 3.3).

Shape. Skewness γ and kurtosis κ (3 for the normal; leptokurtic if $\kappa > 3$) measure asymmetry and fat tails; under $a + bX$ the kurtosis is invariant and the skewness changes sign with b (definition 3.4 and proposition 3.5).

Change of variables. For $Y = g(X)$ monotone, $f_Y(y) = f_X(h(y)) |h'(y)|$ with $h = g^{-1}$ (theorem 3.7).

Lognormal. $Y = e^X$, $X \sim N(\mu, \sigma^2)$: positive, right-skewed, the law of prices; μ, σ belong to $\ln Y$. $f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} e^{-(\ln y - \mu)^2/(2\sigma^2)}$, $\mathbb{E}[Y^k] = e^{k\mu + k^2\sigma^2/2}$, mode < median < mean.

Expected option payoff. $\mathbb{E}[(X - K)^+] = \int_K^\infty [1 - F(x)] dx$ (theorem 3.9). Normal: $(\mu - K)\Phi(d) + \sigma\phi(d)$, $d = (\mu - K)/\sigma$; lognormal: $e^{\mu + \sigma^2/2}\Phi(d + \sigma) - K\Phi(d)$, $d = (\mu - \ln K)/\sigma$, the core of Black-Scholes.

MGF. $M_X(t) = \mathbb{E}[e^{tX}]$, if finite near 0, determines the law and its convergence (theorem 3.14); products for independent sums; $M_{a+bX}(t) = e^{at}M_X(bt)$; $N(\mu, \sigma^2)$: $e^{\mu t + \sigma^2 t^2/2}$; χ_n^2 : $(1 - 2t)^{-n/2}$. The lognormal has every moment but no MGF.

Portfolios and diversification. $\mathbb{E}[a^\top X] = a^\top \mu$, $\text{Var}(a^\top X) = a^\top \Sigma a$ (proposition 3.17); averaging uncorrelated risks divides the variance by n , common (correlated) risk remains.

Conditional normal. $\mathbb{E}[Y | X] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$, $\text{Var}(Y | X) = (1 - \rho^2)\sigma_Y^2$: the regression line (3.2); in general $\mathbb{E}[Y | X]$ is the best predictor of Y given X (proposition 3.19).

Principal components. $p = \Gamma^\top(x - \alpha)$ rotates to uncorrelated coordinates, $\text{Cov } p = \Lambda$, $\text{tr } \Sigma = \sum \lambda_k$ (proposition 3.25).

Heavy tails. A normal law with random variance has $\kappa = 3\mathbb{E}[V^2]/(\mathbb{E}[V])^2 \geq 3$ (proposition 3.29): stochastic volatility makes returns fat-tailed; t and Cauchy laws have power tails.

Limit theorems. Chebyshev $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2)$ gives the LLN (lemma 3.31 and theorem 3.32); CLT $\sqrt{n}(\bar{X}_n - \mu) \rightarrow N(0, \sigma^2)$ (theorem 3.34): the error shrinks like $\sigma/\sqrt{n}$. Without a finite mean (Cauchy) averaging does not help.

Stein's lemma. For jointly normal Y, Y^* : $\text{Cov}(g(Y), Y^*) = \mathbb{E}[g'(Y)]\text{Cov}(Y, Y^*)$ (lemma 3.36); it turns the first-order condition of expected utility into a covariance.

Risk-adjusted price. With CARA utility and normal payoffs, $p_a = [\mathbb{E}[\tilde{F}] - A(Q_a + X)\text{Var } \tilde{F}]/(1 + r_f)$ (eq. (3.5)): discounted expectation minus a premium proportional to risk aversion, position and variance.

CAPM. $\mathbb{E}[\tilde{r}_i] = r_f + \beta_i(\mathbb{E}[\tilde{r}_m] - r_f)$, $\beta_i = \text{Cov}(\tilde{r}_i, \tilde{r}_m)/\text{Var } \tilde{r}_m$ (theorem 3.38): only the risk shared with the market portfolio is priced; the rest is diversified away.

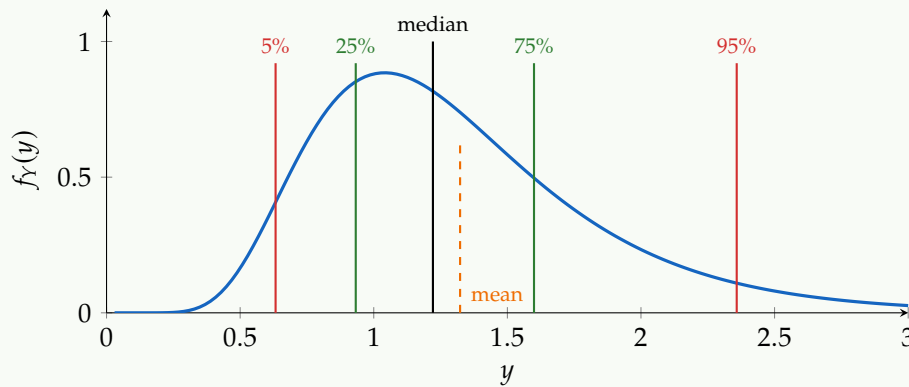
Solutions

Solutions of the exercises and labs of this chapter.

SOLUTION: Exercise 3.1

- (a) $q_p = e^{0.2+0.4z_p}$: 0.633, 0.933, 1.221, 1.600, 2.358 [► 2024 L4 @ 48:58].
- (b) Mode $e^{\mu - \sigma^2} = e^{0.04} = 1.041$; mean $e^{\mu + \sigma^2/2} = e^{0.28} = 1.323$; standard deviation $1.323\sqrt{e^{0.16} - 1} = 0.551$. So mode < median < mean, as in the plot.
- (c) Multiplicatively: $1.221/0.633 = 2.358/1.221 = 1.93 = e^{0.4 \cdot 1.645}$, because $\ln X_1$ is symmetric about μ .

In class the example is read as an asset with an annualised return of about 20% and volatility 0.4. Be careful which return: 0.2 is the expected log-return; the median gross return is $e^{0.2} = 1.221$ (+22.1%), the expected gross return $e^{0.28} = 1.323$ (+32.3%). The gap $\sigma^2/2 = 0.08$ is the convexity correction that reappears as the $-\sigma^2/2$ in the solution of the geometric Brownian motion (chapter 17).



Density of the lognormal(0.2, 0.4²) law with its 5, 25, 50, 75, 95% quantiles (as on slide 16 of lec04_1) and its mean. The mode (1.041, the peak) lies left of the median, the mean right of it.

← back

SOLUTION: Exercise 3.2

- (a) With $z = (x - \mu)/\sigma$, $(\int \phi)^2 = \frac{1}{2\pi} \iint e^{-(u^2+v^2)/2} du dv = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\infty e^{-r^2/2} r dr d\theta = 1$, and $\phi > 0$.
- (b) For $\mu = 0$, $\mathbb{E}[Y^k] = \int e^{kx} \phi(x) dx$; completing the square, $kx - \frac{x^2}{2\sigma^2} = -\frac{(x - k\sigma^2)^2}{2\sigma^2} + \frac{k^2\sigma^2}{2}$, and by (a) the shifted Gaussian integrates to 1: $\mathbb{E}[Y^k] = e^{k^2\sigma^2/2}$. So $\mathbb{E}[Y] = e^{\sigma^2/2}$, $\text{Var } Y = e^{2\sigma^2} - e^{\sigma^2}$; for general μ multiply by e^μ and $e^{2\mu}$.

← back

SOLUTION: Exercise 3.3

- (a) With $m_k = \mathbb{E}[Y^k] = e^{k\mu + k^2\sigma^2/2}$ and $w = e^{\sigma^2}$: $\mathbb{E}[(Y - m_1)^3] = m_3 - 3m_1m_2 + 2m_1^3 = m_1^3(w^3 - 3w + 2) = m_1^3(w - 1)^2(w + 2)$ and $\text{Var } Y = m_2 - m_1^2 = m_1^2(w - 1)$, so $\gamma = (w + 2)\sqrt{w - 1} > 0$, independent of μ .
- (b) For $Z = (X - \mu)/\sigma$, integrate by parts with $z\phi(z) = -\phi'(z)$: $\mathbb{E}[Z^4] = \int z^3 z\phi dz = 3 \int z^2 \phi dz = 3 \mathbb{E}[Z^2] = 3$.

← back

SOLUTION: Exercise 3.4

$(K - x)^+ = \int_{-\infty}^K \mathbf{1}\{x \leq y\} dy$; take expectations and exchange (Tonelli): $\mathbb{E}[(K - X)^+] = \int_{-\infty}^K F(y) dy$. Pointwise $(x - K)^+ - (K - x)^+ = x - K$, so the expectations differ by $\mathbb{E}[X] - K$. Long call plus short put pays $X - K$ in every state: a forward bought at price K .

← back

SOLUTION: Exercise 3.5

- (a) $M(t) = \sum_k e^{tk} \lambda^k e^{-\lambda} / k! = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$, for every t .
- (b) $K(t) = \ln M = \lambda(e^t - 1)$: $K'(0) = \lambda$ (mean), $K''(0) = \lambda$ (variance).
- (c) $\prod_i M_i(t) = e^{(\sum_i \lambda_i)(e^t - 1)}$, the MGF of $\text{Poisson}(\sum \lambda_i)$; conclude by theorem 3.14(i).

← back

SOLUTION: Exercise 3.6

Lognormal: in the text (section 3.5, $h = 1/x$, $t_1 = (\ln x)^2$, $t_2 = \ln x$). Poisson: $\frac{\lambda^k e^{-\lambda}}{k!} = \frac{1}{k!} e^{-\lambda} \exp(k \ln \lambda)$: $h = 1/k!$, $c = e^{-\lambda}$, $w = \ln \lambda$, $t = k$. Exponential: $\lambda e^{-\lambda x} \mathbf{1}\{x \geq 0\}$: $h = \mathbf{1}\{x \geq 0\}$, $c = \lambda$, $w = -\lambda$, $t = x$. (The support must not depend on θ : true in all three.)

← back

SOLUTION: Exercise 3.7

$X_3 = \pm 1$ with probability $\frac{1}{2}$ each, and for $a, c \in \{\pm 1\}$: $\mathbb{P}(X_1 = a, X_3 = c) = \mathbb{P}(X_1 = a, X_2 = c/a) = \frac{1}{4} = \mathbb{P}(X_1 = a)\mathbb{P}(X_3 = c)$; the same for (X_2, X_3) . But $\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = -1) = 0 \neq \frac{1}{8}$: X_3 is a function of the other two.

← back

SOLUTION: Exercise 3.8

$\hat{\Sigma} = \frac{1}{T} X^* X^{*\top} = \frac{1}{T} V D U^\top U D V^\top = V \frac{D^2}{T} V^\top$ with V orthogonal and D^2/T diagonal and decreasing: this is the spectral decomposition, so $\hat{\Gamma} = V$, $\hat{\Lambda} = D^2/T$ (unique up to signs and ties). Then $P = \hat{\Gamma}^\top X^* = V^\top V D U^\top = D U^\top$.

← back

SOLUTION: Exercise 3.9

Conditionally $\mathbb{E}[e^{tX} | V] = e^{t^2 V/2}$, so $M_X(t) = \mathbb{E}[e^{t^2 V/2}] = \frac{1}{1-t^2/2}$ for $t^2/2 < 1$. The Laplace density $\frac{1}{2b} e^{-|x|/b}$ has MGF $1/(1-b^2 t^2)$: here $b = 1/\sqrt{2}$. $\text{Var } X = 2b^2 = 1 = \mathbb{E}[V]$; $\mathbb{E}[X^4] = 24b^4 = 6$, so $\kappa = 6$, and proposition 3.29 gives $3 \mathbb{E}[V^2]/(\mathbb{E}[V])^2 = 3 \cdot 2 = 6$.

← back

SOLUTION: Exercise 3.10

- (a) $\sigma^2/(n\varepsilon^2) \leq 0.01$ gives $n \geq 10,000$.
- (b) $n \approx (2.576 \sigma/\varepsilon)^2 \approx 664$.
- (c) Chebyshev is valid for every distribution with finite variance, hence crude; the CLT value is an approximation for large n . Sharper exponential bounds that are still rigorous (Chernoff, Hoeffding) use MGF-type arguments, as mentioned in class [► 2013 L3 @ 1:01:53].

← back

SOLUTION: Exercise 3.11

$\varphi_{\bar{X}_n}(t) = \varphi(t/n)^n = e^{-n|t|/n} = e^{-|t|}$: the average has exactly the law of one observation. No contradiction: the law of large numbers needs $\mathbb{E}[|X|] < \infty$, and the Cauchy law has no mean. With very heavy tails averaging does not reduce dispersion: one extreme observation dominates the sum; use medians or other robust statistics.

← back

SOLUTION: Exercise 3.12

With ϕ the $N(\mu, \sigma^2)$ density, $(x - \mu)\phi(x) = -\sigma^2 \phi'(x)$, so integrating by parts (the boundary term vanishes under the integrability assumptions) $\mathbb{E}[g(X)(X - \mu)] = -\sigma^2 \int g \phi' = \sigma^2 \int g' \phi = \sigma^2 \mathbb{E}[g'(X)]$. Then $\text{Cov}(g(X), Y) = \text{Cov}(g(X), a + bX + \varepsilon) = b \text{Cov}(g(X), X) = b \mathbb{E}[g'(X)(X - \mu)] = b \sigma^2 \mathbb{E}[g'(X)]$, since ε is independent of X ; and $b \sigma^2 = \rho \tau \sigma = \text{Cov}(X, Y)$.

← back

SOLUTION: Exercise 3.13

- (a) $\mathbb{E}[X] = \mathbb{E}[e^Y] = M_Y(1) = e^{\mu+\sigma^2/2}$ by (3.1).
- (b) $\mathbb{E}[X^2] = M_Y(2) = e^{2\mu+2\sigma^2}$, so $\text{Var } X = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2} = \mu_*^2(e^{\sigma^2} - 1)$.
- (c) Mean and standard deviation of X : $t = 1/12$: 1.0146, 0.0770; $t = 3/12$: 1.0445, 0.1376; $t = 1$: 1.1903, 0.3177.
- (d) $\mathbb{E}[(X - K)^+] = \mathbb{E}[X\mathbf{1}\{X > K\}] - K\mathbb{P}(X > K)$, $\mathbb{P}(X > K) = \mathbb{P}(Y > \ln K) = \Phi(d)$ with $d = (\mu - \ln K)/\sigma$; and $\mathbb{E}[e^Y\mathbf{1}\{Y > \ln K\}] = e^{\mu+\sigma^2/2}\mathbb{P}(Y' > \ln K)$ with $Y' \sim N(\mu + \sigma^2, \sigma^2)$ (complete the square), which is $e^{\mu+\sigma^2/2}\Phi(d + \sigma)$ (also corollary 3.10).
- (e) Expected payoffs in dollars, $K = 100$: 3.83, 7.86, 23.35; $K = 120$: 0.04, 1.09, 11.99 for $T = 1/12, 3/12, 1$. A longer expiry raises both the drift and the dispersion of X , so the expected payoff grows; the out-of-the-money call grows much faster in relative terms, since it needs a large move to pay anything.

← back

SOLUTION: Exercise 3.14

- (a) Wealth after T years is e^Y with $Y \sim N(T\mu, T\sigma^2)$; its 0.1% quantile is $e^{T\mu+z\sigma\sqrt{T}}$ with $z = \Phi^{-1}(0.001) = -3.090$, so $h(T) = 1 - e^{T\mu+z\sigma\sqrt{T}}$. With $\mu = \ln 1.1$, $\sigma = \ln 1.2$: $h = 37.4\%$, 54.3% , 56.3% , 45.9% for $T = 1, 5, 10, 20$. (These are not the author's numbers, which must use another parametrisation; the conclusion below does not depend on it.)
- (b) $T\mu + z\sigma\sqrt{T}$ is minimal at $\sqrt{T} = -z\sigma/(2\mu)$, $T^* = (z\sigma/2\mu)^2 \approx 8.7$ years, where $h \approx 56.5\%$. h increases up to T^* and then decreases, to 0 as $T \rightarrow \infty$, because the drift $T\mu$ beats the spread $\sigma\sqrt{T}$.
- (c) The argument picks horizons on the increasing branch of h , and a 0.1% quantile ignores everything else: the probability of losing at all, $\Phi(-\mu\sqrt{T}/\sigma)$, falls with T , while the size of the worst outcomes grows. Whether time “diversifies” depends on the risk measure (and, for an investor, on utility: Samuelson's point), not on a single quantile.

← back

SOLUTION: Exercise 3.15

- (a) $M(t) = \int_0^\infty \lambda e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda-t}$ for $t < \lambda$.
- (b) $M'(0) = 1/\lambda$, $M''(0) = 2/\lambda^2$: mean $1/\lambda$, variance $1/\lambda^2$.
- (c) $\mathbb{P}(X > t) = e^{-\lambda t}$.
- (d) $\mathbb{P}(X > s+t \mid X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$.
- (e) $\mathbb{P}(\min_i X_i > t) = \prod_i \mathbb{P}(X_i > t) = e^{-n\lambda t}$: exponential with rate $n\lambda$.
- (f) When the first of the three leaves you start, and by memorylessness you and the two others have fresh i.i.d. exponential times. You are last if you are not the first of three to finish (probability $\frac{2}{3}$) and then not the first of two ($\frac{1}{2}$): $\frac{1}{3}$.

← back

SOLUTION: Exercise 3.16

- (a) $\mathbb{P}(\tau = 0) = \mathbb{P}(X_1 > 1) = e^{-\lambda}$.
- (b) $\{\tau \geq n\} = \{S_n \leq 1\}$, so $\mathbb{P}(\tau = n) = F_{S_n}(1) - F_{S_{n+1}}(1) = e^{-\lambda}\lambda^n/n!$: τ is Poisson(λ).

← back

SOLUTION: Exercise 3.17

- (a) $q \geq 0$ and $\int_0^\infty Vp(V) \, dV / \mathbb{E}[V] = 1$.
- (b) $\text{Cov}(g(X), X) = \mathbb{E}[g(X)X] - \mathbb{E}[g(X)]\mathbb{E}[X] = \mathbb{E}[g(X)(X - \mathbb{E}[X])]$.
- (c) Yes: it is $\mathbb{E}[Z] = \mathbb{E}[\mathbb{E}[Z \mid V]]$ for $Z = g(X)(X - \mathbb{E}[X])$.
- (d) $\mathbb{E}[X \mid V] = \mu$ for every V , so $\mathbb{E}[X \mid V] = \mathbb{E}[X]$ and (3) is the same expression as (2).
- (e) Yes. Under Q the pair (V, X) has density $q(V)p(X \mid V) = \frac{V}{\mathbb{E}[V]}p(V)p(X \mid V)$, so $\mathbb{E}\left[g'(X)\frac{V}{\mathbb{E}[V]}\right] = \mathbb{E}^Q[g'(X)]$; and $\text{Var } X = \mathbb{E}[\text{Var}(X \mid V)] = \sigma^2\mathbb{E}[V]$. Then (*) is exactly (**).

← back

Chapter 4

Stochastic Processes I: Random Walks, Markov Chains, Martingales

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L5 (from 1:12:25)	P. Kempthorne	Definition of a martingale; the simple random walk and simulations of its paths; the martingale $S_n^2 - n\sigma^2$.
2024 L6 (first 67 min)	P. Kempthorne	Product and exponential martingales; filtrations, non-anticipating strategies and martingale transforms; stopping times; gambler's ruin (fair and biased) by optional stopping; Markov processes and chains, n -step transitions; credit-rating migration; an up/down Markov chain for AAPL.
2013 L5	C. Lee	Processes as distributions over paths; simple random walk, its properties, gambler's ruin by recursion; Markov chains, transition matrix, stationary distribution via Perron–Frobenius; martingales, stopping times, optional stopping theorem.

Files. Slides `mit18_642_f24_lec05.pdf` (*Stochastic Processes I*, 44 slides in 88 pages of incremental builds); 2013 lecture notes `MIT18_S096F13_lecnote5.pdf`. The R script `MC_Example1.r` used for the AAPL example (slide 44) was distributed on Canvas only and is not among the OCW files. **Problem sets.** 2024 PS2 Problem 1, PS3 Problem 3; 2013 PS2 Problems A-3, A-4, B-5. **Elsewhere.** The first 72 minutes of 2024 L5 (PCA, $\chi^2/t/F$ distributions, LLN, CLT, a one-period CAPM) are in [chapter 3](#); the second half of 2024 L6 (regression) is in [chapter 5](#).

Motivation

A price history is one realisation of a *stochastic process*. Standing at time t we know the past path and want to say something about the future: whether it depends on the past (can prices be predicted?), how the process behaves in the long run, and how likely extreme “boundary” events are [[2013 L5 @ 8:25](#)]. This chapter introduces three families of discrete-time processes that answer these questions in increasing generality and that return, in continuous time, for the rest of the course:

- the *random walk*, the simplest model of a price and the discrete skeleton of Brownian motion ([chapter 11](#));
- *Markov chains*, where the future depends on the past only through the present state (credit ratings, market regimes); they extend the stochastic matrices of [section 2.3](#);

- *martingales*, the mathematical model of a fair game. Discounted prices are martingales under the risk-neutral measure, gains from trading are martingale transforms (discrete stochastic integrals, [chapter 17](#)), and the optional stopping theorem turns hard probability questions into one-line computations.

For a physicist: the random walk is diffusion, a Markov chain is a master equation relaxing to equilibrium, and a martingale is a quantity that is conserved *on average*; optional stopping is the use of such a conservation law to solve a boundary-value problem.

4.1 Stochastic processes

Definition 4.1 (Stochastic process). A stochastic process is a family $\{X_t, t \in \mathcal{T}\}$ of random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by time: $\mathcal{T} = \{0, 1, 2, \dots\}$ (discrete time) or $\mathcal{T} = [0, \infty)$ (continuous time) [[2013 L5 @ 1:03](#)]. Equivalently, each outcome $\omega \in \Omega$ selects a whole *path* $t \mapsto X_t(\omega)$, so a process is a probability distribution on a space of paths [[2013 L5 @ 3:09](#)].

The first reading is the intuitive one (a random value at each time); the second is the one needed to do mathematics: at time t we observe the initial segment of a path drawn once and for all, and we reason about the unobserved rest of it.

4.2 The random walk

Definition 4.3 (Random walk). Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance σ^2 , and let $S_0 = 0, S_n = X_1 + \dots + X_n$. $\{S_n\}$ is a *random walk* with steps X_i . When $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$ it is the (one-dimensional) *simple random walk* (SRW) [[2013 L5 @ 12:35](#)], [[2024 L5 @ 1:14:36](#)].

The path can be drawn as the graph of $n \mapsto S_n$, or as the trajectory of a walker taking unit steps left or right on a line [[2013 L5 @ 13:40](#)]. The 2013 course and the 2024 problem sets also use the *biased* walk ($\mathbb{P}(X = +1) = p \neq \frac{1}{2}$) and walks with a zero step (2024 PS2 Problem 1, [Exercise 4.1](#)).

Proposition 4.4 (Basic properties). For a random walk:

- (i) $\mathbb{E}[S_n] = n\mu$ and $\text{Var}(S_n) = n\sigma^2$; for the SRW $\mathbb{E}[S_n] = 0, \text{Var}(S_n) = n$ [[2024 L5 @ 1:15:36](#)];
- (ii) (independent increments) for $0 \leq k_0 \leq k_1 \leq \dots \leq k_r$ the increments $S_{k_1} - S_{k_0}, \dots, S_{k_r} - S_{k_{r-1}}$ are mutually independent [[2013 L5 @ 20:56](#)];
- (iii) (stationary increments) $S_{k+h} - S_k \stackrel{d}{=} S_h$ for all $k, h \geq 0$ [[2013 L5 @ 22:03](#)];
- (iv) (CLT) $(S_n - n\mu)/(\sigma\sqrt{n}) \rightarrow N(0, 1)$ in distribution.

Properties (ii) and (iii) are exactly those that define Brownian motion in continuous time [[2013 L5 @ 22:03](#)]: the random walk is “the fundamental stochastic process”, and much of [chapter 11](#) is its continuous limit.

The $\sqrt{n}$ scale. By (iv) the SRW at time n is typically at distance of order $\sqrt{n}$ from the origin, not n : although $|S_n|$ could be as large as n , the probability that $|S_n| \leq 2\sqrt{n}$ tends to 0.954

Example 4.2 (Same marginals, different processes) [[2013 L5 @ 4:10](#)]. For $t \geq 0$ consider

- (a) $X_t = t$ with probability 1;
- (b) $X_t = \varepsilon t$ for all t , with a single sign $\varepsilon = \pm 1$ drawn with probability 1/2 each;
- (c) $X_t = \varepsilon_t t$ with an independent sign ε_t for every t .

In (b) and (c) every X_t has the same distribution ($\pm t$ with probability 1/2), but in (b) one observation reveals the entire future, while in (c) the past says nothing about the future beyond $X_t \in \{t, -t\}$ [[2013 L5 @ 7:20](#)]. A process is *not* determined by its one-time marginals: what matters is the joint law of the path.

Proof. (i) Linearity of expectation, and additivity of variance for independent summands.
(ii) The increment over $(k_{i-1}, k_i]$ is a function of the block $X_{k_{i-1}+1}, \dots, X_{k_i}$; functions of disjoint blocks of independent variables are independent.
(iii) $S_{k+h} - S_k = X_{k+1} + \dots + X_{k+h}$ is a sum of h i.i.d. copies of X , like S_h .
(iv) Central limit theorem, [chapter 3](#). $\square$

[► 2013 L5 @ 17:51]. Simulated paths (slides 4–8) become more jagged as n grows and stay mostly inside the envelopes $\pm\sqrt{n}$, $\pm2\sqrt{n}$, $\pm3\sqrt{n}$ [► 2024 L5 @ 1:16:43]; see fig. 4.1. Nevertheless the envelopes are crossed again and again: with probability one the SRW returns to 0 infinitely often, and for every $c > 0$ it exceeds $c\sqrt{n}$ infinitely often [► 2013 L5 @ 18:52] (a consequence of the law of the iterated logarithm, $\limsup_n S_n/\sqrt{2n \ln \ln n} = 1$ a.s., not proved in the course).

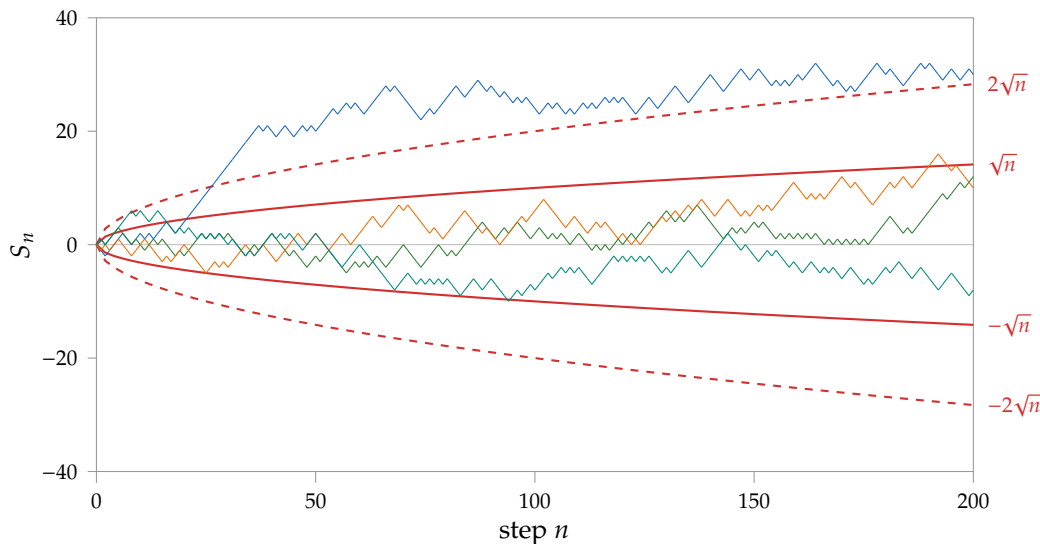


Figure 4.1: Four simulated paths of the simple random walk (200 steps) with the one- and two-standard-deviation envelopes $\pm\sqrt{n}$, $\pm2\sqrt{n}$; slides 4–8 of lec05 show the same picture for up to 10 000 steps [► 2024 L5 @ 1:16:43].

1 COURSE NOTE: Slips in the 2013 lecture

At [► 2013 L5 @ 16:47] the value S_t is said to be approximately normal “with mean 0 and variance $\sqrt{t}$ ”; the variance is t and the *standard deviation* is $\sqrt{t}$, which is the scale meant in the rest of the discussion. (The $1/t$ proposed by a student is the variance of the average S_t/t .)

INTUITION: Diffusion

S_n is the position of a particle receiving independent unit kicks: $\langle S_n^2 \rangle = n$ is Einstein’s $\langle x^2 \rangle = 2Dt$ with $D = \frac{1}{2}$ (unit step, unit time). Rescaling space by $\sqrt{n}$ and time by n , $t \mapsto S_{\lfloor nt \rfloor}/\sqrt{n}$ converges to Brownian motion (Donsker’s theorem, chapter 11); the jagged look of the long simulated paths is the nowhere-differentiability of Brownian paths in the making.

IN FINANCE: Random walks as price models

L. Bachelier (1900), cited in the 2013 notes, modelled prices on the Paris exchange as a random walk; the 2013 notes call the SRW “a (rather inaccurate) model of stock price”. The course refines it in two directions that appear in the problem sets: a drift, calibrated to an expected daily price change (2024 PS2 Problem 1, where an intraday price moves in 78 five-minute steps), and a random walk for the *logarithm* of the price, $\log P_t = \log P_{t-1} + X_t$ (2024 PS3 Problem 3), which keeps prices positive and makes returns, not price changes, i.i.d. Time-varying volatility comes later (chapter 13).

Exercise 4.1 (PS2 (2024), Problem 1: random walk with bias). Let X_n be i.i.d. steps with $\mathbb{P}(X_n = +1) = p$, $\mathbb{P}(X_n = 0) = 1 - (p + q)$, $\mathbb{P}(X_n = -1) = q$, where $p, q > 0$ and $p + q < 1$ (symmetric if $p = q$, biased upwards if $p > q$), and $S_n = \sum_{i=1}^n X_i$, $S_0 = 0$.

- (a) Derive the moment generating function $M_{X_i}(t)$ of a step and the mean and variance of X_i .

- (b) Derive the moment generating function $M_{S_n}(t)$ and the mean and variance of S_n .
- (c) With μ, σ^2 the mean and variance of a step, prove that $Z_n = (S_n - n\mu)/\sqrt{n\sigma^2}$ converges in distribution to $N(0, 1)$.
- (d) Model the intraday price of a stock as such a walk with 78 five-minute steps (the six and a half hours of NYSE trading), with $p = 0.52$ and $q = 0.47$. Which step size gives an expected daily price change of +\$0.50? With that step size, what is the standard deviation of the daily price change?

Exercise 4.2 (lecnote5, Proposition 2.1). Prove the independent- and stationary-increment properties of the random walk (proposition 4.4) directly from the definition, and check which of them survive if the steps are independent but not identically distributed.

4.3 Markov chains

4.3.1 The Markov property

Definition 4.5 (Markov process, Markov chain). A process $\{X_t\}$ is a *Markov process* if for all times $u < s < t$, given the present value X_s , the future value X_t is independent of the past values X_u [► 2024 L6 @ 36:30]:

$$[X_t | X_r, r \leq s] \stackrel{d}{=} [X_t | X_s].$$

A discrete-time Markov process with a finite or countable *state space* S is a *Markov chain*:

$$\mathbb{P}(X_{n+1} = j | X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = \mathbb{P}(X_{n+1} = j | X_n = i) \quad (4.1)$$

for all states and all n [► 2024 L6 @ 39:38].

“The effect of the past on the future is summarised by the current state” [► 2013 L5 @ 33:32]: two paths arriving at the same state at time s have the same future distribution, however they got there [► 2024 L6 @ 38:30].

Exercise 4.3 (PS2 (2013), A-3: identify Markov processes). Which of the following are Markov processes?

- (a) The simple random walk S_t .
- (b) $X_t = |S_t|$.
- (c) $X_0 = 0, X_{t+1} = X_t + (Y_t - \frac{1}{\lambda})$ with Y_t i.i.d. exponential with parameter λ .
- (d) $X_0 = 0, X_{t+1} = X_t + Z_t Z_{t-1} \cdots Z_0$ with Z_t i.i.d. log-normal.
- (e) $X_0 = 0, X_{t+1} = X_t + W_t + W_{t-1} + \cdots + W_0$ with W_t i.i.d. normal.

Example 4.6 (Random walks are Markov). If $S_{n+1} = S_n + X_{n+1}$ with X_{n+1} independent of $(S_0, \dots, S_n)$, then $\mathbb{P}(S_{n+1} \in \cdot | S_0, \dots, S_n)$ is the law of $S_n + X$, a function of S_n only. For the SRW, $\mathbb{P}(S_{n+1} = s | S_0, \dots, S_n) = \frac{1}{2}$ if $s = S_n \pm 1$ and 0 otherwise [► 2013 L5 @ 37:45]. The state space $\mathbb{Z}$ is infinite, so there is no finite transition matrix [► 2013 L5 @ 43:12].

4.3.2 Transition matrices

The one-step transition probabilities $\mathbb{P}(X_{n+1} = j | X_n = i)$ together with the initial distribution $p_i = \mathbb{P}(X_0 = i)$ specify the law of the whole process [► 2024 L6 @ 41:45]. The chain has *stationary* (time-homogeneous) transition probabilities if these do not depend on n

► 2024 L6 @ 42:52]; then they form the *transition matrix*

$$P = (P_{ij})_{i,j \in S}, \quad P_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i), \quad P_{ij} \geq 0, \quad \sum_j P_{ij} = 1,$$

a *row*-stochastic matrix (row i = where you are, column j = where you go) ► 2024 L6 @ 43:58]. All chains below are time-homogeneous.

Proposition 4.7 (Law of the path). $\mathbb{P}(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = p_{i_0} P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} i_n}$.

Theorem 4.8 (n -step transitions, Chapman–Kolmogorov). Let $P_{ij}^{(n)} = \mathbb{P}(X_{m+n} = j \mid X_m = i)$ (independent of m). Then

$$P_{ij}^{(n)} = \sum_{k \in S} P_{ik} P_{kj}^{(n-1)}, \quad P^{(0)} = I, \quad \text{hence} \quad P^{(n)} = P^n, \quad P^{(m+n)} = P^{(m)} P^{(n)}.$$

The distribution of X_n is the row vector $\mathbf{p}^{(n)} = \mathbf{p} P^n$, i.e. $\mathbb{P}(X_n = k) = \sum_j p_j P_{jk}^{(n)}$ ► 2024 L6 @ 52:29].

The 2013 notes decompose on the *last* step instead, $P_{ij}^{(n)} = \sum_k P_{ik}^{(n-1)} P_{kj}$, and the lecture reads the two-step matrix as a matrix product (“sum over all intermediate states”) ► 2013 L5 @ 42:08]. First-step analysis is the more useful habit: it is how hitting probabilities and expected hitting times are computed (section 4.6).

4.3.3 Stationary distributions and convergence

Definition 4.10 (Stationary distribution). A probability (row) vector π on S is *stationary* if $\pi P = \pi$, i.e. $\pi_j = \sum_k \pi_k P_{kj}$ for all j ► 2013 L5 @ 53:49]. If $X_0 \sim \pi$ then $X_n \sim \pi$ for every n ► 2013 L5 @ 54:52]. Equivalently, $\pi^\top$ is an eigenvector of eigenvalue 1 of the column-stochastic $A = P^\top$.

Theorem 4.11 (Convergence to equilibrium). Let S be finite with m states and suppose some power P^k has all entries positive (in particular, if all $P_{ij} > 0$). Then there is a unique stationary distribution π , all $\pi_j > 0$, and for every initial distribution

$$\|\mathbf{p} P^n - \pi\|_1 \leq 2(1 - m\delta)^{\lfloor n/k \rfloor}, \quad \delta = \min_{i,j} (P^k)_{ij} > 0;$$

in particular $P_{ij}^{(n)} \rightarrow \pi_j$ for all i, j : the chain forgets its initial state geometrically fast.

2 COURSE NOTE: Row versus column conventions

The 2024 slides use the row-stochastic P above, acting on *row* vectors of probabilities from the right. The 2013 notes (and section 2.3) use the column-stochastic $A = P^\top$ acting on column vectors: $\pi(t+1) = A\pi(t)$ is the transpose of $\mathbf{p}^{(t+1)} = \mathbf{p}^{(t)} P$. This chapter uses P . In the 2013 lecture the machine example below was first written in the row form and then “flipped” ► 2013 L5 @ 51:41].

Proof. Write $A = \{X_0 = i_0, \dots, X_{n-1} = i_{n-1}\}$ and $B = \{X_n = i_n\}$. Then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$ (joint = marginal $\times$ conditional), $\mathbb{P}(B \mid A) = \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1})$ by the Markov property, and this equals $P_{i_{n-1} i_n}$ by stationarity. Induction on n ► 2024 L6 @ 45:04], ► 2024 L6 @ 48:09]. $\square$

Proof (first-step analysis). Condition on the first step ► 2024 L6 @ 49:16]:

$$\begin{aligned} P_{ij}^{(n)} &= \sum_k \mathbb{P}(X_1 = k \mid X_0 = i) \mathbb{P}(X_n = j \mid X_1 = k, X_0 = i) = \sum_k P_{ik} P_{kj}^{(n-1)} \\ &= \sum_k P_{ik} \mathbb{P}(X_{n-1} = j \mid X_0 = k) = \sum_k P_{ik} P_{kj}^{(n-1)}, \end{aligned}$$

using the Markov property and then stationarity. In matrix form $\mathbf{p}^{(n)} = \mathbf{p} P^{(n-1)}$ ► 2024 L6 @ 51:21]; induction gives P^n . $\square$

Example 4.9 (A machine that breaks) ► 2013 L5 @ 44:15]. A machine is working (W) or broken (B) each day. If working it breaks the next day with probability 0.01; if broken it is repaired by the next day with probability 0.8:

$$P = \begin{pmatrix} 0.99 & 0.01 \\ 0.8 & 0.2 \end{pmatrix}, \quad P^2 = \begin{pmatrix} 0.9881 & 0.0119 \\ 0.952 & 0.048 \end{pmatrix}.$$

What is the state distribution after ten years, $e_W P^{3650}$ ► 2013 L5 @ 46:26]? The lecture “cheats”: after so long the distributions on day 3650 and 3651 should coincide, so the answer π should satisfy $\pi = \pi P$ ► 2013 L5 @ 47:27]. This is the two-state chain of example 2.3 with $p = 0.01$, $q = 0.8$: $\pi = (0.8, 0.01)/0.81 = (0.98765, 0.01235)$ and the second eigenvalue is $1 - p - q = 0.19$, so P^n converges to the matrix with both rows equal to π like 0.19^n and P^{3650} equals it to machine precision. In the long run the machine is broken on 1.23% of the days.

The course's argument (2013 notes, Theorem 3.3, for $k = 1$ [► 2013 L5 @ 49:36]). $A = P^T$ is column-stochastic with positive entries. By [proposition 2.2](#) its eigenvalues satisfy $|\lambda| \leq 1$ and 1 is one of them; by Perron–Frobenius ([theorem 2.14](#)) the dominant eigenvalue is simple, has a positive eigenvector, and all other eigenvalues have $|\lambda| < 1$. The positive eigenvector, normalised, is π , and the other modes of $A^n p^T$ decay (as in the state equations of [section 2.5](#)), so $A^n p^T \rightarrow \pi^T$ [► 2013 L5 @ 52:41].

INTUITION: Relaxation to equilibrium

$p^{(n+1)} = p^{(n)}P$ is a discrete-time master equation. The stationary distribution is its equilibrium and $1 - |\lambda_2|$ (the spectral gap) is the inverse relaxation time, as in [example 2.3](#). The contraction proof has a probabilistic meaning (coupling): run two copies of the chain from different starts; at every block of k steps they have probability at least $m\delta$ of landing in the same state, after which they can be glued together. The initial condition is forgotten as soon as the copies have met.

A self-contained proof (not from the course). For two probability vectors μ, ν let $d = \mu - \nu$, so $\sum_i d_i = 0$. Then, writing $Q = P^k$,

$$\begin{aligned} (dQ)_j &= \sum_i d_i Q_{ij} = \sum_i d_i (Q_{ij} - \delta) \\ &\Rightarrow \|dQ\|_1 \leq \sum_i |d_i| \sum_j (Q_{ij} - \delta) = (1 - m\delta) \|d\|_1 \end{aligned}$$

because $Q_{ij} - \delta \geq 0$ and each row of Q sums to 1; similarly $\|dP\|_1 \leq \|d\|_1$. Hence $\mu \mapsto \mu Q$ is a contraction of the (complete) simplex in the ℓ^1 distance, and by Banach's fixed-point theorem has a unique fixed point π . Since $(\pi P)Q = (\pi Q)P = \pi P$, πP is also a fixed point of Q , so $\pi P = \pi$; any stationary distribution of P is fixed by Q , hence equals π . Finally

$$\begin{aligned} \|pP^n - \pi\|_1 &= \|(p - \pi)P^n\|_1 \\ &\leq (1 - m\delta)^{\lfloor n/k \rfloor} \|p - \pi\|_1, \end{aligned}$$

with $\|p - \pi\|_1 \leq 2$, and $\pi_j = (\pi Q)_j \geq \delta > 0$. $\square$

When the hypotheses fail.

- *Periodicity.* The 2013 notes (Example 3.2) consider the walk $X_{n+1} = X_n \pm 1 \pmod{N}$ on the cycle $\mathbb{Z}_N$, with stationary distribution $\pi_i = 1/N$ (the matrix is doubly stochastic, 2013 PS2 A-4). If N is even the chain alternates between even and odd states, no power of P is positive and $P_{ij}^{(n)}$ does not converge: a stationary distribution need not be a limit (compare the chain $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ in [section 2.3](#)).
- *Reducibility.* If the state space splits into pieces that do not communicate, the stationary distribution need not be unique (2013 PS2 A-4(a)). States i, j *communicate* if each can be reached from the other in some number of steps; a chain whose transition graph is strongly connected is *irreducible* [► 2024 L6 @ 1:03:03]. For a finite irreducible chain the stationary distribution is unique and positive; if moreover it is aperiodic, some P^k is positive and [theorem 4.11](#) applies (standard facts, not proved in the course).
- *Infinite state space.* “A corresponding theorem is not true” (2013 notes): the SRW on $\mathbb{Z}$ has no stationary distribution. Indeed $\pi_j = \frac{1}{2}(\pi_{j-1} + \pi_{j+1})$ makes $j \mapsto \pi_j$ linear; being non-negative on all of $\mathbb{Z}$ it is constant, and a constant cannot sum to 1.

Estimating a chain from data (not discussed in class). Given an observed path with n_{ij} transitions $i \rightarrow j$, [proposition 4.7](#) gives the likelihood $\prod_{i,j} P_{ij}^{n_{ij}}$ (times p_{i_0}). Maximising $\sum_{i,j} n_{ij} \log P_{ij}$ subject to $\sum_j P_{ij} = 1$ with Lagrange multipliers gives the empirical frequencies $\hat{P}_{ij} = n_{ij} / \sum_k n_{ik}$; this is what the R package `markovchain` uses in the case study below.

3 COURSE NOTE: The puzzle left open in the 2013 lecture

For the machine example the lecture sums the two components of $Av = \lambda v$ (A column-stochastic) and gets $\sum_i v_i = \lambda \sum_i v_i$, hence $\lambda = 1$; then notices that the same argument seems to force *every* eigenvalue to be 1, which is absurd, and leaves the point open [► 2013 L5 @ 56:57]. The resolution: the identity only says that $\lambda = 1$ or $\sum_i v_i = 0$. Eigenvectors of the other eigenvalues have components summing to zero (like $(1, -1)^T$ in [example 2.3](#)), so they cannot be normalised to probability vectors. For the Perron–Frobenius eigenvector, which is positive, the sum is non-zero and the argument correctly gives $\lambda_0 = 1$.

Exercise 4.4 (PS2 (2013), A-4).

- Construct a finite Markov chain whose stationary distribution is not unique.
- A chain is *doubly stochastic* if, in addition to $\sum_j p_{ij} = 1$, also $\sum_i p_{ij} = 1$ for every j (the problem phrases this as “each row sums to one” for the transposed matrix A of the 2013 notes). Prove that the uniform distribution $\pi_j = 1/m$ is stationary for a doubly stochastic chain on m states.

Exercise 4.5 (PS2 (2013), B-5: walk with reflecting barriers). A person walks on the line, stepping right with probability b and left with probability $1 - b$ at each period. He starts in one of the positions $1, \dots, m$; a step to 0 (or to $m + 1$) is instantly reflected back to 1 (or to m). Equivalently, in position 1 (position m) he stays put with probability $1 - b$ (probability b).

- Model this as a Markov chain and write its transition matrix.
- Find the stationary distribution.

4.3.4 Markov chains in finance

Credit ratings. Rating agencies (S&P, Moody’s, Fitch) grade corporate and sovereign borrowers from AAA (highest) through AA, A, BBB (investment grade) to BB, B, CCC, CC, C (speculative grade) and D (default) [► 2024 L6 @ 52:29]. The rating determines the interest rate at which a company can borrow by issuing bonds [► 2024 L6 @ 53:29]. Modelling the rating of an issuer as a Markov chain with a one-year *migration matrix*, default being an absorbing state ($P_{DD} = 1$), the probability of default within n years for an issuer rated i today is $(P^n)_{iD}$. Slide 43 shows such a matrix (attributed to the CreditMetrics Technical Document): a AAA issuer keeps its rating over one year with probability 43.78%, while a CCC issuer is upgraded to B with probability 41.53%, stays CCC with 32.46% and defaults with 19.15% [► 2024 L6 @ 54:32].

IN FINANCE: Uses and caveats of rating chains (not discussed in class)

Migration matrices drive credit-portfolio models (CreditMetrics), regulatory capital, and the pricing of credit-risky bonds and credit derivatives (chapters 8 and 23). The Markov assumption is only an approximation: recently downgraded issuers are more likely to be downgraded again (“rating momentum”), and migration rates vary with the business cycle, so the matrix is not time-homogeneous.

CASE STUDY: Up and down days of AAPL (slide 44, MC_Example1.r)

Goal. Check whether the direction of daily price moves has short memory [► 2024 L6 @ 55:39]. **Data.** AAPL daily closing prices P_t downloaded with `quantmod` (2013/14 to September 2024); the chain is fitted on the last four and a half years [► 2024 L6 @ 58:51]. **Method.** A day is Up if $P_t > P_{t-1}$ and Down if $P_t < P_{t-1}$. The two-day state on day t is one of UU, UD, DU, DD (moves on days $t - 1$ and t), which gives a 4-state chain whose transitions are partly forced: from UU one can only go to UU or UD, and so on (fig. 4.2). The package `markovchain` (Spedicato, 2017) estimates the transition matrix by empirical frequencies and draws the transition graph [► 2024 L6 @ 59:56]. The same is repeated with three-day states (8 states) [► 2024 L6 @ 1:02:01]. **Results** (as reported in class; no knitted output is available). Two

4 COURSE NOTE: On the migration table of slide 43

In the widely quoted one-year matrices (S&P, Moody’s, CreditMetrics) roughly 90% of AAA issuers keep their rating; the AAA row on the slide, which sends more than half of the AAA issuers to AA, is unusual. The OCW version marks the table’s source as unknown; use its numbers as an illustration only.

Example 4.12 (Default probabilities from a migration matrix (not from the course)). With three states, investment grade (IG), high yield (HY) and default (D), take

$$P = \begin{pmatrix} 0.97 & 0.025 & 0.005 \\ 0.05 & 0.90 & 0.05 \\ 0 & 0 & 1 \end{pmatrix}.$$

The cumulative default probabilities $(P^n)_{iD}$ are 0.5%, 3.46%, 8.46% for an IG issuer and 5%, 20.75%, 33.87% for an HY issuer at $n = 1, 5, 10$ years; they are not n times the one-year probability, because issuers migrate between grades before defaulting. Since D is absorbing and reachable from everywhere, the unique stationary distribution is e_D : every issuer eventually defaults, and the interesting quantities are transient ones.

up days tend to be followed by a third ($\hat{P}(UU \rightarrow UU) > 0.5$), while after two down days a reversal is more likely ($\hat{P}(DD \rightarrow DD) < 0.5$) [▶ 2024 L6 @ 1:00:56]. In the 8-state chain, three down days are followed by an up day with probability close to 0.6; the largest probability of a down day, 0.510, follows a mixed state with two up days preceded by a down day [▶ 2024 L6 @ 1:05:15]. All states communicate (the graph is strongly connected) [▶ 2024 L6 @ 1:04:05]. **Lesson.** Discretising returns into a few states and fitting a Markov chain is a quick exploratory tool, easy to rerun on other tickers or periods [▶ 2024 L6 @ 1:06:18]. *Our remark:* with about 1100 trading days each of the 8 states is visited roughly 140 times, so an estimated probability near 0.5 has a standard error of about $\sqrt{0.25/140} \approx 0.04$: 0.510 is indistinguishable from 0.5, and even 0.6 is only about two standard errors away, before accounting for having looked at 8 states. Note also that the slide's definitions leave days with $P_t = P_{t-1}$ unclassified; one inequality should be non-strict, as the lecturer points out [▶ 2024 L6 @ 55:39].

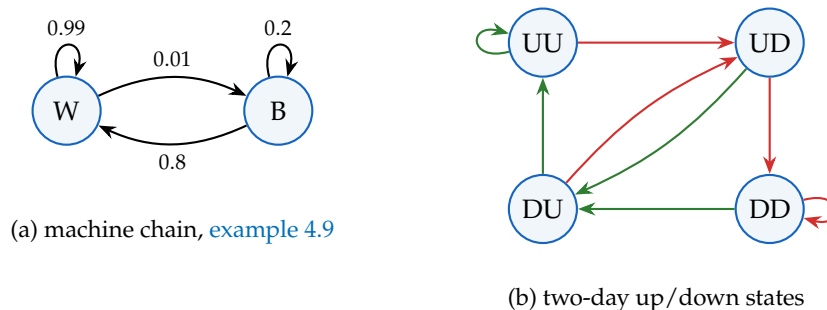


Figure 4.2: (a) The two-state machine chain of the 2013 lecture. (b) Allowed transitions of the two-day states of the AAPL case study: the second letter of today's state is the first letter of tomorrow's; green arrows are followed by an up day, red ones by a down day. The estimated probabilities were shown in class [▶ 2024 L6 @ 1:00:56].

Exercise 4.6 (Time to default (not from the course)). For the rating chain of example 4.12, let Q be the 2×2 block of transitions among the non-default states. Show that the expected number of years before default, starting from IG or HY, is the vector $(I - Q)^{-1}\mathbf{1}$ (condition on the first step), and compute it.

4.4 Martingales

4.4.1 Information and conditional expectation

Let $X_1, X_2, \dots$ be the observed process. The *information set* $\mathcal{F}_n$ collects what is known at time n , i.e. the values of $X_1, \dots, X_n$ (formally, the σ -algebra they generate) [▶ 2024 L6 @ 8:32]. A random variable Y is *known at time n* , written $Y \in \mathcal{F}_n$, iff $Y = f(X_1, \dots, X_n)$ for some function f ; the increasing family $\{\mathcal{F}_n\}$ is a *filtration*, and one writes $\mathbb{E}[Z \mid \mathcal{F}_n] = \mathbb{E}[Z \mid X_1, \dots, X_n]$ [▶ 2024 L6 @ 13:00]. We use the following properties of conditional expectation (chapter 3), valid for integrable Z :

- *taking out what is known:* $\mathbb{E}[YZ \mid \mathcal{F}_n] = Y \mathbb{E}[Z \mid \mathcal{F}_n]$ if $Y \in \mathcal{F}_n$;
- *tower property:* $\mathbb{E}[\mathbb{E}[Z \mid \mathcal{F}_n] \mid \mathcal{F}_m] = \mathbb{E}[Z \mid \mathcal{F}_m]$ for $m \leq n$; in particular $\mathbb{E}[\mathbb{E}[Z \mid \mathcal{F}_n]] = \mathbb{E}[Z]$;

- *independence*: $\mathbb{E}[Z \mid \mathcal{F}_n] = \mathbb{E}[Z]$ if Z is independent of $X_1, \dots, X_n$.

4.4.2 Definition and first properties

Definition 4.13 (Martingale). A sequence $M_n = f_n(X_1, \dots, X_n)$, $n \geq 1$, with M_0 a constant, is a *martingale* with respect to $\{X_n\}$ (or to $\{\mathcal{F}_n\}$) if for all $n \geq 1$

$$\mathbb{E}[|M_n|] < \infty \quad \text{and} \quad \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] = M_{n-1} \quad (4.2)$$

► 2024 L5 @ 1:13:28]. It is a *submartingale* if $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] \geq M_{n-1}$ and a *supermartingale* if $\leq$ (terms not used in class).

Standing at time $n - 1$, the best prediction of the next value is the current value: the conditional expectations of the future are “flat” ► 2024 L6 @ 2:05]. If M_n is the balance of a gambler, the game is *fair*: whatever happened so far, the expected gain of the next round is zero ► 2013 L5 @ 1:01:24]. The balance of a roulette player is a supermartingale, not a martingale: the game is designed so that $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] < M_n$ ► 2013 L5 @ 1:02:24].

Proposition 4.14. If $\{M_n\}$ is a martingale then $\mathbb{E}[M_n \mid \mathcal{F}_m] = M_m$ for all $m \leq n$, and $\mathbb{E}[M_n] = M_0$ for all n . The increments $D_k = M_k - M_{k-1}$ satisfy $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$; if they are square-integrable they are uncorrelated, and $\text{Var}(M_n) = \sum_{k=1}^n \mathbb{E}[D_k^2]$.

4.4.3 Examples

The 2024 slides give four families; each one is used later.

The simulations in class show a few paths growing to 5–10 times the initial wealth, while most lose almost everything ► 2024 L6 @ 5:20]. The calculation behind the pictures (ours): $\mathbb{E}[M_n] = 1$ for every n , but $\log M_n = \sum_k \log X_k$ is a random walk with drift $\mathbb{E}[\log X] = \frac{1}{2} \ln 0.75 = -0.144$ per step, so $M_n \rightarrow 0$ almost surely by the law of large numbers. After 50 steps the median wealth is $0.75^{25} \approx 7.5 \times 10^{-4}$, and $\mathbb{P}(M_{50} \geq 1) = \mathbb{P}(\text{Bin}(50, \frac{1}{2}) \geq 32) \approx 0.032$; the mean 1 is carried by rare paths (the largest possible value is $1.5^{50} \approx 6 \times 10^8$). In the 2013 example $\mathbb{E}[\log X] = \frac{1}{3} \ln 2 - \frac{2}{3} \ln 2 < 0$ as well.

INTUITION: Mean versus typical behaviour

For multiplicative processes the mean and the typical value part ways: $\mathbb{E}[M_n]$ is dominated by exponentially rare paths, much as a partition function can be dominated by rare configurations. The typical growth rate is $\mathbb{E}[\log X]$, not $\log \mathbb{E}[X]$; by Jensen’s inequality $\mathbb{E}[\log X] < \log \mathbb{E}[X] = 0$ unless X is constant. In finance this is the gap between arithmetic and geometric mean returns (“volatility drag”), and the reason for the debate mentioned in class ► 2024 L5 @ 49:47] on whether an investor should maximise the expected *logarithm* of wealth (the Kelly criterion, of which Paul Samuelson was the best-known critic).

5 COURSE NOTE: 2013 versus 2024 definitions

The 2013 notes define a martingale with respect to its own past, $X_t = \mathbb{E}[X_{t+1} \mid X_0, \dots, X_t]$, and omit the integrability condition. The 2024 definition allows the conditioning information to be generated by a different, “driving” sequence X_n (as in the examples below). A martingale with respect to $\{X_n\}$ is also a martingale with respect to its own past, by the tower property, since $M_0, \dots, M_{n-1}$ are functions of $X_1, \dots, X_{n-1}$.

Proof. By the tower property, $\mathbb{E}[M_n \mid \mathcal{F}_m] = \mathbb{E}[\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] \mid \mathcal{F}_m] = \mathbb{E}[M_{n-1} \mid \mathcal{F}_m]$, and induction on $n - m$ (2013 notes, Proposition 4.2). Taking $m = 0$ gives $\mathbb{E}[M_n] = M_0$. For $j < k$, $D_j \in \mathcal{F}_{k-1}$, so $\mathbb{E}[D_j D_k] = \mathbb{E}[D_j \mathbb{E}[D_k \mid \mathcal{F}_{k-1}]] = 0$; hence the variance of the sum is the sum of the variances. ◻

Example 4.15 (Sums: random walks ► 2024 L5 @ 1:13:28]). If X_n are independent with $\mathbb{E}[X_n] = 0$, then $S_n = X_1 + \dots + X_n$ is a martingale: $\mathbb{E}[S_n \mid \mathcal{F}_{n-1}] = S_{n-1} + \mathbb{E}[X_n] = S_{n-1}$. In particular the SRW is a martingale ► 2013 L5 @ 1:01:24]. With non-zero means, the *compensated* walk $S_n - \sum_{k \leq n} \mathbb{E}[X_k]$ is a martingale.

Example 4.16 (Squares: $S_n^2 - n\sigma^2$ ► 2024 L5 @ 1:17:44]). If moreover $\text{Var}(X_n) = \sigma^2$, then $M_n = S_n^2 - n\sigma^2$ is a martingale:

$$\mathbb{E}[S_n^2 \mid \mathcal{F}_{n-1}] = S_{n-1}^2 + 2S_{n-1}\mathbb{E}[X_n] + \mathbb{E}[X_n^2] = S_{n-1}^2 + \sigma^2.$$

Its paths look very different from those of S_n (slides 10–12): bounded below by $-n\sigma^2$, unbounded above, with a long right tail ► 2024 L6 @ 3:09]. It measures how much the walk has spread compared with its expected spread; its continuous-time analogue $B_t^2 - t$ encodes the quadratic variation of Brownian motion, the origin of the extra term in Itô’s formula (chapter 17).

For ± 1 steps, $\phi(\lambda) = pe^\lambda + qe^{-\lambda}$ and $\phi(\lambda) = 1$ is a quadratic equation in e^λ with roots 1 and q/p : hence $(q/p)^{S_n}$ is a martingale (fig. 4.3, left) [► 2024 L6 @ 33:19]. This is the key to the biased gambler's ruin (section 4.6). The exponent λ_0 is the “adjustment coefficient” of actuarial ruin theory, and (4.3) is the discrete ancestor of the exponential martingale $e^{\lambda B_t - \lambda^2 t/2}$ of Brownian motion and of the change of measure behind risk-neutral pricing (chapter 15).

Markov versus martingale. The two notions are independent [► 2013 L5 @ 1:05:32]. The SRW is both. A biased walk is Markov but not a martingale. A martingale that is not Markov (our example): toss a fair coin ξ_1 and let $M_n = \xi_1 + c(\xi_1) \sum_{k=2}^n \xi_k$ with further fair signs ξ_k and step size $c = 1$ if $\xi_1 = +1$, $c = 2$ if $\xi_1 = -1$. It is a martingale (a martingale transform, see below), but $M_3 = -1$ can be reached with either step size, so the law of M_4 given $M_3 = -1$ depends on the earlier path. Processes of neither kind are the rule rather than the exception.

Remark 4.19 (Prices or log-prices?). If $\log P_t$ is a random walk with i.i.d. steps X_t (2024 PS3 Problem 3), the price itself satisfies $\mathbb{E}[P_t | \mathcal{F}_{t-1}] = P_{t-1} \mathbb{E}[e^X]$, so P_t is a martingale iff $\mathbb{E}[e^X] = 1$. For normal steps $\mathbb{E}[e^X] = e^{\mu + \sigma^2/2}$, i.e. iff $\mu = -\sigma^2/2$: a martingale log-price ($\mu = 0$) gives an *upward*-drifting price. This lognormal correction (2024 PS2 Problem 2, chapter 3) is the same $-\sigma^2/2$ that appears in the drift of $\log S_t$ under Black–Scholes.

4.5 Martingale transforms and stopping times

4.5.1 Trading strategies as martingale transforms

Definition 4.20 (Non-anticipating sequence, martingale transform). A sequence $\{A_n, n \geq 1\}$ is *non-anticipating* (or predictable) with respect to $\{\mathcal{F}_n\}$ if $A_n \in \mathcal{F}_{n-1}$ for every n : A_n is decided with the information available at time $n - 1$ [► 2024 L6 @ 9:44]. The *martingale transform* of a martingale $\{M_n\}$ by $\{A_n\}$ is

$$\tilde{M}_n = M_0 + A_1(M_1 - M_0) + A_2(M_2 - M_1) + \cdots + A_n(M_n - M_{n-1}) = M_0 + \sum_{k=1}^n A_k \Delta M_k$$

[► 2024 L6 @ 10:51].

Theorem 4.21 (Martingale transform theorem). If $\{M_n\}$ is a martingale and $\{A_n\}$ is non-anticipating and bounded, $|A_n| \leq C_n$ for constants C_n , then $\{\tilde{M}_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$ [► 2024 L6 @ 13:00].

Example 4.17 (Products [► 2024 L6 @ 4:15]). If $X_n \geq 0$ are independent with $\mathbb{E}[X_n] = 1$ and $M_0 = 1$, then $M_n = X_1 X_2 \cdots X_n$ is a martingale: $\mathbb{E}[M_n | \mathcal{F}_{n-1}] = M_{n-1} \mathbb{E}[X_n] = M_{n-1}$. The 2024 slides take $X_n \in \{0.5, 1.5\}$ with probability $1/2$ each: the wealth of someone who stakes everything, every period, on a fair bet that gains or loses 50% [► 2024 L6 @ 5:20]. The 2013 lecture takes $X_n = 2$ with probability $1/3$ and $X_n = \frac{1}{2}$ with probability $2/3$, again with mean 1 [► 2013 L5 @ 1:03:24].

Example 4.18 (Exponential martingales [► 2024 L6 @ 6:28]). Let Y_n be i.i.d. with moment generating function $\phi(\lambda) = \mathbb{E}[e^{\lambda Y}] < \infty$. Then $X_n = e^{\lambda Y_n} / \phi(\lambda)$ are independent with mean 1, so by example 4.17

$$M_n = \frac{e^{\lambda S_n}}{\phi(\lambda)^n}, \quad S_n = Y_1 + \cdots + Y_n, \quad (4.3)$$

is a martingale for every fixed λ [► 2024 L6 @ 7:30]. If $\lambda_0 \neq 0$ solves $\phi(\lambda_0) = 1$, then $M_n = e^{\lambda_0 S_n}$ is a martingale. Since ϕ is convex with $\phi(0) = 1$ and $\phi'(0) = \mathbb{E}[Y]$, such a root exists (with the sign opposite to $\mathbb{E}[Y]$) as soon as $\mathbb{E}[Y] \neq 0$ and ϕ grows beyond 1 on that side, e.g. for bounded steps taking both signs.

Proof. $\tilde{M}_n$ is a function of $X_1, \dots, X_n$ and $\mathbb{E}[|\tilde{M}_n|] \leq |M_0| + \sum_{k \leq n} C_k (\mathbb{E}[|M_k|] + \mathbb{E}[|M_{k-1}|]) < \infty$. Taking out what is known,

$$\mathbb{E}[\tilde{M}_n - \tilde{M}_{n-1} | \mathcal{F}_{n-1}] = A_n \mathbb{E}[M_n - M_{n-1} | \mathcal{F}_{n-1}] = A_n \cdot 0 = 0$$

□

IN FINANCE: You cannot beat a fair game

Read M_n as the (discounted) price of an asset and A_k as the number of shares held from $k - 1$ to k , chosen using only the information available at $k - 1$ [► 2024 L6 @ 10:51]. Then $\tilde{M}_n - M_0 = \sum_k A_k \Delta M_k$ is the cumulative trading gain: exactly the “holdings times price increment, holdings decided first” sum of (2.1) and the discrete version of the Itô integral $\int \theta_t dS_t$ (chapter 17). The theorem says that if prices are martingales, *every* bounded trading strategy has zero expected gain. This is the content of the fundamental theorem of asset pricing read backwards: absence of arbitrage is equivalent to the existence of a measure $\mathbb{Q}$ under which discounted prices are martingales; in one period this is the relation $P_0 = \beta \mathbb{E}_{\mathbb{Q}}[P_T]$ of theorem 2.9.

4.5.2 Stopping times

Definition 4.22 (Stopping time). A random variable τ with values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is a *stopping time* with respect to $\{\mathcal{F}_n\}$ if $\{\tau \leq n\} \in \mathcal{F}_n$ for every n (equivalently $\{\tau = n\} \in \mathcal{F}_n$) [► 2024 L6 @ 14:04].

Viewing ω as a path, the event $\{\omega : \tau(\omega) \leq n\}$ is decided by the first n steps of the path: at every time we know whether we have already stopped [► 2024 L6 @ 16:16]. A stopping time is a rule to quit a game that looks only at what has happened so far [► 2013 L5 @ 1:09:40].

Definition 4.23 (Stopped and truncated processes). For a stopping time τ , the value at the stopping time is $X_\tau = \sum_{n \geq 0} X_n \mathbf{1}\{\tau = n\}$ (on $\{\tau < \infty\}$) [► 2024 L6 @ 17:24]. Since τ can be infinite, one works with the truncated times $n \wedge \tau = \min(n, \tau)$ [► 2024 L6 @ 18:28] and the *stopped process* $\{X_{n \wedge \tau}, n \geq 0\}$, which follows X until τ and then stays frozen. $X_{n \wedge \tau} = \sum_{k < n} X_k \mathbf{1}\{\tau = k\} + X_n \mathbf{1}\{\tau \geq n\}$ is a function of $X_0, \dots, X_n$.

Example 4.24 (Stopping times and non-stopping times) [► 2013 L5 @ 1:10:49]. In the coin-toss game with balance S_n :

- (a) the first time the balance reaches \$100, $\tau = \min\{n : S_n = 100\}$, is a stopping time; so is the first time it reaches either \$100 or -\$50, and any first hitting time of a set;
- (b) the time of the first peak (the first local maximum of the balance) is *not* a stopping time: to know that n is a peak one must see $S_{n+1} < S_n$ [► 2013 L5 @ 1:11:50];
- (c) one step after the first peak *is* a stopping time, as a student remarks [► 2013 L5 @ 1:12:51].

Proof. Let $A_k = \mathbf{1}\{\tau \geq k\} = 1 - \mathbf{1}\{\tau \leq k - 1\}$. It is bounded and $A_k \in \mathcal{F}_{k-1}$, because $\{\tau \leq k - 1\} \in \mathcal{F}_{k-1}$. The transform telescopes up to the stopping time:

$$M_0 + \sum_{k=1}^n A_k (M_k - M_{k-1}) = M_0 + \sum_{k=1}^{n \wedge \tau} (M_k - M_{k-1}) = M_{n \wedge \tau},$$

which is a martingale by theorem 4.21 [► 2024 L6 @ 20:29]. Then apply proposition 4.14. $\square$

Theorem 4.25 (Stopped martingales are martingales). If $\{M_n\}$ is a martingale and τ a stopping time, then $\{M_{n \wedge \tau}\}$ is a martingale; in particular $\mathbb{E}[M_{n \wedge \tau}] = M_0$ for every n .

Exercise 4.7 (Stopping times (not from the course)). Let σ, τ be stopping times. Show that $\sigma \wedge \tau, \sigma \vee \tau$ and $\tau + 1$ are stopping times, but that $\tau - 1$ in general is not. Deduce

that “one step after the first peak” is a stopping time and “the first peak” is not (example 4.24).

4.5.3 The optional stopping theorem

theorem 4.25 gives $\mathbb{E}[M_{n \wedge \tau}] = M_0$ at every fixed n . The *optional stopping theorem* passes to the limit $n \rightarrow \infty$, where $M_{n \wedge \tau} \rightarrow M_\tau$ on $\{\tau < \infty\}$.

Theorem 4.26 (Optional stopping). Let $\{M_n\}$ be a martingale and τ a stopping time. Then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ in each of the following cases:

- (i) $\tau \leq T$ for a constant T (2013 notes, Theorem 5.3, “Doob’s optional stopping theorem, weak form”) [► 2013 L5 @ 1:14:06];
- (ii) $\tau < \infty$ almost surely and the stopped process is bounded, $|M_{n \wedge \tau}| \leq K$ for all n ;
- (iii) $\mathbb{E}[\tau] < \infty$ and the increments are bounded, $|M_n - M_{n-1}| \leq c$.

“A mortal being has no winning strategy” in a fair game (2013 notes): whatever rule you use to decide when to quit, as long as it does not peek into the future and you cannot play for an unbounded time with unbounded credit, your expected final balance equals your initial one [► 2013 L5 @ 1:06:34], [► 2013 L5 @ 1:15:08]. The hypotheses are not decorations: 2013 notes, Exercise 5.4 (Exercise 4.10), asks why stopping the SRW when it first reaches \$100 does not contradict the theorem although $\mathbb{E}[S_\tau] = 100$.

IN FINANCE: The “martingale” betting system (not discussed in class)

Bet \$1 on a fair coin; after each loss double the stake; stop at the first win. With $A_k = 2^{k-1}$ on $\{\tau \geq k\}$ and τ the first win, the net gain at τ is $2^{\tau-1} - (1 + 2 + \dots + 2^{\tau-2}) = +1$ with certainty, although $\mathbb{E}[\tau] = 2 < \infty$. Condition (iii) fails because the stakes, i.e. the increments of the transformed process, are unbounded, and so does (ii), since the loss before the first win is unbounded. At any fixed horizon n the game is still fair: with probability $1 - 2^{-n}$ you have won \$1, with probability 2^{-n} you have lost $2^n - 1$, and the mean is 0. This doubling scheme is the historical meaning of “martingale” in gambling (the lecturer recalls the other meaning, a horse’s bridle [► 2024 L5 @ 1:18:49]). Its trading version, doubling a losing position to win it back, needs unlimited capital and is a classic route to ruin.

Exercise 4.8 (PS3 (2024), Problem 3: random walk for the log price). Let P_t be an asset price, $S_t = \log P_t$ and $X_t = S_t - S_{t-1} = \log(P_t/P_{t-1})$, with the X_t i.i.d., $\mathbb{E}[X_t] = \mu$ and $\text{Var } X_t = \sigma^2 > 0$.

- (a) For which (μ, σ^2) is $\{S_t\}$ a Markov process? Explain.
- (b) For which (μ, σ^2) is $\{S_t\}$ a martingale? Explain.

6 COURSE NOTE: Erratum: slides 17, 20 and 23 of lec05

- Slide 23 (proof above): “else replace M_n by $M'_n = M_n - M_{n-1}$ ” should read $M'_n = M_n - M_0$ (to reduce to $M_0 = 0$ one subtracts the initial value, not the previous one). In the displayed identity the right-hand side should have n in place of the summation index k : $\sum_{k=1}^n A_k(M_k - M_{k-1}) = M_\tau \mathbf{1}(\tau \leq n-1) + M_n \mathbf{1}(\tau \geq n) = M_{n \wedge \tau}$.
- Slide 20 calls $\{0, 1, 2, \dots\}$ the “state space” of the process; in the definition of a stopping time it plays the role of the *time index set*, in which τ takes values (together with ∞).
- Slide 17: in the special case the sum should read $S_n = \sum_{i=1}^n Y_i$, and λ_0 must be a root of $\phi(\lambda_0) = 1$ other than the trivial root $\lambda_0 = 0$.

Proof. (i) For $n = T$, $M_{T \wedge \tau} = M_\tau$, so $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ by theorem 4.25. (This is the 2013 proof: $M_\tau = M_0 + \sum_{i=0}^{T-1} (M_{i+1} - M_i) \mathbf{1}\{\tau \geq i+1\}$, and each term has zero mean after conditioning on $\mathcal{F}_i$, because $\{\tau \geq i+1\}$ is decided by time i .) (ii) Since $\tau < \infty$, $M_{n \wedge \tau} = M_\tau$ for all $n \geq \tau$, so $M_{n \wedge \tau} \rightarrow M_\tau$ almost surely; by bounded convergence $\mathbb{E}[M_\tau] = \lim_n \mathbb{E}[M_{n \wedge \tau}] = \mathbb{E}[M_0]$. (iii) $|M_{n \wedge \tau}| \leq |M_0| + c\tau$, an integrable bound, so dominated convergence applies as in (ii). $\square$

- (c) Let $S_0 = 0$ and $\tau = \min\{t : S_t > 0.20 \text{ or } S_t < -0.20\}$, the first time the log price rises above 0.20 or falls below -0.20 . If the steps are i.i.d. $N(\mu, \sigma^2)$, which conditions on μ and σ^2 make $\mathbb{P}(S_\tau > 0.20) > \frac{1}{2}$? (The problem statement says “ S_t are i.i.d. Normal”; the steps X_t are meant.)

Exercise 4.9 (A martingale that is not Markov (not from the course)). For the process $M_n = \xi_1 + c(\xi_1) \sum_{k=2}^n \xi_k$ of [section 4.4.3](#), write M_n as a martingale transform, and compute $\mathbb{P}(M_4 = -3 \mid M_3 = -1)$ and $\mathbb{P}(M_4 = -3 \mid M_3 = -1, M_1 = -1)$ to show that $\{M_n\}$ is not Markov.

Exercise 4.10 (lecnote5, Exercise 5.4). In the coin-toss game the gambler stops the first time her balance reaches \$100. By construction $\mathbb{E}[S_\tau] = 100$. Does this contradict the optional stopping theorem? Which hypothesis of [theorem 4.26](#) fails?

4.6 Gambler’s ruin

Two players bet \$1 per round on a fair coin until one of them is broke. Let S_n be my net gain after n rounds, with my bankroll B and my opponent’s bankroll A (A, B positive integers) [[2024 L6 @ 21:35](#)], [[2013 L5 @ 23:03](#)]. The game ends at

$$\tau = \min\{n : S_n = +A \text{ or } S_n = -B\},$$

a stopping time (a first hitting time), and I take all my opponent’s money iff $S_\tau = +A$ [[2024 L6 @ 22:46](#)].

Lemma 4.27. For a walk with steps ± 1 and $\mathbb{P}(X = +1) = p \in (0, 1)$, $\mathbb{P}(\tau > n(A+B)) \leq (1 - p^{A+B})^n$. Hence $\tau < \infty$ almost surely and $\mathbb{E}[\tau] < \infty$.

Proposition 4.28 (Fair gambler’s ruin). For the SRW,

$$\mathbb{P}(S_\tau = +A) = \frac{B}{A+B}, \quad \mathbb{P}(S_\tau = -B) = \frac{A}{A+B}, \quad \mathbb{E}[\tau] = AB.$$

With equal bankrolls each player wins with probability $\frac{1}{2}$, by symmetry [[2013 L5 @ 25:08](#)]. Playing until I win \$100 or lose \$50 ($A = 100$, $B = 50$), I win with probability $1/3$ [[2013 L5 @ 26:08](#)], and the game lasts 5000 rounds on average. In a fair game the richer player is more likely to win everything [[2024 L6 @ 27:05](#)], and the expected duration is the product of the bankrolls [[2024 L6 @ 29:13](#)].

The 2013 route: first-step analysis. Let $f(k)$ be the probability of hitting $+A$ before $-B$ when starting from $k \in \{-B, \dots, A\}$ [[2013 L5 @ 29:19](#)]. Conditioning on the first toss and using the stationarity of the increments,

$$f(k) = \frac{1}{2}f(k+1) + \frac{1}{2}f(k-1) \quad (-B < k < A), \quad f(A) = 1, \quad f(-B) = 0$$

Proof. Split time into blocks of $A+B$ steps. If all steps of a block are $+1$ the walk, wherever it is in $(-B, A)$, leaves the interval during that block. Blocks are independent and each has only $+1$ steps with probability p^{A+B} , so surviving n blocks has probability at most $(1 - p^{A+B})^n$. Then $\mathbb{E}[\tau] = \sum_{m \geq 0} \mathbb{P}(\tau > m) \leq (A+B) \sum_{n \geq 0} (1 - p^{A+B})^n < \infty$. $\square$

Proof. S_n is a martingale and $|S_{n \wedge \tau}| \leq \max(A, B)$, so by [lemma 4.27](#) and optional stopping (b), $0 = \mathbb{E}[S_\tau] = A\mathbb{P}(S_\tau = A) - B(1 - \mathbb{P}(S_\tau = A))$ [[2024 L6 @ 23:52](#)], [[2024 L6 @ 24:54](#)], which gives the first formula. For the second, $M_n = S_n^2 - n$ is a martingale ([example 4.16](#) with $\sigma^2 = 1$), so by [theorem 4.25](#)

$$\mathbb{E}[S_{n \wedge \tau}^2] = \mathbb{E}[n \wedge \tau] \quad \text{for every } n.$$

As $n \rightarrow \infty$ the left side tends to $\mathbb{E}[S_\tau^2]$ by bounded convergence ($S_{n \wedge \tau}^2 \leq \max(A^2, B^2)$) and the right side to $\mathbb{E}[\tau]$ by monotone convergence ($n \wedge \tau \uparrow \tau$). Hence [[2024 L6 @ 29:13](#)]

$$\mathbb{E}[\tau] = \mathbb{E}[S_\tau^2] = A^2 \frac{B}{A+B} + B^2 \frac{A}{A+B} = AB. \quad \square$$

7 COURSE NOTE: Erratum: slides 24–26 of lec05

- “Problem: Solve for $\mathbb{P}(\tau = +A)$ ” (and the title of slide 25) should read $\mathbb{P}(S_\tau = +A)$: τ is a time, S_τ the level hit. A student pointed this out in class and the lecturer confirmed [[2024 L6 @ 26:05](#)].

[► 2013 L5 @ 30:20]. The increments $f(k+1) - f(k)$ are therefore constant: $f(-B+r) = \alpha r$ with $\alpha = 1/(A+B)$, and $f(0) = B/(A+B)$ (2013 notes, Section 2).

INTUITION: Harmonic functions and electrostatics

The recursion says that f is *discrete-harmonic*: its value is the average of its neighbours, i.e. its discrete Laplacian vanishes. Equivalently $f(S_{n \wedge \tau})$ is a martingale, which is why the two methods agree. The hitting probability is the solution of a Dirichlet problem, like the potential between two plates held at potentials 0 and 1, hence linear. In continuous time the same idea links expectations of Brownian functionals to PDEs (backward Kolmogorov and Feynman–Kac equations, used to price options in chapter 15).

Remark 4.29 (A student’s symmetry argument [► 2013 L5 @ 27:11]). For $A = 100$, $B = 50$ a student guessed $1/3$ as follows. Let $g(a, b)$ be the probability of gaining a before losing b . To gain 100 before losing 50, the walk must first reach $+50$ before -50 (probability $\frac{1}{2}$); from $+50$ it must then gain 50 before losing 100, with probability $g(50, 100) = 1 - g(100, 50)$ by the reflection $S \mapsto -S$. So $g = \frac{1}{2}(1 - g)$ and $g = 1/3$. The lecturer was unsure it could be made rigorous; it can: restarting the walk at a stopping time gives again a SRW independent of the past (the *strong Markov property*, not proved in this course).

The biased walk. Now $\mathbb{P}(X = +1) = p$, $\mathbb{P}(X = -1) = q = 1 - p$, $p \neq \frac{1}{2}$ [► 2024 L6 @ 30:16]. First-step analysis still works but is laborious [► 2024 L6 @ 31:16]; the exponential martingale solves the problem in two lines [► 2024 L6 @ 32:19].

Proposition 4.30 (Biased gambler’s ruin). With $r = q/p \neq 1$,

$$\mathbb{P}(S_\tau = +A) = \frac{r^B - 1}{r^{A+B} - 1}.$$

As $p \rightarrow \frac{1}{2}$, $r = 1 + \varepsilon$ with $\varepsilon \rightarrow 0$, and the ratio tends to $B\varepsilon/((A+B)\varepsilon) = B/(A+B)$, recovering the fair case. The expected duration (not derived in class) follows from the martingale $S_n - n(p-q)$ (example 4.15) and optional stopping (c) (bounded increments, $\mathbb{E}[\tau] < \infty$): $\mathbb{E}[S_\tau] = (p-q)\mathbb{E}[\tau]$ (Wald’s identity), so

$$\mathbb{E}[\tau] = \frac{A\mathbb{P}(S_\tau = A) - B\mathbb{P}(S_\tau = -B)}{p - q}.$$

- “ $\lim_{n \rightarrow \infty} \mathbb{E}[S_{n \wedge \tau}] = S_\tau$ w.p. 1” mixes a number and a random variable: the correct statement is $S_{n \wedge \tau} \rightarrow S_\tau$ almost surely (because $\mathbb{P}(\tau = \infty) = 0$), and then $\mathbb{E}[S_\tau] = \lim_n \mathbb{E}[S_{n \wedge \tau}] = 0$ by bounded convergence.
- Slide 26: “ $M_{n \wedge \tau} \leq \max(A^2, B^2) + \tau$, bounded r.v.” is not a valid justification, since τ is not bounded. The limit must be taken separately in the two terms of $M_{n \wedge \tau} = S_{n \wedge \tau}^2 - n \wedge \tau$, with bounded and monotone convergence, as in the proof above.

Proof. By example 4.18, $M_n = r^{S_n}$ is a martingale with $M_0 = 1$. Since $-B \leq S_{n \wedge \tau} \leq A$, $M_{n \wedge \tau}$ lies between r^{-B} and r^A , and optional stopping (b) with lemma 4.27 gives [► 2024 L6 @ 34:22]

$$1 = \mathbb{E}[r^{S_\tau}] = r^A \mathbb{P}(S_\tau = A) + r^{-B} (1 - \mathbb{P}(S_\tau = A)) \implies \mathbb{P}(S_\tau = A) = \frac{r^B - 1}{r^{A+B} - 1} \quad \square$$

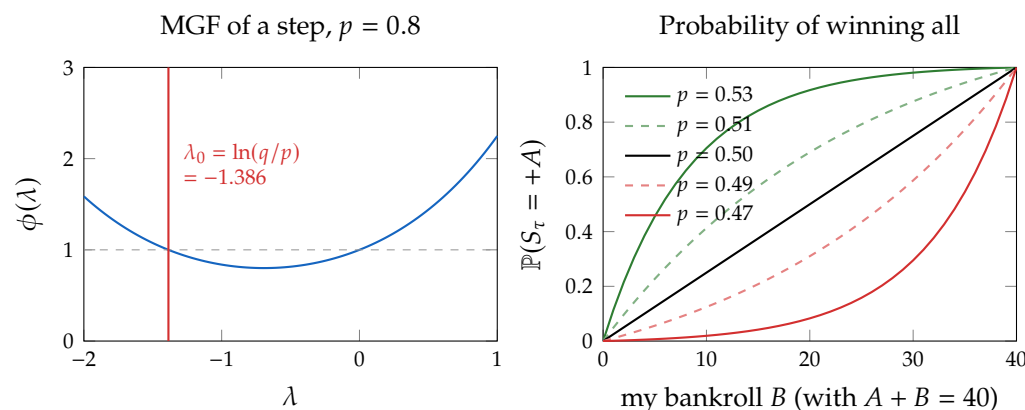


Figure 4.3: Left: the moment generating function $\phi(\lambda) = pe^\lambda + qe^{-\lambda}$ of a ± 1 step equals 1 at $\lambda = 0$ and at $\lambda_0 = \ln(q/p)$, as on slide 28 [► 2024 L6 @ 33:19]. Right: probability of winning the opponent's whole bankroll, [proposition 4.30](#), as a function of my initial capital when the total capital is 40. A bias of a few percent per bet turns the straight line of the fair game into an S-shaped curve.

CASE STUDY: Probability surface in R (slides 30–31)

The slides define `fcn.proba(A,B)` implementing [proposition 4.30](#) with $p = 0.53$, evaluate it on the grid $A, B \in \{1, \dots, 30\}$ with `outer` (after `Vectorize`), and draw the surface (`persp3D`) and its contour lines [► 2024 L6 @ 35:30]. Some values: $A = B = 10$ gives 0.769 and $A = B = 20$ gives 0.917. Against an opponent of unlimited wealth ($A \rightarrow \infty$) the probability of never being ruined is $1 - r^B$: 0.45 for $B = 5$, 0.70 for $B = 10$ and 0.97 for $B = 30$. This is why the contour lines on slide 31 become horizontal for large A .

IN FINANCE: Edges and ruin (numbers ours)

The limits $A \rightarrow \infty$ make the 2013 remark precise: “if one has some advantage over an opponent in some game, then no matter how small that advantage is, he/she will win in the long run”. With $p > q$ the probability of never being ruined against an infinitely rich opponent is $1 - (q/p)^B > 0$; with $p < q$ ruin is certain. The casino is the infinitely rich opponent with the edge: betting \$1 on red at European roulette ($p = 18/37$) with \$100, aiming to reach \$200, succeeds with probability 0.0045 after about 3700 bets on average; aiming for \$20 starting from \$10 succeeds with probability 0.37. For a trader, $p - q$ is the edge per trade and B the capital in units of the bet size: a small edge protects against ruin only if the bets are small relative to the capital.

Exercise 4.11 (lecnote5, Exercise 5.5). For positive integers a, b the player stops the first time the balance equals a or $-b$. Find the distribution of S_τ . (This is [proposition 4.28](#); redo it without looking. The notes write “ $X_\tau = b$ ” for the second outcome; $-b$ is meant.)

Exercise 4.12 (Biased ruin by first-step analysis (not from the course)). For the biased walk let $f(k) = \mathbb{P}(\text{hit } A \text{ before } -B \mid S_0 = k)$. Show that $f(k) = pf(k+1) + qf(k-1)$, solve this linear recursion (characteristic roots 1 and q/p) with $f(A) = 1$, $f(-B) = 0$, and recover [proposition 4.30](#). Then use Wald's identity to show that the expected

8 COURSE NOTE: 2013 versus 2024 treatment

2013 derives the fair ruin probability by first-step analysis and then, at the end of the lecture, re-derives $\mathbb{P} = 1/3$ for $(100, -50)$ as an application of optional stopping, glossing over the fact that the hitting time is unbounded [► 2013 L5 @ 1:16:09]. 2024 builds the machinery first (transforms, stopping times, stopped martingales), then computes the ruin probability, the expected duration and the biased case, all by optional stopping. 2013 has instead the link between stationary distributions and Perron–Frobenius, which 2024 does not discuss.

duration tends to AB as $p \rightarrow \frac{1}{2}$.

4.7 Looking ahead

- *Random walk* $\rightarrow$ *Brownian motion* ([chapter 11](#)): independent stationary increments and the $\sqrt{t}$ scale survive the limit; $S_n^2 - n$ becomes $B_t^2 - t$ and $e^{\lambda S_n} / \phi(\lambda)^n$ becomes $e^{\lambda B_t - \lambda^2 t / 2}$.
- *Martingale transforms* $\rightarrow$ *Itô integrals* ([chapters 17](#) and [18](#)): gains from trading, and the reason why hedged portfolios under the risk-neutral measure are martingales ([chapter 15](#)).
- *Markov chains* $\rightarrow$ *Markov diffusions*: Chapman–Kolmogorov becomes the Fokker–Planck equation, and first-step analysis becomes the backward equation behind option-pricing PDEs.
- *Optional stopping* returns in first-passage problems: barrier options, default times in structural credit models, American-style early exercise.
- The *Poisson process*, the continuous-time counting process of events with exponential waiting times, is not covered in these lectures; 2013 PS2 builds the Poisson distribution from exponential waiting times ([chapter 3](#)).

KEY FORMULAS

Random walk:	$\mathbb{E}[S_n] = n\mu, \quad \text{Var } S_n = n\sigma^2,$ $(S_n - n\mu) / (\sigma\sqrt{n}) \rightarrow N(0, 1)$
Markov chain:	$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = p_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n},$ $P^{(n)} = P^n, \quad p^{(n)} = p P^n$
Stationary dist.:	$\pi P = \pi; \quad P^k > 0 \Rightarrow P_{ij}^{(n)} \rightarrow \pi_j \text{ geometrically}$
Martingale:	$\mathbb{E}[M_n \mathcal{F}_{n-1}] = M_{n-1},$ $\mathbb{E}[M_n \mathcal{F}_m] = M_m, \quad \mathbb{E}[M_n] = M_0$
Examples:	$S_n, \quad S_n^2 - n\sigma^2,$ $\prod_k X_k \ (\mathbb{E}[X] = 1), \quad e^{\lambda S_n} / \phi(\lambda)^n$
Transform:	$\tilde{M}_n = M_0 + \sum_{k \leq n} A_k \Delta M_k,$ $A_k \in \mathcal{F}_{k-1} \Rightarrow \text{martingale}$
Opt. stopping:	$\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ if τ bounded, or $M_{n \wedge \tau}$ bounded, or $\mathbb{E}[\tau] < \infty$ & bounded steps
Fair ruin:	$\mathbb{P}(S_\tau = A) = \frac{B}{A+B}, \quad \mathbb{E}[\tau] = AB$
Biased ruin:	$\mathbb{P}(S_\tau = A) = \frac{(q/p)^B - 1}{(q/p)^{A+B} - 1},$ $\mathbb{E}[\tau] = \frac{\mathbb{E}[S_\tau]}{p-q}$

Chapter 5

Regression Analysis

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L6 (from 1:07:22)	P. Kempthorne	Multiple linear regression: set-up, polynomial, Fourier and autoregressive fits of the high-yield spread, the steps of model fitting, assumptions on the errors.
2024 L8	P. Kempthorne	OLS, hat matrix and geometry; normal linear model via moment-generating functions; t and F tests; prostate-cancer example (standardised covariates, R^2 , diagnostics); Gauss–Markov; GLS.
2024 L11	P. Kempthorne	Recap, $t^2 = F$, GLS, maximum likelihood, M-estimators (robust, quantile), ridge, principal-components regression, lasso; ETF case study; CAPM regressions (GE, all S&P 500 stocks) and tests for a change of regime.
2013 L6	P. Kempthorne	The same theory: OLS, Gauss–Markov with proof, GLS, distribution theory via MGFs and the QR decomposition, maximum likelihood.
Files. Slides <code>mit18_642_f24_lec06_1.pdf</code> (used in L6, L8 and L11); <code>mit18_642_f24_lec06_2.pdf</code> and <code>mit18_642_f24_lec06_3.pdf</code> (polynomial and Fourier fits of the high-yield spread); <code>mit18_642_f24_lec06_4.pdf</code>, the knitted <code>ETF_Casestudy.rmd</code> of <code>mit18_642_f24_etf_casestudy.zip</code> (which also contains <code>ETF_Casestudy_XLB.rmd/.pdf</code>, <code>spyders_symbols_labels.csv</code>, <code>casestudy_1_0_etfs.RData</code>); note <i>Hypothesis Testing in Regression Analysis</i> <code>mit18_642_f24_lec08.pdf</code>; 2013 notes <code>MIT18_S096F13_lecnote6.pdf</code>; 2013 Case Study 1 <code>CaseStudy1.pdf</code> with <code>fm_casestudy_1_0.r</code> and <code>fm_casestudy_1_rcode.r</code>; 2013 Case Study 2 <code>CaseStudy2.pdf</code> with <code>fm_casestudy_2_rcode.r</code>, <code>fm_casestudy_fx_1.r</code> and <code>fred_fxrates.txt</code>. Installation scripts and R instructions are described in chapter B. Problem sets. 2013 PS3 (all problems). Least squares via the SVD (2013 PS1, B-4) is in chapter 2. Not published. The note on CAPM regressions for all S&P 500 stocks shown in 2024 L11; only its GE part is in <code>lec08</code> (see section 5.11.2).		

Motivation

Linear regression is the most used statistical tool in finance: every beta, hedge ratio, factor exposure, CAPM test or “is this strategy’s alpha real?” question is a regression. The mathematics has two layers, and this chapter develops both:

- *Geometry* (no probability): least squares is the orthogonal projection of the data vector $\mathbf{y} \in \mathbb{R}^n$ on the column space of the design matrix X ([section 5.2](#)). This is linear algebra from [chapter 2](#).

- *Statistics*: under assumptions on the errors, the projection has optimality properties (Gauss–Markov, [section 5.3](#)) and known sampling distributions ([section 5.5](#)), which give t and F tests ([section 5.7](#)), including tests for a change of regime ([section 5.8](#)).

The rest is about what to do when the assumptions fail: diagnostics ([section 5.9](#)), robust and quantile estimators ([section 5.6](#)), and shrinkage ([section 5.10](#)). The course’s message, repeated in every lecture, is methodological: propose a model, fit it, *check the assumptions*, revise [► 2024 L6 @ 1:18:03]. The case studies (sector ETFs, CAPM for GE and the S&P 500, the Chinese yuan) show what this looks like on real data.

5.1 The linear regression model

Data and goals. We observe n cases $i = 1, \dots, n$; for each, a *response* (dependent variable) y_i and p *explanatory* (independent) variables $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})^\top$ [► 2024 L6 @ 1:07:22]. In finance y_i may be a stock return and $\mathbf{x}_i$ market or macroeconomic factors [► 2013 L6 @ 2:04]. The goals are prediction, causal inference (which factors really drive y ?), approximation of an unknown function, and the discovery or validation of functional relationships [► 2024 L6 @ 1:08:24].

Definition 5.1 (Linear model). The *general linear model* specifies the conditional distribution of y_i given $\mathbf{x}_i$ as

$$y_i = \hat{y}_i + \varepsilon_i, \quad \hat{y}_i = \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_p x_{i,p},$$

with regression parameters $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ constant over the cases and a residual (error) ε_i varying over the cases. In matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} x_{1,1} & \cdots & x_{1,p} \\ \vdots & \ddots & \vdots \\ x_{n,1} & \cdots & x_{n,p} \end{pmatrix} \in \mathbb{R}^{n \times p} :$$

cases are rows, variables are columns [► 2013 L6 @ 29:24]. An intercept is included by taking a first column of ones.

“Linear” means linear in $\boldsymbol{\beta}$. The columns of $\mathbf{X}$ can be any functions of the underlying variables [► 2024 L6 @ 1:10:32][► 2013 L6 @ 6:19]:

- polynomials, $x_{i,j} = x_i^j$ (a truncated Taylor expansion);
- Fourier terms, $x_{i,j} = \sin(jx_i)$ or $\cos(jx_i)$, for cyclical behaviour;
- time-series regressions, where i indexes time and the regressors include lagged values $y_{i-1}, y_{i-2}, \dots$ of the response (autoregressions, [chapter 9](#)) [► 2013 L6 @ 7:24].

What matters is that $\hat{y}_i$ is linear in the parameters; this is what makes the estimation a problem of linear algebra.

LAB: high-yield spread, functions of time against persistence

Question. Can the *high-yield spread* be described by a function of calendar time, and what is actually predictable in it? (Case study of [lec06_2](#) and [lec06_3](#).)

Data. `data/fm_hyspread.dat`: daily spread 2010–2017, Moody’s Baa corporate yield

1 COURSE NOTE: Erratum in the slides

Slides 3 and 7 of [lec06_1](#) (and the 2013 notes) write the last term as $\beta_{i,p} x_{i,p}$: it must be $\beta_p x_{i,p}$, since the parameters do not depend on the case; Kempthorne corrects this in class [► 2024 L6 @ 1:09:30]. In the same slides the bottom-right entry of $\mathbf{X}$ is printed $x_{p,n}$ instead of $x_{n,p}$.

minus the 10-year US Treasury yield, in percentage points [► 2024 L6 @ 1:11:33](FRED series BAA10Y; 2001 days). Lower-rated borrowers pay more than the Treasury, so it is always positive.

In Python (numpy, least squares with `np.linalg.lstsq`):

1. Plot the spread. Regress it on powers of time, from the constant to an 8th-order polynomial, and record R^2 (rescale time to $[-1, 1]$ first: why?).
2. Regress it on Fourier terms $\sin(ju), \cos(ju)$, $j = 1, \dots, J$, with u running from 0 to 2π over the sample, for $J = 1, \dots, 4$.
3. Regress today's spread on yesterday's, $y_t = a + by_{t-1} + \varepsilon_t$. Compare the three R^2 , look at the fitted curves at the ends of the sample, and say which model you would use to forecast tomorrow.

Reference script: `labs/ch05_hyspread.py` (-plot draws the fits).

[Course results →](#)

Steps for fitting a model [► 2024 L6 @ 1:14:47][► 2013 L6 @ 8:24].

1. Propose a model: the response and its *scale* (e.g. log prices, so that errors are percentage moves [► 2013 L6 @ 9:25]), the explanatory variables (including functions of them), and assumptions on the distribution of ε .
2. Define a criterion for judging estimators (least squares, maximum likelihood, robust, Bayes, methods for missing data) [► 2013 L6 @ 23:07].
3. Characterise the best estimator under that criterion and compute it.
4. Check the assumptions of step 1: residual analysis (the errors ε_i are unobservable, the residuals $\hat{\varepsilon}_i = y_i - \hat{y}_i$ are their proxies), influence diagnostics, outlier detection [► 2013 L6 @ 26:16].
5. If necessary modify the model and/or the assumptions and go back to step 1.

Step 5 is where the work is on hard problems: the model should be tailored to the process, not the other way round [► 2013 L6 @ 12:31].

Definition 5.2 (Assumptions on the errors). In increasing generality [► 2024 L6 @ 1:15:54] [► 2013 L6 @ 14:38]:

- *Normal linear model*: $\varepsilon_1, \dots, \varepsilon_n$ i.i.d. $N(0, \sigma^2)$.
- *Gauss–Markov*: $\mathbb{E}[\varepsilon] = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 I_n$, i.e. zero mean, constant variance (homoscedastic), uncorrelated. Uncorrelated is weaker than independent [► 2013 L6 @ 15:40].
- *Generalised Gauss–Markov*: $\mathbb{E}[\varepsilon] = \mathbf{0}$, $\text{Cov}(\varepsilon) = \sigma^2 \Sigma$ with a general positive definite Σ (correlated and/or heteroscedastic errors).
- *Non-Gaussian errors*: Laplace, Pareto, or *contaminated normal*: a fraction $1 - \delta$ of the ε_i i.i.d. $N(0, \sigma^2)$ and a fraction δ from some contaminating distribution.

Correlated residuals arise naturally when a straight line is fitted to a curved relationship (residuals are systematically above, then below the line) or when unobserved factors persist in time [► 2013 L6 @ 20:58]. Generalised Gauss–Markov is the typical situation of time-series regressions, and heavy-tailed errors that of asset returns [► 2024 L6 @ 1:16:57].

INTUITION: A physicist's reading

This is curve fitting with error bars. Least squares with equal error bars is χ^2 minimisation; Generalised Gauss–Markov is the case of a full covariance matrix of the measurements; the five steps are the usual loop “model, fit, look at the residuals (pulls), improve the model”. The one difference from a lab: in finance there is no repeated experiment, and the “error bars” themselves must be estimated from the same data.

5.2 Ordinary least squares and its geometry

5.2.1 Normal equations

The *least-squares criterion* is [► 2024 L8 @ 1:06]

$$Q(\beta) = \sum_{i=1}^n (y_i - x_i^\top \beta)^2 = (y - X\beta)^\top (y - X\beta) = \|y - X\beta\|^2,$$

and the *ordinary least-squares* (OLS) estimate $\hat{\beta}$ minimises it. Differentiating,

$$\frac{\partial Q}{\partial \beta_j} = \sum_{i=1}^n 2(-x_{i,j})(y_i - x_i^\top \beta) = -2X_{[j]}^\top (y - X\beta), \quad \frac{\partial Q}{\partial \beta} = -2X^\top (y - X\beta),$$

where $X_{[j]}$ is the j -th column of X [► 2013 L6 @ 33:40]. The Hessian is $\partial^2 Q / \partial \beta \partial \beta^\top = 2X^\top X$, positive semi-definite since $v^\top X^\top X v = \|Xv\|^2 \geq 0$: Q is convex, so every stationary point is a global minimum [► 2024 L8 @ 5:16][► 2013 L6 @ 32:34]. Setting the gradient to zero gives the *normal equations*

$$X^\top (y - X\hat{\beta}) = \mathbf{0} \iff X^\top X\hat{\beta} = X^\top y \implies \hat{\beta} = (X^\top X)^{-1} X^\top y. \quad (5.1)$$

$X^\top X$ is invertible iff $Xv = \mathbf{0} \implies v = \mathbf{0}$, i.e. iff X has *full column rank* p [► 2024 L8 @ 7:33]: no explanatory variable may be a linear combination of the others [► 2013 L6 @ 37:50]. Without full rank $\hat{\beta}$ is not unique, but the fitted values are (see below), which is enough for pure prediction [► 2013 L6 @ 37:50].

Remark 5.3. Kempthorne jokes that there is nothing “normal” about the normal equations [► 2024 L8 @ 7:33]. A useful way to remember the name: (5.1) says that the residual vector is *normal* (perpendicular) to every column of X .

5.2.2 The hat matrix

The fitted values are $\hat{y} = X\hat{\beta} = Hy$ with the $n \times n$ *hat matrix* [► 2024 L8 @ 8:36][► 2013 L6 @ 39:58]

$$H = X(X^\top X)^{-1} X^\top,$$

and the residuals are $\hat{\varepsilon} = y - \hat{y} = (I_n - H)y$.

Proposition 5.4 (Properties of the hat matrix). Let $X \in \mathbb{R}^{n \times p}$ have full column rank.

(i) $H^\top = H$ and $H^2 = H$; the same holds for $I_n - H$.

Proof. (i) $H^\top = X[(X^\top X)^{-1}]^\top X^\top = H$ because $X^\top X$ is symmetric;

$$H^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1} X^\top = H$$

[► 2024 L8 @ 10:41]; $(I - H)^2 = I - 2H + H^2 =$

- (ii) H is the orthogonal projection onto $\text{col}(X)$, and $I_n - H$ the orthogonal projection onto $\text{col}(X)^\perp$.
- (iii) $\text{tr} H = \text{rank } H = p$ and $\text{tr}(I_n - H) = n - p$.
- (iv) $X^\top \hat{\varepsilon} = \mathbf{0}$, hence $\hat{\varepsilon} \perp \hat{\mathbf{y}}$ and $\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\hat{\varepsilon}\|^2$.

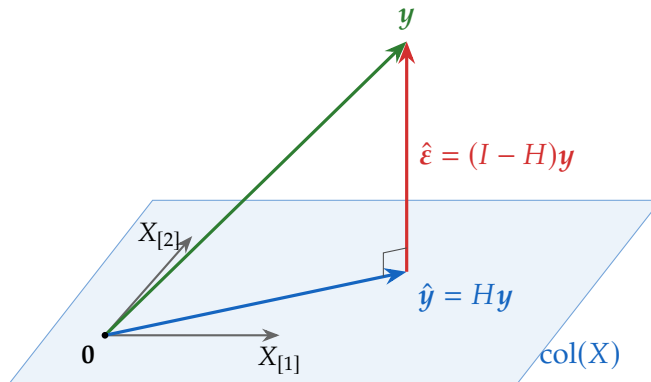


Figure 5.1: Least squares in $\mathbb{R}^n$: $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ on the column space of X (spanned here by two columns), and the residual is perpendicular to it [► 2024 L8 @ 13:50] [► 2013 L6 @ 42:08].

INTUITION: Two pictures of the same fit

In the usual scatter plot we look at n points in $\mathbb{R}^p$ and a fitted hyperplane. In fig. 5.1 we look at *one* point $\mathbf{y}$ in $\mathbb{R}^n$ (one axis per case) and a p -dimensional subspace: least squares drops the perpendicular [► 2013 L6 @ 42:08]. The second picture is the one in which all the distribution theory below becomes transparent: the residual lives in an $(n - p)$ -dimensional space, which is why “ $n - p$ degrees of freedom” appear everywhere.

Exercise 5.1 (PS3 (2013), Problem 1(a)–(c): the hat matrix).

- (a) Prove that H is a projection matrix: $H^\top = H$ and $H^2 = H$.
- (b) The diagonal element H_{ii} is the *leverage* of case i . Show that $\partial \hat{y}_i / \partial y_i = H_{ii}$.
- (c) Show that the average of the H_{ii} is p/n . Hint: $\text{tr}(AB) = \text{tr}(BA)$.

5.2.3 R^2

Assume from now on that X contains a column of ones (an intercept). Then $\mathbf{1}^\top \hat{\varepsilon} = 0$ by (5.1): residuals sum to zero and $\bar{\hat{\mathbf{y}}} = \bar{\mathbf{y}}$. Subtracting $\bar{\mathbf{y}}\mathbf{1}$ (which lies in $\text{col}(X)$) from both sides of Pythagoras gives the *analysis of variance*

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{ESS}} + \underbrace{\sum_i \hat{\varepsilon}_i^2}_{\text{RSS}}.$$

Definition 5.5 (Coefficient of determination). $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$ (“multiple

$I - H$ [► 2024 L8 @ 11:45].

(ii) If $\mathbf{v} = X\mathbf{a} \in \text{col}(X)$ then $H\mathbf{v} = X(X^\top X)^{-1}X^\top X\mathbf{a} = \mathbf{v}$; if $\mathbf{w} \perp \text{col}(X)$ then $X^\top \mathbf{w} = \mathbf{0}$ and $H\mathbf{w} = \mathbf{0}$. Since $\mathbb{R}^n = \text{col}(X) \oplus \text{col}(X)^\perp$, H is the orthogonal projection, and $I - H$ the complementary one.

(iii) $\text{tr} H = \text{tr}((X^\top X)^{-1}X^\top X) = \text{tr} I_p = p$, using $\text{tr}(AB) = \text{tr}(BA)$; for a projection, trace = rank.

(iv) is (5.1); then $\hat{\mathbf{y}}^\top \hat{\varepsilon} = \hat{\beta}^\top X^\top \hat{\varepsilon} = 0$ and Pythagoras [► 2024 L8 @ 14:54]. $\square$

R -squared”), and the *adjusted* $R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p)}{\text{TSS}/(n-1)}$, which penalises the number of parameters.

Proposition 5.6. *With an intercept, $R^2 = \text{Corr}(\mathbf{y}, \hat{\mathbf{y}})^2$, the squared sample correlation between observed and fitted values* [► 2024 L8 @ 59:31].

Proof. Let $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\tilde{\hat{\mathbf{y}}} = \hat{\mathbf{y}} - \bar{\hat{y}}\mathbf{1}$. Since $\hat{\boldsymbol{\varepsilon}} \perp \hat{\mathbf{y}}$ and $\hat{\boldsymbol{\varepsilon}} \perp \mathbf{1}$, $\tilde{\mathbf{y}}^\top \tilde{\hat{\mathbf{y}}} = (\hat{\mathbf{y}} + \hat{\boldsymbol{\varepsilon}})^\top \tilde{\hat{\mathbf{y}}} = \|\tilde{\hat{\mathbf{y}}}\|^2$, so $\text{Corr}^2 = \|\tilde{\hat{\mathbf{y}}}\|^4 / (\|\tilde{\mathbf{y}}\|^2 \|\tilde{\hat{\mathbf{y}}}\|^2) = \text{ESS}/\text{TSS}$. □

For simple regression ($p = 2$, one slope) this is the familiar squared correlation of x and y , and $\hat{\beta}_1 = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / \sum_i (x_i - \bar{x})^2 = \widehat{\text{Cov}}(x, y) / \widehat{\text{Var}}(x)$, the beta of [definition 1.2](#). It is worth calibrating the eye: the daily CAPM regression of GE on the market in [section 5.11.2](#) has $R^2 = 0.30$, a clearly visible but very noisy linear cloud [► 2024 L11 @ 59:45].

Exercise 5.2 (not from the course: R^2 with orthogonal regressors). Let $\mathbf{z}_1, \dots, \mathbf{z}_m$ be centred and mutually orthogonal, and regress $\mathbf{y}$ on $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_m$.

- Show that the multiple R^2 equals the sum of the R^2 's of the m simple regressions of $\mathbf{y}$ on $\mathbf{1}, \mathbf{z}_j$, and that each slope is the same in the simple and in the multiple regression.
- Check this on the ETF numbers ($0.545 + 0.006 + 0.079 + 0.012$ vs 0.642) and explain why the order of the principal components by variance says nothing about their R^2 .

5.2.4 The QR decomposition

Numerically one never inverts $X^\top X$. The 2013 lecture [► 2013 L6 @ 1:08:26] uses the QR decomposition

$$\mathbf{X} = \mathbf{Q}\mathbf{R}, \quad \mathbf{Q} \in \mathbb{R}^{n \times p}, \quad \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p, \quad \mathbf{R} \in \mathbb{R}^{p \times p} \text{ upper triangular,}$$

obtained by Gram–Schmidt on the columns of $\mathbf{X}$ [► 2013 L6 @ 1:09:26]. Column by column, $\mathbf{X}_{[1]} = \mathbf{Q}_{[1]}r_{11}$, $\mathbf{X}_{[2]} = \mathbf{Q}_{[1]}r_{12} + \mathbf{Q}_{[2]}r_{22}, \dots$, so

$$r_{11} = \|\mathbf{X}_{[1]}\|, \quad \mathbf{Q}_{[1]} = \frac{\mathbf{X}_{[1]}}{r_{11}};$$

$$r_{12} = \mathbf{Q}_{[1]}^\top \mathbf{X}_{[2]}, \quad r_{22} = \|\mathbf{X}_{[2]} - r_{12}\mathbf{Q}_{[1]}\|, \quad \mathbf{Q}_{[2]} = \frac{\mathbf{X}_{[2]} - r_{12}\mathbf{Q}_{[1]}}{r_{22}}; \dots$$

(each column is orthogonalised against the previous ones) [► 2013 L6 @ 1:10:32]. Plugging $\mathbf{X} = \mathbf{Q}\mathbf{R}$ into the OLS formulas [► 2013 L6 @ 1:11:35]:

$$\hat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{y}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 \mathbf{R}^{-1}(\mathbf{R}^{-1})^\top, \quad \mathbf{H} = \mathbf{Q}\mathbf{Q}^\top.$$

$\hat{\boldsymbol{\beta}}$ is computed by back-substitution in $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{Q}^\top \mathbf{y}$. The advantage (not stressed in class): the condition number of $\mathbf{X}^\top \mathbf{X}$ is the square of that of $\mathbf{X}$, so forming $\mathbf{X}^\top \mathbf{X}$ loses twice as many digits when the regressors are nearly collinear. R's `lm()` works through a QR decomposition (the `qr` component of a fitted `lm` object). The SVD route of 2013 PS1 B-4 ([chapter 2](#)) is the most stable of all and is the one used for ridge and PC regression in [section 5.10](#).

5.2.5 Reparametrisation and standardised covariates

If $\mathbf{X}' = \mathbf{X}\mathbf{G}$ with $\mathbf{G} \in \mathbb{R}^{p \times p}$ invertible, then $\text{col}(\mathbf{X}') = \text{col}(\mathbf{X})$, so $\mathbf{H}$, $\hat{\mathbf{y}}$ and $\hat{\boldsymbol{\varepsilon}}$ are unchanged and the new parameter is $\boldsymbol{\beta}' = \mathbf{G}^{-1}\boldsymbol{\beta}$ (2013 PS3, [Exercise 5.7](#)). The important special case is

standardisation [► 2024 L8 @ 54:05]: replace each covariate by its z-score $(X_{[j]} - \bar{x}_j \mathbf{1})/s_j$. Then

- fitted values, residuals, R^2 , F statistic, and the t statistics and p -values of the slopes are identical [► 2024 L8 @ 57:24];
- the slope coefficient becomes $s_j \beta_j$: the effect on y of a *one-standard-deviation* move of x_j , which makes coefficients of variables in different units comparable [► 2024 L8 @ 58:27];
- the intercept changes meaning: it becomes $\bar{y}$, the prediction at the average covariate.

In the prostate example of [section 5.9](#) the coefficient of `lcavol` goes from 0.5435 to 0.6080 (its training-sample standard deviation is 1.119) and that of `svi` from 0.877 to 0.349, while the t values stay 5.206 and 2.949.

5.3 The Gauss–Markov theorem

Definition 5.7 (Gauss–Markov assumptions). The data $(\mathbf{y}, X)$ follow a linear model satisfying the Gauss–Markov assumptions if $\mathbf{y}$ is an observation of a random vector $\mathbf{Y}$ with [► 2024 L8 @ 1:11:00][► 2013 L6 @ 45:22]

$$\mathbb{E}[\mathbf{Y} \mid X, \boldsymbol{\beta}] = X\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y} \mid X, \boldsymbol{\beta}) = \sigma^2 I_n \quad (\sigma^2 > 0),$$

equivalently $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$. Everything is conditional on X , which is treated as fixed.

For a random vector $\mathbf{Y}$ with mean $\boldsymbol{\mu}$ and covariance Σ and a constant matrix A ,

$$\mathbb{E}[A\mathbf{Y}] = A\boldsymbol{\mu}, \quad \text{Cov}(A\mathbf{Y}) = A \Sigma A^\top$$

[► 2013 L6 @ 1:12:37]. With $A = (X^\top X)^{-1} X^\top$ this gives, under Gauss–Markov,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = (X^\top X)^{-1} X^\top X \boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}. \quad (5.2)$$

Proof. (2013 notes) [► 2013 L6 @ 51:42]. Without loss of generality $c_{p+1} = 0$; write $\mathbf{c} = (c_1, \dots, c_p)^\top$. Then $\hat{\theta} = \mathbf{c}^\top \hat{\boldsymbol{\beta}} = \mathbf{c}^\top (X^\top X)^{-1} X^\top \mathbf{y} \equiv \mathbf{d}^\top \mathbf{y}$ with $\mathbf{d} = X(X^\top X)^{-1} \mathbf{c} \in \text{col}(X)$, and $\mathbb{E}[\hat{\theta}] = \mathbf{c}^\top \boldsymbol{\beta}$ by (5.2). Let $\tilde{\theta} = \mathbf{b}^\top \mathbf{y}$ be any linear unbiased estimator and $\mathbf{f} = \mathbf{b} - \mathbf{d}$, so $\tilde{\theta} = \hat{\theta} + \mathbf{f}^\top \mathbf{y}$. Unbiasedness of both gives $0 = \mathbb{E}[\mathbf{f}^\top \mathbf{y}] = \mathbf{f}^\top X \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^p$, hence $X^\top \mathbf{f} = \mathbf{0}$: $\mathbf{f} \perp \text{col}(X)$, in particular $\mathbf{f} \perp \mathbf{d}$. Therefore

$$\text{Var}(\tilde{\theta}) = \text{Var}(\mathbf{d}^\top \mathbf{y}) + \text{Var}(\mathbf{f}^\top \mathbf{y}) + 2 \text{Cov}(\mathbf{d}^\top \mathbf{y}, \mathbf{f}^\top \mathbf{y}) = \text{Var}(\hat{\theta}) +$$

with equality iff $\mathbf{f} = \mathbf{0}$ [► 2013 L6 @ 53:43]. $\square$

Theorem 5.8 (Gauss–Markov). For known constants $c_1, \dots, c_{p+1}$ let $\theta = c_1 \beta_1 + \dots + c_p \beta_p + c_{p+1}$ and $\hat{\theta} = c_1 \hat{\beta}_1 + \dots + c_p \hat{\beta}_p + c_{p+1}$ with the OLS estimates. Then $\hat{\theta}$ is

- unbiased;
- linear in $\mathbf{y}$, $\hat{\theta} = \sum_i b_i y_i$ with constants b_i depending on X only.

Under the Gauss–Markov assumptions it has the smallest variance among all linear unbiased estimators of θ : it is the best linear unbiased estimator (BLUE) [► 2024 L8 @ 1:12:05] [► 2013 L6 @ 46:25].

Remarks.

- The theorem is very general [► 2024 L8 @ 1:13:07]: with $c = x_0$ it says that $x_0^T \hat{\beta}$ is the best linear unbiased estimate of the mean response at a new point x_0 ; with $c = x_1 - x_2$, of the difference of mean responses of two cases. Since it holds for every c , it is equivalent to the matrix statement $\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta}) \geq 0$ for every linear unbiased $\tilde{\beta}$.
- No normality is needed. Econometrics texts call this the BLUE property of OLS [► 2013 L6 @ 54:45].
- “Best” only among *linear unbiased* estimators. A biased estimator (ridge, [section 5.10](#)) or a non-linear one (robust M-estimators, [section 5.6](#)) can have a smaller mean squared error, especially with heavy-tailed errors.

Design: where to put the x 's [► 2024 L8 @ 27:37]. $\text{Var}(\hat{\beta}_j) = \sigma^2 C_{jj}$ with $C_{jj} = [(X^T X)^{-1}]_{jj}$. If we control X (an experiment), we should make C_{jj} small. If $X^T X$ is diagonal, $C_{jj} = 1/\|X_{[j]}\|^2$: the longer the column, the more precise the estimate [► 2024 L8 @ 29:50]. For a straight line $y = \beta_0 + \beta_1 x$,

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2} :$$

points far from $\bar{x}$ pin the slope down [► 2024 L8 @ 30:55]. In general the precision is governed by the eigenvalues of $X^T X$ (optimal experimental design) [► 2024 L8 @ 31:59].

INTUITION: Lever arms

A physicist measuring a spring constant spreads the loads over the widest possible range: an error δy at a point at distance d from the centroid tilts the line by $\delta y/d$. The same “lever arm” will reappear as the *leverage* H_{ii} of a case ([section 5.9](#)): the points that pin the line down are also the ones that can pull it away.

Exercise 5.3 (not from the course: optimal design of a line fit). For $y = \beta_0 + \beta_1 x + \varepsilon$ under Gauss–Markov with n even and $x_i \in [-1, 1]$ at your choice:

- derive $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ with $S_{xx} = \sum_i (x_i - \bar{x})^2$;
- find the design minimising $\text{Var}(\hat{\beta}_1)$ and compute the leverages H_{ii} for it;
- explain why this design cannot detect curvature, and how the curvature test of `residualPlots()` would behave.

5.4 Generalised least squares

Suppose now $Y = X\beta + \varepsilon$ with $\mathbb{E}[\varepsilon] = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 \Sigma$, where σ^2 is an unknown scale and Σ is a *known* $n \times n$ positive definite matrix giving the relative variances and the correlations of the observations [► 2024 L8 @ 1:13:07][► 2013 L6 @ 54:45]. By the spectral theorem ([theorem 2.11](#)) $\Sigma = U\Lambda U^T$ has the symmetric positive definite inverse square root $\Sigma^{-1/2} = U\Lambda^{-1/2}U^T$. Transform the data [► 2024 L8 @ 1:15:23][► 2013 L6 @ 55:47]:

$$Y^* = \Sigma^{-1/2}Y, \quad X^* = \Sigma^{-1/2}X, \quad \varepsilon^* = \Sigma^{-1/2}\varepsilon : \\ Y^* = X^*\beta + \varepsilon^*, \quad \text{Cov}(\varepsilon^*) = \sigma^2 \Sigma^{-1/2} \Sigma \Sigma^{-1/2} = \sigma^2 I_n.$$

The transformed model satisfies Gauss–Markov with the *same* β , so by [theorem 5.8](#) the BLUE is OLS on the starred data:

$$\hat{\beta}_{\text{GLS}} = (X^{*T} X^*)^{-1} X^{*T} Y^* = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y, \quad \text{Cov}(\hat{\beta}_{\text{GLS}}) = \sigma^2 (X^T \Sigma^{-1} X)^{-1}. \quad (5.3)$$

2 COURSE NOTE: Erratum in the 2013 proof

Slide 17 of [lecnote6](#) writes $0 = \mathbb{E}[f^T y] = d^T \mathbb{E}[y] = f^T (X\beta)$; the middle term should be $f^T \mathbb{E}[y]$. The key point, easy to miss, is that unbiasedness must hold *for every* β , which is what forces $f \perp \text{col}(X)$.

Equivalently $\hat{\beta}_{\text{GLS}}$ minimises the Mahalanobis-type criterion $(y - X\beta)^\top \Sigma^{-1}(y - X\beta)$. If $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ this is *weighted least squares* with weights $1/\sigma_i^2$: noisy cases are down-weighted [► 2024 L8 @ 1:16:24][► 2013 L6 @ 57:55]. The main source of non-diagonal Σ in finance is time dependence: errors at nearby dates are more correlated than errors far apart [► 2024 L8 @ 1:14:15]; e.g. AR(1) errors have $\Sigma_{ij} \propto \rho^{|i-j|}$ (chapter 9).

INTUITION: Whitening

$\chi^2 = r^\top C^{-1}r$ with the covariance matrix C of correlated measurements is exactly the GLS criterion. Multiplying by $\Sigma^{-1/2}$ “whitens” the noise: in the new coordinates the noise is isotropic and the plain Euclidean projection of fig. 5.1 is optimal again.

Remark 5.9 (Not discussed in class). If the errors are correlated but OLS is used anyway, $\hat{\beta}$ is still unbiased, but it is no longer best and its true covariance is the “sandwich” $\sigma^2(X^\top X)^{-1}X^\top \Sigma X(X^\top X)^{-1}$, not $\sigma^2(X^\top X)^{-1}$: the standard errors printed by `lm()` are wrong. In practice Σ is unknown; it is estimated (feasible GLS) or the standard errors are corrected (heteroscedasticity- and autocorrelation-consistent errors).

5.5 The normal linear model: distribution theory

5.5.1 Multivariate normal vectors via MGFs

For an n -variate random vector Y the moment-generating function is $M_Y(t) = \mathbb{E}[e^{t^\top Y}]$, $t \in \mathbb{R}^n$ [► 2024 L8 @ 21:10]. In the normal linear model the Y_i are independent $N(\mu_i, \sigma^2)$ with $\mu = X\beta$ (independent but not identically distributed [► 2013 L6 @ 57:55]), so the expectation of the product factorises [► 2024 L8 @ 22:13][► 2013 L6 @ 1:02:09]:

$$M_Y(t) = \prod_{i=1}^n \mathbb{E}[e^{t_i Y_i}] = \prod_{i=1}^n e^{t_i \mu_i + \frac{1}{2} t_i^2 \sigma^2} = \exp\left(t^\top \mu + \frac{1}{2} t^\top \Sigma t\right), \quad \Sigma = \sigma^2 I_n.$$

Definition 5.10 (Multivariate normal). $Y \sim N_n(\mu, \Sigma)$, with $\mu \in \mathbb{R}^n$ and Σ symmetric positive semi-definite, if $M_Y(t) = \exp(t^\top \mu + \frac{1}{2} t^\top \Sigma t)$ for all t . Then $\mathbb{E}[Y] = \mu$ and $\text{Cov } Y = \Sigma$; if $\Sigma > 0$ the density is $(2\pi)^{-n/2}(\det \Sigma)^{-1/2} \exp(-\frac{1}{2}(y - \mu)^\top \Sigma^{-1}(y - \mu))$. Since an MGF finite around 0 determines the distribution, recognising this form is enough to identify a normal vector.

Proposition 5.11 (Linear maps of normal vectors). If $Y \sim N_n(\mu, \Sigma)$ and $A \in \mathbb{R}^{m \times n}$ then $Z = AY \sim N_m(A\mu, A\Sigma A^\top)$ [► 2024 L8 @ 31:59][► 2013 L6 @ 1:14:49]. In particular every sub-vector, and every component, is normal; and two jointly normal sub-vectors with zero cross-covariance are independent.

Applying this with $A = (X^\top X)^{-1}X^\top$ [► 2024 L8 @ 24:21]: $t = A^\top \tau = X(X^\top X)^{-1}\tau$ gives $t^\top \mu = \tau^\top \beta$ and $t^\top (\sigma^2 I) t = \tau^\top \sigma^2 (X^\top X)^{-1} \tau$, so $\hat{\beta} \sim N_p(\beta, \sigma^2 (X^\top X)^{-1})$ and $\hat{\beta}_j \sim N(\beta_j, \sigma^2 C_{jj})$ [► 2024 L8 @ 26:31][► 2013 L6 @ 1:04:19].

3 COURSE NOTE: Vocabulary

In the 2024 L8 video Kempthorne often says “multinomial” [► 2024 L8 @ 20:07]: he means *multinormal*, i.e. multivariate normal. Nothing in this chapter involves the multinomial distribution.

Proof. $M_Z(\tau) = \mathbb{E}[e^{\tau^\top AY}] = M_Y(A^\top \tau) = \exp(\tau^\top A\mu + \frac{1}{2} \tau^\top A\Sigma A^\top \tau)$. For a component take $A = e_j^\top$ [► 2013 L6 @ 1:07:25]. If $Z = (Z_1, Z_2)$ is normal with $\text{Cov}(Z_1, Z_2) = 0$, the quadratic form in the exponent splits and the joint MGF is the product $M_{Z_1}(\tau_1)M_{Z_2}(\tau_2)$, which characterises independence [► 2024 L8 @ 37:20]. ◻

5.5.2 The fundamental theorem

Definition 5.12 (χ^2 , t and F distributions). If $Z_1, \dots, Z_v$ are i.i.d. $N(0, 1)$, $\sum_k Z_k^2 \sim \chi_v^2$ [► 2013 L6 @ 1:16:57]. If $Z \sim N(0, 1)$ and $V \sim \chi_v^2$ are independent, $Z/\sqrt{V/v} \sim t_v$ (Student) [► 2024 L8 @ 38:21]. If $V_1 \sim \chi_{v_1}^2$ and $V_2 \sim \chi_{v_2}^2$ are independent, $(V_1/v_1)/(V_2/v_2) \sim F_{v_1, v_2}$ [► 2024 L8 @ 45:38]. Note $t_v^2 \sim F_{1, v}$.

Theorem 5.13 (Normal linear regression). Let $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $X \in \mathbb{R}^{n \times p}$ of rank p and $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 I_n)$. Then [► 2024 L8 @ 33:00][► 2013 L6 @ 1:15:54]:

- (i) $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y} \sim N_p(\boldsymbol{\beta}, \sigma^2 (X^T X)^{-1})$;
- (ii) $\hat{\boldsymbol{\varepsilon}} = (I_n - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2 (I_n - H))$, a degenerate normal: $X^T \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ identically [► 2024 L8 @ 34:06];
- (iii) $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ are independent;
- (iv) $RSS = \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}} \sim \sigma^2 \chi_{n-p}^2$; hence $\hat{\sigma}^2 = RSS/(n-p)$ is unbiased;
- (v) for each j ,

$$\hat{t}_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{C_{jj}}} \sim t_{n-p}, \quad C_{jj} = [(X^T X)^{-1}]_{jj}. \quad (5.4)$$

The unbiasedness of $\hat{\sigma}^2$ does not need normality [► 2024 L8 @ 36:15]: under Gauss–Markov, $\mathbb{E}[\hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}] = \text{tr Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2 \text{tr}(I_n - H) = \sigma^2(n-p)$ by [proposition 5.4\(iii\)](#). The $n-p$ appears because the residuals satisfy p exact linear constraints $X^T \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$: there are not n independent error terms [► 2024 L8 @ 35:14].

INTUITION: Why Student’s t has fat tails

In $\hat{t}_j$ the unknown σ is replaced by the random $\hat{\sigma}$. When $\hat{\sigma}$ happens to be small, $|\hat{t}_j|$ is large; the denominator fluctuates around 1 (in units of σ) and this mixing over scales fattens the tails [► 2024 L8 @ 40:28]. W. S. Gosset, a statistician at the Guinness brewery, noticed that standardised means of quality samples of size four were far more variable than $N(0, 1)$ predicted; the brewery did not let him publish under his name, hence “Student” [► 2024 L8 @ 41:29]. For $n-p \gtrsim 30$ the difference from the normal is small; with $n-p = 63$ (prostate example) the two-sided 5% critical value is 2.00 instead of 1.96.

INTUITION: Degrees of freedom, geometrically

RSS/σ^2 is the squared length of a standard Gaussian vector projected on an $(n-p)$ -dimensional subspace: fitting p parameters “uses up” p dimensions. The familiar

4 COURSE NOTE: Errata in the MGF slides

- Slide 17 of [lec06_1](#): the MGF of $\hat{\beta}_1$ is obtained with $\boldsymbol{\tau} = t(1, 0, \dots, 0)^T$, not $t(1, 0, \dots, 1)^T$.
- Slides 14–16 write the exponent as $\mathbf{t}^T \mathbf{u} + \dots$: $\mathbf{u}$ is the mean vector $\boldsymbol{\mu}$.
- Slide 18 (“Theorem*”) assumes rank $A = m \leq n$. The result holds for every A if degenerate normals (singular Σ , no density) are allowed, as in [definition 5.10](#); this is needed right away, because the stacked matrix $\begin{pmatrix} (X^T X)^{-1} X^T \\ I-H \end{pmatrix}$ used for independence has $n+p$ rows but rank n , and $\text{Cov}(\hat{\boldsymbol{\varepsilon}})$ is singular.

Proof. (i) was shown above.

(ii) $\text{Cov}((I-H)\mathbf{y}) = \sigma^2(I-H)(I-H)^T = \sigma^2(I-H)$ and $(I-H)X\boldsymbol{\beta} = \mathbf{0}$.

(iii) $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ are jointly normal (both linear in $\mathbf{y}$) with $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (X^T X)^{-1} X^T (\sigma^2 I)(I-H) = \sigma^2 (X^T X)^{-1} (X^T - X^T H) = \mathbf{0}$ because $X^T H = X^T$; apply [proposition 5.11](#) (this is the “joint MGF = product of MGFs” argument of the 2024 slides).

(iv) Following the 2013 notes [► 2013 L6 @ 1:16:57]: let $X = QR$ and complete the columns of Q to an orthonormal basis of $\mathbb{R}^n$ by $W \in \mathbb{R}^{n \times (n-p)}$ (continue Gram–Schmidt on $[X | I_n]$). Then $A = \begin{pmatrix} Q^T \\ W^T \end{pmatrix}$ is orthogonal and

$$\mathbf{z} = A\mathbf{y} = \begin{pmatrix} \mathbf{z}_Q \\ \mathbf{z}_W \end{pmatrix} \sim N_n\left(\begin{pmatrix} Q^T Q R \boldsymbol{\beta} \\ W^T Q R \boldsymbol{\beta} \end{pmatrix}, \sigma^2 A A^T\right) = N_n\left(\begin{pmatrix} R \boldsymbol{\beta} \\ \mathbf{0}_{n-p} \end{pmatrix}, \right),$$

So $\mathbf{z}_Q \sim N_p(R\boldsymbol{\beta}, \sigma^2 I_p)$ and $\mathbf{z}_W \sim N_{n-p}(\mathbf{0}, \sigma^2 I_{n-p})$ are independent. Now $\hat{\boldsymbol{\beta}} = R^{-1} Q^T \mathbf{y} = R^{-1} \mathbf{z}_Q$, and since $Q Q^T + W W^T = A^T A = I_n$, $\hat{\boldsymbol{\varepsilon}} = (I - Q Q^T) \mathbf{y} = W W^T \mathbf{y} = W \mathbf{z}_W$. Hence $RSS = \|W \mathbf{z}_W\|^2 = \|\mathbf{z}_W\|^2$ is σ^2 times a sum of $n-p$ independent squared standard normals. (This also re-proves (iii): $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ are functions of the independent blocks $\mathbf{z}_Q$ and $\mathbf{z}_W$.)

(v) $(\hat{\beta}_j - \beta_j)/(\sigma \sqrt{C_{jj}}) \sim N(0, 1)$ by (i), independent of $RSS/\sigma^2 \sim \chi_{n-p}^2$ by (iii)–(iv); dividing, σ cancels and the ratio is (5.4). $\square$

5 COURSE NOTE: Erratum: the missing square root

Slide 20 of [lec06_1](#) and slide 34 of [lecnote6](#) write the denominator of $\hat{t}_j$ as $\hat{\sigma} C_{jj}$. It must be $\hat{\sigma} \sqrt{C_{jj}}$, the estimated standard deviation of $\hat{\beta}_j$; Kempthorne corrects it at the start of L11 [► 2024 L11 @ 3:16]. Also, slide 38 of [lecnote6](#) writes $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$ for $(X^T X)^{-1} X^T \mathbf{y}$.

$1/(n-1)$ in the sample variance is the case $p = 1$ (projection on $\mathbf{1}$), and χ^2/ndf in a fit is $\hat{\sigma}^2/\sigma^2$.

A $(1 - \alpha)$ confidence interval for β_j follows immediately: $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} \hat{\sigma} \sqrt{C_{jj}}$, where $\hat{\sigma} \sqrt{C_{jj}}$ is the *standard error* $\text{se}(\hat{\beta}_j)$ printed by R.

5.6 Maximum likelihood and M-estimation

5.6.1 Maximum likelihood in the normal model

The *likelihood* is the joint density of the data seen as a function of the unknown parameters, $L(\boldsymbol{\beta}, \sigma^2) = p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2)$; maximum likelihood (ML) estimates make the observed data “most likely” [► 2024 L11 @ 10:53] [► 2013 L6 @ 1:20:05]. For the normal linear model

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}}, \quad \log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} Q(\boldsymbol{\beta})$$

[► 2024 L11 @ 13:02]. Maximise in two steps: $\boldsymbol{\beta}$ enters only through $-Q(\boldsymbol{\beta})$, so $\hat{\boldsymbol{\beta}}_{\text{ML}} = \hat{\boldsymbol{\beta}}_{\text{OLS}}$ for every σ^2 ; then $\partial \log L / \partial \sigma^2 = -\frac{n}{2\sigma^2} + \frac{Q(\hat{\boldsymbol{\beta}})}{2\sigma^4} = 0$ gives

$$\hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}, \quad \mathbb{E}[\hat{\sigma}^2]_{\text{ML}} = \frac{n-p}{n} \sigma^2 \quad (\text{biased})$$

[► 2024 L11 @ 14:08]. So least squares is also “doing something decent” from the likelihood point of view when the errors are Gaussian [► 2013 L6 @ 1:21:11]. ML estimators are asymptotically optimal (smallest variance as $n \rightarrow \infty$) for any correctly specified model [► 2024 L11 @ 11:58]: if the error density were known and non-Gaussian, ML would use it.

5.6.2 Generalised M-estimators

Definition 5.14 (Generalised M-estimator). For data $(\mathbf{y}, X)$ and the model $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, choose $\hat{\boldsymbol{\beta}}$ to minimise $Q(\boldsymbol{\beta}) = \sum_{i=1}^n h(y_i, \mathbf{x}_i, \boldsymbol{\beta}, \sigma^2)$ [► 2024 L11 @ 15:14]. The function h defines the estimator:

- (i) least squares: $h = (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$;
- (ii) least absolute deviations (“MAD”): $h = |y_i - \mathbf{x}_i^\top \boldsymbol{\beta}|$;
- (iii) maximum likelihood: $h = -\log p(y_i \mid \boldsymbol{\beta}, \mathbf{x}_i, \sigma^2)$ (so Laplace errors give LAD);
- (iv) robust M-estimator: $h = \chi(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})$ with χ even and increasing on $(0, \infty)$;
- (v) quantile estimator, for fixed $\tau \in (0, 1)$: $h = \rho_\tau(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})$ with the *check loss*

$$\rho_\tau(u) = \begin{cases} \tau |u|, & u \geq 0, \\ (1 - \tau) |u|, & u < 0. \end{cases}$$

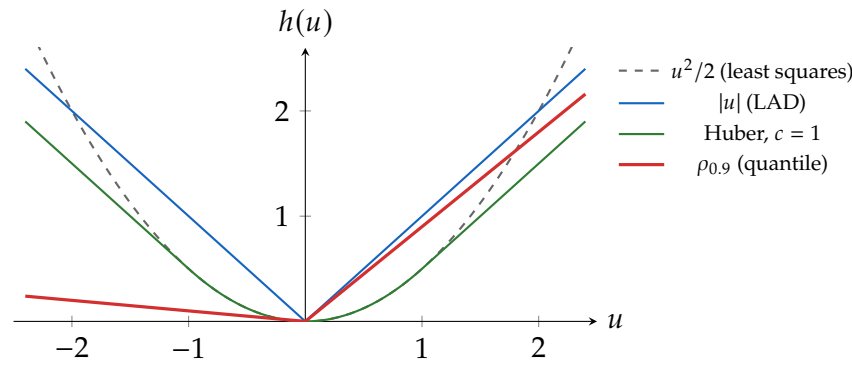


Figure 5.2: Loss functions of M-estimators. Quadratic loss lets large residuals dominate; LAD and Huber grow linearly; the check loss $\rho_{0.9}$ penalises positive residuals nine times more than negative ones, so the fit is pushed up to the 90th percentile.

Robust estimators were designed for the contaminated-normal situation: normal errors most of the time, with a small fraction of gross errors [► 2024 L11 @ 17:30]. When the error distribution is unknown they protect the fit from a few extreme cases, which in finance means crash days. A standard example (not detailed in class) is Huber's loss, quadratic for $|u| \leq c$ and linear beyond.

Proposition 5.15 (The check loss estimates quantiles). Let $y_1, \dots, y_n \in \mathbb{R}$ and $f(c) = \sum_i \rho_\tau(y_i - c)$. Then f is convex and piecewise linear, and c minimises f iff

$$\#\{i : y_i < c\} \leq \tau n \leq \#\{i : y_i \leq c\},$$

i.e. iff c is a sample τ -quantile. For $\tau = \frac{1}{2}$ this is the median, and $\rho_{1/2} = \frac{1}{2}|u|$, so $\tau = \frac{1}{2}$ gives LAD [► 2024 L11 @ 18:37].

Proof. Each term is convex in c . The right derivative of $\rho_\tau(y_i - c)$ is $-\tau$ if $y_i > c$ and $1 - \tau$ if $y_i \leq c$, so $f'_+(c) = -\tau \#\{y_i > c\} + (1 - \tau) \#\{y_i \leq c\} = \#\{y_i \leq c\} - \tau n$; similarly $f'_-(c) = \#\{y_i < c\} - \tau n$. A convex function is minimised exactly where $f'_-(c) \leq 0 \leq f'_+(c)$. $\square$

With c replaced by $x_i^\top \beta$, the minimiser estimates the conditional τ -quantile of y given x (quantile regression): for $\tau = 0.9$, the upper decile of outcomes [► 2024 L11 @ 19:41]. The objective is convex (“a bowl with straight edges”) but not differentiable, so there are no normal equations to solve; one moves downhill using one-sided derivatives, or solves a linear programme [► 2024 L11 @ 20:41].

IN FINANCE: Heavy tails in practice

In the GE CAPM regression of section 5.11.2 the histogram of residuals is compared with two normal fits [► 2024 L11 @ 1:00:53]: the ML fit (sample mean and standard deviation) and a robust one, whose scale is estimated from the interquartile range ($\hat{\sigma}_{\text{rob}} = \text{IQR}/1.349$, since the IQR of $N(0, \sigma^2)$ is 1.349σ) [► 2024 L11 @ 1:03:05]. The ML curve is too wide in the centre and too thin in the tails. A sharper check uses *fitted percentiles* $u_i = \Phi((\hat{\varepsilon}_i - \hat{\mu})/\hat{\sigma})$, which should be uniform on $(0, 1)$ if the model is right (probability integral transform) [► 2024 L11 @ 1:04:09]: with the ML scale they pile up near 0.5, with the robust scale they are uniform except in the extreme 1% on each side [► 2024 L11 @ 1:05:11]. Quantile regression is the natural tool for Value-at-Risk (chapter 19).

Exercise 5.4 (not from the course: quantile regression as maximum likelihood). Show that minimising $\sum_i \rho_\tau(y_i - x_i^\top \beta)$ is maximum likelihood for errors with the *asymmetric*

Laplace density $f(u) = \frac{\tau(1-\tau)}{s} \exp(-\rho_\tau(u)/s)$, and check that f is a density whose τ -quantile is 0. Which estimator of [definition 5.14](#) is ML for Laplace errors?

5.7 Hypothesis testing

5.7.1 Individual coefficients

To test $H_0 : \beta_j = b$ against $H_1 : \beta_j \neq b$ use $\hat{t} = (\hat{\beta}_j - b)/\text{se}(\hat{\beta}_j)$, which is t_{n-p} under H_0 by [\(5.4\)](#); reject when $|\hat{t}| > c$ for a critical value c [\[► 2024 L11 @ 5:26\]](#). The p -value is the probability, computed under H_0 , of a statistic at least as extreme as the one observed, $P(|T_{n-p}| \geq |\hat{t}|)$ [\[► 2024 L8 @ 49:56\]](#). The coefficient table of R's `summary(lm(...))` has columns Estimate ($\hat{\beta}_j$), Std. Error ($\hat{\sigma}\sqrt{C_{jj}}$), t value (test of $b = 0$: the “signal-to-noise ratio”) and $\text{Pr}(> |t|)$ [\[► 2024 L8 @ 48:48\]](#). Conventionally $p < 0.05$ is “significant” and $p < 0.01$ “highly significant”. The case $b = 0$ asks whether variable j can be dropped from the model [\[► 2024 L11 @ 5:26\]](#).

5.7.2 Nested models and the F test

Lemma 5.17 (Quadratic forms of normal vectors). Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$ and let P_1, P_2 be orthogonal projections of ranks r_1, r_2 with $P_1 P_2 = 0$ and $P_1 \boldsymbol{\mu} = P_2 \boldsymbol{\mu} = \mathbf{0}$. Then $\mathbf{Y}^\top P_k \mathbf{Y} / \sigma^2 \sim \chi_{r_k}^2$ ($k = 1, 2$), and the two quadratic forms are independent.

Theorem 5.18 (F test of a linear hypothesis). In the normal linear model let $A \in \mathbb{R}^{q \times p}$ have rank q and consider $H_0 : A\boldsymbol{\beta} = \mathbf{0}$. Let RSS be the residual sum of squares of the unconstrained fit and RSS_0 that of the least-squares fit under the constraint. Under H_0

$$F = \frac{(\text{RSS}_0 - \text{RSS})/q}{\text{RSS}/(n-p)} \sim F_{q, n-p}.$$

The most common case is $A = [0 \ I_q]$: $H_0 : \beta_{k+1} = \dots = \beta_p = 0$ with $k = p - q$, i.e. the sub-model using only the first k columns of X is adequate [\[► 2024 L8 @ 43:35\]](#)[\[► 2024 L11 @ 6:33\]](#).

Both numerator and denominator estimate σ^2 when H_0 is true [\[► 2024 L11 @ 6:33\]](#); under the alternative the numerator is inflated by $\|(H - H_0)\boldsymbol{\mu}\|^2/q$, so large F is evidence against

Example 5.16 (Is GE's beta equal to one? ([lec08](#), section 1)). Daily excess returns of GE on the market excess return, $n = 1676$ days (2017–2023): $\hat{\alpha} = -0.00054$ (se 0.00051, $p = 0.285$), $\hat{\beta} = 1.0830$ (se 0.0400), $R^2 = 0.304$, $\hat{\sigma} = 0.0207$. Testing $\beta = 0$ is pointless (a stock is obviously correlated with the market, $t = 27.1$); the interesting null is $\beta = 1$, “same systematic risk as the market” [\[► 2024 L11 @ 59:45\]](#):

$$\hat{t} = \frac{1.0830 - 1}{0.0400} = 2.075, \quad p = P(|T_{1674}| \geq 2.075) = 0.038,$$

so GE's beta is (just) significantly above one at the 5% level [\[► 2024 L11 @ 1:07:25\]](#).

6 COURSE NOTE: Errata in [lec08](#), section 1

On page 3 both hypotheses are labelled H_0 ; the alternative is $H_1 : \beta_j \neq 0$. On page 4, “ $(\hat{\beta}_2 - \beta_2) \sim N(\beta_2, sd(\hat{\beta}_2))$ ” should read $\hat{\beta}_2 - \beta_2 \sim N(0, sd(\hat{\beta}_2)^2)$, i.e. $\hat{\beta}_2 \sim N(\beta_2, sd^2)$.

Proof. Write $P_k = U_k U_k^\top$ with $U_k \in \mathbb{R}^{n \times r_k}$ having orthonormal columns (spectral theorem: the eigenvalues of a projection are 0 and 1). Then $U_k^\top \mathbf{Y} \sim N_{r_k}(U_k^\top \boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $U_k^\top \boldsymbol{\mu} = U_k^\top P_k \boldsymbol{\mu} = \mathbf{0}$, and $\mathbf{Y}^\top P_k \mathbf{Y} = \|U_k^\top \mathbf{Y}\|^2$. Moreover $U_1^\top U_2 = U_1^\top (P_1 P_2) U_2 = 0$, so $\text{Cov}(U_1^\top \mathbf{Y}, U_2^\top \mathbf{Y}) = \sigma^2 U_1^\top U_2 = 0$ and the two Gaussian vectors are independent ([proposition 5.11](#)). $\square$

Proof. The constrained fitted values range over $L_0 = \{X\boldsymbol{\beta} : A\boldsymbol{\beta} = \mathbf{0}\}$, a subspace of $\text{col}(X)$ of dimension $p - q$ (X is injective). The constrained least-squares fit is the projection $H_0 \mathbf{y}$ on L_0 , so $\text{RSS}_0 = \mathbf{y}^\top (I - H_0) \mathbf{y}$ and $\text{RSS}_0 - \text{RSS} = \mathbf{y}^\top (H - H_0) \mathbf{y}$. Since $L_0 \subset \text{col}(X)$, $HH_0 = H_0 H = H_0$; hence $H - H_0$ is the orthogonal projection on $\text{col}(X) \ominus L_0$ (rank q) and $(I - H)(H - H_0) = 0$. Under H_0 , $\boldsymbol{\mu} = X\boldsymbol{\beta} \in L_0$, so $(H - H_0)\boldsymbol{\mu} = (I - H)\boldsymbol{\mu} = \mathbf{0}$. By [lemma 5.17](#), $(\text{RSS}_0 - \text{RSS})/\sigma^2 \sim \chi_q^2$ and $\text{RSS}/\sigma^2 \sim \chi_{n-p}^2$ independently, and the ratio of the two over their degrees of freedom is $F_{q, n-p}$. $\square$

H_0 [► 2024 L8 @ 44:36]. An affine hypothesis $A\beta = b$ reduces to this one by subtracting $X\beta_b$ (any β_b with $A\beta_b = b$) from y .

Proposition 5.19 (Wald form; $t^2 = F$). For $H_0 : A\beta = b$,

$$RSS_0 - RSS = (A\hat{\beta} - b)^T [A(X^T X)^{-1} A^T]^{-1} (A\hat{\beta} - b).$$

In particular, for a single coefficient ($A = e_j^T$, $q = 1$) $RSS_0 - RSS = (\hat{\beta}_j - b)^2 / C_{jj}$ and the F statistic equals $\hat{t}^2$ [► 2024 L11 @ 7:39].

R's `car::linearHypothesis()` implements tests of linear hypotheses in this F form [► 2024 L11 @ 1:08:29]. For GE (example 5.16) it gives $F = 4.304 = 2.075^2$ ($p = 0.038$) for $\beta = 1$ and $F = 1.144 = 1.070^2$ ($p = 0.285$) for $\alpha = 0$: the same tests as the t statistics.

Overall F and ANOVA. With an intercept, the test that all $p - 1$ slopes vanish has $RSS_0 = TSS$, so

$$F = \frac{R^2 / (p - 1)}{(1 - R^2) / (n - p)}.$$

This is the last line of `summary.lm`: $F = 13.4$ on $(8, 63)$ degrees of freedom for the prostate data ($R^2 = 0.630$) and $F = 271.9$ on $(4, 607)$ for the ETF regression of section 5.11 ($R^2 = 0.642$). Adding blocks of variables one at a time and recording the drop in RSS gives the sequential *analysis-of-variance table*, derived through the QR decomposition in 2013 PS3 (Exercise 5.12).

IN FINANCE: Statistical versus practical significance; multiple testing

With thousands of daily observations, tiny effects become “significant”: in the ETF case study a principal component explaining 0.6% of the variance of the response has $p = 0.052$ [► 2024 L11 @ 45:48]. Conversely, when the same test is run on hundreds of stocks (is $\alpha = 0$ for each S&P 500 stock?), about 5% of them will be significant at the 5% level even if every null is true [► 2024 L11 @ 1:16:47]. A handful of significant alphas is therefore no evidence against the CAPM, and “the best of 500 back-tests” is not a strategy.

Exercise 5.5 (not from the course: $t^2 = F$ on real numbers).

- From proposition 5.19, show directly that for $H_0 : \beta_j = b$ the F statistic equals $\hat{t}^2$.
- Using only the GE output of example 5.16 (estimate, standard error, $n - p = 1674$, $RSS = 0.71819$), reproduce the RSS 0.72003 of the model constrained by $\beta = 1$ printed by `linearHypothesis`.

7 COURSE NOTE: Erratum: denominator degrees of freedom

Slide 20 of lec06_1 gives $df_2 = n - k$ for the sub-model F test. It is $df_2 = n - p$, the residual degrees of freedom of the full model, as in lec08 (section 2) and in the formula on the same slide.

Proof. Minimising $\|y - X\beta\|^2$ subject to $A\beta = b$ with a Lagrange multiplier gives $\hat{\beta}_0 = \hat{\beta} - (X^T X)^{-1} A^T M^{-1} (A\hat{\beta} - b)$, $M = A(X^T X)^{-1} A^T$. As $X(\hat{\beta} - \hat{\beta}_0) \in \text{col}(X) \perp \hat{\epsilon}$, Pythagoras gives $RSS_0 = RSS + (\hat{\beta} - \hat{\beta}_0)^T X^T X (\hat{\beta} - \hat{\beta}_0)$, and substituting, $(\hat{\beta} - \hat{\beta}_0)^T X^T X (\hat{\beta} - \hat{\beta}_0) = (A\hat{\beta} - b)^T M^{-1} (A\hat{\beta} - b)$. For $q = 1$, dividing by $\hat{\sigma}^2$ gives $(\hat{\beta}_j - b)^2 / (\hat{\sigma}^2 C_{jj}) = \hat{t}^2$. □

5.8 Testing for a change of regime

Regressions on time series are fitted over long samples, but the relationship may change: a firm changes business, a currency changes its peg. lec08 (section 3) shows how to test whether a regression is the same in two periods [► 2024 L11 @ 1:09:33].

Set-up. Order the n cases in time; period A is $i = 1, \dots, n_A$, period B the remaining $n_B = n - n_A$. Partition $\mathbf{y} = (\mathbf{y}_A, \mathbf{y}_B)$, $\mathbf{X} = \begin{pmatrix} \mathbf{X}_A \\ \mathbf{X}_B \end{pmatrix}$ and allow different parameters:

$$\mathbf{y}_A = \mathbf{X}_A \boldsymbol{\beta} + \boldsymbol{\varepsilon}_A, \quad \mathbf{y}_B = \mathbf{X}_B \boldsymbol{\gamma} + \boldsymbol{\varepsilon}_B, \quad \boldsymbol{\varepsilon}_A, \boldsymbol{\varepsilon}_B \text{ independent, } N(\mathbf{0}, \sigma^2 \mathbf{I}),$$

with $H_0 : \boldsymbol{\gamma} = \boldsymbol{\beta}$ (no change) against $H_1 : \boldsymbol{\gamma} \neq \boldsymbol{\beta}$ [► 2024 L11 @ 1:10:37].

Approach 1: separate regressions. In block form the unconstrained model is

$$\begin{pmatrix} \mathbf{y}_A \\ \mathbf{y}_B \end{pmatrix} = \begin{pmatrix} \mathbf{X}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_B \end{pmatrix} \begin{pmatrix} \boldsymbol{\beta} \\ \boldsymbol{\gamma} \end{pmatrix} + \boldsymbol{\varepsilon}.$$

The normal equations decouple, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}_A^T \mathbf{X}_A)^{-1} \mathbf{X}_A^T \mathbf{y}_A$, $\hat{\boldsymbol{\gamma}} = (\mathbf{X}_B^T \mathbf{X}_B)^{-1} \mathbf{X}_B^T \mathbf{y}_B$ (separate OLS fits) and $\text{RSS} = \text{RSS}_A + \text{RSS}_B$, with $n - 2p$ residual degrees of freedom. The constrained model is the pooled regression $\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with residual sum RSS_0 . By [theorem 5.18](#) ($q = p$ constraints $\boldsymbol{\gamma} - \boldsymbol{\beta} = \mathbf{0}$, $2p$ parameters)

$$F = \frac{(\text{RSS}_0 - \text{RSS}_A - \text{RSS}_B)/p}{(\text{RSS}_A + \text{RSS}_B)/(n - 2p)} \sim F_{p, n-2p} \quad \text{under } H_0. \quad (5.5)$$

In econometrics this is the *Chow test* (the name is not used in the course).

Approach 2: parametrise the change [► 2024 L11 @ 1:11:39]. Let $\boldsymbol{\delta} = \boldsymbol{\gamma} - \boldsymbol{\beta}$. Then

$$\begin{pmatrix} \mathbf{y}_A \\ \mathbf{y}_B \end{pmatrix} = \begin{pmatrix} \mathbf{X}_A & \mathbf{0} \\ \mathbf{X}_B & \mathbf{X}_B \end{pmatrix} \begin{pmatrix} \boldsymbol{\beta} \\ \boldsymbol{\delta} \end{pmatrix} + \boldsymbol{\varepsilon}, \quad H_0 : \boldsymbol{\delta} = \mathbf{0}.$$

The new design is the old one times $\mathbf{G} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ acting on the parameters, so it has the same column space: same fit, same RSS, same F ([section 5.2.5](#)). The advantage is that the individual t statistics of $\hat{\boldsymbol{\delta}}$ tell *which* coefficients changed. In R this is the model with interactions between the regressors and a period indicator, $\mathbf{y} \sim \mathbf{x} * \text{ind.periodB}$.

CASE STUDY: Did GE's CAPM change? (lec08, section 4)

Choosing the split. Fit the CAPM regression of [example 5.16](#) to 2017–2023 and plot the cumulative sum of the residuals standardised to equal variance, $\hat{\varepsilon}_i / \sqrt{1 - H_{ii}}$ ([section 5.9](#)). Under the model this should wander like a driftless random walk; instead it falls steadily to a minimum at 2018-12-27 and then rises [► 2024 L11 @ 1:12:44]. Period A: up to 2018-12-27 (500 days); period B: after.

Fit with interactions (`Ra.RF ~ Mkt.RF * ind.periodB`, $n = 1676$):

	Estimate	Std. error	t
intercept (α_A)	-0.00288	0.00092	-3.13
Mkt.RF (β_A)	0.774	0.110	7.02
ind.periodB (δ_α)	0.00334	0.00110	3.04
Mkt.RF:ind.periodB (δ_β)	0.354	0.118	2.99

So $\hat{\alpha}_B = 0.00046$ and $\hat{\beta}_B = 1.128$: the alpha turns from negative to (slightly) positive and the beta rises from 0.77 to 1.13 [► 2024 L11 @ 1:13:44]. The F test of $\delta_\alpha = \delta_\beta = 0$ (`anova(lmfit0, lmfit0a)` or `linearHypothesis`): RSS drops from 0.71819 to 0.71020, $F = \frac{0.0079821/2}{0.71020/1672} = 9.40$ on $(2, 1672)$ degrees of freedom, $p = 8.8 \times 10^{-5}$.

8 COURSE NOTE: Erratum in lec08, section 3.1

The F statistic of Approach 1 is printed with denominator $\text{RSS}/(n - p)$. The unconstrained model has $2p$ parameters, so it must be $\text{RSS}/(n - 2p)$ as in (5.5); the text right below the formula (and section 3.2) correctly states $df_2 = n - 2p$.

Lesson. CAPM parameters are not constant over long periods; practitioners estimate them on *rolling windows*, which raises the question of the window length ▶ 2024 L11 @ 1:13:44].

Open issues (lec08, section 5). The test assumes i.i.d. normal errors with the same variance in both periods. If the variances differ, fit the two periods separately by maximum likelihood and use a generalised likelihood-ratio test instead of F . If the errors are dependent (check with the Durbin–Watson, Ljung–Box, turning-point, difference-sign or McLeod–Li tests), time-series models are needed (chapter 9). If they are non-normal, large-sample theory still makes $\hat{\beta}$ approximately normal and F tests are replaced by χ^2 tests.

CASE STUDY: Exchange-rate regimes of the Chinese yuan (2013 Case Study 2)

Question. The yuan (CNY) was pegged to the US dollar until July 2005; China then announced a managed float against an undisclosed *basket* of currencies, with daily moves up to 0.3%. What is the implicit basket, does it change, and how fast does the yuan move against it?

Data and method. `fm_casestudy_fx_1.r` downloads from FRED the daily dollar exchange rates listed in `fred_fxrates.txt` (23 currencies plus trade-weighted dollar indices, 1999–2013). `fm_casestudy_2_rcode.r` re-expresses eight of them in Swiss francs (a stable numeraire outside the basket), so that the dollar itself becomes a regressor, and regresses daily log changes of CNY/SFR on those of USD, JPY, EUR, GBP, KRW, MYR and THB against the franc (the Frankel–Wei regression).

Results.

- 2001 – June 2005: coefficient on USD 1.000 (se 0.0005, $t = 1839$), all others insignificant, $R^2 = 0.9999$: a perfect dollar peg.
- July–December 2005: USD 0.19 (not significant), Korean won 0.18 ($t = 4.9$), Malaysian ringgit 0.75 ($t = 5.1$), $R^2 = 0.949$: the peg is gone.
- 2006–2013, year by year: USD weight 0.87–0.97 (t from 25 to 97), with small, changing, sometimes significant weights on MYR, KRW, EUR, GBP or THB; R^2 between 0.93 and 0.998.
- The intercept is a daily drift against the implied basket; annualised as $(1 + \hat{\alpha})^{252} - 1$ it is negative in almost every year (−2.9% in 2006, −6.1% in 2007, −7.3% in 2008, +0.2% in 2009, then −0.5% to −4.8%): fewer yuan per franc than the basket implies, i.e. a steady managed *appreciation* of the yuan.
- Diagnostics for 2012: no heteroscedasticity, but residual tails heavier than normal.

Lesson. A regression coefficient can identify an exchange-rate regime, and fitting it period by period reveals regime changes; the structural break of July 2005 is visible at a glance. `fm_casestudy_fx_1.r` also loads the R package `fxregime`, which implements this regression together with formal tests for changes of regime.

9 COURSE NOTE: Choosing the break from the data (not discussed in class)

Here the break date was chosen *after* looking at the cumulative residuals, i.e. at the most favourable point for rejection. The $F_{2,1672}$ distribution assumes the date is fixed in advance, so the reported p -value is too optimistic. With an unknown break date one uses the supremum of the F statistics over candidate dates (Andrews' sup- F test) or CUSUM tests, which formalise exactly the cumulative-residual plot used above. The conclusion for GE would very likely survive, given $p \approx 10^{-4}$.

Exercise 5.6 (not from the course: the Chow test).

- Show that the residual sum of squares of the block-diagonal design $\begin{pmatrix} X_A & 0 \\ 0 & X_B \end{pmatrix}$ is $RSS_A + RSS_B$.
- Show that $\begin{pmatrix} X_A & 0 \\ X_B & X_B \end{pmatrix}$ has the same column space, and relate the two parameter vectors.

- (c) For GE, compute $\hat{\alpha}_B$, $\hat{\beta}_B$ and the F statistic from the numbers in the case study of [section 5.8](#). Why can you not obtain the standard error of $\hat{\beta}_B$ from that table alone?

5.9 Regression diagnostics

Step 4 of the modelling loop. By [theorem 5.13\(ii\)](#) the residuals are correlated and have unequal variances, $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - H_{ii})$, even when the errors are i.i.d. [► 2024 L8 @ 1:01:33](#). Diagnostics correct for this.

Leverage. H_{ii} is the *leverage* of case i . Since $\hat{y}_i = \sum_j H_{ij}y_j$, $\partial \hat{y}_i / \partial y_i = H_{ii}$: the fraction of its own fitted value that a case dictates (2013 PS3). From $H = H^2 = H^T$, $H_{ii} = \sum_j H_{ij}^2 \geq H_{ii}^2$, so $0 \leq H_{ii} \leq 1$; with an intercept $H_{ii} \geq 1/n$, and $\sum_i H_{ii} = p$, so the average is p/n [► 2024 L8 @ 1:05:44](#). If $H_{ii} = 1$ the fitted line passes through the point: that case alone determines its fit [► 2024 L8 @ 1:04:44](#). With an intercept,

$$H_{ii} = \frac{1}{n} + (x_i - \bar{x})^T (\mathcal{X}^T \mathcal{X})^{-1} (x_i - \bar{x}),$$

where $\mathcal{X}$ is the matrix of centred covariates (2013 PS3, [Exercise 5.7](#)): leverage grows with the Mahalanobis distance of x_i from the centroid of the covariates (dividing by $n - 1$ turns $\mathcal{X}^T \mathcal{X}$ into the sample covariance matrix). High leverage is a property of the x 's only. R's `influence.measures()` flags cases with $H_{ii} > 3p/n$.

Residuals. The *standardised* residual (R: `rstandard`) is $r_i = \hat{\varepsilon}_i / (\hat{\sigma} \sqrt{1 - H_{ii}})$; the *studentised* (deleted) residual (R: `rstudent`, or `studres` in MASS) uses the estimate $\hat{\sigma}_{(i)}$ computed without case i , $t_i = \hat{\varepsilon}_i / (\hat{\sigma}_{(i)} \sqrt{1 - H_{ii}})$, and has exactly a t_{n-p-1} distribution under the normal model, which makes outliers easy to judge [► 2024 L8 @ 1:02:36](#).

Proposition 5.20 (Case deletion). Let $\hat{\beta}_{(i)}$, $\hat{y}_{(i)} = X\hat{\beta}_{(i)}$ and $\hat{\sigma}_{(i)}^2$ be computed with case i deleted. Then

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i \hat{\varepsilon}_i}{1 - H_{ii}}, \quad \hat{\sigma}_{(i)}^2 = \hat{\sigma}^2 + \frac{1}{n - p - 1} \left(\hat{\sigma}^2 - \frac{\hat{\varepsilon}_i^2}{1 - H_{ii}} \right),$$

and Cook's distance is

$$CD_i = \frac{\|\hat{y} - \hat{y}_{(i)}\|^2}{p \hat{\sigma}^2} = \frac{\hat{\varepsilon}_i^2}{p \hat{\sigma}^2} \frac{H_{ii}}{(1 - H_{ii})^2} = \frac{r_i^2}{p} \frac{H_{ii}}{1 - H_{ii}}.$$

The proofs, via the Sherman–Morrison–Woodbury formula (deleting a row is a rank-one update of $X^T X$), are 2013 PS3 Problem 2 ([Exercises 5.9](#) and [5.10](#)). Nothing needs to be refitted: all leave-one-out quantities come from one fit. The last form of CD_i says *influence* = *outlyingness* $\times$ *leverage*: a large residual at a low-leverage point, or a high-leverage point on the line, is harmless; both together move the fit. Since $\|\hat{y} - \hat{y}_{(i)}\|^2 = (\hat{\beta} - \hat{\beta}_{(i)})^T X^T X (\hat{\beta} - \hat{\beta}_{(i)})$, CD_i measures how far $\hat{\beta}$ moves in the metric of its own confidence ellipsoid: values comparable with the median of $F_{p, n-p}$ (about 1) are large [► 2024 L8 @ 1:06:44](#).

Graphical tools [► 2024 L8 @ 1:08:51](#). In R (stats and car):

<code>plot(fit)</code>	residuals vs fitted (flat band), normal Q–Q, scale–location ($\sqrt{ r_i }$ vs fitted), residuals vs leverage with Cook contours
<code>influence.measures()</code>	leave-one-out stats: DFBETAS, DFFITS, COVRATIO, Cook’s distance, hat values
<code>qqPlot()</code>	Q–Q plot of studentised residuals against t quantiles, with confidence band [► 2024 L8 @ 1:03:44]
<code>influencePlot()</code>	studentised residuals vs hat values, bubble size $\propto$ Cook’s distance
<code>residualPlots()</code>	residuals vs each regressor; curvature test = t test of added squared term; Tukey test = same with $\hat{y}^2$ [► 2024 L8 @ 1:09:53]
<code>leveragePlots()</code>	added-variable plots: effect of each regressor after adjusting for others
<code>durbinWatsonTest()</code>	autocorrelation of residuals (time-ordered data)

CASE STUDY: Prostate cancer data (lec06_1, slides 21–33)

Data. Not financial, deliberately: the prostate data of the faraway package (used in Hastie, Tibshirani and Friedman’s *Elements of Statistical Learning*), 97 patients, response `lpsa` (log PSA), eight covariates (log cancer volume `lcavol`, log prostate weight, age, ..., seminal vesicle invasion `svi`). A random 75% training sample ($n = 72$, `set.seed(1)`) is used. The experience transfers: only the variable names change from one regression problem to another [► 2024 L8 @ 46:42].

Fit ($p = 9$ with intercept): $R^2 = 0.630$, $R^2_{\text{adj}} = 0.583$, $\hat{\sigma} = 0.724$ on 63 degrees of freedom, $F = 13.4$. Significant: `lcavol` ($t = 5.21$) and `svi` ($t = 2.95$); `lweight` borderline ($p = 0.07$). The pairs plot already shows the strong `lpsa`–`lcavol` relation [► 2024 L8 @ 48:48]. Standardising the covariates leaves every t value unchanged (section 5.2.5) [► 2024 L8 @ 57:24].

Diagnostics. Studentised residuals: unimodal, symmetric, within the Q–Q band except case 39 (–3.06). Hat values: cases 32 (0.41) and 74 (0.33) exceed $3p/n = 0.375$ or come close, against an average $p/n = 0.125$. Cook’s distances of the flagged cases are only 0.07–0.13. Curvature tests: none significant (smallest $p = 0.10$ for `svi`; Tukey test $p = 0.69$) [► 2024 L8 @ 1:09:53].

Lesson. A satisfactory model: no curvature, no dominant case; the scale–location plot hints at mild non-constancy of the variance.

Exercise 5.7 (PS3 (2013), Problem 1(d)–(e): invariance and leverage). (d) Prove that H is unchanged if X is replaced by $X' = XG$ for any invertible $p \times p$ matrix G . (e) Let X be $n \times (p + 1)$ with a first column of ones and p covariates, and let G be the identity except for the first row, which is $(1, -\bar{x}_1, \dots, -\bar{x}_p)$, with $\bar{x}_j = \frac{1}{n} \sum_i x_{i,j}$. By (d) the models with X and $X' = XG$ have the same fitted values and residuals.

- If $\beta = (\beta_0, \dots, \beta_p)^\top$ is the parameter for X , show that $\beta' = G^{-1}\beta$ is the parameter for X' ; find G^{-1} and the elements of β' explicitly.
- Show that $X'^\top X' = \begin{pmatrix} n & \mathbf{0}_p^\top \\ \mathbf{0}_p & X^\top X \end{pmatrix}$, where X has entries $x_{i,j} - \bar{x}_j$.
- Prove $H_{ij} = \frac{1}{n} + (x_i - \bar{x})^\top [X^\top X]^{-1} (x_j - \bar{x})$, where $x_i = (x_{i,1}, \dots, x_{i,p})^\top$. Hence the leverage H_{ii} increases with the (squared) Mahalanobis distance between x_i and $\bar{x}$.

5.10 Shrinkage and principal-components regression

When regressors are many or strongly correlated (the four index ETFs below have pairwise correlations 0.77–0.95), $X^T X$ is ill-conditioned and $\text{Cov}(\hat{\beta}) = \sigma^2(X^T X)^{-1}$ is huge in some directions. Three remedies, all expressed through the SVD, were presented in L11.

Standardisation first [► 2024 L11 @ 23:47]. Centre $\mathbf{y}$ and the columns of X and scale the columns to unit variance; the intercept is then estimated by $\bar{y}$. Otherwise a penalty on $\|\beta\|$ would depend on the units: measuring a covariate in cents instead of dollars would make its coefficient 100 times smaller and almost free [► 2024 L11 @ 24:49]. Below X is the standardised $n \times p$ matrix (no intercept column) with thin SVD $X = UDV^T$, $U^T U = I_p$, V orthogonal, $D = \text{diag}(d_1 \geq \dots \geq d_p)$ (theorem 2.12), and $c_j = \mathbf{u}_j^T \mathbf{y}$.

5.10.1 Ridge regression

Definition 5.21 (Ridge). For a complexity parameter $\lambda \geq 0$ [► 2024 L11 @ 22:46],

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} (\|\mathbf{y} - X\beta\|^2 + \lambda \|\beta\|^2) = (X^T X + \lambda I_p)^{-1} X^T \mathbf{y}.$$

The gradient condition is $-2X^T(\mathbf{y} - X\beta) + 2\lambda\beta = \mathbf{0}$; $X^T X + \lambda I$ is invertible for every $\lambda > 0$ even when X is rank-deficient. The estimate is linear in $\mathbf{y}$ and equals OLS for $\lambda = 0$. With the SVD [► 2024 L11 @ 28:04]:

$$\hat{\beta}^{\text{ridge}} = V(D^2 + \lambda I)^{-1} D U^T \mathbf{y}, \quad \hat{\mathbf{y}}^{\text{ridge}} = U D(D^2 + \lambda I)^{-1} D U^T \mathbf{y} = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} c_j \mathbf{u}_j,$$

whereas $\hat{\mathbf{y}}^{\text{LS}} = U U^T \mathbf{y} = \sum_j c_j \mathbf{u}_j$. Ridge shrinks the component of the fit along $\mathbf{u}_j$ by the factor $d_j^2/(d_j^2 + \lambda)$: little along the directions in which the regressors vary a lot (large d_j , the leading principal components), much along the directions of little variation [► 2024 L11 @ 29:04]. It is biased, $\mathbb{E}\hat{\beta}^{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T X \beta$, but has smaller variance; λ is chosen by cross-validation (`cv.glmnet` in the case study).

Proposition 5.22 (Ridge as a Bayes estimate). In the normal linear model with prior $\beta \sim N_p(\mathbf{0}, \tau^2 I_p)$, the posterior mode (which, the posterior being Gaussian, is also its mean) is $\hat{\beta}^{\text{ridge}}$ with $\lambda = \sigma^2/\tau^2$ [► 2024 L11 @ 25:52].

The isotropic prior says that a priori no direction of $\mathbb{R}^p$ is preferred [► 2024 L11 @ 26:58]; a strong prior (small τ , large λ) pulls all coefficients towards zero.

Exercise 5.8 (not from the course: ridge as augmented least squares).

- Show that $\hat{\beta}^{\text{ridge}}$ is the OLS estimate for the augmented data $\tilde{\mathbf{y}} = (\mathbf{y}, \mathbf{0}_p)$, $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}$.
- Using the SVD, show that $\|\hat{\beta}^{\text{ridge}}\|$ decreases in λ and compute $\text{Cov}(\hat{\beta}^{\text{ridge}})$ under Gauss–Markov.
- Show that ridge has smaller variance than OLS in every direction: $\text{Cov}(\hat{\beta}) - \text{Cov}(\hat{\beta}^{\text{ridge}}) \geq \mathbf{0}$.

10 COURSE NOTE: Erratum and notation in the ridge slides

Slide 41 of **lec06_1** rescales the columns by “ $S^{-1/2}X$ ” with $S = \frac{1}{n} \text{diag}(X^T X)$: since S is $p \times p$ this must be $X S^{-1/2}$ (multiplication on the right scales columns). On the same slide $H^* = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T$ is the *centring* matrix, not the hat matrix.

Proof. The posterior density is proportional to likelihood $\times$ prior,

$$\exp(-\|\mathbf{y} - X\beta\|^2/(2\sigma^2)) \exp(-\|\beta\|^2/(2\tau^2));$$

maximising it means minimising $\|\mathbf{y} - X\beta\|^2 + (\sigma^2/\tau^2)\|\beta\|^2$. $\square$

5.10.2 Principal-components regression

The principal components of the (centred) regressors are $\mathbf{z}_j = X\mathbf{v}_j$, where $\mathbf{v}_j$ are the eigenvectors of the sample covariance $S = \frac{1}{n}X^\top X = \frac{1}{n}VD^2V^\top$ [► 2024 L11 @ 30:05]. They are orthogonal, $\mathbf{z}_j = d_j\mathbf{u}_j$, with sample variance d_j^2/n , and $\mathbf{z}_1$ is the unit-norm combination of the columns of X with the largest variance (chapter 7). PC regression fixes $m \leq p$ and regresses $\mathbf{y}$ on $Z_m = [\mathbf{z}_1 \cdots \mathbf{z}_m] = U_m D_m$ [► 2024 L11 @ 31:05]:

$$\hat{\mathbf{y}} = (Z_m^\top Z_m)^{-1} Z_m^\top \mathbf{y}, \quad \hat{y}_j = \frac{\mathbf{z}_j^\top \mathbf{y}}{d_j^2}, \quad \hat{\boldsymbol{\beta}}^{\text{PC}} = V_m \hat{\mathbf{y}}, \quad \hat{\mathbf{y}}^{\text{PC}} = U_m U_m^\top \mathbf{y} = \sum_{j=1}^m c_j \mathbf{u}_j.$$

Orthogonality makes each $\hat{y}_j$ a separate simple regression, and

$$\text{Cov}(\hat{\boldsymbol{\beta}}^{\text{PC}}) = \sigma^2 V_m D_m^{-2} V_m^\top \leq \sigma^2 V D^{-2} V^\top = \text{Cov}(\hat{\boldsymbol{\beta}}) :$$

variance is reduced by dropping the noisiest directions [► 2024 L11 @ 32:13].

Comparison [► 2024 L11 @ 33:19]. The three fits use the same building blocks $c_j \mathbf{u}_j$:

$$\hat{\mathbf{y}}^{\text{LS}} = \sum_{j=1}^p c_j \mathbf{u}_j, \quad \hat{\mathbf{y}}^{\text{ridge}} = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} c_j \mathbf{u}_j, \quad \hat{\mathbf{y}}^{\text{PC}} = \sum_{j=1}^m c_j \mathbf{u}_j.$$

Least squares keeps all, ridge shrinks smoothly, PC regression keeps the first m and discards the rest.

INTUITION: Spectral filters

This is the classic regularisation of an ill-posed inverse problem: ridge is Tikhonov regularisation (a smooth filter $d^2/(d^2 + \lambda)$ on the singular values), PC regression is the truncated SVD (a sharp cut-off). Both trade a little bias for a large reduction of the noise amplified by small singular values.

A caveat, visible in the ETF case study [► 2024 L11 @ 46:50]: the principal components are ranked by how much the *regressors* vary, not by how much they explain $\mathbf{y}$. A low-variance component can be an important predictor, so keeping “the first m ” may throw away the useful direction.

5.10.3 The lasso

Definition 5.23 (Lasso).

$$\hat{\boldsymbol{\beta}}^{\text{lasso}} = \arg \min_{\boldsymbol{\beta}} \left(\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p |\beta_j| \right),$$

equivalently: minimise $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ subject to $\|\boldsymbol{\beta}\|_1 \leq t$ [► 2024 L11 @ 33:19]. The constraint binds only for $t < t_0 = \|\hat{\boldsymbol{\beta}}^{\text{LS}}\|_1$; the *regularisation path* is $\hat{\boldsymbol{\beta}}^{\text{lasso}}$ as a function of $\lambda \geq 0$ (or of $t \in [0, t_0]$).

The L_1 ball is a diamond (a cross-polytope in $\mathbb{R}^p$) with corners on the axes. The elliptical contours of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$, centred at $\hat{\boldsymbol{\beta}}^{\text{LS}}$, typically first touch it at a corner, where some coefficients are exactly zero [► 2024 L11 @ 35:28](fig. 5.3). The lasso therefore does *variable*

selection: as λ grows, variables leave the model one after the other [► 2024 L11 @ 36:31]. There is no closed form; the path is computed by coordinate descent (`glmnet`).

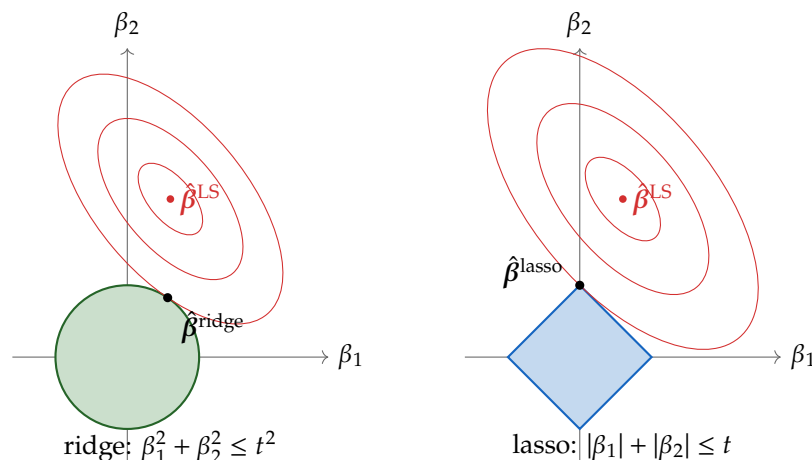


Figure 5.3: Constrained forms of ridge and lasso (same least-squares contours, $t = 1$). The solution is where the smallest contour touches the constraint region: a generic point of the circle for ridge, the corner $\beta_1 = 0$ of the diamond for the lasso [► 2024 L11 @ 34:22]. (Contours computed for a quadratic form with axis ratio 2:1.)

Remark 5.24 (Orthonormal design; not discussed in class). If $X^T X = I_p$, then $\|y - X\beta\|^2 = \|y - X\hat{\beta}\|^2 + \|\beta - \hat{\beta}\|^2$ with $\hat{\beta} = X^T y$, and both problems decouple:

$$\hat{\beta}_j^{\text{ridge}} = \frac{\hat{\beta}_j}{1 + \lambda}, \quad \hat{\beta}_j^{\text{lasso}} = \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda/2)_+.$$

Ridge rescales every coefficient; the lasso subtracts a constant and sets the small ones to exactly zero (“soft thresholding”).

Regularised models reappear in the 2013 lecture on regularised pricing and risk models (chapter 20), in machine learning (chapter 16), and PCA in chapter 7.

5.11 Case studies

5.11.1 Sector ETFs regressed on index ETFs

CASE STUDY: ETF case study (lec06_4 = ETF_Casestudy.pdf, sector XLP)

Data. Adjusted daily prices from Yahoo (2000-01-03 to 2024-09-19) of the 13 SPDR ETFs listed in `spyders_symbols_labels.csv`: four index ETFs (SPY: S&P 500, MDY: mid-caps, QQQ: Nasdaq-100, DIA: Dow Jones) and nine sector ETFs (XLB materials, XLV health care, XLP consumer staples, XLY consumer discretionary, XLE energy, XLF financials, XLI industrials, XLK technology, XLU utilities). Converted to weekly log returns (1289 weeks; saved in `casestudy_1_0_etfs.RData`); the analysis uses 2013–2024 ($n = 612$ weeks). Response: XLP (annualised mean 10.0%, volatility 13.7%, against 13.7% and 16.1% for SPY).

OLS [► 2024 L11 @ 42:42]: $\hat{y} = 0.0002 + 1.007 \text{SPY} - 0.368 \text{MDY} - 0.314 \text{QQQ} + 0.400 \text{DIA}$, all slopes with $|t| > 4.3$; $R^2 = 0.642$, $\hat{\sigma} = 0.0114$ (weekly), $F = 271.9$. The mixed

signs are a symptom of the collinear regressors (index correlations 0.77–0.95): XLP loads positively on the market but less on its growth and mid-cap parts. Diagnostics: influential weeks 2016-11-11 (studentised residual -4.8), 2020-03-06 ($+4.4$), 2022-05-20 (-5.8); the COVID week 2020-03-20 has hat value 0.135 against $p/n = 0.008$. The curvature tests are all significant at 5% ($p \approx 0.02$ – 0.04): some non-linearity in the market response.

z-scores. Regressing on standardised index returns reproduces every t value and R^2 ; only the scale of the coefficients and the intercept change.

PCA of the regressors [► 2024 L11 @ 43:42]: the four standardised index returns have PC variances in proportions 90.9%, 6.5%, 2.3%, 0.3% (cumulative 97.4% with two, 99.7% with three). PC1 has equal loadings (0.48–0.52, the “market”), PC2 contrasts QQQ ($+0.79$) with MDY and DIA.

PC regressions [► 2024 L11 @ 44:48]: simple regressions of XLP on each PC have $R^2 = 0.545, 0.006$ ($p = 0.052$), 0.079 and 0.012 . Their sum is 0.642 , the R^2 of the full regression, because the PCs are orthogonal (Exercise 5.2). The third component, with only 2.3% of the regressor variance, explains 13 times more of XLP than the second: dropping it (PC123 would keep it, but a “first two” rule would not) changes the answer. Using PCs 1–3 gives coefficients very different from OLS; using the PCs with $|t| > \sqrt{2}$ (all four here) reproduces OLS exactly [► 2024 L11 @ 51:03].

Ridge and lasso (glmnet, λ by 10-fold cross-validation over 10^{-5} – 10^3) [► 2024 L11 @ 47:55]: coefficient traces shrink to zero, not monotonically for ridge; along the lasso path the third variable (QQQ) leaves first, then MDY, then DIA [► 2024 L11 @ 50:01]. With the CV-optimal λ both fits are almost identical to OLS ($R^2 = 0.6417$ ridge, 0.6417 lasso vs 0.6418): with $n = 612$ and 4 regressors there is little to regularise [► 2024 L11 @ 51:03].

XLB version (ETF_Casestudy_XLB.pdf, materials): $R^2 = 0.805$, loadings SPY 0.85, MDY 0.50, QQQ -0.38 , DIA not significant ($t = 0.43$); PC1 alone gives $R^2 = 0.73$. The week 2020-03-20 is now very influential (Cook’s distance 0.99), and all curvature tests are highly significant ($p < 0.001$).

Remark on the code. Cumulative returns are plotted as $\prod_t (1 + r_t)$ although r_t are log returns; the correct growth factor is $\exp(\sum_t r_t)$. The difference (about $-r_t^2/2$ per week) only affects the plots.

IN FINANCE: Why regress a sector on the indices?

The fitted coefficients are a *hedge*: holding XLP and shorting 1.007 SPY, -0.368 MDY, ... leaves (in sample) only the residual, i.e. the sector-specific part of the return, with most of the market risk removed [► 2024 L11 @ 40:39]. The same regression, run with a hedge fund’s returns as the response and liquid instruments as regressors, is the basis of *hedge-fund replication*: if most of a fund’s return is explained by tradable factors, those can be bought cheaply [► 2024 L11 @ 41:39]. (Regressing an index on its sectors would instead give $R^2 \approx 1$, as noted by a student [► 2024 L11 @ 39:39].)

5.11.2 CAPM regressions

The CAPM (Sharpe 1964, Lintner 1965; theory in chapter 10) states that in an efficient market in which all investors hold mean–variance efficient portfolios and can borrow and lend at the risk-free rate r_f ,

$$\mathbb{E}[R_j] = r_f + \beta_j(\mathbb{E}[R_M] - r_f), \quad \beta_j = \frac{\text{Cov}(R_j, R_M)}{\text{Var}(R_M)},$$

where R_M is the return of the market portfolio [► 2024 L11 @ 54:12]. With excess returns $R_{j,t}^* = R_{j,t} - r_{f,t}$ and $R_{M,t}^*$ observed over time, fit

$$R_{j,t}^* = \alpha_j + \beta_j R_{M,t}^* + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \text{ white noise,}$$

the alpha–beta regression of [definition 1.2](#) [► 2024 L11 @ 55:18]. Taking expectations, the CAPM holds iff $\alpha_j = 0$ [► 2024 L11 @ 56:27]. Two natural tests: $\alpha_j = 0$ (is the asset efficiently priced?) and $\beta_j = 1$ (does it carry more or less systematic risk than the market?) [► 2024 L11 @ 1:06:14].

CASE STUDY: CAPM and macro factors for GE and XOM, 2000–2013 (2013 Case Study 1)

Data. `fm_casestudy_1_0.r` downloads daily prices of BAC, GE, JDSU, XOM and the S&P 500 (Yahoo) and from FRED the constant-maturity Treasury yields (3 months, 1, 5, 10 years), Moody’s Aaa and Baa yields and the WTI oil price, for 2000-01-03 to 2013-05-31 (3373 days). `fm_casestudy_1_rcode.r` computes daily log returns and the daily risk-free return $\log(1 + r_{3M} \Delta t / 360)$ with Δt the number of calendar days since the previous close (money-market convention).

GE: $\hat{\alpha} = -0.00013$ ($t = -0.56$), $\hat{\beta} = 1.184$ ($t = 66.6$), $R^2 = 0.568$: consistent with CAPM, with more market risk than average. Influence: 129 of 3372 days exceed the leverage threshold; the plot of hat values over time shows they are concentrated in the 2008 crisis, when market moves were extreme.

Adding oil (daily log change of WTI): for GE the oil coefficient is -0.034 ($t = -3.6$): GE falls when oil rises. For XOM, simple CAPM $\hat{\beta} = 0.830$, $R^2 = 0.451$ (lower than GE: an oil company tracks the market less); with oil, $\hat{\beta} = 0.782$ and oil coefficient $+0.132$ ($t = 16.1$), R^2 up to 0.491. The high-leverage days are those with extreme market or oil moves, far from the centre of the (R_M^*, oil) scatter (the Mahalanobis interpretation of 2013 PS3).

Lesson. The market factor explains half of a large stock’s daily variance; a second factor can be highly significant while adding little R^2 .

CASE STUDY: CAPM for GE and for the S&P 500 stocks (2024 L11)

GE 2017–2023 (lec08; excess return `Ra.RF` regressed on the market excess return `Mkt.RF` of the Fama–French data): see [example 5.16](#) and the regime test of [section 5.8](#). The residuals are clearly non-normal [► 2024 L11 @ 1:00:53]; see the box on heavy tails in [section 5.6](#).

All S&P 500 stocks (note shown in class, not published) [► 2024 L11 @ 1:14:45]: the same regression for the 380–400 constituents with a long enough history, grouped by sector. R^2 ranges over $(0, 1)$, typically above 0.2. Alphas vary by sector (positive for construction and computer technology, negative for conglomerates) [► 2024 L11 @ 1:15:47], but the p -values of $\alpha_j = 0$ are above 0.05 for all but a handful of stocks, roughly the 5% expected by chance: the CAPM is not rejected stock by stock [► 2024 L11 @ 1:16:47]. Higher betas tend to go with higher alphas [► 2024 L11 @ 1:17:53]. Betas are strongly sector-dependent: consumer staples lowest (the ten lowest betas, 0.37 down to 0.18, are mostly staples), technology and consumer discretionary highest [► 2024 L11 @ 1:18:59]. The largest significant alpha, ENPH (energy), is 0.003 per day, i.e. $(1.003)^{252} - 1 \approx 113\%$ per year [► 2024 L11 @ 1:20:01].

Next step. Multi-factor models with the Fama–French factors, already downloaded with the data [► 2024 L11 @ 1:22:11]([chapter 7](#)).

IN FINANCE: Using alpha and beta

Beta is the hedge ratio against the index and the measure of systematic risk that the CAPM says is rewarded; alpha is the part of the return not explained by market exposure (definition 1.2). In practice betas are estimated on rolling windows (section 5.8) and often shrunk towards 1, e.g. the widely quoted “adjusted beta” $\frac{2}{3}\hat{\beta} + \frac{1}{3}$ (not discussed in class): the same bias–variance trade as ridge regression, with the prior centred at the market beta.

5.12 Differences between the 2013 and 2024 lectures

KEY FORMULAS

OLS:	$\hat{\beta} = (X^T X)^{-1} X^T y = R^{-1} Q^T y,$ $H = X(X^T X)^{-1} X^T = Q Q^T, \quad \hat{\varepsilon} = (I - H)y$
Geometry:	$H^2 = H = H^T, \quad \text{tr } H = p,$ $X^T \hat{\varepsilon} = \mathbf{0}, \quad \text{TSS} = \text{ESS} + \text{RSS}, \quad R^2 = \text{Corr}(y, \hat{y})^2$
Gauss–Markov:	$\mathbb{E} \hat{\beta} = \beta, \quad \text{Cov } \hat{\beta} = \sigma^2 (X^T X)^{-1};$ $c^T \hat{\beta}$ is BLUE
GLS:	$\hat{\beta}_{\text{GLS}} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y,$ $\text{Cov} = \sigma^2 (X^T \Sigma^{-1} X)^{-1}$
Normal model:	$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$ $\perp\!\!\!\perp \text{RSS} \sim \sigma^2 \chi_{n-p}^2, \quad \hat{\sigma}^2 = \frac{\text{RSS}}{n-p}$
t test:	$\hat{t}_j = (\hat{\beta}_j - b) / (\hat{\sigma} \sqrt{C_{jj}}) \sim t_{n-p}, \quad C_{jj} = [(X^T X)^{-1}]_{jj}$
F test:	$F = \frac{(\text{RSS}_0 - \text{RSS})/q}{\text{RSS}/(n-p)} \sim F_{q, n-p}$ $\text{RSS}_0 - \text{RSS} = (A\hat{\beta} - b)^T [A(X^T X)^{-1} A^T]^{-1} (A\hat{\beta} - b)$
Chow test:	$F = \frac{(\text{RSS}_0 - \text{RSS}_A - \text{RSS}_B)/p}{(\text{RSS}_A + \text{RSS}_B)/(n-2p)}$ $\sim F_{p, n-2p}$
ML:	$\hat{\beta}_{\text{ML}} = \hat{\beta}, \quad \hat{\sigma}_{\text{ML}}^2 = \text{RSS}/n$
Diagnostics:	$\text{Var } \hat{\varepsilon}_i = \sigma^2 (1 - H_{ii}), \quad \text{CD}_i = \frac{r_i^2}{p} \frac{H_{ii}}{1 - H_{ii}},$ $\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i \hat{\varepsilon}_i}{1 - H_{ii}}$
Shrinkage:	$\hat{y}^{\text{ridge}} = \sum_j \frac{d_j^2}{d_j^2 + \lambda} c_j u_j,$ $\hat{y}^{\text{PC}} = \sum_{j \leq m} c_j u_j,$ $\lambda = \sigma^2 / \tau^2 \text{ (Bayes)}$
CAPM:	$R_{j,t}^* = \alpha_j + \beta_j R_{M,t}^* + \varepsilon_{j,t}, \quad H_0 : \alpha_j = 0$

11 COURSE NOTE: 2013 L6 vs 2024 L6, L8, L11

The core (OLS, hat matrix, Gauss–Markov, GLS, normal distribution theory, ML, M-estimators) is the same material in both years, with nearly identical slides. **2013** spends one lecture on it and adds the QR decomposition and the full proofs of the Gauss–Markov and distribution theorems [► 2013 L6 @ 1:08:26]; its case studies are the CAPM with oil (Case Study 1) and the yuan regimes (Case Study 2). **2024** spreads the topic over parts of three lectures and adds real-data diagnostics (prostate data), hypothesis testing and regime-change tests (**lec08**), ridge, PC and lasso regression, the ETF case study and the S&P 500 CAPM study. The 2013 lecture ran out of time before M-estimation [► 2013 L6 @ 1:21:11]; it is covered in the notes and in 2024 L11.

Exercises

Standalone exercises follow the section they practise; the connected series (2013 PS3, Problems 2 and 3, which build on one another) are kept together here.

From 2013 Problem Set 3

Throughout, $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $X \in \mathbb{R}^{n \times p}$ of full rank $p \leq n$, $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{y}$, $H = X(X^\top X)^{-1}X^\top$, $\hat{\boldsymbol{\varepsilon}} = (I_n - H)\mathbf{y}$.

Exercise 5.9 (PS3 (2013), Problem 2(a): Sherman–Morrison–Woodbury). Let A be a $p \times p$ symmetric matrix of rank p and a, b two $q \times p$ matrices of rank $q < p$. Prove that, whenever the inverses exist,

$$(A + a^\top b)^{-1} = A^{-1} - A^{-1}a^\top(I_q + bA^{-1}a^\top)^{-1}bA^{-1}.$$

Exercise 5.10 (PS3 (2013), Problem 2(b)–(d): case-deletion influence). (b) Using Sherman–Morrison–Woodbury, show that the least-squares estimate with case i deleted is $\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - (X^\top X)^{-1}x_i\hat{\varepsilon}_i/(1 - H_{ii})$, where $x_i^\top$ is the i -th row of X and $\hat{\varepsilon}_i = y_i - x_i^\top \hat{\boldsymbol{\beta}}$. (c) Cook's distance is $CD_i = \|\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(i)}\|^2/(p\hat{\sigma}^2)$ with $\hat{\mathbf{y}}_{(i)} = X\hat{\boldsymbol{\beta}}_{(i)}$. Show that $CD_i = \frac{\hat{\varepsilon}_i^2}{p\hat{\sigma}^2} \cdot \frac{H_{ii}}{(1 - H_{ii})^2}$. (d) Let $\hat{\sigma}_{(i)}^2$ be the unbiased estimate of σ^2 with case i deleted. Show that $\hat{\sigma}_{(i)}^2 = \hat{\sigma}^2 + \frac{1}{n-p-1}(\hat{\sigma}^2 - \frac{\hat{\varepsilon}_i^2}{1-H_{ii}})$.

Exercise 5.11 (PS3 (2013), Problem 3(a)–(b): constrained least squares via QR). Consider $H_0 : \beta_{k+1} = \dots = \beta_p = 0$. Let $X_I = [X_{[1]} \dots X_{[k]}]$, $\hat{\boldsymbol{\beta}}_I = (X_I^\top X_I)^{-1}X_I^\top \mathbf{y}$ and $\hat{\boldsymbol{\beta}}_0 = (\hat{\boldsymbol{\beta}}_I, \mathbf{0}_{p-k})$.

- (a) Show that $\hat{\boldsymbol{\beta}}_0$ minimises $(\mathbf{y} - X\boldsymbol{\beta})^\top(\mathbf{y} - X\boldsymbol{\beta})$ subject to $\beta_j = 0$ for $j > k$.
- (b) With $X = QR$, show that $X_I = Q_I R_I$ where Q_I is the first k columns of Q and R_I the upper-left $k \times k$ block of R ; verify $\hat{\boldsymbol{\beta}}_I = R_I^{-1}Q_I^\top \mathbf{y}$ and $\hat{\mathbf{y}}_I = H_I \mathbf{y}$ with $H_I = Q_I Q_I^\top$.

Exercise 5.12 (PS3 (2013), Problem 3(c)–(d): sequential ANOVA). Let $A = \begin{pmatrix} Q^\top \\ W^\top \end{pmatrix}$ be orthogonal as in the proof of [theorem 5.13](#) and $\mathbf{z} = A\mathbf{y} = (z_1, \dots, z_n)^\top$. (c) Prove that $\mathbf{y}^\top \mathbf{y} = \sum_{i=1}^n z_i^2$, $\hat{\mathbf{y}}^\top \hat{\mathbf{y}} = \sum_{i=1}^p z_i^2$, $\hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \sum_{i=p+1}^n z_i^2$, and for the constrained model $\hat{\mathbf{y}}_I^\top \hat{\mathbf{y}}_I = \sum_{i=1}^k z_i^2$, $\hat{\boldsymbol{\varepsilon}}_I^\top \hat{\boldsymbol{\varepsilon}}_I = \sum_{i=k+1}^n z_i^2$. (d) Using $\mathbf{z}_Q \sim N_p(R\boldsymbol{\beta}, \sigma^2 I_p)$ independent of $\mathbf{z}_W \sim N_{n-p}(\mathbf{0}, \sigma^2 I_{n-p})$, deduce that $SS_{\text{ERROR}} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} \sim \sigma^2 \chi_{n-p}^2$; that under H_0 $SS_{\text{REG}(k+1, \dots, p|1, \dots, k)} = \hat{\mathbf{y}}^\top \hat{\mathbf{y}} - \hat{\mathbf{y}}_I^\top \hat{\mathbf{y}}_I = \sum_{i=k+1}^p z_i^2 \sim \sigma^2 \chi_{p-k}^2$, independent of SS_{ERROR} ; and that $\hat{F} = \frac{SS_{\text{REG}}/(p-k)}{SS_{\text{ERROR}}/(n-p)} \sim F_{p-k, n-p}$. Arrange the results in the ANOVA table with rows “regression on $1, \dots, k$ ” ($\hat{\mathbf{y}}_I^\top \hat{\mathbf{y}}_I$, k d.f.), “regression on $k+1, \dots, p$ adjusting for $1, \dots, k$ ” ($p-k$ d.f., mean square MS_0 , $\hat{F} = MS_0/MS_{\text{Error}}$), “error” ($n-p$ d.f.) and “total” ($\mathbf{y}^\top \mathbf{y}$, n d.f.).

Solutions

Solutions of the exercises and labs of this chapter.

SOLUTION: Lab: high-yield spread, functions of time against persistence

Course results. Higher orders follow the two big waves (peaks around 2012 and early 2016) better and better, but the polynomials swing wildly at the ends of the sample, and nobody believes the spread is an 8th-order polynomial in calendar time [► 2024 L6 @ 1:12:37]. In class a

third model was shown (not in the files): the *autoregression* of today’s spread on yesterday’s, whose fit is almost indistinguishable from the data, with $R^2 = 0.995$ [► 2024 L6 @ 1:13:44]. (In lec06_3 the legend colours do not match the curves: the 1-cycle fit is drawn in blue, not green.)
With the FRED data (the course’s exact window is not given, so the numbers differ a little):

model	poly 1	poly 4	poly 8	Fourier 2	Fourier 4	AR(1)
R^2	0.18	0.51	0.81	0.59	0.82	0.997

Rescaling time avoids powers of numbers like 2017, whose columns are almost collinear (an ill-conditioned $X^T X$). The AR(1) slope is $b = 0.9997$: the spread is close to a random walk.
Lesson. A good in-sample fit is not a model. Deterministic functions of time only describe the past; the autoregression exploits *persistence* (tomorrow’s spread is close to today’s), which is what is actually predictable. But with $b \approx 1$ its R^2 says little: “tomorrow = today” already has $R^2 \approx 0.997$; the honest comparison is on the daily *changes* ([chapter 9](#)).

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Chapter 6

Linear Rates: Products and Models

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L7	A. Gunstensen (Mizuho)	Linear rates products and the size of their markets; LIBOR and its replacement by SOFR; discounting = funding; deposits, SOFR futures and interest-rate swaps; swap valuation and par swap rates; yield-curve construction (knot points, CDF and spline interpolation, calibration, FOMC curves, turns); delta ladders and gamma; P&L attribution; clearing and the CCP basis; electronic trading and price making; Q&A on the rates outlook, research roles and municipal bonds.

Files. Slides *Linear Rates Models mit18_642_f24_lec07.pdf* (45 slides). Slide 39 (fast model pricing) and slides 40–44 (bonds) were skipped in class for lack of time [► 2024 L7 @ 1:14:29]; they are covered in [sections 6.10](#) and [6.11](#). The data file *fedbondyielddata_20230911.csv* and the notebook *feddataanalysis.ipynb* are listed on OCW together with Lecture 9 (PCA of Treasury yield changes) and are treated in [chapter 7](#). **Problem sets.** None; all exercises (each after the section it practises) are *not from the course*. **2013.** No counterpart: the 2013 course has no lecture on linear rates products. The dynamics of the whole curve are the subject of the 2013 HJM lecture ([chapter 24](#)).

Motivation

Bonds, deposits, futures and swaps are the largest and most liquid markets in the world, and their payoffs are fixed or linear in interest rates: pricing them requires no model of rate *dynamics*, only today's *discount curve*. [Chapter 1](#) took that curve as given (a flat rate, or yields read off a screen, [fig. 1.3](#)). This chapter asks the practitioner's questions: which rate should one discount with, which instruments reveal the curve, how does one fill the gaps between a few dozen quotes, how does one hedge and explain the daily P&L of a book of swaps? The lecturer announced a deliberately "non-mathy" talk: a house is designed by thinking about rooms and how they connect, not by spending most of the time on hammers [► 2024 L7 @ 1:01]. The mathematics is elementary (linear algebra, interpolation, Taylor expansions); what matters is the set of conventions and design choices, and the consequences of each choice for hedging. The curve built here is the input of every rates-option model; its PCA is in [chapter 7](#) and its stochastic evolution in [chapter 24](#).

6.1 Linear products and their markets

Definition 6.1 (Linear product). A rates product is *linear* if its payoff is either fixed or a linear function of observed interest rates [► 2024 L7 @ 2:06]: (vanilla) bonds, certificates of deposit, interest-rate futures, forward rate agreements, interest-rate swaps. There is no optionality (no caps, swaptions, callable or range-accrual features).

The PV of a linear product is a linear function of discount factors and forward rates, so it is fixed by today's curve alone: no volatility enters, except through small corrections (futures margining, averaging, payment delays) discussed below. Why should a quant care [► 2024 L7 @ 3:07]?

- **Size and liquidity** (slide 4). Bonds outstanding about \$128T (ICMA, 2020), interest-rate swaps \$573T notional (BIS, 2023), SOFR futures open interest above \$10T. Daily volumes: US Treasuries about \$900B, swaps \$800B, SOFR futures about 4 million contracts. In a market this size, a tiny margin is a large business [► 2024 L7 @ 3:07].
- **Hedging and inputs.** Linear products hedge the complex ones, and their curves are inputs of the complex models: one builds up from the simple stuff [► 2024 L7 @ 4:08].
- **Post-2008 complexity.** Before the crisis vanilla modelling was considered trivial. Funding, clearing and margining (sections 6.3 and 6.9) made even “\$1 paid in ten years” hard to value, with no unique answer [► 2024 L7 @ 4:08]. Regulation (including model-risk management) ended much of the business in opaque, highly levered securitisations that had been sold to buyers who could not assess them [► 2024 L7 @ 5:08].

6.2 Interest rates and reference rates

6.2.1 Simple interest and accrual

An interest rate is the rate at which one earns (or pays) interest on a principal [► 2024 L7 @ 7:10]. Money-market instruments use *simple* interest:

$$I = N r \Delta t, \quad \text{PV}(X \text{ paid after } \Delta t) = \frac{X}{1 + r_d \Delta t}, \quad (6.1)$$

where Δt is the *accrual fraction* of the period and r_d a discounting rate. Example from class: \$1000 in one year discounted at 5% is worth $1000/1.05 = \$952.38$ today [► 2024 L7 @ 8:11]. The interest rate r of an investment is easy to observe (the rate a bank advertises on a CD); the discounting rate r_d is not, and it depends on who is doing the discounting (section 6.3).

6.2.2 LIBOR

The *London Interbank Offered Rate* was created in the late 1960s as a reference for syndicated bank lending: corporate loans are mostly floating-rate, resetting every one or three months, and a common index was needed [► 2024 L7 @ 9:13]. Every day a panel of about 20 banks reported the rate at which they could borrow unsecured for various terms; the rate was a trimmed average of the submissions [► 2024 L7 @ 10:16]. The first swaps (a cross-currency swap involving IBM in 1981, then single-currency interest-rate swaps) were built on it, and by LIBOR's cessation some \$350–400T of swaps referenced it [► 2024 L7 @ 11:16].

Its weaknesses surfaced after 2008 (slide 7):

1 COURSE NOTE: Day counts (not discussed in class)

Δt is computed from dates with a *day-count convention*: ACT/360 (actual days/360; SOFR, money markets, the floating legs of USD swaps), ACT/365F, 30/360 (bond-style), ACT/ACT (Treasuries). Dates themselves are moved off holidays by a *business-day convention* (e.g. “modified following”). The lecturer's advice to job candidates: after a year on a swaps desk you will know *date arithmetic*, and at least 90% of all problems with swaps are date problems [► 2024 L7 @ 34:51]. The number of days between two dates has no unique answer; it depends on the convention.

- it was a *quoted*, not a transacted rate: submitters estimated where they *would* borrow;
- it was *unsecured*, and after Lehman almost nobody lent unsecured term money: some tenors saw a handful of real transactions in six months while a rate had to be published every day [▶ 2024 L7 @ 13:26];
- hence it could be manipulated. A book can have millions of dollars of P&L per basis point (bp = 0.01%) of a fixing, and a group of traders coordinated with their banks' submitters to nudge the fixings; this led to prison sentences and multi-billion-dollar fines [▶ 2024 L7 @ 12:21].

Regulators (led from London) required a replacement “anchored in actual transactions and hard to manipulate” [▶ 2024 L7 @ 13:26]. USD LIBOR ended in June 2023; sterling earlier, Canada and Mexico around 2024 [▶ 2024 L7 @ 18:40]. The migration of mortgages, bonds, loans and hundreds of trillions of swaps was enormous and tedious, comparable to the euro conversion and Y2K [▶ 2024 L7 @ 15:31].

IN FINANCE: Why banks liked an unsecured index

A bank's floating-rate loans pay LIBOR plus a fixed spread. When the banking system is under stress, bank credit spreads widen and LIBOR rises, so the bank's *income* rises exactly when its own funding becomes expensive: a natural hedge (a “right-way” feature) [▶ 2024 L7 @ 17:40], [▶ 2024 L7 @ 19:44]. A secured rate contains no bank credit risk and lacks this property. The flip side: LIBOR-based hedges of instruments that depend on government rates suffer when the LIBOR–Treasury spread blows out. In 2011–12 this cost the lecturer's employer of the time a write-off of several billion dollars on structured notes (\$8.4B quoted in class), most of which came back later [▶ 2024 L7 @ 14:31].

6.2.3 SOFR

A *repo* (repurchase agreement) is a collateralised loan: I deliver \$100M of Treasuries and receive, say, \$98M cash, agreeing to buy the bonds back the next day; the interest paid is the *repo rate* (reverse repo for the other side) [▶ 2024 L7 @ 16:35]. This is a real market with real volume (about \$800B a day according to slide 8).

Definition 6.2 (SOFR). The *Secured Overnight Financing Rate* is the main USD reference rate: a volume-weighted median of the rates on overnight repo transactions collateralised by US Treasuries, published every business day [▶ 2024 L7 @ 15:31].

SOFR has three inconvenient features [▶ 2024 L7 @ 16:35]: it is a *daily* rate, while nobody wants to pay interest daily; a term rate built from it is known only at the *end* of the period (set in arrears), while borrowers like to know their payment in advance; and it is *secured*, while banks prefer to lend unsecured (previous box) [▶ 2024 L7 @ 17:40].

From daily fixings to term rates [▶ 2024 L7 @ 20:49]. Over an accrual period of d days, let r_i be the SOFR fixing of business day i , applying for d_i calendar days ($d_i = 3$ for a Friday fixing, which covers the weekend), $d = \sum_{i=1}^n d_i$. The two standard term rates (slide 9) are

$$R^{\text{comp}} = \frac{360}{d} \left[\prod_{i=1}^n \left(1 + \frac{r_i d_i}{360} \right) - 1 \right], \quad R^{\text{avg}} = \frac{1}{d} \sum_{i=1}^n r_i d_i. \quad (6.2)$$

R^{comp} is exactly what one earns by rolling an overnight deposit (interest on interest), and $R^{\text{comp}} \geq R^{\text{avg}}$ when rates are non-negative (Exercise 6.1). Most USD products pay

2 COURSE NOTE: Erratum (spoken)

In class SOFR is expanded as “Securitized Overnight Funding Rate” and described as an average. The name is *Secured Overnight Financing Rate*, computed by the Federal Reserve Bank of New York (published since April 2018, selected in 2017 by the Fed-convened Alternative Reference Rates Committee) as a volume-weighted *median* of transaction rates. Similarly “OAS rate” in the transcript at 8:11 should read OIS (overnight index swap) rate, as on slide 6.

compounded SOFR: it is simpler to calibrate, and it needs no correction for rate volatility, which an average does [► 2024 L7 @ 21:49]. The reason is [proposition 6.9](#) below: a compounded leg is replicated exactly by rolling a deposit.

Other rates [► 2024 L7 @ 21:49]–[► 2024 L7 @ 26:16](slide 10).

- **Term SOFR:** 1-, 3-, 6- and 12-month rates published by the CME by fitting a model of daily rates to SOFR futures prices and averaging the fitted daily rates over the term [► 2024 L7 @ 24:02]. It is set in advance, like LIBOR, and popular in corporate loans, whose borrowers do not want to handle daily fixings. To prevent it from becoming a new LIBOR, the Fed restricts dealers to trading it only to hedge their loan books. The market is therefore one-way, and term-SOFR swaps trade 4–5 bp away from vanilla SOFR swaps, 20–30 times the bid–offer [► 2024 L7 @ 25:09].
- **Fed Funds:** the older unsecured overnight interbank rate, still widely used; plus Prime, DISCO (a Fannie Mae discount-note rate, the lecturer’s favourite name), MMD (a municipal benchmark) and many others [► 2024 L7 @ 26:16].

Example 6.3 (Compounding vs averaging (not from class)). A 91-day period with a fixing every day ($d_i = 1$): 5.30% for 40 days, then 5.05% after a 25 bp FOMC cut. Then $R^{\text{avg}} = 5.1599\%$ and $R^{\text{comp}} = 5.1933\%$: a 3.3 bp difference. It is “a few basis points running”, irrelevant for a retail borrower but twenty times the bid–offer of a swap market maker.

Exercise 6.1 (not from the course).

- Show that $R^{\text{comp}} \geq R^{\text{avg}}$ in (6.2) whenever all $r_i \geq 0$.
- For a constant rate r fixed every day for n days ($d_i = 1$), show that $R^{\text{comp}} - r = \frac{(n-1)r^2}{720} + O(r^3)$. Evaluate for $r = 5.2\%$, $n = 91$ and compare with the example of [section 6.2](#).

6.3 Discounting equals funding

Which rate should discount a future cash flow [► 2024 L7 @ 26:16]? Assume (slides 11–12) we own a fixed payment F at time T , worth $P(t)$ today, and that we bought it with borrowed money: in this world nobody pays out of pocket, positions are funded. Consider one day, from t_0 to t_1 , with no market move [► 2024 L7 @ 27:21]:

$$\text{interest paid on the funding: } I(t_1) = P(t_0) \frac{r_b}{360},$$

$$\text{accretion of the claim: } P(t_1) = P(t_0) \left(1 + \frac{r_d}{360}\right), \quad \text{i.e. } \Delta P = P(t_0) \frac{r_d}{360},$$

where r_b is our borrowing rate and r_d the rate used to discount F .

Proposition 6.4 (Discounting rate = funding rate). *Let an agent borrow and lend at the rate r_b over $[t, T]$ (n days, daily compounding, rates constant as on the slides). The only price of F paid at T that gives the agent no riskless trade is $P(t) = F(1 + r_b/360)^{-n}$: future cash flows must be discounted at the funding rate, $r_d = r_b$.*

Remark 6.5. If the agent borrows at r_b but can only lend at $r_l < r_b$, the argument only gives the band $F(1 + r_b/360)^{-n} \leq P \leq F(1 + r_l/360)^{-n}$. Different agents have different funding rates (the lecturer’s example: borrowers at 4%, 6% and 8% depending on their credit) and therefore different values for the same contract: there is no single price [► 2024 L7 @ 9:13]. The lecturer’s one-sentence summary of the lecture: *funding and discounting are the same thing* [► 2024 L7 @ 28:24].

Proof. The funded position has a daily P&L $P(r_d - r_b)/360$, known in advance. If $r_d > r_b$, buying the claim with borrowed money earns a riskless carry; if $r_d < r_b$, selling it and lending the proceeds does. Summing over the holding period, the terminal P&L of “borrow P , buy the claim” is $F - P(1 + r_b/360)^n$, which vanishes for every holding period only if P is discounted at r_b [► 2024 L7 @ 28:24]. $\square$

INTUITION: A stationary state

Think of the funded position as a closed system in which “nothing happens”: the claim drifts up at r_d , the loan at r_b . Zero P&L in the absence of market moves (a stationary state) requires equal drifts. The discount curve is not a law of nature but the rate at which the holder’s money actually grows or costs; compare [section 1.7.2](#), where the same argument with a single universal rate gave $1/(1+r)$.

IN FINANCE: Funding desks and OIS discounting

Banks do not let each trader manage the funding of a thousand counterparty relationships: a central funding desk (treasury) lends to all desks at a generic internal rate and manages the funding bases and small collateral exposures itself [► 2024 L7 @ 9:13]. For collateralised trades the relevant funding rate is the rate paid on the collateral, an overnight rate: hence the market standard of discounting with an OIS (now SOFR) curve [► 2024 L7 @ 8:11]. The costs of funding uncollateralised trades are priced as funding valuation adjustments, part of the XVA family of [chapter 8](#).

6.4 The instruments that reveal the curve

To build a curve we start from what trades actively and is easy to observe (slide 13): whatever the model, it must reprice these instruments, because they can be traded [► 2024 L7 @ 28:24]. The workhorses are cash deposits (CDs), interest-rate futures and swaps [► 2024 L7 @ 29:26]. We write each price as a function of the discount function $Z(t)$ (value today of \$1 paid at t ; Λ_t in [chapter 1](#)), then invert.

6.4.1 Deposits and certificates of deposit

Short-term money can be invested at a fixed simple rate $r(t)$ for maturities up to about a year (CDs, interbank deposits; also repo and Treasury bills, not discussed in class). Investing 1 today returns $1 + r(t)\Delta$ at t , so (slide 14)

$$1 = [1 + r(t)\Delta]Z(t) \quad \implies \quad Z(t) = \frac{1}{1 + r(t)\Delta} \quad [\text{► 2024 L7 @ 29:26}]. \quad (6.3)$$

Deposits are well observed but awkward for a swaps desk: they are *funded* (a \$100M CD requires \$100M of cash and balance sheet), whereas derivatives need little cash up front and allow more leverage [► 2024 L7 @ 30:32].

6.4.2 Forward rates and FRAs

A floating payment that sets in the future can be replicated with fixed flows [► 2024 L7 @ 39:04]. Let the rate apply to $[t_1, t_2]$ with accrual fraction Δ , and suppose zero-coupon bonds maturing at t_1 and t_2 can be traded.

Proposition 6.6 (Implied forward rate). *The fair (zero-PV) rate fixed today for lending N from t_1 to t_2 is*

$$f(t_1, t_2) = \frac{1}{\Delta} \left(\frac{Z(t_1)}{Z(t_2)} - 1 \right). \quad (6.4)$$

This is [proposition 1.13](#) of [chapter 1](#) with $F_{t_1 t_2} = f\Delta$.

Proof. Lending N at t_1 and receiving $N(1 + f\Delta)$ at t_2 has present value $-NZ(t_1) + Nf\Delta Z(t_2) + NZ(t_2)$. Setting it to zero gives $1 + f\Delta = Z(t_1)/Z(t_2)$. If the market rate were higher, lend forward and short the replicating zeros (buy the t_1 zero, sell the t_2 zero) for a riskless profit, and vice versa. $\square$

FRAs (not discussed in class). A forward rate agreement on notional N and strike K pays at the end of the period $N(R - K)\Delta$, where R is the rate fixed for $[t_1, t_2]$ (in the LIBOR world usually settled at t_1 as $N(R - K)\Delta/(1 + R\Delta)$, the same value). By [proposition 6.6](#) its value today is $N(f - K)\Delta Z(t_2)$ and the fair strike is $K = f$. A swap is a strip of FRAs sharing one strike.

6.4.3 SOFR futures

Short-rate futures are among the most liquid contracts in every currency [▶ 2024 L7 @ 30:32](#). The US contracts trade at the CME (slide 15) [▶ 2024 L7 @ 31:35](#):

- **3-month SOFR futures** settle on the *compounded* SOFR over a period running from one *IMM date* to the next (third Wednesday of March, June, September, December; the dates come from the old International Monetary Market in Chicago);
- **1-month SOFR futures** settle on the *average* SOFR over a calendar month.

Prices are quoted as $F = 100 - R$, with R the settlement rate in percent: a 5% rate gives 95, a 6% rate 94 [▶ 2024 L7 @ 22:58](#). The convention makes the contract behave like a bond: when rates fall (the market “rallies”), the price rises [▶ 2024 L7 @ 31:35](#). Futures prices estimate forward rates up to a small *convexity adjustment* (slide 15):

$$f(t_1, t_2) = 100 - F - \text{Cvx}, \quad \text{Cvx} > 0. \quad (6.5)$$

Origin of the adjustment [▶ 2024 L7 @ 32:39](#). Futures are margined daily: a contract entered at no cost (apart from an initial margin) pays the holder the daily price change. A long position gains when the price rises, i.e. when rates fall, and the gain can then be reinvested only at the lower rate; when rates rise, the losses must be funded at the higher rate. This adverse correlation between the margin flows and the rate at which they are carried makes a long futures position slightly worse than a long FRA, so the futures price is lower and the futures rate $100 - F$ higher than the forward rate. The adjustment can be estimated from a model, or simply fitted together with swaps and futures prices.

Remark 6.7 (Size of the adjustment (not discussed in class)). Daily settlement makes the futures rate the expectation of R under the risk-neutral (bank-account) measure, $R^{\text{fut}} = \mathbb{E}^Q[R]$, while the forward rate is its expectation under the t_2 -forward measure, whose density with respect to Q is $D/\mathbb{E}^Q[D]$ with $D = \exp(-\int_0^{t_2} r_s ds)$ (change of numeraire, [chapter 15](#)). Hence

$$R^{\text{fut}} - f = \mathbb{E}^Q[R] - \frac{\mathbb{E}^Q[DR]}{\mathbb{E}^Q[D]} = -\frac{\text{Cov}^Q(D, R)}{\mathbb{E}^Q[D]} > 0,$$

because high rates come with low discount factors. In the Ho–Lee model with normal short-rate volatility σ , $\text{Cvx} \approx \frac{1}{2}\sigma^2 t_1 t_2$ (continuous compounding; e.g. Hull’s *Options, Futures, and Other Derivatives*). With $\sigma = 1\%$ per year: 0.6 bp for a contract one year out, 13 bp at five years, 51 bp at ten years. It is negligible at the front and essential at the back, which is why long-dated curves use swaps rather than futures.

Exercise 6.2 (not from the course).

- Explain, without formulas, why the convexity adjustment makes the futures rate *higher* than the forward rate, and what would change if margin balances earned a fixed rate instead of the overnight rate.

3 COURSE NOTE: Erratum: slide 20

The slide states the problem for a period $[t_1, t_2]$ but writes the equations with t_0, t_1 : “ $PV = -NZ(t_0) + Nf\Delta Z(t_1) + NZ(t_1) = 0$ ” and $f = \frac{1}{\Delta}(\frac{Z(t_0)}{Z(t_1)} - 1)$; it also says “borrow N at t_0 ”. The consistent statement is (6.4): the start of the accrual period plays the role of t_0 on the slide, the end that of t_1 .

- (b) With $\sigma = 0.8\%$ per year, evaluate $\frac{1}{2}\sigma^2 t_1 t_2$ for 3-month contracts expiring in 2, 5 and 10 years. From which maturity does it exceed a bid–offer of 0.25 bp?

6.4.4 Interest-rate swaps

Definition 6.8 (Vanilla interest-rate swap). An agreement between two counterparties to exchange periodic cash flows on a notional N until maturity (slide 16) [► 2024 L7 @ 33:47]: the *fixed leg* pays a rate K set on the trade date, the *floating leg* pays a floating index observed periodically (today compounded SOFR). The *payer* pays fixed and receives floating; the *receiver* does the opposite. Notionals are not exchanged.

Swaps are the most liquid rates derivative. They convert fixed-rate into floating-rate exposure and vice versa, are off balance sheet, cost little up front, and can be customised to any schedule a counterparty will accept [► 2024 L7 @ 33:47].

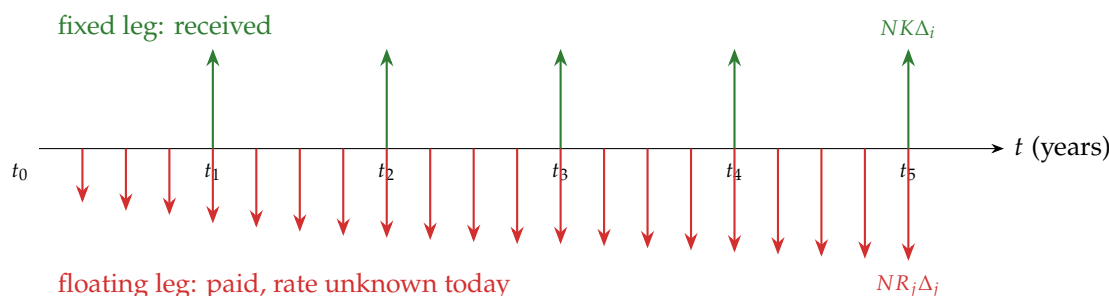


Figure 6.1: Cash flows of a 5-year receiver swap with annual fixed payments $NK\Delta_i$ and a floating leg drawn with quarterly payments $NR_j\Delta_j$ (USD SOFR swaps actually pay the compounded rate annually on both legs). The floating amounts are unknown today; with an upward-sloping forward curve they are expected to grow.

Rolling out the cash flows. The logic is simple but error-prone (slide 17) [► 2024 L7 @ 35:54]: today → trade date; + spot rule (typically two business days) → effective date; + tenor → unadjusted maturity; step back by the payment frequency → unadjusted roll dates; + business-day adjustment → accrual dates; + reset rule → rate-fixing dates; + payment rule → payment dates. Every step depends on holiday calendars, which change (Juneteenth, a state funeral, new national holidays), so models must treat them as data. The standard calendar source lists 294 distinct dollar calendars (every exchange and business centre has its own) [► 2024 L7 @ 37:00]. With the wrong calendar you misprice, and by the winner's curse you get the trade only when the mispricing is against you.

6.5 Swap valuation

Let the fixed leg pay at $t_1 < \dots < t_n$ with accrual fractions Δ_i and the floating leg at $s_1 < \dots < s_m$ with fractions Δ_j , both starting at $t_0 = s_0$.

Fixed leg ▶ 2024 L7 @ 37:00]. Discount each known payment (slide 18):

$$PV_{\text{fix}} = \sum_{i=1}^n K \Delta_i N Z(t_i) = N K A, \quad A = \sum_{i=1}^n \Delta_i Z(t_i), \quad (6.6)$$

where A is the *annuity* (or level, or PV01 of the fixed leg per unit rate).

Floating leg ▶ 2024 L7 @ 38:04]. Replace unknown rates by forwards (slide 19), $PV_{\text{float}} = \sum_j f_j \Delta_j N Z(s_j)$. This needs two curves: an *index* (projection) curve for the forwards f_j and a *discount* curve for the $Z(s_j)$. For vanilla SOFR swaps both are the SOFR curve, so there is only one curve. For compounded SOFR the replication is exact:

Proposition 6.9 (Floating leg). Assume the discount curve is the SOFR curve (*proposition 6.4* with SOFR as the funding rate) and that each floating period pays $NR_j \Delta_j$ at s_j , where R_j is SOFR compounded over $[s_{j-1}, s_j]$ as in (6.2). Then the value of period j is $N [Z(s_{j-1}) - Z(s_j)] = N f_j \Delta_j Z(s_j)$ with f_j the forward rate (6.4), and

$$PV_{\text{float}} = N [Z(s_0) - Z(s_m)]. \quad (6.7)$$

Proof. Invest N at s_{j-1} in an overnight deposit and roll principal and interest every day. At s_j it has become $N \prod_i (1 + r_i \Delta_i / 360) = N(1 + R_j \Delta_j)$, whatever the fixings. Hence the payment equals “value of the rolled deposit at s_j ” minus N . The first part costs N at s_{j-1} , worth $NZ(s_{j-1})$ today; the second is a fixed amount worth $NZ(s_j)$. The equality with $N f_j \Delta_j Z(s_j)$ is (6.4). Summing over j , the sum telescopes. □

INTUITION: A floating-rate note is worth par

Adding the notional at the end, $PV_{\text{float}} + NZ(s_m) = NZ(s_0)$: a note paying the funding rate is worth par at each reset, because its coupons are exactly what the money would have earned anyway. So a receiver swap is “long a fixed-coupon bond, short a floating-rate note”: a bond position financed at the floating rate. This is also why an arithmetic *average* of SOFR needs a correction: it is not what a rolled deposit earns, so the static replication fails and a small, volatility-dependent adjustment appears
▶ 2024 L7 @ 21:49].

Corollary 6.10 (Par swap rate). A swap traded at zero PV (the market standard) has the par swap rate ▶ 2024 L7 @ 40:06]

$$S = \frac{\sum_{j=1}^m f_j \Delta_j Z(s_j)}{\sum_{i=1}^n \Delta_i Z(t_i)} = \frac{Z(t_0) - Z(t_n)}{A}. \quad (6.8)$$

The first form (slide 21) is a weighted average of forward rates; the second holds under the single-curve assumptions of *proposition 6.9*. For a spot-starting swap $Z(t_0) \approx 1$.

Par swap rates are quoted on every broker screen, and they are what the curve must reproduce. With annual payments, $\Delta_i = 1$ and $Z(t_0) = 1$, (6.8) is exactly the par-bond equation used for bootstrapping in *section 1.8*: the swap rate is the coupon of a par bond priced off the SOFR curve.

Mark-to-market and DV01. After inception, let S_t and A_t be the par rate and annuity of the remaining schedule. Then

$$V_{\text{payer}} = N [(Z(t_0) - Z(t_n)) - KA] = N A_t (S_t - K), \quad V_{\text{receiver}} = -V_{\text{payer}}, \quad (6.9)$$

and to first order (holding A_t fixed) a 1 bp rise in the par rate changes the payer’s value by $DV01 \approx N A_t \times 10^{-4}$.

IN FINANCE: Why swaps were invented, and why they trade at par

A comment from the audience [► 2024 L7 @ 41:14]: in the late 1980s swaps boomed because highly rated companies could borrow long-term fixed cheaply, while weaker companies could only borrow short-term floating. Each borrowed where it had a comparative advantage and they swapped the payments, both ending up with cheaper funding of the type they wanted [► 2024 L7 @ 42:14] (Exercise 6.8). Today swaps are traded at (or within a bid–offer of 0.1–0.2 bp around) zero PV, i.e. at the par rate. Off-market swaps appear inside structured transactions, where the upfront value is accounted for elsewhere or is where the P&L is taken [► 2024 L7 @ 41:14].

Example 6.11 (Valuing swaps (not from class)). With the curve of example 6.16 below: a 5-year receiver swap on \$100M struck at $K = 4.00\%$, when the 5-year par rate is 3.40% and $A_5 = 4.5178$, is worth $100M \times 0.006 \times 4.5178 = \$2.71M$. A new 10-year par swap on \$100M has $A_{10} = 8.3195$ and a DV01 of about \$83,200 per bp.

Remark 6.12 (Two curves in the LIBOR era). With LIBOR the index (unsecured, bank credit) and the discount rate (collateral, OIS) differed, so forwards came from a LIBOR curve and discount factors from an OIS curve, and the telescoping (6.7) failed (“multi-curve” pricing). The SOFR transition has largely brought vanilla USD swaps back to a single curve.

Exercise 6.3 (not from the course). (a) Show that a note paying compounded SOFR in arrears on N , plus N at maturity, is worth N at every reset date when discounting is on the SOFR curve.

(b) Deduce (6.7) again.

(c) USD SOFR swaps pay two business days after the end of each period. Write the value of one floating period with this payment lag and explain why the telescoping is no longer exact.

Exercise 6.4 (not from the course). A company has a 5-year \$50M floating-rate loan at SOFR+150 bp and wants to fix its cost.

(a) Should it pay or receive fixed on a swap?

(b) With the 5-year par rate at 3.40% , what all-in fixed rate does it lock in (ignore day-count and payment-date differences)?

(c) One year later the 4-year par rate is 3.90% and $A_4 = 3.60$. What is the swap’s value to the company, and where does the offsetting loss appear?

6.6 Building the yield curve

The yield curve is the core valuation model of a rates desk: a function $t \mapsto Z(t)$ that feeds all other pricing [► 2024 L7 @ 42:14]. It is built in three steps (slide 22) [► 2024 L7 @ 43:15]:

1. parameterise it by *knot points* $\{(t_k, Z(t_k))\}_{k=1}^N$;
2. choose an *interpolation* to get $Z(t)$ for $t \notin \{t_k\}$ (say a 4.5-year discount factor from annual knots);
3. *calibrate* the $Z(t_k)$ to market quotes.

6.6.1 Knot points

The simplest and most common choice: the maturities of the calibration instruments (1, 2, 3, 5, 7, 10-year swaps give knots at 1, 2, 3, 5, 7, 10 years) [► 2024 L7 @ 43:15]. Other

choices are possible (e.g. FOMC meeting dates, [section 6.6.4](#)), but the knots must be chosen together with the interpolation and the instruments: bad choices give unstable hedges [[2024 L7 @ 44:16](#)]. Asked why the knots are not chosen to minimise interpolation error (e.g. Chebyshev nodes), the lecturer answers that one wants a one-to-one link between knots and calibration instruments: with knots at 5, 10, 15, ... years and instruments concentrated at the short end, the curve would be over-determined in one region and under-determined in another. Moreover the hedges come out in terms of the input instruments, which must be liquid and tradeable [[2024 L7 @ 45:19](#)].

6.6.2 Interpolation: local versus global

Definition 6.13 (CDF interpolation). Constant daily forward (CDF) interpolation assumes that the daily forward rate f_d , defined by $Z(d+1) = Z(d)/(1 + f_d/360)$, is constant between consecutive knots [[2024 L7 @ 44:16](#)].

Proposition 6.14 (CDF = log-linear discount factors). Under CDF interpolation, for $t_k \leq t \leq t_{k+1}$,

$$\ln Z(t) = \frac{t_{k+1} - t}{t_{k+1} - t_k} \ln Z(t_k) + \frac{t - t_k}{t_{k+1} - t_k} \ln Z(t_{k+1}), \quad (6.10)$$

equivalently the product $t r_z(t)$ of time and continuously compounded zero rate $r_z(t) = -\ln Z(t)/t$ is linear between knots (slide 23). The interpolation is local: $Z(t)$ depends only on the two neighbouring knots.

Proof. $\ln Z(d) = \ln Z(t_k) - \sum_{d'=t_k}^{d-1} \ln(1 + f_{d'}/360)$ is linear in the number of days when $f_{d'}$ is constant; matching the two endpoints gives (6.10), and $t r_z(t) = -\ln Z(t)$. In continuous time this is a piecewise-constant instantaneous forward $f(t) = -d \ln Z / dt$. $\square$

CDF is very fast, local, and produces the characteristic staircase forward curve ([fig. 6.2](#)): the simplest, least smooth method [[2024 L7 @ 46:21](#)].

Definition 6.15 (Cubic spline). Fit on each interval $[t_k, t_{k+1}]$ a cubic

$$x(t) = \sum_{j=0}^3 \alpha_{kj} (t - t_k)^j,$$

where x is the interpolated quantity (typically the zero rate), and impose continuity of x, x', x'' at the interior knots (slide 24) [[2024 L7 @ 46:21](#)]. Two end conditions (e.g. $x'' = 0$ at both ends, a *natural* spline) close the system: $4(N-1)$ coefficients for $2(N-1)$ interpolation conditions, $2(N-2)$ smoothness conditions and 2 end conditions.

The spline is reasonably fast and very smooth, but *global*: the curve near an 8-year point depends on all knots, because moving a distant knot changes the shape everywhere through the smoothness constraints [[2024 L7 @ 47:24](#)]. The lecturer's advice to future rate quants is to keep a laminated card reading "fast, local hedges or smooth, global hedges": one can be anywhere in between, but not have both. He has argued about this with traders for 30 years. Traders want an 8-year trade to be hedged with 7- and 10-year swaps only, not with 5- or 15-year ones. Among smooth methods the cubic spline is the worst in this respect, precisely because it is the smoothest. Tension splines and B-splines sit in between, and much interpolation technology from computer graphics could be used [[2024 L7 @ 46:21](#)](slide 29).

4 COURSE NOTE: Erratum: figure on slide 23

The chart next to the CDF bullets is titled "Yield", but a step function is what CDF produces for the forward rate. The zero yield $r_z(t) = \frac{1}{t} \int_0^t f(s) ds$ is continuous (of the form $a + b/t$ on each interval). Read the vertical axis as the daily forward rate.

5 COURSE NOTE: Erratum: slide 24

The slide writes $x_i(t) = \sum_{j=0}^3 \alpha_{ij} (t - t_i)^j$, using i both for the interval and for the summation. The correct form is the one above: interval index k , power index j .

INTUITION: The spline as an elastic beam

A natural cubic spline minimises the bending energy $\int x''(t)^2 dt$ among all interpolants: it is the draftsman’s elastic ruler pinned at the knots. Moving one pin bends the whole ruler. For equally spaced knots the disturbance decays by the factor $2 - \sqrt{3} \approx 0.27$ per knot *with alternating sign*, the root of $\lambda^2 + 4\lambda + 1 = 0$ in the tridiagonal system for the second derivatives. This is exactly the ringing visible in the spline hedges of [fig. 6.3](#). CDF corresponds to a “string” of independent segments in $\ln Z$: no stiffness, no propagation, no smoothness.

6.6.3 Calibration

For each calibration instrument write the pricing error as a function of the curve (slide 25) [▶ 2024 L7 @ 48:27](#):

$$E_i(Z) = P_i(Z) - P_i^{\text{obs}}, \quad i = 1, \dots, N.$$

- With N knots and N instruments solve $E_i = 0$ exactly with any root finder. When the interpolation is local and instruments are sorted by maturity, each new instrument introduces a single new unknown and the system can be *bootstrapped* one equation at a time (as in [section 1.8](#)).
- Otherwise minimise $E = \sum_i E_i(Z)^2$, possibly with extra terms, e.g. smoothness penalties on an under-specified curve, or accept any curve that prices every instrument within its bid–offer (which is indistinguishable from an exact fit) [▶ 2024 L7 @ 49:28](#).

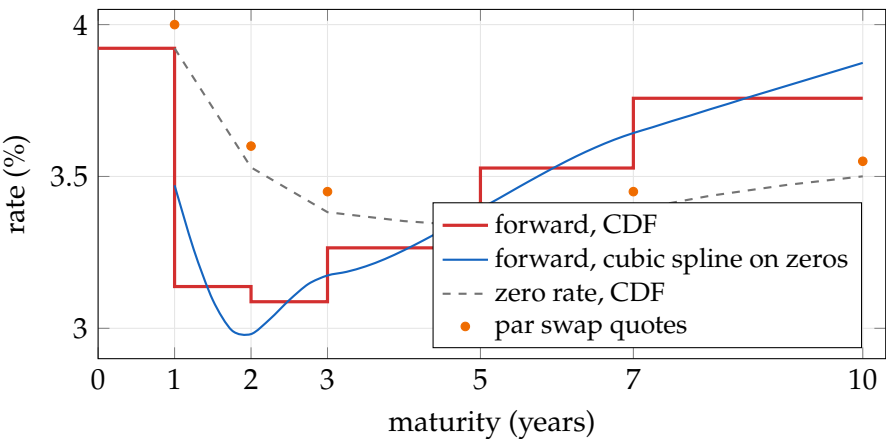


Figure 6.2: Curves implied by the quotes of [example 6.16](#). CDF gives a staircase of forward rates with jumps at the knots; the spline (drawn from the first knot; below 1 year both methods coincide) gives a smooth forward curve that dips below the CDF forwards around 2 years. Both reprice the six quotes exactly. Continuously compounded rates; the 1-year zero rate is $\ln 1.04 = 3.922\%$.

Exercise 6.5 (not from the course). Using the quotes of [example 6.16](#):

- bootstrap Z_1, Z_2, Z_3 by hand;
- show that CDF gives $Z_4 = \sqrt{Z_3 Z_5}$, write the scalar equation for Z_5 and solve it numerically;
- compute the one-year forward rates $f(k - 1, k)$, $k = 1, \dots, 5$, and the 4-year par rate;
- prove [proposition 6.14](#) directly from the daily compounding definition, and show that the forward curve jumps at a knot unless the two adjacent segments have equal

Example 6.16 (Bootstrapping a small swap curve (not from class)). Annual-pay par swap rates ($\Delta_i = 1$, single curve, day counts ignored): 1Y 4.00%, 2Y 3.60%, 3Y 3.45%, 5Y 3.40%, 7Y 3.45%, 10Y 3.55%. For $T \leq 3$ every payment date is a knot and (6.8) gives the recursion $Z_T = (1 - S_T \sum_{k < T} Z_k)/(1 + S_T)$. For 5Y, CDF gives $Z_4 = \sqrt{Z_3 Z_5}$ and Z_5 solves the scalar equation $1 - Z_5 = S_5(Z_1 + Z_2 + Z_3 + \sqrt{Z_3 Z_5} + Z_5)$; similarly for 7Y and 10Y. Result:

knot t_k (years)	1	2
par rate (%)	4.00	3.60
$Z(t_k)$	0.96154	0.93184
zero rate r_z (% , cont.)	3.922	3.530
forward on $(t_{k-1}, t_k]$ (%)	3.922	3.138

A natural cubic spline on zero rates, calibrated to the same six quotes, reprices them exactly too, but it implies a 4-year par rate of 3.405% against 3.419% for CDF. For a maturity that is not quoted, the choice of interpolation moves the price by more than a basis point, about ten times the bid–offer.

slopes of $\ln Z$.

6.6.4 Central-bank meeting curves, turns and data-driven curves

FOMC curves. The history of daily SOFR (slide 26) is not smooth at all: it looks like a staircase [► 2024 L7 @ 49:28]. It is flat between FOMC meetings and jumps when the Fed changes its policy rate: near zero after March 2020, a sequence of steps up in 2022–23, a plateau around 5.3% until the 50 bp cut of September 2024 (“the chart needs a 50 bp drop at the right edge”) [► 2024 L7 @ 50:32]. The spike at the left edge is a repo squeeze related to tax-payment flows (September 2019). This suggests CDF interpolation with knots at the meeting dates: daily rates are constant between meetings, which is now almost universal in the industry [► 2024 L7 @ 50:32]. Two complications (slides 27–28):

- there are 8 FOMC meetings a year but only 4 quarterly futures, so the problem is under-determined. One adds 1-month futures or smoothness constraints, or uses *FOMC swaps* (meeting-to-meeting OIS), which now trade and directly price the rate expected after a given meeting [► 2024 L7 @ 51:34];
- the meeting calendar is known only about two years ahead, and instruments are sparse beyond the front end. Hence a *hybrid* curve: CDF on meeting dates at the front, spline at the back, which works well in practice [► 2024 L7 @ 50:32].

Turns [► 2024 L7 @ 52:37]. Banks report balance sheets and return on assets at quarter- and year-ends and want small balance sheets on those dates, so borrowing across year-end costs a premium: a spike in the forward curve every year. Curves include these *turns* explicitly. Some hedge funds trade pairs of swaps ending just before and just after year-end, isolating the turn and exploiting whoever models it badly; in size this is real money [► 2024 L7 @ 53:40].

Data-driven curves [► 2024 L7 @ 53:40]. A USD curve has about 50 inputs of very different liquidity; perhaps half a dozen are super-liquid. One can estimate historical factors (curve shapes) by PCA (chapter 7), drive the first few factors from the most liquid instruments and build the curve from them. This works well for bond curves (slide 29).

6.7 Risk: PV01, delta ladders and gamma

On a desk, “risk” first means: how much do I make or lose if rates move [► 2024 L7 @ 54:42]?

Definition 6.17 (PV01 and delta ladder). The *PV01* (or *DV01*) of a position is its sensitivity to a 1 bp move of the curve inputs, computed by a two-sided finite difference (slide 30)

$$\frac{\partial \text{PV}}{\partial r} \approx \frac{\text{PV}(r + \epsilon) - \text{PV}(r - \epsilon)}{2\epsilon}, \quad \epsilon = 1 \text{ bp}.$$

Bumping each curve input x_k separately gives the vector of partial PV01s, the *delta ladder* (partial deltas, bucketed risk, delta strip) $\delta_k = \partial \text{PV} / \partial x_k$.

The ladder is what the desk hedges and what risk limits are set on. It is computed by brute force (bump the input, rebuild the curve, revalue) or analytically, but the concept is the same [► 2024 L7 @ 54:42]. If the inputs are the calibration quotes, the ladder translates directly into hedges:

Proposition 6.18 (Hedging with the calibration instruments). *Let the curve be calibrated exactly to par swaps with quotes $q = (q_1, \dots, q_N)$ and annuities A_k . A payer swap on instrument k , struck at today's quote, has $\partial PV_k / \partial q_j = A_k \delta_{kj}$ per unit notional. Hence a portfolio with ladder $\delta_k = \partial V / \partial q_k$ is first-order neutral to every quote after adding payer swaps with notionals*

$$x_k = -\frac{\delta_k}{A_k}, \quad k = 1, \dots, N.$$

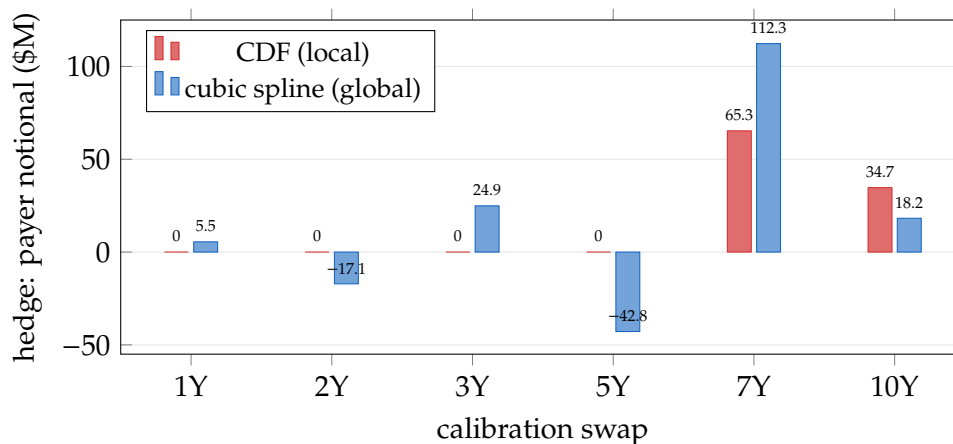


Figure 6.3: Hedge of a \$100M 8-year receiver swap with the six calibration swaps, $x_k = -\delta_k / A_k$ from [proposition 6.18](#), for the two interpolations of [example 6.16](#). Both hedges total about \$100M, but the spline one rings with alternating signs through the whole curve (“fast, local” versus “smooth, global” [▶ 2024 L7 @ 47:24](#)).

Gamma [▶ 2024 L7 @ 55:47](#). Second-order risk (gamma, or convexity) matters more or less depending on the book. Slide 31 defines it as the change in delta for a 1 bp move:

$$\Gamma = \frac{PV(r + \epsilon) - 2PV(r) + PV(r - \epsilon)}{\epsilon} = \epsilon \frac{\partial^2 PV}{\partial r^2} + O(\epsilon^3).$$

Note the single ϵ in the denominator: this is $\Delta(r + \frac{\epsilon}{2}) - \Delta(r - \frac{\epsilon}{2})$ with $\Delta = \partial PV / \partial r$, i.e. the second derivative times the bump, not the second derivative itself. The full matrix of cross gammas $\partial^2 PV / \partial x_j \partial x_k$ needs bumps in pairs, of order N^2 revaluations: 2,500 for a 50-input curve, too expensive. Desks use a parallel gamma, a factor (PCA) gamma, or scenario analysis instead [▶ 2024 L7 @ 55:47](#).

Practical issues [▶ 2024 L7 @ 56:49](#)(slide 32).

- **Bump size:** 1 bp is a reporting convention. Monte Carlo pricers need larger bumps, or the noise swamps the difference (the “noisy delta” problem of [chapter 1](#)); PDE pricers show grid effects for large bumps; too large a bump creates unnatural curve shapes.

Proof. Because the calibrated curve reprices instrument k at par rate q_k for every q , (6.9) gives $PV_k(q) = (q_k - K)A_k(q)$. Differentiating, $\partial_{q_j} PV_k = \delta_{kj}A_k + (q_k - K)\partial_{q_j} A_k$, and the second term vanishes at $K = q_k$. The total first-order sensitivity to q_j is $\delta_j + x_j A_j = 0$. $\square$

Example 6.19 (An 8-year swap hedged with 7- and 10-year swaps (not from class)). Take the curve of [example 6.16](#) and a \$100M 8-year receiver swap at its par rate (3.492% with CDF). Its parallel PV01 is $-\$68,835$ per bp, equal to $-NA_8 \times 10^{-4}$ with $A_8 = 6.883$. The ladder with respect to the six quotes depends on the interpolation ([fig. 6.3](#)):

- **CDF:** $-\$39,982$ on the 7Y quote, $-\$28,840$ on the 10Y, a few dollars elsewhere. The hedge is to pay fixed on $\$65.3M$ 7Y and $\$34.7M$ 10Y: exactly what traders want;
- **cubic spline:** $-533, +3,244, -6,969, +19,314, -68,766, -15,138$ dollars per bp on 1Y, $\dots$, 10Y. The hedge requires positions of alternating sign in every maturity, including $\$112M$ of 7Y and $-\$43M$ of 5Y: a “smeared hedge” (slide 32).

In both cases the ladder sums to the parallel PV01.

- *Sum of partials vs parallel shift*: by the chain rule they agree to first order, as in the example; the discrepancy comes from cross-gamma terms at finite ϵ . Sequential bumps are one remedy.
- *Interpolation*: the ladder of a position is a property of position *and* curve model. Local interpolation concentrates it near the maturity, smooth interpolation smears it, which is a source of endless arguments [► 2024 L7 @ 56:49].

Exercise 6.6 (not from the course).

- Show that the gamma of slide 31 equals $\epsilon \partial^2 PV / \partial r^2 + O(\epsilon^3)$.
- For a 10-year zero-coupon bond with continuously compounded rate 4% and notional \$100M, compute the PV01 and this gamma for $\epsilon = 1$ bp.
- How many revaluations does a full central-difference cross-gamma matrix need for N curve inputs?
- Why does a Monte Carlo pricer need a larger ϵ (chapter 1)?

6.8 P&L attribution

Every morning a desk starts with positions and a market, and every evening it has made or lost money: why [► 2024 L7 @ 56:49]? Skill, market direction, client flow earning the bid-offer? The attribution is also one of the best checks of the model, and in some jurisdictions a regulatory requirement. Total P&L is known exactly (slide 33), $PnL(t_0, t_1) = PV(t_1) - PV(t_0)$. It is split into three parts (slide 34) [► 2024 L7 @ 57:58]:

- **Portfolio change**: new trades, unwinds, novations, resets, exercises. Conceptually simple, mechanically the hardest (booking systems, operational flows).
- **Time**, or *theta*: the change in value if the market “does not move”.
- **Market moves**: to first order $\sum_i \frac{\partial PV}{\partial x_i} \Delta x_i$ (the linear term on slide 34), plus $\frac{1}{2} \sum_{i,j} \Gamma_{ij} \Delta x_i \Delta x_j$ and higher terms.

What remains is the *unexplained* P&L. A large or systematic unexplained part signals missing risk factors or a bad model. The order in which the three steps are applied matters (the decomposition is path-dependent), so conventions must be fixed and documented.

What does “the market did not move” mean? [► 2024 L7 @ 58:59]. The curve is a function of maturity, and when the calendar moves one must decide what is held fixed. There is more than one defensible answer (not detailed in class):

- *Forwards realised*: yesterday’s forward curve for fixed calendar dates is today’s curve, $Z_{t_1}(T) = Z_{t_0}(T)/Z_{t_0}(t_1)$. Every funded position then earns exactly the funding rate: theta is pure carry.
- *Constant curve by tenor*: the zero (or par) rate for each time-to-maturity is unchanged. On an upward-sloping curve a bond “rolls down” to lower yields and earns carry *plus* roll-down (Exercise 6.7).

For a physicist: the first convention is Lagrangian (follow each cash flow), the second Eulerian (sit at a fixed point of the maturity axis). Whatever the choice, the desk and its controllers must use the same one.

Exercise 6.7 (not from the course). Annual-compounding zero rates are 3% (1 year) and 4% (2 years).

- Price a 2-year zero-coupon bond.
- Value it one year later if the forwards are realised, and if the curve by tenor is unchanged.
- Split the one-year return in (b) into carry and roll-down, and relate each to the two theta conventions of [section 6.8](#).

6 COURSE NOTE: The spoken P&L example

In class the market term is illustrated with a long position of \$1B of bonds, a PV01 of “\$10 million” and gains of \$50M for a 5 bp move [[▶ 2024 L7 @ 58:59](#)]. The numbers were improvised: \$1B of 10-year bonds has a PV01 of roughly \$0.8–0.9M, so 5 bp is worth about \$4M, and a long bond position gains when rates *fall*.

6.9 Clearing, margin and the CCP basis

A swap starts near zero PV, but as markets move its value drifts away from zero and becomes a credit exposure: if a counterparty that owes me \$10M on a swap defaults, I lose it (slide 35) [[▶ 2024 L7 @ 1:00:00](#)]. Lehman, a huge swap dealer, made this concrete in 2008: its counterparties had to replace hundreds of billions of hedges at once, bid-offers widened about twentyfold for a month or two, and the one large dealer still trading normally captured the flow [[▶ 2024 L7 @ 1:01:03](#)]. After the crisis regulators made *central clearing* mandatory for vanilla swaps: once two parties agree a trade, both assign their side to a central counterparty (CCP: LCH in London, CME, JSCC in Japan) and post margin to it, so they no longer face each other’s credit [[▶ 2024 L7 @ 1:01:03](#)].

IN FINANCE: Fewer, larger risks

Clearing replaces many small bilateral risks by concentrated exposure to a few enormous, deeply interconnected CCPs. A failure of one of them would be a systemic catastrophe, which is why members post margin and default-fund contributions and CCPs are heavily capitalised [[▶ 2024 L7 @ 1:02:09](#)].

The CCP basis [[▶ 2024 L7 @ 1:02:09](#)]. A dollar owed by LCH is not the same as a dollar owed by CME: positions at different CCPs cannot be offset operationally or for collateral. From about 2017 the fixed rate for paying fixed at CME exceeded that at LCH. On a broker screen for 26 June 2017 (slide 36; source Tradition) the CME–LCH spread grew with maturity, from 0.10 bp at 1 year through 1.05 (5Y), 2.45 (10Y) and 3.40 (30Y) to 3.55 bp at 50 years, in a market with a bid-offer of about 0.2 bp [[▶ 2024 L7 @ 1:03:10](#)]. It persisted for years (recently it has changed sign). Why not receive at CME, pay at LCH and retire [[▶ 2024 L7 @ 1:03:10](#)]?

- **Double margin:** the two offsetting trades sit at different CCPs, so initial margin (1–2% of notional, quoted in class) is posted twice. Its funding cost is the main driver [[▶ 2024 L7 @ 1:04:16](#)].
- **Netting:** CME also clears futures, so a dealer long futures and short swaps there gets partial margin offsets that the arbitrage would destroy [[▶ 2024 L7 @ 1:04:16](#)].
- **Preferences:** large US asset managers, one-way payers of fixed, preferred CME, a US-jurisdiction CCP not controlled by dealers, over LCH [[▶ 2024 L7 @ 1:05:21](#)]. (LCH Group is today majority-owned by the London Stock Exchange Group, with banks as minority shareholders; the perception quoted in class concerns its governance.)

The general lesson: margin and capital requirements let spreads persist between instruments that would otherwise be arbitrated. To price correctly one must include how the trade is margined, how much capital it consumes, and every other cost that used to be ignored. Estimating these costs and pushing them into the upfront price has been one of

the largest quant efforts of the last 15 years, made harder by the fact that many are portfolio effects that must be allocated to individual trades [► 2024 L7 @ 1:06:21](see [chapter 8](#)).

Exercise 6.8 (not from the course). Firm A can borrow fixed at 4.00% or floating at SOFR+0.30%; firm B at 5.20% fixed or SOFR+1.00%. A wants floating, B fixed.

- (a) Show that A has an absolute advantage in both markets but a comparative advantage in fixed.
- (b) Design a swap through a bank earning 0.10% that splits the remaining gain equally between A and B.
- (c) Which risk does each firm take on in exchange (think of [section 6.9](#))?

6.10 Electronic trading and price making

Electronic trading is one of the hottest areas for quants [► 2024 L7 @ 1:07:23]. As a rule of thumb, equities are about ten years ahead of FX, and FX ten years ahead of fixed income. Slide 37 (Flow Traders data, 2023) shows the share of electronic trading: above 90% for US cash equities and equity options, European ETFs and FX spot; about 80% for European government bonds and 65% for US Treasuries; 35–45% for investment-grade corporates; 22–27% for high yield; about 12% for munis and 10% for emerging-market bonds. Small trades go electronic first: \$1M of Treasuries does not move a market trading \$1T a day, whereas \$500M does and needs care [► 2024 L7 @ 1:08:24].

Participants [► 2024 L7 @ 1:10:26]. *Principal trading firms* (high-frequency shops) have no customers, trade their own capital and are extremely fast; the lecturer met a physics PhD hired to optimise the microwave links between Chicago and New York, and realised that banks will never beat them at latency [► 2024 L7 @ 1:11:26]. *Dealers* (banks) are fast (microseconds, not nanoseconds) and have customers, who pay the bid–offer. *Customers* want efficient execution, a view of the full market and confidence that they are not being taken advantage of [► 2024 L7 @ 1:12:26].

Price making (slide 38) [► 2024 L7 @ 1:12:26]. Start from a mid price, from liquidity providers for actively traded instruments or from a model for off-the-run ones, then *skew* it according to:

- the current position: be aggressive on risk-reducing trades, conservative on risk-increasing ones;
- the client tier: a valuable client relationship justifies a better price [► 2024 L7 @ 1:13:29];
- *client toxicity*: the classic toxic client sends the same request to ten dealers at once. Whoever wins finds nine others hedging the same way, and the hedge has moved away (winner’s curse) [► 2024 L7 @ 1:13:29];
- market volatility (widen when the market may move before the hedge is done) and short-term prediction models (market microstructure).

All of this requires tightly integrated systems. A credit check that answers after the client has gone is a lost trade [► 2024 L7 @ 1:14:29].

Fast model pricing (slide 39, not presented in class). Model prices must be recomputed in microseconds. One linearises the output around the last full valuation,

$$r(t) \approx r(t_0) + \sum_{i=1}^n \frac{\partial r}{\partial x_i} \Delta x_i,$$

splits the drivers into fast ones (fed through the linearisation in real time) and slow ones, and periodically recomputes the full model. The slide omits the index on Δx ; it must be Δx_i .

6.11 Bonds revisited

Slides 40–44 were not presented in class; they extend [sections 1.8](#) and [1.8.4](#).

Why bonds are harder than swaps. Government bonds are the most liquid fixed-income instruments in many markets. G20 government bonds are usually treated as default-free, while emerging-market bonds need a default model. Despite their apparent simplicity bonds are harder than swaps: bonds are not fungible (each issue has its own coupon, maturity, liquidity, repo specialness), whereas all swaps are generated from one curve (slide 40).

Price–yield with f payments a year (slide 41). Bonds trade and settle on price, but yield is the better comparison across coupons and maturities. With n remaining periods, coupon C_i per period per unit notional and f coupons per year,

$$P_D(y) = \sum_{i=1}^n \frac{C_i}{(1 + y/f)^i} + \frac{1}{(1 + y/f)^n},$$

which reduces to (1.4) for $f = 1$. The slide’s chart shows the familiar picture: premium when the yield is below the coupon, discount when above.

Proposition 6.20 (Modified and Macaulay duration). Let $D_{\text{mod}} = -\frac{1}{P} \frac{\partial P}{\partial y}$ and let $D_{\text{Mac}} = \sum_i \frac{i}{f} \frac{CF_i(1+y/f)^{-i}}{P}$ be the PV-weighted average payment time in years. Then $D_{\text{Mac}} = D_{\text{mod}} (1 + y/f)$ (slide 42).

Proof. $\partial_y (1 + y/f)^{-i} = -\frac{i}{f} (1 + y/f)^{-i-1}$, so
 $-P'(y) = (1 + y/f)^{-1} \sum_i \frac{i}{f} CF_i (1 + y/f)^{-i} =$
 $P D_{\text{Mac}} / (1 + y/f).$ $\square$

(The slide writes the factor as $1 + y/n$ with n the compounding frequency, called f on the previous slide.) Duration is “not generic but good to gain insight”: it assumes a flat move of a single yield, while swap risk is a ladder.

Bonds against the swap curve: the z-spread (slide 43). Swap and bond books hedge each other, so their risks must be aggregated. Pricing a bond off the swap (SOFR) curve, $P'_D = \sum_i C_i Z(t_i) + Z(t_n)$, will not reproduce its market price; the *z-spread* s is the parallel shift of the continuously compounded discount rates that does:

$$P_D = \sum_{i=1}^n C_i Z(t_i) e^{-st_i} + Z(t_n) e^{-st_n}.$$

It measures the bond’s credit, liquidity and supply premium relative to swaps, and it lets the bond’s rate risk be expressed as a delta ladder on the swap curve plus a single spread

risk.

Bond flavours (slide 44): government, asset-backed, corporate, municipal, high yield and more, each with its own valuation and risk nuances. *Municipal bonds* were discussed in class [▶ 2024 L7 @ 1:08:24]: a huge number of small and heterogeneous issuers (states, towns, school boards, toll-road and bridge projects) and interest exempt from federal income tax. At a tax rate τ an investor is indifferent between a muni yielding y_m and a taxable bond yielding y_T when $y_m = (1 - \tau)y_T$, so munis yield about 70% of comparable taxable bonds [▶ 2024 L7 @ 1:09:26]. Only taxpayers benefit, so the market is retail: the lecturer's former electronic muni platform (acquired by TD) did 5,000–10,000 trades a day with an average size around \$28,000, absurdly small by government-bond standards but profitable in volume. Emerging-market bonds are idiosyncratic (payment terms, credit support) and still mostly traded by phone [▶ 2024 L7 @ 1:10:26].

Example 6.21 (z-spread and DV01 (not from class)). A 5-year bond with 4% annual coupons trades at 101.50. Off the curve of [example 6.16](#) it would be worth $P'_D = 102.71$; the z-spread is $s = 25.6$ bp. Its yield is 3.666%, $D_{\text{Mac}} = 4.633$, $D_{\text{mod}} = 4.469$, and its DV01 is $D_{\text{mod}} P \times 10^{-4} = 0.0454$ per 100 of face value.

IN FINANCE: From the Q&A

Rates outlook [▶ 2024 L7 @ 1:15:32]. The Fed balances inflation, pushed up by COVID-era stimulus and by the energy shock after Russia's invasion of Ukraine, against employment. High rates discourage borrowing (business start-ups, and above all housing, the largest purchase in most lives), and a cutting cycle had just begun. *Two kinds of fixed-income research* [▶ 2024 L7 @ 1:16:39]: client-facing economics research makes forecasts and trade recommendations; internal quant research on pricing and hedging makes no predictions at all. It only needs to know how to hedge ("if I knew where rates are going, I would be on my Caribbean island"). *Muni analytics* [▶ 2024 L7 @ 1:17:42]: revenue bonds depend on project cash flows (toll roads lost their revenue in COVID); all munis depend on future tax policy, since the value of the exemption moves with the tax rate ([Exercise 6.10](#)); and the practical problem is *sourcing*: finding who holds the bonds a client wants, which is why munis are still voice-driven [▶ 2024 L7 @ 1:18:43].

KEY FORMULAS

Simple interest, deposit:	$I = Nr\Delta t,$ $Z(t) = 1/(1 + r(t)\Delta)$
SOFR term rates:	$R^{\text{comp}} = \frac{360}{d} \left[\prod_i (1 + \frac{r_i d_i}{360}) - 1 \right],$ $R^{\text{avg}} = \frac{1}{d} \sum_i r_i d_i$
Discounting:	$r_d = r_b$ (discount at the funding rate)
Forward rate:	$f(t_1, t_2) = \frac{1}{\Delta} (Z(t_1)/Z(t_2) - 1)$
Futures:	$f = 100 - F - \text{Cvx},$ $\text{Cvx} \approx \frac{1}{2} \sigma^2 t_1 t_2$ (Ho-Lee)
Swap legs:	$\text{PV}_{\text{fix}} = NK \sum_i \Delta_i Z(t_i) = NKA,$ $\text{PV}_{\text{float}} = N [Z(t_0) - Z(t_n)]$
Par rate, MTM:	$S = (Z(t_0) - Z(t_n))/A,$ $V_{\text{payer}} = NA(S - K), \quad \text{DV01} \approx NA \cdot 10^{-4}$
CDF interpolation:	$\ln Z$ linear between knots $\iff$ $f(t)$ piecewise constant
Calibration:	$E_i = P_i(Z) - P_i^{\text{obs}} = 0 \quad \text{or} \quad \min \sum_i E_i^2$
PV01, gamma:	$\frac{\text{PV}(r+\epsilon) - \text{PV}(r-\epsilon)}{2\epsilon},$ $\Gamma = \frac{\text{PV}(r+\epsilon) - 2\text{PV}(r) + \text{PV}(r-\epsilon)}{\epsilon^2}$
Hedge with inputs:	$x_k = -\delta_k / A_k$
P&L explain:	$\text{PnL} = \text{portfolio} + \Theta \Delta t$ $+ \sum_i \delta_i \Delta x_i + \frac{1}{2} \sum_{ij} \Gamma_{ij} \Delta x_i \Delta x_j + \text{unexplained}$
Bonds:	$D_{\text{Mac}} = D_{\text{mod}}(1 + y/f),$ $P_D = \sum_i C_i Z(t_i) e^{-st_i} + Z(t_n) e^{-st_n}$
Munis:	$y_m = (1 - \tau) y_T$

Exercise 6.9 (not from the course). Using the bond of the example in [section 6.11](#) (DV01 0.0454 per 100) and the 5-year swap annuity $A_5 = 4.5178$:

- which swap and which notional make \$10M face of the bond DV01-neutral?
- Show that for small s , $s \approx (P'_D - P_D) / \sum_i t_i \text{CF}_i Z(t_i)$, and check it against $s = 25.6 \text{ bp}$.
- Which risk remains after the hedge?

Exercise 6.10 (not from the course).

- At which tax rate is a 2.8% muni equivalent to a 4% taxable bond?
- A 10-year muni is priced to yield 70% of a 4% taxable yield. If investors come to expect the tax rate to fall from 30% to 20%, estimate the price change with a modified duration of 8, all else equal. Connect this to the Q&A remark on tax policy.

Chapter 7

Principal Component Analysis and Factor Models

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L9	S. Andreev (Two Sigma)	PCA as a rotation of coordinates; toy examples, robustness, PCA versus OLS; demeaning and variance normalisation; eigenvalue diagnostics, in- and out-of-sample variance explained; PCA on financial time series; US Treasury curve (level, slope, curvature, regime indicators); PC-neutral curves and flies, leverage.
2013 L15	P. Kempthorne	Linear factor models as cross-sectional and time-series regressions; Sharpe's single-index model, macroeconomic and fundamental (BARRA, Fama–French) models; factor-mimicking portfolios; statistical factor models: factor analysis (maximum likelihood, EM, likelihood-ratio test) and PCA; case study on Treasury yields.

Files. Slides `mit18_642_f24_lec09.pdf` (the European-indices example, slides 26–34, was not covered in class); Jupyter notebook `feddataanalysis.ipynb` with the data file `fedbondyielddata_20230911.csv` (distributed in 2024 week 5); readings `mit18_642_f24_pset_analysis.pdf` (P. Kempthorne, *Principal Components Analysis in Finance: Modeling the Dynamics of Interest Rates*, 2016) and `mit18_642_f24_pset_rcode.pdf` (its R code and a 5-year PCA of Treasury yields); 2013 slides `MIT18_S096F13_lecnote15.pdf` and case study `MIT18_S096F13_CaseStudy7.pdf`. **Problem sets.** 2024 PS2 Problem 3 and PS3 Problem 4; 2013 PS7 (all problems).

Motivation

Financial data come in large and highly redundant blocks: the yields of thirty Treasury maturities, the returns of five hundred stocks, a dozen European equity indices. When the 10-year yield rises the 5- and 30-year yields almost always rise too, so thirty numbers per day carry perhaps three numbers' worth of information. This chapter develops two complementary ways of exploiting the redundancy:

- **Principal component analysis (PCA):** a rotation of coordinates such that the first few axes carry most of the variance. It is model-free (unsupervised learning, no dependent variable), pure linear algebra, and fast [► 2024 L9 @ 3:05].
- **Factor models:** the statistical model $x_t = \alpha + Bf_t + \epsilon_t$ with a few common factors and uncorrelated asset-specific noise. It reduces the $m(m+1)/2$ entries of a covariance matrix to $O(mK)$ parameters, and is the backbone of risk models and of the CAPM/APT view of returns [► 2013 L15 @ 0:00].

The 2024 lecture is the practitioner's view (a fixed-income quant at Two Sigma: workflow, pitfalls, and how PCA turns into tradeable portfolios); the 2013 lecture is the statistician's view (models, estimation, tests). Both analyse daily changes of US Treasury yields and find the same three factors: *level*, *slope* and *curvature*. The mathematics needed is the spectral theorem, the SVD and Perron–Frobenius of [chapter 2](#) ([theorems 2.11](#), [2.12](#) and [2.14](#)) and least squares ([chapter 5](#)).

7.1 Principal components of a random vector

Let $\mathbf{x} = (x_1, \dots, x_p)^\top$ be a random vector with $\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}$ and $\text{Cov}[\mathbf{x}] = \Sigma$. For any weight vector $\mathbf{w}$, $\text{Var}(\mathbf{w}^\top \mathbf{x}) = \mathbf{w}^\top \Sigma \mathbf{w} \geq 0$: Σ is symmetric positive semi-definite ([section 2.6](#); 2024 PS2 Problem 3). If $\mathbf{x}$ are asset returns and $\mathbf{w}$ portfolio weights, $\mathbf{w}^\top \Sigma \mathbf{w}$ is the portfolio variance. By the spectral theorem ([theorem 2.11](#))

$$\Sigma = \Gamma \Lambda \Gamma^\top = \sum_{k=1}^p \lambda_k \boldsymbol{\gamma}_k \boldsymbol{\gamma}_k^\top, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0, \quad \Gamma^\top \Gamma = I_p.$$

7.1.1 Variance maximisation

Definition 7.1 (Principal components). The first principal direction is a unit vector $\mathbf{w}_1$ maximising $\text{Var}(\mathbf{w}^\top \mathbf{x}) = \mathbf{w}^\top \Sigma \mathbf{w}$ subject to $\|\mathbf{w}\| = 1$. The k -th principal direction $\mathbf{w}_k$ maximises the same variance subject to $\|\mathbf{w}\| = 1$ and $\mathbf{w} \perp \mathbf{w}_1, \dots, \mathbf{w}_{k-1}$ [[2013 L15 @ 1:07:08](#)]. The principal component variables (scores, PC factors) are $p_k = \mathbf{w}_k^\top (\mathbf{x} - \boldsymbol{\mu})$; the entries of $\mathbf{w}_k$ are the loadings of PC k , and $\lambda_k / \sum_j \lambda_j$ is the fraction of variance explained.

Theorem 7.2 (PCA is the spectral decomposition of Σ). With the notation above:

- (i) the k -th principal direction can be taken to be the eigenvector $\boldsymbol{\gamma}_k$, and $\text{Var}(p_k) = \lambda_k$ (if the eigenvalues are distinct, $\mathbf{w}_k = \pm \boldsymbol{\gamma}_k$);
- (ii) $\mathbf{p} = \Gamma^\top (\mathbf{x} - \boldsymbol{\mu})$ has $\mathbb{E}[\mathbf{p}] = \mathbf{0}$ and $\text{Cov}[\mathbf{p}] = \Lambda$: the PCs are uncorrelated;
- (iii) $\mathbf{x} = \boldsymbol{\mu} + \Gamma \mathbf{p} = \boldsymbol{\mu} + \sum_{k=1}^p p_k \boldsymbol{\gamma}_k$;
- (iv) total variance is preserved: $\sum_i \text{Var}(x_i) = \text{tr } \Sigma = \text{tr } \Lambda = \sum_k \lambda_k = \sum_k \text{Var}(p_k)$.

Equivalently, with a Lagrange multiplier, $\nabla_{\mathbf{w}} [\mathbf{w}^\top \Sigma \mathbf{w} - \lambda (\mathbf{w}^\top \mathbf{w} - 1)] = 0$ gives $\Sigma \mathbf{w} = \lambda \mathbf{w}$: the stationary points of the Rayleigh quotient $\mathbf{w}^\top \Sigma \mathbf{w} / \mathbf{w}^\top \mathbf{w}$ are the eigenvectors and the stationary values the eigenvalues ([section 2.6](#)).

INTUITION: A rigid rotation of the data cloud

$\mathbf{x} \mapsto \Gamma^\top (\mathbf{x} - \boldsymbol{\mu})$ moves the origin to the centre of mass and rotates the axes; distances are unchanged [[2013 L15 @ 1:02:46](#)]. Andreev's toy example: a one-dimensional data set embedded in the plane along the x -axis, the diagonal or the y -axis looks two-dimensional, but PCA immediately returns one direction with all the variance and a second with none [[2024 L9 @ 5:11](#)]; add a little noise in the perpendicular direction and λ_2 becomes small but non-zero [[2024 L9 @ 7:22](#)]. PCA returns two things: *directions* (eigenvectors) and their *importance* (eigenvalues). For a physicist, Σ is the second-moment tensor of the cloud of points (unit masses); the inertia tensor $I = \text{tr}(\Sigma) \mathbb{K} - \Sigma$

1 COURSE NOTE: Notation across the sources

The 2024 slides use a data matrix X of size $N \times p$ (rows = observations) and loadings $\mathbf{w}_k, W$; the 2013 slides use m assets, T periods, a data matrix of size $m \times T$ (columns = periods) and eigenvectors $\boldsymbol{\gamma}_k, \Gamma$; the 2016 reading uses a tenors $\times$ days matrix A . Here: p variables and N observations with $X \in \mathbb{R}^{N \times p}$ in the PCA sections, m assets and T periods in the factor-model sections, and $\boldsymbol{\gamma}_k, \Gamma$ for eigenvectors (loadings) throughout.

Proof. (i) Expand $\mathbf{w} = \sum_j c_j \boldsymbol{\gamma}_j$; then $\|\mathbf{w}\|^2 = \sum_j c_j^2 = 1$ and $\mathbf{w}^\top \Sigma \mathbf{w} = \sum_j \lambda_j c_j^2 \leq \lambda_1$, with equality for $\mathbf{c} = \mathbf{e}_1$ (and only for $\mathbf{c} = \pm \mathbf{e}_1$ if $\lambda_1 > \lambda_2$). Inductively, if $\mathbf{w}_j = \boldsymbol{\gamma}_j$ for $j < k$, the constraint $\mathbf{w} \perp \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_{k-1}$ forces $c_1 = \dots = c_{k-1} = 0$, so $\mathbf{w}^\top \Sigma \mathbf{w} = \sum_{j \geq k} \lambda_j c_j^2 \leq \lambda_k$, attained at $\boldsymbol{\gamma}_k$.
(ii) $\mathbb{E}[\mathbf{p}] = \Gamma^\top (\mathbb{E}[\mathbf{x}] - \boldsymbol{\mu}) = \mathbf{0}$ and $\text{Cov}[\mathbf{p}] = \Gamma^\top \Sigma \Gamma = \Gamma^\top \Gamma \Lambda \Gamma^\top \Gamma = \Lambda$ [[2013 L15 @ 1:01:42](#)].
(iii) Multiply $\mathbf{p} = \Gamma^\top (\mathbf{x} - \boldsymbol{\mu})$ by $\Gamma = (\Gamma^\top)^\top$.
(iv) $\text{tr}(\Gamma \Lambda \Gamma^\top) = \text{tr}(\Lambda \Gamma^\top \Gamma) = \text{tr } \Lambda$ [[2013 L15 @ 1:09:11](#)]. $\square$

has the same eigenvectors, so the principal directions are the *principal axes of inertia*, PC1 being the axis with the smallest moment of inertia (the direction along which the cloud is most elongated).

Remark 7.3 (Uniqueness). If γ_k is an eigenvector, so is $-\gamma_k$: the sign of each PC is arbitrary (the 2024 slides: “sign does not matter”), and different software may return different signs for the same data. If $\lambda_k = \lambda_{k+1}$, any orthonormal basis of the eigenspace is equally valid: the directions are not determined at all. This is the mathematical root of the robustness problems of [section 7.2.4](#).

7.1.2 Geometry: best subspace, and PCA versus regression

Proposition 7.5 (Best K -dimensional approximation). Let $V \in \mathbb{R}^{p \times K}$ have orthonormal columns, so that VV^T is the orthogonal projection onto their span. Then

$$\mathbb{E} \|(x - \mu) - VV^T(x - \mu)\|^2 = \text{tr} \Sigma - \text{tr}(V^T \Sigma V) \geq \sum_{j>K} \lambda_j,$$

with equality for $V = \Gamma_K = [\gamma_1 \cdots \gamma_K]$. The first K PCs span the K -dimensional subspace that minimises the mean squared perpendicular distance of the data from it.

For a sample this is the Eckart–Young statement of [section 2.7](#): the truncated SVD of the centred data matrix is its best rank- K approximation.

PCA is not regression [\[► 2024 L9 @ 8:28\]](#). OLS of y on x minimises *vertical* squared distances, OLS of x on y *horizontal* ones, PCA *perpendicular* ones ([fig. 7.1](#)). For standardised variables with correlation ρ the three lines through the centre have slopes ρ , $1/\rho$ and 1 [\[► 2024 L9 @ 9:31\]](#). OLS presupposes a dependent variable (a causal or predictive direction); PCA treats all coordinates symmetrically and says nothing about causality. Use regression to predict y from x , PCA to describe the joint correlation structure [\[► 2024 L9 @ 10:34\]](#). Two further differences: PCA is equivariant under rotations of the data (rotate the cloud, the PCs rotate with it) [\[► 2024 L9 @ 11:38\]](#), but *not* under rescaling of individual variables, whereas OLS predictions are unaffected by changes of units. Hence the covariance-versus-correlation question of [section 7.2.3](#).

Example 7.4 (Two variables with equal variances). For $\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ with $\rho > 0$: $\lambda_{1,2} = \sigma^2(1 \pm \rho)$ with $\gamma_1 = \frac{1}{\sqrt{2}}(1, 1)^T$ (the sum) and $\gamma_2 = \frac{1}{\sqrt{2}}(1, -1)^T$ (the difference), whatever ρ is; PC1 explains the fraction $(1 + \rho)/2$. As $\rho \rightarrow 0$ the eigenvalues merge: an isotropic “blob” has no preferred direction and moving a single data point can swing the estimated PC1 by any angle [\[► 2024 L9 @ 12:44\]](#). (The general bivariate case is 2013 PS7 Problem 1.)

Proof. With $y = x - \mu$, $\|(I - VV^T)y\|^2 = y^T(I - VV^T)y$ since $I - VV^T$ is a projection; taking expectations gives $\text{tr} \Sigma - \text{tr}(V^T \Sigma V)$. Now $\text{tr}(V^T \Sigma V) = \sum_j \lambda_j c_j$ with $c_j = \|V^T \gamma_j\|^2$. Each $c_j \in [0, 1]$ (a projection does not increase length) and $\sum_j c_j = \text{tr}(V^T \Gamma \Gamma^T V) = \text{tr}(V^T V) = K$. Hence

$$\sum_j \lambda_j c_j - \sum_{j \leq K} \lambda_j = \sum_{j \leq K} \lambda_j (c_j - 1) + \sum_{j > K} \lambda_j c_j \leq \lambda_K \left(\sum_{j \leq K} (c_j - 1) + \sum_{j > K} 1 \right) = 0$$

and $V = \Gamma_K$ gives $c_j = 1$ for $j \leq K$, 0 otherwise. $\square$

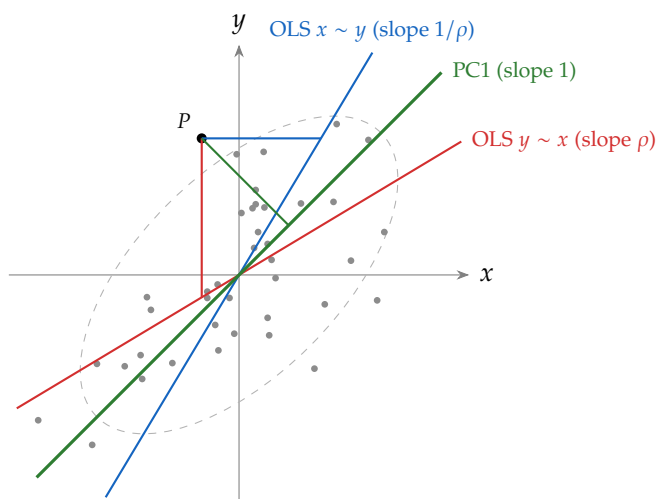


Figure 7.1: Forty standardised points with sample correlation $\rho = 0.6$ and the 2σ ellipse. For the point P the three segments are the distances minimised by OLS of y on x (vertical), OLS of x on y (horizontal) and PCA (perpendicular); cf. slide 8 of lec09 [► 2024 L9 @ 9:31].

7.1.3 Positive correlations and the market factor

If all covariances are positive, $\Sigma_{ij} > 0$ (typical for yields of different maturities or for stocks in one market), the Perron–Frobenius theorem (theorem 2.14) applies to Σ : λ_1 is simple and γ_1 can be chosen with *all entries positive* (2024 PS2 Problem 3). PC1 is then a long-only portfolio: the “market” or “level” factor, and all other PCs, being orthogonal to a positive vector, have loadings of both signs: they are long/short spreads.

Exercise 7.1 (PS2 (2024), Problem 3: principal components, special case). A portfolio holds n assets with fixed weights $w = (w_1, \dots, w_n)^T$; over a given horizon the assets have rates of return $X = (X_1, \dots, X_n)^T$ with mean μ and covariance $\Sigma = (\sigma_{ij})$, and the portfolio return is $Y = w^T X$.

- Show that $\mathbb{E}[Y] = w^T \mu$ and $\text{Var}[Y] = w^T \Sigma w$.
- Prove that Σ is positive semi-definite, for any random variables $X_1, \dots, X_n$.
- Suppose all entries of Σ are positive. Prove that the largest eigenvalue $\lambda_{\max}$ has multiplicity one and an eigenvector v that can be chosen with all components positive.
- Write the first principal component variable PC1 in terms of X , μ and v . What are its mean and variance?

Exercise 7.2 (not from the course). Prove the claims of example 7.6: for $\Sigma = \sigma^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^T]$ find all eigenvalues and eigenvectors, and the fraction of variance explained by PC1. For which ρ is Σ positive semi-definite? Relate the result to theorem 2.14 when $\rho > 0$.

Example 7.6 (Equicorrelated assets (not discussed in class)). For p assets with common variance σ^2 and common correlation $\rho > 0$, $\Sigma = \sigma^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^T]$. Then $\lambda_1 = \sigma^2(1 + (p - 1)\rho)$ with $\gamma_1 = 1/\sqrt{p}$ (the equal-weighted portfolio), and $\lambda_2 = \dots = \lambda_p = \sigma^2(1 - \rho)$. PC1 explains $(1 + (p - 1)\rho)/p \rightarrow \rho$ as $p \rightarrow \infty$: even a modest average correlation produces a dominant first component in a large universe, while the remaining $p - 1$ eigenvalues are degenerate, so the individual higher PCs are not identified (remark 7.3).

7.2 Sample PCA, SVD and practical choices

7.2.1 Data matrix and singular value decomposition

Stack N observations (days) of p variables (tenors, assets) in $X \in \mathbb{R}^{N \times p}$, row i being $\mathbf{x}_i^\top$. With $\bar{\mathbf{x}} = X^\top \mathbf{1}/N$, the centred matrix and sample covariance are

$$X_c = X - \mathbf{1}\bar{\mathbf{x}}^\top = (I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top)X, \quad S = \frac{1}{N-1}X_c^\top X_c$$

(the 2013 slides divide by N ; it does not matter for the eigenvectors) ▶ 2013 L15 @ 41:21].

Proposition 7.7 (PCA via the SVD). Let $X_c = UD\hat{\Gamma}^\top$ be a thin SVD: $U \in \mathbb{R}^{N \times p}$ with $U^\top U = I_p$, $D = \text{diag}(d_1 \geq \dots \geq d_p \geq 0)$, $\hat{\Gamma} \in \mathbb{R}^{p \times p}$ orthogonal. Then:

- (i) $S = \hat{\Gamma}\hat{\Lambda}\hat{\Gamma}^\top$ with $\hat{\lambda}_k = d_k^2/(N-1)$: the right singular vectors are the loadings;
- (ii) the $N \times p$ matrix of PC scores $P = X_c\hat{\Gamma} = UD$ has zero column means and orthogonal columns; column k (the time series of PC k) has sample variance $\hat{\lambda}_k$;
- (iii) $X_c = P\hat{\Gamma}^\top = \sum_k d_k \mathbf{u}_k \hat{\gamma}_k^\top$, and truncation after K terms is the best rank- K approximation of X_c ;
- (iv) $\text{rank } X_c \leq \min(N-1, p)$: with $N \leq p$ observations at most $N-1$ eigenvalues are non-zero.

Andreev computes the covariance matrix and diagonalises it (“we need the covariance matrix for other things anyway”); the SVD gives the same result without forming S ▶ 2024 L9 @ 18:02], ▶ 2013 L15 @ 1:06:05]. Numerically the SVD is preferable because forming $X_c^\top X_c$ squares the condition number; in cost both routes are $O(Np^2)$ (plus $O(p^3)$ for the eigen-decomposition).

Exercise 7.3 (lecnote15, slide 32: PCA via the SVD). Let X^* be the $m \times T$ row-centred data matrix, $\hat{\Sigma}_x = \frac{1}{T}X^*X^{*\top}$, and $X^* = VDU'$ its SVD with V ($m \times m$) orthogonal, U ($T \times m$) with orthonormal columns and $D = \text{diag}(d_1 \geq \dots \geq d_m \geq 0)$. Show that $\hat{\Lambda} = \frac{1}{T}D^2$, $\hat{\Gamma} = V$, and that the matrix of sample principal components is $P = \hat{\Gamma}'X^* = DU'$.

7.2.2 A worked example: six days of Treasury yields

The 2016 reading (pset_analysis) runs PCA on the smallest possible data set: FRED constant-maturity yields of 9 Treasury securities (3 months to 20 years) on the first six business days of 2001. Daily changes in basis points, minus their row means, form the 9×5 matrix A of table 7.1 (tenors $\times$ days, the transpose of X_c , so the roles of U and $\hat{\Gamma}$ are swapped: loadings are the *left* singular vectors of A).

Proof. (i) $X_c^\top X_c = \hat{\Gamma}DU^\top UD\hat{\Gamma}^\top = \hat{\Gamma}D^2\hat{\Gamma}^\top$.
(ii) $P = UD\hat{\Gamma}^\top\hat{\Gamma} = UD$, so $P^\top P = D^2$; $\mathbf{1}^\top P = \mathbf{1}^\top X_c\hat{\Gamma} = \mathbf{0}$ because $\mathbf{1}^\top X_c = \mathbf{0}$.
(iii) is theorem 2.12 and the Eckart–Young theorem.
(iv) $\mathbf{1}^\top X_c = \mathbf{0}$ means the N rows of X_c are linearly dependent. $\square$

2 COURSE NOTE: Erratum: slides 10–11 of lec09

- “ $\Sigma = X^\top X$ is the covariance matrix” needs the factor $1/N$ (or $1/(N-1)$); without it the SVD relation reads $X^\top X = WM^\top MW^\top$ with $M = \text{diag}(\sqrt{(N-1)\lambda_k})$, not $\text{diag}(\sqrt{\lambda_k})$.
- “Computation of the covariance matrix is $O(N^2p^2)$ ” is wrong: $X^\top X$ costs $O(Np^2)$, the same order as the thin SVD; the advantage of the SVD is accuracy, not asymptotic cost.
- “ W is the eigenvectors of X ”: W holds the right singular vectors of X , i.e. the eigenvectors of $X^\top X$.
- Slide 11 calls $\Sigma = W\Lambda W^\top$ the “SVD decomposition”: it is the spectral decomposition (for a PSD matrix it coincides with the SVD).

3 COURSE NOTE: Erratum: dates in Table 1 of pset_analysis

Table 1 of the reading labels the yield columns Jan 03, ..., Jan 10 and the changes Jan 04, 05, 06, 07, 10; but 6–7 January 2001 were a Saturday and a Sunday. The data are those of pages 5–6 of the reading: yields on Jan 02, 03, 04, 05, 08, 09 and changes dated Jan 03, 04, 05, 08, 09, as used in table 7.1.

Table 7.1: Six-day example: centred yield changes A (bp) and the PCA of `pset_analysis`. Last rows: PC variances $\hat{\lambda}_k = d_k^2/(n-1)$, $n = 5$ days, and fractions of variance.

Tenor	A: centred daily changes (bp)					s_j	loadings		
	Jan 03	Jan 04	Jan 05	Jan 08	Jan 09		PC1	PC2	PC3
3M	-5.4	-19.4	-12.4	19.6	17.6	17.70	0.38	0.53	-0.48
6M	-4.6	-14.6	-12.6	14.4	17.4	15.03	0.34	0.44	-0.05
1Y	1.0	-14.0	-14.0	9.0	18.0	14.12	0.36	0.26	0.23
2Y	9.6	-10.4	-16.4	2.6	14.6	13.13	0.35	-0.03	0.46
3Y	13.4	-10.6	-17.6	1.4	13.4	13.99	0.37	-0.13	0.43
5Y	18.6	-11.4	-15.4	-0.4	8.6	14.03	0.35	-0.29	0.12
7Y	20.8	-11.2	-14.2	0.8	3.8	13.92	0.32	-0.37	-0.23
10Y	20.8	-12.2	-11.2	-0.2	2.8	13.37	0.30	-0.38	-0.35
20Y	14.6	-7.4	-7.4	0.6	-0.4	8.99	0.18	-0.28	-0.36
$\hat{\lambda}_k$ (bp ²): 1323.9, 397.2, 34.2, 1.4, 0; total 1756.7							75.4%	22.6%	1.9%

The singular values are $d = (72.77, 39.86, 11.70, 2.39, 0)$. Observations:

- The total variance 1756.7 equals the sum of the nine tenor variances s_j^2 ([theorem 7.2\(iv\)](#)).
- PC5 has zero variance: five centred columns span at most four dimensions ([proposition 7.7\(iv\)](#)). With real data one uses hundreds of days, yet the reading notes the “curious fact” that the loadings look almost the same as in the five-day toy.
- PC1 has positive, roughly equal loadings (a level shift); its standard deviation, 36.4 bp, exceeds that of every single tenor (at most 17.7 bp), because a unit-norm vector with nine positive entries of about 1/3 adds up nine highly correlated moves.
- PC2 is short-minus-long (a change of slope), PC3 is negative at the two ends and positive in the middle (curvature; the overall sign is arbitrary). The daily scores (in bp) are $d_k v_k$:

$$\text{PC1} = (27.2, -37.6, -41.0, 17.2, 34.2) \text{ (up, down, down, up, up),}$$

$$\text{PC2} = (-31.7, -4.5, 3.1, 18.5, 14.6).$$

7.2.3 Centring and scaling: covariance or correlation?

The typical workflow [[2024 L9 @ 21:09](#)]:

1. transform the data to be as close as possible to multivariate normal (mean and covariance then describe the whole distribution: PCA only sees second moments);
2. subtract the mean of each variable and possibly scale it to unit variance;
3. compute the $p \times p$ covariance (or correlation) matrix, symmetric and PSD by construction;
4. diagonalise, order by decreasing eigenvalue, keep the first K components.

Centring is not optional. PCA rotates about the origin [► 2024 L9 @ 25:28]. Feed it raw prices (large positive numbers) and PC1 simply points from the origin to the centre of the cloud, capturing nothing about co-movement [► 2024 L9 @ 26:31]. Slide 13 of lec09 shows the same cloud before and after centring. For daily yield changes the mean is tiny (a fraction of a basis point per day), so centring changes little, but it remains good practice [► 2024 L9 @ 44:13]. In time series the *rolling* mean can itself be unstable, one more reason to work with changes rather than levels.

Scaling is a modelling choice [► 2024 L9 @ 22:10]. Standardising $z_i = (x_i - \mu_i)/\sigma_i$ turns PCA of Σ into PCA of the correlation matrix $R = D_\sigma^{-1} \Sigma D_\sigma^{-1}$, $D_\sigma = \text{diag}(\sigma_1, \dots, \sigma_p)$. Then

- the total variance is $\text{tr } R = p$ and every variable enters on the same footing;
- the eigenvectors of R are *not* the rescaled eigenvectors of Σ (section 7.1.2: PCA is not scale-equivariant);
- without scaling, a variable with much larger variance dominates PC1.

Andreev's toy example: the same data set looks one-dimensional (strongly elongated) in raw units and like a structureless blob after normalising both axes [► 2024 L9 @ 23:12]. Rule of thumb [► 2024 L9 @ 24:24]: if the coordinates have different units (temperature in Seattle versus pressure in Atlanta), standardise; if they have the same units and their relative size is meaningful (grades on two quizzes, yield changes in bp), covariance PCA keeps useful information about relative volatility. For Treasury yield changes Andreev uses covariance PCA [► 2024 L9 @ 45:18]; how to normalise is "very much the art" and sometimes the research question itself. For 2001–2005 Treasury changes the two choices give PC1/PC2/PC3 shares of 84.9/9.2/2.9% (covariance) and 77.1/15.7/3.5% (correlation); 2024 PS3 Problem 4 asks why.

For equity indices the slides use log returns $r_t = \ln(p_t/p_{t-1})$, which are closer to normal than price changes; in that example covariance and correlation PCA give almost the same loadings because the index variances are similar.

7.2.4 How many components, and can we trust them?

PCA always returns an answer, even for pure noise [► 2024 L9 @ 12:44]. Andreev's advice: look at the *eigenvalues*, not only at the eigenvectors [► 2024 L9 @ 13:44]. A component is worth using if its eigenvalue is well separated from the next one, ideally by an order of magnitude; with rolling PCA, plot the eigenvalue (or variance-explained) time series on a log scale and check that the lines never cross [► 2024 L9 @ 14:50]. If PC1 and PC2 swap places under small changes of the data, there is no stable structure and the PCA output "obfuscates rather than clarifies" [► 2024 L9 @ 15:54]. The standard display is the *scree plot* of $\hat{\lambda}_k$ against k (fig. 7.2).

INTUITION: Degenerate perturbation theory (not discussed in class)

Why do nearly equal eigenvalues make eigenvectors unstable? Estimation noise perturbs the covariance, $\hat{\Sigma} = \Sigma + \delta\Sigma$ with $\delta\Sigma = O(N^{-1/2})$. First-order perturbation theory, exactly as for a Hamiltonian in quantum mechanics, gives

$$\delta\lambda_k = \gamma_k^\top \delta\Sigma \gamma_k, \quad \delta\gamma_k = \sum_{j \neq k} \frac{\gamma_j^\top \delta\Sigma \gamma_k}{\lambda_k - \lambda_j} \gamma_j.$$

Eigenvalues are robust, but eigenvectors rotate by an amount inversely proportional to the gap $\lambda_k - \lambda_j$; for a degenerate level the eigenvectors are fixed by the perturbation, not by the data. Andreev’s “move one point and watch PC1 swing” demonstration [► 2024 L9 @ 11:38] is this formula at work.

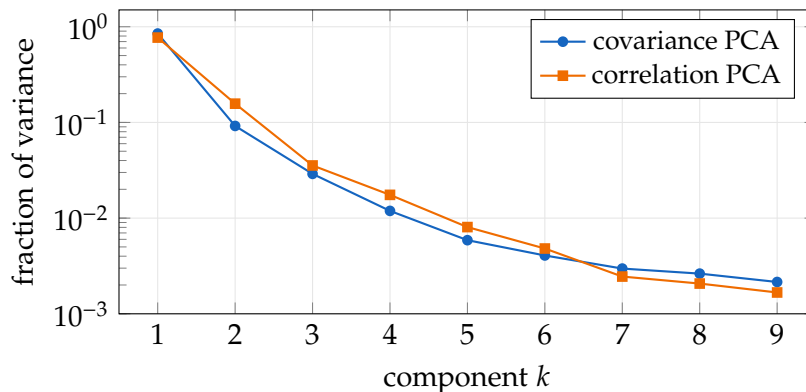


Figure 7.2: Scree plot (log scale) of the PCA of daily changes of 9 Treasury yields, 2001–2005 (2013 Case Study 7; 2024 PS3 Problem 4). The first three eigenvalues are separated by roughly an order of magnitude each, the pattern Andreev wants to see [► 2024 L9 @ 53:42]; standardising moves variance from PC1 to PC2.

Effective dimension. Slide 22 of lec09 (after M. Del Giudice, 2020) summarises a spectrum in one number. With $\pi_k = \lambda_k / \sum_j \lambda_j$,

$$\text{ED} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} = \frac{1}{\sum_k \pi_k^2} \in [1, p],$$

the “effective number of independent directions” (the inverse participation ratio, familiar from localisation physics): ED = 1 for a rank-one matrix such as the 4×4 matrix of ones, 2 for two decoupled blocks of perfectly correlated variables, p for the identity. For the 2001–2005 Treasury covariance PCA, $\text{ED} \approx 1.37$.

Out-of-sample variance explained [► 2024 L9 @ 20:07]. In finance PCA is calibrated on the past and applied to the future. Let W be the loadings estimated on a sample X and let a new sample Y have covariance $\Sigma^Y = W^Y \Lambda^Y (W^Y)^T$. The PCs defined by W have, in the new sample, covariance matrix

$$W^T \Sigma^Y W = D^T \Lambda^Y D, \quad D = (W^Y)^T W \text{ (orthogonal),}$$

$$\text{share of PC } k = \frac{(D^T \Lambda^Y D)_{kk}}{\text{tr } \Lambda^Y} = \frac{\sum_j D_{jk}^2 \lambda_j^Y}{\sum_j \lambda_j^Y}. \quad (7.1)$$

Since $\sum_j D_{jk}^2 = 1$, the out-of-sample variance of calibrated PC k is a weighted average of the new eigenvalues: if the directions have rotated ($D \neq I$), variance leaks from the top factors to lower ones, and by [proposition 7.5](#) the first K calibrated PCs can never explain more than the first K PCs of the new sample itself. The off-diagonal entries of $D^T \Lambda^Y D$ show that calibrated PCs are no longer uncorrelated out of sample [► 2024 L9 @ 53:42]: in the Treasury example the correlations between next-day PC1, PC2, PC3 moves are a few

percent (slide 19: 7.1% between PC1 and PC2), small enough for the factors to be called stable [► 2024 L9 @ 54:47].

Exercise 7.4 (not from the course). In (7.1) show that $\sum_k (D^\top \Lambda^Y D)_{kk} = \text{tr } \Lambda^Y$ and that, for every K , $\sum_{k \leq K} (W^\top \Sigma^Y W)_{kk} \leq \sum_{k \leq K} \lambda_k^Y$. When does equality hold for $K = 1$?

4 COURSE NOTE: A verbal slip

At [▶ 2024 L9 @ 19:07] the speaker says that the “product of eigenvalues” is the variance of the data set; the slide (and [theorem 7.2\(iv\)](#)) has the correct statement: total variance = $\text{tr } \Sigma$ = sum of the eigenvalues. (The product is $\det \Sigma$, the “generalised variance”.)

7.2.5 PCA on financial time series

What is special about finance is that the data are *time series* [▶ 2024 L9 @ 27:32]: each day is one observation and the dimension is the number of variables observed that day. Consequences [▶ 2024 L9 @ 28:32]:

- **Model changes, not levels** [▶ 2024 L9 @ 29:32]: levels (prices, yields) are non-stationary and not centred, while P&L depends on changes after the bet is placed.
- **Hyperparameters** [▶ 2024 L9 @ 31:40]: the length of the look-back window, the weighting of old versus recent observations (rolling window, exponential weights), the set of variables included. More history or more variables is not always better: the marginal data may be noisier or less representative of where the bets will be placed.
- **Sampling frequency** [▶ 2024 L9 @ 33:46]: daily data are the default (long, clean histories); intraday data add observations but also microstructure noise. Volatility scales like $\sqrt{\text{time}}$, so the sampling interval sets its units: annualise daily numbers with $\sqrt{252}$ before comparing [▶ 2024 L9 @ 34:51] ([chapter 13](#)).
- **Rolling PCA**: re-estimating every day turns loadings and eigenvalues into time series. Slowly varying loadings are reassuring; long stable periods interrupted by jumps signal regime changes [▶ 2024 L9 @ 30:33].

7.3 The Treasury yield curve: level, slope and curvature

The market. The US Treasury funds the debt by regular auctions of bills and of 2, 3, 5, 7, 10, (since 2020) 20 and 30-year notes and bonds [▶ 2024 L9 @ 38:00]; the most liquid points also trade as CME futures (TU $\approx$ 1.5y, FV $\approx$ 4y, TY $\approx$ 7y, UXY $\approx$ 10y, US $\approx$ 15y, WN $\approx$ 25y equivalent maturities). The market is the deepest in the world and allows large leverage [▶ 2024 L9 @ 35:55]; the Federal Reserve publishes free, high-quality fitted curves [▶ 2024 L9 @ 36:58]. All these securities are loans to the same borrower, and their prices depend on the expected path of the short rate set by the Fed: yield changes of different maturities are therefore very highly correlated [▶ 2024 L9 @ 39:04]. In the 2024 data (par yields, daily changes) the correlation is 98% between 2y and 3y, 81% between 2y and 10y, 68% between 2y and 30y and 92% between 10y and 30y (slide 17) [▶ 2024 L9 @ 40:07]: “whenever you see a correlation matrix with high concentrations of high correlations, it is a prime candidate for PCA”. The 2013 case study finds the same pattern for 2001–2005: correlations above 0.9 near the diagonal, decaying to 0.24 between 3 months and 20 years [▶ 2013 L15 @ 1:15:26]. Typical daily volatilities are 4–7 bp [▶ 2013 L15 @ 1:14:24].

The yield curve ([section 1.8](#), [fig. 1.3](#)) is usually upward sloping but can be inverted, as in 2023–2024 [▶ 2024 L9 @ 41:11]; the Fed sets its short end (the 50 bp cut of September 2024 moved the leftmost point), the rest is market expectation [▶ 2024 L9 @ 43:12]. A movie of the curve would show “strings flopping around”: mostly moving up and down, a little tilting, a little bending [▶ 2024 L9 @ 42:12].

The three factors. Running PCA on daily yield changes gives, in every period and data set ([fig. 7.3](#), [table 7.2](#)):

- **PC1 = level:** all loadings of the same sign; yields move up or down together. It explains roughly 85–95% of the variance [► 2024 L9 @ 47:24], [► 2013 L15 @ 1:19:47].
- **PC2 = slope:** loadings negative at the short end and positive at the long end (or vice versa): steepening versus flattening; 5–10% [► 2024 L9 @ 48:24].
- **PC3 = curvature:** same sign at both ends, opposite sign in the “belly”: a second difference; 1–3% [► 2013 L15 @ 1:20:51].

The top three factors explain 97–99% of the daily variance.

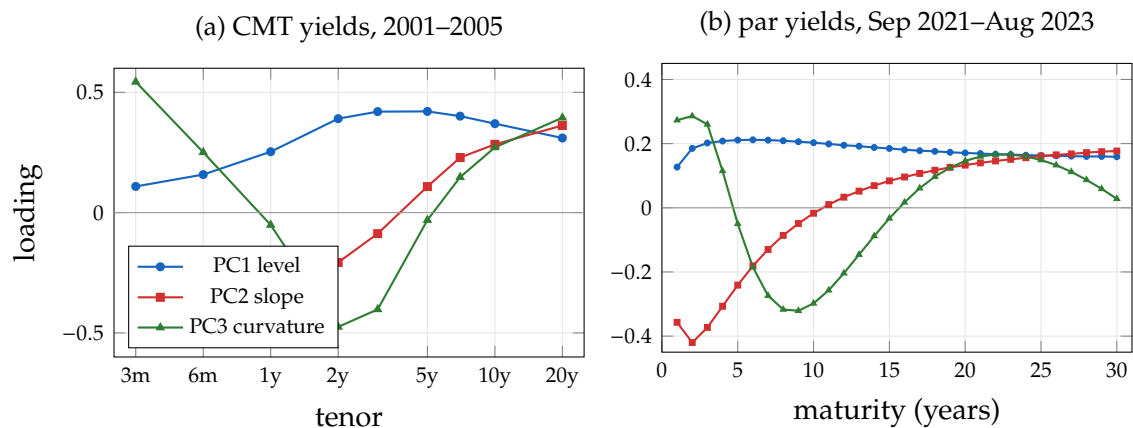


Figure 7.3: Loadings of the first three PCs of daily yield changes (covariance PCA). (a) 9 FRED constant-maturity yields, 2001–2005, princomp output of 2013 Case Study 7. (b) Fed par yields at maturities 1–30 years, 504-day window ending 1 September 2023, computed from fedbondyielddata_20230911.csv as in feddataanalysis.ipynb (signs fixed so that the 20–30y loadings sum to a positive number); PC1–PC3 explain 89.2%, 8.3% and 1.0%. Compare slide 18 of lec09.

Table 7.2: Fraction of the variance of daily yield changes explained by the first three PCs.

Data	Source	PC1	PC2	PC3	top 3
9 CMT yields, 6 days of 2001	2016 reading	75.4	22.6	1.9	99.9
9 CMT yields, 2001–2005, covariance	Case Study 7	84.9	9.2	2.9	97.0
9 CMT yields, 2001–2005, correlation	PS7 (2013), PS3 (2024)	77.1	15.7	3.5	96.3
9 CMT yields, 2009–2013, covariance	Case Study 7	88.6	6.8	1.6	96.9
30 par yields, 2-year window to Sep 2023	computed from the 2024 data	89.2	8.3	1.0	98.5

INTUITION: Modes of a vibrating string

Level, slope and curvature are the analogues of the first normal modes of a string: 0, 1 and 2 sign changes (nodes). This is partly a mathematical consequence of any smooth, strongly and positively correlated curve (not discussed in class): for the correlation kernel $\rho_{ij} = e^{-\kappa|\tau_i - \tau_j|}$ on the same nine tenors with $\kappa = 0.1$, the first four eigenvectors have exactly 0, 1, 2, 3 sign changes and explain 66%, 15%, 9%, 4% (my computation; such kernels are “oscillation matrices” in the sense of Gantmacher and Krein). The economics lies in the *details*: the precise shapes, the explained variances and how they change over time. The same holds for any Karhunen–Loève expansion of a random

field with a smooth covariance.

Reading the loadings as regime indicators [▶ 2024 L9 @ 49:29]. The PC1 loading is small at the very short end and levels off beyond a few years: short rates are anchored by the Fed, long rates by expectations. The maturity at which PC1 “levels off” (in the notebook: the first tenor with loading above 0.15) was 1–3 years until the financial crisis, rose to 5–7 years when the Fed promised to keep rates low for long (quantitative easing, forward guidance), fell back, jumped again with the COVID response and decreased during the 2022 tightening [▶ 2024 L9 @ 50:33], [▶ 2024 L9 @ 1:14:43]. Similarly the 2-year PC1 loading collapsed after 2012 (the Fed was not moving) and rebounded when policy became uncertain; the *location of the belly* of PC3 (usually near 10y in the US) shifts with regimes [▶ 2024 L9 @ 51:34]. Andreev’s macro reading of curvature [▶ 2024 L9 @ 1:17:51]: the front end is driven by the central bank, the intermediate part by inflation expectations, the long end by growth expectations; investors also differ by segment (speculators and short-term funding at 2y, corporate issuance at 5–10y, pension funds and insurers matching long liabilities at the long end) [▶ 2024 L9 @ 1:18:57]. In an emerging market such as Mexico the belly sits nearer 5 years, reflecting shorter investment horizons. Once the level and slope are hedged out, these small but structured features emerge from “the big noise of rates going up and down” [▶ 2024 L9 @ 1:19:59].

The last PC is interesting too [▶ 2013 L15 @ 1:21:57]. In the 2013 case study PC9, with the smallest variance (0.8 bp/day), is essentially the 7-year yield against a weighted average of the 5- and 10-year: the most stable linear combination of yields, i.e. the best hedge of the 7-year point with its neighbours. Cumulating the scores shows the history of the factors [▶ 2013 L15 @ 1:23:00]: over 2001–2005 cumulative PC1 fell from 0 to about –8 and came back (rates fell, then rose), cumulative PC2 rose to about 6 and fell back (the curve steepened, then flattened), curvature moved much less [▶ 2013 L15 @ 1:24:00].

IN FINANCE: Nelson–Siegel–Svensson curves (not discussed in class)

The Fed file `fedbondyieldldata_20230911.csv` is the Gurkaynak–Sack–Wright data set: daily zero-coupon yields (SVENY01–30, continuously compounded), par yields (SVENPY), instantaneous and one-year forward rates (SVENF, SVEN1F) from 1961, and the parameters BETA0–BETA3, TAU1, TAU2 of the fitted Svensson curve

$$y(n) = \beta_0 + \beta_1 \frac{1 - e^{-n/\tau_1}}{n/\tau_1} + \beta_2 \left[\frac{1 - e^{-n/\tau_1}}{n/\tau_1} - e^{-n/\tau_1} \right] + \beta_3 \left[\frac{1 - e^{-n/\tau_2}}{n/\tau_2} - e^{-n/\tau_2} \right]$$

(I checked that this reproduces SVENY to four decimals). The three Nelson–Siegel basis functions are a parametric version of level (β_0), slope (β_1 , which dies out with maturity) and curvature (β_2 , a hump); β_3 adds a second hump. Because the published curves are smooth fits, their PCA loadings are smoother than those of raw bond yields.

CASE STUDY: Reading `pset_analysis` and its R code `pset_rcode`: five years of yields

Goal. Reproduce the six-day example and extend it to 2001–2005. **Data.** FRED constant-maturity yields DGS3M0, ..., DGS20 (script `fm_freddata.r`), 1247 daily changes. **Method.** `svd()` of the centred days×tenors matrix, $\hat{\lambda}_k = d_k^2/(n-1)$, bar plots of loadings, daily and cumulative PC1/PC2 series, `pairs()` and interactive 3D plots of (PC1, PC2, PC3); then the built-in `princomp()` and `screplot()`, which give the

same result up to the signs of the loadings ([remark 7.3](#)). **Results.** Total variance 0.0306 (%²); PC1/PC2/PC3 explain 84.8/9.3/2.9%; PC1 loadings 0.11 (3m) rising to 0.42 (3y–5y) and 0.31 (20y); cumulative PC1 falls over the first half of the period and recovers, cumulative PC2 (the spread) rises then falls. **Lesson.** Five days already reveal the structure that five years confirm. *Two small inconsistencies:* in the five-year section the text says n is “the number of columns (days)” but the days are now rows ($\hat{\lambda}_k = d_k^2 / (\text{NROW}(X) - 1)$ in the code); and the text calls PC2 “shorter minus longer” although in this SVD its loadings are negative at the short end, i.e. long minus short. Signs are arbitrary.

CASE STUDY: `feddataanalysis.ipynb`: Andreev’s notebook on the Fed data

Data. Par yields SVENPY01–30 from the first day with a 30-year yield (25 Nov 1985) to 1 Sep 2023, 9430 days, daily differences. **Method** (Python/pandas): 504-day (2-year) rolling covariance of yield changes, `numpy.linalg.eigh` each day, signs normalised so that the 20–30y loadings sum to a positive number; variance explained per factor (area and log plots); PC1 loading stability for 2, 5, 10, 30y; “maturity for PC1 leveling” (first tenor with loading > 0.15), PC2 pivot and PC3 belly; next-day yield changes projected on today’s loadings (the out-of-sample check ([7.1](#))); out-of-sample variance explained with loadings two years old; a PC1/PC2-neutral 5s7s10s fly and a mean-reversion strategy on it. **Results.** Correlation matrix of slide 17 (reproduced exactly); out-of-sample correlations between next-day PC1, PC2, PC3 moves: 4.0% (PC1/PC2), –0.2% (PC1/PC3), –9.0% (PC2/PC3) (the slide shows 7.1%, –1.6%, –7.3% from another run); strategy Sharpe ratio 0.10 (slide: 0.09). **Lesson.** The last cell repeats the strategy with the signal *not* delayed by one day and gets spectacular, unrealistic returns: a textbook demonstration of look-ahead bias. Andreev recommends the notebook as a template for a final project [[2024 L9 @ 1:11:29](#)].

7.4 PC portfolios: curves, flies and hedging

7.4.1 Loadings as portfolio weights

Why care about PC2 and PC3 if PC1 explains 90% of the variance? Because they are where relative-value trading happens [[2024 L9 @ 55:49](#)]. The six liquid tenors are so correlated that trading any of them outright is essentially a bet on PC1 [[2024 L9 @ 56:51](#)]; PCA manufactures from them a handful of *uncorrelated* portfolios, i.e. more independent bets [[2024 L9 @ 57:54](#)].

Positions in bonds are measured in *dv01*: \$1 dv01 of a bond is the notional whose value changes by \$1 for a 1 bp change of its yield ([section 1.8.4](#)). Let h_i be the dv01 held in tenor i (positive = long the bond, which gains when the yield falls). To first order the daily P&L in dollars is

$$\text{PnL} = -\mathbf{h}^\top \Delta \mathbf{y} = - \sum_k (\mathbf{h}^\top \boldsymbol{\gamma}_k) p_k, \quad \Delta \mathbf{y} \text{ in bp},$$

using $\Delta \mathbf{y} = \sum_k p_k \boldsymbol{\gamma}_k$ ([theorem 7.2\(iii\)](#), centring neglected). The *exposure* of the portfolio to PC k is $e_k = \mathbf{h}^\top \boldsymbol{\gamma}_k$. Buying “the PC1 portfolio” means holding every bond in proportion to its PC1 loading [[2024 L9 @ 58:59](#)]; a portfolio with $e_1 = 0$ is *PC1-neutral*, one with $e_1 = e_2 = 0$ is PC1/PC2-neutral.

PC-neutral curves (“PC1 residuals”). Given two tenors a, b , the portfolio $h = e_a - \alpha e_b$ (\$1 dv01 of a against α dv01 of b) is PC1-neutral iff

$$\gamma_{a1} - \alpha \gamma_{b1} = 0 \iff \alpha = \frac{\gamma_{a1}}{\gamma_{b1}}, \tag{7.2}$$

where γ_{ik} is the loading of tenor i on PC k . Example: the 2s10s *steepener*, short \$1 dv01 of 10-year, long α dv01 of 2-year: it profits when the curve steepens and is insensitive to parallel (PC1) moves. Such curve trades exist in the market as standard products [► 2024 L9 @ 1:00:00].

PC-neutral flies (“PC2 residuals”). Given three tenors, $h = \alpha_a e_a + \alpha_b e_b - e_c$ is neutral to PC1 and PC2 iff

$$\begin{pmatrix} \gamma_{a1} & \gamma_{b1} \\ \gamma_{a2} & \gamma_{b2} \end{pmatrix} \begin{pmatrix} \alpha_a \\ \alpha_b \end{pmatrix} = \begin{pmatrix} \gamma_{c1} \\ \gamma_{c2} \end{pmatrix}. \tag{7.3}$$

Example: the 2s10s30s “belly” fly buys \$1 dv01 of 10-year and sells α_2 of 2-year and α_{30} of 30-year: a pure bet on curvature.

7.4.2 Leverage, and why higher PCs are the interesting ones

A PC2 portfolio moves perhaps a third as much as the PC1 portfolio per unit notional; scaling it up three times gives the same daily standard deviation and a return stream uncorrelated with the market direction [► 2024 L9 @ 1:01:03]. Far from being irrelevant, the higher PCs are “where a lot of the money is made” [► 2024 L9 @ 1:03:07]; beyond PC3, however, the leverage needed becomes too expensive. The strategy requires a liquid market and cheap financing (repo); regulators (the CFTC and others) watch the resulting leveraged long/short Treasury positions as a possible systemic risk [► 2024 L9 @ 1:02:05]. In equities PC1 is much less dominant, so less leverage is needed; market-neutral long/short equity funds are precisely “hedging out PC1” [► 2024 L9 @ 1:04:09].

Pros of PC-residual portfolios	Cons (slide 21)
Bets unrelated to the main market move (PC1).	Transaction cost per unit of volatility is much higher.
More effectively uncorrelated bets (diversity).	Weights are model-dependent and change over time.
Low volatility per notional: a mispricing may have a better signal-to-noise ratio.	Much larger notionals, i.e. leverage and balance sheet; financing (overnight repo) can dry up in stress.

Model risk. The hedge ratios are only as good as the loadings. When COVID hit and the Fed made an emergency cut, the curve dynamics changed overnight [► 2024 L9 @ 1:07:21]; a leveraged PC2 position computed with stale loadings is no longer PC1-neutral and its P&L is dominated by unintended level risk, while there is no history yet to recalibrate on [► 2024 L9 @ 1:08:21]. Possible answers: implied (option) markets, or calibrating on past periods of a similar regime (e.g. all Fed cutting cycles), with its own pitfalls [► 2024 L9 @ 1:09:25]. Event days also matter for risk: daily yield volatility is typically 5–6 bp, but the market may price twice that on the day of a major inflation release [► 2024 L9 @ 1:06:18].

5 COURSE NOTE: Erratum: the dv01 box on slide 20 of 1ec09

The box defines “buy \$1 dv01 of bond” as the notional such that the position *earns* \$1 when the yield goes *up* 1 bp. A long bond position loses when yields rise; the examples on the same slide (a steepener sells the 10-year bond, a belly fly buys the 10-year) use the standard convention adopted here. The notebook avoids the issue by computing returns directly in yield space (+weights· Δy), i.e. as positions in yields rather than in bonds; any sign convention works if used consistently.

6 COURSE NOTE: Erratum: slide 20 of 1ec09

The fly system on the slide has $W_{b,0}$ in the lower-right entry; it must be $W_{b,1}$ (the PC2 loading of tenor b), as in (7.3). (The slides index PCs from 0.)

Example 7.8 (Hedge ratios and volatilities from the 2001–2005 PCA). Rebuild $\Sigma = \Gamma \Lambda \Gamma^T$ from the loadings and standard deviations of 2013 Case Study 7 (this reproduces the case-study volatilities and correlations to three digits) and measure risk as the standard deviation of daily P&L per \$1 dv01 of the reference bond:

Trade (PC-neutrality)	Weights (dv01)
2s10s curve (PC1)	–1 of 10y, +0.947 of 2y
2s5s10s fly (PC1, PC2)	+1 of 5y, –0.423 of 2y
5s7s10s fly (PC1, PC2)	+1 of 7y, –0.367 of 5y

The 5s7s10s fly is the notebook’s trade and is close to PC9 of the case study. To run it at the risk of an outright 7-year position one needs about 6.4 times the dv01 on the belly, plus the offsetting wings.

Diversity of PC portfolios. Slide 23 computes the effective dimension of the correlation matrix of the returns of PC portfolios: PC1 alone 1, PC1+PC2 1.99, PC1+PC2+PC3 2.97 (almost exactly uncorrelated also out of sample), and 5.28 when the residual portfolios left after removing three PCs are added. Higher-order residuals also tend to *mean-revert*. The notebook's strategy [► 2024 L9 @ 1:10:25]: build the PC1/PC2-neutral 5s7s10s fly every day from the current loadings; signal $s = z$ -score of its cumulative return against an exponentially weighted mean (half-life 100 days) and standard deviation (half-life 60 days), delayed by one day; position = $-s$. The Sharpe ratio is only about 0.1: the point is the method, not the strategy.

7.4.3 Hedging a portfolio against PC1

For an equity portfolio with weights V on p indices (returns, so $\text{PnL} = V^T r$), the exposure to PC n is $V^T \gamma_n$. Projecting out the first component,

$$V' = V - (V^T \gamma_1) \gamma_1, \quad V'^T \gamma_1 = 0, \quad \text{Var}(V'^T r) = V'^T \Sigma V' = V^T \Sigma V - \lambda_1 (V^T \gamma_1)^2,$$

the last identity because $(I - \gamma_1 \gamma_1^T) \Sigma (I - \gamma_1 \gamma_1^T) = \Sigma - \lambda_1 \gamma_1 \gamma_1^T$. If the expected return ("alpha") of the strategy is roughly orthogonal to PC1, hedging keeps the alpha and removes most of the risk: higher Sharpe ratio.

CASE STUDY: European stock indices (slides 26–34; not covered in class [► 2024 L9 @ 1:12:30])

Data. Daily closes of AEX, DAX, FTSE MIB, IBEX, OMX, Euro Stoxx 50 and CAC, 4 Jan 2010 to 16 Sep 2016 (same time zone and currency), rebased to 100 EUR and converted to demeaned log returns. Correlations 72–98%. **PCA.** PC1 explains 89% with nearly equal loadings: a "Euro stocks" factor. PC2 (about 5%, half of the rest) separates North (Netherlands, Germany, Sweden: positive) from South (Italy, Spain: negative), France near zero. Covariance and correlation PCA agree. Rolling 6-month PCA: top three factors explain 90–98% throughout, PC1 loadings stable, PC2 fairly stable. **Hedging example** (thesis: German stocks outperform, 2010, target 100m EUR annualised volatility): buying the DAX (vol 20%) needs 500m and returned 75m (Sharpe 0.75); the PC1-neutral portfolio $e_{\text{DAX}} - 0.377 \gamma_1$ (weights +0.86 DAX, about -0.14 each other index) has vol 6.7%, needs 1,500m and returned about 230m (Sharpe 2.3). **Other uses:** North-versus-South trades; Frankfurt opens before Madrid, so a PCA model can price Spain by proxy just before its open. *Slide typos:* the rebasing date is 4 Jan 2010 (not "1/4/2016"), and the legend entry "Swiss" on slides 27 and 31 is the Euro Stoxx 50.

IN FINANCE: Limitations (slide 35)

PCA always gives an answer, so test robustness to the window and to the set of variables; it works best on data close to multivariate normal; and it implies no causality. To predict a dependent variable, use regression (chapter 5). Andreev's closing advice [► 2024 L9 @ 1:22:03]: data that are hard to get (illiquid markets, emerging-market curves) are where fewer competitors look, hence where quantitative modelling may still pay.

7 COURSE NOTE: Differences between slides and speech

At [► 2024 L9 @ 1:10:25] the speaker describes the strategy as "momentum/mean reversion for PC1"; slide 24 and the notebook apply mean reversion to the PC1/PC2-neutral fly.

8 COURSE NOTE: Erratum: slide 32 of lec09

"The stdev of the resulting portfolio = $V'^T \Sigma V'$ "; that is the variance; the standard deviation is $\sqrt{V'^T \Sigma V'}$.

7.5 Linear factor models

7.5.1 Definition and covariance structure

Definition 7.9 (Linear factor model). For m assets observed over T periods [► 2013 L15 @ 1:06], [► 2013 L15 @ 3:13]:

$$x_{i,t} = \alpha_i + \beta_{1,i}f_{1,t} + \cdots + \beta_{K,i}f_{K,t} + \epsilon_{i,t}, \quad \mathbf{x}_t = \boldsymbol{\alpha} + B\mathbf{f}_t + \boldsymbol{\epsilon}_t, \quad (7.4)$$

with intercepts $\boldsymbol{\alpha} \in \mathbb{R}^m$, factor loadings $B \in \mathbb{R}^{m \times K}$ (row i is $\beta_i^\top$), common factors $\mathbf{f}_t \in \mathbb{R}^K$, $K \ll m$, and specific factors $\boldsymbol{\epsilon}_t$. Assumptions [► 2013 L15 @ 5:19]: $\{\mathbf{f}_t\}$ is covariance stationary with $\mathbb{E}[\mathbf{f}_t] = \boldsymbol{\mu}_f$, $\text{Cov} \mathbf{f}_t = \Omega_f$; $\{\boldsymbol{\epsilon}_t\}$ is white noise with $\mathbb{E}[\boldsymbol{\epsilon}_t] = \mathbf{0}$, $\text{Cov} \boldsymbol{\epsilon}_t = \Psi = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$, $\text{Cov}(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_{t'}) = \mathbf{0}$ for $t \neq t'$; and $\text{Cov}(\mathbf{f}_t, \boldsymbol{\epsilon}_{t'}) = \mathbf{0}$ for all t, t' .

The key assumption is that Ψ is *diagonal*: all co-movement between assets goes through the factors.

Proposition 7.10 (Moments of the factor model). $\mathbb{E}[\mathbf{x}_t | \mathbf{f}_t] = \boldsymbol{\alpha} + B\mathbf{f}_t$, $\text{Cov}[\mathbf{x}_t | \mathbf{f}_t] = \Psi$, and unconditionally

$$\mathbb{E}[\mathbf{x}_t] = \boldsymbol{\alpha} + B\boldsymbol{\mu}_f, \quad \Sigma_x = \text{Cov}[\mathbf{x}_t] = B\Omega_f B^\top + \Psi. \quad (7.5)$$

Proof. By the law of total covariance, $\text{Cov}[\mathbf{x}_t] = \mathbb{E}[\text{Cov}(\mathbf{x}_t | \mathbf{f}_t)] + \text{Cov}(\mathbb{E}[\mathbf{x}_t | \mathbf{f}_t]) = \Psi + B\Omega_f B^\top$; directly, $\text{Cov}(B\mathbf{f}_t + \boldsymbol{\epsilon}_t) = B\Omega_f B^\top + \Psi + 2\text{Cov}(B\mathbf{f}_t, \boldsymbol{\epsilon}_t)$ and the cross term vanishes [► 2013 L15 @ 7:25]. $\square$

Parameter count [► 2013 L15 @ 8:27]. An unrestricted covariance has $m(m+1)/2$ parameters; the factor structure has mK (loadings) $+ K(K+1)/2$ (Ω_f) $+ m$ (Ψ) [► 2013 L15 @ 10:39]. For $m = 500$ stocks and $K = 10$ factors: 125,250 versus 5,555. Moreover each new asset adds only $K+1$ parameters. This is why factor models are used for portfolio optimisation, where the (inverse) covariance matrix drives the allocation (chapter 10), and for risk management, where scenarios are naturally expressed as factor moves [► 2013 L15 @ 16:54].

Three regression views (chapter 5). *Cross-sectional*: fix t ; (7.4) is a regression of the m returns on the columns of B with “coefficients” $\mathbf{f}_t$ [► 2013 L15 @ 4:18]. *Time series*: fix asset i ; $x_i = \mathbf{1}_T \alpha_i + F\beta_i + \epsilon_i$ with $F \in \mathbb{R}^{T \times K}$ the factor history and $\text{Cov} \epsilon_i = \sigma_i^2 I_T$: an ordinary regression satisfying the Gauss–Markov assumptions [► 2013 L15 @ 11:41]. *Multivariate*: stacking all assets, $X = \mathbf{1}_T \boldsymbol{\alpha}^\top + FB^\top + E$ with $X = [x_1 | \cdots | x_m] \in \mathbb{R}^{T \times m}$ [► 2013 L15 @ 12:45] (the slides write B for the $K \times m$ matrix of the time-series form, the transpose of the cross-sectional B). Whether factors or loadings are observed defines the three families below [► 2013 L15 @ 0:00].

7.5.2 Identification

When the factors are latent, the model is not identified [► 2013 L15 @ 42:40]: for any invertible $H \in \mathbb{R}^{K \times K}$, $\mathbf{f}_t^* = H\mathbf{f}_t$ and $B^* = BH^{-1}$ give the same $\mathbf{x}_t$, with $\boldsymbol{\mu}_f^* = H\boldsymbol{\mu}_f$ and $\Omega_f^* = H\Omega_f H^\top$ [► 2013 L15 @ 43:49]. One uses this freedom to normalise [► 2013 L15 @ 44:50]:

- *orthonormal factors*: if $\Omega_f = \Gamma_f \Lambda_f \Gamma_f^\top$, take $H = \Lambda_f^{-1/2} \Gamma_f^\top$, so that

$$\Omega_f^* = \Lambda_f^{-1/2} \Gamma_f^\top \Gamma_f \Lambda_f \Gamma_f^\top \Gamma_f \Lambda_f^{-1/2} = I_K;$$

- *zero-mean factors*: absorb $B\boldsymbol{\mu}_f$ into the intercept, $\boldsymbol{\alpha}^* = \boldsymbol{\alpha} + B\boldsymbol{\mu}_f$.

Then $\Sigma_x = BB^T + \Psi$ [► 2013 L15 @ 46:00]. An indeterminacy remains: $B \mapsto BQ$ with Q orthogonal leaves BB^T unchanged. This rotational freedom is exploited to make factors interpretable (section 7.7.1).

7.6 Macroeconomic and fundamental factor models

7.6.1 Observed factors: Sharpe's single-index model

The simplest factor model [► 2013 L15 @ 13:46]: excess returns regressed on the excess return $R_{M,t}$ of the market index,

$$x_{i,t} = \alpha_i + \beta_i R_{M,t} + \epsilon_{i,t}, \quad \Sigma_x = \sigma_M^2 \beta \beta^T + \Psi, \quad \sigma_M^2 = \text{Var}(R_{M,t}).$$

The covariance between two different assets is $\beta_i \beta_j \sigma_M^2$ [► 2013 L15 @ 14:48]: a rank-one matrix plus a diagonal, trivial to build even for thousands of securities [► 2013 L15 @ 15:53]. β_i measures the systematic risk of asset i (the CAPM of chapter 10 adds the equilibrium restriction $\alpha_i = 0$).

Estimation [► 2013 L15 @ 17:57]. The time-series regression of each asset satisfies Gauss–Markov, so OLS gives BLUE estimates $(\hat{\alpha}_i, \hat{\beta}_i)$ (and the MLE under normality); unbiased residual variances are $\hat{\sigma}_i^2 = \hat{\epsilon}_i^T \hat{\epsilon}_i / (T - 2)$; $\hat{\sigma}_M^2$ is the sample variance of R_M ; and $\hat{\Sigma}_x = \hat{\sigma}_M^2 \hat{\beta} \hat{\beta}^T + \hat{\Psi}$.

INTUITION: What PCA sees in a single-index world (not discussed in class)

If all specific variances are equal, $\Psi = \psi I$, then $\Sigma_x = \sigma_M^2 \beta \beta^T + \psi I$ has top eigenvector $\beta / \|\beta\|$ with eigenvalue $\sigma_M^2 \|\beta\|^2 + \psi$, and all other eigenvalues equal to ψ . PCA recovers the market factor exactly and finds nothing else (a degenerate remainder, remark 7.3). With unequal σ_i^2 the recovery is only approximate: this is the difference between PCA and factor analysis (section 7.7.2).

7.6.2 Macroeconomic multifactor models

Any observable series is a candidate factor, judged by how much it contributes to explaining returns and covariances [► 2013 L15 @ 19:00]: market risk, inflation (CPI, PPI, commodity prices), industrial production, money growth, interest rates, housing starts, unemployment. Chen, Roll and Ross (1986) used not the levels but the *surprises*: unanticipated changes, computed as innovations of a vector autoregression fitted to the macro variables (chapter 9) [► 2013 L15 @ 20:03], [► 2013 L15 @ 21:07]. Since F is observed, each asset's loadings are again time-series OLS estimates, with $\hat{\sigma}_i^2 = \hat{\epsilon}_i^T \hat{\epsilon}_i / (T - K - 1)$, $\hat{\Omega}_f$ the sample covariance of the factors and $\hat{\Sigma}_x = \hat{B} \hat{\Omega}_f \hat{B}^T + \hat{\Psi}$ [► 2013 L15 @ 22:10]. If residual variances differ or residuals are correlated, generalised least squares is the efficient estimator [► 2013 L15 @ 23:15].

7.6.3 Fundamental factor models

Here the *loadings* are observed asset attributes (industry, market capitalisation, dividend yield, book-to-price, earnings-to-price, ...) and the factor realisations are estimated. This is the BARRA approach of Barr Rosenberg [► 2013 L15 @ 24:17], still the standard of commercial risk models.

9 COURSE NOTE: Erratum: slide 22 of lecnote15

The slide takes $H = \Gamma \Lambda^{-1/2}$; then $H \Omega_f H^T = \Gamma \Lambda^{-1/2} \Gamma \Lambda \Gamma^T \Lambda^{-1/2} \Gamma^T$, which is not I_K in general. The correct choice is $H = \Lambda^{-1/2} \Gamma^T$ (or $Q \Lambda^{-1/2} \Gamma^T$ for any orthogonal Q).

10 COURSE NOTE: Erratum: slide 11 of lecnote15

The estimator of the multifactor covariance is written $\hat{\sigma}_M^2 \hat{B} \hat{\Omega}_f \hat{B}^T + \hat{\Psi}$; the factor $\hat{\sigma}_M^2$ is a leftover of the single-index slide. Correct: $\hat{\Sigma}_x = \hat{B} \hat{\Omega}_f \hat{B}^T + \hat{\Psi}$ (the factor variances are already inside $\hat{\Omega}_f$).

Industry factor model [► 2013 L15 @ 29:35]. Split the m stocks into K industries and let $\beta_{i,k} = 1$ if stock i is in industry k , 0 otherwise: B is a known, time-invariant matrix of indicators [► 2013 L15 @ 30:37]. Oil stocks move together, utilities (regulated, low volatility) move together and differently. For each t , estimate f_t by the cross-sectional regression of x_t on B [► 2013 L15 @ 31:40]:

$$\hat{f}_t = (B^T B)^{-1} B^T x_t, \quad B^T B = \text{diag}(m_1, \dots, m_K), \quad (7.6)$$

where m_k is the number of stocks in industry k : $\hat{f}_{k,t}$ is simply the *average return of industry k* [► 2013 L15 @ 32:43]. Then $\hat{e}_t = x_t - B\hat{f}_t$, $\hat{\Omega}_f$ and $\hat{\Psi}$ are sample (co)variances of the $\hat{f}_t$ and of the residuals, and $\hat{\Sigma} = B\hat{\Omega}_f B^T + \hat{\Psi}$ [► 2013 L15 @ 33:48].

Heteroscedasticity. Stocks have different specific variances, so OLS in (7.6) is inefficient; the GLS estimator weights each stock by the inverse of its noise,

$$\hat{f}_t = (B^T \Psi^{-1} B)^{-1} B^T \Psi^{-1} x_t,$$

with Ψ estimated in a first pass. Kempthorne stresses the analogy [► 2013 L15 @ 34:53]: mean-variance optimisation weights assets by $\Sigma^{-1}\mu$ (reward penalised by variance), GLS weights observations by the inverse noise variance: in both cases noisy inputs must count less.

Factor-mimicking portfolios [► 2013 L15 @ 35:56]. The estimate $\hat{f}_t = Hx_t$ is linear in the returns [► 2013 L15 @ 37:00], so each row of H is a portfolio of the m assets. Normalising the row weights to sum to one (one unit of capital) gives a tradeable portfolio whose return tracks the factor [► 2013 L15 @ 39:04].

Proposition 7.11 (Mimicking portfolios have unit exposure). Let $H = (B^T \Psi^{-1} B)^{-1} B^T \Psi^{-1}$ and let h_k^T be its k -th row. Then $HB = I_K$, so under (7.4) with $\alpha = 0$ the portfolio h_k has return $f_{k,t} + h_k^T \epsilon_t$: unit exposure to factor k and zero to the others. Among all portfolios with these exposures ($B^T h = e_k$) it has the smallest specific variance $h^T \Psi h$ (not discussed in class).

In the industry model with OLS, h_k is the equal-weighted portfolio of industry k .

Fama–French factors [► 2013 L15 @ 25:23]. Fama and French turned attributes into factor returns by sorting. At each t , rank the stocks by an attribute (market cap for *size*, book-to-market for *value versus growth*) [► 2013 L15 @ 26:25], split into quintiles, and define the factor as the return of the equal-weighted top quintile minus the bottom quintile (a zero-cost hedge portfolio) [► 2013 L15 @ 27:32]. Then estimate each stock's loadings on these factors by time-series regressions. In the published factors the sign is chosen so that the factor has a positive average return, e.g. small minus big (SMB) and high minus low book-to-market (HML) [► 2013 L15 @ 28:35]. Value stocks are cheap relative to book equity; growth stocks are expensive because the price anticipates future growth.

7.7 Statistical factor models

Now neither factors nor loadings are observed: only the returns $x_1, \dots, x_T$ [► 2013 L15 @ 40:18]. Both methods below model the sample covariance matrix.

11 COURSE NOTE: Erratum: slide 17 of lecnote15

" $\hat{\Sigma} = B' \hat{\Omega}_f B + \hat{\Psi}$ " has the transposes swapped ($B' \hat{\Omega}_f B$ is not even defined, B being $m \times K$); correct: $\hat{\Sigma} = B \hat{\Omega}_f B^T + \hat{\Psi}$.

Proof. $HB = (B^T \Psi^{-1} B)^{-1} B^T \Psi^{-1} B = I_K$. For the minimum, the Lagrangian $\frac{1}{2} h^T \Psi h - \ell^T (B^T h - e_k)$ gives $h = \Psi^{-1} B \ell$ and $B^T \Psi^{-1} B \ell = e_k$, i.e. $h = \Psi^{-1} B (B^T \Psi^{-1} B)^{-1} e_k = h_k$ (Gauss–Markov in portfolio language). $\square$

7.7.1 Factor analysis by maximum likelihood

Take the normalised model with Gaussian variables [► 2013 L15 @ 47:02]: $f_t \sim N_K(\mathbf{0}, I_K)$, $\epsilon_t \sim N_m(\mathbf{0}, \Psi)$, independent over time, so that $x_t \sim N_m(\alpha, \Sigma_x)$ i.i.d. with $\Sigma_x = BB^\top + \Psi$ [► 2013 L15 @ 48:05]. The log-likelihood is [► 2013 L15 @ 49:05]

$$\ell(\alpha, \Sigma_x) = -\frac{Tm}{2} \ln(2\pi) - \frac{T}{2} \ln \det \Sigma_x - \frac{1}{2} \sum_{t=1}^T (x_t - \alpha)^\top \Sigma_x^{-1} (x_t - \alpha), \quad (7.7)$$

to be maximised over α, B, Ψ subject to $\Sigma_x = BB^\top + \Psi$. The maximiser of α is $\bar{x}$, and then $\ell = -\frac{T}{2} [m \ln 2\pi + \ln \det \Sigma_x + \text{tr}(\Sigma_x^{-1} S)]$ with S the (divisor- T) sample covariance.

The EM algorithm [► 2013 L15 @ 51:06]. The likelihood is complicated because the factors are hidden; if they were observed, estimating B and Ψ would be ordinary regression. The EM algorithm (Dempster, Laird and Rubin 1977) alternates the two: estimate the hidden variables given the parameters (E-step), re-estimate the parameters as if the hidden variables were known (M-step); each iteration increases the likelihood [► 2013 L15 @ 52:10]. For centred data (not discussed in class; I verified numerically that the likelihood increases monotonically):

$$\begin{aligned} \text{E: } \beta &= B^\top \Sigma_x^{-1}, \quad \mathbb{E}[f_t | x_t] = \beta x_t, \quad \mathbb{E}[f_t f_t^\top | x_t] = I - \beta B + \beta x_t x_t^\top \beta^\top; \\ \text{M: } B_{\text{new}} &= S \beta^\top (I - \beta B + \beta S \beta^\top)^{-1}, \quad \Psi_{\text{new}} = \text{diag}(S - B_{\text{new}} \beta S). \end{aligned}$$

Time dependence can be added through the linear state-space framework of the Kalman filter (chapter 9).

Factor scores [► 2013 L15 @ 53:16]. With $\hat{\alpha}, \hat{B}, \hat{\Psi}$ in hand, estimate each f_t by the cross-sectional GLS regression of $x_t - \hat{\alpha}$ on $\hat{B}$: $\hat{f}_t = (\hat{B}^\top \hat{\Psi}^{-1} \hat{B})^{-1} \hat{B}^\top \hat{\Psi}^{-1} (x_t - \hat{\alpha})$. These estimated factor series can feed other models, e.g. statistical factors of commodity returns used to explain stocks [► 2013 L15 @ 54:17], and can be rescaled into factor-mimicking portfolios.

Rotations and interpretation [► 2013 L15 @ 56:23]. Factor analysis was born in psychometrics: Spearman's analysis of intelligence tests, where one looks for the "dimensions of intelligence". The orthogonal indeterminacy $B \mapsto BQ$ is used to choose rotated loadings that are easier to interpret (e.g. the varimax criterion, which pushes each loading towards 0 or $\pm$ large) [► 2013 L15 @ 57:30].

How many factors? The likelihood-ratio test [► 2013 L15 @ 58:36]. Compare the unrestricted MLE ($\tilde{\alpha} = \bar{x}, \tilde{\Sigma} = S$) with the K -factor MLE:

$$\begin{aligned} \text{LR}(K) &= 2[\ell(\tilde{\alpha}, \tilde{\Sigma}) - \ell(\hat{\alpha}, \hat{B}\hat{B}^\top + \hat{\Psi})] \approx \chi_\nu^2, \\ \nu &= \frac{m(m+1)}{2} - \left[mK + m - \frac{K(K-1)}{2} \right] = \frac{(m-K)^2 - (m+K)}{2}, \end{aligned}$$

where the K -factor model has $mK + m$ parameters minus $K(K-1)/2$ for the rotational freedom. H_0 (K factors suffice) is rejected for large LR. For $m = 9$: $\nu = 12, 6, 1$ for $K = 3, 4, 5$, and $\nu < 0$ for $K = 6$, so at most five factors can be fitted (R's `factanal` also applies Bartlett's small-sample correction). Unlike PCA, the ML solution is scale-equivariant: analysing covariances or correlations gives equivalent answers (not discussed in class).

7.7.2 PCA as an approximate factor model; the principal factor method

12 COURSE NOTE: Erratum: slide 24 of lecnote15

The log-likelihood is written with $-\frac{TK}{2} \log(2\pi) - \frac{K}{2} \log |\Sigma|$; taking the logarithm of the likelihood on slide 23 gives $-\frac{Tm}{2} \ln(2\pi) - \frac{T}{2} \ln |\Sigma_x|$, as in (7.7).

Partition the PCA of Σ_x into the first K and the remaining $m - K$ components [► 2013 L15 @ 1:03:46]:

$$x = \alpha + \Gamma_1 p_1 + \Gamma_2 p_2 = \alpha + Bf + \epsilon, \quad B = \Gamma_1, \quad f = p_1, \quad \epsilon = \Gamma_2 p_2.$$

This looks like a factor model (and $B = \Gamma_1 \Lambda_1^{1/2}$, $f = \Lambda_1^{-1/2} p_1$ gives orthonormal factors), but $\text{Cov } \epsilon = \Gamma_2 \Lambda_2 \Gamma_2^\top$ is in general *not diagonal* [► 2013 L15 @ 1:04:59]: the “specific” parts are correlated across assets. PCA is an approximate factor model, “principal components”, not “principal factors”. The *principal factor method* fixes this by iterating PCA on a reduced covariance:

1. PCA of the sample covariance, $\hat{\Sigma}_x = \hat{\Gamma} \hat{\Lambda} \hat{\Gamma}^\top$;
2. initial estimates $\tilde{B}_0 = \hat{\Gamma}_{(K)} \hat{\Lambda}_{(K)}^{1/2}$ (first K columns and the leading $K \times K$ block), $\tilde{\Psi}_0 = \text{diag}(\hat{\Sigma}_x) - \text{diag}(\tilde{B}_0 \tilde{B}_0^\top)$;
3. eigen-decompose the reduced matrix $\hat{\Sigma}_x - \tilde{\Psi}_0$ (which under the model equals $BB^\top$, of rank K) and repeat step 2 with its eigenvectors and eigenvalues, obtaining $\tilde{B}_1, \tilde{\Psi}_1$;
4. iterate until $\tilde{\Psi}_s$ stops changing; use the last estimates.

	PCA	Factor analysis (ML)
Model	none: exact rotation of the data	$\Sigma_x = BB^\top + \Psi$, Ψ diagonal, Gaussian likelihood
Output	ordered orthogonal directions, variances	loadings up to rotation, specific variances
Residuals	correlated across variables	uncorrelated by assumption
Scale	covariance and correlation PCA differ	scale-equivariant
Number of factors	scree plot, variance explained	likelihood-ratio test
Computation	one SVD	iterative (EM)

CASE STUDY: 2013 Case Study 7: factor modelling of Treasury yields

Data. FRED constant-maturity yields, 9 tenors (3m to 20y), 3 Jan 2000 to 31 May 2013; daily changes (in percentage points) over two periods [► 2013 L15 @ 1:11:14]: 2001–2005 and 2009–May 2013 (the section title says “2009–2012”, the code uses 2009–2013). **2001–2005** [► 2013 L15 @ 1:12:18]: all mean changes negative (rates fell on balance) [► 2013 L15 @ 1:13:23]; daily volatilities 3.8 bp (3m) to 7.0 bp (3y); correlations decreasing away from the diagonal. PCA (princomp): 84.9%, 9.2%, 2.9% [► 2013 L15 @ 1:18:43]; PC1 a weighted average peaking at 3–5y, PC2 long minus short, PC3 curvature, PC9 the 7y-versus-5y/10y hedge; the PC scores are exactly uncorrelated. Factor analysis (factanal): 4 factors rejected ($\chi^2 = 26.8$, 6 df, $p = 0.0002$), 5 factors rejected at 5% ($\chi^2 = 4.33$, 1 df, $p = 0.037$), 6 factors cannot be fitted ($\nu < 0$): no reduced-dimension model passes the test. **2009–2013:** short rates at zero, so the 3m–1y yields hardly move (3m volatility 1.25 bp versus 6.6 bp at 7–20y); PCA 88.6%, 6.8%, 1.6%, with PC1 loadings near zero at the short end; PC2 now opposes long and *medium* tenors. Factor analysis: 3 factors rejected ($\chi^2 = 231$, 12 df), 4 factors accepted ($\chi^2 = 5.2$, 6 df, $p = 0.52$); the 3m yield has a large specific variance (uniqueness 0.65). **Lesson** [► 2013 L15 @ 1:25:03]: the structure depends on the period (monetary regime), so the calibration window matters, and the fitted factors are only a starting point for modelling their dynamics (as in 2013 PS7 Problem 4). (Several uniquenesses equal 0.005, the lower bound used

13 COURSE NOTE: Erratum: slide 32 of lecnote15

For the SVD $X^* = VDU'$ of the $m \times T$ centred matrix, U must be $T \times m$ with orthonormal columns, $U'U = I_m$ (the slide says “ U : ($m \times T$) row-orthonormal, $UV' = I_m$ ”, which is dimensionally inconsistent with $X^* = VDU'$). With this correction the exercise on the slide (Exercise 7.3) works out.

by factanal: near-“Heywood” cases, where a variable is almost entirely common factor; not discussed in class.)

KEY FORMULAS

PCA:	$\Sigma = \Gamma \Lambda \Gamma^T, \quad p = \Gamma^T(x - \mu), \quad \text{Cov } p = \Lambda,$ $x = \mu + \sum_k p_k \gamma_k$
Variational:	$\gamma_1 = \arg \max_{\ w\ =1} w^T \Sigma w,$ $\text{share}_k = \lambda_k / \text{tr } \Sigma$
Best subspace:	$\min_V \mathbb{E}[\ (I - VV^T)(x - \mu)\ ^2]$ $= \sum_{j>K} \lambda_j \quad (V = \Gamma_K)$
Sample PCA:	$X_c = U D \hat{\Gamma}^T, \quad \hat{\lambda}_k = d_k^2 / (N - 1), \quad P = X_c \hat{\Gamma} = U D$
Correlation PCA:	$R = D_\sigma^{-1} \Sigma D_\sigma^{-1}, \quad \text{tr } R = p$
Out of sample:	$W^T \Sigma^Y W = D^T \Lambda^Y D,$ $D = (W^Y)^T W; \quad \text{ED} = (\sum \lambda)^2 / \sum \lambda^2$
PC exposures:	$e_k = h^T \gamma_k;$ curve $\alpha = \gamma_{a1} / \gamma_{b1};$ fly: 2×2 system (7.3)
PC1 hedge:	$V' = V - (V^T \gamma_1) \gamma_1,$ $\text{Var} = V^T \Sigma V - \lambda_1 (V^T \gamma_1)^2$
Factor model:	$x_t = \alpha + B f_t + \epsilon_t,$ $\Sigma_x = B \Omega_f B^T + \Psi \quad (\Psi \text{ diagonal})$
Single index:	$\Sigma_x = \sigma_M^2 \beta \beta^T + \Psi$
Factor estimates:	$\hat{f}_t = (B^T \Psi^{-1} B)^{-1} B^T \Psi^{-1} x_t,$ $HB = I_K$
LR test df:	$\nu = \frac{1}{2} [(m - K)^2 - (m + K)]$

14 COURSE NOTE: 2013 versus 2024

Both years analyse daily changes of US Treasury yields and find level, slope and curvature. 2013 (Kempthorne, statistics) uses FRED constant-maturity yields at 9 tenors, fixed 5-year windows, and adds the factor-model and factor-analysis machinery; 2024 (Andreev, practice) uses the Fed’s fitted par yields at 30 maturities with a rolling 2-year window, and focuses on stability, out-of-sample behaviour and PC-neutral trading. The 2024 course has no lecture on macroeconomic or fundamental factor models; their portfolio-theory side reappears in [chapter 10](#).

Exercise 7.5 (not from the course).

- Count the free parameters of the K -factor model $\Sigma_x = B B^T + \Psi$ and derive the degrees of freedom ν of the likelihood-ratio test.
- Check the values of ν reported in 2013 Case Study 7 and explain why a 6-factor model cannot be fitted to 9 yields.
- For the single-index model with $\Psi = \psi I$, show that PCA recovers $\beta / \|\beta\|$ as PC1; what happens if the σ_i^2 differ?

Exercises

Standalone exercises follow the section they practise. The exercises that build on one another stay together here: PS3 Problem 4 with the exercises that reuse its data (PS7 Problem 4 and the PC-neutral hedges), and PS7 Problems 1–2.

From 2024 Problem Set 3

Exercise 7.6 (PS3 (2024), Problem 4; also PS7 (2013), Problem 3: PCA of yield changes). Daily changes of the 9 FRED constant-maturity Treasury yields over 2001–2005 are analysed twice, with the sample covariance matrix and with the sample correlation matrix. The standard deviations of the PC variables are

	1	2	3	4	5	6	7	8	9
covariance	0.1618	0.0532	0.0299	0.0192	0.0135	0.0112	0.0096	0.0090	0.0081
correlation	2.6338	1.1892	0.5652	0.3971	0.2694	0.2080	0.1485	0.1366	0.1225

(proportions of variance 84.9, 9.2, 2.9 % and 77.1, 15.7, 3.5 % for the first three) and the first three loading vectors are

Tenor	covariance PCA			correlation PCA		
	PC1	PC2	PC3	PC1	PC2	PC3
3M	0.1088	0.5225	0.5434	−0.2083	0.6293	0.5831
6M	0.1581	0.4817	0.2508	−0.2830	0.5154	−0.0654
1Y	0.2531	0.4107	−0.0514	−0.3369	0.2698	−0.4066
2Y	0.3905	0.2071	−0.4750	−0.3625	0.0064	−0.4005
3Y	0.4197	0.0873	−0.4022	−0.3664	−0.0665	−0.2719
5Y	0.4206	−0.1082	−0.0311	−0.3670	−0.1619	0.0036
7Y	0.4009	−0.2289	0.1475	−0.3608	−0.2298	0.1435
10Y	0.3697	−0.2837	0.2710	−0.3525	−0.2656	0.2554
20Y	0.3101	−0.3621	0.3944	−0.3290	−0.3338	0.4125

(a) PC1 loadings are all positive in the covariance case and all negative in the correlation case. Is the difference in sign meaningful? (Does the eigen-decomposition change if an eigenvector is multiplied by -1 ?) (b) The PC1 loadings range more widely in the covariance case; the loading on the least variable change (3M) is larger in magnitude in the correlation case, and those on the most variable changes are smaller. Explain why, recalling that PC1 is the unit-norm weighted combination with the largest variance and that in the correlation case each yield change is scaled to unit variance. (c) Interpret the first three PCs of the correlation case and compare with the covariance case. (d) If the goal is to model the dynamics of the whole term structure across all tenors, argue why correlation PCA might be preferred. (e) (2024 only) Compute the total variance of the correlation-case PCs. Is it an integer? Explain.

From 2013 Problem Set 7

Exercise 7.7 (PS7 (2013), Problem 1: PCA of a bivariate vector). Let $\mathbf{X} = (X_1, X_2)^T$ have mean $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)^T$ and covariance $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$.

- Compute the eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$ of Σ .
- Compute orthonormal eigenvectors γ_1, γ_2 .

15 COURSE NOTE: On the data of the previous exercise

The problem sets say the changes are “in basis point units”; the numbers are in percentage points (a PC1 standard deviation of 0.1618 is 16 bp per day, the tenor volatilities of Case Study 7 are 0.04–0.07). The 2013 version prints correlation-case standard deviations about 0.04% smaller (their squares sum to 8.993 instead of 9, a factor $(N-1)/N$ with $N \approx 1247$ days, presumably a different divisor convention); the loadings coincide up to signs.

- (c) Verify $\Sigma = \sum_i \lambda_i \gamma_i \gamma_i^\top$.
- (d) For $p_i = \gamma_i^\top (X - \alpha)$ prove $\mathbb{E}[p_i] = 0$, $\text{Var}(p_i) = \lambda_i$ and $\text{Cov}(p_1, p_2) = 0$.
- (e) Are the eigenvectors in (b) unique? If not, how are different solutions related?

Exercise 7.8 (PS7 (2013), Problem 2: PCA as a distance-preserving affine map). A distance-preserving affine map of the plane can be written $T(X) = \Phi^\top (X - \mu)$ with $\Phi = [\phi_1 : \phi_2]$ orthogonal and $\mu \in \mathbb{R}^2$: it moves the origin to μ and rotates the axes.

- (a) Find the unit vector ϕ_1 maximising $\text{Var}(\phi_1^\top X)$ and show that the maximum is λ_1 of the previous exercise.
- (b) Find the unit vector ϕ_2 minimising $\text{Var}(\phi_2^\top X)$ and show that the minimum is λ_2 .
- (c) Give an equivalent definition of PCA in terms of affine transformations of the variables and their variances and expectations.

Exercise 7.9 (PS7 (2013), Problem 4: dynamics of the PC scores). A VAR(3) with constant (chapter 9) was fitted to the scores of the first three PCs of the correlation-matrix analysis of Exercise 7.6 ($T = 1245$ days). Estimates and t -values:

Regressor	eq. Comp.1		eq. Comp.2		eq. Comp.3	
	est.	t	est.	t	est.	t
Comp.1, lag 1	0.0527	1.86	-0.0203	-1.65	-0.0029	-0.48
Comp.2, lag 1	-0.1340	-2.08	0.1382	4.93	-0.0143	-1.03
Comp.3, lag 1	-0.1322	-1.00	-0.0833	-1.45	0.0163	0.58
Comp.1, lag 2	-0.0530	-1.87	-0.0072	-0.59	-0.0135	-2.23
Comp.2, lag 2	-0.0542	-0.84	0.0370	1.32	-0.0037	-0.27
Comp.3, lag 2	0.1825	1.39	-0.0195	-0.34	-0.0523	-1.86
Comp.1, lag 3	-0.0169	-0.60	-0.0085	-0.69	-0.0019	-0.32
Comp.2, lag 3	0.0130	0.21	-0.0403	-1.46	-0.0030	-0.22
Comp.3, lag 3	0.0830	0.63	0.0148	0.26	-0.0712	-2.53
constant	-0.0114	-0.15	0.0121	0.38	0.0002	0.01
R^2 ; F -test p -value	0.013	0.054	0.029	$4 \cdot 10^{-5}$	0.013	0.059

Residual standard errors are 2.60, 1.13, 0.56; residual correlations are below 0.025 in absolute value; the largest root of the characteristic polynomial has modulus 0.438. (a)–(c) Interpret the estimated equations for Comp.1, Comp.2 and Comp.3. (d) For each equation, is there evidence of mean reversion or of momentum in the PC variable? Explain.

Additional exercises

Exercise 7.10 (not from the course). Using the covariance-PCA loadings of Exercise 7.6:

- (a) find the PC1-neutral 2s10s steepener and check that its exposure to PC2 is non-zero;

- (b) find the PC1/PC2-neutral 2s5s10s fly (buy \$1 dv01 of 5y) and compute its exposure to PC3;
- (c) prove the variance formula for the PC1-hedged portfolio $V' = V - (V^\top \gamma_1) \gamma_1$ and show that it never exceeds $\text{Var}(V^\top r)$.

Chapter 8

Counterparty Risk, Margin and CVA

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L10	J. Shepherd	<i>Counterparty Risk Optimization</i> (Quantile Technologies, part of LSEG). Types of risk and central clearing; coherent risk measures, VaR and expected shortfall (historical and model-building estimates, time scaling, back-testing, Rockafellar–Uryasev); variation and initial margin, ISDA SIMM; multilateral optimisation of initial margin in a network of banks, numerical issues, “nice” and “fair” solutions.
2013 L26	Y. Tang	<i>Introduction to Counterparty Credit Risk</i> (Morgan Stanley, given remotely). CVA, DVA, funding risk, wrong-way risk (Berkshire Hathaway’s index puts), the CVA conundrum; enterprise-level derivatives modelling; martingale measures, martingale testing, resampling and interpolation. Last minutes: closing remarks of the instructors (section 8.8).

Files. Slides `mit18_642_f24 lec10.pdf` (51 slides); 2013 slides `MIT18_S096F13_lecnote26.pdf` (27 slides). **Problem sets.** No problem set of either year covers this chapter; all exercises are marked “not from the course”. **Not covered in class.** 2024: slide 17 (FRTB) was skipped, slides 26–28 (SIMM practicalities) and 48 (observed savings) only shown; 2013: slides 21–25 (extinguisher trade, “bigger picture”) were not reached for lack of time (summarised in [section 8.4](#)).

Motivation

An over-the-counter (OTC) derivative is a bilateral promise. Over the life of a swap sometimes you owe your counterparty money and sometimes it owes you; if it defaults while it owes you, you lose part of the receivable. Before 2008 this *counterparty credit risk* was a side issue. During the crisis the mark-to-market losses it caused were larger than the losses from actual defaults [► [2013 L26 @ 6:22](#)], and the regulatory response (Basel III capital for CVA, mandatory central clearing, margin rules for uncleared derivatives) changed the whole derivatives business. The two guest lectures show the two sides of this change:

- **2013 (Y. Tang, Morgan Stanley):** *pricing* counterparty risk. The credit valuation adjustment (CVA) and its relatives (DVA, funding), wrong-way risk, and why all of them need models of the whole portfolio, not of one trade at a time.
- **2024 (J. Shepherd, Quantile/LSEG):** *mitigating* it with collateral. Variation margin, initial margin computed as a value-at-risk, and how a network of banks can reduce the total initial margin they post by agreeing on new trades, found by convex optimisation.

The mathematics: option-like expectations under the pricing measure, coherent risk measures (VaR and expected shortfall), quadratic forms of Gaussian vectors, convex and mixed-integer optimisation, and change of numeraire. VaR is treated in more depth, from the 2013 lecture on the topic, in [chapter 19](#).

8.1 Counterparty risk in OTC derivatives

Types of risk. Trading derivatives creates several risks [[▶ 2024 L10 @ 3:11](#)]: *market* risk (values move with the market), *credit* risk (a debtor does not pay an obligation), *operational and legal* risk (people, systems and events; the hardest to model, because it is mostly people making mistakes), *liquidity* risk (a transaction cannot be executed at the market price, typically because the position is so large that selling it moves the price, or outgoing cash flows cannot be funded), and *counterparty* risk: the potential loss on the market value of transactions when the counterparty defaults.

Counterparty versus credit risk. Suppose you buy protection (a credit default swap, CDS) on a JPMorgan bond from Goldman Sachs. You have *credit* risk to JPMorgan: the value of the CDS depends on JPMorgan defaulting. You have *counterparty* risk to Goldman Sachs: whether it will actually pay what the CDS is worth [[▶ 2024 L10 @ 4:13](#)]. For a corporate bond the economics are similar, but the risk to the issuer is called *issuer* risk [[▶ 2013 L26 @ 3:10](#)]. Counterparty risk has two components [[▶ 2013 L26 @ 2:04](#)]:

- *default risk*: at default we recover only a fraction R of what we are owed;
- *mark-to-market (MTM) risk*: even if nothing is due for ten years, a widening of the counterparty's credit spread lowers the value of the trade today, because a buyer of the trade would pay less for it [[▶ 2013 L26 @ 3:10](#)].

What makes counterparty risk harder than bond credit risk is that the amount at risk is *random and two-sided*: the value of a swap can be positive or negative, and it depends on all trades with the same counterparty.

Mitigation. The standard tools (slide 4 of 1ec10) are [[▶ 2024 L10 @ 5:15](#)]: *netting* of offsetting payments; *hedging* with new trades; *collateralisation* (collecting margin, a kind of insurance payment against the counterparty's default); contractual clauses such as periodic resets of the mark-to-market value or break clauses; and *central clearing counterparties* (CCPs).

Central clearing. A CCP steps into the middle of bilateral trades by *novation*: the trade between A and B is replaced by two trades, A with the CCP and the CCP with B [[▶ 2024 L10 @ 6:15](#)]. By itself this only changes the counterparty; the gain comes from *multilateral netting*, since all of a firm's positions now face the same counterparty and offset ([fig. 8.1](#)) [[▶ 2024 L10 @ 7:20](#)]. As a rough rule, total counterparty risk scales like $\sum_{\text{counterparties}} \|\text{position}\|$, and netting reduces the number of terms. Standard (vanilla) interest-rate swaps must be cleared by regulation, while more exotic products such as swaptions are not eligible, so every bank ends up with a mixture of cleared and bilateral trades [[▶ 2024 L10 @ 8:25](#)].

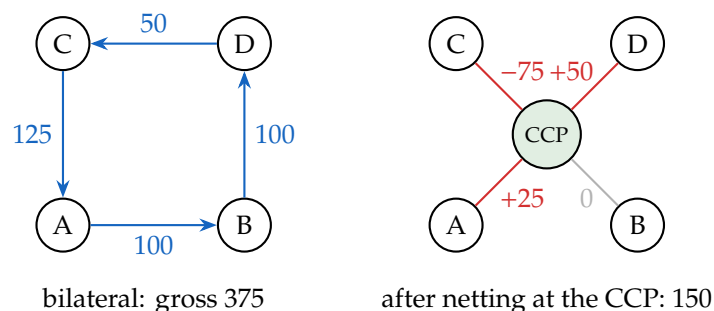


Figure 8.1: Multilateral netting at a CCP (slide 5 of 1ec10). Left: four bilateral positions; an arrow into a node counts +, out of a node -. Right: after novation each party faces only the CCP and its positions net: A has $+125 - 100 = +25$, B has $+100 - 100 = 0$ (no counterparty risk left), C has $+50 - 125 = -75$, D has $+100 - 50 = +50$. The CCP itself is flat.

IN FINANCE: What a CCP costs (not discussed in class)

Clearing concentrates risk in the CCP, which protects itself by collecting initial margin and contributions to a default fund from its members. Splitting a portfolio between cleared and bilateral trades also destroys netting *between* the two sets. This is why “move more trades into the CCP” is a genuine design choice in the optimisation of [section 8.6](#), not an automatic improvement.

8.2 Exposure, netting and collateral

Fix a counterparty and a *netting set* (all trades covered by one master agreement, whose values are offset at default). Let $V(t)$ be the value *to us* of the netting set computed with a counterparty-default-free model, and $C(t)$ the value of the collateral we hold ($C < 0$ if we have posted).

Definition 8.1 (Exposure profiles). The *exposure* at time t is $E(t) = (V(t) - C(t))^+$, the amount we would lose (before recovery) if the counterparty defaulted at t . Then:

- *expected exposure* $EE(t) = \mathbb{E}[E(t)]$, and *expected positive exposure* $EPE = \frac{1}{T} \int_0^T EE(t) dt$, called *mean positive exposure* (MPE) in the 2013 slides;
- *negative exposure* $(V(t) - C(t))^- = \max(C(t) - V(t), 0)$, what we owe; its mean is the MNE of the 2013 slides;
- *potential future exposure* $PFE_q(t)$, the q -quantile of $E(t)$ (typically $q = 95\%$ or 99%), used by banks to set credit limits (not discussed in class).

For pricing (CVA) the expectation is under the pricing measure $\mathbb{Q}$; for risk management under $\mathbb{P}$.

The positive part is the key non-linearity: if $V > 0$ at default we recover only RV , but if $V < 0$ we still owe the full amount to the defaulted counterparty. There is no “default gain”, and this asymmetry makes the loss an *option-like* payoff [► 2013 L26 @ 19:10]. [Figure 8.2](#) shows the typical shapes: the uncertainty of $V(t)$ grows like $\sqrt{t}$ by diffusion, while for a swap the remaining cash flows, and with them the sensitivity to rates, shrink linearly to maturity.

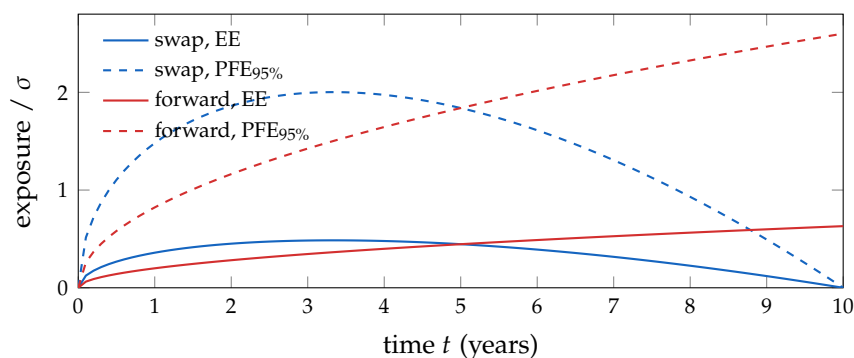


Figure 8.2: Stylised exposure profiles (not from the course). If $V(t) \sim N(0, s(t)^2)$ and there is no collateral, then $EE(t) = s(t)/\sqrt{2\pi} \approx 0.40 s(t)$ and $PFE_{95\%}(t) = 1.645 s(t)$. Swap: $s(t) = \sqrt{t}(1 - t/T)$ with $T = 10$ (diffusion times remaining duration, peak at $T/3$). FX forward: $s(t) = 0.5\sqrt{t}$ (a single exchange at maturity).

Proposition 8.2 (Netting reduces exposure). For trades with values $V_1, \dots, V_n$ in one netting set, $(\sum_i V_i)^+ \leq \sum_i V_i^+$ pathwise; hence $EE_{net}(t) \leq \sum_i EE_i(t)$.

Proof. $x \mapsto x^+$ is subadditive: $(x+y)^+ \leq x^+ + y^+$ because the right side is $\geq x+y$ and ≥ 0 . Induct on n and take expectations. $\square$

If one trade goes from 0 to +\$100M and an offsetting trade with the same counterparty goes to -\$100M, the netted default loss is zero [► 2013 L26 @ 18:05]. The price of counterparty risk is therefore *not* a sum of per-trade quantities: to price it for one new trade you must know every other trade facing the same counterparty. Ordinary derivative pricing never needs this [► 2013 L26 @ 7:29].

Collateral. Under a *Credit Support Annex* (CSA) to the master agreement the parties exchange collateral. *Variation margin* (VM) is exchanged daily and equals the current value, $VM(t) = V(t)$; the CSA also fixes thresholds below which no collateral is called, minimum transfer amounts and eligible assets (these details were not discussed in class, apart from the thresholds mentioned on the SIMM slides). With perfect daily VM the exposure is no longer the *level* of V but its *increment* between the last margin call the defaulter honoured and the moment we have closed out or replaced the trades. That increment is covered by *initial margin* (section 8.5).

INTUITION: From $\sqrt{T}$ to $\sqrt{MPOR}$

For a diffusive value process the typical size of $V(t)$ after 5 years is $\sigma\sqrt{5}$, while the typical move over a 10-business-day close-out period is $\sigma\sqrt{10/250}$, about 9% of it. Collateralisation replaces a large, slowly growing exposure by a small, constantly renewed one. The price is liquidity: margin must be paid in cash or high-quality securities, often at short notice.

Exercise 8.1 (Netting and correlation (not from the course)). Two trades with the same counterparty have values $V_1, V_2 \sim N(0, \sigma^2)$ at time t , with correlation ρ , and there is no collateral.

- Compute EE with and without netting and show that the ratio is $\sqrt{(1+\rho)/2}$.
- What happens for $\rho = -1$ and $\rho = 1$?
- Explain why the CVA of a new trade depends on the existing portfolio (the

“incremental CVA”).

8.3 CVA, DVA and funding: pricing counterparty risk

8.3.1 A puzzle

Tang opened with a question [► 2013 L26 @ 8:32]. You have an uncollateralised swap with value 0 on day one (ignoring counterparty risk). Later its value is +\$100M, and then the counterparty unexpectedly defaults with 50% recovery, so you receive \$50M in cash. Over the life of the trade, have you made \$50M, lost \$50M, or are you flat?

A dealer has *lost* \$50M [► 2013 L26 @ 12:45]. Its market risk is hedged, for example by an offsetting swap with another dealer or an exchange (or the trade is simply intermediated back to back). When the client trade went from 0 to +100, the hedge went from 0 to −100, and the hedge counterparty is solvent and must be paid in full. The dealer receives 50 and pays 100. In general the loss at default is $(1 - R)$ times the exposure. A CVA desk that had sold the trading desk protection on the counterparty (and itself bought CDS protection) would have made the trading desk whole [► 2013 L26 @ 13:53].

8.3.2 Credit valuation adjustment

Definition 8.3 (CVA, 2013 slides). The *credit valuation adjustment* is the price of counterparty credit risk, an adjustment to the value given by a counterparty-default-free model or a broker quote [► 2013 L26 @ 4:13]. With τ the counterparty’s default time, T the final maturity of the netting set, R_τ the recovery rate, $V_{\tau-}, C_{\tau-}$ the portfolio value and collateral just before default, and $\beta_t = \exp(\int_0^t r_u du)$ the money-market account,

$$\text{CVA}_{\text{MPE}} = -\mathbb{E}_0^{\mathbb{Q}} \left[\mathbf{1}_{\{\tau \leq T\}} \frac{(V_{\tau-} - C_{\tau-})^+ (1 - R_\tau)}{\beta_\tau} \right] \quad [\text{► 2013 L26 @ 15:57}]. \quad (8.1)$$

The adjusted value of the portfolio is $V_0 + \text{CVA}_{\text{MPE}}$.

The minus sign matters in practice: CVA on receivables is a cost, the amount charged to the counterparty. Theoretical papers often define CVA as a positive number and subtract it; that is fine as long as you are consistent [► 2013 L26 @ 14:55]. In (8.1), $\mathbf{1}_{\{\tau \leq T\}}$ says there is no counterparty risk if the counterparty defaults after the portfolio has matured; $(V - C)^+$ is the exposure; $(1 - R)$ turns it into a loss given default; and β_τ discounts [► 2013 L26 @ 17:01].

Proposition 8.4 (CVA as an integral of expected exposure (not derived in class)). Assume τ is independent of the market variables (no wrong-way risk, section 8.3.5), has a deterministic hazard rate $h(t)$ with survival function $S(t) = \exp(-\int_0^t h)$, and R is constant. Then

$$\text{CVA} = -(1 - R) \int_0^T \text{EE}^*(t) h(t) S(t) dt, \quad \text{EE}^*(t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{(V_t - C_t)^+}{\beta_t} \right].$$

If moreover h and r are constant, $\text{EE}(t) \approx \text{EPE}$ is roughly flat, and the CDS spread obeys the “credit triangle” $s \approx (1 - R)h$, then

$$\text{CVA} \approx -\text{EPE} \times s \times D, \quad D = \int_0^T e^{-(r+h)t} dt = \frac{1 - e^{-(r+h)T}}{r + h}, \quad (8.2)$$

Proof. Let $g(t) = \mathbb{E}^{\mathbb{Q}}[(V_t - C_t)^+ / \beta_t]$. Since τ is independent of the market, $\mathbb{E}[\mathbf{1}_{\{\tau \leq T\}}(1 - R)(V_{\tau-} - C_{\tau-})^+ / \beta_\tau] = (1 - R)\mathbb{E}[\mathbf{1}_{\{\tau \leq T\}}g(\tau)]$ (paths are continuous at a fixed time, so $V_{t-} = V_t$ a.s.), and τ has density $h(t)S(t)$. For the approximation, put $\text{EE}^*(t) = e^{-rt}\text{EPE}$ and $hS(t) = he^{-ht}$, then use $(1 - R)h = s$. $\square$

where D is the risky duration (annuity) of the counterparty. This is the “back-of-the-envelope” formula of the slides: $-MPE \times CDS \text{ par spread} \times \text{duration}$ [► 2013 L26 @ 13:53].

The wider the spread (default more likely) and the longer the duration (more time to default), the larger the charge [► 2013 L26 @ 14:55].

INTUITION: CVA is an option on a basket

$(V_\tau - C_\tau)^+$ is the payoff of a call on the netting set with strike equal to the collateral, exercised at the random time τ . CVA is a default-probability-weighted strip of such options. Since a counterparty trades many products (rates, FX, credit, equity, commodities, mortgages), it is an option on a *basket of cross-asset derivatives* [► 2013 L26 @ 20:11]. Pricing it needs a joint simulation of all risk factors of all trades: this is what Tang calls enterprise-level modelling (section 8.7), and the reason his group at Morgan Stanley was called Cross Asset Modeling.

IN FINANCE: CVA in the bank

CVA changes the true economic price of a trade, so it is charged to clients and hedged dynamically by a CVA desk, which buys CDS protection and hedges the market factors that drive V . It also enters regulatory capital: Basel III added a CVA capital charge, so trades with large CVA consume more capital and lower the return on equity [► 2013 L26 @ 5:17]. In 2008, CVA mark-to-market losses were larger than the losses from actual defaults [► 2013 L26 @ 6:22].

Exercise 8.2 (CVA from exposure (not from the course)).

- Derive proposition 8.4 and the approximation (8.2).
- For the exposure profile $EE(t) = \sigma\sqrt{t}(1 - t/T)/\sqrt{2\pi}$ of fig. 8.2, with $r = 0$ and constant hazard rate h , compute CVA to first order in h and compare with $EPE \cdot s \cdot T$.
- Explain qualitatively how wrong-way risk changes the result.

8.3.3 DVA and first-to-default

The same logic applied to *our own* default gives a *liability* CVA [► 2013 L26 @ 21:23]. With $\bar{\tau}$ our default time and $\bar{R}$ our recovery, the slides write

$$CVA_{MNE} = -\mathbb{E}_0^Q \left[\mathbf{1}_{\{\bar{\tau} \leq T\}} \frac{(V_{\bar{\tau}-} - C_{\bar{\tau}-})^- (1 - \bar{R}_{\bar{\tau}})}{\beta_{\bar{\tau}}} \right] = \mathbb{E}_0^Q \left[\mathbf{1}_{\{\bar{\tau} \leq T\}} \frac{(C_{\bar{\tau}-} - V_{\bar{\tau}-})^+ (1 - \bar{R}_{\bar{\tau}})}{\beta_{\bar{\tau}}} \right] \geq 0.$$

This is a *benefit*: if we default while owing money, we pay only a fraction of it. It is usually called *debit valuation adjustment* (DVA) [► 2013 L26 @ 22:27].

Should the two adjustments know about each other? If we default first, the counterparty’s later default no longer matters to us, and vice versa. A *bilateral* CVA puts first-to-default indicators $\mathbf{1}_{\{\tau \leq \min(\bar{\tau}, T)\}}$ and $\mathbf{1}_{\{\bar{\tau} \leq \min(\tau, T)\}}$ into the two terms (not written in class). Tang’s view is that for pricing this refinement does not need to be modelled; he gave references on the slides rather than discussing it [► 2013 L26 @ 23:30].

Example 8.5 (Size of CVA). A 5-year uncollateralised swap book with EPE = \$5M, counterparty spread $s = 100$ bp, $R = 40\%$ (so $h = 1.67\%$) and $r = 3\%$: $D = (1 - e^{-0.0467 \cdot 5})/0.0467 = 4.46$ and $CVA \approx -5 \cdot 0.01 \cdot 4.46 = -\0.22 M. If the spread doubles, the CVA roughly doubles: a mark-to-market loss with no default at all.

1 COURSE NOTE: Sign convention on slide 6 of lecnote26

For the displayed formula to be a benefit, $(x)^-$ must be read as $\min(x, 0) \leq 0$, as Tang explains (“the positive sign becomes a negative sign”). With the other common convention $x^- = \max(-x, 0)$ the formula would carry the opposite sign. The right-hand side above avoids the ambiguity.

IN FINANCE: Why DVA is controversial (not discussed in class)

DVA means that a bank books a gain when its own credit worsens, and it can hardly be hedged, since a bank cannot sell protection on itself. Regulators therefore do not let DVA gains count towards capital, even though accounting standards include DVA in fair value.

8.3.4 Funding and liquidity risk

Second example [► 2013 L26 @ 23:30]: same trade, value +\$100M, uncollateralised, with market risk and counterparty risk properly hedged. Is anything left? Students proposed interest-rate risk (hedged by assumption) and “key-man” risk [► 2013 L26 @ 25:40]. Tang’s answer is *funding liquidity risk* [► 2013 L26 @ 25:40]:

- hedges are typically futures or collateralised trades. When our uncollateralised receivable is worth +100 the hedge is worth −100, and we must post 100 of cash *now*, while the receivable arrives only later. That cash must be borrowed at our unsecured funding spread [► 2013 L26 @ 26:47];
- the amount depends on market moves (*contingent* liquidity risk), and failing to raise it is how firms die: “you may end up like Lehman” [► 2013 L26 @ 27:51];
- symmetrically, an uncollateralised *payable* generates a funding *benefit* (we receive collateral on its hedge), which can partially offset the funding need of the receivables [► 2013 L26 @ 27:51];
- further risks: unexpected tail events and (equity) capital [► 2013 L26 @ 28:54].

In the spirit of (8.2), a first estimate of the funding cost is $-s_F \times EPE \times D$ and of the benefit $+s_F \times |ENE| \times D$, with s_F our own funding spread (a common simplification, not from the lecture).

IN FINANCE: The XVA family (terminology not used in class)

The adjustments to the default-free price are nowadays collectively called XVA: CVA (counterparty default), DVA (own default), FVA (funding of uncollateralised positions, split into cost FCA and benefit FBA), KVA (cost of regulatory capital held against the trade, Tang’s “capital”), and MVA (cost of funding the *initial margin* of section 8.5 over the life of the trade). The 2013 lecture covered CVA, DVA, funding and capital. The 2024 lecture is about the world that the margin rules created, in which CVA is smaller because trades are collateralised and the cost has moved into margin; its reading list cites J. Gregory, *The xVA Challenge*.

8.3.5 Wrong-way risk

Definition 8.6 (Wrong-way risk). *Wrong-way risk* (WWR) is present when exposure to a counterparty is positively correlated with its probability of default: the counterparty owes us more exactly when it is more likely to default [► 2013 L26 @ 33:02]. (*Right-way* risk is the opposite.) Then the independence assumption of proposition 8.4 fails, and the relevant exposure is the *conditional* one, $CVA = -(1 - R) \int_0^T \mathbb{E}^Q[(V_t - C_t)^+ / \beta_t \mid \tau = t] d\mathbb{Q}(\tau \leq t)$, which is larger than $EE^*(t)$ under WWR.

CASE STUDY: Berkshire Hathaway's index puts (2013 slides 8–9)

The trade ▶ 2013 L26 @ 29:54]. Selling a put below the current price is an alternative to buying the stock: similar downside, no upside beyond the premium, and a steady income if repeated (“name your price and get paid for it”). Around 2005–2006, as Tang recalls, Berkshire Hathaway sold long-dated (10–15 year) puts on four indices (S&P 500, FTSE 100, Euro Stoxx 50, Nikkei 225) and collected about \$4bn of premium *without posting collateral*. It was one of the largest derivative cash outflows of its time for the dealers, who were the buyers ▶ 2013 L26 @ 30:55]. Had Berkshire posted collateral, the 2008 sell-off would have required many billions of it (see page 16 of Berkshire's 2012 shareholder letter, cited on the slides).

The dealer's risks ▶ 2013 L26 @ 31:56]. Handling long-dated equity options was not the difficulty; many dealers could not handle the *counterparty* side: CVA on a large uncollateralised receivable; *wrong-way* CVA, because in a sell-off the puts gain value while an insurer with large equity holdings becomes more likely to default ▶ 2013 L26 @ 33:02]; and *wrong-way funding*, because the dealer paid \$4bn up front and had to fund its collateralised hedges exactly in a crisis ▶ 2013 L26 @ 34:06]. The dealer therefore charged CVA, wrong-way CVA, funding and wrong-way funding costs. It then had to risk-manage them: a static CDS hedge (fine for a bond) leaves *gap risk* because the exposure changes over time ▶ 2013 L26 @ 35:11]. Part of the risk was sold to investors in Berkshire's bonds through a *credit-linked note* paying a higher coupon, and through tranching portfolio protection ▶ 2013 L26 @ 36:17].

Berkshire's side ▶ 2013 L26 @ 37:18]. Selling puts is selling insurance on the equity market; the premium is day-one cash, a cheap source of funding (Buffett compared it to borrowing at about 1%); and there is no cash outflow until maturity. So Berkshire suffered mark-to-market losses in 2008 but no liquidity drain.

Feedback ▶ 2013 L26 @ 38:25]. In the crisis, dealers' CVA hedging meant buying more and more CDS protection on Berkshire, which widened Berkshire's CDS spread beyond what fundamentals suggested. Bond investors then demanded higher coupons, raising Berkshire's funding cost, so Berkshire approached dealers about posting collateral after all: “no free lunch” ▶ 2013 L26 @ 39:30].

Who loses? Tang's answer: the end buyers of the puts, the hedgers who pay the premium and gain nothing if the market does not fall far enough ▶ 2013 L26 @ 40:33].

8.3.6 The CVA conundrum

To hedge CVA on counterparty A you buy credit protection on A from counterparty B. Now you have counterparty risk to B, so you buy protection on B from C, and so on: an infinite series ▶ 2013 L26 @ 41:41]. This also undermines the theory: arbitrage pricing is replication with hedge instruments, and here replication needs infinitely many of them ▶ 2013 L26 @ 43:52]. The practical way to stop the series at the first term is to buy the protection on A *fully collateralised*, typically from a dealer under a CSA: collateral is settled as the value moves, much like daily settlement of futures, so the residual risk to B is negligible ▶ 2013 L26 @ 43:52].

8.4 Risk measures: value at risk and expected shortfall

Initial margin is a risk measure of a portfolio's value change, and to optimise it we must know which risk measures are well behaved. This section follows the first half of the 2024 lecture.

8.4.1 Coherent risk measures

A risk measure should capture both the *size* of potential losses and their *probability* [► 2024 L10 @ 8:25]. Below L denotes the *loss* of a portfolio over a horizon (a random variable; gains are negative losses) and $\rho(L) \in \mathbb{R}$ its risk.

Definition 8.7 (Coherent risk measure, Artzner et al. 1999). ρ is *coherent* if for all losses L, L_1, L_2 :

- (i) normalised: $\rho(0) = 0$;
- (ii) positively homogeneous: $\rho(\lambda L) = \lambda \rho(L)$ for $\lambda \geq 0$;
- (iii) translation invariant: $\rho(L + a) = \rho(L) + a$ for a deterministic amount $a \in \mathbb{R}$;
- (iv) monotone: $L_1 \leq L_2$ in all scenarios $\Rightarrow \rho(L_1) \leq \rho(L_2)$;
- (v) subadditive: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$ [► 2024 L10 @ 9:29].

Subadditivity is the important one: merging two portfolios should not create risk, and it may reduce it through diversification [► 2024 L10 @ 10:31].

8.4.2 VaR and ES

Definition 8.8 (VaR and ES). For a confidence level $\beta \in (0, 1)$ and a horizon (implicit in L):

$$\text{VaR}_\beta(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L \leq \ell) \geq \beta\},$$

$$\text{ES}_\beta(L) = \frac{1}{1 - \beta} \int_\beta^1 \text{VaR}_u(L) du.$$

If L has a continuous density f_L then $\int_{-\infty}^{\text{VaR}_\beta} f_L = \beta$ and

$$\text{ES}_\beta(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\beta] = \frac{1}{1 - \beta} \int_{\text{VaR}_\beta}^{\infty} x f_L(x) dx.$$

VaR answers “what is the loss that I am β sure not to exceed over the next 10 days?” (e.g. \$10M at $\beta = 99\%$); ES answers “if I am in the worst $1 - \beta$ of cases, how much do I lose on average?” [► 2024 L10 @ 10:31]. Clearly $\text{ES}_\beta \geq \text{VaR}_\beta$, since ES averages quantiles $\text{VaR}_u \geq \text{VaR}_\beta$ [► 2024 L10 @ 12:35]. VaR was popularised by JPMorgan in the 1990s and ES appeared around 2000; both are used heavily by regulators and in counterparty risk.

2 COURSE NOTE: Slides 21–25 of lecnote26: the extinguisher (not presented)

Tang ran out of time and referred students to the slides [► 2013 L26 @ 1:16:34]. The question: normally a dealer *charges* CVA; how can it structure a trade in which it *pays* the counterparty for the latter’s default risk? Answer: a trade with positive value to the dealer that increases with the counterparty’s default probability, namely an *extinguisher* (zero-recovery) swap, which knocks out at zero value if either party defaults. It is worth something when the dealer expects to have more future *payables* than *receivables*. Example: an emerging-market sovereign issues a USD bond and swaps it into local currency with a dealer (receives USD, pays local currency). Because of the interest-rate differential (the FX forward), the correlation between FX and credit, and the jump of the FX rate at default, the dealer expects to owe money on the swap; the extinguisher cancels this debt at the sovereign’s default. In effect the dealer has bought protection on the sovereign from the sovereign itself, and can sell it on to a third party. Issues: it reduces recovery for the other creditors; the default event must reference the underlying bond; there are legal and franchise risks; and why would the sovereign default, knowing it would lose its receivable on the swap? Slide 25 places all this in the “bigger picture” of a bank’s objectives: revenue, risk management (market, counterparty, liquidity, capital) and return on capital, with the non-linear portfolio effects requiring enterprise-level models.

3 COURSE NOTE: Erratum: slide 6 of lec10

The slide states homogeneity for all α ; it holds only for $\alpha \geq 0$ (doubling a position doubles its risk, but reversing it does not negate the risk). It also writes translation invariance as $\rho(X + A) = \rho(X) + \alpha$ for “a deterministic portfolio with guaranteed return α ”. With X a loss, as in the rest of the slides, adding a guaranteed *gain* α *lowers* the risk: $\rho(X - \alpha) = \rho(X) - \alpha$; equivalently, adding a deterministic *loss* a raises it by a , as in (iii) above. Finally, the reference list dates Artzner–Delbaen–Eber–Heath to 2002; the paper appeared in *Mathematical Finance* in 1999, the year quoted in class.

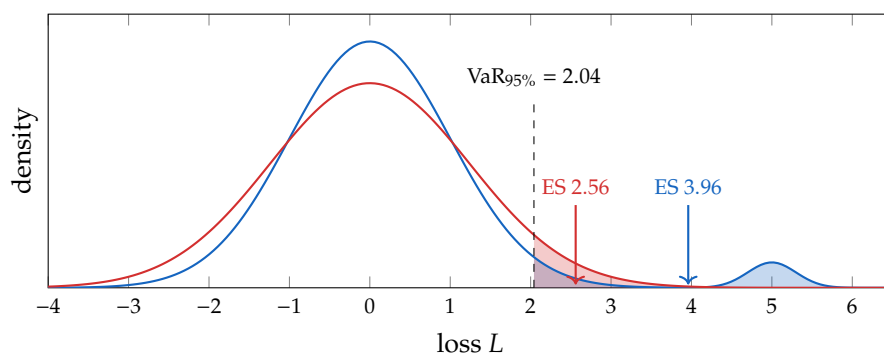


Figure 8.3: Same VaR, different tails (in the spirit of slide 8 of lec10; numbers not from the course). Red: $N(0, 1.24^2)$. Blue: $97\% N(0, 1) + 3\% N(5, 0.3^2)$, a book that usually makes small gains but occasionally loses a lot. Both have $\text{VaR}_{95\%} = 2.04$, but the blue $\text{ES}_{95\%}$ is 3.96 against 2.56: VaR does not see what lies beyond the quantile.

Fat tails. VaR is blind to the shape of the tail beyond the quantile (fig. 8.3) [► 2024 L10 @ 12:35]. Such a distribution is not artificial: a desk that sells out-of-the-money options collects a small premium most of the time and occasionally suffers a large loss when the options end in the money [► 2024 L10 @ 14:37].

VaR is not subadditive. Slide 9 [► 2024 L10 @ 14:37]: two independent portfolios A and B each lose 100 with probability 4% and 0 otherwise. For each, $\text{VaR}_{0.95} = 0$ (the 4% tail lies beyond the quantile) and $\text{ES}_{0.95} = (4 \cdot 100 + 1 \cdot 0)/5 = 80$. The sum A+B loses 200 w.p. 0.16%, 100 w.p. 7.68%, 0 w.p. 92.16%, so $\text{VaR}_{0.95} = 100 > 0 + 0$ while $\text{ES}_{0.95} = (0.16 \cdot 200 + 4.84 \cdot 100)/5 = 103.2 \leq 160$ [► 2024 L10 @ 15:39]. ES is subadditive in general (it is coherent; the slides point to a paper with seven different proofs), but proving it takes some work. Conversely, for normal-like (elliptical) distributions VaR is subadditive, so in practice the failure is less dramatic than such contrived examples suggest [► 2024 L10 @ 16:45]. For optimisation, however, it is fatal, because of the following fact.

Proposition 8.9 (Subadditivity and convexity). *Let ρ be positively homogeneous. Then ρ is subadditive iff it is convex.*

Proof. If ρ is subadditive, then for $\lambda \in [0, 1]$: $\rho(\lambda L_1 + (1 - \lambda)L_2) \leq \rho(\lambda L_1) + \rho((1 - \lambda)L_2) = \lambda \rho(L_1) + (1 - \lambda)\rho(L_2)$. If ρ is convex: $\rho(L_1 + L_2) = 2\rho(\frac{1}{2}L_1 + \frac{1}{2}L_2) \leq \rho(L_1) + \rho(L_2)$. $\square$

Without homogeneity the equivalence fails: $\sqrt{|x|}$ is subadditive but not convex. The slides (slide 33) state “convexity is equivalent to sub-additivity”; the proposition makes precise when this is true. Since VaR is not subadditive, it is not convex as a function of the portfolio, and non-convex functions are much harder to minimise than convex ones [► 2024 L10 @ 16:45].

Spectral risk measures. A risk measure can be described by the weight it puts on each quantile, $M_\phi(L) = \int_0^1 \phi(u) \text{VaR}_u(L) du$ with $\phi \geq 0$, $\int \phi = 1$ [► 2024 L10 @ 17:45]. VaR puts all its weight on $u = \beta$ (a delta function); ES puts a flat weight $1/(1 - \beta)$ on $u > \beta$; an exponential spectral measure uses $\phi(u) \propto e^{-(1-u)/\gamma}$, weighting extreme losses most. M_ϕ is coherent iff ϕ is non-decreasing (Acerbi, 2002): a sensible risk-averse measure never gives worse losses less weight. This fails for VaR, whose weight drops to zero beyond β .

INTUITION: Quantile versus conditional mean

VaR is a quantile, like a median: it does not move if you drag the extreme tail further out. ES is the first moment of the tail, so it does. Convexity is the property that makes a minimisation problem a bowl with one bottom (the same role convexity of the free energy plays for thermodynamic stability); a risk measure without it can have many local minima in portfolio space.

Exercise 8.3 (VaR is not subadditive (not from the course)).

- Verify all the numbers of the A, B, A+B example (VaR and ES at 95%).
- Replace the 4% loss probability by p and find the range of p for which $\text{VaR}_{0.95}(A + B) > \text{VaR}_{0.95}(A) + \text{VaR}_{0.95}(B)$.
- Show that for jointly normal losses VaR is subadditive at any $\beta \geq \frac{1}{2}$.

8.4.3 Normal model and time scaling

Proposition 8.10 (Normal VaR and ES). If $L \sim N(\mu, \sigma^2)$ and $z_\beta = N^{-1}(\beta)$, with φ the standard normal density, then

$$\text{VaR}_\beta = \mu + \sigma z_\beta, \quad \text{ES}_\beta = \mu + \sigma \frac{\varphi(z_\beta)}{1 - \beta} = \mu + \sigma \frac{e^{-z_\beta^2/2}}{\sqrt{2\pi}(1 - \beta)} \quad [\text{2024 L10 @ 18:45}].$$

Proof. $L = \mu + \sigma Z$ and quantiles are equivariant under increasing affine maps. For ES, $\mathbb{E}[Z \mid Z \geq z] = \frac{1}{1 - \Phi(z)} \int_z^\infty x \varphi(x) dx = \frac{\varphi(z)}{1 - \Phi(z)}$, since $x\varphi(x) = -\varphi'(x)$. $\square$

Numerically, $z_{0.99} = 2.326$ and $\varphi(z_{0.99})/0.01 = 2.665$, while $\varphi(z_{0.975})/0.025 = 2.338$. Hence under normality the 99% VaR ($\mu + 2.326\sigma$) and the 97.5% ES ($\mu + 2.338\sigma$) are almost the same number: this is how the regulators' switch from 99% VaR to 97.5% ES was calibrated (slide 17).

Square root of time. If daily changes ΔP_i are i.i.d. $N(0, \sigma^2)$, the T -day change has standard deviation $\sigma\sqrt{T}$, so T -day VaR $= \sqrt{T} \times$ one-day VaR, and the same for ES [2024 L10 @ 19:47]. Real returns are autocorrelated: a bad day tends to be followed by another.

Proposition 8.11 (Autocorrelated daily changes). If (ΔP_i) is stationary with variance σ^2 and $\text{Corr}(\Delta P_i, \Delta P_{i+k}) = \rho^k$ (an AR(1) structure), then

$$\text{Var}\left(\sum_{i=1}^T \Delta P_i\right) = \sigma^2 \left[T + 2 \sum_{k=1}^{T-1} (T - k) \rho^k \right],$$

and for normal changes with zero mean the ratio of T -day to one-day VaR (or ES) is the square root of the bracket. For $T = 2$ it is $\sqrt{2(1 + \rho)}$, as on slide 11.

Proof. Expand the variance of the sum: there are T diagonal terms and $2(T - k)$ pairs at lag k . $\square$

	$T = 1$	2	5	10	50	250
$\rho = 0$	1.00	1.41	2.24	3.16	7.07	15.81
$\rho = 0.05$	1.00	1.45	2.33	3.31	7.43	16.62
$\rho = 0.1$	1.00	1.48	2.42	3.46	7.80	17.47
$\rho = 0.2$	1.00	1.55	2.62	3.79	8.62	19.35

The table of slide 11 is reproduced exactly by the formula above. For the EURO STOXX the observed lag-one autocorrelation is about 0.1 (between 0.05 and 0.15 for other indices) [► 2024 L10 @ 20:54], so the usual 10-day rule $\sqrt{10} = 3.16$ underestimates the true factor 3.46 by about 10%; yet almost everybody uses $\sqrt{T}$ [► 2024 L10 @ 21:59]. (The observed higher-lag autocorrelations on the slide are close to 0 rather than ρ^k ; with only a lag-one correlation the factor is $\sqrt{10 + 2 \cdot 9 \cdot 0.1} = 3.44$, practically the same.)

Exercise 8.4 (Normal model and FRTB (not from the course)).

- Prove proposition 8.10.
- Find the confidence level β' at which the normal ES equals the normal 99% VaR, and compare it with the 97.5% used by FRTB.
- With $\delta = (3000, 4000, 2000, 1000)$ and the covariance matrix of slide 16 (rows (1.02, 0.97, 0.80, 0.94), (0.97, 1.08, 0.82, 0.96), (0.80, 0.82, 0.89, 0.76), (0.94, 0.96, 0.76, 0.96), all $\times 10^{-4}$), reproduce $\sigma_P = 96.6$, VaR = 225, ES = 257, and compute the 10-day values with and without the autocorrelation correction for $\rho = 0.1$.

8.4.4 Estimating VaR and ES

CASE STUDY: Historical simulation on four European indices (slides 12–15)

Data [► 2024 L10 @ 21:59]. Daily closes of IBEX 35 (Spain), CAC 40 (France), DAX (Germany) and EURO STOXX 50, 2006–2023, containing two stress periods (the 2008 crisis and COVID in 2020). All four are in euros, so there is no FX conversion; holidays differ across countries: days when all markets were closed were dropped, otherwise the missing prices were forward-filled, which creates zero returns [► 2024 L10 @ 23:01]. (Shepherd’s aside: in the industry a surprising share of time goes into holiday calendars and FX.)

Method [► 2024 L10 @ 24:02]. With v_i^j the level of index j on day i and positions $\delta = (3000, 4000, 2000, 1000)$ in CAC, DAX, IBEX, EURO STOXX, the P&L of day i is $\Delta P_i = \sum_j \delta_j (v_i^j - v_{i-1}^j) / v_{i-1}^j$. For a window of $n = 500$ days (two years), sort the losses: the 99% VaR is the 5th worst (1% of 500) and the 99% ES the mean of the 4 worst; the 97.5% VaR is the 12th worst and the ES the mean of the 11 worst [► 2024 L10 @ 27:07]. (Some include the VaR scenario itself in the average; the market convention is to exclude it [► 2024 L10 @ 28:10].)

Results (slides 14–15; checked against the listed 12 largest losses):

window	VaR ₉₉	ES ₉₉	VaR _{97.5}	ES _{97.5}
2022–2023 (recent)	320	377	227	303
2020–2021 (COVID)	450	815	306	534
2015–2016 (calm)	297	477	246	348
2007–2008 (crisis)	448	689	370	505

Lesson. The choice of the historical window matters far more than any other modelling choice [► 2024 L10 @ 26:07]. This is why regulators added *stressed VaR*: besides the most recent window, VaR is also computed on a fixed period known to be stressful (2008 or COVID) [► 2024 L10 @ 27:07].

Model building (variance–covariance). Assume index returns $\Delta x_i \sim N(0, \sigma_i^2)$ with correlations ρ_{ij} , covariance $C_{ij} = \rho_{ij}\sigma_i\sigma_j$, and a linear portfolio $\Delta P = \sum_i \delta_i \Delta x_i$ with sensitivities $\delta_i = \partial P / \partial x_i$ [► 2024 L10 @ 28:10]. Then $\Delta P \sim N(0, \sigma_P^2)$ with

$$\sigma_P^2 = \sum_{i,j} \rho_{ij} \delta_i \delta_j \sigma_i \sigma_j = \delta^T C \delta, \quad \text{VaR}(T, \beta) = \sqrt{T} z_\beta \sigma_P, \quad \text{ES}(T, \beta) = \sqrt{T} \frac{\varphi(z_\beta)}{1 - \beta} \sigma_P.$$

With the 2022–2023 daily covariance matrix of slide 16 (entries about 10^{-4} , i.e. daily volatilities of about 1%) one gets $\sigma_P = 96.6$, $\text{VaR}(1, 99\%) = 225$ and $\text{ES}(1, 99\%) = 257$ [► 2024 L10 @ 29:13], well below the historical 320 and 377 for the same period: the empirical distribution has fatter tails than the normal [► 2024 L10 @ 30:15].

8.4.5 Back-testing

A VaR model can be tested by counting *exceptions*, days on which the realised loss exceeded the VaR [► 2024 L10 @ 30:15]. If the model is right, exceptions are independent Bernoulli trials with probability $p = 1 - \beta$, so the number N of exceptions in n days is $\text{Bin}(n, p)$ and

$$\mathbb{P}(N \geq m) = \sum_{k=m}^n \binom{n}{k} p^k (1-p)^{n-k}.$$

The rule of thumb is to reject the model if this probability is below 5% [► 2024 L10 @ 31:18]. In the example the normal $\text{VaR}(1, 99\%) = 225$ was exceeded on 12 of the 500 days, while 5 were expected.

Back-testing VaR is easy; back-testing ES is not, and this is the main reason VaR is still used despite its flaws [► 2024 L10 @ 32:23]. ES is a property of the whole tail of a distribution, while each day provides a single draw: one realised loss cannot be compared with a tail mean without assumptions on the shape of the tail [► 2024 L10 @ 1:17:27]. Regulation moved from VaR (Basel I, 10-day 99%) to stressed VaR (Basel 2.5) to ES at 97.5% with variable liquidity horizons (FRTB, the Fundamental Review of the Trading Book). Banks still back-test the VaR: they build one distributional model, compute capital from its ES, and accept the ES if the VaR of the same model back-tests well [► 2024 L10 @ 1:16:23]. The “model” is the distributional assumption; VaR and ES are two numbers read off it [► 2024 L10 @ 1:18:28].

Exercise 8.5 (Back-testing (not from the course)).

- Compute $\mathbb{P}(N \geq 12)$ for $N \sim \text{Bin}(500, 0.01)$ exactly and with the Poisson approximation.
- Find the largest number of exceptions accepted at the 5% level.
- Why is the in-sample back-test of the example biased in favour of the model?
- Explain why a single observed loss cannot “test” an ES forecast, and propose a test that uses the ES forecasts and realised losses of many days.

8.4.6 Expected shortfall as an optimisation problem

Let $x \in X \subset \mathbb{R}^n$ be a portfolio, $y \in \mathbb{R}^m$ the random market change with density $p(y)$, and $f(x, y)$ the loss. Write $\psi(x, \alpha) = \mathbb{P}(f(x, y) \leq \alpha)$, so $\text{VaR}_\beta(x) = \min\{\alpha : \psi(x, \alpha) \geq \beta\}$.

4 COURSE NOTE: Erratum: ES formula on slide 16 of lec10

The slide writes $\text{ES}(T, \beta) = \sqrt{T} e^{-N^{-1}(\beta) \sigma_P} / (\sqrt{2\pi}(1 - \beta))$. The exponent must be $-[N^{-1}(\beta)]^2/2$, as on slide 11 and in proposition 8.10. The quoted ES = 257 was computed with the correct formula (2.665×96.6); the misprinted one would give 376.

5 COURSE NOTE: Erratum: the back-test on slide 16 and in class

The slide’s formula has $(1 - p)^k$; the binomial probability needs $(1 - p)^{n-k}$. More importantly, for $n = 500$, $m = 12$, $p = 0.01$ the correct value is $\mathbb{P}(N \geq 12) = 0.52\%$, not 5.2%. By the stated 5% rule the normal model is therefore *rejected* (even at the 1% level), whereas in class it was accepted as “not overly aggressive” [► 2024 L10 @ 1:20:34]. At the 5% level the model is accepted with up to 9 exceptions ($\mathbb{P}(N \geq 9) = 6.7\%$) and rejected from 10 on ($\mathbb{P}(N \geq 10) = 3.1\%$). The rejection also agrees with the comparison with the historical VaR (225 against 320). Note that this back-test is in-sample (covariance and exceptions from the same window), which if anything favours the model.

Theorem 8.12 (Rockafellar–Uryasev). *Let*

$$F_\beta(x, \alpha) = \alpha + \frac{1}{1-\beta} \int_{\mathbb{R}^m} [f(x, y) - \alpha]^+ p(y) dy.$$

If $\psi(x, \cdot)$ is continuous, then $F_\beta(x, \cdot)$ is convex and continuously differentiable, $\text{VaR}_\beta(x)$ is a minimiser, and $\min_\alpha F_\beta(x, \alpha) = \text{ES}_\beta(x)$. Consequently

$$\min_{x \in X} \text{ES}_\beta(x) = \min_{(x, \alpha) \in X \times \mathbb{R}} F_\beta(x, \alpha),$$

and if $f(\cdot, y)$ is convex and X convex, this is a convex problem [► 2024 L10 @ 33:27].

With S equally likely scenarios y_s (historical or simulated) and linear losses $f(x, y_s) = -x^\top r_s$, the problem becomes a *linear program*:

$$\min_{x, \alpha, u} \alpha + \frac{1}{(1-\beta)S} \sum_{s=1}^S u_s \quad \text{s.t.} \quad u_s \geq -x^\top r_s - \alpha, \quad u_s \geq 0, \quad x \in X,$$

“basically a linear problem” [► 2024 L10 @ 34:27]. No such representation exists for VaR. So ES is harder to back-test but much easier to optimise, and for margin optimisation the latter matters more.

Exercise 8.6 (ES as a linear program (not from the course)).

- Complete the proof of [theorem 8.12](#) for a discrete distribution with S equally likely scenarios, taking βS to be an integer, and check that $\min_\alpha F$ is the average of the $(1-\beta)S$ largest losses.
- Write the LP that minimises the 97.5% ES of a long-only fully invested portfolio of the four indices over the 500 historical scenarios of 2022–2023.
- Why can the same trick not be used for VaR?

Proof. $\alpha \mapsto [f - \alpha]^+$ is convex, so F_β is convex in α , and

$$\frac{\partial F_\beta}{\partial \alpha} = 1 - \frac{\mathbb{P}(f > \alpha)}{1-\beta} = 1 + \frac{\psi(x, \alpha) - 1}{1-\beta} = \frac{\psi(x, \alpha) - \beta}{1-\beta},$$

which vanishes at $\alpha_\beta = \text{VaR}_\beta(x)$. There

$$\int [f - \alpha_\beta]^+ p dy = \int_{f \geq \alpha_\beta} f p dy - \alpha_\beta (1 - \psi(x, \alpha_\beta)) = (1-\beta) \text{ES}_\beta(x)$$

so $F_\beta(x, \alpha_\beta) = \alpha_\beta + \text{ES}_\beta - \alpha_\beta = \text{ES}_\beta$ (slides 18–20). If f is jointly convex in (x, α) , so is $[f - \alpha]^+$ and hence F_β ; minimising first over α and then over x is the same as minimising jointly. $\square$

8.5 Margin: variation margin, initial margin and ISDA SIMM

8.5.1 Why margin

Consider a bank A that is long a swap with C and short an offsetting swap with B (slide 21). A receives cash flows $\{T_i\}$ and pays $\{S_i\}$ on each swap, typically every three or six months; before the next payment date the counterparty may default and not pay [► 2024 L10 @ 35:33]. To mitigate this the parties exchange *variation margin* daily, equal to the current value V of the trades [► 2024 L10 @ 36:34].

If C defaults it stops paying VM. A must replace or unwind the trade with C, which takes some days; meanwhile its hedge with B keeps moving and A keeps paying VM to B. If the value of the defaulted trade to A rises in the meantime, A has to pay more for the replacement than the collateral it holds, and that is a loss [► 2024 L10 @ 37:35]. *Initial margin* covers this loss.

Definition 8.13 (Margin). *Margin is collateral posted by one counterparty to the one that collects it, to cover potential losses on derivatives (slide 22).*

- *Variation margin* (VM) covers the current present value.
- *Initial margin* (IM) covers the potential change of value between the counterparty’s

default and the time $t = \delta t$ at which the surviving party can exit the position. The period δt is the *margin period of risk* (MPOR).

IM is sized as a VaR (or ES) of the value change over the MPOR [► 2024 L10 @ 38:39].

Slide 23 shows it as a picture: a value path up to the default, a fan of possible paths over the MPOR, and a distribution at close-out whose upper tail is covered by the IM collected and whose lower tail is the IM posted. IM is held in segregated accounts and cannot be used for trading, so it is a real cost (the MVA of [section 8.3](#)) [► 2024 L10 @ 1:12:07].

There are two kinds of IM [► 2024 L10 @ 38:39]. *Cleared* IM is set by each CCP's own model and has existed for decades. *Bilateral* IM on uncleared trades was introduced by the post-crisis rules (Basel III/BCBS-IOSCO) and phased in from 2016; the industry standard model is ISDA's *Standard Initial Margin Model* (SIMM).

Exercise 8.7 (Tang's puzzle with collateral (not from the course)). Redo the puzzle of [section 8.3](#) (value 0 $\rightarrow$ +100, default with $R = 50\%$) when the trade is under a CSA with daily VM and zero threshold, the counterparty stops paying VM at default, and the value moves from 100 to 104 during a 10-day MPOR. What is the loss with and without an initial margin of 5? What part of the original \$50M loss does VM remove, and what part does IM remove?

8.5.2 The SIMM as a delta-gamma VaR

SIMM is an exposure-based proxy for a 10-business-day ($\delta t = 14/365$ in calendar time), 99% VaR [► 2024 L10 @ 39:41]. Let $P(t)$ be the portfolio value to the surviving party, the counterparty defaulting at $t = 0$, with $VM(0) = P(0)$. Ignoring path dependence, cash flows in $(0, \delta t)$ and discounting, and letting only the market state $\mathcal{M}$ move by $\Delta \mathcal{M}$ over the MPOR, a second-order expansion gives

$$P(\delta t) \approx P(0) + \delta^\top \Delta \mathcal{M} + \frac{1}{2} \Delta \mathcal{M}^\top \gamma \Delta \mathcal{M},$$

with δ the vector of deltas and γ the (symmetric) gamma matrix. The loss is the *increase* $P(\delta t) - P(0)$, since the collateral is frozen at $P(0)$, and IM should satisfy $\mathbb{P}(P(\delta t) - P(0) < \text{IM}) = 0.99$.

Proposition 8.14 (Moments of the delta-gamma P&L). If $\Delta \mathcal{M} \sim N(0, C)$, then $\Delta P = \delta^\top \Delta \mathcal{M} + \frac{1}{2} \Delta \mathcal{M}^\top \gamma \Delta \mathcal{M}$ has

$$\mathbb{E}[\Delta P] = \frac{1}{2} \text{tr}(\gamma C), \quad \text{Var}(\Delta P) = \delta^\top C \delta + \frac{1}{2} \text{tr}((\gamma C)^2).$$

Approximating ΔP by a normal with these moments, $\text{IM} \approx \frac{1}{2} \text{tr}(\gamma C) + 2.326 \sqrt{\delta^\top C \delta + \frac{1}{2} \text{tr}((\gamma C)^2)}$.

Without gamma this is exactly the model-building VaR of [section 8.4.4](#) with $T = 10$, $\beta = 0.99$: $\text{IM} = 2.326 \sqrt{10} \sqrt{\delta^\top C_1 \delta}$ for a daily covariance C_1 [► 2024 L10 @ 39:41]. The gamma (convexity) terms are the “two additional terms” of the slide: SIMM is “basically VaR”.

Exercise 8.8 (Delta-gamma margin (not from the course)).

- Prove $\text{Var}(X^\top \gamma X) = 2 \text{tr}((\gamma C)^2)$ for $X \sim N(0, C)$ and symmetric γ .
- One risk factor with daily variance c , delta δ and gamma γ : write mean and

Proof. Write $X = \Delta \mathcal{M}$. $\mathbb{E}[X^\top \gamma X] = \sum_{ij} \gamma_{ij} C_{ij} = \text{tr}(\gamma C)$. By Isserlis' (Wick's) theorem, $\mathbb{E}[X_i X_j X_k X_l] = C_{ij} C_{kl} + C_{ik} C_{jl} + C_{il} C_{jk}$, so

$$\mathbb{E}[(X^\top \gamma X)^2] = (\text{tr} \gamma C)^2 + 2 \text{tr}(\gamma C \gamma C), \quad \text{Var}(X^\top \gamma X) = 2 \text{tr}((\gamma C)^2)$$

the factor $\frac{1}{4}$ gives $\frac{1}{2} \text{tr}((\gamma C)^2)$. The cross term $\text{Cov}(\delta^\top X, X^\top \gamma X)$ involves third moments of a centred Gaussian and vanishes. $\square$

6 COURSE NOTE: Caveat on slide 25 of lec10

The slide gives $\text{IM} \approx 2.326 \sqrt{10} \sqrt{\delta^\top C \delta} + \frac{1}{2} \sqrt{10} \text{tr}(\gamma C) + \lambda \sqrt{\frac{1}{2} \text{tr}((\gamma C)^2)}$. Two remarks.

variance of the 10-day ΔP and check how each term scales with the horizon.

- (c) For a short option position ($\gamma < 0$ in the value to the surviving party) discuss whether the normal approximation over- or underestimates the 99% quantile.

- Splitting the square root of a sum into a delta term and a curvature term, with a separately calibrated multiplier λ , is a modelling choice (SIMM treats delta and curvature margins separately), not a consequence of the derivation; moreover ΔP is not Gaussian, so λ absorbs the skew of the quadratic term.
- The time scaling is inconsistent: if C is the *daily* covariance, the 10-day covariance is $10C$, so the mean term is $\frac{1}{2} \text{tr}(\gamma 10C) = 5 \text{tr}(\gamma C)$, not $\frac{1}{2} \sqrt{10} \text{tr}(\gamma C)$, and the last term scales by $\sqrt{10}$, not by 1. Deltas scale like $\sqrt{T}$, gammas like T .

8.5.3 SIMM in practice

The formulation above has too many risk factors (so C , δ , γ are huge) and γ is hard to compute. SIMM therefore simplifies further (slides 26–27): curvature is proxied from *vega*, $\gamma \propto \mathcal{V}/(T\sigma_N)$; risk factors are grouped into *buckets* (e.g. currencies) and *risk classes* (IR, FX, equity, commodity, qualifying and non-qualifying credit); sensitivities are delta, vega and curvature; the results are aggregated up a tree with prescribed correlations, and the *product classes* (rates/FX, credit, equity, commodity) are simply added with no netting between them. Additional terms handle thresholds and concentration, and when several regulators apply the posted IM is the maximum over them.

Calibration [► 2024 L10 @ 40:43]. The correlations and risk weights, which cover all asset classes, are recalibrated on historical data. In the lecture (October 2024) SIMM 2.7, due in early December 2024, was expected to be the first calibration without the 2008 or COVID periods in its window, cutting SIMM IM by roughly 5–20% and releasing on the order of \$40bn of margin. When the 2020 shocks arrived, the annual calibration made SIMM underestimate risk for more than a year; ISDA, which owns the model and runs the calibration with the banks, then moved to semi-annual recalibration [► 2024 L10 @ 1:13:11].

Large positions (slide 28). Big positions take longer to unwind, which is liquidity risk again. One approach scales the MPOR with the position size, $\Delta T = \Delta T_0 \max(1, N/N_0)$; since IM grows like $N\sqrt{\Delta T}$, beyond N_0 one gets $\text{IM} \propto N^{3/2}$. Another adds a concentration add-on $W(\delta) \delta^T C \delta$ with W increasing (e.g. piecewise linear). Both keep the IM convex in the position.

8.6 Multilateral optimisation of initial margin

This is the main topic of the 2024 lecture and the business of Quantile.

8.6.1 The problem

Consider parties $\mathcal{P} = \{A, B, C, D, E, \dots\}$ (banks) with bilateral portfolios against each other and cleared portfolios at one or more CCPs, all posting IM (slide 29) [► 2024 L10 @ 41:48]. The goal is to find a set of new *hedge trades* between pairs of parties that reduces the total IM in the system (slide 30) [► 2024 L10 @ 42:50]. There are no hedge variables with the CCP: one can only book trades between two banks, but if the chosen instrument is clearable (a vanilla swap) the trade is cleared and changes the cleared IM instead of the bilateral one [► 2024 L10 @ 43:55].

Definition 8.15 (Margin minimisation problem, slide 31). Let $\mathcal{H}$ be the set of hedge instruments, $\mathcal{M}$ the market risk factors, $D_{p,q,m}^0$ the initial sensitivity of party p 's portfolio facing q to factor m , and $D_{h,p,q,m}$ the sensitivity of one unit of hedge h . The

unknowns are the amounts $x_{h,p,q}$ of hedge h that p trades with q . Solve

$$\begin{aligned}
 \min_x \quad & \sum_{p \in \mathcal{P}} \left(\sum_{q \neq p} \text{SIMM}_{p,q}(D_{p,q}) + \sum_{c \in \text{CCP}} \text{IM}_{p,c}(D_{p,c}) \right) \\
 \text{s.t.} \quad & D_{p,q,m} = D_{p,q,m}^0 + \sum_{h \in \mathcal{H}} x_{h,p,q} D_{h,p,q,m} & \forall m \in \mathcal{M}, \\
 & x_{h,p,q} = -x_{h,q,p} & \forall h, p \neq q \quad (\text{symmetry}), \\
 & \sum_q x_{h,p,q} = 0 & \forall h, p \quad (\text{flatness}), \\
 & T_{p,q,m}^- \leq \sum_h x_{h,p,q} D_{h,p,q,m} \leq T_{p,q,m}^+ & \forall m, p \neq q \quad (\text{risk limits}).
 \end{aligned}$$

The objective is the total margin in the system: bilateral SIMM summed over all pairs, plus cleared IM summed over CCPs [► 2024 L10 @ 43:55]. The new risk is the old risk plus the sum of the hedges times their risk [► 2024 L10 @ 44:59]. The constraints:

- **Symmetry:** if p sells something to q , q must buy exactly that from p . This was the constraint the banks worried about most: telling Goldman to buy 100 from JPMorgan while telling JPMorgan to sell 200 to Goldman would only be discovered when the trade is booked [► 2024 L10 @ 46:02].
- **Flatness** (cash-flow neutrality): whatever p sells to one party it buys from others, so each party's total position in every instrument, and hence its market risk, is unchanged. Only *counterparty* risk is reshuffled, and nobody makes or loses money from the system itself. It is not strictly necessary, but relaxing it makes everything much harder [► 2024 L10 @ 47:02].
- **Risk limits:** each party can bound how much the risk to each factor against each counterparty may change [► 2024 L10 @ 48:06].

7 COURSE NOTE: Notation on slide 31

The slide writes $\text{IM}_{p,ccp}(D_{p,q})$; the argument should be the cleared sensitivity $D_{p,ccp}$. The risk update is written for bilateral risk only; for clearable hedge types $h \in \mathcal{H}_c$ the update goes to the CCP, $D_{p,c,m} = D_{p,c,m}^0 + \sum_{h \in \mathcal{H}_c} \sum_q x_{h,p,q} D_{h,p,q,m}$, which is how cleared IM enters the optimisation [► 2024 L10 @ 42:50].

8.6.2 Toy model: the median rule

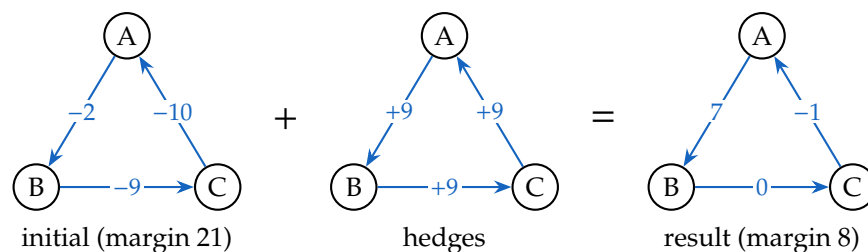


Figure 8.4: The toy model of slide 32. An arrow $p \rightarrow q$ labelled x means that p 's position against q is x (and q 's against p is $-x$). Moving $t = 9$ units around the cycle $A \rightarrow B \rightarrow C \rightarrow A$ satisfies symmetry and flatness by construction and reduces the total $|x|$ from $2 + 9 + 10 = 21$ to $7 + 0 + 1 = 8$.

Proposition 8.16 (Median rule for a triangle). *Three parties, one risk factor, one hedge instrument, bilateral margin proportional to $|\text{net position}|$ (the one-factor SIMM $2.326\sqrt{10} \sigma|D|$). Let a_1, a_2, a_3 be the initial positions oriented around the cycle $A \rightarrow B \rightarrow C \rightarrow A$. Then every hedge satisfying symmetry and flatness adds the same amount t to all three oriented edges, and the total margin $\sum_k |a_k + t|$ is minimised at $t = -\text{median}(a_1, a_2, a_3)$.*

Proof. Flatness at A gives $x_{AB} + x_{AC} = 0$, i.e. $x_{AB} = x_{CA}$ by symmetry; at B, $x_{BA} + x_{BC} = 0$, i.e. $x_{BC} = x_{AB}$. So $x_{AB} = x_{BC} = x_{CA} = t$. The function $g(t) = \sum_k |a_k + t|$ is convex and piecewise linear with slope $\#\{k : a_k + t > 0\} - \#\{k : a_k + t < 0\}$, which changes sign at $t = -\text{median}(a)$. $\square$

In the example of slide 32, $a = (-2, -9, -10)$, the median is -9 , and moving 9 units around the triangle turns the positions into $(7, 0, -1)$, cutting the margin from 21 to 8

[► 2024 L10 @ 48:06]. Any network can be decomposed into triangles, and one can think of the optimisation as moving the median around each triangle. Since edges are shared between triangles and there are additional constraints, this is only a heuristic and not the real solution method [► 2024 L10 @ 49:07].

INTUITION: Hedges are circulations on a graph

Think of $x_{h,i,j}$ as a *flow* on the complete graph of parties: symmetry says it is an antisymmetric edge function, and flatness says its divergence vanishes at every node (Kirchhoff's current law). Divergence-free flows are circulations, i.e. the cycle space of the graph, of dimension $E - V + 1$; for n fully connected parties this is $\binom{n}{2} - n + 1 = \binom{n-1}{2}$, spanned by triangles. For a triangle it is one number, the t above. The optimisation moves risk around loops without creating or destroying it at any node.

8 COURSE NOTE: Erratum: result panel of slide 32

In the "result" triangle the labels on the A-B edge are -7 next to A and 7 next to B. Adding the "initial" (-2 at A, 2 at B) and "hedges" (9 at A, -9 at B) panels gives +7 at A and -7 at B. The margin, which depends only on $|x|$, is unaffected.

Exercise 8.9 (Toy margin optimisation (not from the course)).

- Four parties on a cycle $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ with oriented positions $a = (3, -5, 8, -1)$: find the optimal circulation t and the margin saving.
- Now add the chord A-C with position 2: parametrise all hedges satisfying symmetry and flatness (the cycle space has dimension 2) and minimise the total $\sum |x|$.
- For the fairness table of [section 8.6](#), check the least-squares criterion and propose a different fairness criterion (e.g. max-min of relative savings) that gives a different answer on some example.

8.6.3 Solving it in practice

In reality one writes down the problem and hands it to a numerical solver [► 2024 L10 @ 50:07]. Some lessons:

- **Convexity is everything.** Minimising ES is piecewise linear, SIMM is *nearly* convex and concentration add-ons are *potentially* convex (slide 33). Using general non-linear solvers did not work well; it is much easier to approximate the objective by a genuinely convex function and use a mature convex solver. Do not write your own [► 2024 L10 @ 51:07]. How hard a problem is depends on its structure and sparsity much more than on its size.
- **Modelling languages** separate the statement of a problem from the numerics. Quantile uses Gurobi; the slide's example is the integer program $\max 5x_1 + 8x_2$ s.t. $x_1 + x_2 \leq 5$, $3x_1 + 7x_2 \leq 25$, $x_i \in \mathbb{N}$, solved in 10 ms with optimum 31 at $(3, 2)$ [► 2024 L10 @ 52:08].
- **Size.** The FX-delta problem (slide 36): 25 parties, 30 currencies, cleared and uncleared non-deliverable forwards (NDFs), one short maturity, about 18,000 variables ($25 \cdot 24$ ordered pairs $\times$ 30 currencies) and 40,000 constraints (pairwise risk limits, pairwise IM-increase limits, notional efficiency, symmetry, flatness, lot sizes), with SIMM plus cleared IM as a convex objective [► 2024 L10 @ 53:14]. NDFs are chosen because each has exposure to exactly one risk factor, one currency pair.
- **Linear algebra.** Interior-point methods take Newton steps: they find the zero of the gradient, and each step requires solving with the Hessian, naively $O(n^3) \approx 10^{13}$ flops for $n \sim 40,000$ [► 2024 L10 @ 54:18]. In practice one does far better: the matrix is positive semidefinite (Cholesky), very sparse (well under 0.1% non-zero) and structured. A

few global rows (symmetry) and party rows (flatness) sit on top of a block-diagonal part, the 36,000 risk constraints of individual (p, q) pairs, which is purely diagonal when each hedge has one risk factor. This *Dantzig–Wolfe* (block-angular) structure is exploited by the solvers [► 2024 L10 @ 55:18].

- **Presolve** removes redundant rows ($x_1 \geq 1$ is implied by $x_1 \geq 2$) and here halved the problem; the barrier method then converged in 58 iterations and 11 s, thanks to the quadratic convergence of Newton’s method [► 2024 L10 @ 59:29].

Nearly parallel risks. The real difficulties are numerical [► 2024 L10 @ 56:22]. Two trades with almost identical risk, e.g. exposures $3E$ and E to two risk factors, give nearly singular systems (slide 39):

$$\begin{pmatrix} 1 & 1 \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{3} \end{pmatrix} \text{ has the solutions } y_1 = 1 - x_1, \quad \text{but} \quad \begin{pmatrix} 1 & 1 \\ 0.3333 & 0.333 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 1 \\ b \end{pmatrix}$$

has the unique solution $(0, 1)$ for $b = 0.333$, $(1, 0)$ for $b = 0.3333$, and $(-110, 111)$ for $b = 0.3$ [► 2024 L10 @ 57:25]. The determinant is -3×10^{-4} and the condition number about 7×10^3 , so rounding in the fourth digit moves the solution by $O(1)$ or even by hundreds. With a sign constraint $x_i \geq 0$ the last system is declared *infeasible* purely because of numerical noise [► 2024 L10 @ 58:28]. Most of the practical work is detecting and removing such situations, not solving convex problems in general.

Results. On a 5-party and a 20-party system with one risk factor and no constraints, almost all margin disappears, and every party ends up either long or short the factor against all its counterparties, never both: all risk has been netted (slides 41–42) [► 2024 L10 @ 1:00:32]. With real constraints (slide 43) the picture is less clean. The empty diagonal is “margin against oneself”, and a block in the middle turned out to be the affiliates of one large bank, which do not exchange margin with each other [► 2024 L10 @ 1:01:35]. A second application (slide 44) is interest-rate margin: 15 parties, USD/EUR/GBP, swaptions and cleared swaps, about 5000 variables and 10,000 constraints, where each trade loads on several risk factors (a swap expiring in 12 years is exposed to the 10Y and 20Y swap rates).

Exercise 8.10 (Rounding and integrality (not from the course)). For the Gurobi example, $\max 5x_1 + 8x_2$ subject to $x_1 + x_2 \leq 5$, $3x_1 + 7x_2 \leq 25$, $x_1, x_2 \geq 0$:

- solve the LP relaxation (optimum 32.5 at $(2.5, 2.5)$);
- show that rounding the LP solution gives either an infeasible point or a poor one;
- find the integer optimum 31 by enumeration. Relate this to the rounding of notionals in [section 8.6](#).

Exercise 8.11 (Nearly parallel risks (not from the course)). For $A_\varepsilon = \begin{pmatrix} 1 & 1 \\ 1/3+\varepsilon & 1/3 \end{pmatrix}$ solve $A_\varepsilon(x, y)^\top = (1, b)^\top$ and show that $x = (b - 1/3)/\varepsilon$. Compute the condition number of A_ε for small ε and explain the three solutions of slide 39. How would you regularise the problem (e.g. a small penalty $\eta\|x\|^2$)?

8.6.4 Nice and fair solutions

A solution must be *feasible* (all constraints hold) and *optimal* (a good IM saving); in practice it must also be *nice* and *fair* [► 2024 L10 @ 1:02:40].

Nice. Traders want round notionals (e.g. multiples of a million), few trades (*notional efficiency*), trade packages, and possibly more trades moved into CCPs for reasons beyond IM. The worst headache came from NDFs: a USD notional and a EUR notional define the FX rate as their ratio, to 15 digits, but banks’ booking systems keep only 6–8 decimals, so the notionals had to be chosen as round dollar amounts that give exact euro amounts at a 4- or 6-decimal rate [► 2024 L10 @ 1:03:41]. Rounding cannot be done afterwards. If a party sells 8 to one counterparty and buys 4 from each of two others, rounding to the nearest 10 gives 10, 0, 0 and breaks flatness. Rounding must be part of the optimisation, which makes it a *mixed-integer* problem, much harder: the FX rounding step of slide 46 had about 10^4 integer variables and took about a minute of branch-and-bound [► 2024 L10 @ 1:04:43].

Fair. What does fairness mean for an optimiser [► 2024 L10 @ 1:05:47]? In the four-party example of slide 47 three solutions give the same total saving:

party	scenario 1	scenario 2	scenario 3	single-party best
A	4	0	2	4
B	4	4	4	4
C	4	4	4	4
D	0	4	2	4

Scenarios 1 and 2 leave one party with nothing; scenario 3 splits the saving, but one could still ask why B and C get their full 4 [► 2024 L10 @ 1:06:47]. One proposal is a two-step algorithm. First optimise for each party alone, keeping all constraints, to get the best achievable IM_p^{best} ; then solve $\min \sum_p (IM_p - IM_p^{best})^2$ [► 2024 L10 @ 1:07:53]. Here the squared shortfalls are 16, 16 and $4 + 4 = 8$, so the algorithm picks scenario 3 [► 2024 L10 @ 1:09:00]. Fairness has no established definition yet; Shepherd considers it the most interesting open problem in this area.

Network effect. Across real runs, both the saving and the saving as a fraction of the initial IM grow with the number of participating parties (slide 49): bigger networks benefit everyone [► 2024 L10 @ 1:10:04].

IN FINANCE: Optimisation versus compression (not discussed in class)

Portfolio compression terminates redundant offsetting trades (e.g. circles of swaps A–B–C–A) and replaces them with fewer trades of smaller gross notional and the same net risk. The IM optimisation of this lecture instead *adds* new trades that redistribute risk between pairs of counterparties. Both keep every party’s market risk unchanged (the flatness constraint) and exploit the same cycle structure of the network.

8.7 Enterprise-level modelling and martingale tools

The second half of the 2013 lecture turns to the models behind CVA.

8.7.1 Trade-level versus enterprise-level models

A *trade-level* model prices each trade independently (PV and Greeks), and the portfolio PV and Greeks are the sums: this is what “derivatives model” usually means [► 2013 L26 @ 44:53].

Enterprise-level models handle the quantities that are *not* linear in the trades: CVA, funding, capital (risk-weighted assets), liquidity [► 2013 L26 @ 45:57]. They must simulate all trades and all market factors of a portfolio consistently, with a proper joint distribution, while reusing the trade-level models to value each trade at future dates.

There is also feedback to the trade-level model [► 2013 L26 @ 47:02]. A cancellable swap facing a counterparty close to default has an exercise boundary that ignores counterparty credit; cancelling early reduces or removes the CVA. In practice the CVA desk pays the trading desk the CVA saving, which the trade model can take as an input to its exercise decision [► 2013 L26 @ 48:05].

Putting models together is numerically delicate: some trades are priced by PDE, some by simulation, and a joint simulation introduces errors [► 2013 L26 @ 50:14]. Tang's remedy is built on martingales: the arbitrage-free price of every traded quantity is a martingale under a suitable measure, and those conditions can be tested and enforced.

8.7.2 Martingale measures and change of numeraire

By Harrison and Pliska, if Y_t is the price of a traded asset with no intermediate cash flows and $N_t > 0$ is a *numeraire* (another such price), then Y_t/N_t is a martingale under the measure $\mathbb{P}_N$ associated with N [► 2013 L26 @ 52:18]. The standard pairs (slides 13–15; see chapters 15, 23 and 24) are:

numeraire	measure	martingale
$\beta(t) = e^{\int_0^t r_u du}$	risk-neutral $\mathbb{Q}$	$Y_t/\beta(t)$
$P(t, T)$ (zero-coupon bond)	T -forward $\mathbb{P}_T$	forward price $F_Y(t, T) = Y_t/P(t, T)$
$P(t, T_{i+1})$	$\mathbb{P}_{T_{i+1}}$	forward LIBOR $L_i(t) = \frac{1}{\delta_i} \left(\frac{P(t, T_i)}{P(t, T_{i+1})} - 1 \right)$
$A_{i,N}(t) = \sum_{j=i+1}^N \delta_{j-1} P(t, T_j)$	annuity $\mathbb{P}_{A_{i,N}}$	swap rate $S_{i,N}(t) = \frac{P(t, T_i) - P(t, T_N)}{A_{i,N}(t)}$
$P^D(t, T)$ (domestic)	$\mathbb{P}_{T,D}$	FX forward $F_{FX}(t, T) = \frac{S_{FX}(t) P^F(t, T)}{P^D(t, T)}$

The pattern is always the same: write the quantity as a ratio of two traded assets [► 2013 L26 @ 55:28]. LIBOR is (a multiple of) a portfolio of two bonds divided by the bond $P(t, T_{i+1})$ [► 2013 L26 @ 53:23]; the swap rate is a portfolio of two bonds divided by the annuity [► 2013 L26 @ 54:27]; the FX forward is interest-rate parity, the foreign bond converted into domestic currency divided by the domestic bond.

Proposition 8.17 (Change of numeraire, slide 16). Let $N_1, N_2 > 0$ be numeraires and Y_T an $\mathcal{F}_T$ -measurable payoff. Then

$$\mathbb{E}_t^{N_1}[Y_T] = \frac{N_2(t)}{N_1(t)} \mathbb{E}_t^{N_2} \left[Y_T \frac{N_1(T)}{N_2(T)} \right], \quad \text{i.e.} \quad \frac{d\mathbb{P}_{N_1}}{d\mathbb{P}_{N_2}} \Big|_{\mathcal{F}_T} = \frac{N_1(T)/N_1(0)}{N_2(T)/N_2(0)}.$$

With $N_1 = P(\cdot, T_i)$, $N_2 = P(\cdot, T_{i+1})$ and $N_1(T)/N_2(T) = 1 + \delta_i L_i(T)$ for $T \leq T_i$: $\mathbb{E}_t^{T_i}[Y_T] = \mathbb{E}_t^{T_{i+1}}[Y_T(1 + \delta_i L_i(T))]/(1 + \delta_i L_i(t))$. This is the change of measure behind the LIBOR market model [► 2013 L26 @ 56:29].

Credit: the survival measure [► 2013 L26 @ 58:38]. By analogy with the swap rate one expects the forward CDS par spread $\tilde{S}_j(t, T_i, T_{i+1})$ of name j to be a martingale under a “risky annuity” measure. The catch is that the risky annuity $\tilde{A}_j$ (the value per unit spread of the premium leg, which pays only while the name survives, i.e. a portfolio of zero-

Proof. The price X_t of the claim paying $X_T = Y_T N_1(T)$ does not depend on the measure used to compute it [► 2013 L26 @ 57:32]: $X_t = N_1(t) \mathbb{E}_t^{N_1}[X_T/N_1(T)] = N_2(t) \mathbb{E}_t^{N_2}[X_T/N_2(T)]$. Substitute X_T and divide by $N_1(t)$. $\square$

recovery defaultable bonds) drops to zero at default, and a numeraire must stay positive [► 2013 L26 @ 59:40]. Schönbucher's solution is to ignore the default states [► 2013 L26 @ 1:00:45]: since $\mathbf{1}_{\{t < \tau_j\}} \tilde{S}_j \tilde{A}_j$ is the value of the premium leg, a traded asset, $\mathbf{1}_{\{t < \tau_j\}} \tilde{S}_j \tilde{A}_j / \beta(t)$ is a $\mathbb{Q}$ -martingale, and the *survival measure*

$$\left. \frac{d\mathbb{P}_{\tilde{A}_j}}{d\mathbb{Q}} \right|_t = \frac{\mathbf{1}_{\{t < \tau_j\}} \tilde{A}_j(t, T_i, T_{i+1})}{\beta(t) \tilde{A}_j(0, T_i, T_{i+1})}$$

makes $\tilde{S}_j(t, T_i, T_{i+1})$ a martingale [► 2013 L26 @ 1:03:55]. $\mathbb{P}_{\tilde{A}_j}$ is absolutely continuous with respect to $\mathbb{Q}$ but *not equivalent* (it gives zero mass to default before t). One difficulty is traded for another, but the second is easier, provided what happens at default is modelled separately [► 2013 L26 @ 1:01:49].

Exercise 8.12 (Change of numeraire (not from the course)).

- Prove [proposition 8.17](#) and deduce that $L_i(t)$ is a $\mathbb{P}_{T_{i+1}}$ -martingale.
- Show that the swap rate $S_{i,N}(t)$ is a martingale under the annuity measure and that it is a weighted average of forward LIBORs with weights $\delta_j P(t, T_{j+1}) / A_{i,N}(t)$.
- Verify [proposition 8.18](#) and show by an example that ordinary linear interpolation in maturity T is not a martingale in general.

8.7.3 Martingale testing, resampling and interpolation

- Martingale testing:** check in the numerical implementation that known martingale relations such as $F(s) = \mathbb{E}_s[F(t)]$ hold; usually they fail slightly [► 2013 L26 @ 1:04:58].
- Martingale resampling:** transform the simulated variables (linearly or log-linearly) so that they satisfy the relations *exactly*, keeping a variance metric fixed. The quadratic resampling of slide 19 is $X = \frac{\text{Std}(X)}{\text{Std}(X_0)} (X_0 - \mathbb{E}[X_0]) + \mathbb{E}[X]$, with X_0 the raw simulated values, $\mathbb{E}[X]$ the martingale target and $\text{Std}(X)$ the target dispersion [► 2013 L26 @ 1:05:59]. This is moment matching, familiar from variance reduction in Monte Carlo.
- Martingale interpolation:** models simulate only a few tenors (say 1Y and 5Y LIBOR). Interpolating a missing tenor (3Y) linearly in maturity does *not* preserve the martingale property; interpolating linearly using the time- s values as the independent variable does [► 2013 L26 @ 1:07:01].

Proposition 8.18 (Martingale interpolation). Let $M(\cdot, T_1)$ and $M(\cdot, T_2)$ be martingales, $M(s, T) = \mathbb{E}_s[M(t, T)]$ for $s \leq t$, and let the target $M(s, T_3)$ be known at a reference time s (e.g. $s = 0$, from today's curve). Define for $t \geq s$

$$M(t, T_3) = \frac{M(s, T_3) - M(s, T_2)}{M(s, T_1) - M(s, T_2)} M(t, T_1) + \frac{M(s, T_1) - M(s, T_3)}{M(s, T_1) - M(s, T_2)} M(t, T_2).$$

Then $\mathbb{E}_s[M(t, T_3)] = M(s, T_3)$ [► 2013 L26 @ 1:10:06].

Proof. The weights are $\mathcal{F}_s$ -measurable and sum to 1. Taking $\mathbb{E}_s$ replaces $M(t, T_k)$ by $a_k = M(s, T_k)$, giving $[(a_3 - a_2)a_1 + (a_1 - a_3)a_2] / (a_1 - a_2) = a_3$ [► 2013 L26 @ 1:11:10]. $\square$

Equivalently, $M(t, T_3)$ is read off the straight line through the points $(a_1, M(t, T_1))$ and $(a_2, M(t, T_2))$ at the abscissa a_3 : linear interpolation with the “right” independent variable.

The LIBOR market model [► 2013 L26 @ 1:12:13]. Martingale modelling proceeds in three steps: find the measure under which a quantity is a martingale, represent the martingale as a stochastic integral (log-normal, CEV or stochastic volatility), then change to a common measure [► 2013 L26 @ 1:13:17]. For forward LIBORs, $dL_i/L_i = \lambda_i(t) dW_t^{T_{i+1}}$ under $\mathbb{P}_{T_{i+1}}$. Under the terminal measure $\mathbb{P}_{T_N}$ (slide 20),

$$\frac{dL_i(t)}{L_i(t)} = - \sum_{j=i+1}^{N-1} \frac{\lambda_i \lambda_j \delta_j L_j(t)}{1 + \delta_j L_j(t)} \rho_{ij} dt + \lambda_i dW_t^{T_N, L_i}, \quad \rho_{ij} = \text{Corr}(d \ln L_i, d \ln L_j),$$

the full-dimensional model with one Brownian motion per LIBOR and the correlations as inputs [► 2013 L26 @ 1:15:33]. The drift follows from [proposition 8.17](#) and Girsanov's theorem ([chapter 17](#)); interest-rate models of this kind are the subject of [chapter 24](#).

8.8 Closing remarks of the 2013 course

KEY FORMULAS

Exposure:	$E(t) = (V(t) - C(t))^+,$ $EE(t) = \mathbb{E}[E(t)], \quad (\sum_i V_i)^+ \leq \sum_i V_i^+$
CVA:	$CVA = -\mathbb{E}^Q[\mathbf{1}_{\{\tau \leq T\}}(1 - R)(V_{\tau-} - C_{\tau-})^+ / \beta_\tau] \approx -EPE \cdot s \cdot D$
DVA:	$DVA = \mathbb{E}^Q[\mathbf{1}_{\{\bar{\tau} \leq T\}}(1 - \bar{R})(C_{\bar{\tau}-} - V_{\bar{\tau}-})^+ / \beta_{\bar{\tau}}]$
VaR, ES:	$VaR_\beta = \inf\{\ell : \mathbb{P}(L \leq \ell) \geq \beta\},$ $ES_\beta = \frac{1}{1-\beta} \int_\beta^1 VaR_u du$
Normal:	$VaR_\beta = \mu + z_\beta \sigma,$ $ES_\beta = \mu + \sigma \varphi(z_\beta) / (1 - \beta),$ $\sigma_P^2 = \delta^T C \delta$
Time scaling:	factor $\sqrt{T + 2 \sum_{k=1}^{T-1} (T - k) \rho^k}$ ($\sqrt{T}$ if $\rho = 0$)
Back-test:	$\mathbb{P}(N \geq m) = \sum_{k \geq m} \binom{n}{k} p^k (1 - p)^{n-k}$
Rockafellar–Uryasev:	$ES_\beta = \min_\alpha \left\{ \alpha + \frac{1}{1-\beta} \mathbb{E}[(L - \alpha)^+] \right\}$
Delta–gamma IM:	$\mathbb{E}[\Delta]P = \frac{1}{2} \text{tr}(\gamma C),$ $\text{Var } \Delta P = \delta^T C \delta + \frac{1}{2} \text{tr}((\gamma C)^2)$
Margin optim.:	$\min \sum_p (\sum_q \text{SIMM}_{p,q} + \sum_c \text{IM}_{p,c}),$ $x_{h,p,q} = -x_{h,q,p}, \quad \sum_q x_{h,p,q} = 0$
Change of numeraire:	$\mathbb{E}_t^{N_1}[Y_T] = \frac{N_2(t)}{N_1(t)} \mathbb{E}_t^{N_2}[Y_T N_1(T) / N_2(T)]$

9 COURSE NOTE: Tang's conclusion and the end of 18.S096 (2013)

Tang, who presented from New Jersey because snow had grounded his flight [► 2013 L26 @ 0:00], concluded that banks need enterprise-level models for the non-linear portfolio effects (CVA, funding, liquidity, capital), built on top of trade-level models with martingale testing, resampling and interpolation. These risks have received much more attention since the crisis [► 2013 L26 @ 1:16:34].

The instructors then closed the course [► 2013 L26 @ 1:17:37]. A list of past final-paper topics was on the course site: mostly Black–Scholes, its extensions and manipulations of the Black–Scholes equation, one statistical analysis of commodity data (recommended as a direction), and numerical and Monte Carlo projects. Students were asked to reflect on what they had learned and on what they would like to explore next. P. Kempthorne described the course as deliberately challenging, meant to give both the mathematical foundations and a real exposure to their applications through practitioners; he said the final papers he had read so far were good and that the faculty wanted to remain a resource after the course [► 2013 L26 @ 1:19:45]. The class would be repeated the following fall in a slightly different form, students were invited to suggest topics (one value of the course being access to speakers from the frontier of the industry), and to fill in the course review [► 2013 L26 @ 1:20:45].

10 COURSE NOTE: 2013 versus 2024

The two lectures reflect a decade of regulation. In 2013 many OTC trades were uncollateralised, and the focus was pricing and hedging CVA, DVA and funding at portfolio level. By 2024 standard swaps are centrally cleared and bilateral trades between banks carry daily VM and regulatory IM, so counterparty risk is managed mainly through collateral: the questions are how much margin (a VaR/ES) and how to reduce it (optimisation), and the cost of counterparty risk has partly turned into the cost of funding margin. Both speakers stress the same point: the relevant quantities are non-linear functions of the *whole* portfolio, or even the whole network.

11 COURSE NOTE: Reading suggested on the slides

2024: S. Boyd and L. Vandenberghe, *Convex Optimization* (the slide misspells the first name as “Steven”); J. Gregory, *The xVA Challenge*; J. Hull, *Risk Management and Financial Institutions*; Artzner et al. (1999) on coherent risk measures; Rockafellar and Uryasev on optimising CVaR; the BCBS–IOSCO margin framework and the ISDA SIMM 2.6 methodology; Embrechts and Wang, “Seven proofs for the subadditivity of expected shortfall”; “Subadditivity re-examined: the case for Value-at-Risk”. 2013: Harrison–Kreps (1979), Harrison–Pliska (1981), Jamshidian (1997) on LIBOR and swap market models, Schönbucher (2003) on survival measures, Tang and Li (2007), Tang and Williams (2010) on funding benefit and cost, Zhu and Pykhtin (2007), “A guide to modeling counterparty credit risk”.

Chapter 9

Time Series Analysis

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L12	P. Kempthorne	Stationary processes; from prices to log returns (S&P 500, Amazon, crude oil, 10-year yield); sample autocorrelations and the Box–Pierce test; Wold representation theorem; lag operator, impulse response; ARMA, $AR(p)$ and its stationarity condition, $AR(1)$, Yule–Walker equations, $MA(q)$; differencing. The lecture ends at slide 22 of 29.
2013 L8 (from 26:19)	P. Kempthorne	The same univariate theory (the first 26 minutes finish regression, chapter 5); adds linear-trend reversion versus stochastic trend, ARIMA, maximum likelihood (LIML/FIML) and the information criteria AIC, BIC, HQ.
2013 L11 (from 28:35)	P. Kempthorne	Multivariate time series: cross-covariance matrices, multivariate Wold, $VAR(p)$ and its $VAR(1)$ form, equation-by-equation least squares and why it is optimal, maximum likelihood, model selection; VAR of US macroeconomic series (Case Study 5). The first 28 minutes are GARCH, chapter 13 .
2013 L12	P. Kempthorne	The fitted macro VAR; cointegration, vector error-correction models, energy futures and crack spreads; linear state-space models (time-varying CAPM, time-varying regression, AR, MA, ARMA); the Kalman filter.
Files. Slides mit18_642_f24_lec11–12.pdf (29 slides, 73 pages with overlays); 2013 slides MIT18_S096F13_lecnote8.pdf, lecnote11.pdf, lecnote12.pdf; 2013 Case Study 3 CaseStudy3.pdf (time series of the US 10-year Treasury yield, in R) and Case Study 5 CaseStudy5.pdf (VAR models of macroeconomic series, in R). Problem sets. 2024 PS4 Problems 5–6 (Problems 1–4 are on Brownian motion, chapter 11); 2013 PS4 (all six problems); 2013 PS6 Problems 1–2 (Problems 3–5 are portfolio theory, chapter 10). Standalone exercises follow the section they practise; the connected ones are at the end of the chapter. Not published or not covered. Slide 5 of the 2024 lecture points to TimeSeries4plots.pdf and TimeSeries4acfplots.pdf, which are not in the download, so the exploratory plots of section 9.1.1 are described from the spoken commentary. The crack-spread cointegration note of 2013 L12 exists only in the video. Slides 23–29 of 2024 (trend reversion, ARIMA, maximum likelihood, model selection, Dickey–Fuller test) were not reached in class; 2013 L8 covered them up to model selection, and the Dickey–Fuller slide was not discussed in either year.		

Motivation

Prices, yields, spreads and macroeconomic indicators are observed *in time*, and the observation today carries information about tomorrow. Chapter 4 modelled the simplest

case, a random walk with independent steps. Real data have more structure: yield changes are weakly autocorrelated, a credit spread is pulled back to a long-run level, an interest rate reacts to unemployment and inflation with a delay, two energy prices drift apart but never far. Time-series analysis supplies the vocabulary:

- *stationarity* and the autocorrelation function, to say what “no structure left” means;
- the *Wold theorem* and the *ARMA* family, to say what structure is possible in a stationary series, and *differencing* / unit-root tests to deal with the series that are not (sections 9.1 to 9.4);
- VAR models and *cointegration*, for several series at once (sections 9.5 and 9.6);
- *state-space* models and the *Kalman filter*, for hidden quantities such as a time-varying beta or a “true” price observed with noise (section 9.7).

The mathematics is linear algebra and least squares (chapters 2 and 5): every model in this chapter is fitted by regressing the series on its own past, and the stationarity conditions are statements about the eigenvalues of a matrix. The case studies (10-year Treasury yield and a three-variable macro VAR) show what the workflow looks like on real data.

9.1 Stationarity and autocorrelation

A *time series* is a stochastic process $\{X_t, t \in T\}$ of random variables indexed by the times of observation, discrete ($T = \{1, 2, \dots\}$, the case used almost throughout) or continuous ($T = [0, \infty)$) [► 2024 L12 @ 0:00]. Its probabilistic behaviour is fixed by the finite-dimensional distributions $p(x_{t_1}, \dots, x_{t_m})$ for every finite set of times [► 2024 L12 @ 1:09][► 2013 L8 @ 27:23].

Definition 9.1 (Strict stationarity). $\{X_t\}$ is *strictly stationary* if $p(x_{t_1+\tau}, \dots, x_{t_m+\tau}) = p(x_{t_1}, \dots, x_{t_m})$ for all τ, m and $(t_1, \dots, t_m)$: the joint law is invariant under time translation [► 2024 L12 @ 2:10].

Definition 9.2 (Covariance stationarity and ACF). $\{X_t\}$ is *covariance (weakly, second-order) stationary* if, for all t ,

$$\mathbb{E}[X_t] = \mu, \quad \text{Var } X_t = \sigma_X^2 < \infty, \quad \text{Cov}(X_t, X_{t+\tau}) = \gamma(\tau)$$

depend at most on the lag τ [► 2024 L12 @ 5:24][► 2013 L8 @ 29:32]. The *autocorrelation function* (ACF) is $\rho(\tau) = \gamma(\tau)/\gamma(0)$ [► 2024 L12 @ 7:27].

Two facts are used constantly. First, $\gamma(-\tau) = \gamma(\tau)$ and $|\rho(\tau)| \leq 1$. Second, for every n the Toeplitz matrix $[\gamma(i-j)]_{i,j=1}^n$ is the covariance matrix of $(X_1, \dots, X_n)$, hence positive semi-definite; not every even function γ is an autocovariance function. Strict stationarity together with a finite variance implies covariance stationarity; the converse fails in general but holds for Gaussian processes, whose law is determined by the first two moments.

INTUITION: Time-translation invariance

Strict stationarity is the statistician’s *equilibrium*: the process looks the same whichever window one watches, like the noise of a system that has relaxed to a steady state. It is what makes estimation possible, because a single path can then be averaged over time in place of an ensemble average (the ergodic hypothesis). Covariance stationarity asks only that the two-point correlation function depend on time differences, $C(t, t') = \gamma(t - t')$;

the Fourier transform of γ is the power spectrum, which the course does not use but which explains why cycles show up as oscillations in $\rho(\tau)$ (section 9.3.1). Parameters of a stationary model are constants that can be estimated [► 2024 L12 @ 3:12][► 2013 L8 @ 28:28].

9.1.1 Making financial series stationary

Prices are not stationary: they trend, and the size of their fluctuations grows with their level. The standard remedy is to model the *log return*. Let Y_t be a positive price and

$$X_t = \log \frac{Y_t}{Y_{t-1}} = \log(1 + R_t), \quad R_t = \frac{Y_t - Y_{t-1}}{Y_{t-1}}, \quad Y_t = Y_0 \exp(X_1 + \cdots + X_t), \quad (9.1)$$

so Y_t is an exponential of a random walk with steps X_t , and the modelling assumption is that $\{X_t\}$ is covariance stationary [► 2024 L12 @ 9:33]. For small returns $X_t \approx R_t - \frac{1}{2}R_t^2$. Taking logs turns exponential growth into linear growth and stabilises the variance, and differencing removes the linear trend; X_t is then a candidate stationary series.

The lecture illustrated this on four series (the plots themselves are not in the download) [► 2024 L12 @ 8:29]:

- *S&P 500*, roughly 2011–2024. The index trends up with sharp COVID drops, so it is not stationary in any window; the monthly log returns fluctuate around a level close to zero with constant spread. Weekly and daily returns keep a stable mean, but the volatility is clearly not constant (extreme in 2020) [► 2024 L12 @ 11:42]. A fitted normal density underestimates both the mass near the mean and the tails: the returns are *leptokurtic* (“slender”), as in section 3.9.1. Volatility clustering is the subject of chapter 13.
- *Amazon*. The price shows strong exponential growth; on the log scale the variability is more homogeneous, and the monthly and weekly log returns look stationary with a better normal fit than the index. Daily returns show a few spikes that no Gaussian model would produce [► 2024 L12 @ 15:56].
- *Crude oil futures*. In April 2020 the front-month contract went *negative*; a holder of 100 contracts woke up owing the broker 100 times a negative price, and some brokers could not even report positions because their systems did not allow negative prices. The logarithm is undefined there, so the lecture’s plots exclude those days [► 2024 L12 @ 18:00]. Log returns require positive prices (spreads and rates can be modelled directly).
- *10-year US Treasury yield*, from above 3% to below 1% after COVID and rising again. Strictly, “log return” is the wrong name for a percentage change of a yield, but the same transformation gives unimodal, roughly symmetric increments, and the normal model fits better than for equity returns [► 2024 L12 @ 20:05].

Modelling Y_t as an exponential random walk means that $S_t = X_1 + \cdots + X_t$ is a sum of steps. In the simplest model the steps are i.i.d. (section 4.2); the models of this chapter let them depend on their own past [► 2024 L12 @ 23:18].

9.1.2 Sample autocorrelation and tests of whiteness

Given observations $x_1, \dots, x_T$ the *sample autocorrelation at lag k* is

$$\hat{r}_k = \frac{\sum_{t=1}^{T-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2} \quad (9.2)$$

[► 2024 L12 @ 24:25], an estimate of $\rho(k)$. If the series is white noise, then $\hat{r}_k$ is approximately $N(0, 1/T)$ (Bartlett); the lecture writes the variance as $1/(T - k)$, which is the same for $T \gg k$ [► 2024 L12 @ 25:35]. Hence the test of $H_0 : \rho(k) = 0$ rejects at the 5% level when

$$|\hat{r}_k| > 1.96\sqrt{\frac{1}{T-k}}, \quad (9.3)$$

which is the pair of dashed lines that R's acf draws around zero [► 2024 L12 @ 26:42]. For a daily series with $T = 500$ the bands are ± 0.088 ; a single autocorrelation of 0.09 is already significant. The lecture saw a large negative monthly autocorrelation at lag 1 for the S&P 500 and, on weekly data, significant negative correlations around lag 5, which a student read as *mean reversion* (price changes that are too high tend to be reversed) [► 2024 L12 @ 27:42]. Amazon showed no significant autocorrelation at any frequency, crude oil a negative lag-2 and positive lag-3 weekly pattern, and Treasury yields strong low-lag dependence [► 2024 L12 @ 29:50].

Testing many lags at once. If $\rho(1) = \dots = \rho(K) = 0$ the $\sqrt{T}\hat{r}_j$ are asymptotically independent $N(0, 1)$, so

$$\text{BP} = T \sum_{j=1}^K \hat{r}_j^2 \approx \chi_K^2, \quad (9.4)$$

the *Box–Pierce* statistic; reject the joint null when BP exceeds the χ_K^2 quantile (18.31 for $K = 10$ at 5%) [► 2024 L12 @ 36:10]. Two refinements, not discussed in class: the Ljung–Box variant $T(T + 2) \sum_j \hat{r}_j^2 / (T - j)$ has a better small-sample χ^2 approximation; and when the test is applied to the residuals of a fitted $\text{ARMA}(p, q)$ model the degrees of freedom drop to $K - p - q$.

IN FINANCE: Residual whiteness as a model check

A time-series model writes $Y_t = \hat{Y}_t + e_t$. It is *adequate* when the residuals have mean zero, constant variance and no autocorrelation, i.e. they are white noise: nothing predictable is left [► 2024 L12 @ 31:59]. If in addition they are Gaussian, the model delivers not only point forecasts but valid *prediction intervals* [► 2024 L12 @ 34:03]. The ACF of the residuals, and the Box–Pierce test on it, are therefore the standard diagnostic: a model whose residuals still show significant autocorrelation “is not done yet” [► 2024 L12 @ 38:24]. The flip side, developed in [chapter 13](#), is that uncorrelated does not mean independent: returns can be white noise while their squares are strongly autocorrelated.

Exercise 9.1 (Box–Pierce by hand (not from the course)). A daily return series with $T = 500$ has $\hat{r}_1 = 0.09$, $\hat{r}_2 = -0.05$, $\hat{r}_3 = 0.04$, $\hat{r}_4 = 0.02$, $\hat{r}_5 = -0.06$.

- Which lags are individually significant at 5%?
- Compute BP for $K = 5$ and compare with χ_5^2 (critical value 11.07).
- Explain how the two answers can disagree and which you would report.
- If the numbers came from the residuals of an $\text{ARMA}(1,1)$ fit, what would the degrees of freedom be?

9.2 The Wold representation theorem

Theorem 9.3 (Wold representation). Every zero-mean covariance-stationary series $\{X_t\}$ can be written

$$X_t = V_t + S_t, \quad S_t = \sum_{i=0}^{\infty} \psi_i \eta_{t-i}, \quad (9.5)$$

where $\psi_0 = 1$, $\sum_i \psi_i^2 < \infty$; $\{\eta_t\}$ is white noise, i.e. $\mathbb{E}[\eta_t] = 0$, $\mathbb{E}[\eta_t^2] = \sigma^2$, $\mathbb{E}[\eta_t]\eta_s = 0$ for $t \neq s$; $\{V_t\}$ is linearly deterministic (a linear combination of its own past with constant coefficients predicts it exactly); and $\mathbb{E}[\eta_t V_s] = 0$ for all t, s [► 2024 L12 @ 38:24][► 2013 L8 @ 31:40].

The theorem says that once the deterministic part is removed, every stationary series is a moving average of white noise, possibly of infinite order. The white noise $\eta_t = X_t - P(X_t | X_{t-1}, X_{t-2}, \dots)$ is the *innovation*: the part of X_t that cannot be predicted linearly from the past, the “news” arriving at t [► 2013 L11 @ 43:32]. An example of a linearly deterministic process is a sinusoid, $V_t = A_1 \cos \omega t + A_2 \sin \omega t$ with random A_1, A_2 : it obeys the exact recursion $V_t = 2 \cos \omega V_{t-1} - V_{t-2}$ (from $\cos \omega t = 2 \cos \omega \cos \omega(t-1) - \cos \omega(t-2)$), so two observations fix it for ever; sums of sinusoids of different frequencies (a Fourier series) are also linearly predictable [► 2024 L12 @ 40:30].

INTUITION: What the theorem does and does not say

The proof shows why the result holds: η_t is the component of X_t orthogonal to the past, and expanding X_t in the orthogonal family $\eta_t, \eta_{t-1}, \dots$ is a Gram–Schmidt procedure. The innovations are only *uncorrelated*, not independent: the theorem describes the best *linear* prediction, and a process such as a GARCH model (chapter 13) is covered by it although its innovations are dependent through their variances. Covariance stationarity of the sum S_t needs only $\sum \psi_i^2 < \infty$; that is why the condition appears [► 2024 L12 @ 50:07].

Sketch of proof. Work in the Hilbert space L^2 of square-integrable random variables with inner product $\langle U, W \rangle = \mathbb{E}[UW]$. Let H_t be the closed linear span of $\{X_s : s \leq t\}$ and P_H the orthogonal projection on H . Define $\eta_t = X_t - P_{H_{t-1}} X_t$. Stationarity makes the time shift a unitary map, so $\sigma^2 = \mathbb{E}[\eta_t^2]$ does not depend on t (we assume $\sigma^2 > 0$; otherwise X is itself deterministic). Since $\eta_s \in H_s \subset H_{t-1}$ for $s < t$ and $\eta_t \perp H_{t-1}$, the η_t are uncorrelated. Put $\psi_j = \mathbb{E}[X_t \eta_{t-j}]/\sigma^2$, independent of t ; then $\psi_0 = \mathbb{E}[X_t \eta_t]/\sigma^2 = \mathbb{E}[\eta_t^2]/\sigma^2 = 1$, because $X_t - \eta_t \in H_{t-1}$. The projection of X_t on $\text{span}\{\eta_t, \dots, \eta_{t-j+1}\}$ is $\sum_{j < j} \psi_j \eta_{t-j}$, and Bessel’s inequality gives $\sigma^2 \sum_j \psi_j^2 \leq \text{Var } X_t < \infty$.

Hence $S_t = \sum_{j \geq 0} \psi_j \eta_{t-j}$ converges in L^2 , and $V_t = X_t - S_t$ is orthogonal to every η_s with $s \leq t$. One checks that $V_t \in \bigcap_s H_s$, the *remote past* of the process, so V_t is determined exactly, by a linear combination, from arbitrarily remote values: it is linearly deterministic. □

9.2.1 Wold’s theorem as a modelling recipe

The theorem suggests how to specify a stationary model for data $x_0, \dots, x_T$ [► 2024 L12 @ 44:41][► 2013 L8 @ 34:48]. Fix a number p of lags to represent the linearly deterministic part, and take an estimation window of size n (after de-trending so that the series has a constant mean level). Set

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} 1 & y_0 & y_{-1} & \cdots & y_{-(p-1)} \\ 1 & y_1 & y_0 & \cdots & y_{-(p-2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & y_{n-1} & y_{n-2} & \cdots & y_{n-p} \end{pmatrix},$$

so the first p rows reach back before the start of the response. The linear projection on p lags is the least-squares fit

$$\hat{\mathbf{y}}^{(p)} = \mathbf{Z}(\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y} = \hat{P}(\mathbf{Y}_t | \mathbf{Y}_{t-1}, \dots, \mathbf{Y}_{t-p}), \quad \hat{\boldsymbol{\varepsilon}}^{(p)} = \mathbf{y} - \hat{\mathbf{y}}^{(p)},$$

the hat matrix of proposition 5.4. The residual series is then modelled as a moving average $\boldsymbol{\varepsilon}_t^{(p)} = \sum_i \psi_i \eta_{t-i}$, giving estimates of the ψ_i and of the innovations η_t [► 2024 L12 @ 45:41]. Two diagnostics decide whether to stop:

- the residual should be orthogonal to lags of $\mathbf{Y}$ beyond p ; if it is not, increase p ;

- the estimated innovations should look like white noise (section 9.1.2); if not, change the specification of the moving average or add deterministic regressors [► 2024 L12 @ 47:50].

As $p \rightarrow \infty$ the projection converges to $P(Y_t | Y_{t-1}, Y_{t-2}, \dots)$, but a sample of size n supports only $p/n \rightarrow 0$, so useful models have a modest p or a moving average with a parsimonious parametrisation [► 2024 L12 @ 48:55][► 2013 L8 @ 38:55]. This is exactly why ARMA models exist (section 9.3).

9.2.2 Lag operator, impulse response, invertibility

The *lag* (backshift) operator L shifts a series back one period: $LX_t = X_{t-1}$, $L^0 = \text{id}$, $L^n X_t = X_{t-n}$; its inverse $L^{-n} X_t = X_{t+n}$ shifts forward [► 2024 L12 @ 52:09][► 2013 L8 @ 40:01]. Polynomials in L multiply and factorise like ordinary polynomials in a complex variable z , and the Wold representation reads

$$X_t = \psi(L)\eta_t + V_t, \quad \psi(L) = \sum_{i \geq 0} \psi_i L^i. \quad (9.6)$$

The *impulse response function* is $\text{IR}(j) = \partial X_t / \partial \eta_{t-j} = \psi_j$, the effect of a shock j periods ago on the process today [► 2024 L12 @ 54:20]; graphed against j it displays delayed responses. If the Federal Reserve chairman changes interest rates and the variable of interest is GNP, ψ_j is the effect j quarters later [► 2013 L8 @ 43:11]. The *long-run cumulative response* is $\sum_j \psi_j = \psi(1)$, the level to which the process eventually moves after one unit innovation [► 2024 L12 @ 56:28][► 2013 L8 @ 45:11].

If $\psi(L)$ is *invertible*, i.e. there is $\psi^{-1}(L) = \sum_i \psi_i^* L^i$ with $\psi^{-1}(L)\psi(L) = L^0$, then for $V_t = 0$

$$\psi^{-1}(L)X_t = \eta_t \iff X_t = \sum_{i \geq 1} \pi_i X_{t-i} + \eta_t, \quad \pi_i = -\psi_i^*, \quad (9.7)$$

an infinite-order autoregression [► 2024 L12 @ 57:31][► 2013 L8 @ 46:17]. The coefficients of the least-squares projection on p lags in section 9.2.1 converge to these π_i [► 2013 L8 @ 47:20]. The basic tool for inverting is the geometric series: for a complex number λ with $|\lambda| > 1$,

$$(1 - \lambda^{-1}L)^{-1} = \sum_{i \geq 0} \lambda^{-i} L^i, \quad (9.8)$$

which converges (on bounded stationary series) precisely because the coefficients λ^{-i} decay [► 2024 L12 @ 1:06:02].

9.3 ARMA models

Definition 9.4 (ARMA(p, q)). $\{X_t\}$ follows an *autoregressive moving-average* model of orders (p, q) if

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \dots + \phi_p(X_{t-p} - \mu) + \eta_t + \theta_1\eta_{t-1} + \dots + \theta_q\eta_{t-q}, \quad (9.9)$$

where $\{\eta_t\}$ is white noise $\text{WN}(0, \sigma^2)$. With the polynomials $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ and $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ this reads

$$\phi(L)(X_t - \mu) = \theta(L)\eta_t, \quad \text{and} \quad X_t = \mu + \psi(L)\eta_t, \quad \psi(L) = \frac{\theta(L)}{\phi(L)}$$

is the Wold decomposition, when $\phi(L)$ is invertible [► 2024 L12 @ 59:35][► 2013 L8 @ 46:17].

The sign convention of the moving-average part is “ $+\theta_j$ ” (as in R); some texts use “ $-\theta_j$ ”. ARMA models are *parsimonious*: $p + q$ parameters generate an infinite Wold expansion [► 2024 L12 @ 1:00:39]. We first look at the two pure cases.

9.3.1 Autoregressive models AR(p)

Set $q = 0$: $\phi(L)(X_t - \mu) = \eta_t$, that is,

$$X_t = c + \sum_{j=1}^p \phi_j X_{t-j} + \eta_t, \quad c = \mu \phi(1) = \mu \left(1 - \sum_j \phi_j\right), \quad (9.10)$$

a linear regression of the series on its own lags, with white-noise errors [► 2024 L12 @ 1:01:47] [► 2013 L8 @ 50:34]. (Taking expectations of (9.10) gives $\mu = c + \mu \sum \phi_j$, whence $c = \mu \phi(1)$.)

Theorem 9.5 (Stationarity of AR(p)). The equation $\phi(L)(X_t - \mu) = \eta_t$ has a covariance-stationary solution of the form $X_t - \mu = \sum_{i \geq 0} \psi_i \eta_{t-i}$ if and only if every root λ_j of the characteristic equation $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p = 0$ lies outside the closed unit disc, $|\lambda_j| > 1$, $j = 1, \dots, p$ [► 2024 L12 @ 1:03:51][► 2013 L8 @ 52:35].

Remark 9.6 (Roots and eigenvalues). The condition is often stated on the reciprocals $1/\lambda_j$, the roots of $z^p - \phi_1 z^{p-1} - \dots - \phi_p = 0$, which are the eigenvalues of the *companion matrix*

$$F = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix};$$

stationarity is $|1/\lambda_j| < 1$ for all j . Software conventions differ: R’s vars package reports the moduli of the eigenvalues (stationary if all < 1), whereas polyroot returns roots of $\phi(z)$ (stationary if all > 1) [► 2013 L12 @ 1:08]. The proof of the correspondence is given for the vector case in [proposition 9.11](#).

Proof. Factorise $\phi(z) = \prod_{j=1}^p (1 - z/\lambda_j)$ (the constant term is 1). If all $|\lambda_j| > 1$, each factor is inverted by the geometric series (9.8), whose coefficients λ_j^{-i} decay geometrically, and the product $\phi^{-1}(L) = \prod_j (1 - L/\lambda_j)^{-1}$ of absolutely summable series is absolutely summable. Thus $\psi(L) = \phi^{-1}(L)$ has $\sum |\psi_i| < \infty$ and $X_t = \mu + \psi(L)\eta_t$ is covariance stationary [► 2024 L12 @ 1:05:00]. Conversely, a root with $|\lambda| < 1$ makes the coefficients λ^{-i} grow, so the expansion diverges (the process is explosive), and a root on the circle $|\lambda| = 1$ gives a unit root (the random walk is the case $p = 1, \lambda = 1$), whose variance grows with t ; in neither case is there a stationary solution driven by the past. $\square$

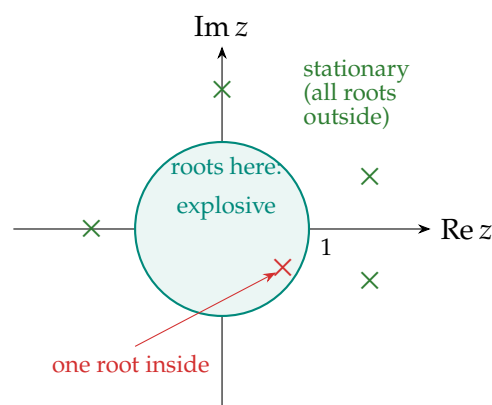


Figure 9.1: Roots λ_j of $\phi(z) = 0$ in the complex plane. Green crosses: a stationary AR model, all roots outside the unit circle. Red cross: a single root inside makes the process explosive.

AR(1)

For $X_t - \mu = \phi(X_{t-1} - \mu) + \eta_t$ the characteristic equation is $1 - \phi z = 0$ with root $\lambda = 1/\phi$: the process is covariance stationary iff $|\phi| < 1$ [► 2024 L12 @ 1:08:12]. Iterating, $X_t - \mu = \sum_{j \geq 0} \phi^j \eta_{t-j}$ (the Wold decomposition, $\psi_j = \phi^j$), and therefore

$$\mathbb{E}[X_t] = \mu, \quad \gamma(0) = \sigma_X^2 = \frac{\sigma^2}{1 - \phi^2}, \quad \gamma(j) = \phi^j \sigma_X^2, \quad \rho(j) = \phi^j \quad (9.11)$$

[► 2024 L12 @ 1:08:12][► 2013 L8 @ 56:46]. (Multiply $X_t - \mu = \phi(X_{t-1} - \mu) + \eta_t$ by $X_{t-j} - \mu$ and take expectations, or sum the geometric series.) The variance exceeds the innovation variance σ^2 for every $\phi \neq 0$, and depends on ϕ only through ϕ^2 : positive and negative autocorrelation inflate it equally [► 2013 L8 @ 57:50]. The regimes:

- $0 < \phi < 1$: *exponential mean reversion* to μ , all correlations positive;
- $-1 < \phi < 0$: *oscillating exponential mean reversion*, correlations alternate in sign [► 2024 L12 @ 1:09:15];
- $\phi = 1$: the Wold decomposition does not exist, X_t is a random walk, not stationary;
- $|\phi| > 1$: explosive.

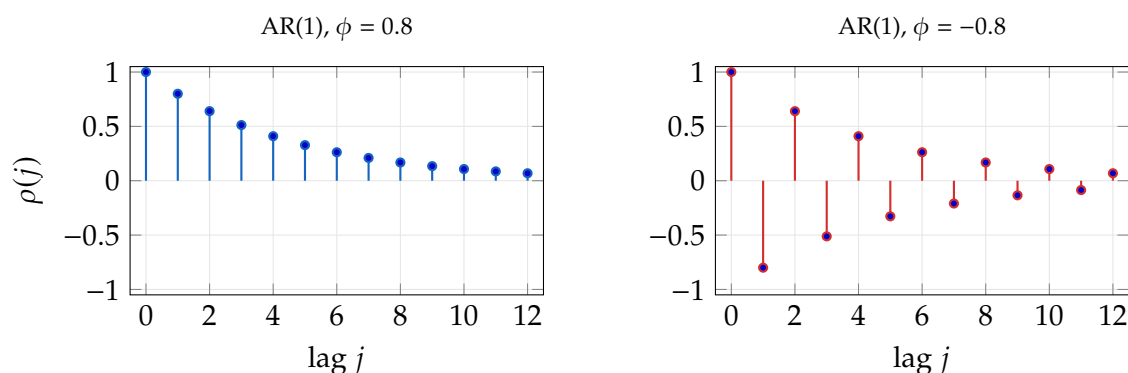


Figure 9.2: Theoretical autocorrelation $\rho(j) = \phi^j$ of an AR(1): geometric decay for $\phi > 0$, geometric decay with alternating sign for $\phi < 0$.

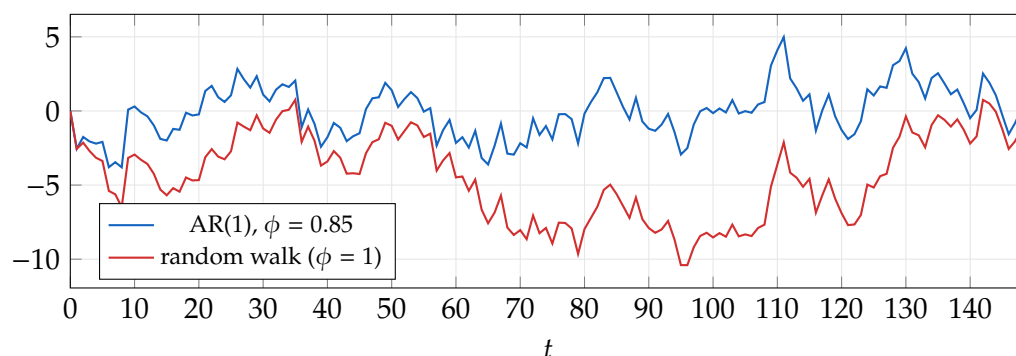


Figure 9.3: Two paths driven by the same simulated white noise (seed 3). The AR(1) with $\phi = 0.85$ keeps returning to $\mu = 0$; the random walk wanders (its variance grows like $t\sigma^2$).

IN FINANCE: AR(1) in practice: mean reversion and Vasicek

The AR(1) is the discrete-time version of the Ornstein–Uhlenbeck (Vasicek) process $dX = -\kappa(X - \mu)dt + \sigma_0 dW$ (chapter 18): sampled every h , the exact transition is an AR(1) with $\phi = e^{-\kappa h}$ and noise variance $\sigma_0^2(1 - e^{-2\kappa h})/(2\kappa)$, consistent with $\gamma(0) = \sigma_0^2/(2\kappa)$ in (9.11). The half-life of a shock is $h \ln 2 / (-\ln \phi)$. Candidates for such a model are variables that should not wander too far: interest rates, interest-rate spreads, real exchange rates and valuation ratios (dividend-to-price, earnings-to-price) [► 2024 L12 @ 1:10:15][► 2013 L8 @ 58:54]. The caveat of the 2013 lecture: this is usually true over short periods; over long ones the level μ to which the series reverts drifts, so the model must be re-estimated [► 2013 L8 @ 59:59].

Yule–Walker equations

Multiply the AR(p) equation $X_t - \mu = \sum_i \phi_i(X_{t-i} - \mu) + \eta_t$ by $X_{t-j} - \mu$ and take expectations. Since $\mathbb{E}[\eta_t(X_{t-j} - \mu)] = 0$ for $j \geq 1$ and $= \sigma^2$ for $j = 0$ (the noise is uncorrelated with the past),

$$\gamma(j) = \sum_{i=1}^p \phi_i \gamma(j-i) + \sigma^2 \delta_{j0}, \quad j = 0, 1, 2, \dots \quad (9.12)$$

[► 2024 L12 @ 1:11:16][► 2013 L8 @ 1:02:01](the slide misprints the second lag as X_{t-1} ; the lecturer corrected it aloud). Dividing by $\gamma(0)$, $\rho(j) = \sum_i \phi_i \rho(j-i)$ for $j \geq 1$. The equations $j = 1, \dots, p$ form the Yule–Walker system

$$\begin{pmatrix} \gamma(1) \\ \gamma(2) \\ \vdots \\ \gamma(p) \end{pmatrix} = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix}, \quad (9.13)$$

using $\gamma(-j) = \gamma(j)$. The matrix is Toeplitz (constant along diagonals), a question the 2013 lecturer put to the class [► 2013 L8 @ 1:03:02]. The zeroth equation then gives the noise variance

$$\sigma^2 = \gamma(0) - \sum_{j=1}^p \phi_j \gamma(j). \quad (9.14)$$

Estimation. Replace the $\gamma(j)$ by sample autocovariances $\hat{\gamma}(j)$ and solve (9.13): these are the Yule–Walker estimates [► 2024 L12 @ 1:13:20]. They are an application of the *method of moments*: equate sample and theoretical moments, and solve for the parameters. This is the tool of choice when the likelihood is hard to write down or to optimise; econometrics is rich in such applications [► 2024 L12 @ 1:14:24][► 2013 L8 @ 1:05:09]. (Two remarks beyond the course: the system is solved efficiently by the Levinson–Durbin recursion, and with the divisor T in $\hat{\gamma}$ the fitted AR model is always stationary.)

Partial autocorrelation

The *partial autocorrelation* at lag k , $\alpha(k)$, is the correlation between X_t and X_{t-k} after removing the linear effect of the intermediate values $X_{t-1}, \dots, X_{t-k+1}$ [► 2024 L12 @ 28:49][► 2013 L12 @ 4:11]. Equivalently, it is the last coefficient ϕ_{kk} of the best linear predictor of X_t from k lags, the coefficient of X_{t-k} in the regression of X_t on $X_{t-1}, \dots, X_{t-k}$ [► 2013 L12 @ 5:13]: the “incremental correlation” of adding one more lag. Together the ACF and PACF identify the order of a model:

	ACF	PACF
AR(p)	decays geometrically or as a damped sinusoid	cuts off after lag p
MA(q)	cuts off after lag q	decays
ARMA(p, q)	decays	decays

The AR row follows because the regression of an AR(p) on more than p lags has a zero coefficient on the extra ones; the MA row is proved in [section 9.3.2](#).

CASE STUDY: 10-year Treasury yield: ACF and PACF (2013 Case Study 3)

The case study downloads daily constant-maturity yields DGS10 from FRED (3 356 non-missing days, January 2000 to May 2013; missing values are bond-market holidays), builds weekly (700) and monthly (161) closing series with `to.weekly`, `to.monthly`, and plots ACF and PACF at each frequency. The high first-order autocorrelation of the levels suggests a unit root at every frequency. The Augmented Dickey–Fuller tests of [section 9.4.2](#) confirm it (p-values 0.057, 0.105 and 0.272 for daily, weekly and monthly levels, none rejecting a unit root at 5%), and reject it for the first differences (p-values 0.01 at all frequencies, the smallest value R reports). For the monthly differences y_0 (160 values) the table of sample ACF and PACF is

lag	1	2	3	4
ACF	0.035	−0.211	0.079	−0.010
PACF	0.035	−0.213	0.100	−0.068
last coeff. of AR(k) fit by <code>lm</code>	0.036	−0.217	0.104	−0.072

The last row confirms the definition of the PACF: it is (nearly) the coefficient of the last lag in a regression on k lags. The small differences arise because `lm` conditions on the first k cases, whereas `acf` uses all available pairs at each lag. Only the lag-2 coefficient is significant ($t = -2.76$ in the AR(2) fit).

Example 9.7 (Yule–Walker with two lags). Suppose the sample autocorrelations of a series are $\hat{\rho}(1) = 0.5$ and $\hat{\rho}(2) = 0.2$. The $p = 2$ system $\begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.2 \end{pmatrix}$ has determinant 0.75 and solution $\phi_1 = (0.5 - 0.5 \cdot 0.2)/0.75 = 8/15 \approx 0.533$ and $\phi_2 = (0.2 - 0.25)/0.75 = -1/15 \approx -0.067$. The fitted process is stationary: $\phi_1 + \phi_2 = 0.467 < 1$, $\phi_2 - \phi_1 = -0.6 < 1$, $|\phi_2| < 1$ ([section 9.3.1](#)). On a simulated AR(2) with $\phi = (1, -0.5)$ and $n = 2000$ the Yule–Walker estimates were (1.017, −0.515) with $\hat{\sigma}^2 = 0.960$ (true value 1).

AR(2): roots, correlations and cycles

The AR(2) model $X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \eta_t$ (2013 PS4 Problem 4, exercise [9.14](#)) is the simplest process that can oscillate. Write $\phi(z) = 1 - \phi_1 z - \phi_2 z^2 = (1 - G_1 z)(1 - G_2 z)$, where $G_i = 1/\lambda_i$ are the roots of $G^2 - \phi_1 G - \phi_2 = 0$. Then

$$G_{1,2} = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2},$$

and the roots are real and distinct if $\phi_1^2 + 4\phi_2 > 0$, real and coincident if $\phi_1^2 + 4\phi_2 = 0$, and a complex-conjugate pair if $\phi_1^2 + 4\phi_2 < 0$. Both $|G_i| < 1$ if and only if the parameters lie in the *stationarity triangle*

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad |\phi_2| < 1.$$

The autocorrelations obey $\rho(k) = \phi_1 \rho(k-1) + \phi_2 \rho(k-2)$ with $\rho(0) = 1$ and $\rho(1) = \phi_1/(1 - \phi_2)$ (the first Yule–Walker equation). This is a linear difference equation whose solutions are combinations of powers of the roots, $\rho(k) = C_1 G_1^k + C_2 G_2^k$ for $G_1 \neq G_2$, and the two initial

conditions give

$$\rho(k) = \frac{G_1(1 - G_2^2)G_1^k - G_2(1 - G_1^2)G_2^k}{(G_1 - G_2)(1 + G_1G_2)}. \quad (9.15)$$

For real roots the correlations decay geometrically in magnitude. For complex roots write $G_{1,2} = d e^{\pm i\omega}$ with $d = \sqrt{-\phi_2}$ and $\omega = \arccos(\phi_1/(2d))$; then

$$\rho(k) = d^k \frac{\sin(\omega k + F)}{\sin F}, \quad \tan F = \frac{1 - \phi_2}{1 + \phi_2} \tan \omega, \quad (9.16)$$

a damped cosine with damping factor d , frequency $f_0 = \omega/2\pi$ (period $1/f_0$) and phase F (taken in the same quadrant as ω). The 2013 problem set states the same result with $|\phi_1|$ and a factor $[\text{sgn } \phi_1]^k$. A series that oscillates around its mean with a typical period of $2\pi/\omega$ periods is the signature of complex roots.

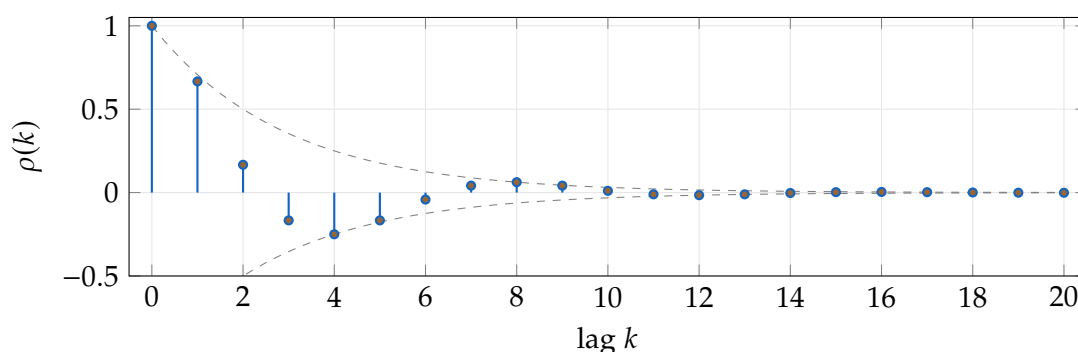


Figure 9.4: Autocorrelation of the AR(2) with $\phi = (1, -0.5)$: a damped cosine of period 8 under the envelope $\pm(1/\sqrt{2})^k$ (dashed).

Example 9.8 (An AR(2) with a period of eight steps). Take $\phi_1 = 1$, $\phi_2 = -0.5$. Then $\phi_1^2 + 4\phi_2 = -1 < 0$, the roots of $\phi(z)$ are $1 \pm i$ (modulus $\sqrt{2} > 1$: stationary), $d = 1/\sqrt{2}$, $\omega = \arccos(1/\sqrt{2}) = \pi/4$: period exactly 8, and $\tan F = 3$. The autocorrelations are $\rho = (1, 0.667, 0.167, -0.167, -0.25, -0.167, -0.042, 0.042, \dots)$ and the variance is $\gamma(0) = \sigma^2(1 - \phi_2)/[(1 + \phi_2)((1 - \phi_2)^2 - \phi_1^2)] = 2.4\sigma^2$, equal to $\sigma^2 \sum_j \psi_j^2$ for the Wold weights $\psi = (1, 1, 0.5, 0, -0.25, -0.25, \dots)$ (checked numerically). Note $\psi(1) = 1/(1 - \phi_1 - \phi_2) = 2$: a unit shock eventually raises the level by 2 (fig. 9.4).

CASE STUDY: A cycle in the monthly yield changes (2013 Case Study 3)

Fitting AR(2) by least squares to the monthly differences of the 10-year yield gives $\hat{\phi}_1 = 0.038$, $\hat{\phi}_2 = -0.217$ (standard errors 0.079, 0.079), $R^2 = 0.048$. The roots of the characteristic polynomial are the complex pair $0.088 \pm 2.145i$, of squared modulus $4.61 > 1$: the fitted model is stationary. With complex roots the series is cyclical, with period $2\pi/\arccos(|\hat{\phi}_1|/(2\sqrt{-\hat{\phi}_2})) = 4.107$, that is, “just over 4 months”; I recomputed this value (and the roots) numerically from the printed coefficients. The function `ar()` selects the order with the AIC (section 9.4.4): `order = 7`. In the AR(7) fit the significant lags are 2 ($t = -3.10$) and 7 ($t = -2.02$), $R^2 = 0.123$; the smallest squared modulus of the characteristic roots is 1.4027 (the case study calls it the “smallest root modulus”: the modulus is $\sqrt{1.4027} = 1.184$), still outside the unit circle. Regression diagnostics of the AR(7) fit (`influence.measures`) flag two high-leverage months, October and November 2008, “the heart of the financial crisis” (section 5.9).

Exercise 9.2 (PS6 (2013), Problem 2: an AR(2) as a difference of two MA processes). With $\{\eta_t\}$ i.i.d. $\text{WN}(0, \sigma^2)$ let $w_t = 5(1 - 0.5L)^{-1}\eta_t$, $v_t = 4(1 - 0.4L)^{-1}\eta_t$ and $x_t = w_t - v_t$.

- Solve for the coefficients ψ_i in $x_t = \eta_t + \sum_{i \geq 1} \psi_i \eta_{t-i}$.
- Prove $\{x_t\}$ is an AR(2) process.
- Solve for ϕ_1, ϕ_2 in $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \eta_t$.

- (d) Prove that any stationary AR(2) process can be written as the difference of two (possibly infinite-order) moving-average processes driven by the same $\{\eta_t\}$.

9.3.2 Moving-average models MA(q)

Set $p = 0$: $X_t - \mu = \theta(L)\eta_t = \eta_t + \theta_1\eta_{t-1} + \cdots + \theta_q\eta_{t-q}$ [► 2024 L12 @ 1:14:24][► 2013 L8 @ 1:06:12]. Because η is white noise, only overlapping terms contribute:

$$\begin{aligned} \mathbb{E}[X_t] &= \mu, & \gamma(0) &= \sigma^2(1 + \theta_1^2 + \cdots + \theta_q^2), \\ \gamma(j) &= \begin{cases} \sigma^2(\theta_j + \theta_{j+1}\theta_1 + \cdots + \theta_q\theta_{q-j}), & 1 \leq j \leq q, \\ 0, & j > q, \end{cases} \end{aligned} \quad (9.17)$$

with $\theta_0 = 1$ [► 2024 L12 @ 1:16:26]. The ACF of an MA(q) *cuts off* at lag q (a complete finite-memory process: X_t and X_{t+j} share no noise term when $j > q$), which is the identification rule of the table above.

Invertibility. The process is *invertible* if all roots of $\theta(z) = 0$ lie outside the closed unit disc; then $\theta^{-1}(L)$ exists by (9.8) and $\eta_t = \theta^{-1}(L)(X_t - \mu)$ is an infinite autoregression of the observations [► 2024 L12 @ 1:15:24]. Invertibility matters for two reasons. Identification: for MA(1), $\rho(1) = \theta/(1 + \theta^2)$, which has maximum $1/2$ at $\theta = 1$ and is unchanged under $\theta \mapsto 1/\theta$; for instance $\theta = 0.25$ and $\theta = 4$ both give $\rho(1) = 4/17 = 0.235$, and if σ^2 is replaced by $\theta^2\sigma^2$ the variance $\gamma(0) = \sigma^2(1 + \theta^2)$ is also unchanged: two different parameter sets with the same second-order structure, hence indistinguishable from the ACF. Interpretation: only the invertible representation has η_t equal to the one-step forecast error of X_t given its past, which is the Wold innovation; one therefore always fits the invertible member of the pair.

Exercise 9.3 (PS4 (2013), Problem 5: the MA(1) process). $X_t = \eta_t + \theta\eta_{t-1}$.

- Derive the ACF.
- Compute the first-order autocorrelation for $\theta = 0.5$ and $\theta = 2.0$.
- Solve the formula for ρ_1 for θ in terms of ρ_1 ; is the solution unique?
- For each of the two models decide whether the process is invertible and, if so, write it as an infinite-order autoregression.

9.3.3 ARMA(p, q) and the ARMA(1,1) model

A general ARMA(p, q) is covariance stationary when $\phi(z)$ has all roots outside the unit disc and is invertible when $\theta(z)$ does. (If ϕ and θ share a root the factors cancel and the model is redundant, a point to remember when over-fitting.) The Wold weights are the power-series coefficients of $\theta(z)/\phi(z)$, and the autocovariances satisfy the Yule–Walker recursion $\gamma(j) = \sum_i \phi_i \gamma(j - i)$ for $j > q$.

For ARMA(1,1), $X_t - \phi X_{t-1} = \phi_0 + \eta_t + \theta\eta_{t-1}$ (2013 PS4 Problem 6 and 2024 PS4 Problem 6, exercise 9.4), matching the powers in $\psi(L) = (1 + \theta L)/(1 - \phi L)$ gives $\psi_0 = 1$

and $\psi_j = \phi^{j-1}(\phi + \theta)$ for $j \geq 1$. Then $\mathbb{E}[X_t] = \phi_0/(1 - \phi)$ and, summing $\sum_j \psi_j^2$,

$$\begin{aligned}\gamma(0) &= \sigma^2 \frac{1 + 2\phi\theta + \theta^2}{1 - \phi^2}, \\ \rho(1) &= \frac{(1 + \phi\theta)(\phi + \theta)}{1 + 2\phi\theta + \theta^2} = \phi + \frac{\theta\sigma^2}{\gamma(0)}, \\ \rho(k) &= \phi \rho(k-1), \quad k \geq 2.\end{aligned}\tag{9.18}$$

Compare with the AR(1), where $\rho(k) = \phi^k$ from the very first lag: the ARMA(1,1) also decays geometrically at rate ϕ , but from lag 2 on, and the first correlation $\rho(1)$ is a free parameter that need not equal ϕ . When $\phi < 0$ the ACF of an AR(1) alternates in sign from lag 1; for an ARMA(1,1) it may be positive at lag 1 and alternate afterwards. Its distinctive pattern is *small first correlation but long memory*: with $\phi = 0.9$, $\theta = -0.85$ one finds $\rho(1) = 0.061$ and $\rho(10) = 0.024$, while an AR(1) with $\phi = 0.9$ has $\rho(1) = 0.9$. An economic index following such a process would show almost no day-to-day autocorrelation together with a slow, persistent decay. This structure arises for the *squared* residuals of a GARCH(1,1) model, which follow an ARMA(1,1) with $\phi = \alpha_1 + \beta_1$ and $\theta = -\beta_1$ ([chapter 13](#), [► 2013 L11 @ 5:15](#)). The comparison with MA(1) is the reverse: an MA(1) has a single non-zero correlation and then nothing (2024 PS4 Problem 6(e)).

Exercise 9.4 (PS4 (2013), Problem 6; PS4 (2024), Problem 6: ARMA(1,1)). For the covariance-stationary $(1 - \phi L)X_t = \phi_0 + (1 + \theta L)\eta_t$:

- Prove $\mu = \phi_0/(1 - \phi)$.
- Prove $\text{Var } X_t = \gamma(0) = \sigma^2 \left[1 + \frac{(\phi + \theta)^2}{1 - \phi^2} \right]$ (equivalently $\sigma^2(1 + 2\phi\theta + \theta^2)/(1 - \phi^2)$).
- Prove that $\rho_1 = \phi + \theta\sigma^2/\gamma(0)$ and $\rho_k = \phi\rho_{k-1}$ for $k > 1$.
- Compare the ACF with that of the AR(1) with the same ϕ_0, ϕ : which pattern of the ACF is impossible for an AR(1)? What behaviour would an economic index following such a process show?
- (2024 only) Compare with the ACF of an MA(1) with the same parameters; what is the pattern of an MA(1) and how does it change for MA(q), $q > 1$? Try to obtain (b)–(c) directly from the covariance recursion rather than via the Wold weights used in [section 9.3.3](#).

9.4 Non-stationarity: differencing, ARIMA and unit-root tests

Many economic series behave like random walks. Box and Jenkins advocate removing trending behaviour by *differencing* [► 2024 L12 @ 1:17:28](#) [[► 2013 L8 @ 1:07:14](#)]. With $\Delta = 1 - L$,

$$\Delta X_t = X_t - X_{t-1}, \quad \Delta^2 X_t = (1 - L)^2 X_t = X_t - 2X_{t-1} + X_{t-2}, \quad \Delta^k = (1 - L)^k = \sum_{j=0}^k \binom{k}{j} (-L)^j.$$

A linear trend $a + bt$ becomes the constant b under Δ ; a quadratic trend is removed by Δ^2 [► 2024 L12 @ 1:18:30](#) [[► 2013 L8 @ 1:08:20](#)]. In practice one uses $k \leq 2$. Modelling ΔX_t or $\Delta^2 X_t$ as stationary means modelling the dynamics of the slope, or of the curvature, of the series, with a constant level and variability [► 2024 L12 @ 1:19:32](#).

9.4.1 Two kinds of trend

Trend reversion. Let $X_t = a + bt + \xi_t$, with $\xi_t = \phi \xi_{t-1} + \eta_t$, $|\phi| < 1$ [► 2013 L8 @ 1:09:21]. Then $\mathbb{E}[X_t] = a + bt$ (not constant, hence not stationary), $\text{Var } X_t = \sigma^2/(1 - \phi^2)$, and

$$\Delta X_t = b + \xi_t - \xi_{t-1} = b + (1 - L)\xi_t = b + (1 - L)(1 - \phi L)^{-1}\eta_t,$$

which is stationary. A shock to ξ dies out and X returns to the deterministic line.

Stochastic trend. The pure *integrated process* of order one, $I(1)$, is $X_t = X_{t-1} + \eta_t$, that is, $\Delta X_t = \eta_t$ [► 2013 L8 @ 1:10:22]. Given X_0 , $X_t = X_0 + TS_t$ with $TS_t = \sum_{j=1}^t \eta_j$ (the *stochastic trend*), a random walk with white-noise steps, and

$$\begin{aligned} \text{Var } X_t &= t\sigma^2, & \text{Cov}(X_t, X_{t-j}) &= (t-j)\sigma^2, \\ \text{Corr}(X_t, X_{t-j}) &= \frac{t-j}{\sqrt{t(t-j)}} = \sqrt{1 - \frac{j}{t}} \longrightarrow 1 \quad (t \rightarrow \infty), \end{aligned} \quad (9.19)$$

so the process is not stationary: variance and covariances depend on t , and consecutive values become almost perfectly correlated. The trend cannot be forecast (the best forecast at any horizon is the last value, with error variance growing like $h\sigma^2$).

The two models differ where it matters for finance: in the trend-reversion model the long-run cumulative response $\psi(1)$ of ΔX_t to a shock is $\psi(1) = (1 - 1)/(1 - \phi) = 0$, the level effect is temporary; for the stochastic trend $\Delta X_t = \eta_t$ has $\psi(1) = 1$, the shock is permanent. Prices are usually modelled as the second kind (remark 4.19), returns as stationary. Differencing a trend-reversion series (“over-differencing”) creates an MA factor $(1 - L)$ with a unit root, which is not invertible; differencing a stochastic trend is the correct treatment.

Definition 9.9 (ARIMA(p, d, q)). $\{X_t\}$ is *integrated ARMA*, ARIMA(p, d, q), if $\Delta^d X_t$ is stationary (and lower-order differencing is not) and follows an ARMA(p, q) model (2024 slide 25, not discussed in class; [► 2013 L8 @ 1:11:26]).

The practical questions ([► 2013 L8 @ 1:12:26]) are: determine the order d of differencing needed to remove deterministic or stochastic trends (section 9.4.2); estimate the parameters (section 9.4.3); choose among competing (p, d, q) (section 9.4.4).

9.4.2 Testing for a unit root: the Dickey–Fuller test

Suppose $X_t = \phi X_{t-1} + \eta_t$ with η_t white noise. The *Dickey–Fuller* (DF) test has

$$H_0 : \phi = 1 \text{ (unit root, non-stationary)} \quad \text{against} \quad H_1 : |\phi| < 1 \text{ (stationary)}.$$

Fit the AR(1) by least squares and form $t_{\hat{\phi}=1} = (\hat{\phi} - 1)/\text{se}(\hat{\phi})$. Under H_1 , $\sqrt{T}(\hat{\phi} - \phi) \xrightarrow{d} N(0, 1 - \phi^2)$ (the standard AR(1) result). Under H_0 the estimator is *superconsistent*, converging at the rate $1/T$, and $T(\hat{\phi} - 1)$, like the t statistic, has a non-standard distribution, the Dickey–Fuller distribution (2024 slide 28, not discussed in either year). It can be derived in a few lines. For $X_t = X_{t-1} + \eta_t$ and $X_0 = 0$,

$$\hat{\phi} - 1 = \frac{\sum_t X_{t-1} \eta_t}{\sum_t X_{t-1}^2}, \quad \sum_{t=1}^T X_{t-1} \eta_t = \frac{1}{2} \left(X_T^2 - \sum_{t=1}^T \eta_t^2 \right),$$

because $X_t^2 = X_{t-1}^2 + 2X_{t-1}\eta_t + \eta_t^2$ telescopes. Dividing the numerator by $T\sigma^2$ and the denominator by $T^2\sigma^2$, and using $X_{\lfloor rT \rfloor}/(\sigma\sqrt{T}) \rightarrow W(r)$ (Donsker's theorem; the random walk converges to Brownian motion, [chapter 11](#)),

$$T(\hat{\phi} - 1) \xrightarrow{d} \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W(r)^2 dr} = \frac{\int_0^1 W dW}{\int_0^1 W^2 dr}, \quad (9.20)$$

the last equality being Itô's formula $\int_0^1 W dW = \frac{1}{2}(W(1)^2 - 1)$ ([chapter 17](#)). The distribution is skewed to the left, so the usual normal or t critical values reject the unit root too often. Simulating 20 000 random walks of length 500, the 5% critical value of $t_{\phi=1}$ is -1.95 (no intercept) and -2.89 (with an intercept; the asymptotic value is about -2.86), compared with -1.645 for the standard normal; the 1% values are -2.55 and -3.46 .

The test is often *augmented* (ADF, Said–Dickey): lagged differences $\Delta X_{t-1}, \dots, \Delta X_{t-k}$ are added to the regression to absorb serial correlation, and k is chosen by an information criterion. The null hypothesis is *non-stationarity*, so failing to reject does not prove a unit root: the power is low against ϕ near 1. Other tests: Phillips–Perron (PP) and Elliott–Rothenberg–Stock (ERS) for a unit-root null, and KPSS, whose null is stationarity [[2013 L8 @ 1:12:26](#)]. The 10-year Treasury yield is the standard illustration ([section 9.3.1](#)): the ADF test does not reject a unit root for the daily, weekly or monthly *levels* and rejects it for the *differences*. The same test was used in the crack-spread example ([section 9.6.4](#)).

Exercise 9.5 (Dickey–Fuller by simulation (not from the course)).

- Simulate 10^4 random walks of length 500 and compute $t_{\phi=1}$ from the regression of X_t on X_{t-1} with and without intercept; check the 5% quantiles -1.95 and -2.89 .
- Repeat with an AR(1) with $\phi = 0.95$ and estimate the power of the test at the 5% level; explain why non-rejection on a persistent yield series is weak evidence for a unit root.
- Verify the identity $\sum_{t=1}^T X_{t-1}\eta_t = \frac{1}{2}(X_T^2 - \sum \eta_t^2)$ and show that its expectation is zero; then estimate $\mathbb{E}[T(\hat{\phi} - 1)]$ by simulation and explain why the ratio in (9.20) is nevertheless biased downwards (the denominator is positively correlated with the numerator's negative part).

9.4.3 Estimation of ARMA models

For Gaussian innovations $\eta_t \sim N(0, \sigma^2)$ i.i.d., the parameters are estimated by maximum likelihood ([section 5.6](#)) (2024 slide 26; [[2013 L8 @ 1:12:26](#)]).

- AR(p).** Conditioning on the first p observations, the log-likelihood is $\ell = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=p+1}^T (x_t - c - \sum_j \phi_j x_{t-j})^2$ with $n = T - p$: conditional maximum likelihood is least squares of x_t on its p lags ([section 9.2.1](#)). This is the *limited-information* (LIML) method, and the resulting $\hat{\sigma}^2 = \text{RSS}/n$ is biased downwards by the same degrees-of-freedom argument as in regression [[2013 L8 @ 6:10](#)]. The *full-information* (FIML) method adds the log-density of the first p values under their stationary distribution, which matters when T is small or the roots are close to the unit circle.
- MA and ARMA.** The likelihood is not a regression, because the past innovations are not observed. LIML sets the first q innovations to zero and computes the others

recursively from the data; the exact likelihood is obtained by writing the model in state-space form and applying the Kalman filter (section 9.7), the *prediction-error decomposition*. The maximisation is numerical, and stationarity and invertibility constrain the parameters; the standard trick is to reparametrise so that the optimiser works on an unconstrained scale (logarithms, logits), as in the GARCH fits of chapter 13.

- *Method of moments*: Yule–Walker (section 9.3.1) for AR.

9.4.4 Model selection: AIC, BIC, HQ

Fit all ARMA(p, q) with $0 \leq p \leq p_{\max}$, $0 \leq q \leq q_{\max}$, let $\tilde{\sigma}^2(p, q)$ be the Gaussian MLE of σ^2 , and choose (p, q) to minimise (2024 slide 27; [► 2013 L8 @ 1:13:33])

$$\begin{aligned} \text{AIC}(p, q) &= \log \tilde{\sigma}^2(p, q) + 2 \frac{p+q}{n}, & (\text{Akaike}) \\ \text{BIC}(p, q) &= \log \tilde{\sigma}^2(p, q) + \log n \frac{p+q}{n}, & (\text{Bayes / Schwarz}) \\ \text{HQ}(p, q) &= \log \tilde{\sigma}^2(p, q) + 2 \log(\log n) \frac{p+q}{n}. & (\text{Hannan–Quinn}) \end{aligned} \quad (9.21)$$

Because the maximised Gaussian log-likelihood is $\hat{\ell} = -\frac{n}{2}(\log(2\pi\tilde{\sigma}^2) + 1)$, each criterion is $-2\hat{\ell}/n$ plus a penalty proportional to the number of parameters (up to an additive constant); the lecture ends on the question of what penalty is appropriate [► 2013 L8 @ 1:14:33]. Adding one parameter lowers $\log \tilde{\sigma}^2$ and raises the penalty. The larger model is preferred iff $2\Delta\hat{\ell} > c_n$, the likelihood-ratio statistic (asymptotically χ_1^2 under the smaller model) exceeds c_n , with $c_n = 2$ (AIC), $\log n$ (BIC) or $2 \log \log n$ (HQ):

	AIC	BIC	HQ
threshold c_n for $n = 100$	2	4.61	3.05
equivalent p -value	0.157	0.032	0.081
threshold c_n for $n = 500$	2	6.21	3.65
equivalent p -value	0.157	0.013	0.056

(For χ_1^2 the AIC threshold 2 corresponds to $|t| > \sqrt{2} = 1.41$.) So AIC is a liberal criterion, which tends to overfit but is the right choice if the true model is infinite-dimensional and one wants good forecasts; BIC assumes the true model has a finite number of parameters that it will find with probability tending to one; HQ makes the same finite-model assumption but with a penalty growing as slowly as consistency allows [► 2013 L8 @ 1:15:39]. The same ideas were used for regression in chapter 5 and are used in Case Study 3 (ar()) minimises the AIC over $p \leq 22$ and picks $p = 7$, with $p = 2$ at a relative AIC of 1.61) and for VAR models below.

9.5 Multivariate time series and VAR models

We now observe m series together, $\mathbf{X}_t = (X_{1,t}, \dots, X_{m,t})^\top \in \mathbb{R}^m$: several exchange rates, a basket of stocks, or macroeconomic variables [► 2013 L11 @ 30:44]. $\{\mathbf{X}_t\}$ is covariance stationary if every component is [► 2013 L11 @ 31:47].

9.5.1 Second-order structure

With $\mu = \mathbb{E}[X_t]$ define the *covariance matrix* $\Gamma_0 = \mathbb{E}[(X_t - \mu)(X_t - \mu)^\top]$ (symmetric, with the variances of the components on the diagonal), the *lag- k cross-covariance matrix* $\Gamma_k = \mathbb{E}[(X_t - \mu)(X_{t-k} - \mu)^\top]$, whose (j, j') entry is $\text{Cov}(X_{j,t}, X_{j',t-k})$, and the *correlation matrices* $R_k = D^{-1/2}\Gamma_k D^{-1/2}$ with $D = \text{diag}(\Gamma_0)$ [► 2013 L11 @ 31:47]. The matrix Γ_0 is symmetric, but Γ_k for $k \neq 0$ is *not*, and $\Gamma_{-k} = \Gamma_k^\top$: the lags of series j' may be correlated with series j without the converse [► 2013 L11 @ 36:04]. If $[\Gamma_k]_{jj'} \neq 0$ for some $k > 0$ one says that $X_{j'}$ *leads* X_j (its past carries information about X_j); if each leads the other there is *feedback* [► 2013 L11 @ 37:08]. This is the idea of *Granger causality* (Granger and Engle received the Nobel prize for it): lags of one variable help predict another. It is predictability, not causation in a philosophical sense.

The *multivariate Wold theorem* holds verbatim with matrices: $X_t = V_t + \sum_{k \geq 0} \Psi_k \eta_{t-k}$ with $\Psi_0 = I_m$, $\sum_k \Psi_k \Psi_k^\top$ convergent, V_t linearly deterministic, and η_t multivariate white noise: $\mathbb{E}[\eta_t] = 0$, $\text{Var } \eta_t = \Sigma$ (positive semi-definite), $\text{Cov}(\eta_t, \eta_{t-k}) = 0$ for $k \neq 0$, uncorrelated with V [► 2013 L11 @ 39:18]. The innovation is $\eta_t = X_t - \mathbb{E}[X_t | \mathcal{F}_{t-1}]$ in the Gaussian case, where $\mathcal{F}_{t-1}$ is the information available before time t : what is new at t [► 2013 L11 @ 42:25].

9.5.2 VAR(p) models and their stationarity

Definition 9.10 (VAR(p)). $\{X_t\}$ follows a *vector autoregression* of order p if

$$X_t = C + \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \cdots + \Phi_p X_{t-p} + \eta_t, \quad (9.22)$$

with $C \in \mathbb{R}^m$, $\Phi_1, \dots, \Phi_p$ real $m \times m$ matrices and $\{\eta_t\}$ multivariate white noise $N(0, \Sigma)$ [► 2013 L11 @ 45:39].

Component j is an AR(p) of its own lags *plus* lag regressors on all the other series: $X_{j,t} = c_j + \sum_{k=1}^p [\Phi_k]_{jj} X_{j,t-k} + \sum_{j' \neq j} \sum_{k=1}^p [\Phi_k]_{jj'} X_{j',t-k} + \eta_{j,t}$: exchange rates that co-move, macro variables that respond to each other [► 2013 L11 @ 46:43].

VAR(1) form. Any VAR(p) is a VAR(1) in higher dimension [► 2013 L11 @ 48:51]. Stack

$$Z_t = (X_t^\top, X_{t-1}^\top, \dots, X_{t-p+1}^\top)^\top \in \mathbb{R}^{mp}.$$

Then

$$Z_t = D + A Z_{t-1} + F_t, \quad A = \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I_m & 0 & \cdots & 0 & 0 \\ 0 & I_m & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_m & 0 \end{pmatrix}, \quad D = \begin{pmatrix} C \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad F_t = \begin{pmatrix} \eta_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (9.23)$$

with A the $(mp \times mp)$ *companion matrix* [► 2013 L11 @ 50:56].

Proposition 9.11 (Stationarity of a VAR). *The following are equivalent:*

- (i) all eigenvalues of A have modulus less than 1;
- (ii) all roots of $\det(I_m - \Phi_1 z - \cdots - \Phi_p z^p) = 0$ lie outside the closed unit disc $|z| \leq 1$.

Under (i)–(ii) the VAR has the unique stationary solution $X_t - \mu = \sum_{k \geq 0} \Psi_k \eta_{t-k}$, where Ψ_k

Proof. Let $\lambda \neq 0$ be an eigenvalue of A with eigenvector $v = (v_1, \dots, v_p)$. The block rows of $Av = \lambda v$ say $\sum_j \Phi_j v_j = \lambda v_1$ and $v_{j-1} = \lambda v_j$ for $j = 2, \dots, p$, so $v_j = \lambda^{p-j} v_p$. Substituting, $(\lambda^p I - \sum_j \Phi_j \lambda^{p-j}) v_p = 0$ with $v_p \neq 0$, i.e. $\det(I - \sum_j \Phi_j \lambda^j) = 0$ at $\lambda = 1/\lambda$; the converse is the same computation backwards. Thus the eigenvalues of A are the reciprocals of the roots, and (i) $\Leftrightarrow$ (ii).

is the upper-left $m \times m$ block of A^k , and

$$\mu = \mathbb{E}[X_t] = (I_m - \Phi_1 - \dots - \Phi_p)^{-1} C. \quad (9.24)$$

The characteristic polynomial $\det(I_m - \Phi_1 z - \dots - \Phi_p z^p)$ has degree mp , hence mp roots [► 2013 L11 @ 53:06]; the mean-adjusted series satisfies $X_t - \mu = \sum_j \Phi_j (X_{t-j} - \mu) + \eta_t$ [► 2013 L11 @ 55:13]. The scalar case $m = 1$ is [theorem 9.5](#), and its proof-by-eigenvalue is [remark 9.6](#).

Exercise 9.6 (PS6 (2013), Problem 1: moments of a VAR(1)). Let $X_t = (X_{1,t}, X_{2,t})^\top$ follow $X_{1,t} = 0.3 + 0.8X_{1,t-1} + \eta_{1,t}$, $X_{2,t} = 0.2 + 0.6X_{1,t-1} + 0.4X_{2,t-1} + \eta_{2,t}$, with $(\eta_{1,t}, \eta_{2,t})^\top$ i.i.d. $N(0, \Sigma)$, $\Sigma = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$. Compute

- $\mu = \mathbb{E}[X_t]$,
- $\Gamma(0) = \text{Cov } X_t$,
- $\Gamma(1) = \text{Cov}(X_t, X_{t-1})$, and
- derive a formula for $\Gamma(h)$, $h \geq 1$. (See [example 9.12](#) for the method.)

Exercise 9.7 (Stationarity of a VAR (not from the course)).

- Write the companion matrix of the univariate AR(2) $X_t = 0.5X_{t-1} + 0.3X_{t-2} + \eta_t$, find its eigenvalues (0.852, -0.352) and the roots of $\phi(z)$; verify $\lambda = 1/z$.
- For the VAR(1) with $A = \begin{pmatrix} 0.9 & 0.4 \\ 0 & 0.5 \end{pmatrix}$ compute Γ_0 from $\text{vec } \Gamma_0 = (I - A \otimes A)^{-1} \text{vec } \Sigma$ for $\Sigma = I$ and the impulse responses Ψ_k ; which shock has the longest-lasting effect on which variable?

If (i) holds, $Z_t - \mu_Z = \sum_{k \geq 0} A^k F_{t-k}$ converges because the spectral radius of A is < 1 (so $\|A^k\|$ decays geometrically), and reading off the first block gives the expansion. Taking expectations in (9.22) gives $\mu = C + \sum \Phi_j \mu$. $\square$

Example 9.12 (Moments of a VAR(1)). For $X_t = C + AX_{t-1} + \eta_t$ with stationary A , let $\tilde{X}_t = X_t - \mu$. Then $\Gamma_0 = A\Gamma_0A^\top + \Sigma$, and with the vectorisation of [section 9.5.3](#) this is solved by $\text{vec } \Gamma_0 = (I - A \otimes A)^{-1} \text{vec } \Sigma$; further, $\Gamma_h = A^h \Gamma_0$ for $h \geq 0$. Take $A = \begin{pmatrix} 0.5 & 0.2 \\ -0.3 & 0.6 \end{pmatrix}$, $C = (1, 0.5)^\top$, $\Sigma = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}$. The eigenvalues of A are $0.55 \pm 0.24i$ (modulus $0.6 < 1$), $\mu = (1.923, -0.192)^\top$, $\Gamma_0 = \begin{pmatrix} 1.443 & 0.244 \\ 0.244 & 0.847 \end{pmatrix}$ and $\Gamma_1 = A\Gamma_0 = \begin{pmatrix} 0.770 & 0.291 \\ -0.287 & 0.435 \end{pmatrix}$, which is not symmetric (checked against the iteration $\Gamma \leftarrow A\Gamma A^\top + \Sigma$). This is the pattern that 2013 PS6 Problem 1 asks for with different numbers.

9.5.3 VAR as a multivariate regression: estimation

Given n observations $x_1, \dots, x_n$ and p pre-sample values, define for each component j the regression $y^{(j)} = Z\beta^{(j)} + \varepsilon^{(j)}$, where $y^{(j)} = (x_{j,1}, \dots, x_{j,n})^\top$ and the same $n \times (mp + 1)$ design matrix Z has rows $(1, x_{t-1}^\top, \dots, x_{t-p}^\top)$ [► 2013 L11 @ 57:18]. The $\beta^{(j)}$ contain the j th rows of $C, \Phi_1, \dots, \Phi_p$. Stacking columns,

$$Y = ZB + E, \quad Y = [y^{(1)} \dots y^{(m)}], \quad B = [\beta^{(1)} \dots \beta^{(m)}], \quad (9.25)$$

a multivariate regression, called *seemingly unrelated regressions* (SUR) in econometrics because the m equations are linked only through the correlation Σ of their errors [► 2013 L11 @ 59:41].

Component-wise least squares. Fit each equation separately, $\hat{\beta}^{(j)} = (Z^\top Z)^{-1} Z^\top y^{(j)}$, that is $\hat{B} = (Z^\top Z)^{-1} Z^\top Y$, take the residual matrix $\hat{E} = Y - Z\hat{B} = (I_n - Z(Z^\top Z)^{-1} Z^\top)Y$, and estimate the innovation covariance by $\hat{\Sigma} = \hat{E}^\top \hat{E} / (n - mp - 1)$ (unbiased entry by entry; the lecture notes divide by $n - pm$) [► 2013 L11 @ 1:04:58]. Treating the equations separately looks wasteful, since the errors are correlated. It is not:

Theorem 9.13 (Optimality of component-wise OLS). For a VAR(p) with unrestricted coefficient matrices,

- component-wise OLS equals the generalised least-squares (GLS) estimator for the whole system with any innovation covariance Σ , hence is BLUE (Gauss–Markov, [theorem 5.8](#),

Proof. Let $\otimes$ be the Kronecker product ($A \otimes B$ is the block matrix with blocks $a_{ij}B$; $(A \otimes B)^\top = A^\top \otimes B^\top$ and $(A \otimes B)(C \otimes D) = AC \otimes BD$) and vec the operator that stacks the columns of a matrix into one vector [► 2013 L11 @ 1:09:04]. Put $y = \text{vec } Y$, $X = I_m \otimes Z$, $\beta^* = \text{vec } B$, $\varepsilon = \text{vec } E$. Then $y = X\beta^* + \varepsilon$ with $\text{Cov } \varepsilon = \Sigma \otimes I_n$, and

and [section 5.4](#));

- (ii) with Gaussian innovations it is the maximum-likelihood estimator of B , and the MLE of Σ is $\tilde{\Sigma} = \hat{E}^T \hat{E} / n$ [▶ 2013 L11 @ 1:06:58](#)].

Order selection and asymptotics. As in the univariate case, fit $\text{VAR}(p)$ for $0 \leq p \leq p_{\max}$ and choose p by

$$\begin{aligned} \text{AIC}(p) &= \log |\tilde{\Sigma}(p)| + 2 \frac{pm^2}{n}, & \text{BIC}(p) &= \log |\tilde{\Sigma}(p)| + \log n \frac{pm^2}{n}, \\ \text{HQ}(p) &= \log |\tilde{\Sigma}(p)| + 2 \log(\log n) \frac{pm^2}{n} \end{aligned} \quad (9.26)$$

[▶ 2013 L11 @ 1:17:24](#). (The slides print $-\log |\tilde{\Sigma}(p)|$, a sign misprint: $\log |\tilde{\Sigma}|$ is the log generalised variance, which decreases as p increases, while the penalty increases; compare [eq. \(9.21\)](#).) For a stationary VAR whose innovations have bounded fourth moments and are independent over time, $Q = \text{plim } Z^T Z / n$ exists and is non-singular and $\sqrt{n}(\hat{\beta}^* - \beta^*) \rightarrow N(0, \Sigma \otimes Q^{-1})$; least-squares and Gaussian maximum-likelihood estimates are asymptotically identical (2013 notes, last slide).

9.5.4 Impulse responses and a macroeconomic VAR

The matrices Ψ_k of the Wold expansion are the *impulse response functions*: $[\Psi_k]_{jj'}$ is the effect on variable j , k periods later, of a unit innovation in variable j' [▶ 2013 L11 @ 1:21:42](#). Because Σ is not diagonal the innovations are correlated, and R's `irf` plots the *orthogonalised* responses (shocks made uncorrelated through a Cholesky factor of Σ , which depends on the ordering of the variables; a point the course does not discuss).

CASE STUDY: VAR of US macroeconomic series (2013 Case Study 5 and 2013 L11–L12)

In the 1980s Litterman and Sims used vector autoregressions of nine macroeconomic aggregates (Treasury-bill rate, M1, GNP deflator, real GNP, business fixed investment, unemployment, trade-weighted dollar, S&P 500, commodity price index) to forecast the economy and to assess the effect of policy variables such as the Federal Funds rate. The case study downloads monthly series from FRED with `quantmod` (UNRATE, FEDFUNDS, TB3MS, CPIAUCSL, M1SL, GDPDEF, GDP, GPDI, TWEXBMTH, and the S&P 500 from Yahoo) and fits VAR models with the `vars` package to three of them: the unemployment rate, the Federal Funds rate and the CPI, over 1960–2000 (492 months) [▶ 2013 L11 @ 1:18:29](#)].

- **Levels.** `VARselect` (up to 12 lags) returns $p = 12, 5, 2, 12$ for AIC, HQ, SC (=BIC), FPE. The VAR(2) chosen by SC has characteristic-root moduli 1.002, 0.986, 0.952, 0.468, 0.331, 0.084: the largest exceeds 1, so the fitted model is *non-stationary* [▶ 2013 L12 @ 0:00](#). (Here “roots” are eigenvalues of the companion matrix, stationary when < 1 ; [remark 9.6](#).)
- **Differences.** Removing the random-walk component by first differences, the ACF and PACF show strong self-correlation and some cross-correlation between unemployment and Fed funds [▶ 2013 L12 @ 2:11](#). `VARselect` gives $p = 12, 3, 3, 12$; the VAR(3) has all root moduli < 0.84 (stationary). The dependent variable is the monthly change; the lecture’s reading of the Fed funds equation [▶ 2013 L12 @ 6:15](#):

$$\Delta \text{FF}_t = -0.71 \Delta \text{UN}_{t-1} + 0.37 \Delta \text{FF}_{t-1} - 0.18 \Delta \text{FF}_{t-2} + \dots \quad (t = -4.9, 8.0, -3.7).$$

$(\Sigma \otimes I_n)^{-1} = \Sigma^{-1} \otimes I_n$. The GLS estimator is $[X^T(\Sigma^{-1} \otimes I_n)X]^{-1}X^T(\Sigma^{-1} \otimes I_n)y$ with

$$X^T(\Sigma^{-1} \otimes I_n)X = (I \otimes Z^T)(\Sigma^{-1} \otimes I_n)(I \otimes Z) = \Sigma^{-1} \otimes Z^T Z,$$

so it equals $[\Sigma \otimes (Z^T Z)^{-1}](\Sigma^{-1} \otimes Z^T)y = [I_m \otimes (Z^T Z)^{-1}Z^T]y = \text{vec}((Z^T Z)^{-1}Z^T Y)$: Σ cancels and one recovers component-wise OLS [▶ 2013 L11 @ 1:11:06](#). This proves (i). For (ii), the Gaussian log-likelihood is

$$\ell(B, \Sigma) = -\frac{nm}{2} \log 2\pi - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \text{tr}[\Sigma^{-1}(Y - ZB)^T(Y - ZB)]$$

[▶ 2013 L11 @ 1:13:12](#). Writing $Y - ZB = \hat{E} + Z(\hat{B} - B)$ and using $Z^T \hat{E} = 0$, the trace equals $\text{tr}[\Sigma^{-1} \hat{E}^T \hat{E}] + \text{tr}[\Sigma^{-1}(\hat{B} - B)^T Z^T Z(\hat{B} - B)]$, and the second term is ≥ 0 ; so for *every* Σ the maximum over B is at $\hat{B}$ (a *concentrated* likelihood, because $\hat{B}$ does not depend on Σ [▶ 2013 L11 @ 1:15:16](#)). The remaining function is $-\frac{n}{2}[\log |\Sigma| + \text{tr}(\Sigma^{-1}S)]$ with $S = \hat{E}^T \hat{E} / n$. Let $u_1, \dots, u_m > 0$ be the eigenvalues of $\Sigma^{-1/2} S \Sigma^{-1/2}$; then $\log |\Sigma| + \text{tr}(\Sigma^{-1}S) = \log |S| + \sum_i (u_i - \log u_i) \geq \log |S| + m$, since $u - \log u \geq 1$, with equality iff all $u_i = 1$, i.e. $\Sigma = S$. (This is the Anderson–Olkin result quoted in the lecture [▶ 2013 L11 @ 1:16:19](#).) $\square$

If unemployment rises, the Fed lowers rates the next period; a rate change tends to be followed by another in the same direction (the Fed does not want to shock the economy), and since $0.37 x_{t-1} - 0.18 x_{t-2} = 0.19 x_{t-1} + 0.18 (x_{t-1} - x_{t-2})$ the negative second lag is an *acceleration* term [► 2013 L12 @ 8:21]. The CPI equation is dominated by its own lags (0.42, $t = 9.5$, and 0.29 at lag 3, $t = 6.7$), the unemployment equation by lags 2 and 3 of itself; R^2 are 0.12 (unemployment), 0.23 (Fed funds), 0.49 (CPI); the residual correlation between unemployment and Fed funds is -0.20 .

- *Impulse responses* (bootstrap 95% bands, 100 runs) [► 2013 L11 @ 1:22:42]. Levels VAR: after a rise in unemployment the Fed funds rate and the CPI decline (less employment, less pressure on prices); after a rise in the Fed funds rate unemployment and the CPI increase; after a rise in the CPI the Fed funds rate increases (consistent with policy against inflation). Differenced VAR: unemployment up $\Rightarrow$ Fed funds down over the next quarters; Fed funds up $\Rightarrow$ a modest increase in inflation, which persists for several quarters (the large third-lag partial autocorrelation of the CPI); CPI up $\Rightarrow$ modestly higher unemployment and higher Fed funds.

Three variables are far too few for a structural model of the economy: the lecture recommends ten to twelve [► 2013 L12 @ 5:13].

9.6 Cointegration

Differencing removes non-stationarity but discards information about the *levels*. Cointegration is the framework that keeps it [► 2013 L12 @ 10:30].

9.6.1 Definition and examples

A process $\{X_t\}$ is *integrated of order d* , $I(d)$, if $(1 - L)^d X_t$ is stationary. If it has a VAR representation $\Phi(L)X_t = \eta_t$ then $\Phi(L) = (1 - L)^d \Phi^*(L)$ with Φ^* describing the stationary VAR($p - d$) of $\Delta^d X_t$ [► 2013 L12 @ 12:43]. The point is that all the components may be $I(1)$ without the vector being “jointly” $I(1)$: linear combinations of the levels may be stationary [► 2013 L12 @ 14:47].

Definition 9.14 (Cointegration). An m -vector $\{X_t\}$ of $I(1)$ series is *cointegrated* if there is $\beta \in \mathbb{R}^m$, $\beta \neq 0$, with $\beta^T X_t$ an $I(0)$ (stationary) process [► 2013 L12 @ 15:49]. The vector β is defined up to scale; normalising $\beta_1 = 1$, $\beta^T X_t = u_t$ says $x_{1,t} = -(\beta_2 x_{2,t} + \dots + \beta_m x_{m,t}) + u_t$: a *long-run equilibrium relationship* $x_1 = -(\beta_2 x_2 + \dots)$ with a stationary *disequilibrium error* (cointegration residual) u_t [► 2013 L12 @ 16:49].

IN FINANCE: Where cointegration appears

Examples listed in the lecture [► 2013 L12 @ 16:49]: the term structure of interest rates (the level of rates is non-stationary, the *spreads* between maturities are stationary: expectations hypothesis); purchasing-power parity (exchange rate and price levels of two countries: otherwise there would be arbitrage); money demand (money, income, prices, interest rates); covered interest-rate parity (spot and forward rates); the law of one price (identical or equivalent assets that must be valued alike: spot and futures, the same asset on two venues). In trading, a cointegrated pair is the basis of *convergence* (pairs) trades: two prices that wander individually but whose spread mean-reverts.

9.6.2 The error-correction (VECM) representation

Let X_t follow a VAR(p) (9.22) and be $I(1)$. Subtract X_{t-1} from both sides, then add and subtract $(\Phi_1 - I)X_{t-2}$, then $(\Phi_2 + \Phi_1 - I)X_{t-3}$, and so on [► 2013 L12 @ 27:08]. The result is

$$\Delta X_t = C + \Pi X_{t-1} + \Gamma_1 \Delta X_{t-1} + \cdots + \Gamma_{p-1} \Delta X_{t-p+1} + \eta_t, \quad (9.27)$$

$$\Pi = \Phi_1 + \cdots + \Phi_p - I_m, \quad \Gamma_k = - \sum_{j=k+1}^p \Phi_j,$$

the *vector error-correction model* (VECM) [► 2013 L12 @ 30:14]. (Proof: for $p = 2$, $X_t = C + \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \eta_t$ is written $X_{t-2} = X_{t-1} - \Delta X_{t-1}$ in the last term, giving $\Delta X_t = C + (\Phi_1 + \Phi_2 - I)X_{t-1} - \Phi_2 \Delta X_{t-1} + \eta_t$; the general case follows by induction.) Everything in (9.27) is $I(0)$ except possibly ΠX_{t-1} , which must therefore be stationary too: it contains the cointegrating combinations. Since X has unit roots, Π is singular. Three cases [► 2013 L12 @ 32:19]:

- rank $\Pi = 0$, $\Pi = 0$: no cointegration; model the differences by a stationary VAR($p - 1$);
- rank $\Pi = m$: X is itself stationary (no unit roots);
- $0 < r = \text{rank } \Pi < m$: $\Pi = \alpha\beta^\top$ with α, β of size $m \times r$ and full rank r . The r columns of β are linearly independent cointegrating vectors; the matrix α holds the *adjustment* (error-correction) coefficients: the process is pulled back towards the equilibrium $\beta^\top X = 0$ at speed α .

The factorisation is not unique: $\alpha\beta^\top = (\alpha G)(\beta G^{-\top})^\top$ for any invertible $r \times r$ matrix G . The stationary directions form an r -dimensional space; a choice of coordinates in it is convention [► 2013 L12 @ 34:22].

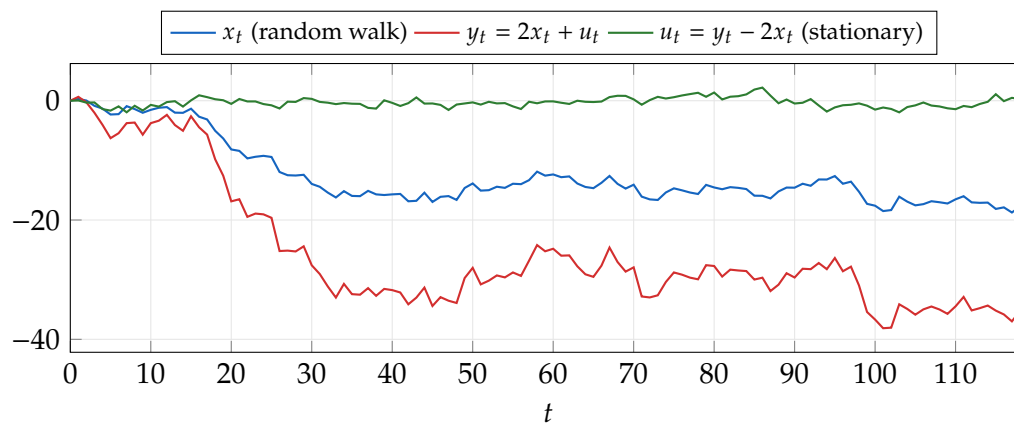


Figure 9.5: A simulated cointegrated pair (seed 7). Each of x and y wanders like a random walk, but $y - 2x$ stays around zero.

Example 9.15 (A cointegrated pair). Let $x_t = x_{t-1} + \varepsilon_{1t}$ (random walk), $u_t = 0.7u_{t-1} + \varepsilon_{2t}$ and $y_t = 2x_t + u_t$. Both x and y are $I(1)$, but $y_t - 2x_t = u_t$ is stationary: $\beta = (-2, 1)^\top$. In error-correction form $\Delta x_t = \varepsilon_{1t}$ and $\Delta y_t = 2\Delta x_t + u_t - u_{t-1} = -0.3(y_{t-1} - 2x_{t-1}) + 2\varepsilon_{1t} + \varepsilon_{2t}$. Hence $\Delta X_t = \alpha\beta^\top X_{t-1} + \text{noise}$ with $\alpha = (0, -0.3)^\top$, and $\Pi = \alpha\beta^\top = \begin{pmatrix} 0 & 0 \\ 0.6 & -0.3 \end{pmatrix}$ has rank one. In the VAR(1) in levels the matrix $I + \Pi$ has eigenvalues 1 (the common stochastic trend) and 0.7 (the stationary spread). Only y reacts to a disequilibrium: x is “weakly exogenous” (the row of α for x is zero). Figure 9.5 shows a simulated path.

Exercise 9.8 (Cointegration and the VECM (not from the course)). For the pair of example 9.15

- rewrite the system as a VAR(1) in levels and verify that $I + \Pi$ has eigenvalues 1 and 0.7.
- Show that Π is not invertible and identify α and β ; check that another factorisation $(\alpha G)(\beta G^{-\top})^\top$ gives the same Π .

- (c) Simulate the system, regress y_t on x_t and apply an ADF test to the residuals (Engle–Granger); do the same for two independent random walks and comment on “spurious regression”.
- (d) What changes in the VECM if both ε_{1t} and u_{t-1} enter the equation of x_t ?

9.6.3 Estimation and tests for the cointegration rank

Sims, Stock and Watson (1990) and Park and Phillips (1989) prove that least squares applied to the original VAR in *levels* gives consistent estimates with the same asymptotic distributions as maximum likelihood, and the reduced-rank constraints on Π hold asymptotically: no new tool is needed for estimation [► 2013 L12 @ 34:22]. For inference on the rank r , Johansen (1991) developed maximum likelihood by reduced-rank regression for Gaussian innovations, following Banerjee and Hendry’s use of the concentrated likelihood [► 2013 L12 @ 37:43]. It gives likelihood-ratio tests of the number of cointegrating vectors: Johansen’s *trace* statistic $-T \sum_{i>r} \log(1 - \hat{\lambda}_i)$ (a sum over the smallest eigenvalues, testing rank $\leq r$) and his *maximum-eigenvalue* statistic $-T \log(1 - \hat{\lambda}_{r+1})$ (rank r against $r + 1$), where the $\hat{\lambda}_i$ are squared sample canonical correlations between ΔX_t and X_{t-1} after adjusting for the lagged differences (the slides describe them as the eigenvalues of $\hat{\Pi}$; I give the standard definition). A cruder route, not in the course, is the Engle–Granger two-step method: regress one series on the others and apply an ADF test to the residual (with different critical values).

9.6.4 Energy futures and crack spreads

The lecture’s example of cointegration uses futures on the CME from the Energy Information Administration: WTI crude oil (CL, quoted in dollars per barrel), RBOB gasoline and heating oil (both in cents per gallon), from 2006 to 2013, first-month contracts. A refinery “cracks” crude oil into gasoline, heating oil, jet fuel; multiplying the product prices by 42 gallons per barrel puts inputs and outputs in the same units, and the products stay above crude, so the *crack spreads* (product minus crude) represent the value added of refining [► 2013 L12 @ 18:54]. The class listed what drives the spread: cost of refining, supply and demand, seasonality (heating oil dearer in winter), and the refiner’s profit [► 2013 L12 @ 23:02]; the spreads look stationary [► 2013 L12 @ 25:05]. The case note (not in the download) tests the levels with the Augmented Dickey–Fuller test: p-values 0.164 for the first crude contract and 0.327 for heating oil, so a unit root cannot be rejected [► 2013 L12 @ 38:47]. In the Johansen procedure the test of rank 0 is almost significant at the 10% level and the test of rank ≤ 1 is not, so the rank is at most one, with cointegrating vector (1, 1.3, -1.7) on crude, RBOB and heating oil, so that “crude plus gasoline minus heating oil” is the stationary combination (ignoring seasonality) [► 2013 L12 @ 41:55]. The first-month contract was chosen because refiners hedge on the futures market, so the analysis addresses hedging refinery risk; longer maturities would be used depending on what one hedges [► 2013 L12 @ 44:07]. The video jumps here between explanations, so the numbers are reported as spoken; commodity models are the subject of [chapter 21](#).

9.7 Linear state-space models and the Kalman filter

Many models of economics and finance, including ARMA models, regressions with time-varying coefficients and models with hidden factors, can be written in one common

form that has an exact and cheap recursive solution, the Kalman filter [► 2013L12@45:11].

9.7.1 The model

Let $y_t \in \mathbb{R}^k$ be the observation at time t , $s_t \in \mathbb{R}^m$ the (hidden) state, $\varepsilon_t \in \mathbb{R}^k$ the observation error and $\eta_t \in \mathbb{R}^n$ the state innovation. The *linear Gaussian state-space model* is

$$s_{t+1} = T_t s_t + R_t \eta_t, \quad \eta_t \sim N(0, Q_t); \quad y_t = Z_t s_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t), \quad (9.28)$$

the *state* (transition) equation and the *observation* (measurement) equation, with i.i.d. noises independent of one another; T_t ($m \times m$) is the transition matrix, R_t ($m \times n$, often columns of the identity) selects which components are shocked, Z_t ($k \times m$) the observation matrix, Q_t, H_t positive definite [► 2013L12@47:18]. Jointly, $(s_{t+1}; y_t) = \begin{pmatrix} T_t \\ Z_t \end{pmatrix} s_t + u_t$ with $u_t = (R_t \eta_t; \varepsilon_t) \sim N(0, \Omega_t)$, $\Omega_t = \text{blockdiag}(R_t Q_t R_t^\top, H_t)$ [► 2013L12@49:25]. Often the model is time-invariant. The state is not unique: replacing s_t by $M s_t$ (M invertible) and T_t by $M T_t M^{-1}$, Z_t by $Z_t M^{-1}$ gives an equivalent model [► 2013L12@1:02:59].

9.7.2 Examples

Time-varying CAPM. Extend the CAPM regression (section 5.11.2) so that alpha and beta drift as Gaussian random walks:

$$r_t = \alpha_t + \beta_t r_{m,t} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2); \quad \alpha_{t+1} = \alpha_t + \xi_t, \quad \beta_{t+1} = \beta_t + \zeta_t,$$

with $\xi_t \sim N(0, \sigma_\xi^2)$, $\zeta_t \sim N(0, \sigma_\zeta^2)$ and ε, ξ, ζ mutually independent, r_t the excess return of the asset and $r_{m,t}$ that of the market [► 2013L12@48:22]. In state-space form $s_t = (\alpha_t, \beta_t)^\top$, $T_t = R_t = I_2$, $Q = \text{diag}(\sigma_\xi^2, \sigma_\zeta^2)$, $Z_t = (1, r_{m,t})$ (a known 1×2 row), $H = \sigma_\varepsilon^2$ [► 2013L12@52:35].

Regression with time-varying coefficients. $y_t = x_t^\top \beta_t + \varepsilon_t$ with $\beta_{j,t+1} = \beta_{j,t} + \eta_{j,t}$, $\eta_{j,t} \sim N(0, \sigma_j^2)$, is $s_t = \beta_t$, $T_t = R_t = I_p$, $Z_t = x_t^\top$, $Q = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$, $H = \sigma^2$ [► 2013L12@54:37]. If all $\sigma_j^2 = 0$ the coefficients are constant and one obtains ordinary linear regression; applying the filter of the next subsection observation by observation is then *recursive least squares*, updating the estimate as data arrive [► 2013L12@55:37].

AR(p). With $s_t = (y_t, y_{t-1}, \dots, y_{t-p+1})^\top$, $s_{t+1} = F s_t + e_1 \eta_t$, where F is the companion matrix of remark 9.6, $e_1 = (1, 0, \dots, 0)^\top$, and $y_t = e_1^\top s_t$ (no measurement error, $\varepsilon_t = 0$) [► 2013L12@56:43].

ARMA(p, q) (Harvey's representation). Older estimation methods for moving-average models were hard to implement; the state-space form made exact maximum likelihood routine [► 2013L12@59:48]. Put $m = \max(p, q + 1)$, $\phi_j = 0$ for $j > p$, $\theta_j = 0$ for $j > q$, $\theta_0 = 1$; then $\{y_t\}$ is ARMA($m, m - 1$). Define the state by $s_{1,t} = y_t$ and

$$s_{k,t} = \sum_{i=k}^m \phi_i y_{t+k-1-i} + \sum_{j=k-1}^{m-1} \theta_j \eta_{t+k-1-j}, \quad k = 2, \dots, m.$$

Rewriting the model equation in terms of $s_{1,t}$ and $s_{2,t}$ and repeating gives

$$s_{t+1} = T s_t + R \eta_{t+1}, \quad y_t = e_1^\top s_t, \quad T = \begin{pmatrix} \phi_1 & 1 & 0 & \cdots & 0 \\ \phi_2 & 0 & 1 & & \vdots \\ \vdots & & & \ddots & 0 \\ \phi_{m-1} & 0 & \cdots & 0 & 1 \\ \phi_m & 0 & \cdots & 0 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 1 \\ \theta_1 \\ \vdots \\ \theta_{m-1} \end{pmatrix} \quad (9.29)$$

[► 2013 L12 @ 1:01:55]. Check for ARMA(1,1) ($m = 2$): the first row of $s_{t+1} = T s_t + R \eta_{t+1}$ is $y_{t+1} = \phi y_t + s_{2,t} + \eta_{t+1}$ and the second is $s_{2,t+1} = \theta \eta_{t+1}$, so $s_{2,t} = \theta \eta_t$ and $y_{t+1} = \phi y_t + \eta_{t+1} + \theta \eta_t$, as required. I also verified numerically that for ARMA(2,1) with $\phi = (0.5, -0.3)$, $\theta = 0.4$ the state-space variances ($\gamma(0) = 1.893$, $\gamma(1) = 1.036$ in units of σ^2 , from the discrete Lyapunov equation $P = T P T^\top + R R^\top$) agree with the Wold-weight sums. The MA(q) is the case $p = 0$. (The lecture notes give a different MA(q) form, with the state $(\eta_{t-1}, \dots, \eta_{t-q})$ and $y_t = \sum_j \theta_j s_j + \eta_t$; there the measurement error and the state innovation are the *same* variable, which violates the independence assumption of (9.28), so I use the ARMA form above.)

9.7.3 The Kalman filter

Let $\mathcal{F}_t = \{y_1, \dots, y_t\}$. The *Kalman filter* is the recursive computation of the conditional laws of the state and of the next observation given the information so far [► 2013 L12 @ 1:07:03]. Because everything is Gaussian these are determined by means and covariances:

$$\begin{aligned} s_{t|t} &= \mathbb{E}[s_t | \mathcal{F}_t], & s_{t|t-1} &= \mathbb{E}[s_t | \mathcal{F}_{t-1}], & y_{t|t-1} &= \mathbb{E}[y_t | \mathcal{F}_{t-1}], \\ P_{t|t} &= \text{Cov}(s_t | \mathcal{F}_t), & P_{t|t-1} &= \text{Cov}(s_t | \mathcal{F}_{t-1}), & F_t &= \text{Cov}(y_t | \mathcal{F}_{t-1}), \end{aligned}$$

where the lecture writes $\Omega_s(t | t)$, $\Omega_s(t | t-1)$, $\Omega_y(t | t-1)$ [► 2013 L12 @ 1:09:12]. The *innovation* (one-step forecast error) is $v_t = y_t - y_{t|t-1} = y_t - Z_t s_{t|t-1}$ [► 2013 L12 @ 1:10:12]. The filter alternates two steps.

1. **Prediction** [► 2013 L12 @ 1:11:18]. From $(s_{t|t}, P_{t|t})$ predict the next state and observation:

$$\begin{aligned} s_{t+1|t} &= T_t s_{t|t}, & P_{t+1|t} &= T_t P_{t|t} T_t^\top + R_t Q_t R_t^\top, \\ y_{t+1|t} &= Z_{t+1} s_{t+1|t}, & F_{t+1} &= Z_{t+1} P_{t+1|t} Z_{t+1}^\top + H_{t+1}. \end{aligned} \quad (9.30)$$

2. **Correction (filtering)** [► 2013 L12 @ 1:12:21]. When y_t is observed, update:

$$s_{t|t} = s_{t|t-1} + G_t v_t, \quad P_{t|t} = P_{t|t-1} - G_t F_t G_t^\top, \quad G_t = P_{t|t-1} Z_t^\top F_t^{-1}, \quad (9.31)$$

G_t being the *filter gain*. The uncertainty on the state decreases by the information in the innovation ([► 2013 L12 @ 1:13:23]). (The slides write the gain with the index $(t-1 | t)$; it must be $(t | t-1)$.)

Two further operations reuse the same quantities. *Forecasting* h steps ahead iterates the prediction step without corrections [► 2013 L12 @ 1:14:24]. *Smoothing* improves the estimate of a past state using the whole sample $\mathcal{F}_T$, by the backward recursion $s_{t|T} = s_{t|t} + S_t(s_{t+1|T} - s_{t+1|t})$ with $S_t = P_{t|t} T_t^\top P_{t+1|t}^{-1}$ (and the analogous recursion for the covariance) [► 2013 L12 @ 1:14:24].

Likelihood. By the chain rule $p(y_1, \dots, y_T; \theta) = p(y_1; \theta) \prod_{t \geq 2} p(y_t | \mathcal{F}_{t-1}; \theta)$, and $y_t | \mathcal{F}_{t-1} \sim N(y_{t|t-1}, F_t)$. So the filter delivers the exact log-likelihood of all parameters θ in

Derivation of the correction step. Conditionally on $\mathcal{F}_{t-1}$ the pair (s_t, y_t) is jointly Gaussian with means $(s_{t|t-1}, Z_t s_{t|t-1})$ and covariance blocks

T_t, Z_t, R_t, Q_t, H_t as a by-product (*prediction-error decomposition*) [► 2013 L12 @ 1:15:25]:

$$\ell(\theta) = -\frac{kT}{2} \log 2\pi - \frac{1}{2} \sum_{t=1}^T \log |F_t| - \frac{1}{2} \sum_{t=1}^T v_t^\top F_t^{-1} v_t. \quad (9.32)$$

It is maximised by Newton-type methods or by the EM algorithm of Dempster, Laird and Rubin (section 7.7.1). As $T \rightarrow \infty$ the MLE is consistent and asymptotically normal with covariance the inverse Fisher information (2013 notes, last slide). The initial state is modelled as $N(s_{1|0}, P_{1|0})$ with pre-specified moments: a large variance (diffuse) if one knows nothing, or the stationary distribution of the state if the model is stationary [► 2013 L12 @ 1:16:25]. The observations are jointly Gaussian and the recursions provide a full specification of the model: means and covariances characterise the whole law.

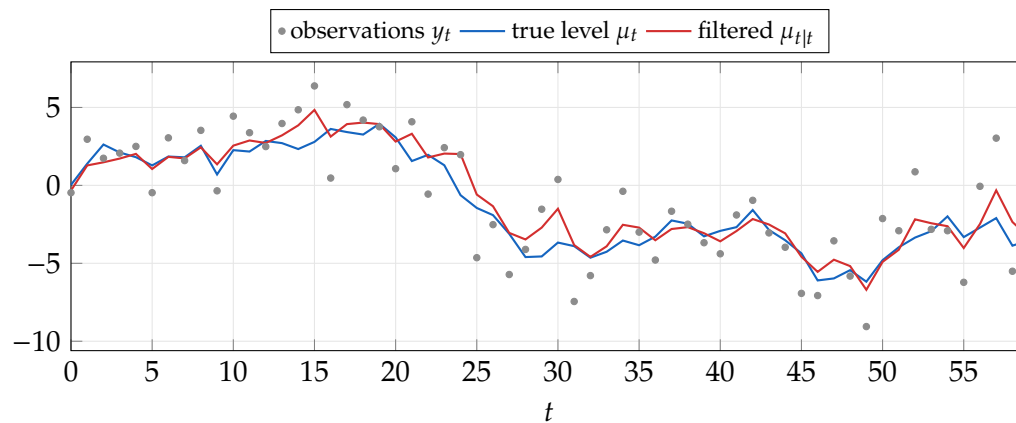


Figure 9.6: Local-level model with $Q = 1, H = 4$ (seed 11). The filter (red) tracks the hidden level (blue) much more closely than the raw observations (dots).

IN FINANCE: Where state-space models appear

Time-varying betas and alphas (the CAPM example above) and, in general, any regression whose coefficients are believed to drift; dynamic factor models, where the factors of chapter 7 follow their own time-series model; a “true” price observed through noisy quotes, as in the USD/RUB forward-point example of chapter 1, where five brokers quote at random and a Kalman filter estimates the underlying price [► 2013 L1 @ 57:41]; and exact maximum-likelihood estimation of ARMA models (section 9.4.3). The filter is cheap: each step costs a few matrix products, and it delivers the likelihood as well as the forecast.

Exercise 9.9 (Kalman filter by hand (not from the course)). Take the local-level model with $Q = 1, H = 4$ and initial state $N(0, 10)$.

- Run the filter for the observations $y_1 = 1.5, y_2 = 0.8, y_3 = 2.2$: compute $G_t, \mu_{t|t}, P_{t|t}$.
- Compute the two-step forecast of y_5 and its variance.
- Show that $P_{t|t-1}$ converges monotonically to $(Q + \sqrt{Q^2 + 4QH})/2$.
- Show that $G_t \rightarrow 0$ as $Q \rightarrow 0$ and interpret it (the level is constant and the filter reduces to the sample mean).

Var $s_t = P_{t|t-1}$, Cov(s_t, y_t) = $P_{t|t-1} Z_t^\top$ (because ε_t is independent of s_t), Var $y_t = F_t = Z_t P_{t|t-1} Z_t^\top + H_t$. Conditioning a Gaussian on part of its coordinates (section 3.7) gives the mean $s_{t|t-1} + P_{t|t-1} Z_t^\top F_t^{-1} (y_t - Z_t s_{t|t-1})$ and the covariance $P_{t|t-1} - P_{t|t-1} Z_t^\top F_t^{-1} Z_t P_{t|t-1} = P_{t|t-1} - G_t F_t^{-1} G_t^\top$, which are (9.31). The prediction step is the linear transformation of a Gaussian by the state equation. □

Example 9.16 (The local-level model: Kalman filter, EWMA and ARIMA(0,1,1)). The simplest model with a hidden state is $y_t = \mu_t + \varepsilon_t, \mu_{t+1} = \mu_t + \eta_t$: a slowly moving true level observed with noise. Here $T = Z = R = 1, Q = \text{Var } \eta, H = \text{Var } \varepsilon$. The recursions (9.30)–(9.31) become $P_{t+1|t} = P_{t|t} + Q, G_t = P_{t|t-1}/(P_{t|t-1} + H)$ and $m_t = (1 - G_t) m_{t-1} + G_t y_t$, where $m_t = \mu_{t|t}$. At the steady state $P = P_{t|t-1}$ solves $P = (1 - K)P + Q$ with $K = P/(P + H)$, that is $P^2 - QP - QH = 0$:

$$P = \frac{Q + \sqrt{Q^2 + 4QH}}{2}, \quad K = \frac{P}{P + H}, \quad P_{t|t} = (1 - K)P$$

For $Q = 1, H = 4$: $P = (1 + \sqrt{17})/2 = 2.562$, $K = 0.390$, $P_{t|t} = 1.562$, while the raw observation has error variance $H = 4$ (a Monte Carlo over 2000 paths gives a filter mean-squared error of 1.558). The filter is an *exponentially weighted moving average* (EWMA) of the observations with weight K and decay factor $1 - K = 0.610$. The same object appears in ARIMA language: $\Delta y_t = \eta_{t-1} + \varepsilon_t - \varepsilon_{t-1}$ has $\gamma(0) = Q + 2H, \gamma(1) = -H$, so $\rho(1) = -H/(Q + 2H) = -4/9$, an MA(1) whose invertible parameter is $\theta = -0.6096$ (from $\theta/(1 + \theta^2) = \rho(1)$), equal to $-(1 - K)$. Thus the local-level model is ARIMA(0,1,1), and the steady-state Kalman filter is the exponentially weighted forecast (sections 9.3.3 and 9.4). The same recursion applied to squared returns is the RiskMetrics volatility estimator, a GARCH(1,1) with $\alpha_0 = 0$ and $\alpha_1 + \beta_1 = 1$ (chapter 13) [► 2013 L11 @ 27:33]. Figure 9.6 shows a simulated path.

Exercise 9.10 (Time-varying beta (not from the course)). In the state-space model of the time-varying CAPM, write out the Kalman recursions for $s_t = (\alpha_t, \beta_t)^\top$.

- Show that if $\sigma_\varepsilon = \sigma_\zeta = 0$ the recursion reproduces recursive least squares for the simple regression.
- Explain how the ratio $\sigma_\zeta^2/\sigma_\varepsilon^2$ sets the effective window length of the estimate, and relate it to the local-level example.
- With daily data for one stock and the S&P 500, estimate σ_ζ^2 by maximising (9.32).

9.8 Notes on the sources

INTUITION: How the pieces fit together

The chapter is one idea seen from four sides. Stationarity plus the Wold theorem say that a stationary series is a linear filter of white noise. ARMA models parametrise that filter with few parameters; the AR part is a regression on the past (so least squares of [chapter 5](#) applies) and the roots of its polynomial decide stability. Non-stationary series are brought to that setting by differencing, and unit-root tests decide when. In several dimensions the polynomial becomes a matrix (VAR), stability is a condition on the eigenvalues of the companion matrix, and cointegration is the case where a rank-deficient matrix Π ties non-stationary series together. State-space models recast all of it as a hidden Markov-type Gaussian system, in which the Kalman filter is the exact recursive regression on the past. Forward links: volatility (GARCH is ARMA for squared returns, [chapter 13](#)), portfolio theory ([chapter 10](#)), continuous time (the AR(1) is the Ornstein–Uhlenbeck process, [chapter 18](#); the Dickey–Fuller limit uses Brownian motion, [chapter 11](#)).

1 COURSE NOTE: 2013 versus 2024, and errata

- **Overlap.** 2024 L12 and 2013 L8 use the same slides for univariate theory ([lec11–12](#) vs. [lecnote8](#)); the 2013 lecture goes further (trend reversion, ARIMA, estimation, model selection, with the information-criteria discussion at the end). Multivariate series, cointegration, state space and the Kalman filter are 2013-only.
- **Slide misprints.**
 - Yule–Walker slide: the second lag is written X_{t-1} instead of X_{t-2} (corrected aloud, [[▶ 2024 L12 @ 1:12:16](#)]), and the sum in the estimator of σ^2 starts at $j - 1$ instead of $j = 1$.
 - 2013 L11, model selection for VAR: the criteria are printed with $-\log |\tilde{\Sigma}(p)|$; it should be $+\log |\tilde{\Sigma}(p)|$ ([eq. \(9.26\)](#)).
 - 2013 L12, Kalman filter: the gain G_t is written with covariances at $(t - 1 | t)$; it should be $(t | t - 1)$.
 - The notes divide $\tilde{\Sigma}$ for the VAR by $n - pm$; the unbiased divisor is $n - mp - 1$ when an intercept is estimated.
- **Case Study 3.** The printed “smallest root modulus 1.4027” is the smallest *squared* modulus ([section 9.3.1](#)). The AR(2) cycle of “just over four months” is reproduced (4.107).
- **Conventions.** $\text{Var}(\hat{r}_k) \approx 1/(T - k)$ (2024) versus $1/T$ (Bartlett): equivalent for $T \gg k$. R’s `vars` reports moduli of companion eigenvalues, whereas `polyroot` returns roots of $\phi(z)$ ([remark 9.6](#)). “Log returns” of yields is a loose name for relative changes of the yield.
- **Not in the download.** The plots [TimeSeries4plots.pdf](#), [TimeSeries4acfplots.pdf](#) and the crack-spread cointegration note; the numbers of [section 9.6.4](#) are as spoken.

KEY FORMULAS

Stationarity:	$\mathbb{E}[X_t] = \mu, \quad \gamma(\tau) = \text{Cov}(X_t, X_{t+\tau}), \quad \rho = \gamma/\gamma(0)$
White noise test:	$\text{BP} = T \sum_{j \leq K} \hat{r}_j^2 \sim \chi_K^2, \quad \hat{r}_k > 1.96/\sqrt{T}$
Wold:	$X_t = V_t + \psi(L)\eta_t, \quad \psi_0 = 1, \quad \sum \psi_j^2 < \infty$
ARMA:	$\phi(L)(X_t - \mu) = \theta(L)\eta_t, \quad \psi(L) = \theta(L)/\phi(L)$
Stationary if:	roots of $\phi(z)$ outside $ z \leq 1$ ($\Leftrightarrow$ eigenvalues of companion matrix $ \lambda < 1$)
AR(1):	$\gamma(0) = \sigma^2/(1 - \phi^2), \quad \rho(j) = \phi^j$
Yule–Walker:	$\gamma(j) = \sum_{i \leq p} \phi_i \gamma(j - i) + \sigma^2 \delta_{j0}$
MA(q):	$\gamma(0) = \sigma^2 \sum_{i \leq q} \theta_i^2, \quad \gamma(j) = 0 \quad (j > q)$
ARMA(1,1):	$\gamma(0) = \sigma^2 \frac{1+2\phi\theta+\theta^2}{1-\phi^2}, \quad \rho(k) = \phi\rho(k-1) \quad (k \geq 2)$
Differencing:	$\Delta = 1 - L; \text{ARIMA}(p, d, q) : \Delta^d X \sim \text{ARMA}(p, q)$
Dickey–Fuller:	$T(\hat{\phi} - 1) \rightarrow \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W^2 dr} \text{ under } \phi = 1$
Information criteria:	$\log \tilde{\sigma}^2 + c_n \frac{p+q}{n}, \quad c_n = 2, \log n, 2 \log \log n$
VAR(p):	$\mathbf{X}_t = \mathbf{C} + \sum_j \Phi_j \mathbf{X}_{t-j} + \boldsymbol{\eta}_t, \quad \boldsymbol{\mu} = (\mathbf{I} - \sum \Phi_j)^{-1} \mathbf{C}$
VAR(1) moments:	$\Gamma_0 = A\Gamma_0 A^\top + \Sigma, \quad \Gamma_h = A^h \Gamma_0$
Estimation:	$\hat{\mathbf{B}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{Y}, \quad \tilde{\Sigma} = \hat{\mathbf{E}}^\top \hat{\mathbf{E}}/n$
VECM:	$\Delta \mathbf{X}_t = \mathbf{C} + \Pi \mathbf{X}_{t-1} + \sum_k \Gamma_k \Delta \mathbf{X}_{t-k} + \boldsymbol{\eta}_t$ $\Pi = \sum_j \Phi_j - \mathbf{I} = \alpha \beta^\top, \quad \text{rank } \Pi = r$
State space:	$\mathbf{s}_{t+1} = \mathbf{T} \mathbf{s}_t + \mathbf{R} \boldsymbol{\eta}_t, \quad \mathbf{y}_t = \mathbf{Z} \mathbf{s}_t + \boldsymbol{\varepsilon}_t$
Kalman:	$\mathbf{s}_{t t} = \mathbf{s}_{t t-1} + \mathbf{G}_t \mathbf{v}_t, \quad \mathbf{G}_t = \mathbf{P}_{t t-1} \mathbf{Z}^\top \mathbf{F}_t^{-1}$ $\mathbf{P}_{t t} = \mathbf{P}_{t t-1} - \mathbf{G}_t \mathbf{F}_t \mathbf{G}_t^\top, \quad \mathbf{P}_{t+1 t} = \mathbf{T} \mathbf{P}_{t t} \mathbf{T}^\top + \mathbf{R} \mathbf{Q} \mathbf{R}^\top$

Exercises

Standalone exercises follow the section they practise. 2013 PS4 Problems 1–4 form a connected series (Problem 1(f) uses Problem 4, and Problems 2–4 build on one another), so they stay together here.

From 2013 Problem Set 4 (and 2024 Problem Set 4, Problems 5–6)

The 2013 PS4 has six problems on univariate models; 2024 PS4 Problem 5 is 2013 Problem 1(a)–(e), and 2024 Problem 6 is 2013 Problem 6 plus a part (e).

Exercise 9.11 (PS4 (2013), Problem 1; PS4 (2024), Problem 5: covariance-stationary AR(2)). Let $X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \eta_t$, $\eta_t \sim \text{WN}(0, \sigma^2)$, be covariance stationary.

- Show that $\mu = \mathbb{E}[X_t] = \phi_0/(1 - \phi_1 - \phi_2)$.
- Show that $\gamma(k) = \phi_1 \gamma(k-1) + \phi_2 \gamma(k-2)$, $k \geq 1$.
- Show that $\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}$.
- The Yule–Walker equations for $k = 1, 2$ are $\rho_1 = \phi_1 + \phi_2 \rho_1$ and $\rho_2 = \phi_1 \rho_1 + \phi_2$

(using $\rho_0 = 1, \rho_{-k} = \rho_k$): solve for ρ_1, ρ_2 in terms of ϕ_1, ϕ_2 .

(e) Solve for ϕ_1, ϕ_2 in terms of ρ_1, ρ_2 .

(f) (2013 only) Derive complete formulas for $\rho_k, k > 2$, using Problem 4(b) below.

Exercise 9.12 (PS4 (2013), Problem 2: difference equations of an AR(p)). Let $\phi(L)X_t = \phi_0 + \eta_t$ be an AR(p) with $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$, and consider the homogeneous equation $\phi(L)g(t) = 0$.

(a) Show $\mu = \phi_0/(1 - \sum_i \phi_i)$, so $\mu = 0$ iff $\phi_0 = 0$ (“de-meaning” removes the constant).

(b) For $\gamma(t) = \text{Cov}(X_t, X_0)$ prove $\phi(L)\gamma(t) = 0$ for all $t \geq 1$.

(c) Same for $\rho(t) = \text{Corr}(X_t, X_0)$.

(d) Given $x_0, \dots, x_t$, let $\hat{x}_t(h) = \mathbb{E}[X_{t+h} \mid x_t, \dots, x_0]$ for $h > 0$ and $\hat{x}_t(h) = x_{t+h}$ for $h \leq 0$; show $\hat{x}_t(h) = \phi_0 + \sum_i \phi_i \hat{x}_t(h-i)$ for every $h > 0$.

(e) For the de-meaned process show that the forecast function satisfies $\phi(L)\hat{x}_t(h) = 0$ in h .

Exercise 9.13 (PS4 (2013), Problem 3: solutions of the difference equations). Consider $\phi(L)g(t) = 0, t = 0, 1, 2, \dots$

(a) Show that $g(t) = C e^{-\lambda t}$ solves it if $z = e^{-\lambda}$ is a root of $\phi(z) = 1 - \sum_i \phi_i z^i = 0$.

(b) Show that $g(t) = C G^t$ solves it if G^{-1} is a root.

(c) If the p roots $z_i = G_i^{-1}$ are distinct, show that $g(t) = \sum_i C_i G_i^t$ is a solution.

(d) If all roots are outside the unit circle, show every such solution is bounded; since the autocovariance is a solution, this is the necessary and sufficient condition for covariance stationarity.

Exercise 9.14 (PS4 (2013), Problem 4: the AR(2) process).

(a) Solve for the two roots z_1, z_2 of the characteristic equation and give the conditions under which they are distinct and real, coincident and real, or complex conjugates.

(b) Prove $\rho_k = C_1 G_1^k + C_2 G_2^k$ with $G_i = z_i^{-1}$; solve for C_1, C_2 when the roots are distinct and show that $\rho_k = \frac{G_1(1 - G_2^2)G_1^k - G_2(1 - G_1^2)G_2^k}{(G_1 - G_2)(1 + G_1 G_2)}$.

(c) For distinct real roots show that the autocovariance decreases geometrically in magnitude.

(d) Under what conditions on ϕ_1 is it always positive, and when does it alternate in sign?

(e) *Optional.* For complex roots write $G_{1,2} = d e^{\pm i 2\pi f_0}$ and show $\rho_k = [\text{sgn}(\phi_1)]^k |d|^k \sin(2\pi f_0 k + F) / \sin F$ with $|d| = \sqrt{-\phi_2}, 2\pi f_0 = \arccos(|\phi_1| / (2\sqrt{-\phi_2}))$ and $\tan F = \frac{1+d^2}{1-d^2} \tan(2\pi f_0)$.

Chapter 10

Portfolio Theory and Portfolio Management

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2013 L14	P. Kempthorne	Markowitz mean–variance analysis (two assets, then m assets with Lagrange multipliers); the risk-free asset, the market portfolio, Tobin’s separation theorem and the capital market line; von Neumann–Morgenstern utility; portfolio constraints (long-only, holding, turnover, benchmark, tracking error, factor, integer); estimation of means and covariances; alternative risk measures (MAD, semi-variance, VaR, CVaR, coherent measures, skewness and kurtosis). Live demonstration of Case Study 6 (simulation and sector ETFs).
2013 L16	J. Xia	Blackboard lecture: goals of portfolio management, the return–volatility map of asset classes, portfolio theory through special cases of two assets, Sharpe ratio, beta, the Kelly formula, the 60/40 portfolio, risk parity and leverage, the “free lunch” of rebalancing, crowd synchronisation (Millennium Bridge, metronomes) and the limits of the theory. Long question session.
2024 L13	J. Xia	The same programme in a slide version with new material from the speaker’s own research: the endowment model, the loss of meaning of volatility as risk, the gain–loss ratio $(G - L)/(G + L)$ for sizing, a feedback model of crowd behaviour and the origin of power laws. Questions on taxes, expected versus worst loss, stop losses.

Files. 2024 slides [mit18_642_f24_lec13.pdf](#) (38 pages). 2013 slides [MIT18_S096F13_lecnote14.pdf](#) (34 pages) and [lecnote16.pdf](#) (2 pages: only the outline, the lecture was given on the blackboard); 2013 Case Study 6 [CaseStudy6.pdf](#) (portfolio theory with US sector ETFs, 14 pages); the R output [Smltn_TwoAst.pdf](#) (simulation of two-asset portfolios, 6 pages) and the four sector-ETF output files [ETF_pridA_15.pdf](#), [ETF_pridA_30.pdf](#) (2009–2013, maximum allocation 15% and 30%), [ETF_pridB_15.pdf](#), [ETF_pridB_30.pdf](#) (2003–2006); the names “periodA” and “periodB” are those of the case study text. **Problem sets.** 2013 PS6 Problems 3–5 (two-asset and m -asset minimum-variance portfolios, equicorrelation); Problems 1–2 of the same set are time series ([chapter 9](#)). Standalone exercises follow the section they practise; the connected ones (Problems 3–5) are at the end of the chapter. **Not covered or not available.** The R code that produced the case-study output is not in the download (only its printed output), so the numbers of [section 10.14](#) are quoted, and checked by an independent computation from the printed input tables. The video shown by the speaker in both years and the 2013 blackboard are not in the notes; [lecnote16](#) contains only the outline of the lecture. The second half of 2013 L14 mentions “sector market-neutral models” with no data ([section 10.14](#)).

Motivation

Every investor faces the same question: given a list of things one could hold, how much of each? The 2013 and 2024 lectures answer it twice. Kempthorne (2013 L14) gives the mathematical theory that earned Harry Markowitz a Nobel prize: describe a portfolio by its expected return and its variance, and find the portfolios that offer the largest return for a given variance. The theory is a piece of applied linear algebra ([chapter 2](#)): a quadratic form to be minimised under linear constraints, solved with Lagrange multipliers, whose solution is governed by the inverse covariance matrix Σ^{-1} . Xia (2013 L16, 2024 L13), who has managed money for a bank and for the Harvard endowment, then asks what survives in practice: the inputs are estimated with large errors, volatility is a poor measure of risk, and markets are populated by agents who react to each other.

The chapter follows this arc.

- [section 10.1](#): the objectives, constraints and horizon that define a portfolio problem.
- [sections 10.2 to 10.4](#): Markowitz theory for two assets, for m assets, and with a riskless asset (efficient frontier, tangency portfolio, capital market line), with complete derivations; [section 10.5](#) links it to beta and the CAPM ([theorem 3.38](#)).
- [sections 10.6 and 10.7](#): diversification, rebalancing, leverage and risk parity.
- [sections 10.8 and 10.9](#): utility theory as the foundation of mean–variance analysis, and sizing by the Kelly criterion and by Xia’s gain–loss ratio.
- [sections 10.10 to 10.12](#): constraints, estimation error, and other risk measures.
- [section 10.13](#): the “beyond Markowitz” ideas of the 2024 lecture (crowds, power laws).
- [section 10.14](#): the 2013 case study and the notes on sources and errata.

Notation. The slides call the vector of expected returns α ; here it is μ , to keep α for the regression intercept of [definition 1.2](#). Vectors are columns, $\mathbf{1} = (1, \dots, 1)^\top$, Σ is the (positive definite) covariance matrix of the returns R of m risky assets and $\sigma_i^2 = \Sigma_{ii}$.

10.1 The portfolio construction problem

Both Xia lectures start with the same exercise: each student writes on a sheet the composition of a hypothetical portfolio (\$10,000 in 2024; in 2013 anything from a \$1,000 saving to a fund’s start-up capital), in percent, in five minutes and without being told the goal [[► 2024 L13 @ 3:14](#)][[► 2013 L16 @ 3:10](#)]. The answers become data for the rest of the hour. In 2024 they were mostly index funds (S&P 500, QQQ), Treasuries and cash, one option strategy, no cryptocurrency (in 2021 the class had picked Dogecoin) [[► 2024 L13 @ 9:26](#)]; in 2013 they ranged from bonds and S&P index funds to timberland, rare coins and lottery tickets [[► 2013 L16 @ 10:16](#)].

10.1.1 Objective, loss tolerance, horizon

Before choosing what to hold one must say what the portfolio is *for*. The considerations listed by Xia are:

- a *return objective* (10%, 50%, 100%?) and a *time horizon*, which he deliberately left out of the exercise [[► 2024 L13 @ 4:15](#)];

- a *loss tolerance*: how much of \$10,000 can be lost before one stops, and drawdown limits [► 2024 L13 @ 15:54];
- an *edge*: why should this investment beat the alternatives [► 2024 L13 @ 5:17];
- *constraints*: liquidity and spending needs, exclusions (fossil fuels, tobacco), inflation, absolute versus relative returns against a benchmark (the excess return being the *alpha* of [definition 1.2](#)) [► 2024 L13 @ 14:48];
- *asset–liability matching*: after 2008 many asset owners learned that when the portfolio draws down the liabilities dominate the balance sheet and spending must be cut [► 2024 L13 @ 18:01];
- *peer pressure and career risk*: managers are compared quarter to quarter although the owner’s horizon is long [► 2024 L13 @ 19:01].

The 2013 version adds the individual’s life cycle: income peaks around age 50, spending follows a different curve, and the portfolio must bridge the gap; the same question is asked of a pension fund, an endowment and a trading desk (how much capital to allocate to each trader, or to each of 30 strategies in a new hedge fund) [► 2013 L16 @ 14:32]. In 2024 the objective is written (slide 7) as $\max R/\sigma$ subject to constraints [► 2024 L13 @ 14:48].

10.1.2 Layers of decisions and the risk–return map

The slides separate three layers [► 2024 L13 @ 7:22]: choice of *markets and instruments*; *data, signals, models, factors, predictions and strategies*; and finally *sizing, capital allocation, portfolio construction, optimisation and risk management*. Only the last is the topic of this chapter. It comes down to comparing investments through their return and their risk, first taken to be the volatility [► 2024 L13 @ 8:25]. On the map with volatility on the horizontal axis (2024 slide 6, 2013 blackboard), cash sits on the vertical axis with zero volatility and a positive return (over 4% in October 2024, [► 2024 L13 @ 11:35]); bonds have a little more of both, stocks more, an index fund less volatility than a single stock, commodities more, private equity and venture capital (illiquid, with a wide range of outcomes) are far up to the right [► 2024 L13 @ 12:39][► 2013 L16 @ 24:06]. Two extremes anchor the picture: a lottery ticket has a return near -100% and a small volatility, and the fair coin flip has zero return and 100% volatility [► 2013 L16 @ 23:02]. A rational investor discards every point that has another point above and to the left of it, which is the idea of the efficient frontier [► 2013 L16 @ 26:13].

IN FINANCE: The endowment model

A university endowment is a perpetual portfolio. It spends about 5% of its value each year, and inflation is about 3%, so the nominal return target is about 8% (more precisely $1.05 \times 1.03 - 1 \approx 8.2\%$), over horizons “longer than ten years” [► 2024 L13 @ 21:07]. The slide adds that liquidity must cover 30–40% of the annual operating budget, and Xia says that roughly 40% of a university’s budget is now paid from endowment spending, as research grants shrink [► 2024 L13 @ 22:12]. An 8% target is not easy to obtain consistently when bonds yield little, so endowments hold a long list of strategies [► 2024 L13 @ 23:15]: cash, government and corporate bonds and credit, macro and relative-value hedge funds, quantitative funds (trend following/CTA, statistical arbitrage), long/short equity, multi-manager platforms, private equity, real estate and natural resources, and newer assets (crypto, intellectual property, legal claims). The “endowment model” is characterised by external rather than internal managers, public and private markets, generalists and specialists, absolute and relative returns, mostly active management with some passive beta, and peer comparison [► 2024 L13 @ 25:20]. Its central skill is

manager selection (“can you predict fund performance, luck or skill?”). Endowments are tax-exempt, so taxes are ignored; a family office would tilt to private equity to defer realised gains [► 2024 L13 @ 1:13:53].

10.1.3 The classic problem

Assume the objectives and the loss tolerance are given and that one can predict the return, volatility and correlation matrix of every investment. What fraction should go into each? The classic problem is to maximise the portfolio return and minimise its risk [► 2024 L13 @ 27:30]. Three observations of the lecture frame the rest of the chapter [► 2024 L13 @ 28:30]. First, the problem is one of *sizing* at every level: for asset classes it is asset allocation, and because ten to fifteen classes make optimisation hard, practitioners often reduce them to three to five *risk factors* (equity, bonds, inflation, currency, credit; see [chapter 7](#)). Second, risk management is the same problem seen from the other side, with the added concerns of concentration, illiquidity and unwanted risks. Third, everything depends on predictions (“capital market assumptions”) which are unreliable ([section 10.11](#)).

10.2 Two assets: diversification and the feasible set

Consider two risky assets with returns R_1, R_2 , expected returns μ_1, μ_2 , volatilities σ_1, σ_2 and correlation ρ , and the fully invested portfolios with weights $(1 - w, w)$, $0 \leq w \leq 1$ (no short sales) [► 2013 L14 @ 5:21][► 2024 L13 @ 30:39]:

$$\begin{aligned} R_w &= (1 - w)R_1 + wR_2, & \mu_w &= (1 - w)\mu_1 + w\mu_2, \\ \sigma_w^2 &= (1 - w)^2\sigma_1^2 + w^2\sigma_2^2 + 2w(1 - w)\rho\sigma_1\sigma_2. \end{aligned} \quad (10.1)$$

The *feasible set* is the curve $\{(\sigma_w, \mu_w) : 0 \leq w \leq 1\}$ in the volatility–return plane [► 2013 L14 @ 7:28]. Since μ_w is affine in w and σ_w^2 is quadratic, σ_w^2 is a quadratic function of μ_w : the curve is one branch of a hyperbola (a parabola in the (μ, σ^2) plane) [► 2013 L14 @ 27:32]. Its shape is governed by ρ , illustrated by the special cases of both Xia lectures.

- $\rho = 1$: $\sigma_w = (1 - w)\sigma_1 + w\sigma_2$ is linear in w , hence so is μ as a function of σ : a straight segment between the two assets and no diversification gain [► 2024 L13 @ 31:39] [► 2013 L16 @ 34:34].
- $\rho = -1$: $\sigma_w = |(1 - w)\sigma_1 - w\sigma_2|$, two segments meeting on the return axis at $w = \sigma_1/(\sigma_1 + \sigma_2)$, where the portfolio has *zero* variance [► 2024 L13 @ 32:39][► 2013 L14 @ 17:06]. A portfolio without risk must earn the riskless rate, otherwise there is arbitrage; hedges of a security by a future exploit exactly this [► 2013 L14 @ 18:06].
- $\rho = 0$: the curve bends to the left of the straight line, and the variance cannot be reduced to zero.
- $\sigma_1 = 0$ (asset 1 is cash with return r_0): $\sigma_w = w\sigma_2$ and $\mu_w = r_0 + (\mu_2 - r_0)\sigma_w/\sigma_2$, a straight line with slope $(\mu_2 - r_0)/\sigma_2$, the *capital allocation line* of [section 10.4](#) [► 2024 L13 @ 33:41] [► 2013 L16 @ 35:37].

Proof. Differentiating (10.2): $\frac{1}{2} \frac{d\sigma_w^2}{dw} = -(1-w)\sigma_1^2 + w\sigma_2^2 + (1-2w)\rho\sigma_1\sigma_2$, which vanishes at w^* ; the second derivative $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ is positive, so this is the global minimum. Substituting gives σ^{*2} (use $1-w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\dots)$). The numerator of w^* is ≥ 0 iff $\rho \leq \sigma_1/\sigma_2$, and that of $1-w^*$ is ≥ 0 iff $\rho \leq \sigma_2/\sigma_1$. $\square$

Proposition 10.1 (Minimum-variance portfolio of two assets). Let $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 > 0$ (that is, not $\rho = 1$ with $\sigma_1 = \sigma_2$). The variance (10.2) is minimised over $w \in \mathbb{R}$ at

$$w^* = \frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}, \quad \sigma^{*2} = \frac{\sigma_1^2\sigma_2^2(1 - \rho^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}.$$

The weight w^* lies in $[0, 1]$ (the minimum is attained by a long-only portfolio) iff $\rho \leq \min(\sigma_1/\sigma_2, \sigma_2/\sigma_1)$; otherwise the minimum over $[0, 1]$ is at the endpoint with smaller volatility. If $\sigma_1 = \sigma_2 = \sigma$ then $w^* = \frac{1}{2}$ and $\sigma^{*2} = \sigma^2(1 + \rho)/2$.

For $\rho = 0$ the formula gives $w^* = \frac{1/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}$: the assets are weighted *inversely to their variances*, as computed by Kempthorne on the board [► 2013 L14 @ 12:52](Exercise 10.10 extends it). Two consequences of proposition 10.1 explain the pictures of the lectures.

- Adding a little of the riskier asset 2 to asset 1 *reduces* the variance as long as $\rho < \sigma_1/\sigma_2$ (the derivative of σ_w^2 at $w = 0$ is $2\sigma_1(\rho\sigma_2 - \sigma_1)$, negative iff $\rho < \sigma_1/\sigma_2$). Even for $\rho = 0$ diversification lowers the risk and, when asset 2 has the higher return, also *raises* the return: every portfolio to the left of the minimum-variance point is dominated [► 2013 L14 @ 14:57]. For ρ larger than the ratio of volatilities the minimum is the safer asset alone.
- The equal-volatility, zero-correlation case gives $\sigma^* = \sigma/\sqrt{2}$: “a significant reduction of the risk by half-and-half” [► 2013 L16 @ 31:31]; for three such assets it is $\sigma/\sqrt{3}$ [► 2013 L16 @ 41:28], and for m assets $\sigma/\sqrt{m}$ (Exercise 10.11).

Example 10.2 (The two assets of the 2013 simulation). Take $\mu = (0.15, 0.25)$, $\sigma = (0.20, 0.30)$, the values printed in the plots of Case Study 6 (section 10.14 explains why these are not the values read out in class). Then $\sigma_1/\sigma_2 = 2/3$ and

ρ	-1	-0.4	0	0.4	0.8
w^* (asset 2)	0.400	0.360	0.308	0.195	-0.235
σ^* (long-only)	0	0.130	0.166	0.192	0.200
μ^* (long-only)	0.190	0.186	0.181	0.170	0.150

For $\rho = 0.8 > 2/3$ and $\rho = 1$ the unconstrained w^* is negative (a short position in asset 2), and the long-only minimum is asset 1 alone. This is why in the last panel of `Smltn_TwoAst` (correlation 0.8) the minimum-variance portfolio (cyan curve) coincides with asset 1. All values were computed from proposition 10.1.

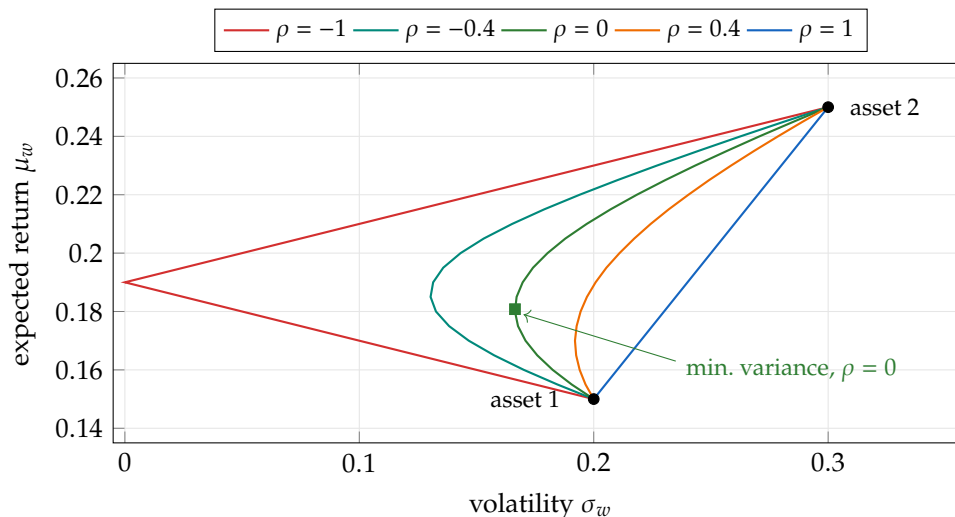


Figure 10.1: Feasible set of the portfolios $(1-w)R_1 + wR_2$, $0 \leq w \leq 1$, for $\mu = (0.15, 0.25)$, $\sigma = (0.20, 0.30)$ and several correlations (example 10.2; compare slides 15–16 of lec13 and the top-right panels of Sm1tn_TwoAst). For $\rho = -1$ the curve reaches the return axis. The lower part of each curve, below the minimum-variance point, is dominated.

INTUITION: Correlation as an angle

Write $\sigma_w^2 = |(1-w)r_1 + wr_2|^2$ with vectors r_1, r_2 of lengths σ_1, σ_2 and angle θ , $\cos \theta = \rho$: the volatility of a portfolio is the *length* of a weighted sum of two vectors. Perfect correlation means parallel vectors (lengths add), $\rho = -1$ antiparallel (lengths cancel), and $\rho = 0$ orthogonal (Pythagoras, $\sigma_w^2 = (1-w)^2\sigma_1^2 + w^2\sigma_2^2$). Diversification is the triangle inequality being strict. The minimum-variance portfolio is the foot of the perpendicular from the origin to the segment joining the tips of the two vectors.

10.3 Markowitz mean–variance optimisation

Now let there be m risky assets with return vector R , $\mathbb{E}[R] = \mu$, $\text{Cov } R = \Sigma$ (positive definite) in a *single period*. A portfolio is a vector w of fractions of the wealth invested in each asset, $w^T \mathbf{1} = 1$ (one unit of capital). Its return $R_w = w^T R$ has

$$\mu_w = w^T \mu, \quad \sigma_w^2 = w^T \Sigma w \quad (10.2)$$

(proposition 3.17) [► 2013 L14 @ 3:11]. Investors prefer higher μ_w and lower σ_w^2 , so portfolios are compared through the pair (μ_w, σ_w^2) [► 2013 L14 @ 23:14]. Markowitz’s idea (1952) is to pose this as a convex optimisation problem.

Definition 10.3 (Markowitz problem I). For a target return μ_0 , choose $w \in \mathbb{R}^m$ (short positions allowed) to

$$\text{minimise } \frac{1}{2} w^T \Sigma w \quad \text{subject to} \quad w^T \mu = \mu_0, \quad w^T \mathbf{1} = 1.$$

The factor $\frac{1}{2}$ only simplifies the derivatives [► 2013 L14 @ 24:14]. The objective is strictly convex ($\Sigma > 0$) and the constraints are linear, so a stationary point of the Lagrangian is the unique global minimum [► 2013 L14 @ 25:20].

Theorem 10.4 (Efficient portfolios without a riskless asset). *Put*

$$a = \mu^\top \Sigma^{-1} \mu, \quad b = \mu^\top \Sigma^{-1} \mathbf{1}, \quad c = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad \Delta = ac - b^2.$$

If μ is not a multiple of $\mathbf{1}$ then $\Delta > 0$ and problem I has the unique solution

$$w_0 = \lambda_1 \Sigma^{-1} \mu + \lambda_2 \Sigma^{-1} \mathbf{1}, \quad \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \mu_0 \\ 1 \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} c\mu_0 - b \\ a - b\mu_0 \end{pmatrix}, \quad (10.3)$$

whose variance is

$$\sigma_0^2 = w_0^\top \Sigma w_0 = \frac{c\mu_0^2 - 2b\mu_0 + a}{\Delta} = \frac{1}{c} + \frac{c}{\Delta} \left(\mu_0 - \frac{b}{c} \right)^2. \quad (10.4)$$

Proof. The Lagrangian is $L = \frac{1}{2} w^\top \Sigma w + \lambda_1 (\mu_0 - w^\top \mu) + \lambda_2 (1 - w^\top \mathbf{1})$ [► 2013 L14 @ 24:14]. Setting $\partial L / \partial w = \Sigma w - \lambda_1 \mu - \lambda_2 \mathbf{1} = \mathbf{0}$ gives $w = \lambda_1 \Sigma^{-1} \mu + \lambda_2 \Sigma^{-1} \mathbf{1}$. Substituting in the two constraints gives the linear system $\mu_0 = a\lambda_1 + b\lambda_2$, $1 = b\lambda_1 + c\lambda_2$, that is (10.3) [► 2013 L14 @ 26:26]. The matrix $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is the Gram matrix of $\mu, \mathbf{1}$ for the inner product $\langle x, y \rangle = x^\top \Sigma^{-1} y$; its determinant Δ is positive by the strict Cauchy–Schwarz inequality, because μ and $\mathbf{1}$ are linearly independent. Finally $\sigma_0^2 = w_0^\top \Sigma w_0 = \lambda_1 w_0^\top \mu + \lambda_2 w_0^\top \mathbf{1} = \lambda_1 \mu_0 + \lambda_2$, which equals (10.4); completing the square in μ_0 gives the second form. $\square$

Thus σ_0^2 is a parabola in the target return [► 2013 L14 @ 27:32]: the smallest variance obtainable with mean μ_0 , and the set of all pairs (σ_0, μ_0) is a hyperbola in the volatility–return plane.

Corollary 10.5 (Global minimum-variance portfolio and the frontier). *The parabola (10.4) is smallest at $\mu_0 = b/c$, where*

$$w_g = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad \sigma_g^2 = \frac{1}{c}, \quad \mu_g = \frac{b}{c}.$$

The efficient frontier is the upper branch $\mu_0 \geq b/c$ (portfolios below it are dominated by the portfolio with the same variance and larger return); the whole curve is a hyperbola with asymptotes $\mu = b/c \pm \sqrt{\Delta/c} \sigma$. Along the efficient branch μ is an increasing concave function of σ , and no portfolio lies to the left of the hyperbola.

10.3.1 The three equivalent problems

The slides pose the same optimisation in three ways (slide 7) [► 2013 L14 @ 28:36]:

- **Problem I:** minimise the variance for a given mean μ_0 ;
- **Problem II:** maximise $w^\top \mu$ for a given variance σ_0^2 , with $w^\top \mathbf{1} = 1$;
- **Problem III:** for a risk-aversion parameter $\lambda \geq 0$ maximise $w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$ with $w^\top \mathbf{1} = 1$.

All three have the same first-order conditions, and the *efficient frontier* is the trace of the solutions when μ_0 (I), σ_0^2 (II) or λ (III) varies. Problem III can be solved in one line, and it shows the structure of the frontier. The condition $\mu - \lambda \Sigma w - \gamma \mathbf{1} = \mathbf{0}$ (with multiplier γ for the budget) gives $w = \lambda^{-1} \Sigma^{-1} (\mu - \gamma \mathbf{1})$, and $\mathbf{1}^\top w = 1$ fixes $\gamma = (b - \lambda)/c$:

$$w(\lambda) = w_g + \frac{1}{\lambda} v, \quad v = \Sigma^{-1} \left(\mu - \frac{b}{c} \mathbf{1} \right), \quad \mathbf{1}^\top v = 0. \quad (10.5)$$

Every efficient portfolio is the minimum-variance portfolio plus a multiple $1/\lambda$ of one fixed zero-cost (long/short) portfolio v . Hence $\mu = b/c + (a - b^2/c)/\lambda$ and $\sigma^2 = 1/c + (a - b^2/c)/\lambda^2$, since $v^\top \Sigma v = a - b^2/c$ and $w_g^\top \Sigma v = 0$; eliminating λ recovers (10.4). Consequently any two distinct efficient portfolios span the whole frontier (the *two-fund theorem*, not stated in class). The risk aversion λ is the Arrow–Pratt index of section 10.8: a very risk-averse investor ($\lambda \rightarrow \infty$) holds w_g .

Exercise 10.1 (not from the course: the frontier).

- Verify that w_g of [corollary 10.5](#) minimises $w^\top \Sigma w$ under $w^\top \mathbf{1} = 1$ and that the frontier (10.4) has asymptotes $\mu = b/c \pm \sqrt{\Delta/c} \sigma$.
- Show that the difference of two efficient portfolios is proportional to v of (10.5) and deduce the two-fund theorem.
- Show that, if $r_0 < \mu_g$, the tangency portfolio maximises the Sharpe ratio over the fully invested portfolios, and find the tangency point of the line through $(0, r_0)$ with the hyperbola.

Example 10.6 (Nine sector ETFs, unconstrained (computed from the rounded case-study inputs)). Take the annualised means, volatilities and correlations of the nine SPDR sector ETFs for 2009–2013 printed in Case Study 6 ([table 10.2](#)). Then $a = 3.44$, $b = 11.44$, $c = 88.04$, $\Delta = 171.9$. The global minimum-variance portfolio has $\sigma_g = 1/\sqrt{c} = 10.7\%$ and $\mu_g = b/c = 13.0\%$, better in volatility than every single ETF (the lowest is consumer staples, 12%, but its mean is 15%), and the asymptotic slope is $\sqrt{\Delta/c} = 1.40$. But the weights, in the order XLB, XLV, XLP, XLY, XLE, XLF, XLI, XLK, XLU, are

$$w_g \approx (0.09, 0.14, 0.94, -0.17, -0.15, -0.02, -0.27, 0.26, 0)$$

with 94% in consumer staples and a 27% short position in industrials: an unconstrained optimum is not what an investor would hold. This is the reason for the constraints of [section 10.10](#).

10.4 A riskless asset: tangency portfolio and capital market line

Add a riskless asset 0 with constant return $R_0 \equiv r_0$ (zero variance, uncorrelated with everything) [[2013 L14 @ 31:55](#)]. The investor puts w in the risky assets and $1 - w^\top \mathbf{1}$ in the riskless one, which may be negative (borrowing at r_0). There is no budget constraint left on w , and

$$\mu_w = w^\top \mu + (1 - w^\top \mathbf{1})r_0 = r_0 + w^\top (\mu - r_0 \mathbf{1}), \quad \sigma_w^2 = w^\top \Sigma w. \quad (10.6)$$

Geometric picture. Two assets are enough: combining the riskless asset with a risky portfolio P gives the line through $(0, r_0)$ and (σ_P, μ_P) ; the higher the slope, the better. The best line is the tangent to the efficient frontier of the risky assets [[2024 L13 @ 35:50](#)] [[2013 L14 @ 33:01](#)], which lies above the frontier because we can now hold a fraction of a risky portfolio in cash [[2013 L14 @ 35:05](#)]. The mathematics confirms it.

Theorem 10.7 (Optimal portfolios with a riskless asset; Tobin separation). Let $\eta = \mu - r_0 \mathbf{1}$ (excess returns) and $d^2 = \eta^\top \Sigma^{-1} \eta = a - 2r_0 b + r_0^2 c$. The problem $\min \frac{1}{2} w^\top \Sigma w$ subject to $\mu_w = \mu_0$ has the solution

$$w_0 = \lambda_1 \Sigma^{-1} \eta, \quad \lambda_1 = \frac{\mu_0 - r_0}{d^2}, \quad \sigma_0 = \frac{|\mu_0 - r_0|}{d}. \quad (10.7)$$

Assume $\mathbf{1}^\top \Sigma^{-1} \eta = b - r_0 c > 0$ (i.e. $r_0 < \mu_g$) and define the market (tangency) portfolio

$$w_M = \frac{\Sigma^{-1} \eta}{\mathbf{1}^\top \Sigma^{-1} \eta}, \quad \mu_M = \frac{a - r_0 b}{b - r_0 c}, \quad \sigma_M = \frac{d}{b - r_0 c}. \quad (10.8)$$

Then every optimal portfolio with $\mu_0 > r_0$ invests in the risky assets in the proportions of w_M : it holds the fraction

$$m_w = w_0^\top \mathbf{1} = \frac{\mu_0 - r_0}{\mu_M - r_0} \quad (10.9)$$

of its wealth in w_M and $1 - m_w$ in the riskless asset [[2013 L14 @ 41:30](#)]. Its return and risk are $\mu_P = r_0 + m_w(\mu_M - r_0)$ and $\sigma_P = m_w \sigma_M$.

Corollary 10.8 (Capital market line and price of risk). The optimal portfolios lie on the half-line, the capital market line (CML)

$$\mu_P = r_0 + \frac{\mu_M - r_0}{\sigma_M} \sigma_P, \quad \sigma_P \geq 0, \quad (10.10)$$

which is tangent to the efficient frontier at M . Its slope $(\mu_M - r_0)/\sigma_M = d$ is the Sharpe

Proof. The Lagrangian $L = \frac{1}{2} w^\top \Sigma w + \lambda_1 [(\mu_0 - r_0) - w^\top \eta]$ gives $\Sigma w = \lambda_1 \eta$, i.e. $w = \lambda_1 \Sigma^{-1} \eta$, and the constraint $w^\top \eta = \mu_0 - r_0$ gives $\lambda_1 d^2 = \mu_0 - r_0$ [[2013 L14 @ 35:05](#)]. The variance is $\lambda_1^2 \eta^\top \Sigma^{-1} \eta = \lambda_1^2 d^2 = (\mu_0 - r_0)^2 / d^2$. Since $w_0 = \lambda_1 (b - r_0 c) w_M$, the risky weights are proportional to w_M with total weight $m_w = \lambda_1 (b - r_0 c)$; (10.8) follows from $\mu_M = w_M^\top \mu$, $\sigma_M^2 = \eta^\top \Sigma^{-1} \eta / (\mathbf{1}^\top \Sigma^{-1} \eta)^2$, and $\mu_M - r_0 = d^2 / (b - r_0 c)$ gives (10.9). $\square$

ratio of the market portfolio, the “market price of risk”: the compensation per unit of volatility. Investors of all risk appetites hold the same risky portfolio and differ only in how much of it (Tobin, 1958) [► 2013 L14 @ 43:42]. If borrowing at r_0 is possible, $m_w > 1$ (i.e. $\mu_0 > \mu_M$) gives points of the CML beyond M , a leveraged market portfolio [► 2013 L14 @ 46:52].

Definition 10.9 (Sharpe ratio). For a portfolio with return R_P , the Sharpe ratio is $SR_P = (\mu_P - r_0)/\sigma_P$ [► 2024 L13 @ 36:54][► 2013 L16 @ 44:21].

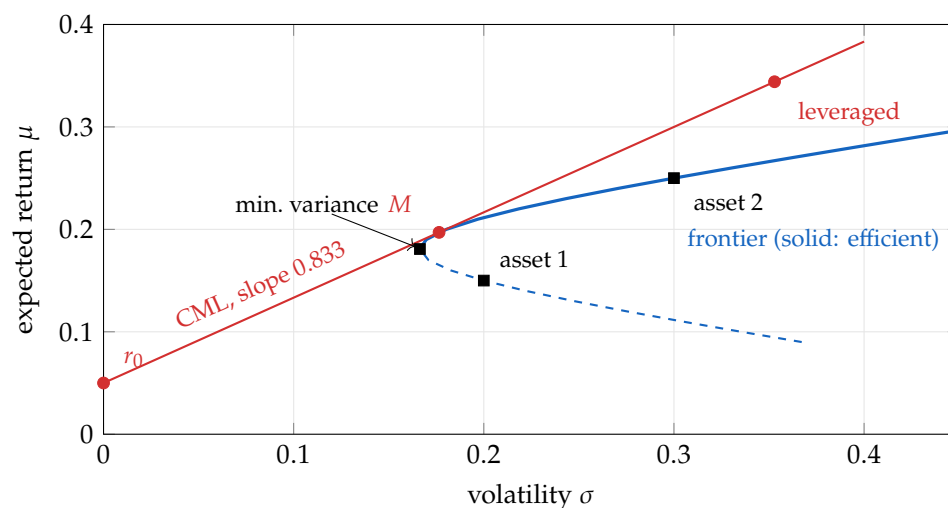
Proposition 10.10 (The tangency portfolio maximises the Sharpe ratio). For every risky portfolio $w \neq 0$, $(w^\top \eta)/\sqrt{w^\top \Sigma w} \leq d$, with equality iff w is a positive multiple of $\Sigma^{-1}\eta$, i.e. of w_M . The Sharpe ratio does not change when a portfolio is mixed with the riskless asset or levered at rate r_0 .

Proof. Cauchy–Schwarz for the inner product defined by Σ :

$$(w^\top \eta)^2 = (w^\top \Sigma \cdot \Sigma^{-1} \eta)^2 \leq (w^\top \Sigma w)(\eta^\top \Sigma^{-1} \eta).$$

Scaling $w \rightarrow tw$ multiplies $\mu - r_0$ and σ by t . ◻

The last statement is the answer to Xia’s question in 2013 (“when you lever up, does the Sharpe ratio change?”): with $w_1 = -1$ in cash and $w_2 = 2$ in the risky asset the volatility is $2\sigma_2$ and the excess return $2(R_2 - R_1)$, so the ratio $(R_2 - R_1)/\sigma_2$ is the same at every point of the line [► 2013 L16 @ 56:12]; the 2024 slide 18 draws the same picture, $\sigma = 2\sigma_p$, $R = 2R_p - R_f$ [► 2024 L13 @ 38:56]. Leverage moves the investor along the line, it does not improve it.



Example 10.11 (Two assets and a riskless rate). Take $\mu = (0.15, 0.25)$, $\Sigma = \text{diag}(0.04, 0.09)$ (uncorrelated assets of example 10.2) and $r_0 = 0.05$. Then $\eta = (0.10, 0.20)$, $\Sigma^{-1}\eta = (2.5, 2.22)$, $w_M = (0.529, 0.471)$, $\mu_M = 19.7\%$, $\sigma_M = 17.6\%$ and $d = \sqrt{0.10^2/0.04 + 0.20^2/0.09} = 0.833$; the individual Sharpe ratios are only 0.50 and 0.67. An investor who wants $\mu_0 = 12\%$ puts $(0.12 - 0.05)/(0.197 - 0.05) = 47.6\%$ of the wealth in M and 52.4% in cash; one who wants $\mu_0 = 20\%$ borrows 2% of the wealth. Twice levered ($m_w = 2$) gives $\mu = 34.4\%$ at $\sigma = 35.3\%$, still on the line (fig. 10.2).

Figure 10.2: Frontier of the two assets of example 10.2 for $\rho = 0$ with short sales allowed (upper branch solid, dominated branch dashed) and the capital market line of example 10.11 with $r_0 = 5\%$, tangent at M . Points on the CML beyond M are levered.

INTUITION: Why one fund is enough

The direction $\Sigma^{-1}\eta$ solves the normal equations $\Sigma w = \eta$, the same form as $X^\top X \hat{\beta} = X^\top y$ of chapter 5: it is the portfolio whose covariances with the assets are proportional to their excess returns, which is exactly the CAPM relation (10.11) of section 10.5. Once the best “direction” in asset space is fixed, scaling it up or down is only a question of appetite, like rescaling a normal mode; the riskless asset supplies the one extra degree of freedom (the length) that the fully invested constraint of section 10.3 did not leave.

10.5 Beta, the security market line and the CAPM

The mean–variance solution also contains the CAPM, by a route different from the expected-utility derivation of [section 3.11](#). This derivation is not given in either lecture, but it explains the slides on α, β (2024 slide 17, 2013 L16).

Proposition 10.12 (Security market line from the tangency portfolio). *Let M be the tangency portfolio (10.8). For every asset i (and every portfolio),*

$$\mu_i - r_0 = \beta_i(\mu_M - r_0), \quad \beta_i = \frac{\text{Cov}(R_i, R_M)}{\sigma_M^2}. \quad (10.11)$$

Under the assumptions of Sharpe and Lintner every investor holds M , so M is the market portfolio of all risky assets and (10.11) is the CAPM ([theorem 3.38](#)). The regression of $R_i - r_0$ on the market excess return has slope β_i and, by (10.11), zero intercept: in the language of [definition 1.2](#)

$$R_P - r_0 = \alpha + \beta(R_{BM} - r_0) + \varepsilon, \quad \beta = \frac{\text{Cov}(R_P, R_{BM})}{\sigma_{BM}^2} = \rho \frac{\sigma_P}{\sigma_{BM}}$$

(2024 slide 17 [[▶ 2024 L13 @ 37:54](#)]), where a nonzero α measures a departure from the security market line, and $\sigma_P^2 = \beta^2 \sigma_{BM}^2 + \sigma_\varepsilon^2$ splits the risk into a systematic and an idiosyncratic part (regression, [section 5.11.2](#)). In the two-asset special case of the lectures (riskless asset plus one risky asset 2, $\sigma_P = w_2 \sigma_2$) the correlation with asset 2 is 1 and $\beta = \sigma_P / \sigma_2 = w_2$ [[▶ 2013 L16 @ 1:02:34](#)]: the 60/40 investor asked how much equity beta the portfolio has needs the same computation with both assets ([section 10.7](#)).

IN FINANCE: Alpha, beta and the Sharpe ratio in practice

Beta is cheap to buy (an index fund); alpha is what a manager is paid for, and the Sharpe ratio SR measures return per unit of total risk while the *information ratio* $\alpha / \sigma_\varepsilon$ measures skill per unit of active risk (not discussed in class). A portfolio manager benchmarked to an index is judged on α and on the tracking error ([section 10.10](#)); a fund with a high Sharpe ratio can be levered to any desired volatility, so Sharpe ratios are the fair basis for comparing funds of different volatilities.

1 COURSE NOTE: Naming and the case $r_0 > \mu_g$

Kempthorne calls w_M “the market portfolio” because it is the *fully invested* optimal portfolio in the sense of $w_M^T \mathbf{1} = 1$ [[▶ 2013 L14 @ 39:15](#)]; the same portfolio is called the “tangency portfolio” elsewhere. That it coincides with the cap-weighted market of all assets is a statement of *equilibrium* (Sharpe 1964, Lintner 1965), proved in [theorem 3.38](#) and used in [section 10.5](#); here it is only the name of a solution. The slides implicitly assume $b - r_0 c > 0$; if instead $r_0 > \mu_g$ the tangency to the frontier touches the lower, dominated branch and w_M has a negative Sharpe ratio; the optimal portfolio for $\mu_0 > r_0$ is then $-w_M$ scaled (short the “market”), which is unrealistic and signals an unreasonable r_0 . The slides also assume that the borrowing and lending rates coincide; with a higher borrowing rate the CML has a kink at M and only the segment up to M is tangent to the frontier (not discussed in class). The key papers listed on slide 16 of [lecnote14](#) are Markowitz (1952), Tobin (1958), Sharpe (1964), Lintner (1965) and Fama (1970) [[▶ 2013 L14 @ 47:54](#)]; Kempthorne remarks that virtually all of them have a Nobel prize, and that Fama had been awarded it two weeks earlier [[▶ 2013 L14 @ 48:55](#)].

Proof. By definition $\Sigma w_M = \eta / (\mathbf{1}^T \Sigma^{-1} \eta)$, i.e. $\text{Cov}(R_i, R_M) = [\Sigma w_M]_i = \eta_i \kappa$ with $\kappa = 1 / (\mathbf{1}^T \Sigma^{-1} \eta)$. Multiplying by $w_{M,i}$ and summing gives $\sigma_M^2 = \kappa(\mu_M - r_0)$, hence $\eta_i = \text{Cov}(R_i, R_M) / \kappa = \text{Cov}(R_i, R_M)(\mu_M - r_0) / \sigma_M^2$. $\square$

10.6 Diversification and rebalancing: the “free lunch”

10.6.1 How much can diversification remove?

For the equally weighted portfolio of n assets, with $\bar{\sigma}^2$ the average variance and $\bar{c}$ the average pairwise covariance,

$$\text{Var}(R_{\text{ew}}) = \frac{1}{n^2} \left(\sum_i \sigma_i^2 + \sum_{i \neq j} \Sigma_{ij} \right) = \frac{\bar{\sigma}^2}{n} + \left(1 - \frac{1}{n}\right) \bar{c}. \quad (10.12)$$

The first term (idiosyncratic risk) vanishes like $1/n$, the second does not: what survives is the average covariance, the *systematic* risk that no diversification removes (example 7.6 for the spectral version). For uncorrelated assets of equal volatility, $\sigma/\sqrt{n}$; Exercises 10.11 and 10.12 ask for the minimum-variance portfolios in the uncorrelated, equicorrelated and general cases.

10.6.2 Rebalancing

Xia’s second illustration of diversification in both years is a two-year, two-asset example [► 2024 L13 @ 40:00][► 2013 L16 @ 52:00]. Asset 1 doubles in year 1 and halves in year 2; asset 2 halves and then doubles. Alone, each asset ends where it started (return 0). Start with 50% in each [► 2024 L13 @ 41:06].

- Year 1: $0.5 \cdot (+100\%) + 0.5 \cdot (-50\%) = +25\%$, so the value is 1.25, of which 1.00 (80%) is now asset 1 and 0.25 (20%) asset 2 [► 2024 L13 @ 42:13].
- *Without rebalancing*, year 2 gives $0.8 \cdot (-50\%) + 0.2 \cdot (+100\%) = -20\%$, and the two-year return is $1.25 \times 0.8 - 1 = 0$: no gain from a diversified portfolio of perfectly negatively correlated assets.
- *With rebalancing* to 50/50 at the end of year 1 (sell 0.375 of asset 1, buy asset 2: 0.625 each), year 2 gives $0.5 \cdot (-50\%) + 0.5 \cdot (+100\%) = +25\%$: two years at +25%, total $1.25^2 = 1.5625$ [► 2024 L13 @ 42:13][► 2013 L16 @ 54:05].

Rebalanced, the portfolio has *zero* volatility and a steady 25% return, “a straight line”.

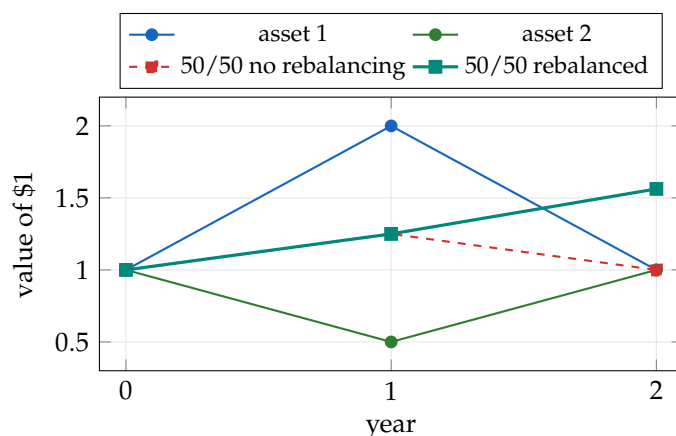


Figure 10.3: The two-year example of Xia (2024 slide 20, 2013 L16): each asset ends at 1, the buy-and-hold 50/50 portfolio also, the rebalanced one at 1.5625.

Two objections are discussed in class [► 2024 L13 @ 43:16]. Is this not a bet on mean reversion, losing money in a trend? Yes: rebalancing sells the winner and buys the loser,

which pays when returns alternate and loses when they trend (the case of [section 2.9](#), where daily rebalancing to equal weights was not obviously better). The rule of thumb is therefore *consistency of views*: the equal weights expressed the belief that the two assets have the same distribution and the same risk; unless the view has changed, letting the weights drift means holding a non-equal portfolio by accident. “The only free lunch is diversification, but you have to rebalance to eat it.”

Proposition 10.13 (Excess growth of a constant mix; not discussed in class). *Let asset i follow a geometric Brownian motion with $dS_i/S_i = \mu_i dt + \sigma_i dW_i$ and $\text{Cov}(dW_i, dW_j) = \rho_{ij} dt$. Asset i alone grows at the logarithmic rate $\mu_i - \frac{1}{2}\sigma_i^2$. A portfolio rebalanced continuously to the weights w has*

$$g(w) = w^\top \mu - \frac{1}{2} w^\top \Sigma w, \quad g(w) - \sum_i w_i (\mu_i - \frac{1}{2} \sigma_i^2) = \frac{1}{2} \left(\sum_i w_i \sigma_i^2 - w^\top \Sigma w \right) \geq 0. \quad (10.13)$$

The gain is the “volatility harvesting” or “diversification return”. Two uncorrelated assets with $\sigma = 20\%$ and the same μ , kept at 50/50, grow at $\mu - 1\%$ per year, while each alone grows at $\mu - 2\%$: the rebalanced portfolio gains $\sigma^2/4 = 1\%$ per year over the average of its components. The example above is the extreme case: $\rho = -1$, growth rates of 0% for each asset, 25% for the mix.

Exercise 10.2 (not from the course: rebalancing). Two independent assets have geometric Brownian motions with $\sigma_1 = 30\%$, $\sigma_2 = 10\%$ and the same μ . Using [proposition 10.13](#), find the constant-mix weight that maximises the log growth rate and the value of the “excess growth” at that weight. Compare with equal weights.

Proof. The wealth V of the constant mix satisfies $dV/V = w^\top \mu dt + w^\top \sigma dW$ (its return is $w^\top$ times the vector of returns, [chapter 17](#)), so by Itô’s formula $d \log V = (w^\top \mu - \frac{1}{2} w^\top \Sigma w) dt + \dots$. The inequality is $\text{Var}(\sum_i w_i R_i) \leq \sum_i w_i \text{Var} R_i$, i.e. convexity of the variance. $\square$

10.7 Leverage, the 60/40 portfolio and risk parity

Two-asset portfolios are important in practice, because managers are often benchmarked to a mix of equities and bonds, the most famous being the 60/40 portfolio: 60% in equities and 40% in bonds. Any fund is still compared with it [[2013 L16 @ 41:28](#)].

10.7.1 Weights are not risks

Equities are much more volatile than bonds, so a 60/40 portfolio is really an equity portfolio in risk terms. The idea of *risk parity* (the name was coined in 2005 by Edward Qian, as Xia recalls) is to allocate *risk* rather than capital: equal risk from each component, instead of weights by market value [[2013 L16 @ 49:54](#)][[2024 L13 @ 38:56](#)]. Before 2000, Xia recalls, a passive 60/40 was the norm; the 2000 equity peak and bond rally and the 2008 crash, when government bonds rallied while equities fell, made investors question it [[2013 L16 @ 48:48](#)]. To define the risk contribution of an asset one needs a decomposition of $\sigma_P = \sqrt{w^\top \Sigma w}$; it comes from Euler’s theorem, since σ_P is homogeneous of degree one in w .

Definition 10.14 (Risk contributions). The *risk contribution* of asset i to a portfolio w is

$$\text{RC}_i = w_i \frac{\partial \sigma_P}{\partial w_i} = \frac{w_i (\Sigma w)_i}{\sigma_P} = \frac{w_i \text{Cov}(R_i, R_P)}{\sigma_P}, \quad \sum_i \text{RC}_i = \sigma_P.$$

A portfolio has *equal risk contributions* (risk parity, ERC) if $\text{RC}_i = \text{RC}_j$ for all i, j , that is $w_i (\Sigma w)_i = \text{const}$.

The identity $\sum_i RC_i = \sigma_P$ is Euler’s theorem, or simply $\sum_i w_i(\Sigma w)_i = w^T \Sigma w = \sigma_P^2$.

Proof. (i) If $w_1\sigma_1 = w_2\sigma_2 =: \kappa$ then $w_1(\Sigma w)_1 = w_1(w_1\sigma_1^2 + w_2\rho\sigma_1\sigma_2) = \kappa^2(1 + \rho)$ and likewise for asset 2.
(ii) With $\Sigma = DCD$, $D = \text{diag}(\sigma_i)$, $C = (1 - \rho)I + \rho\mathbf{1}\mathbf{1}^T$, and $w_i = \kappa/\sigma_i$, i.e. $Dw = \kappa\mathbf{1}$: $w_i(\Sigma w)_i = w_i\sigma_i(CDw)_i = \kappa^2(C\mathbf{1})_i = \kappa^2(1 + (m - 1)\rho)$, independent of i .
(iii) $\mu_i - r_0 = s\sigma_i$ gives $\eta = sD\mathbf{1}$ and $\Sigma^{-1}\eta = sD^{-1}C^{-1}\mathbf{1} = sD^{-1}\mathbf{1}/(1 + (m - 1)\rho)$, proportional to $1/\sigma_i$. $\square$

Proposition 10.15 (Risk parity for two assets and for equal correlations).

- (i) For two assets with volatilities σ_1, σ_2 and any correlation $\rho > -1$ the risk-parity weights are $w_1 : w_2 = 1/\sigma_1 : 1/\sigma_2$ (inverse volatility).
- (ii) The same holds for m assets with equal pairwise correlations, whatever the volatilities.
- (iii) If, in addition, all assets have the same Sharpe ratio, this risk-parity portfolio is the tangency portfolio w_M of [theorem 10.7](#).

That is the “levered risk parity” of 2024 slide 19: a levered bond allocation to achieve an equal risk contribution from equities and bonds, so that the leverage is what buys back the return. Because Sharpe ratios are invariant to leverage ([proposition 10.10](#)), the gain comes from choosing the point of maximal Sharpe ratio.

IN FINANCE: Limits of risk parity

Risk parity uses *no return forecasts*, only volatilities and correlations; this is attractive because, a remark from the audience (apparently by Kempthorne, whom the transcript does not name) put it, returns cannot be estimated accurately even with more data, whereas volatilities can ([section 10.11](#)) [▶ 2013 L16 @ 1:25:00](#). But it does depend on a forecast of volatility from history: asked why risk-parity funds lost heavily in May–June 2013, Xia explained that they overweight low-volatility bonds, which sold off when the Fed chairman announced the tapering of quantitative easing [▶ 2013 L16 @ 1:14:26](#). He then asked whether this episode proves the approach wrong, or the 2008 crisis proves it right, and concluded that the evidence is inconclusive: one is “driving while looking in the rear-view mirror” [▶ 2013 L16 @ 1:15:30](#). Leverage carries liquidity risk, not treated in the lecture [▶ 2013 L16 @ 59:21](#).

INTUITION: Leverage and beta

Levering a portfolio by a factor ℓ multiplies its beta to every factor, and its volatility, by ℓ , and the excess return by ℓ : the Sharpe ratio is unchanged, like the direction of a vector when it is rescaled. For $\sigma_P = w_2\sigma_2$ in the two-asset special case, $\beta = \sigma_P/\sigma_2$ [▶ 2013 L16 @ 1:03:38](#): “a higher volatility of the portfolio means a higher beta to the risky asset”.

Exercise 10.3 (not from the course: risk parity). Two assets have volatilities 16% and 6% and correlation 0.3.

- (a) Find the risk-parity weights and check that the risk contributions of [definition 10.14](#) are equal.
- (b) What fraction of the risk of the 60/40 portfolio comes from the first asset?

Example 10.16 (60/40 versus risk parity, hypothetical numbers). Let equities have volatility 15% and bonds 5%, uncorrelated, and let both have a Sharpe ratio of 0.3 (excess returns 4.5% and 1.5%). These numbers are *not* from the course; they are chosen so that the volatility ratio is 3, which reproduces the 25/75 equity/bond risk-parity split quoted in class [▶ 2013 L16 @ 59:21](#). Then:

portfolio (eq./bond)	σ_P	excess ret.	Sharpe
60/40	9.2%	3.3%	0.358
risk parity 25/75	5.3%	2.25%	0.424
risk parity levered 1.74×	9.2%	3.9%	0.424

The 60/40 portfolio takes 95% of its risk from equities. Risk parity has a higher Sharpe ratio (here it is the tangency portfolio by [proposition 10.15\(iii\)](#), $0.3\sqrt{2} = 0.424$) but a lower return; to get it back, one *levers* it: borrowing 74% of the capital at the riskless rate brings the volatility up to that of the 60/40 portfolio and the excess return to 3.9% against 3.3% [▶ 2013 L16 @ 56:12](#) [▶ 2024 L13 @ 40:00](#). Values computed with the formulas above.

- (c) By what factor must the risk-parity portfolio be levered to have the volatility of the 60/40 portfolio?

10.8 Utility theory: why mean and variance?

Markowitz reduces a portfolio to its mean and variance. When is that legitimate? Kempthorne's answer is the *von Neumann–Morgenstern theory* of decisions under uncertainty: a rational agent has a utility function u of wealth and chooses the portfolio P maximising the *expected utility* $\mathbb{E}[u(W)]$ of the terminal wealth $W = W_0(1 + R_P)$ [► 2013 L14 @ 48:55]. Sensible preferences give $u' > 0$ (more is better) and $u'' < 0$ (decreasing marginal utility, risk aversion) [► 2013 L14 @ 52:01]. The Arrow–Pratt coefficients are

$$\lambda_A(W) = -\frac{u''(W)}{u'(W)} \quad (\text{absolute risk aversion}),$$

$$\lambda_R(W) = -\frac{W u''(W)}{u'(W)} \quad (\text{relative}).$$

(Slide 19 of 1ecnote14; the same λ_A appears in [section 3.11](#), where the CARA case is treated.)

	$u(W)$	$\lambda_A(W)$	$\lambda_R(W)$	remark
linear	$a + bW, b > 0$	0	0	risk neutral
quadratic	$W - \frac{1}{2}\lambda W^2, W < 1/\lambda$	$\frac{\lambda}{1 - \lambda W}$	$\frac{\lambda W}{1 - \lambda W}$	mean–variance is exact
exponential	$1 - e^{-\lambda W}$	λ	λW	CARA
power	$W^{1-\lambda}, 0 < \lambda < 1$	λ/W	λ	CRRA
logarithmic	$\ln W$	$1/W$	1	CRRA, Kelly

Table 10.1: The utility functions of slide 20 of 1ecnote14 [► 2013 L14 @ 54:08], with their risk-aversion coefficients (computed here). Quadratic utility has *increasing* absolute risk aversion and is not increasing beyond $W = 1/\lambda$.

10.8.1 From expected utility to mean–variance

Proposition 10.17 (Three routes to mean–variance analysis).

- For any smooth u , expanding around $w_* = \mathbb{E}[W]$ gives to second order $\mathbb{E}[u(W)] \approx u(w_*) + \frac{1}{2}u''(w_*)\text{Var } W$, i.e. up to a positive factor and a constant, $\mathbb{E}[W] - \frac{1}{2}\lambda_A(w_*)\text{Var } W$ [► 2013 L14 @ 53:06].
- For quadratic utility the expected utility is exactly $\mu_W - \frac{\lambda}{2}(\mu_W^2 + \sigma_W^2)$, a function of the mean and variance of W alone [► 2013 L14 @ 55:09].
- If $W \sim N(\mu_W, \sigma_W^2)$, for any increasing concave u , $\mathbb{E}[u(W)]$ is increasing in μ_W and decreasing in σ_W^2 , so that only the pair (μ_W, σ_W) matters and the efficient frontier is the set of candidates for the optimum; for exponential utility, $\mathbb{E}[u(W)] = 1 - \exp\{-\lambda(\mu_W - \frac{\lambda}{2}\sigma_W^2)\}$.

Proof. (i) Taylor's formula at w_* , with $\mathbb{E}[W - w_*] = 0$: the first-order term drops, and $u'' = -\lambda_A u'$.
(ii) Compute $\mathbb{E}[W - \frac{\lambda}{2}W^2]$.
(iii) Write $\mathbb{E}[u(W)] = \int u(x)\varphi(x; \mu_W, \sigma_W^2) dx$; the Gaussian density satisfies the heat equation $\partial_{\sigma^2}\varphi = \frac{1}{2}\partial_x^2\varphi$ and $\partial_\mu\varphi = -\partial_x\varphi$, so integrating by parts, $\partial_{\sigma^2}\mathbb{E}[u] = \frac{1}{2}\mathbb{E}[u''(W)] < 0$ and $\partial_\mu\mathbb{E}[u] = \mathbb{E}[u'(W)] > 0$. For the exponential, use the moment generating function of the normal (eq. (3.1)). $\square$

INTUITION: Variance as the heat-equation time

For a Gaussian, the variance plays the role of the time in the diffusion equation: $\mathbb{E}[u(W)]$ is u smoothed by a Gaussian of width σ_W , and smoothing a concave function can only lower its value.

So mean–variance analysis is the right thing to do if the utility is quadratic, *or* if returns are Gaussian (Kempthorne, 2013: “the imagination that returns are Gaussian”), at least to first orders otherwise [► 2013 L14 @ 55:09]. With exponential utility and normal returns, wealth $W = W_0(1 + \mathbf{w}^\top \mathbf{R})$ has certainty equivalent $W_0(1 + \mathbf{w}^\top \boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}^\top \Sigma \mathbf{w})$ with $\lambda = \lambda_A W_0$, exactly *problem III* of [section 10.3](#); with a riskless asset the solution is $\mathbf{w} = \lambda^{-1} \Sigma^{-1} \boldsymbol{\eta}$, i.e. the fraction $m_w = (\mu_M - r_0)/(\lambda \sigma_M^2)$ of the wealth in the market portfolio (from [theorem 10.7](#)): the risk aversion fixes only *how much*, in accord with Tobin’s separation.

Exercise 10.4 (not from the course: CARA utility). An investor with $u(W) = 1 - e^{-AW}$ and initial wealth W_0 can invest in m assets with normal returns and in a riskless asset. Show that the optimal amounts invested in the risky assets are $A^{-1} \Sigma^{-1} \boldsymbol{\eta}$ (in dollars) and identify the fraction of the wealth held in the tangency portfolio.

10.8.2 Higher moments

When returns are not Gaussian one goes on in the Taylor series (slides 32–33 of [lecnote14](#)) [► 2013 L14 @ 56:11]:

$$\mathbb{E}[u(W)] = u(w_*) + \frac{1}{2} u''(w_*) \text{Var } W + \frac{1}{3!} u'''(w_*) m_3 + \frac{1}{4!} u^{(4)}(w_*) m_4 + \dots, \quad m_k = \mathbb{E}[(W - w_*)^k],$$

with m_3 and m_4 the third and fourth central moments (the slides call them “Skew” and “Kurtosis”). For the utilities of [table 10.1](#) (exponential, power, log) $u''' > 0$ and $u^{(4)} < 0$: investors like positive skewness and dislike large fourth moments. The corresponding optimisation $\max \mathbb{E}[R_P] - \lambda_1 \text{Var } R_P + \lambda_2 m_3 - \lambda_3 m_4$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ has no closed form; the slides mention multi-objective methods and *polynomial goal programming*, and warn that skewness and kurtosis are hard to estimate (they are sensitive to outliers and need large samples).

10.9 Sizing: the Kelly criterion and the gain–loss ratio

10.9.1 Kelly

The *Kelly formula* is mentioned by Xia and, at the end of the lecture, by Kempthorne [► 2013 L16 @ 45:21]. Consider repeated independent bets in which a stake gains g per unit staked with probability p and loses l per unit with probability $q = 1 - p$. Betting a fraction f of the wealth each time multiplies the wealth by $1 + fg$ or $1 - fl$, and by the law of large numbers ([section 3.10](#)) $\frac{1}{n} \log W_n \rightarrow G(f) = p \log(1 + fg) + q \log(1 - fl)$ almost surely. The fraction that maximises the long-run growth rate is

$$f^* = \frac{p}{l} - \frac{q}{g} = \frac{pg - ql}{gl}, \quad (10.14)$$

since $G'(f) = pg/(1 + fg) - ql/(1 - fl) = 0$ gives $pg(1 - fl) = ql(1 + fg)$, i.e. $f gl(p + q) = pg - ql$. For an even-money bet ($g = l = 1$), $f^* = p - q = 2p - 1$: with probability $\frac{1}{2}$ one bets nothing, with $p = 1$ everything, and with $p = 0$ one bets everything *on the other side*,

as Xia says [► 2013 L16 @ 45:21]. It is expected-log utility (table 10.1), the “Kelly criterion” of Kelly, Shannon and Thorp, criticised by Samuelson in the debate recalled in the case of section 3.11 and in *Fortune’s Formula*; the co-lecturer, speaking from the audience, stresses that, unlike the single-period theory of this chapter, it deals with *multi-period* investment [► 2013 L16 @ 1:26:07]. For one asset with Gaussian excess return μ and variance σ^2 per period the growth rate of a fraction f is $r_0 + f\mu - \frac{1}{2}f^2\sigma^2$ (proposition 10.13), maximised at $f^* = \mu/\sigma^2$, the Sharpe ratio divided by the volatility (not discussed in class); Kelly is problem III with $\lambda = 1$.

Exercise 10.5 (not from the course: Kelly). A bet wins $g = 2$ per unit staked with probability $p = 0.6$ and loses $l = 1$ otherwise.

- Compute the Kelly fraction and the optimal growth rate $G(f^*)$.
- What growth rate does the half-Kelly fraction $f^*/2$ give, as a percentage of $G(f^*)$?
- Show that for small edges half-Kelly gives 3/4 of the optimal growth rate.

10.9.2 Xia’s gain–loss ratio

Xia argues that volatility is a poor risk measure because it is symmetric; a long out-of-the-money call *wants* high volatility, a short option position wants low volatility [► 2024 L13 @ 47:28], and the Sharpe ratio inherits the flaw; the Sortino ratio separates the downside but does not tell how much to invest [► 2024 L13 @ 48:30]. He proposes to summarise the distribution of the profit and loss R of an investment (per period, or of the terminal outcome) by the *expected gain* and *expected loss*, both positive numbers,

$$G = \mathbb{E}[R^+], \quad L = \mathbb{E}[R^-], \quad \text{so} \quad G - L = \mathbb{E}[R], \quad G + L = \mathbb{E}[|R|],$$

and to rank investments by the ratio

$$\frac{G - L}{G + L} = 1 - \frac{2L}{G + L} = \frac{\mathbb{E}[R]}{\mathbb{E}[|R|]} \in [-1, 1], \quad (10.15)$$

which he interprets directly as a position size (2024 slide 24) [► 2024 L13 @ 48:30]. In the (L, G) plane every investment is a point; the 45° line is $G = L$ (zero expected return) and the ratio is the slope of the ray from the origin: one wants points far above the diagonal (slide 24: “ $G \gg L$ ”) and maximises G for given L , which replaces the volatility budget by a *loss* budget [► 2024 L13 @ 50:37]. For a discrete payoff, G and L are sums of probability times payoff over winning and losing outcomes; in the continuous case they are integrals of the payoff against the density over $R > 0$ and $R < 0$, or against a target return other than 0 [► 2024 L13 @ 50:37]. Estimation needs only “half distributions”, no more work than a mean and a standard deviation [► 2024 L13 @ 52:41]; the gain and loss of a combination of two investments depends on how they move together and is found by Monte Carlo simulation of the joint distribution [► 2024 L13 @ 54:54]. The 2024 *Investment Game* of Exercise 1.1 is the empirical version: for the chosen stock, sum the daily gains and losses over two months and compute $(G - L)/(G + L)$; picking a very volatile stock raises $G - L$ but also the denominator, so the ratio ranks investments of different volatilities on a common scale [► 2024 L13 @ 51:37].

Proposition 10.18 (Properties of the ratio (10.15); not proved in class).

- For an even-money bet ($R = \pm 1$ with probabilities p, q) the ratio is $p - q$, the Kelly fraction (10.14); for $g \neq l$ the two differ (for $g = 2, l = 1, p = \frac{1}{2}$: Kelly $f^* = 0.25$, ratio

Proof. (i) $G = p, L = q, G + L = 1$. For the Kelly value use (10.14).

(ii) For $R = m + sZ$, $\mathbb{E}[|R|] = s[2\varphi(S) + S(2\Phi(S) - 1)]$ (fold the normal), and $h'(S) = 2\Phi(S) - 1$, so $(S/h)' = (h - Sh')/h^2 = 2\varphi(S)/h^2 > 0$.

(iii) $G = 1, L = 0$; for $N(1, 1)$, $h(1) = 1.1666$ and

$= 1/3$). So “the same as Kelly for binary bets” (slide 24) holds for even-money bets, or as an approximation.

(ii) If $R \sim N(m, s^2)$ the ratio equals $S/h(S)$ with $S = m/s$ and $h(S) = 2\varphi(S) + S(2\Phi(S) - 1)$, an increasing function of the Sharpe ratio: for symmetric returns it gives the same ranking as the Sharpe ratio.

(iii) For skewed payoffs the two rankings differ: $R \equiv 2$ w.p. $\frac{1}{2}$ and 0 w.p. $\frac{1}{2}$ has mean 1 and standard deviation 1, so does $N(1, 1)$, and the ratios are 1 (never a loss) and 0.857.

$$1/h(1) = 0.857.$$

□

Exercise 10.6 (2024 Investment Game and the gain–loss ratio (ungraded)). The 2024 lecture (slide 36) repeats the game of [Exercise 1.1](#): \$10,000 in one stock or ETF from the close of 13 September to the close of 15 November, one optional switch on 11 October, and the final calculation of G , L and $(G - L)/(G + L)$ from the daily profits and losses. Carry it out on a two-month window of your choice (for example for the sector ETFs of [table 10.2](#)), compute the Sharpe ratio of the same daily P&L, and compare the ranking of two investments by the two measures; see [proposition 10.18](#).

Exercise 10.7 (not from the course: gain–loss ratio). Investment A pays +3 with probability 0.4 and −1 with probability 0.6. Investment B pays +1 or −1 with probability 0.5 each plus a fixed 0.3. Compute for each the mean, the standard deviation, the Sharpe ratio, and $(G - L)/(G + L)$; comment on the difference in rankings and on the Kelly fractions.

10.10 Portfolio constraints

Real portfolios are subject to constraints, which Kempthorne lists (slides 22–25 of [lecnote14](#)) [[▶ 2013 L14 @ 57:16](#)]. With w the target portfolio, w_0 the current one, $x = w - w_0$ the trade and w_B a benchmark portfolio:

- **Long-only:** $w_j \geq 0$. **Holding constraints:** $L_j \leq w_j \leq U_j$, for instance a cap on each position; in a medium-frequency equity strategy, a position must not exceed a day’s trading volume, or it could not be liquidated [[▶ 2013 L14 @ 58:20](#)].
- **Turnover:** $|x_j| \leq U_j$ for each asset and $\sum_j |x_j| \leq U_*$ overall, because one cannot trade arbitrarily much in one period.
- **Benchmark exposure:** $\sum_i |w_i - w_{B,i}| \leq U_B$, to remain close to an index (S&P 500, NASDAQ-100, Russell) while trying to beat it [[▶ 2013 L14 @ 59:30](#)].
- **Tracking error:** $TE_P = R_P - R_B = (w - w_B)^T R$, so $\text{Var}(TE_P) = (w - w_B)^T \Sigma (w - w_B) \leq \bar{\sigma}_{TE}^2$ [[▶ 2013 L14 @ 1:00:31](#)].
- **Factor constraints:** for a factor model $R_{i,t} = \alpha_i + \sum_k \beta_{i,k} f_{k,t} + \epsilon_{i,t}$ ([section 7.5](#)), the exposure to factor k is $\sum_i \beta_{i,k} w_i$ and one imposes $|\sum_i \beta_{i,k} w_i| \leq U_k$; many strategies neutralise all exposures, $\sum_i \beta_{i,k} w_i = 0$ [[▶ 2013 L14 @ 1:01:33](#)].
- **Minimum transaction or holding size and integer constraints:** equities trade in round lots and one share of a stock worth several hundred dollars is a significant amount compared with one of a \$50 stock, which matters for small portfolios but not for large ones [[▶ 2013 L14 @ 1:02:33](#)].

Constraints of the first four types are linear or quadratic in w : slide 25 writes general linear constraints $A_w w \leq u_w$, $A_x x \leq u_x$, $A_B(w - w_B) \leq u_B$ and quadratic ones $w^T Q_w w \leq q_w$,

$\mathbf{x}^\top \mathbf{Q}_x \mathbf{x} \leq q_x, (\mathbf{w} - \mathbf{w}_B)^\top \mathbf{Q}_B (\mathbf{w} - \mathbf{w}_B) \leq q_B$. The optimisation is then the same as before with additional Lagrange multipliers, that is a *convex quadratic programme* [► 2013 L14 @ 1:03:33]. Integer constraints make it non-convex.

Proposition 10.19 (Structure of the constrained solution; not discussed in class).

Consider the problem I with additional linear inequality constraints.

- (i) The optimal value $\sigma_0^2(\mu_0)$ is never smaller than the unconstrained one (10.4), and it increases as constraints are added or tightened.
- (ii) The Karush–Kuhn–Tucker conditions with a fixed set of active constraints are linear in $(\mathbf{w}, \text{multipliers})$ and in μ_0 ; therefore the optimal weights are piecewise affine in the target return μ_0 , and the pieces end when a new constraint becomes active or a multiplier changes sign.

Proof. (i) A smaller feasible set has a larger minimum.

(ii) With the active constraints imposed as equalities, the first-order conditions are the linear system of theorem 10.4 enlarged by the active rows, whose solution is affine in the right-hand side $(\mu_0, 1, \text{bounds})$; strict convexity makes it unique. □

That is exactly what the plots of the case study show (section 10.14): weights that grow along straight lines in the target return, kinks when a weight reaches its cap, and new assets entering. Below the first kink (only the constraints $w_j \geq 0$ active) the solution is a *scaled* fixed portfolio: “they enter in the same proportion”, the de-levered tangency portfolio of theorem 10.7 computed on the assets that enter, i.e. the *long-only* tangency portfolio, not the unconstrained one, which has short positions (Case Study 6, and Kempthorne [► 2013 L14 @ 1:09:02]).

IN FINANCE: Benchmarked management

A manager measured against an index does not solve problem I but the same quadratic problem in the *active weights* $\mathbf{x}_B = \mathbf{w} - \mathbf{w}_B$: maximise $\mathbf{x}_B^\top \boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{x}_B^\top \boldsymbol{\Sigma} \mathbf{x}_B$, the tracking-error variance being the risk. The efficient frontier is then drawn in the plane (tracking error, active return), and its slope is the information ratio. Factor-neutrality constraints remove market, sector or style bets so that the return is pure stock selection.

10.11 Estimation error and the fragility of the optimiser

Everything above takes $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ as known, “the problem where we assume basically certainty” [► 2013 L14 @ 2:07]. In practice they are estimated, and the estimates are noisy [► 2013 L14 @ 20:08]. Xia’s first objection to modern portfolio theory is exactly that: the capital market assumptions of return, volatility and correlation are not reliable, and the mean–variance solution is very sensitive to them, is “unstable” and has “unlimited” solutions unless bounded by artificial constraints [► 2024 L13 @ 45:22].

10.11.1 How noisy is a sample mean?

Suppose returns are i.i.d. Gaussian with annual mean μ and volatility σ and are observed for T years, at any frequency. The sample mean has standard error $\sigma/\sqrt{T}$ independent of the sampling frequency, since only the total time span matters, whereas the sample volatility, computed from n increments, has relative standard error $1/\sqrt{2n}$ (section 3.10, chapter 13). For 500 weekly returns ($T = 9.6$ years, $n = 500$):

	true value	standard error	sample values in Smltn_TwoAst
mean of asset 1	15%	$\sigma_1/\sqrt{T} = 6.4\%$	0.085–0.21
mean of asset 2	25%	$\sigma_2/\sqrt{T} = 9.7\%$	0.177–0.343
vol. of asset 1	20%	3.2% of itself (0.6%)	0.191–0.200
vol. of asset 2	30%	3.2% of itself (0.9%)	0.287–0.291

This is what Kempthorne shows by simulation: sample means may be far from the population values while volatilities and correlations are estimated much better [► 2013 L14 @ 21:08], and what the audience remark of 2013 L16 says about returns versus volatilities [► 2013 L16 @ 1:25:00]. In finance “more data” means longer periods (the mean does not improve by sampling faster), while regime changes limit how far back one can go: the choice of the estimation period is itself an issue [► 2013 L14 @ 1:20:06]. With the five-year sector-ETF sample of Case Study 6 the mean of a sector with $\sigma = 20\%$ has a standard error of 9%, larger than most differences between the sector means (10% to 24%).

10.11.2 What the optimiser does with the errors

Because $w \propto \Sigma^{-1}\eta$ and highly correlated assets make Σ nearly singular, errors in the means are amplified in the weights. On the rounded nine-ETF inputs of [table 10.2](#) (computed here), the tangency portfolio for $r_0 = 0$ is extremely leveraged, and *very small changes of the means, of one percentage point per asset*, change it a lot:

weights	XLB	XLV	XLP	XLY	XLE	XLF	XLI	XLK	XLU	$\sum \Delta w $
base means	−0.32	0.59	0.58	1.49	−0.06	−0.36	−0.45	−0.10	−0.35	
XLY mean +1 pp	−0.34	0.63	0.53	1.73	−0.05	−0.39	−0.54	−0.18	−0.38	0.58
±1 pp draw A	−0.32	0.75	0.28	1.19	−0.09	−0.35	−0.29	0.07	−0.24	1.24
±1 pp draw B	−0.57	0.68	0.42	2.07	−0.05	−0.55	−0.19	−0.33	−0.48	1.91

(Draw A is +, +, −, −, +, +, +, +, + pp and draw B −, −, −, +, −, −, +, −, − pp for XLB to XLU; the weights sum to 1.) For comparison, the sum of absolute weights is 4.3: a ±1pp error in each mean, well inside the statistical error, moves 29% and 44% of that gross exposure (draws A and B). In this respect the caps of [section 10.10](#) act as regularisation, “artificial constraints” that bound the solutions [► 2024 L13 @ 45:22]. The minimum-variance portfolio w_g does not use μ at all, which is why it is popular.

10.11.3 Remedies

The slides list the standard alternatives [► 2013 L14 @ 1:20:06] (slide 27 of [lecnote14](#)): the sample means and covariance are the least-squares, unbiased and, under Gaussian assumptions, maximum likelihood estimates, but one may instead use *exponentially weighted moving averages* ([chapter 13](#)), *dynamic factor models*, and optimisation with a simple model for Σ , notably Sharpe’s *single-index model* $R_i = \alpha_i + \beta_i R_m + \epsilon_i$ (with uncorrelated ϵ_i), which gives $\Sigma = \sigma_m^2 \beta \beta^\top + \text{diag}(\sigma_{\epsilon,i}^2)$, or the *constant correlation model*, $\Sigma_{ij} = \bar{\rho} \sigma_i \sigma_j$. A factor model estimates $2m + 1$ parameters instead of $m(m + 1)/2$ (for $m = 500$ stocks, 1001 instead of 125,250), and “results in more precise inputs to the optimisation” [► 2013 L14 @ 1:20:06]; see [section 7.5](#) and [chapter 20](#) (shrinkage, regularised risk models) and [section 5.10](#). The returns are the weak point: forecasting them is the actual job of an active manager, as an audience member observed in 2013 [► 2013 L16 @ 1:22:59].

IN FINANCE: Why the frontier is estimated with errors: “error maximisation”

An optimiser puts most weight on the assets whose estimated means are too high and whose estimated variances are too low, which is precisely where estimation errors are largest: the out-of-sample performance of the mean–variance solution is typically worse than the in-sample frontier suggests (not discussed in class).

10.12 Other risk measures

Volatility is only one measure of risk, and Kempthorne closes 2013 L14 by showing how the machinery extends [► 2013 L14 @ 1:20:06](slides 29–33). Let $R_P = w^\top R$.

- **Mean absolute deviation**, $MAD = \mathbb{E}[|w^\top(R - \mu)|]$, leads to a linear programme with linear or quadratic constraints. Where the variance is the natural scale for Gaussian returns, MAD is natural for heavier-tailed, e.g. exponential (Laplace), returns [► 2013 L14 @ 1:21:13].
- **Semi-variance**, $\mathbb{E}[\min(R_P - \mathbb{E}[R_P], 0)^2]$, the downside variance, introduced by Markowitz himself: it controls downside rather than upside risk [► 2013 L14 @ 1:21:13]. Its square root, relative to a target, is the denominator of the Sortino ratio [► 2024 L13 @ 48:30].
- **Value at risk**, $\text{VaR}_{1-\epsilon}(R_P) = \min\{r : \mathbb{P}(R_P \leq -r) \leq \epsilon\}$ with $\epsilon = 0.05, 0.01, 0.001$, popularised by JP Morgan’s RiskMetrics [► 2013 L14 @ 1:22:18]. It is standard for cash instruments (stocks, bonds) but, as Kempthorne says, for derivatives with nonlinear payoffs “things get complicated” [► 2013 L14 @ 1:24:25].
- **Conditional VaR** (expected shortfall), $\text{CVaR}_{1-\epsilon}(R_P) = \mathbb{E}[-R_P \mid -R_P \geq \text{VaR}_{1-\epsilon}(R_P)]$, the average loss beyond the VaR, which is where VaR is silent; expected to enter bank regulation [► 2013 L14 @ 1:23:22]. The optimisation of CVaR is a linear programme (Rockafellar and Uryasev, 2000, cited on the slide).

Definitions with the loss convention, the coherence axioms and the estimation of VaR and ES are in [section 8.4](#) and [chapter 19](#); here only the portfolio-theory consequences are needed.

Proposition 10.20 (VaR is not subadditive; variance is not monotone; not stated in class). *Two independent positions each lose 100 with probability 4% and otherwise 0. At the 95% level, $\text{VaR} = 0$ for each (the loss occurs with probability 4% < 5%), but for the portfolio of both $\mathbb{P}(\text{loss} > 0) = 1 - 0.96^2 = 7.84\% > 5\%$, so $\text{VaR}_{95\%} = 100 > 0 + 0$: diversification increases VaR. The expected shortfalls are 80 for each and 103.2 for the portfolio (≤ 160), as CVaR is subadditive. Likewise the variance violates monotonicity: $R = 0$ and $R' \in \{0, 1\}$ with probability $\frac{1}{2}$ each satisfy $R \leq R'$ but $\text{Var } R = 0 < \text{Var } R' = \frac{1}{4}$, whereas monotonicity requires $s(R) \geq s(R')$.*

(Checked by direct computation: $\text{ES}_{95\%}(\text{portfolio}) = (0.16\% \cdot 200 + 4.84\% \cdot 100) / 5\% = 103.2$.)

IN FINANCE: Expected loss, worst loss, stop losses, drawdowns

Asked for the best risk measure, Xia answers the *expected loss* L , arguing that gains and losses are usually not symmetric; one should also know the worst case but cannot size a portfolio on it (one would never invest) [► 2024 L13 @ 1:14:54]. Stop losses on trading platforms are set tighter than the expected loss (cut the position by half after a small

2 COURSE NOTE: Erratum: translation invariance on slide 31 of [lecnote14](#)

The slide, written for *returns*, defines a coherent measure by: monotonicity ($R_P \leq R_{P'}$, w.p. 1 $\Rightarrow s(R_P) \geq s(R_{P'})$), subadditivity, positive homogeneity, and “translational invariance” $s(R_P + a) \leq s(R_P) - a$. The standard axiom (Artzner–Delbaen–Eber–Heath 1999) is an equality, $s(R_P + a) = s(R_P) - a$: adding a sure

loss, another half after the next), which gives the trader an asymmetric, option-like payoff and forces a distribution of the trades that fits it; and hedge-fund managers, who take a share of profits but not of the losses, hold “a free call option” on the fund [► 2024 L13 @ 1:16:58]. The loss tolerance of [section 10.1](#) is often expressed as a *maximum drawdown* $\max_{s \leq t} (\max_{u \leq s} V_u - V_s) / \max_{u \leq s} V_u$, the largest peak-to-trough fall of the portfolio value V (not defined in class; see the drawdowns of the concentrated portfolio in [section 2.9](#)).

gain a lowers the risk by exactly a (in the loss convention of [definition 8.7](#), $\rho(L + a) = \rho(L) + a$). The slide’s “coherent” verdicts (variance not, VaR not, CVaR yes) are correct [► 2013 L14 @ 1:24:25].

Exercise 10.8 (not from the course: VaR and CVaR). Modify [proposition 10.20](#) to three independent positions with default probability 3% and loss 100: compute the 95% VaR and expected shortfall of one position and of the portfolio of three, and check subadditivity of ES.

10.13 Beyond Markowitz: crowds and power laws (2024)

In 2024 Xia devotes the second half of the lecture to three ideas from his own research (slides marked “sections 4–7 highlight my research work”): the loss of meaning of volatility as risk ([section 10.9](#)), the unpredictability of markets because of *crowd behaviour*, and the origin of *power-law* distributions. They are not tested, and are heuristic, but they answer his “MPT issue number 2”: how do we predict the future from the past [► 2024 L13 @ 54:54]? The scientific method (observe, quantify, extract a pattern, build a model, predict, repeat and calibrate, control) applies to physics and engineering, but in finance the historical patterns may not be repeatable because humans adapt and herd [► 2024 L13 @ 56:01]. In 2013 the same point was made with videos and no formulas: a bridge whose walkers synchronise their steps, and a bank of metronomes on a moving plate that synchronise [► 2013 L16 @ 1:05:51]; in 2024 with a video of birds [► 2024 L13 @ 58:03] and the London Millennium Bridge [► 2024 L13 @ 59:05]. The lesson for portfolio management: an investor who finds the “optimal” strategy finds that everyone else implements it too, and the collective system is not optimal but fragile; individual managers also have an incentive to herd, since underperforming the peers is punished more than being wrong with the crowd (career risk) [► 2013 L16 @ 1:10:12]. Participants change the market they are trying to optimise [► 2013 L16 @ 1:12:19].

10.13.1 A feedback model of crowd behaviour

Slide 28 (drawn on slide 34 with heterogeneous agents) describes the interaction loop of N agents, an external force E and an observation O (the price) [► 2024 L13 @ 1:00:05]:

$$dS_i = C_i dE + B_i dO + \varepsilon_i, \quad dO = \sum_{i=1}^N A_i dS_i, \quad (10.16)$$

where S_i is the action of agent i , A_i its power (influence on the observation), C_i its response to the external force, B_i its *reactive function* (how strongly it reacts to the observation) and ε_i noise. In the rational state B_i takes low values $\{B_L\}$; when dO moves a lot, agents switch to a reactive state with high values $\{B_H\}$, and N_H counts them [► 2024 L13 @ 1:01:08]. The *order parameter*

$$R(t) = \frac{|\sum_i dS_i(t)|}{\sum_i |dS_i(t)|} \in [0, 1]$$

is 1 if all agents act in the same direction (synchronised) and near 0 if they are not [► 2024 L13 @ 1:01:08]. The slides plot R against N_H/N (rising steeply from ≈ 0.1 at $N_H = 0$ to 1 at $N_H = N$) and against the noise amplitude (falling from 1 to near 0: noise desynchronises, as in coupled oscillators) [► 2024 L13 @ 1:02:15], and a simulated “bubble”: an external force that rises, then falls and oscillates makes the number of reactive agents rise and fall and the observation follow; if the force stays high forever, all agents end up reactive and the observation explodes, a *tipping point* when $A \times B_{\max} > 1$ (slide 30) [► 2024 L13 @ 1:03:21].

Proposition 10.21 (Mean-field reading of the tipping point; my derivation, not shown in class). Suppose all agents react with the same $B_i = B$. Substituting the first equation of (10.16) into the second,

$$\begin{aligned} dO &= \sum_i A_i C_i dE + B \left(\sum_i A_i \right) dO + \sum_i A_i \varepsilon_i, \quad \text{so} \\ dO &= \frac{N}{1 - \bar{A}B}, \quad N = \sum_i A_i C_i dE + \sum_i A_i \varepsilon_i, \quad \bar{A} = \sum_i A_i. \end{aligned}$$

The response of the price to any external push or noise is amplified by the factor $1/(1 - \bar{A}B)$, which is finite and positive iff $\bar{A}B < 1$ and diverges as $\bar{A}B \rightarrow 1$: with $\bar{A}B \geq 1$ the loop gain exceeds 1 and the feedback is unstable. The switch from B_L to B_H (a higher B when many agents are reactive) moves the system towards the threshold, which is the tipping point of slide 30.

This is the analogue of the gain of a linear amplifier with positive feedback (Barkhausen criterion), or of the critical coupling of a mean-field ferromagnet, $\chi = 1/(1 - J\bar{A})$, the susceptibility diverging at the critical temperature. The lecture presents the mechanism qualitatively; the numerical scenarios on the slides are simulations whose parameters are not given.

10.13.2 Power laws

Why are wealth, city sizes, venture-capital returns or viral posts so unequal, unlike heights? [► 2024 L13 @ 1:05:33]. The 80–20 rule, the Matthew effect and “winner takes all” describe distributions that are *scale-free*: the top 20% of the people own 80% of the wealth, and within them again the top 20% own 80%, and so on. In symbols $p(bx) = g(b)p(x)$ for all $b > 0$ holds (with smooth p) iff p is a power law, $p(x) = Cx^{-a}$ (with $g(b) = b^{-a}$; the “only if” is the standard functional-equation argument, $p(bx) = g(b)p(x)$ forces $g(bb') = g(b)g(b')$) [► 2024 L13 @ 1:05:33]. The three usual descriptions are related as follows (slide 33) [► 2024 L13 @ 1:09:39]:

$$\begin{aligned} \text{Pareto (tail): } \mathbb{P}[X > x] &\sim x^{-k}, & \text{Zipf (rank } r\text{): } \mathbb{E}[x_{(r)}] &\sim r^{-b}, \\ \text{density: } f(x) &= Cx^{-a}, & a &= 1 + k = 1 + \frac{1}{b}. \end{aligned}$$

(Differentiating the tail gives $a = 1 + k$; the r -th largest of N satisfies $r/N = \mathbb{P}[X > x_{(r)}]$, so $x_{(r)} \sim r^{-1/k}$ and $b = 1/k$.) A Gaussian has no such structure; heights are Gaussian because they are set once, whereas in social phenomena “the rich get richer”: agents have *different powers* $A(i, t)$ and gain power by betting in the right direction of the observation, so the power of the crowd concentrates in a few “super agents” (governments and central banks, the Fed being the example named in class; in the old days, some hedge funds) whose actions dominate the system [► 2024 L13 @ 1:08:38]. He also mentions entropy and stability of such systems as a topic “beyond the scope”.

IN FINANCE: What the crowd material means for practice

Xia's conclusions of the 2024 lecture (slide 35) [► 2024 L13 @ 1:11:45]: portfolio management is about *sizing*; clarify objectives and know the loss tolerance; diversification helps but requires rebalancing; volatility is not risk and the Sharpe ratio captures neither skewness nor sizing (use the gain–loss ratio); capital market assumptions are not reliable; crowd behaviour can push markets to extremes; watch the powerful super agents; understand the key drivers of the portfolio. The 2013 version ends similarly: “solving the problem” is not done, since *when you participate in the market you change the market*, and quantitative finance “is not like solving physics problems” [► 2013 L16 @ 1:11:12]. A questioner asked whether this is game theory; Xia agreed that it has a lot to do with it, but that the markets have no clearly defined rules [► 2013 L16 @ 1:13:22]. Whether a computer can beat a human in investment, he “cannot confidently say it will not happen”; a colleague added that managers also manage people [► 2013 L16 @ 1:18:37].

Example 10.22 (The 80–20 rule and the Pareto exponent). If wealth has the tail $\mathbb{P}[X > x] = (x_m/x)^k$ with $k > 1$, the top fraction q of the population holds the fraction $q^{1-1/k}$ of the total wealth (integrate x over the tail). The 80–20 rule, $0.2^{1-1/k} = 0.8$, gives $k = \ln 5 / \ln 4 = 1.161$, $a = 2.16$, $b = 0.861$; and the top 20% of the top 20% hold 80% of the 80%, i.e. 64% of all wealth (top 4%), the self-similarity of the lecture. Note k close to 1: the mean is barely finite (for $k \leq 1$ it is infinite).

Exercise 10.9 (not from the course: crowd model and power laws).

- In [proposition 10.21](#), take N agents with $A_i = A/N$ and $B_i \in \{B_L, B_H\}$, a fraction x in the reactive state; write the loop gain and find the critical x for the tipping point.
- For a Pareto tail $\mathbb{P}[X > x] = (x_m/x)^k$, $k > 1$, prove that the top fraction q holds $q^{1-1/k}$ of the total, and compute k for a “90–10” rule.

10.14 Case study: simulation and US sector ETFs (2013 Case Study 6)

CASE STUDY: CaseStudy6.pdf, Smltn_TwoAst.pdf, ETF_prid*.pdf

Goal. Kempthorne's live demonstration of the theory in R [► 2013 L14 @ 8:32]:

- the feasible set of two assets for several correlations;
- mean–variance optimal portfolios of the nine US sector ETFs, with a riskless asset, with the constraint that no asset exceeds 30% or 15% of the capital.

Part 1: two-asset simulation. Two Gaussian assets with $\mu = (0.15, 0.25)$, $\sigma = (0.20, 0.30)$ (see the erratum below), 500 weekly returns simulated for $\rho = -0.8, -0.4, 0, 0.4, 0.8$. Each page of Smltn_TwoAst has four panels: cumulative returns of the two assets (blue: asset 2, green: asset 1); the feasible set of [section 10.2](#); the scatter plot of the weekly returns; and the same cumulative returns with the cumulative return of the minimum-variance portfolio (cyan) [► 2013 L14 @ 8:32]. In the PDF as distributed only the two asset points are drawn in the second panel and the scatter panel is blank, so the curves of [fig. 10.1](#) were recomputed. The sample statistics printed above the panels are:

ρ	−0.8	−0.4	0	0.4	0.8
sample means	0.115, 0.341	0.144, 0.343	0.183, 0.334	0.210, 0.321	0.085, 0.177
sample vols	0.191, 0.291	0.194, 0.291	0.200, 0.289	0.200, 0.287	0.196, 0.287
sample corr.	−0.784	−0.359	0.031	0.395	0.791

Sample volatilities and correlations are close to the population values; the sample

means are off by about one standard error (section 10.11); for $\rho = 0.8$ both are well below the population values (0.085 and 0.177 against 0.15 and 0.25), an unlucky draw, and the cumulative-return curves of the last panel look very different from the others. Lessons observed live: with $\rho \leq 0.4$ the minimum-variance portfolio has a return and a volatility better than asset 1 alone; with $\rho = -1$ the variance would be zero [► 2013 L14 @ 11:47]; every portfolio to the left of the minimum-variance point is suboptimal [► 2013 L14 @ 14:57].

Part 2: nine sector ETFs. The SPDR sector funds XLB (materials), XLV (health care), XLP (consumer staples), XLY (consumer discretionary), XLE (energy), XLF (financials), XLI (industrials), XLK (technology) and XLU (utilities), for two periods: 2009–2013 (“periodA”, up to a week before the lecture; 251 observations on the time axis, i.e. weekly data over about 4.8 years, an inference: the frequency is not stated) and 2003–2006 (“periodB”, 209 observations). table 10.2 gives the annualised return and volatility of each. The correlations are high, with the average pairwise correlation 0.74 in 2009–2013 (range 0.57–0.93) and 0.55 in 2003–2006 (range 0.16–0.85): there is much less diversification available in the aftermath of the crisis.

Method. For a grid of 101 target returns (“target return index” in the plots) the R script minimises the variance of a portfolio of the nine ETFs and the riskless asset (return ≈ 0 : the frontier of plot 4 starts at the origin), subject to full investment, $w_j \geq 0$ and $w_j \leq 30\%$ or 15% . It prints the allocations for three target volatilities and draws six plots: 1 risk–return scatter with the riskless asset; 2 optimal allocations versus target return (lines); 3 the same as a stacked bar chart (dark red at the top is cash); 4 the efficient frontier over the ETFs; 5 the frontier with the target portfolio for a 10% volatility (circle); 6 cumulative returns of the ETFs with that optimal portfolio in bold blue [► 2013 L14 @ 1:04:54]–[► 2013 L14 @ 1:11:11].

ETF	sector	2009–2013		2003–2006	
		return	vol.	return	vol.
XLB	materials	0.16	0.24	0.16	0.18
XLV	health care	0.17	0.15	0.07	0.12
XLP	consumer staples	0.15	0.12	0.08	0.09
XLY	consumer discretionary	0.24	0.21	0.14	0.14
XLE	energy	0.15	0.25	0.26	0.20
XLF	financials	0.13	0.33	0.15	0.13
XLI	industrials	0.18	0.23	0.15	0.14
XLK	technology	0.18	0.19	0.12	0.18
XLU	utilities	0.10	0.16	0.19	0.13

Table 10.2: Annualised return and volatility of the nine sector ETFs printed in CaseStudy6. (The correlation matrices are printed there as well; the computations of example 10.6 and of this section use them.)

Results (2009–2013). With a 30% cap and a target volatility of 9.9% the printed optimum is XLV 15.7%, XLP 30%, XLY 22.6%, cash 31.7%, for a return of 12.4%; for a target volatility of 15.3% it is fully invested: XLV 30%, XLP 22.7%, XLY 30%, XLK 17.3%, for a return

3 COURSE NOTE: Independent check of the case study

The R code is not available, so the printed allocations were re-derived from the printed means, volatilities and correlations (all rounded to two decimals) by maximising the return at a given volatility, with $r_0 = 0$, $w_j \in [0, c]$ and full investment or less, with a general-purpose solver. At a target volatility of 15.3% (30% cap) it gives XLV 30%, XLP 22.9%, XLY 30%, XLK 17.1% and a return of 18.8% (printed: 30, 22.7, 30, 17.3, 18.7); at 9.9% it gives XLV 21.8%, XLP 30%, XLY 18.7%, cash 29.5% and 12.7% (printed: 15.7, 30, 22.6, 31.7, 12.4). The sets of assets, the order in which the caps bind and the frontiers agree with the printed ones; the few percentage points of difference in XLV and XLY come from the rounding of the inputs (a half-point change in a mean moves weights by several points), a small illustration of the sensitivity of section 10.11. For the 15% cap at 9.9% volatility the recomputation gives XLV, XLP, XLY, XLK at 15%, XLU 3.6%, XLI 1.6%, cash 34.8% (printed: XLU 6.9%, cash 33.1%, return 11.7% in both).

of 18.7%; for 0.9% volatility it holds 93.5% in cash. The comments of the case study, confirmed by the plots [► 2013 L14 @ 1:07:59]–[► 2013 L14 @ 1:11:11]:

- As the target return rises from zero only XLY, XLP and XLV enter, in *fixed proportions* (the de-levered long-only tangency portfolio, [proposition 10.19](#); the unconstrained one of [section 10.11](#) is very different); allocations grow linearly (plot 2).
- The cap binds first for consumer staples; XLY and XLV are increased; when all three are at 30% technology (higher return) enters and the allocations to XLY and XLP fall to make room. No allocation is ever given to financials.
- In the efficient frontier of plot 4, every ETF except consumer discretionary is dominated by an optimal portfolio, and the frontier shows the CML-like straight segment up to the first kink.
- With a 15% cap the constraints bind sooner: XLV, XLP, XLY, XLK, then XLU and XLI, and finally XLB; the frontier is *below* the 30% one ([proposition 10.19\(i\)](#)), because to reach a higher return one has to hold assets with a lower return per unit of risk [► 2013 L14 @ 1:12:14].
- In 2003–2006 the optimal portfolios are completely different: utilities, energy, financials, consumer staples and industrials, XLU being the first to hit the cap, and technology never enters (15% cap: XLP, XLY, XLE, XLF, XLI, XLU at the cap, XLB in addition at the top). The ranking of sectors in one period says little about the next.

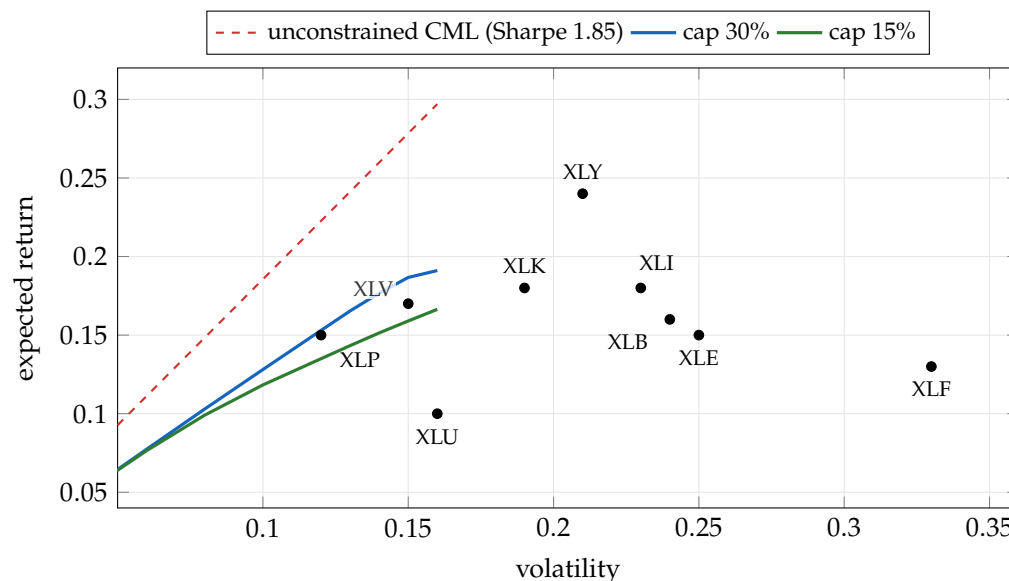


Figure 10.4: Nine sector ETFs, 2009–2013 (inputs of [table 10.2](#) and the printed correlations; the risk-free rate is taken to be 0). Dots: individual ETFs. Blue and green: efficient frontier with cash and long-only positions capped at 30% and 15% (plots 4 of `ETF_pridA_30` and `ETF_pridA_15`, recomputed here). The dashed line is the CML of the *unconstrained* problem, with a Sharpe ratio of 1.85, reached only with large short positions ([section 10.11](#)).

IN FINANCE: Sector market-neutral models and diversification

In the second demonstration of 2013 L14 Kempthorne replaces the ETFs by hedge-fund style strategies: multi-factor pricing models trading long/short, market-neutral, within each of the nine sectors, over five years [► 2013 L14 @ 1:15:29]. Their total returns are

modest (20–60% over five years) but they are *little correlated* with each other, so the optimiser gets a large benefit from combining them; the model for the utilities sector receives the largest weight, then industrials and energy, and a target volatility of 10% is reached [► 2013 L14 @ 1:17:35]. The data behind the plots are not in the files, so only this qualitative description is given. The lesson is the one of [section 10.2](#): low correlation makes the frontier bend to the left, and diversification is worth most for uncorrelated bets.

10.15 Notes on the sources, differences between the years and errata

KEY FORMULAS

Portfolio:	$\mu_w = \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_w^2 = \mathbf{w}^\top \Sigma \mathbf{w}, \quad \mathbf{w}^\top \mathbf{1} = 1$
Two assets:	$\sigma_w^2 = (1-w)^2 \sigma_1^2 + w^2 \sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$ $w^* = \frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}, \quad \sigma^{*2} = \frac{\sigma_1^2 \sigma_2^2 (1-\rho^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
Constants:	$a = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}, \quad b = \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}, \quad c = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad \Delta = ac - b^2$
Optimal weights:	$\mathbf{w}_0 = \lambda_1 \Sigma^{-1} \boldsymbol{\mu} + \lambda_2 \Sigma^{-1} \mathbf{1}, \quad (\lambda_1, \lambda_2) = \frac{1}{\Delta}(c\mu_0 - b, a - b\mu_0)$
Frontier:	$\sigma_0^2 = \frac{c\mu_0^2 - 2b\mu_0 + a}{\Delta} = \frac{1}{c} + \frac{c}{\Delta}(\mu_0 - \frac{b}{c})^2$
Min. variance:	$\mathbf{w}_g = \frac{\Sigma^{-1} \mathbf{1}}{c}, \quad \sigma_g^2 = \frac{1}{c}, \quad \mu_g = \frac{b}{c}$
Risk aversion:	$\mathbf{w}(\lambda) = \mathbf{w}_g + \frac{1}{\lambda} \Sigma^{-1}(\boldsymbol{\mu} - \frac{b}{c} \mathbf{1})$
With riskless asset:	$\mathbf{w}_0 = \frac{\mu_0 - r_0}{d^2} \Sigma^{-1} \boldsymbol{\eta}, \quad \boldsymbol{\eta} = \boldsymbol{\mu} - r_0 \mathbf{1}, \quad d^2 = \boldsymbol{\eta}^\top \Sigma^{-1} \boldsymbol{\eta}$
Tangency portfolio:	$\mathbf{w}_M = \frac{\Sigma^{-1} \boldsymbol{\eta}}{\mathbf{1}^\top \Sigma^{-1} \boldsymbol{\eta}}, \quad \mu_M - r_0 = \frac{d^2}{b - r_0 c}, \quad \sigma_M = \frac{d}{b - r_0 c}$
Tobin, CML:	$m_w = \frac{\mu_0 - r_0}{\mu_M - r_0}, \quad \mu_P = r_0 + \frac{\mu_M - r_0}{\sigma_M} \sigma_P$
Sharpe ratio:	$SR = \frac{\mu_P - r_0}{\sigma_P} \leq d$ (max. at $\mathbf{w}_M$), leverage-invariant
SML:	$\mu_i - r_0 = \beta_i(\mu_M - r_0), \quad \beta_i = \frac{\text{Cov}(R_i, R_M)}{\sigma_M^2} = \rho_{iM} \frac{\sigma_i}{\sigma_M}$
Equal weights:	$\text{Var } R_{ew} = \frac{\bar{\sigma}^2}{n} + (1 - \frac{1}{n})\bar{c}$
Constant mix:	$g(\mathbf{w}) = \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}$ (log growth)
Risk contribution:	$RC_i = \frac{w_i(\Sigma \mathbf{w})_i}{\sigma_P}, \quad \sum_i RC_i = \sigma_P$; 2 assets: $w_i \propto 1/\sigma_i$
Utility:	$\lambda_A = -\frac{u''}{u'}, \quad \lambda_R = -\frac{Wu''}{u'}; \quad \mathbb{E}[u] \approx u(w_*) + \frac{1}{2} u'' \text{Var } W$ CARA+normal: $CE = \mu_W - \frac{\lambda}{2} \sigma_W^2$
Kelly:	$f^* = \frac{p}{l} - \frac{q}{s}; \quad \text{Gaussian: } f^* = \frac{\mu}{\sigma^2}$
Gain–loss ratio:	$\frac{G-L}{G+L} = \frac{\mathbb{E}[R]}{\mathbb{E}[R]} \in [-1, 1], \quad G = \mathbb{E}[R^+], \quad L = \mathbb{E}[R^-]$
Constraints:	$\text{TE}^2 = (\mathbf{w} - \mathbf{w}_B)^\top \Sigma (\mathbf{w} - \mathbf{w}_B), \quad \sum_i \beta_{ik} w_i = 0$
Estimation:	$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{T}}, \quad \frac{\text{se}(\hat{\sigma})}{\sigma} \approx \frac{1}{\sqrt{2n}}$
Crowd loop:	$dO = \frac{\sum A_i C_i dE + \sum A_i \varepsilon_i}{1 - \bar{A}\bar{B}}$ (mean field)
Power law:	$\mathbb{P}[X > x] \sim x^{-k}, \quad f \sim x^{-a}, \quad a = 1 + k = 1 + \frac{1}{b}$

4 COURSE NOTE: 2013 versus 2024

The 2013 lectures are split: Kempthorne gives the mathematics (his lecture has a full derivation, a case study, utility theory, constraints and risk measures), Xia the practice. The 2024 lecture has only Xia, and it is a different lecture from his 2013 one. It keeps the special cases of two assets (slides 14–20; Xia says he skips the mathematics that he covered in earlier years [► 2024 L13 @ 40:00]) and the rebalancing example, and it adds: the endowment model, MPT issue 1 (volatility as risk) with the gain–loss ratio, MPT issue 2 (predicting the future) with the crowd model, and power laws. It drops the Kelly formula, the beta discussion and the Millennium Bridge and metronome videos of 2013 (2024 shows a video of birds and mentions the bridge). The 2024 slides point to the 2013 video of Xia’s lecture as “one of the top five most viewed courses” of OpenCourseWare [► 2024 L13 @ 1:06]. What Kempthorne derives (Markowitz problem, tangency portfolio, Tobin, utility, constraints, risk measures) is not covered in 2024 by anyone: the derivations of this chapter follow [lecnote14](#).

5 COURSE NOTE: Erratum: parameters of the two-asset simulation

Kempthorne says at the start of the demonstration that asset 1 has mean 15% and volatility 25%, asset 2 mean 20% and volatility 30% [► 2013 L14 @ 5:21], and page 2 of [CaseStudy6](#) prints the same numbers (with typos: “ $E(R_1) = 0.20 = \alpha_2$ ” and “ $\sqrt{\text{Var } R_1} = 0.30 = \sigma_2$ ” should refer to R_2 , and “ $\sigma_w^2 = \text{Var}(R_2)$ ” should be $\text{Var}(R_w)$). But the plots of [Smltn_TwoAst](#) are titled “mean returns 0.15, 0.25, asset volatilities 0.20, 0.30”, Kempthorne himself reads “mean returns 15%, 25%, volatility 20 to 30” when he shows them [► 2013 L14 @ 10:44], and the sample statistics (0.19–0.20 and 0.29 for the volatilities, cumulative return ≈ 3.3 for asset 2, i.e. about 0.34 per year over 9.6 years) fit these values, not the printed ones. This chapter uses the values of the plots.

6 COURSE NOTE: Errata and remarks on CaseStudy6

- Section 2.3 refers to “Plot 4 in the two files [ETF_S_1_periodA_30.pdf](#) and [ETF_S_1_periodA_30.pdf](#)”: the second should be the 15% file ([ETF_pridA_15](#)); the case study cites the file names of the original R output ([ETFS_1_periodA_30](#)), which differ from

Exercises

Standalone exercises follow the section they practise. PS6 Problems 3–5 are a connected series (Problem 4 starts “Now $m > 2$ assets” and extends Problem 3, Problem 5 continues them), so they stay together here.

From 2013 Problem Set 6 (Problems 3–5; Problems 1–2 are in [chapter 9](#))

In all three problems the portfolio w is fully invested, $\sum_j w_j = 1$, with no short sales, $w_j \geq 0$.

Exercise 10.10 (PS6 (2013), Problem 3: two risky assets). Two assets with returns $R = (R_1, R_2)^T$, means (α_1, α_2) , covariance $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ and a portfolio $R_w = w_1R_1 + w_2R_2$.

- Prove that $\text{Var}(R_w) \leq \max(\sigma_1^2, \sigma_2^2)$ for every such portfolio.
- For $\sigma_1 = \sigma_2$ and $\rho = 0$ find the minimum-variance portfolio w^* and its variance.
- For $\sigma_1 = \sigma_2$ and arbitrary ρ find w^* and $\text{Var}(R_{w^*})$, and graph the minimal variance as a function of $\rho \in [-1, 1]$.

Exercise 10.11 (PS6 (2013), Problem 4: $m > 2$ assets, equal volatilities). Now $m > 2$ assets with covariance matrix $\Sigma = (\rho_{ij}\sigma_i\sigma_j)$.

- Prove that $\text{Var}(R_w) \leq \max_j \sigma_j^2$ for every long-only fully invested portfolio.
- If $\sigma_1 = \dots = \sigma_m$ and all $\rho_{ij} = 0$ ($i \neq j$), find the minimum-variance portfolio, express its variance as a function of m and find its limit as $m \rightarrow \infty$.
- If $\sigma_1 = \dots = \sigma_m$ and $\rho_{ij} \equiv \rho$ for all $i \neq j$ (equicorrelation): first show that $\rho \geq -1/(m-1)$ (hint: $\mathbf{1}$ is an eigenvector of Σ ; compute its eigenvalue and impose that Σ be positive semi-definite). Then find the minimum-variance portfolio and its variance, graph the variance as a function of $\rho \in [-\frac{1}{m-1}, 1]$, find the limit as $m \rightarrow \infty$ and compare it with (b), commenting on the ability to diversify away risk by adding equicorrelated assets.

Exercise 10.12 (PS6 (2013), Problem 5: unequal variances and the general case).

- For $m = 2$, $\sigma_1 \neq \sigma_2$ and $\rho = 0$, find the minimum-variance portfolio and its variance.
- For $m > 2$ and $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$ (arbitrary variances, zero correlations) find the minimum-variance portfolio and its variance, and express the variance in terms of m and $\tilde{\sigma}^2 = \left(\frac{1}{m} \sum_j \sigma_j^{-2}\right)^{-1}$.
- For $m > 2$ and an arbitrary positive definite Σ , find the minimum-variance portfolio and its variance.

the OCW names `ETF_pridA_30`.

- Section 3.3 (2003–2006, 15% cap) says both “technology never enters the optimal allocation” and “the allocations to XLK increase until its limit is reached”: the second bullet is copied from Section 2.3; the printed table has XLK at 0.000 throughout.
- The bars of plot 3 use very similar colours for consumer discretionary and energy (both green) and the line plots repeat colours (XLB and XLU are both black in the first plot of each file), so the ETFs are identified through the tables.

7 COURSE NOTE: Slides and lectures: small corrections

- lecnote14**, slide 19 writes $\mathbb{E}[u(W)] \propto \mathbb{E}[W - \frac{1}{2}\lambda(W - w_*)^2] \approx \mathbb{E}[W] - \frac{1}{2}\lambda \text{Var } W$: the $\propto$ hides an additive constant and the positive factor $u'(w_*)$; the content is that of [proposition 10.17\(i\)](#).
- Slide 31, translation invariance: see the erratum in [section 10.12](#).
- Slide 24 of **lec13** asserts the ratio $(G - L)/(G + L)$ is “the same as the Kelly criterion for binary bets”: true only for even-money bets ([proposition 10.18\(i\)](#)).
- In 2013 L16 Xia says that after year 1 the winning asset “becomes 75” when rebalancing: the correct amounts are 62.5 in each asset ([section 10.6](#)); the 2024 slide has it right.
- The transcript of 2013 L14 says Fama “was just awarded, what, two weeks ago”: the 2013 economics prize (Fama, Hansen, Shiller) was announced on 14 October 2013, ten days before the lecture of 24 October (the case-study date).
- In 2013 L16 two remarks made “from the audience” (about estimating returns versus volatility, and about the Kelly criterion, Shannon and Samuelson) are by a co-lecturer, in all likelihood Kempthorne; the transcript does not identify the speaker.
- lecnote16** contains only the outline (five items: portfolio construction, portfolio theory in special cases, risk parity, limitations, what we learned): Xia used a single slide and the blackboard, so 2013 L16 has no formulas apart from those reconstructed here from the transcript and from the 2024 slides.

Chapter 11

Stochastic Processes II: Brownian Motion

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L14	P. Kempthorne	History; Brownian motion with diffusion coefficient σ^2 : increments, Markov property, covariance; transition density and the diffusion equation; Brownian motion as limit of normalised random walks (invariance principle); reflection principle, maximum, first hitting time; reflected and absorbed Brownian motion; Brownian bridge and the empirical distribution function; Brownian motion with drift; gambler's ruin by infinitesimal one-step analysis; non-differentiability; quadratic variation; log-price quadratic variation; Laplace increments at exponential times; filtrations and adapted processes.
2013 L17	C. Lee	Continuous-time processes and why their law is hard to describe; the existence theorem for standard Brownian motion; random-walk limit; Brown, Einstein and Bachelier; zero crossings, $\sqrt{t}$ envelope, nowhere differentiability; distribution of the maximum by the reflection principle; quadratic variation and $(dB)^2 = dt$; Brownian motion with drift; <i>why</i> $S_t \neq e^{B_t}$, Taylor expansion and the first appearance of Itô's lemma.

Files. 2024 slides `mit18_642_f24_lec14_1.pdf` (77 pages with overlays, 20 slides) and the figure decks `lec14_2` (two-dimensional Brownian motion, 201 animation frames), `lec14_3` (normal densities $p(x | t)$, $t = 1, \dots, 10$), `lec14_4` (limiting random walk, 201 frames), `lec14_5` (reflected Brownian motion, three trials), `lec14_6` ($\mathbb{P}[M(t) > x]$), `lec14_7` (hitting-time density), `lec14_8` (gambler's ruin with drift, four panels), `lec14_9` (quadratic variation, 201 frames); 2013 note `MIT18_S096F13_lecnote17.pdf`. **Problem sets.** 2024 PS4 Problems 1–2 (conditional moments and scaling of Brownian motion; Problems 3–4 are the maximum-likelihood estimation of a geometric Brownian motion, [chapter 13](#)); 2024 PS5 Problems 1–2 (Lévy processes: Brownian motion with drift and the Poisson process, with simulations; Problems 3–5 are volatility estimation, [chapter 13](#)). 2013 PS8 Problem A-2 is 2024 PS4 Problem 1 and Problem B-1 is on two-dimensional Brownian motion; PS8 belongs to [chapter 17](#), see the course note in [section 11.8](#). Standalone exercises follow the section they practise. **Not covered.** The construction of Brownian motion (the existence theorem is only quoted in both years), the proofs of Donsker's theorem and of nowhere differentiability, and the law of the iterated logarithm. The R code that generated the figure decks is not part of the download; the figures of this chapter are recomputed from formulas.

Motivation

Chapter 4 studied random walks, martingales and stopping times in discrete time. Prices, however, change at every instant a trade takes place, and the mathematics of derivative pricing ([chapters 15](#), [17](#) and [18](#)) is carried by a *continuous-time* model. Its basic building block is *Brownian motion*, the continuous-time random walk. It was found three times, in three unrelated places: by the botanist Robert Brown (1827), watching pollen grains

jitter in water; by Louis Bachelier (1900), as a model of prices on the Paris exchange; by Albert Einstein (1905), who explained Brown's observation by the bombardment of the grain by water molecules and obtained the diffusion law $\langle x^2 \rangle = 2Dt$ [► 2024 L14 @ 0:00] [► 2013 L17 @ 18:21]. Norbert Wiener, an MIT mathematician, gave it its rigorous foundation in 1923, which is why it is also called the *Wiener process* [► 2024 L14 @ 1:04][► 2013 L17 @ 13:43].

The reason for its centrality is a *universality* statement, made precise in [section 11.2](#): any random walk with finite-variance steps, watched from far away, looks like Brownian motion. For a physicist this is the central limit theorem in disguise; for a trader it says that a price built from many small, roughly independent trades is described by two numbers, a drift and a volatility. This chapter develops the calculus of *sample paths* that the next chapters use:

- the definition and the elementary properties (Gaussian, Markov, martingale), and the diffusion equation ([section 11.1](#));
- Brownian motion as the scaling limit of the random walk ([section 11.2](#));
- the strange geometry of the paths: continuous but nowhere differentiable, with *finite quadratic variation* $[B]_T = T$, the fact that later becomes Itô's lemma ([section 11.3](#));
- the reflection principle, giving the law of the running maximum and of the first hitting time ([section 11.4](#)), with reflected and absorbed Brownian motion;
- Brownian motion with drift and the gambler's ruin problem, geometric Brownian motion, and why the price is *not* e^{Bt} ([sections 11.5](#) and [11.6](#));
- further processes built from B : bridge, random-time sampling, two dimensions, Lévy processes, filtrations ([section 11.7](#)).

11.1 Definition and elementary properties

11.1.1 Brownian motion

In discrete time the law of a random walk is specified through its increments: "each step is ± 1 with probability $\frac{1}{2}$ ". In continuous time there is no next step to describe, so one lists *properties* of increments over all intervals and asserts that a process with these properties exists [► 2013 L17 @ 4:10].

Definition 11.1 (Standard Brownian motion). A stochastic process $\{B(t), t \geq 0\}$ is a *standard Brownian motion* (Wiener process) if

- (i) $B(0) = 0$ and $t \mapsto B(t)$ is continuous (with probability one);
- (ii) (stationary Gaussian increments) for $0 \leq s < t$, $B(t) - B(s) \sim N(0, t - s)$;
- (iii) (independent increments) for $t_1 \leq t_2 \leq t_3 \leq t_4$ the increments $B(t_2) - B(t_1)$ and $B(t_4) - B(t_3)$ are independent; more generally the increments over non-overlapping intervals are mutually independent [► 2024 L14 @ 3:08][► 2013 L17 @ 7:21].

The 2024 slides call σ^2 the *diffusion coefficient* and write $B(t + \Delta t) - B(t) \sim N(0, \sigma^2 \Delta t)$ already in the definition; from the slide on drift onwards B is standard and σB is used. We follow the second convention: *Brownian motion with diffusion coefficient σ^2* is $\sigma B(t)$, and one may start it at any level x by adding x ("Brownian motion started at x ") [► 2024 L14 @ 9:36]. Property (ii) is the one to remember: over a time span h the process moves by a centred normal variable whose *variance is the time elapsed*, so its typical size is $\sqrt{h}$ [► 2013 L17 @ 12:40].

Proposition 11.2 (Gaussian characterisation). *A process with continuous paths and $B(0) = 0$ is a standard Brownian motion if and only if it is a centred Gaussian process with covariance function $\text{Cov}(B(s), B(t)) = \min(s, t)$.*

Note the meaning of the covariance: $B(s)$ and $B(t)$ are correlated to the extent that the two time windows $[0, s]$ and $[0, t]$ overlap, $\text{Corr}(B(s), B(t)) = \sqrt{s/t}$ for $s < t$ [► 2024 L14 @ 8:34].

Proposition 11.3 (Elementary properties). *Let B be a standard Brownian motion and $\mathcal{F}_t = \sigma(B(u), u \leq t)$.*

- (i) Mean and variance. $\mathbb{E}[B(t)] = 0$, $\text{Var } B(t) = t$: the standard deviation grows like $\sqrt{t}$ [► 2024 L14 @ 6:28].
- (ii) Markov property. For $h > 0$, $B(t+h) = B(t) + [B(t+h) - B(t)]$ where the bracket is $N(0, h)$ and independent of $\mathcal{F}_t$; hence the law of $B(t+h)$ given $\mathcal{F}_t$ is $N(B(t), h)$, which depends on the past only through $B(t)$ [► 2024 L14 @ 5:17].
- (iii) Martingales. $B(t)$, $B(t)^2 - t$ and, for every real λ , $\exp(\lambda B(t) - \frac{1}{2}\lambda^2 t)$ are martingales with respect to $\{\mathcal{F}_t\}$.
- (iv) Symmetries. $-B$; the shifted process $B(t_0 + t) - B(t_0)$; the rescaled process $c^{-1/2}B(ct)$ for $c > 0$; and the time-inverted process $tB(1/t)$ ($t > 0$, and 0 at $t = 0$) are again standard Brownian motions.

Property (c) is the continuous-time image of [examples 4.15, 4.16](#) and [4.18](#): the same three martingales that appeared for the random walk (sum, square minus $n\sigma^2$, exponential) survive the limit. The variance t that appears in $B^2 - t$ is the *quadratic variation* of [section 11.3](#). The martingale $\exp(\lambda B - \lambda^2 t/2)$ is the ancestor of the change of measure behind risk-neutral pricing ([chapter 15](#)).

INTUITION: The physicist's reading

Standard Brownian motion is the position of a free overdamped particle whose velocity has been eliminated: the Langevin equation $\dot{x} = \sqrt{2D} \xi(t)$ with white noise ξ has

1 COURSE NOTE: Existence, and why the definition is indirect

Both lecturers stress that (i)–(iii) are a *list of demands* on a probability law on the space of continuous paths, and that the theorem “such a law exists” is hard and not proved in the course [► 2013 L17 @ 6:20]. The sample space is the space of paths $\omega : [0, \infty) \rightarrow \mathbb{R}$ itself, with $\mathbb{P}$ a measure on it (Wiener measure); a path is an outcome, and there is no “density at ω ” [► 2013 L17 @ 5:18]. One standard route (Wiener 1923; Lévy 1948) builds B as a uniform limit of random piecewise-linear functions obtained by refining a dyadic grid; another is Donsker’s theorem below, where B is *defined* as the limit of rescaled random walks (this is how the 2024 lecture introduces it, [► 2024 L14 @ 17:22]). What (i)–(iii) do determine is the joint law of $(B(t_1), \dots, B(t_m))$ for every finite set of times, by writing $B(t_k)$ as a sum of independent normal increments; continuity of the paths is the extra input that makes the law unique on path space.

Proof. ($\Rightarrow$) Take $s \leq t$. Then $B(t) = B(s) + [B(t) - B(s)]$ with independent summands, so $\mathbb{E}[B(s)B(t)] = \mathbb{E}[B(s)^2] + \mathbb{E}[B(s)]\mathbb{E}[B(t) - B(s)] = s + 0$ [► 2024 L14 @ 7:33]. Any vector $(B(t_1), \dots, B(t_m))$ is a linear image of independent normal increments, hence jointly Gaussian. ($\Leftarrow$) For $s < t$ the variable $B(t) - B(s)$ is centred normal with variance $t - 2s + s = t - s$. For $s_1 < t_1 \leq s_2 < t_2$, $\text{Cov}(B(t_1) - B(s_1), B(t_2) - B(s_2)) = t_1 - t_1 - s_1 + s_1 = 0$ (write out the four minima); jointly Gaussian and uncorrelated implies independent. $\square$

Proof. (i) is property (ii) of [definition 11.1](#) with $s = 0$.

(ii) is property (iii) of [definition 11.1](#) together with the definition of $\mathcal{F}_t$.

(iii) For $s < t$, using independence of $B(t) - B(s)$ from $\mathcal{F}_s$: $\mathbb{E}[B(t) \mid \mathcal{F}_s] = B(s)$; $\mathbb{E}[B(t)^2 \mid \mathcal{F}_s] = B(s)^2 + 2B(s)\mathbb{E}[B(t) - B(s)] + \mathbb{E}[(B(t) - B(s))^2] = B(s)^2 + (t - s)$; and with the moment generating function of a normal variable, $\mathbb{E}[e^{\lambda(B(t)-B(s))}] = e^{\lambda^2(t-s)/2}$, so $\mathbb{E}[e^{\lambda B(t) - \lambda^2 t/2} \mid \mathcal{F}_s] = e^{\lambda B(s)} e^{\lambda^2(t-s)/2} e^{-\lambda^2 t/2} = e^{\lambda B(s) - \lambda^2 s/2}$.

(iv) Each candidate is continuous (at $t = 0$ for the inverted one use $B(u)/u \rightarrow 0$, which is the law of large numbers for B), centred Gaussian, and one checks that its covariance is $\min(s, t)$; then apply [proposition 11.2](#). For the rescaling, $\text{Cov}(c^{-1/2}B(cs), c^{-1/2}B(ct)) = c^{-1} \min(cs, ct) = \min(s, t)$; this is the content of 2024 PS4 Problem 2 ([Exercise 11.1](#)); for the inversion, $st \min(1/s, 1/t) = \min(t, s)$. $\square$

solution $x(t) = \sqrt{2D} B(t)$, so $\langle x^2 \rangle = 2Dt$ (Einstein's law) and $\sigma^2 = 2D$. The white noise $\xi = \dot{B}$ does not exist as a function (section 11.3); only its integral does. The law of the whole path is the *Wiener measure*, the measure of the Euclidean path integral of a free particle, $\propto \exp(-\frac{1}{2} \int_0^t \dot{x}^2 ds) \prod dx(s)$, formal but useful; the transition density below is its propagator (the heat kernel), and $\min(s, t)$ is the Green function of $-d^2/dt^2$ with Dirichlet condition at 0.

IN FINANCE: Bachelier's model

Bachelier's thesis modelled the price itself as $S(t) = S(0) + \mu t + \sigma B(t)$, "arithmetic" Brownian motion; the 2013 note credits him with the first use of the process to value stocks and options [► 2013 L17 @ 13:43]. Its defect is that the price becomes negative with positive probability, so equities are modelled through $\log S$ or through relative changes (section 11.6). The arithmetic model survives where levels can be negative or small compared with their moves, in particular for interest rates and spreads, where "normal volatility" is quoted in basis points per year.

11.1.2 Transition density and the diffusion equation

Let B^x be Brownian motion started at x , i.e. $x + B$. For $\tau > 0$ and diffusion coefficient σ^2 , the density of $B^x(s + \tau)$ given $B^x(s) = x$ is the Gaussian

$$p(y, \tau | x) = \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\left(-\frac{(y-x)^2}{2\sigma^2\tau}\right), \quad (11.1)$$

that is, $\mathbb{P}[B(t) \leq c | B(s) = x] = \Phi((c-x)/(\sigma\sqrt{t-s}))$ in terms of the standard normal cdf Φ and pdf φ [► 2024 L14 @ 9:36].

Proposition 11.4 (Diffusion equation). *The kernel (11.1) satisfies*

$$\frac{\partial p}{\partial \tau} = \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial y^2}, \quad \int_{-\infty}^{\infty} p(y, \tau | x) dy = 1 \quad (\tau > 0), \quad p(y, \tau | x) \xrightarrow{\tau \downarrow 0} \delta(y-x), \quad (11.2)$$

and (11.2) has no other solution with these properties. The same holds in the variable x (backward equation).

The semigroup property (Chapman–Kolmogorov, theorem 4.8) is the convolution identity $\int p(z, \tau_2 | y) p(y, \tau_1 | x) dy = p(z, \tau_1 + \tau_2 | x)$, the statement that a sum of independent normals is normal with added variances. In other disciplines (11.2) is the *heat equation*, in population dynamics the *Fokker–Planck* or *Kolmogorov forward equation* [► 2024 L14 @ 17:22]; it returns in the valuation of derivatives (chapter 15), where the Black–Scholes PDE is a backward equation of this type.

Proof. Put $u = y - x$. Then $\partial_\tau p = p(-\frac{1}{2\tau} + \frac{u^2}{2\sigma^2\tau^2})$ and $\partial_y^2 p = p(-\frac{1}{\sigma^2\tau} + \frac{u^2}{\sigma^4\tau^2})$, so $\partial_\tau p = \frac{\sigma^2}{2} \partial_y^2 p$. Integrating (11.2) over y gives $\partial_\tau \int p = \frac{\sigma^2}{2} [\partial_y p]_{-\infty}^{\infty} = 0$: probability is conserved [► 2024 L14 @ 16:17]. The Fourier transform in y turns the equation into $\partial_\tau \hat{p} = -\frac{1}{2}\sigma^2 k^2 \hat{p}$, whose solution with $\hat{p}(k, 0) = e^{-ikx}$ is $e^{-ikx - \sigma^2 k^2 \tau/2}$, the transform of (11.1); this gives uniqueness among tempered solutions [► 2024 L14 @ 17:22]. The backward version follows from p depending on $y - x$ only. $\square$

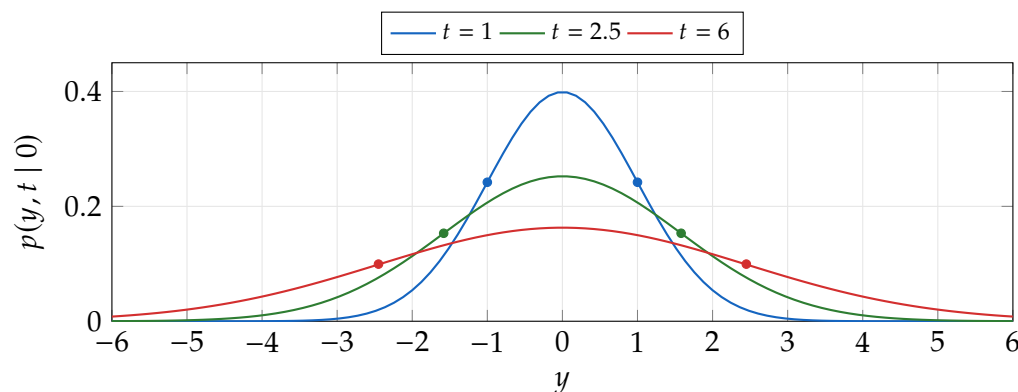


Figure 11.1: Diffusion of the density of $B(t)$ (the panels of lec14_3, which shows $t = 1, 1.5, \dots, 10$). The density drops at the centre and rises in the tails; the dots mark the inflection points $y = \pm\sqrt{t}$, where the curvature changes sign, so that $\partial_t p < 0$ inside and > 0 outside [► 2024 L14 @ 14:08].

Exercise 11.1 (PS4 (2024), Problem 2: Brownian scaling). Let B be a standard Brownian motion. Prove:

- (a) if $B(0) = 0$, then for every $t > 0$, $\{B(st), s \geq 0\} \stackrel{d}{=} \{t^{1/2}B(s), s \geq 0\}$ as processes;
- (b) more precisely, the two families have the same finite-dimensional distributions:
for $s_1 < \dots < s_n$, $(B(s_1t), \dots, B(s_nt)) \stackrel{d}{=} (t^{1/2}B(s_1), \dots, t^{1/2}B(s_n))$.

Exercise 11.2 (not from the course: time inversion). Prove that $W(t) = tB(1/t)$ ($t > 0$), $W(0) = 0$, is a standard Brownian motion (check that it is centred Gaussian with covariance $\min(s, t)$; for continuity at 0 use $B(u)/u \rightarrow 0$ as $u \rightarrow \infty$). Deduce that $B(t)/t \rightarrow 0$ as $t \rightarrow \infty$ from the continuity of W at 0.

11.2 Brownian motion as the limit of random walks

Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = 0$, $\text{Var } X_i = 1$, and $S_n = X_1 + \dots + X_n$ (section 4.2). By the central limit theorem (theorem 3.34), $\mathbb{P}[S_n/\sqrt{n} \leq c] \rightarrow \Phi(c)$, i.e. $\mathbb{P}[S_n \leq x] \approx \Phi(x/\sqrt{n})$ [► 2024 L14 @ 17:22]. To speak about a *path* rather than one time, rescale space by $\sqrt{n}$ and time by n .

Definition 11.5 (Normalised random walk). $B_n(t) = S_{\lfloor nt \rfloor} / \sqrt{n}$ for $t \geq 0$, where $\lfloor nt \rfloor$ is the largest integer $\leq nt$; the process jumps at $t = k/n$ by $X_k/\sqrt{n}$ and is constant in between [► 2024 L14 @ 19:29]. The piecewise-linear version, which joins the points $(k/n, S_k/\sqrt{n})$ (the version of the 2013 note, “connect the dots”), is continuous and differs from it by at most $\max_{k \leq nT} |X_k|/\sqrt{n}$ on $[0, T]$ [► 2013 L17 @ 14:56].

Theorem 11.6 (Donsker’s invariance principle). As $n \rightarrow \infty$ the process B_n converges in distribution (weakly, as a random element of the space of paths on $[0, T]$ with the uniform metric) to a standard Brownian motion B , whatever the distribution of the steps X_i [► 2024 L14 @ 20:44]. In particular, for every continuous functional F of the path (e.g. $\sup_{t \leq 1}$, $\int_0^1 (\cdot)^2 dt$, the value at time 1), $F(B_n) \xrightarrow{d} F(B)$.

2 COURSE NOTE: Erratum in the 2013 note

lecnote17 defines the piecewise-linear path by $Z(t/n) = Y_t$ for a walk with variance-one steps; the rescaling by $1/\sqrt{n}$ is missing (the values would be of order $\sqrt{n}$ and no limit could exist). The correct definition is $Z(t/n) = Y_t/\sqrt{n}$, as in the 2024 slides.

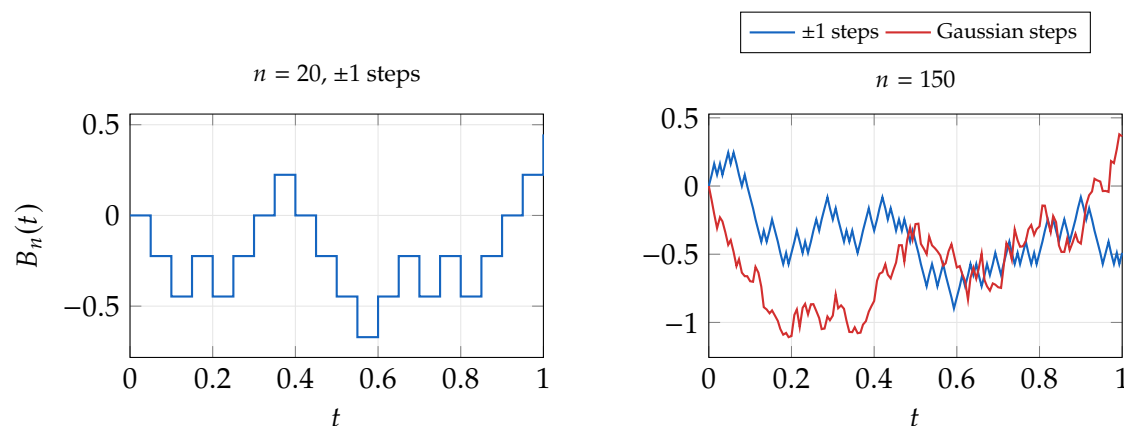


Figure 11.2: Normalised random walks $B_n(t) = S_{\lfloor nt \rfloor} / \sqrt{n}$ (two independent simulations, seeds fixed). Already at $n = 150$ the Bernoulli and the Gaussian walk are visually indistinguishable, the point of the animation lec14_4 (there up to $n = 2000$) [► 2024 L14 @ 21:46].

What is proved here. Finite-dimensional convergence. For $0 \leq t_1 < \dots < t_m$ the increments $B_n(t_j) - B_n(t_{j-1})$ are sums of $\lfloor nt_j \rfloor - \lfloor nt_{j-1} \rfloor \sim n(t_j - t_{j-1})$ i.i.d. steps divided by $\sqrt{n}$, over disjoint blocks, hence independent; by the central limit theorem each converges to $N(0, t_j - t_{j-1})$, and by independence jointly, which is exactly the joint law of the increments of B [► 2013 L17 @ 17:19]. This gives $\text{Var } B(t) = \lim_n \text{Var } B_n(t) = t$ and $\mathbb{E}[B(t)] = 0$, the mean and variance of the slides [► 2024 L14 @ 19:29]. *Tightness*, that no path of B_n oscillates wildly between grid times so that the limit is a continuous process, requires a maximal inequality (Kolmogorov or Doob) and is not proved in the course; it is what turns convergence of the finite-dimensional laws into convergence of laws on path space [► 2013 L17 @ 6:20]. □

INTUITION: Universality: the CLT as a fixed point

Only the first two moments of the steps survive in the limit; all the rest (skewness, kurtosis, lattice structure) is washed out at rate $1/\sqrt{n}$. This is the renormalisation-group picture: $S_n/\sqrt{n}$ is the coarse-graining of S_n by a factor n , and the Gaussian is the fixed point of this flow, with σ^2 the only relevant coupling. The consequence for modelling is what the lecture emphasised: one may use Brownian motion when the microscopic steps are non-Gaussian, provided they have finite variance and are not too dependent [► 2024 L14 @ 22:47]. The 2013 lecture gives the two physical examples: the pollen grain receives a huge number of tiny left/right pushes from water molecules (Einstein), and a stock price is pushed up or down by a huge number of tiny buy and sell orders [► 2013 L17 @ 17:19]. The limit fails when the variance is infinite (stable Lévy processes with jumps) or when the steps are strongly dependent (long memory, fractional Brownian motion).

IN FINANCE: Sampling frequency and the model

An analyst observes prices on a grid of spacing h : if the increments are like a random walk, the sampled process at fine resolution looks like Brownian motion (theorem 11.6 with $n \sim 1/h$) [► 2013 L17 @ 17:19]. Two warnings. At very short horizons (ticks) market microstructure (bid-ask bounce) makes the increments negatively correlated, so the limit is not Brownian; and price jumps have heavy tails, so “finite variance” can fail (section 3.9.1). Universality is a statement about the *long-horizon* behaviour of the sum of many small steps.

Where the theorem is used. Because $F(B_n) \rightarrow F(B)$ for continuous F , asymptotic distributions of statistics of random walks are the distributions of functionals of Brownian motion. For the simple random walk one even has an exact discrete analogue of the maximum formula of section 11.4: $\mathbb{P}[\max_{k \leq n} S_k \geq a] = 2\mathbb{P}[S_n > a] + \mathbb{P}[S_n = a]$ (checked exactly for $n = 100, a = 10$: both sides equal 0.3197), so $\mathbb{P}[\max_{k \leq n} S_k \geq a\sqrt{n}] \rightarrow 2(1 - \Phi(a))$, the Brownian result. Similarly $n^{-2} \sum_k S_{k-1}^2 \rightarrow \int_0^1 B(t)^2 dt$ and $n^{-1} \sum_k S_{k-1} X_k \rightarrow \int_0^1 B dB$, which is how the Dickey–Fuller limit of eq. (9.20) (section 9.4.2) is obtained; the Kolmogorov–Smirnov statistic of section 11.7.1 is another example.

11.3 Sample paths: roughness and quadratic variation

11.3.1 Size, zeros and the envelope

Almost every Brownian path has the following features (2013 lecture): it *crosses the t -axis infinitely often*, in fact it has zeros accumulating at $t = 0$ and at every later zero, and it neither drifts off nor stays near zero; at time t_0 it is $N(0, t_0)$, so it stays “around” the curves $\pm\sqrt{t}$ (typical size $\sqrt{t_0}$, never 100 times that in practice) [► 2013 L17 @ 25:48]. The precise version of “it does not deviate too much” is the law of the iterated logarithm (not proved in the course): $\limsup_{t \rightarrow \infty} B(t)/\sqrt{2t \ln \ln t} = 1$ almost surely, so the $\sqrt{t}$ envelope is exceeded infinitely often, but only by slowly growing factors (compare [section 4.2](#)).

11.3.2 Non-differentiability

The path is continuous by definition, but it is nowhere differentiable, which was “really surprising” in the words of the 2013 lecture, and matters because the whole of classical calculus is built on derivatives [► 2013 L17 @ 27:48].

Proof of the first statement. The heuristic of the slides [► 2024 L14 @ 1:02:33]: the difference quotient $D(\Delta t) = [B(t_0 + \Delta t) - B(t_0)]/\Delta t$ is $N(0, 1/\Delta t)$, so for every fixed $M < \infty$, $\mathbb{P}(|D| \leq M) = \mathbb{P}(|N(0, 1)| \leq M\sqrt{\Delta t}) \leq M\sqrt{2\Delta t/\pi} \rightarrow 0$: the slope does not settle down; it is of order $(\Delta t)^{-1/2}$ (equivalently $\Delta B = O_P(\sqrt{\Delta t}) \gg \Delta t$). To make this a proof, note that a finite derivative at t_0 forces, for some integers K, m , $|B(t_0 + s) - B(t_0)| \leq Ks$ for all $0 < s < 1/m$; the event $E_{K,m}$ so defined satisfies $\mathbb{P}(E_{K,m}) \leq \mathbb{P}(|B(t_0 + \delta) - B(t_0)| \leq K\delta) \leq K\sqrt{2\delta/\pi}$ for every $\delta < 1/m$, hence $\mathbb{P}(E_{K,m}) = 0$; take the countable union over K, m [► 2013 L17 @ 43:15]. The statement for all t_0 simultaneously needs a covering argument and is quoted. □

Proposition 11.7 (No derivative). Fix $t_0 \geq 0$. With probability one, B is not differentiable at t_0 . In fact, with probability one there is no t_0 at which B is differentiable (Paley–Wiener–Zygmund, 1933).

11.3.3 Quadratic variation

Let $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$ be a partition of $[0, T]$ with mesh $|\Pi| = \max_j(t_j - t_{j-1})$ and define the Π -quadratic variation [► 2024 L14 @ 1:03:39]

$$QV_\Pi([0, T]) = \sum_{j=1}^n [B(t_j) - B(t_{j-1})]^2.$$

3 COURSE NOTE: Erratum in the 2013 proof of non-differentiability

The written proof (Lecnote17, Proposition 2.4) bounds the event by $\mathbb{P}(M(\delta) \leq A\delta)$ for the maximum $M(\delta) = \max_{s \leq \delta} B(s)$, and computes it as “ $2(1 - \Phi(A\sqrt{\delta}))$ ”, which tends to zero”. That value tends to 1, not 0. By [proposition 11.11](#) the correct value is $\mathbb{P}(M(\delta) \leq A\delta) = 2\Phi(A\sqrt{\delta}) - 1$, which does tend to 0; the lecture itself stumbles over the direction of the inequality at this point [► 2013 L17 @ 46:23]. Both versions are statements for a fixed t_0 ; the slide of 2024 (“not differentiable at t , for any $t \geq 0$ ”) is likewise a statement for each fixed t . The argument above avoids the maximum altogether.

Proof. Write $\Delta_j = t_j - t_{j-1}$ and $\Delta_j B = B(t_j) - B(t_{j-1}) \sim N(0, \Delta_j)$, independent over j . Then $\mathbb{E}[(\Delta_j B)^2] = \Delta_j$, $\text{Var}[(\Delta_j B)^2] = 3\Delta_j^2 - \Delta_j^2 = 2\Delta_j^2$

(the fourth moment of $N(0, \Delta_j)$ is $3\Delta_j^2$), so

$$\mathbb{E}[Q]V_\Pi = \sum_j \Delta_j = T, \quad \text{Var } QV_\Pi = 2 \sum_j \Delta_j^2 \leq 2|\Pi| \sum_j \Delta_j = 2T|\Pi|$$

For equally spaced points $QV_\Pi \stackrel{d}{=} (T/n)\chi_n^2$ [► 2024 L14 @ 1:08:03], mean T and variance $2T^2/n$. Chebyshev's inequality gives convergence in probability; if $\sum_k |\Pi_k| < \infty$ then $\sum_k \mathbb{P}(|QV_{\Pi_k} - T| > \varepsilon) \leq \sum_k 2|\Pi_k|T/\varepsilon^2 < \infty$ and Borel–Cantelli gives almost sure convergence. $\square$

Theorem 11.8 (Quadratic variation of Brownian motion). $QV_\Pi([0, T]) \rightarrow T$ as $|\Pi| \rightarrow 0$ in L^2 (hence in probability), and almost surely along any sequence of partitions with $\sum_k |\Pi_k| < \infty$, for instance the dyadic partitions [► 2024 L14 @ 1:06:57][► 2013 L17 @ 47:24].

The theorem says that the *sum of squared increments is not random and does not depend on the path*: $[B]_T = T$ for almost every path [► 2024 L14 @ 1:06:57]. Compare a continuously differentiable function f : by the mean value theorem $\sum_j (f(t_j) - f(t_{j-1}))^2 = \sum_j f'(s_j)^2 \Delta_j^2 \leq \max |f'|^2 |\Pi| T \rightarrow 0$, so its quadratic variation is zero [► 2013 L17 @ 48:29]. Brownian paths “bounce back and forth too much”: the squared moves do not vanish as the grid is refined. Two corollaries.

Corollary 11.9 (Infinite total variation). Almost surely $\sum_j |B(t_j) - B(t_{j-1})| \rightarrow \infty$ as $|\Pi| \rightarrow 0$.

So one cannot define $\int f dB$ pathwise as a Riemann–Stieltjes integral, and the stochastic (Itô) integral of chapter 17 is needed.

The rule of thumb is written

$$(dB)^2 = dt, \quad dB \, dt = 0, \quad (dt)^2 = 0, \quad (11.3)$$

where $dB \sim \sqrt{dt}$ carries the whole effect at second order: the increments of a Brownian path are of order $\sqrt{\Delta t}$, their squares of order Δt (summing to a finite number), while a product $\Delta B \, \Delta t$ is $O(\Delta t^{3/2})$ and vanishes on summation [► 2013 L17 @ 53:49]. This is the reason

4 COURSE NOTE: Erratum and remarks on the proof in the notes and slides

- The 2013 proof invokes the law of large numbers for the sum of n i.i.d. variables $(B(t_{i+1}) - B(t_i))^2$ of mean T/n ; the summands change with n (a triangular array), so the classical law does not apply verbatim, and the mean and variance computation above (Chebyshev) is the clean way; the quantity converges in L^2 and, for nested partitions, almost surely. The lecture's own computation had a slip corrected by a student (the mean of X_i^2 is T/n , not T^2/n^2) [► 2013 L17 @ 56:07].
- Slide 19 of lec14_1 (PDF pages 73–74) says “ $QV([0, T]) = 1$ for virtually every path”; the value is T (it equals 1 only for $T = 1$), as the claim on the same slide states.
- On the slide of the Brownian bridge (slide 13, section 11.7.1) “Moments of $B(t)$ ” should read “moments of $X(t)$ ”.
- Slide 2 dates Einstein's diffusion paper to 1904; it appeared in 1905 (as in the 2013 note and in our Motivation).
- The diffusion equation on slide 6 (PDF page 15) is written $\partial_t p = \frac{1}{2} \partial_y^2 p$, which is the case $\sigma = 1$ announced there (“easy to verify for $s = 0$ and $\sigma = 1$ ”); for a general diffusion coefficient the factor is $\frac{1}{2} \sigma^2$, proposition 11.4.

Proof. $\sum_j \Delta_j B^2 \leq \max_j |\Delta_j B| \cdot \sum_j |\Delta_j B|$. The left side tends to $T > 0$ and, by uniform continuity of the path, $\max_j |\Delta_j B| \rightarrow 0$; hence the sum of absolute increments diverges. $\square$

why the Taylor expansion of a function of B has a second-order term (section 11.6).

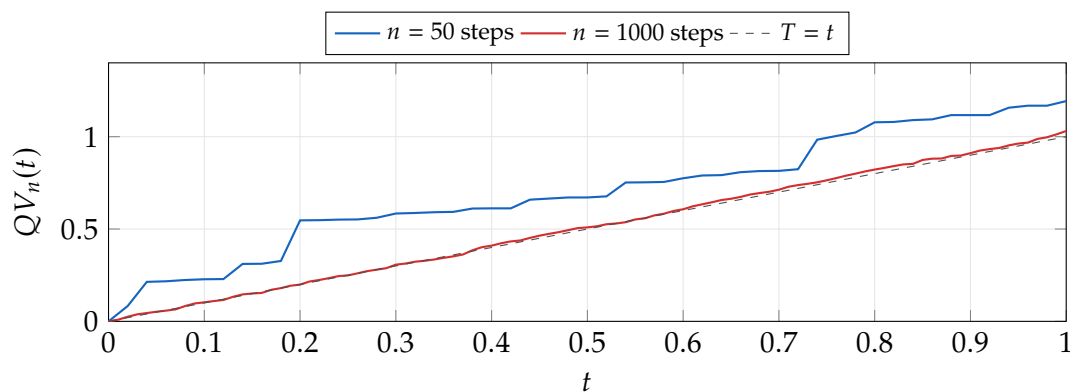


Figure 11.3: Cumulative quadratic variation $QV_n(t) = \sum_{k \leq nt} (\Delta B_k)^2$ of a Gaussian random walk $\Delta B_k = X_k/\sqrt{n}$, at two resolutions. At $t = 1$ the values are 1.19 and 1.03 (theory: standard deviation $\sqrt{2/n}$, i.e. 0.20 and 0.045); the path converges to the straight line t , as in lec14_9 (there n up to 2000) [► 2024 L14 @ 1:08:03].

IN FINANCE: Realised variance and stochastic volatility

If the log price is $\log(P_t/P_0) = \mu t + \sigma B(t)$, the sum of squared log returns over a period $[0, T]$ (the *realised variance*) estimates $\sigma^2 T$ *without knowing the drift and without error as the sampling becomes finer*: with n returns its relative standard deviation is $\sqrt{2/n}$ (about 8.9% for $n = 252$ daily returns of one year, so $8.9\%/2 \approx 4.5\%$ in volatility terms). This is the theoretical basis for measuring volatility by high-frequency data (chapter 13). The lecture proposes plotting the cumulative squared returns of an asset against time: for a Brownian model the graph is a straight line of slope σ^2 ; slope changes and curvature reveal periods of low and high volatility and suggest a model of dynamic volatility [► 2024 L14 @ 1:10:21]. Drift, by contrast, cannot be learned by refining the sampling: $\hat{\mu}$ has standard error $\sigma/\sqrt{T}$ whatever n is (2024 PS4 Problem 3, chapter 13).

11.4 The reflection principle, the maximum and hitting times

Suppose Brownian motion models a stock price during a trading day (9:30 to 16:00). Traders care about the *highest and lowest* values reached, not only the closing value [► 2013 L17 @ 31:00]: barrier options pay if a level is touched, a stop-loss triggers when it is reached, a firm defaults when its value hits a threshold. The two objects are the running maximum and the first hitting time. For a standard Brownian motion B and levels $a > 0$ define

$$M(t) = \max_{0 \leq u \leq t} B(u), \quad \tau_a = \min\{u \geq 0 : B(u) = a\}. \quad (11.4)$$

The maximum exists because a continuous function attains its maximum on the compact set $[0, t]$ [► 2024 L14 @ 28:05][► 2013 L17 @ 31:00]. Since the path is continuous and starts at $0 < a$, hitting a before t and having a maximum $\geq a$ are the same event:

$$\{\tau_a \leq t\} = \{M(t) \geq a\}. \quad (11.5)$$

τ_a is a *stopping time* (definition 4.22): whether $\tau_a \leq t$ is decided by the path up to time t [► 2024 L14 @ 24:53][► 2013 L17 @ 33:04].

Lemma 11.10 (Reflection principle). Let $\tilde{B}(u) = B(u)$ for $u \leq \tau_a$ and $\tilde{B}(u) = a - (B(u) - a) = 2a - B(u)$ for $u > \tau_a$: the path is reflected in the level a from the first hitting time on. Then $\tilde{B}$ is again a standard Brownian motion [► 2024 L14 @ 24:53].

The proof uses the *strong Markov property*: at the stopping time τ_a the process $B(\tau_a + u) - B(\tau_a)$ is a new Brownian motion independent of the past $\mathcal{F}_{\tau_a}$; the Markov property of [proposition 11.3\(ii\)](#) still holds when the fixed time s is replaced by a stopping time, because τ_a can be approximated by discrete stopping times ([theorem 4.25](#)) (2013 note: “the distribution of $B(t) - B(\tau_a)$ is not affected by the fact that we conditioned on $\tau_a < t$ ”) [► 2013 L17 @ 35:54]. Since $-B$ is also a Brownian motion, gluing the old path up to τ_a to the reflected new increment $-(B(\tau_a + u) - a)$ gives a process with the same law. The lemma is the statement that after touching the level the path is equally likely to go up or down [► 2024 L14 @ 24:53]; [fig. 11.4](#) shows one path and its reflection (the deck lec14_5 shows it for $x = 50$ and 10 000 steps).

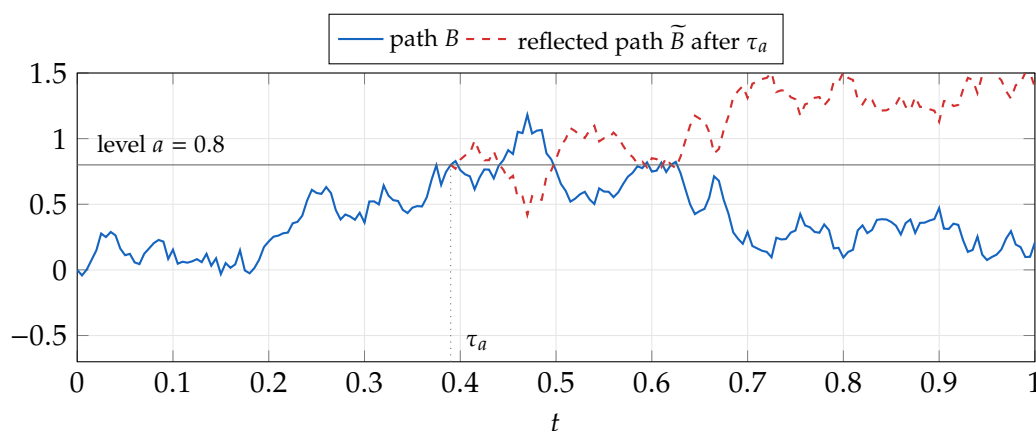


Figure 11.4: A simulated Brownian path (200 steps, $a = 0.8$) and its reflection in the level a after the first hitting time τ_a . The two paths have the same probability; the original ends below a , its reflection above.

Proposition 11.11 (Distribution of the maximum). For $t > 0$ and $a > 0$,

$$\mathbb{P}[M(t) \geq a] = 2\mathbb{P}[B(t) > a] = 2\left[1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right]. \quad (11.6)$$

Equivalently $M(t) \stackrel{d}{=} |B(t)|$, with density $\sqrt{2/(\pi t)} e^{-m^2/(2t)}$ on $m > 0$, and $\mathbb{E}[M(t)] = \sqrt{2t/\pi}$.

Remark 11.12 (Joint law of maximum and endpoint; not discussed in class). The same reflection gives, for $m > 0$ and $x \leq m$, $\mathbb{P}[M(t) \geq m, B(t) \leq x] = \mathbb{P}[B(t) \geq 2m - x] = 1 - \Phi((2m - x)/\sqrt{t})$ (reflect the part of the path after τ_m : the event $\{B(t) \leq x\}$ becomes $\{\tilde{B}(t) \geq 2m - x\}$). For $x = m$ this is (11.6). By symmetry, the minimum satisfies $\mathbb{P}[\min_{u \leq t} B(u) \leq -a] = 2[1 - \Phi(a/\sqrt{t})]$ as well.

Proof. Two views, both used in the lectures. **Path pairing** [► 2024 L14 @ 28:05]. Let $A_{\text{END}} = \{B(t) > a\}$ and $A_{\text{MAX}} = \{M(t) \geq a\}$. By continuity $A_{\text{END}} \subset A_{\text{MAX}}$ (a path ending above a has crossed it, so $\tau_a < t$). Every $\omega_0 \in A_{\text{END}}$ is paired with its reflection ω_1 ([lemma 11.10](#)), which ends below a but has still maximum $\geq a$, and has the same probability; conversely every path in $A_{\text{MAX}} \setminus A_{\text{END}}$ arises this way. Hence $\mathbb{P}[A_{\text{MAX}}] = \mathbb{P}[A_{\text{MAX}} \cap A_{\text{END}}] + \mathbb{P}[A_{\text{MAX}} \cap A_{\text{END}}^c] = \mathbb{P}[A_{\text{END}}] + \mathbb{P}[A_{\text{END}}]$ (the event $B(t) = a$ has probability 0). **Conditioning on τ_a** [► 2013 L17 @ 34:09]. Given $\tau_a < t$, the strong Markov property says $B(t) - B(\tau_a) = B(t) - a$ is symmetric, so $\mathbb{P}[B(t) > a \mid \tau_a < t] = \mathbb{P}[B(t) < a \mid \tau_a < t] = \frac{1}{2}$. Therefore $\mathbb{P}[M(t) \geq a] = \mathbb{P}[\tau_a < t] = 2\mathbb{P}[B(t) > a, \tau_a < t] = 2\mathbb{P}[B(t) > a]$, the last step because $B(t) > a$ already implies $\tau_a < t$. The rest follows from $\mathbb{P}[B(t) > a] = 1 - \Phi(a/\sqrt{t})$, by differentiating in a , and by $\mathbb{E}[M] = \int_0^\infty \mathbb{P}[M > a] da$. □

IN FINANCE: Barrier events: a path is twice as likely to touch a level as to end beyond it

Let $\log(S_t/S_0) = \sigma B(t)$ (no drift, $\sigma = 30\%$ per year). The probability that the price is below $S_0 e^{-0.2} \approx 0.82 S_0$ at the end of one year is $\Phi(-0.2/0.3) = 25.2\%$; the probability that the stock *touches* that level at some time during the year is $2\Phi(-0.2/0.3) = 50.5\%$ ([proposition 11.11](#) applied to $-B$). For a knock-out or knock-in barrier option, or a stop-loss, the touching probability is what matters, and “double the terminal probability” is the first-order rule; a drift or discrete (say daily) monitoring changes the number, the latter by lowering it (a path can cross between two observations without being seen) [[2013 L17 @ 31:00](#)]. Range-based volatility estimators combine the maximum and the minimum of the log price over a day: with the range $R = \max - \min$ of a driftless Brownian path one has $\mathbb{E}[R^2] = 4 \ln 2 \cdot \sigma^2 t$ (Parkinson 1980, derived from the joint law of [remark 11.12](#); it is the subject of 2024 PS5 Problem 3 and of [chapter 13](#)); in the 2013 lecture Kempthorne, from the audience, remarks that he uses the range of the process as a more precise measure of volatility than close-to-close prices [[2013 L17 @ 39:05](#)].

Exercise 11.3 (not from the course: reflection and the minimum). Let $m(t) = \min_{u \leq t} B(u)$. Show $\mathbb{P}[m(t) \leq -a] = 2\mathbb{P}[B(t) < -a]$ and that $M(t) - B(t) \stackrel{d}{=} M(t) \stackrel{d}{=} |B(t)|$ at a fixed time t (hint: use [remark 11.12](#) or the time-reversed path $B(t-u) - B(t)$). What is the probability that a driftless log price with $\sigma = 25\%$ never falls below -10% during one year?

11.4.1 First hitting time

By (11.5) and [proposition 11.11](#),

$$\mathbb{P}[\tau_a \leq t] = 2 \left[1 - \Phi \left(\frac{a}{\sqrt{t}} \right) \right], \quad f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi}} t^{-3/2} e^{-a^2/(2t)}, \quad 0 < t < \infty, \quad (11.7)$$

the density being the derivative in t (using $\frac{d}{dt} \Phi(at^{-1/2}) = -\frac{1}{2} at^{-3/2} \varphi(at^{-1/2})$) [[2024 L14 @ 31:35](#)]. This is the Lévy distribution.

Proposition 11.13 (Properties of τ_a).

- (i) $\tau_a < \infty$ almost surely: a Brownian path reaches every level.
- (ii) $\mathbb{E}[\tau_a] = \infty$: the tail is $\mathbb{P}[\tau_a > t] = 2\Phi(a/\sqrt{t}) - 1 \sim a\sqrt{2/(\pi t)}$ and $\int^\infty t^{-1/2} dt = \infty$.
- (iii) The density is unimodal with mode at $t = a^2/3$, and its median is $a^2/(\Phi^{-1}(3/4))^2 \approx 2.198 a^2$.
- (iv) Scaling: $\tau_a \stackrel{d}{=} a^2 \tau_1$.

Proof. (i) $\mathbb{P}[\tau_a \leq t] \rightarrow 2[1 - \Phi(0)] = 1$ as $t \rightarrow \infty$.
(ii) as stated.
(iii) $\frac{d}{dt} \ln f = -\frac{3}{2t} + \frac{a^2}{2t^2} = 0$ at $t = a^2/3$; the median solves $2[1 - \Phi(a/\sqrt{t})] = \frac{1}{2}$, i.e. $a/\sqrt{t} = \Phi^{-1}(3/4) = 0.6745$.
(iv) from (11.7), which depends on $a/\sqrt{t}$ only. $\square$

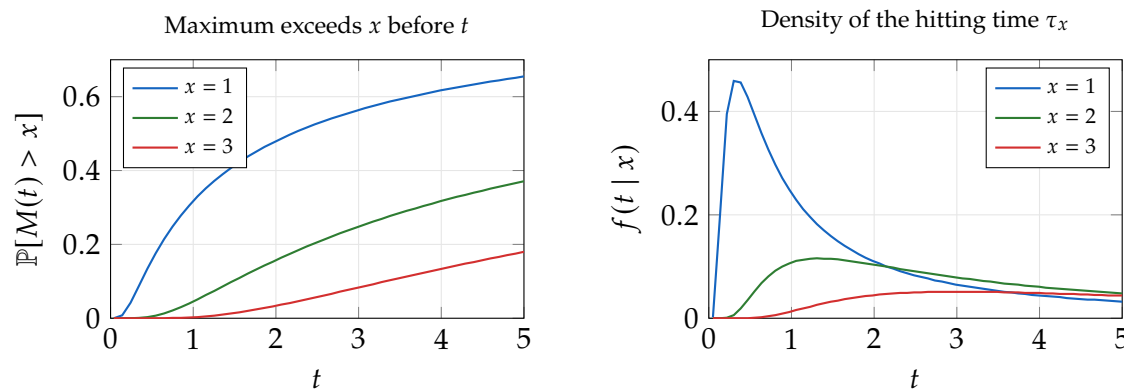


Figure 11.5: Left: $\mathbb{P}[M(t) > x] = 2[1 - \Phi(x/\sqrt{t})]$ (file 1ec14_6); at $t = 5$ the values are 0.655, 0.371, 0.180. Right: the hitting-time density (11.7) (file 1ec14_7); the peak of $x = 1$ is at $t = 1/3$ with height 0.463, and levels farther away are reached later, with a much more spread-out distribution ▶ 2024 L14 @ 31:35].

INTUITION: A path reaches everything, but takes an infinite mean time

Recurrence and infinite mean go together: Brownian motion in one dimension reaches every level with probability one, but the waiting time has the heavy tail $t^{-1/2}$. It is the continuous version of the simple random walk, which returns to 0 a.s. while the expected return time is infinite (section 4.6, theorem 4.26: optional stopping fails for τ_a because $B(\tau_a) = a \neq 0 = \mathbb{E}[B(0)]$ and the stopped process is not uniformly bounded). The scaling $\tau_a \stackrel{d}{=} a^2 \tau_1$ is diffusion again: time to travel a grows like a^2 .

Exercise 11.4 (not from the course: a hitting time with infinite mean). Show $\mathbb{P}[\tau_a < \infty] = 1$ and $\mathbb{E}[\tau_a] = \infty$ from (11.7). Why does optional stopping ($\mathbb{E}[B(\tau_a)] = 0$) fail? Use $\tau_a \stackrel{d}{=} a^2 \tau_1$ to show that the median of τ_a scales like a^2 ; compute the median of τ_1 .

11.4.2 Reflected and absorbed Brownian motion

Two modifications of B appear in the lecture ▶ 2024 L14 @ 33:44].

Definition 11.14 (Reflected and absorbed Brownian motion). *Reflected Brownian motion* at zero is $R(t) = |B(t)|$ (the path is folded onto the positive half line). *Absorbed Brownian motion* started at $x > 0$ is $A(t) = x + B(t)$ for $t \leq \tau$ and $A(t) = 0$ for $t > \tau$, where τ is the first hitting time of 0.

Proposition 11.15 (Moments of R and law of A).

- (i) $\mathbb{E}[R(t)] = \sqrt{2t/\pi}$ and $\text{Var } R(t) = (1 - 2/\pi)t$.
- (ii) The law of $A(t)$ has an atom at 0 of mass $2[1 - \Phi(x/\sqrt{t})]$ (the probability of having been absorbed), and on $y > 0$ the density

$$p_A(y, t | x) = p(y, t | x) - p(-y, t | x) = \frac{1}{\sqrt{2\pi t}} \left[e^{-(y-x)^2/2t} - e^{-(y+x)^2/2t} \right]. \quad (11.8)$$

Check: $\int_0^\infty p_A dy = \Phi(x/\sqrt{t}) - [1 - \Phi(x/\sqrt{t})] = 1 - 2[1 - \Phi(x/\sqrt{t})]$, so density plus atom

Proof. (i) $R(t) \stackrel{d}{=} M(t)$ by proposition 11.11, so $\mathbb{E}[R] = \sqrt{2t/\pi}$; $\mathbb{E}[R^2] = \mathbb{E}[B^2] = t$ gives the variance.
(ii) Absorption at 0 from x is the hitting of the level $-x$ by B from 0, so $\mathbb{P}[\tau \leq t] = 2[1 - \Phi(x/\sqrt{t})]$ by (11.7). For $y > 0$, $\mathbb{P}[x + B(t) \in dy, \tau \leq t]$ is, by reflection in 0, the density of a path ending at $-y$, that is $p(-y, t | x)$; subtracting this from the free density gives (11.8) (the *method of images*, with an image charge of opposite sign at $-x$). ◻

sum to 1; and $\mathbb{E}[A(t)] = x$ for all t (absorbed Brownian motion is the martingale $B(t \wedge \tau)$, the loss of mass to 0 being compensated by the surviving paths). *Not to be confused:* the slides use the word “reflected” both for the folded process $|B|$ and, in the section on the reflection principle, for the path reflected *in a level after the hitting time*; only the second is used in [lemma 11.10](#).

IN FINANCE: A price that can go bankrupt

The lecture proposes absorbed Brownian motion as the price of an asset which is positive as long as the firm is viable, but stays at 0 once bankrupt [[▶ 2024 L14 @ 34:50](#)]. Structural credit models (Merton, Black–Cox; cf. [chapter 8](#)) do exactly this: the probability of default before t is a hitting probability, and the survival probability with σ -volatile assets started x above the barrier is $2\Phi(x/(\sigma\sqrt{t})) - 1$, a straightforward application of (11.7).

Exercise 11.5 (not from the course: absorbed Brownian motion). For absorbed Brownian motion A started at $x > 0$ ([definition 11.14](#)) show that (11.8) and the mass $2[1 - \Phi(x/\sqrt{t})]$ at 0 sum to 1, and that $\mathbb{E}[A(t)] = x$. Is $A(t)^2 - t \wedge \tau$ a martingale?

11.5 Brownian motion with drift and the gambler’s ruin problem

Definition 11.16 (Brownian motion with drift). $X(t) = \mu t + \sigma B(t)$ for a standard Brownian motion B , a drift $\mu \in \mathbb{R}$ and a volatility $\sigma > 0$ [[▶ 2024 L14 @ 43:19](#)] [[▶ 2013 L17 @ 47:24](#)]. X has continuous paths, independent increments and $X(t) - X(s) \sim N(\mu(t-s), \sigma^2(t-s))$, so $\mathbb{E}[X(t)] = \mu t$ and $\text{Var } X(t) = \sigma^2 t$; it reduces to B for $\mu = 0, \sigma = 1$.

Proposition 11.17 (Drift wins in the long run). For every $\varepsilon > 0$, with probability tending to one $(\mu - \varepsilon)t \leq X(t) \leq (\mu + \varepsilon)t$ as $t \rightarrow \infty$; indeed $X(t)/t \rightarrow \mu$ (almost surely). Yet at moderate times the noise $\sigma\sqrt{t}$ dominates the drift μt as long as $t < (\sigma/\mu)^2$.

Proof. $X(t)/t - \mu = \sigma B(t)/t$ has variance $\sigma^2/t \rightarrow 0$ (convergence in probability; the almost sure version is the law of large numbers for B). The two terms μt and $\sigma\sqrt{t}$ are equal at $t = (\sigma/\mu)^2$. $\square$

The last remark of the proposition is the reason why drifts are so hard to estimate: for a stock with $\mu = 10\%$ and $\sigma = 30\%$ the time scale is $(\sigma/\mu)^2 = 9$ years, and for it the estimate $\hat{\mu} = X(T)/T$ has standard error $\sigma/\sqrt{T}$ ([chapter 13](#)).

One-step analysis. Look at the increment $\Delta X = X(t + \Delta t) - X(t) = \mu\Delta t + \sigma\Delta B$ with $\Delta B \sim N(0, \Delta t)$ independent of the past. Then

$$\mathbb{E}[\Delta]X = \mu \Delta t, \quad \mathbb{E}[(\Delta X)^2] = \sigma^2 \Delta t + (\mu \Delta t)^2 = \sigma^2 \Delta t + o(\Delta t), \quad \mathbb{E}[|\Delta X|^c] = o(\Delta t) \quad (c > 2) \quad (11.9)$$

[[▶ 2024 L14 @ 44:20](#)]: to order Δt , only the mean and the variance of the increment matter. If f is twice differentiable, Taylor’s formula and (11.9) give

$$\mathbb{E}[f(x + \Delta X)] = f(x) + \left(\mu f'(x) + \frac{1}{2} \sigma^2 f''(x) \right) \Delta t + o(\Delta t), \quad (11.10)$$

so $\mathcal{L}f = \mu f' + \frac{1}{2} \sigma^2 f''$ is the *generator* of the process. This expansion is why the lecture cares about the powers of the increment: the value of a derivative on an underlying X is a function $f(X)$, and its dynamics are read from the Taylor series [[▶ 2024 L14 @ 47:36](#)]. It is the seed of Itô’s lemma ([section 11.6](#)).

11.5.1 Gambler's ruin for Brownian motion

Let X be a Brownian motion with drift μ and volatility σ , started at $x \in (a, b)$, and let $\tau = \min\{u : X(u) = a \text{ or } b\}$. Find

$$u(x) = \mathbb{P}[X(\tau) = b \mid X(0) = x],$$

the probability of reaching the upper level first. This generalises the random walk of [section 4.6](#) to continuous time and continuous state [► 2024 L14 @ 48:45](#).

Infinitesimal derivation (the lecture). Wait a short time Δt so small that hitting a level in the meantime is negligible. By the Markov property, $u(x) = \mathbb{E}[u(x + \Delta X)]$; expanding with [\(11.10\)](#),

$$\begin{aligned} u(x) &= u(x) + (\mu u'(x) + \tfrac{1}{2}\sigma^2 u''(x))\Delta t + o(\Delta t) \\ \implies \mu u'(x) + \tfrac{1}{2}\sigma^2 u''(x) &= 0, \quad u(a) = 0, \quad u(b) = 1 \end{aligned}$$

[► 2024 L14 @ 51:55](#). This is a first-order equation in $v = u'$: $v'/v = -2\mu/\sigma^2$, so $u'(x) = C e^{-2\mu x/\sigma^2}$ and

$$u(x) = \frac{g(x) - g(a)}{g(b) - g(a)}, \quad g(x) = e^{-2\mu x/\sigma^2} \quad (\mu \neq 0); \quad u(x) = \frac{x - a}{b - a} \quad (\mu = 0) \quad (11.11)$$

[► 2024 L14 @ 56:16](#). For $\mu = 0$, $g(x) = x$ is the other solution of the equation (the general solution is $Ax + B$); with no drift, the probability of success is linear in the distance travelled, exactly as for the fair gambler ([proposition 4.28](#)).

Rigorous derivation by a martingale. The infinitesimal argument assumes that u is twice differentiable, which one does not want to take for granted. The proof by optional stopping needs nothing of the kind. By [proposition 11.3\(iii\)](#) with $\lambda = \theta\sigma$, the process $e^{\theta X(t)} = e^{\theta\mu t + \theta\sigma B(t)}$ is a martingale iff $\theta\mu = -\theta^2\sigma^2/2$, i.e. $\theta = -2\mu/\sigma^2$. Hence $M(t) = g(X(t)) = e^{-2\mu X(t)/\sigma^2}$ is a martingale, exactly the continuous analogue of r^{S_n} with $r = q/p$ of [example 4.18](#). The stopped martingale $M(t \wedge \tau)$ is bounded, and $\tau < \infty$ a.s., so optional stopping ([theorem 4.26\(ii\)](#)) gives $g(x) = \mathbb{E}[g(X(\tau))] = u g(b) + (1 - u) g(a)$, which is [\(11.11\)](#). It is also the exact limit of the discrete formula [proposition 4.30](#): for steps $\pm\sigma\sqrt{h}$ with $p = \frac{1}{2}(1 + \mu\sqrt{h}/\sigma)$, one has $r = q/p \approx e^{-2\mu\sqrt{h}/\sigma}$, and $r^{S_n} \rightarrow e^{-2\mu X/\sigma^2}$. A numerical check (not in the course): for $\mu = 0.5$, $\sigma = 1$, $a = 0$, $b = 3$, $x = 1$ formula [\(11.11\)](#) gives $u = 0.66524$, and the exact ruin probability of the biased walk with $h = 10^{-4}$ is 0.66524.

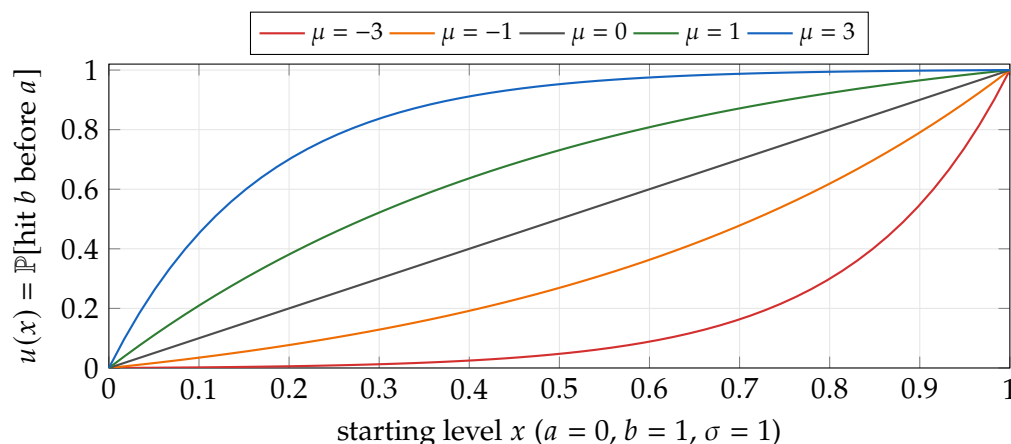


Figure 11.6: Probability of hitting $b = 1$ before $a = 0$ as a function of the starting point, for several drifts (the four panels of 1ec14_8 show $b = 3$ and $b = 1$ with $\mu = \pm(10^{-4}, 0.5, \dots, 3)$). With no drift the probability is the straight line x ; a positive drift bends the curve above it (the barrier b is reached first from almost everywhere), a negative drift below it
▶ 2024 L14 @ 1:00:20].

Proposition 11.18 (Consequences of (11.11); not discussed in class).

- (i) Symmetry: $u_\mu(x) + u_{-\mu}(a + b - x) = 1$.
- (ii) If $\mu > 0$, the probability that X ever falls to a level d below its start is $e^{-2\mu d/\sigma^2}$, so the running minimum of X is exponentially distributed with rate $2\mu/\sigma^2$; if $\mu \leq 0$ every level below is reached with probability one.
- (iii) If $\mu = 0$, $\mathbb{E}[\tau] = (x - a)(b - x)/\sigma^2$.

Proof. (i) reflect the interval, $x \mapsto a + b - x$, which changes the sign of the drift and swaps the two barriers.

(ii) Let $b \rightarrow \infty$ in $1 - u = (g(b) - g(x))/(g(b) - g(a))$: for $\mu > 0$, $g(b) \rightarrow 0$ and the limit is $g(x)/g(a) = e^{-2\mu(x-a)/\sigma^2}$; for $\mu < 0$, $g(b) \rightarrow \infty$ and the limit is 1.

(iii) $X(t)^2 - \sigma^2 t$ is a martingale (proposition 11.3(iii)); optional stopping gives $\sigma^2 \mathbb{E}[\tau] = \mathbb{E}[X(\tau)^2] - x^2 = \frac{(b-x)a^2 + (x-a)b^2}{b-a} - x^2 = (x-a)(b-x)$. $\square$

IN FINANCE: How likely is a losing streak? (numbers not from the course)

Take the parameters used in 2024 PS5 for a simulation, $\mu = 0.30$ and $\sigma = 0.40$, for the *log* value of a strategy (the drift already includes any $-\sigma^2/2$ correction). The probability that its value ever falls by 0.2 (about 18% in level terms) below its running starting point, however long one waits, is $e^{-2 \cdot 0.30 \cdot 0.2 / 0.16} = e^{-0.75} = 47\%$, even though the strategy has a positive expected return of 30% per year. Positive drift protects in the long run but not from drawdowns: the minimum of the path is exponential with mean $\sigma^2/(2\mu) = 0.27$.

Exercise 11.6 (PS4 (2024), Problem 1; PS8 (2013), Problem A-2: conditional moments).

- (a) Let $B(t)$ be a standard Brownian motion and $t > s \geq 0$ fixed. Compute $\mathbb{E}[B(t) | B(s)]$ and $\text{Var}[B(t) | B(s)]$.
- (b) Let $X(t)$ be a Brownian motion with drift μ and volatility $\sigma = 2$ (the 2013 version has no volatility). Compute $\mathbb{E}[X(t) | X(s)]$ and $\text{Var}[X(t) | X(s)]$.
- (c) For a standard Brownian motion and fixed reals $t > s > 0$, compute $\mathbb{E}[\exp(\sigma(B(t) - B(s)))]$, for a real constant σ (the multiplier of the increment inside the exponential).

Exercise 11.7 (not from the course: exit time and ruin). For $X = \mu t + \sigma B$ with $\mu \neq 0$ compute $\mathbb{E}[\tau]$ for the exit time of (a, b) (hint: the martingale $X(t) - \mu t$ and (11.11)). Check that the limit $\mu \rightarrow 0$ is $(x - a)(b - x)/\sigma^2$.

11.6 Geometric Brownian motion and the first look at Itô's lemma

The price of a stock cannot be $S_0 + \sigma B_t$: it would become negative. What is reasonable, the lecture argues, is that the *percentage* change is normal, $\Delta S/S = \mu \Delta t + \sigma \Delta B$, written $dS_t = S_t(\mu dt + \sigma dB_t)$ [► 2013 L17 @ 1:00:19]. The natural question, asked in the 2013 lecture, is whether the solution of $dS = S dB$ is $S_t = e^{B_t}$, as in classical calculus; the answer is no [► 2013 L17 @ 1:01:24]. Here is why.

The Taylor argument. Let f be a smooth function (in the 2013 lecture a call payoff, $f(B_T) = (B_T - s_0)^+$, or the price of any derivative on the underlying) and B a Brownian motion [► 2013 L17 @ 1:03:29]. If B were differentiable, $df(B_t) = f'(B_t) \frac{dB}{dt} dt$; it is not [► 2013 L17 @ 1:06:36], so one keeps the increment dB and writes Taylor's formula $f(B_{t+x}) - f(B_t) = f'(B_t) \Delta B + \frac{1}{2} f''(B_t) \Delta B^2 + \dots$ with $\Delta B = B_{t+x} - B_t$ [► 2013 L17 @ 1:07:39]. In calculus the term ΔB^2 is negligible; here, by (11.3), $\Delta B^2 \approx \Delta t$ is of the same order as the time step, so it cannot be dropped:

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt \quad (\text{Itô's lemma, heuristic form}) \quad (11.12)$$

[► 2013 L17 @ 1:11:55]. The lecturer's summary: if one thing is to be remembered it is (11.12), if two, that $(dB)^2 = dt$ [► 2013 L17 @ 1:14:05]. Proofs and generalisations (functions of t and X , stochastic integrals) are in chapter 17.

Application to the exponential. With $f(x) = e^x$, (11.12) gives $de^{B_t} = e^{B_t} dB_t + \frac{1}{2} e^{B_t} dt$: the process e^B has a *drift* $\frac{1}{2} S dt$ that $dS = S dB$ does not have. Indeed $\mathbb{E}[e^{B_t}] = e^{t/2}$ (moment generating function of the normal), whereas $dS = S dB$ means that S is a martingale with constant mean S_0 ; the same computation is the exponential martingale of proposition 11.3(iii), and the correction $-\frac{1}{2}\sigma^2 t$ in the exponent is the same lognormal correction as in remark 4.19. The solution of $dS = S(\mu dt + \sigma dB)$ is therefore

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t\right], \quad (11.13)$$

geometric Brownian motion (GBM): $\log S_t$ is a Brownian motion with drift $\mu - \frac{1}{2}\sigma^2$ and volatility σ , so S_t is lognormal (section 3.3) with

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}, \quad \text{Var } S_t = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

The mechanism in discrete time (our check). The discrete product of one-step relative changes, $\prod_i (1 + \sigma \Delta_i B)$ over a grid of mesh h , has mean exactly $\prod (1 + 0) = 1$ and equals $\exp(\sum_i \ln(1 + \sigma \Delta_i B)) = \exp(\sigma B_t - \frac{1}{2}\sigma^2 \sum_i (\Delta_i B)^2 + \dots)$; the sum of squares converges to t by theorem 11.8, so the product converges to $\exp(\sigma B_t - \frac{1}{2}\sigma^2 t)$: the extra factor $e^{-\sigma^2 t/2}$ is the quadratic variation at work. In a simulation with 2×10^5 paths of $400 \pm \sqrt{h}$ steps ($t = 1$, $\sigma = 1$) the mean of the product is 1.005, whereas the mean of e^{B_1} is 1.656 (theory: 1 and $e^{1/2} = 1.649$).

IN FINANCE: Arithmetic versus geometric growth

With $\mu = 8\%$ and $\sigma = 20\%$ the expected value of an investment grows as $e^{0.08 \cdot 10} = 2.23$ over ten years, but the median as $e^{(0.08-0.02)10} = 1.82$, and the probability of ending below the initial value is $\Phi(-0.06 \cdot \sqrt{10}/0.2) = 17\%$. The gap between the mean and the typical path grows with $\sigma^2 t$, exactly as for the multiplicative martingales of [section 4.4](#) (volatility drag). GBM is the model of Black, Scholes and Merton ([chapter 15](#)); Brownian motion with drift, on the other hand, is the model of the *logarithm* of the price, which is why the gambler's-ruin formulas of [section 11.5.1](#) apply to log-prices and log-levels.

Exercise 11.8 (not from the course: geometric Brownian motion). For S_t in (11.13) compute $\mathbb{E}[S_t]$, $\text{Var } S_t$, and $\mathbb{P}[S_t < S_0]$. With $\mu = 0.08$, $\sigma = 0.2$ find the horizon at which the probability of a loss falls below 5%. Show that $\mathbb{E}[e^{\lambda B(t)}] = e^{\lambda^2 t/2}$ and use it to explain why e^B is not a martingale.

11.7 Processes built from Brownian motion

11.7.1 The Brownian bridge and the empirical distribution function

Definition 11.19 (Brownian bridge). $X(t) = B(t) - tB(1)$, $0 \leq t \leq 1$ ▶ 2024 L14 @ 35:57.

The path is pinned at both ends: $X(0) = 0 = X(1)$, since it is B minus the straight line from 0 to $B(1)$ that it would have followed on average. It is a centred Gaussian process with

$$\text{Var } X(t) = t(1-t), \quad \text{Cov}(X(s), X(t)) = \min(s, t) - st \quad (s, t \in [0, 1]), \quad (11.14)$$

from $\text{Cov}(X(s), X(t)) = s - st - st + st$ for $s \leq t$. Since $\text{Cov}(X(t), B(1)) = t - t = 0$ and the pair is jointly Gaussian, X is independent of $B(1)$; consequently X has the law of Brownian motion *conditioned* to return to 0 at time 1 (the endpoint is irrelevant for the shape of the fluctuations). The variance is largest in the middle, $\frac{1}{4}$ at $t = \frac{1}{2}$.

Where it comes from: the empirical distribution function. Let $U_1, \dots, U_n$ be i.i.d. uniform on $(0, 1)$ and $\hat{F}_n(t) = \#\{U_i \leq t\}/n$. Each indicator $\mathbf{1}\{U_i \leq t\}$ is Bernoulli with mean t and variance $t(1-t)$, so $\mathbb{E}[\hat{F}_n(t)] = t$ and $\text{Var } \hat{F}_n(t) = t(1-t)/n$; by the central limit theorem $\sqrt{n}[\hat{F}_n(t) - t] \rightarrow N(0, t(1-t))$ ▶ 2024 L14 @ 38:05. The whole process $\sqrt{n}[\hat{F}_n(t) - t]$ converges to the Brownian bridge (Donsker, 1952): variance $t(1-t)$ and covariance $\min(s, t) - st$ are those of (11.14), checked directly on the multinomial counts.

IN FINANCE: Testing a return distribution: probability integral transform and Kolmogorov–Smirnov

If $Y_1, \dots, Y_n$ are i.i.d. with a *continuous* cdf F , then $U_i = F(Y_i)$ are i.i.d. uniform ([proposition 3.3](#)), so the statistic $D_n = \sup_y |\hat{F}_n(y) - F(y)| = \sup_t |\hat{F}_n^U(t) - t|$ has a distribution that does not depend on F : $\sqrt{n}D_n \rightarrow \sup_t |X(t)|$, whose law is $\mathbb{P}[\sup |X| \leq x] = 1 - 2 \sum_{k \geq 1} (-1)^{k-1} e^{-2k^2 x^2}$ (Kolmogorov); the 5% critical value is 1.358. To test whether a set of returns is compatible with an assumed model, transform the data by the model's cdf and test uniformity ▶ 2024 L14 @ 41:10. (If the parameters of F are estimated from the same data, the critical values are smaller, as in the Lilliefors correction.) This is a model-selection tool; residual diagnostics of [chapter 9](#) play the same role for time series.

11.7.2 Sampling at random times: Laplace increments

Observe B at $t = 0, 1, 2, \dots$: the increments are i.i.d. $N(0, 1)$ (with drift and volatility, $N(\mu, \sigma^2)$) [► 2024 L14 @ 1:13:28]. Suppose instead the observation times are random, with i.i.d. exponential gaps $T \sim \text{Exp}(\lambda)$ (rate λ) independent of B [► 2024 L14 @ 1:14:35]. Then the increment $B(T)$ has density

$$f(x) = \int_0^\infty \lambda e^{-\lambda t} \frac{e^{-x^2/2t}}{\sqrt{2\pi t}} dt = \sqrt{\frac{\lambda}{2}} e^{-\sqrt{2\lambda}|x|}, \quad (11.15)$$

the *Laplace* (two-sided exponential) density, instead of a bell curve: a *normal variance mixture* in the sense of [proposition 3.29](#), $B(T) = \sqrt{T} Z$, with variance $\mathbb{E}[T] = 1/\lambda$ and excess kurtosis $3\mathbb{E}[T^2]/(\mathbb{E}[T])^2 - 3 = 3$ (checked by numerical integration of (11.15) at two points). Its tails are heavier than the normal ones, and the lecture reports that a Laplace model fits S&P 500 returns much better [► 2024 L14 @ 1:15:44]. The economic reading: the useful clock for prices is not calendar time but a random *business time* (trades, information arrivals), and the process is a Brownian motion *subordinated* to a random clock. Time-changed and stochastic-volatility models take this idea further ([chapter 13](#)).

Exercise 11.9 (not from the course: the Laplace mixture). Derive (11.15) (hint: $\int_0^\infty t^{-1/2} e^{-\lambda t - x^2/2t} dt = \sqrt{\pi/\lambda} e^{-\sqrt{2\lambda}|x|}$), verify that the density integrates to one, and compute its variance and excess kurtosis.

11.7.3 Two-dimensional Brownian motion

The lecture opens with the animation of a particle moving in the plane, where each coordinate takes independent Gaussian steps, no direction being preferred [► 2024 L14 @ 1:04]. Formally $\mathbf{B}(t) = (B_1(t), B_2(t))$ with independent standard Brownian motions has density $(2\pi t)^{-1} e^{-|\mathbf{y}|^2/2t}$, invariant under rotations; the distance $|\mathbf{B}(t)|$ has the Rayleigh law $\mathbb{P}[|\mathbf{B}(t)| < \varepsilon] = 1 - e^{-\varepsilon^2/(2t)}$ (for $t = \varepsilon = 1$: 39.3%) and $\mathbb{E}[|\mathbf{B}(t)|] = \sqrt{\pi t/2}$. The path in the plane is fractal-looking and its two projections are the one-dimensional processes of the rest of the chapter (file lec14_2, 2000 steps). Correlated Brownian motions, obtained from independent ones by a matrix ([chapter 2](#)), are the basis of models with several risk factors.

11.7.4 Filtrations and adapted processes

Let X be a stochastic process, for instance a price. The *filtration* $\{\mathcal{F}_t\}$ generated by X is the increasing family of σ -fields of events determined by $\{X(u), 0 \leq u \leq t\}$: $\mathcal{F}_t \subset \mathcal{F}_{t'}$ for $t < t'$; it is the information available at time t [► 2024 L14 @ 1:16:44]. A process Y is *adapted* to X if $Y(t)$ depends only on $\{X(u), u \leq t\}$, equivalently $\mathcal{G}_t \subset \mathcal{F}_t$ for the filtration $\mathcal{G}$ of Y : the transformation is not forward-looking. For example if $Y(t)$ is the cash flow of a strategy that trades the security X , adaptedness says that one cannot trade on tomorrow's price [► 2024 L14 @ 1:17:48]. Adaptedness is what makes the gains from trading a martingale when X is one: the continuous-time analogue of the martingale transform of [theorem 4.21](#) (with constant-mean processes, martingales, “applied again and again” in the lecture [► 2024 L14 @ 1:18:55]) is the stochastic integral $\int Y dX$ of [chapter 17](#), and the price of an option is a conditional expectation with respect to $\mathcal{F}_t$ ([chapter 15](#)). All the martingale statements of [proposition 11.3](#) hold with respect to the filtration of B .

11.7.5 Lévy processes: Brownian motion and the Poisson process

2024 PS5 opens with a general definition (Problems 1–2, [Exercises 11.10](#) and [11.11](#)); the topic was not lectured on.

Definition 11.20 (Lévy process). $\{L(t), t \geq 0\}$ is a *Lévy process* if

- (i) $L(0) = 0$;
- (ii) stationary increments: $L(t) - L(s) \stackrel{d}{=} L(t - s)$ for $s \leq t$;
- (iii) independent increments over disjoint intervals;
- (iv) continuity in probability: $\lim_{s \rightarrow t} \mathbb{P}[|L(t) - L(s)| > \varepsilon] = 0$ for all $\varepsilon > 0$ and t .

(The problem set indexes t over all of $\mathbb{R}$; only $t \geq 0$ is used.) It generalises Definition [11.1](#), which asks in addition that the increments be normal and the paths continuous: continuity in probability is much weaker (it does not forbid jumps at unpredictable times). Two examples are the subject of the problem set. *Brownian motion with drift* ([definition 11.16](#)) is a Lévy process: its increments are $N(\mu h, \sigma^2 h)$. The *Poisson process* N with intensity λ has independent increments with $N(t) - N(s) \sim \text{Poisson}(\lambda(t - s))$; it is continuous in probability, since $\mathbb{P}[N(t) - N(s) \geq 1] = 1 - e^{-\lambda(t-s)} \rightarrow 0$, although *every* path is a pure-jump staircase. The two are the extreme cases of the Lévy–Itô decomposition (not covered): every Lévy process is the sum of a drift, a Brownian motion, and a superposition of jumps (compensated when small).

	paths	increment over h	quadratic variation on $[0, T]$
$X = \mu t + \sigma B$	continuous, nowhere differentiable	$N(\mu h, \sigma^2 h)$	$\sigma^2 T$, deterministic
Poisson N , rate λ	piecewise constant, unit jumps	$\text{Poisson}(\lambda h)$	$N(T)$ (each jump adds 1^2), random

Table 11.1: Brownian motion with drift and the Poisson process, the two building blocks of jump–diffusion models.

INTUITION: Two universal limits

Sums of many small independent increments give Brownian motion ([theorem 11.6](#)). Sums of many *rare* unit increments give the Poisson process: a walk that steps up by 1 with probability λ/n at each of n sub-intervals has $\text{Bin}(n, \lambda/n) \rightarrow \text{Poisson}(\lambda)$ increments (the law of small numbers). The compensated process $N(t) - \lambda t$ is a martingale with variance λt , and $(N(t) - \lambda t)/\sqrt{\lambda}$ looks Brownian when the rate is large, a bridge between the two limits. In finance the two blocks together give the *jump–diffusion* of Merton (1976): Brownian noise for the day-to-day movements, Poisson jumps for crashes, defaults and announcements. Realised variance ([section 11.3](#)) measures both: the quadratic variation of the diffusive part plus the sum of the squared jumps.

Exercise 11.10 (PS5 (2024), Problem 1: Brownian motion with drift as a Lévy process). Let W be a Brownian motion with drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$.

- (a) Prove that W is a Lévy process in the sense of [definition 11.20](#).

- (b) Simulate the path of W with $T = 1$ year, $\mu = 0.30$, $\sigma = 0.40$ and $m = 252$ increments, $t_i = iT/m$, and plot five simulated paths (the process values at the time increments).

Exercise 11.11 (PS5 (2024), Problem 2: the Poisson process). A stochastic process $\{N(t), t \geq 0\}$ is a Poisson process of intensity (jump rate) λ if for any $0 = t_0 < t_1 < \dots < t_n$ the increments $N(t_k) - N(t_{k-1})$ are independent, and $N(t) - N(s)$ is Poisson with mean $\lambda(t - s)$ for all $0 \leq s < t$.

- (a) Prove that N is a Lévy process.
- (b) Simulate the path of a Poisson process with $T = 1$ year, $\lambda = 24$ events per year and $m = 252$ increments, $t_i = iT/m$, and plot five simulated paths.

11.8 Looking ahead and the two lectures compared

- $(dB)^2 = dt$ and (11.12) are the beginning of *stochastic calculus*: Itô integrals $\int \theta dB$ for adapted θ , Itô's lemma for $f(t, X_t)$, geometric Brownian motion as the solution of an SDE (chapters 17 and 18), Girsanov's theorem and risk-neutral pricing (chapter 15).
- The martingale $e^{\lambda B - \lambda^2 t/2}$ and the gambler's-ruin martingale $e^{-2\mu X/\sigma^2}$ are the continuous-time versions of example 4.18; the reflection principle and hitting-time density are the working tools for barrier options and structural credit models.
- The diffusion equation (11.2) is the seed of the Black–Scholes PDE, the Fokker–Planck equation and the Feynman–Kac formula of the next chapters.
- The Dickey–Fuller distribution of section 9.4.2 is the law of $\int B dB / \int B^2 dt$.

5 COURSE NOTE: 2013 and 2024 compared; problem-set overlap

Both years cover the same core: definition, random-walk limit, non-differentiability, reflection principle and the maximum, quadratic variation, drift. They differ in emphasis. 2013 (C. Lee) is rigorous about the definition (existence theorem, path space), proves non-differentiability by the maximum, and spends the second half of the lecture on *quadratic variation as the reason for Itô's lemma*, ending with the question $S_t = e^{B_t}$?; he does not treat the bridge, Laplace increments, reflected and absorbed Brownian motion, the hitting-time density, or gambler's ruin with drift. 2024 (P. Kempthorne) is more applied and computational: transition density and diffusion equation, the normalised random walk simulated for Bernoulli and Gaussian steps, path pairing for the reflection principle, hitting-time density, the bridge and the empirical distribution function, gambler's ruin by infinitesimal analysis, quadratic variation of log-prices, subordination and adapted processes; it stops before Itô's lemma, which is in the next lecture (chapter 17). The quadratic-variation theorem is stated for arbitrary partitions in 2024 and proved for equal spacing in 2013; theorem 11.8 above proves it in general. Problem sets: 2024 PS4 Problem 1 is 2013 PS8 Problem A-2 (same three parts, with $\sigma = 2$ in part (b) in 2024); 2013 PS8 Problem B-1 (density and rotation invariance of a two-dimensional Brownian motion, and $\mathbb{P}[B(t) \in D]$ for a disc D) belongs to chapter 17, together with the rest of PS8; section 11.7.3 gives the results it needs. 2024 PS4 Problems 3–4 (MLE for the geometric Brownian motion) are treated in chapter 13.

KEY FORMULAS

Brownian motion:	$B(0) = 0, B(t) - B(s) \sim N(0, t - s)$ independent $\text{Cov}(B(s), B(t)) = \min(s, t)$
Martingales:	$B(t), B(t)^2 - t, e^{\lambda B(t) - \lambda^2 t/2}$
Density:	$p(y, \tau x) = \frac{1}{\sqrt{2\pi\sigma^2\tau}} e^{-(y-x)^2/(2\sigma^2\tau)}$ $\partial_\tau p = \frac{\sigma^2}{2} \partial_y^2 p$
Donsker:	$B_n(t) = S_{[nt]}/\sqrt{n} \Rightarrow B(t) \quad (\mathbb{E}[X] = 0, \text{Var } X = 1)$
Scaling:	$\{B(ct)\} \stackrel{d}{=} \{\sqrt{c} B(t)\}$
Paths:	continuous, nowhere differentiable; $\sum (\Delta B)^2 \rightarrow T, (\text{d}B)^2 = \text{d}t$
Maximum:	$\mathbb{P}[M(t) \geq a] = 2\mathbb{P}[B(t) > a] = 2[1 - \Phi(a/\sqrt{t})]$
Hitting time:	$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi}} t^{-3/2} e^{-a^2/(2t)}, \quad \mathbb{E}[\tau_a] = \infty$
Reflected:	$\mathbb{E}[B(t)] = \sqrt{2t/\pi}, \quad \text{Var } B(t) = (1 - \frac{2}{\pi})t$
Absorbed at 0:	$p_A(y) = p(y, t x) - p(-y, t x)$ $\mathbb{P}[A(t) = 0] = 2[1 - \Phi(x/\sqrt{t})]$
Bridge:	$X(t) = B(t) - tB(1), \quad \text{Cov}(X(s), X(t)) = \min(s, t) - st$
Drift:	$X = \mu t + \sigma B, \quad \mathcal{L}f = \mu f' + \frac{1}{2}\sigma^2 f''$
Ruin:	$\mathbb{P}[X(\tau) = b X(0) = x] = \frac{e^{-2\mu x/\sigma^2} - e^{-2\mu a/\sigma^2}}{e^{-2\mu b/\sigma^2} - e^{-2\mu a/\sigma^2}}$ $(= \frac{x-a}{b-a} \text{ if } \mu = 0)$
Itô (heuristic):	$\text{d}f(B) = f'(B) \text{d}B + \frac{1}{2}f''(B) \text{d}t$
GBM:	$S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma B_t}, \quad \mathbb{E}[S_t] = S_0 e^{\mu t}$
Laplace:	$B(T), T \sim \text{Exp}(\lambda) : f(x) = \sqrt{\lambda/2} e^{-\sqrt{2\lambda} x }$

Chapter 12

Data Science and AI for Biomedical Portfolios

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L18	A. W. Lo (MIT Sloan; guest)	Biomedicine at an inflection point (gene therapy); the “valley of death”; risk versus uncertainty; Sharpe ratio and the Madoff example; one anti-cancer drug as an investment; a portfolio of 150 independent projects; rare diseases, correlation and BridgeBio; a fusion “mega fund”; questions (patents, expertise, COVID-19 vaccines).

Files. Transcript only: Lecture 18 - Applying Data Science and Artificial Intelligence to Managing Biomedical Portfolios.txt. The same text exists as 18642-lecture-18-version-5_transcript.pdf in 18.642-fall-2024/static_resources. There are no slides, no problem set and no other file for this lecture; everything said about the slides below is therefore unknown, and the chapter is built on the spoken text. **Problem sets.** None. The exercises (each after the section it practises) are all *not from the course*. **Not from the lecture.** Every derivation marked “Reconstruction” (correlated projects, binomial and beta-binomial counts, securitisation, ML for trial success) is my own mathematics behind the numbers quoted by Lo; the numerics were checked by a short Python script. The 2024 L1 slides list a past final-project paper, *A Megafund Approach to Pediatric Oncology Drug Development*, which uses exactly this framework.

Motivation

Prof. Lo is a financial economist (director of the MIT Laboratory for Financial Engineering, principal investigator at CSAIL) [► 2024 L18 @ 0:00]. His message is not about biology: it is that a *financing structure* can change the risk-reward profile of an investment, and that the required mathematics is a few lines of probability. A single drug-development project is a lottery ticket that nobody wants: costly, slow, and very likely to fail. A portfolio of many such tickets whose outcomes are only weakly correlated has a much better ratio of expected return to risk, and that is what makes it fundable. The chapter follows the lecture and then adds the mathematics that the lecture only quotes:

- the Sharpe ratio as the quantity to be raised, and the two ways to raise it (section 12.2);
- the single-project calculation with its numbers, and the $\sqrt{N}$ law for independent projects (sections 12.3 and 12.4);
- *what breaks when projects are correlated*: the variance of a sum, the diversification limit, counts of successes, and parameter uncertainty as a hidden common factor (sections 12.5 and 12.6); this is the point the lecturer stresses with rare diseases and with the fusion question;

- a sketch of securitisation (debt and equity tranches) and of predicting trial success from data ([sections 12.7 and 12.8](#));
- the second example of the lecture, fusion energy, and a numerical erratum ([section 12.9](#)).

Portfolio variance is developed in general in [chapter 10](#); the covariance formula for sums is in [chapter 3](#).

12.1 The setting: a valley of death

The lecture opens with two clinical stories, both single-gene diseases treated by a one-time gene therapy (a correct copy of the gene is carried by a virus into the cells): Leber's congenital amaurosis, a blindness caused by one DNA "typo" [[▶ 2024 L18 @ 4:07](#)], and Canavan disease, in which the myelin insulation of nerve cells degrades and children rarely live beyond about ten years [[▶ 2024 L18 @ 6:13](#)]. Lo uses them, together with the "convergence" of life sciences, physical sciences and engineering and the various "omics" [[▶ 2024 L18 @ 13:36](#)], to argue that science is advancing fast. The bottleneck he identifies is *economics* [[▶ 2024 L18 @ 14:39](#)].

The specific obstacle is the *valley of death*: the gap between academic experiments and the first human clinical trials, where money is hard to raise [[▶ 2024 L18 @ 15:42](#)]. He explains it with the distinction, familiar in economics, between

- *risk*: randomness whose probability law can be parametrised (normal, gamma, χ^2), and
- *uncertainty*: "unknown unknowns", with no data from which to assess probabilities [[▶ 2024 L18 @ 16:47](#)].

Drug development has three features that investors must digest: it costs hundreds of millions to billions of dollars per approved drug, the probability of success is low, and the process takes 10 to 15 years [[▶ 2024 L18 @ 17:50](#)]. Compared with writing an app, all three are extreme.

INTUITION: Risk versus uncertainty, in physicist's terms

Risk is a stochastic model with known parameters (a Hamiltonian with a heat bath at known temperature); uncertainty is not knowing the Hamiltonian. [section 12.6](#) shows that the two are not unrelated: uncertainty about a parameter that all projects share acts exactly like a common random field coupling them, and it limits diversification just as correlation does.

12.2 Sharpe ratio: what investors want

Lo's classroom exercise shows four cumulative-return curves, unnamed, from 2008 onward, and asks which one to choose [[▶ 2024 L18 @ 18:54](#)]. The four assets are revealed as Treasury bills (safe, about zero real return), the S&P 500, Pfizer, and the Fairfield Sentry fund, the feeder of Madoff's Ponzi scheme, which was chosen by most because it looked like the best trade-off [[▶ 2024 L18 @ 20:04](#)]. The moral is that investors want a high expected return per unit of risk, formalised by the Sharpe ratio.

Definition 12.1 (Sharpe ratio). For an investment with return R , risk-free return r_f over the same horizon, the *Sharpe ratio* is

$$SR = \frac{\mathbb{E}[R] - r_f}{\sqrt{\text{Var } R}}.$$

The lecture quotes $SR \approx 0.33$ for the S&P 500 and 0.34 for Pfizer, and an “order of magnitude higher” figure, on paper, for the Ponzi scheme [► 2024 L18 @ 22:07]. Lo adds that the Sharpe ratio of biomedical investing has been declining for decades, not because the numerator is unattractive but because risk and uncertainty grow faster as the science becomes more sophisticated [► 2024 L18 @ 23:07].

There are, jokes Lo, two ways to increase a ratio: raise the numerator or lower the denominator (or both) [► 2024 L18 @ 24:11]. Better science and engineering act on both. What is under-used in biomedicine is *new financing and business models*, and this is where mathematicians can contribute: constructing structures whose risk–reward trade-off differs from that of the underlying assets [► 2024 L18 @ 25:14].

IN FINANCE: Why the Madoff curve fooled people

A smooth, upward curve has a tiny sample standard deviation, so its Sharpe ratio is enormous; a Sharpe ratio that looks too good is itself a warning sign. In the lecture the point is behavioural (“we all want high-yielding, low-risk assets”) and it sets up the target: not a lottery ticket, but a pooled structure with a respectable Sharpe ratio [► 2024 L18 @ 22:07].

12.3 One anti-cancer drug as an investment

Lo poses the question as a fund. Raise \$200 million, wait 10 years, and with probability 95% return nothing [► 2024 L18 @ 25:14]. The back-of-the-envelope numbers for an oncology project are [► 2024 L18 @ 26:19]:

- cost of the clinical trials: about \$200 million, up front;
- 10 years to complete them;
- probability of success: a bit under 5% historically (3.4% in oncology), rounded to $p = 5\%$;
- if successful, revenues of \$2 billion per year in years 11–20 (the patent lifetime is 20 years, so the exclusivity runs until then), equivalent to a single payment of \$12.3 billion in year 10.

The equivalence is a present-value computation (section 1.7.2). The lecture does not state the discount rate, but the numbers are reproduced by $r = 10\%$:

$$2 \times \sum_{k=1}^{10} \frac{1}{1.1^k} = 2 \times \frac{1 - 1.1^{-10}}{0.1} = 2 \times 6.1446 = 12.29 \approx 12.3 \quad (\text{billion, at year 10}). \quad (12.1)$$

(This is a *Reconstruction*: only the \$12.3 billion is quoted.)

Lo’s conclusion is that few investors take this bet [► 2024 L18 @ 28:23]. The scaling $\sqrt{10}$ used for the volatility is the usual rule for independent yearly increments; how the expected return is annualised is a convention (the simple average $207.5\%/10 \approx 21\%$ would give

another number), so the pair (12%, 423.5%) is best taken as the lecturer's stated input to the next step.

Exercise 12.1 (not from the course). Verify (12.1): find the discount rate for which \$2 billion in each of years 11–20 is worth \$12.3 billion at year 10, and compute the value at year 0 (in units of the \$200M cost) of the project of [example 12.2](#), using its expected payoff.

12.4 A portfolio of independent projects

Instead of one project, invest in $N = 150$ at once: this needs $150 \times \$200\text{M} = \30 billion, a sum that Lo defers ("assume we have it") [[2024 L18 @ 29:26](#)]. If the projects are independent and identically distributed the expected return of the portfolio equals that of each project, while the standard deviation is divided by $\sqrt{N}$: $423.5\%/\sqrt{150} = 34.6\%$ [[2024 L18 @ 30:30](#)]. The Sharpe ratio is then about $12\%/34.6\% \approx 0.35$, ten times the single-project value.

Proposition 12.3 (Reconstruction: equal-weight portfolio of N i.i.d. projects). Let $R_1, \dots, R_N$ be i.i.d. with mean μ and variance σ^2 (returns per dollar invested) and let $R_P = \frac{1}{N} \sum_i R_i$. Then $\mathbb{E}[R_P] = \mu$, $\text{Var } R_P = \sigma^2/N$ and $\text{SR}_N = \sqrt{N} \text{SR}_1$.

With the numbers of [example 12.2](#): $\text{SR}_1 = 0.0283$, $\text{SR}_{150} = 0.0283\sqrt{150} = 0.347$. Lo phrases the step as "the variance of the sum is the sum of the variances" and "the risk goes down by the square root of the number of projects"; the second phrase refers to the standard deviation [[2024 L18 @ 29:26](#)].

INTUITION: Central-limit bookkeeping

Ten thousand independent spins have relative fluctuations of order 10^{-2} of their mean: this is the same $1/\sqrt{N}$ law that makes pressure well defined. The economics of "a \$30 billion fund" is the statement that the fund is a large- N system whose fluctuations are small, provided the couplings between the components are small; the rest of the chapter is about that proviso.

Lo then asks where \$30 billion would come from, and answers that if the fund is attractive enough, small investors can supply it in aggregate (part of a retirement account or of a summer-internship salary) [[2024 L18 @ 30:30](#)]. This is the second half of the strategy: the same mathematics that makes the fund attractive also justifies broad retail participation, and [section 12.7](#) sketches how such capital can be structured.

12.5 Correlation: the real constraint

The i.i.d. assumption is flagged by Lo himself ("I'll come back to that assumption later") [[2024 L18 @ 29:26](#)]. He returns to it with *rare diseases* [[2024 L18 @ 31:30](#)]. There are roughly 12,000 of them, each with a small patient population, but more than 30 million Americans are affected in aggregate. Development is comparatively cheap because trials can be small (he mentions a Leber's-amaurosis approval on 30–40 patients) and the target is a known gene [[2024 L18 @ 32:31](#)]. Above all the diseases are so different that the successes and failures of different programmes are barely correlated. To see why this matters, recall the variance of a sum.

Example 12.2 (Reconstruction: mean and volatility of the single project). Let G be the gross ten-year return, $G = 12.3/0.2 = 61.5$ with probability $p = 0.05$ and 0 otherwise. Then

$$\mathbb{E}[G] = pG_0 = 3.075, \quad \text{Var } G = G_0^2 p(1-p), \quad \sqrt{\text{Var } G} =$$

with $G_0 = 61.5$. The annualised expected return is $3.075^{1/10} - 1 \approx 11.9\%$, the "about 12%" of the lecture. The annualised standard deviation, obtained by scaling the ten-year standard deviation by $\sqrt{10}$, is $13.40/\sqrt{10} = 4.24$, that is 424%, the "423.5%" of the lecture [[2024 L18 @ 28:23](#)] (the small difference comes from rounding of the inputs). The Sharpe ratio with $r_f \approx 0$ is $0.12/4.24 \approx 0.03$, "close to zero": hardly anyone would invest.

Proof. Linearity of $\mathbb{E}$ gives the mean. For the variance, independence gives $\text{Var}(\sum_i R_i) = \sum_i \text{Var } R_i = N\sigma^2$, and dividing the sum by N divides the variance by N^2 . $\square$

Proposition 12.4 (Variance of a portfolio of N correlated projects). Let $R_1, \dots, R_N$ have common variance σ^2 and common pairwise correlation ρ . Then

$$\text{Var } R_P = \frac{1}{N^2} \left(\sum_i \text{Var } R_i + \sum_{i \neq j} \text{Cov}(R_i, R_j) \right) = \sigma^2 \left(\frac{1}{N} + \frac{N-1}{N} \rho \right), \quad (12.2)$$

so that

$$\text{SR}_N = \frac{\text{SR}_1}{\sqrt{1/N + (1 - 1/N)\rho}} \xrightarrow{N \rightarrow \infty} \frac{\text{SR}_1}{\sqrt{\rho}}. \quad (12.3)$$

This is the statement of the lecture: with N projects there are $N(N-1)/2$ pairwise covariances, “a lot of them”, and in a typical investment portfolio they are almost all positive [► 2024 L18 @ 33:36]. Only when they vanish does the risk fall as $1/\sqrt{N}$. The relevant limit of (12.2) is the *undiversifiable* floor $\sigma^2 \rho$: no number of projects removes the common component.

Table 12.1: Reconstruction: standard deviation and Sharpe ratio of an equal-weight portfolio for the single-project inputs of example 12.2 (mean 12%, standard deviation 423.5%, $r_f = 0$), computed from (12.2).

ρ	$N = 15$		$N = 150$	
	std. dev.	SR	std. dev.	SR
0	109%	0.11	34.6%	0.35
0.01	117%	0.10	54.6%	0.22
0.02	124%	0.10	69.0%	0.17
0.05	143%	0.08	100%	0.12
0.10	169%	0.07	138%	0.09

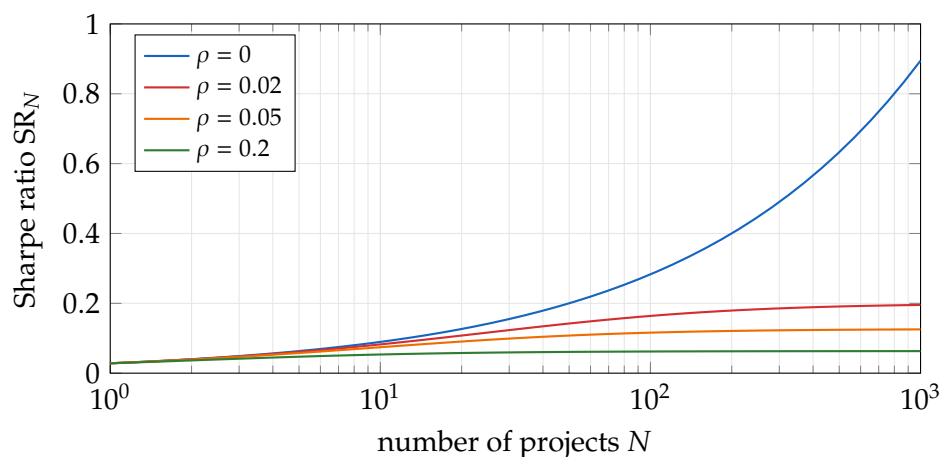


Figure 12.1: Reconstruction: Sharpe ratio of an equal-weight portfolio of N projects with pairwise correlation ρ , from (12.3) with $\text{SR}_1 = 0.0283$. For $\rho = 0$ it grows as $\sqrt{N}$; for $\rho > 0$ it saturates at $\text{SR}_1/\sqrt{\rho}$.

Proof. The double sum $\sum_{i \neq j}$ has $N(N-1)$ ordered pairs, each equal to $\rho\sigma^2$. Hence $\text{Var } R_P = \frac{1}{N^2} (N\sigma^2 + N(N-1)\rho\sigma^2)$, which is (12.2). The Sharpe ratio uses the same expected return, whence (12.3). $\square$

1 COURSE NOTE: How many projects, how much money?

Lo’s claim is that because rare-disease programmes are nearly uncorrelated, one needs “10 or 20” projects rather than 150, hence \$400–500 million rather than \$30 billion [► 2024 L18 @ 34:38]. That is not obtained by scaling the \$200 million per project of the oncology example: 15 such projects would cost \$3 billion. It also assumes that rare-disease projects are cheaper and have a different profile than the oncology numbers used above, which the lecture does not quantify. The reported input is a Monte Carlo simulation with industry numbers published in his paper, giving about 22% per year and a Sharpe ratio above 1 [► 2024 L18 @ 35:38]; the parameters are not in the lecture, so table 12.1 should *not* be read as a reproduction of that result: with the oncology inputs and $N = 15$ a Sharpe ratio of 0.11 is all one gets, even with independence.

IN FINANCE: The BridgeBio story

A former student of Lo (a biotech venture capitalist) reran the simulations with his own parameters, then with further modifications; they ran about 35 versions over six months, after which he quit his job to found a company [► 2024 L18 @ 36:43]. BridgeBio Pharma came out of stealth in 2017 with a private-equity investor (KKR) taking part in a biotech for the first time; it raised about \$40 million after a friends-and-family round, then \$135 million, then \$300 million (about \$400–500 million in total), had about 15 projects, went public and raised a further \$750 million, and traded at about \$4.6 billion of market capitalisation on the day of the lecture [► 2024 L18 @ 37:45]. By his account about 20 projects were in the clinic, two drugs had been approved in 2021 and the Canavan gene therapy shown earlier belongs to the company [► 2024 L18 @ 39:52]. Lo's remark, worth keeping: finance cannot turn a bad drug into a good one, but the wrong financing can destroy a good one; science comes first [► 2024 L18 @ 41:57]. (Facts about the company are as stated in class; they were not checked.)

Exercise 12.2 (not from the course). With $R_P = \frac{1}{N} \sum_i R_i$ and general covariance matrix Σ , the variance is $\frac{1}{N^2} \mathbf{1}^T \Sigma \mathbf{1}$ (chapter 10). Show that for the common-correlation matrix $\Sigma = \sigma^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^T]$ this reproduces (12.2), and find its eigenvalues. Which eigenvector carries the “undiversifiable” part?

12.6 Counting successes

For a project financed at year 0 the natural random variable is a Bernoulli success indicator $X_i \in \{0, 1\}$ with $\mathbb{P}(X_i = 1) = p$. This section is a *Reconstruction* (not from the course) of what the number of successes looks like when the indicators are independent or correlated.

Independent projects. $K = \sum_i X_i \sim \text{Bin}(N, p)$. The probability of at least one success is $1 - (1 - p)^N$: with $p = 5\%$ it is 54% for $N = 15$ and 99.95% for $N = 150$ (the chance of nothing is $0.95^{150} = 4.6 \times 10^{-4}$). If every success returns $G_0 = 61.5$ times a single project's cost and every project has the same cost, the portfolio (total cost N units) returns $G_0 K$ units, so the portfolio is at least at break-even when $K \geq N/G_0$. For $N = 150$ this means $K \geq 3$ ($150/61.5 = 2.44$), which has probability 98.2%; for $N = 15$ it means $K \geq 1$, probability 54%.

Correlated outcomes. Two Bernoulli variables with the same p have $\text{Cov}(X_i, X_j) = \rho p(1 - p)$, and then

$$\text{Var } K = Np(1 - p)(1 + (N - 1)\rho), \quad (12.4)$$

by the same counting argument as in proposition 12.4: $\rho > 0$ inflates the fluctuations of K by $1 + (N - 1)\rho$ compared with the binomial. The distribution of K also acquires a fat right tail and a fat mass at 0. Two standard ways to produce correlated Bernoulli indicators:

- (a) *One-factor Gaussian model* (reconstruction; the model behind many credit-risk portfolios). Let $Z_i = \sqrt{a} Y + \sqrt{1 - a} \varepsilon_i$ with Y, ε_i i.i.d. $N(0, 1)$ and declare project i a success if $Z_i < c = \Phi^{-1}(p)$. Given the common factor Y the projects are independent with success probability $q(Y) = \Phi((c - \sqrt{a} Y)/\sqrt{1 - a})$, so $\mathbb{P}(K \geq k) = \mathbb{E}_Y [\mathbb{P}(\text{Bin}(N, q(Y)) \geq k)]$, a one-dimensional integral. The Bernoulli correlation ρ is smaller than the latent correlation a (it is 0.026 for $a = 0.1$ and 0.058 for $a = 0.2$).

(b) *Uncertainty about p* (beta-binomial). The lecturer’s distinction between risk and uncertainty has an exact counterpart. Suppose the projects share an unknown success rate $\Pi \sim \text{Beta}(\alpha, \beta)$ with $\mathbb{E}[\Pi] = p$ and, given Π , are independent Bernoulli(Π). Then $\text{Cov}(X_i, X_j) = \text{Var } \Pi$ and

$$\rho = \frac{\text{Var } \Pi}{p(1-p)} = \frac{1}{\alpha + \beta + 1}. \tag{12.5}$$

Indeed $\text{Var } \Pi = p(1-p)/(\alpha + \beta + 1)$ for a beta distribution. The parameter $\alpha + \beta$ is the “effective sample size” of the prior information about p : if the historical rate (3.4% against 5%) were known to only about 50 observations, $\alpha + \beta = 49$ and $\rho = 0.02$, a correlation that no amount of diversification can remove (table 12.1 with $\rho = 0.02$).

Table 12.2: Reconstruction: probability that the portfolio of N projects with $p = 5\%$ has at least k successes, independent case ($\rho = 0$), one-factor Gaussian model with latent correlation a and beta-binomial ($\alpha + \beta = 49, \rho = 0.02$). $N = 150, k = 3$ is break-even in the sense of the text.

	$N = 15, k \geq 1$	$N = 150, k \geq 1$	$N = 150, k \geq 3$	$N = 150, k \geq 4$
independent	0.537	0.9995	0.982	0.945
Gaussian, $a = 0.1$	0.482	0.968	0.821	0.730
Gaussian, $a = 0.2$	0.432	0.899	0.689	0.601
Gaussian, $a = 0.5$	0.300	0.616	0.419	0.368
beta-binomial	0.491	0.970	0.838	0.753

INTUITION: What correlation does to the tails

The mean number of successes (7.5 out of 150) is unchanged; what changes is the probability of the unlucky scenario in which *everything fails together*. That scenario is invisible in the average and decisive for someone who has to repay creditors. This is why the lecture insists on the near-independence of rare diseases [► 2024 L18 @ 33:36] and why the portfolio manager’s key parameter is a correlation, not a mean.

Exercise 12.3 (not from the course). Prove (12.4). Then compute the standard deviation of K for $N = 150, p = 0.05$ and $\rho \in \{0, 0.02, 0.05\}$ and compare it with the mean 7.5.

Exercise 12.4 (not from the course). Show that in the beta-binomial model of section 12.6(b), $\rho = 1/(\alpha + \beta + 1)$, and that after observing s successes in n historical trials of a common process the posterior correlation between two future projects is $1/(\alpha + \beta + n + 1)$. What sample size n makes it fall below 0.005 starting from $\alpha + \beta = 49$?

Exercise 12.5 (not from the course). Simulate the one-factor Gaussian model of section 12.6(a) for $N = 150, p = 0.05, a = 0.1$ and estimate $\mathbb{P}(K \geq 3)$ by Monte Carlo with 10^5 paths; compare with table 12.2.

12.7 From portfolio to structure: a securitisation sketch

In the lecture the \$30 billion is obtained by appeal to the investing public [► 2024 L18 @ 30:30] and, for fusion, to retail money [► 2024 L18 @ 57:45]. The natural instrument, which Lo calls a

“mega fund” [► 2024 L18 @ 54:42] and which the past final project of 2024 L1 develops for paediatric oncology, is a pool of projects funded by tranches of different seniority. He does not describe tranches in this lecture; the following is a generic *Reconstruction*.

The moral of [example 12.5](#) is that securitisation transforms one risk profile into several (the same principle as in structured credit), and that a rating of the senior tranche is a statement about the joint law of the project outcomes. The correlation input is exactly what the two rare-disease and fusion arguments in the lecture are about.

Exercise 12.6 (not from the course). In [example 12.5](#) let the senior debt be only a fraction f of the \$30 billion, with face value $30f \times 1.05^{10}$. Find, for the independent model, the smallest f such that the senior tranche is repaid with probability at least 0.99. (The rest of the capital is equity and absorbs the first losses.)

Example 12.5 (Reconstruction: one senior debt tranche on the 150-project pool). Pool of $N = 150$ projects, all financed at year 0 by \$30 billion, with K successes and terminal cash flow $V = 12.3K$ (billion dollars) at year 10. Suppose the whole \$30 billion is raised as zero-coupon debt at 5%, so the face value due at year 10 is $30 \times 1.05^{10} = 48.9$. It is repaid if and only if $12.3K \geq 48.9$, i.e. $K \geq 4$. With independent projects $\mathbb{P}(K \geq 4) = 0.945$; in the one-factor model the same probability is 0.73 for $a = 0.1$ and 0.60 for $a = 0.2$ ([table 12.2](#)). A structure with a junior equity tranche that absorbs the first losses, and only a fraction of the capital raised as senior debt, would push the senior repayment probability far higher; but the equity holders take the correlation risk, and the example shows that its price depends mostly on ρ , not on N .

12.8 Where the data science and AI enter

Despite the title, the lecture says almost nothing about machine learning: artificial intelligence appears in passing as a competitor for investors' money [► 2024 L18 @ 1:16:35], and Lo's affiliation with CSAIIL opens the talk [► 2024 L18 @ 0:00]. His actual “data science” is a *simulation* of the portfolio with parameters supplied by industry experts [► 2024 L18 @ 35:38]. What follows is a short *Reconstruction* (not from the course) of how the data would enter a real model.

- Success probability from features.* Model the outcome $Y_i \in \{0, 1\}$ of project i with covariates x_i (indication, mechanism, phase, sponsor, biomarker) as $\mathbb{P}(Y_i = 1 | x_i) = \sigma(w^T x_i)$, with $\sigma(u) = 1/(1 + e^{-u})$, and fit w by minimising the log-loss $-\sum_i [Y_i \log \sigma(w^T x_i) + (1 - Y_i) \log(1 - \sigma(w^T x_i))]$ with a penalty on $\|w\|$ (regularisation, [chapter 20](#); more general classifiers, [chapter 16](#)). The output is a project-specific p_i in place of the pooled $p = 5\%$; the portfolio mean is then $\sum_i p_i G_i$.
- Shrinkage and uncertainty.* With few historical trials, the estimate of p is imprecise, and a Bayesian update of a $\text{Beta}(\alpha, \beta)$ prior with s successes in n trials gives $\text{Beta}(\alpha + s, \beta + n - s)$ with mean $(\alpha + s)/(\alpha + \beta + n)$. By ([12.5](#)) the residual uncertainty about p is the common factor that limits diversification; more data lowers ρ .
- Dependence between projects.* Correlations (shared mechanisms, shared regulators, shared macro conditions for funding) are estimated from the same features, e.g. by treating shared attributes as factors, as in [chapter 7](#); the sample correlation of a handful of binary outcomes is very noisy, so the structure of the factors matters more than the numbers.

2 COURSE NOTE: What the lecture actually contains

No data set, code or slide of the lecture is available. The only quantitative statements are those reproduced in [sections 12.3](#) and [12.4](#): \$200M, 10 years, 5%, \$12.3B, 12%, 423.5%, 150, 34.6%, Sharpe ≈ 0.34 –0.35, the rare-disease reduction to 10–20 projects and \$400–500M, and a Sharpe ratio above 1 from a Monte Carlo of a published paper. Everything else in this chapter is derived here.

12.9 Second example: a fusion mega fund

The second half of the lecture applies the same idea to energy. The argument is that the “energy transition” is not a transition: consumption of oil, gas and coal is still growing as more people join the middle class, so low-carbon, on-demand sources are needed in addition [► 2024 L18 @ 42:02]. The candidates listed are fusion, fission and geothermal; fusion promises more energy per kilogram of fuel than fission and no long-lived waste, but requires the three Lawson conditions of temperature, pressure (density) and confinement time [► 2024 L18 @ 47:17]. Lo points to a 20-tesla magnet built at MIT with Commonwealth Fusion Systems in 2021 and a laboratory reaction with net energy gain [► 2024 L18 @ 49:23].

His question, asked with Dennis Whyte, is whether finance can pay for fusion. The idea is to *decompose* “a commercially viable power plant” into a portfolio of enabling technologies (sustained plasma at about 100 million degrees, ultra-clean vacuum systems, magnetic confinement, and so on), each with a value of its own, for instance a vacuum system also useful in drug development, and to treat the portfolio as in the biomedical case, with perhaps 150 components [► 2024 L18 @ 51:34]. He adds a board of scientific and business advisers whose participation would follow once serious capital is deployed [► 2024 L18 @ 54:42].

A student asks whether the components would not be highly correlated. The answer is twofold: complete reactors from different companies are correlated, but not perfectly, because they use different approaches (magnetic versus inertial); and, more importantly, technologies such as heating, vacuum and magnet shaping have little to do with each other [► 2024 L18 @ 56:44]. In the language of [proposition 12.4](#): the argument is that ρ is small, and it should be treated as the key empirical input, since the floor $SR_1/\sqrt{\rho}$ is not eliminated by adding components that share a common “fusion is feasible” factor (compare (12.5)).

Exercise 12.7 (not from the course). Erratum check: with the numbers of [section 12.9](#), what fraction of the 130 million households must contribute \$1,500 to raise \$30 billion? What one-time contribution per household would be needed from 3% of them?

12.10 The rest of the lecture

The Harvey Lodish story [► 2024 L18 @ 58:45]. As a young faculty member in 1983 Lodish was approached to help develop a treatment for Gaucher’s disease, a genetic enzyme deficiency; the company became Genzyme, which got the first enzyme-replacement therapy approved in 1991 and was sold to Sanofi for \$20 billion in 2014. In 2002 his unborn grandson was diagnosed with the same disease, and the drug saved him. The point Lo draws is that finance need not be zero-sum: “doing well by doing good” [► 2024 L18 @ 1:02:57].

Questions.

- *Patents* [► 2024 L18 @ 1:05:05]: a patent gives exclusive use of the compound for 20 years (which is why revenues run over years 11–20 in [section 12.3](#)); afterwards anyone can manufacture cheaply, like aspirin. Producing the pill is cheap; the expensive part is the trials that lead to approval.
- *How to find and evaluate opportunities* [► 2024 L18 @ 1:07:08]: expertise. Neil Kumar (BridgeBio’s co-founder, a chemical-engineering PhD) worked at a consultancy on pharma strategy, then at a biotech venture firm, which had passed on single-drug projects as too risky; the simulations gave him the conviction to fund them himself.
- *How can students invest?* [► 2024 L18 @ 1:08:11]: BridgeBio is public, at very high risk; private companies are harder; MIT is rich in opportunities for investors; Lo also teaches health-care financing (15.482, with an online version in MIT’s Open Learning Library) [► 2024 L18 @ 1:11:21].
- *Crises* [► 2024 L18 @ 1:15:32]: COVID-19 temporarily removed the valley of death: the time between the sequencing of the virus and the first human injection was 63 days, against 3–5 years normally; afterwards the valley returned because vaccines are not attractive to investors compared with, for example, AI [► 2024 L18 @ 1:17:39].

3 COURSE NOTE: Erratum: the retail-funding arithmetic

Lo asks for a one-time \$1,500 from each of 3% of 130 million households, “more than \$30 billion” [► 2024 L18 @ 56:44]. The product is

$$130 \times 10^6 \times 0.03 \times 1500 = \$5.85 \times 10^9,$$

about \$5.9 billion, not \$30 billion; reaching \$30 billion at \$1,500 each needs about 15% of households. (He also says once that fusion is “when we split an atom”: that is fission; fusion joins nuclei. Both are slips of speech in the transcript; the intended meaning is clear.)

KEY FORMULAS

- Sharpe ratio: $SR = (\mathbb{E}[R] - r_f) / \sqrt{\text{Var } R}$.
- Single project (lecture): cost \$200M, 10 years, $p = 5\%$, payoff \$12.3B; $\mathbb{E}[G] = 3.075$ per dollar, $\sqrt{\text{Var } G} = 61.5\sqrt{p(1-p)} = 13.4$ over 10 years (424% per year after dividing by $\sqrt{10}$).
- N i.i.d. projects: $\text{Var } R_P = \sigma^2/N$, $SR_N = \sqrt{N} SR_1$.
- Common correlation ρ : $\text{Var } R_P = \sigma^2(\frac{1}{N} + \frac{N-1}{N}\rho)$, $SR_N \rightarrow SR_1/\sqrt{\rho}$.
- Successes: $\text{Var } K = Np(1-p)(1 + (N-1)\rho)$; $\mathbb{P}(K \geq 1) = 1 - (1-p)^N$ if independent.
- Unknown $p \sim \text{Beta}(\alpha, \beta)$: $\rho = 1/(\alpha + \beta + 1)$.

Chapter 13

Volatility Modeling

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L19	P. Kempthorne	Definition and annualisation of volatility; historical estimators (simple, exponential, RiskMetrics); geometric Brownian motion and maximum likelihood; Garman–Klass and the other open-high-low-close (OHLC) estimators; case study of the S&P 500 and of ARKK (naive, ARIMA and seasonal ARIMA models of rolling volatility); Poisson jump diffusion; Laplace law; ARCH, GARCH, maximum likelihood, stylised facts, t innovations, euro/dollar case study.
2013 L9 (from 19:52)	P. Kempthorne	The same programme (slides <code>lecnote9</code>): historical volatility, time scales in GBM (trading versus calendar days), Garman–Klass, jump diffusion, ARCH, GARCH, fits to the euro/dollar rate. The first 20 minutes finish univariate time series (information criteria, Dickey–Fuller test, Case Study 3), see chapter 9 .
2013 L11 (first 28 min)	P. Kempthorne	Recap and completion: GARCH(1, 1) as ARMA for squared returns, long-run variance, likelihood and parameter constraints, residual diagnostics, t innovations and their degrees of freedom, value at risk, volatility clustering, mean reversion and persistence, RiskMetrics. The rest of the lecture is multivariate time series (chapter 9).
Files. 2024 slides <code>mit18_642_f24_lec17_1.pdf</code> (“Volatility Modeling”, 31 slides, 36 pages with overlays) and <code>mit18_642_f24_lec17_2.pdf</code> (35 pages, R case study on the S&P 500); zip <code>mit18_642_f24_rstudio_pset5.zip</code> (case studies of the S&P 500 and of ARKK, R Markdown source and knitted PDF); 2013 slides <code>MIT18_S096F13_lecnote9.pdf</code> (31 pages), 2013 Case Study 4 <code>CaseStudy4.pdf</code> (volatility of exchange-rate returns, in R). The case study loads a workspace built by <code>fm_casestudy_fx_1.r</code> from the Federal Reserve data <code>fred_fxrates.txt</code> (setup files: appendix B). Problem sets. 2024 PS4 Problems 3–4 (Problems 1–2 and 5–6 are in chapter 11 and chapter 9); 2024 PS5 Problems 3–5 (Problems 1–2, Lévy processes, are in chapter 11); 2013 PS5, all four problems (its Problems 1–2 are the same as 2024 PS4 Problems 3–4). Standalone exercises follow the section they practise; the connected ones are at the end of the chapter. Not published or not covered. The papers of Garman–Klass, Parkinson and Yang–Zhang, and the RiskMetrics technical document, are distributed on Canvas or on the web and are not in the download; the auxiliary R function file <code>test_vol1b.r</code> used by Case Study 4 is also missing, so the two-time-scale fits of section 13.2 are described from the printed output. The 2024 lecture reached the end of the slides; slide 31 (extensions of GARCH) is only a list of names. Stochastic-volatility and implied-volatility models are named but not developed in either year.		

Motivation

Prices move, and *how much* they move is the central risk quantity of finance: it sets the width of a confidence band for tomorrow's P&L ([chapter 19](#)), the price of an option ([chapter 15](#)), the size of a margin requirement ([chapter 8](#)) and the weights of a portfolio ([chapter 10](#)). Two facts make it interesting.

- Volatility is *not observed*. The price is; the standard deviation of its increments has to be estimated, and the quality of the estimate depends dramatically on the information used (closing prices only, or the whole intraday range) and on the model assumed (constant volatility, or volatility that moves).
- Volatility is *not constant*. Calm and turbulent weeks alternate, so that squared returns are strongly autocorrelated although returns themselves are not: white noise that is not independent ([section 9.1.2](#)). The models of Engle and Bollerslev, ARCH and GARCH, put this in one line and, with one extra parameter, produce fat tails, mean reversion of volatility and a forecasting formula.

For a physicist the situation is that of a noisy detector whose noise level drifts: one measures a *scale parameter*, one has to choose between averaging over a long window (small statistical error, large bias if the scale drifts) and a short one, and the statistics of the scale itself becomes a subject of study. The chapter follows the lectures: historical estimators ([section 13.1](#)), the constant-volatility benchmark ([section 13.2](#)), range-based estimators and the S&P 500 case study ([section 13.3](#)), mixtures for fat tails ([section 13.4](#)), ARCH and GARCH ([sections 13.5](#) and [13.6](#)), estimation and the euro/dollar case study ([section 13.7](#)), and a summary of stylised facts ([section 13.8](#)).

13.1 Volatility and its historical estimation

13.1.1 Definition and annualisation

The lectures define *volatility* as the annualised standard deviation of the change in price (or value) of a financial security [[2024 L19 @ 1:00](#)][[2013 L9 @ 21:57](#)]. It is a risk measure and an input to option pricing. Given prices P_t the *log return* over one period is $R_t = \log(P_t/P_{t-1})$; log returns add over successive periods whereas percentage returns multiply [[2024 L19 @ 4:13](#)], as in [eq. \(9.1\)](#). If $\{R_t\}$ is covariance stationary [[2024 L19 @ 5:16](#)], the *period volatility* is $\sigma = \sqrt{\text{Var } R_t}$, and the *annualised volatility* is

$$\text{vol} = \sqrt{N} \sigma, \quad N = 252 \text{ (daily), } 52 \text{ (weekly), } 12 \text{ (monthly)}, \quad (13.1)$$

where 252 is the number of business days in a year [[2024 L19 @ 2:00](#)][[2013 L9 @ 25:10](#)]. The justification is the variance of a sum: the return over a year is the sum of N period returns, and if these are *uncorrelated* with a common variance, $\text{Var}(\sum_{t=1}^N R_t) = N\sigma^2$. A daily standard deviation of 1% is thus a volatility of $1\% \times \sqrt{252} = 15.9\%$. Scaling all horizons to one year makes assets and maturities comparable. Orders of magnitude quoted in 2013: bond yields move by less than 5 percentage points a year, currencies by about 10%, equity indices by 30%–40% in a nervous year [[2013 L9 @ 23:03](#)].

Remark 13.1 (What the square-root rule needs). Only *uncorrelatedness* is used, not independence, so (13.1) holds for the *unconditional* variance of ARCH and GARCH returns, which are uncorrelated but dependent ([proposition 13.10](#)). It fails when returns are autocorrelated ([proposition 8.11](#)) and, as shown in [section 13.6.5](#), it fails for

the *conditional* variance over a horizon: after a turbulent day the volatility expected over the next month is smaller than $\sqrt{21}$ times today's volatility.

13.1.2 Sample volatility and its precision

Given $R_1, \dots, R_T$ the sample estimate is

$$\hat{\sigma} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2}, \quad \bar{R} = \frac{1}{T} \sum_{t=1}^T R_t, \quad (13.2)$$

which uses the unbiased estimator of the variance, hence the divisor $T-1$ [► 2024 L19 @ 5:16]. How precise is it?

Proposition 13.2 (Precision of the sample volatility). Let R_t be i.i.d. with kurtosis κ (definition 3.4). Then $\hat{\sigma}^2$ is asymptotically normal with $\text{Var } \hat{\sigma}^2 \approx (\kappa - 1)\sigma^4/T$. For Gaussian returns the result is exact in the form $(T-1)\hat{\sigma}^2/\sigma^2 \sim \chi_{T-1}^2$ (section 3.9), so $\text{Var } \hat{\sigma}^2 = 2\sigma^4/(T-1)$, and by the delta method

$$\frac{\text{sd}(\hat{\sigma})}{\sigma} \approx \frac{1}{\sqrt{2(T-1)}}.$$

With $T = 21$ daily returns (one month) the relative standard error of $\hat{\sigma}$ is 15.8%: a monthly estimate of 12% annual volatility is uncertain by about ± 1.9 percentage points; with $T = 252$ it is 4.5%. For a heavy-tailed series with $\kappa = 6$ the variance of $\hat{\sigma}^2$ is 2.5 times larger than in the Gaussian case. (This is the quantitative content of the “trade-off” below and of 2013 PS5 Problem 4.)

Proof. Put $\mu = \mathbb{E}[R]$ and $Y_t = (R_t - \mu)^2$, i.i.d. with mean σ^2 and variance $\mathbb{E}[(R - \mu)^4] - \sigma^4 = (\kappa - 1)\sigma^4$. Then $\hat{\sigma}^2 = \frac{1}{T-1} [\bar{Y} - (\bar{R} - \mu)^2]$, and $(\bar{R} - \mu)^2 = O(1/T)$ does not affect the leading term, so the central limit theorem applied to $\bar{Y}$ gives $\text{Var } \hat{\sigma}^2 \approx (\kappa - 1)\sigma^4/T$. For $\kappa = 3$ this is $2\sigma^4/T$. The χ^2 statement is the classical result for the sample variance of a normal sample, and $\text{Var } \chi_{T-1}^2 = 2(T-1)$. Finally $\hat{\sigma} = g(\hat{\sigma}^2)$ with $g(x) = \sqrt{x}$, $g'(\sigma^2) = 1/(2\sigma)$, so $\text{Var } \hat{\sigma} \approx \text{Var}(\hat{\sigma}^2)/(4\sigma^2) = \sigma^2/(2(T-1))$. $\square$

Exercise 13.1 (PS5 (2013), Problem 4: are year-to-year differences in volatility significant?). The annual sample variances of the daily log returns of the S&P 500 close are (year: daily variance, number of days) 2006: 3.981×10^{-5} , 251; 2007: 1.019×10^{-4} , 251; 2008: 6.677×10^{-4} , 253; 2009: 2.950×10^{-4} , 252; 2010: 1.295×10^{-4} , 252; 2011: 2.164×10^{-4} , 252; 2012: 6.459×10^{-5} , 250. Model the returns of one year as an i.i.d. sample from $N(\mu, \sigma^2)$ and let $\hat{\sigma}^2$ be the unbiased sample variance.

- Prove that $\hat{\sigma}^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2$.
- Invert the χ^2 percentiles $q_{0.025}, q_{0.975}$ to obtain a two-sided 95% confidence interval for σ^2 (the problem gives the percentiles for 249 to 252 degrees of freedom): compute it for 2008, express it as an annualised volatility ($\sqrt{253} \sigma$), and say whether the sample volatility of any other year falls inside it.
- Under the null hypothesis of equal daily variances in two years, show that $S = \hat{\sigma}_{2008}^2 / \hat{\sigma}_{2007}^2$ has an F distribution (state its degrees of freedom); the problem gives $q_{0.025} = 0.7804$, $q_{0.975} = 1.2815$ for $F_{252,250}$. Compute S for 2008 against 2007, find the P -value (the α at which the test is on the boundary), and repeat for 2008 against 2006.

13.1.3 Historical predictors: simple and exponential averages

To *predict* volatility one needs a single-period sample estimate $\hat{\sigma}_t^2$ for each period t . If t indexes months and the data are daily, it is the sample variance of the daily returns of month t ; if t indexes days and the data are daily, it is simply R_t^2 (the mean is negligible); with

intraday data the daily value is the sum of squared intraday returns (the *realised variance*)

[► 2024 L19 @ 7:28][► 2013 L9 @ 24:06]. Four predictors of σ_{t+1}^2 are then defined [► 2024 L19 @ 6:23]

[► 2013 L9 @ 27:15]:

$$\text{historical average: } \tilde{\sigma}_{t+1}^2 = \frac{1}{t} \sum_{j=1}^t \hat{\sigma}_j^2 \quad (\text{all data}),$$

$$\text{simple moving average: } \tilde{\sigma}_{t+1}^2 = \frac{1}{m} \sum_{j=0}^{m-1} \hat{\sigma}_{t-j}^2 \quad (\text{last } m),$$

$$\text{exponential moving average (EMA): } \tilde{\sigma}_{t+1}^2 = (1 - \beta)\hat{\sigma}_t^2 + \beta \tilde{\sigma}_t^2, \quad 0 \leq \beta \leq 1 \quad (\text{all data}), \quad (13.3)$$

$$\text{EWMA over the last } m: \quad \tilde{\sigma}_{t+1}^2 = \frac{\sum_{j=0}^{m-1} \beta^j \hat{\sigma}_{t-j}^2}{\sum_{j=0}^{m-1} \beta^j} \quad (\text{last } m).$$

The EMA is a recursion and can be applied over the whole history. Unrolling it gives the same weights as the last line for $m = \infty$.

Proof. Iterate the recursion n times: $\tilde{\sigma}_{t+1}^2 = (1 - \beta) \sum_{j < n} \beta^j \hat{\sigma}_{t-j}^2 + \beta^n \tilde{\sigma}_{t+1-n}^2$, and let $n \rightarrow \infty$. The mean age is $(1 - \beta) \sum_j j \beta^j = \beta / (1 - \beta)$. The weight falls to one half when $\beta^j = \frac{1}{2}$. For independent $\hat{\sigma}_t^2$ with variance $2\sigma^4$ the variance of the weighted sum is $2\sigma^4 \sum_j w_j^2 = 2\sigma^4(1 - \beta)^2 / (1 - \beta^2)$, and a simple average of n observations has variance $2\sigma^4/n$. $\square$

Proposition 13.3 (Weights of the exponential average). *If the recursion (13.3) runs from the remote past, $\tilde{\sigma}_{t+1}^2 = (1 - \beta) \sum_{j \geq 0} \beta^j \hat{\sigma}_{t-j}^2$. The weights sum to 1; the mean age of the data is $\beta / (1 - \beta)$, the half-life of a weight is $\ln 2 / \ln(1/\beta)$, and if the $\hat{\sigma}_t^2 = R_t^2$ come from i.i.d. zero-mean Gaussian returns, $\tilde{\sigma}^2$ is unbiased with*

$$\text{Var } \tilde{\sigma}_{t+1}^2 = 2\sigma^4 \frac{1 - \beta}{1 + \beta}, \quad \text{i.e. an effective sample size } n_{\text{eff}} = \frac{1 + \beta}{1 - \beta}.$$

	$\beta = 0.90$	$\beta = 0.94$	$\beta = 0.97$
mean age (days)	9.0	15.7	32.3
half-life (days)	6.6	11.2	22.8
n_{eff} (days)	19	32.3	65.7

ASIDE: RiskMetrics and the lecturer

Kempthorne was in the Sloan School's sponsoring research centre at the time.

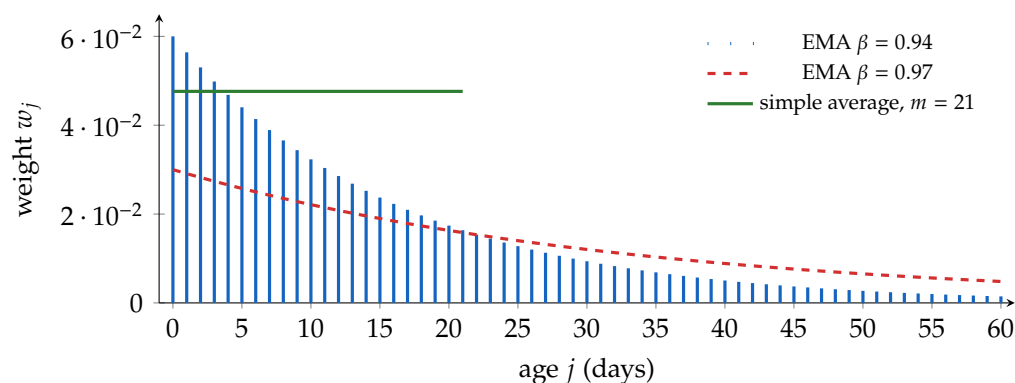


Figure 13.1: Weights given to the squared return j days ago. The simple average of 21 days is flat and then cuts off, so that a spike is kept for exactly 21 days and then dropped; the exponential averages fade out smoothly.

IN FINANCE: RiskMetrics

The exponential average is the workhorse of the RiskMetrics methodology that J.P. Morgan published in the 1990s [► 2024 L19 @ 10:39]. Its technical document tabulates decay factors for foreign exchange, zero-coupon yields, equity indices and other assets, chosen by comparing the mean squared error of the resulting volatility forecasts: they are generally well above 0.9, more like 0.95, which says that volatility changes slowly [► 2024 L19 @ 11:41]; the 2013 lecture translates this into an averaging period of roughly 45 to 90 days [► 2013 L9 @ 26:12]. (The published values are $\beta = 0.94$ for daily and 0.97 for monthly data; this is not stated in class.) The point of a benchmark is not optimality but agreement: everybody knows what is measured [► 2024 L19 @ 12:44]. The methodology is adopted without a long-run mean of volatility, which is why it corresponds to $\alpha_0 = 0$, $\alpha_1 + \beta_1 = 1$ in a GARCH(1, 1) (section 13.6.7).

13.1.4 Regression forecasts, trade-offs and evaluation

A more flexible predictor regresses the target on the last p single-period estimates, $\tilde{\sigma}_{t+1}^2 = \gamma_{1,t} \hat{\sigma}_t^2 + \gamma_{2,t} \hat{\sigma}_{t-1}^2 + \cdots + \gamma_{p,t} \hat{\sigma}_{t-p+1}^2 + u_t$, fitted on all data or on a rolling window of the last m [► 2024 L19 @ 9:36]. It resembles but is not the autoregression of the series $\hat{\sigma}_t^2$, since the coefficients are re-estimated as the window moves.

Every choice (window length m , decay β , number of lags) balances two effects: more data reduces the statistical error of the estimate (proposition 13.2), but data close to time t describe the current volatility better if it changes [► 2024 L19 @ 9:36][► 2013 L9 @ 28:15]. The choice is made *out of sample*, separately for asset classes, sampling frequencies and forecast horizons, with error measures such as the mean squared error, the mean absolute error and the mean absolute percentage error

$$\text{MSE} = \frac{1}{n} \sum e_t^2, \quad \text{MAE} = \frac{1}{n} \sum |e_t|, \quad \text{MAPE} = \frac{100}{n} \sum \left| \frac{e_t}{y_t} \right|, \quad e_t = y_t - \hat{y}_t, \quad (13.4)$$

[► 2024 L19 @ 9:36][► 2013 L9 @ 29:17]. (MAPE requires a target that never comes close to zero.)

1 COURSE NOTE: Erratum in the slides

Slide 5 of lec17_1 (and slide 7 of the 2013 deck) writes the coefficient of the second lag as $\gamma_{1,t}$, the same symbol as that of the first lag; it should be $\gamma_{2,t}$, as above.

13.2 The Gaussian benchmark: geometric Brownian motion

13.2.1 The model and its returns

In continuous time the standard model of a price is the *geometric Brownian motion*

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad (13.5)$$

with σ the volatility and μ the mean return per unit time, W a standard Brownian motion: its increments over an interval are $N(0, \text{length})$ and increments over disjoint intervals are independent [► 2024 L19 @ 12:44][► 2013 L9 @ 31:23]. Dividing by S shows that the relative increment is a drift plus a random walk with Gaussian steps. The theory is the subject of [chapters 11](#) and [17](#); what is needed here is its consequence for sampled data. If the price is observed at $t_0 < t_1 < \dots < t_n$ with $\Delta_j = t_j - t_{j-1}$, the log returns $R_j = \log(S(t_j)/S(t_{j-1}))$ are *independent* with

$$R_j \sim N(\mu_* \Delta_j, \sigma^2 \Delta_j), \quad \mu_* = \mu - \frac{1}{2} \sigma^2, \quad (13.6)$$

that is, $\log S(t)$ is a Brownian motion with drift μ_* and volatility σ [► 2024 L19 @ 14:00][► 2013 L9 @ 33:36]. The drift of the *log* price differs from the drift of the price by $-\sigma^2/2$: expanding $\log S$ to second order, $d \log S = dS/S - \frac{1}{2} (dS/S)^2$, and $(dW)^2 = dt$ gives the extra term [► 2024 L19 @ 15:05](the same lognormal correction as in [remark 4.19](#); the derivation is [chapter 17](#)). In working with stochastic differential equations one must not stop at first order.

13.2.2 Maximum likelihood, and what more data can and cannot buy

For equally spaced data with $\Delta_j \equiv 1$ the log-likelihood of $R_1, \dots, R_n$ is that of i.i.d. normals, and the maximum-likelihood estimators are [► 2024 L19 @ 15:05][► 2013 L9 @ 34:41]

$$\hat{\mu}_* = \bar{R} = \frac{1}{n} \sum_{t=1}^n R_t, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^n (R_t - \bar{R})^2, \quad (13.7)$$

the sample mean and the sample variance with divisor n (not $n-1$); the derivation is 2013 PS5 Problem 1(a) ([Exercise 13.10](#)). If the Δ_j vary, “exercise” ([Exercise 13.11](#)) [► 2024 L19 @ 17:18]. A question with a striking answer is what happens when the number n of observations in a fixed window $[0, T]$ grows, that is when the sampling frequency increases [► 2013 L9 @ 35:46]. The exercises show the following. The maximum-likelihood estimator of the drift is always the cumulative return divided by the length of the period, so its variance is σ^2/T *whatever the sampling*: no amount of intraday data helps to measure a drift. The estimator of the variance, on the other hand, becomes arbitrarily precise, $\text{Var } \hat{\sigma}^2 = O(1/n)$, and even for irregular sampling the variance of the estimator is the same for every spacing of the same $n+1$ points (2024 PS4 Problem 4(c)). Physically: over a fixed observation time one can determine the noise amplitude of a diffusion as well as one wishes, but the mean velocity only as well as $\sigma/\sqrt{T}$. This asymmetry underlies the estimators of [section 13.3](#): extra information inside the day sharpens σ but not μ .

13.2.3 Which clock? Trading time versus calendar time

Case Study 4 of 2013 fits (13.6) to daily EUR/USD and GBP/USD returns from 1999 to September 2013 (3 708 returns) on two time scales [► 2013 L9 @ 36:48]. In *trading time* every return is one period, $\Delta_j = 1$. In *clock (calendar) time* Δ_j is the number of calendar days between the two quotes, so that the Monday return, which spans a weekend, has a variance three times larger.

◇ **BEYOND THE CORE: The fitted numbers**

The annualised estimates printed by the case study are

	trading days		clock time	
	$\hat{\mu}_*$ (per year)	$\hat{\sigma}$	$\hat{\mu}_*$ (per day)	$\hat{\sigma}$
EUR/USD	0.00926	0.1025	2.5×10^{-5}	0.1228
GBP/USD	-0.00185	0.0946	-5.1×10^{-6}	0.1119

The volatility depends on the model of time: 10.25% against 12.2% for the euro

► 2013 L9 @ 40:02].

◇ **BEYOND THE CORE: Diagnostics: Q–Q plot and fitted percentiles**

The diagnostic used to decide is the *normal Q–Q plot* (sorted returns against the expected order statistics $x_j = \mathbb{E}[X_{[j]}]$ of a standard normal sample; a straight line means Gaussian) and the histogram of the *fitted percentiles* $\hat{F}(R_j)$, which is uniform on $(0, 1)$ if the model is right ► 2013 L9 @ 42:08]. Both show *leptokurtic* returns (slender centre, fat tails): the extreme fitted percentile bins are over-populated, the middle under-populated ► 2013 L9 @ 45:14]. In calendar time the 1% percentile is exceeded about as often as it should, so the tail fit is slightly better, but not the centre ► 2013 L9 @ 46:20].

INTUITION: Subordinated clocks

If a Brownian motion looks wrong when observed against calendar time, it may still be a Brownian motion against another clock $\tau(t)$: $\log S(t) = B(\tau(t))$. Such processes are called *subordinated*. Natural clocks are cumulative volume or number of trades, a proxy for the flow of information reaching the market ► 2013 L9 @ 41:04]. In physics the same idea appears as “operational time” in anomalous diffusion. The Laplace law of section 13.4.2 is the simplest example, with an exponentially distributed clock.

13.3 Range-based estimators from open, high, low and close

13.3.1 The setting and the notion of efficiency

A daily *bar* reports four prices: open O , high H , low L and close C . Close-to-close volatility uses one number per day and ignores the rest ► 2024 L19 @ 18:22]► 2013 L9 @ 47:20]. Garman and Klass (1980) asked how much precision the other three prices buy. Work with $X(t) = \log S(t)$, a Brownian motion *without drift* ($\mu_* = 0$) with variance σ^2 per day, and let $f \in (0, 1)$ be the fraction of the day during which the market is closed ► 2024 L19 @ 19:27]. Day 0 closes at $t = 0$ (price C_0), day 1 opens at $t = f$ (O_1) and closes at $t = 1$ (C_1), and

$$H_1 = \max_{f \leq t \leq 1} X(t), \quad L_1 = \min_{f \leq t \leq 1} X(t),$$

all in logarithms. The three increments $O_1 - C_0 \sim N(0, f\sigma^2)$, $C_1 - O_1 \sim N(0, (1-f)\sigma^2)$ and their sum $C_1 - C_0 \sim N(0, \sigma^2)$ describe the overnight gap, the trading session and the whole day; the first two are independent.

Every estimator below is an unbiased estimator of σ^2 , and estimators are compared by their variance.

Definition 13.4 (Efficiency). The *efficiency* of an unbiased estimator $\hat{\sigma}^2$ of σ^2 is $\text{eff}(\hat{\sigma}^2) = \text{Var}(\hat{\sigma}_0^2)/\text{Var}(\hat{\sigma}^2)$, where $\hat{\sigma}_0^2 = (C_1 - C_0)^2$ is the close-to-close estimator ► 2024 L19 @ 25:52]. An estimator of efficiency e based on n days is as precise as the close-to-close estimator

based on en days.

Since $C_1 - C_0 = \sigma Z$ with $Z \sim N(0, 1)$, $\hat{\sigma}_0^2 = \sigma^2 Z^2$ with $\sigma^2 Z^2 \sim \sigma^2 \chi_1^2$ has mean σ^2 and variance $\sigma^4 \text{Var} \chi_1^2 = 2\sigma^4$ [► 2024 L19 @ 21:32][► 2013 L9 @ 50:30]. The same holds for the two normalised gap and session estimators,

$$\hat{\sigma}_1^2 = \frac{(O_1 - C_0)^2}{f}, \quad \hat{\sigma}_2^2 = \frac{(C_1 - O_1)^2}{1-f}, \quad \mathbb{E}[\hat{\sigma}_i^2] = \sigma^2, \quad \text{Var} \hat{\sigma}_i^2 = 2\sigma^4 \quad (i = 0, 1, 2), \quad (13.8)$$

because $(O_1 - C_0)/\sqrt{f}$ and $(C_1 - O_1)/\sqrt{1-f}$ are again σZ [► 2024 L19 @ 22:41][► 2013 L9 @ 51:33].

Lemma 13.5 (Combining independent unbiased estimators). Let V_1, V_2 be independent unbiased estimators of the same parameter with variances v_1, v_2 . Among the estimators $aV_1 + (1-a)V_2$ the variance is minimal for $a = v_2/(v_1 + v_2)$ (inverse-variance weighting), it equals $v_1 v_2 / (v_1 + v_2)$, and efficiencies add: $\text{eff} = \text{eff}_1 + \text{eff}_2$.

Proof. $\text{Var} = a^2 v_1 + (1-a)^2 v_2$ is a parabola with minimum at $a = v_2/(v_1 + v_2)$, where it equals $v_1 v_2 / (v_1 + v_2)$. Hence $1/\text{Var} = 1/v_1 + 1/v_2$, and efficiency is $2\sigma^4$ times $1/\text{Var}$. $\square$

Since $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$ are independent [► 2024 L19 @ 23:44], the average $\hat{\sigma}_*^2 = \frac{1}{2}\hat{\sigma}_1^2 + \frac{1}{2}\hat{\sigma}_2^2$ has $\text{Var} = \sigma^4$ and efficiency 2: the opening price alone doubles the information of the close. In 2013 the lecturer stressed the reading of this number: the opening price buys as much as a second day of data [► 2013 L9 @ 52:40]. Weighting the gap and the session equally is right because both are σZ in law, whatever the value of f (which drops out of (13.8)).

13.3.2 Parkinson's high-low estimator

Parkinson (1980) proposed to use the range $H - L$ of the Brownian path over the day [► 2024 L19 @ 25:52][► 2013 L9 @ 55:46]:

$$\hat{\sigma}_3^2 = \frac{(H_1 - L_1)^2}{4 \ln 2} \quad (f = 0). \quad (13.9)$$

It is unbiased because for a driftless Brownian motion of variance σ^2 on a unit interval, $\mathbb{E}[(H - L)^2] = 4 \ln 2 \sigma^2$. The proof needs the joint law of the maximum and the minimum, obtained by the reflection principle (Feller, 1951) and not reproduced here; a check by simulation is given below. The same law gives the fourth moment $\mathbb{E}[(H - L)^4] = 9\zeta(3)\sigma^4$, whence $\text{Var} \hat{\sigma}_3^2 = (9\zeta(3) - 16 \ln^2 2)\sigma^4 / (16 \ln^2 2) = 0.4073 \sigma^4$. A single day's range thus carries more than four days of close-to-close information: "the high and low have a lot of information in them" [► 2024 L19 @ 26:55]. Note that the estimator is positive whenever the range is positive, even on a day with a zero close-to-close return.

Exercise 13.2 (PS5 (2024), Problem 3: the extreme-value estimator of Parkinson (1980)). Read M. Parkinson, "The extreme value method for estimating the variance of the rate of return", *J. Business* 53(1) (1980) 61–65. (b) Explain the definition of the estimator D_x of Section II. (c) Explain the definition of D_l of Section III. (d) Using the results of Section IV, explain why the improvement obtained with D_l as an estimate of the variance V (Section V) could matter particularly in studies of the dependence of V on time and on price, if there is any. (Part (a), "read the article", has no answer.) The estimator of section 13.3.2 is the one in question.

2 COURSE NOTE: Numerical discrepancy in the slides: the efficiency of Parkinson's estimator

Slide 10 of lec17_1 (and slide 13 of the 2013 deck) states $\text{eff}(\hat{\sigma}_3^2) \approx 5.2$. The exact value from the moments above is $2/0.4073 = 4.91$, and a simulation of 2×10^5 Brownian paths (Brownian-bridge extremes, 400 steps) gives $\text{Var} \hat{\sigma}_3^2 = 0.4074$ and efficiency 4.91. The lecture slides also state $a \approx 0.17$ as the optimal weight for the combination $\hat{\sigma}_4^2 = a\hat{\sigma}_1^2 + (1-a)\hat{\sigma}_3^2$ of the gap estimator and the Parkinson estimator; by lemma 13.5 the optimal weight is $1/(1 + \text{eff}_3) = 0.169$ with the exact value (0.161 with 5.2), and the combination has efficiency $1 + 4.91 = 5.91$, not 6.2. So $a \approx 0.17$ is consistent with the exact value and 6.2 with the quoted 5.2; the statistical conclusions are unaffected, and the

13.3.3 Garman–Klass, and the “best analytic” estimator

figures 7.4 and 8.4 below are reproduced exactly.
This is my check, not a statement of the course.

Garman and Klass (1980) found the estimator of minimal variance among those that are *scale invariant*, i.e. that scale as λ^2 when the price is multiplied by λ , hence expressible in the normalised prices $u = H - O$, $d = L - O$, $c = C - O$ (logarithms) [► 2024 L19 @ 27:57] [► 2013 L9 @ 56:47]:

$$\hat{\sigma}_{**}^2 = 0.511 (u - d)^2 - 0.019 \{c(u + d) - 2ud\} - 0.383 c^2, \quad \text{eff}(\hat{\sigma}_{**}^2) \approx 7.4. \quad (13.10)$$

The derivation minimises a quadratic form over the coefficients of an analytic scale-invariant combination: “some quadratic forms and minimising them”. The coefficients are close to those of the simpler estimator used in practice

$$\hat{\sigma}_{\text{GK}}^2 = \frac{1}{2}(u - d)^2 - (2 \ln 2 - 1) c^2, \quad 2 \ln 2 - 1 = 0.386, \quad (13.11)$$

which is the one implemented in the case study (formula of [section 13.3.6](#) below, with $\ln(H/L) = u - d$, $\ln(C/O) = c$). It is unbiased because $\mathbb{E}[\frac{1}{2}(H - L)^2] = 2 \ln 2 \sigma^2$ (from $\mathbb{E}[(H - L)^2] = 4 \ln 2 \sigma^2$) and $\mathbb{E}[c^2] = \sigma^2$: the two terms give $(2 \ln 2) - (2 \ln 2 - 1) = 1$ times σ^2 . This explains where the mysterious constant $2 \ln 2 - 1$ comes from: it is the coefficient that cancels the bias of the range term. Simulation: efficiency 7.40 for both (13.10) and (13.11). Why does one bother with efficiency? With an efficient estimator one can use a shorter window and still be precise, which allows tracking changes of volatility with less lag [► 2024 L19 @ 30:03] [► 2013 L9 @ 58:56].

For $0 < f < 1$ the opening price may differ from the previous close, and the *composite* estimator combines the overnight gap with the best analytic estimator of the session, normalised by its length $1 - f$:

$$\hat{\sigma}_{\text{GK}}^2 = a \frac{(O_1 - C_0)^2}{f} + (1 - a) \frac{\hat{\sigma}_{**}^2}{1 - f}, \quad a \approx 0.12, \quad \text{eff} \approx 8.4. \quad (13.12)$$

Both terms are unbiased for σ^2 and independent (disjoint increments), with efficiencies 1 and 7.4, so [lemma 13.5](#) gives the optimal weight $a = 1/(1 + 7.4) = 0.119 \approx 0.12$ and the efficiency $1 + 7.4 = 8.4$ [► 2024 L19 @ 31:13] [► 2013 L9 @ 58:56], exactly the values of the slides.

♦ BEYOND THE CORE: Gap plus range

(In the same way the composite of the gap and the session range, whose second term is $(H - L)^2/(4 \ln 2 (1 - f))$, has $a = 0.17$ and efficiency 5.9.)

Exercise 13.3 ((not from the course): unbiasedness of a range estimator). Using $\mathbb{E}[(H - L)^2] = 4 \ln 2 \sigma^2$ and $\mathbb{E}[c^2] = \sigma^2$, verify that the Garman–Klass estimator (13.11) is unbiased, and that with a drift $\mu_* \neq 0$ the close-to-close estimator c^2 has expectation $\sigma^2 + \mu_*^2$ (units of a day). Why does this not contradict the estimator being unbiased for a driftless process?

13.3.4 Rogers–Satchell and Yang–Zhang

♦ BEYOND THE CORE: Estimators robust to drift and opening jumps

All estimators so far assume no drift, and the composite one assumes that the only jump is the opening gap. Two later refinements are listed in the slides as extensions and used in the case study.

- *Rogers–Satchell* (1991) allow a non-zero drift (but no opening jump):

$$\hat{\sigma}_{\text{RS}}^2 = u(u - c) + d(d - c) = \ln \frac{H}{C} \ln \frac{H}{O} + \ln \frac{L}{C} \ln \frac{L}{O}. \quad (13.13)$$

It is unbiased for every drift. (Simulation, with a drift of 1.5σ per day, far larger than any real one: mean of (13.13) = $1.002\sigma^2$, whereas close-to-close gives $3.26\sigma^2$, Parkinson $1.85\sigma^2$ and (13.11) $1.30\sigma^2$.) At zero drift its efficiency is 5.97, below that of Garman–Klass: robustness against drift is paid for with variance.

- *Yang–Zhang* (2000) allow both a drift and opening gaps. Over n days,

$$\hat{\sigma}_{\text{YZ}}^2 = \hat{\sigma}_o^2 + k \hat{\sigma}_c^2 + (1 - k) \hat{\sigma}_{\text{RS}}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}, \quad (13.14)$$

where $\hat{\sigma}_o^2$ is the sample variance of the overnight log returns $\ln(O_i/C_{i-1})$, $\hat{\sigma}_c^2$ the sample variance of the open-to-close returns $\ln(C_i/O_i)$ and $\hat{\sigma}_{\text{RS}}^2$ the mean of (13.13) over the n days [► 2024 L19 @ 31:13]. It is a weighted average of the overnight, open-to-close and Rogers–Satchell variances with the weights that minimise the estimation error, and, because the first two are *sample* variances, it does not depend on the drift. For $n = 21$, $k = 0.139$. It is the version recommended “these days” [► 2024 L19 @ 31:13].

3 COURSE NOTE: Names and efficiencies in the case study

The R function `volatility()` of the package `TTR` (through `tidyquant`) offers six estimators: `close`, `parkinson`, `garman.klass`, `rogers.satchell`, `gk.yz` and `yang.zhang`. The one labelled “Garman–Klass Yang and Zhang” (`gk.yz`) is (13.11) plus the squared overnight return $\ln^2(O_i/C_{i-1})$, i.e. the composite (13.12) with unit weights and no normalisation by f and $1 - f$; it is unbiased if the overnight variance is proportional to f . The case study writes that it “has an efficiency of 8 times” the close-to-close estimator. Under the model of section 13.3.1 its efficiency is $2/(2f^2 + (1 - f)^2 2/7.4)$: it is 8.4 at its best, for $f \approx 0.11$, 7.4 for $f = 0$, 3.5 for $f = 0.5$ and only 1.8 for an index that is closed 17.5 hours out of 24 ($f = 0.73$) if overnight variance were proportional to closed time (which is not so: the overnight variance of equity indices is a much smaller share, and this is why Yang–Zhang re-weight; not discussed in class). Parameter names: `TTR`’s `alpha=1.34` enters as $k = (\alpha - 1)/(\alpha + (n + 1)/(n - 1))$, identical to (13.14).

13.3.5 Checking the efficiencies by simulation

◆ BEYOND THE CORE: Monte Carlo check

The claims of the previous subsections can be verified with a Monte Carlo experiment: 2×10^5 Brownian paths on $[0, 1]$ ($\sigma = 1$, $f = 0$), with the high and low of each of 400 steps drawn exactly from the Brownian bridge so that discretisation does not bias the range.

estimator	mean, $\mu_* = 0$	efficiency, $\mu_* = 0$	mean, $\mu_* = 1.5\sigma$
close-to-close c^2	1.00	1.00	3.26
Parkinson (13.9)	1.00	4.91	1.85
Rogers–Satchell (13.13)	1.00	5.97	1.00
Garman–Klass (13.11)	1.00	7.40	1.30
best analytic (13.10)	1.00	7.40	1.29

With real data the high and low are recorded from discrete trades and quotes, so range estimators are biased downwards; this is not discussed in the course.

13.3.6 Case study: volatility of the S&P 500 and of ARKK (2024)

CASE STUDY: Historical volatility with R: S&P 500 (lecture) and ARKK (problem set)

Goal. Compare the six estimators on real bars and ask which is more *predictable* [► 2024 L19 @ 32:16]. **Lesson.** The efficient estimator (Yang–Zhang) is also the more predictable, the estimates of the same asset differ noticeably with the estimator, and volatility levels differ across assets by an order of magnitude; the conclusions above are only descriptive.

◆ BEYOND THE CORE: The S&P 500 and ARKK case study in detail

Data. Daily OHLC bars from Yahoo Finance through `tidyquant::tq_get`: the S&P 500 index from 2014-01-02 to 2024-11-04 (2 729 days) and the ARKK exchange-traded fund (Cathie Wood) from 2014-10-31 to 2024-10-30 (2 516 days) ▶ 2024 L19 @ 34:26]. **Method.** Rolling windows of $n = 21$ days and $N = 252$, with the formulas

$$\begin{aligned}\hat{\sigma}_{\text{cc}} &= \sqrt{\frac{N}{n-2} \sum_{i=1}^{n-1} (r_i - \bar{r})^2}, \quad r_i = \ln \frac{C_{i+1}}{C_i}, \\ \hat{\sigma}_{\text{P}} &= \sqrt{\frac{N}{4n \ln 2} \sum_{i=1}^n \ln^2 \frac{H_i}{L_i}}, \\ \hat{\sigma}_{\text{GK}} &= \sqrt{\frac{N}{n} \sum_{i=1}^n \left[\frac{1}{2} \ln^2 \frac{H_i}{L_i} - (2 \ln 2 - 1) \ln^2 \frac{C_i}{O_i} \right]}, \\ \hat{\sigma}_{\text{RS}} &= \sqrt{\frac{N}{n} \sum_{i=1}^n \left[\ln \frac{H_i}{C_i} \ln \frac{H_i}{O_i} + \ln \frac{L_i}{C_i} \ln \frac{L_i}{O_i} \right]},\end{aligned}$$

plus `gk.yz` and Yang–Zhang (13.14); *close-to-close* uses the unbiased variance of the $n - 1$ returns in a window of n prices. **Results (S&P 500).** All estimators show the same pattern: annualised volatility mostly between 5% and 25%, around 12%–15% in calm periods, with spikes and a huge one in the COVID crash of spring 2020 (peak close-to-close 0.98, Parkinson 0.53, GK 0.52, Yang–Zhang 0.70) ▶ 2024 L19 @ 34:26]. The range estimators are smoother and different in level: over 2023–24 the Yang–Zhang volatility is generally below the close-to-close one, though sometimes far above it (scatter plot of the pairs) ▶ 2024 L19 @ 36:32]. A close-to-close window keeps a spike until it drops out 21 days later, so it stays elevated longer than the others; estimators that fall faster after a spike are probably better for predicting realised volatility ▶ 2024 L19 @ 38:43].

Modelling the rolling volatility, ≈ 450 days of 2023–24 (forecast / `fpp2` package of Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*; Chapter 3 for the naive method and residual diagnostics, Chapter 8 for automatic ARIMA and seasonal ARIMA) ▶ 2024 L19 @ 40:47]:

S&P 500, 2023–24	model	Ljung–Box Q^* (df, p)	MAPE
close-to-close	naive	31.2 (10, 5×10^{-4})	3.67
	ARIMA(3, 1, 0)	16.2 (7, 0.023)	3.74
	seasonal, [21]	89.1 (34, 8×10^{-7})	(meaningless)
Yang–Zhang	naive	88.1 (10, 10^{-14})	2.31
	ARIMA(1, 1, 1)	9.5 (8, 0.31)	2.23
	ARIMA(2, 1, 3)(2, 0, 1) _[21]	31.8 (34, 0.58)	1.92

The *naive* model $\hat{y}_t = y_{t-1}$ is a random walk (efficient-market flavour: the best forecast is where things are now); its residuals are the first differences and the Ljung–Box test (eq. (9.4)) rejects whiteness ▶ 2024 L19 @ 41:49]. `auto.arima` picks a model by the corrected AIC, $\text{AICc} = \text{AIC} + 2k(k + 1)/(n - k - 1)$ with k parameters (for ARIMA(3, 1, 0): -3303.73 against -3303.64 , $n \approx 450$) ▶ 2024 L19 @ 45:02]; the coefficients are judged by their ratio to the standard error: for close-to-close ARIMA(3, 1, 0) the AR(3) coefficient has $0.126/0.047 \approx 2.7$ (“almost 3”), the AR(2) 1.4 ▶ 2024 L19 @ 47:12]. For Yang–Zhang the fit is an ARIMA(1, 1, 1) with $\hat{\phi} = 0.696$ and $\hat{\theta} = -0.446$, many standard errors from zero, and the residuals are close to white noise, except for a large autocorrelation at *lag* 21, the length of the window ▶ 2024 L19 @ 48:14](explained in proposition 13.6). A *seasonal* ARIMA with period 21 removes it ▶ 2024 L19 @ 50:17]: with a seasonal MA coefficient of -0.78 (13 standard errors) the Ljung–Box test no longer rejects, and the training MAPE falls to 1.92 ▶ 2024 L19 @ 57:58]. The forecast plot shows a point forecast (solid line) and Gaussian prediction intervals that widen with the horizon, with the volatility reverting to a mean level after a low period ▶ 2024 L19 @ 59:01].

ARKK (problem set). The same code gives a very different picture: close-to-close volatility of ARKK ranges from 10% to 120%, with peaks of about 1.2 in 2020 and 1.0 in 2022, three to four times the index. Results: close-to-close naive $Q^* = 35.3$ (df 10), MAPE 3.50; ARIMA(1, 1, 1) with $\hat{\phi} = 0.876$, $\hat{\theta} = -0.799$, MAPE 3.55; Yang–Zhang naive $Q^* = 13.5$ ($p = 0.19$, already white), MAPE

2.01; ARIMA(1, 1, 1) $\hat{\phi} = 0.780$, $\hat{\theta} = -0.697$, MAPE 1.99; seasonal ARIMA(0, 1, 1)(2, 0, 2)_[21] with drift, $Q^* = 37.5$ ($p = 0.44$), MAPE 1.78. For the close-to-close series `auto.arima` on the differences selects no seasonal term at all and the Ljung–Box test still rejects ($Q^* = 124.6$, df 40). The scatter of YZ against close-to-close for ARKK lies around the diagonal, with YZ below it at high volatility.

◇ **BEYOND THE CORE: Why lag 21: the window itself**

Proposition 13.6 (Why the residuals of a rolling volatility have an autocorrelation at the window length). Let x_t be i.i.d. with variance s^2 and let $v_t = \frac{1}{m} \sum_{j=0}^{m-1} x_{t-j}$ be the rolling mean of window m . Then the first differences (the residuals of the naive model) satisfy $\Delta v_t = (x_t - x_{t-m})/m$, so $\rho_{\Delta v}(1) = \dots = \rho_{\Delta v}(m-1) = 0$ and $\rho_{\Delta v}(m) = -\frac{1}{2}$.

This is what the lecture describes as observations *entering and leaving the window* [► 2024 L19 @ 49:15]; a simulation with 4×10^5 i.i.d. chi-squared terms gives $\rho(21) = -0.501$. It also shows that the good performance of the seasonal model is partly mechanical: the term x_{t-m} leaving the window is known from the past, so a model with a seasonal moving-average term can predict it, and the variance of the one-step error falls to that of x_t/m , half of $\text{Var}(\Delta v_t)$; the root-mean-square error should drop by the factor $1/\sqrt{2} = 0.71$, and it drops from 0.0051 to 0.0039 for Yang–Zhang. Part of the “predictability of volatility” in the case study is the predictability of the estimator’s own window, not of the market (our remark).

Exercise 13.4 (PS5 (2024), Problem 4: volatility case studies). The R Markdown files

- Case_Study_Estimating__SP500_Historical_Volatility.Rmd,
- Case_Study_Estimating__ARKK_Historical_Volatility.Rmd

produce the S&P 500 and ARKK case studies (section 13.3.6), which use the naive, ARIMA and seasonal ARIMA methods of Hyndman and Athanasopoulos, *Forecasting: Principles and Practice* (2nd ed.), Chapter 3 (naive method, residual diagnostics) and Chapter 8 (automatic ARIMA).

- Comment on the S&P 500 results you find interesting or significant.
- The same for ARKK.
- Compare the volatility findings for ARKK and the S&P 500.

Exercise 13.5 (PS5 (2024), Problem 5, optional (extra credit): your own stock). Choose a stock symbol and edit the ARKK file to run the same analysis.

- Produce the PDF of the case study.
- Comment on results you find interesting.
- Compare your stock with the S&P 500 and ARKK.

4 COURSE NOTE: Erratum: stale numbers in a note of the slides

Page 31 of `lec17_2` says “the MAPE of the seasonal ARIMA model of the YZ volatility is 1.746 versus 3.58 for the naive model of the close volatility”, but the printed accuracy tables on pages 15 and 30 of the same file give 1.917 and 3.666. The note is hard-coded text left over from another run; in the knitted S&P 500 PDF of the problem-set zip, which is a re-run with 2726 days, the note is generated from the output (1.905 against 3.626). The values quoted above are those of the lecture PDF.

Proof. $\Delta v_t = v_t - v_{t-1} = (x_t - x_{t-m})/m$: all terms but two cancel. The variance is $2s^2/m^2$; for lags $1 \leq h < m$ the two differences $(x_t - x_{t-m})$ and $(x_{t+h} - x_{t+h-m})$ share no term; for $h = m$ they share x_t with opposite signs, covariance $-s^2/m^2$. □

13.4 Fat tails from mixtures: jumps and the Laplace law

Geometric Brownian motion has returns that are exactly normal, in contrast with the histograms of section 13.2.3. Two extensions keep independent returns but give them heavier tails; both are *normal variance mixtures* in the sense of proposition 3.29, which will be used repeatedly: if $X | V \sim N(\mu, \sigma^2 V)$, the kurtosis is $3 \mathbb{E}[V^2]/(\mathbb{E}[V])^2 \geq 3$.

13.4.1 Poisson jump diffusion

The *Poisson jump diffusion* adds to (13.5) a compound Poisson process [► 2024 L19 @ 1:00:08] [► 2013 L9 @ 58:56]:

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t) + \gamma \sigma Z(t) d\pi(t), \quad (13.15)$$

where π is a Poisson process of rate λ (the jumps arrive at random, unpredictable times), the marks $Z(t)$ are i.i.d. $N(0, 1)$ and γ sets the scale of the jumps in units of the diffusion volatility. For the distribution of the return we treat the jump as acting additively on the log-price, the convention of the lecture (the exact jump of log S would be $\log(1 + \gamma \sigma Z)$). Over a period Δ the number N of jumps is Poisson($\lambda\Delta$), and given $N = k$ the centred return is the sum of the Gaussian diffusion increment and k independent Gaussian jumps:

$$R - m \mid N = k \sim N(0, \sigma^2(\Delta + k\gamma^2)).$$

So the return has a *Poisson mixture of Gaussian density* $\sum_{k \geq 0} e^{-\lambda\Delta} \frac{(\lambda\Delta)^k}{k!} \varphi_{\sigma^2(\Delta + k\gamma^2)}(x - m)$ [► 2024 L19 @ 1:02:28][► 2013 L9 @ 1:00:00].

Proposition 13.7 (Variance and kurtosis of the jump diffusion). *The variance of R is $\sigma^2\Delta(1 + \lambda\gamma^2)$ and its kurtosis is*

$$\kappa = 3 + \frac{3\gamma^4\lambda}{\Delta(1 + \lambda\gamma^2)^2} > 3.$$

For $\lambda = 0.3$ per day, $\gamma = 3$ and $\Delta = 1$ day, $\kappa = 8.3$ (simulation: 8.32). The excess kurtosis decays as $1/\Delta$: on longer horizons many jumps are added and the central limit theorem restores normality, in accordance with the empirical observation that monthly returns are closer to Gaussian than daily ones.

◊ BEYOND THE CORE: Estimation by the EM algorithm

Estimation by maximum likelihood is hard because the likelihood is a product of infinite series, but the model has a natural latent variable, the number of jumps in each period; the *EM algorithm* of Dempster, Laird and Rubin then has closed-form steps and returns, as a by-product, the posterior number of jumps per period [► 2024 L19 @ 1:01:19][► 2013 L9 @ 1:04:18](Pickard, Kempthorne and Zakaria, 1987).

A finding reported in class: when fitted to stock returns, the jump rate came out very high, a jump every couple of days, so the model was not detecting rare crashes but mimicking a fat-tailed distribution by a mixture of a plain normal, a normal with one jump, with two jumps, and so on [► 2024 L19 @ 1:02:28].

13.4.2 Brownian motion observed at random times: the Laplace law

If a Brownian motion is observed not at equally spaced times but after random, *exponentially distributed* increments of time, the increments of the process follow the *Laplace* (two-sided exponential) law [► 2024 L19 @ 1:03:32]

$$f(x \mid \mu, b) = \frac{1}{2b} e^{-|x - \mu|/b}, \quad \mathbb{E}[X] = \mu, \quad \text{Var } X = 2b^2, \quad \kappa = 6. \quad (13.16)$$

Proof. Apply [proposition 3.29](#) with $V = \Delta + N\gamma^2$: $\mathbb{E}[V] = \Delta(1 + \lambda\gamma^2)$ and, since $\text{Var } N = \lambda\Delta$, $\text{Var } V = \gamma^4\lambda\Delta$. Then $\kappa = 3(1 + \text{Var } V/(\mathbb{E}[V])^2)$. $\square$

HISTORY: Laplace's first law of errors

Laplace himself introduced it in 1774 as the “first law of errors”.

References. P.-S. Laplace, “Mémoire sur la probabilité des causes par les événements”, *Mémoires de l'Académie royale des sciences présentés par divers savans* 6 (1774) 621–656; S. M. Stigler, *The History of Statistics*, Harvard University Press, 1986, ch. 3.

Proposition 13.8. Let τ be exponential with mean s and $X = \mu + \sigma B_\tau$ with B a standard Brownian motion independent of τ . Then $X \sim \text{Laplace}(\mu, b)$ with $b^2 = \sigma^2 s/2$.

It is a mixture with an exponentially distributed clock, the simplest “subordinated” process (section 13.2.3), and it has $P(|X - \mu| > 4\sqrt{\text{Var } X}) = e^{-4\sqrt{2}} = 0.35\%$ against 6.3×10^{-5} for the normal.

CASE STUDY: Laplace against normal on S&P 500 returns

The lecture fits both laws to $n = 1000$ daily S&P 500 returns, 2014-10-10 to 2018-10-02 [► 2024 L19 @ 1:05:34]. Method of moments for the Laplace law: $\hat{\mu} = \bar{R} = 0.000427$, $\hat{b} = \sqrt{\hat{\text{Var}}/2} = 0.005652$; the implied annualised volatility is $\sqrt{252}\sqrt{2}\hat{b} = 12.69\%$ (a check: $\sqrt{2} \times 0.005652 \times \sqrt{252} = 0.1269$). The histogram is peaked with exponentially decaying tails and the Laplace curve follows it much better than the Gaussian (drawn in red for comparison). The Q - Q plot (sorted observations against theoretical quantiles) is close to a straight line for the Laplace law and clearly curved for the normal [► 2024 L19 @ 1:06:37]. So the Laplace law is a better model of independent daily returns; what neither can explain is that returns are not independent.

Proof. Given τ , $X \sim N(\mu, \sigma^2 \tau)$, so the characteristic function is $\mathbb{E}[e^{iu(X-\mu)}] = \mathbb{E}[e^{-\sigma^2 u^2 \tau/2}] = (1 + \sigma^2 s u^2/2)^{-1}$ (Laplace transform of the exponential law). The Laplace density has characteristic function $e^{iu\mu}/(1 + b^2 u^2)$; the two agree for $b^2 = \sigma^2 s/2$. As a mixture $V = \tau$ has $\mathbb{E}[V^2]/(\mathbb{E}[V])^2 = 2$, whence $\kappa = 6$ (proposition 3.29) and $\text{Var } X = \sigma^2 s = 2b^2$. $\square$

13.5 ARCH models

The models so far have independent returns and hence a volatility that is the same in every period. The data say otherwise (section 13.8): large returns tend to follow large returns. Engle (1982) proposed to let the volatility of period t depend on the returns of the preceding periods [► 2024 L19 @ 1:07:45][► 2013 L9 @ 1:05:24].

Definition 13.9 (ARCH(p)). Let $y_t = \log(S_t/S_{t-1})$ and $\mathcal{F}_t$ the information available at time t . The *autoregressive conditionally heteroskedastic* model of order p is

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2, \quad (13.17)$$

where $\mu_t = \mathbb{E}[y_t | \mathcal{F}_{t-1}]$ is the conditional mean, $\{z_t\}$ are i.i.d. with $\mathbb{E}[z_t] = 0$, $\text{Var } z_t = 1$, independent of $\mathcal{F}_{t-1}$, and $\alpha_0 > 0$, $\alpha_i \geq 0$. Then $\sigma_t^2 = \text{Var}(y_t | \mathcal{F}_{t-1})$ is the *conditional variance*.

The name says it all: *conditional heteroskedasticity* means that the variance given the past changes over time; *autoregressive* refers to its dependence on lagged squares. If ε_{t-1} was large, the variance of ε_t is large and ε_t is likely to be large too. The constraints have a reason: if some α_i were negative, a large past shock could make σ_t^2 negative [► 2013 L9 @ 1:08:30] [► 2024 L19 @ 1:12:05].

◊ **BEYOND THE CORE: Why $\alpha_0 > 0$**

(The slides write $\alpha_j \geq 0$ for $j = 0, \dots, p$, i.e. allow $\alpha_0 = 0$, in which case the variance can shrink to zero and the model degenerates; we take $\alpha_0 > 0$.)

Proposition 13.10 (ARCH innovations are uncorrelated but not independent). Let $\varepsilon_t = \sigma_t z_t$ with $\sigma_t > 0$ $\mathcal{F}_{t-1}$ -measurable, $\mathbb{E}[\sigma_t^2] < \infty$, and z_t as in definition 13.9. Then

- (i) $\mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = 0$ and $\mathbb{E}[\varepsilon_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$;
- (ii) $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$: the innovations form a martingale difference sequence and are uncorrelated;

Proof. (i) z_t is independent of σ_t : $\mathbb{E}[\sigma_t z_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[z_t] = 0$, $\mathbb{E}[\sigma_t^2 z_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$. (ii) For $s < t$, ε_s is $\mathcal{F}_{t-1}$ -measurable, so $\mathbb{E}[\varepsilon_t \varepsilon_s] = \mathbb{E}[\varepsilon_s \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}]] = 0$. (iii) $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-1}^2] = \mathbb{E}[\sigma_t^2 \varepsilon_{t-1}^2] = \alpha_0 \mathbb{E}[\varepsilon^2] + \alpha_1 \mathbb{E}[\varepsilon^4]$, while $\mathbb{E}[\varepsilon_t^2] \mathbb{E}[\varepsilon_{t-1}^2] = (\mathbb{E}[\varepsilon^2])(\alpha_0 + \alpha_1 \mathbb{E}[\varepsilon^2])$; subtract. $\square$

(iii) for ARCH(1) with $\mathbb{E}[\varepsilon_t^4] < \infty$, $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-1}^2) = \alpha_1 \text{Var} \varepsilon_t^2 > 0$: the squares are correlated, so the ε_t are not independent.

This is the precise sense of the remark of [section 9.1.2](#) that residual whiteness is not the end of the story: the ACF of the returns is flat (the martingale difference), that of the squares is not [\[► 2024 L19 @ 1:09:55\]](#). For the S&P 500 the autocorrelations of the squared, de-measured log returns are significantly positive at (about) the first 10 lags [\[► 2024 L19 @ 1:09:55\]](#); for the euro/dollar rate the ACF and PACF of the squares are large and of the returns marginal ([section 13.7.4](#)) [\[► 2013 L9 @ 1:18:02\]](#).

13.5.1 ARCH is an autoregression for the squared returns

Put $u_t = \varepsilon_t^2 - \sigma_t^2$ and add it to both sides of (13.17):

$$\varepsilon_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2 + u_t, \quad \mathbb{E}[u_t | \mathcal{F}_{t-1}] = 0, \quad \text{Var}[u_t | \mathcal{F}_{t-1}] = (\kappa_z - 1)\sigma_t^4, \quad (13.18)$$

where $\kappa_z = \mathbb{E}[z^4]$ (equal to 3 for Gaussian z , so that the conditional variance of u_t is $2\sigma_t^4$) [\[► 2024 L19 @ 1:09:55\]](#)[\[► 2013 L9 @ 1:09:31\]](#). The squared innovations follow an AR(p) whose noise u_t has mean zero and a variance that changes with time. This yields a quick test for ARCH effects.

Lagrange-multiplier test for ARCH effects. To test $H_0 : \alpha_1 = \cdots = \alpha_p = 0$, fit the mean model, take the residuals $\hat{\varepsilon}_t = y_t - \hat{\mu}_t$, regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-p}^2$ (an AR(p) fit to the squared residuals), and compute

$$\text{LM} = nR^2 \stackrel{H_0}{\approx} \chi_p^2, \quad (13.19)$$

with R^2 the coefficient of determination of that regression [\[► 2024 L19 @ 1:09:55\]](#) [\[► 2013 L9 @ 1:10:34\]](#). Under H_0 the squares $\hat{\varepsilon}_t^2$ have no linear dependence on the past and R^2 is small; for $p = 1$, R^2 is the squared sample correlation between $\hat{\varepsilon}_t^2$ and $\hat{\varepsilon}_{t-1}^2$, so $nR^2 \approx n\hat{r}_1^2$ with $\sqrt{n}\hat{r}_1 \rightarrow N(0, 1)$, as in [eq. \(9.4\)](#).

◊ BEYOND THE CORE: McLeod–Li and quasi-maximum likelihood

For general p the statistic is asymptotically equivalent to the Box–Pierce statistic of the squared residuals (the McLeod–Li test; this remark is not from the class). The regression uses least squares, whose assumptions are not those of a Gaussian likelihood; its estimates are *quasi-maximum-likelihood* estimates, consistent but not efficient [\[► 2013 L9 @ 1:11:40\]](#).

IN FINANCE: Volatility clustering and prediction

An ARCH model predicts nothing about the *sign* of tomorrow's return but a lot about its size, and size is what matters for risk: the conditional variance σ_t^2 that it produces is a one-step-ahead volatility forecast that reacts to the latest shock. This is the improvement over the historical averages of [section 13.1.3](#), whose weights are chosen by hand instead of being estimated.

Exercise 13.6 (PS5 (2013), Problem 3: properties of the ARCH(1) model). Let $y_t = \mu_t + \varepsilon_t$, $\varepsilon_t = z_t \sigma_t$, $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$, with z_t i.i.d., $\mathbb{E}[z_t] = 0$, $\text{Var} z_t = 1$, $\mathbb{E}[z_t^3] = 0$, $\mathbb{E}[z_t^4] = \kappa_z$ ($= 3$ if Gaussian). Prove:

(a) $\mathbb{E}[\varepsilon_t^2] = \alpha_0/(1 - \alpha_1)$;

5 COURSE NOTE: Erratum: the noise of the AR equation is not white noise with variance $2\sigma^4$

Slides 21 and 25 of [lec17_1](#) (slides 20 and 24 of 2013) write $\text{Var}[u_t | \mathcal{F}_t] = \text{Var}(\varepsilon_t^2) = 2\sigma_t^4$, and later $u_t \sim \text{WN}(0, 2\sigma^4)$. Two corrections. First, the conditional variance is taken given $\mathcal{F}_{t-1}$ (given $\mathcal{F}_t$ the variable u_t is known), and it equals $(\kappa_z - 1)\sigma_t^4$, i.e. $2\sigma_t^4$ only for Gaussian z_t . Second, σ_t^2 is random, so u_t is not white noise in the strong sense: it is a martingale difference (hence uncorrelated), and it has a constant *unconditional* variance $(\kappa_z - 1)\mathbb{E}[\sigma_t^4]$ only if $\mathbb{E}[\varepsilon_t^4] < \infty$ ([theorem 13.18](#)). The ARMA representation of GARCH below is therefore an ARMA with uncorrelated, not independent, innovations; the second-moment formulas of [section 9.3.3](#) apply verbatim.

- (b) $\mathbb{E}[\varepsilon_t^3] = 0$;
- (c) $\mathbb{E}[\varepsilon_t^4] = \kappa_z \alpha_0^2 (1 + \alpha_1) / [(1 - \alpha_1)(1 - \kappa_z \alpha_1^2)]$;
- (d) which constraints on α_0, α_1 does (c) require for the fourth moment to be finite and stationary;
- (e) the kurtosis of ε_t : for Gaussian z_t , is the unconditional distribution of ε_t heavier-tailed than the Gaussian? (The problem as printed writes the constant κ_z in (c) without naming it; it is the kurtosis of z_t stated just before.)

13.6 GARCH models

13.6.1 Definition and parsimony

Data often need a large ARCH order to reproduce the persistence of volatility (a hint of the infinite-order autoregression of Wold's theorem, [section 9.2](#)). Bollerslev (1986) added lagged *variances* to the right-hand side [► 2024 L19 @ 1:12:05][► 2013 L9 @ 1:14:50][► 2013 L11 @ 1:07].

Definition 13.11 (GARCH(p, q)). With $y_t = \mu_t + \varepsilon_t$ and $\varepsilon_t = \sigma_t z_t$ as in [definition 13.9](#),

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad \alpha_0 > 0, \alpha_i \geq 0, \beta_j \geq 0. \quad (13.20)$$

The most used case is the GARCH(1, 1),

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (13.21)$$

parsimonious (three parameters, plus those of the mean) and adequate for many financial series; in practice the (1, 1) order is often the one selected among the GARCH orders [► 2024 L19 @ 1:13:09][► 2013 L9 @ 1:16:58].

Proposition 13.12 (GARCH(1, 1) is an ARCH(∞) with geometric weights). If $\beta_1 < 1$ and the process is stationary ([theorem 13.15](#)),

$$\sigma_t^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{j \geq 0} \beta_1^j \varepsilon_{t-1-j}^2.$$

Proof. Iterate (13.21): $\sigma_t^2 = \alpha_0 \sum_{j < n} \beta_1^j + \alpha_1 \sum_{j < n} \beta_1^j \varepsilon_{t-1-j}^2 + \beta_1^n \sigma_{t-n}^2$; the last term tends to 0 in expectation because $\mathbb{E}[\sigma_{t-n}^2]$ is finite and constant and $\beta_1^n \rightarrow 0$. $\square$

So the conditional variance is an exponentially weighted average of past squared innovations plus a constant: the same structure as the EMA of [proposition 13.3](#), with the decay factor β_1 and the weight $\alpha_1/(1 - \beta_1)$ on the average. Two parameters (α_1, β_1) replace the ten or more coefficients of a high-order ARCH, which is the parsimony argument of the lecture: the Wold theorem promises that some infinite-order model fits, and GARCH gives a parametrisation of the infinite coefficient sequence by two numbers [► 2013 L9 @ 1:20:10].

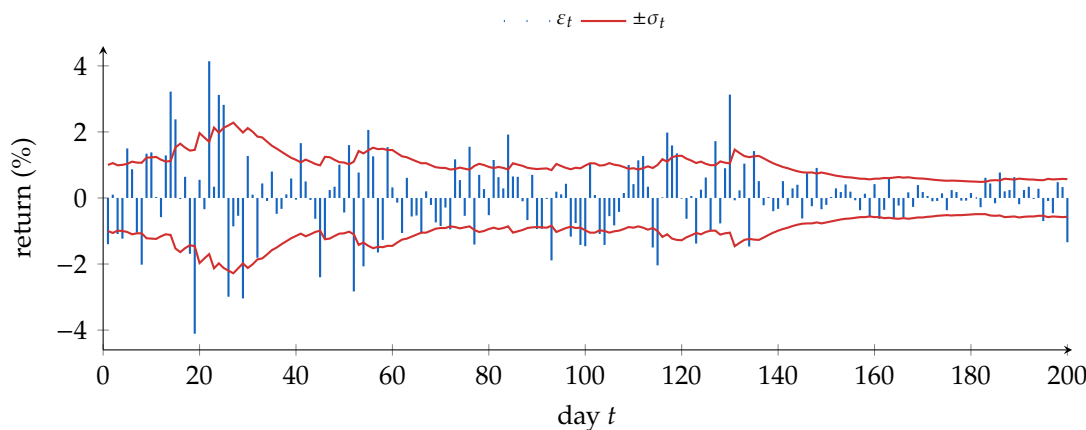


Figure 13.2: A simulated GARCH(1, 1) path, Gaussian z_t , $\alpha_1 = 0.12$, $\beta_1 = 0.85$, long-run volatility 1% per day (200 days). Returns (bars) and the conditional volatility $\pm\sigma_t$: calm and turbulent stretches alternate, and the amplitude of the returns follows σ_t , though the returns are uncorrelated (lag-1 autocorrelation 0.06, of the squares 0.15).

13.6.2 ARMA representation of the squared innovations

Proposition 13.13 (GARCH is ARMA for ε^2). Let $u_t = \varepsilon_t^2 - \sigma_t^2$, so that $\mathbb{E}[u_t | \mathcal{F}_{t-1}] = 0$. A GARCH(p, q) satisfies, with $\alpha_i = 0$ for $i > p$ and $\beta_j = 0$ for $j > q$,

$$\varepsilon_t^2 = \alpha_0 + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) \varepsilon_{t-i}^2 + u_t - \sum_{j=1}^q \beta_j u_{t-j}, \quad (13.22)$$

an ARMA($\max(p, q), q$) for ε_t^2 . For the GARCH(1, 1): $\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \varepsilon_{t-1}^2 + u_t - \beta_1 u_{t-1}$, an ARMA(1, 1) with $\phi = \alpha_1 + \beta_1$ and $\theta = -\beta_1$.

Proof. From $\sigma_{t-j}^2 = \varepsilon_{t-j}^2 - u_{t-j}$ substitute in (13.20): $\varepsilon_t^2 - u_t = \alpha_0 + \sum_i \alpha_i \varepsilon_{t-i}^2 + \sum_j \beta_j (\varepsilon_{t-j}^2 - u_{t-j})$; collect terms. $\square$

This is the statement announced in [section 9.3.3](#) [► 2024 L19 @ 1:13:09][► 2013 L11 @ 4:15].

◊ **BEYOND THE CORE: A slip in the 2013 transcript**

(the transcript of 2013 writes $-\beta_1 u_t$ for $-\beta_1 u_{t-1}$).

With the AR polynomial $A(L) = 1 - \sum_i (\alpha_i + \beta_i) L^i$ and the MA polynomial $B(L) = 1 - \sum_j \beta_j L^j$ one has

$$A(L) \varepsilon_t^2 = \alpha_0 + B(L) u_t \quad [\text{► 2013 L11 @ 6:19}].$$

◊ **BEYOND THE CORE: The root condition**

Lemma 13.14 (Stationarity of the ARMA). For $c_i = \alpha_i + \beta_i \geq 0$ the roots of $A(z) = 1 - \sum_i c_i z^i$ lie outside the closed unit disc iff $\sum_i c_i < 1$.

Proof. If $\sum c_i < 1$, then for $|z| \leq 1$, $|\sum c_i z^i| \leq \sum c_i < 1$, so $A(z) \neq 0$. If $\sum c_i \geq 1$, then $A(0) = 1 > 0 \geq A(1)$, and A is continuous, so it has a root in $(0, 1]$. $\square$

Consequently the ARMA of (13.22) is covariance stationary iff $\sum (\alpha_i + \beta_i) < 1$; for the (1, 1) case, $\alpha_1 + \beta_1 < 1$ [► 2024 L19 @ 1:13:09][► 2013 L11 @ 6:19]. When $\mathbb{E}[\varepsilon^4] < \infty$ ([theorem 13.18](#)) the noise u_t has constant variance and is uncorrelated, and the ARMA formulas of [section 9.3.3](#) give the autocorrelation function of the squares:

$$\rho_{\varepsilon^2}(1) = \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}, \quad \rho_{\varepsilon^2}(k) = (\alpha_1 + \beta_1)^{k-1} \rho_{\varepsilon^2}(1), \quad k \geq 2. \quad (13.23)$$

◆ **BEYOND THE CORE: Checking the formula**

(Insert $\phi = \alpha_1 + \beta_1$, $\theta = -\beta_1$ in eq. (9.18): $1 + \phi\theta = 1 - \alpha_1\beta_1 - \beta_1^2$, $\phi + \theta = \alpha_1$, $1 + 2\phi\theta + \theta^2 = 1 - 2\alpha_1\beta_1 - \beta_1^2$.) The lag-one correlation does not depend on the kurtosis of z_t . For $\alpha_1 = 0.10$, $\beta_1 = 0.85$: $\rho(1) = 0.179$, $\rho(5) = 0.146$ (simulation, 2×10^6 days: 0.181 and 0.146); for $\alpha_1 = 0.05$, $\beta_1 = 0.85$ one gets $\rho(1) = 0.061$ with slow decay, the “small first correlation but long memory” example of section 9.3.3.

The slow geometric decay at rate $\alpha_1 + \beta_1$, near 1, is what the significant autocorrelations of squared returns at many lags (the case studies) require, and what a low-order ARCH cannot produce.

13.6.3 Stationarity and long-run variance

Theorem 13.15 (Stationarity of the GARCH(1, 1)). Let $\alpha_0 > 0$ and $\phi = \alpha_1 + \beta_1$. There is a covariance stationary solution of (13.21) iff $\phi < 1$, and then

$$\sigma_*^2 := \mathbb{E}[\varepsilon_t^2] = \mathbb{E}[\sigma_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}. \quad (13.24)$$

More generally, a GARCH(p, q) is covariance stationary iff $\sum_i (\alpha_i + \beta_i) < 1$, with $\sigma_*^2 = \alpha_0 / [1 - \sum_{i \leq \max(p,q)} (\alpha_i + \beta_i)]$.

Taking expectations of (13.21) is how the lecture obtains the *long-run variance* $\sigma_*^2 = \alpha_0 + (\alpha_1 + \beta_1)\sigma_*^2$ [► 2024 L19 @ 1:14:13]. Conversely $\alpha_0 = (1 - \alpha_1 - \beta_1)\sigma_*^2$: one can parametrise the model by $(\sigma_*^2, \alpha_1, \beta_1)$, and σ_*^2 is then simply the unconditional variance of the returns (which the sample variance estimates).

13.6.4 Persistence, mean reversion and half-life

Subtracting σ_*^2 from (13.21) written with $\alpha_0 = (1 - \phi)\sigma_*^2$ gives the *mean-reverting form*

$$\sigma_t^2 - \sigma_*^2 = \alpha_1(\varepsilon_{t-1}^2 - \sigma_*^2) + \beta_1(\sigma_{t-1}^2 - \sigma_*^2), \quad \varepsilon_t^2 - \sigma_*^2 = \phi(\varepsilon_{t-1}^2 - \sigma_*^2) + u_t - \beta_1 u_{t-1}, \quad (13.25)$$

the second being the ARMA of proposition 13.13 [► 2024 L19 @ 1:14:13][► 2013 L11 @ 25:25]. It says that the deviation of the squared return from its long-run average is a fraction ϕ of the previous deviation plus noise: a discrete Ornstein–Uhlenbeck process (an AR(1), section 9.3.1; the continuous-time version is chapter 18) with MA(1) noise. Because $0 < \phi < 1$ volatility *reverts to the mean*: the excess of variance over σ_*^2 shrinks by the factor ϕ per period [► 2013 L11 @ 26:27]. The sum $\alpha_1 + \beta_1$ is therefore called the *persistence* of the model [► 2013 L11 @ 27:33]; the *half-life* of a variance shock is $\ln \frac{1}{2} / \ln \phi$ periods:

persistence ϕ	0.90	0.95	0.97	0.99	0.997
half-life (days)	6.6	13.5	22.8	69	231

In the ARMA language the persistence is the AR root; the parameters α_1 (reaction to news) and β_1 (memory) have distinct roles: for the same ϕ a larger α_1 gives a more jagged volatility with larger kurtosis (theorem 13.18), a larger β_1 a smoother one.

Proof. Put $a_t = \alpha_1 z_t^2 + \beta_1$, an i.i.d. sequence with $\mathbb{E}[a_t] = \phi$; then (13.21) reads $\sigma_t^2 = \alpha_0 + a_{t-1}\sigma_{t-1}^2$, where a_{t-1} is independent of σ_{t-1}^2 . *Necessity.* If a stationary solution with $\mathbb{E}[\sigma_t^2] = m < \infty$ exists, taking expectations gives $m = \alpha_0 + \phi m$, so $m = \alpha_0 / (1 - \phi)$, which is positive only if $\phi < 1$. *Sufficiency.* Iterating n times, $\sigma_t^2 = \alpha_0(1 + a_{t-1} + a_{t-1}a_{t-2} + \dots + a_{t-1} \dots a_{t-n+1}) + a_{t-1} \dots a_{t-n} \sigma_{t-n}^2$. If $\phi < 1$ the series with $n = \infty$ has non-negative terms and expectation $\alpha_0 \sum_k \phi^k = \alpha_0 / (1 - \phi) < \infty$, hence converges almost surely and in L^1 , and it defines a stationary solution (a function of $z_{t-1}, z_{t-2}, \dots$); the remainder term has expectation $\phi^n \mathbb{E}[\sigma_{t-n}^2] \rightarrow 0$, so this is the only solution with finite mean. For the general (p, q) case take expectations in (13.20) (with $\mathbb{E}[\varepsilon_t^2] = \mathbb{E}[\sigma_t^2]$) and use lemma 13.14 for the equivalence with the ARMA polynomial [► 2013 L11 @ 7:23]. □

6 COURSE NOTE: Erratum: typo in the general long-run variance; constraints

Slide 26 of lec17_1 (slide 25 of 2013) writes $\sigma_*^2 = \alpha_0 + (\sum_1^p \alpha_i + \sum_1^q \beta_i)\sigma_*^2$; the last coefficient should read β_j . Slide 27 has a typo in the constraint $\sum \alpha_u$ for $\sum \alpha_i$. The strict stationarity of the process is a weaker condition than covariance stationarity (Nelson, 1990; not discussed in class): the GARCH(1, 1) has a strictly stationary solution iff $\mathbb{E}[\ln(\alpha_1 z_t^2 + \beta_1)] < 0$, and by Jensen $\mathbb{E}[\ln a_t] \leq \ln \phi$, so $\phi < 1$ suffices but $\phi \geq 1$ is not excluded: the model may be strictly stationary with infinite variance.

IN FINANCE: Persistence of the euro/dollar volatility

The Gaussian GARCH(1, 1) fitted to EUR/USD returns in the 2013 case study (section 13.7.4) has $\hat{\alpha}_1 = 0.0278$, $\hat{\beta}_1 = 0.9696$, $\hat{\omega} = 1.19 \times 10^{-7}$: persistence 0.9973, half-life 259 trading days (about a year), long-run variance $\hat{\omega}/(1 - \hat{\phi}) = 4.46 \times 10^{-5}$, i.e. a daily volatility 0.668% and an annual 10.6%, remarkably close to the 10.25% of the constant-volatility fit of section 13.2.3. The case study concludes that the model is “very close to being non-stationary” and that there is practically no tendency to revert to a specific volatility level within a horizon of a few months [► 2013 L11 @ 27:33]: volatility is a slowly wandering quantity, a feature that risk managers exploit by using an exponential average with a slow decay.

13.6.5 Forecasting the variance

Theorem 13.16 (Variance forecasts of a GARCH(1, 1)). Let $\phi = \alpha_1 + \beta_1 < 1$. For every $h \geq 1$,

$$\mathbb{E}[\varepsilon_{t+h}^2 | \mathcal{F}_t] = \mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t] = \sigma_*^2 + \phi^{h-1}(\sigma_{t+1}^2 - \sigma_*^2), \quad (13.26)$$

where $\sigma_{t+1}^2 = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2$ is known at time t . The expected total variance of the next H periods is

$$\mathbb{E}\left[\sum_{h=1}^H \varepsilon_{t+h}^2 \middle| \mathcal{F}_t\right] = H\sigma_*^2 + (\sigma_{t+1}^2 - \sigma_*^2) \frac{1 - \phi^H}{1 - \phi}. \quad (13.27)$$

Two consequences. The forecast decays geometrically to the long-run variance at the rate ϕ per period: in the short run the variance moves incrementally, in the long run it returns to σ_*^2 [► 2013 L11 @ 26:27]. And, since the ε_t are uncorrelated, (13.27) is the variance of the H -period return (the conditional variance of a sum of martingale differences is the sum of the conditional variances of the terms), so the annualised volatility expected over the next H days is $\sqrt{(252/H) \times}$ the right-hand side of (13.27).

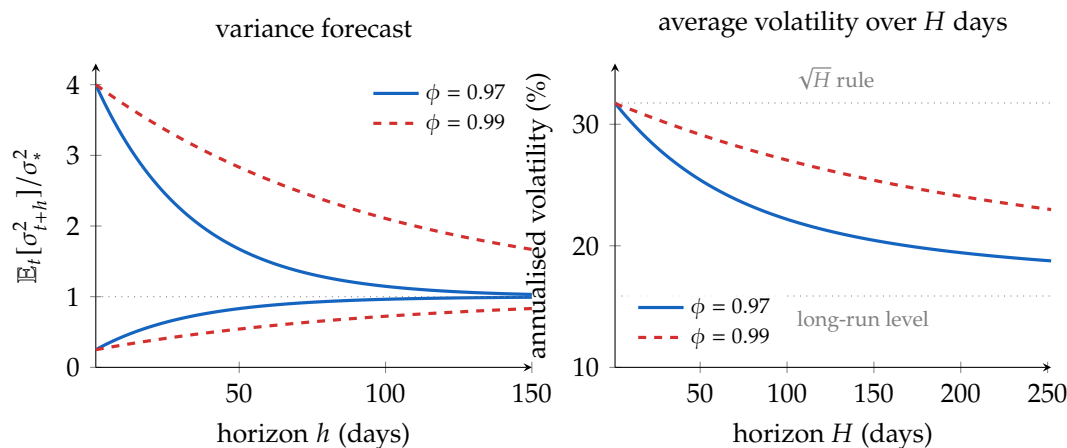


Figure 13.3: Left: expected conditional variance $\mathbb{E}_t[\sigma_{t+h}^2]$ in units of σ_*^2 after a shock ($\sigma_{t+1} = 2\sigma_*$) and after a calm day ($\sigma_{t+1} = \sigma_*/2$), for two persistences, from (13.26). Right: annualised volatility expected over the next H days after the shock of example 13.17, from (13.27); the dotted lines are the $\sqrt{H}$ extrapolation of today's volatility and the long-run level 15.9%.

Proof. By the tower property $\mathbb{E}[\varepsilon_{t+h}^2 | \mathcal{F}_t] = \mathbb{E}[\mathbb{E}[\varepsilon_{t+h}^2 | \mathcal{F}_{t+h-1}] | \mathcal{F}_t] = \mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t]$. Write m_h for the deviation $\mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t] - \sigma_*^2$. Taking $\mathbb{E}[\cdot | \mathcal{F}_t]$ in (13.25) for $h \geq 2$ gives $m_h = \alpha_1 m_{h-1} + \beta_1 m_{h-1} = \phi m_{h-1}$, and $m_1 = \sigma_{t+1}^2 - \sigma_*^2$; induct. The second formula is the geometric sum $\sum_{h=1}^H \phi^{h-1}$. □

Example 13.17 (After a turbulent day). Let $\sigma_* = 1\%$ per day (annualised 15.9%), $\alpha_1 = 0.06$, $\beta_1 = 0.91$, so $\phi = 0.97$ and the half-life is 22.8 days, and let today's news raise the forecast for tomorrow to $\sigma_{t+1} = 2\%$ (31.7% annualised). Then the annualised volatility expected over the next H days is 31.7% for $H = 1$ and

H (days)	5	10	21
GARCH(1, 1)	31.0%	30.2%	28.6%
$\sqrt{H}$ rule from σ_{t+1}	31.7%	31.7%	31.7%

For a 21-day horizon the standard deviation of the return is 8.26% against $\sqrt{21} \times 2\% = 9.17\%$ from the naive scaling, an overstatement of 11%; for a year the naive rule is wrong by a factor 1.7. This is the failure of the square-root-of-time rule announced in remark 13.1, and it is relevant for the multi-day value at risk of chapter 19 (cf. proposition 8.11, where the discrepancy has the opposite sign, being due to autocorrelation of returns).

Exercise 13.7 ((not from the course): forecasting). For a GARCH(1, 1) with $\alpha_1 = 0.08$, $\beta_1 = 0.90$, long-run daily volatility 1%, and $\sigma_{t+1} = 1.6\%$, compute the expected annualised volatility over the next 21 and 252 days from (13.27), and the half-life of a variance shock. Check by simulation.

13.6.6 Fourth moment and kurtosis

GARCH reproduces the fat tails of returns even when the innovations z_t are Gaussian, and one can say by how much.

Theorem 13.18 (Kurtosis of the GARCH(1, 1)). Let $\kappa_z = \mathbb{E}[z_t^4] < \infty$, $\phi = \alpha_1 + \beta_1 < 1$, and assume $\Delta := 1 - \phi^2 - (\kappa_z - 1)\alpha_1^2 > 0$. Then $\mathbb{E}[\varepsilon_t^4] < \infty$ and the kurtosis of ε_t is

$$\kappa_\varepsilon = \frac{\mathbb{E}[\varepsilon_t^4]}{(\mathbb{E}[\varepsilon_t^2])^2} = \kappa_z \frac{1 - \phi^2}{1 - \phi^2 - (\kappa_z - 1)\alpha_1^2} > \kappa_z. \quad (13.28)$$

If $\Delta \leq 0$ the fourth moment is infinite. For Gaussian z the condition is $(\alpha_1 + \beta_1)^2 + 2\alpha_1^2 < 1$, i.e. $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$.

The mechanism is that of [proposition 3.29](#): given the past, ε_t is normal with variance σ_t^2 , and this variance fluctuates, so the marginal law is a mixture of Gaussians with kurtosis $\kappa_z \mathbb{E}[\sigma^4]/(\mathbb{E}[\sigma^2])^2 \geq \kappa_z$, strictly unless σ_t is constant [[2013 L11 @ 24:19](#)]; the slides describe GARCH as a “stochastic mixture of Gaussians with higher kurtosis”. For $\beta_1 = 0$ one recovers the ARCH(1) formulas of 2013 PS5 Problem 3 ([Exercise 13.6](#)): $\mathbb{E}[\varepsilon^4] = \kappa_z \alpha_0^2(1 + \alpha_1)/[(1 - \alpha_1)(1 - \kappa_z \alpha_1^2)]$, finite iff $\alpha_1 < 1/\sqrt{3} = 0.577$ for Gaussian z .

♦ **BEYOND THE CORE: Kurtosis for typical parameters**

(α_1, β_1)	ϕ	$1 - \phi^2 - 2\alpha_1^2$	kurtosis (Gaussian z)
(0.05, 0.90)	0.95	0.0925	3.16
(0.10, 0.85)	0.95	0.0775	3.77 (simulation 3.79)
(0.0278, 0.9696), EUR/USD fit	0.9973	0.0038	4.22
(0.10, 0.88)	0.98	0.0196	6.06
(0.09, 0.90)	0.99	0.0037	16.1
(0.12, 0.87)	0.99	-0.0089	∞

Persistence close to 1 makes the fourth moment barely exist, and then sample kurtosis is an unreliable estimator of the theoretical one: sample fourth moments of such a process converge very slowly.

♦ **BEYOND THE CORE: Power-law tails**

(A tail-index refinement, not discussed: $\mathbb{P}(|\varepsilon| > x) \sim x^{-\xi}$ with ξ solving $\mathbb{E}[a_t^{\xi/2}] = 1$, so GARCH tails are power laws, as in the empirical “inverse cubic” law of [section 3.9.1](#).)

13.6.7 The RiskMetrics limit: IGARCH

The RiskMetrics exponential average with decay factor β is the GARCH(1, 1) with $\alpha_0 = 0$, $\alpha_1 = 1 - \beta$, $\beta_1 = \beta$, that is $\phi = 1$:

$$\sigma_{t+1}^2 = (1 - \beta) \varepsilon_t^2 + \beta \sigma_t^2, \quad (13.29)$$

Proof. With $a_t = \alpha_1 z_t^2 + \beta_1$ as before, $\mathbb{E}[a_t] = \phi$ and $\mathbb{E}[a_t^2] = \alpha_1^2 \kappa_z + 2\alpha_1\beta_1 + \beta_1^2 = \phi^2 + (\kappa_z - 1)\alpha_1^2$, so $\Delta = 1 - \mathbb{E}[a_t^2]$. Finite fourth moment iff $\mathbb{E}[a^2] < 1$: the series of [theorem 13.15](#) has terms $\alpha_0 \prod_{j \leq k} a_{t-j}$ of L^2 norm $\alpha_0(\mathbb{E}[a^2])^{k/2}$, summable iff $\mathbb{E}[a^2] < 1$ (Minkowski’s inequality gives sufficiency); if $\mathbb{E}[\sigma^4] < \infty$ and stationary, the identity below forces $\mathbb{E}[a^2] < 1$. Squaring $\sigma_t^2 = \alpha_0 + a_{t-1}\sigma_{t-1}^2$ and using independence of a_{t-1} and σ_{t-1}^2 ,

$$\mathbb{E}[\sigma^4] = \alpha_0^2 + 2\alpha_0\phi \sigma_*^2 + \mathbb{E}[a^2] \mathbb{E}[\sigma^4] \implies \mathbb{E}[\sigma^4] = \frac{\alpha_0^2(1 + \phi)}{(1 - \phi)\Delta},$$

because $\alpha_0^2 + 2\alpha_0\phi \cdot \alpha_0/(1 - \phi) = \alpha_0^2(1 + \phi)/(1 - \phi)$. Since $\varepsilon_t = \sigma_t z_t$ with z_t independent of σ_t , $\mathbb{E}[\varepsilon^4] = \kappa_z \mathbb{E}[\sigma^4]$ and $\kappa_\varepsilon = \kappa_z \mathbb{E}[\sigma^4]/\sigma_*^4 = \kappa_z \frac{(1 + \phi)(1 - \phi)^2}{(1 - \phi)\Delta} = \kappa_z \frac{1 - \phi^2}{\Delta} > \kappa_z$. $\square$

an *integrated* GARCH (IGARCH). It has no long-run variance, and by (13.26) with $\phi = 1$ its forecast is flat, $\mathbb{E}_t[\sigma_{t+h}^2] = \sigma_{t+1}^2$ for all h , so it scales volatility with $\sqrt{H}$ [► 2013 L11 @ 27:33]. The lecturer describes the advantage: one tracks a possibly non-stationary volatility without presuming that a long-run average consistent with the past exists.

◊ **BEYOND THE CORE: Two caveats on IGARCH**

Two caveats, not discussed: as a *generating* process the IGARCH with $\alpha_0 = 0$ is degenerate, since $\ln \sigma_t^2 = \ln \sigma_0^2 + \sum_{j \leq t} \ln a_j$ has i.i.d. increments of strictly negative mean $\mathbb{E}[\ln a] < \ln \mathbb{E}[a] = 0$ (Jensen), so $\sigma_t^2 \rightarrow 0$ almost surely; for $\alpha_1 = 0.06$ and Gaussian z the drift is $\mathbb{E}[\ln a] = -0.0032$ per day, slow enough to be invisible in a sample of a few hundred days. And the recursion (13.29) is exactly the steady-state Kalman filter of a local-level model (section 9.7), which is a better way to think of it: a *filter* for the current level of volatility, not a model of its evolution.

13.7 Estimation of ARCH and GARCH models

13.7.1 The likelihood

A GARCH model specifies the law of y_t *given the past*, and this is exactly what the likelihood needs. Write $\theta = (c, \alpha_0, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)$ for the parameters, $y_t = c + \varepsilon_t$ (a constant mean; section 13.7.4 uses an AR mean), and $\varepsilon_t = \sigma_t z_t$. The joint density factorises into conditional densities (the chain rule of probability) [► 2024 L19 @ 1:14:13] [► 2013 L11 @ 8:26].

Proposition 13.19 (Gaussian log-likelihood of a GARCH model). *If z_t is $N(0, 1)$ then $y_t \mid \mathcal{F}_{t-1} \sim N(c, \sigma_t^2)$, where $\sigma_t^2 = \sigma_t^2(\theta)$ is computed recursively from (13.20) and the data, and*

$$\ell(\theta) = \sum_{t=1}^T \log p(y_t \mid \mathcal{F}_{t-1}; \theta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2} \right], \quad \varepsilon_t = y_t - c. \quad (13.30)$$

Two points.

◊ **BEYOND THE CORE: Starting values**

First, the recursion needs starting values (ε_0^2 and σ_0^2): common choices are the sample variance for both, or the long-run variance $\alpha_0/(1 - \phi)$; the effect fades out geometrically at rate β_1 (proposition 13.12) but is felt for a long time when $\beta_1 \approx 1$ (not discussed in class).

Second, the likelihood is *not* a quadratic function of θ and has no closed-form maximiser (unlike the regression of chapter 5); the maximum is found numerically. The two terms have a clear meaning: $\log \sigma_t^2$ penalises a large predicted variance, $\varepsilon_t^2/\sigma_t^2$ penalises a variance too small for the return that occurred, and the maximum balances them, exactly as in the Gaussian estimate of the scale of an i.i.d. sample (where $\sigma_t^2 \equiv \hat{\sigma}^2$ gives (13.7)).

Constraints and reparametrisation. The parameters are constrained: $\alpha_0 > 0$, $\alpha_i, \beta_j \geq 0$ and $\sum_i \alpha_i + \sum_j \beta_j < 1$ (theorem 13.15) [► 2013 L11 @ 9:30]. The lecturer's remark is practical: optimisation algorithms behave well on *unconstrained* parameters, so one transforms them, the logarithm for a positive parameter, the logit $\log \frac{x}{1-x}$ for a parameter in $(0, 1)$ [► 2013 L11 @ 10:39]. For the GARCH(1, 1) one can write

$$\alpha_0 = e^a, \quad \alpha_1 = \phi s, \quad \beta_1 = \phi(1 - s), \quad \phi = \frac{1}{1 + e^{-u}}, \quad s = \frac{1}{1 + e^{-v}},$$

with $(a, u, v) \in \mathbb{R}^3$ free; then $\alpha_0 > 0$, $\alpha_1, \beta_1 > 0$ and $\alpha_1 + \beta_1 = \phi \in (0, 1)$ hold automatically. (This particular map is our example.)

Proof. $p(y_1, \dots, y_T) = \prod_t p(y_t \mid y_{t-1}, \dots, y_1)$ and, given $\mathcal{F}_{t-1}$, both c and σ_t^2 are known numbers, so the conditional law is the stated normal. Take logarithms. $\square$

◆ **BEYOND THE CORE: Variance targeting**

A second trick, not discussed, is *variance targeting*: replace α_0 by $(1 - \phi)\hat{s}^2$, with $\hat{s}^2$ the sample variance of the returns, which removes one parameter and enforces the long-run level (13.24).

Remark 13.20 (Estimation problems). If $\alpha_1 = 0$ the parameter β_1 disappears from the model (since σ_t^2 is constant), so the likelihood is flat in β_1 and the estimators of α_1, β_1 are badly behaved near this boundary; and when $\alpha_1 + \beta_1 \approx 1$ the asymptotic theory is delicate as well. Standard errors in section 13.7.4 are those printed by the software. When z_t is not Gaussian the Gaussian likelihood (13.30) is still a sensible objective: its maximiser is a *quasi-maximum-likelihood* estimator, which is consistent under conditions weaker than normality (Bollerslev and Wooldridge, 1992; not discussed) but no longer efficient, and it is the natural counterpart of the remark on the LM test in section 13.5.1.

Exercise 13.8 ((not from the course): estimation). Simulate 2000 returns of a Gaussian GARCH(1, 1) with $(\alpha_0, \alpha_1, \beta_1) = (2 \times 10^{-6}, 0.08, 0.90)$, write the log-likelihood (13.30) in the reparametrisation of section 13.7.1 and maximise it; compare the estimates with the truth over 200 replications and comment on the bias and dispersion of $\hat{\alpha}_1 + \hat{\beta}_1$.

13.7.2 Innovations with the Student t distribution

The Gaussian assumption leaves two symptoms in the residuals of the case study: standardised residuals with values six standard deviations from the mean, and a bend of the Q–Q plot at the extremes [► 2013 L11 @ 14:47]. A natural heavier-tailed alternative is the Student t law. Because $\text{Var } z_t$ must be 1, one takes the *standardised* t with $\nu > 2$ degrees of freedom:

$$z = \sqrt{\frac{\nu-2}{\nu}} T_\nu, \quad f_\nu(z) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \quad (13.31)$$

where T_ν has the usual t density $\propto (1 + x^2/\nu)^{-(\nu+1)/2}$ [► 2013 L11 @ 19:08]. The scaling is needed because $\text{Var } T_\nu = \nu/(\nu-2)$; this is why the case-study code multiplies by $\sqrt{(\nu-2)/\nu}$.

Proposition 13.21 (Moments and tails of the t law). Let $T_\nu = Z/\sqrt{V/\nu}$ with $Z \sim N(0, 1)$ independent of $V \sim \chi_\nu^2$. Then $\text{Var } T_\nu = \nu/(\nu-2)$ for $\nu > 2$; the kurtosis is finite for $\nu > 4$ and equals

$$\kappa_t = 3 + \frac{6}{\nu-4} \quad (\nu > 4),$$

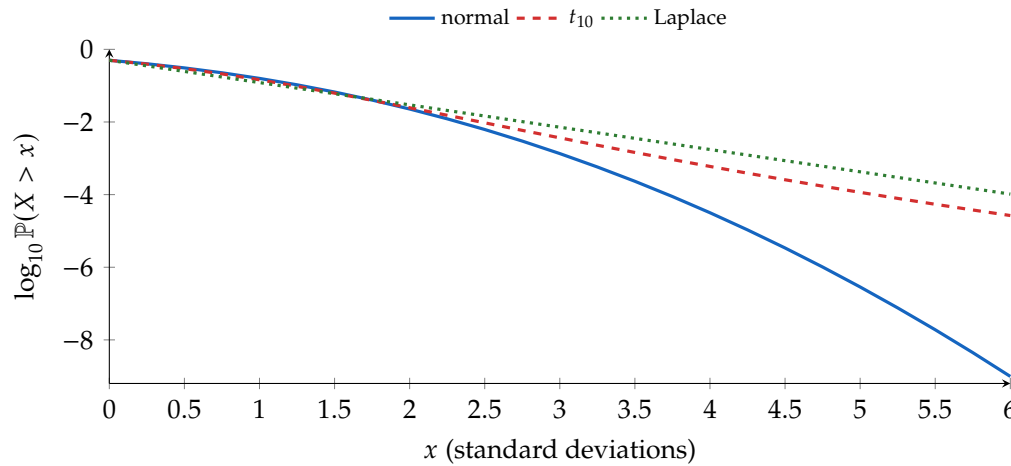
and $\mathbb{P}(|T_\nu| > x) \sim c_\nu x^{-\nu}$ as $x \rightarrow \infty$: a power-law tail, with only the moments of order $< \nu$ finite.

Proof. Conditionally on V , $T_\nu \sim N(0, \nu/V)$, so T_ν is a normal variance mixture (proposition 3.29) with mixing variable ν/V . Now $\mathbb{E}[V^{-1}] = 1/(\nu-2)$ and $\mathbb{E}[V^{-2}] = 1/((\nu-2)(\nu-4))$ for the χ_ν^2 law (they follow from $\mathbb{E}[V^k] = 2^k \Gamma(\nu/2 + k)/\Gamma(\nu/2)$ for $k = -1, -2$, and are infinite for $\nu \leq 2$, resp. $\nu \leq 4$). Hence $\text{Var } T_\nu = \nu \mathbb{E}[V^{-1}] = \nu/(\nu-2)$, $\mathbb{E}[T_\nu^4] = 3\nu^2 \mathbb{E}[V^{-2}] = 3\nu^2/((\nu-2)(\nu-4))$ and $\kappa_t = \mathbb{E}[T_\nu^4]/(\text{Var } T_\nu)^2 = 3(\nu-2)/(\nu-4) = 3 + 6/(\nu-4)$. The tail follows from the density (13.31), which decays like $|z|^{-(\nu+1)}$. □

So $\nu = 10$ has $\kappa_t = 4$, $\nu = 5$ has 9, $\nu = 30$ has 3.23; as $\nu \rightarrow \infty$ the law tends to the normal. The value $\nu = 30$ (or 25–40, the answers of the students in class) is when the density and the distribution function are visually indistinguishable from the normal [► 2013 L11 @ 19:08]. What differs is the *far tail*, best seen on a logarithmic scale [► 2013 L11 @ 20:13], fig. 13.4.

7 COURSE NOTE: What “a six-sigma move” means

The 2013 lecture says that with 10 degrees of freedom a six-standard-deviation move “occurs about 1 in 10 000 times” [► 2013 L11 @ 21:15]. The figure corresponds to $|T_{10}| > 6$ for the *unstandardised* t_{10} ($\mathbb{P} = 1.3 \times 10^{-4}$, i.e. one time in 7 600). In units of the standard deviation of the



fitted law, $|z| > 6$ means $|T_{10}| > 6/\sqrt{0.8} = 6.7$, with probability 5.3×10^{-5} (one in 19 000; one-sided 2.7×10^{-5}); for the normal it is 2.0×10^{-9} . The order of magnitude and the message are unchanged: with fat tails such moves are not impossible, one per few decades of daily data.

Figure 13.4: One-sided tail probability of three laws with unit variance, against the distance x from the mean in standard deviations. Up to about $x = 2$ the three curves are close, and the t_{10} is even *below* the normal near $x = 1.6$ (its centre is more peaked, “leptokurtic”); beyond $x \approx 2.5$ the normal falls off super-exponentially, the Laplace exponentially and the t_{10} as a power law. At $x = 6$ the normal has 10^{-9} , the t_{10} 2.7×10^{-5} and the Laplace 10^{-4} .

Likelihood with t innovations. With $\varepsilon_t = \sigma_t z_t$ and the density (13.31), the conditional density of y_t is $f_v(\varepsilon_t/\sigma_t)/\sigma_t$, and

$$\ell(\theta, \nu) = T \log \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\pi(\nu-2)}} - \frac{1}{2} \sum_{t=1}^T \log \sigma_t^2 - \frac{\nu+1}{2} \sum_{t=1}^T \log \left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2} \right). \quad (13.32)$$

Compared with (13.30), the quadratic penalty $\varepsilon_t^2/\sigma_t^2$ is replaced by a logarithm that grows slowly: an extreme return is punished much less, so the fit is less driven by a few outliers (the volatilities of the two fits in the case study are nearly identical for this reason and because the parameters of σ_t^2 are largely determined by the bulk of the data). The degrees of freedom ν is estimated like any other parameter, most simply by *profiling*: fit the model for a grid of ν and compare the maximised log-likelihoods [► 2013 L11 @ 23:17] [► 2024 L19 @ 1:19:31]. The tail heaviness is inherited by the returns twice, from z_t and from the fluctuation of σ_t : by theorem 13.18 the GARCH(1, 1) with t_ν innovations has kurtosis $\kappa_z(1 - \phi^2)/[1 - \phi^2 - (\kappa_z - 1)\alpha_1^2]$ with $\kappa_z = 3 + 6/(\nu - 4)$.

Exercise 13.9 ((not from the course): kurtosis with t innovations). For which (α_1, β_1) is the kurtosis of a GARCH(1, 1) with t_6 innovations finite? Compare with the Gaussian case for $\alpha_1 + \beta_1 = 0.95$. Prove that the maximum α_1 for an ARCH(1) with fourth moment decreases from $1/\sqrt{3}$ to $1/\sqrt{\kappa_z}$.

13.7.3 Diagnostics and model choice

After fitting, the standardised residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$ should behave like i.i.d. draws of the assumed law with unit variance. Hence [► 2024 L19 @ 1:14:13][► 2013 L11 @ 11:39]:

- $\hat{z}_t$ should be uncorrelated (Ljung–Box/Box–Pierce test of eq. (9.4)) and, the point of the exercise, $\hat{z}_t^2$ should be uncorrelated: the GARCH structure should have absorbed the dependence of the squares.

- Normality (or fit of the assumed t): normal Q–Q plot, the Jarque–Bera test $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24] \approx \chi_2^2$ (skewness $\hat{S}$, kurtosis $\hat{K}$), the Shapiro–Wilk test, the test based on the uniformity of the fitted percentiles $\hat{F}(y_t)$ (as in [section 13.2.3](#)) and the Kolmogorov–Smirnov distance between empirical and fitted distribution functions.
- Model selection among orders and among innovation laws by the Akaike and Bayesian information criteria ([eq. \(9.21\)](#)); a model with more parameters, such as a free ν , must gain enough log-likelihood to pay for them.

(Strictly the residual tests use estimated parameters, so their null distributions are only approximate; not discussed.)

13.7.4 Case study: EUR/USD returns, ARCH and GARCH fits (2013)

CASE STUDY: EUR/USD returns with ARCH, GARCH and t GARCH models

Lesson. Volatility clustering is strongly present, a parsimonious GARCH(1, 1) describes it better than a high-order ARCH, the Gaussian assumption is inadequate in the tails but the volatility estimates are robust to it, and the t law with about 10 degrees of freedom fits the standardised residuals.

◊ BEYOND THE CORE: The EUR/USD case study in detail

Data. The daily returns of the euro in dollars, from the Federal Reserve series used in [section 13.2.3](#) (3 708 log returns, 1999 to September 2013). The lecture of 2024 shows the same pages of the note for the dollar–pound series; the numbers below are those of the note, which uses EUR/USD [[▶ 2013 L9 @ 1:16:58](#)][[▶ 2024 L19 @ 1:15:15](#)]. **Step 1: dependence in the squares.** The ACF and PACF of the returns show only marginal dependence, those of the *squared* returns have highly significant, slowly decaying autocorrelations and a very significant PACF at low lags, which suggests an autoregression for the squares, i.e. ARCH [[▶ 2013 L9 @ 1:18:02](#)][[▶ 2024 L19 @ 1:15:15](#)]. **Step 2: Gaussian ARCH(p) and GARCH(1, 1) by maximum likelihood** (R function `tseries::garch`):

model	$\hat{\alpha}_0$	$\hat{\alpha}_1$	other	JB	LB p	floor
ARCH(1)	4.05×10^{-5}	0.0285		626.6	0.79	10.1%
ARCH(2)	3.62×10^{-5}	0.0264	$\hat{\alpha}_2 = 0.1001$	459.2	0.76	9.6%
ARCH(10)	2.06×10^{-5}	0.0035	$\sum_i \hat{\alpha}_i = 0.504$	368.4	0.85	7.2%
GARCH(1, 1)	1.31×10^{-7}	0.0278	$\hat{\beta}_1 = 0.9692$	77.1	0.0043	3.3%

(JB is the Jarque–Bera statistic of the residuals, LB p the Ljung–Box p -value for $\hat{z}^2$; the last column is the annualised lower bound of the fitted volatility, $\sqrt{252\hat{\alpha}_0}$ for ARCH, $\sqrt{252\hat{\alpha}_0/(1 - \hat{\beta}_1)}$ for GARCH; the estimated α_i of ARCH(10) are all significant except the first and the last, and the sum 0.504 is the persistence.) The volatility paths of the four fits [[▶ 2013 L11 @ 2:09](#)][[▶ 2013 L9 @ 1:19:09](#)] show the following. (i) Every ARCH model has a *hard lower bound* $\sqrt{\hat{\alpha}_0}$, lower for higher orders, which is fixed by the sample; (ii) the GARCH(1, 1) with three parameters is smoother and explores a much wider range of levels than the ten-parameter ARCH(10) [[▶ 2013 L9 @ 1:17:21](#)][[▶ 2024 L19 @ 1:17:21](#)]; (iii) its persistence is $\hat{\alpha}_1 + \hat{\beta}_1 = 0.9970$: it is “very close to being non-stationary”, with the long-run variance $\hat{\alpha}_0/(1 - \hat{\phi}) = 4.34 \times 10^{-5}$ (daily volatility 0.66%, 10.5% annual). Residual checks: the Jarque–Bera test rejects normality for all fits, less strongly for GARCH(1, 1); the Ljung–Box tests of the squared residuals no longer reject for the ARCH fits and marginally still reject ($p = 0.004$) for the GARCH(1, 1).

Step 3: AR mean, t innovations. A mean model is fitted first: an AR(7) is chosen by AIC, with only the seventh lag significant ($t = 2.0$, $R^2 = 0.004$), and residual standard error 0.00645. In the AR(7)–GARCH(1, 1) fit of the package `rugarch` [[▶ 2013 L11 @ 13:44](#)]:

innovations	$\hat{\omega}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	persistence
Gaussian	1.192×10^{-7}	0.02777	0.96956	0.99733
t_{10} (fixed)	1.285×10^{-7}	0.02920	0.96810	0.99730

The Q–Q plot of the AR(7) residuals (constant variance) bends strongly at both ends, with values up to 6 standard deviations [► 2013 L11 @ 14:47]; those of the Gaussian GARCH residuals bend much less but still bend beyond ± 2.5 [► 2013 L11 @ 15:54], and those of the t_{10} fit are along the line. The two fitted volatility paths are practically identical (differences at most 0.0002 in the daily σ_t , about 0.3 percentage points in annualised volatility), and the persistence is the same to four digits [► 2013 L11 @ 16:59]. **Step 4: choosing ν .** Fitting for $\nu = 4, \dots, 30$ with the parameters of the mean and of σ_t^2 re-estimated each time gives the profile log-likelihood below (the note prints it with the sign of a negative log-likelihood, and plots “Minus Log Likelihood”); its maximum is at $\nu = 10$ [► 2013 L11 @ 23:17](the 2024 lecture reads 9 from the same plot [► 2024 L19 @ 1:19:31]: the difference is 0.16 log-likelihood units).

ν	4	5	6	8	10	12	15	20	30
$\ell_{\max} - \ell(\nu)$	33.1	15.3	7.1	1.0	0	0.6	2.4	5.6	10.3

◊ **BEYOND THE CORE: A confidence interval for ν**

The profile above is one number per ν from the note. Our own reading, not made in class: since $2[\ell_{\max} - \ell(\nu)]$ is approximately χ^2_1 , the values of ν with $\ell_{\max} - \ell < 1.92$ form an approximate 95% confidence interval, $\nu \in [8, 14]$; the Gaussian limit is rejected (for $\nu = 30$, $2 \times 10.3 = 20.7$, and the normal, $\nu = \infty$, is worse still).

8 COURSE NOTE: The 9 degrees of freedom and “spherical symmetry”

In 2024 the lecturer read 9 degrees of freedom off the plot and gave a heuristic explanation: a t law arises as the projection of a spherically symmetric law in a space of dimension equal to the degrees of freedom, so $\nu \approx 9$ would mean that the volatility is stable over about nine periods [► 2024 L19 @ 1:20:34]. Take this as a suggestive statement only. What is true and used above is different: t_ν is a Gaussian mixture with an inverse-chi-squared variance (proposition 13.21), i.e. the return is Gaussian with a variance that is random and slowly changing, which is the same idea as the GARCH itself. In the case study the estimate of ν is also very flat around its maximum (interval [8, 14] above), so 9 and 10 are equivalent.

13.7.5 Value at risk with t innovations

The conditional distribution of the next return is $c + \sigma_{t+1}z$ with z standardised. The one-day value at risk at level p (the loss exceeded with probability p ; the notion of chapter 19) is

$$\text{VaR}_p = -(\hat{\mu}_{t+1} + q_p \hat{\sigma}_{t+1}), \quad q_p = \sqrt{\frac{\nu-2}{\nu}} t_{\nu,p} \text{ (Student)}, \quad q_p = \Phi^{-1}(p) \text{ (Gaussian)}, \quad (13.33)$$

with $t_{\nu,p}$ the p -quantile of T_ν [► 2013 L11 @ 18:05]. The lecture’s observation is that at the usual levels of 2.5% or 5% the two multipliers are barely different, and that the difference appears in the far tail [► 2013 L11 @ 19:08]:

level p	5%	2.5%	1%	0.1%
Gaussian, $-q_p$	1.645	1.960	2.326	3.090
standardised t_{10} , $-q_p$	1.621	1.993	2.472	3.706
ratio	0.99	1.02	1.06	1.20

At 5% the t quantile is actually *smaller*, because the standardised t has a more peaked centre (fig. 13.4). In the case study the 2.5% limits of the AR(7)–GARCH(1, 1)– t_{10} model were exceeded by 107 of the 3 708 returns on the lower side (2.89%) and by 92 on the upper side (2.48%), against 92.7 expected on each side [► 2013 L11 @ 27:33].

◊ **BEYOND THE CORE: Is 2.89% compatible with 2.5%?**

(With this frequency the lower exceedances are consistent with 2.5%: the one-sided binomial p -value is 0.076; our computation.)

IN FINANCE: From GARCH to a risk number

A risk desk fits a GARCH(1,1) (or applies RiskMetrics) to its P&L, obtains $\hat{\sigma}_{t+1}$, multiplies by a quantile and reports the VaR; multi-day VaR uses the forecasts of [section 13.6.5](#) rather than the $\sqrt{H}$ rule. What can go wrong: the volatility estimate reacts with a lag to a regime change; the quantile is model dependent far in the tail; and the independence of z_t is assumed. The backtest is the exceedance count of the preceding paragraph.

9 COURSE NOTE: A detail of the case-study code

In the R code the limits are first computed around the fitted conditional mean and then *overwritten* by limits around the constant long-term mean $\hat{c}_0/(1 - \sum \hat{\phi}_i)$ of the AR part (the second assignment replaces the first). The reported exceedance frequencies are therefore those of the second version. Since the conditional mean varies very little ($R^2 = 0.4\%$) the difference is immaterial, but a VaR should be centred on $\hat{\mu}_{t+1}$ as in [\(13.33\)](#).

13.8 Stylised facts and extensions

The lecture ends with a list of empirical regularities of returns and volatility, attributed to Engle, Bollerslev and Nelson (1994) [\[► 2024 L19 @ 1:21:36\]](#)[\[► 2013 L11 @ 24:19\]](#):

stylised fact	evidence in this chapter	ingredient of the models
returns nearly uncorrelated, squares correlated	ACF/PACF of the euro returns and their squares; S&P 500 squares significant at about 10 lags	ARCH: ε_t martingale difference, ε_t^2 an AR (proposition 13.10)
volatility clustering	large ε_t^2 follow large ones (GARCH path of fig. 13.2)	$\alpha_1 > 0$, persistence ϕ near 1
fat tails, leptokurtosis	Q-Q plots; kurtosis of the euro residuals; Laplace better than normal	GARCH is a stochastic mixture of Gaussians; t innovations
mean reversion and persistence of volatility	$\hat{\alpha}_1 + \hat{\beta}_1 = 0.997$, half-life about a year	$\sigma_t^2 - \sigma_*^2$ decays at rate ϕ
volatility depends on the estimator and on the time scale	trading against clock time; range estimators	sections 13.2.3 and 13.3

The slides close with a list of extensions, only named [\[► 2013 L11 @ 28:35\]](#): EGARCH (Nelson, 1991/1992), TGARCH (Glosten, Jagannathan and Runkle, 1993), PGARCH (Ding, Engle and Granger), GARCH-in-mean, and non-Gaussian innovations (Student t , generalised error distribution, GED). The lecturer’s advice is to master the specification under Gaussian and t assumptions first; the extensions are reading for a final paper [\[► 2013 L11 @ 28:35\]](#) [\[► 2024 L19 @ 1:21:36\]](#).

◊ BEYOND THE CORE: The extensions in formulas

To make the names concrete (the formulas are not in the lectures):

GJR/TGARCH: $\sigma_t^2 = \alpha_0 + (\alpha_1 + \gamma \mathbf{1}_{\{\varepsilon_{t-1} < 0\}}) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$,

EGARCH: $\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha |z_{t-1}| + \gamma z_{t-1}$,

GARCH-in-mean: $y_t = c + \lambda \sigma_t^2 + \varepsilon_t$.

The parameter γ captures the *leverage effect*, the observation that a negative return raises future volatility more than a positive return of the same size (a fall in the price of a firm raises its leverage); EGARCH has no positivity constraints because it models $\log \sigma_t^2$. GARCH-in-mean lets the expected return depend on the risk. All these keep the GARCH(1,1) recursion as their core.

IN FINANCE: Realised, implied and stochastic volatility

Volatility is also *implied* by option prices: the number that, inserted into a formula such as that of [chapter 15](#), reproduces the market price. The implied volatility is a forward-looking quantity and typically exceeds the subsequently realised one (a risk premium); the models of this chapter estimate the *physical* volatility from past returns. Neither year develops implied volatility or *stochastic-volatility* models (where σ_t has its own noise, not a function of the past returns as in GARCH); we only mention them, in the words of the course notes, as topics for further study.

INTUITION: Physicist's summary

A GARCH(1, 1) is a damped memory kernel for the energy of the noise: σ_t^2 is a leaky integrator of past ε^2 with a relaxation time $-1/\ln \phi$, driven by the noise itself (multiplicative noise). Everything in the chapter follows from this: the stationary level is the balance of injection and decay, the fat tails are the fluctuations of the local noise temperature, and IGARCH (RiskMetrics) is the same integrator with no leak.

Summary of results

KEY FORMULAS

Annualised volatility:	$\sqrt{N}\sigma$, $N = 252/52/12$; $\text{sd}(\hat{\sigma})/\sigma \approx 1/\sqrt{2(T-1)}$
EWMA:	$\tilde{\sigma}_{t+1}^2 = (1-\beta)\hat{\sigma}_t^2 + \beta\tilde{\sigma}_t^2$, $n_{\text{eff}} = \frac{1+\beta}{1-\beta}$ half-life $\ln 2/\ln(1/\beta)$, mean age $\beta/(1-\beta)$
GBM returns:	$R_j \sim N(\mu_*\Delta_j, \sigma^2\Delta_j)$, $\mu_* = \mu - \frac{1}{2}\sigma^2$
GBM MLE ($\Delta_j = 1$):	$\hat{\mu}_* = \bar{R}$, $\hat{\sigma}^2 = \frac{1}{n} \sum (R_j - \bar{R})^2$
Efficiencies ($\mu_* = 0$):	close-to-close 1, Parkinson 4.9, Rogers–Satchell 6.0 Garman–Klass 7.4, with opening jump 8.4
Parkinson:	$\hat{\sigma}^2 = (H - L)^2/(4 \ln 2)$
Garman–Klass:	$\hat{\sigma}^2 = \frac{1}{2}(u - d)^2 - (2 \ln 2 - 1)c^2$
Jump diffusion:	$\text{Var } R = \sigma^2\Delta(1 + \lambda\gamma^2)$, $\kappa = 3 + \frac{3\gamma^4\lambda}{\Delta(1+\lambda\gamma^2)^2}$
Laplace:	$f = \frac{1}{2b}e^{- x-\mu /b}$, $\text{Var} = 2b^2$, $\kappa = 6$
GARCH(1, 1):	$\sigma_t^2 = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \beta_1\sigma_{t-1}^2$
ARMA form:	$\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1)\varepsilon_{t-1}^2 + u_t - \beta_1u_{t-1}$
Stationarity, long run:	$\phi = \alpha_1 + \beta_1 < 1$, $\sigma_*^2 = \frac{\alpha_0}{1-\phi}$
Forecast:	$\mathbb{E}_t[\sigma_{t+h}^2] = \sigma_*^2 + \phi^{h-1}(\sigma_{t+1}^2 - \sigma_*^2)$ half-life $\ln(1/2)/\ln \phi$
ACF of squares:	$\rho(1) = \frac{\alpha_1(1-\alpha_1\beta_1-\beta_1^2)}{1-2\alpha_1\beta_1-\beta_1^2}$, $\rho(k) = \phi^{k-1}\rho(1)$
Kurtosis:	$\kappa_\varepsilon = \kappa_z \frac{1-\phi^2}{1-\phi^2-(\kappa_z-1)\alpha_1^2}$ (finite iff the denominator is positive)
Gaussian log-likelihood:	$\ell = -\frac{1}{2} \sum_t [\log 2\pi + \log \sigma_t^2 + \varepsilon_t^2/\sigma_t^2]$
Standardised t :	$z = \sqrt{\frac{v-2}{v}}T_v$, $\kappa_t = 3 + \frac{6}{v-4}$
Value at risk:	$\text{VaR}_p = -(\mu_{t+1} + q_p\sigma_{t+1})$
RiskMetrics/IGARCH:	$\alpha_0 = 0$, $\alpha_1 = 1 - \beta$, $\beta_1 = \beta$ ($\phi = 1$)

Exercises

Standalone exercises follow the section they practise. 2013 PS5 Problems 1–2 are a connected series (Problem 2 generalises the equal sampling intervals of Problem 1 and they share the set-up below), so they stay together here.

From 2013 Problem Set 5 (its Problems 1 and 2 are Problems 3 and 4 of 2024 Problem Set 4)

In the first two problems $X(t)$ follows $dX/X = \mu dt + \sigma dW$ and the price is sampled on $[0, T]$, so that the log returns $Y_i = \log(X_i/X_{i-1})$ are independent $N(\mu_*h_i, \sigma^2h_i)$ with $\mu_* = \mu - \frac{1}{2}\sigma^2$ (this is given).

Exercise 13.10 (PS5 (2013), Problem 1; PS4 (2024), Problem 3: equal sampling intervals). Sample at equal increments $h = T/n$, $t_i = ih$.

- (a) Show that the maximum-likelihood estimators are $\hat{\mu}_* = \frac{1}{n} \sum Y_i$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (Y_i - \hat{\mu}_*)^2$ (the 2024 version normalises by h : $\hat{\mu}_* = \frac{1}{hn} \sum Y_i$, $\hat{\sigma}^2 = \frac{1}{hn} \sum (Y_i - h\hat{\mu}_*)^2$).
- (b) Derive the distribution of $\hat{\mu}_*$, with its mean and variance.
- (c) Derive the distribution of $\hat{\sigma}^2$, with its mean and variance.
- (d) Let $n \rightarrow \infty$ on the fixed period: what are the limiting distributions of $\hat{\mu}_{*,n}$ and $\hat{\sigma}_n^2$?
- (e) An estimator sequence $\hat{\theta}_n$ is weakly consistent if $\lim_n \mathbb{P}(|\hat{\theta}_n - \theta| > \epsilon) = 0$ for every $\epsilon > 0$ (the 2013 text writes the condition with a small typo; the 2024 hint is Markov's inequality). Which of the two sequences is weakly consistent?

Exercise 13.11 (PS5 (2013), Problem 2; PS4 (2024), Problem 4: variable sampling intervals). Now the increments $h_i > 0$ are arbitrary with $\sum_i h_i = T$.

- (a) Derive the MLE of μ_* and its distribution for a fixed set $\{h_i\}$.
- (b) Derive the MLE of σ^2 and its distribution.
- (c) With only $n + 1$ price points allowed (including X_0 and X_T), prove that: the MLE of σ^2 depends on the spacing, but its *variance* is the same for every spacing; and the MLE of μ_* is the same, with the same variance, for every spacing. (In the 2024 statement the parts of Problem 4 are mislabelled 3(a)–3(c).)

Chapter 14

Event Contracts and a Regulated Exchange

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L20	T. Mansour (Kalshi), guest; introduced by P. Kempthorne	Why an exchange for events; the three-year regulatory road and the lawsuit against the regulator; betting market versus exchange; arbitrage against other assets; cash collateral and margin; who trades (institutions, retail); revenue model; market makers and quoted spreads; career advice. Questions from students and from J. Xia and V. Strela.
Files. None: only the video transcript exists for this lecture (no slides, no notes, no problem set). Everything attributed to the lecture below is taken from the spoken words, and the tags [► 2024 L20 @ 0:00] point into the video. The company facts (volumes, fees, dates, investors) are the speaker’s own statements in late 2024 and are reported as such, without independent verification. Not from the course. The mathematics of event contracts (Sections 14.3 to 14.7) is a reconstruction added for these notes: the lecture contains no formulas. Every such section says so, and all exercises (each after the section it practises) are marked “(not from the course)”.		

Motivation

Most of this course prices contracts whose payoff is a smooth function of an underlying: a bond, a forward, a call. The guest lecture is about the most primitive contract one can imagine, one that pays a fixed amount if a stated event happens and nothing otherwise. The founder of Kalshi describes the idea that started the company: if there are markets for equities, credit, commodities and currencies, why not a market on *any question about the future* [\[► 2024 L20 @ 4:13\]](#)? The lecture itself is a story (the idea, the regulator, the product, the business), so the notes do two things. Sections [14.1](#) and [14.2](#) report faithfully what was said. Sections [14.3](#) to [14.7](#) supply the mathematics a physicist would want next to the story: an event contract is a digital option, its price is a (risk-neutral) probability, related contracts must obey no-arbitrage constraints, a market maker earns a spread and pays for adverse selection, a trader can size a bet with the Kelly rule, and a market forecast can be scored and compared with the information behind it. It is a short chapter; the finance content is

light and the value is in seeing the one-period model of [section 2.4](#) at its smallest scale.

14.1 The story of the exchange (as told in class)

The speaker. Tarek Mansour took this course as an MIT student, worked at Goldman Sachs, Citadel and Bridgewater, and about three years before the lecture co-founded Kalshi with Luana Lopes Lara, also from MIT [[▶ 2024 L20 @ 0:00](#)]. He credits the course for showing that every mathematical concept has a financial application, and recalls a term paper on reinforcement learning applied to the stock market and the use of Markov chains on a trading desk [[▶ 2024 L20 @ 39:55](#)].

The idea. In 2016 many market participants wanted a hedge against Brexit and had only proxies: stocks, options, other traditional instruments, indirect and expensive because traded over the counter [[▶ 2024 L20 @ 3:12](#)]. The proposal: a financial market that prices any event of economic, social or cultural relevance. He argued that such a market could be the largest of all, simply because it is broader than any other [[▶ 2024 L20 @ 4:13](#)].

The regulatory road. Such markets had historically been illegal or unregulated in the US. The founders phoned essentially every securities and derivatives lawyer in the country; the answers ranged from laughter to “the CFTC will never allow it” [[▶ 2024 L20 @ 5:15](#)]. They kept going because none of the objections was, in his words, intellectually satisfying: manipulation is a solvable problem, as it was for stock markets, and the “gambling” objection does not distinguish event bets from grain futures, where some trade to hedge and some to speculate [[▶ 2024 L20 @ 6:22](#)]. It took two and a half to three years to get the exchange and the clearinghouse regulated, after which Kalshi launched what he calls the first legal prediction market in the US [[▶ 2024 L20 @ 6:22](#)].

The lawsuit. The CFTC did not allow the company to list an election market. Kalshi sued its own regulator, and, according to the speaker, won “earlier this fall” (2024), which made it possible for the first time in a century to trade the US election legally; the significance is that the ruling widens the universe of things counted as financial instruments [[▶ 2024 L20 @ 7:24](#)]. He reports more than one billion (dollars) of volume in the last month, plans to add categories (finance, sports, “every piece of news”) and to make the contracts tradable from brokerage accounts such as Schwab, Robinhood and Fidelity [[▶ 2024 L20 @ 8:26](#)]. The company had raised more than \$100 million from several investors [[▶ 2024 L20 @ 10:33](#)]. After the election the volume stayed higher than before it; categories mentioned as growing are crypto, economics, cabinet appointments, weather and entertainment [[▶ 2024 L20 @ 21:03](#)].

14.2 How the market is organised (as told in class)

Betting market versus exchange. A student asks how this differs from a betting market [[▶ 2024 L20 @ 11:33](#)]. The answer has two parts. First, structure: it is an *exchange* with an order book, counterparties trade against each other, there is no bookie who sets prices, and an institution hedging Brexit would never accept bookie prices, while an over-the-counter quote from a bank is negotiated bilaterally and, in his view, poor [[▶ 2024 L20 @ 11:33](#)]. Second, nature of the risk: rolling a die at a casino creates an artificial risk for entertainment, whereas Brexit is a *natural* risk that some people carry and may wish to transfer, by hedging, or increase, by speculating [[▶ 2024 L20 @ 12:36](#)]. That, he says, is why it is a financial market. He also expects betting markets and exchanges eventually to converge.

Arbitrage. Asked about arbitrage between betting markets and Kalshi, he says Kalshi

1 COURSE NOTE: Dates and legal status

Statements such as “first legal prediction market”, “first time in 100 years” and the outcome of the lawsuit are the speaker’s account at the date of the lecture. The notes neither confirm nor extend them; for anything with legal weight consult primary sources.

is the only legal venue in the US, so the arbitrage that matters is against traditional assets: during the election many hedge funds traded the election market against crypto and the S&P 500 [► 2024 L20 @ 13:39]. Section 14.4 formalises what such cross-asset and cross-contract consistency means.

Collateral and margin. In a futures market a participant posts a small fraction of the notional, say 1% to 10%, with the clearinghouse, which then settles gains and calls for more margin after losses [► 2024 L20 @ 18:57]. Kalshi is *fully cash-collateralised*: the money is put up front, positions are marked to market in real time, and a participant cannot lose more than was posted [► 2024 L20 @ 17:57]. Leverage is expected to be introduced, at the earliest at the end of the following year, under strict regulatory requirements aimed at preventing anything like 2008; for products with strong correlation some collateral is already returned [► 2024 L20 @ 18:57]. The economics of the last statement is explored in [Exercise 14.3](#).

Who trades and how the business earns. By volume, 30–40% is institutional and the rest retail, with an expected drift toward 50/50. Institutions trade much more volume but can be charged less; retail trades less and can be charged more, and the speaker calls the institution-first model of the large exchanges a strategic mistake [► 2024 L20 @ 22:04]. The company was not profitable at the time, by choice: the strategy is growth first [► 2024 L20 @ 26:17]. Revenue sources listed:

- trading fees, of the order of \$1–\$2 per \$100 traded;
- a separate market-making arm, competing with other firms, among them SIG;
- in the future, data, currently free and open [► 2024 L20 @ 27:20].

Market makers. Kalshi imposes obligations on market makers; his example is the election market, where they had to quote a spread of \$0.01 with at least \$1 million on each side. The tight spread meant that a very large trade (he mentions a hypothetical \$100 million) could be executed within about one cent of slippage [► 2024 L20 @ 28:22]. Incentives are given at the start of a market, after which flow and the bid–ask spread sustain the market maker [► 2024 L20 @ 29:23]. In the CFTC world, he adds, there is little competition between exchanges (CME has interest-rate swaps and the S&P, ICE energy), and he expects Kalshi to dominate event-related products [► 2024 L20 @ 29:23].

Advice. The closing part of the lecture is career advice: perseverance, taking more risk, not over-weighting authority figures, taking courses that connect theory with practice [► 2024 L20 @ 30:27]. It contains nothing to derive and is not summarised further.

14.3 Event contracts as digital options

Definition 14.1 (Event contract). Let A be an event decided at a settlement date T . The *event contract* on A (“Yes” contract) pays $\mathbf{1}_A$ units of cash at T : one unit if A occurs and zero otherwise. The “No” contract pays $\mathbf{1}_{A^c} = 1 - \mathbf{1}_A$.

This is an Arrow–Debreu security for the two-state space $\{A, A^c\}$: in the one-period model of [section 2.4](#) it is the asset whose payoff vector is $(1, 0)$. If the price is π_A today and the cash account earns rate r over $[0, T]$, the fundamental theorem [theorem 2.9](#) says that absence of arbitrage is equivalent to the existence of a pricing measure $\mathbb{Q}$ with

$$\pi_A = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A] = e^{-rT} \mathbb{Q}(A), \quad 0 \leq \pi_A \leq e^{-rT}. \quad (14.1)$$

Since exchange contracts are cash-collateralised and run for weeks or months at interest rates of a few per cent, the discount factor is close to 1 and the quoted price in $[0, 1]$ is read

2 COURSE NOTE: Reconstruction

From here to Section 14.7 the material is *not from the course*: the lecture states no formulas. It is written to connect what Mansour describes with the tools of [chapter 3](#) and [section 2.4](#).

as a probability, $\pi_A \approx \mathbb{Q}(A)$. (Whether the collateral earns interest is a product detail the lecture does not mention; we ignore it and set $r = 0$ from now on.)

Proposition 14.2 (Event contract as a digital option). Let S_T be a terminal price and $C(K)$ the price of a European call with strike K and payoff $(S_T - K)^+$. A contract paying $\mathbf{1}_{\{S_T > K\}}$ (a “digital” or “binary” call) has price

$$D(K) = -\frac{\partial C}{\partial K}(K) \quad (r = 0),$$

the negative slope of the call price as a function of strike.

Proof. The call spread that is long one call at K and short one at $K + h$ pays $(S_T - K)^+ - (S_T - K - h)^+$, which equals 0 for $S_T \leq K$, h for $S_T \geq K + h$, and lies between 0 and h in between. Dividing by h and letting $h \rightarrow 0$ the payoff converges pointwise to $\mathbf{1}_{\{S_T > K\}}$ (at $S_T = K$ it is a matter of convention). Its price is $(C(K) - C(K + h))/h \rightarrow -C'(K)$. $\square$

So event contracts are the elementary bricks out of which vanilla options are built: a call is the integral of digitals, $C(K) = \int_K^\infty D(k) dk$ (for $r = 0$). Chapter 15 computes $D(K)$ in the Black–Scholes model. What is new in an event exchange is that the event need not be a function of a traded price at all (weather, a cabinet appointment, an election), so there is no underlying with which to replicate; the price is then set by trading alone.

INTUITION: Physicist’s reading

A contract on A measures $\mathbf{1}_A$, a projector in the algebra of observables, and the price is its “expectation value” in the state $\mathbb{Q}$. The set of admissible prices for several events is the set of expectation values of a commuting family of projectors, that is, the convex hull of the vertices $\mathbf{1}_\omega$. Section 14.4 is just a description of that polytope.

IN FINANCE: Risk-neutral versus real-world probability

By (14.1) the price gives $\mathbb{Q}(A)$, not the true probability $\mathbb{P}(A)$. If the pricing kernel is $m = d\mathbb{Q}/d\mathbb{P}$ then $\mathbb{Q}(A) = \mathbb{E}^\mathbb{P}[m\mathbf{1}_A]$. Example: $\mathbb{P}(A) = 0.5$ and $m(A) = 0.8$, $m(A^c) = 1.2$ (the event is a “good” state in which one is willing to pay less for a payoff, e.g. a payoff correlated with market gains); then $\mathbb{E}^\mathbb{P}[m] = 1$ and the price is $0.5 \cdot 0.8 = 0.40$, well below the true 0.50. For an event that is unrelated to aggregate wealth, m is constant, $\mathbb{Q} = \mathbb{P}$, and the price is the physical probability; this is the case in which prediction markets can be read directly as forecasts. The lecture does not discuss this point; it does say that hedge funds traded the election contracts against crypto and the S&P [► 2024 L20 @ 13:39], which is exactly a bet that the election outcome is correlated with those assets.

Exercise 14.1 (not from the course). In the Black–Scholes model with $r = 0$, $C(K) = S\Phi(d_1) - K\Phi(d_2)$ (chapter 15), where Φ is the standard normal distribution function. Show that $D(K) = -C'(K) = \Phi(d_2)$, and interpret it as $\mathbb{Q}(S_T > K)$.

14.4 No-arbitrage constraints between contracts

Let Ω be a finite set of mutually exclusive, exhaustive outcomes (e.g. which candidate wins) and, with $r = 0$, suppose there are contracts $\mathbf{1}_\omega$ for each $\omega \in \Omega$ with prices π_ω .

Proposition 14.3 (Prices sum to one). If there is no arbitrage and contracts on all $\omega \in \Omega$ trade at the same price for buying and selling, then $\sum_\omega \pi_\omega = 1$. More generally, for events $A \subseteq B$: $0 \leq \pi_A \leq \pi_B \leq 1$, and for disjoint A, B : $\pi_{A \cup B} = \pi_A + \pi_B$.

Proof. A portfolio holding one of every contract pays $\sum_\omega \mathbf{1}_\omega = 1$ in every state. If its cost $\sum \pi_\omega$ were less than 1 it would be an arbitrage (buy it), if more than 1 the reverse portfolio would be (sell it). The other statements follow the same way: $\mathbf{1}_B - \mathbf{1}_A \geq 0$, so its price cannot be negative,

With a bid–ask spread the equality becomes inequalities. Let a_ω be the best ask (price at which one buys) and b_ω the best bid. Buying all contracts costs $\sum a_\omega$ and pays exactly 1; selling all yields $\sum b_\omega$. Hence, before fees,

$$\sum_{\omega} b_{\omega} \leq 1 \leq \sum_{\omega} a_{\omega} \quad (14.2)$$

is necessary for absence of arbitrage. Similarly the “No” contract on A is the complement of the “Yes”: buying No at a^{No} is equivalent to selling Yes at price $1 - a^{\text{No}}$, so on a well-organised book $b^{\text{Yes}} = 1 - a^{\text{No}}$ and $a^{\text{Yes}} = 1 - b^{\text{No}}$: Yes and No are two views of the same order book.

Proposition 14.5 (Monotone thresholds and the implied density). *Let X be a real quantity (an inflation print, say) and $P(k)$ the price of “ $X > k$ ”. Then P is non-increasing in k , and $P(k) - P(k') = \mathbb{Q}(k < X \leq k') \geq 0$ for $k < k'$. A market with a ladder of thresholds $k_1 < \dots < k_n$ therefore implies a discrete risk-neutral distribution for X .*

Remark 14.6. Applying this to $X = S_T$ recovers [proposition 14.2](#): $P(K) = D(K) = -C'(K)$, so $-P'(k)$ is the risk-neutral density of S_T , a fact used in option-pricing theory for the implied distribution.

IN FINANCE: Portfolio margin and correlated products

With full cash collateral a seller of a contract that pays 1 must lock $1 - \pi$ on top of the premium π received, so the locked amount per sold contract equals the maximum possible payout, 1. Consider sold contracts on two disjoint, exhaustive events A and B (so $B = A^c$): exactly one of the two contracts pays, so the joint liability is 1, not 2. Collateral of 1 suffices, and 1 can be released. This is the elementary form of the statement, heard in the lecture, that where two products are strongly correlated some collateral is returned [\[▶ 2024 L20 @ 20:01\]](#). (The lecture gives no formula; the general rule is that the collateral of a portfolio is its worst-case loss $-\min_{\omega} \text{payoff}(\omega)$ minus premiums, a linear program over the finite state space.)

Exercise 14.2 (not from the course). Four candidates have asks 0.40, 0.30, 0.15, 0.10 and bids 0.38, 0.28, 0.13, 0.08. Is there an arbitrage without fees? With a fee of 1 cent per contract bought, how much must the sum of asks fall below 1 for buying one of each to be profitable?

Exercise 14.3 (not from the course). Two independent events A, B with $\mathbb{P} = \frac{1}{2}$ each. Under full cash collateral, a trader sells Yes on both at price 0.5. Compute the locked collateral and the maximum loss of the portfolio. Do the same when $B = A^c$ (compare [section 14.4](#)). Explain why the difference in the required collateral is due to the correlation.

and $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B$ by linearity. $\square$

Example 14.4 (Three candidates). Asks of 0.48, 0.36, 0.19 on three exhaustive candidates sum to $1.03 \geq 1$ (allowed), while bids of 0.47, 0.35, 0.18 sum to 1.00. Suppose instead the asks were 0.45, 0.36, 0.16: they sum to $0.97 < 1$. Buying one of each costs 0.97 and pays 1 for sure, a profit of 3 cents per set, before fees. Fees of the order of \$1–\$2 per \$100 traded [\[▶ 2024 L20 @ 27:20\]](#) (which is 1 to 2 cents per dollar of price paid, so ≈ 1 to 2 cents on a 0.97 cost) would eat most of this margin: this is why the prices need not be exactly consistent, and an arbitrage is real only if it survives costs.

Proof. $\{X > k'\} \subseteq \{X > k\}$; apply [proposition 14.3](#). $\square$

14.5 Market making, spread and adverse selection

The lecture says that market makers commit to a tight spread (one cent, with \$1 million a side, in the election market) and are paid by flow and by the spread [\[▶ 2024 L20 @ 28:22\]](#). A minimal model that shows what this spread has to cover is the following (a one-period version of the Glosten–Milgrom model, *not from the course*).

Model. An event A has prior probability $\frac{1}{2}$ (and $r = 0$), so the outcome $V = \mathbf{1}_A \in \{0, 1\}$. A market maker quotes a bid b and an ask a and faces one arriving trader. With probability α the trader is *informed*: he knows V and buys if $V = 1$, sells if $V = 0$. With probability $1 - \alpha$ the trader is uninformed and buys or sells with probability $\frac{1}{2}$ each. The market maker breaks even on each side (competition), that is, sets each quote equal to the conditional expectation of V given the trade.

Proof. $\mathbb{P}(\text{buy} \mid V=1) = \alpha + \frac{1-\alpha}{2}$ and $\mathbb{P}(\text{buy} \mid V=0) = \frac{1-\alpha}{2}$. By Bayes with prior $\frac{1}{2}$, $a = \mathbb{E}[V \mid \text{buy}] = \frac{\alpha + (1-\alpha)/2}{\alpha + (1-\alpha)/2} = \frac{1+\alpha}{2}$. By symmetry $b = \mathbb{E}[V \mid \text{sell}] = \frac{1-\alpha}{2}$. $\square$

Proposition 14.7 (Spread equals the informed fraction). In this model $a = \frac{1+\alpha}{2}$, $b = \frac{1-\alpha}{2}$, so the spread is $a - b = \alpha$, and the market maker's expected profit is zero.

The model captures only adverse selection. Two further costs, both standard, complete the picture: *inventory risk* (a market maker who has sold n Yes contracts is short $n\mathbf{1}_A$ and carries the variance $n^2 p(1-p)$ until she hedges, by buying back or by trading a correlated asset, as the funds arbitraging the election against the S&P did [► 2024 L20 @ 13:39]) and *fees*. A market maker's spread over a lifetime of the contract must cover both, which is why exchanges incentivise it at the start of a market, when informed fraction is high and volume low [► 2024 L20 @ 29:23].

INTUITION: Where the spread comes from

The spread is not a fee for making the trade, but insurance against the person on the other side knowing more than you. Doubling the informed fraction doubles the spread. Near settlement, when everyone knows the answer, $\alpha \rightarrow 1$ and both quotes converge to 0 and 1: the market “goes to the boundary” and quoting a tight two-sided market stops being possible.

Example 14.8. For $\alpha = 0.1$: $a = 0.55$, $b = 0.45$. Against an uninformed trader (probability 0.9) the market maker gains, per unit, half the spread, 0.05, on average (she sells at 0.55 a contract worth 0.5 in expectation). Against an informed trader (probability 0.1) she loses 0.45: she sells at 0.55 a contract that pays 1. The expected profit is $0.9(0.05) - 0.1(0.45) = 0$, confirming [proposition 14.7](#). If the required spread is one cent, as in the election market, the model says the market maker believed that $\alpha \approx 0.01$: at most one trade in a hundred was against someone who knew the answer, and large trades could be absorbed because the fraction of informed flow was so small.

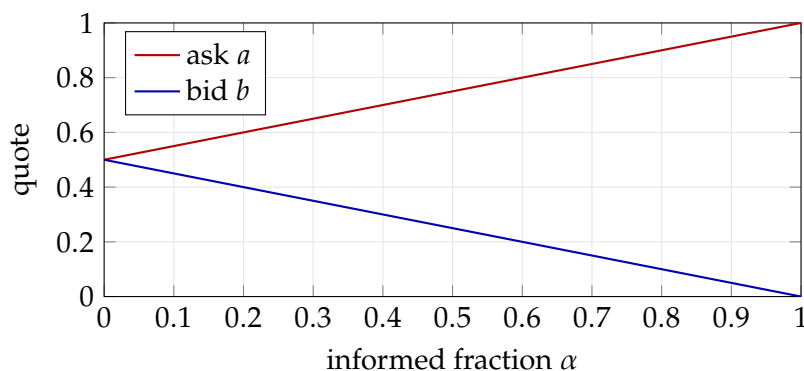


Figure 14.1: Quotes of the break-even market maker of [proposition 14.7](#). The gap between the lines is the informed fraction α .

Exercise 14.4 (not from the course). Generalise [proposition 14.7](#) to a prior $\mathbb{P}(V = 1) = \pi_0 \neq \frac{1}{2}$ and derive the ask and bid. Show that the spread is α times a factor depending on π_0 , and identify the factor.

14.6 Sizing a bet: the Kelly criterion

A participant who believes $\mathbb{P}(A) = p$ while the contract trades at $c < 1$ must decide how much to buy. Let wealth be W and invest a fraction $f \in [0, 1)$ of it in Yes contracts at price c (so the number of contracts is fW/c). Ignoring fees, wealth after settlement is

$$W(1 + fb) \text{ if } A, \quad W(1 - f) \text{ if } A^c, \quad b = \frac{1 - c}{c},$$

where b is the net odds per unit staked.

Proof. $g'(f) = \frac{pb}{1+fb} - \frac{1-p}{1-f} = 0$ gives $pb(1-f) = (1-p)(1+fb)$, i.e. $f(pb + (1-p)b) = pb - (1-p)$, so $f^* = p - \frac{1-p}{b} = p - \frac{(1-p)c}{1-c} = \frac{p-c}{1-c}$. Concavity: $g'' < 0$ since both logarithms are concave. $\square$

Proposition 14.9 (Kelly fraction for a binary contract). The expected log growth $g(f) = p \ln(1 + fb) + (1 - p) \ln(1 - f)$ is strictly concave and maximised at

$$f^* = \frac{p - c}{1 - c} \quad (\text{if } p > c; f^* = 0 \text{ otherwise}).$$

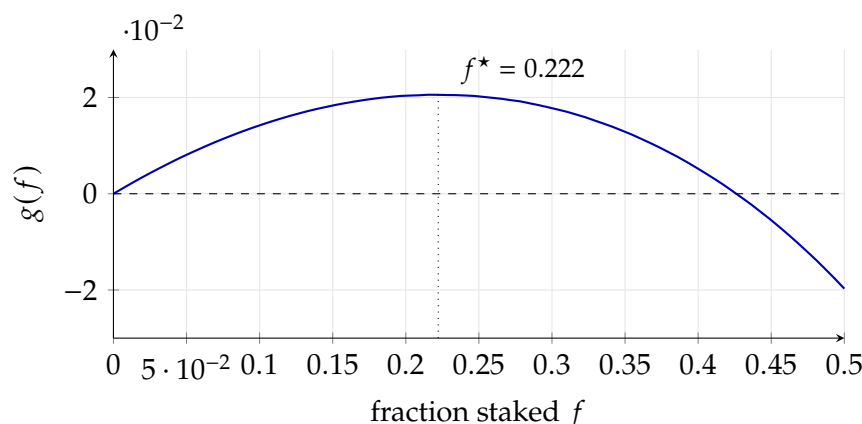


Figure 14.2: Expected log growth for $c = 0.55$, $p = 0.65$ (example 14.10). The curve crosses zero at about twice the Kelly fraction.

Example 14.10. Let $c = 0.55$ and $p = 0.65$: $b = 0.818$, $f^* = 0.10/0.45 = 0.222$ and $g(f^*) = 0.0206$ per bet (about 2% of wealth per bet, in the log sense). Staking half of that, $f = 0.111$, still gives $g = 0.0153$ (three quarters of the optimal growth) with a much smaller variance, whereas staking double, $f = 0.444$, gives $g = -0.0041 < 0$: an *over-bet* with a positive edge loses money in the long run. (Numbers checked numerically.)

IN FINANCE: Edge and price

Only the gap $p - c$ matters, scaled by the maximum loss $1 - c$: an event contract is the cleanest setting for Kelly, because the outcome is a genuine Bernoulli variable and the odds are read directly from the price. The estimate of p is itself uncertain, and f^* is sensitive to it ($\partial f^* / \partial p = 1/(1 - c)$), which is why practitioners use a fraction of Kelly.

Exercise 14.5 (not from the course). A trader believes $p = 0.70$ and can buy Yes at $c = 0.60$, paying a fee of 0.01 per contract at purchase. Compute the Kelly fraction with and without the fee (treat the fee as an increase of the price, $c \rightarrow 0.61$). By how much does the fee reduce the optimal log growth $g(f^*)$?

14.7 Scoring rules, calibration and aggregation

Brier score. A forecaster reports $q \in [0, 1]$ for an event with outcome $\mathbf{1}_A$, and is scored by $(q - \mathbf{1}_A)^2$ (lower is better). If the true probability is p , the expected score is

$$\mathbb{E}(q - \mathbf{1}_A)^2 = p(1 - q)^2 + (1 - p)q^2 = p(1 - p) + (q - p)^2.$$

It is minimised at $q = p$: the Brier score is a *proper* scoring rule, honesty is optimal. For $p = 0.6$ the expected scores are 0.24 at $q = 0.6$, 0.25 at $q = 0.5$ and 0.28 at $q = 0.8$ (verified numerically). The irreducible term $p(1 - p)$ is the variance of the event; only the excess $(q - p)^2$ measures forecasting error. A market price is a forecast, and it is *calibrated* if among all events priced at q a fraction q occur; comparing a collection of prices with outcomes in this way is how one tests the claim, implicit in the lecture, that an exchange for events makes the world “smarter about the future” [► 2024 L20 @ 32:38].

Aggregation of information. Suppose each of n traders holds an independent private signal s_i about A , with likelihood ratio $\lambda_i = \mathbb{P}(s_i | A) / \mathbb{P}(s_i | A^c)$. By Bayes’ rule in odds form,

$$\frac{\mathbb{P}(A | s_1, \dots, s_n)}{\mathbb{P}(A^c | s_1, \dots, s_n)} = \frac{\mathbb{P}(A)}{\mathbb{P}(A^c)} \prod_i \lambda_i,$$

so in *log-odds* the evidence adds: $\text{logit } p = \text{logit } p_0 + \sum_i \ln \lambda_i$, with $\text{logit } p = \ln \frac{p}{1-p}$. With prior $\frac{1}{2}$ and $\lambda = (2, 3, 1.5)$ the posterior odds are 9 and $p = 0.9$, higher than any single trader would say (each alone would have $2/3, 3/4, 3/5$). The purpose of a market is to make prices carry the pooled log-odds, and trading is the mechanism: each trader whose belief differs from the price trades, and by [proposition 14.9](#) the size of the trade grows with the edge, so the price moves toward the belief-weighted pool. That this works in practice is an empirical claim, not a theorem: it needs enough liquidity, the market-maker obligations of the previous sections, and traders who cannot manipulate at little cost (the “manipulation” objection heard by the founders [► 2024 L20 @ 6:22]).

KEY FORMULAS

- Event contract: pays $\mathbf{1}_A$; price $\pi_A = e^{-rT} \mathbb{Q}(A) \approx \mathbb{Q}(A)$ (eq. (14.1)); digital call $D(K) = -\partial C / \partial K$.
- No arbitrage: $\sum_\omega \pi_\omega = 1$; with a spread, $\sum b_\omega \leq 1 \leq \sum a_\omega$ (eq. (14.2)); thresholds: $P(k)$ non-increasing.
- Yes and No: $b^{\text{Yes}} = 1 - a^{\text{No}}$.
- Market maker with informed fraction α (prior $\frac{1}{2}$): $a = \frac{1+\alpha}{2}$, $b = \frac{1-\alpha}{2}$, spread α .
- Kelly for a contract bought at c with belief p : $f^\star = (p - c) / (1 - c)$.
- Brier: $\mathbb{E}[(q - \mathbf{1}_A)^2] = p(1 - p) + (q - p)^2$.
- Independent evidence: $\text{logit } p = \text{logit } p_0 + \sum_i \ln \lambda_i$.
- From the lecture: cash collateral 100%, fee of order \$1–\$2 per \$100, election quotes \$0.01 wide with \$1M per side.

Exercise 14.6 (not from the course). Show that the logarithmic score $-\ln q$ (if A) and $-\ln(1 - q)$ (if A^c) is also proper, and compute its expected value for true probability p , in terms of the binary entropy and the Kullback–Leibler divergence.

Chapter 15

Black–Scholes Formula and Risk-Neutral Valuation

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L21	V. Strela	Update of the yield curve after the September 2024 Fed meeting; two-horse race (the bookie who sets odds from the pool); forwards, calls, puts; one-step replication of a forward and of a call; risk-neutral probability; lognormal dynamics, why $dW \sim \sqrt{dt}$, heuristic Itô formula; replication argument and the Black–Scholes PDE; boundary conditions and the closed-form formula; the risk-neutral density; put–call parity on Bloomberg screens (AAPL, IBM) and the implied-volatility smile; the digital option and the call spread.
2013 L19	V. Strela	The same material (nearly the same slides), with the discrete one-step market in more detail (bond units b , the interest-rate quiz on the forward), suggestions for final-paper topics (heat equation, formula by expectation), an IBM option screen of March 2006, put–call parity, and the digital option as an interview question.

Files. Slides `mit18_642_f24 lec21.pdf` (26 slides) and 2013 slides `MIT18_S096F13_lecnote19.pdf` (25 slides); no problem set, case study or code accompanies this lecture in either year. **Not recorded.** 2024 L22, *The Spectrum of Systematic Trading Strategies in Liquid Instruments* (R. Garon, Millennium Management), has no video and no slides: the guest did not grant permission to publish them, so there is nothing to summarise here; the next recorded lecture is L23 (machine learning, [chapter 16](#)). **Exercises.** None of the course problem sets deals with this lecture; the exercises (each after the section it practises) are all marked “(not from the course)”.

Motivation

Chapter 2 priced a call in a one-period market with two states and found that the real-world probability of “up” did not matter ([example 2.6](#)). This chapter takes that idea to its limit. The question is: *what is the price today of a payoff $f(S_T)$ that is fixed by the price of a stock at a future date T ?* The answer, due to Black, Scholes and Merton (1973), is

- *model-dependent but preference-free:* once one specifies how the stock moves, the price is determined, and neither the expected return of the stock nor the risk appetite of investors enters;
- *constructive:* the price is the cost of a self-financing portfolio in stock and cash that replicates the payoff, and that portfolio is also the hedge;
- *an expectation:* the price equals a discounted expectation of the payoff under a fictitious *risk-neutral* measure $\mathbb{Q}$, under which every traded asset grows at the risk-free rate;
- *a PDE:* the same price solves the Black–Scholes equation, a heat equation in disguise.

The lecture is deliberately placed *before* the stochastic-calculus lectures ([chapters 17](#) and [18](#)): Itô's formula is introduced heuristically, in the way a physicist would expand a function of a Brownian path, and the rigorous statements are deferred. Read this chapter as the *application* that motivates the formalism of the next ones, and as the continuous-time counterpart of [section 2.4](#). Sections [15.2–15.4](#) are the discrete world, [sections 15.5](#) and [15.6](#) the passage to continuous time, [sections 15.7](#) to [15.9](#) the two routes to the formula, and [sections 15.10](#) to [15.13](#) what one does with it in practice.

15.1 Where the lecture starts: the yield curve

As promised in the first lecture, the class opens with an update of the US Treasury curve of [fig. 1.3](#). Slide 2 overlays the curve of 3 September 2024 (just before the course began) on that of 15 November 2024 [[▶ 2024 L21 @ 0:00](#)]. The Fed had cut in the meantime by 75 basis points in total (in the calendar: 50 bp in September and 25 bp in November, a detail not spoken in class), and the short end of the curve moved down by about that amount while the long end moved by only about 50 bp [[▶ 2024 L21 @ 1:07](#)]: not a parallel shift but a steepening.

The lecturer then puts the current shape next to three historical snapshots [[▶ 2024 L21 @ 2:14](#)]: the curves of 2007 (just before the crisis) and 2019 were both *inverted* and a recession followed, while by September 2021 the curve had a normal shape and the recession had been short. He calls this “non-scientific prediction”: a serious analysis would decompose the curves with principal components ([chapter 7](#)), and the term structure is, in his words, in the centre of finance because the discount factors of [section 1.7.2](#) are read off it. For the present chapter only one consequence matters: discounting uses a rate r , which the lecture takes *constant*. Everything that follows is an exercise in a single-rate world; making r stochastic is the subject of the short-rate and HJM models of [chapter 24](#).

15.2 Setting the odds: a bookie's version of hedging

The lecture introduces risk-neutral valuation with an example that has no mathematics in it [[▶ 2024 L21 @ 4:21](#)][[▶ 2013 L19 @ 0:00](#)]. A bookie knows that horse 1 wins a race with probability 20% and horse 2 with probability 80%. The public does not know this, and bets \$10 000 on horse 1 and \$50 000 on horse 2 [[▶ 2024 L21 @ 5:24](#)][[▶ 2013 L19 @ 1:01](#)]; the pool is \$60 000 (a commission, say 1%, is what the bookie keeps in the end and is ignored below). The bookie must announce the odds “ k to one”: a winning stake of s returns $s + ks$.

Odds from the “true” probabilities: 4 to 1. Odds 4 : 1 equal the true ratio 0.8 : 0.2. If horse 1 wins, the bookie collects \$60 000 and pays $10\,000 + 4 \cdot 10\,000 = \$50\,000$: profit \$10 000. If horse 2 wins, he pays $50\,000 + \frac{1}{4} \cdot 50\,000 = \$62\,500$: loss \$2 500 [[▶ 2024 L21 @ 6:28](#)]. The expected profit is $0.2 \cdot 10\,000 + 0.8 \cdot (-2\,500) = 0$, but only on average: the bookie carries the risk of the one race that is run.

Odds from the pool: 5 to 1. Now ignore the true probabilities and set the odds by the money staked, $50\,000/10\,000 = 5$. If horse 1 wins the bookie pays $10\,000 + 5 \cdot 10\,000 = \$60\,000$; if horse 2 wins he pays $50\,000 + \frac{1}{5} \cdot 50\,000 = \$60\,000$. Either way he pays exactly what he collected [[▶ 2024 L21 @ 8:37](#)][[▶ 2013 L19 @ 3:06](#)]: the book is *hedged* and any commission is

riskless profit. The pool proportions

$$q_1 = \frac{10\,000}{60\,000} = \frac{1}{6}, \quad q_2 = \frac{50\,000}{60\,000} = \frac{5}{6}$$

play the role of a *risk-neutral* probability: whichever the true probabilities (p_1, p_2) , the bookie's expected profit with odds set from q is zero and its variance is zero. The price of a bet on horse 1 (its stake, relative to the payout) is $q_1 = 1/6$, not $p_1 = 1/5$.

INTUITION: What the example teaches

The bookie prices by *what is already priced by the market* (the pool) and not by what he believes. The numbers q_i are not forecasts: they are the prices of the two Arrow–Debreu securities of [definition 2.7](#) (“\$1 if horse i wins”) normalised to sum to one. Three lessons carry over to derivatives [▶ 2024 L21 @ 9:42](#):

- an arbitrage-free price is the cost of a portfolio that reproduces the payoff in every outcome;
- that cost can be written as an expectation, but with the “pool” probabilities q and not the true ones;
- the pool probabilities need not coincide with reality, and so a model of real-world probabilities is not needed to price and hedge.

Exercise 15.1 (not from the course: the bookie). Pools of A on horse 1 and B on horse 2. Show that odds set to B/A to 1 on horse 1 (and A/B to 1 on horse 2) give the bookie the same profit whichever horse wins, and identify the “risk-neutral probabilities” q_i . Compute the standard deviation of the bookie's profit in the example if the odds are set from the true probabilities $(0.2, 0.8)$, and compare with the pool odds.

1 COURSE NOTE: Transcript slips

In the 2013 transcript the bookie's payout on the second horse is given as “\$12.25” for the quarter of \$50 000; it is \$12 500 (so the total payout is \$62 500 and the loss \$2 500, as on the slide). Also in 2013, to the quiz “which interest rate does the forward strike imply?” a student answers “2.5”, which the lecturer first repeats and then corrects on the fly: $80 \rightarrow 88.41$ is about 10% over the two years of [section 15.3](#), i.e. 5% per year continuously compounded, $80 e^{0.05 \cdot 2} = 88.41$ [▶ 2013 L19 @ 16:42](#).

15.3 Forwards, calls and puts

Throughout, S_t is the price at time t of a stock (no dividends), T the *expiry* of the contract, K its *strike* and $\tau = T - t$ the time to expiry. The three simplest derivatives pay, at T [▶ 2024 L21 @ 10:47](#) [▶ 2013 L19 @ 4:06](#),

$$\text{forward: } S_T - K, \quad \text{call: } (S_T - K)^+, \quad \text{put: } (K - S_T)^+,$$

with $x^+ = \max(x, 0)$. A *forward contract* is an obligation to buy at T at the price K fixed today; K is chosen so that the contract costs nothing to enter [▶ 2024 L21 @ 11:54](#) [▶ 2013 L19 @ 5:07](#). A *call option* is the right, not the obligation, to buy at K , and a *put option* the right to sell at K [▶ 2024 L21 @ 12:55](#). The call is insurance against the stock going up if one is short, and the put is insurance against a fall: its owner is protected below K . Unlike a forward, an option costs a premium today, because it can only win. A call (put) is *in the money* if $S > K$ ($S < K$), *at the money* if $S = K$, and *out of the money* otherwise; the lecture always means *European* options, exercisable only at T (American options, exercisable any time, have no closed form and are computed on trees or finite-difference grids).

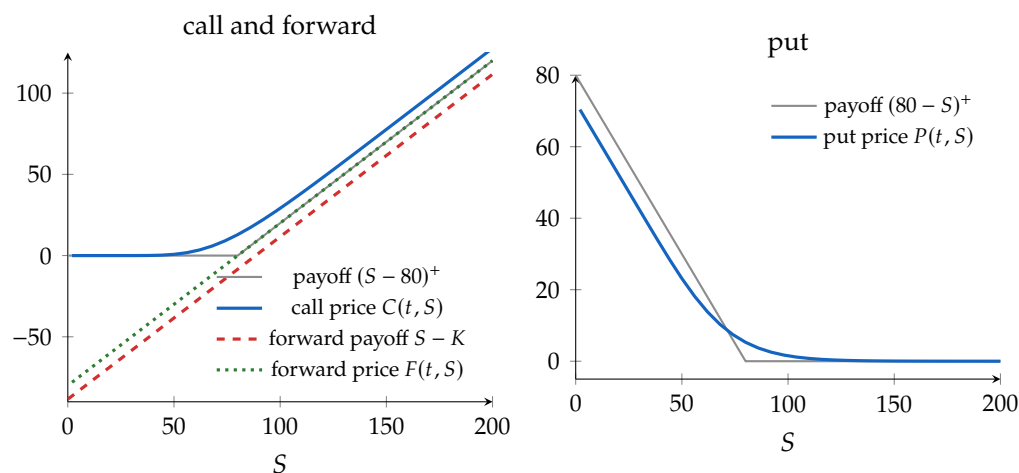


Figure 15.1: Payoffs at expiry (grey or dashed) and prices two years before expiry (coloured), as functions of the stock price, for $K = 80$, $r = 5\%$, $\sigma = 20\%$, $\tau = 2$ years (call, put) and $K = 88.41 = 80e^{0.1}$ (forward), the numbers of slides 3–6. The volatility is not stated on the slides; 20% reproduces the digital curve of slide 26 (section 15.11). The price curves are the Black–Scholes formula derived below; the forward price is $S - Ke^{-r\tau}$.

Figure 15.1 anticipates the goal of the lecture [► 2024 L21 @ 13:55]: to compute the curved lines, the prices today of the options, and the straight line for the forward. It also states the three messages of the lecture, which are exactly the content of the box below [► 2024 L21 @ 15:01][► 2013 L19 @ 9:22].

KEY FORMULAS

Three messages of the lecture (slide 7).

- Given the current price of the stock and a model for its dynamics, the price of a derivative is not uncertain.
- The price does not depend on the risk preferences of market participants (nor on the expected return of the stock).
- The mathematics of stochastic processes computes the price and the risks (hedge ratios) of the derivative.

15.4 One step: replication and the risk-neutral probability

The lecture builds the theory in a market with one period of length dt and two states [► 2024 L21 @ 16:04][► 2013 L19 @ 11:27]. It is the market of section 2.4.1 with continuous compounding: the stock moves from S_0 to S_1 (up, with real-world probability p) or $S_2 < S_1$ (down), a riskless account grows from B to $Be^{r dt}$, and a derivative f pays f_1 or f_2 .

15.4.1 The forward

A forward on the stock pays $S - K_0$ at dt and costs nothing today. The naive approach sets the price equal to the expected payoff with the real-world p , $p(S_1 - K_0) + (1 - p)(S_2 - K_0) = 0$, i.e. $K_0 = pS_1 + (1 - p)S_2$ (equal to $\frac{1}{2}(S_1 + S_2)$ if $p = \frac{1}{2}$) [► 2024 L21 @ 17:05][► 2013 L19 @ 13:31]. It is wrong, because nobody knows p , and it is exploitable, as follows [► 2024 L21 @ 19:08][► 2013 L19 @ 14:33].

Proposition 15.1 (Forward price). *In the absence of arbitrage, $K_0 = S_0 e^{r dt}$: the strike of a zero-cost forward is the price of the stock capitalised at the risk-free rate, whatever the real-world dynamics.*

So the stock itself (financed with a loan) replicates the forward: it has exactly the same payoff, and no assumption on p , S_1 , S_2 or on how the stock “really” behaves was used. In the lecture’s example, $S_0 = 80$, $T = 2$ years and $K = 88.41$: $88.41/80 = e^{0.1}$, so $r = 5\%$ continuously compounded [► 2013 L19 @ 16:42]. With $r = 0$ one gets $K_0 = S_0$, and one can still ask for a probability that gives the same price as the expectation: the number q with $K_0 = qS_1 + (1 - q)S_2$ satisfies

$$q = \frac{S_0 e^{r dt} - S_2}{S_1 - S_2} \in (0, 1) \quad \text{iff} \quad S_2 < S_0 e^{r dt} < S_1, \quad qS_1 + (1 - q)S_2 = S_0 e^{r dt}. \quad (15.1)$$

This q (called p_n or p_{rn} on the slides, q in 2013) is the *risk-neutral probability*: under it the stock earns exactly the risk-free rate on average [► 2024 L21 @ 23:25]. The condition $0 < q < 1$ is the no-arbitrage condition of [proposition 2.5](#) with $1 + r_f T$ replaced by $e^{r dt}$.

15.4.2 A general claim

For a call (or any claim) the stock alone is no longer enough, but a mix of stock and cash is. Look for a stock position a (in units of stock) and a cash amount B (in the riskless account, growing to $Be^{r dt}$) with [► 2024 L21 @ 25:30]–[► 2024 L21 @ 28:33][► 2013 L19 @ 19:51]

$$f_1 = aS_1 + Be^{r dt}, \quad f_2 = aS_2 + Be^{r dt}. \quad (15.2)$$

Subtracting, $a = \frac{f_1 - f_2}{S_1 - S_2}$, and then $B = e^{-r dt} \frac{S_1 f_2 - S_2 f_1}{S_1 - S_2}$. By no arbitrage $f_0 = aS_0 + B$ [► 2024 L21 @ 29:42][► 2013 L19 @ 20:55], and substituting one finds

$$f_0 = e^{-r dt} [q f_1 + (1 - q) f_2], \quad q = \frac{S_0 e^{r dt} - S_2}{S_1 - S_2}, \quad (15.3)$$

the same q as in (15.1). Indeed $aS_0 + B = e^{-r dt} \left[\frac{(f_1 - f_2)S_0 e^{r dt} + S_1 f_2 - S_2 f_1}{S_1 - S_2} \right] = e^{-r dt} \left[f_1 \frac{S_0 e^{r dt} - S_2}{S_1 - S_2} + f_2 \frac{S_1 - S_0 e^{r dt}}{S_1 - S_2} \right]$ [► 2024 L21 @ 31:51][► 2013 L19 @ 21:59]. This is (2.4) of [section 2.4.1](#) again.

Proposition 15.2 (One-step risk-neutral pricing). *In the one-step market with $S_2 < S_0 e^{r dt} < S_1$, every claim f is replicable, and its unique arbitrage-free price is $f_0 = e^{-r dt} \mathbb{E}_Q[f_1]$ with $Q = (q, 1 - q)$ given by (15.1). Equivalently, both $e^{-rt} S_t$ and $e^{-rt} f_t$ are martingales under Q (a two-point statement of the first fundamental theorem, [theorem 2.9](#)).*

The price of the call is thus computed by taking expectations of its payoff, $C_0 = e^{-r dt} q(S_1 - K)$ if $S_2 < K < S_1$ (slide 12) [► 2024 L21 @ 31:51]. The bookie’s pool proportions play the role of q ; a physicist would say that Q is the measure that makes the *discounted* price a martingale, i.e. an unbiased random walk.

INTUITION: Replicating portfolio versus riskless portfolio

Equation (15.2) says that the cash-plus-stock portfolio $aS + B$ has the same value as f in both states. Turn it around [► 2024 L21 @ 32:52]: the portfolio “one unit of the derivative, minus a units of stock” pays $f_i - aS_i = Be^{r dt}$ in both states: it is *riskless*, and must

Proof. At time 0, borrow S_0 from the bank, buy the stock and sell the forward with strike K_0 : the net cost is zero. At dt deliver the stock in exchange for K_0 and repay $S_0 e^{r dt}$: the payoff is $K_0 - S_0 e^{r dt}$ in both states, a certain amount that cost nothing. If it were positive this would be an arbitrage; if negative, the opposite position would be [► 2024 L21 @ 20:11] [► 2013 L19 @ 15:35]. Hence $K_0 = S_0 e^{r dt}$. □

Example 15.3 (Numbers of [example 2.6](#) in continuous compounding). Take $S_0 = 100$, $S_1 = 130$, $S_2 = 90$, $K = 100$ and $e^{r dt} = 1.03$ (the value used in [example 2.6](#)). Then $a = 30/40 = 0.75$, $B = -0.75 \cdot 90/1.03 = -65.53$, $q = (103 - 90)/40 = 0.325$, and $f_0 = 0.75 \cdot 100 - 65.53 = 0.325 \cdot 30/1.03 = 9.466$, in agreement with [example 2.6](#).

therefore grow at the risk-free rate. For a physicist this is a statement that the hedged combination has zero “noise”, i.e. $a = \Delta f / \Delta S$ is a discrete derivative that cancels the fluctuation. In continuous time the same argument is used in the form “riskless portfolio” (section 15.7). Which of the two forms is more convenient depends on the problem ▶ 2024 L21 @ 32:52].

The real-world probability p never appears.

Exercise 15.2 (not from the course: one-step market). $S_0 = 50$, $S_1 = 60$, $S_2 = 42$, $r \, dt = \ln 1.02$.

- Compute q , check the no-arbitrage condition and find the fair forward strike.
- Price a call ($K = 50$) and a put ($K = 50$) by replication and by $\mathbb{E}_Q$ and verify parity.
- Let the real-world probability of “up” be $p = 0.9$: compute $e^{-r \, dt} \mathbb{E}_P[\text{call}]$ and the profit of a trader who sells the call at that price and hedges with a shares.

15.5 Many steps: the binomial tree and its limit

This section is not spelled out in the lecture, but it is the bridge between the one-step market of section 15.4 and the continuous model of the slides (the second bullet of 2013 slide 21, “tree methods, equivalent to explicit scheme”); it is also the way most exotic options are still priced.

Split $[0, T]$ into n steps of length $dt = T/n$. In each step the stock is multiplied by u or by d ; the Cox–Ross–Rubinstein (CRR) choice is

$$u = e^{\sigma\sqrt{dt}}, \quad d = \frac{1}{u} = e^{-\sigma\sqrt{dt}}, \quad q = \frac{e^{r \, dt} - d}{u - d}. \quad (15.4)$$

Each node of the tree is a copy of the one-step market, so proposition 15.2 applies at every node and prices are computed by *backward induction* from the payoff at the leaves,

$$f(S, t) = e^{-r \, dt} [q f(uS, t + dt) + (1 - q) f(dS, t + dt)], \quad f(S, T) = \text{payoff}(S). \quad (15.5)$$

Unrolling it, $f(S_0, 0) = e^{-rT} \mathbb{E}_Q[\text{payoff}(S_T)]$ with $S_T = S_0 u^J d^{n-J}$ and $J \sim \text{Bin}(n, q)$. Since the number q was fixed by the requirement that the *discounted* stock be a martingale, this is the multi-period FTAP of theorem 2.9 and the martingale $e^{-rt} S_t$ of chapter 4.

Proposition 15.4 (Two limits of the tree). Let $dt \rightarrow 0$ with (15.4).

- Distribution.** $q = \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma} \sqrt{dt} + O(dt)$ and $\ln(S_T/S_0) \Rightarrow N((r - \frac{1}{2}\sigma^2)T, \sigma^2 T)$. The price of a call converges to $e^{-rT} \mathbb{E}_Q[(S_T - K)^+]$ with S_T lognormal, i.e. to the Black–Scholes value of section 15.9.
- Equation.** If f is smooth, (15.5) implies $f_t + \frac{1}{2}\sigma^2 S^2 f_{SS} + rS f_S - rf = o(1)$, the Black–Scholes PDE of section 15.7.

Proof. (i) Expand $u = 1 + \sigma\sqrt{dt} + \frac{1}{2}\sigma^2 dt + O(dt^{3/2})$, $d = 1 - \sigma\sqrt{dt} + \frac{1}{2}\sigma^2 dt + O(dt^{3/2})$, so $u - d = 2\sigma\sqrt{dt} + O(dt)$ and $e^{r \, dt} - d = \sigma\sqrt{dt} + (r - \frac{1}{2}\sigma^2)dt + O(dt^{3/2})$, which gives the expansion of q . A step of $\ln S$ is $\sigma\sqrt{dt} \xi$ with $\xi = \pm 1$, $\mathbb{P}_Q(\xi = 1) = q$; its mean is $\sigma\sqrt{dt}(2q - 1) = (r - \frac{1}{2}\sigma^2)dt + O(dt^{3/2})$ and its variance $\sigma^2 dt [1 - (2q - 1)^2] = \sigma^2 dt (1 + O(dt))$. Summing $n = T/dt$ independent steps, the mean tends to $(r - \frac{1}{2}\sigma^2)T$ and the variance to $\sigma^2 T$; the steps are bounded by $\sigma\sqrt{dt} \rightarrow 0$, so the Lindeberg central limit theorem (chapter 3) applies. The payoff is continuous with linear growth and $\mathbb{E}_Q[S_T] = S_0 e^{rT}$ exactly for every n , which supplies the uniform integrability needed to pass to the limit in the expectation. (ii) By the definition of q , $q(u - 1) + (1 - q)(d - 1) = e^{r \, dt} - 1$ exactly, and $q(u - 1)^2 + (1 - q)(d - 1)^2 =$

Table 15.1 and fig. 15.2 show the convergence for the option of fig. 15.1 at the money: $S_0 = K = 80$, $T = 2$, $r = 5\%$, $\sigma = 20\%$, for which the formula of section 15.9 gives $C = 12.9014$. The error decays like $1/n$ but alternates in sign with the parity of n , because at the money the strike sits on a node for even n and in the middle of a lattice cell for odd n : the column $n \cdot \text{error}$ is stable near -2.26 for even n and near $+1.73$ for odd n (e.g. $n = 51, 101, 1001$), so the oscillation does not die out, only its amplitude shrinks. The

single-step value 14.51 is more than 12% off.

n	tree price	error	$n \cdot \text{error}$	q
1	14.5106	+1.6092	+1.61	0.6132
2	11.8731	-1.0283	-2.06	0.5775
5	13.2506	+0.3492	+1.75	0.5481
10	12.6786	-0.2228	-2.23	0.5338
50	12.8564	-0.0451	-2.25	0.5150
100	12.8789	-0.0226	-2.26	0.5106
1000	12.8992	-0.0023	-2.26	0.5034
5000	12.9010	-0.0005	-2.27	0.5015

Table 15.1: CRR tree price of the European call $S_0 = K = 80$, $T = 2$, $r = 5\%$, $\sigma = 20\%$ (Black-Scholes: 12.9014). Computed for these notes.

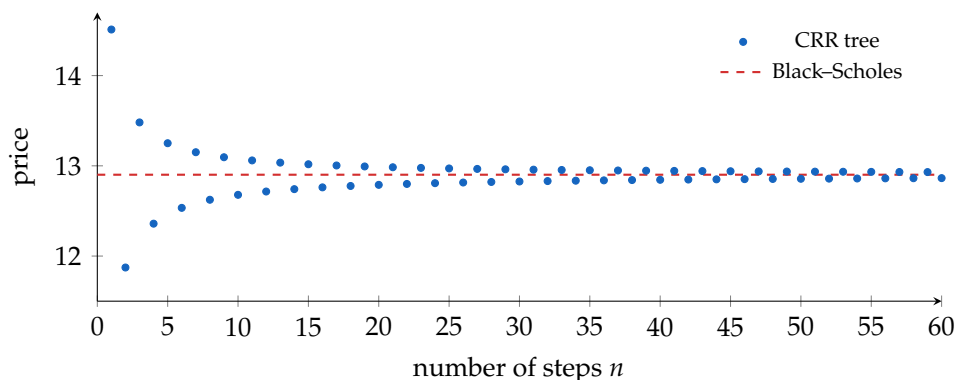


Figure 15.2: Convergence of the binomial price to the Black-Scholes value (same parameters as table 15.1); the alternation between odd and even n is visible.

INTUITION: Why $\sigma\sqrt{dt}$ and where $-\frac{1}{2}\sigma^2$ comes from

The steps of the tree are not independent of the time step: if the space step were $\propto dt$ the process would freeze, and if it were larger than $\sqrt{dt}$ the variance would explode. The unique scaling that leaves a non-trivial limit is the diffusive one, $\text{step} \sim \sqrt{dt}$, as in Donsker's theorem (theorem 11.6); this is what the lecturer proves in the next section by counting variances. The drift correction $-\frac{1}{2}\sigma^2$ in the exponent is the same lognormal correction as in remark 4.19: the tree has $\mathbb{E}_Q[S_T] = S_0 e^{rT}$ (the martingale property is exact), but $\ln S_T$ has mean $(r - \frac{1}{2}\sigma^2)T$ because $\mathbb{E}[e^X] = e^{\mathbb{E}[X] + \text{Var } X/2}$ for a normal variable.

Exercise 15.3 (not from the course: the tree).

- Show that the CRR tree prices a forward $(S_T - K)$ exactly, for every n .
- Write a program that prices the call of table 15.1 and reproduces the $1/n$ error and the odd-even oscillation; then choose the strike lattice-aligned ($K = S_0 u^k$) and observe the effect.
- Show that with $u = e^{\sigma\sqrt{dt} + \nu dt}$, $d = e^{-\sigma\sqrt{dt} + \nu dt}$ (drift ν in the tree) one still

$\sigma^2 dt (1 + O(\sqrt{dt}))$. Taylor-expanding $f(uS, t + dt)$ and $f(dS, t + dt)$ to second order in $(u - 1)S$, $(d - 1)S$ and first order in dt and inserting in (15.5) gives $e^{r dt} f = f + f_t dt + (e^{r dt} - 1)S f_S + \frac{1}{2}\sigma^2 S^2 f_{SS} dt + o(dt)$; subtract f and divide by dt . $\square$

converges to Black–Scholes provided q is defined by the martingale condition.

15.6 Lognormal dynamics and a heuristic Itô formula

15.6.1 The model

The lecture models the stock by the *lognormal* (geometric Brownian motion, eq. (11.13)) dynamics [► 2024 L21 @ 33:56][► 2013 L19 @ 22:04]

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (15.6)$$

where μ is the (real-world) drift, σ the *volatility*, and dW the increment of a Brownian motion: normal, with $\mathbb{E}[dW] = 0$ and $\text{sd}(dW) = \sqrt{dt}$, independent for disjoint time intervals [► 2013 L19 @ 23:04]. The proportional change dS/S has a deterministic part and a noise part. The lecture says clearly that the choice is an *example*: the method needs only that the stock has some dynamics, and other choices (stochastic volatility, jumps) lead to other equations of the same type [► 2024 L21 @ 33:56].

15.6.2 Why the standard deviation is $\sqrt{dt}$

In the 2024 lecture Strela stops to answer a question that “always bothers” him [► 2024 L21 @ 36:07]. Take a random walk $W_n = \sum_{i=1}^n \xi_i$ with steps $\xi_i = \pm 1$ with probability $\frac{1}{2}$, uncorrelated: $\mathbb{E}[W_n] = 0$ (a martingale) and $\text{Var } W_n = \sum_i \mathbb{E}[\xi_i^2] = n$ [► 2024 L21 @ 37:10]. Now put the n steps in a finite time T , $dt = T/n$, and let each step have size $\pm\delta$: $\text{Var } W = n\delta^2 = T\delta^2/dt$. For the limit $dt \rightarrow 0$ to be neither 0 nor ∞ one needs δ^2/dt to stay finite, i.e. $\delta \propto \sqrt{dt}$ [► 2024 L21 @ 40:26]. That is the content of Donsker’s theorem (theorem 11.6); the same argument fixed the size of the steps of the tree (15.4).

15.6.3 Taylor expansion with a $\sqrt{dt}$ noise

Let $f(S, t)$ be smooth. In a deterministic world the second-order Taylor formula is $df = f_t dt + f_S dS + \frac{1}{2} f_{tt} dt^2 + f_{tS} dt dS + \frac{1}{2} f_{SS} dS^2 + \dots$ [► 2024 L21 @ 43:35]. With dS from (15.6) the terms have very different sizes [► 2024 L21 @ 44:45]: $dt dS \sim dt^{3/2}$ and dt^2 are negligible, but

$$(dS)^2 = \sigma^2 S^2 (dW)^2 + O(dt^{3/2}), \quad (dW)^2 \sim dt,$$

is of the *same* order as dt and must be kept. One writes $(dW)^2 = dt$ (eq. (11.3)) and obtains, to order dt , [► 2024 L21 @ 45:45]

$$df = \left(f_t + \mu S f_S + \frac{1}{2} \sigma^2 S^2 f_{SS} \right) dt + \sigma S f_S dW, \quad (15.7)$$

Itô’s formula for a function of S and t (slide 15). The lecturer calls it a “quick and dirty derivation” [► 2024 L21 @ 47:46]; the rigorous statement, with the Itô integral and the proof, is in chapter 17.

INTUITION: Physicist’s reading

Equation (15.7) is the Taylor expansion of a function of a Brownian path, with the “ordinary” second-order term promoted to first order because the path is fractal

2 COURSE NOTE: A student’s objection

A student asks how $(dW)^2 = dt$ can hold when dW is random [► 2024 L21 @ 55:16]. The answer given (by the lecturer and by the instructor, P. Kempthorne) is that the notation is sloppy but reflects a theorem: the *quadratic*

($\Delta W \sim \sqrt{\Delta t}$). Its physical avatar is the Langevin equation with multiplicative noise interpreted in the Itô (left-point) sense: $\frac{1}{2}\sigma^2 S^2 f_{SS}$ is the “noise-induced drift” that a Stratonovich convention would absorb in the definition of the drift (chapter 17). The same extra term makes $\ln S$ drift at $\mu - \frac{1}{2}\sigma^2$ rather than μ : taking $f = \ln S$ in (15.7) gives $d \ln S = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW$, and hence eq. (11.13), $S_T = S_t \exp[(\mu - \frac{1}{2}\sigma^2)\tau + \sigma(W_T - W_t)]$.

variation of Brownian motion over $[0, t]$ is exactly t , $\sum_j (\Delta_j W)^2 \rightarrow t$ almost surely as the mesh goes to zero (theorem 11.8) [► 2024 L21 @ 56:21]. Each squared increment is random, of mean dt and standard deviation $\sqrt{2} dt$, but their *sum* over $n \sim 1/dt$ intervals has mean t and standard deviation $\sqrt{2t} dt \rightarrow 0$: a law of large numbers. Inside an integral $(dW)^2$ can therefore be replaced by dt .

15.7 The Black–Scholes equation

We now do in continuous time what section 15.4.2 did in one step [► 2024 L21 @ 47:46] [► 2013 L19 @ 23:04]. **Assumptions** (implicit in the lecture): the stock follows (15.6) with constant μ, σ ; the interest rate r is constant and the same for borrowing and lending; the stock pays no dividend; trading is continuous, without transaction costs, with unrestricted short selling and fractional positions; the claim is European.

15.7.1 Derivation by replication

Let a portfolio hold $a_t = a(S_t, t)$ shares of the stock and a cash amount B_t in the money-market account, which grows as dB (due to interest) $= rB dt$ [► 2024 L21 @ 49:52]. Its value is $V = aS + B$. The portfolio is *self-financing*: no money is added or withdrawn, so the change of value is only due to price moves and interest,

$$dV = a dS + rB dt = (a\mu S + rB) dt + a\sigma S dW. \quad (15.8)$$

(On the slides: $df = a dS + dB$; the lecture writes $dB = rB dt$ and $B = f - aS$, the same thing.) We want the portfolio to *replicate* the claim: $V_t = f(S_t, t)$ at all times, with $f(S, T)$ the payoff. Equate (15.7) and (15.8) [► 2024 L21 @ 51:59][► 2013 L19 @ 25:12].

Noise terms (coefficients of dW): $\sigma S f_S = a\sigma S$, so

$$a = \frac{\partial f}{\partial S} \quad (\text{the delta}) \quad [\text{► 2024 L21 @ 57:27}]. \quad (15.9)$$

Drift terms (coefficients of dt), with $B = f - aS = f - S f_S$ from $V = f$: $f_t + \mu S f_S + \frac{1}{2}\sigma^2 S^2 f_{SS} = \mu S f_S + r(f - S f_S)$. The real-world drift μ cancels and one is left with

$$\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} + rS \frac{\partial f}{\partial S} - rf = 0 \quad (15.10)$$

[► 2024 L21 @ 58:33][► 2013 L19 @ 26:20], the *Black–Scholes equation* (slide 17), a backward parabolic PDE with final condition $f(S, T) = \text{payoff}(S)$. It is the result of Black and Scholes (1973); Scholes and Merton received the 1997 Nobel prize in economics (Black had died in 1995), and all three were associated with MIT [► 2024 L21 @ 1:01:47]. If the claim traded at any price other than V_t , buying the cheaper and selling the dearer of f and V would be an arbitrage, which is why $f = V$.

The lecture derives the same equation in the “riskless portfolio” form [► 2024 L21 @ 49:52]. Let $\Pi = f - aS$ with a held fixed over the interval $[t, t + dt]$. By (15.7), $d\Pi = df - a dS = (f_t + \frac{1}{2}\sigma^2 S^2 f_{SS} + (f_S - a)\mu S) dt + (f_S - a)\sigma S dW$. Choosing $a = f_S$ kills both μ and the noise: $d\Pi = (f_t + \frac{1}{2}\sigma^2 S^2 f_{SS}) dt$ is deterministic, hence must equal $r\Pi dt = r(f - S f_S) dt$ by no arbitrage, which is (15.10) again. The delta-hedged position (long the option, short f_S shares) earns the risk-free rate.

KEY FORMULAS

What the equation says (slide 18; ▶ 2024 L21 @ 1:00:39).

- Every tradable claim on S , whatever its payoff, satisfies (15.10); the payoff enters only through the final (and boundary) conditions.
- The real-world drift μ does not appear: only σ and r do. (Compare section 2.4.1: the probability of “up” does not appear.)
- The replicating portfolio is also a *hedging strategy*: to hedge a short position in the claim, hold $a = \partial f / \partial S$ shares and put the rest, $f - S f_S$, in the bank. For a traded, liquid stock this is what a derivatives desk does all day ▶ 2013 L19 @ 28:31].
- With another model for S the equation changes (different coefficients, possibly more state variables), but the derivation is identical.

15.7.2 Final and boundary conditions

As for any PDE one needs conditions ▶ 2024 L21 @ 1:03:55] ▶ 2013 L19 @ 30:40]. The *final* condition is the payoff, $C(S, T) = (S - K)^+$ for a call and $P(S, T) = (K - S)^+$ for a put. The *boundary* conditions follow from the economic sense of the options ▶ 2024 L21 @ 1:04:56] ▶ 2013 L19 @ 31:49]:

$$\text{call: } C(0, t) = 0, \quad C(S, t) \sim S - Ke^{-r(T-t)} \quad (S \rightarrow \infty), \quad (15.11)$$

$$\text{put: } P(0, t) = Ke^{-r(T-t)}, \quad P(S, t) \rightarrow 0 \quad (S \rightarrow \infty). \quad (15.12)$$

The stock price $S = 0$ is absorbing for geometric Brownian motion (a stock worth zero stays worthless), so the call is worthless and the put is a certain payment of K at T , worth $Ke^{-r(T-t)}$ today. For $S \rightarrow \infty$ the call is exercised almost surely and is worth the forward, $S - Ke^{-r(T-t)}$.

15.7.3 Reduction to the heat equation

Slide 18 states that “by a change of variables” (15.10) becomes the heat equation $u_\tau = u_{xx}$; the 2024 lecturer calls finding the change of variables a good exercise ▶ 2024 L21 @ 1:02:51], and in 2013 he proposed it as a final-paper topic ▶ 2013 L19 @ 30:40]. Here it is (the slide’s normalisation $u_\tau = u_{xx}$ differs from ours by a rescaling $\tau \rightarrow \sigma^2 \tau / 2$).

Put $\tau = T - t$ (so $\partial_t = -\partial_\tau$) and $y = \ln S$ (so $S f_S = f_y$ and $S^2 f_{SS} = f_{yy} - f_y$). Then (15.10) reads

$$f_\tau = \frac{1}{2} \sigma^2 f_{yy} + \left(r - \frac{1}{2} \sigma^2\right) f_y - r f.$$

Remove the discounting with $f = e^{-r\tau} w$, which gives $w_\tau = \frac{1}{2} \sigma^2 w_{yy} + \left(r - \frac{1}{2} \sigma^2\right) w_y$, and remove the drift with a *Galilean shift*, $x = y + \left(r - \frac{1}{2} \sigma^2\right) \tau$, $w(y, \tau) = u(x, \tau)$: since $w_\tau = u_\tau + \left(r - \frac{1}{2} \sigma^2\right) u_x$, $w_y = u_x$, $w_{yy} = u_{xx}$, one gets

$$\begin{aligned} f(S, t) &= e^{-r\tau} u(x, \tau), & x &= \ln S + \left(r - \frac{1}{2} \sigma^2\right) \tau, \\ \frac{\partial u}{\partial \tau} &= \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2}, & u(x, 0) &= \text{payoff}(e^x). \end{aligned} \quad (15.13)$$

(One checks it on exponentials: $f = e^{-r\tau} e^{\alpha x + \sigma^2 \alpha^2 \tau / 2}$ satisfies (15.10) for every α ; verified symbolically for these notes.) By proposition 11.4 the solution with initial datum u_0 is $u(x, \tau) = \int p(y, \tau | x) u_0(y) dy$, the Gaussian kernel (11.1) with variance $\sigma^2 \tau$: that is, $u(x, \tau) = \mathbb{E}[u_0(x + \sigma \sqrt{\tau} Z)]$ with $Z \sim N(0, 1)$. Written in the original variables,

$$f(S, t) = e^{-r\tau} \mathbb{E}[\text{payoff}(S e^{(r - \sigma^2/2)\tau + \sigma \sqrt{\tau} Z})], \quad (15.14)$$

3 COURSE NOTE: Erratum: the call at infinity

Slide 19 of 2024 (slide 18 of 2013) states $C(\infty, t) \cong S$. To leading order this is right, but the correct asymptotics is (15.11): the call behaves as the *forward* $S - Ke^{-r(T-t)}$, as the 2013 lecturer says in words (“discounted S minus K ”) ▶ 2013 L19 @ 31:49]. The put condition $P(0, t) = Ke^{-r(T-t)}$ is correct on both sets of slides.

which is exactly the risk-neutral expectation of [section 15.8](#): the two routes to the price (PDE and expectation) give the same answer, as they must (this is the Feynman–Kac formula, [chapter 18](#)).

INTUITION: Diffusion of a payoff backwards in time

The pricing PDE is the heat equation run *backwards*: the payoff at T is the “initial temperature profile” and the price today is that profile after diffusing for a time τ with diffusivity $\sigma^2/2$, in the frame of the log-price moving with velocity $r - \frac{1}{2}\sigma^2$ and with a decay $e^{-r\tau}$ (discounting). The hockey stick smoothed by a Gaussian is precisely the curved price line of [fig. 15.1](#); the smoothing width $\sigma\sqrt{\tau}$ is the size of the “typical move” of the log-price until expiry, so an option far from the strike is the payoff, and an option near the strike is worth more than its intrinsic value because the price may still cross K (the “time value” or convexity value).

Exercise 15.4 (not from the course: the heat equation).

- Verify the change of variables of [section 15.7.3](#) by direct substitution.
- Solve $u_\tau = \frac{1}{2}\sigma^2 u_{xx}$ for $u(x, 0) = \mathbf{1}\{x > \ln K\}$ and show that $f = e^{-r\tau}\Phi(d_2)$ (the digital).
- The slide’s normalisation is $u_\tau = u_{xx}$: find the rescaling of τ .

15.8 Risk-neutral valuation

The second route to the price is the one the lecturer wants to stress: “mostly about risk-neutral pricing, and Black–Scholes is an illustration of the method” [[2024 L21 @ 0:00](#)]. In discrete time the price was $f_0 = e^{-r\Delta t}\mathbb{E}_Q[f_1]$ with the special probability q . The continuous-time statement is [[2024 L21 @ 1:06:01](#)][[2013 L19 @ 33:55](#)]

$$f_t = e^{-r(T-t)} \mathbb{E}_Q[f_T | S_t], \quad \text{where under } \mathbb{Q}: \quad dS = rS dt + \sigma S dW^\mathbb{Q}, \quad (15.15)$$

so that $S_t = e^{-r(T-t)}\mathbb{E}_Q[S_T | S_t]$: under $\mathbb{Q}$ the *discounted* stock price $e^{-rt}S_t$ is a martingale (slides 13, 21).

How to find $\mathbb{Q}$. The lecturer’s argument [[2024 L21 @ 1:07:01](#)]: take the expectation of (15.6) (the dW term has mean zero), $d\mathbb{E}[S_t] = \mu\mathbb{E}[S_t]dt$, so $\mathbb{E}[S_T] = S_t e^{\mu(T-t)}$ for the real-world dynamics. In the risk-neutral world the stock must grow at the risk-free rate, $\mathbb{E}_Q[S_T] = S_t e^{r(T-t)}$ (this was the property $qS_1 + (1-q)S_2 = S_0 e^{r\Delta t}$ of the tree), which forces $\mu \rightarrow r$. The volatility is *unchanged*: the size of the fluctuations is a pathwise property (the quadratic variation, [theorem 11.8](#)), the same under every equivalent measure, whereas the drift can be changed at will. Changing only the drift of a Brownian motion is the content of Girsanov’s theorem ([chapter 18](#)): $W_t^\mathbb{Q} = W_t + \lambda t$ with $\lambda = (\mu - r)/\sigma$, the *market price of risk* (the Sharpe ratio of the stock; this identification is not made in the lecture). In this language

$$dS = \mu S dt + \sigma S dW = rS dt + \sigma S (dW + \lambda dt) = rS dt + \sigma S dW^\mathbb{Q}. \quad (15.16)$$

The lognormal density of the terminal price under $\mathbb{Q}$ is (slide 21)

$$p_\mathbb{Q}(S_T | S_t) = \frac{1}{S_T \sigma \sqrt{2\pi(T-t)}} \exp \left[-\frac{(\ln(S_T/S_t) - (r - \sigma^2/2)(T-t))^2}{2\sigma^2(T-t)} \right], \quad (15.17)$$

the density of the lognormal variable [eq. \(11.13\)](#) with $\mu \rightarrow r$ [[2024 L21 @ 1:09:15](#)]. Pricing a claim then means “integrate the payoff against this density and discount”, instead of solving a PDE [[2024 L21 @ 1:09:15](#)]; this is easy for a lognormal density and typically not for other models. The equivalence of [\(15.15\)](#) with the PDE is the Feynman–Kac formula ([chapter 18](#)), already visible in [\(15.14\)](#).

INTUITION: Why “neutral”

There is no claim that investors are risk-neutral. Real-world prices already contain risk premia: $\mu > r$ is the reward for bearing the risk of the stock. The claim is that the *price of a derivative relative to the stock* does not depend on those premia because the derivative can be replicated by the stock and the bank; so one may compute the price in any world where the stock earns r , and the simplest is the one where nobody cares about risk. Changing $\mu \rightarrow r$ moves the drift of the log-price, not its width: a physicist would say that $\mathbb{Q}$ is the same diffusion in a different, tilted, “potential” (the tilt is the drift), and the Radon–Nikodym factor $\exp(-\lambda W_T - \frac{1}{2}\lambda^2 T)$ is a Boltzmann-like reweighting of the paths.

IN FINANCE: Where the approach is used

For a desk the two languages are complementary. The PDE (finite-difference grids) is the tool for low-dimensional problems with early exercise (American options), barriers and local-vol models; the expectation (Monte Carlo) for high-dimensional or path-dependent payoffs; trees are equivalent to an explicit finite-difference scheme ([proposition 15.4\(ii\)](#)) [[2024 L21 @ 1:04:56](#)][[2013 L19 @ 34:59](#)]. Slide 22 of 2024 lists what the whole business uses: PDE, numerical methods, stochastic calculus, simulation, statistics, “much, much more”.

4 COURSE NOTE: Erratum: the density on slide 21 (2024) and 20 (2013)

The pre-factor of the printed density is wrong in both years. In 2024 it reads $S_t/(\sigma S_T \sqrt{2\pi T})$, in 2013 $1/(\sigma S \sqrt{2\pi T})$. The correct normalisation, which makes the density integrate to one in S_T , is $1/(S_T \sigma \sqrt{2\pi(T-t)})$, as in [\(15.17\)](#): the S_T in the denominator is the Jacobian of $S_T = e^X$ and the time is the *time to expiry* $T - t$, as in the exponent, not T . The exponent on the slide is correct.

Example 15.5 (The wrong expectation). For $S_0 = K = 80$, $T = 2$, $r = 5\%$, $\sigma = 20\%$ the correct price is 12.90. Suppose the stock has real-world drift $\mu = 12\%$. Discounting the *real-world* expectation of the payoff at the risk-free rate gives $e^{-rT} \mathbb{E}[(S_T - K)^+] = 22.17$, a 72% overestimate: it is not an arbitrage price, because one cannot replicate the option with it. A Monte Carlo estimate under $\mathbb{Q}$ with $2 \cdot 10^5$ paths gives 12.955 ± 0.040 (one standard error), consistent with 12.901. (Computed for these notes.)

15.9 The Black–Scholes formula

Theorem 15.6 (Black–Scholes formula). In the model [\(15.6\)](#), with $\tau = T - t$, the European call and put are worth

$$C(S, t) = S \Phi(d_1) - K e^{-r\tau} \Phi(d_2) = e^{-r\tau} [e^{r\tau} S \Phi(d_1) - K \Phi(d_2)], \quad (15.18)$$

$$P(S, t) = K e^{-r\tau} \Phi(-d_2) - S \Phi(-d_1) = e^{-r\tau} [K \Phi(-d_2) - e^{r\tau} S \Phi(-d_1)], \quad (15.19)$$

where Φ is the standard normal cdf and

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau} = \frac{\ln(S/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \quad (15.20)$$

The right-hand forms are the ones on slide 20 [[2024 L21 @ 1:04:56](#)][[2013 L19 @ 32:52](#)]. In the lecture the formula is quoted as “the solution of the PDE in terms of the error function, which has to be tabulated”; here are two derivations.

This is also [corollary 3.11\(ii\)](#) with $\mu_{\ln} = \ln S + (r - \frac{1}{2}\sigma^2)\tau$ and $\sigma_{\ln} = \sigma\sqrt{\tau}$, exactly as anticipated in the finance box of that chapter; and it is the two-line computation suggested in 2013 as a final-paper topic [[2013 L19 @ 34:59](#)].

Proof 1: risk-neutral expectation. By [\(15.15\)](#), $C = e^{-r\tau} \mathbb{E}[(S_T - K)^+]$ with $S_T = S e^{(r - \sigma^2/2)\tau + \sigma\sqrt{\tau}Z}$, $Z \sim N(0, 1)$ (the marginal of [\(15.17\)](#)). The event $\{S_T > K\}$ is $\{Z > -d_2\}$ because $\ln(S_T/K) = \sigma\sqrt{\tau}(Z + d_2)$. So

$$\mathbb{E}[(S_T - K)^+] = \mathbb{E}[S_T \mathbf{1}\{Z > -d_2\}] - K \mathbb{P}(Z > -d_2), \quad \mathbb{P}(Z$$

For the first term use the identity, for real b, c ,

$$\mathbb{E}[e^{bZ} \mathbf{1}\{Z > c\}] = \int_c^\infty e^{bz} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = e^{b^2/2} \int_c^\infty \frac{e^{-(z-b)^2/2}}{\sqrt{2\pi}} dz$$

(completing the square). With $b = \sigma\sqrt{\tau}$, $c = -d_2$ one has $b - c = d_2 + \sigma\sqrt{\tau} = d_1$ and $\mathbb{E}[S_T \mathbf{1}\{Z > -d_2\}] = S e^{(r-\sigma^2/2)\tau} e^{\sigma^2\tau/2} \Phi(d_1) = S e^{r\tau} \Phi(d_1)$. Multiplying by $e^{-r\tau}$ gives (15.18). The put follows in the same way with $\{Z < -d_2\}$, or from theorem 15.8 below. $\square$

Proof 2: solving the PDE. By (15.14) the solution of (15.10) with final condition $(S - K)^+$ and the boundary conditions (15.11) is the same Gaussian integral as in Proof 1; uniqueness of the bounded-growth solution of the heat equation (proposition 11.4) shows that it is the solution. One can also verify directly that (15.18) solves (15.10), using the identity $S\varphi(d_1) = K e^{-r\tau} \varphi(d_2)$ (section 15.12), and that $C \rightarrow (S - K)^+$ as $\tau \rightarrow 0$ (for $S > K$, $d_{1,2} \rightarrow +\infty$; for $S < K$, $d_{1,2} \rightarrow -\infty$). $\square$

Example 15.7 (The at-the-money call of the slides). For $S = K = 80$, $\tau = 2$, $r = 5\%$, $\sigma = 20\%$: $d_1 = 0.14/0.2828 = 0.4950$, $d_2 = 0.2121$, $\Phi(d_1) = 0.6897$, $\Phi(d_2) = 0.5840$, $S\Phi(d_1) = 55.175$, $K e^{-r\tau} \Phi(d_2) = 72.39 \cdot 0.5840 = 42.274$ and $C = 12.901$. By parity the put is 5.288. The two terms are the price of receiving the stock if $S_T > K$ and of paying K if $S_T > K$, the number $\Phi(d_2) = 0.584$ being the risk-neutral probability of finishing in the money.

5 COURSE NOTE: On the parameters of the slides

The slides label the plots with $S = 80$, $K = 80$, $T = 2$ years (forward: $K = 88.41$) and never state r or σ . The forward strike identifies $r = 5\%$ (continuously compounded, $80e^{0.1} = 88.41$), and the digital of slide 26 tends to $0.905 = e^{-0.1}$ for large S and takes the value ≈ 0.53 at $S = 80$; with $\sigma = 20\%$ formula (15.23) gives 0.528. All figures in this chapter that reproduce the slides use $r = 5\%$, $\sigma = 20\%$; this identification is ours.

INTUITION: Reading the formula

- $e^{-r\tau} \Phi(d_2)$ = price of the *cash-or-nothing* (digital) option, $\Phi(d_2) = \mathbb{Q}(S_T > K)$; $S\Phi(d_1)$ is the price of the *asset-or-nothing* option, which pays S_T if $S_T > K$. $\Phi(d_1)$ is a probability too, but under the measure in which the stock, not the bank, is the numeraire: the probability that the option is exercised as seen by a holder of the stock.
- As $\sigma\sqrt{\tau} \rightarrow 0$ the option becomes the forward $(S - K e^{-r\tau})^+$: the time value is the smoothing of the hockey stick over a width $\sigma\sqrt{\tau}$ (fig. 15.1).
- At the money forward ($S e^{r\tau} = K$), $d_1 = -d_2 = \frac{1}{2}\sigma\sqrt{\tau}$ and $C = S [2\Phi(\frac{1}{2}\sigma\sqrt{\tau}) - 1] \approx 0.4 S \sigma\sqrt{\tau}$, the Bachelier rule of thumb quoted in chapter 3 (not discussed in class).
- The price depends on S, K, τ, r, σ but not on μ . The only quantity that is not observed is σ , which is what traders quote instead of a price (section 15.13).

Exercise 15.5 (not from the course: the formula by expectation). Derive the put formula (15.19) directly from $P = e^{-r\tau} \mathbb{E}_{\mathbb{Q}}[(K - S_T)^+]$ and check parity. Evaluate the call and the put for $S = 100$, $K = 90$, $\tau = 0.5$, $r = 3\%$, $\sigma = 25\%$. Show that C is increasing and convex in S , increasing in σ and τ , and that $(S - K e^{-r\tau})^+ \leq C \leq S$ (the no-arbitrage bounds).

Exercise 15.6 (not from the course: risk-neutral drift). Apply (15.7) to $e^{-rt} S_t$ to show that $d(e^{-rt} S_t) = (\mu - r)e^{-rt} S_t dt + \sigma e^{-rt} S_t dW$ and, with $W_t^{\mathbb{Q}} = W_t + \lambda t$, that it is a martingale for $\lambda = (\mu - r)/\sigma$. Compute $\mathbb{E}_{\mathbb{P}}[C_T]$ and $\mathbb{E}_{\mathbb{Q}}[C_T]$ for the call of example 15.7 with $\mu = 12\%$ and explain why $e^{-rT} \mathbb{E}_{\mathbb{P}}[C_T]$ is not the price.

15.10 Put–call parity: a model-free replication

The lecture illustrates replication with a relation that requires *no* assumption on the stock dynamics ▶ 2024 L21 @ 1:10:22 | ▶ 2013 L19 @ 38:07].

Theorem 15.8 (Put–call parity). For European options on a non-dividend-paying asset with the same strike K and expiry T , in the absence of arbitrage,

$$C(t) - P(t) = S_t - Ke^{-r(T-t)}. \quad (15.21)$$

If (15.21) failed, say $C - P > S - Ke^{-r\tau}$, one would sell the call, buy the put and the stock, and borrow $Ke^{-r\tau}$, locking in the difference at no risk. The remark of the lecture is that the relation is *independent of the dynamics*: one can price a put from the observed calls without a model, which matters in the real market where the stock is *not* lognormal [► 2024 L21 @ 1:12:31][► 2013 L19 @ 38:07]. It is also the reason why the implied volatility of a call and of a put with the same strike and expiry must coincide.

Proof. The payoffs at T satisfy $(S_T - K)^+ - (K - S_T)^+ = S_T - K$ (the “hockey stick” and its reflection add up to a straight line [► 2024 L21 @ 1:11:25]). So buying a call and selling a put replicates a forward with strike K *statically*, i.e. with no trading before T : it is a forward contract, and by [proposition 15.1](#) (applied over $[t, T]$) it is worth $S_t - Ke^{-r(T-t)}$ (stock, minus the bond that pays K). The two portfolios have the same payoff in every state, so they have the same price. [► 2024 L21 @ 1:12:31] □

15.10.1 Parity on Bloomberg screens

The slides check the relation on real quotes (mid = average of bid and ask). The tables below use numbers read from the screens for these notes; a reading error of a cent is possible.

K	call mid	put mid	$C - P$	$S - Ke^{-r\tau}$	difference
75	6.900	0.300	6.600	6.555	+0.045
80	2.950	1.300	1.650	1.583	+0.067
85	0.750	4.150	-3.400	-3.390	-0.010
90	0.125	8.800	-8.675	-8.362	-0.313

Table 15.2: IBM options expiring 22 April 2006, screen of 8 March 2006 (2013 slide 23), $S = 81.14$; the discount uses $r = 4.5\%$ and $\tau = 45/365$ (our assumptions: the slide does not show a rate, so this is a rough choice, and dividends are ignored). Equity options are American, and the deep in-the-money put ($K = 90$, wide spread) carries an early-exercise premium that parity does not include.

K	call mid	put mid	$C - P$	$S - PV(D) - Ke^{-r\tau}$	difference
90	23.63	0.26	23.37	23.14	+0.23
100	14.00	0.77	13.23	13.15	+0.08
110	6.10	2.83	3.27	3.17	+0.10
120	1.62	8.35	-6.73	-6.81	+0.08

Table 15.3: AAPL options expiring 20 January 2017 ($\tau = 81$ days), screen of 31 October 2016 (2024 slide 24), $S = 113.54$, $r = 0.83\%$ and the dividend of \$0.57 shown on the screen (its present value is subtracted from S : parity for a dividend-paying stock is $C - P = S - PV(D) - Ke^{-r\tau}$, “not discussed” in class).

On slide 24 the lecturer plots the call, the put and their difference against strike: the difference is a straight line of slope $-e^{-r\tau} \approx -1$, while the implied volatility of the same options is a U-shaped curve (see [section 15.13](#)). A straight line is what parity requires, and “if it wouldn’t be, people would jump on it immediately” [► 2024 L21 @ 1:13:36]. In 2013 the same idea is shown on the IBM screen of 2006 (slide 23), at the end of the lecture: far from the strike a call is worth $S - K$ ($S - K = 81.14 - 55 = 26.14$, against a quoted mid of 26.50),

and between strikes the prices change by 5 when the strike changes by 5 [► 2013 L19 @ 36:03].

IN FINANCE: Replication as a way of thinking

The lecturer's closing message [► 2024 L21 @ 1:14:40][► 2013 L19 @ 48:38]: "what you try to do is look at the payout of a portfolio and replicate it with the instruments at hand, or at least approximate it, because if you did, you know the price now, or you know an arbitrage". Conversely one builds a complicated portfolio of listed options and looks for a mispricing with respect to a replicating one: nowadays this is "practically impossible" for vanilla options, but was done "25 years ago" with more complex options in FX (a story he leaves to J. Xia). The hedge of a book of exotics is usually dynamic and needs rebalancing [► 2013 L19 @ 49:42]; static replications like parity and the call spread below are the exceptions, and the most robust.

Exercise 15.7 (not from the course: put-call parity). Prove parity when the stock pays a known cash dividend D at $t_D \in (t, T)$: $C - P = S - De^{-r(t_D-t)} - Ke^{-r\tau}$. Using the numbers of table 15.2 choose an interest rate r that best fits the four strikes by least squares, and comment on the residuals. Design an arbitrage (list the trades and the guaranteed profit) if the call in the $K = 100$ row of table 15.3 were quoted at 14.50 instead of 14.00.

15.11 Digital options and call spreads

The lecture ends with a challenge [► 2024 L21 @ 1:15:43], which in 2013 was posed as an interview question [► 2013 L19 @ 43:24]: *price and hedge a digital (all-or-nothing) option*, which pays 1 if $S_T > K$ and 0 otherwise, given that the prices of calls of *all strikes* are known. Slide 26 of 2024 compares the Black–Scholes digital with a "call spread" built from the same AAPL data.

Replication. A call spread that is long a call with strike $K - \varepsilon$ and short a call with strike $K + \varepsilon$ pays $(S_T - K + \varepsilon)^+ - (S_T - K - \varepsilon)^+$, a ramp from 0 (for $S_T \leq K - \varepsilon$) to 2ε (for $S_T \geq K + \varepsilon$). A student proposes it and the lecturer confirms: $\pm \frac{1}{2}$ around K gives a ramp up to 1, $\pm \frac{1}{4}$ needs two spreads, and in general the payoff of $\frac{1}{2\varepsilon}$ spreads is a ramp of unit height and width 2ε [► 2013 L19 @ 44:31]. Letting $\varepsilon \rightarrow 0$, the digital price is

$$D(K) = \lim_{\varepsilon \rightarrow 0} \frac{C(K - \varepsilon) - C(K + \varepsilon)}{2\varepsilon} = -\frac{\partial C}{\partial K} \quad (15.22)$$

whatever the model [► 2013 L19 @ 46:37], since $C(K) = e^{-r\tau} \mathbb{E}_{\mathbb{Q}}[(S_T - K)^+]$ and $\partial_K(S_T - K)^+ = -\mathbf{1}\{S_T > K\}$; this is the Breeden–Litzenberger relation of corollary 3.10 ($D = e^{-r\tau} \mathbb{Q}(S_T > K)$), while $\partial_K^2 C = e^{-r\tau} p_{\mathbb{Q}}(K)$). The lecturer notes that the answer is a derivative of the call price with respect to K , and that it should not be surprising: the derivative of a ramp is a step [► 2013 L19 @ 46:37]; the same idea works for any payoff, e.g. a box. In practice the dealer sells the digital to a client and hedges with two liquid calls on the exchange, and digitals are less liquid than calls [► 2013 L19 @ 47:37].

In Black–Scholes the digital costs

$$D = e^{-r\tau} \Phi(d_2), \quad (15.23)$$

which for $S = K = 80$, $\tau = 2$, $r = 5\%$, $\sigma = 20\%$ is 0.5284; the call spread with half-width $\varepsilon = 10, 5, 2, 1, 0.1$ costs 0.5291, 0.5286, 0.52846, 0.52843, 0.528423 (computed for these

notes). For large S , $D \rightarrow e^{-r\tau} = 0.905$, the value of a certain \$1 at T , as in the left graph of slide 26 [► 2024 L21 @ 1:16:44].

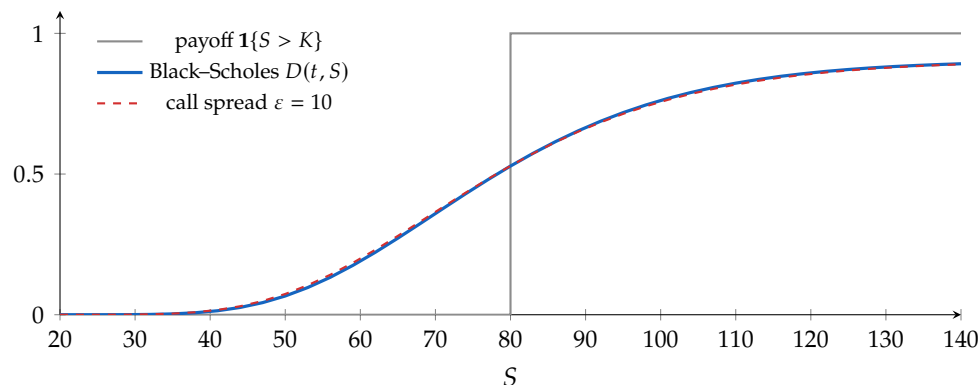


Figure 15.3: The digital option ($K = 80$, $\tau = 2$, $r = 5\%$, $\sigma = 20\%$) and the call spread with strikes 70 and 90. The spread is already close to the digital price; for $\varepsilon \rightarrow 0$ it converges to it.

With market data. The right graph of slide 26 is the call spread built from the AAPL quotes (slide 24) as a function of $K - S$. From the numbers of table 15.3, $(C(110) - C(120))/10 = 0.448$ and $(C(105) - C(125))/20 = 0.451$ estimate the digital with $K = 115$ (probability of finishing in the money, discounted). A flat-volatility Black-Scholes digital with the implied volatility of the $K = 115$ call (19.8%) would give only 0.413: the difference is caused by the *skew* of the smile ($\partial\sigma/\partial K < 0$ there) and is a first-principles example of “the world is not Black-Scholes” (section 15.13). The calculation is ours (not in class).

INTUITION: Static versus dynamic replication

Any smooth payoff $g(S_T)$ is a continuous portfolio of calls, $g(S_T) = g(0) + g'(0)S_T + \int_0^\infty g''(K)(S_T - K)^+ dK$ (a Taylor formula with integral remainder in K ; not discussed in class), so its price is a fixed combination of call prices with no model at all. The Black-Scholes dynamic hedge is what one needs when the instruments are *not* available at all strikes.

Exercise 15.8 (not from the course: digital options).

- Prove $D = -\partial C / \partial K$ for any model where $C(K) = e^{-r\tau} \mathbb{E}_Q[(S_T - K)^+]$.
- Show that a call spread with strikes K and $K + \varepsilon$, scaled by $1/\varepsilon$, *super-replicates* the digital, so that the dealer’s ask for the digital is $\leq [C(K) - C(K + \varepsilon)]/\varepsilon$, and find the sub-replicating spread giving a bid.
- In a model with a smile, $\sigma = \sigma(K)$, show that $D = e^{-r\tau} \Phi(d_2) - \mathcal{V} \sigma'(K)$ and explain the sign of the correction for the AAPL data of table 15.5.

15.12 Greeks and hedging

The replication argument produced one hedge ratio, $a = \partial f / \partial S$ (15.9). Traders call the derivatives of the price with respect to its inputs the *Greeks*; they measure the risks of a position (“the mathematical apparatus allows one to compute the current price of a derivative and its risks”, slide 7). The lecture only uses the delta; the rest of this section is

not discussed in class but follows from [theorem 15.6](#) and completes the picture.

Proposition 15.9 (Greeks of the European call). With $\varphi = \Phi'$ the standard normal density and $\tau = T - t$,

$$\begin{aligned}\Delta &= \frac{\partial C}{\partial S} = \Phi(d_1), & \Gamma &= \frac{\partial^2 C}{\partial S^2} = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}, & \mathcal{V} &= \frac{\partial C}{\partial \sigma} = S\varphi(d_1)\sqrt{\tau}, \\ \Theta &= \frac{\partial C}{\partial t} = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2), & \rho &= \frac{\partial C}{\partial r} = K\tau e^{-r\tau}\Phi(d_2).\end{aligned}\quad (15.24)$$

For the put, $\Delta_P = \Delta - 1$ and $\Gamma, \mathcal{V}$ are the same; $\Theta_P = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}\Phi(-d_2)$ and $\rho_P = -K\tau e^{-r\tau}\Phi(-d_2)$. The price satisfies (15.10) in the form $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta = rC$.

Numbers for $S = K = 100$, $\tau = 1$, $r = 5\%$, $\sigma = 20\%$ ($d_1 = 0.35$, $d_2 = 0.15$; call 10.4506, put 5.5735; checked against finite differences and against the PDE, residual 10^{-16} , for these notes):

	Δ	Γ	$\mathcal{V}$ (per unit σ)	Θ (per year)	ρ (per unit r)
call	0.6368	0.01876	37.52	-6.414	53.23
put	-0.3632	0.01876	37.52	-1.658	-41.89

A one-point (1%) rise in volatility adds about 0.375 to the price, and one day of time decay of the call costs about $6.41/365 = 0.018$.

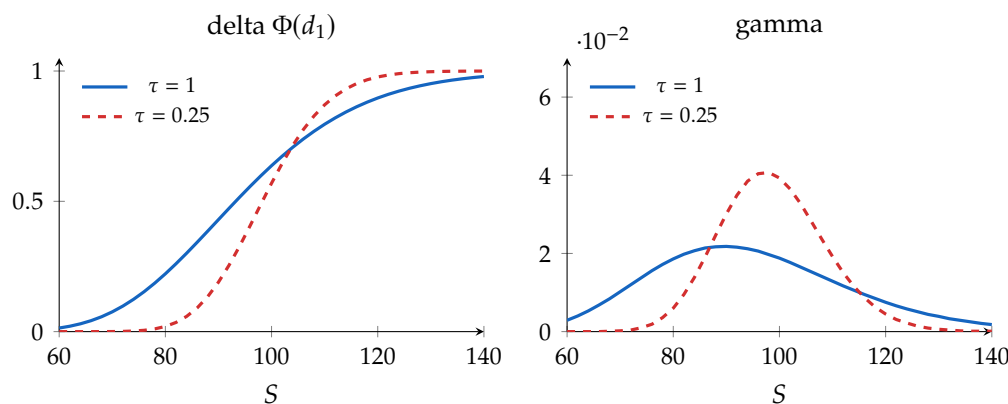


Figure 15.4: Delta and gamma of a call with $K = 100$, $r = 5\%$, $\sigma = 20\%$. Near expiry delta approaches the step function of the payoff and gamma concentrates around the strike, which makes short-dated at-the-money options the hardest to hedge.

Delta hedging. To be short one call and hedged one holds $\Delta = \Phi(d_1)$ shares (about 64 shares per 100 calls in the example above) and a negative cash position; the combination has no first-order sensitivity to S . The residual risk is second order: over a small move δS the hedged book changes by $\Theta \delta t + \frac{1}{2}\Gamma \delta S^2$, and the PDE $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r(C - S\Delta)$ says that, on average, the gain from gamma ($\frac{1}{2}\Gamma \delta S^2 \approx \frac{1}{2}\Gamma \sigma^2 S^2 \delta t$) pays for the time decay: a long-gamma position bleeds theta and earns it back through movement, exactly at the rate that makes the hedged book earn r . This is the origin of the trader's statement that an option trade is a bet on realised versus implied volatility ([chapter 13](#)).

Proof. First the identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$: since $d_2 = d_1 - \sigma\sqrt{\tau}$, $\varphi(d_2) = \varphi(d_1)\exp(\sigma\sqrt{\tau}d_1 - \frac{1}{2}\sigma^2\tau)$ and $\sigma\sqrt{\tau}d_1 = \ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau$, so $\varphi(d_2) = \varphi(d_1)(S/K)e^{r\tau}$. Now $\partial_S d_1 = \partial_S d_2 = 1/(S\sigma\sqrt{\tau})$ and $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)]\partial_S d_1 = \Phi(d_1)$, the bracket vanishing by the identity. Differentiating once more gives Γ . For $\mathcal{V}$: $\partial_\sigma C = S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$, by the identity and $d_1 - d_2 = \sigma\sqrt{\tau}$. The theta and rho are computed in the same way (or theta from the PDE, once Δ, Γ are known); the put values follow from parity $P = C - S + Ke^{-r\tau}$. $\square$

rebalancings n	mean error	std. dev.	std. dev. $\times \sqrt{n}$
1	-0.208	6.109	6.11
2	-0.088	4.379	6.19
4	-0.092	3.225	6.45
12	-0.040	1.902	6.59
52	-0.014	0.927	6.69
252	-0.005	0.426	6.76

Table 15.4: Error of discrete delta hedging of a short call ($S_0 = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$, price 10.4506 received) in a market whose real-world drift is $\mu = 10\% \neq r$: terminal value of the hedge portfolio minus the payoff, $2 \cdot 10^4$ simulated paths, hedge ratio $\Phi(d_1)$ recomputed at n equally spaced dates. Computed for these notes.

What discrete rebalancing costs. The derivation (15.8) assumed that the hedge is adjusted *continuously*. Table 15.4 shows what happens with n rebalancings: the hedge error has a small mean (below 0.21 here, and shrinking with n) and standard deviation proportional to $n^{-1/2}$ (the last column is roughly constant), i.e. proportional to $\sqrt{dt}$: the risk is not eliminated for a finite n , only made small, and every rebalancing has a transaction cost, so the desk chooses a finite frequency [► 2013 L19 @ 28:31] (“every time dt we will have to rebalance”).

IN FINANCE: What breaks the assumptions

Discrete hedging and transaction costs (above); jumps and gaps, for which no hedge in the stock alone is perfect and the market is incomplete (section 2.4: a non-replicable claim has an interval of arbitrage-free prices) [► 2024 L21 @ 1:14:40]; stochastic volatility, so that σ itself needs hedging with other options; funding and credit spreads (chapter 8); dividends and borrow costs. Each is a chapter of its own in a derivatives library; Black–Scholes remains the language in which prices are *quoted*.

Exercise 15.9 (not from the course: Greeks). Prove $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, derive $\Delta, \Gamma, \mathcal{V}, \Theta$ and check that they satisfy the Black–Scholes PDE. For an at-the-money-forward option show that $C \approx 0.4 S \sigma \sqrt{\tau}$; prove in general that $\mathcal{V} = S^2 \sigma \tau \Gamma$. What is the Δ of a call with S very large? Of a call very close to expiry, at the money?

Exercise 15.10 (not from the course: delta hedging by simulation). Reproduce table 15.4: simulate GBM with drift μ , sell a call at the Black–Scholes price, rebalance the delta at n dates, and record the terminal error. Check that the distribution does not depend on μ (for large n), that its width scales as $n^{-1/2}$, and add a proportional transaction cost $c|\Delta a|S$ to find the n that minimises the cost-plus-risk for your favourite risk penalty. What changes if you hedge with a wrong σ (say 15% when the true one is 20%)?

15.13 Implied volatility and the smile

The formula has one input that is not observed, σ . Since $\partial C / \partial \sigma = \mathcal{V} > 0$ and $C \rightarrow (S - Ke^{-r\tau})^+$ as $\sigma \rightarrow 0$, $C \rightarrow S$ as $\sigma \rightarrow \infty$, the map $\sigma \mapsto C_{BS}(\sigma)$ is a strictly increasing bijection from $(0, \infty)$ onto $((S - Ke^{-r\tau})^+, S)$; hence every market price in that interval

determines a unique *implied volatility* σ_{imp} (found numerically, e.g. by Newton’s method with the vega as derivative). Parity implies that call and put of the same strike and expiry have the same σ_{imp} .

In Black–Scholes σ is one constant for the stock, so the implied volatility should be the *same for all strikes and expiries*. The lecture points out that it is not [► 2024 L21 @ 1:13:36] [► 2013 L19 @ 37:04]: plotted against strike it is a curve, a “smile” (in 2013 the word is “skewed, or rather smiled”), shown in the lower right graph of slide 24. Table 15.5 and fig. 15.5 report the numbers of that screen.

K	call mid	screen implied vol. (%)	recomputed (%)
90	23.63	32.53	33.12
100	14.00	23.71	24.13
110	6.10	20.47	20.69
120	1.62	19.06	19.08
130	0.31	19.41	19.37
140	0.09	21.45	21.79

Table 15.5: AAPL calls of table 15.3 ($S = 113.54$, $\tau = 81$ days). “Recomputed” inverts the European Black–Scholes formula with $S - PV(D)$, $r = 0.83\%$ (our computation); the small differences from the Bloomberg values reflect the American-style and dividend conventions of the screen, and are largest for deep in-the-money and far out-of-the-money quotes.

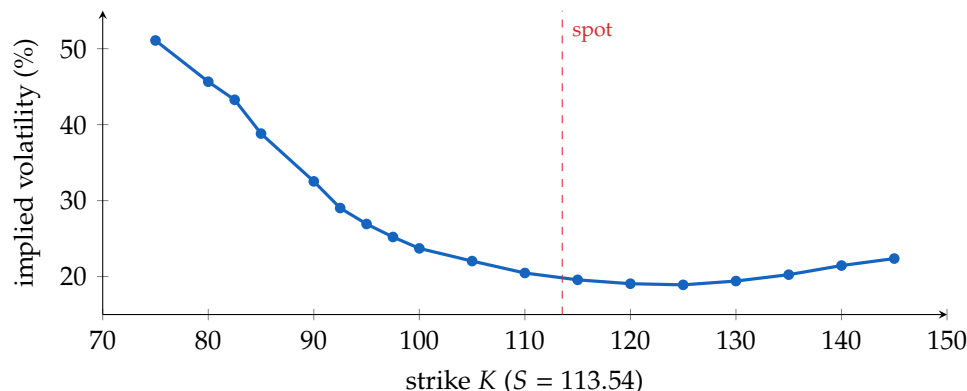


Figure 15.5: Implied volatility of the AAPL calls, screen of 31 October 2016 (values read from 2024 slide 24; the quote at $K = 87.5$ is missing). A constant-volatility model would give a horizontal line.

The curve decreases steeply for low strikes and turns up slightly for high ones: options in the wings are relatively *expensive*. Two consequences. First, the lognormal distribution is not the distribution the market uses: returns have heavier tails and skew (section 3.9.1), volatility moves with the price, and the wings price in crash risk. Second, prices of *non-vanilla* claims cannot be read off one volatility number: as in the digital of section 15.11, the skew enters. The lecturer’s conclusion [► 2024 L21 @ 1:14:40] is that Black–Scholes is a “nice example” of the method, but more complicated dynamics are needed for stocks: *stochastic volatility* (the volatility itself follows an SDE; chapters 13 and 18) or *jump-diffusion*. With jumps “life becomes more complicated”: the market is incomplete and martingale

techniques for continuous paths no longer suffice. The risk-neutral *method* is unchanged: find the measure(s) $\mathbb{Q}$ under which discounted traded assets are martingales, and price by discounted expectation; what changes is that (for stochastic volatility and jumps) $\mathbb{Q}$ is no longer unique ([theorem 2.9](#), part 2), and calibration to the observed option prices (the smile) is what selects it.

Exercise 15.11 (not from the course: implied volatility). Implement Newton's method for σ_{imp} with the vega as derivative, starting from $\sigma_0 = \sqrt{2|\ln(S/K) + r\tau|/\tau}$. Recompute the last column of [table 15.5](#). Why does the iteration fail for deep out-of-the-money options and what is a safer alternative (bisection, Brent)?

15.14 The two lectures compared, and errata

KEY FORMULAS

One step and tree (continuous compounding, up/down factors u, d):

$$q = \frac{e^{r\,dt} - d}{u - d}, \quad qu + (1 - q)d = e^{r\,dt}, \quad f_0 = e^{-r\,dt} [qf_u + (1 - q)f_d], \quad a = \frac{f_u - f_d}{S_u - S_d},$$

$$\text{CRR: } u = e^{\sigma\sqrt{dt}} = 1/d, \quad q = \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{dt} + O(dt).$$

Model and PDE:

$$dS = \mu S\,dt + \sigma S\,dW, \quad df = (f_t + \mu S f_S + \frac{1}{2}\sigma^2 S^2 f_{SS})\,dt + \sigma S f_S\,dW,$$

$$f_t + \frac{1}{2}\sigma^2 S^2 f_{SS} + rS f_S - rf = 0, \quad f(S, T) = \text{payoff}, \quad a = f_S.$$

Risk-neutral valuation:

$$f_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}}[f_T], \quad dS = rS\,dt + \sigma S\,dW^{\mathbb{Q}}, \quad W_t^{\mathbb{Q}} = W_t + \frac{\mu - r}{\sigma}t.$$

Black–Scholes formula ($\tau = T - t$):

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1),$$

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad C - P = S - Ke^{-r\tau}.$$

Greeks:

$$\Delta_C = \Phi(d_1), \quad \Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}, \quad \mathcal{V} = S\varphi(d_1)\sqrt{\tau}, \quad S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2),$$

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta = rC.$$

Digital and call spread: $D = e^{-r\tau}\Phi(d_2) = -\partial C/\partial K = \lim_{\varepsilon \rightarrow 0} \frac{C(K - \varepsilon) - C(K + \varepsilon)}{2\varepsilon}.$

6 COURSE NOTE: 2013 versus 2024

The core slides are the same: the 2024 deck is the 2013 deck (V. Strela was the speaker in both) with five changes.

- *Opening.* 2024 starts with the update of the yield curve ([section 15.1](#)); 2013 goes straight to the bookie.
- *More derivation.* The 2024 lecture spends about 15 minutes on why $\text{sd}(dW) = \sqrt{dt}$ and on the Taylor expansion behind Itô's formula and on a student's objection to $(dW)^2 = dt$ ([section 15.6](#)); 2013 assumes the students know Itô from the lecture that precedes it ("did you talk about Itô already? OK, cool").
- *Discrete market.* 2013 keeps the number b of bond units and B_0 separate (its slide 10 has $b = \frac{S_1 f_2 - S_2 f_1}{(S_1 - S_2)B_0 e^{r\,dt}}$), 2024 uses the cash amount B directly; the risk-neutral probability is called q in 2013 and p_{rn} in 2024.
- *Examples.* 2024 replaces the March 2006 IBM screen (2013 slide 23) with two recent Bloomberg screens (AAPL 2016 with the implied-volatility smile, and IBM 2015), adds the put–call parity plot and the real-data call spread (slide 26); 2013 discusses parity and the smile only orally, at the end, and treats the digital as an interview question with the audience answering.
- *Suggestions.* 2013 explicitly proposes as final-paper topics the reduction to the heat equation and the derivation of the formula as a lognormal expectation [[2013 L19 @ 30:40](#)], both carried out above; 2024 announces the next two (guest) lectures, one of them by J. Hull on machine learning ([chapter 16](#)).

In 2013 the lecturer also remarks that the two-year rate was then about 32.5 basis points, so that the 5% of the example is illustrative and not realistic [[2013 L19 @ 17:43](#)].

7 COURSE NOTE: Errata found in the material

- *Density of S_T* (2024 slide 21, 2013 slide 20): the pre-factor should be $1/(S_T \sigma \sqrt{2\pi(T-t)})$; see [section 15.8](#).
- *Boundary condition of the call* (slide 19, 2013 slide 18): $C(S, t) \rightarrow S - Ke^{-r(T-t)}$ as $S \rightarrow \infty$, not only $\cong S$; see [section 15.7](#).

- *Spoken slips* (2013 transcript): the bookie's payout "\$12.25" is \$12 500 ([section 15.2](#)). (2024 transcript): "our strike has to be 0" after the argument for zero interest rates means $K = S_0$ ($K = S_0 e^{r \, dt}$ in general) [[▶ 2024 L21 @ 22:20](#)]; " S_2 less than 0" in the call diagram is $S_2 < K$ [[▶ 2024 L21 @ 26:32](#)].

Chapter 16

Introduction to Machine Learning

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L23	J. Hull (Rotman School, Toronto; guest)	What machine learning is; unsupervised, supervised and reinforcement learning; training/validation/test sets and overfitting; neural networks, activation functions, gradient descent, backpropagation; two applications (learning the Black-Scholes-Merton price, explaining the movements of the implied-volatility surface); reinforcement learning, exploration versus exploitation, the game of Nim, Q-learning, deep Q-learning; hedging derivatives by reinforcement learning; questions from the class.
Files. Slides <code>mit18_642_f24_lec23.pdf</code> (40 slides, no overlays); transcript <code>Lecture 23 - Introduction to Machine Learning.txt</code> (about 75 minutes). Problem sets. None; all exercises are marked “(not from the course)”. Not covered in 2013. The 2013 course had no machine-learning lecture. Not in the lecture. The bias-variance decomposition, the convergence analysis of gradient descent, the derivation of backpropagation, <i>k</i>-means and the Bellman equation are <i>not</i> derived in the lecture, which is a survey by an applied researcher. They are derived here so that the slides can be read as mathematics; each such passage is marked (<i>not discussed in class</i>). The numerical experiments marked “our reconstruction” were run for these notes and are <i>not</i> Hull’s results.		

Motivation

Every chapter so far fitted a model whose functional form was decided in advance: a linear factor model (chapters 5 and 7), an ARMA process (chapter 9), a Gaussian return distribution (chapter 3). Machine learning starts from the opposite end. One has a large data set and a flexible family of functions with many parameters, and asks the computer to find the function that predicts well *on data it has not seen*. The mathematics is that of the earlier chapters, used with a different emphasis:

- least squares becomes one loss function among many, minimised by *gradient descent* instead of by a closed formula (section 5.1);
- overfitting, which in section 5.10 was tamed by a ridge penalty chosen by cross-validation, becomes the central practical issue, handled by a train/validation/test split and by stopping the optimiser early;

- the flexible family is the *neural network*, a composition of linear maps and fixed nonlinearities whose gradient is computed by the chain rule (“backpropagation”);
- decisions taken in sequence, such as hedging an option over its life, lead to *reinforcement learning*, dynamic programming without a model.

John Hull, known for his derivatives textbook, gave this lecture in 2024 and uses his own examples: a network that imitates the Black–Scholes–Merton formula, a network that predicts how implied volatility moves when the stock moves, and reinforcement-learning hedging. [Section 16.1](#) gives the vocabulary; [section 16.2](#) develops the theory of generalisation; [section 16.3](#) treats clustering; [sections 16.4](#) and [16.5](#) develop gradient descent and neural networks; [sections 16.6](#) and [16.7](#) are the two finance applications; and [section 16.8](#) covers reinforcement learning. The regularised models of [chapter 20](#) are the classical counterpart of what is said here about overfitting.

16.1 What machine learning is

16.1.1 Vocabulary, and three kinds of learning

Hull introduces machine learning (ML) as a branch of artificial intelligence: give a program a lot of data and let it find relationships between variables and make predictions [\[► 2024 L23 @ 7:17\]](#). Some techniques date to the 1950s; what changed is computer speed and the falling cost of storing data. The terminology differs from statistics, and the lecture insists that this is a real obstacle to newcomers [\[► 2024 L23 @ 11:39\]](#)–[\[► 2024 L23 @ 12:44\]](#).

Statistics	Machine learning	In these notes
independent variables, regressors	<i>features</i>	$x \in \mathbb{R}^d$
dependent variable, response	<i>target</i> (or <i>label</i> for a class)	y
coefficients β	<i>weights</i> (and <i>biases</i>)	θ
fitting, estimation	<i>training</i>	minimise the loss
sample, one pass over the data	<i>epoch</i> (one pass of the optimiser)	
nonlinear link / basis function	<i>activation function</i>	φ
step size of a numerical optimiser	<i>learning rate</i>	η
tuning parameter (e.g. ridge λ)	<i>hyperparameter</i>	$\lambda, \alpha, \varepsilon$

Table 16.1: A small dictionary. Hull’s book has a ten-page glossary; these are the terms used in the lecture.

The lecture classifies algorithms in three families [\[► 2024 L23 @ 11:39\]](#).

Unsupervised learning looks for structure in features alone, e.g. divides customers into clusters [\[► 2024 L23 @ 14:53\]](#). “No supervision” means that nobody tells the algorithm what to predict [\[► 2024 L23 @ 13:47\]](#). See [section 16.3](#).

Supervised learning predicts a numerical target (*regression*) or a class (*classification*) from features. Linear regression ([chapter 5](#)) is the classical example. See [sections 16.2](#) and [16.5](#).

Reinforcement learning learns a rule for a *sequence* of decisions in an environment that changes unpredictably. See [section 16.8](#).

16.1.2 Digitisation versus learning; statistics versus ML

Hull’s periodisation of industrial revolutions (steam, then electricity and mass production, then computers and digitisation, roughly 1950–2000, and now the fourth, AI) is used to separate two things [► 2024 L23 @ 7:17]. If loan officers apply known rules, a computer can *automate* them, which is digitisation. If the rules are unknown, ML can *infer* them from past applications and decisions; and one can go a step further and improve the rules by using data on how the loans turned out [► 2024 L23 @ 8:24]–[► 2024 L23 @ 9:24].

The second contrast concerns workflow [► 2024 L23 @ 9:24]–[► 2024 L23 @ 10:29]. Traditional statistics: formulate a hypothesis, collect data, test. Machine learning: collect data, try different models, find patterns or build a predictive tool.

INTUITION: The physicist’s reading

A physicist knows both modes: one fits a theory-driven model (Hooke’s law) and one also explores a data set for unexpected structure. What makes the second mode legitimate is precisely what the lecture goes on to stress, namely that the model is judged on data that played no role in selecting it (the validation and test sets of [section 16.2](#)). Searching a data set for patterns without such a safeguard is what physicists call the look-elsewhere effect: with enough candidate models, one always fits. ML’s answer is procedural (sample splitting), where classical hypothesis testing corrects the significance level.

16.2 Supervised learning: generalisation and the bias–variance trade-off

16.2.1 Loss, training set, validation set, test set

Definition 16.1 (Supervised learning problem). Given pairs (x_i, y_i) , $i = 1, \dots, n$ (features and target), choose a function h_θ in a parametrised family $\mathcal{H}$ so as to make the *empirical risk*

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, h_\theta(x_i))$$

small, where ℓ is a loss function. The goal is a small *expected* loss on new pairs, not on the n given ones.

The choice $\ell(y, \hat{y}) = (y - \hat{y})^2$ with a linear $h_\theta(x) = \beta^\top x$ is ordinary least squares ([section 5.1](#)); other losses (absolute, check loss) are the M-estimators of [section 5.6](#). Hull’s lecture uses squared error (“mse”) and, in one plot, absolute error (see the erratum in [section 16.6](#)).

Because there is no hypothesis, the data are divided into three sets [► 2024 L23 @ 15:58]–[► 2024 L23 @ 17:03], typically about 60%, 20% and 20%.

- the *training set* determines the parameters θ of each candidate model;
- the *validation set* compares candidate models (and picks hyperparameters such as the degree of a polynomial or the number of layers);
- the *test set* is kept aside and used once, to report the out-of-sample performance of the model finally chosen [► 2024 L23 @ 18:06].

The rule of thumb is to increase the complexity of the model until performance on the validation set starts to deteriorate [► 2024 L23 @ 18:06]. The three-way split is the simple, data-rich analogue of the cross-validation used in [section 5.10](#) to choose the ridge parameter, and the validation error plays the role of the criterion that [chapter 20](#) will minimise in a more structured setting.

Hull's salary example [► 2024 L23 @ 19:09]–[► 2024 L23 @ 21:21]. Salaries in a profession are regressed on age with polynomials (slide 11, whose raw data are not published). A fifth-degree polynomial follows the training points closely but does poorly on the validation points (*overfitting*); a straight line misses the rise-and-fall pattern (*underfitting*); a quadratic, in which salary rises and then tails off with age, is the best model. [Figure 16.1](#) reproduces the mechanism, not the data.

16.2.2 The bias–variance decomposition (not discussed in class)

Overfitting and underfitting are two faces of one identity. Fix a point x and write $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var } \varepsilon = \sigma^2$, ε independent of the training sample; $\hat{h}$ is the fitted function, a random function through the training sample. Expanding the square around $\mathbb{E}[\hat{h}(x)]$ (the cross terms vanish by independence and by $\mathbb{E}[\hat{h} - \mathbb{E}[\hat{h}]] = 0$),

$$\mathbb{E}[(y - \hat{h}(x))^2] = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{(f(x) - \mathbb{E}[\hat{h}(x)])^2}_{\text{bias}^2} + \underbrace{\text{Var } \hat{h}(x)}_{\text{variance}}. \quad (16.1)$$

For a concrete family the last two terms can be computed exactly. Take the linear-in-parameters case of [section 5.1](#): a fixed design, $y = f + \varepsilon \in \mathbb{R}^n$, $\text{Cov } \varepsilon = \sigma^2 I$, and the fit $\hat{y} = Hy$ with H the hat matrix (a rank- p orthogonal projection, [proposition 5.4](#)); p is the number of fitted parameters (for a polynomial of degree d , $p = d + 1$).

Proposition 16.2 (Training and test error of a projection fit). Let $b^2 = \frac{1}{n} \|(I - H)f\|^2$ be the squared bias averaged over the design points. Then

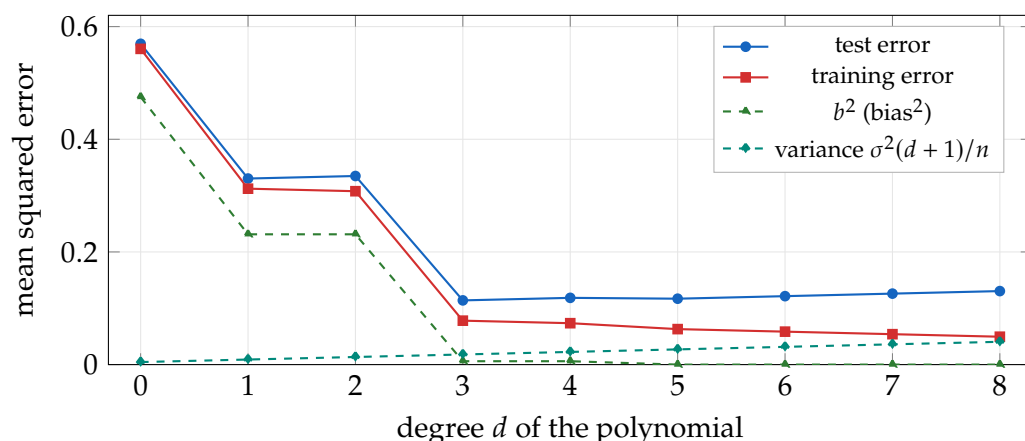
$$\underbrace{\frac{1}{n} \mathbb{E}[\|y - \hat{y}\|^2]}_{\text{training error}} = \sigma^2 + b^2 - \frac{p}{n} \sigma^2, \quad \underbrace{\frac{1}{n} \mathbb{E}[\|y^* - \hat{y}\|^2]}_{\text{test error at the same } x_i, \text{ fresh noise}} = \sigma^2 + b^2 + \frac{p}{n} \sigma^2,$$

and the variance term of (16.1), averaged over the design points, is $\frac{p}{n} \sigma^2$.

The gap test – training = $2\sigma^2 p/n$ is the *optimism* of the training error (it is the correction in Mallows' C_p). More parameters lower the bias and raise the variance linearly in p ; the training error decreases with p *always*, so it cannot choose the model. This is why the lecture selects on a validation set.

Proof. Since $I - H$ is a projection of trace $n - p$: $y - \hat{y} = (I - H)f + (I - H)\varepsilon$, and the cross term has zero mean, so $\mathbb{E}[\|y - \hat{y}\|^2] = \|(I - H)f\|^2 + \sigma^2 \text{tr}(I - H) = nb^2 + \sigma^2(n - p)$. For the fit error, $\hat{y} - f = -(I - H)f + H\varepsilon$ gives $\mathbb{E}[\|\hat{y} - f\|^2] = nb^2 + \sigma^2 p$; the variance is $\mathbb{E}[\|H\varepsilon\|^2] = \sigma^2 \text{tr } H = p\sigma^2$. Finally $y^* - \hat{y} = \varepsilon^* + (f - \hat{y})$ with ε^* independent of $\hat{y}$, so the test error is $\sigma^2 + \frac{1}{n} \mathbb{E}[\|\hat{y} - f\|^2]$. $\square$

Example 16.3 (Polynomials of degree d fitted to $\sin 2\pi x$). Take $n = 20$ equispaced points $x_i \in [0, 1]$, $f(x) = \sin(2\pi x)$, $\sigma = 0.3$ (our reconstruction; Hull's salary data are not available). The bias b^2 is the residual of the projection of f on the polynomials, computed numerically (Legendre basis, python); the variance is exactly $\sigma^2(d + 1)/n$. Test error and training error follow from [proposition 16.2](#): the test error is minimal at $d = 3$ (value 0.114, of which $\sigma^2 = 0.09$ is noise), and it then rises slowly with d , while the training



error keeps falling (0.078 at $d = 3$, 0.050 at $d = 8$).

Figure 16.1: Expected training and test errors for polynomial regression of degree d ($n = 20$, $\sigma^2 = 0.09$; our reconstruction of the mechanism of slide 11). The test error is $\sigma^2 + b^2 + \sigma^2(d+1)/n$; the training error is $\sigma^2 + b^2 - \sigma^2(d+1)/n$. The minimum of the test error (degree 3) is the model the validation set would select; the training error alone would choose the largest degree.

IN FINANCE: Overfitting is the everyday enemy of backtests

Financial data are short and noisy (a few thousand daily returns, a low signal-to-noise ratio). A model with many free parameters will always fit the past; the validation and test sets must respect time order, otherwise information from the future leaks into the fit (chapter 9). The lecture's advice to keep a test set untouched until the end is, for a trading strategy, the discipline of an out-of-sample period used once.

Remark 16.4 (Forward reference). The classical route to a controlled bias–variance trade-off in a regression is not to shrink the number of parameters but the size of the coefficients: ridge regression, whose effective number of parameters is $\sum_j d_j^2 / (d_j^2 + \lambda) < p$ (section 5.10). Chapter 20 uses the same idea to calibrate pricing and risk models.

Exercise 16.1 (not from the course: optimism of any linear smoother). Let $\hat{\mathbf{y}} = S\mathbf{y}$ for a fixed matrix S (ridge regression, section 5.10, is an example with $S = X(X^\top X + \lambda I)^{-1}X^\top$), $\mathbf{y} = \mathbf{f} + \boldsymbol{\varepsilon}$, $\text{Cov } \boldsymbol{\varepsilon} = \sigma^2 I$. Show that $\frac{1}{n}\mathbb{E}[\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2] - \frac{1}{n}\mathbb{E}[\|\mathbf{y} - \hat{\mathbf{y}}\|^2] = \frac{2\sigma^2}{n} \text{tr } S$, where $\mathbf{y}^* = \mathbf{f} + \boldsymbol{\varepsilon}^*$ is an independent copy, so that $\text{tr } S$ plays the role of the number of parameters in proposition 16.2. Evaluate $\text{tr } S$ for ridge in terms of the singular values.

16.3 Unsupervised learning: clustering

The lecture spends one slide on unsupervised learning [► 2024 L23 @ 14:53]–[► 2024 L23 @ 15:58]: give the software a list of customers with, say, ten attributes (annual spend, number of products bought, location, ...) and ask it to group them so that customers in a cluster have similar feature values. The value is in finding clusters one had not thought of. No algorithm is named. The standard one is k -means, derived here (*not discussed in class*).

Definition 16.5 (k -means). Given points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and K , choose an assignment

$c : \{1..n\} \rightarrow \{1..K\}$ and centres $\mu_1, \dots, \mu_K$ to minimise

$$J(c, \mu) = \sum_{i=1}^n \|x_i - \mu_{c(i)}\|^2.$$

Proof. For fixed centres, J is a sum over i of terms each minimised by the nearest centre, so step 1 cannot increase J . For a fixed assignment and one cluster C_k , $\sum_{i \in C_k} \|x_i - m\|^2$ is a convex quadratic in m with gradient $2 \sum_{i \in C_k} (m - x_i)$, which vanishes at the mean, so step 2 cannot increase J . There are finitely many assignments and J is non-increasing, so the sequence of assignments cannot cycle except through equal values, and the algorithm stops. $\square$

Proposition 16.6 (Lloyd’s algorithm). *Alternate*

1. $c(i) \leftarrow \arg \min_k \|x_i - \mu_k\|^2$ and
2. $\mu_k \leftarrow$ mean of the points assigned to k .

Each step does not increase J , so J decreases monotonically and the algorithm stops after finitely many steps at a local minimum (in general not the global one).

INTUITION: Link with principal components

k -means minimises a within-cluster sum of squares, exactly as PCA minimises a residual sum of squares over subspaces (section 7.1.2): PCA compresses to a linear subspace, k -means to K points. It is a known result (Ding and He, 2004; not proved here) that a continuous relaxation of the k -means objective is solved by the leading principal components of the data, so PCA is often used to reduce the features before clustering. Unlike PCA, J has many local minima; one restarts from different initial centres.

Example 16.7 (Six points). Take $(0, 0), (0, 1), (1, 0)$ and $(5, 5), (5, 6), (6, 5)$ with initial centres $(0, 0)$ and $(5, 5)$. The assignment is already stable, the means are $(\frac{1}{3}, \frac{1}{3})$ and $(\frac{16}{3}, \frac{16}{3})$, and $J = \frac{4}{3} + \frac{4}{3} = \frac{8}{3} \approx 2.667$ (checked in python).

Exercise 16.2 (not from the course: k -means).

- (a) Show that the mean minimises $\sum_{i \in C} \|x_i - m\|^2$ and that the minimum equals $\frac{1}{2|C|} \sum_{i, j \in C} \|x_i - x_j\|^2$.
- (b) Take the four points $(0, 0), (4, 0), (0, 1), (4, 1)$ and $K = 2$. Show that “left pair/right pair” and “bottom pair/top pair” are both stable under Lloyd’s algorithm, and compute J for each (1 and 16): a local minimum need not be a global one.

16.4 Optimisation: gradient descent

16.4.1 The algorithm and Hull’s example

Training a model means minimising $\hat{R}(\theta)$, a function of many parameters, for which there is no closed formula (least squares is the exception). Hull describes the procedure in words [► 2024 L23 @ 31:08]: choose starting values and a *learning rate*, compute the gradient, which shows the direction in which the parameters can be improved, take a step against it, and repeat. In symbols,

$$\theta_{k+1} = \theta_k - \eta \nabla \hat{R}(\theta_k). \quad (16.2)$$

The first slide on the subject (slide 16) minimises the one-variable function $y = x^2 - 8x + 20$ starting at $x_0 = 1$ [► 2024 L23 @ 34:17]; the minimum is at $x = 4$ (there $y = 4$). The gradient is $2x - 8 = -6$ at x_0 , and the iterates printed on the slide, 2.2 and 2.92, correspond to a learning rate $\eta = 0.2$: $x_1 = 1 + 0.2 \cdot 6 = 2.2$, $x_2 = 2.2 + 0.2 \cdot 3.6 = 2.92$.

16.4.2 Convergence, learning rate, scaling (not discussed in class)

Near a minimum every smooth function is a quadratic, so the quadratic case explains the behaviour of the method. Let $f(\theta) = \frac{1}{2}(\theta - \theta^*)^T A(\theta - \theta^*) + f^*$ with A symmetric positive definite, eigenvalues $0 < \lambda_{\min} \leq \dots \leq \lambda_{\max}$. Then $\nabla f = A(\theta - \theta^*)$ and (16.2) gives, for the error $e_k = \theta_k - \theta^*$,

$$e_{k+1} = (I - \eta A)e_k, \quad e_k = (I - \eta A)^k e_0.$$

In the eigenbasis of A each component is multiplied by $(1 - \eta\lambda_j)$ at every step. Hence:

Proposition 16.8 (Gradient descent on a quadratic). *The iteration converges for every e_0 iff $|1 - \eta\lambda_j| < 1$ for all j , i.e. $0 < \eta < 2/\lambda_{\max}$. The error is multiplied by $r(\eta) = \max_j |1 - \eta\lambda_j|$ per step; r is minimal, and equal to $\frac{\kappa-1}{\kappa+1}$ with $\kappa = \lambda_{\max}/\lambda_{\min}$ the condition number, at $\eta = 2/(\lambda_{\max} + \lambda_{\min})$. For $\eta > 1/\lambda_{\max}$ the components along the top eigenvector change sign at each step (oscillation).*

For Hull's example $A = 2$ (one dimension), so $\lambda = 2$, $e_k = (1 - 2\eta)^k e_0 = (0.6)^k \cdot (-3)$ for $\eta = 0.2$: convergence is geometric, with $1 - 2\eta = 0.6$. The stability condition is $\eta < 1$; $\eta = \frac{1}{2}$ reaches the minimum in one step; $\eta > \frac{1}{2}$ overshoots and alternates around 4; $\eta > 1$ diverges (fig. 16.2). This is Hull's warning about the learning rate [► 2024 L23 @ 36:26]–[► 2024 L23 @ 37:28]: too small, and one never reaches the bottom in reasonable time; too big, and the iterates oscillate from one side of the valley to the other.

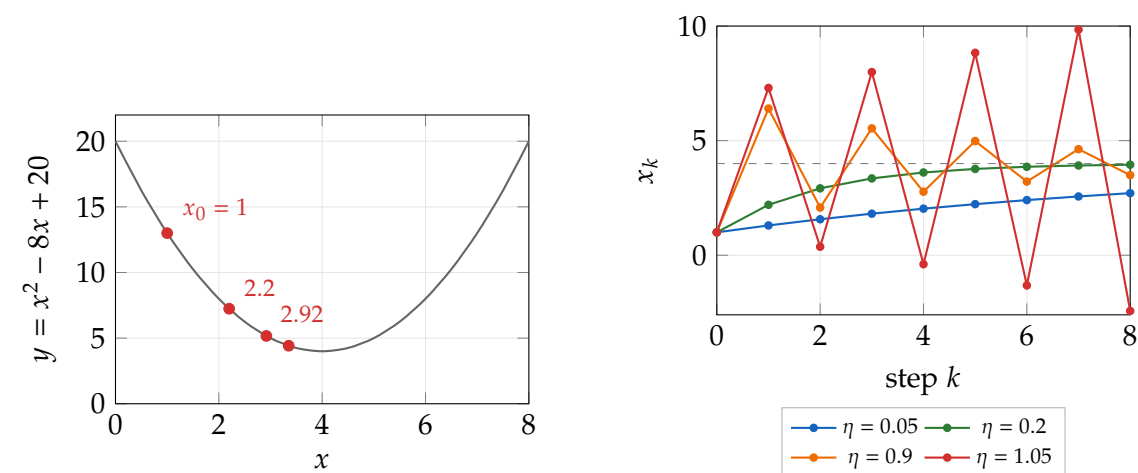


Figure 16.2: Left: the first three steps of gradient descent from $x_0 = 1$ with $\eta = 0.2$ on the function of slide 16. Right: the iterates for four learning rates ($x_{k+1} - 4 = (1 - 2\eta)(x_k - 4)$; dashed line: the minimiser $x = 4$). $\eta = 0.05$ is too small, $\eta = 0.9$ oscillates but converges, and $\eta = 1.05$ diverges.

The role of κ explains Hull's third remark, that the method *works best when the variables have been scaled* [► 2024 L23 @ 37:28]. For a least-squares loss $\frac{1}{2}\|y - X\beta\|^2$ the matrix A is $X^T X$, and its condition number is the ratio of the squared singular values of X . A regressor measured in units 100 times larger changes the corresponding diagonal entry of $X^T X$ by

1 COURSE NOTE: A slip in the spoken example

In the spoken introduction Hull says the function is " $6x^2 - 8x + 20$ " with minimiser $x = 4$ [► 2024 L23 @ 34:17]. That cannot be right: he also says the gradient is $2x - 8$, which is the derivative of $x^2 - 8x + 20$, the function on the slide. (For $6x^2 - 8x + 20$ the minimiser would be $2/3$.) The slide is correct. Also, on slide 16 the vertical lines of the picture lie at about 1, 2.4, 2.9, 3.3; the labels 2.2 and 2.92 in the text should be trusted, since 2.2 and 2.92 are exactly what (16.2) gives with $\eta = 0.2$.

Proof. The first two statements are the geometric-series criterion applied to each eigen-component. The maximum of $|1 - \eta\lambda|$ over $[\lambda_{\min}, \lambda_{\max}]$ is attained at an end point; r is decreasing in η while $1 - \eta\lambda_{\min}$ dominates and increasing when $\eta\lambda_{\max} - 1$ dominates, so the minimum is where $1 - \eta\lambda_{\min} = \eta\lambda_{\max} - 1$, which gives the stated η and $r = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}}$. □

10^4 and can make κ huge; then the stable step size is dictated by the stiff direction while the flat direction crawls (the rate is $\frac{\kappa-1}{\kappa+1} \approx 1 - 2/\kappa$). Standardising the columns, as done before ridge in [section 5.10](#), is the remedy in both cases.

Remark 16.9 (Not covered: stochastic gradients). With n in the millions one computes the gradient of (16.2) on a random subset (*mini-batch*) of the data at each step, and an *epoch* is one pass over all batches. Hull uses “epoch” for one training iteration [[▶ 2024 L23 @ 41:44](#)]; in his examples the whole training set is used at every step of an epoch, as far as the slides say. The lecture also mentions “clever ways” of setting the learning rate [[▶ 2024 L23 @ 37:28](#)] without naming them (adaptive schemes such as Adam belong there).

Exercise 16.3 (not from the course: gradient descent on an ill-conditioned quadratic).

Let $f(\boldsymbol{\theta}) = \frac{1}{2}(\theta_1^2 + 10\theta_2^2)$.

- For which η does gradient descent converge?
- Find the η that minimises the contraction factor and this factor.
- Starting at $(1, 1)$, how many steps are needed to bring $\|\boldsymbol{\theta}\|$ below 10^{-3} with that η , and with $\eta = 0.05$?
- Rescale $\theta_2 = \theta'_2/\sqrt{10}$ and repeat: what has changed?

16.4.3 Early stopping is a form of ridge regression (not discussed in class)

Hull’s stopping rule, “continue until the validation results diverge from the training results” [[▶ 2024 L23 @ 36:26](#)], has a precise meaning in the linear model. Let $X = UDV^T$ be the thin SVD as in [section 5.10](#) and run (16.2) on $\frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ from $\boldsymbol{\beta}_0 = \mathbf{0}$. Along the j -th right singular vector the error contracts by $(1 - \eta d_j^2)$ per step, so after k steps the fitted values are

$$\hat{\mathbf{y}}_k = \sum_{j=1}^p [1 - (1 - \eta d_j^2)^k] c_j \mathbf{u}_j, \quad \text{ridge: } \hat{\mathbf{y}}^{\text{ridge}} = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} c_j \mathbf{u}_j, \quad c_j = \mathbf{u}_j^T \mathbf{y}.$$

Both filters go from 0 (for $d_j \rightarrow 0$) to 1 (for large d_j): directions of large variance are fitted first, small-variance directions last and only if we keep going. The filter $1 - e^{-\eta k d^2}$ (the small- η limit of the first) and $d^2/(d^2 + \lambda)$ agree qualitatively with $\lambda \approx 1/(\eta k)$. For $\eta = 0.05$ and $d^2 \in \{0.1, 1, 10\}$ the two filters are (0.049, 0.401, 0.999) for gradient descent after $k = 10$ steps and (0.048, 0.333, 0.833) for ridge with $\lambda = 2$; after $k = 200$ steps and $\lambda = 0.1$ they are (0.633, 1.000, 1.000) and (0.500, 0.909, 0.990) (python). Stopping early therefore shrinks small-variance directions like a ridge penalty, and the number of epochs is the hyperparameter chosen on the validation set instead of λ .

16.5 Neural networks

16.5.1 Architecture

A supervised model that predicts the output from the input “in one go” (linear regression is the model to keep in mind) is replaced by a sequence of stages [[▶ 2024 L23 @ 22:21](#)]–[▶ 2024 L23 @ 24:34](#): the input layer holds the known variables, $v_{0,1}, v_{0,2}, \dots$; a first hidden layer $v_{1,1}, v_{1,2}, \dots$ is computed from it, a second hidden layer from the first, and so on up

to the output layer (slide 13). Each neuron is a weighted combination of all the neurons of the previous layer, to which an *activation function* is applied [► 2024 L23 @ 25:41].

Definition 16.10 (Feedforward neural network). Let $L \geq 1$, layer widths $m_0 = d, m_1, \dots, m_L$, weight matrices $W^{(l)} \in \mathbb{R}^{m_l \times m_{l-1}}$, bias vectors $\mathbf{b}^{(l)} \in \mathbb{R}^{m_l}$ and activation functions $\varphi_l : \mathbb{R} \rightarrow \mathbb{R}$ applied componentwise. With $\mathbf{a}^{(0)} = \mathbf{x}$,

$$\mathbf{z}^{(l)} = W^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad \mathbf{a}^{(l)} = \varphi_l(\mathbf{z}^{(l)}), \quad l = 1, \dots, L,$$

and the network output is $h_\theta(\mathbf{x}) = \mathbf{a}^{(L)}$, $\theta = \{W^{(l)}, \mathbf{b}^{(l)}\}_l$. The parameters are the *weights* [► 2024 L23 @ 26:48]–[► 2024 L23 @ 27:51].

Hull’s slide writes a neuron as a “weighted average” followed by f ; the constant term $\mathbf{b}^{(l)}$ (the bias) is left implicit. The number of parameters is the sum of $m_l(m_{l-1} + 1)$; for the Black–Scholes network below (5 inputs, three hidden layers of 20 neurons, one output) it is $20 \cdot 6 + 2 \cdot 20 \cdot 21 + 21 = 981$, to be estimated from 6,000 observations. In the volatility-surface network (3 inputs) it is 941.

Activation functions (slide 14) [► 2024 L23 @ 28:54]–[► 2024 L23 @ 30:01]: identity $f(y) = y$; sigmoid $f(y) = 1/(1 + e^{-y}) \in (0, 1)$; hyperbolic tangent $f(y) = (e^{2y} - 1)/(e^{2y} + 1) \in (-1, 1)$; ReLU $f(y) = \max(y, 0)$. Two identities are used below:

$$\sigma'(y) = \sigma(y)(1 - \sigma(y)) \leq \frac{1}{4}, \quad \tanh y = \frac{e^{2y} - 1}{e^{2y} + 1} = 2\sigma(2y) - 1, \quad (\tanh)' = 1 - \tanh^2.$$

(The slide’s formula for $\tanh$ is correct; both identities were confirmed numerically.) If all activations are the identity the network collapses to $\mathbf{x} \mapsto W^{(L)} \dots W^{(1)} \mathbf{x} + \text{const}$, a linear model, which is Hull’s remark [► 2024 L23 @ 28:54]: it is why the nonlinearity is essential, and why the identity is nevertheless used on the *last* layer, where it lets the output range over $\mathbb{R}$. A sigmoid output would confine the prediction to $(0, 1)$, which is impossible for an option price [► 2024 L23 @ 39:36].

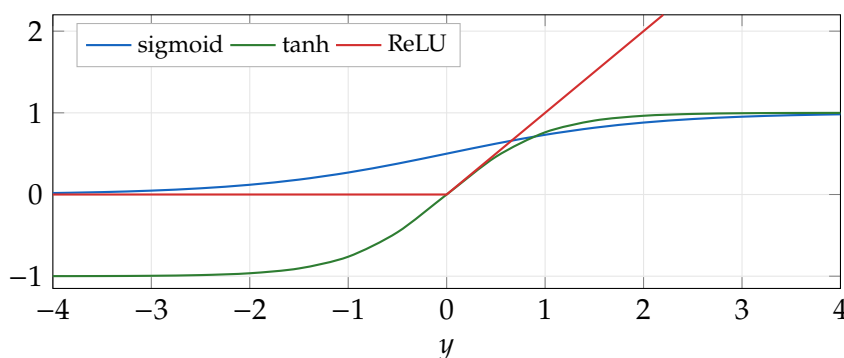


Figure 16.3: The activation functions of slide 14 (the ReLU is cut at the upper edge of the frame).

Remark 16.11 (Not discussed: what the hidden layers buy). A network with one hidden layer of sufficiently many sigmoid (or tanh) units can approximate any continuous function on a compact set to arbitrary accuracy (universal approximation theorem, Cybenko 1989, Hornik 1991); the theorem says nothing about how many units are needed or how to find the weights. The lecture’s point is more modest: several stages make the model “a lot more flexible” than one written-down equation [► 2024 L23 @ 24:34].

A network with no hidden layer, an identity output and squared loss is the linear regression of [chapter 5](#); with a sigmoid output and the cross-entropy loss $\ell = -y \log p - (1 - y) \log(1 - p)$, $p = \sigma(\mathbf{w}^\top \mathbf{x})$, it is logistic regression, the natural model for the loan-acceptance example of [section 16.1](#), with the remarkably simple gradient $\partial \ell / \partial \mathbf{z} = p - y$ (from $\sigma' = \sigma(1 - \sigma)$).

16.5.2 Backpropagation

Gradient descent needs $\partial \ell / \partial \theta$ for *every* weight, and there may be many. The lecture says only that each output is a function of a function of a function, so the gradient follows from the chain rule, with shortcuts that avoid computing each derivative from scratch [[2024 L23 @ 32:16](#)–[2024 L23 @ 33:16](#)], and that this is called backpropagation [[2024 L23 @ 37:28](#)]. The derivation is short (*not discussed in class*).

For one training pair and loss $\ell = \frac{1}{2} \|\mathbf{a}^{(L)} - \mathbf{y}\|^2$ define the *error signals* $\delta^{(l)} = \partial \ell / \partial \mathbf{z}^{(l)}$.

Proposition 16.12 (Backpropagation).

$$\delta^{(L)} = (\mathbf{a}^{(L)} - \mathbf{y}) \odot \varphi'_L(\mathbf{z}^{(L)}), \quad \delta^{(l)} = (\mathbf{W}^{(l+1)\top} \delta^{(l+1)}) \odot \varphi'_l(\mathbf{z}^{(l)}),$$

$$\frac{\partial \ell}{\partial \mathbf{W}^{(l)}} = \delta^{(l)} \mathbf{a}^{(l-1)\top}, \quad \frac{\partial \ell}{\partial \mathbf{b}^{(l)}} = \delta^{(l)},$$

where $\odot$ is the componentwise product. One forward pass (to store the $\mathbf{z}^{(l)}, \mathbf{a}^{(l)}$) and one backward pass give the whole gradient at a cost proportional to the number of weights.

Proof. The last formula is the chain rule for $\mathbf{a}^{(L)} = \varphi_L(\mathbf{z}^{(L)})$. Since $\mathbf{z}^{(l+1)}$ depends on $\mathbf{z}^{(l)}$ only through $\mathbf{a}^{(l)} = \varphi_l(\mathbf{z}^{(l)})$, $\delta_j^{(l)} = \sum_i \delta_i^{(l+1)} \partial z_i^{(l+1)} / \partial z_j^{(l)} = \sum_i \delta_i^{(l+1)} W_{ij}^{(l+1)} \varphi'_l(z_j^{(l)})$. Finally $z_i^{(l)} = \sum_j W_{ij}^{(l)} a_j^{(l-1)} + b_i^{(l)}$ gives $\partial \ell / \partial W_{ij}^{(l)} = \delta_i^{(l)} a_j^{(l-1)}$ and $\partial \ell / \partial b_i^{(l)} = \delta_i^{(l)}$. $\square$

The name is the direction of the recursion: the error signal is propagated from the output layer to the input layer, and each layer's derivative reuses the one above. (The formulas were checked against a finite-difference derivative for a $3 \rightarrow 4 \rightarrow 1$ sigmoid network: the derivative of a weight agrees to 8 digits.) For sigmoid layers $\varphi' = a(1 - a) \leq \frac{1}{4}$, so the product of φ' along many layers shrinks the signal, the *vanishing-gradient* problem, one reason why ReLU ($\varphi' \in \{0, 1\}$) is popular in deep networks. Hull reports that all the activation functions he tried behaved similarly well, and that sigmoid worked best in his two applications [[2024 L23 @ 1:12:20](#)].

Exercise 16.4 (not from the course: backpropagation). A network has one input, one hidden layer with two tanh neurons, and a linear output, $h = w_1 \tanh(u_1 x + c_1) + w_2 \tanh(u_2 x + c_2) + d$, with loss $\ell = \frac{1}{2} (h - y)^2$. Using [proposition 16.12](#), compute $\partial \ell / \partial u_1$, $\partial \ell / \partial c_1$, $\partial \ell / \partial w_1$ and check with a finite-difference in python for random values of the parameters.

16.6 Application: learning the Black–Scholes–Merton price

CASE STUDY: A network that imitates Black–Scholes–Merton (slides 18–21)

Goal. Test whether a network can recover a known pricing function from noisy data [[2024 L23 @ 38:30](#)]. **Data.** 10,000 European call prices from the Black–Scholes–Merton formula, with parameters drawn at random: $S \in [40, 60]$, $K \in [0.5S, 1.5S]$, $r \in [0, 5\%]$, $\sigma \in [10\%, 40\%]$, $T \in [0.25, 2]$ years. To each price a normal error with mean 0 and standard deviation 0.15 is added. The split is 6,000 training, 2,000 validation, 2,000 test observations [[2024 L23 @ 40:41](#)]. **Method.** Three hidden layers of 20 neurons,

sigmoid activation everywhere except the last layer, where the identity is used because a sigmoid output would lie in $(0, 1)$ and prices do not [► 2024 L23 @ 39:36]. Training error and validation error are recorded as the number of epochs grows. **Result.** The validation error stops improving and slowly worsens after a while while the training error keeps decreasing (slide 19); on the 50-epoch moving average (slide 20) the minimum is found and training is stopped after 2,575 epochs [► 2024 L23 @ 40:41]–[► 2024 L23 @ 41:44]. With only 10,000 observations the network imitates the formula very well, in spite of the noise [► 2024 L23 @ 42:46]. **Lesson.** The noise variance cannot be fitted away: the best possible mean squared error is $0.15^2 = 0.0225$ and the flexible model reaches it only if it stops before it starts fitting the noise. Early stopping is the safeguard (section 16.4).

Our reconstruction. To see the mechanics we repeated the experiment (numpy, hand-written backpropagation as in proposition 16.12, inputs standardised, full-batch Adam with learning rate 10^{-2} ; the slides do not say which optimiser was used). After 4,000 epochs the test-set mean squared error against the noisy prices is 0.0259 against a floor of 0.0225, and the root-mean-square distance between the network and the *noise-free* Black–Scholes–Merton price is 0.054 (the prices range up to about 25). At 4,000 epochs the validation error was still decreasing, so no overfitting phase was reached in this short run; Hull’s 2,575 epochs are specific to his optimiser and learning rate, not a universal number.

IN FINANCE: Why imitate a formula that one already has?

Hull’s answer [► 2024 L23 @ 42:46]–[► 2024 L23 @ 44:53]: some derivatives have no formula and are valued by Monte Carlo simulation, which can take minutes, too slow with a client on the phone or for scenario analysis, where a portfolio is revalued thousands of times. The recipe is

1. run the simulation many times to build a table relating price to inputs, without adding any noise;
2. train a network to reproduce the table;
3. price by a forward pass through the network, which is nearly instantaneous.

The Black–Scholes exercise is the test bed, because there the right answer is known. The price paid is the training set: the surrogate is reliable only inside the range of inputs it was trained on.

16.7 Application: how the implied-volatility surface moves

The second application [► 2024 L23 @ 45:53]–[► 2024 L23 @ 50:07] concerns hedging with the delta of chapter 15 and the volatility surface of chapter 13. The standard delta hedge takes $\partial c / \partial S$ with the implied volatility held fixed. If the implied volatility tends to move with the asset price (when the stock rises the firm is less levered, so volatility falls [► 2024 L23 @ 46:57]), the hedge that minimises the variance of the hedged position is different.

Proposition 16.13 (Minimum-variance delta (not discussed in class)). *Let an option have value $f(S, \sigma)$ with $\delta = \partial f / \partial S$ and $v = \partial f / \partial \sigma$ (vega), and to first order $\Delta f \approx \delta \Delta S + v \Delta \sigma$ over a short interval. Write $\Delta \sigma = b \Delta S + \epsilon$ with $b = \text{Cov}(\Delta \sigma, \Delta S) / \text{Var}(\Delta S)$ the regression slope*

2 COURSE NOTE: Erratum: the axis of slides 19–20 is not the mean squared error

The titles of slides 19 and 20 say “mse”, but the vertical axis is labelled *Mean Absolute Error* and the curves level off near 0.12. For a noise $N(0, 0.15^2)$ the smallest attainable mean absolute error is $0.15\sqrt{2/\pi} = 0.1197$, while the smallest attainable mean squared error is 0.0225. So the plotted quantity is the mean *absolute* error, and it is the title that is wrong (the transcript also says only “mse”). Slide 26, by contrast, does show a mean squared error (values around 6.5×10^{-5}).

Proof. $\Delta f - h \Delta S = (\delta + vb - h) \Delta S + v \epsilon$, the two terms are uncorrelated, so the variance is $(\delta + vb - h)^2 \text{Var}(\Delta S) + v^2 \text{Var} \epsilon$, minimal at $h = \delta + vb$. □

(chapter 5) and ϵ uncorrelated with ΔS . The number of shares h minimising $\text{Var}(\Delta f - h\Delta S)$ is

$$\delta_{\text{MV}} = \delta + \nu b.$$

Hull and White (2017) proposed an analytic model for $\mathbb{E}[\Delta\sigma \mid \Delta S]$ (its form is not on the slides). The lecture's neural network replaces it by a learned function [► 2024 L23 @ 48:01]: *inputs* the daily asset-price return, the moneyness (measured by delta) and the time to maturity; *output* the change in implied volatility. For a call, $\nu > 0$ and typically $b < 0$, so $\delta_{\text{MV}} < \delta$.

CASE STUDY: S&P 500 options (slides 24–27)

Data. S&P 500 call options, 2014–2019, 100 options sampled at random per day, 125,700 observations, split 60/20/20% (75,420, 25,140, 25,140). **Method.** Three hidden layers of 20 neurons; several activation functions tried, sigmoid best [► 2024 L23 @ 48:01]. Training stopped after 5,826 epochs, where the validation mse turned up (slide 26: mse of about 6.5×10^{-5} on the training and 6.8×10^{-5} on the validation set). **Results.** On the test set the network gave only a modest 11% improvement over the Hull–White analytic model [► 2024 L23 @ 49:02]. But when the VIX index of day t was added as a feature, a further improvement of over 60% was found: the behaviour of the surface differs in high-volatility and low-volatility environments [► 2024 L23 @ 49:02]–[► 2024 L23 @ 50:07]. The slides do not define “improvement”; presumably it is the relative reduction of the mean squared error of the predicted volatility change. **Lesson.** The 60% is a result one is unlikely to find by “playing around” [► 2024 L23 @ 50:07]: the network was used to *discover* that the regime matters, and the regime variable was then added by hand.

16.8 Reinforcement learning

16.8.1 The model and the exploration–exploitation trade-off

Reinforcement learning (RL) finds a strategy for a *series* of decisions in a changing environment [► 2024 L23 @ 50:07]–[► 2024 L23 @ 51:11]; software of this kind beats the best human chess and Go players. The general model (slide 30) has a state S_t , an action A_t taken in it, a reward R_{t+1} and a new state S_{t+1} [► 2024 L23 @ 52:21]: $S_0 \xrightarrow{A_0} (R_1, S_1) \xrightarrow{A_1} (R_2, S_2) \cdots$.

Definition 16.14 (Value of an action). Fix a policy π (a rule $S \mapsto A$). With discount factor $\gamma \in [0, 1]$ the *return* is $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$, and $Q^\pi(s, a) = \mathbb{E}[G_t \mid S_t = s, A_t = a]$ is the value of taking a in s and following π afterwards. The optimal $Q^* = \max_\pi Q^\pi$ satisfies the Bellman equation

$$Q^*(s, a) = \mathbb{E} \left[R_{t+1} + \gamma \max_{a'} Q^*(S_{t+1}, a') \mid S_t = s, A_t = a \right].$$

The Bellman equation is dynamic programming (the backward induction of chapter 15 on a tree) but written for an unknown environment, so that the expectation is replaced by observed trials. The lecture does not use these formulas; in Nim below $\gamma = 1$ and the only reward is ± 1 at the end.

At each trial the agent must choose between *exploitation*, taking the best action found so far, and *exploration*, taking a random one to learn something new [► 2024 L23 @ 53:22]. With probability ϵ it explores and with probability $1 - \epsilon$ it exploits (ϵ -greedy); ϵ starts at 1 and is multiplied at each trial by a decay factor such as 0.995 (a hyperparameter set by trial and

error, slide 31), so that after enough trials the agent exploits only [► 2024 L23 @ 54:25]. With the decay 0.9995 used in the Nim tables, $\varepsilon_n = 0.9995^n$ is 0.607, 0.082, 0.0067 after 1,000, 5,000, 10,000 trials.

16.8.2 Nim and the updating rule

The game (slide 32) [► 2024 L23 @ 54:25]–[► 2024 L23 @ 55:31]: a pile of n matches, two players alternately take 1, 2 or 3; whoever must take the last match loses. Hull takes $n = 8$, reward +1 for a win and −1 for a loss, an opponent who plays at random, state S = matches left when it is the agent's turn, action A = number taken, and $Q(S, A)$ initialised at 0.

Proposition 16.15 (Perfect play in Nim). *The player to move with s matches loses against perfect play iff $s \equiv 1 \pmod{4}$.*

This is the “ $4n + 1$ ” strategy of the lecture [► 2024 L23 @ 56:34]–[► 2024 L23 @ 57:39]: five matches is lost, nine is lost, and so on. Starting from 8 the right move is to take 3. The point of the exercise is to see whether a machine that knows nothing finds it [► 2024 L23 @ 57:39].

Proof. Induction on s . With $s = 1$ the mover must take the last match and loses. If $s \neq 1$, i.e. $s - 1 \equiv 1, 2, 3 \pmod{4}$ ($s \geq 2$), the mover takes $(s - 1) \bmod 4 \in \{1, 2, 3\}$ and leaves $4m + 1$ matches to the opponent, who loses by the induction hypothesis. If $s \equiv 1$ and $s > 1$, whatever $a \in \{1, 2, 3\}$ the mover takes leaves $s - a \equiv 1 - a \not\equiv 1 \pmod{4}$, a winning position for the opponent. $\square$

Updating (slides 33–34) [► 2024 L23 @ 58:40]–[► 2024 L23 @ 1:00:45]. In the first trial of the slides the agent takes 1, the opponent 3, the agent 1, the opponent 3, and the agent wins. The pairs (8, 1) and (4, 1) visited are updated with the rule

$$Q(S, A)^{\text{new}} = Q(S, A)^{\text{old}} + \alpha [G - Q(S, A)^{\text{old}}], \quad (16.3)$$

where α (e.g. 0.05) is a hyperparameter: with $G = 1$ and $Q^{\text{old}} = 0$ the new value is 0.05, as on the slide. It is an exponentially weighted moving average of the gains, and a stochastic-approximation step for the mean of G . There are two choices for the gain G :

- the final reward of the trial (*Monte Carlo method*): $G = R_{\text{end}} = \pm 1$;
- the value of the next state under the best decision, $G = R + \gamma \max_{a'} Q(S', a')$ (*temporal-difference learning*, or *Q-learning*), which is the Bellman equation with the expectation replaced by one observation.

Monte Carlo waits until the game is over and needs the whole trajectory. Temporal-difference learning *bootstraps*: it updates from a neighbouring estimate immediately. Under ε -greedy play the Monte Carlo average estimates the value of the current exploratory policy, whereas the temporal-difference target estimates the value of the greedy one, which is why the latter learns faster while ε is large. (That Q-learning converges to Q^* if α decays suitably and every pair is visited infinitely often is a theorem of Watkins and Dayan; not shown here.)

Example 16.16 (Exact solution against a random opponent (our reconstruction)). Against an opponent who picks uniformly among the legal moves, the optimal values follow from backward induction, $Q^*(s, a) = \frac{1}{n} \sum_b V^*(s - a - b)$ with $V^*(s) = \max_a Q^*(s, a)$, $V^*(0)$ read as +1 when the opponent took the last match and $Q^*(s, s) = -1$ when the agent did. The values (python) are:

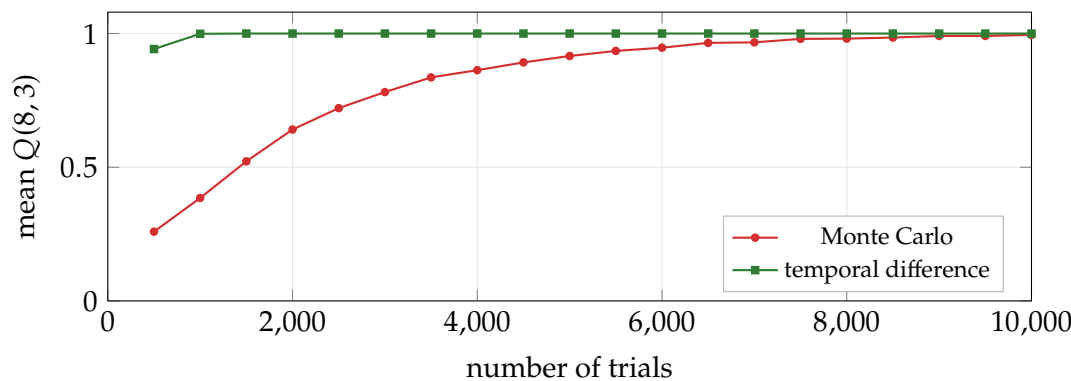


Figure 16.4: Learning Nim with $\varepsilon_0 = 1$, decay 0.9995, $\alpha = 0.05$ (parameters of slides 35–36). The curves are the value of the winning move $Q(8, 3)$, averaged over 100 independent runs of our own simulation. The temporal-difference rule learns the right value in about a thousand trials, the Monte Carlo rule in about ten times more (compare the tables of slides 35 and 36).

In our simulation of 200 independent runs, the fraction of runs in which the greedy move at $s = 8$ is 3 after 1,000, 5,000, 10,000 trials is 62.5%, 95%, 100% for the Monte Carlo rule and 100% at all three horizons for temporal-difference learning. This matches the lecture’s remark that with Monte Carlo the machine is not yet sure after 1,000 trials, is fairly sure after 5,000 and certain after 10,000, and that temporal difference is faster [► 2024 L23 @ 1:01:52]–[► 2024 L23 @ 1:02:54]. Hull adds that when the opponent also learns, moves 1 and 2 from 8 turn negative [► 2024 L23 @ 1:03:55]: against a perfect opponent $Q(8, 1)$ and $Q(8, 2)$ are -1 , because 7 and 6 leave a winning position.

Exercise 16.5 (not from the course: Nim).

- Verify [proposition 16.15](#) for $s \leq 13$ by listing the winning moves.
- Verify by hand the exact values $Q^*(5, \cdot) = (\frac{1}{3}, \frac{1}{3}, 0)$ and $Q^*(7, \cdot) = (\frac{7}{9}, 1, \frac{1}{3})$ of [example 16.16](#).
- For which starting number of matches $n \leq 13$ is there a first move that wins even if the opponent plays perfectly?

Exercise 16.6 (not from the course: one Q -learning step). In Nim with $\alpha = 0.05$, $\gamma = 1$, let $Q(8, 3) = 0.5$ and $Q(4, \cdot) = (0.1, 0.2, 0.4)$. A trial goes $8 \xrightarrow{3} 5 \rightarrow 4$ (opponent takes 1). Compute the updated $Q(8, 3)$ by

- the temporal-difference rule and
- the Monte Carlo rule, if the trial ends with a win.

$a \backslash s$	1	2	3	4	5	6	7	8
1	-1	1	0	$\frac{1}{3}$	$\frac{1}{3}$	1	$\frac{7}{9}$	$\frac{7}{9}$
2		-1	1	0	$\frac{1}{3}$	$\frac{1}{3}$	1	$\frac{7}{9}$
3			-1	1	0	$\frac{1}{3}$	$\frac{1}{3}$	1

At $s = 8$ the best move is 3 with value 1 (the opponent, left with 5 and moving at random, always ends in a lost position for himself, since the agent then plays perfectly), while 1 and 2 give $7/9 \approx 0.778$, positive because the opponent is not perfect. Compare slides 35–36: after 10,000 trials the slide’s row $s = 8$ reads (0.674, 0.793, 1.000) for Monte Carlo and (0.813, 0.910, 1.000) for temporal-difference, so the action 3 is found and the other two values are noisy estimates of 0.778. Column 7 is 0 everywhere in the slides, correctly: after the agent’s first move and the opponent’s reply at most 6 matches remain, so state 7 is never met. Some entries, e.g. $Q(3, 1) = 0.618$ in the temporal-difference table against the exact 0, are stale: once ε is small the agent rarely tries a bad action again, and its value keeps the noise of the few early trials.

16.8.3 Deep Q -learning

When there are many states or actions the table is filled only partly [► 2024 L23 @ 1:03:55]–[► 2024 L23 @ 1:05:00], so Q is estimated as a function by a network $Q_\theta(S, A)$, trained to minimise the sum of squared differences between the network and the target [► 2024 L23 @ 1:05:00](slide 37):

$$L(\theta) = \sum_{\text{observed } (S, A, R, S')} \left(R + \gamma \max_{a'} Q_\theta(S', a') - Q_\theta(S, A) \right)^2,$$

where, as in (16.3), the target is held fixed when differentiating (a “semi-gradient”; common implementations keep a lagged copy of θ inside the target for stability; not discussed in class). This is nonlinear least squares, i.e. sections 16.4 and 16.5 with a moving target. The table of Nim is the special case with one parameter per pair (S, A) and a constant step: (16.3) is one gradient step on the square $\frac{1}{2}(G - Q(S, A))^2$ with learning rate α .

16.8.4 Hedging derivatives by reinforcement learning

Hull’s research question [▶ 2024 L23 @ 1:05:00]–[▶ 2024 L23 @ 1:06:05]: the traditional hedge uses the Greek letters, i.e. derivatives of the portfolio value with respect to a variable over a very short horizon. RL instead looks several periods ahead and chooses today’s hedge to work well over the whole horizon. **Findings** (slide 39) [▶ 2024 L23 @ 1:06:05]–[▶ 2024 L23 @ 1:10:14]:

- for vanilla options it does about as well as Greek-letter hedging when there are no transaction costs, but it *saves transaction costs*, in some cases significantly;
- for exotic options such as barrier options it produces better results even with no transaction costs;
- the user *chooses the objective function*: e.g. VaR_{95} or CVaR_{95} of the hedging loss, which the Greeks cannot do;
- it is robust and works well in stressed periods; JP Morgan is the main competitor doing this research;
- disadvantage: it is much more computationally demanding, minutes rather than nothing for a Greek, but computers are getting faster.

INTUITION: Why looking ahead saves costs (own illustration, not from the lecture)

Hull’s example [▶ 2024 L23 @ 1:07:09]: the delta is 0.5, rises to 0.7, and there is a 50% chance that at the next hedging date it will be back at 0.5. With a proportional cost c per share traded, rebalancing at once costs $0.2c$ now and, with probability $\frac{1}{2}$, another $0.2c$ later: expected cost $0.3c$. Waiting costs 0 now and $0.2c$ only if the delta stays at 0.7: expected cost $0.1c$. The price of waiting is a hedge that is 0.2 shares short for one period, which adds a variance $0.2^2 \text{Var}(\Delta S) = 0.04 \text{Var}(\Delta S)$ to the hedged position. Waiting is better when $0.2c$ exceeds the value put on this variance, i.e. when transaction costs are large, which is Hull’s conclusion [▶ 2024 L23 @ 1:08:13]. A Greek-letter hedger, who looks one period ahead, always rebalances; a multi-period agent can trade the two effects off.

Exercise 16.7 (not from the course: minimum-variance delta). A call has $\delta = 0.60$ and vega $\nu = 25$ (price change per unit change of σ in absolute terms, e.g. 0.01 of volatility is 0.25). Suppose over a day $\Delta\sigma = -0.008 \Delta S + \epsilon$ (ΔS in dollars). Compute δ_{MV} . How does the variance of the hedged position change when $h = \delta_{\text{MV}}$ replaces $h = \delta$, if $\text{Var}(\Delta S) = 1$ and $\text{Var}(\epsilon) = 10^{-5}$?

16.9 The questions at the end

The remaining minutes are questions [▶ 2024 L23 @ 1:11:20]–[▶ 2024 L23 @ 1:15:24]. Asked whether the choice of activation matters, Hull answers that sigmoid has worked well, perhaps by chance, that the alternatives tried gave results close to each other, and that the

3 COURSE NOTE: VaR and CVaR: a misstatement in the transcript

Explaining the objective functions, Hull says that with VaR_{95} one wants “the 95th percentile point of the loss distribution to be as high as possible” [▶ 2024 L23 @ 1:09:13]. A risk manager wants it *as low as possible*: the objective is to *minimise* the 95th percentile of the loss (as for the CVaR_{95} described in the next sentence, the average loss beyond that percentile, “as low as possible”). The definitions and the properties of both measures are in chapter 19.

size of the model (neurons per layer, number of layers) matters more. Asked whether a two-layer network can be extended to three layers without starting again, he believes one must start from scratch. (This is the practice in his examples; in the wider literature weights of a smaller network are sometimes reused as an initialisation, not discussed.) He points to the Python code for the book's examples on his Toronto website [► 2024 L23 @ 1:13:20], and, asked how the objective function is chosen, says that it depends on the problem: to match an exotic option price one minimises the difference between the network and target price [► 2024 L23 @ 1:14:20]–[► 2024 L23 @ 1:15:24].

IN FINANCE: Summary: where the tools sit in the course

- *Regression and shrinkage* (chapter 5): squared loss, linear model, ridge penalty; the network is the nonlinear generalisation and early stopping is the ridge penalty in disguise (section 16.4).
- *PCA* (chapter 7): dimension reduction on the features, also the linear cousin of clustering.
- *Volatility and Black–Scholes* (chapters 13 and 15): the network learns the pricing function and the response of the surface; the minimum-variance delta is a regression coefficient.
- *Regularised models* (chapter 20): the finance-industry treatment of the same bias–variance balance, for pricing and risk model calibration.
- *VaR* (chapter 19): the loss functionals used as objectives in RL hedging.

KEY FORMULAS

Empirical risk:	$\hat{R}(\theta) = \frac{1}{n} \sum_i \ell(y_i, h_\theta(x_i))$
Bias–variance:	$\mathbb{E}[(y - \hat{h})^2] = \sigma^2 + (f - \mathbb{E}[\hat{h}])^2 + \text{Var } \hat{h}$
Projection fit:	test = $\sigma^2 + b^2 + \frac{p}{n}\sigma^2$, train = $\sigma^2 + b^2 - \frac{p}{n}\sigma^2$
k-means:	$J = \sum_i \ x_i - \mu_{c(i)}\ ^2$ (Lloyd: monotone decrease)
Gradient descent:	$\theta_{k+1} = \theta_k - \eta \nabla \hat{R}(\theta_k)$
Quadratic case:	$e_{k+1} = (I - \eta A)e_k$; $0 < \eta < \frac{2}{\lambda_{\max}}$; $r_{\min} = \frac{\kappa-1}{\kappa+1}$
Early stopping:	filter $1 - (1 - \eta d^2)^k \leftrightarrow \text{ridge } \frac{d^2}{d^2 + \lambda}$, $\lambda \approx \frac{1}{\eta k}$
Network layer:	$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}$, $a^{(l)} = \varphi(z^{(l)})$
Activations:	$\sigma(y) = \frac{1}{1+e^{-y}}$, $\sigma' = \sigma(1 - \sigma)$, $\tanh y = 2\sigma(2y) - 1$
Backpropagation:	$\delta^{(l)} = (W^{(l+1)\top} \delta^{(l+1)}) \odot \varphi'(z^{(l)})$ $\partial \ell / \partial W^{(l)} = \delta^{(l)} a^{(l-1)\top}$
Min-variance delta:	$\delta_{\text{MV}} = \delta + v \text{Cov}(\Delta\sigma, \Delta S) / \text{Var}(\Delta S)$
Bellman:	$Q^*(s, a) = \mathbb{E}[R + \gamma \max_{a'} Q^*(S', a')]$
Q update:	$Q \leftarrow Q + \alpha[G - Q]$ $G = R_{\text{end}} \text{ (MC)}, \quad G = R + \gamma \max_{a'} Q(S', a') \text{ (TD)}$
Exploration:	$\varepsilon_n = \varepsilon_{n-1} \cdot \text{decay}$, $\varepsilon_n = 0.9995^n$

Chapter 17

Stochastic Calculus (Itô Integral and Itô's Lemma)

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2024 L24	P. Kempthorne	Brownian motion with drift and the one-step analysis; filtrations; the Itô integral built from constant, step and adapted step integrands; the worked case $\int B \, dB$; the general integral, its martingale property, the isometry and the Riemann-sum (left-point) representation; Gaussian integrals; Itô's formula for $f(B_t)$ and $f(t, B_t)$, integrals by anti-derivatives, the PDE condition for martingales, the exponential martingale, ruin with drift; standard processes (last slide of the lecture).
2024 L25 (to 18:00)	P. Kempthorne	Recap of Itô's formula; geometric Brownian motion as an SDE, the log price and the drift correction $\mu - \frac{1}{2}\sigma^2$, mean and variance of the price, annualised return, simulation of 50 paths. (From 19:01 the lecture derives the Black–Scholes equation and solves the heat equation; that part belongs to chapters 15 and 18 , only the derivation by Itô's lemma is summarised here.)
2013 L18	C. Lee	Taylor expansion and $(dB)^2 = dt$; Itô's lemma for $f(t, B_t)$ and for $f(t, X_t)$; examples B^2 , $e^{\mu t + \sigma B}$, geometric Brownian motion without drift; integration as the inverse of differentiation, left-point sums, adapted processes; normality, isometry and martingale property; equivalent measures, Radon–Nikodým derivative and Girsanov's theorem (with a comment by P. Kempthorne at 1:10:39).

Files. Slides [mit18_642_f24 lec19_1.pdf](#) (26 slides, 84 pages with overlays; this is Lecture 24, not 19); [mit18_642_f24 lec19_2.pdf](#) (7 pages: the R simulation of Brownian motion with drift, of e^W and of the cumulative average log return); 2013 note [MIT18_S096F13_lecnote18.pdf](#) (7 pages).

Problem sets. 2013 PS8, all problems (A-1 to A-5, B-1 to B-3); Problem A-2 also opens 2024 PS4 and is treated in [chapter 11](#), where its solution is developed. 2024 PS5 (Lévy processes, simulations) belongs to [chapter 11](#); there is no 2024 problem set on stochastic calculus. Each exercise follows the section it practises. **Not covered in class.** 2024 does not treat change of measure, 2013 does not construct the integral (the isometry is stated without proof); the construction of [section 17.3](#), the proof sketch of Itô's lemma with control of the remainder, the multi-dimensional formula and the Stratonovich integral are additions of these notes (marked as such).

Motivation

A stock price is not differentiable in time, and a Brownian path is not differentiable either ([chapter 11](#)). Yet finance needs to speak about *integrals* such as the gain $\int \theta_t \, dS_t$ of a trading strategy that holds θ_t units of an asset whose price is S_t , about the *differential* of an

option price $G(S_t, t)$ when S is random, and about the solution of $dS = \mu S dt + \sigma S dB$. Ordinary calculus fails because the squared increments of B do not vanish faster than dt : $(dB)^2 = dt$ (theorem 11.8 and eq. (11.3)). Stochastic calculus is the calculus that keeps this one extra term and is otherwise as close to ordinary calculus as possible. In physical language: it is the calculus of a particle driven by white noise, and the extra term is the Itô (or “noise-induced”) drift that a physicist has to decide about when writing a Langevin equation.

- The *Itô integral* $\int_0^t f dB$ evaluates the integrand at the *left* end of each small interval, so that the integrand never looks into the future (sections 17.2 and 17.3); the price is that classical rules change, e.g. $\int_0^t B dB = \frac{1}{2}(B_t^2 - t)$ (section 17.4), which is the identity the Dickey–Fuller distribution of section 9.4.2 is built on.
- *Itô’s lemma* is the chain rule of this calculus, $df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt$ (section 17.5). It converts every question about functions of Brownian motion into a drift and a martingale part (section 17.6) and solves geometric Brownian motion (section 17.7).
- With several noise sources the same idea gives the product rule and the multi-dimensional formula (section 17.8); and Girsanov’s theorem (section 17.9) says that a drift can be removed by changing the probability measure, which is the mathematical heart of risk-neutral pricing (chapter 15).

We use the background of chapter 4 (martingales, conditional expectation) and chapter 11 (definition of Brownian motion, quadratic variation, filtrations). The stochastic differential equations built on this calculus are the subject of chapter 18.

17.1 From Brownian motion with drift to Itô processes

Let B be a standard Brownian motion and put

$$X(t) = \mu t + \sigma B(t), \quad t \geq 0, \quad (17.1)$$

Brownian motion with drift μ and variance parameter σ^2 (definition 11.16) [► 2024 L24 @ 1:05]. The increments are independent and $\text{Var}[X(t) - X(s)] = \sigma^2|t - s|$. The lecture looks at a small step, $X(t + \Delta t) - X(t) = \mu\Delta t + \sigma\Delta B$ with $\Delta B = B(t + \Delta t) - B(t)$ [► 2024 L24 @ 4:09]; the increment ΔX is *exactly* $N(\mu\Delta t, \sigma^2\Delta t)$ and

$$\mathbb{E}[(\Delta X)^2] = \sigma^2\Delta t + (\mu\Delta t)^2 = \sigma^2\Delta t + o(\Delta t), \quad \mathbb{E}[|\Delta X|^c] = o(\Delta t) \quad (c > 2) \quad (17.2)$$

[► 2024 L24 @ 6:19]: only the *second* moment of the increment is of the order of the time step, all higher ones are negligible. (The lecture slide writes $\mathbb{E}[(\Delta X)^c]$ without absolute value; for odd c that is a different quantity, but it is also $o(\Delta t)$.) This is the whole content of “ $(dB)^2 = dt$ ” in probabilistic language.

In differential notation (17.1) reads $dX = \mu dt + \sigma dB$, and its solution is

$$X_t = X_0 + \int_0^t \mu ds + \int_0^t \sigma dB_s = X_0 + \mu t + \sigma(B_t - B_0) \quad (17.3)$$

[► 2024 L24 @ 7:24]. The differential form is only shorthand for the integral form. The lecture wants to allow a drift and a volatility that depend on time and on the state, $dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dB_t$: an *Itô process*, the model for a price, an interest rate or the value of a derivative that depends on time to maturity and on the level of the underlying

[► 2024 L24 @ 9:32]. The second integral in (17.3) is then no longer a trivial difference of the endpoints; a student asks exactly this and the lecture answers that Itô integrals are the missing definition [► 2024 L24 @ 10:36].

INTUITION: Why no ordinary integral: white noise

If B were differentiable, $\xi = dB/dt$ would be a noise and $dX = \mu dt + \sigma \xi dt$ a Langevin equation. The path of B is nowhere differentiable, and by [corollary 11.9](#) its total variation is infinite, so $\int f dB$ cannot be defined path by path as a Riemann–Stieltjes integral. Physicists write $\langle \xi(t)\xi(s) \rangle = \delta(t-s)$: ξ is a generalised function, and a Langevin equation with *multiplicative* noise $\sigma(X)\xi$ is ambiguous until one says at which point of the small interval σ is evaluated. Stochastic calculus resolves the ambiguity by a rule; the Itô rule is “at the beginning” ([sections 17.2 and 17.4](#)), the alternative “at the middle” gives the Stratonovich integral that obeys the ordinary chain rule.

17.2 Filtrations, adapted processes and integrals of step processes

17.2.1 The probability set-up

As in the lecture, let $(\Omega, \{\mathcal{F}_t\}, \mathbb{P})$ be the probability model of the Brownian motion: Ω the set of paths ω , $\mathcal{F}_t$ the σ -field of events determined by the path up to time t , i.e. by $\{B_s, s \leq t\}$ (the information available at t ; the filtration was introduced in [section 11.7.4](#)), and $B_t = B_t(\omega)$ a specific path [► 2024 L24 @ 12:42]. Below B is a Brownian motion *with respect to* this filtration: $B_t - B_s$ is $N(0, t-s)$ and independent of $\mathcal{F}_s$ for $s < t$.

Definition 17.1 (Adapted process). A process $\{\Delta_t\}$ is *adapted* (to B , or to the filtration $\{\mathcal{F}_t\}$) if for every t the random variable Δ_t depends only on $\{B_s : s \leq t\}$, i.e. is $\mathcal{F}_t$ -measurable [► 2013 L18 @ 40:10].

Adapted means *non-anticipating*: a strategy that decides the position at t from what has been observed until t . The 2013 lecture gives examples for a process X [► 2013 L18 @ 42:26]: $\Delta_t = X_t$ is adapted; $\Delta_t = \min\{X_t, c\}$ is adapted; $\Delta_t = X_{t+1}$ is not (it looks one unit ahead); $\Delta_t = \max_{0 \leq s \leq T} X_s$, for a fixed $T > t$, is not (it uses the future until T). In the same spirit $\Delta_t = X_{t/2}$ is adapted and $\Delta_t = X_{2t}$ is not. If τ is a stopping time ([definition 4.22](#)), the *stopped* process $X_{t \wedge \tau}$ is adapted; the note writes X_τ , see the erratum in [section 17.10](#).

IN FINANCE: Trading strategies must be adapted

If S_t is a price and Δ_t the number of units held at t , the strategy makes sense only if Δ is adapted: one cannot buy more just before a jump that has not yet happened. The 2013 note’s example is the strategy $\Delta_t = \pm 1$ (buy or sell one share): it is admissible only if the sign is decided from past information [► 2013 L18 @ 39:05]. The discrete counterpart is the daily P&L of [eq. \(2.1\)](#), where the holdings $q(t+1)$ are chosen at the end of day t : the sum $\sum_t q(t+1)\Delta p(t)$ is a left-point sum.

Exercise 17.1 (PS8 (2013), Problem A-3: adapted processes). Let X_t be a given stochastic process. Which of the following processes Y_t are adapted to X_t ?

- (a) $Y_t = X_{T-t}$ for $t \leq T$, for a fixed T .
- (b) $Y_t = \max_{0 \leq s \leq 2t} X_s$.

- (c) $Y_t = |\{i \in [0, t] : X_i \geq 0\}|$ (the amount of time in $[0, t]$ during which X is non-negative; the printed set-notation is ambiguous).

17.2.2 Integrals of step processes

The lecture builds the integral $I(f) = \int_0^T f \, dB$ in stages of increasing generality [► 2024 L24 @ 19:15]. For each path ω one wants $I(f)(\omega)$, a *random variable* with a mean and a variance.

- (a) *Constant integrand* $f \equiv 1$: $I(f) = \int_0^T dB_s = B_T - B_0 = B_T$, with mean 0 and variance T [► 2024 L24 @ 21:17]. For an indicator $f = \mathbf{1}(a < s \leq b)$ the integral is $B_b - B_a$, with variance $b - a$ [► 2024 L24 @ 25:29].
- (b) *Deterministic step function* $f(s) = \sum_{i=0}^{n-1} a_i \mathbf{1}(t_i < s \leq t_{i+1})$ with constants a_i and $0 = t_0 < t_1 < \dots < t_n = T$:

$$I(f) = \sum_{i=0}^{n-1} a_i (B_{t_{i+1}} - B_{t_i}), \quad \mathbb{E}[I(f)] = 0, \quad \text{Var } I(f) = \sum_i a_i^2 (t_{i+1} - t_i) \quad (17.4)$$

[► 2024 L24 @ 26:33]. The a_i can be read as the number of shares held during $(t_i, t_{i+1}]$ of an asset following Brownian motion, and $I(f)$ as the gain or loss. The slides write the last time as $t_{n+1} = T$; with n intervals it is $t_n = T$.

- (c) *Adapted step process*: the levels $a_i = a_i(\omega)$ are random, $\mathcal{F}_{t_i}$ -measurable and with $\mathbb{E}[a_i^2] < \infty$ [► 2024 L24 @ 28:40]. The integral is again (17.4), now with random a_i [► 2024 L24 @ 31:54].

Case (c) is the only new idea, and it is the one that fixes the meaning of the integral: *the level a_i is known at the left end t_i of the interval on which it is used*. We call such an $f = \sum_i a_i \mathbf{1}(t_i < s \leq t_{i+1})$ a *simple process* (“non-anticipatory” in the lecture) and write $I_t(f) = \sum_i a_i (B_{t \wedge t_{i+1}} - B_{t \wedge t_i})$ for the integral up to time t .

Proposition 17.2 (Simple integrands: mean, martingale, isometry). *Let f be a simple process. Then $I_t(f)$ is linear in f , and*

- (i) $\{I_t(f)\}_{t \leq T}$ is a martingale, in particular $\mathbb{E}[I_t(f)] = 0$;
- (ii) (Itô isometry) $\mathbb{E}[I_T(f)^2] = \mathbb{E} \int_0^T f(s)^2 \, ds = \sum_i \mathbb{E}[a_i^2] (t_{i+1} - t_i)$; more generally $\mathbb{E}[(I_t - I_s)^2 | \mathcal{F}_s] = \mathbb{E}[\int_s^t f^2 \, du | \mathcal{F}_s]$.

The isometry is the identity that carries everything that follows: *the variance of a stochastic integral is the integrated mean square of the integrand*. It is the quadratic variation $(dB)^2 = dt$ at work: the cross terms cancel because increments are independent, and each squared increment is replaced by its mean Δt_i .

Proof. Write $\Delta_i B = B_{t_{i+1}} - B_{t_i}$, independent of $\mathcal{F}_{t_i}$ with mean 0 and variance $\Delta t_i = t_{i+1} - t_i$. (i) Take $s < t$; refining the grid we may assume s and t are grid points. Then $I_t - I_s = \sum_{s \leq t_i < t} a_i \Delta_i B$ and, for each such i , by the tower property, $\mathbb{E}[a_i \Delta_i B | \mathcal{F}_s] = \mathbb{E}[a_i \mathbb{E}[\Delta_i B | \mathcal{F}_{t_i}] | \mathcal{F}_s] = 0$, because a_i is $\mathcal{F}_{t_i}$ -measurable and $\Delta_i B$ is independent of $\mathcal{F}_{t_i}$ with mean 0 (this is the answer given in the lecture to a student who asked how the mean can vanish when a_i is random [► 2024 L24 @ 35:04]). (The products are integrable since $\mathbb{E}[|a_i \Delta_i B|] \leq (\mathbb{E}[a_i^2] \Delta t_i)^{1/2}$.) (ii) Expand $I_T^2 = \sum_i a_i^2 (\Delta_i B)^2 + 2 \sum_{i < j} a_i \Delta_i B a_j \Delta_j B$. For $i < j$ the factor $a_i \Delta_i B a_j$ is $\mathcal{F}_{t_j}$ -measurable and $\Delta_j B$ is independent of it with mean 0, so the cross terms have mean 0. For the diagonal terms a_i and $\Delta_i B$ are independent, so $\mathbb{E}[a_i^2 (\Delta_i B)^2] = \mathbb{E}[a_i^2] \Delta t_i$. The conditional version is the same computation given $\mathcal{F}_s$. □

17.3 The Itô integral in general

17.3.1 Construction (not given in class)

The lecture states the definition and refers to Steele's book, chapter 6, for "some very strong mathematical arguments", adding that it is "not really a fun read" [► 2024 L24 @ 43:35]. The idea is short enough to record, because it is what the isometry is for. Let $\mathcal{H}^2$ be the set of processes $f(\omega, s)$, $0 \leq s \leq T$, that are adapted ($f(\cdot, s)$ is $\mathcal{F}_s$ -measurable for every s) and satisfy

$$\|f\|^2 := \mathbb{E} \int_0^T f(\omega, s)^2 ds = \int_{\Omega} \left[\int_0^T f(\omega, s)^2 ds \right] d\mathbb{P}(\omega) < \infty \quad (17.5)$$

[► 2024 L24 @ 45:40]. This is the squared norm of $L^2(d\mathbb{P} \times dt)$, the space of integrands. The steps are:

1. *Approximation.* Every $f \in \mathcal{H}^2$ is the limit, in this norm, of simple processes f_n (for a continuous adapted bounded f take $f_n = \sum f(t_i) \mathbf{1}(t_i < s \leq t_{i+1})$ on finer and finer grids and use bounded convergence; the general case is a density argument).
2. *Cauchy property.* By linearity and [proposition 17.2\(ii\)](#), $\mathbb{E}[(I(f_n) - I(f_m))^2] = \|f_n - f_m\|^2 \rightarrow 0$: the integrals $I(f_n)$ form a Cauchy sequence in $L^2(d\mathbb{P})$.
3. *Definition.* $I(f) := \lim_n I(f_n)$ in $L^2(d\mathbb{P})$; the limit does not depend on the approximating sequence (interlace two sequences and use step 2).

This is why the isometry $\|I(f)\|_{L^2(d\mathbb{P})} = \|f\|_{L^2(d\mathbb{P} \times dt)}$ (the slide writes the two *squared* norms, and $\text{Var } I(f) = \|I(f)\|^2$) is highlighted in the lecture: it is an isometry between the space of integrands and the space of integrals, it "formalises the convergence of sequences of Itô integrals" and lets one speak about closed subspaces and limits [► 2024 L24 @ 51:02] [► 2013 L18 @ 49:10]. The integral up to $t < T$ is the integral of the truncated integrand $f \mathbf{1}(0 \leq s \leq t)$ [► 2024 L24 @ 46:46].

Theorem 17.4 (Properties of the Itô integral). Let $f, g \in \mathcal{H}^2$ and $X_t = \int_0^t f(\omega, s) dB_s$.

- (i) *Linearity:* $\int (af + bg) dB = a \int f dB + b \int g dB$.
- (ii) *Martingale:* $\{X_t\}$ is a martingale with respect to $\{\mathcal{F}_t\}$ (it has a continuous version), $\mathbb{E}[X_t] = 0$ and $\mathbb{E}[X_{t+k} | \mathcal{F}_t] = X_t$ [► 2024 L24 @ 47:51] [► 2013 L18 @ 52:22].
- (iii) *Isometry:* $\mathbb{E}[X_t^2] = \mathbb{E} \int_0^t f^2 ds$ and $\text{Var}[X_{t+k} | \mathcal{F}_t] = \mathbb{E}[\int_t^{t+k} f^2 ds | \mathcal{F}_t]$ [► 2024 L24 @ 48:57].
- (iv) *Deterministic integrands are Gaussian:* if $f = f(s)$ is a deterministic (continuous) function, $\{X_t\}$ is a mean-zero Gaussian process with independent increments and covariance

$$\text{Cov}(X_s, X_t) = \int_0^{s \wedge t} f(u)^2 du$$

[► 2024 L24 @ 56:28] [► 2013 L18 @ 46:59].

The sum $I(f) = \sum a_i \Delta_i B$ of a *deterministic* integrand is a sum of independent normals, and the theorem says that this survives in the limit: the 2013 lecture presents it as "the sum of normal variables is normal" [► 2013 L18 @ 46:59]. For a *random* adapted integrand X_t is a martingale with a known variance but in general *not* Gaussian; the standard example is $\int_0^t B dB = \frac{1}{2}(B_t^2 - t)$ ([section 17.4](#)), which is a shifted χ^2 variable, skewed to the right.

Example 17.3 (The step approximation of $\int B dB$). Take the grid $t_i = iT/n$ and $a_i = B_{t_i}$ (adapted, with $\mathbb{E}[a_i^2] = t_i$). Then $I = \sum_{i=0}^{n-1} B_{t_i}(B_{t_{i+1}} - B_{t_i})$ has mean 0 and, by the isometry,

$$\text{Var } I = \sum_i \mathbb{E}[B_{t_i}^2] \Delta t_i = \sum_{i=0}^{n-1} \frac{iT}{n} \cdot \frac{T}{n} = \frac{T^2(n-1)}{2n} \xrightarrow{n \rightarrow \infty} \frac{T^2}{2}$$

[► 2024 L24 @ 33:58]. The limit $n \rightarrow \infty$ is the lecture's fourth case and is the subject of [section 17.4](#).

Proof. (i)–(iii) hold for simple processes ([proposition 17.2](#)), and both sides are continuous for L^2 -limits: L^2 -limits preserve expectations, second moments and conditional expectations ($\mathbb{E}[\cdot | \mathcal{F}_s]$ is a contraction on L^2). (iv) For a step function $X_t = \sum a_i(B_{t \wedge t_{i+1}} - B_{t \wedge t_i})$ is a linear combination of independent Gaussian increments, hence Gaussian, and L^2 -limits of Gaussian variables are Gaussian (characteristic functions converge). The covariance follows from the isometry, $\text{Cov}(X_s, X_t) = \mathbb{E}[X_s(X_s + (X_t - X_s))] = \mathbb{E}[X_s^2]$ for $s \leq t$ because the increment $X_t - X_s$ is orthogonal to X_s . □

17.3.2 The Riemann-sum representation and the left point

Theorem 17.5 (Riemann representation). For a continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ and the partitions $t_i = iT/n$ of $[0, T]$,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(B_{t_{i-1}}) (B_{t_i} - B_{t_{i-1}}) = \int_0^T f(B_s) dB_s$$

in probability [► 2024 L24 @ 55:23] [► 2013 L18 @ 33:47].

Sketch. The sum is $I(f_n)$ for the simple process $f_n(s) = f(B_{t_{i-1}})$ on $(t_{i-1}, t_i]$. If f is bounded, $f_n(s) \rightarrow f(B_s)$ for every s by continuity of the path, so $\|f_n - f(B)\|^2 \rightarrow 0$ by bounded convergence and $I(f_n) \rightarrow I(f(B))$ in L^2 by the isometry. For unbounded f one truncates f where the path exceeds a level and lets the level grow: the paths are bounded on $[0, T]$, so the truncation is inactive with probability close to one (hence convergence in probability only). $\square$

In ordinary Riemann integration it does not matter which point of each small interval is used; *here it does*. Evaluating at the right end, or at the middle, gives different limits, because $\sum (\Delta B)^2$ does not vanish [► 2013 L18 @ 35:48]; the explicit case is in [section 17.4](#). The lecture also mentions the general statement that any intermediate point $t_i^* \in [t_{i-1}, t_i]$ gives the same limit for a deterministic continuous f (*Gaussian integrals*, slide 12, after Steele) [► 2024 L24 @ 57:30]; this is right, because for deterministic f the difference between two choices is a sum of independent terms of variance $\sum (f(t_i^*) - f(t_{i-1}))^2 \Delta t_i \rightarrow 0$.

IN FINANCE: The left point is the no-lookahead rule

The Itô rule is the mathematical form of “you cannot see the future”: the position over $(t_i, t_{i+1}]$ is decided at t_i . The 2013 lecture says that this intuition is *built into the theory*, which is why it fits finance so well; the same sum with the position taken from the right end would use the price move itself to decide how much to hold [► 2013 L18 @ 37:01]. The martingale property of [theorem 17.4\(ii\)](#) is the statement that the gains of an adapted strategy in a fair game (a martingale price) are themselves a fair game, the continuous-time version of the martingale transform of [theorem 4.21](#).

Remark 17.6 (Integrability conditions; not discussed in class). The last slide of L24 defines a *standard process* $X_t = x_0 + \int_0^t a(\omega, s) ds + \int_0^t b(\omega, s) dB_s$ with a, b adapted and $\mathbb{P}(\int_0^t |a| ds < \infty) = \mathbb{P}(\int_0^t b^2 ds < \infty) = 1$, “otherwise the integrals may explode” [► 2024 L24 @ 1:21:11]. The weaker condition $\int_0^T b^2 ds < \infty$ almost surely (instead of finite expectation) still defines $\int b dB$ (by stopping when $\int b^2$ exceeds a level), but the result is then only a *local* martingale: mean zero and the isometry may fail. Whenever we assert a martingale property below we check $\mathbb{E} \int f^2 < \infty$.

17.4 The basic computation: $\int B dB$

Classical calculus would give $\int_0^T B dB = \frac{1}{2} B_T^2$. The lecture’s fourth case computes the limit of the step approximations of [example 17.3](#) and finds something else [► 2024 L24 @ 37:10].

Theorem 17.7 (Ito integral of Brownian motion). For every $T > 0$,

$$\int_0^T B_s dB_s = \frac{1}{2} B_T^2 - \frac{1}{2} T \quad \text{almost surely.} \quad (17.6)$$

Proof. For a partition $0 = t_0 < \dots < t_n = T$ use the algebraic identity, “trivial but incredibly useful” in the lecture,

$$B_{t_i} (B_{t_{i+1}} - B_{t_i}) = \frac{1}{2} (B_{t_{i+1}}^2 - B_{t_i}^2) - \frac{1}{2} (B_{t_{i+1}} - B_{t_i})^2,$$

which is $b(c-b) = \frac{1}{2}(c^2-b^2) - \frac{1}{2}(c-b)^2$. Summing

So the correction $-\frac{1}{2}T$ “drags” on the integral because of the quadratic variation [► 2024 L24 @ 42:33]; (17.6) also appears in 2013 as the example of a ‘violation’ of the fun-

damental theorem of calculus [► 2013 L18 @ 18:49]. Checks that do not need the proof: the mean is $\mathbb{E}[\frac{1}{2}(B_T^2 - T)] = 0$, as required for a martingale (theorem 17.4(ii)); the variance is $\frac{1}{4} \text{Var}(B_T^2) = \frac{1}{4} \cdot 2T^2 = \frac{1}{2}T^2$, which is the isometry value $\int_0^T s \, ds$ of example 17.3 [► 2024 L24 @ 35:04]; the random variable $\frac{1}{2}(B_T^2 - T)$ is a shifted, rescaled χ_1^2 , hence *not* Gaussian (skewness $2\sqrt{2} \approx 2.83$). The 2013 note verifies the martingale property directly by computing $\mathbb{E}[\int_{t_1}^{t_2} B \, dB \mid \mathcal{F}_{t_1}] = 0$ (lecnote18, Example 2.4).

Corollary 17.8 (Used in the Dickey–Fuller distribution). *For a standard Brownian motion W , $\int_0^1 W \, dW = \frac{1}{2}(W(1)^2 - 1)$. This is the numerator of the limiting distribution of the unit-root estimator $T(\hat{\phi} - 1)$ in section 9.4.2 (there obtained from the discrete telescoping identity $X_t^2 = X_{t-1}^2 + 2X_{t-1}\eta_t + \eta_t^2$, the finite- n version of the proof above).*

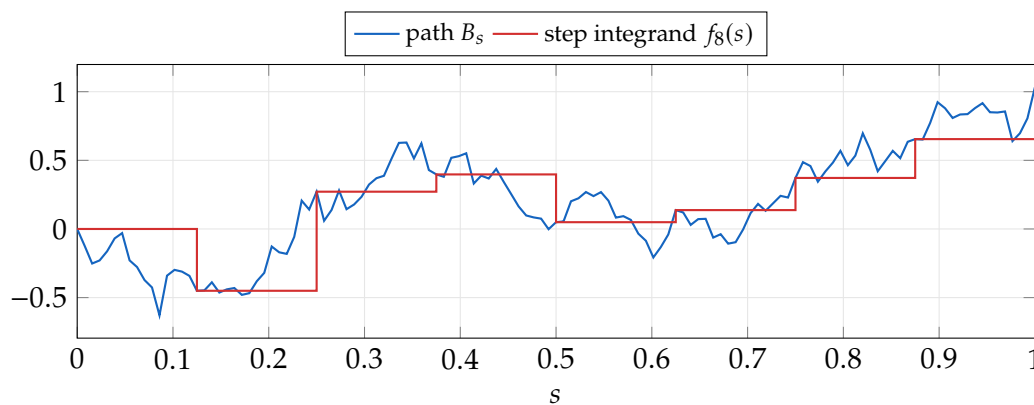


Figure 17.1: The step approximation of example 17.3: a simulated path of B on $[0, 1]$ (blue) and the adapted step integrand $f_8(s) = B_{t_i}$ on $(t_i, t_{i+1}]$ (red, $n = 8$). Each step is frozen at the value of the path at its *left* end.

17.4.1 Left, middle and right points: Itô, Stratonovich, backward

Evaluate the integrand at another point of each interval and the limit changes; for $\int B \, dB$ this can be done exactly. With $\Delta_i B = B_{t_{i+1}} - B_{t_i}$:

$$\begin{aligned} \text{left point (Itô):} \quad & \sum_i B_{t_i} \Delta_i B = \frac{1}{2} B_T^2 - \frac{1}{2} \sum_i (\Delta_i B)^2 \rightarrow \frac{1}{2} B_T^2 - \frac{1}{2} T, \\ \text{mid-point (Stratonovich):} \quad & \sum_i \frac{1}{2} (B_{t_i} + B_{t_{i+1}}) \Delta_i B = \frac{1}{2} \sum_i (B_{t_{i+1}}^2 - B_{t_i}^2) = \frac{1}{2} B_T^2 \quad \text{exactly,} \\ \text{right point (backward):} \quad & \sum_i B_{t_{i+1}} \Delta_i B = \frac{1}{2} B_T^2 + \frac{1}{2} \sum_i (\Delta_i B)^2 \rightarrow \frac{1}{2} B_T^2 + \frac{1}{2} T. \end{aligned}$$

The three limits differ by $\pm \frac{1}{2}T$, the quadratic variation, fig. 17.2. The mid-point rule, the *Stratonovich* integral $\int B \circ dB = \frac{1}{2} B_T^2$, obeys the ordinary rules of calculus (chain rule without the second-order term); it is the limit of physically smooth noise, which is why it is the convention of physicists who start from a Langevin equation with a finite correlation time. The price is that it is not a martingale and the integrand looks into the future: the integrand is evaluated at a point that is not yet known. The right-point sums are the “equivalent version” of the 2013 remark, in which one gets $df = f'(B) dB - \frac{1}{2} f''(B) dt$, i.e. formally $(dB)^2 = -dt$ [► 2013 L18 @ 37:01]; this is correct once the integral is understood as the

over i the first term telescopes:

$$\sum_{i=0}^{n-1} B_{t_i} (B_{t_{i+1}} - B_{t_i}) = \frac{1}{2} B_T^2 - \frac{1}{2} \sum_{i=0}^{n-1} (B_{t_{i+1}} - B_{t_i})^2$$

[► 2024 L24 @ 40:21]. By theorem 11.8 the last sum, the quadratic variation of the partition, tends to T as the mesh goes to zero (in L^2 , hence in probability; almost surely for dyadic partitions), and by theorem 17.5 the left side tends to $\int_0^T B \, dB$ [► 2024 L24 @ 42:33]. $\square$

backward one (from the identity above, $\int B \, d^{\text{back}} B = \frac{1}{2} B^2 + \frac{1}{2} t$, so $d(\frac{1}{2} B^2) = B \, d^{\text{back}} B - \frac{1}{2} dt$). The lecture calls it “just cool stuff” and moves on; Itô’s is the convention for finance.

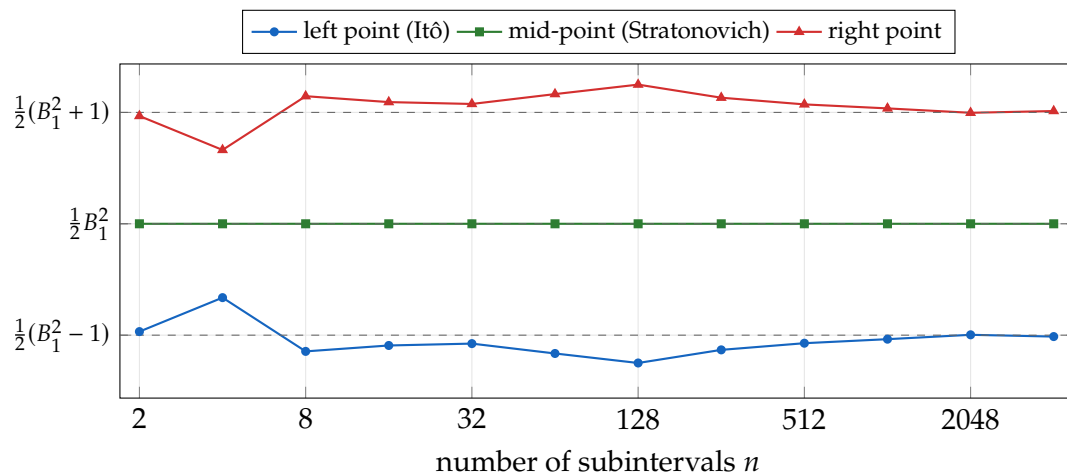


Figure 17.2: One simulated Brownian path on $[0, 1]$ with $B_1 = 1.032$, discretised at n equally spaced points ($n = 2, \dots, 4096$, sub-sampled from a single fine path). The left-point sums converge (slowly, at rate $n^{-1/2}$) to the Itô value $\frac{1}{2}(B_1^2 - 1) = 0.032$, the mid-point sums are *exactly* the Stratonovich value $\frac{1}{2}B_1^2 = 0.532$, and the right-point sums tend to $\frac{1}{2}(B_1^2 + 1) = 1.032$; the three limits are the tick labels of the vertical axis.

Exercise 17.2 (PS8 (2013), Problem B-2: Itô isometry). Use the Itô isometry to calculate $\mathbb{E}[X_t^2]$, where $X_t = \int_0^t B_s \, dB_s$.

Exercise 17.3 (not from the course: Stratonovich). Show that the mid-point sums $\sum_i \left[\frac{1}{2}(B_{t_i} + B_{t_{i+1}}) \right]^2 \Delta_i B$ converge to $\frac{1}{3} B_T^3$ as the mesh tends to zero, and compare with the left-point (Itô) limit $\int_0^T B^2 \, dB = \frac{1}{3} B_T^3 - \int_0^T B_s \, ds$ from (17.11). (Hint: compare each term with $\frac{1}{3}(B_{t_{i+1}}^3 - B_{t_i}^3)$.)

17.5 Itô’s lemma

17.5.1 One variable

Theorem 17.9 (Itô’s formula for $f(B_t)$). If $f : \mathbb{R} \rightarrow \mathbb{R}$ has a continuous second derivative, then for every $t \geq 0$

$$f(B_t) = f(0) + \int_0^t f'(B_s) \, dB_s + \frac{1}{2} \int_0^t f''(B_s) \, ds, \quad (17.7)$$

written in shorthand $df(B_t) = f'(B_t) \, dB_t + \frac{1}{2} f''(B_t) \, dt$.

[▶ 2024 L24 @ 57:30][▶ 2024 L25 @ 0:00][▶ 2013 L18 @ 0:00] The first integral is a stochastic integral (mean zero, a martingale when $\mathbb{E} \int f'(B)^2 < \infty$), the second an ordinary Riemann integral: a random *adapted drift* [▶ 2024 L24 @ 58:36]. It is the chain rule with one extra term, the *Itô correction* $\frac{1}{2} f'' \, dt$. (Both years also record the classical formula, obtained if dB/dt existed, $df = f'(B) \frac{dB}{dt} dt$, which “makes no sense”, and the attempt $df = f' \, dB$ that “at least

makes sense" but "does not quite work" [lecnote18](#), §1.)

INTUITION: Where the extra term comes from

Expand $f(B + \Delta B) - f(B) = f' \Delta B + \frac{1}{2} f'' \Delta B^2 + \frac{1}{6} f''' \Delta B^3 + \dots$. Since $\Delta B \sim \sqrt{\Delta t}$, the terms are of order $\Delta t^{1/2}, \Delta t, \Delta t^{3/2}, \dots$: after summing $1/\Delta t$ steps, the first one is a random martingale, the second contributes a finite drift (its fluctuations are negligible, as shown above), and all others vanish. This is the 2013 argument: " dt^2 and $dt dB$ are insignificant, the only term that matters is $dB^2 = dt$ ", with the heuristic $dB \approx \sqrt{dt}$ [[2013 L18 @ 4:34](#)]. A physicist recognises it: taking expectations kills the martingale, $d\mathbb{E}[f(B_t)]/dt = \frac{1}{2}\mathbb{E}[f''(B_t)]$, i.e. $\frac{1}{2}\partial_x^2$ is the generator of Brownian motion and the density of B_t solves the diffusion equation of [proposition 11.4](#). Itô's formula is the pathwise version of the heat equation.

17.5.2 Functions of time and space; the multiplication table

For $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$ (continuous derivatives f_t, f_x, f_{xx}) the Taylor expansion has the second-order terms $\frac{1}{2}f_{tt}\Delta t^2 + f_{tx}\Delta t\Delta B + \frac{1}{2}f_{xx}\Delta B^2$; only the last one survives [[2024 L24 @ 1:06:15](#)] [[2024 L25 @ 1:03](#)][[2013 L18 @ 2:17](#)]. We summarise this in the *multiplication table* of Itô calculus

$$(dB)^2 = dt, \quad dB dt = 0, \quad (dt)^2 = 0, \quad (17.8)$$

(the same as [eq. \(11.3\)](#); the products $dB dt \sim dt^{3/2}$ and dt^2 are negligible against dt).

Theorem 17.10 (Itô's formula for $f(t, B_t)$). For $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$,

$$f(t, B_t) - f(0, 0) = \int_0^t \left[f_t + \frac{1}{2}f_{xx} \right](s, B_s) ds + \int_0^t f_x(s, B_s) dB_s, \quad (17.9)$$

i.e. $df = (f_t + \frac{1}{2}f_{xx}) dt + f_x dB$.

[[2024 L24 @ 1:08:20](#)][[2013 L18 @ 7:37](#)] The proof is that of [theorem 17.9](#) with the Taylor expansion in two variables (the additional terms $f_t \Delta t$ sum to a Riemann integral). The 2013 lecture notes that the second variable is just a placeholder: one writes f_x for the derivative in the second argument and then substitutes $x = B_t$ [[2013 L18 @ 9:48](#)].

Theorem 17.11 (Itô's formula for standard processes). Let $dX_t = a_t dt + b_t dB_t$, with a, b adapted and $\int_0^t |a| ds < \infty$, $\int_0^t b^2 ds < \infty$ almost surely (a standard process, [remark 17.6](#)). For $f \in C^{1,2}$,

$$df(t, X_t) = f_t dt + f_x dX_t + \frac{1}{2}f_{xx} (dX_t)^2 = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx} \right) dt + b_t f_x dB_t, \quad (17.10)$$

where $(dX_t)^2 = b_t^2 dt$ by (17.8).

[[2024 L24 @ 1:21:11](#)][[2024 L25 @ 0:00](#)][[2013 L18 @ 11:53](#)]

The lecture writes the last result for a "Markov" process $dX = \mu(X_t, t) dt + \sigma(X_t, t) dB_t$ and $Y_t = G(X_t, t)$ as

$$dG = \left[\mu G_x + G_t + \frac{1}{2}\sigma^2 G_{xx} \right] dt + \sigma G_x dB_t$$

[[2024 L24 @ 1:22:14](#)][[2024 L25 @ 5:16](#)]; this is (17.10) with $a = \mu(X, t)$, $b = \sigma(X, t)$.

Proof (sketch with the error control; the lecture only Taylor-expansions)
Take a partition $t_i = it/n$ and telescope, $f(B_t) - f(0) = \sum_i [f(B_{t_{i+1}}) - f(B_{t_i})]$ [[2024 L24 @ 59:45](#)]. Taylor's theorem with the Lagrange remainder gives, with $\Delta_i B = B_{t_{i+1}} - B_{t_i}$ and some ξ_i between B_{t_i} and $B_{t_{i+1}}$,

$$f(B_{t_{i+1}}) - f(B_{t_i}) = f'(B_{t_i})\Delta_i B + \frac{1}{2}f''(B_{t_i})(\Delta_i B)^2 + \frac{1}{6}f'''(\xi_i)(\Delta_i B)^3$$

Sum over i and treat the three sums. (i) $\sum f'(B_{t_i})\Delta_i B \rightarrow \int_0^t f'(B) dB$ by [theorem 17.5](#).

(ii) Write $(\Delta_i B)^2 = \Delta t_i + [(\Delta_i B)^2 - \Delta t_i]$. The sum $\frac{1}{2}\sum f''(B_{t_i})\Delta t_i$ is a Riemann sum of the continuous function $s \mapsto f''(B_s)$ and tends to $\frac{1}{2}\int_0^t f''(B_s) ds$. The fluctuation $D_n = \frac{1}{2}\sum f''(B_{t_i})[(\Delta_i B)^2 - \Delta t_i]$ is a sum of martingale differences ($\Delta_i B$ is independent of $\mathcal{F}_{t_i}$ and $\mathbb{E}[(\Delta_i B)^2 - \Delta t_i] = 0$), so for bounded f''

$$\begin{aligned} \mathbb{E}[D_n^2] &= \frac{1}{4} \sum_i \mathbb{E}[f''(B_{t_i})^2] \text{Var}[(\Delta_i B)^2] = \frac{1}{2} \sum_i \mathbb{E}[f''(B_{t_i})^2] \Delta t_i \\ &\leq \frac{1}{2} \|f''\|_\infty^2 t \max_i \Delta t_i \rightarrow 0, \end{aligned}$$

using $\text{Var}[(\Delta_i B)^2] = 2\Delta t_i^2$. This is the point where " $(dB)^2 = dt$ " enters: the squared increment can be replaced by its mean because the difference has variance of order Δt^2 , whose sum is negligible.

(iii) The remainder is bounded by $\frac{1}{6}\omega(\max_i |\Delta_i B|) \sum (\Delta_i B)^2$, where ω is the modulus of continuity of f'' on the (compact) range of the path; the sum is bounded in probability (it converges to t) and $\max_i |\Delta_i B| \rightarrow 0$ by uniform continuity of the path, so the remainder tends to 0 in probability. For unbounded f'' one stops the path when it leaves a large interval. Collecting the three limits gives (17.7). $\square$

Idea. As above, $df = f_t dt + f_x dX + \frac{1}{2}f_{xx}(dX)^2$, and substituting $dX = a dt + b dB$, $(dX)^2 = a^2 dt^2 + 2ab dt dB + b^2 dB^2 = b^2 dt$: only $b^2 dB^2$ survives, the extra term is $\frac{1}{2}b^2 f_{xx} dt$. The 2013 lecture stresses that "it all started from one fact, the quadratic variation" and that the formula looks complicated only be-

17.5.3 Integrals by anti-derivatives

Solving (17.7) for the stochastic integral gives the lecture's method to compute $\int f(B_s) dB_s$: if $F' = f$ (choose $F(0) = 0$), then

$$\int_0^t f(B_s) dB_s = F(B_t) - \frac{1}{2} \int_0^t f'(B_s) ds \quad (17.11)$$

[► 2024 L24 @ 1:01:58][► 2024 L25 @ 1:03]. For $F(x) = x$ ($f = 1, f' = 0$) it returns B_t ; for $F(x) = \frac{1}{2}x^2$ it returns $\frac{1}{2}B_t^2 - \frac{1}{2}t$, the result of [section 17.4](#) [► 2024 L24 @ 1:05:13]. Another example (ours): for $F = f = e^x - 1$,

$$\int_0^t e^{B_s} dB_s = e^{B_t} - 1 - \frac{1}{2} \int_0^t e^{B_s} ds,$$

an exact identity between a martingale and a functional of the path, checked pathwise in a simulation below.

Remark 17.12 (Stratonovich form; not discussed in class). If one wants a chain rule without the second-order term, use the mid-point integral. The two are related by $\int \sigma(X) \circ dB = \int \sigma(X) dB + \frac{1}{2} \int \sigma'(X)\sigma(X) dt$ (for X a solution of the Itô equation with volatility σ): for $\sigma(x) = x, X = B$, this is $\int B \circ dB = \int B dB + \frac{1}{2}t$, which is the pair of formulas of [section 17.4.1](#). Consequently a Langevin equation with multiplicative noise has a *noise-induced drift* $\frac{1}{2}\sigma\sigma'$ if read in Itô's convention and none in Stratonovich's; in finance the Itô reading is used because prices must be martingales when there is no drift.

Exercise 17.4 (PS8 (2013), Problem A-4: Itô's formula). Use Itô's formula to compute the differentials of the following (B_t is a Brownian motion):

- (a) $f(t, B_t) = B_t^3$;
- (b) $\sin B_t$;
- (c) $\cos(t^3 + B_t^2)$;
- (d) $e^{B_t^2}$;
- (e) $f(t, B_t) = \int B_t^2 dB_t$;
- (f) $f(t, B_t) = \int B_t dt$;
- (g) $f(t, X_t) = X_t^2$, where $dX_t = \mu dt + \sigma dB_t$ with μ, σ constants. (Parts (e) and (f) are printed as indefinite integrals; interpret them as the processes $\int_0^t B_s^2 dB_s$ and $\int_0^t B_s ds$.)

Exercise 17.5 (PS8 (2013), Problem B-3: integration by parts).

- (a) Prove the integration by parts formula for the Itô integral, $\int_0^t h(s) dB_s = h(t)B_t - \int_0^t h'(s)B_s ds$ (for a deterministic differentiable h).
- (b) Using (a), prove that for each fixed T the random variable $\int_0^T B_s ds$ is normally distributed.

cause one forgets where the terms come from [► 2013 L18 @ 14:58]. The note's proof (Theorem 1.1) has a stray "... $dt dB_T$ " for the discarded terms, which are $ab f_{xx} dt dB$ and $\frac{1}{2}a^2 f_{xx}(dt)^2$. The rigorous version needs the same estimates as in [theorem 17.9](#) with $f''(B_{t_i})$ replaced by $f_{xx}(t_i, X_{t_i})b_{t_i}^2$. □

1 COURSE NOTE: Erratum: the Itô integral is not a Gaussian process

Slide 13 of [lec19_1](#) (and the spoken commentary [► 2024 L24 @ 58:36]) describes $\int_0^t f'(B_s) dB_s$ in (17.7) as a "zero-mean adaptive Gaussian process". Only the zero mean (indeed the martingale property) is correct; for a random integrand the integral is in general not Gaussian, as the example $\int_0^t B dB = \frac{1}{2}(B_t^2 - t)$ shows. Gaussianity holds only for *deterministic* integrands ([theorem 17.4\(iv\)](#)).

2 COURSE NOTE: Erratum: slide 20 and the upper limit

Slide 20 of [lec19_1](#) defines the standard process by $X_t = x_0 + \int_0^t a ds + \int_0^t b dB_s$, with the upper limit T in the stochastic integral; it should be t , as in the first integral and as in the definition on slide 9.

17.6 Examples: martingales, moments, exponentials

17.6.1 Martingales from the PDE condition

Equation (17.9) shows at once when $f(t, B_t)$ is a martingale: the dt -term must vanish.

Theorem 17.17 (PDE condition for martingales). Let $f \in C^{1,2}$ satisfy the backward heat equation

$$f_t + \frac{1}{2} f_{xx} = 0, \quad (17.14)$$

and suppose $\mathbb{E} \int_0^T (f_x(s, B_s))^2 ds < \infty$. Then $M_t = f(t, B_t) = f(0, 0) + \int_0^t f_x dB$ is a (square-integrable) martingale with $\mathbb{E}[|M_T - M_0|^2] = \mathbb{E} \int_0^T f_x^2 ds$.

[► 2024 L24 @ 1:10:26][► 2024 L25 @ 2:06][► 2013 L18 @ 54:35] This is the connection “martingales $\leftrightarrow$ PDEs” of the lecture: a solution of the PDE gives a martingale, and conversely the martingale condition reads $f_t = -\frac{1}{2} f_{xx}$ [► 2024 L24 @ 1:11:31]. Both years extract the general principle: in an SDE written as “something $\times dt$ plus something $\times dB$ ”, the first term is the drift (the trend) and the second the noise; a process with no dt -term is a martingale, a process with a drift is not [► 2013 L18 @ 54:35]. The 2013 lecture points at the use in pricing: a hedged portfolio with no noise term must have zero drift (relative to the risk-free rate), otherwise it would be an arbitrage [► 2013 L18 @ 56:45], made precise in chapter 15.

Examples (checked by direct differentiation): $f = x$; $f = x^2 - t$ ($f_t = -1$, $f_{xx} = 2$); $f = x^3 - 3tx$ ($f_t = -3x$, $f_{xx} = 6x$); $f = x^4 - 6tx^2 + 3t^2$ ($f_t = -6x^2 + 6t$, $f_{xx} = 12x^2 - 12t$); in general $f = t^{n/2} H_n(x/\sqrt{t})$ with the Hermite polynomials (own remark). And the exponential martingale, the lecture’s Example 4: $f(t, x) = e^{\sigma x - \sigma^2 t/2}$ has $f_t = -\frac{1}{2} \sigma^2 f$ and $\frac{1}{2} f_{xx} = \frac{1}{2} \sigma^2 f$, so (17.14) holds; and $\mathbb{E} \int_0^t \sigma^2 f(s, B_s)^2 ds = \sigma^2 \int_0^t e^{\sigma^2 s} ds < \infty$ ($\mathbb{E}[e^{2\sigma B_s - \sigma^2 s}] = e^{\sigma^2 s}$), so $f(t, B_t)$ is a martingale with $\mathbb{E}[f(t, B_t)] = f(0, 0) = 1$ for all t [► 2024 L24 @ 1:12:36]. The variable $f(t, B_t) = e^{\sigma B_t} e^{-\sigma^2 t/2}$ is a scaled lognormal, and $\mathbb{E}[e^{\sigma B_t}] = e^{\sigma^2 t/2}$ (moment generating function, section 3.3; it was a problem-set question in the class) [► 2024 L24 @ 1:14:45].

Remark 17.18 (The integrability condition matters; not discussed in class). A solution of (17.14) with a stochastic integral that is only a local martingale, i.e. with $\mathbb{E} \int f_x^2 = \infty$, need not be a martingale. The condition of the theorem is the analogue for continuous time of the bounded-increments condition in optional stopping (section 4.5).

17.6.2 Brownian motion with drift and ruin

Let $X_t = \mu t + \sigma B_t$ and $\tau = \inf\{t : X_t = A \text{ or } X_t = -B'\}$ with $A, B' > 0$ (the lecture writes B for the lower barrier; we write B' to avoid a clash with the Brownian motion). We want $\mathbb{P}(X_\tau = A)$ [► 2024 L24 @ 1:15:46]. Look for h such that $M_t = h(X_t)$ is a martingale, so that the stopped process $M_{t \wedge \tau}$ is one too, with boundary values $h(A) = 1$, $h(-B') = 0$. Then $\mathbb{P}(X_\tau = A) = \mathbb{E}[M_\tau] = M_0 = h(0)$ [► 2024 L24 @ 1:16:48]. By (17.10), $dh(X_t) = (\mu h' + \frac{1}{2} \sigma^2 h'')(X_t) dt + \sigma h'(X_t) dB_t$, so the condition is $\mu h' + \frac{1}{2} \sigma^2 h'' = 0$; the lecture obtains it through $f(t, x) = h(\mu t + \sigma x)$ and (17.14), where $f_t = \mu h'$ and $f_{xx} = \sigma^2 h''$ [► 2024 L24 @ 1:19:04]. It is a first-order equation for h' , $h'' = -(2\mu/\sigma^2)h'$, whose solution with the boundary conditions is

$$h(y) = \frac{e^{-2\mu y/\sigma^2} - e^{2\mu B'/\sigma^2}}{e^{-2\mu A/\sigma^2} - e^{2\mu B'/\sigma^2}}, \quad \mathbb{P}(X_\tau = A) = h(0)$$

[► 2024 L24 @ 1:21:11]. This is the formula (11.11) of section 11.5.1 with $a = -B'$, $b = A$ and starting point 0; there it was obtained by the infinitesimal argument and by optional stopping applied to $e^{-2\mu X/\sigma^2}$, which is the exponential martingale (17.13) with $\theta = -2\mu/\sigma^2$.

Example 17.13 (The square). $f(x) = x^2$: $d(B_t^2) = 2B_t dB_t + dt$ [► 2013 L18 @ 18:49] [► 2024 L25 @ 1:03]. Hence $B_t^2 - t = 2 \int_0^t B dB$ is a martingale and $\mathbb{E}[B_t^2] = t$, in agreement with proposition 11.3(iii). For X with $dX = \mu dt + \sigma dB$ the same formula gives $d(X^2) = (2\mu X + \sigma^2) dt + 2\sigma X dB$; the lecture takes $\mu = 0$, $\sigma = 1$, $G = x^2$ and concludes $\int_0^t X dX = \frac{1}{2}(X_t^2 - t)$ [► 2024 L25 @ 1:03].

Example 17.14 (Moments of Brownian motion). For $f(x) = x^n$, $d(B^n) = nB^{n-1} dB + \frac{1}{2}n(n-1)B^{n-2} dt$. Taking expectations (the stochastic integral has mean zero because $\mathbb{E} \int B^{2n-2} ds < \infty$), $m_n(t) = \mathbb{E}[B_t^n]$ satisfies $m_n(t) = \frac{1}{2}n(n-1) \int_0^t m_{n-2}(s) ds$. Starting from $m_2(s) = s$ this gives $m_4(t) = \frac{1}{2} \cdot 4 \cdot 3 \int_0^t s ds = 3t^2$ and $m_6(t) = \frac{1}{2} \cdot 6 \cdot 5 \int_0^t 3s^2 ds = 15t^3$. In general $\mathbb{E}[B_t^{2k}] = (2k-1)!! t^k$, odd moments vanish. A Monte Carlo check with 2×10^4 paths of 1000 steps gave $\mathbb{E}[B_1^4] \approx 2.99$ (theory 3).

Example 17.15 (Exponentials and geometric Brownian motion without drift). Let $f(t, x) = \exp(\mu t + \sigma x)$. Then $f_t = \mu f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$ and

$$df(t, B_t) = (\mu + \frac{1}{2}\sigma^2) f dt + \sigma f dB_t \quad (17.12)$$

[► 2013 L18 @ 22:03]. (The student’s question at that point, why $\frac{1}{2}$ multiplies σ^2 and not σ'^2 : in the general formula (17.10) the σ is the volatility of the argument process X , here $X = B$ has volatility 1; the σ of (17.12) sits inside the function f .) The model question of the 2013 lecture: $S_t = e^{\sigma B_t}$ is not the solution of $dS = \sigma S dB$ (a process with no drift), because by (17.12) with $\mu = 0$ it has drift $\frac{1}{2}\sigma^2 S$ [► 2013 L18 @ 24:18]. The drift is removed by choosing $\mu = -\frac{1}{2}\sigma^2$: the process

$$S_t = \exp(\sigma B_t - \frac{1}{2}\sigma^2 t) \quad \text{satisfies} \quad dS_t = \sigma S_t dB_t, \quad S_0 = 1 \quad (17.13)$$

[► 2013 L18 @ 26:24] [► 2024 L24 @ 1:12:36], and is a martingale with $\mathbb{E}[S_t] = 1$ (both years; the lecture of 2024 obtains it as the case $f(t, x) = e^{\sigma x - \sigma^2 t/2}$ with $f_t = -\frac{1}{2}\sigma^2 f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$). Compare example 4.18: this is its continuous-time limit. In discrete form the statement is the exact identity $S_t - 1 = \sigma \int_0^t S dB$; we checked it pathwise for $\sigma = 1$ with left-point sums: the root-mean-square difference between $S_1 - 1$ and $\sum S_i \Delta_i B$ over 2000 paths is 0.295, 0.092, 0.028, 0.009 for $n = 10, 10^2, 10^3, 10^4$ steps, decreasing like $n^{-1/2}$ (the strong order of the Euler scheme, see chapter 18).

Example 17.16 (Time and a drift: $t^2 + X^2$). Let $dX = \mu dt + \sigma dB$ and $f(t, x) = t^2 + x^2$. Then $f_t = 2t$, $f_x = 2x$, $f_{xx} = 2$ and

$$df(t, X_t) = 2t dt + 2X_t dX_t + (dX_t)^2 = (2t + 2\mu X_t + \sigma^2) dt + 2\sigma X_t dB_t$$

Itô's lemma gives the same result in one line, and shows why $\mu h' + \frac{1}{2}\sigma^2 h'' = 0$ is the right equation: the operator $\mu\partial_x + \frac{1}{2}\sigma^2\partial_x^2$ is the generator of X .

Exercise 17.6 (PS8 (2013), Problem A-1: identify martingales). Which of the following are martingales (with respect to their natural filtration)?

- (a) A simple random walk.
- (b) $X_t = |S_t|$ where S_t is a simple random walk.
- (c) $X_0 = 0$ and $X_{t+1} = X_t + (Y_t - \frac{1}{\lambda})$ for $t \geq 0$, where the Y_t are i.i.d. exponential random variables with parameter λ .
- (d) $X_0 = 0$ and $X_{t+1} = X_t + Z_t Z_{t-1} \cdots Z_0$ for $t \geq 0$, where the Z_t are i.i.d. lognormal.
- (e) $X_t = B_t^{(1)} B_t^{(2)}$, the product of two independent Brownian motions.

Exercise 17.7 (PS8 (2013), Problem A-2). This problem (conditional mean and variance of Brownian motion, of Brownian motion with drift, and $\mathbb{E} \exp(\sigma(B(t) - B(s)))$) is 2024 PS4 Problem 1 and is stated with the other Brownian-motion exercises in [Exercise 11.6](#).

3 COURSE NOTE: Erratum in [lecnote18](#), Example 1.3(iv)

The note writes $df = 2t \, dt + 2X_t \, dX_t + (dX_t)^2 = 2t \, dt + 2(\mu \, dt + \sigma \, dB_t) + \sigma^2 \, dt = (2t + 2\mu + \sigma^2) \, dt + 2\sigma \, dB_t$: the factor X_t in $2X_t \, dX_t$ has been dropped in the second step. The correct result is the one above.

17.7 Geometric Brownian motion

The standard model of a price is that its *relative* change is a Brownian motion with drift,

$$dP_t = \mu P_t \, dt + \sigma P_t \, dB_t, \quad \mu \in \mathbb{R}, \sigma > 0, \quad (17.15)$$

[► 2024 L25 @ 5:16][► 2013 L18 @ 25:22]. Since P_t appears on both sides one is tempted to think that $\ln P_t$ has differential dP/P ; Itô's lemma shows it does not. Apply (17.10) to $G(P, t) = \ln P$, with $G_P = 1/P$, $G_t = 0$, $G_{PP} = -1/P^2$ and $a = \mu P$, $b = \sigma P$:

$$d \ln P_t = \left[\frac{1}{P} \mu P + \frac{1}{2} \left(-\frac{1}{P^2} \right) \sigma^2 P^2 \right] dt + \frac{1}{P} \sigma P \, dB_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma \, dB_t \quad (17.16)$$

[► 2024 L25 @ 6:19][► 2024 L24 @ 1:22:14]. The log price is a Brownian motion with drift $\mu - \frac{1}{2}\sigma^2$ and variance rate σ^2 : the drift on the log scale has a *drag* $-\frac{1}{2}\sigma^2$ due to the quadratic variation [► 2024 L25 @ 6:19]. Integrating,

$$P_t = P_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right]. \quad (17.17)$$

Conversely, (17.17) solves (17.15): apply (17.12) with drift parameter $\mu - \frac{1}{2}\sigma^2$, which gives the drift $(\mu - \frac{1}{2}\sigma^2 + \frac{1}{2}\sigma^2)P = \mu P$. (That there is no other solution is the uniqueness theorem of [chapter 18](#); [eq. \(11.13\)](#) derived the same solution.) Equation (17.13) is the special case $\mu = 0$ and $P_0 = 1$, i.e. the martingale.

17.7.1 Distribution, mean and annualised return

Over $(t, T]$, with $\tau = T - t$, the log return is a sum of independent normal increments,

$$\ln \frac{P_T}{P_t} \sim N \left(\left(\mu - \frac{1}{2} \sigma^2 \right) \tau, \sigma^2 \tau \right), \quad \mathbb{E}[P_T | P_t] = P_t e^{\mu \tau}, \quad \text{Var}[P_T | P_t] = P_t^2 e^{2\mu \tau} (e^{\sigma^2 \tau} - 1) \quad (17.18)$$

[► 2024 L25 @ 7:23], by the lognormal moments ([section 3.3](#)); the slide's " $\text{Var}[P_T | P_t]$ " should read $\text{Var}[P_T | P_t]$. The parameter μ is the *expected rate of return* (mean growth of the

price), while the typical (median) growth rate is $\mu - \frac{1}{2}\sigma^2$. The lecture's numerical example: $\mu = 0.30$ per year, $\sigma = 0.40$, $\tau = \frac{1}{2}$ year. Then

$$\mathbb{E}[P_T | P_t] = P_t e^{0.15} = 1.1618 P_t, \quad \text{sd}[P_T | P_t] = 1.1618 \sqrt{e^{0.08} - 1} P_t = 0.3353 P_t$$

► 2024 L25 @ 9:32: an expected gain of 16.2% rather than the 15% one would obtain by naively halving μ (the convexity of the exponential, Jensen), and a standard deviation larger than a naive reading of σ suggests. The continuously compounded annualised rate of return $r = \frac{1}{\tau} \ln(P_T/P_t)$ satisfies

$$r \sim N\left(\mu - \frac{1}{2}\sigma^2, \frac{\sigma^2}{\tau}\right) \quad (17.19)$$

► 2024 L25 @ 11:37. Hence over long horizons the realised rate of return converges (almost surely, by the strong law $B_T/T \rightarrow 0$) to the constant $\mu - \frac{1}{2}\sigma^2$, and the variance of the average rate goes to zero as $1/\tau$: the growth of the price is “at a constant rate with vanishing variability” ► 2024 L25 @ 12:40. Note that this constant is $\mu - \frac{1}{2}\sigma^2$, *not* μ : if $\mu < \frac{1}{2}\sigma^2$ then $P_t \rightarrow 0$ almost surely although $\mathbb{E}[P_t] = P_0 e^{\mu t}$ grows (own remark; for $\mu = 0.05$, $\sigma = 0.4$ the typical rate is -3% per year). The mean is carried by rare, enormous outcomes.

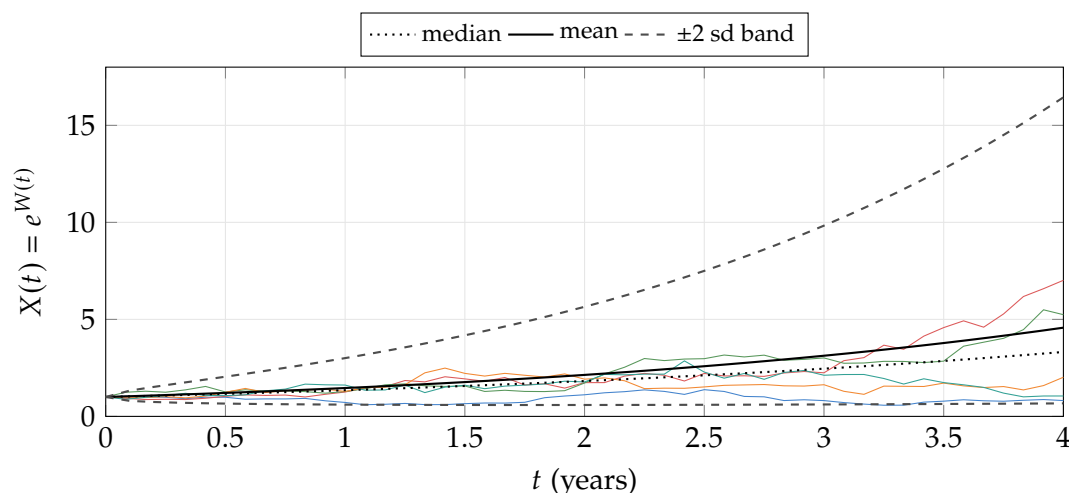


Figure 17.3: The simulation of lec19_2 in the setting of the lecture (► 2024 L25 @ 13:42): $W(t) = 0.30t + 0.40B(t)$ on $[0, 4]$ and $X = e^W$, five paths (different random numbers from the lecture's, $m = 1008$ steps of which every 21st is drawn). Dotted: the median $e^{0.30t}$; solid: the mean $e^{(0.30+0.08)t}$ (the drift of X in (17.15) is $0.30 + \frac{1}{2} \cdot 0.40^2$); dashed: the band $\exp(0.30t \pm 2 \cdot 0.40\sqrt{t})$, the image of the ± 2 standard-deviation band of W . The band is far above the typical path, as in the lecture's picture.

The simulation. The lecture simulates 5 and then 50 paths, draws the mean and ± 2 standard-deviation bands of W and their exponentials, and then the *cumulative average log return* $R(t) = \ln[X(t)/X(0)]/t = W(t)/t$, whose bands are $\mu \pm 2\sigma/\sqrt{t}$ ► 2024 L25 @ 13:42. Two observations of the lecture:

- among the 50 exponentials some paths reach 10–20 times the initial value, “extraordinary returns” of the type of a few stocks over some years, consistent with a plain geometric Brownian motion ► 2024 L25 @ 15:50;
- $R(t)$ is very dispersed for short horizons and stabilises around the constant level as

4 COURSE NOTE: Two conventions for μ in the lecture

Slide 25 of lec19_1 and lec19_2 simulate W , a Brownian motion with drift 0.30, volatility 0.40 and $T = 4$ years, and then $X = e^W$. In this convention 0.30 is the drift of the *log* price, and X solves (17.15) with $\mu = 0.30 + \frac{1}{2}(0.40)^2 = 0.38$, so that $\mathbb{E}[X_4] = e^{1.52} = 4.57$ and median $X_4 = e^{1.2} = 3.32$. In the numerical example of the previous slides 0.30 is the μ of (17.15): then $\mathbb{E}[P_T]/P_t = e^{0.30\tau}$ and the median is $e^{0.22\tau}$. Both are right, but they are different processes; the lecture does not remark on it.

the horizon grows [► 2024 L25 @ 16:54].

Our own run (numpy, not the R code of lec19_2, so the paths differ) of 50 paths: $\mathbb{P}(X_4 > 10) = 1 - \Phi((\ln 10 - 1.2)/0.8) = 8.4\%$, so 4.2 of 50 paths are expected above 10; the run had one (maximum 17.1), and its sample mean of X_4 was 3.48, well below the theoretical 4.57, because the mean is dominated by rare large paths and 50 paths are too few to contain them; the median 2.79 versus 3.32. The average log return $R(4)$ had mean 0.245 and standard deviation 0.183 over the 50 paths (theory 0.30 and $0.40/2 = 0.20$).

IN FINANCE: Itô's lemma and the Black–Scholes equation

The second half of L25 (from 19:01) derives the pricing equation with the tools of this chapter; the argument is developed in [chapters 15](#) and [18](#), here is the core. Let P follow (17.15) and let $G_t = G(P_t, t)$ be the price of a derivative that depends only on the asset price and time. By (17.10),

$$dG = \left(G_P \mu P + G_t + \frac{1}{2} \sigma^2 P^2 G_{PP} \right) dt + G_P \sigma P dB,$$

[► 2024 L25 @ 19:01]. Short one derivative and hold G_P units of the asset, $V = -G + G_P P$. The noise terms cancel in an infinitesimal time (the dB of G is $G_P \sigma P dB$, that of $G_P P$ is the same), leaving $dV = -(G_t + \frac{1}{2} \sigma^2 P^2 G_{PP}) dt$, a riskless change [► 2024 L25 @ 22:15]. By no arbitrage a riskless portfolio earns the risk-free rate r , $dV = rV dt$ [► 2024 L25 @ 24:16], whence the *Black–Scholes equation*

$$G_t + rP G_P + \frac{1}{2} \sigma^2 P^2 G_{PP} - rG = 0, \quad (17.20)$$

valid for *any* contingent claim; different claims differ by the terminal condition at maturity, e.g. $G(P, T) = (P - K)^+$ for a call [► 2024 L25 @ 25:20]. The delta G_P changes with time and the hedge must be rebalanced: this is the answer given in class to the student's question about the changing amount of asset [► 2024 L25 @ 29:28]. Note that the drift μ of the asset has disappeared from (17.20): this is the key fact behind risk-neutral pricing ([section 17.9](#)).

17.8 Several dimensions, covariation and the product rule

This section is not covered in the lectures (only the two-dimensional Brownian motion of 2013 PS8 Problem B-1 touches it, see [section 11.7.3](#)); it is the natural completion of the formalism and is marked as our addition.

17.8.1 Covariation

For two processes the multiplication table (17.8) generalises: the *covariation* of X and Y over $[0, t]$ is $[X, Y]_t = \lim \sum_i (\Delta_i X)(\Delta_i Y)$ as the mesh goes to zero (in probability). Its differential is written $dX dY$. If $dX = a dt + b dB$ and $dY = a' dt + b' dB$ (same B) then $dX dY = bb' dt$. For *independent* Brownian motions $B^{(1)}, B^{(2)}$,

$$dB^{(1)} dB^{(2)} = 0, \quad \text{because} \quad \mathbb{E} \left[\left(\sum_i \Delta_i B^{(1)} \Delta_i B^{(2)} \right)^2 \right] = \sum_i \Delta t_i^2 \rightarrow 0 \quad (17.21)$$

(the cross terms vanish by independence and mean zero; the diagonal terms are $\mathbb{E}[\Delta B^{(1)2}] \mathbb{E}[\Delta B^{(2)2}] = \Delta t^2$). If $B^{(2)} = \rho B^{(1)} + \sqrt{1 - \rho^2} \tilde{B}$ with $\tilde{B}$ independent of $B^{(1)}$ (correlated

Brownian motions, [chapter 2](#)), then $dB^{(1)}dB^{(2)} = \rho dt$. A numerical check (2×10^5 paths, 200 steps, $t = 1$): the root-mean-square of $\sum \Delta B^{(1)}\Delta B^{(2)}$ was 0.0707, the theoretical $1/\sqrt{200} = 0.0707$.

17.8.2 Itô's formula in several variables

Theorem 17.19 (Multi-dimensional Itô formula; not discussed in class). Let $B = (B^{(1)}, \dots, B^{(m)})$ be independent Brownian motions and $X = (X_1, \dots, X_n)$ with $dX_i = a_i dt + \sum_{k=1}^m b_{ik} dB^{(k)}$ (standard processes). For $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n)$,

$$df(t, X_t) = \left[f_t + \sum_i a_i f_{x_i} + \frac{1}{2} \sum_{i,j} (bb^T)_{ij} f_{x_i x_j} \right] dt + \sum_{i,k} b_{ik} f_{x_i} dB_t^{(k)}, \quad (17.22)$$

i.e. $df = f_t dt + \sum_i f_{x_i} dX_i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} dX_i dX_j$ with $dX_i dX_j = (bb^T)_{ij} dt$.

The proof is the Taylor expansion to second order with (17.8) and (17.21). The two consequences we need:

Corollary 17.20 (Product rule and integration by parts). For standard processes X, Y , $d(XY) = X dY + Y dX + dX dY$. In particular for h deterministic and continuously differentiable, $d(hB) = h dB + B h' dt$.

Proof. Take $f(x, y) = xy$: $f_{xx} = f_{yy} = 0$, $f_{xy} = 1$ in (17.22). In the second statement $dh dB = h' dt dB = 0$. $\square$

The extra term $dX dY$ is absent in classical calculus. For $X = Y = B$ it gives $d(B^2) = 2B dB + dt$, [example 17.13](#); for correlated Brownian motions $d(B^{(1)}B^{(2)}) = B^{(1)}dB^{(2)} + B^{(2)}dB^{(1)} + \rho dt$, so $B^{(1)}B^{(2)} - \rho t$ is a martingale (it is a sum of two Itô integrals with $\mathbb{E} \int B^2 < \infty$); for $\rho = 0$ the product of independent Brownian motions is a martingale.

17.8.3 Two- and n -dimensional Brownian motion

Let $B = (B^{(1)}, \dots, B^{(n)})$ be an n -dimensional Brownian motion (independent coordinates); for $n = 2$ this is the process of 2013 problem B-1. Apply (17.22) to a function $f(x)$ of $x \in \mathbb{R}^n$:

$$df(B_t) = \nabla f(B_t) \cdot dB_t + \frac{1}{2} \Delta f(B_t) dt, \quad \Delta = \sum_i \partial_{x_i}^2. \quad (17.23)$$

- *Harmonic functions.* If $\Delta f = 0$ then $f(B_t)$ is a (local) martingale: the martingales of an n -dimensional Brownian motion are the harmonic functions. The generator is $\frac{1}{2}\Delta$, as in the heat kernel of [section 11.7.3](#).
- *The squared distance.* For $f = |x|^2$, $d|B|^2 = 2B \cdot dB + n dt$, so $|B_t|^2 - nt$ is a martingale and $\mathbb{E}[|B_t|^2] = nt$ (for $n = 2$: $2t$; in the physicist's language $\langle r^2 \rangle = 2dDt$ with diffusion coefficient $D = \frac{1}{2}$ in $d = n$ dimensions).
- *The radial part (Bessel process).* For $f = |x|$: $\nabla f = x/|x|$, $\Delta f = (n-1)/|x|$, so $R_t = |B_t|$ satisfies $dR = dW + \frac{n-1}{2R} dt$ with $dW = \frac{B}{|B|} \cdot dB$, and $(dW)^2 = \sum_i B_i^2/|B|^2 dt = dt$; by Lévy's characterisation (not proved here) W is a Brownian motion. The drift $(n-1)/(2R)$ is the "centrifugal" push of a random walk in the plane away from the origin. As a check for $n = 2$: $\mathbb{E}[R_t] = \sqrt{\pi t/2}$ from the Rayleigh law of [section 11.7.3](#), and $\frac{d}{dt}\mathbb{E}[R_t] = \mathbb{E}\frac{1}{2R_t} = \frac{1}{2}\sqrt{\pi/(2t)}$, the derivative of $\sqrt{\pi t/2}$ (using $\mathbb{E}1/R_t = \sqrt{\pi/(2t)}$).

Exercise 17.8 (PS8 (2013), Problem B-1: two-dimensional Brownian motion). Let $Y(t) = (X_1(t), X_2(t))$ where X_1, X_2 are independent Brownian motions (a 2-dimensional Brownian motion).

- (a) For fixed t , find the probability density function of $Y(t)$.
- (b) Let $D_\rho = \{x \in \mathbb{R}^2 : |x| < \rho\}$ and compute $\mathbb{P}(Y(t) \in D_\rho)$.

Exercise 17.9 (not from the course: exit from a disc). For a 2-dimensional Brownian motion started at the origin, let $\tau = \inf\{t : |B_t| = \rho\}$. Use $|B_t|^2 - 2t$ (section 17.8) and optional stopping (section 4.5) to find $\mathbb{E}[\tau]$.

Exercise 17.10 (not from the course: discounted price). Let $dP = rP dt + \sigma P dB$. Show that $d(e^{-rt}P_t) = \sigma e^{-rt}P_t dB_t$ using the product rule, so that the discounted price is a martingale, and compute $\text{Var } P_T$.

17.9 Change of measure and Girsanov's theorem

This is the last topic of the 2013 lecture; 2024 did not treat it [► 2013 L18 @ 58:55]. The question is: *can a process with drift be viewed as a process without drift?* Both are probability distributions over paths, and a typical path of the first drifts along the line μt while a typical path of the second stays around the axis [► 2013 L18 @ 1:00:57]. The answer of Girsanov's theorem is yes, at the price of changing the probability measure, and the consequence is that “almost any question about Brownian motion with drift may be rephrased as a parallel question about standard Brownian motion” (the quotation from Steele in the 2013 note).

17.9.1 Equivalent measures and the Radon–Nikodým derivative

Two probability measures $\mathbb{P}, \tilde{\mathbb{P}}$ on the same space Ω are *equivalent* if $\mathbb{P}(A) > 0 \iff \tilde{\mathbb{P}}(A) > 0$ for all events A : they agree about which events are impossible, although not on their probabilities [► 2013 L18 @ 1:03:11]. In the finite example of the lecture, $\Omega = \{1, 2, 3\}$, $\mathbb{P} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $\tilde{\mathbb{P}} = (\frac{2}{3}, \frac{1}{6}, \frac{1}{6})$: for $A = \{1, 2\}$ one has $\mathbb{P}(A) = \frac{2}{3}$ and $\tilde{\mathbb{P}}(A) = \frac{5}{6}$, different, but both positive (the transcript garbles the numbers of $\tilde{\mathbb{P}}$; these are consistent with the values $\frac{2}{3}, \frac{5}{6}$ quoted). If instead $\tilde{\mathbb{P}} = (\frac{2}{3}, \frac{1}{3}, 0)$ the measures are not equivalent: $\mathbb{P}(\{3\}) = \frac{1}{3}$ but $\tilde{\mathbb{P}}(\{3\}) = 0$ [► 2013 L18 @ 1:06:22]. Equivalent measures are related by a positive density:

Theorem 17.21 (Radon–Nikodým; quoted). $\tilde{\mathbb{P}}$ is equivalent to $\mathbb{P}$ if and only if there is a random variable $Z > 0$ with $\mathbb{E}[Z] = 1$ such that $\tilde{\mathbb{P}}(A) = \mathbb{E}[Z \mathbf{1}_A]$ for all A ; then $\tilde{\mathbb{E}}[Y] = \mathbb{E}[ZY]$. One writes $Z = d\tilde{\mathbb{P}}/d\mathbb{P}$, the Radon–Nikodým derivative.

[► 2013 L18 @ 1:04:17] In the finite example $Z = (2, \frac{1}{2}, \frac{1}{2})$. A mapping of paths in general destroys equivalence: the law of the *square* $B(t)^2$ is not equivalent to that of $B(t)$, because B^2 never takes negative values and B does [lecnote18, §3 (this is essentially PS8 Problem A-5)].

Exercise 17.11 (PS8 (2013), Problem A-5: non-equivalent laws). Let $B(t)$ be a Brownian motion. Prove that the probability distributions of $B(t)$ and of $B(t)^2$ are not equivalent (two probability distributions $\mathbb{P}$ and $\mathbb{Q}$ are equivalent if for every set X , $\mathbb{P}(X) > 0$ if and only if $\mathbb{Q}(X) > 0$).

17.9.2 Girsanov's theorem

Theorem 17.22 (Girsanov, simple version). Let X be a Brownian motion with drift μ (and unit variance) on $[0, T]$ under $\mathbb{P}$, and define $\tilde{\mathbb{P}}$ by

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = Z = \exp\left(-\mu X_T + \frac{1}{2}\mu^2 T\right). \quad (17.24)$$

Then $\tilde{\mathbb{P}}$ is a probability measure, equivalent to $\mathbb{P}$, and under $\tilde{\mathbb{P}}$ the process X is a standard Brownian motion without drift.

The path picture of a physicist: with $\mathbb{P}[x(\cdot)] \propto e^{-\frac{1}{2} \int (\dot{x} - \mu)^2}$ and $\tilde{\mathbb{P}}[x(\cdot)] \propto e^{-\frac{1}{2} \int \dot{x}^2}$, the ratio is $e^{\mu(x_T - x_0) - \mu^2 T/2} = 1/Z$: reweighting paths by an exponential in the endpoint (an exponential tilting, or Cameron–Martin) shifts the drift.

Theorem 17.23 (Girsanov, general version). Let B be a Brownian motion under $\mathbb{P}$ on $[0, T]$, θ a bounded adapted process, and

$$Z_T = \exp\left(-\int_0^T \theta_u dB_u - \frac{1}{2} \int_0^T \theta_u^2 du\right), \quad \tilde{\mathbb{P}} = Z_T \cdot \mathbb{P}. \quad (17.25)$$

Then $Y_t = B_t + \int_0^t \theta_u du$ is a Brownian motion under $\tilde{\mathbb{P}}$.

[► 2013 L18 @ 1:10:39]

For constant $\theta = \mu$: $Z_T = e^{-\mu B_T - \mu^2 T/2}$ and $Y = B + \mu t$ is Brownian under $\tilde{\mathbb{P}}$; equivalently $B = Y - \mu t$ is a Brownian motion with drift $-\mu$ under $\tilde{\mathbb{P}}$. The general theorem lets the drift removed be random (adapted). The 2013 lecture states the simple version and emphasises how to use it: an expectation of a functional of a process with drift under $\mathbb{P}$ becomes an expectation of a functional of an undrifted process under $\tilde{\mathbb{P}}$, $\mathbb{E}[F] = \tilde{\mathbb{E}}[F/Z]$ (the lecture writes “Z tilde”, as it swaps the roles of the two measures) [► 2013 L18 @ 1:14:50]. A martingale is a fair game; the change of measure converts an unfair game into a fair one, and this is why it matters for pricing.

IN FINANCE: Risk-neutral measure; not discussed in class

Let the real-world price follow (17.15) and let r be the risk-free rate. Take the constant $\theta = (\mu - r)/\sigma$ (the market price of risk) and $Y = B + \theta t$, a Brownian motion under $\tilde{\mathbb{P}} = \mathbb{Q}$. Then $dP = \mu P dt + \sigma P(dY - \theta dt) = rP dt + \sigma P dY$: under $\mathbb{Q}$ the price grows at the risk-free rate, and $e^{-rt} P_t$ is a martingale. The derivative price is the discounted $\mathbb{Q}$ -expectation of the payoff (chapter 15); this is the same reason why μ disappeared from (17.20). Both measures are equivalent, so “impossible under one” means “impossible under the other”, which is exactly what no-arbitrage requires (theorem 2.9).

Remark 17.24 (Why equivalence is a surprise). The two lectures record the same astonishment [► 2013 L18 @ 1:09:36]: paths with and without drift look completely different, since as $t \rightarrow \infty$ one hugs the line μt and the other the axis. Both statements are true: on

Proof. Take $0 = t_0 < \dots < t_n = T$ and the increments $x_i = X_{t_i} - X_{t_{i-1}}$; under $\mathbb{P}$ they are independent $N(\mu\Delta_i, \Delta_i)$ with $\Delta_i = t_i - t_{i-1}$, with joint density $\prod_i (2\pi\Delta_i)^{-1/2} e^{-(x_i - \mu\Delta_i)^2/2\Delta_i}$. Since $Z = \prod_i e^{-\mu x_i + \mu^2 \Delta_i/2}$, the density of the increments under $\tilde{\mathbb{P}}$, i.e. of $Z d\mathbb{P}$, is the product of

$$(2\pi\Delta_i)^{-1/2} \exp\left[-\frac{(x_i - \mu\Delta_i)^2}{2\Delta_i} - \mu x_i + \frac{\mu^2 \Delta_i}{2}\right] = (2\pi\Delta_i)^{-1/2} \exp$$

because $-\frac{(x - \mu\Delta)^2}{2\Delta} = -\frac{x^2}{2\Delta} + \mu x - \frac{1}{2}\mu^2 \Delta$. This is a probability density (so $\mathbb{E}[Z] = 1$) and says that the increments are independent $N(0, \Delta_i)$ under $\tilde{\mathbb{P}}$. The finite-dimensional distributions of X are those of Brownian motion, and the paths are continuous. $\square$

5 COURSE NOTE: Erratum: the sign in the note's Theorem 3.2

lecnote18 (Theorem 3.2) states $d\tilde{\mathbb{P}}/d\mathbb{P} = e^{-\mu\omega(T) - \mu^2 T/2}$ for X a Brownian motion with drift μ under $\mathbb{P}$. The correct exponent is $-\mu X_T + \frac{1}{2}\mu^2 T$, as in (17.24). With the sign of the note, $\mathbb{E}[Z] = \mathbb{E}[e^{-\mu X_T - \mu^2 T/2}] = e^{-\mu^2 T/2} e^{-\mu^2 T/2} = e^{-\mu^2 T} \neq 1$ (as $\mathbb{E}[e^{-\mu X_T}] = e^{-\mu^2 T + \mu^2 T/2}$), so Z would not be a density (Monte Carlo, $\mu = 0.5$, $T = 1$: $0.7788 = e^{-0.25}$ against 1.0000 for the correct sign; with the correct sign $\mathbb{E}[Z \mathbf{1}_{X_T > 1}] = 0.1586 = 1 - \Phi(1)$ and $\mathbb{E}[ZX_T^2] = 1.00$, the values of a standard normal). The general theorem (theorem 17.23) below, which the note states correctly, is consistent with it (take $\theta = \mu$ and start from $X - \mu t$).

Proof sketch (not given in class). By (17.10), $dZ_t = -\theta_t Z_t dB_t$, so Z is a positive martingale with $Z_0 = 1$, $\mathbb{E}[Z_T] = 1$ (the integrand is square-integrable since θ is bounded), and by Bayes $\tilde{\mathbb{E}}[G | \mathcal{F}_s] = \mathbb{E}[Z_t G | \mathcal{F}_s]/Z_s$. For real λ put $N_t = Z_t \exp(\lambda Y_t - \frac{1}{2}\lambda^2 t)$. Then $\ln N_t = \int_0^t (\lambda - \theta) dB - \frac{1}{2} \int_0^t (\lambda - \theta)^2 du$ (the terms $\lambda \int \theta$ and $-\frac{1}{2}\lambda^2 t$ complete the square), so N is again a martingale of the form (17.25). Then for $s < t$

$$\tilde{\mathbb{E}}[e^{\lambda(Y_t - Y_s)} | \mathcal{F}_s] = \frac{\mathbb{E}[N_t | \mathcal{F}_s] e^{\lambda^2 t/2} e^{-\lambda Y_s}}{Z_s} = e^{\lambda^2(t-s)/2}.$$

This is the moment generating function of $N(0, t-s)$ for every λ , so $Y_t - Y_s$ is $N(0, t-s)$ under $\tilde{\mathbb{P}}$ and independent of $\mathcal{F}_s$: Y is a Brownian motion. $\square$

a *finite* interval $[0, T]$ the measures are equivalent, the density Z being an explicit, finite, positive number; on $[0, \infty)$ they are mutually singular, since $\{\lim X_t/t = \mu\}$ has probability 1 under $\mathbb{P}$ and 0 under $\tilde{\mathbb{P}}$ (strong law of large numbers), and $Z_T = e^{-\mu X_T + \mu^2 T/2} \rightarrow 0$ almost surely under $\mathbb{P}$ as $T \rightarrow \infty$.

Remark 17.25 (The discrete picture). P. Kempthorne's comment during the 2013 lecture [► 2013 L18 @ 1:10:39]: a gambler's fortune against a casino with unfavourable odds is a random walk with a negative drift; taking out the expected loss turns it into a martingale, to which martingale theory (convergence theorems) applies. In the discrete case the change of measure is a likelihood ratio: a walk with up-probability $p = \frac{1}{2}(1 + \varepsilon)$ against the fair one has, over n steps with $S_n = \sum \xi_i$, $\prod_i [(2p)^{-(1+\xi_i)/2} (2q)^{-(1-\xi_i)/2}] = \exp(-\varepsilon S_n + \frac{1}{2}n\varepsilon^2 + O(n\varepsilon^3))$; with steps $\pm\sqrt{h}$ and $\varepsilon = \mu\sqrt{h}$ this tends to $e^{-\mu X_T + \mu^2 T/2}$, the density (17.24) (own computation).

Exercise 17.12 (not from the course: Girsanov by simulation). For $\mu = 0.5$, $T = 1$ let $X_T \sim N(\mu, 1)$ under $\mathbb{P}$ and $Z = e^{-\mu X_T + \mu^2/2}$. Compute (exactly, and by simulation) $\mathbb{E}[Z]$, $\mathbb{E}[Z \mathbf{1}_{\{X_T > 1\}}]$ and $\mathbb{E}[Z X_T^2]$, and explain the results using theorem 17.22.

17.10 Errata and the two lectures compared

Looking ahead. Chapter 18 takes (17.10) to solve stochastic differential equations: existence and uniqueness, the Ornstein–Uhlenbeck and Vasicek models, and the numerical (Euler–Maruyama) scheme whose strong order $n^{-1/2}$ we saw in example 17.15; the Feynman–Kac formula links the PDE condition of theorem 17.17 to expectations. Chapter 15 uses (17.20) and the change of measure of section 17.9 to price options.

KEY FORMULAS

Table:	$(dB)^2 = dt, \quad dB \, dt = 0, \quad (dt)^2 = 0, \quad dB^{(1)} dB^{(2)} = \rho \, dt$
Itô integral:	$\mathbb{E} \int_0^t f \, dB = 0, \quad \mathbb{E} \left(\int_0^t f \, dB \right)^2 = \mathbb{E} \int_0^t f^2 \, ds$ $\int_0^t B \, dB = \frac{1}{2}(B_t^2 - t) \quad (\text{left point})$
Itô's lemma:	$df(t, X_t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x \, dB, \quad dX = a \, dt + b \, dB$
Product rule:	$d(XY) = X \, dY + Y \, dX + dX \, dY$
Martingale test:	$f_t + \frac{1}{2} f_{xx} = 0 \Rightarrow f(t, B_t) \text{ martingale} \quad (\text{e.g. } e^{\sigma B - \sigma^2 t/2}, B^2 - t)$
GBM:	$dP = \mu P \, dt + \sigma P \, dB \Rightarrow P_t = P_0 e^{(\mu - \sigma^2/2)t + \sigma B_t}$ $\mathbb{E}[P_t] = P_0 e^{\mu t}, \quad r = \frac{1}{t} \ln \frac{P_t}{P_0} \sim N\left(\mu - \frac{1}{2}\sigma^2, \sigma^2/t\right)$
Girsanov:	$Z = e^{-\mu X_T + \mu^2 T/2} \quad (X \text{ has drift } \mu \text{ under } \mathbb{P})$ $\Rightarrow X \text{ is driftless under } Z \cdot \mathbb{P}$

6 COURSE NOTE: 2013 and 2024 compared

Order of presentation. 2013 (C. Lee) starts from the *differentiation*: Taylor expansion, $(dB)^2 = dt$ and the lemma for $f(t, B_t)$ and $f(t, X_t)$ come first; the integral is *defined as the inverse* of differentiation (“a very stupid definition” whose existence is left open), and only afterwards is it linked to left-point Riemann sums. 2024 (P. Kempthorne) goes the other way, from integrals of step processes through the isometry to the limit and then derives the formula from Taylor's theorem; the two constructions agree. *Content only in 2013:* adapted processes with examples, the “normal” theorem for deterministic integrands, the martingale-versus-drift dictionary and its use in arbitrage arguments, change of measure and Girsanov. *Content only in 2024:* the explicit step-process cases including $\int B \, dB$ as a limit, the isometry as an isometry of L^2 spaces, Gaussian integrals with covariance, the anti-derivative method, the PDE condition and the ruin problem, geometric Brownian motion with numerical example and simulation, the derivation of the Black–Scholes equation. *Common:* both give the isometry and the martingale property without proof, both stop at the level of “reasonable functions” (the 2013 note writes $g \in L^2$; the 2024 slides $\mathbb{E} \int f^2 < \infty$).

7 COURSE NOTE: Further errata and misprints

- **lecnote18** Theorem 2.3 writes the integrability condition as “ $\int_0^t g^2(t, B_t) \, dt \, dB_t < \infty$ ”; it should be $\mathbb{E} \int_0^t g(s, B_s)^2 \, ds < \infty$, the condition (17.5).
- **lecnote18** Example 2.2(iv) claims that X_τ is an adapted process for a stopping time τ ; X_τ is a single random variable, not

a process. The intended statement is that the stopped process $t \mapsto X_{t \wedge T}$ is adapted.

- **lecnote18** Example 1.3(i) calls $\mu t + \sigma B_t$ Brownian motion with “variance σ^2 ”; the variance parameter is σ^2 .
- Slide 7 of **lec19_1** defines the step process with a last time $t_{n+1} = T$ (it is $t_n = T$); slide 10 writes $\text{Var}[I(f)] = \|I(f)\|_{L^2}^2$, where the norm has to be squared (the slide says “(squared) norm” below).
- In the 2013 lecture the differential is stated aloud as $df = f' dB + f'' dt$ [**► 2013 L18 @ 1:11**]: the coefficient of the last term is $\frac{1}{2}$, correctly written in the note.
- The 2024 transcript (spoken) says that the expectation of the increment powers $c > 2$ is “comparable to” $\sigma^2 \Delta t$ [**► 2024 L24 @ 7:24**]; the slide correctly says $o(\Delta t)$.

Chapter 18

Stochastic Differential Equations

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2024 L25 (from 19:01)	P. Kempthorne	The first 18 minutes (recap of Itô’s formula, geometric Brownian motion) are in chapter 17 . Then: the Black–Scholes equation by hedging (slides 2–4, also chapters 15 and 17); the heat equation: linearity and superposition, the Dirac initial condition, general initial data, the indicator example, solution by similarity/invariance and by step functions (slides 5–12); SDEs: Lipschitz and growth conditions, existence and uniqueness, coefficient matching for geometric Brownian motion and its “paradox”, the Ornstein–Uhlenbeck process and Vasicek’s model, the product-process solution, systems of SDEs and the position–velocity example (slides 13–20); numerical methods listed on slide 21.
2013 L21	C. Lee	The goal of an SDE and the existence/uniqueness theorem; coefficient matching for geometric Brownian motion and (by a product guess) for the Ornstein–Uhlenbeck process; numerical methods: finite differences for ODEs, Monte Carlo along a fixed Brownian path, the tree method; the heat equation, superposition, the Dirac delta and the Gaussian kernel, the link with the Black–Scholes equation.
Files. Slides mit18_f24 lec24.pdf (21 slides, this is Lecture 25, not 24; slides 2–4 Black–Scholes, 5–12 heat equation, 13–20 SDEs, 21 numerical methods and references); 2013 note MIT18_S096F13 lecnote21.pdf (7 pages). Problem sets. 2013 PS9 (pset9.pdf : Problems A-1, A-2, A-3; there is no Part B), and Exercise 3.2 of lecnote21 . 2024 has no problem set on this topic. The 2013 problem set (due 12/5/13, “no need to turn in”) has only Part A, which “straightforwardly follows from the definition”. Each exercise follows the section it practises. Additions of these notes (marked as such where they occur): the proof of existence and uniqueness by Picard iteration, generators, the Feynman–Kac formula, the Fokker–Planck equation, the exact AR(1) sampling of the Ornstein–Uhlenbeck process, the Vasicek bond price, the Euler–Maruyama and Milstein convergence experiments, and the CIR process. The lectures state only that the Black–Scholes PDE and expectations are two faces of one object; the Feynman–Kac formula that proves it is here.		

Motivation

[Chapter 17](#) taught us to differentiate functions of Brownian motion. This chapter runs the calculus backwards, exactly as an ordinary differential equation is the reverse of differentiation: *given* the local rules “expected change per unit time = μ , variance of the change per unit time = σ^2 ”, find the process. A stochastic differential equation (SDE) is

the law of motion of a random quantity: a stock price, a short interest rate, the velocity of a particle in a fluid. Four questions organise the chapter.

- *Does a solution exist, and is it unique?* Yes under a Lipschitz and a growth condition (section 18.1).
- *Can we solve it in closed form?* Rarely, but the two most important examples can: geometric Brownian motion, the model of a price, and the Ornstein–Uhlenbeck (Vasicek) process, the model of a mean-reverting quantity such as an interest rate (sections 18.2 and 18.3). Sampled at discrete times the Ornstein–Uhlenbeck process is exactly the AR(1) model of chapter 9.
- *How does an SDE connect to partial differential equations?* Through the generator of the process: the Feynman–Kac formula turns a PDE into an expectation and back, and the Fokker–Planck equation is the PDE for the density (section 18.5). Both lectures end with the simplest of these PDEs, the heat equation, which is the Black–Scholes equation in disguise (section 18.6).
- *What if there is no closed form?* Simulate: the Euler–Maruyama scheme, Monte Carlo, trees (section 18.7).

For a physicist most of this is familiar under other names: an SDE with additive noise is a Langevin equation, the Ornstein–Uhlenbeck process is the velocity of a Brownian particle, and the Fokker–Planck equation and Feynman–Kac formula are the Schrödinger-type equation and the path integral of the same diffusion. We use Itô’s formula (17.10) of chapter 17 throughout.

18.1 Stochastic differential equations: existence and uniqueness

18.1.1 The definition

Let B be a standard Brownian motion and $\mu(t, x), \sigma(t, x)$ two given functions of time and space. The SDE

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 = x_0 \quad (18.1)$$

[► 2024 L25 @ 1:04:34][► 2013 L21 @ 0:00] is a shorthand for an integral equation, exactly as in Itô’s formula. A *solution* is a continuous adapted process X such that for all $t \geq 0$

$$X_t = x_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s, \quad (18.2)$$

the first integral being an ordinary (pathwise) integral and the second an Itô integral (chapter 17). Note the difference with all of chapter 17: there the integrands were given adapted processes; here the integrand $\sigma(s, X_s)$ depends on the unknown X itself, so the object to be found appears on both sides.

INTUITION: Reading an SDE

Over a short time h , (18.1) says $X_{t+h} - X_t \approx \mu(t, X_t)h + \sigma(t, X_t)(B_{t+h} - B_t)$: conditionally on the present the change is normal with mean μh and variance $\sigma^2 h$. So μ is the local drift (mean velocity) and σ^2 the local variance rate, the *diffusion coefficient* of a physicist ($\sigma^2 = 2D$). In the language of chapter 17 the equation is a Langevin equation $\dot{X} = \mu(t, X) + \sigma(t, X)\xi_t$ with white noise $\xi = \dot{B}$, in the Itô convention: the noise multiplies σ evaluated at the *start* of each step.

The 2013 lecture stresses that this is the reverse of what had been done so far [► 2013 L21 @ 0:00]: before, a function was differentiated to obtain an equation; now the equation is given and the process has to be found by going backwards. Even for ordinary or partial differential equations one does not expect a closed form; what one *can* always say is whether a solution exists.

18.1.2 Existence and uniqueness

Theorem 18.1 (Existence and uniqueness). Fix $T > 0$ and suppose the coefficients of (18.1) satisfy, for some constant K and all $t \in [0, T]$, $x, y \in \mathbb{R}$, the space-variable Lipschitz condition

$$|\mu(t, x) - \mu(t, y)|^2 + |\sigma(t, x) - \sigma(t, y)|^2 \leq K|x - y|^2 \quad (18.3)$$

and the spatial growth condition

$$|\mu(t, x)|^2 + |\sigma(t, x)|^2 \leq K(1 + |x|^2). \quad (18.4)$$

Then there is a continuous adapted solution X on $[0, T]$ with $\sup_{0 \leq t \leq T} \mathbb{E}[X_t^2] < \infty$. Moreover if X and Y are two continuous solutions with this L^2 bound, then $\mathbb{P}(X_t = Y_t \text{ for all } t \in [0, T]) = 1$.

[► 2024 L25 @ 1:04:34][► 2013 L21 @ 2:13] Both lectures state the theorem without proof, refer to the literature (Steele; Oksendal), and comment on the hypotheses. The Lipschitz condition says that μ and σ cannot change too fast when the space variable moves a little: at most linearly in the distance [► 2013 L21 @ 4:27]. The growth condition says that the coefficients cannot blow up too fast at infinity [► 2024 L25 @ 1:05:34][► 2013 L21 @ 5:31]. “Essentially unique” means unique up to a null set: the two solutions have the same paths with probability one [► 2024 L25 @ 1:06:36]. The 2013 speaker adds the advice that matters in practice: do not worry about the technical conditions; *a solution exists, but do not expect a closed form* [► 2013 L21 @ 5:31].

Sketch by Picard iteration (not given in class).

Put $X_t^{(0)} = x_0$ and $X_t^{(n+1)} = x_0 + \int_0^t \mu(s, X_s^{(n)}) ds + \int_0^t \sigma(s, X_s^{(n)}) dB_s$. Using $(a + b)^2 \leq 2a^2 + 2b^2$, Cauchy–Schwarz for the ds integral, the Itô isometry (theorem 17.4) for the noise integral, and (18.3) (which contains the Lipschitz bound of μ and of σ separately), the differences $\varphi_n(t) = \mathbb{E}[|X_t^{(n+1)} - X_t^{(n)}|^2]$ satisfy

$$\varphi_n(t) \leq C \int_0^t \varphi_{n-1}(s) ds, \quad C = 2K(T + 1),$$

hence $\varphi_n(t) \leq \varphi_0(T)(Ct)^n/n!$ by induction, and $\varphi_0(T) < \infty$ by (18.4). The series $\sum_n \sqrt{\varphi_n}$ converges, so $X^{(n)}$ is Cauchy in L^2 uniformly in t ; its limit solves (18.2). (Continuity of paths needs Doob’s maximal inequality; we skip it.) For uniqueness, if X, Y are two L^2 -bounded solutions then $D(t) = \mathbb{E}[|X_t - Y_t|^2]$ satisfies $D(t) \leq C \int_0^t D(s) ds$; Gronwall’s lemma gives $D \equiv 0$, so $X_t = Y_t$ a.s. for each t , and by continuity for all t simultaneously. $\square$

Example 18.2 (Why both conditions matter; not discussed in class).

- (a) $\dot{x} = x^2$, $x(0) = 1$ has the solution $x(t) = 1/(1-t)$, which explodes at $t = 1$: $\mu(x) =$

The two conditions are *sufficient, not necessary*, and each has its own job: the Lipschitz condition prevents non-uniqueness and the growth condition prevents explosion. Two ordinary differential equations (the case $\sigma = 0$) show what goes wrong. Under only *local* Lipschitz continuity one still has uniqueness and existence up to an explosion time. The

solution of (18.1) under the conditions of [theorem 18.1](#) is a *Markov process* (quoted): given X_t , the future depends on the past only through the present, because the future noise is independent of $\mathcal{F}_t$. This is what makes the generator of [section 18.5](#) meaningful.

Exercise 18.1 (PS9 (2013), Problem A-1: verify solutions). Verify that the given process solves the given SDE (use Itô's formula, [chapter 17](#)).

- (a) $X_t = \exp(B_t)$ solves $dX_t = \frac{1}{2}X_t dt + X_t dB_t$.
- (b) $X_t = B_t/(1+t)$ solves $dX_t = -\frac{1}{1+t}X_t dt + \frac{1}{1+t} dB_t$.
- (c) $X_t = \sin B_t$ solves $dX_t = -\frac{1}{2}X_t dt + \sqrt{1-X_t^2} dB_t$.

Exercise 18.2 (PS9 (2013), Problem A-2: a cube-root SDE). Let $a > 0$ and $dX_t = \frac{1}{3}X_t^{1/3} dt + X_t^{2/3} dB_t$, $X_0 = a$. Show that $X_t = (a^{1/3} + \frac{1}{3}B_t)^3$, $t \geq 0$, solves it. (Where do the conditions of [theorem 18.1](#) fail? See [example 18.2](#).)

x^2 is locally Lipschitz but violates (18.4).

- (b) $\dot{x} = 3x^{2/3}$, $x(0) = 0$ has the two solutions $x \equiv 0$ and $x = t^3$: $x^{2/3}$ is not Lipschitz at 0. The stochastic analogue is the SDE of PS9 Problem A-2, $dX = \frac{1}{3}X^{1/3} dt + X^{2/3} dB$ (real cube roots): with $X_0 = a > 0$ the process $X_t = (a^{1/3} + B_t/3)^3$ is a solution, and it is the only one *until* X first hits 0, where the coefficients stop being Lipschitz. For $a = 0$ both $X \equiv 0$ and $X_t = (B_t/3)^3$ solve the equation (apply Itô's formula to $f(B) = (B/3)^3$): uniqueness fails exactly where (18.3) fails.

1 COURSE NOTE: Misprints in the 2013 note, Theorem 1.1

[lecnote21](#) introduces the equation “for given functions a and b ” although the coefficients are called μ, σ ; writes the solution as

“ $X(T) = \int_0^T \dots$ ”, omitting the initial value x_0 (it is (18.2) with $t = T$); and leaves the placeholder “such that (L2 bound)”, which the 2024 slide 13 spells out as $\sup_{0 \leq t \leq T} \mathbb{E}[X_t^2] < \infty$, as in [theorem 18.1](#).

2 COURSE NOTE: Erratum in Problem A-1(c)

Itô's formula gives $d \sin B_t = -\frac{1}{2} \sin B_t dt + \cos B_t dB_t$, and $\sqrt{1 - \sin^2 B_t} = |\cos B_t|$, which equals $\cos B_t$ only while $\cos B_t \geq 0$, i.e. up to the first time $|B|$ reaches $\pi/2$. The stated equation therefore holds until that time; afterwards it holds only in law, with $d\tilde{B}_t = \text{sgn}(\cos B_t) dB_t$ (again a Brownian motion, by Lévy's characterisation). The verification asked in the problem is exactly the computation above.

18.2 Solving by coefficient matching: geometric Brownian motion

18.2.1 The method

Both lectures show the one systematic recipe they have: *guess that the solution is a function* $X_t = f(t, B_t)$ of time and the driving Brownian motion, differentiate by Itô's formula (17.9), and match coefficients with the SDE [► 2024 L25 @ 1:07:37][► 2013 L21 @ 6:34]. For $f \in C^{1,2}$,

$$dX_t = \left(f_t + \frac{1}{2} f_{xx} \right) (t, B_t) dt + f_x(t, B_t) dB_t.$$

Comparing with (18.1) (and using that $X_t = f(t, B_t)$) we need

$$\mu(t, f) = f_t + \frac{1}{2} f_{xx}, \quad \sigma(t, f) = f_x. \quad (18.5)$$

These are two equations for *one* unknown function, so the method can only work for special SDEs; the 2013 lecturer warns that it solves only a few SDEs and that most of the time it will not apply [► 2013 L21 @ 6:34]. It is a systematic way to *guess*, and in the end the guess is verified by Itô's formula.

This is the formula (17.17) that [chapter 17](#) obtained from the logarithm; here the same answer comes from a different logic, and by [theorem 18.1](#) it is *the* solution: the coefficients $\mu x, \sigma x$ are Lipschitz and of linear growth. The 2013 lecturer is candid about the status of the method [► 2013 L21 @ 11:01]: we already knew the answer and merely fitted it; the point of the exercise is the algorithm, which works when one cannot guess f .

Example 18.3 (Geometric Brownian motion).

Let $\mu, \sigma > 0$ be constants and $dX = \mu X dt + \sigma X dB$, $X_0 = x_0 > 0$. Then (18.5) reads $\mu f = f_t + \frac{1}{2} f_{xx}$ and $\sigma f = f_x$. The second equation is an ODE in x : $f_x/f = \sigma$, so $f(t, x) = e^{\sigma x + g(t)}$ [► 2024 L25 @ 1:08:48] [► 2013 L21 @ 7:45]. Inserting in the first,

$$\mu f = g'(t) f + \frac{1}{2} \sigma^2 f \implies g'(t) = \mu - \frac{1}{2} \sigma^2, \quad g(t) = (\mu - \frac{1}{2} \sigma^2) t$$

the factor f dividing out

18.2.2 The lognormal law and the “paradox”

By (18.6), $\ln X_t \sim N(\ln x_0 + \mu_* t, \sigma^2 t)$ with $\mu_* = \mu - \frac{1}{2}\sigma^2$: X_t is lognormal with $\mathbb{E}[X_t] = x_0 e^{\mu t}$ (the $e^{-\sigma^2 t/2}$ from the drift cancels $\mathbb{E}[e^{\sigma B_t}] = e^{\sigma^2 t/2}$) [► 2024 L25 @ 1:11:00]. The details, the numerical example of the 2024 lecture ($\mu = 0.3$, $\sigma = 0.4$, half a year: $\mathbb{E}[P_T]/P_t = 1.1618$, standard deviation $0.335 P_t$) and the annualised return are in [section 17.7.1](#).

The lecture then points to a seeming paradox [► 2024 L25 @ 1:12:02]. Write $\ln X_t = \ln x_0 + t(\mu_* + \sigma B_t/t)$. Since $B_t/t \rightarrow 0$ almost surely (strong law of large numbers for Brownian motion),

$$0 < \mu < \frac{1}{2}\sigma^2 \implies \mathbb{E}[X_t] = x_0 e^{\mu t} \rightarrow \infty \quad \text{while} \quad X_t \rightarrow 0 \text{ a.s., so } \mathbb{P}(X_t < c) \rightarrow 1 \quad \forall c > 0. \quad (18.7)$$

The expected price grows exponentially and yet every path goes to zero. The resolution is that the expectation of a lognormal is dominated by rare, extraordinarily large values. The typical (median) path grows like $x_0 e^{\mu_* t}$, which is decreasing here. A numerical illustration ($\mu = 5\%$, $\sigma = 40\%$, so $\mu_* = -3\%$ per year; own computation):

t (years)	$\mathbb{E}[X_t]/x_0 = e^{\mu t}$	median $e^{\mu_* t}$	$\mathbb{P}(X_t < x_0)$
10	1.65	0.74	0.59
100	148	0.050	0.77
1000	5.2×10^{21}	9×10^{-14}	0.99

The lecturer connects this with markets: some stocks do extraordinarily well and are noticed, others drift towards zero, and on average there is growth [► 2024 L25 @ 1:13:03]. (The critical case $\mu = \frac{1}{2}\sigma^2$ is different again: $\ln X_t$ then has no drift and oscillates between $\pm\infty$.)

INTUITION: Ensemble average versus time average

A physicist would say that the multiplicative process is *non-ergodic*: the ensemble average $\mathbb{E}[X_t] \propto e^{\mu t}$ and the growth rate of a single realisation, the time average $\lim_{t \rightarrow \infty} \frac{1}{t} \ln X_t = \mu - \frac{1}{2}\sigma^2$, differ. The gap $\frac{1}{2}\sigma^2$ is the volatility drag, the Itô correction of [chapter 17](#): an investor who holds the asset for a long time experiences μ_* , not μ . (The same quantity, $\mu - \frac{1}{2}\sigma^2$, is the growth rate the Kelly criterion maximises; not discussed in class.)

18.3 Linear SDEs: the Ornstein–Uhlenbeck and Vasicek processes

18.3.1 The model

Geometric Brownian motion has no memory of a level: a price that has doubled has no reason to come back. Many quantities do have a preferred level (an interest rate, a credit spread, the velocity of a molecule). Replace the constant drift by a restoring force proportional to the distance from that level:

Definition 18.4 (Ornstein–Uhlenbeck process). For $\kappa > 0$ (speed of mean reversion), $m \in \mathbb{R}$ (long-run level) and $\sigma > 0$ the Ornstein–Uhlenbeck (OU) process solves

$$dX_t = -\kappa (X_t - m) dt + \sigma dB_t, \quad X_0 = x_0. \quad (18.8)$$

[► 2024 L25 @ 1:09:56][► 2013 L21 @ 9:55]. The initial condition $f(0, 0) = x_0$ fixes $e^c = x_0$, hence

$$X_t = x_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t\right]. \quad (18.6)$$

3 COURSE NOTE: Erratum: the standard deviation of $\ln X_t$

Slide 15 writes the lognormal parameters as $\mu_* = (\mu - \sigma^2/2)t$ and $\sigma_* = \sigma t$, and the lecturer says aloud “our standard deviation is σt ” [► 2024 L25 @ 1:12:02]. The variance of σB_t is $\sigma^2 t$, so the standard deviation is $\sigma_* = \sigma\sqrt{t}$ (equivalently the variance is $\sigma^2 t$, as in the transcript at 7:23). The slide’s claim about $\mathbb{E}[X_t] = x_0 e^{\mu t}$ and the paradox are correct. The paradox needs $0 < \mu < \sigma^2/2$: for $\mu \leq 0$ the mean does not grow, and there is nothing surprising in $X_t \rightarrow 0$.

[▶ 2024 L25 @ 1:14:09][▶ 2013 L21 @ 13:16]The drift $-\kappa(X - m)$ is negative when $X > m$ and positive when $X < m$ (slide 16): a force that pulls X back to m , proportional to the distance; the local variability σ is constant, unlike geometric Brownian motion (example 18.3), where the noise scales with the level [▶ 2013 L21 @ 13:16]. The two lectures use different letters:

	speed	level	noise	source
these notes, ch. 9	κ	m (μ)	σ (σ_0)	
2024 slides 16–18	α	μ	σ	
2013 note and lecture (level 0)	α	0	σ	
2013 PS9 A-3, $dR = (\alpha - \beta R) dt + \sigma dB$	β	α/β	σ	

(The symbol μ is kept for the *drift* of geometric Brownian motion, so the level of the OU process is called m here; chapter 9 writes μ for it and σ_0 for the noise.)

18.3.2 Solution by the product-process guess

Take first $m = 0$, $dX = -\kappa X dt + \sigma dB$. Coefficient matching with $f(t, B_t)$ fails here [▶ 2013 L21 @ 13:16]: (18.5) would require $f_x = \sigma$, so $f = \sigma x + g(t)$, and then $\mu(t, f) = -\kappa(\sigma x + g)$ has to equal g' , which depends on x . The lectures change the ansatz to a *product process*

$$X_t = a(t)\left(x_0 + \int_0^t b(s) dB_s\right), \quad a(0) = 1, \tag{18.9}$$

with a, b differentiable deterministic functions [▶ 2024 L25 @ 1:15:12][▶ 2013 L21 @ 14:18]. (The 2013 lecturer admits that he cannot motivate the guess beyond experience [▶ 2013 L21 @ 15:34]; the reason it works is explained below.) Differentiate with the product rule (corollary 17.20):

$$dX_t = a'(t)\left(x_0 + \int_0^t b dB\right) dt + a(t)b(t) dB_t = \frac{a'(t)}{a(t)} X_t dt + a(t)b(t) dB_t.$$

There is *no extra term*: a is deterministic and of finite variation, so $da d(\int b dB) = a' dt dB = 0$; the 2013 lecturer makes this subtle point explicit, noting that once a process is written as an integral, differentiating it returns the integrand and nothing more [▶ 2013 L21 @ 16:37]. Matching with $dX = -\kappa X dt + \sigma dB$ gives

$$\frac{a'(t)}{a(t)} = -\kappa, \quad a(t)b(t) = \sigma \implies a(t) = e^{-\kappa t}, \quad b(t) = \sigma e^{\kappa t}$$

[▶ 2024 L25 @ 1:17:17][▶ 2013 L21 @ 18:42]. For a general level m apply this to $Y = X - m$ (which has the same differential and $Y_0 = x_0 - m$).

Proposition 18.5 (Solution of the OU equation). *The solution of (18.8) is*

$$X_t = m + (x_0 - m) e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dB_s. \tag{18.10}$$

The *integrating factor* makes the same computation transparent (own remark). Since $e^{\kappa t}$ is deterministic, the product rule gives $d(e^{\kappa t}(X_t - m)) = e^{\kappa t}(dX_t + \kappa(X_t - m) dt) = \sigma e^{\kappa t} dB_t$, and integrating from 0 to t yields (18.10). This is the stochastic version of the method for a linear ODE $\dot{x} = -\kappa x + f(t)$, and it also solves the linear SDE with time-dependent coefficients $dX = (-\kappa(t)X + f(t)) dt + \sigma(t) dB$ (the Hull–White extension) by replacing $e^{\kappa t}$

HISTORY: From Brownian particles to interest rates

Uhlenbeck and Ornstein (1930) used this equation for the velocity v of a Brownian particle: Langevin’s equation $m_p v' = -\gamma v + \sqrt{2\gamma k_B T} \xi$ has the form (18.8) with $m = 0$, $\kappa = \gamma/m_p$, $\sigma^2 = 2\gamma k_B T/m_p^2$; the stationary variance $\sigma^2/(2\kappa) = k_B T/m_p$ is the equipartition theorem. The lecture says the model was first used for the velocity of gas molecules whose average is m [▶ 2024 L25 @ 1:14:09] [▶ 2013 L21 @ 14:18]. Vasicek (1977) used it for the short interest rate, with m the target or equilibrium rate (for instance the rate targeted by the central bank) [▶ 2024 L25 @ 1:15:12].

References. P. Langevin, *C. R. Acad. Sci. Paris* 146, 530 (1908); G. E. Uhlenbeck and L. S. Ornstein, “On the theory of the Brownian motion”, *Phys. Rev.* 36, 823 (1930); O. Vasicek, “An equilibrium characterization of the term structure”, *J. Financial Economics* 5, 177–188 (1977).

with $e^{\int_0^t \kappa}$. That this is a solution can also be checked directly; verifying the formula for the Vasicek form of the model is PS9 Problem A-3 (Exercise 18.3).

Exercise 18.3 (PS9 (2013), Problem A-3: the Vasicek model). (Vasicek interest-rate model.) Prove that $R(t) = e^{-\beta t} R(0) + \frac{\alpha}{\beta} (1 - e^{-\beta t}) + \sigma e^{-\beta t} \int_0^t e^{\beta s} dB_s$ solves the SDE $dR(t) = (\alpha - \beta R(t)) dt + \sigma dB_t$.

18.3.3 Properties

The stochastic integral in (18.10) has a deterministic integrand, so it is a Gaussian random variable with mean 0 and, by the Itô isometry, variance $\sigma^2 \int_0^t e^{-2\kappa(t-s)} ds$ (chapter 17). This gives:

Proposition 18.6 (Moments, law, stationarity). Let X solve (18.8). Then X is a Gaussian process and

$$\mathbb{E}[X_t] = m + (x_0 - m)e^{-\kappa t} \rightarrow m, \quad (18.11)$$

$$\text{Var } X_t = \sigma^2 \int_0^t e^{-2\kappa(t-s)} ds = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t}) \rightarrow \frac{\sigma^2}{2\kappa}, \quad (18.12)$$

$$\text{Cov}(X_s, X_t) = \frac{\sigma^2}{2\kappa} (e^{-\kappa(t-s)} - e^{-\kappa(t+s)}), \quad 0 \leq s \leq t. \quad (18.13)$$

As $t \rightarrow \infty$, $X_t \rightarrow N(m, \sigma^2/(2\kappa))$. If $X_0 \sim N(m, \sigma^2/(2\kappa))$ is independent of B , then $X_t \sim N(m, \sigma^2/(2\kappa))$ for all t : the process is strictly stationary, with autocorrelation $\rho(\tau) = e^{-\kappa|\tau|}$.

Proof. (18.11) and (18.12) were shown above ([► 2024 L25 @ 1:17:17], slide 18). For the covariance, $s \leq t$: $\text{Cov}(X_s, X_t) = \sigma^2 e^{-\kappa(t+s)} \mathbb{E}[\int_0^s e^{\kappa u} dB_u \int_0^t e^{\kappa u} dB_u] = \sigma^2 e^{-\kappa(t+s)} \int_0^s e^{2\kappa u} du$, since the part of the second integral beyond s is independent of the first, and this is (18.13). If X_0 is independent with variance $\sigma^2/(2\kappa)$, then $\text{Var } X_t = e^{-2\kappa t} \sigma^2/(2\kappa) + \sigma^2(1 - e^{-2\kappa t})/(2\kappa) = \sigma^2/(2\kappa)$ and the mean stays m ; the covariance becomes $\frac{\sigma^2}{2\kappa} e^{-\kappa(t-s)}$, a function of $t - s$; a Gaussian process with stationary mean and covariance is strictly stationary. $\square$

[► 2024 L25 @ 1:17:17][► 2013 L21 @ 19:56] This is the point of slide 18 and of the transcript: unlike Brownian motion, whose variance grows like t , the OU process has a *limiting distribution* with constant mean and variance; it is ergodic (time averages converge to the averages under the stationary law), so the long-run distribution does not depend on the starting point [► 2024 L25 @ 1:17:17]. The mean decays exponentially to m with relaxation time $1/\kappa$; the *half-life* of a deviation is $\ln 2/\kappa$.

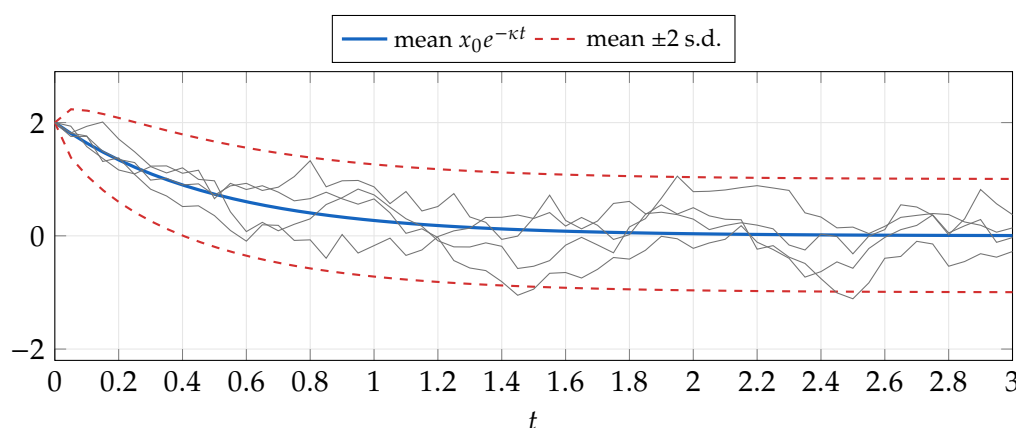


Figure 18.1: Four exact simulated paths of the OU process ($\kappa = 2$, $m = 0$, $\sigma = 1$, $x_0 = 2$; seed 11) with the mean and the band mean ± 2 standard deviations from (18.11)–(18.12). The paths are pulled to $m = 0$ within a time of order $1/\kappa = 0.5$ and then fluctuate with standard deviation $\sigma/\sqrt{2\kappa} = 0.5$ around it; compare the growing fan of Brownian paths.

IN FINANCE: Vasicek's model and its limits

In Vasicek's model the short rate R satisfies $dR = (\alpha - \beta R) dt + \sigma dB$: this is (18.8) with $\kappa = \beta$, $m = \alpha/\beta$, and (18.10) is exactly the formula of PS9 Problem A-3. Its virtues are tractability (Gaussian rates, closed-form bond prices, section 18.5) and an economic story (rates return to an equilibrium level). Its defects: rates can become negative (the stationary law is normal; with $\sigma/\sqrt{2\kappa} = 1\%$ around $m = 4\%$ the probability is $\Phi(-4) \approx 3 \times 10^{-5}$, but not zero), the volatility does not depend on the level, and a single factor cannot fit an arbitrary yield curve; the Hull–White extension makes m a function of time to fit the curve, and section 18.8 gives a positive alternative (CIR). Historical mean reversion does not survive over decades: the level to which rates revert itself moves (chapter 9).

18.3.4 Sampling an OU process: the exact AR(1)

Chapter 9 claims that the AR(1) is the discrete-time version of the OU process, with autoregressive coefficient $\phi = e^{-\kappa h}$ and noise variance $\sigma^2(1 - e^{-2\kappa h})/(2\kappa)$. Here is the derivation (an addition of these notes).

Proposition 18.7 (Exact discretisation). Let X solve (18.8) and $h > 0$. For every $t \geq 0$,

$$X_{t+h} = m + \phi(X_t - m) + \varepsilon_{t+h}, \quad \phi = e^{-\kappa h}, \quad \varepsilon_{t+h} = \sigma \int_t^{t+h} e^{-\kappa(t+h-s)} dB_s, \quad (18.14)$$

where $\varepsilon_{t+h} \sim N(0, \sigma_\varepsilon^2)$ with

$$\sigma_\varepsilon^2 = \sigma^2 \frac{1 - e^{-2\kappa h}}{2\kappa} = \frac{\sigma^2}{2\kappa} (1 - \phi^2), \quad (18.15)$$

independent of $\mathcal{F}_t$. Hence X_{ih} , $i = 0, 1, 2, \dots$, is exactly the Gaussian AR(1) process $X_{i+1} - m = \phi(X_i - m) + \varepsilon_{i+1}$ with i.i.d. noise; its stationary variance is $\sigma_\varepsilon^2/(1 - \phi^2) = \sigma^2/(2\kappa)$ and its autocorrelation $\phi^j = e^{-\kappa jh}$ is the sampled continuous autocorrelation.

Proof. Apply the integrating factor on $[t, t+h]$ instead of $[0, t]$: $d(e^{\kappa u}(X_u - m)) = \sigma e^{\kappa u} dB_u$, so $e^{\kappa(t+h)}(X_{t+h} - m) - e^{\kappa t}(X_t - m) = \sigma \int_t^{t+h} e^{\kappa u} dB_u$; multiply by $e^{-\kappa(t+h)}$ to get (18.14). The noise is a Gaussian Wiener integral of a deterministic function over $[t, t+h]$, so it is independent of $\mathcal{F}_t$ and normal with variance $\sigma^2 \int_t^{t+h} e^{-2\kappa(t+h-s)} ds = \sigma^2(1 - e^{-2\kappa h})/(2\kappa)$ by isometry. Since $1 - \phi^2 = 1 - e^{-2\kappa h}$, the stationary variance of the AR(1) is $\sigma_\varepsilon^2/(1 - \phi^2) = \sigma^2/(2\kappa)$, the value of (18.12) at $t = \infty$, independent of h , as it must be. $\square$

The intercept of the AR(1) in the usual form $X_{i+1} = c + \phi X_i + \varepsilon$ is $c = m(1 - \phi)$, and the half-life of a shock, $h \ln 2/(-\ln \phi) = \ln 2/\kappa$, is the one quoted in section 9.3.1. Three practical consequences:

- *Estimation.* Fit an AR(1) to data sampled every h (chapter 9) and read off the continuous-time parameters: $\hat{\kappa} = -\ln \hat{\phi}/h$, $\hat{m} = \hat{c}/(1 - \hat{\phi})$, $\hat{\sigma}^2 = 2\hat{\kappa} \hat{\sigma}_\varepsilon^2/(1 - \hat{\phi}^2)$. This needs $0 < \hat{\phi} < 1$: an AR(1) with $\phi \leq 0$ is not the sampling of any OU process (it would oscillate around the mean). Example: monthly data ($h = 1/12$) with $\hat{\phi} = 0.98$ gives $\hat{\kappa} = 12 \cdot 0.0202 = 0.24$ per year, half-life 2.9 years.
- *The exact scheme is preferable to Euler.* The Euler–Maruyama step $X_{i+1} = X_i - \kappa(X_i - m)h + \sigma\sqrt{h}Z$ is also an AR(1), but with $\phi_E = 1 - \kappa h$ and $\sigma_E^2 = \sigma^2 h$: both are the first-order Taylor expansions of (18.14)–(18.15), and its stationary variance is $\sigma^2/(\kappa(2 - \kappa h))$, not $\sigma^2/(2\kappa)$; the scheme is not even stable when $\kappa h > 2$. With $\kappa = 2$, $h = 0.25$ (own computation): exact $\phi = 0.607$, noise variance $0.158 \sigma^2$, stationary variance $0.25 \sigma^2$; Euler $\phi = 0.5$, noise variance $0.25 \sigma^2$, stationary variance $0.333 \sigma^2$, a third too large.
- *Simulation.* Paths of the OU process on any grid are generated exactly (no discretisation error) by iterating (18.14); this was done for fig. 18.1. The same holds for

geometric Brownian motion, $X_{t+h} = X_t \exp((\mu - \sigma^2/2)h + \sigma\sqrt{h}Z)$; for most SDEs there is no such scheme (section 18.7).

Exercise 18.4 (not from the course: OU covariance and estimation).

- (a) Derive (18.13) and check it against (18.12) at $s = t$.
- (b) Show that the OU process is the only stationary Gaussian Markov process with continuous paths, up to parameters (hint: $\rho(\tau)$ must satisfy $\rho(\tau + \tau') = \rho(\tau)\rho(\tau')$).
- (c) Daily data ($h = 1/252$) of a spread give an AR(1) fit with $\hat{\phi} = 0.995$ and residual standard deviation 0.02. Compute $\hat{\kappa}$, the half-life in days and the stationary standard deviation of the spread.

Exercise 18.5 (not from the course: exact versus Euler for OU). Find the stationary variance of the Euler–Maruyama chain $X_{i+1} = X_i - \kappa(X_i - m)h + \sigma\sqrt{h}Z_i$ and compare it with $\sigma^2/(2\kappa)$. For which h is the chain stationary? By what factor is the variance wrong for $\kappa h = 0.5$? Write a short program that simulates both schemes and compares the empirical variance.

18.4 Systems of SDEs; position and velocity

Several quantities driven by several noises are described by a *vector SDE* [► 2024 L25 @ 1:18:22]

$$d\mathbf{X}_t = \boldsymbol{\mu}(t, \mathbf{X}_t) dt + \boldsymbol{\sigma}(t, \mathbf{X}_t) d\mathbf{B}_t, \quad \mathbf{X}_0 = \mathbf{x}_0, \quad (18.16)$$

where $\mathbf{X} \in \mathbb{R}^m$, $\boldsymbol{\mu}$ is a vector of drifts $\mu_1, \dots, \mu_m$, $\mathbf{B} = (B^{(1)}, \dots, B^{(m)})$ are *independent* Brownian motions and $\boldsymbol{\sigma} = [\sigma_{ij}]$ is an $m \times m$ matrix (slide 19). The noises in the components are correlated in general: the increments over dt have covariance matrix $\boldsymbol{\sigma}\boldsymbol{\sigma}^\top dt$, which is what the lecturer means by “a covariance matrix reflecting how increments of the components are related” [► 2024 L25 @ 1:18:22]. Existence and uniqueness hold under the same Lipschitz and growth conditions with $|\cdot|$ read as Euclidean and matrix norms (quoted).

18.4.1 Linear systems

When $\boldsymbol{\mu} = -A(\mathbf{x} - \mathbf{m})$ and $\boldsymbol{\sigma} = \Sigma$ are constant, the integrating factor e^{At} solves the system exactly as in the scalar case (own extension):

$$\mathbf{X}_t = \mathbf{m} + e^{-At}(\mathbf{x}_0 - \mathbf{m}) + \int_0^t e^{-A(t-s)}\Sigma d\mathbf{B}_s, \quad \text{Cov}(\mathbf{X}_t) = \int_0^t e^{-Au}\Sigma\Sigma^\top e^{-A^\top u} du. \quad (18.17)$$

The process is stationary in the limit iff all eigenvalues of A have positive real part; the limiting covariance C_∞ solves the Lyapunov equation $AC_\infty + C_\infty A^\top = \Sigma\Sigma^\top$ (scalar case: $C_\infty = \sigma^2/(2\kappa)$). Sampled every h the process is *exactly* the VAR(1) of chapter 9 (section 9.5), $\mathbf{X}_{i+1} - \mathbf{m} = \Phi(\mathbf{X}_i - \mathbf{m}) + \boldsymbol{\varepsilon}_{i+1}$ with $\Phi = e^{-Ah}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \int_0^h e^{-Au}\Sigma\Sigma^\top e^{-A^\top u} du$; the stationarity condition of the VAR (eigenvalues of Φ inside the unit disc) is precisely $\text{Re } \lambda(A) > 0$.

18.4.2 The example of the slides: position and velocity

Slide 20 puts a mean-reverting velocity V and the position X it moves in one system

[► 2024 L25 @ 1:19:23]:

$$dX_t = V_t dt + \sigma_1 dB_t^{(1)}, \quad dV_t = -\kappa(V_t - m) dt + \sigma_2 dB_t^{(2)}, \quad (18.18)$$

that is (18.16) with $\mu = (v, -\kappa(v - m))$ and $\sigma = \text{diag}(\sigma_1, \sigma_2)$. The velocity is an OU process, and the position is its running integral plus independent noise; the position itself is *not* mean-reverting ($A = \begin{pmatrix} 0 & -1 \\ 0 & \kappa \end{pmatrix}$ has a zero eigenvalue). The slide only sets up the system; we compute its most instructive statistic (own computation). Take $m = 0$, $\sigma_1 = 0$, $X_0 = 0$ and $V_0 \sim N(0, \sigma^2/(2\kappa))$ stationary, $\sigma = \sigma_2$. Then $X_t = \int_0^t V_s ds$ and

$$\text{Var } X_t = 2 \int_0^t (t-s) C(s) ds = \frac{\sigma^2}{\kappa^2} \left(t - \frac{1 - e^{-\kappa t}}{\kappa} \right), \quad C(s) = \frac{\sigma^2}{2\kappa} e^{-\kappa s}. \quad (18.19)$$

Two regimes: for $\kappa t \ll 1$, $\text{Var } X_t \approx \frac{\sigma^2}{2\kappa} t^2 = \langle v^2 \rangle t^2$ (*ballistic* motion: the velocity has no time to change); for $\kappa t \gg 1$, $\text{Var } X_t \approx \frac{\sigma^2}{\kappa^2} t = 2Dt$ with the diffusion coefficient $D = \int_0^\infty C(s) ds = \sigma^2/(2\kappa^2)$ (*diffusive* motion, Einstein's law; this is the Green–Kubo formula). A Monte Carlo check with $\kappa = \sigma = 1$, $t = 3$ over 2×10^5 paths gives 2.046 against the exact 2.050 of (18.19). Independent position noise σ_1 simply adds $\sigma_1^2 t$.

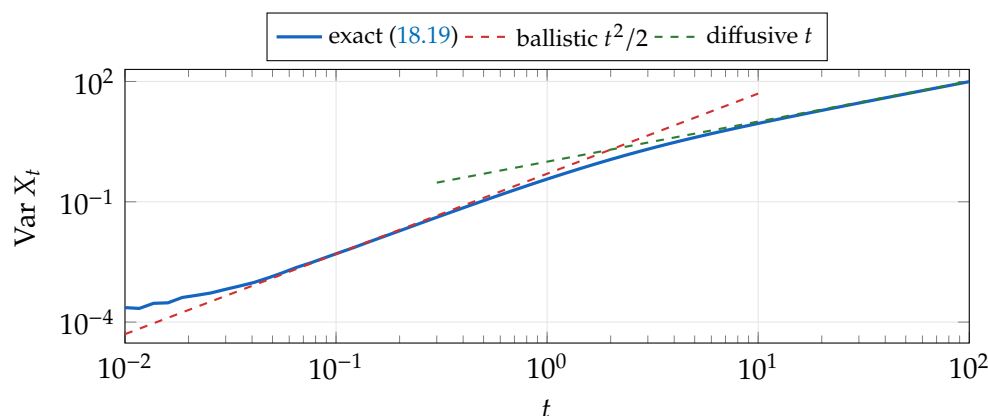


Figure 18.2: Variance of the position, (18.19) with $\kappa = \sigma = 1$: slope 2 at short times, slope 1 at long times, crossover at $t \sim 1/\kappa$.

INTUITION: How Brownian motion arises

Let $\kappa \rightarrow \infty$ with $\sigma^2/\kappa^2 = 2D$ fixed: the velocity forgets its past infinitely fast and $X_t \rightarrow \sqrt{2D} B_t$. Brownian motion is the overdamped limit of the position of a particle with a mean-reverting velocity, the description of [section 11.1.1](#). In finance the same structure appears when a factor (say a trend, or a stochastic mean) drives an asset whose own noise is small.

IN FINANCE: Mean-reverting spreads; not discussed in class

A relative-value (“pairs”) trade models a spread $S_t = \ln P_t^1 - \beta \ln P_t^2$ between two related assets as an OU process. Estimating the AR(1) coefficient of the sampled spread gives κ and the half-life of [section 18.3.4](#); the stationary standard deviation $\sigma/\sqrt{2\kappa}$ sets the scale of the signal (trade when $|S - m|$ exceeds a multiple of it). If $\hat{\phi}$ is close to 1 the

spread is indistinguishable from a random walk (unit root, [chapter 9](#)) and the strategy has no edge: the OU model is only as good as the mean reversion is real.

Exercise 18.6 (not from the course: position and velocity). For the system (18.18) with $m = 0, \sigma_1 = 0$:

- (a) derive (18.19) without assuming stationarity of V_0 , taking $V_0 = v_0$ deterministic;
- (b) show that the diffusion coefficient in the long-time limit is $D = \sigma^2/(2\kappa^2)$ and relate it to γ and T in the physical parametrisation of the intuition box in [section 18.3](#);
- (c) show that X_t converges in law to $\sqrt{2D} B_t$ in the limit $\kappa \rightarrow \infty$ with $\sigma^2/\kappa^2 = 2D$ fixed.

18.5 Generators, Feynman–Kac and the Kolmogorov equations

This section is an addition of these notes: the lectures use the two facts below without proving them (2024 slides 2–3 derive the PDE by hedging [[2024 L25 @ 24:16](#)]; both years say that the Black–Scholes PDE is the heat equation, whose solution is the payoff integrated against a Gaussian [[2024 L25 @ 32:40](#)][[2013 L21 @ 54:18](#)], but neither proves the equivalence with an expectation), and [chapters 15](#) and [17](#) defer them to here.

18.5.1 The generator

For the diffusion (18.1) define the second-order differential operator

$$\mathcal{L}_t f(x) = \mu(t, x) f'(x) + \frac{1}{2} \sigma^2(t, x) f''(x). \quad (18.20)$$

It is the *generator* of X (the expansion (11.10) in [chapter 11](#) is the same statement at the level of one small step). Itô's formula (17.10) for $f(t, X_t)$ reads, in this notation,

$$f(t, X_t) = f(0, x_0) + \int_0^t (\partial_s + \mathcal{L}_s) f(s, X_s) ds + \int_0^t \sigma \partial_x f(s, X_s) dB_s. \quad (18.21)$$

The last term is a martingale (under an integrability condition, [theorem 17.4](#)); so $f(t, X_t) - \int_0^t (\partial_s + \mathcal{L}_s) f ds$ is a martingale. This is Dynkin's formula, and [theorem 17.17](#) is its case $\mathcal{L} = \frac{1}{2} \partial_x^2$: $f(t, B_t)$ is a martingale when $f_t + \frac{1}{2} f_{xx} = 0$. The generators of our examples:

$$\text{BM: } \frac{1}{2} \partial_x^2; \quad \text{GBM: } \mu x \partial_x + \frac{1}{2} \sigma^2 x^2 \partial_x^2; \quad \text{OU: } -\kappa(x - m) \partial_x + \frac{1}{2} \sigma^2 \partial_x^2.$$

18.5.2 The Feynman–Kac formula

Proof. Fix (t, x) and let $D_s = \exp(-\int_t^s r(v, X_v) dv)$, a process of finite variation with $dD_s = -r(s, X_s) D_s ds$. By the product rule ([corollary 17.20](#); $dD dX = 0$) and Itô's formula (18.21),

$$\begin{aligned} d(D_s u(s, X_s)) &= D_s \left[u_s + \mu u_x + \frac{1}{2} \sigma^2 u_{xx} - r u \right](s, X_s) ds + \\ &= D_s \sigma u_x(s, X_s) dB_s, \end{aligned}$$

the bracket vanishing by (18.22). Under the integrability assumption $D_s u(s, X_s)$ is a martin-

Theorem 18.8 (Feynman–Kac). Let $u \in C^{1,2}([0, T] \times \mathbb{R}) \cap C([0, T] \times \mathbb{R})$ solve the terminal value problem

$$\frac{\partial u}{\partial t} + \mu(t, x) \frac{\partial u}{\partial x} + \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 u}{\partial x^2} - r(t, x) u = 0, \quad u(T, x) = g(x), \quad (18.22)$$

with r continuous and bounded below, and assume $\mathbb{E} \int_t^T (e^{-\int_t^s r} \sigma u_x(s, X_s))^2 ds < \infty$. Then

$$u(t, x) = \mathbb{E} \left[\exp \left(- \int_t^T r(s, X_s) ds \right) g(X_T) \mid X_t = x \right], \quad (18.23)$$

where X solves (18.1) started at x at time t .

gale on $[t, T]$, hence $u(t, x) = \mathbb{E}[D_t u(t, X_t)] = \mathbb{E}[D_T u(T, X_T) \mid X_t = x] = \mathbb{E}[D_T g(X_T) \mid X_t = x]$. $\square$

Read the formula in both directions: a PDE solved *backwards* from the terminal condition g is a conditional expectation of the payoff, discounted along the path; conversely every such expectation solves the PDE (under regularity conditions, quoted). Special cases:

- $r = 0, \mu = 0, \sigma = 1$: $u_t + \frac{1}{2} u_{xx} = 0$, $u(t, x) = \mathbb{E}[g(x + B_{T-t})]$: the backward heat equation of [theorem 17.17](#). With $\tau = T - t$ it is the heat equation of [section 18.6](#).
- *Black–Scholes*. Take for X the price under the risk-neutral measure, $dS = rS dt + \sigma S dW$ ([section 17.9](#)), a constant rate r and g the payoff. Then (18.22) is $u_t + rSu_S + \frac{1}{2} \sigma^2 S^2 u_{SS} - ru = 0$, the Black–Scholes equation ([17.20](#)) that slides 2–3 of the lecture derive by hedging [► 2024 L25 @ 24:16](#), and (18.23) is the price $e^{-r(T-t)} \mathbb{E}_Q[g(S_T) \mid S_t]$, the risk-neutral valuation of [chapter 15](#) (eq. (15.15)). Two proofs, one equation: the hedging argument shows that the price *must* satisfy the PDE; Feynman–Kac shows that the risk-neutral expectation *does*, and the PDE with its terminal condition has one solution. The real-world drift μ of S disappears because the expectation is taken under $\mathbb{Q}$, where the drift is r (Girsanov’s theorem, [theorem 17.23](#)).

Example 18.9 (Second moment of the OU process). Let X be the OU process (18.8) and $g(x) = x^2$, $r = 0$, $\tau = T - t$. From (18.10)–(18.12), $u(t, x) = \mathbb{E}[X_T^2 \mid X_t = x] = (m + (x - m)e^{-\kappa\tau})^2 + \sigma^2 \frac{1 - e^{-2\kappa\tau}}{2\kappa}$. Feynman–Kac says this solves $u_t - \kappa(x - m)u_x + \frac{1}{2} \sigma^2 u_{xx} = 0$ with $u(T, x) = x^2$; substituting confirms it (checked symbolically for these notes). Similarly for geometric Brownian motion $u = x^p \exp[(p\mu + \frac{1}{2}p(p-1)\sigma^2)\tau]$ solves the PDE with $g = x^p$.

IN FINANCE: The Vasicek bond price

A zero-coupon bond paying 1 at T has price $P(t, T) = \mathbb{E}_Q[e^{-\int_t^T R_s ds} \mid R_t = r]$ ([chapter 6](#)). By Feynman–Kac (with $\mu(r) = \kappa(\theta - r)$, the OU drift under $\mathbb{Q}$, and $r(t, x) = x$, $g = 1$) it solves $P_t + \kappa(\theta - r)P_r + \frac{1}{2} \sigma^2 P_{rr} - rP = 0$, $P(T, r) = 1$. The affine ansatz $P = \exp[a(\tau) - b(\tau)r]$, $\tau = T - t$, gives $-a' + b'r - \kappa(\theta - r)b + \frac{1}{2} \sigma^2 b^2 - r = 0$ for all r , hence

$$b' = 1 - \kappa b, \quad a' = -\kappa\theta b + \frac{1}{2} \sigma^2 b^2, \quad a(0) = b(0) = 0$$

and, with $b = (1 - e^{-\kappa\tau})/\kappa$,

$$P(t, T) = \exp \left[-b(\tau)r + (b(\tau) - \tau) \left(\theta - \frac{\sigma^2}{2\kappa^2} \right) - \frac{\sigma^2 b(\tau)^2}{4\kappa} \right]. \quad (18.24)$$

The yield $-\ln P/\tau$ tends to $\theta - \sigma^2/(2\kappa^2)$ for long maturities, below the level θ by a convexity term. Example ($\mathbb{Q}$ -parameters $\kappa = 0.5$, $\theta = 4\%$, $\sigma = 1\%$, $r_0 = 3\%$, $\tau = 5$): $P = 0.8343$, yield 3.62%, long yield 3.98%; a Monte Carlo average of $e^{-\int^R}$ over 2×10^5 exactly sampled paths gives 0.83429 ± 0.00006 (own computation; the formula is verified symbolically to satisfy the PDE). Under the real-world measure the drift is $\kappa(\theta_{\mathbb{P}} - r)$; with a constant market price of risk λ , Girsanov’s theorem ([theorem 17.23](#))

replaces the level by $\theta = \theta_{\mathbb{P}} - \sigma\lambda/\kappa$ and leaves κ, σ unchanged. This is a first example of the short-rate models of [chapter 24](#) (not discussed in class).

Exercise 18.7 (not from the course: Feynman–Kac).

- For geometric Brownian motion $dS = rS dt + \sigma S dW$ and payoff $g(S) = S^p$ compute $u(t, S) = e^{-r\tau} \mathbb{E}[S_T^p | S_t = S]$ and verify (18.22).
- Check that $u = S$ (the stock) and $u = e^{-r\tau}$ (a bond) solve the Black–Scholes equation, and explain in words what this says.
- For the OU process solve $u_t - \kappa(x - m)u_x + \frac{1}{2}\sigma^2 u_{xx} = 0$, $u(T, x) = x$, with the ansatz $u = A(\tau)x + B(\tau)$ and compare with the mean (18.11).

Exercise 18.8 (not from the course: Vasicek bond). Derive the ODEs for $a(\tau), b(\tau)$ in [section 18.5.2](#), solve them and confirm (18.24). Compute the yield curve for $\tau = 1, 2, 5, 10, 30$ years for $\kappa = 0.5$, $\theta = 4\%$, $\sigma = 1\%$, $r_0 = 3\%$. Show that the yield is a decreasing function of σ at long maturities and interpret it.

18.5.3 Transition densities: backward and forward equations

Let $p(s, x; t, y)$ be the density of X_t given $X_s = x$. Feynman–Kac with $r = 0$ and g arbitrary says $u(s, x) = \mathbb{E}[g(X_t) | X_s = x] = \int g(y)p(s, x; t, y) dy$ solves $u_s + \mathcal{L}_s u = 0$; hence p satisfies the *Kolmogorov backward equation* in (s, x) . There is also an equation in the *forward* variables (t, y) :

Proposition 18.10 (Fokker–Planck equation). *If the density $p(t, y)$ of X_t is smooth and decays fast enough,*

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial y} [\mu(t, y)p] + \frac{1}{2} \frac{\partial^2}{\partial y^2} [\sigma^2(t, y)p] =: \mathcal{L}_t^* p. \quad (18.25)$$

The Fokker–Planck equation is a conservation law, $\partial_t p + \partial_y J = 0$ with probability current $J = \mu p - \frac{1}{2} \partial_y (\sigma^2 p)$: the drift transports probability and the noise diffuses it. It is the equation a physicist derives from the Langevin equation (Kramers–Moyal expansion), and it is the equation $u_t = \lambda u_{xx}$ of the two lectures, with $\lambda = \sigma^2/2$, when $\mu = 0$ and σ is constant. The backward equation is the adjoint of the forward one: it describes how the *payoff* is smoothed, the forward one how the *density* spreads. Examples:

- The OU transition density is the Gaussian $N(m + (x - m)e^{-\kappa\tau}, \sigma^2(1 - e^{-2\kappa\tau})/(2\kappa))$ (from (18.14)); it satisfies (18.25) in (τ, y) , $\partial_\tau p = \kappa \partial_y [(y - m)p] + \frac{1}{2} \sigma^2 \partial_y^2 p$, and the backward equation in (τ, x) (checked symbolically for these notes).
- Stationary solution: $\partial_t p = 0$ and zero current, $\mu p = \frac{1}{2} (\sigma^2 p)'$; for OU, $p'/p = -2\kappa(y - m)/\sigma^2$, so $p_{\text{st}} \propto e^{-\kappa(y - m)^2/\sigma^2}$, the normal law $N(m, \sigma^2/(2\kappa))$ of [proposition 18.6](#). In general, for a drift $-U'(y)$ and constant σ , $p_{\text{st}} \propto e^{-2U(y)/\sigma^2}$: a Boltzmann distribution at effective temperature $\sigma^2/2$.
- Geometric Brownian motion: the lognormal density of [chapter 15](#) (eq. (15.17)).

Proof. For a smooth test function g with compact support, taking expectations in (18.21) (the martingale term has mean zero) gives $\frac{d}{dt} \mathbb{E}[g(X_t)] = \mathbb{E}[L_t g(X_t)] = \int (\mu g' + \frac{1}{2} \sigma^2 g'') p dy$. Integrating by parts once in the first term and twice in the second (no boundary terms) turns this into $\int g \mathcal{L}_t^* p dy$, while the left side is $\int g \partial_t p dy$. Since g is arbitrary, (18.25) follows. $\square$

Exercise 18.9 (not from the course: Fokker–Planck).

- For $dX = -U'(X)dt + \sigma dB$ with $U(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2$ (double well) find the stationary density and show it is normalisable.
- Derive the Fokker–Planck equation of geometric Brownian motion and check that the lognormal density solves it.
- For $\sigma^2 = 0.5$ compute the ratio of the stationary density at a well bottom $x = 1$ to that at the barrier top $x = 0$.

18.6 The heat equation

The 2013 lecture ends and the 2024 lecture begins its second half with the simplest partial differential equation of all, the diffusion (heat) equation [► 2013 L21 @ 42:14][► 2024 L25 @ 32:40]

$$\frac{\partial u}{\partial t} = \lambda \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x), \quad x \in \mathbb{R}, t > 0, \quad (18.26)$$

with $\lambda > 0$ (2024 slide 5; the 2013 note takes $\lambda = 1$). The reason to study it in a finance course is that the Black–Scholes equation (17.20) reduces to it by a change of variables [► 2013 L21 @ 43:17][► 2024 L25 @ 32:40], made explicit in section 15.7.3: the payoff is the initial condition, and the price today is the payoff diffused for a time $T - t$. Its physical model is a long, thin, perfectly insulated bar, $u(x, t)$ the temperature at position x [► 2024 L25 @ 36:58][► 2013 L21 @ 43:17]. The heat equation of chapter 11 (proposition 11.4) is the same equation for the transition density of Brownian motion, and by the Fokker–Planck equation (18.25) $\lambda = \sigma^2/2$.

18.6.1 Superposition and the Gaussian kernel

- Linearity.** If u_1, u_2 solve (18.26), so does $u_1 + u_2$; more generally a superposition $\int_{-\infty}^{\infty} u_s(x, t) c(s) ds$ of solutions indexed by s is a solution, if one may differentiate under the integral sign [► 2024 L25 @ 39:01][► 2013 L21 @ 44:20]. The strategy: solve the “easy” initial value problems and superpose [► 2024 L25 @ 39:01].
- The easy problem: a Dirac delta.** Put all the heat at one point, $u(x, 0) = \delta(x)$ ($\delta = 0$ for $x \neq 0$, $\int \delta = 1$; Dirac’s “function” that a physicist knows well, and that the lecturer introduces with a note on Dirac himself [► 2024 L25 @ 40:05][► 2013 L21 @ 46:28]). Its solution is the Gaussian density with mean 0 and variance $2\lambda t$,

$$u_\delta(x, t) = \frac{1}{\sqrt{4\pi\lambda t}} e^{-x^2/(4\lambda t)}, \quad (18.27)$$

which tends to δ as $t \rightarrow 0$ [► 2024 L25 @ 34:54][► 2013 L21 @ 47:38]. (For $\lambda = \frac{1}{2}$ it is the standard $N(0, t)$ density of Brownian motion, slide 7; for $\lambda = 1$ it is $\frac{1}{\sqrt{2\pi t}} e^{-x^2/4t}$, the 2013 note’s Observation 2.) Direct substitution shows that it solves (18.26): $\partial_t u_\delta = \lambda \partial_x^2 u_\delta$ (checked symbolically).

- General initial data.** Since $u_0(x) = \int \delta(x - s) u_0(s) ds$, superposing translated kernels gives

$$u(x, t) = \int_{-\infty}^{\infty} u_\delta(x - s, t) u_0(s) ds = \mathbb{E} \left[u_0(x + \sqrt{2\lambda} B_t) \right] \quad (18.28)$$

[► 2024 L25 @ 43:19][► 2013 L21 @ 47:38]. For a “reasonable” u_0 (say bounded and continuous, or of moderate growth such as a call payoff, since the kernel has Gaussian tails)

Example 18.11 (Indicator of an interval). Let $u_0 = 1_{(a,b)}$ [► 2024 L25 @ 48:39]. Then by (18.28), with $X_t = x + \sqrt{2\lambda} B_t \sim N(x, 2\lambda t)$,

$$u(x, t) = \mathbb{P}(a < X_t < b) = \Phi\left(\frac{b-x}{\sqrt{2\lambda t}}\right) - \Phi\left(\frac{a-x}{\sqrt{2\lambda t}}\right) \quad (18.29)$$

[► 2024 L25 @ 50:44], slide 9 for $\lambda = \frac{1}{2}$. Figure 18.3 shows the initial box melting into a bell curve.

the derivatives $\partial_t u_\delta, \partial_x^2 u_\delta$ are integrable and can be taken inside the integral, so u solves the equation, and $u(x, t) \rightarrow u_0(x)$ as $t \rightarrow 0$ [► 2024 L25 @ 45:27][► 2013 L21 @ 52:06]. The 2013 remark that the integral “need not be well defined” for arbitrary u_0 is the statement that the class of admissible initial data must be restricted. The form on the right is the probabilistic reading: *the solution at (x, t) is the expectation of the initial datum at the random point reached by a Brownian particle started at x* . In the lecture’s picture, heat is a cloud of particles released at one point, each moving as a Brownian motion, and their density is normal [► 2013 L21 @ 47:38]. This is Feynman–Kac (theorem 18.8) for Brownian motion.

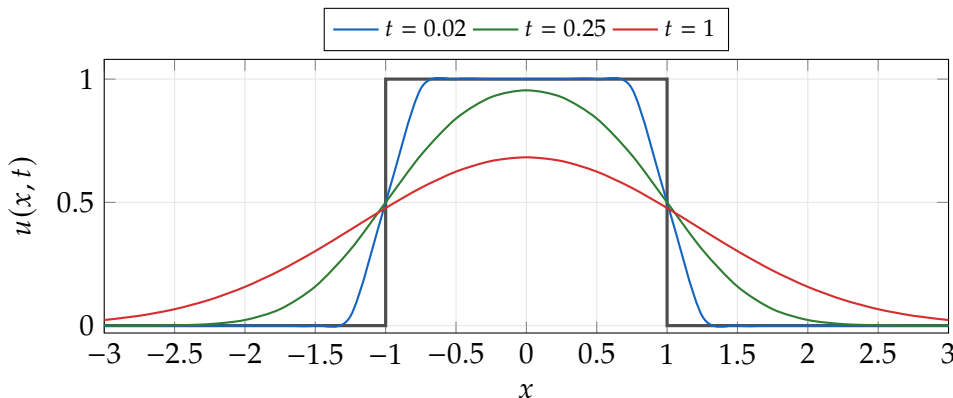


Figure 18.3: Solution (18.29) of the heat equation with $\lambda = \frac{1}{2}$ for $u_0 = \mathbf{1}_{(-1,1)}$ (thick black): at time t the profile has width of order $\sqrt{t}$ and its maximum falls. Each curve is the probability that a Brownian particle started at x is in $(-1, 1)$ at time t .

18.6.2 Solution by similarity and invariance

The 2024 lecture derives the same solution without the kernel [► 2024 L25 @ 51:49]. Rescale time and space, $\tau = \alpha t$ and $y = \beta x$, and put $v(y, \tau) = u(x, t)$. The chain rule gives $u_t = \alpha v_\tau$, $u_x = \beta v_y$, $u_{xx} = \beta^2 v_{yy}$, so (18.26) becomes $\alpha v_\tau = \lambda \beta^2 v_{yy}$: for $\alpha = \beta^2$, $v_\tau = \lambda v_{yy}$, the same equation [► 2024 L25 @ 52:53]. The heat equation is invariant under the scaling $t \rightarrow \alpha t$, $x \rightarrow \sqrt{\alpha} x$ (diffusive scaling). One then looks for solutions with the same invariance, $u(x, t) = u(\sqrt{\alpha} x, \alpha t)$ for all α ; taking $\alpha = 1/t$,

$$u(x, t) = u(x/\sqrt{t}, 1) = h(x/\sqrt{t}),$$

a function of one variable [► 2024 L25 @ 56:00]. Substituting $y = x/\sqrt{t}$ in (18.26): $u_t = -\frac{y}{2t} h'(y)$ and $u_{xx} = h''(y)/t$, so $-\frac{1}{2} y h' = \lambda h''$, i.e. $\frac{d}{dy} \ln h'(y) = -\frac{y}{2\lambda}$ [► 2024 L25 @ 57:04]. Integrating twice,

$$h'(y) = c e^{-y^2/(4\lambda)}, \quad h(y) = \Phi\left(\frac{y}{\sqrt{2\lambda}}\right)$$

(a Gaussian density with variance 2λ , whose antiderivative is the normal distribution function; the constants are fixed by $h(-\infty) = 0$, $h(+\infty) = 1$) [► 2024 L25 @ 58:08]. Hence

$$u(x, t) = \Phi\left(\frac{x}{\sqrt{2\lambda t}}\right) \quad \longrightarrow \quad H(x) = \begin{cases} 1 & x > 0 \\ \frac{1}{2} & x = 0 \\ 0 & x < 0 \end{cases} \quad (t \rightarrow 0), \quad (18.30)$$

so it solves the heat equation with a *step* initial condition [► 2024 L25 @ 59:14]. Translating, $\Phi((x - a)/\sqrt{2\lambda t})$ has a step at a ; the difference of two of them has the initial condition $\mathbf{1}_{(a,b)}$ (with value $\frac{1}{2}$ at the endpoints), which reproduces (18.29) [► 2024 L25 @ 1:01:25]. Every step function, and by limits every reasonable initial condition, is a combination of steps [► 2024 L25 @ 1:03:32].

Remark 18.12 (The kernel is the derivative of the step solution). Since ∂_x commutes with the equation, $\partial_x u$ solves it whenever u does, and $\partial_x H = \delta$. Indeed $\partial_x \Phi(x/\sqrt{2\lambda t}) = u_\delta(x, t)$ of (18.27): the two derivations of the lecture (delta function, then similarity for the step) are one, in this way (own remark). The 2013 note leaves the derivation of (18.27) as its Exercise 3.2, via $\xi = x/\sqrt{t}$ and $U(\xi) = t^{1/2}u(x, t)$ (Exercise 18.10).

Exercise 18.10 (lecnote21, Exercise 3.2: derivation of the heat kernel). Derive $u_\delta(x, t) = \frac{1}{2\sqrt{\pi t}}e^{-x^2/(4t)}$, the solution of $u_t = u_{xx}$ with $u(x, 0) = \delta(x)$, by using the variable $\xi = x/\sqrt{t}$ and the function $U(\xi) = t^{1/2}u(x, t)$, and restating the heat equation as an ODE for U .

18.7 Numerical solution of SDEs

Most SDEs, like most PDEs, have no closed-form solution, and the closed-form methods above fail for most of them [► 2013 L21 @ 19:56]. Both lectures list the numerical options, finite-difference methods, Monte Carlo methods and tree methods (2024 slide 21, without detail); the 2013 lecture explains them [► 2013 L21 @ 20:56].

18.7.1 Finite differences for an ODE

The idea of the finite-difference method is Taylor’s formula on a grid. The lecture’s example is $u'(x) = 5u(x) + 2$, $u(0) = 0$, to be evaluated at $x = 1$ [► 2013 L21 @ 22:04]. Chop $[0, 1]$ into steps of size h and use $u((i + 1)h) \approx u(ih) + h u'(ih) = u(ih) + h(5u(ih) + 2)$, which computes each value from the previous one [► 2013 L21 @ 23:07]. With $h = \frac{1}{2}$: $u(\frac{1}{2}) \approx 1$, $u(1) \approx \frac{9}{2}$ (the note’s Example 2.1). The exact solution is $u = \frac{2}{5}(e^{5x} - 1)$, $u(1) = 58.97$: with two steps the answer is off by a factor 13, and it improves only as $h \rightarrow 0$ (own computation of the scheme $u_n = \frac{2}{5}[(1 + 5h)^n - 1]$):

h	1/2	1/10	1/100	1/1000	exact
$u(1)$	4.50	22.67	52.20	58.23	58.97

The error is proportional to h (first order): this is the *Euler method*, and the theorem quoted in the lecture is that for a reasonable equation the approximation converges to the true value as $h \rightarrow 0$ [► 2013 L21 @ 25:16]. The same idea applies to functions of two variables on a grid, filled layer by layer from the boundary values [► 2013 L21 @ 26:18]. The lecture also warns about the cost/accuracy trade-off in choosing h [► 2013 L21 @ 39:02]: a smaller h gives a better answer and costs more time; and a grid fine enough for $X(1)$ may be too coarse for intermediate times close to 0 where only a few steps have been taken.

4 COURSE NOTE: 2013 and 2024 on the heat equation; notation and misprints

2013 states the kernel and the superposition and asserts the link with Black-Scholes; the reduction is proposed as a final project [► 2013 L21 @ 54:18](carried out in section 15.7.3). 2024 derives the solution by similarity and adds the step-function construction. Normalisations differ: 2013 has $u_t = u_{xx}$ ($\lambda = 1$), 2024 slide 5 has general λ and slides 8–9 specialise to $\lambda = \frac{1}{2}$, i.e. Brownian motion with unit variance rate. The 2013 note writes the initial condition sometimes as $u(0, x)$ and sometimes as $u_0(x, 0)$; the arguments of u appear in both orders ($u(x, t)$ and $u(0, x)$), which is only a notational slip. In the transcript of 2024 the argument of Φ is spoken as “ y over $2\sqrt{2\lambda}$ ” [► 2024 L25 @ 58:08]; the slide (and the calculation above) has $y/\sqrt{2\lambda}$.

18.7.2 Why it does not apply directly, and the fix: Monte Carlo along a fixed path

For the SDE (18.1) the analogue of $u' = f(u)$ has a random right-hand side: the point at time $2h$ could have been reached from many points at time h , and it is unknown which [► 2013 L21 @ 27:20]. The fix is to *fix a Brownian path*: once a sample path B is drawn, every increment $\Delta B_i = B_{(i+1)h} - B_{ih}$ is known and the same Taylor step can be used [► 2013 L21 @ 28:21]:

$$X_{i+1} = X_i + \mu(t_i, X_i)h + \sigma(t_i, X_i)\Delta B_i, \quad t_i = ih, \quad \Delta B_i = \sqrt{h}Z_i, \quad Z_i \sim N(0, 1) \text{ i.i.d.} \quad (18.31)$$

This is the *Euler–Maruyama scheme*. Then repeat with many independent paths (the *Monte Carlo simulation*): each gives one value $X_T^{(k)}$, and the empirical distribution of these values approximates the law of X_T [► 2013 L21 @ 31:37]. Any intermediate time can be read off the same paths, subject to the accuracy remark above. The steps of the note, *sample a path, solve along it, repeat*, are exactly this. (The obstacle is the underlying randomness of the Brownian motion; once a path is fixed it disappears [► 2013 L21 @ 33:43].)

Definition 18.13 (Strong and weak convergence; not discussed in class). A scheme with step $h = T/n$ and output X_T^h has *strong order* q if $\mathbb{E}[|X_T - X_T^h|] \leq Ch^q$ (paths are close: the two are driven by the same B), and *weak order* w if $|\mathbb{E}[f(X_T)] - \mathbb{E}[f(X_T^h)]| \leq Ch^w$ for smooth f (laws are close).

Euler–Maruyama has strong order $\frac{1}{2}$ and weak order 1 under the conditions of theorem 18.1 and smooth coefficients (quoted). The order $\frac{1}{2}$ is the price of the noise: over one step the increment is of size $\sqrt{h}$. The *Milstein* scheme adds the next term of the Itô–Taylor expansion, $X_{i+1} = X_i + \mu h + \sigma \Delta B_i + \frac{1}{2}\sigma\sigma'(\Delta B_i^2 - h)$, and has strong order 1 (the correction is 0 when σ is constant, so for OU Euler–Maruyama already has order 1 in the strong sense). Finally the Monte Carlo error itself, of order $N^{-1/2}$ in the number N of paths, adds to the discretisation error; the two should be balanced.

CASE STUDY: Convergence of Euler–Maruyama for geometric Brownian motion

Goal. Check the orders with a case where the exact solution (18.6) is known. *Set-up.* $dX = \mu X dt + \sigma X dB$, $\mu = 0.3$, $\sigma = 0.4$, $x_0 = 1$, $T = 1$ (the parameters of the 2024 example), $N = 2 \times 10^5$ Brownian paths generated on the finest grid ($n = 256$) and coarsened by summing increments, so that all schemes and the exact solution see the *same* path. Code (own; NumPy):

```
import numpy as np
rng = np.random.default_rng(1)
mu, sg, T, N, nmax = 0.3, 0.4, 1.0, 200000, 256
dW = rng.standard_normal((N, nmax)) * np.sqrt(T / nmax)
for n in (4, 8, 16, 32, 64, 128, 256):
    h = T / n
    d = dW.reshape(N, n, nmax // n).sum(axis=2) # coarse increments
    X = np.ones(N); M = np.ones(N)
    for i in range(n):
        X = X * (1 + mu*h + sg*d[:, i]) # Euler-Maruyama
        M = M * (1 + mu*h + sg*d[:, i] + 0.5*sg**2*(d[:, i]**2 - h)) # Milstein
    exact = np.exp((mu - sg**2/2)*T + sg*d.sum(axis=1))
    print(n, abs(X-exact).mean(), abs(M-exact).mean(), (1+mu*h)**n - np.exp(mu*T))
```

Result (columns: steps n , strong error of Euler, strong error of Milstein, weak error of $\mathbb{E}[X_T]$, which is deterministic here: $\mathbb{E}[X_T^h] = (1 + \mu h)^n$):

n	Euler, $\mathbb{E}[X_T - X_T^h]$	Milstein	$ \mathbb{E}[X_T^h] - \mathbb{E}[X_T] $
4	5.9×10^{-2}	3.3×10^{-2}	1.4×10^{-2}
16	3.0×10^{-2}	8.7×10^{-3}	3.7×10^{-3}
64	1.5×10^{-2}	2.2×10^{-3}	9.5×10^{-4}
256	7.6×10^{-3}	5.5×10^{-4}	2.4×10^{-4}

Halving h divides the Euler strong error by $\sqrt{2}$ (fitted slopes 0.49–0.50), the Milstein strong error and the weak error by 2 (slopes 0.94 rising to 1.00): strong order $\frac{1}{2}$ (Euler), 1 (Milstein) and weak order 1, as stated. The same run shows why the distinction matters for finance: for *prices*, i.e. expectations, the weak order is what counts, and Euler is enough; for path-dependent hedging error one needs paths, i.e. strong convergence. [Figure 18.4](#) plots the errors. *Lesson.* A convergence experiment against a known solution is the way to validate a simulator before using it where no closed form exists.

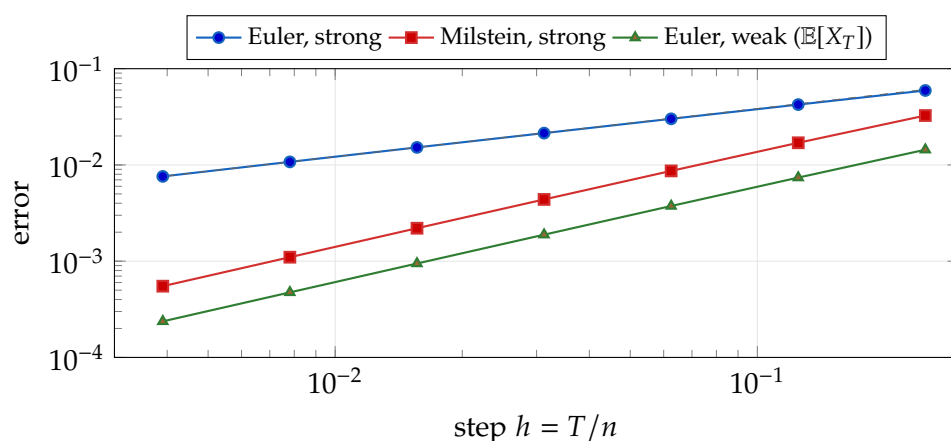


Figure 18.4: Errors of [section 18.7.2](#) against the step size: Euler strong error (slope $\frac{1}{2}$, the dashed line is a slope- $\frac{1}{2}$ reference), Milstein strong error and the weak error of $\mathbb{E}[X_T]$ (both slope 1).

Exercise 18.11 (not from the course: convergence experiment). Repeat [section 18.7.2](#) for the OU process (where the exact solution is (18.14)) and for the SDE $dX = \sin X \, dt + \cos X \, dB$ (no closed form: use a very fine Euler solution as reference). Fit the strong and weak orders of Euler–Maruyama and Milstein. Why does Milstein give no improvement for OU?

18.7.3 The tree method

Trees use the other picture of Brownian motion, as the limit of a random walk ([chapter 11](#), Donsker) [► 2013 L21 @ 34:48]: replace ΔB_i in (18.31) by $\pm\sqrt{h}$ with probability $\frac{1}{2}$ each. Then X_T takes only finitely many values, and their probabilities are known exactly: after n steps of a recombining tree the node with j up-moves has probability $\binom{n}{j}2^{-n}$ (in the lecture’s picture with $n = 5$ steps, 1, 5, 10, 10, 5, 1 over 32) [► 2013 L21 @ 36:54]. So one *computes* the expectation instead of sampling it: no statistical error. For geometric Brownian motion the multiplicative form $X_{i+1} = X_i(1 + \mu h \pm \sigma\sqrt{h})$ does recombine, and (18.31) is Euler–Maruyama with two-point increments. Example (own computation;

payoff $(X_T - 1)^+$, same parameters, exact value $x_0 e^{\mu T} \Phi(d_1) - K \Phi(d_2) = 0.41012$ with $d_1 = [\ln(x_0/K) + (\mu + \sigma^2/2)T]/(\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$:

steps n	16	64	256	1024
Euler-type tree $X(1 + \mu h \pm \sigma\sqrt{h})$, error	-5.9×10^{-3}	-1.2×10^{-3}	-2.2×10^{-4}	-7.5×10^{-5}
log-tree $x_0 e^{(\mu - \sigma^2/2)T + \sigma\sqrt{h}(2j-n)}$, error	-1.1×10^{-3}	4×10^{-5}	1.0×10^{-4}	5×10^{-6}

The errors are not monotone for the second tree (the strike sits at a different place relative to the nodes as n changes, a known feature of the binomial method for a kinked payoff) but tend to 0 with n . The 2013 speaker adds that accuracy can be increased in two ways, with more possibilities at each step (a trinomial tree with steps $+1, 0, -1$ also converges to Brownian motion) or with smaller time steps [► 2013 L21 @ 41:06], and remarks that he does not know which of the methods is used in practice, but that Monte Carlo and trees seem the more important ones [► 2013 L21 @ 36:54]. A comment from the audience in the 2013 lecture gives the practical criterion [► 2013 L21 @ 40:02]: for *path-independent* claims (the payoff depends on X_T only, so its law suffices), Monte Carlo is natural; for *path-dependent* problems, in particular early exercise (American options), a tree is preferable because at every node one can condition on the situation reached and compare exercise value with continuation value.

IN FINANCE: Pricing by simulation and by PDE

Monte Carlo pricing is Feynman–Kac (18.23) evaluated by simulation: simulate S under the risk-neutral dynamics $dS = rS dt + \sigma S dW$ (section 17.9), average the discounted payoff $e^{-rT} g(S_T)$ over N paths, with standard error $O(N^{-1/2})$ independent of dimension. This is the method of choice for high-dimensional or path-dependent payoffs; PDE methods (finite differences, section 18.6 and chapter 15) and trees are faster in one or two dimensions and handle early exercise naturally; the cost of a grid grows exponentially with dimension. Which drift to simulate is the point of section 18.5: the risk-free rate r , not the real-world μ (which matters for risk management and backtests, chapter 19).

18.8 Beyond the lectures: square-root and other diffusions

The two examples solved in class have linear coefficients, hence Lipschitz. Two extensions show what changes when they are not (an addition of these notes; none of this is in the lectures).

The CIR process. The Cox–Ingersoll–Ross model $dX = \kappa(\theta - X) dt + \sigma\sqrt{X} dB$ is a mean-reverting process whose noise vanishes at 0, so it stays nonnegative: the natural model of a positive interest rate or a variance (the Heston model of chapter 13 uses it for the variance). The coefficient $\sigma\sqrt{x}$ is not Lipschitz at 0, so theorem 18.1 does not apply, but $x \mapsto \sqrt{x}$ is Hölder- $\frac{1}{2}$ and the Yamada–Watanabe theorem (quoted) still gives a unique solution. Boundary behaviour is governed by the *Feller condition*: if $2\kappa\theta \geq \sigma^2$ the process never reaches 0; otherwise it hits 0 and is reflected. The linear drift makes the moments tractable: by Dynkin’s formula (18.21), $d\mathbb{E}[X_t] = \kappa(\theta - \mathbb{E}[X_t]) dt$, so

$$\mathbb{E}[X_t] = \theta + (x_0 - \theta)e^{-\kappa t}, \quad \text{Var } X_t = x_0 \frac{\sigma^2}{\kappa} (e^{-\kappa t} - e^{-2\kappa t}) + \theta \frac{\sigma^2}{2\kappa} (1 - e^{-\kappa t})^2$$

(the variance from the second-moment equation $\frac{d}{dt}\mathbb{E}[X^2] = (2\kappa\theta + \sigma^2)\mathbb{E}[X] - 2\kappa\mathbb{E}[X^2]$). Unlike OU the law is not normal: X_t is c times a noncentral χ^2 variable with $d = 4\kappa\theta/\sigma^2$ degrees of freedom, $c = \sigma^2(1 - e^{-\kappa t})/(4\kappa)$ and noncentrality $x_0 e^{-\kappa t}/c$, which allows exact sampling. Check ($\kappa = 1.5$, $\theta = 0.04$, $\sigma = 0.2$, $x_0 = 0.09$, $t = 1$; Feller ratio $2\kappa\theta/\sigma^2 = 3$): the formulas give mean 0.05116 and variance 7.379×10^{-4} ; 2×10^6 exact samples give 0.05114 and 7.375×10^{-4} (own computation). The Euler scheme (18.31) may produce negative values, for which $\sqrt{X}$ is undefined; the standard cure is to truncate ($\sqrt{X^+}$), and the exact sampler above avoids the issue. In the same family, the squared radius $|B|^2$ of an n -dimensional Brownian motion (section 17.8) satisfies $dZ = n dt + 2\sqrt{Z} dW$, a CIR process without mean reversion (the squared Bessel process).

Other state-dependent volatilities. The choice $\sigma(x) = \sigma x^\beta$ (CEV) interpolates between a normal model ($\beta = 0$, the OU/Bachelier world) and geometric Brownian motion ($\beta = 1$); local- and stochastic-volatility models, in which σ is a function of the price or itself a diffusion, lead to SDEs that must be treated numerically (chapter 13). The theory of this chapter (existence, generator, Feynman–Kac, Fokker–Planck, Euler–Maruyama) applies to all of them.

Exercise 18.12 (not from the course: CIR moments). Derive the moment equations of the CIR process from Dynkin’s formula and solve for $\mathbb{E}[X_t]$. Verify the stationary mean θ and variance $\theta\sigma^2/(2\kappa)$, and check the Feller condition for a short-rate CIR model with $\kappa = 0.3$, $\theta = 3\%$, $\sigma = 0.1$.

18.9 The two lectures compared, errata

5 COURSE NOTE: 2013 and 2024 compared

Common: existence and uniqueness under the Lipschitz and growth conditions (stated without proof); coefficient matching for geometric Brownian motion; the Ornstein–Uhlenbeck process solved by the product-process guess $a(t)(x_0 + \int b dB)$; the heat equation solved by superposition of Gaussians, with the link to Black–Scholes. *Only in 2013:* the goal of an SDE as the reverse of differentiation; finite differences for ODEs with the example $u' = 5u + 2$, the fixed-path Monte Carlo idea, the tree method with its binomial weights, the comments on trinomial trees and path dependence; the Dirac delta and Gaussian kernel as *given* (derivation left as Exercise 3.2). *Only in 2024:* the Black–Scholes equation by hedging (slides 2–4; the 2013 counterpart is a different lecture), the L^2 bound in the theorem, the lognormal “paradox” of geometric Brownian motion, the properties of the OU process (limit law, stationary start, the case $m \neq 0$), Vasicek’s model, systems of SDEs and the position–velocity example, the derivation of the heat solution by similarity/invariance and step functions; the numerical methods are only listed (slide 21). Neither year mentions the Feynman–Kac formula, Girsanov’s theorem for SDEs, the CIR process or the Euler–Maruyama order of convergence.

6 COURSE NOTE: Errata and misprints

- 2024 slide 15: $\sigma_* = \sigma t$ should be $\sigma\sqrt{t}$ (section 18.2).
- lecnote21, Theorem 1.1 and the

KEY FORMULAS

SDE:	$dX = \mu(t, X) dt + \sigma(t, X) dB, \quad X_t = x_0 + \int_0^t \mu ds + \int_0^t \sigma dB$
Existence, uniqueness:	$ \mu(x) - \mu(y) ^2 + \sigma(x) - \sigma(y) ^2 \leq K x - y ^2,$ $ \mu ^2 + \sigma ^2 \leq K(1 + x ^2) \Rightarrow \exists! \text{ solution, } \sup_t \mathbb{E}[X_t^2] < \infty$
Coefficient matching:	$X = f(t, B) : \mu(t, f) = f_t + \frac{1}{2} f_{xx}, \quad \sigma(t, f) = f_x$
GBM:	$dX = \mu X dt + \sigma X dB \Rightarrow X_t = x_0 e^{(\mu - \sigma^2/2)t + \sigma B_t}, \quad \mathbb{E}[X_t] = x_0 e^{\mu t}$ $\mu < \frac{1}{2} \sigma^2 : X_t \rightarrow 0 \text{ a.s.; } \text{std. dev. of } \ln X_t = \sigma \sqrt{t}$
OU:	$dX = -\kappa(X - m) dt + \sigma dB \Rightarrow X_t = m + (x_0 - m)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dB_s$ $\mathbb{E}[X_t] = m + (x_0 - m)e^{-\kappa t}, \quad \text{Var } X_t = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}), \quad X_\infty \sim N(m, \frac{\sigma^2}{2\kappa})$ $\text{Cov}(X_s, X_t) \rightarrow \frac{\sigma^2}{2\kappa} e^{-\kappa t-s } \text{ (stationary), } \text{half-life } \ln 2 / \kappa$
Exact AR(1):	$X_{t+h} - m = \phi(X_t - m) + \varepsilon, \quad \phi = e^{-\kappa h}, \quad \text{Var } \varepsilon = \frac{\sigma^2}{2\kappa}(1 - \phi^2)$ $\hat{\kappa} = -\ln \hat{\phi} / h, \quad \hat{\sigma}^2 = 2\hat{\kappa} \hat{\sigma}_\varepsilon^2 / (1 - \hat{\phi}^2)$
Vector OU:	$X_t = m + e^{-At}(x_0 - m) + \int_0^t e^{-A(t-s)} \Sigma dB_s,$ $\Phi = e^{-Ah}$
Generator:	$\mathcal{L} = \mu \partial_x + \frac{1}{2} \sigma^2 \partial_x^2, \quad df = (\partial_t + \mathcal{L})f dt + \sigma f_x dB$
Feynman–Kac:	$u_t + \mathcal{L}u - ru = 0, \quad u(T, \cdot) = g \iff u = \mathbb{E}[e^{-\int_t^T r} g(X_T) X_t = x]$
Fokker–Planck:	$\partial_t p = -\partial_y(\mu p) + \frac{1}{2} \partial_y^2(\sigma^2 p), \quad p_{\text{st}} \propto e^{-2U/\sigma^2} \quad (\mu = -U')$
Vasicek bond:	$P = e^{-br + (b - \tau)(\theta - \sigma^2/2\kappa^2)},$ $-\sigma^2 b^2 / 4\kappa, \quad b = \frac{1 - e^{-\kappa\tau}}{\kappa}$
Heat equation:	$u_t = \lambda u_{xx} \Rightarrow u = \int \frac{e^{-(x-s)^2/4\lambda t}}{\sqrt{4\pi\lambda t}} u_0(s) ds,$ $u = \mathbb{E}[u_0(x + \sqrt{2\lambda} B_t)]$ $u_0 = \mathbf{1}_{(a,b)} : u = \Phi\left(\frac{b-x}{\sqrt{2\lambda t}}\right) - \Phi\left(\frac{a-x}{\sqrt{2\lambda t}}\right)$
Euler–Maruyama:	$X_{i+1} = X_i + \mu h + \sigma \sqrt{h} Z_i; \text{ strong order } \frac{1}{2}, \text{ weak order } 1$
Milstein:	$+ \frac{1}{2} \sigma \sigma' (\Delta B_i^2 - h); \text{ strong order } 1$

paragraph before it: “for given functions a and b ” should be μ, σ ; the integral equation lacks x_0 ; “(L2 bound)” is a placeholder for $\sup_t \mathbb{E}[X_t^2] < \infty$; “Lipshictz” is spelled so in both the note and slide 13.

- **lecnote21**, §2.2 says that for a known path B_t “the equation now becomes a PDE”; for a fixed path it is a recursion in one variable (an ODE-type finite-difference step), as (18.31) shows; the tree and Monte Carlo descriptions write the equation as $df(t, B_t) = g dB + h dt$, which is the form of a function of Brownian motion rather than of a general SDE.
- The 2013 transcript’s remark that OU has “a drift term . . . minus exponential” at 13:16 is a slip: the drift is *linear*, $-\kappa X$; the exponentials appear only in the solution.
- In the 2024 transcript $\Phi(y/2\sqrt{2\lambda})$ is spoken; the slide has $\Phi(y/\sqrt{2\lambda})$, which is right (section 18.6).
- PS9 Problem A-1(c): see the note after Exercise 18.1. Everything else in the slides (Black–Scholes derivation, heat kernel, OU moments) was re-derived above and found correct; the numerics of this chapter were checked in Python and, where indicated, symbolically.

Part II

Additional topics from 2013 (18.S096)

Chapter 19

Value at Risk Models

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2013 L7	K. Abbott (Morgan Stanley, Firm Risk Management)	What a risk-management group is for; one-asset VaR as a 1% order statistic; returns instead of levels; the normal multipliers 2.33 and 1.645; fat tails; fixed-income positions through the PV01; covariance and correlation, portfolio VaR as $x^T \Sigma x$ (variance–covariance method); data and weighting choices; Monte Carlo VaR through the eigen-decomposition of the covariance matrix (the “Gaussian copula”). Historical simulation, marginal VaR and back-testing are on the slides only: Abbott announced that he would skip historical simulation.
Files. Slides MIT18_S096F13_lecnote7.pdf (115 pages, about 100 distinct slides). Problem sets. None. Abbott mentions a homework on plotting distributions that he usually sets before this lecture; it is not among the course files. (The t distribution that a student asks about [► 2013 L7 @ 26:29] belongs to chapter 13, where the 2013 problem set on volatility is treated.) Overlap with other chapters. Coherent risk measures, VaR and expected shortfall, the normal formulas, the square-root-of-time rule, the estimation of VaR by historical simulation and back-testing with the binomial test are in section 8.4 (from the 2024 lecture); VaR with GARCH and t innovations and the estimation of volatility are in section 13.7.5 and chapter 13. This chapter does not repeat them; it treats what the 2013 lecture adds: the <i>three methodologies</i> (variance–covariance, historical simulation, Monte Carlo) side by side, the practitioner’s choices behind each, decompositions of VaR by position, non-linear positions, and the practice of back-testing and stress testing. Sections or paragraphs marked “(not discussed in class)” are our additions.		

Motivation

A trading firm holds thousands of positions and must be able to answer, every morning and in a few numbers, “how much can we lose tomorrow?” Ken Abbott, who helped set up the risk group of a large bank in the mid-1980s and later became the operating officer of firm risk management at Morgan Stanley, describes the job of risk management as making sure that management knows the risk profile of the firm, protecting it from unacceptable concentrations of risk, and ensuring “no surprises” [► 2013 L7 @ 2:05]. He uses two analogies:

a *spotlight*, which points at any risk one wants to look at, and a *colouring book*, in which one then decides whether what is lit is good, bad, too big or too small [► 2013 L7 @ 5:22]. The second half of the job is communication: the chairman is often not a quant, so a complex idea has to be told in plain prose and simple pictures [► 2013 L7 @ 3:11].

Value at risk (VaR) is the number that came out of this need. The methods are elementary statistics dressed in linear algebra:

- a VaR is an *order statistic* of the distribution of tomorrow's profit and loss (section 19.1);
- positions in bonds are mapped to yield sensitivities before they enter (section 19.2);
- under normality the portfolio VaR is $z\sqrt{\mathbf{x}^T \Sigma \mathbf{x}}$, the *variance-covariance* method (section 19.3);
- the same order statistic can be read off historical data (*historical simulation*, section 19.4) or off simulated data drawn with the covariance structure of the past (*Monte Carlo*, section 19.5);
- all three approximate the P&L of non-linear positions (section 19.6), and all three are checked against realised P&L by *back-testing* and completed by stress tests (section 19.7).

The physicist will recognise most of the mathematics: a quadratic form in a covariance matrix, a diagonalisation, a random-number generator. What is new is the vocabulary and the long list of practical choices, each of which changes the number. Abbott's recurring warning is that whenever a modelling assumption moves the answer materially, "sirens should go off" [► 2013 L7 @ 57:48].

19.1 One asset: VaR as an order statistic

Throughout, VaR is measured on the *loss* $L = -\Delta P$ over a horizon that in this lecture is one day, as in definition 8.8: for a level β (typically 99% or 95%), VaR_β is the loss that is exceeded with probability $1 - \beta$. Abbott states it in words: the worst 1% of the possible outcomes [► 2013 L7 @ 9:34]. The object is an *order statistic*, "the maximum or the minimum of a set of observations" in the words of the professor [► 2013 L7 @ 8:29], here an intermediate one: the value below which a fraction $1 - \beta$ of the ordered outcomes lies. Order statistics have a large literature, so once VaR is recognised as one, its distribution and its estimation error are known [► 2013 L7 @ 9:34], as section 19.4 will use.

19.1.1 Returns, not levels

The starting point is a time series of prices, say the Mexican IPC stock index over 1995–96 (slide 7). Prices are close to a random walk (chapter 4): the best forecast of tomorrow's level is today's level, the series is not stationary, and a histogram of *levels* is a meaningless multimodal blob that depends on where the walk happened to wander [► 2013 L7 @ 12:48]. The histogram of one-day *returns* (percentage changes, log changes or, in some markets, absolute changes) is unimodal and roughly symmetric, with the familiar bell shape (slides 8–9) [► 2013 L7 @ 16:00]. This is the whole reason for working with returns: they are the increments of the walk, approximately stationary with a mean and a variance that do not drift, and it is for them that a distributional assumption can be made. It is the same step from a Brownian motion to its increments that the physicist takes when computing a diffusion coefficient.

Proposition 19.1 (One-asset variance method). Let a position of value V have a one-day return $r \sim N(0, \sigma^2)$. Then for $\beta \in (0, 1)$

$$\text{VaR}_\beta = z_\beta \sigma V, \quad z_\beta = \Phi^{-1}(\beta),$$

with $z_{0.99} = 2.326$ and $z_{0.95} = 1.645$. The recipe is: collect prices, compute returns, estimate the variance $\hat{\sigma}^2 = \frac{1}{n-1} \sum_i (r_i - \bar{r})^2$, take the square root, multiply by $z_\beta V$ (slide 14) [► 2013 L7 @ 30:40].

The multipliers are *one-sided*: the 99% VaR leaves 1% in a single tail, hence 2.33 and not the 1.96 of a two-sided 95% confidence interval, which leaves 2.5% in each tail [► 2013 L7 @ 10:38]. A student asks about short positions; in this model the answer is that the same number serves for a long or a short position because the density is symmetric, but “getting your signs right” is one of Abbott’s recurrent themes [► 2013 L7 @ 24:21]. He also uses the normal quantile as a heuristic on real, skewed data: “what is a two or three standard-deviation move?” means something to a manager even when the empirical distribution is lopsided [► 2013 L7 @ 24:21].

Remark 19.2 (Eight exceptions are not evidence of fat tails; our computation). The sample has about 520 daily returns (the 8 exceedances are 1.5% of about 530). If the normal VaR were exactly right, the number of exceedances would be $\text{Bin}(520, 0.01)$ with mean 5.2 and standard deviation 2.3, and $\mathbb{P}(N \geq 8) = 15\%$. The likelihood-ratio test of [section 19.7](#) gives $p = 0.25$. The data are compatible with the normal model at the 99% level: eight exceedances against five are noise. The heavy tails of index returns are real (they are visible in the higher-frequency data of [chapter 13](#)), but this count does not show it, which is a small lesson in the very point Abbott stresses next.

Proof. $L = -Vr \sim N(0, \sigma^2 V^2)$ and $\mathbb{P}(L \leq z_\beta \sigma V) = \Phi(z_\beta) = \beta$. This is [proposition 8.10](#) with $\mu = 0$; the mean is dropped because for a one-day horizon it is small against σ . □

Example 19.3 (Mexican IPC, slides 12–13). The variance of the daily IPC returns from January 1995 to December 1996 is 0.000324, so $\hat{\sigma} = 0.018012 = 1.8012\%$. Then $2.33 \times 1.8012\% = 4.1968\%$ (with $z_{0.99} = 2.3263$ one gets 4.190%): on a position of MXP 100 the loss exceeds MXP 4.2 only about 1% of the days [► 2013 L7 @ 22:18]. In fact the IPC lost more than 4.2% on 8 days of the sample, “about 1.5%” of the days, which Abbott reads as a sign of fat tails: large moves are more likely than the normal law implies [► 2013 L7 @ 26:29].

19.1.2 Everything is estimated with error

“Every time you estimate something you estimate it with error” [► 2013 L7 @ 17:01]: the *risk* is 10 means it is neither 10 000 nor 0 [► 2013 L7 @ 18:03]. The sample variance follows a chi-squared law: for i.i.d. normal returns $(n-1)\hat{\sigma}^2/\sigma^2 \sim \chi_{n-1}^2$ [► 2013 L7 @ 19:07] ([section 3.9](#)). Hence it cannot be negative, has a long right tail, and by [section 13.1.2](#) the relative standard error of $\hat{\sigma}$ is $1/\sqrt{2(n-1)}$: 3.1% for the 520 observations above, 4.5% for one year of data ($n = 252$). The VaR multiplies $\hat{\sigma}$, so it inherits this error before any model error. Abbott distinguishes the *standard deviation* of a data set from the *standard error* of an estimate [► 2013 L7 @ 20:10]. The empirical quantile has a larger error still ([proposition 19.9](#)).

IN FINANCE: Clean your data

“Step 2b: graph your data” [► 2013 L7 @ 29:37]. Everybody’s data contain errors, missing values, stale quotes, and Abbott’s rule is that one may be fired for incompetence but not for using someone else’s bad data. Sources named on the slides are the Federal Reserve (the H.15 series for yields and H.10 for exchange rates, free and going back decades), Bloomberg (the cleanest in his experience, precisely because every user reports errors) and commercial vendors (slide 46) [► 2013 L7 @ 51:31]. The same applies to software: he tells the story of a programmer who claimed that a matrix square root multiplied by itself reproduced the covariance matrix “to 16 decimals”, when the code was in fact passing a float to a fixed-point variable [► 2013 L7 @ 43:13]. The other 2013 case studies ([chapters 5](#) and [9](#)) do the same first step in R.

1 COURSE NOTE: Spoken claim on the frequency of large moves

Warning against extrapolating the normal law, Abbott says that a three-standard-deviation move occurs “once every 10 000 observations” [► 2013 L7 @ 25:25]. For a normal variable $\mathbb{P}(|Z| > 3) = 0.27\%$, i.e. one day in 370 (one in 741 for one tail), while $\mathbb{P}(|Z| > 4) = 6.3 \times 10^{-5}$ is one in 15 800 (one in 31 600 for one tail). The figure quoted lies between the two, and is presumably meant for four standard deviations or for a mixture of both. The remark about eight standard deviations is right in substance: $\mathbb{P}(|Z| > 8) = 1.2 \times 10^{-15}$, one day in 8×10^{14} , about 3×10^{12} years of daily data. Under a normal model such a move “does not happen”, so when it happens it is the model that has failed, not the market that was unlucky.

Exercise 19.1 (not from the course: estimation error of a quantile). Using [proposition 19.9](#), find the number of days of history needed to estimate the 99% VaR of a normal loss with a relative standard error of 5%; and the same for the 99.9% VaR. Repeat for a Laplace loss (density $\frac{1}{2b}e^{-|x|/b}$, quantile $b \ln \frac{1}{2(1-\beta)}$). (Answer for the first part: about 1 000 days.)

19.2 Bonds: from yield changes to price changes

Equities, currencies and commodities are quoted in prices, and the previous section applies directly. The time series available for bonds are *yields* (slide 16) [► 2013 L7 @ 31:42]. To turn a yield move into a money move one needs the slope of the price–yield curve at the current yield: a “poor man’s Jacobian”, in Abbott’s words, valid for small moves because the curve is non-linear and one uses its tangent [► 2013 L7 @ 31:42].

Definition 19.4 (PV01 (DV01, PVBP)). The *PV01* of a position is the change of its value for a rise of the yield by one basis point (1 bp = 0.01%). Its sign is negative for a long bond. For a bond of price P and modified duration D ([proposition 1.10](#)), $PV01 \approx -D P \times 10^{-4}$.

If σ_{bp} denotes the standard deviation of the daily yield change in basis points, the P&L of the position is $\approx PV01 \cdot \Delta y_{bp}$ and the variance method of [proposition 19.1](#) gives

$$VaR_{\beta} = z_{\beta} |PV01| \sigma_{bp}. \quad (19.1)$$

Abbott’s slides express the yield volatility as the volatility σ_y of the *relative* changes $\Delta y/y$ (which is the “percentage change” of a yield): then $\sigma_{bp} = 100 y\% \sigma_y$ with $y\%$ the yield in percent. Slide 17 writes this as position $\times PV01 \times \text{close} \times \sigma_{\text{yield}} \times 2.33 \times 100$ [► 2013 L7 @ 33:48]. For equities, currencies and commodities the analogue is position $\times \sigma_{\text{price}} \times 2.33$.

IN FINANCE: PV01, duration and units

Duration is the (present-value weighted) average time of the cash flows, measured in years; the PV01 is a money amount per basis point per unit of face value. A duration of 7.9 years means a PV01 of about \$790 per \$1 million of face value, “roughly the same digits but different units” [► 2013 L7 @ 33:48]. Bond traders think in PV01, insurance companies and portfolio managers who must match long-dated liabilities think in duration [► 2013 L7 @ 34:51]. “Be careful of the units: this is one of the easiest errors to make” (slide 20) is the most repeated sentence of the lecture. The market quotes yields, so risk is quoted per basis point: nobody computes $(702 - 701)/702$ in his head [► 2013 L7 @ 34:51]. See [section 1.8.4](#) and, for delta ladders by maturity bucket, [section 6.7](#).

Credit spread. For a corporate or asset-backed bond the yield is a risk-free rate plus a credit spread (slide 24: $6\% = 4.5\% + 1.5\%$ in Abbott’s example). One separates the interest-rate PV01 from the spread PV01 (CSPV01) and treats the two as different risk factors with their own volatilities and correlations; slide 23 states that the two coincide when the recovery rate is zero and are otherwise different, and lists calculated spreads, CDS and asset-swap spreads as sources [► 2013 L7 @ 36:52]. At spreads of 1 000–1 500 bp the spread is mainly a default probability, and models of default, loss given default and recovery take over from the smooth sensitivity used here.

Example 19.5 (USD 100 of a 10-year Treasury, slides 18–22). The bond has coupon 4.644% (semi-annual), ten years to maturity and yield equal to the coupon, so that its price is 100 (slide 21). Repricing at 4.654% gives 99.920765, hence $PV01 = -0.07923$ per 100 (slide 19; our recomputation gives -0.079235). The volatility of the relative changes of the yield is 1.312% per day. The daily yield volatility is therefore $4.644 \times 100 \times 0.01312 = 6.09$ bp, and

$$VaR_{99\%} = 2.33 \times 0.07923 \times 6.093 = 1.1248 \text{ per 100 of face value}$$

(slide 22: 1.12479). The 99% one-day loss is thus 1.12% of the position: a daily yield volatility of 6.1 bp times a duration of 7.9 years is 0.48% of price, and $2.33 \times 0.48\% = 1.12\%$.

2 COURSE NOTE: Erratum: slides 18–22, the ten-year example

- Slide 22 writes $VaR = -0.07923 \times 0.04644 \times .01312 \times 2.33 \times 100 = 1.12479$. With the yield entered as 0.04644 the product is 0.011248, not 1.12479: the factor 100 that converts percent points of yield to basis points has to be 10^4 if the yield is a fraction, or the “close” has to be entered in percent (4.644), as the value 1.12479 shows and as slide 17 requires.

Exercise 19.2 (not from the course: PV01). A bond has a semi-annual coupon of 3%, five years to maturity, and yield 3%. Compute its price at yields 3% and 3.01%, the PV01 per 100 face value, the modified and Macaulay durations, and the one-day 99% VaR of a position of face value 10^7 if the daily volatility of the relative changes of the yield is 1.5%. Compare PV01 with $-D_{\text{mod}} P \cdot 10^{-4}$.

- Slide 21 says that yield and coupon should be divided by 100 and writes “4.644% = .0644”; it should read .04644.
- The volatility on slide 18 is for the period “since 1/1/03” and the DV01 is computed “at the 12/31/96 yield”, while the Excel extract of slide 21 uses the date 12/27/2006: the dates are inconsistent (presumably 1/1/03 to 12/27/06, and 12/31/06).
- The cell labelled “Duration @ Yield=Coupon” (slide 21) is 7.9273; that is the *modified* duration, which is what the spreadsheet function returns. The Macaulay duration of this bond is 8.111 years, and $8.111/(1 + 0.04644/2) = 7.927$. The definition on slide 20 (“weighted average time to cash flows”) is that of the Macaulay duration.

19.3 The variance–covariance method

19.3.1 Covariance, correlation and the quadratic form

For two assets with returns a, b the sample covariance is $\hat{\sigma}_{ab} = \frac{1}{n-1} \sum_i (a_i - \bar{a})(b_i - \bar{b})$ (slide 25) and the correlation is $\hat{\rho}_{ab} = \hat{\sigma}_{ab}/(\hat{\sigma}_a \hat{\sigma}_b)$. Covariance says by how much and in which direction b moves when a moves away from its mean; it is not unit-free (measure the fertiliser in tons or in ounces and it changes), whereas correlation is a pure number in $[-1, 1]$, “an index of linearity” [► 2013 L7 @ 39:01]. If the covariances are known the correlations follow; the converse needs the variances too: $\Sigma = D R D$ with $D = \text{diag}(\sigma_1, \dots, \sigma_n)$ and R the correlation matrix [► 2013 L7 @ 41:07]. Both matrices are symmetric; Σ has the variances on the diagonal (the covariance of a variable with itself) and R has ones. Abbott convinces himself of $\text{Var}(a + b) = \text{Var } a + \text{Var } b + 2 \text{Cov}(a, b)$ on a spreadsheet: for the numbers of slide 27, $0.333059 + 0.469281 + 2 \times 0.349660 = 1.501660$, equal to the variance of $a + b$ computed directly [► 2013 L7 @ 42:10]. (This is an algebraic identity of the sample covariance, as Cov is bilinear, so the check tests the spreadsheet rather than the statistics, which is the point of checking.)

19.3.2 Portfolio VaR

With several assets one writes the variance of a combination with x units of a and y units of b as $x^2 \text{Var } a + y^2 \text{Var } b + 2xy \text{Cov}(a, b)$ and so on for three, four, n assets (slides 31–32), a pattern that becomes unmanageable “very fast” unless one uses matrices [► 2013 L7 @ 45:18]. Sum all the entries of the covariance matrix and one gets the variance of the portfolio holding one unit of each asset (slides 35–36); the general statement is [proposition 3.17](#).

Proposition 19.6 (Variance–covariance (delta-normal) VaR). Let $\mathbf{x} \in \mathbb{R}^n$ be the position vector, x_i being the P&L of the portfolio per unit return of risk factor i (in money units; for a bond it is the PV01 and the factor is the yield change in basis points), and let the vector of one-day factor changes $\mathbf{r}$ be $N(\mathbf{0}, \Sigma)$. Then $\Delta P = \mathbf{x}^T \mathbf{r} \sim N(0, \sigma_P^2)$ with

$$\sigma_P^2 = \mathbf{x}^T \Sigma \mathbf{x} = \sum_{i,j} x_i x_j \Sigma_{ij}, \quad \text{VaR}_\beta = z_\beta \sqrt{\mathbf{x}^T \Sigma \mathbf{x}} \quad [\text{► 2013 L7 @ 47:22}].$$

Consequently $\text{VaR}_\beta \leq z_\beta \sum_i |x_i| \sqrt{\Sigma_{ii}}$, the sum of the stand-alone VaRs, with equality iff the P&L contributions $x_i r_i$ are pairwise perfectly positively correlated.

Proof. A linear combination of jointly normal variables is normal with variance $\mathbf{x}^T \Sigma \mathbf{x}$ ([propositions 3.17](#) and [3.21](#)); the level follows as in [proposition 19.1](#). The bound is the Cauchy–Schwarz inequality, $|\Sigma_{ij}| \leq \sqrt{\Sigma_{ii} \Sigma_{jj}}$, applied to each term: $\mathbf{x}^T \Sigma \mathbf{x} \leq (\sum_i |x_i| \sqrt{\Sigma_{ii}})^2$. □

Written with the correlation matrix, $\sigma_P^2 = \mathbf{y}^T R \mathbf{y}$ with $y_i = x_i \sigma_i$ the stand-alone one-standard-deviation P&L of factor i : the standard deviations add in quadrature, corrected by the correlations. The physicist’s reading: σ_P is the length of $\mathbf{x}$ in the metric Σ ; uncorrelated risks add like independent errors, perfectly correlated risks add like lengths, and negative correlations cancel.

IN FINANCE: Signs, units and the position vector

Nearly every mistake in this method is a sign or a unit [► 2013 L7 @ 43:13]: “like physics, only worse, because if you get the units wrong in physics you blow up, and here it just costs money”. A short gold position is entered as a negative number. The exchange rate dollar/yen is quoted in yen per dollar (say 95): a dollar investor who owns yen loses when the quote rises to 100, so the position enters as -100 , whereas if the covariance matrix measures yen in dollars per yen (0.0105) the same position is $+100$ [► 2013 L7 @ 48:22]. For a bond, similarly, the PV01 of a long position is negative because the yield is the factor [► 2013 L7 @ 49:25]. Positions must all be in the same currency and in the same scale (units or millions), and the covariance matrix must use the same convention for returns (percent or log changes). Volatilities and correlations must be for the same differencing interval as the VaR horizon.

Exercise 19.3 (not from the course: two assets). Two positions with daily volatilities $\sigma_1 = 1\%$, $\sigma_2 = 2\%$ (P&L per unit of position value), correlation ρ , and values $V_1 = 100$, $V_2 = 50$ (long both).

- Write the 99% VaR as a function of ρ using [proposition 19.6](#); for which ρ is it equal to the sum of the stand-alone VaRs?
- Find the position V_2 (possibly negative) that minimises the VaR for given V_1 and ρ , and show that it is the minimum-variance hedge $V_2 = -\rho(\sigma_1/\sigma_2)V_1$.
- Compute the component VaRs of the original portfolio for $\rho = 0.5$ and check that they add up to the VaR.

Example 19.7 (Three currencies, slides 35–42). Take positions of -100 in CAD, -50 in CHF and -25 in DEM (USD amounts, written in parentheses on slide 39; on the sign convention see the box below) and the covariance matrix of the daily changes

$$\Sigma = \begin{pmatrix} 0.000037 & -0.000018 & -0.000017 \\ -0.000018 & 0.000321 & 0.000257 \\ -0.000017 & 0.000257 & 0.000227 \end{pmatrix}.$$

The daily volatilities are 0.61%, 1.79% and 1.51%, and the correlations $\rho_{\text{CAD,CHF}} = -0.17$, $\rho_{\text{CAD,DEM}} = -0.19$, $\rho_{\text{CHF,DEM}} = 0.95$ (our computation; unsurprisingly the two European currencies move together). Then

$$\mathbf{x}^T \Sigma \mathbf{x} = 1.691875, \quad \sigma_P = 1.3007, \quad 2.33 \sigma_P = 3.0307$$

(slide 40; $2.3263 \sigma_P = 3.026$ with the exact quantile). The stand-alone VaRs are 1.417, 2.087 and 0.878, adding to 4.382: the diversification benefit is 1.35, a third of the sum. Almost none of it comes from the imperfect correlation of CHF and DEM (with $\rho = 0.95$ and positions of the same sign their two stand-alone VaRs add to 2.97 against 2.94 together, a benefit of 0.03); the remaining 1.32 comes from the negative correlation of CAD with both.

19.3.3 Where does the risk come from? Absolute, incremental and component VaR

A firm wants to attribute the portfolio VaR to desks or positions. The slides define (slides 59–62) the *absolute VaR* of a bucket as its stand-alone VaR (the position vector with all other entries set to zero; equivalently its sensitivity times its volatility) and the *marginal* (in today’s language *incremental*) VaR as the difference between the portfolio VaR and the VaR with that bucket removed, to see “natural hedges” and the effect of hedge portfolios [► 2013 L7 @ 49:25]. For the example: removing the CAD position lowers VaR by 0.096 only, removing CHF by 1.508 and removing DEM by 0.709. These numbers add up to 2.31, not to 3.03: incremental VaRs depend on what else is in the book and are not additive. The slides later (77–80, see [section 19.4](#)) use another notion, the *component VaR*, which is additive.

Proposition 19.8 (Component (Euler) VaR; not discussed in class). In the variance–covariance method $\text{VaR}(\mathbf{x}) = z_\beta \sqrt{\mathbf{x}^T \Sigma \mathbf{x}}$ is positively homogeneous of degree one, so by Euler’s theorem

$$\text{VaR} = \sum_i x_i \frac{\partial \text{VaR}}{\partial x_i} = \sum_i C_i, \quad C_i = z_\beta \frac{x_i (\Sigma \mathbf{x})_i}{\sqrt{\mathbf{x}^T \Sigma \mathbf{x}}} = z_\beta \frac{\text{Cov}(x_i r_i, \mathbf{x}^T \mathbf{r})}{\sigma_P}.$$

For the currency example $C = (0.425, 1.852, 0.753)$, adding to 3.031; the CAD position, nearly uncorrelated with the rest, contributes little, and $\partial \text{VaR} / \partial x_i$ is the change of VaR per unit added of position i . A position with negative C_i is a hedge of the rest of the book. Component VaRs are what one compares with limits by desk; the stand-alone VaRs are

Proof. $\partial \sigma_P / \partial x_i = (\Sigma \mathbf{x})_i / \sigma_P$ and $\sum_i x_i (\Sigma \mathbf{x})_i = \mathbf{x}^T \Sigma \mathbf{x} = \sigma_P^2$. Also $(\Sigma \mathbf{x})_i = \text{Cov}(r_i, \mathbf{x}^T \mathbf{r})$. □

larger and add to $4.382 > 3.031$.

Exercise 19.4 (not from the course: incremental versus component VaR). In the variance-covariance method, show that the incremental VaR of position i , $\text{VaR}(x) - \text{VaR}(x - x_i e_i)$, is at most the component VaR C_i of [proposition 19.8](#). (Hint: $x \mapsto \sqrt{x^\top \Sigma x}$ is a convex function.) Verify it on the numbers of [example 19.7](#): $(0.096, 1.508, 0.709)$ against $(0.425, 1.852, 0.753)$. When are the two equal?

19.3.4 The choices behind the covariance matrix

The lecture then goes through the assumptions that hide in Σ (slides 45–56). We list them with Abbott’s opinions.

- **How much history.** Rule of thumb for time series: at least 100 observations [[2013 L7 @ 53:37](#)]. Regulators require at least a year; firms use one, two or five years. Slide 47 adds that the window should be weighed against data availability, changes of pricing regime (Brazil 1995, the introduction of the euro, the 2008 crisis) and market anomalies that traders might exploit.
- **Weighting.** Equal weights, or weights that decay exponentially into the past: the covariance is

$$\hat{\sigma}_{xy} = \frac{\sum_{i=0}^{n-1} \omega^i (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=0}^{n-1} \omega^i},$$

with $i = 0$ today, $i = 1$ yesterday, and ω near 0.95 [[2013 L7 @ 55:42](#)]. The weight on the m most recent observations is $(1 - \omega^m)/(1 - \omega^n) \approx 1 - \omega^m$; for $\omega = 0.94$, $m = 76$ observations carry 99.1% of the weight (99% needs $m = 74.4$), as Abbott says [[2013 L7 @ 56:44](#)]. This is the RiskMetrics estimator; its effective sample size $(1 + \omega)/(1 - \omega) \approx 32$ observations and its relation to GARCH are in [sections 13.1.3](#) and [13.6.7](#). Abbott’s opinion is that the choice of ω has a material effect on the measured volatility (slide 51 compares a rolling six-month window with $\omega = 0.92, \dots, 0.98$) and that any assumption with such an effect calls for extreme care: one can get whichever volatility one wants (“lies, damn lies, and statistics”). He prefers equal weights out of Occam’s razor: not knowing how much more last week matters than three weeks ago, he does not pretend to [[2013 L7 @ 53:37](#)]. Slide 49 lists the case for the other side: it smooths the volatility as positions age and reflects a heuristic that recent information matters more.

- **Missing data.** Fill gaps by linear interpolation, the previous day’s value, a Brownian bridge (a Monte Carlo between the two neighbours) or regression on other series [[2013 L7 @ 54:41](#)]; this is also what protects against non-positive-definite matrices ([section 19.5](#)).
- **Update frequency.** Daily updating of Σ is overkill; Abbott’s firm updates weekly, “and that is overkill, but tell that to my regulators” [[2013 L7 @ 55:42](#)]. Slide 53: changes of volatility are common in short-term euro-deposit and Japanese rates, changes of correlation are hard to estimate.
- **Percentage or log changes.** $r_{\text{pct}} = e^{r_{\text{log}}} - 1 = r_{\text{log}} + \frac{1}{2}r_{\text{log}}^2 + \dots$, so the two differ by second-order terms that matter only at the ends of the distribution: a 4% log move is a 4.08% percentage move. Abbott uses percentage changes for convenience, although prices are closer to lognormal; the log form is what one simulates, since it keeps prices positive [[2013 L7 @ 30:40](#)]. Even yields, which the lognormal model keeps positive, have gone negative (slide 54).

- **Differencing interval.** Daily data are the norm. One can scale up a daily covariance matrix, but slide 55 warns of asynchronicity (markets close at different times) and serial correlation, whose effect on the $\sqrt{T}$ rule is quantified in [proposition 8.11](#).
- **Bucketing.** A book cannot carry a risk factor per position; positions are mapped to a set of risk indices (equal-VaR, duration-weighted, hedge-based buckets, slides 52 and 69) at a loss of granularity that should not cost accuracy if done well.
- **Stability of correlations.** A one-day horizon makes correlations less ambiguous, and short horizons mean that linear approximations are less of a problem, but correlations tend to “swing” from neutral to directional when markets are under stress (slide 37): the diversification benefit of [example 19.7](#) is what disappears first in a crisis.

Abbott’s summary of the method is the flow chart of slide 44: market data from several sources; returns; tests (graphs, “2 standard deviations”); covariance matrix; positions and PV01s in a vector; the multiplication $\mathbf{x}^T \Sigma \mathbf{x}$; and then portfolio VaR, absolute and marginal sub-portfolio VaRs and scenario analysis [▶ 2013 L7 @ 51:31](#). Up to the late 1990s this was how it was done: in 1996 about 80% of the European banks used the variance–covariance method [▶ 2013 L7 @ 50:25](#). Twenty years earlier the calculation was squeezed into the 256 columns of a spreadsheet [▶ 2013 L7 @ 48:22](#).

19.4 Historical simulation

Abbott skipped this part in class (“historical simulation is Monte Carlo without the Monte Carlo part” [▶ 2013 L7 @ 6:26](#)); it is on slides 63–85, which we summarise.

19.4.1 The method

Historical simulation marks the *current* portfolio to market using the changes of the risk factors observed on each day of a past window, ranks the resulting hypothetical P&L numbers and reads off the order statistic (slide 63). With n historical scenarios $\mathbf{r}_1, \dots, \mathbf{r}_n$ (vectors of factor returns, or of changes measured in standard deviations), current factor values $\mathbf{S}$ and a valuation function V ,

$$\Delta P_j = V(\mathbf{S} \odot (\mathbf{1} + \mathbf{r}_j)) - V(\mathbf{S}) \approx \mathbf{x}^T \mathbf{r}_j, \quad \text{VaR}_\beta = -\Delta P_{(k)}, \quad k = (1 - \beta)n,$$

$\Delta P_{(k)}$ being the k -th smallest P&L. The steps of slide 67 are: collect position data; map to risk indices; compute sensitivities (deltas, gammas); collect historical data; compute returns; shock the portfolio with the returns; compute P&L; rank and take the order statistic. The horizontal layout of slide 75 keeps a P&L column for each desk and for the total, so the desk VaR is the k -th worst of the desk column and the total VaR the k -th worst of the total column. The rank k depends on the significance level and on the data available:

history	1 year	2 years	3 years	4 years
observations n	250	500	750	1 000
99%: worst $k = 0.01n$	2.5 (2nd–3rd)	5	7.5 (7th–8th)	10
95%: worst $k = 0.05n$	12.5	25	37.5	50

(the 99% row is slide 75; the 95% row is ours). It is “probably sensible” to compute the 1%, 5% and 10% levels together, and to save the detail, because the P&L at the tail can be driven

by very few positions (slide 73). The method is *nonparametric*: it uses no distribution, so skewness, fat tails and the actual dependence between factors, including credit-spread jumps, are in the data; it works equally for long and short positions; aggregation is trivial (add the P&L columns) (slides 64 and 82). The weaknesses (slide 65): it needs a lot of data, and history is missing exactly for new products; it assumes that the future is drawn from the same distribution as the window (stationarity) — slide 66 calls it a limited Monte Carlo, and in statistical language it is a bootstrap in which the past scenarios are resampled with equal probability (slide 66); and the tail estimate rests on very few points. The last point deserves a proof.

Proposition 19.9 (Standard error of an empirical quantile; not discussed in class). Let $L_1, \dots, L_n$ be i.i.d. losses with distribution function F and density f , continuous and positive at $q = F^{-1}(\beta)$, and let $\hat{q}_n$ be the empirical β -quantile. Then, as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{q}_n - q) \Rightarrow N\left(0, \frac{\beta(1-\beta)}{f(q)^2}\right).$$

For the standard normal at $\beta = 99\%$, $f(q) = \varphi(2.326) = 0.0267$, so $\text{sd}(\hat{q}_n) = 0.236/\sqrt{n/250}$: 10% of the VaR with one year of data, 5% with four years ($n = 1\,000$), 3% with ten. A heavy tail is worse, because $f(q)$ is smaller: for the standardised Student t_4 , $f(q) = 0.0123$ and the error is more than twice as large. [fig. 19.1](#) shows the sampling distribution in a simulation.

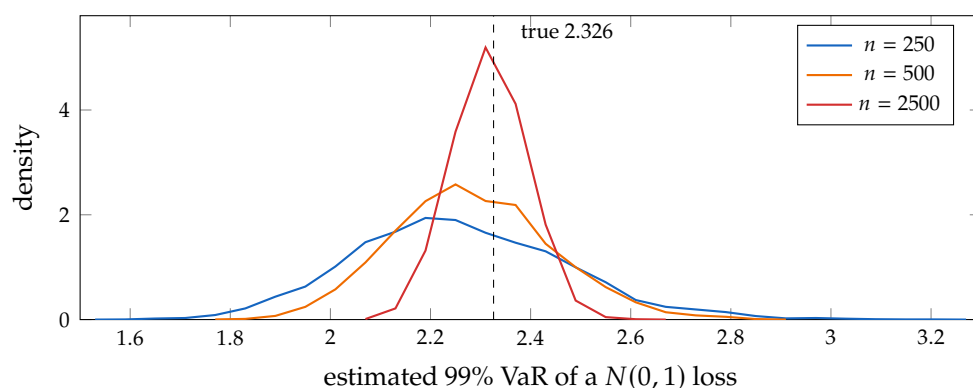


Figure 19.1: Historical-simulation estimates of the 99% VaR of a standard normal loss: sampling distribution over 5 000 independent samples of $n = 250, 500$ and $2\,500$ days (histograms with bin width 0.06; estimator: linearly interpolated empirical quantile). Standard deviations 0.213, 0.158 and 0.073 against the asymptotic 0.236, 0.167 and 0.075 of [proposition 19.9](#); means 2.258, 2.284 and 2.318, so a short window also biases the VaR downwards. Simulation by the authors.

19.4.2 Marginal VaR in historical simulation

Slides 77–81 attribute a historical-simulation VaR along an organisational hierarchy. The VaR engine produces P&L *vectors*, one entry per date (here 1 043 dates, four years of history), and the vector of a parent unit is the sum of the vectors of its children, $y = \sum_i x_i$ (Firm = Equities + Fixed income).

Proposition 19.10 (Variance–covariance ratio allocation). Let $y = \sum_{i=1}^n x_i$ be the parent

Sketch. Let $\hat{F}_n$ be the empirical distribution function. $n\hat{F}_n(q) \sim \text{Bin}(n, \beta)$, so $\hat{F}_n(q) - \beta$ has variance $\beta(1-\beta)/n$ and, by the central limit theorem, is asymptotically normal. By definition $\hat{F}_n(\hat{q}_n) \approx \beta$, and $\hat{F}_n(\hat{q}_n) - \hat{F}_n(q) \approx F(\hat{q}_n) - F(q) \approx f(q)(\hat{q}_n - q)$ for large n (the empirical process is uniformly close to F ; this step is the Bahadur representation of the quantile). Hence $\hat{q}_n - q \approx -(\hat{F}_n(q) - \beta)/f(q)$, which has the stated variance. $\square$

Proof. $\text{Var } y = \text{Cov}(y, y) = \text{Cov}(y, \sum_i x_i)$ by bilinearity (slides 78–79 write $\sigma_y = \sum_i E[(y - \bar{y})(x_i - \bar{x}_i)]/\sigma_y$). Divide by $\text{Var } y$ and multiply

$P\&L$ and x_i the child $P\&L$ s, as random variables over the historical dates. Then

$$\sigma_y^2 = \sum_{i=1}^n \text{Cov}(y, x_i), \quad \text{MVar}_i = \frac{\text{Cov}(y, x_i)}{\text{Var } y} \text{VaR}_y,$$

so that $\sum_i \text{MVar}_i = \text{VaR}_y.$

by VaR_y .

□

Variance is used as a proxy for risk to split a number that is a quantile. In the variance–covariance model of [proposition 19.6](#) the two are the same thing, MVar_i being the component VaR of [proposition 19.8](#). In a general empirical setting the exact Euler contribution of VaR is a conditional expectation, $\mathbb{E}[x_i \mid y = \text{VaR}]$, which is a noisy average of the very few scenarios near the quantile, and the ratio allocation is a smoothed substitute (a remark of ours). The example on slide 81 has firm VaR 81 411 with children marginals 2 822, 14 273 and 64 316 (sum 81 411); the second child’s stand-alone VaR is 29 990 against a marginal of 14 273, and the first’s 14 481 against 2 822: the diversifying units carry much less than their stand-alone VaR. The marginals are relative to the immediate parent in the hierarchy, and the five children of the largest unit sum to 69 020 against a stand-alone parent VaR of 72 457; the slide attributes the gap to a different “golden run” of the data, although the text claims that the children’s marginals add to the parent’s stand-alone VaR.

19.4.3 The ratio of the 95% to the 99% VaR

Slide 85 shows the ratio of the 95% to the 99% VaR of a portfolio over the decade 2001–2011 (the slide does not say whose), with the normal-law value 1.65/2.33 marked in red. The ratio $z_{0.95}/z_{0.99} = 0.7071$ is, to three digits, $1/\sqrt{2}$. Before 2008 the ratio fluctuates between about 0.63 and 0.74, around the normal value; from the end of 2008 it drops to 0.40–0.65. The slide has no comment, but the ratio measures tail thickness: for heavier tails the 99% quantile grows faster than the 95% one. For a standardised Student t_ν (variance one) the ratio is:

ν	3	4	6	10	30	100	∞
q_{95}	1.359	1.507	1.587	1.621	1.640	1.644	1.645
q_{99}	2.622	2.649	2.566	2.472	2.374	2.340	2.326
q_{95}/q_{99}	0.518	0.569	0.618	0.656	0.691	0.702	0.707

so that a ratio of 0.5–0.6 corresponds to a tail like that of a t_3 – t_4 (the table is ours; historical simulation adds sampling noise, [proposition 19.9](#)). The lesson, again ours: the ratio is a free diagnostic of whether the normal multiplier of [proposition 19.6](#) can be trusted, and after 2008 it could not.

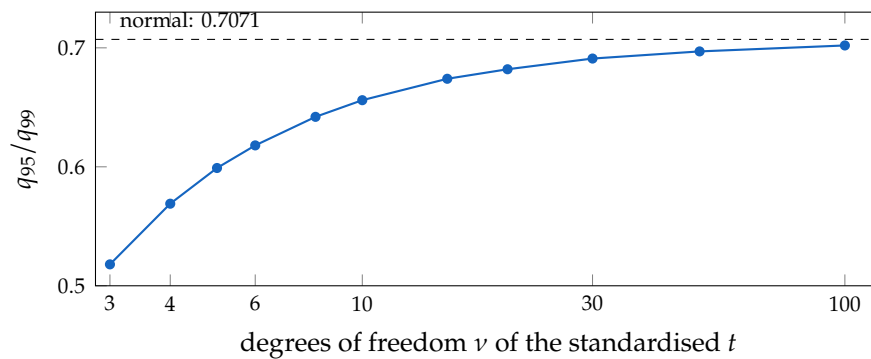


Figure 19.2: The ratio of the 95% and 99% quantiles for standardised Student- t losses as a function of the degrees of freedom (the observable of slide 85).

Exercise 19.5 (not from the course: the 95/99 ratio). Show that for a Laplace distribution the ratio q_{95}/q_{99} does not depend on the scale, and evaluate it: $\ln 10/\ln 50 = 0.589$. Compare with the table of [section 19.4](#): which Student- t has the same ratio? What does the ratio become for a Pareto tail $\mathbb{P}(L > x) \sim x^{-\alpha}$, and how would you estimate α from the ratio observed on slide 85?

19.5 Monte Carlo simulation

Instead of the closed form of [proposition 19.6](#), one can simulate: draw, say, 10 000 possible outcomes for tomorrow, reprice, sort, and take the 100th worst for the 99% level (the 500th for 95%) [[2013 L7 @ 1:00:53](#)]. It is again an order statistic, but now of a sample that we generate, so the assumptions can be relaxed at will: the distribution need not be normal (a t , a customised law, mean reversion, jumps) [[2013 L7 @ 1:01:58](#)]. The method, named during the Manhattan Project after games of chance, needs only that the system be described by probability density functions (slide 86). Its financial uses (slide 87) are path-dependent products, the risk of convex portfolios in hedge analysis, portfolios with strong jump risk (insurance), credit exposure — and VaR. The seven steps of slide 89, drawn as a diagram on slide 88, are:

1. generate uniform random numbers;
2. convert them to standard normals;
3. build the covariance matrix;
4. factor it;
5. create correlated random shocks;
6. use the shocks to reprice;
7. sort the P&L and read the order statistic.

19.5.1 Random numbers

Truly random numbers cannot be produced by a computer; they come from an external source such as radioactive decay [[2013 L7 @ 1:03:03](#)]. What software provides are *pseudo-random* numbers, the output of a deterministic algorithm that “look independent and uniform” (slide 90; see Knuth), and Abbott recalls, as a young man, seeing two computers

print the same “random” sequence (the effect of an identical seed) [► 2013 L7 @ 1:04:09]. *Quasi-random* (low-discrepancy) sequences impose uniformity on purpose, so that a finite draw does not leave a region empty; they are harder to implement (slide 90). Reproducibility is a virtue for an audit trail: fix the seed.

From uniform to normal. A uniform variable U on $(0, 1)$ becomes a standard normal by $Z = \Phi^{-1}(U)$ (the probability integral transform of [proposition 3.3](#), slide 91) [► 2013 L7 @ 1:05:13]. Slide 91 also points to the Box–Muller method of *Numerical Recipes*, which avoids inverting Φ :

$$Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2), \quad Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2)$$

are independent $N(0, 1)$ if U_1, U_2 are independent uniform (proof, not given in class: if Z_1, Z_2 are i.i.d. $N(0, 1)$ then in polar coordinates the angle is uniform on $(0, 2\pi)$, independent of the radius, and $\mathbb{P}(Z_1^2 + Z_2^2 > s) = e^{-s/2}$, so that $U_1 = e^{-(Z_1^2 + Z_2^2)/2}$ is uniform; invert). Abbott admits that today he trusts the built-in generators of Excel and MATLAB [► 2013 L7 @ 1:05:13], and that “I kind of violate my own rules” [► 2013 L7 @ 1:05:13].

19.5.2 Correlated shocks

Draw independent standard normals and give them the covariance structure of the market. Everything rests on a “square root” of Σ .

Proposition 19.11 (Correlated normal shocks). Let Σ be a symmetric positive semidefinite matrix and $\mathbf{Z} \sim N(\mathbf{0}, I_n)$ a column vector. If A is any real matrix with $AA^T = \Sigma$, then $\mathbf{r} = A\mathbf{Z} \sim N(\mathbf{0}, \Sigma)$. Two standard choices:

(i) eigenvalue decomposition: $\Sigma = E\Lambda E^T$ with E orthogonal and $\Lambda = \text{diag}(\lambda_i)$, $\lambda_i \geq 0$ ([theorem 2.11](#)); $A = E\Lambda^{1/2}$;

(ii) Cholesky: $\Sigma = LL^T$ with L lower triangular, which exists for Σ positive definite; $A = L$.

The matrix A is not unique: AQ works for every orthogonal Q , and all choices give the same distribution.

Proof. $\mathbf{r}$ is a linear function of a Gaussian vector and $\text{Cov } \mathbf{r} = A I A^T = \Sigma$, as in [proposition 3.21](#). For (i), $AA^T = E\Lambda^{1/2}\Lambda^{1/2}E^T = \Sigma$. $\square$

Written for a row vector of shocks (as in a spreadsheet, where each row is a draw) the recipe is $\mathbf{r}^T = \mathbf{Z}^T \Lambda^{1/2} E^T$ [► 2013 L7 @ 1:10:40]. The lecture calls $\Lambda^{1/2} E^T$ the “transformation matrix”, “a prism”: uncorrelated white noise goes in and correlated noise comes out [► 2013 L7 @ 1:12:42]. For two assets the Cholesky factor is explicit,

$$L = \begin{pmatrix} \sigma_1 & 0 \\ \rho\sigma_2 & \sigma_2\sqrt{1-\rho^2} \end{pmatrix}, \quad r_1 = \sigma_1 Z_1, \quad r_2 = \sigma_2(\rho Z_1 + \sqrt{1-\rho^2} Z_2),$$

which shows why $|\rho| \leq 1$ is necessary. Slides 94–95 list the trade-off: Cholesky is computationally simpler and used by several packages but requires positive *definiteness*; the eigen-decomposition is more complex, more robust (it works for semidefinite matrices) and is what Abbott uses. He notes that this construction is the vaunted *Gaussian copula*, which most people treat as a black box although anyone with a little statistics can open it [► 2013 L7 @ 1:07:21]. Precisely, the joint law of $\mathbf{r}$ is the multivariate normal, whose copula is the Gaussian copula with correlation matrix R ; the same eigen-decomposition on the correlation matrix is the principal-component analysis of [chapter 7](#). Abbott adds in passing that the method “can really only take two distributions” [► 2013 L7 @ 1:18:06]; presumably he means the normal and the Student t , the two elliptical laws that share this construction (with one extra step, see [section 19.5.4](#)).

Exercise 19.6 (not from the course: correlated normals).

- Derive the Cholesky factor of $\Sigma = \begin{pmatrix} 4 & 2.4 \\ 2.4 & 9 \end{pmatrix}$ and give the two-line simulation recipe.
- Show that $A = E\Lambda^{1/2}E^\top$ (the symmetric square root) also satisfies $AA^\top = \Sigma$ and relate it to $E\Lambda^{1/2}$ of [proposition 19.11](#).
- For three assets with $\rho_{12} = \rho_{23} = \rho$, show that ρ_{13} must lie in $[2\rho^2 - 1, 1]$, and find the eigenvalues of the matrix of [example 19.12](#).
- Prove the Box–Muller formulas.

19.5.3 When the covariance matrix is not positive semidefinite

A covariance matrix must be positive semidefinite, “another of those high-school-maths things that suddenly matter” [▶ 2013 L7 @ 1:08:33](#). If different entries are estimated from different amounts of data (say 100 observations for most pairs and 25 for one of them), the estimated matrix can have a *negative eigenvalue*; then $\Lambda^{1/2}$ is imaginary and the simulation cannot even start. Abbott reports that, in the 1990s, nobody he asked (including a chairman of a statistics department) knew why he was getting negative eigenvalues; the cure is to fill in the missing data so that every entry uses the same sample [▶ 2013 L7 @ 1:09:38](#). A minimal example shows that it is not a matter of rounding.

Other remedies than filling in the data exist (replacing the matrix by the nearest positive-semidefinite correlation matrix, or setting the negative eigenvalues to zero and rescaling); the slide only lists “precision, negative eigenvalues, computation time, data storage, error audit trail” as the potential problems (slide 97).

19.5.4 Checking the simulation, and why simulate at all

Abbott’s test of his spreadsheet: 10 series of yield changes (Treasury, AAA, AA, ...) with about 800 observations; a 10×10 correlation matrix, for instance 0.83 between the government 10-year and the AAA 10-year; the transformation matrix; 1 000 rows of independent normals multiplied by it [▶ 2013 L7 @ 1:13:47](#). The correlations of the simulated data are close to but not identical with the original ones, “because of sampling error”: with 1 000 draws the standard deviation of an estimated correlation is $(1 - \rho^2)/\sqrt{1\,000}$ ([section 19.5.2](#)), for example 0.02–0.03 for $\rho = 0.5$ [▶ 2013 L7 @ 1:17:02](#). With a million draws the match is much closer. Slide 96 lists the ways of checking: re-create the covariance matrix from the simulated data; test the variances (an F or χ^2 test); test the correlations; Box’s M test for the equality of covariance matrices. “It is difficult to make small mistakes” (slide 96): a wrong factorisation gives grossly wrong correlations.

Why draw a million scenarios when the data set has a few hundred? Because VaR looks at the tail: with 100 observations there is one point in the 1% tail, which says little about how fast the distribution drops off, whereas a simulation can put a billion points there [▶ 2013 L7 @ 1:17:02](#). The price is the assumption of the generating distribution (slide 98 lists “multivariate normality, serial correlation, skewness, kurtosis”, and the need to adjust for mean reversion and jump diffusion). The simulation error of the estimated quantile is [proposition 19.9](#) with n replaced by the number N of draws, which is at our disposal: $N = 10^4$ gives a standard error of 0.037 standard deviations of the P&L, $N = 10^6$ one of 0.004.

3 COURSE NOTE: Erratum: slides 95 and 96

- Slide 95 states $R\Lambda^{1/2}E' \sim N(0, \Sigma)$ where R is called an $n \times 1$ vector of normals; the product as written needs R to be a *row* ($1 \times n$) vector, as in the spreadsheet; for a column vector the formula is $E\Lambda^{1/2}R$.
- Slide 96 gives, as a test of correlations, $\hat{\rho} \sim N(\rho_0, (1 - \rho_0^2)/n)$. The large-sample variance of a sample correlation from n normal pairs is $(1 - \rho_0^2)^2/n$, i.e. standard deviation $(1 - \rho_0^2)/\sqrt{n}$; equivalently, Fisher’s $\text{artanh } \hat{\rho}$ is $N(\text{artanh } \rho_0, 1/(n - 3))$. For $\rho_0 = 0.95$ and $n = 1\,000$ the standard deviation is 0.003, not 0.01.
- The variance test on the same slide, $\hat{\sigma}_o^2/\sigma^2 \sim F(m - 1, k - 1)$, is a comparison of two independent sample variances from samples of sizes m and k (original and simulated data).

Example 19.12 (An inconsistent correlation matrix; not from the lecture). Let three assets have $\rho_{12} = \rho_{23} = 0.9$ and $\rho_{13} = 0$, each perfectly plausible if estimated separately:

$$R = \begin{pmatrix} 1 & 0.9 & 0 \\ 0.9 & 1 & 0.9 \\ 0 & 0.9 & 1 \end{pmatrix}, \quad \text{eigenvalues } 1 - 0.9\sqrt{2} = -0.273,$$

For the direction $v = (1, -\sqrt{2}, 1)/2$, $v^\top R v = -0.273$: a portfolio with a negative “variance”, hence an imaginary VaR. In general $\rho_{12} = \rho_{23} = \rho$ forces $\rho_{13} \in [2\rho^2 - 1, 1]$: indeed $\det R = -(\rho_{13} - 1)(\rho_{13} + 1 - 2\rho^2)$, which must be non-negative (the geometric statement is that three angles $\arccos \rho_{12}$, $\arccos \rho_{23}$, $\arccos \rho_{13}$ between unit vectors satisfy a triangle inequality). For $\rho = 0.9$ the lower bound is 0.62, and $\rho_{13} = 0$ is impossible.

CASE STUDY: Three methods, one portfolio; our numerical experiment

Take the three-currency portfolio of [example 19.7](#) ($\sigma_P = 1.3007$) and compute the one-day 99% VaR in four ways. Simulations use [proposition 19.11](#) with the eigen-decomposition; the fat-tailed version replaces the normal shock by a multivariate Student t_4 with the *same* covariance matrix, $\mathbf{r} = \sqrt{(\nu - 2)/\nu} \mathbf{AZ} / \sqrt{W/\nu}$ with $W \sim \chi^2_\nu$ independent of $\mathbf{Z}$ (a normal variance mixture, [proposition 3.29](#)).

method	99% VaR	standard error	99.9% VaR
variance–covariance (proposition 19.6), $z_{0.99}\sigma_P$	3.026	—	4.020
Monte Carlo, normal, $N = 10^4$ (mean of 300 runs)	3.021	0.049	—
Monte Carlo, normal, Cholesky instead, $N = 10^4$	3.022	0.045	—
Monte Carlo, normal, $N = 10^6$ (one run)	3.026	≈ 0.005	—
t_4 shocks, exact	3.446	—	6.598
t_4 shocks, Monte Carlo $N = 10^4$ (mean of 300)	3.429	0.102	—
t_4 shocks, Monte Carlo $N = 2 \cdot 10^6$ (one run)	3.453	≈ 0.01	—

The normal Monte Carlo converges to the closed form, as it must (the closed form *is* the limit); the standard error agrees with $0.037 \sigma_P = 0.049$ from [proposition 19.9](#). The two factorisations are statistically indistinguishable. With the same covariance matrix but heavy tails the 99% VaR is 14% higher and the 99.9% VaR 64% higher: the variance is a summary that hides the tail, and only the simulation (or the history) can see it. Note also that the Monte Carlo error with 10^4 draws, 0.05, is already small against this model risk, 0.4.

19.6 Non-linear positions

The variance–covariance method assumes that the P&L is linear in the risk factors, $\Delta P = \mathbf{x}^\top \mathbf{r}$. This is exact for stocks and currencies, and for bonds it holds through the PV01 for small moves ([section 19.2](#)); for options it is the delta approximation. Slide 70 states the rule of thumb: “for a daily VaR, delta is probably enough; gamma is needed for a long holding period or for volatile markets”, and that convexity is not a big problem for bonds. Historical simulation and Monte Carlo can do better: they need a way to compute the P&L for each scenario, either by *exact repricing* of every instrument with the full model (accurate, computationally heavy) or from a *parametric representation* of the price (a Taylor expansion, or a grid of precomputed values interpolated between scenarios, “pricing grids”, slide 84), which can be very accurate but where cross-partial terms create noise (slides 83 and 99). Slide 83 illustrates it with the P&L of an option book as a function of a shock: exact curve, coarse grid, refined grid.

The following example, ours, shows that gamma can matter even for one day. Let the risk factor return be $x \sim N(0, 1)$ in units of its daily standard deviation, and let the P&L be

$$\Delta P(x) = \delta x + \frac{1}{2}\Gamma x^2, \quad \delta > 0, \Gamma < 0$$

(a short-gamma position, for example a written option; δ is the P&L of a one-sigma move to first order, and the units are those of δ). The three approximations below use the same δ and Γ .

Proposition 19.13 (Exact VaR of a delta–gamma position; ours). For $\Gamma < 0$ the loss $L = -\Delta P = \frac{1}{2}|\Gamma|x^2 - \delta x$ satisfies $L \leq \ell$ iff $x_- \leq x \leq x_+$ with $x_{\pm} = (\delta \pm \sqrt{\delta^2 + 2|\Gamma|\ell})/|\Gamma|$. The exact VaR_{β} is the unique root ℓ of $\Phi(x_+) - \Phi(x_-) = \beta$. For $\delta = 0$, $\text{VaR}_{\beta} = \frac{1}{2}|\Gamma|\chi_{1,\beta}^2$, with $\chi_{1,\beta}^2$ the β -quantile of the χ_1^2 law (6.635 for $\beta = 99\%$).

Proof. Solve the quadratic inequality; the loss is increasing in ℓ so the equation has one root. For $\delta = 0$, $L = \frac{1}{2}|\Gamma|x^2$ and $x^2 \sim \chi_1^2$. $\square$

	delta-normal $z_{0.99} \delta$	delta–gamma, normal moments	exact $\text{VaR}_{99\%}$ (proposition 19.13)	exact $\text{ES}_{99\%}$ (simulation)
$\delta = 1, \Gamma = -0.5$	2.33	2.72	3.68	4.46
$\delta = 0, \Gamma = -0.5$	0	1.07	1.66	2.11

The “delta–gamma normal” column matches the mean $\frac{1}{2}\Gamma$ and variance $\delta^2 + \frac{1}{2}\Gamma^2$ of the P&L to a normal law (see proposition 8.14 for the moments of such a quadratic form, and section 15.12 for the Greeks). The delta-normal VaR is 37% too low for the first position and simply zero for the delta-neutral one, which nevertheless loses 1.66 one day in a hundred. Even the moment-matched normal misses by a quarter, because the loss is a skewed, bounded-below variable ($\frac{1}{2}|\Gamma|x^2$ is a χ_1^2), and so does not resemble a normal law in the tail; the exact quantile comes from the distribution of the quadratic form, or from simulation.

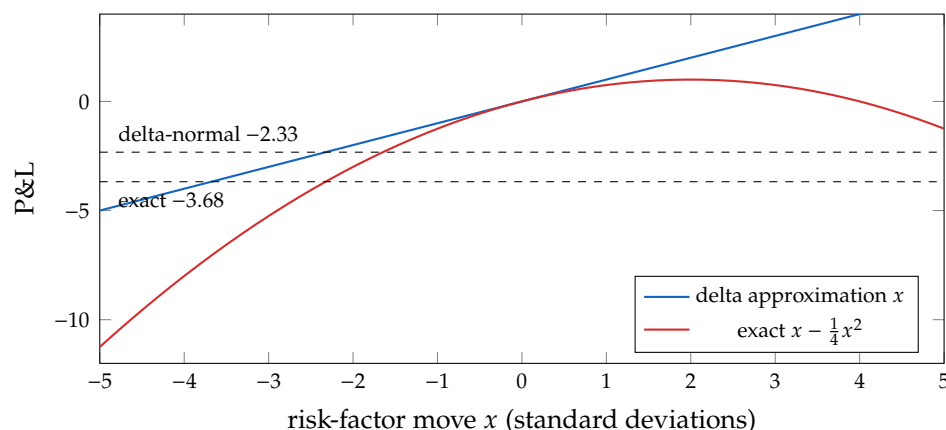


Figure 19.3: P&L of the short-gamma position $\delta = 1, \Gamma = -0.5$ against the risk-factor move. The tangent (delta) sees a loss of 2.33 at the 99% move $x = -2.33$; the true P&L there is -3.68 , which is also the true 99% VaR (the other root $x_+ = 6.33$ of proposition 19.13 is negligible). For a delta-neutral position the loss is even in x and large moves of either sign lose.

Exercise 19.7 (not from the course: gamma). For the position $\Delta P = \delta x + \frac{1}{2}\Gamma x^2$ with $x \sim N(0, 1)$, $\delta = 0$ and $\Gamma < 0$, show that $\text{VaR}_{\beta} = \frac{1}{2}|\Gamma|\chi_{1,\beta}^2$, and that the expected shortfall is $\text{ES}_{\beta} = \frac{|\Gamma|}{2(1-\beta)}(2z_{\beta}\varphi(z_{\beta}) + (1 - \beta))$, where φ is the standard normal density and z_{β} the $\frac{1+\beta}{2}$ -quantile. Evaluate for $\Gamma = -0.5, \beta = 99\%$ and compare with the value 2.11 of the table. (Hint: $\mathbb{E}[x^2 \mid |x| > z] = 1 + z\varphi(z)/(1 - \Phi(z))$.)

19.7 Back-testing, stress tests and use of VaR

19.7.1 Why a bank needs it

The first paragraph of the motivation has an accounting side that Abbott insists on [► 2013 L7 @ 13:55]. A balance sheet has assets on one side, financed by borrowing or by the firm's own money (equity); the ratio of the two is the leverage. Every loss reduces the equity, and when it reaches zero the firm is bankrupt. Risk management asks how large a one-day loss can be, with some level of certainty, so that the equity is enough to absorb it [► 2013 L7 @ 14:57]: *that* is the capital that supports a position. VaR is the number that says how much is needed.

IN FINANCE: The uses of a VaR number

Position limits ("how much is too much?" [► 2013 L7 @ 4:13]), capital, and the daily report to management. Abbott's caveats, all on slides: "VaR numbers in isolation are worthless"; one must compute them every day and compare them (slide 74); the actual P&L, or changes in the risk indices, must be used in the comparison; repricing is preferred to sensitivities; and VaR must be used together with *stress testing* and *scenario analysis*. Slide 76 is a diagram of the data flow: back-office systems provide positions and sensitivities, data providers provide history, and the VaR engine (the "1% worst") feeds limit reports, stress tests, historical scenarios and ad-hoc analyses. Historical scenarios (the 2008 crisis, Brazil in 1995, the introduction of the euro, slide 47) are the natural stress tests: revalue today's book with the moves of a known bad day, an "automatic scenario analysis" (slide 82). "90% of risk management is knowing what you own and where you own it" (slide 68).

19.7.2 Back-testing

A *back-test* compares the VaR computed at time t with the realised P&L at $t + 1$ (slide 101). Reasons: it is required by regulators; it is a reality check on the calculation; it finds errors; it detects changes of the risk profile. The counting test (the binomial law of the number of exceedances, with rejection at 5%) is in [section 8.4.4](#); we add its likelihood-ratio version, which is the one used in the literature.

Proposition 19.14 (Kupiec's unconditional-coverage test; not discussed in class). *If the VaR at level β is correct and exceedances are independent, then in n days the number N of exceedances is $\text{Bin}(n, p)$ with $p = 1 - \beta$. The likelihood-ratio statistic of $H_0: p = 1 - \beta$ against a free exceedance probability, $\hat{p} = N/n$,*

$$\text{LR} = -2 \ln \frac{(1-p)^{n-N} p^N}{(1-\hat{p})^{n-N} \hat{p}^N}$$

is asymptotically χ_1^2 under H_0 (Wilks' theorem), and one rejects when it exceeds 3.84 (5%) or 6.63 (1%).

Applications: the IPC example of [remark 19.2](#) ($N = 8$, $n = 520$): $\text{LR} = 1.31$, p -value 0.25, not rejected. The 12 exceedances in 500 days of the counting test in [section 8.4.4](#): $\text{LR} = 7.11$, $p = 0.008$, rejected, consistent with the 0.5% tail probability found there. Five exceedances in 250 days: $\text{LR} = 1.96$, $p = 0.16$: not rejected, although the expected count is 2.5: a year of data cannot tell a good 99% model from a moderately bad one.

IN FINANCE: The regulatory traffic light; not discussed in class

The Basel back-test of the 1990s uses $n = 250$ days of 99% VaR. With $p = 1\%$, $\mathbb{P}(N \leq 4) = 89.2\%$, $\mathbb{P}(N \leq 9) = 99.97\%$ and $\mathbb{P}(N \geq 10) = 0.025\%$ (our computation), so the *green zone* is 0–4 exceptions, the *yellow zone* 5–9 and the *red zone* 10 or more; the capital multiplier applied to the VaR increases from 3 in the green zone through the yellow zone to 4 in the red zone (from memory of the 1996 amendment; the reader should check the figures against the Basel text). The 10-day, 99% horizon is obtained from one-day figures by $\sqrt{10}$, with the caveats of [proposition 8.11](#). The later tightening to stressed VaR and expected shortfall is in [section 8.4.4](#).

Exercise 19.8 (not from the course: back-test). A desk reports a 99% one-day VaR and records 7 exceedances in 250 days. Compute the likelihood-ratio statistic of [proposition 19.14](#) and its p -value, the exact binomial tail probability $\mathbb{P}(N \geq 7)$, and the Basel zone. Do the two tests agree? Why is a test with 250 days not powerful against a true exceedance probability of 2%?

19.7.3 Practical issues

Slides 102–103 list the practical issues: back-tests are done down to the desk level, where every outlier must be documented (“need to document outliers”), but one often needs to aggregate, and aggregation hides spikes; *split books* (a position whose P&L is shared across desks) are a problem; whether one compares with front-office or back-office P&L, the timing of P&L recognition, re-organisations, data maintenance, the frequency of updates, and whether upside jumps should count as exceptions (a VaR is one-sided). Slides 104–105 (repeated as 113–114) plot the daily revenue of a business as bars with the 99% and 95% VaR as lines below the axis. In the first, the revenue stays within the lines except for a few isolated days, as it should; in the second the revenue is calm for a long period and then, after a break, the bars fall far outside the VaR lines, which adapt only slowly. The slides on the next pages (106–111, without axis labels or commentary in the lecture) show more series of this kind, of a P&L band with a horizontal level: they illustrate that the scale of the P&L can change abruptly (a position is cut, a new product starts from zero, a data error produces a spike) and that the VaR, the limit, or the data process reacts with a delay.

Slide 112 is the cleanest example of a regime change. It shows the spread of a five-year US credit-card asset-backed security over about four years ending in 2008: for three years it stays at about 0.3%, and its daily moves at a few basis points; from the summer of 2007 the spread rises to about 4% and the daily moves reach tens of basis points, with the old percentile bands (dashed lines) ending up far inside the new distribution. A 99% VaR estimated from an equally weighted window of one or two years, as Abbott prefers, must fail on such a series for the length of the window, and the exponentially weighted estimate of [section 19.3.4](#) fails for about $1/(1 - \omega)$ days: the price of simplicity, stated in Abbott’s own terms.

19.7.4 Abbott's principles, in brief

INTUITION: A physicist's summary

A VaR is the location of a quantile of a distribution that one never sees and estimates from few data, using a Gaussian approximation whose parameters carry error and whose tails are wrong. The three methods differ in what they take as given: the variance–covariance method a linear map and a normal law (two parameters, a matrix); historical simulation nothing but the stationarity of the past (a sample); Monte Carlo a generating distribution (a model). Each buys a benefit at a cost: speed and transparency against tails; realism against sample size; flexibility against model risk. Abbott's practical advice reads like a laboratory notebook: use the simplest estimator that you understand, look at the data before you trust it, test every step (a matrix square root squared must return the matrix), record units and signs, and treat every number as an estimate with an error bar. Then keep a second, independent estimator, a stress test, to see what the first one cannot.

Summary of results

KEY FORMULAS

One asset (normal):	$\text{VaR}_\beta = z_\beta \sigma V, \quad z_{0.99} = 2.326, \quad z_{0.95} = 1.645$
Bond:	$\text{VaR}_\beta = z_\beta \text{PV01} \sigma_{\text{bp}}, \quad \sigma_{\text{bp}} = 100 y\% \sigma_{\Delta y/y}$
Portfolio:	$\text{VaR}_\beta = z_\beta \sqrt{\mathbf{x}^\top \Sigma \mathbf{x}}, \quad \Sigma = \text{DRD}$
Component VaR:	$C_i = z_\beta \frac{x_i (\Sigma \mathbf{x})_i}{\sqrt{\mathbf{x}^\top \Sigma \mathbf{x}}}, \quad \sum_i C_i = \text{VaR}$
EWMA covariance:	$\hat{\sigma}_{xy} = \frac{\sum_i \omega^i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i \omega^i}, \quad \text{weight on last } m \text{ obs.} \approx 1 - \omega^m$
Hist. simulation:	$\text{VaR}_\beta = -\Delta P_{(k)}, \quad k = (1 - \beta)n; \quad \text{sd}(\hat{q}) \approx \frac{\sqrt{\beta(1 - \beta)/n}}{f(q)}$
Marginal VaR (hist.):	$\text{MVaR}_i = \frac{\text{Cov}(y, x_i)}{\text{Var } y} \text{VaR}_y, \quad \sum_i \text{MVaR}_i = \text{VaR}_y$
Monte Carlo:	$\mathbf{r} = E\Lambda^{1/2}\mathbf{Z} \text{ or } L\mathbf{Z} \sim N(\mathbf{0}, \Sigma), \quad \Sigma = E\Lambda E^\top = LL^\top$
Box–Muller:	$Z_{1,2} = \sqrt{-2 \ln U_1} (\cos, \sin)(2\pi U_2)$
Delta–gamma, $\delta = 0$:	$\text{VaR}_\beta = \frac{1}{2} \Gamma \chi_{1,\beta}^2$
Kupiec:	$\text{LR} = -2 \ln \frac{(1-p)^{n-N} p^N}{(1-N/n)^{n-N} (N/n)^N} \sim \chi_1^2$
Correlation error:	$\text{sd}(\hat{\rho}) \approx (1 - \rho^2)/\sqrt{n}$

4 COURSE NOTE: What the lecture covered and what the slides add

Abbott states at the start that he will “fly through” the material, which is normally two hours of his university course [[► 2013 L7 @ 6:26](#)], and announces that he will skip historical simulation to spend the time on Monte Carlo [[► 2013 L7 @ 58:51](#)]. The spoken lecture therefore corresponds to slides 1–62 (one-asset VaR, fixed income, variance–covariance with data issues and marginal VaR) and 86–97 (Monte Carlo); slides 63–85 (historical simulation, marginal VaR by variance ratio, the 95/99 ratio) and 98–115 (the end of Monte Carlo, back-testing) are not discussed in the recording. He promises to send spreadsheets and papers by email to anyone who asks [[► 2013 L7 @ 7:27](#)], and jokes, at the end, that the only use he has found for trigonometry is the seasonal decomposition of temperature data in sines and cosines [[► 2013 L7 @ 1:20:10](#)]. He teaches a longer version at NYU and other schools. The 2024 course has no counterpart lecture on these methods: the 2024 Lecture 10 ([section 8.4](#)) takes the estimation of VaR and ES by historical simulation as its starting point and concentrates on the mathematics of risk measures, so the two chapters are complementary.

Chapter 20

Regularized Pricing and Risk Models

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2013 L10 (1 h 30)	I. Masyukov (Morgan Stanley)	Guest lecture. Bonds, yield, duration, convexity; interest-rate swaps and the swap rate as an average of forward rates; building a yield curve with splines; bond spread over the swap curve; the cost of hedging a swap book; PCA hedging and its instabilities; a hedging model regularised by smoothness of the forward curve; the pricing/calibration engine and the HJM model; calibration of a two-dimensional volatility surface as an ill-posed inverse problem; SVD analysis, Tikhonov regularisation, truncated SVD; questions on splines, relative-value trading and bond curves.

Files. Slides MIT18_S096F13_lecnote10.pdf (51 pages, no incremental builds). **Problem sets.** None. The exercises (each after the section it practises) are all “(not from the course)”. **Not published.** The talk has no 2024 counterpart. The slides give the numerical tables and figures of the swap-book example, but no data or code; the toy computations of this chapter (a swap-curve Jacobian, a one-dimensional calibration) are my own and are marked as such.

Motivation

Most quantities a bank computes every day are the output of an *inverse problem*. A model has parameters (a yield curve, a volatility surface); a handful of liquid instruments have observed prices; the parameters must be chosen so that the model reproduces the prices, and only then can it be used to value and hedge a book of thousands of trades that are *not* among the calibration instruments. The forward map “parameters $\mapsto$ prices” is smooth and easy; its inverse is not. Small errors in the prices, or small differences between instruments, are amplified into wild oscillations of the parameters, and the model that “fits perfectly” prices the rest of the book badly. The cure, the subject of this guest lecture, is *regularisation*: add to the fit a penalty that says what a sensible solution looks like, and give up an exact fit in exchange for stability.

The chapter follows the lecture. [Section 20.1](#) recalls bonds, swaps and the yield curve. [Section 20.2](#) shows on a toy how calibrating a curve exactly already amplifies a 1 bp change into a ripple of several basis points. [Section 20.3](#) treats the risk side: hedging with a few liquid swaps by PCA and by a smoothness-regularised alternative. [Section 20.4](#) develops the mathematics behind both, singular values, Tikhonov regularisation and truncated SVD,

which is the natural continuation of the shrinkage methods of [section 5.10](#); [section 20.5](#) discusses how to choose the regularisation parameter (not covered in class). [Section 20.6](#) applies everything to the calibration of a volatility surface for the Heath–Jarrow–Morton model ([chapter 24](#)). The linear algebra needed is the SVD of [section 2.7](#).

20.1 Bonds, swaps and the yield curve

20.1.1 Bond price, yield, duration and convexity

A fixed-rate bond pays cash flows $C_1, \dots, C_N$ (coupons, plus the face value in C_N) at times $t_1 < \dots < t_N$ [[2013 L10 @ 2:11](#)]. Cash paid later is worth less, so each cash flow is multiplied by a *discount factor* $\Lambda_i \in (0, 1]$ that decreases with t_i [[2013 L10 @ 5:16](#)], and the fair value is $P = \sum_{i=1}^N \Lambda_i C_i$ [[2013 L10 @ 6:20](#)]. For a bond the price P is not something to be computed: it is the result of a liquid market and is known [[2013 L10 @ 7:20](#)]. What is needed is a *model for discounting*. The simplest one uses a single parameter, the *yield to maturity* y (continuous compounding, one rate for all dates) [[2013 L10 @ 8:27](#)]:

$$\Lambda_i = e^{-yt_i}, \quad P(y) = \sum_{i=1}^N e^{-yt_i} C_i. \quad (20.1)$$

With $C_i > 0$ the map $y \mapsto P(y)$ is strictly decreasing, so price and yield determine each other one-to-one [[2013 L10 @ 9:31](#)]. The yield is therefore not a traded quantity but a *convention* to summarise price and cash flows [[2013 L10 @ 10:34](#)]. Differentiating,

$$\begin{aligned} \frac{dP}{dy} &= - \sum_i t_i e^{-yt_i} C_i, & D &= -\frac{1}{P} \frac{dP}{dy} = \sum_{i=1}^N t_i \pi_i, & \pi_i &= \frac{e^{-yt_i} C_i}{P}, & \sum_i \pi_i &= 1, \\ C &= \frac{1}{P} \frac{d^2 P}{dy^2} = \sum_i t_i^2 \pi_i > 0. \end{aligned}$$

The *duration* D is the present-value-weighted average time of the cash flows (“weighted time”) [[2013 L10 @ 11:40](#)]; normalising by P makes it independent of how many bonds are held. A zero-coupon bond has one weight, so $D = T$; a coupon bond has $D < t_N$, because every $t_i \leq t_N$ and at least one is strictly smaller [[2013 L10 @ 13:47](#)]. Convexity is a weighted mean of t_i^2 , hence positive. These are the definitions of [section 1.8.4](#) in the continuous-compounding convention ([proposition 1.10](#)), and [eq. \(1.5\)](#) is the Taylor expansion $\Delta P/P \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2$.

All of this assumes one y for all dates, hence that the curve moves in *parallel*. That was an acceptable approximation before the crisis, but with a steep curve, where short rates sit near zero and long rates are expected to rise, it is not [[2013 L10 @ 14:55](#)]. [Section 20.1.4](#) replaces y by a full curve.

20.1.2 Interest-rate swaps

In a swap one party pays a fixed rate C and receives a floating rate (three-month LIBOR in the lecture, fixed on the day and known in advance for the accrual period) [[2013 L10 @ 17:02](#)]. With payment dates $T_1 < \dots < T_n$, accrual fractions δ_i (day-count), discount factors Λ_i and

1 COURSE NOTE: Sign and letters in the slides

Slide 8 defines “duration” as $d = \frac{1}{P} \frac{\partial P}{\partial y}$, which is *negative* (the speaker says so: the duration is always negative); here, as in [section 1.8.4](#), $D = -d > 0$. Slide 9 names the second derivative $c = \partial^2 P / \partial y^2$ and then writes its explicit form as “ $d = \sum t_i^2 e^{-yt_i} C_i$ ”: the letter should be c , and this is the *unnormalised* second derivative, $C = c/P$ in the notation used here.

Example 20.1 (Duration explains most of the P&L, convexity the rest (my numbers)). Take $N = 5$ annual payments $C = (4, 4, 4, 4, 104)$ and $y = 4\%$. Then $P = 99.640$, $D = 4.629$ years, $C = 22.42$. If y rises by 50 bp the exact price change is -2.2868% ; the duration

forward rates r_i for the period ending at T_i ,

$$PV_{\text{fixed}} = \sum_i C \delta_i \Lambda_i = C \sum_i w_i, \quad PV_{\text{float}} = \sum_i r_i \delta_i \Lambda_i = \sum_i r_i w_i, \quad w_i = \delta_i \Lambda_i,$$

[► 2013 L10 @ 18:14]. A swap costs nothing to enter, so the fixed rate is set so that the two legs have the same value [► 2013 L10 @ 19:21]:

$$C = \frac{\sum_i r_i w_i}{\sum_i w_i} \quad (\text{same payment frequency on both legs}), \quad (20.2)$$

the *swap rate*, a weighted average of forward rates, which is what traders quote [► 2013 L10 @ 20:27]. If the forward rates are defined by $\Lambda_{i-1} = (1 + r_i \delta_i) \Lambda_i$ (the single-curve convention of section 1.8.5, $\Lambda_0 = 1$), the floating leg telescopes, $PV_{\text{float}} = \sum_i (\Lambda_{i-1} - \Lambda_i) = 1 - \Lambda_n$, and $C = (1 - \Lambda_n) / \sum_i \delta_i \Lambda_i$ (my remark, not on the slides). The fixed leg is a bond coupon stream, which is why a swap can be hedged with a bond [► 2013 L10 @ 20:27], and why swap risk is measured in *buckets* of par rates: the trader wants to know the P&L if the 5-year swap rate moves by 1 bp.

20.1.3 Building a curve

A liquid market gives swap rates for a few maturities, whereas a book of hundreds of thousands of swaps needs a discount factor for every day up to 30–40 years [► 2013 L10 @ 22:35]. Constructing the curve means [► 2013 L10 @ 23:39]:

1. choose the input instruments;
2. choose which quantity is interpolated (daily forward rates, zero rates, discount factors) and the interpolation family (piecewise-constant, linear, tension spline, cubic spline) with its *knots*;
3. *calibrate*: solve for the spline parameters so that every input instrument is repriced at par (PV of the swap equal to zero) [► 2013 L10 @ 24:42]. The lecture's example uses 15 swaps from 1 to 30 years with quotes from 0.33% to 2.67% (the market of a year earlier; table 20.1); the resulting curve of three-month forwards is far from flat, steep in the first five years and with a hump in the 20-year region [► 2013 L10 @ 30:01].

Splines and B-splines. A *cubic spline* is a function that is a cubic polynomial on each interval between knots and has two continuous derivatives at every knot [► 2013 L10 @ 25:48]. The knots in the lecture's picture are 1, 10, 20, 40, 80, 160, 240 (quarters), denser where instruments are dense [► 2013 L10 @ 1:22:09]. All splines with given knots form a vector space, which has a convenient basis of bell-shaped, locally supported functions [► 2013 L10 @ 26:49].

Definition 20.2 (B-splines, Cox–de Boor recursion). For a non-decreasing knot sequence $\tau_0 \leq \tau_1 \leq \dots$ let $B_{i,0}(t) = 1$ if $\tau_i \leq t < \tau_{i+1}$ and 0 otherwise, and

$$B_{i,k}(t) = \frac{t - \tau_i}{\tau_{i+k} - \tau_i} B_{i,k-1}(t) + \frac{\tau_{i+k+1} - t}{\tau_{i+k+1} - \tau_{i+1}} B_{i+1,k-1}(t)$$

(a term with zero denominator is dropped) [► 2013 L10 @ 1:08:13]. Degree $k = 3$ gives cubic B-splines.

term alone gives -2.3146% and adding convexity -2.2865% . For $+200$ bp the figures are -8.824% exactly, -9.258% with duration only and -8.810% with convexity. The trader who explains the day's loss with first derivatives only [► 2013 L10 @ 15:56] is left with an "unexplained" term of order $\frac{1}{2}C(\Delta y)^2$.

Proposition 20.3 (Properties of B-splines). $B_{i,k} \geq 0$; it vanishes outside $[\tau_i, \tau_{i+k+1}]$; $\sum_i B_{i,k}(t) = 1$ on the interior of the knot range; for simple knots $B_{i,k} \in C^{k-1}$. Hence every $f = \sum_i c_i B_{i,k}$ is a spline of degree k and its values lie between the smallest and the largest c_i .

Local support makes the calibration matrix banded and the response to a change of one parameter local; the convex-combination property makes the parameters interpretable as “values of the curve”. This is why the same basis is used for the surface, in two dimensions, in [section 20.6](#).

20.1.4 Bond spread over the curve

Once a swap curve Λ_i exists, a bond can be priced off it with one extra parameter, the *bond spread* s , determined so that the model reproduces the (liquid, known) bond price [[2013 L10 @ 31:01](#)]:

$$P = \sum_{i=1}^N e^{-st_i} \Lambda_i C_i. \quad (20.3)$$

Compared with y the correction is small: a curve level of 3% and a spread of a few basis points, about a hundred times smaller [[2013 L10 @ 32:10](#)]. The gain is consistency: sensitivities of the swap book to the curve inputs are inherited by the bond, and one is no longer bound to the single “parallel” mode of duration. The spread contains credit or liquidity components of the bond, and even a swap-versus-Treasury spread is traded; for a 10-year maturity the most liquid instrument is the bond, then, surprisingly, the spread, and only then the swap itself, so the curve is built from bond yield plus spread rather than from the raw 10-year swap quote [[2013 L10 @ 33:10](#)].

IN FINANCE: Why the curve is built on swaps and not on bonds

Asked why one does not invert the bond curve instead, the speaker gave two reasons [[2013 L10 @ 1:27:24](#)]. A swap starting today always exists, and tomorrow’s swap starts tomorrow, whereas bonds have a fixed maturity: the newest issue (*on-the-run*) is by far the most liquid and older ones (*off-the-run*) trade at different levels, so there is no continuous family of maturities and a bond curve can only be a least-squares fit, not an exact one. The switch of liquidity from one issue to the next creates “roll effects” that make the curve unstable. So the swap desk builds the swap curve and *projects* bonds onto it, by the spread (20.3), since bonds are its preferred hedging instruments.

Proof. Non-negativity and support follow by induction from the recursion, whose coefficients are non-negative on the supports involved. For the partition of unity shift the index in the second term of $\sum_i B_{i,k}$:

$$\sum_i B_{i,k} = \sum_i B_{i,k-1} \left[\frac{t - \tau_i}{\tau_{i+k} - \tau_i} + \frac{\tau_{i+k} - t}{\tau_{i+k} - \tau_i} \right] = \sum_i B_{i,k-1},$$

and the case $k = 0$ is 1. Smoothness: at a simple knot each recursion step raises the differentiability by one, starting from a jump. (Checked numerically: the maximal deviation from 1 is 4×10^{-16} .) $\square$

20.2 Calibrating a curve exactly is already ill-conditioned

The lecture shifts one input of the calibrated curve, the 9-year swap rate, by 1 bp, keeping all the other 14 quotes fixed, and plots the response of the three-month forward rates: it is a ripple, zero before about 5 years and after about 20 years, with a swing of about ± 14 bp around the 9-year point [[2013 L10 @ 36:16](#)]. The curve is exact at every input, so it *must* move in the neighbourhood of the shifted input (the 8- and 10-year par rates are pinned), and it does so by an oscillation much larger than the shift. The speaker’s comment: this is an ill-posed problem, where a small change of the input causes a large change of the output.

A toy where everything is explicit (not in class). Let the instantaneous forward rate be constant on each interval $(T_{i-1}, T_i]$, $T_0 = 0$, between consecutive input maturities, with

value f_i , and let the par rate of the swap maturing at T_i be (ignoring discounting) the average forward rate, $S_i = \frac{1}{T_i} \sum_{k \leq i} (T_k - T_{k-1}) f_k$, i.e. $S = Mf$ with M lower triangular and invertible. Inverting,

$$f_i = \frac{T_i S_i - T_{i-1} S_{i-1}}{T_i - T_{i-1}}, \quad \text{so} \quad J = M^{-1} : \quad \Delta f_i = \frac{T_i \Delta S_i - T_{i-1} \Delta S_{i-1}}{T_i - T_{i-1}}.$$

A shift $\Delta S_j = 1$ bp moves f_j by $+T_j/(T_j - T_{j-1})$ and f_{j+1} by $-T_j/(T_{j+1} - T_j)$: for the 9-year input, +9 bp on (8, 9] and -9 bp on (9, 10]; for 10 years +10 and -5; for 25 years +5 and -5. Each maturity acts as a lever with arm T_j divided by the spacing of the instruments, and no amount of care in the method removes it, because the data (an average up to T_j) constrain only a difference of averages. For the 15 maturities of the lecture the condition number of M is 23.4 (my computation). Figure 20.1 draws the two responses.

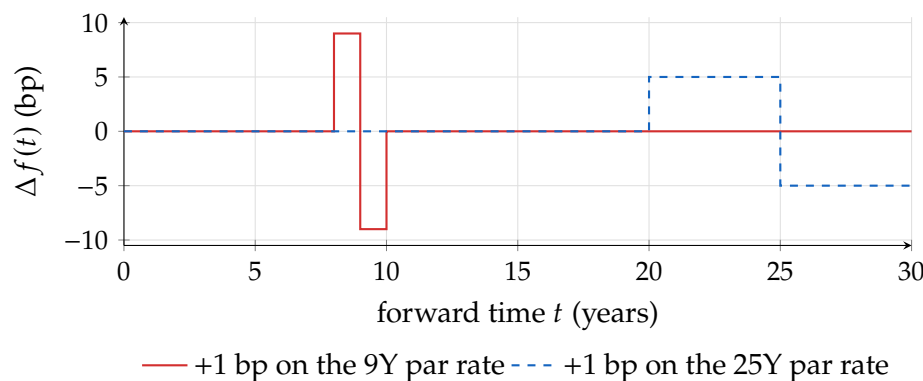


Figure 20.1: Forward-rate response to a 1 bp shift of one par rate in the piecewise-constant toy (my computation): the curve calibrated exactly answers with a ripple whose amplitude is the maturity divided by the spacing of instruments. The lecture's spline curve gives a similar but smoother ripple (► 2013 L10 @ 36:16).

Remark 20.4 (Smoother is not automatically better (my computation)). Replacing the piecewise-constant forwards by a cubic B-spline with 15 coefficients (interior knots by the averaging rule of de Boor) and again calibrating exactly gives a condition number of about 6×10^3 for the calibration matrix, and the response to the same 1 bp shift of the 9-year rate has amplitude of between 10 and 30 bp at 9–10 years, about 70 bp at 12, 150 bp at 16 and 200 bp at 20 years. The exact calibration of a few averaged quantities to a flexible smooth curve is genuinely unstable in the sparse long end; the lecture's curve (ripple confined to ± 14 bp) shows that the design of the interpolation matters.

Exercise 20.1 ((not from the course)).

- For a swap with equally spaced payment dates show $C = (1 - \Lambda_n) / \sum_i \delta_i \Lambda_i$ and that C lies between the smallest and the largest forward rate.
- In the piecewise-constant toy of section 20.2 derive Δf_i for a shift of the j -th par rate and show that its entries are $T_j/(T_j - T_{j-1})$ and $-T_j/(T_{j+1} - T_j)$; what is the largest absolute value over all j for the 15 maturities of the lecture?
- The matrix M is lower triangular: compute its determinant.

20.3 Hedging a swap book: PCA and the smoothness-regularised model

20.3.1 The cost of hedging bucket by bucket

The *risk* of a book is the vector $r \in \mathbb{R}^n$ of P&L per basis point for a shift of each of the n input rates (here $n = 15$ swap maturities); to first order the P&L for a shift u of the inputs is $r^T u$. Table 20.1 shows the lecture's example [► 2013 L10 @ 37:22]. The net risk is tiny (2,266 in total), but the risk in each bucket is large and of both signs. To be insensitive to *every* bucket the trader must trade $\pm r_i$ swaps in each of them, paying half the bid–offer of each instrument; the liquid maturities (yellow in the slide: 1, 2, 5, 10, 30 years) cost 0.10 bp, the others 0.25 bp. The charge of a bucket is $|r_i| \times \text{bid–offer}$, and the total is $\sum_i |r_i| b_i = 3,617,629$ [► 2013 L10 @ 38:25]. The gross exposure, not the net one, is what costs money, so traders never hedge every bucket: they hedge a seven-year risk with the cheaper liquid points around it [► 2013 L10 @ 40:30]. The question is how to do this systematically.

Swap	quote (%)	raw risk	bid–offer (bp)	charge
1Y	0.33	–200,000	0.10	20,000
2Y	0.39	1,330,000	0.10	133,000
3Y	0.49	–200,000	0.25	50,000
4Y	0.64	1,200,000	0.25	300,000
5Y	0.86	–722,450	0.10	72,245
6Y	1.09	–35,255	0.25	8,814
7Y	1.29	–537,430	0.25	134,358
8Y	1.48	–3,850,000	0.25	962,500
9Y	1.64	1,580,000	0.25	395,000
10Y	1.79	288,751	0.10	28,875
12Y	2.04	–401,350	0.25	100,338
15Y	2.29	50,000	0.25	12,500
20Y	2.50	4,000,000	0.25	1,000,000
25Y	2.60	–1,000,000	0.25	250,000
30Y	2.67	–1,500,000	0.10	150,000
total		2,266		3,617,629

Table 20.1: The swap book of the lecture (slide 15; the unit of the raw risk is dollars per basis point, presumably). The charge is $|\text{risk}| \times \text{bid–offer}$; I checked that the column sums to 3,617,629.

20.3.2 Hedging with a few liquid swaps: the general formulation

Let $H \in \mathbb{R}^{n \times k}$ have as columns the risk vectors (in the n buckets) of k liquid hedging instruments; if the same instruments calibrate the curve, a hedge instrument is sensitive only to its own input, and H is made of columns $e_{i(j)}$ of the identity (slide 19). Take $k = 5$: the 1, 2, 5, 10 and 30 year swaps. Let $F \in \mathbb{R}^{n \times m}$ collect market *scenarios* (moves of the inputs) [► 2013 L10 @ 41:37]. Holding a hedge position that transfers risk Hx , the residual

risk is $r - Hx$ and its P&L in scenario f is $f^\top(r - Hx)$, so one chooses

$$x = \arg \min_x \|F^\top(r - Hx)\|^2. \quad (20.4)$$

The vector x (the “translated risk”; the hedge trades are $-x$) is the equivalent risk of the book in the k liquid swaps.

20.3.3 PCA hedging

Take for F the first principal components of historical daily changes of the inputs. If $D = USP^\top$ is the SVD of the data matrix of changes (rows are days), P has as columns the principal components, which are eigenvectors of $D^\top D$ (chapter 7, theorem 2.12) [► 2013 L10 @ 42:50]; keeping k of them, one for each hedging instrument, is a low-rank approximation of the market. In the lecture’s data the first component is the “rates do not move now but move in the future” mode (steep up to 10 years, flat afterwards, because the Fed is holding), the second is a tilt, the third more complex [► 2013 L10 @ 43:57].

With $F = P \in \mathbb{R}^{n \times k}$ the objective (20.4) can be made *exactly* zero, since $P^\top H \in \mathbb{R}^{k \times k}$ is (generically) invertible [► 2013 L10 @ 45:02]:

$$P^\top(r - Hx) = 0 \implies x = (P^\top H)^{-1} P^\top r = R^\top r, \quad R = P(H^\top P)^{-1} \in \mathbb{R}^{n \times k}. \quad (20.5)$$

The matrix R deserves an interpretation (slide 22, [► 2013 L10 @ 47:12]): it is the unique combination $R = PY$ of the components with $H^\top R = I_k$. Its j -th column is the *scenario* (a shift of all 15 inputs, as a combination of principal components) in which the j -th hedging swap moves by 1 bp and the other four do not move at all; and $x_j = r^\top R_{\cdot j}$ is the P&L of the book in that scenario. Rows of R are the weights with which the risk of each bucket is assigned to the liquid swaps. This is the matrix of slide 20, whose rows have the unit vectors at the hedging tenors (a swap hedges itself) and, for example, assigns the 4-year risk mainly to the 2-year and 5-year swaps, with weights 0.60 and 0.70. The translated risks and their charges are in table 20.2 (first block): total charge 442,036 against 3,617,629, a reduction of a factor 8.2.

Weaknesses of PCA hedging [► 2013 L10 @ 48:14].

- The components are tuned to a historical window, and the coefficients of R are not stable, in particular for a short window that reflects the current regime; when they change the hedge must be rebalanced, which costs money.
- The instability comes from the noisy components attached to small singular values, exactly the mechanism of section 20.4.
- PCA is a least-squares approximation and reacts strongly to outliers: a single day on which rates spiked and reverted has an unwarranted influence.
- It overfits history: a model that would have worked for the last three months need not work for the next three.
- It makes *no assumption on the shape of the yield curve*: if the maturity labels of the swaps were scrambled, the computation would produce the same numbers, unscrambled, although 2 years is close to 1 year and lies between 1 and 5 years. Statistical structure without economic structure is the recurrent theme of section 5.10 as well.

2 COURSE NOTE: Sign convention on slides 16 and 19

Slide 16 writes the objective as $\|F^\top(r + Hx)\|^2$ and slides 19–23 use $r - Hx$ (and $x = R^\top r$). I use the latter throughout; the two differ by $x \rightarrow -x$.

3 COURSE NOTE: “Orders of magnitude”

The speaker says that the charge becomes “orders of magnitude smaller”, “something like 400” instead of 3.6 million [► 2013 L10 @ 47:12]. The slide gives 442,036 for the PCA model (and 446,717 for the regularised one below), i.e. about 12% of the raw charge: a very large saving, but one order of magnitude, not several.

20.3.4 A hedging model regularised by smoothness of the forward curve

The alternative [► 2013 L10 @ 50:21]: forget history and impose an *economic assumption*. If nothing is known about what happens in ten years, there is no reason for the forward rate to have a spike at that date, so scenarios should be smooth in forward-rate space [► 2013 L10 @ 51:25]. Let $J \in \mathbb{R}^{q \times n}$ be the Jacobian that translates a shift of the n curve inputs into a shift of q forward rates on a fine grid (for the toy above, $q = n$ and $J = M^{-1}$), and L a smoothness matrix acting on the forward grid, for instance the first-difference matrix, whose rows are $(\dots, 1, -1, \dots)$. We look for the smoothest scenarios consistent with the hedging instruments: for each hedge j ,

$$\min_{\rho} \|LJ\rho\|^2 \quad \text{subject to} \quad H^T \rho = e_j, \quad (20.6)$$

where $\rho = R_{\cdot j}$ is the input scenario (1 bp on hedge j , 0 on the other hedges) and $\|LJ\rho\|^2$ is the roughness of the forward curve it induces. With a soft constraint, penalising $\|H^T \rho - e_j\|^2 + \lambda^2 \|LJ\rho\|^2$ and setting the gradient to zero gives, for all columns at once,

$$R = (HH^T + \lambda^2 M)^{-1} H, \quad M = (LJ)^T (LJ), \quad (20.7)$$

with translated risk $x = R^T r$ as before [► 2013 L10 @ 52:25].

Proof. Lagrange: $2M\rho = H\mu$, so $\rho = M^{-1}H\mu/2$ and $H^T \rho = e_j$ fixes $\mu/2 = (H^T M^{-1}H)^{-1}e_j$. For the limit use the push-through identity $(\lambda^2 M + HH^T)^{-1}H = M^{-1}H(\lambda^2 I + H^T M^{-1}H)^{-1}$ (multiply both sides by $\lambda^2 M + HH^T$ and simplify) and let $\lambda \rightarrow 0$. When M is singular the matrix $HH^T + \lambda^2 M$ is still invertible under the stated condition, and the minimiser of (20.6) is unique on the affine set $H^T \rho = e_j$; I do not prove the convergence in that case. $\square$

Proposition 20.5 (Soft and hard constraint). Let $M = (LJ)^T LJ$ be invertible. Then $H^T M^{-1}H$ is invertible for H of full column rank, the solution of (20.6) is $R^* = M^{-1}H(H^T M^{-1}H)^{-1}$, and $(HH^T + \lambda^2 M)^{-1}H \rightarrow R^*$ as $\lambda \rightarrow 0$. If M is singular the same limit holds, by the usual penalty-method argument, as long as $\ker M \cap \ker H^T = \{0\}$ (as in the toy, where the kernel is the parallel shift).

With $H^T R \approx I$ up to the size of λ (the lecture's matrix has 1.00 and 0.00 on the hedging rows, the smoothness being what fills the other rows) the regularised model produces the columns of slide 24 [► 2013 L10 @ 52:25]. They are *hat-shaped* (peaked at the hedging tenor and decaying smoothly), whereas the PCA columns follow the historical modes; the results are in the second block of table 20.2.

4 COURSE NOTE: Slide 23 leaves out a factor

Slide 23 shows " $R \sim (HH^T + \lambda^2(LJ)^T LJ)^{-1}$ ". As written the matrix is $n \times n$, not $n \times k$, and does not satisfy $H^T R \approx I$: the correct expression, obtained from the two conditions on the same slide, is (20.7), with H on the right. I read " $\sim$ " as "is approximately", the exact constraint being reached for $\lambda \rightarrow 0$.

translated risk on	1Y	2Y	5Y	10Y	30Y	total charge
PCA (slide 20)	−426,757	1,892,770	−1,538,808	−268,764	293,261	442,036
regularised (slide 24)	−487,769	2,082,997	−1,752,958	78,962	64,483	446,717
toy of this chapter	−331,027	1,956,764	−1,873,952	−92,944	343,425	459,811

Table 20.2: Translated risk $x = R^T r$ of the book of [table 20.1](#) on the five liquid swaps, and the charge at 0.1 bp per instrument (the slides print the charge of the PCA row and I computed the others; the toy, my computation, uses $J = M^{-1}$ of [section 20.2](#), first differences for L and $\lambda^2 = 10^{-4}$). Multiplying the printed two-decimal matrices by the raw risk reproduces the slide values up to the rounding of the entries (differences up to about 25,000 per entry, since raw risks reach 4 million).

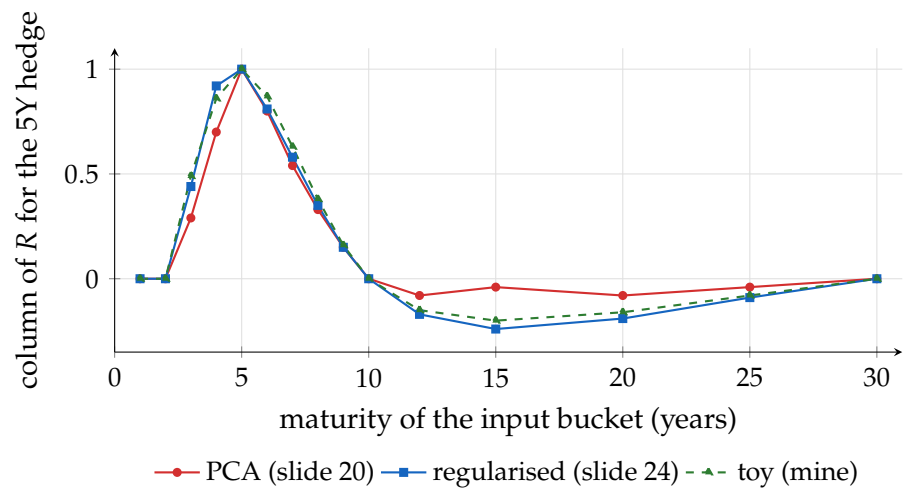


Figure 20.2: The scenario in which the 5-year hedge moves by 1 bp and the other four hedges do not: $j = 5Y$ column of R . Both models are equal to 1 at 5Y and vanish at the other hedging tenors; the regularised column is a smooth hat whose negative lobe (rates at 12–25 years move opposite to keep forwards smooth) is more pronounced than PCA’s; the toy reproduces the shape. Points are the entries of the matrices of slides 20 and 24 (two decimals).

IN FINANCE: Model risk in hedging and relative value

Two hedges with different assumptions cost the same here (about 445k) but distribute the risk differently, e.g. 10Y risk of −268,764 (PCA) versus +78,962 (regularised). Traders have their own view of the dynamics of the market, but always *some* model [\[► 2013 L10 @ 47:12\]](#). The lecture’s argument for the regularised model is that it rests on a *fundamental* assumption (smoothness of forwards) rather than on a window of data [\[► 2013 L10 @ 1:26:24\]](#). The two ideas can also be combined (a PCA regulariser inside the smoothness model).

Exercise 20.2 ((not from the course)). Let $J = I_4$, L the first-difference matrix (3×4) and $H = (e_1, e_4)$ (hedges in buckets 1 and 4) in (20.6). Show that the smoothest scenarios are the linear interpolations $R_1 = (1, \frac{2}{3}, \frac{1}{3}, 0)^T$ and $R_2 = (0, \frac{1}{3}, \frac{2}{3}, 1)^T$, compute the translated risk $x = R^T r$ for $r = (0, 1, 1, 0)^T$, and check that $M = (LJ)^T LJ$ is singular with kernel spanned by $(1, 1, 1, 1)^T$ (and that [proposition 20.5](#) still applies). What would the

second-difference matrix give?

Exercise 20.3 ((not from the course)). Prove the push-through identity $(\lambda^2 M + HH^\top)^{-1}H = M^{-1}H(\lambda^2 I + H^\top M^{-1}H)^{-1}$ used in [proposition 20.5](#), and use it to show that $H^\top R_\lambda \rightarrow I$ with $R_\lambda = (HH^\top + \lambda^2 M)^{-1}H$ and that $H^\top R_\lambda = H^\top M^{-1}H(\lambda^2 I + H^\top M^{-1}H)^{-1}$ (so the deviation from I is of order λ^2).

20.4 Linear inverse problems, the SVD and Tikhonov regularisation

The lecture analyses the calibration problem with the tools of linear algebra [[2013 L10 @ 1:11:29](#)]. This section develops that analysis in full.

20.4.1 The problem and its singular value decomposition

Let $A \in \mathbb{R}^{M \times N}$ be the linearised pricing map (model parameters $x \in \mathbb{R}^N$ to M market observables), and

$$y = Ax + \varepsilon \quad (20.8)$$

the market data, with an error ε (bid–offer, stale quotes, model error) [[2013 L10 @ 1:11:29](#)]. Take the thin SVD ([theorem 2.12](#), [eq. \(2.5\)](#)) $A = \sum_{i=1}^r s_i u_i v_i^\top$ with $s_1 \geq \dots \geq s_r > 0$, $r = \text{rank } A$, orthonormal $u_i \in \mathbb{R}^M$ and $v_i \in \mathbb{R}^N$. In the picture of the slide, A is a rotation, a scaling by s_i and a rotation, and the inverse is the rotation, the scaling by $1/s_i$, the rotation [[2013 L10 @ 1:11:29](#)].

Proposition 20.6 (Minimum-norm least squares). *The vectors minimising $\|Ax - y\|$ are $x = \sum_{i \leq r} \frac{u_i^\top y}{s_i} v_i + z$ with $z \in \ker A$; the one of smallest norm is*

$$x^\dagger = A^\dagger y = \sum_{i=1}^r \frac{u_i^\top y}{s_i} v_i = VS^{-1}U^\top y. \quad (20.9)$$

It equals $(A^\top A)^{-1}A^\top y$ if $r = N$ (more data than parameters), and $A^\top(AA^\top)^{-1}y$, which fits the data exactly, if $r = M < N$.

Proof. Write $x = \sum_j \xi_j v_j + z$ (with $z \perp v_j$). Then $\|Ax - y\|^2 = \sum_{i \leq r} (s_i \xi_i - b_i)^2 + \|y\|^2 - \sum_{i \leq r} b_i^2$, $b_i = u_i^\top y$, minimised for $\xi_i = b_i/s_i$; the component $z \in \ker A$ does not enter and the norm is smallest for $z = 0$. The two closed forms follow by substituting the SVD. $\square$

The slide's formula $x = (A^\top A)^{-1}A^\top y$ is thus valid only when the columns of A are independent; for the surface calibration, where there are fewer instruments than parameters, one needs (20.9) (the SVD form on the same slide is general).

20.4.2 Noise amplification and the condition number

Insert (20.8) into (20.9): $x^\dagger = A^\dagger Ax + \sum_i \frac{u_i^\top \varepsilon}{s_i} v_i$, the second term being the *propagated error* [[2013 L10 @ 1:12:32](#)]:

$$\|\delta x\|^2 = \sum_{i=1}^r \frac{(u_i^\top \varepsilon)^2}{s_i^2}, \quad \mathbb{E}[\|\delta x\|^2] = \sigma^2 \sum_{i=1}^r \frac{1}{s_i^2} \quad \text{if } \text{Cov } \varepsilon = \sigma^2 I.$$

Each mode i of the noise is divided by s_i : a small singular value turns an innocent perturbation of the data into a large variation of the parameters. A scale-free measure is

the condition number $\kappa = s_1/s_r$ [► 2013 L10 @ 1:12:32]: for x in the row space of A ,

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa \frac{\|\varepsilon\|}{\|Ax\|},$$

because $\|\delta x\| \leq \|\varepsilon\|/s_r$ and $\|Ax\| \leq s_1\|x\|$; the bound is attained for $\varepsilon \parallel u_r, x \parallel v_1$. A problem is *well posed* (Hadamard) if a solution exists, is unique and depends continuously on the data; it is *ill-posed* if one of the three fails, and *ill-conditioned* if the discretised problem has a large κ . The lecture's calibration problems are underdetermined (uniqueness fails: fewer instruments than parameters, hence $\ker A \neq \{0\}$) and unstable (small s_i) [► 2013 L10 @ 59:53].

INTUITION: A physicist's reading: the deconvolution analogue

The typical forward map averages (the price of a swaption is an integral of the volatility surface over a region), and averaging damps the fine details of the parameters: their contribution to y is $s_i \xi_i$ with s_i small. Inverting means amplifying back what the data barely contain; when the data have a noise floor σ , all modes with $s_i |\xi_i| \lesssim \sigma$ are lost, and dividing by their s_i turns them into noise. This is the same as the inversion of a heat-kernel smoothing, or an image deconvolution, and the standard fix is the same.

The discrete Picard picture. For noiseless data $y = Ax$ one has $u_i^\top y = s_i v_i^\top x$: the coefficients $|u_i^\top y|$ decay at least as fast as s_i if x is reasonable. With noise, the coefficients level off at the noise floor σ and the ratios $|u_i^\top y|/s_i$ that enter (20.9) grow. Figure 20.3(b) shows this for the toy calibration of section 20.5.

Exercise 20.4 ((not from the course)). Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}$.

- Solve $Ax = y$ for $y = (2, 2.01)^\top$ and for $y' = (2, 2.02)^\top$; by what fraction do data and solution change?
- Compute the singular values and $\kappa(A)$ (about 4×10^2) and check that the change of (a) is compatible with the bound of section 20.4.2.
- Compute the Tikhonov solutions (20.11) for $\lambda = 0.1$ for both y and y' and compare.

20.4.3 Truncated SVD and Tikhonov regularisation

Both methods multiply each term of (20.9) by a *filter factor* $w_i \in [0, 1]$:

$$x_w = \sum_{i=1}^r w_i \frac{u_i^\top y}{s_i} v_i. \quad (20.10)$$

- Truncated SVD (TSVD):** $w_i = 1$ for $i \leq k$ and 0 for $i > k$ [► 2013 L10 @ 1:16:49]. One keeps the well-determined modes and discards the rest, which are the part of the parameter space that the calibration cannot see (the “null space” of the model, in the speaker's words). It is the same idea as the PCA risk model of section 20.3.3 and is principal-components regression in the language of section 5.10. The danger: if the discarded modes do affect the price of the portfolio, the truncation is a bias one pays for.

5 COURSE NOTE: The “noiseless” slide

Slide 46 argues that in the noiseless case “ $A(A^\top A)^{-1}A^\top y = y$ ” so that “the model appears accurate”, but “pricing of the actual portfolio with the model B may be unstable”. The point is right; the printed identity is not: slide 46 writes “ $A^\top(A^\top A)^{-1}A = I$ ”, which is not even conformable. The correct statement is that $AA^\dagger = \sum_{i \leq r} u_i u_i^\top$ is the orthogonal projector onto the range of A , which is the identity on $\mathbb{R}^M$ exactly when $r = M$, i.e. when the instruments are independent, for any $N \geq M$. Then *every* data vector, noise included, is repriced exactly: exact repricing certifies nothing, because it is achieved by fitting the noise. What matters is the price $p = Bx$ of the instruments *not* used in the calibration, and $Bx^\dagger$ inherits the amplified error $B \delta x$. Slide 47 has the same misprint for the regularised repricing matrix, see (20.12).

- *Tikhonov regularisation* [► 2013 L10 @ 1:14:44]:

$$x_\lambda = \arg \min_x \left\{ \|Ax - y\|^2 + \lambda^2 \|x\|^2 \right\} = (A^\top A + \lambda^2 I)^{-1} A^\top y = \sum_{i=1}^r \underbrace{\frac{s_i^2}{s_i^2 + \lambda^2}}_{w_i} \frac{u_i^\top y}{s_i} v_i. \quad (20.11)$$

The gradient of the objective is $2A^\top(Ax - y) + 2\lambda^2 x$; setting it to zero gives the normal equations $(A^\top A + \lambda^2 I)x = A^\top y$, uniquely solvable for every $\lambda > 0$ even if A is rank-deficient (the matrix is positive definite), and $A^\top = VSU^\top$ gives the filter form, since $(A^\top A + \lambda^2 I)^{-1} A^\top = V \operatorname{diag}\left(\frac{s_i}{s_i^2 + \lambda^2}\right) U^\top$. Hence Tikhonov is a *smooth* filter: $w_i \approx 1$ for $s_i \gg \lambda$ and $w_i \approx s_i^2/\lambda^2$ for $s_i \ll \lambda$ (fig. 20.3(a)). The essential point is stated by the speaker: one no longer divides by a small number [► 2013 L10 @ 1:14:44]. Quantitatively, the gain applied to a mode of noise is $\frac{w_i}{s_i} = \frac{s_i}{s_i^2 + \lambda^2} \leq \frac{1}{2\lambda}$ (maximum at $s_i = \lambda$), instead of $1/s_r$ for the exact inverse, and $1/s_k$ for TSVD.

INTUITION: Ridge regression in another language

This is the ridge regression of section 5.10 with $d_j \rightarrow s_i$ and ridge parameter $\lambda_{\text{ridge}} = \lambda^2$ (the lecture writes the penalty as $\|\lambda x\|^2$). The smooth filter $s^2/(s^2 + \lambda^2)$ is the same shrinkage factor, and proposition 5.22 gives its Bayesian reading: a Gaussian prior $x \sim N(0, \tau^2 I)$ with $\lambda^2 = \sigma^2/\tau^2$. Early stopping of gradient descent acts as another filter of the same family (section 16.4).

The price of stability is that the calibration is no longer exact. The fit and the “re-pricing” operator are, by (20.11),

$$Ax_\lambda = H_\lambda y, \quad H_\lambda = A(A^\top A + \lambda^2 I)^{-1} A^\top = \sum_i w_i u_i u_i^\top \neq I, \quad (20.12)$$

the corrected form of the “ $A^\top(A^\top A + \lambda^2 I)^{-1} A \neq I$ ” of slide 47 [► 2013 L10 @ 1:15:49]; the residual is $(I - H_\lambda)y$. Its trace $\operatorname{tr} H_\lambda = \sum_i w_i$ is the *effective number of parameters* used by the fit.

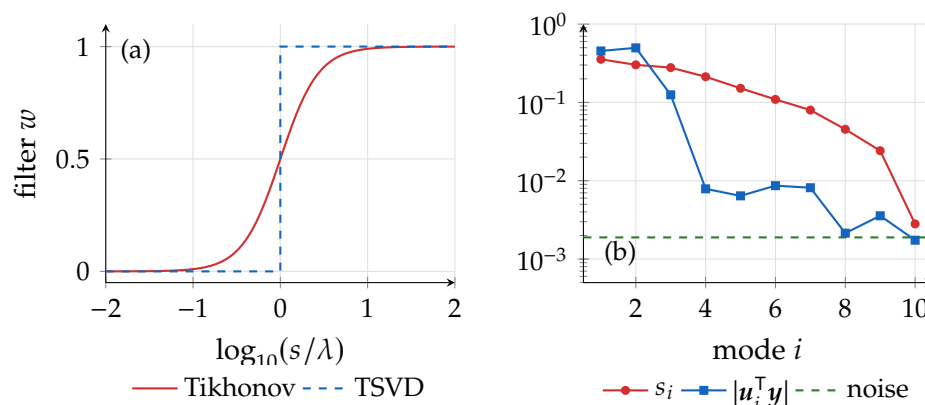


Figure 20.3: (a) Filter factors of Tikhonov ($w = s^2/(s^2 + \lambda^2)$) and of TSVD (cut at $s = \lambda$). (b) Singular values s_i of the toy calibration matrix of section 20.5 (ten instruments, 100 parameters, $\kappa = 126$) and the coefficients $|u_i^\top y|$ of noisy data (1% price noise; my computation). From mode 4 on the coefficients stop tracking s_i and stay within one to four times the noise floor $\|\epsilon\|/\sqrt{M}$ (dashed), whereas s_i keeps falling: the terms $u_i^\top y/s_i$ of (20.9) are then dominated by noise.

Exercise 20.5 ((not from the course)). Prove the following for Tikhonov regularisation with $L = I$:

- (a) $\rho(\lambda) = \|Ax_\lambda - y\|$ is non-decreasing and $\|x_\lambda\|$ non-increasing in λ ;
- (b) as $\lambda \rightarrow 0$, $x_\lambda \rightarrow A^\dagger y$ (proposition 20.6), also when A is rank-deficient;
- (c) as $\lambda \rightarrow \infty$, $x_\lambda \approx A^\top y / \lambda^2$;
- (d) $0 \leq \text{tr } H_\lambda \leq r$ and it decreases from r to 0.

20.4.4 General form: weights, smoothness matrices and several penalties

Penalising the amplitude $\|x\|$ is rarely what one wants. The general problem, with a diagonal weight matrix W and a penalty matrix L , is

$$x_\lambda = \arg \min_x \left\{ \|W(Ax - y)\|^2 + \lambda^2 \|Lx\|^2 \right\} = (A^\top W^2 A + \lambda^2 L^\top L)^{-1} A^\top W^2 y. \quad (20.13)$$

The solution is unique iff $\ker(WA) \cap \ker L = \{0\}$. It is an ordinary least-squares problem for the *augmented system* $\begin{pmatrix} WA \\ \lambda L \end{pmatrix} x \approx \begin{pmatrix} Wy \\ 0 \end{pmatrix}$, which is how it is solved stably in practice (QR, section 5.2.4). Three remarks.

- *Weights.* $W = \text{diag}(1/\sigma_i)$, with σ_i the tolerance of instrument i (its bid-offer): the data term is then a χ^2 and a residual of one tolerance costs one unit. Instruments that are important or liquid get a small tolerance and are fitted more tightly [► 2013 L10 @ 1:07:09].
- *Smoothness instead of amplitude.* With $L = I$ the penalty pulls x towards 0 , which for a volatility surface means a volatility of zero. The speaker's point [► 2013 L10 @ 1:19:00]: ask first what one does *not* like. A perfectly flat surface is acceptable, so the penalty should vanish on it, i.e. L should have the constants in its kernel: the first-difference matrix with rows $(\dots, 1, -1, \dots)$ of slide 41 penalises $\sum_i (x_i - x_{i+1})^2$, a discrete $\int |\nabla x|^2$ (in two dimensions the differences of neighbouring nodes along both axes).
- *No bias on the kernel of L .* If $Ax = y$ exactly and $Lx = 0$, then $x_\lambda = x$: indeed $x_\lambda = (A^\top W^2 A + \lambda^2 L^\top L)^{-1} A^\top W^2 Ax = x - \lambda^2 (A^\top W^2 A + \lambda^2 L^\top L)^{-1} L^\top Lx = x$. Regularisation biases the solution only in the directions it penalises, so the bias is small when the penalty encodes true prior knowledge [► 2013 L10 @ 1:17:52].

Slide 42 combines two penalties, one on the *change* of the surface with respect to a prior v_0 (matrix L_1 acting on $v - v_0 = Bx$) and one of Tikhonov type on the coefficients (matrix L_2), with parameters λ_1, λ_2 (section 20.6.4); this is (20.13) with $\lambda^2 L^\top L \rightarrow \lambda_1^2 (L_1 B)^\top L_1 B + \lambda_2^2 L_2^\top L_2$. The Bayesian reading is the same as for ridge: a Gaussian prior with precision $(\lambda^2/\sigma^2) L^\top L$, a random-walk prior for a difference penalty.

20.4.5 Bias, variance and the meaning of λ

For Tikhonov with $L = I$ the error can be computed exactly. Let $y = Ax + \varepsilon$, $\mathbb{E}[\varepsilon] = 0$, $\text{Cov } \varepsilon = \sigma^2 I$, and $\xi_i = v_i^\top x$. From (20.10), $x_\lambda = \sum_i w_i (\xi_i + u_i^\top \varepsilon / s_i) v_i$, hence

$$\mathbb{E}[\|x_\lambda - x\|^2] = \sum_{i=1}^r \left[\underbrace{(1 - w_i)^2 \xi_i^2}_{\text{bias}^2} + \underbrace{\sigma^2 \frac{w_i^2}{s_i^2}}_{\text{variance}} \right] + \sum_{i>r} \xi_i^2, \quad 1 - w_i = \frac{\lambda^2}{s_i^2 + \lambda^2}, \quad \frac{w_i^2}{s_i^2} = \frac{s_i^2}{(s_i^2 + \lambda^2)^2}. \quad (20.14)$$

6 COURSE NOTE: Notation of the slides

On slide 42 the inverse matrix is called A , which is also the model matrix of slides 44–48 (here $K = (J^\top W^2 J + \dots)^{-1}$ in section 20.6.4); and the penalty matrix L_2 is applied to the volatility vector v on slide 40 but to the coefficients x on slide 42.

(I checked (20.14) by Monte Carlo on a random 8×6 matrix: 0.7747 against 0.7750 from 4×10^4 samples.) The last sum is the part of x in the kernel of A , which no method can recover. Bias grows and variance falls with λ ; at $\lambda = 0$ the variance $\sigma^2 \sum s_i^{-2}$ is the amplification of section 20.4.2. Minimising each term over w_i : $-2(1-w)\xi^2 + 2w\sigma^2/s^2 = 0$ gives

$$w_i^* = \frac{s_i^2 \xi_i^2}{s_i^2 \xi_i^2 + \sigma^2} = \frac{s_i^2}{s_i^2 + \sigma^2 / \xi_i^2}, \quad (20.15)$$

a Wiener filter, i.e. Tikhonov with a mode-dependent $\lambda_i^2 = \sigma^2 / \xi_i^2$: a mode is kept if the signal it contributes to the data, $s_i |\xi_i|$, exceeds the noise σ , and suppressed otherwise. TSVD is the 0/1 version of this filter and a single $\lambda^2 = \sigma^2 / \tau^2$ (with τ^2 a typical size of ξ_i^2) is the ridge–Bayes value of proposition 5.22. The optimum cannot be used as it stands, since ξ_i is unknown, but it says what λ is: a noise-to-signal ratio for the parameters.

Exercise 20.6 ((not from the course)). Take $N = M = 1$: $y = ax + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. Find the Tikhonov estimator, its bias and variance, and the λ that minimises the mean-square error as a function of x ; compare with (20.15). What happens if one substitutes the ridge–Bayes value $\lambda^2 = \sigma^2 / \tau^2$ and the true $|x|$ is much larger or much smaller than τ ?

20.5 Choosing the regularisation parameter (not covered in class)

The lecture calls λ a “small regularisation parameter” and says nothing on how to fix it [► 2013 L10 @ 52:25]. Three standard criteria, then a numerical test. Let $\rho(\lambda) = \|W(Ax_\lambda - y)\|$ be the residual. For $L = I$ and $b_i = u_i^\top y$, $\rho^2 = \sum_{i \leq r} (1 - w_i)^2 b_i^2 + \text{const}$ increases with λ and $\|x_\lambda\|^2 = \sum_i w_i^2 b_i^2 / s_i^2$ decreases (each w_i decreases with λ), so the trade-off is monotone.

- *Discrepancy principle* (Morozov): choose λ with $\rho(\lambda) = \delta$, the size of the data error, $\delta \approx \sigma \sqrt{M}$ (with $W = \text{diag}(1/\sigma_i)$: $\rho^2 \approx M$). Rationale: fitting closer than the noise fits the noise. This is the lecture’s remark that every input has a tolerance and calibrating exactly is pointless [► 2013 L10 @ 1:06:08]; it needs the noise level, here the bid–offer.
- *Generalised cross-validation*. Leave-one-out cross-validation predicts each datum from the others. For a linear smoother $\hat{y} = H_\lambda y$ one has $y_i - \hat{y}_i^{(-i)} = (y_i - \hat{y}_i) / (1 - (H_\lambda)_{ii})$ (proof: the fit computed without datum i is also the minimiser when y_i is replaced by its own prediction $\hat{y}_i^{(-i)}$, since that term is then zero; so $\hat{y}_i^{(-i)} = \sum_{j \neq i} H_{ij} y_j + H_{ii} \hat{y}_i^{(-i)}$). Replacing $(H_\lambda)_{ii}$ by the average $\text{tr } H_\lambda / M$ gives

$$\text{GCV}(\lambda) = \frac{M \rho(\lambda)^2}{(M - \text{tr } H_\lambda)^2},$$

which needs no noise level; it is the counterpart of the cross-validation used for ridge in section 5.10.

- *L-curve*. Plot $\log \rho(\lambda)$ against $\log \|Lx_\lambda\|$; the corner of the “L” separates the regime dominated by noise (small λ , large seminorm) from the regime dominated by bias.

CASE STUDY: A toy calibration with regularisation (my computation; not in class)

Goal. Mimic the calibration of a volatility slice. **Set-up.** A smooth “true” volatility $v(s)$ on a grid of $N = 100$ points, $s \in [0, 10]$ (level 0.19–0.25); $M = 10$ instruments, each an average of v with a Gaussian window (centres 0.5–9.7, widths growing from

0.46 to 1.56, overlapping); prices $y_i = (Av)_i(1 + 0.01z_i)$ with z_i standard normal, so $\delta = \|\varepsilon\| = 0.0060$; $\kappa(A) = 126$ on the ten modes. **Methods.**

- (a) The lecture's "formal" solution: ten piecewise-constant basis functions, square Jacobian, exact inversion.
- (b) Tikhonov (20.13) with the first-difference matrix L of slide 41, $W = I$, on the fine grid.

Results (fig. 20.4).

- (a) Reprices the ten inputs to 5×10^{-17} but the surface swings between 0.002 and 0.40 (root-mean-square error 0.106, roughness $\|Lv\| = 0.64$), and a 1% change in one input moves it by 16% of its level in the worst place: the skyline of the lecture ▶ 2013 L10 @ 1:04:01].
- (b) With λ from GCV (0.44, effective number of parameters $\text{tr } H_\lambda = 6.2$): error 0.0031, roughness 0.0144 (truth 0.0147), the same 1% perturbation moves it by 1.3% of its level. The discrepancy principle gives $\lambda = 0.96$ and error 0.0032; the best possible value (knowing the truth) is 0.0031 at $\lambda = 0.65$. GCV picks a residual of 0.46δ , a mild over-fit, typical of GCV.

Which penalty? With the same tuning, the best error is 0.0167 for $L = I$ (amplitude), 0.0039 for the second difference and 0.0031 for the first difference. TSVD of A (minimum-norm solution) reaches at best 0.018 (with $k = 8$ modes) and 0.063 with all ten: an unsuitable penalty (here: the minimum norm, which pulls the surface towards zero) gives errors five to six times larger than the suitable one, whereas λ can be changed by a factor two with hardly any effect on the error. **Lesson.** Regularisation is a matter of choosing *what* to penalise (prior knowledge of the solution) first, and how much second.

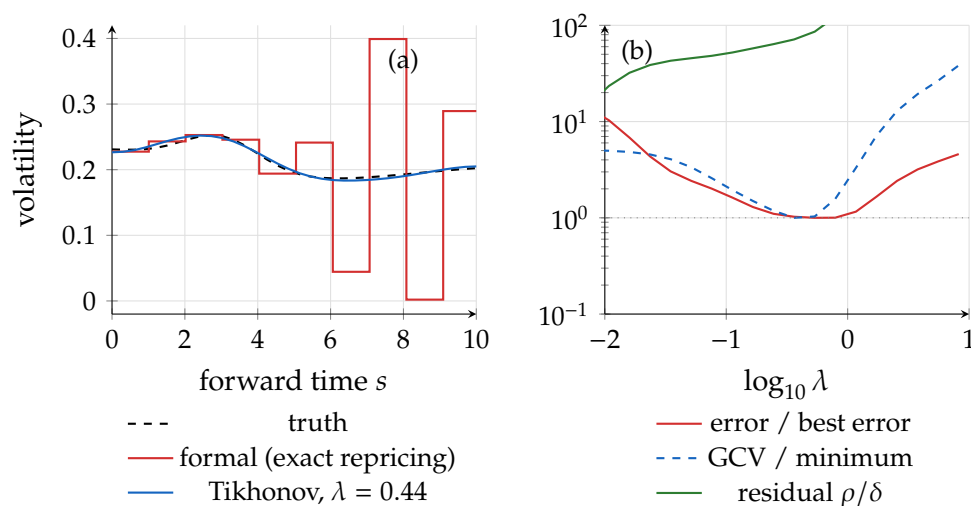


Figure 20.4: The toy calibration of section 20.5 (my computation). (a) The formal exact solution is the "skyline"; the smoothness-regularised one follows the truth to 0.003. (b) Error, GCV and residual against λ (each normalised as indicated; dotted line: $\rho = \delta$, the discrepancy principle). The minima of the error and of GCV are broad and lie within a factor two of each other.

Exercise 20.7 ((not from the course)). Reproduce the toy of [section 20.5](#) (any language): choose your own smooth v on $[0, 10]$, ten Gaussian-window instruments, 1% noise; compute the formal solution, the Tikhonov solution for a grid of λ , the discrepancy and GCV parameters. Vary the penalty ($L = I$, first difference, second difference) and the noise level, and report which choice is most sensitive to λ .

20.6 Calibrating a volatility surface

20.6.1 The pricing engine and the HJM model

The lecture now steps back [[▶ 2013 L10 @ 53:26](#)]. A *pricing engine* takes the trader's portfolio and the model parameters (curves, volatility surface, ...) and returns values and risks. To make the parameters consistent with the market, a *calibration engine* compares the model prices of a set of benchmark instruments with their market prices and adjusts the parameters until they agree (slide 25); only then is the portfolio priced. The example is the Heath–Jarrow–Morton model ([chapter 24](#)), used to price volatility products by Monte Carlo [[▶ 2013 L10 @ 54:35](#)]:

$$df_{t,s} = \mu_{t,s} dt + f_{t,s}^\beta V(t,s) \rho(t,s) \cdot dB_t^{\mathbb{Q}}, \quad (20.16)$$

for the forward rate $f_{t,s}$ observed at calendar time t for forward time s : μ is the drift (fixed by absence of arbitrage, [chapter 24](#)), β the “skew factor” ($\beta = 1$ lognormal, $\beta = 0$ normal model), ρ the correlation and factor structure, $B^{\mathbb{Q}}$ a Brownian motion, and $V(t,s)$ the *volatility surface* that is the subject of the lecture [[▶ 2013 L10 @ 55:41](#)]. The Monte Carlo advances the whole forward curve from one quarter to the next, and every step needs the value of V on the corresponding cell of the triangular grid of (calendar time, forward time) [[▶ 2013 L10 @ 56:47](#)]; the instruments that pin it down, caps and swaptions, are sensitive to *regions* of this triangle ([▶ 2013 L10 @ 57:48](#); a swaption depends on the forward rates that will be observed at the option expiry and on their volatility until then), and the regions of different instruments overlap heavily [[▶ 2013 L10 @ 58:50](#)].

20.6.2 Why the problem is hard

The surface has a value for each of the cells of a triangle with 240 quarters (60 years) on a side, that is $240 \cdot 241/2 = 28,920$ unknowns [[▶ 2013 L10 @ 59:53](#)] (the slide says “~28k”), whereas only 20–50 swaptions and caps are quoted. The problem is underdetermined (the kernel of the forward map has dimension at least 28,870), ill-posed because of the overlapping sensitivities, and its unregularised solution is unstable and noisy [[▶ 2013 L10 @ 1:00:55](#)]. A full $28,920 \times 28,920$ matrix of doubles takes 6.7 GB (my count), which the speaker rules out as “no memory”. And the result must be *smooth*: two similar swaptions in the portfolio should get similar prices, and a spike of volatility in some region of the future is either explained by an economic event or is an artefact [[▶ 2013 L10 @ 1:00:55](#)].

20.6.3 The “formal” solution and why it fails

Represent the surface, flattened to a vector v with one entry per cell, as a prior value plus a combination of basis functions [[▶ 2013 L10 @ 1:01:56](#)]:

$$v = v_0 + Bx, \quad (20.17)$$

7 COURSE NOTE: Counting the elements of the triangle

In the transcript [[▶ 2013 L10 @ 59:53](#)] the number of triangle elements is first said to be “200K” and then a matrix of 28K×28K is discussed; slide 29 gives “~28k”, which is the right count: half of $240^2 = 57,600$ plus the diagonal is 28,920.

where the columns of B are the basis functions (values on the grid) and x the weights. The pricing model gives the sensitivities of the prices of the calibration instruments to a perturbation of the surface along each basis function [► 2013 L10 @ 1:02:59]; working with log-prices, $q_i = \ln(q_i^{\text{mdl}}/q_i^0)$, $q_i^{\text{in}} = \ln(q_i^{\text{mkt}}/q_i^0)$,

$$J_{ij} = \frac{\partial q_i}{\partial x_j}, \quad q \approx Jx. \quad (20.18)$$

The *formal* approach takes as many basis functions as instruments ($N = M$), piecewise constant on the region of sensitivity of each instrument, so that J is square and (“if the basis functions are reasonable”) invertible, and solves $x = J^{-1}q^{\text{in}}$ iteratively [► 2013 L10 @ 1:02:59]. The iteration converges quickly because for at-the-money instruments the price is nearly proportional to the volatility: for the Black–Scholes forward price (chapter 15), $C = F [2N(\sigma\sqrt{T}/2) - 1] = F \sigma\sqrt{T}/\sqrt{2\pi}(1 - \sigma^2 T/24 + \dots)$, so $\ln C \approx \ln \sigma + \text{const}$ and q is nearly linear in the volatility increments (my derivation of the speaker’s remark; the calibrated instruments are swaptions in an HJM Monte Carlo, for which the proportionality is heuristic).

The outcome is an exact fit and a meaningless surface: “Manhattan skyline, with the Hudson river” [► 2013 L10 @ 1:04:01](slide 33, “Exact, but . . . meaningless”). Smoothing the piecewise-constant basis functions gives a better-looking surface but not a good one, and the instability remains: a 1% change in the price of the 5y × 10y swaption changes the adjustment of the surface by 4% (slide 35: one tall isolated “building with an antenna”) [► 2013 L10 @ 1:05:04]. This is what section 20.4.2 predicts: J is square, so $J^\dagger = J^{-1}$ and errors are amplified by up to $\kappa(J)$; the toy of section 20.5 shows the same phenomenon (1% of noise, a factor 16 on the surface).

20.6.4 The regularised design

The lecture turns ill-posedness to advantage with four changes [► 2013 L10 @ 1:06:08](slide 36).

- *Tolerance.* Each input has a bid–offer, so there is no point in calibrating exactly. The fit becomes a weighted least-squares term $\|W(q - q^{\text{in}})\|^2$, W diagonal with small weights on instruments of large bid–offer and large ones on important instruments. A small loss of accuracy buys a large gain of smoothness.
- *Smooth basis functions.* Cubic B-splines (definition 20.2) in each direction, and their tensor products $B_a(t)B_b(s)$ as two-dimensional basis functions [► 2013 L10 @ 1:08:13]. Since B-splines are non-negative and sum to one (proposition 20.3), every combination is a reasonable surface, and its regularity is inherited from the basis (“every basis function makes sense, so any linear combination will be good enough”).
- *More basis functions than instruments* ($N > M$): it is not known in advance which shapes are needed, and the number of unknowns is still tiny compared with the 28,920 cells (with 24 basis functions per direction, 576 coefficients, so the matrix of the normal equations is 576×576 , about 2.7 MB) [► 2013 L10 @ 1:07:09]. The knots are denser at short forward times, where instruments are dense [► 2013 L10 @ 1:22:09].
- *Penalties.* The problem becomes [► 2013 L10 @ 1:09:19]

$$\|W(q - q^{\text{in}})\|^2 \rightarrow \min, \quad \|L_1(v - v_0)\|^2 \rightarrow \min, \quad \|L_2 v\|^2 \rightarrow \min, \quad (20.19)$$

with a penalty L_1 on the *change* of the surface with respect to a prior and a penalty L_2 on the surface itself (slide 40). The gradient penalty is a matrix with rows $(\dots, 1, -1, \dots)$ for every pair of neighbouring cells (slide 41, [► 2013 L10 @ 1:10:20]); on

a rectangular grid of $n_t \times n_s$ nodes it is $L_1 = \begin{pmatrix} D_t \otimes I \\ I \otimes D_s \end{pmatrix}$ with D the one-dimensional first-difference matrices (on the triangle only the pairs of cells that both lie inside are kept).

Combining the three terms with weights λ_1, λ_2 gives (slide 42, ▶ 2013 L10 @ 1:10:20)

$$\begin{aligned} \mathbf{x} &= \arg \min \left\{ \|W(J\mathbf{x} - \mathbf{q}^{\text{in}})\|^2 + \|\lambda_1 L_1 B \mathbf{x}\|^2 + \|\lambda_2 L_2 \mathbf{x}\|^2 \right\} = K J^T W^2 \mathbf{q}^{\text{in}}, \\ K &= (J^T W^2 J + \lambda_1^2 (L_1 B)^T L_1 B + \lambda_2^2 L_2^T L_2)^{-1}, \end{aligned} \quad (20.20)$$

which is (20.13) with the two penalty terms of section 20.4.4 (setting the gradient of the objective to zero gives the normal equations; here $L_1 B \mathbf{x} = \mathbf{v} - \mathbf{v}_0$ so the first penalty acts on the change of surface, and L_2 acts on the coefficients). K is invertible when the penalties together have no kernel in common with WJ ; $\lambda_2 > 0$ with $L_2 = I$ guarantees it. Two computational remarks from the Q&A ▶ 2013 L10 @ 1:24:19]: only $J^T W^2 \mathbf{q}^{\text{in}}$ changes when market quotes tick, whereas the penalty matrices (and, if J is approximately constant, $J^T W^2 J$) can be precomputed, so the calibration runs in real time on live data; and the linear algebra is done in the space of $N \ll 28,920$ coefficients, never on the full surface. When the pricing map $\mathbf{q}(\mathbf{x})$ is not linear, each Newton step solves a regularised problem of this form; the resulting damped iteration is a Levenberg–Marquardt method (not discussed in class).

INTUITION: Where in the chapter the pieces fit

Formula (20.20) is one object seen three ways. As *statistics*: ridge regression (section 5.10) with a design matrix WJ , observation weights, and a prior that surfaces are smooth and close to $\mathbf{v}_0$. As *linear algebra*: the SVD filter (20.10) on the singular values of WJ in the metric of the penalty. As *engineering*: what a trader looks at is the residual: the surface is a smooth fit through the market, and what is left over is information.

Slide 43 shows the calibrated surface: smooth, with a few gentle peaks and a sharper one at short forward times. Asked about a peak, the speaker explains ▶ 2013 L10 @ 1:22:09] that the grid is uniform in quarters whereas the instruments are at 3 months, 6 months, 1, 2, 5, 10 years, so on a logarithmic axis the feature would not look sharp; the surface has more detail where there are more instruments, and no sharper features can be supported elsewhere.

IN FINANCE: Relative value: the calibrated surface as a benchmark

Traders watch the surface as it moves with the live quotes ▶ 2013 L10 @ 1:24:19]. If an instrument trades away from the smooth surface implied by the others (something that “is not typical”, a judgement that “comes with years of experience”) it is a candidate mispricing: a spike is likely to disappear soon, and one trades (a swaption, or a more exotic product with the appropriate exposure) against it. This is *relative value analysis* and, in the speaker’s words, how the desk makes money ▶ 2013 L10 @ 1:25:20]. It works only if the model is robust: a regulariser based on a fundamental principle (smoothness) is more likely to be stable in the future than one estimated from a past window such as PCA ▶ 2013 L10 @ 1:26:24]. The same remarks apply to the swap curve.

IN FINANCE: Interpolation, splines and their limits

To a question on other fitting techniques the speaker answers that “spline” and “interpolation” are the same thing for him: one has limited inputs and wants the curve in between ▶ 2013 L10 @ 1:20:07]. The regularised fit goes beyond interpolation: it does

not pass through the data, which is the essential difference between [section 20.1.3](#) and [section 20.6.4](#).

What regularisation buys, and costs [► 2013 L10 @ 1:17:52]. It gives stability (indispensable for ill-conditioned problems) and a more realistic solution, at the expense of not fitting the data exactly and possibly of a *biased* solution (for example a smoothness constraint makes the result smoother than reality). The bias is controlled by *what* is penalised, in the sense of the “no bias on the kernel of L ” remark in [section 20.4.4](#): open the textbook and one finds “Tikhonov” and penalises the amplitude, which would push the volatility towards zero; a flat surface is acceptable, so one penalises derivatives instead [► 2013 L10 @ 1:19:00]. The slides’ references are the Wikipedia pages on the HJM model, the yield curve, inverse problems, Tikhonov regularisation, the SVD, PCA and B-splines.

KEY FORMULAS

- Bond: $P = \sum_i \Lambda_i C_i$; yield $\Lambda_i = e^{-y t_i}$; duration $D = \sum_i t_i \pi_i$, convexity $C = \sum_i t_i^2 \pi_i$ (20.1); bond spread $P = \sum_i e^{-s t_i} \Lambda_i C_i$ (20.3).
- Swap rate $C = \sum_i r_i w_i / \sum_i w_i$, $w_i = \delta_i \Lambda_i$ (20.2).
- Cox–de Boor: $B_{i,k} = \frac{t - \tau_i}{\tau_{i+k} - \tau_i} B_{i,k-1} + \frac{\tau_{i+k+1} - t}{\tau_{i+k+1} - \tau_{i+1}} B_{i+1,k-1}$, $\sum_i B_{i,k} = 1$.
- PCA hedge: $\mathbf{x} = R^T \mathbf{r}$, $R = P(H^T P)^{-1}$, $H^T R = I$ (20.5); smooth hedge: $R = (HH^T + \lambda^2(LJ)^T L J)^{-1} H$ (20.7).
- Inverse problem $\mathbf{y} = A\mathbf{x} + \boldsymbol{\varepsilon}$: $\mathbf{x}^\dagger = \sum_i \frac{\mathbf{u}_i^T \mathbf{y}}{s_i} \mathbf{v}_i$, error $\sum_i \frac{\mathbf{u}_i^T \boldsymbol{\varepsilon}}{s_i} \mathbf{v}_i$, $\kappa = s_1/s_r$ (20.9).
- Filters $\mathbf{x}_w = \sum_i w_i \frac{\mathbf{u}_i^T \mathbf{y}}{s_i} \mathbf{v}_i$: Tikhonov $w_i = \frac{s_i^2}{s_i^2 + \lambda^2}$, $\mathbf{x}_\lambda = (A^T A + \lambda^2 I)^{-1} A^T \mathbf{y}$; TSVD $w_i = \mathbf{1}_{i \leq k}$ (20.11); gain $\leq 1/(2\lambda)$; repricing $H_\lambda = \sum_i w_i \mathbf{u}_i \mathbf{u}_i^T \neq I$.
- General form $\min \|W(A\mathbf{x} - \mathbf{y})\|^2 + \lambda^2 \|L\mathbf{x}\|^2$: $\mathbf{x} = (A^T W^2 A + \lambda^2 L^T L)^{-1} A^T W^2 \mathbf{y}$; no bias on $\ker L$ (20.13).
- MSE = $\sum_i [(1 - w_i)^2 \xi_i^2 + \sigma^2 w_i^2 / s_i^2] + \sum_{i>r} \xi_i^2$; optimal $\lambda_i^2 = \sigma^2 / \xi_i^2$ (20.14)–(20.15).
- Choosing λ : discrepancy $\rho(\lambda) = \delta$; GCV = $M\rho^2 / (M - \text{tr } H_\lambda)^2$.
- Surface calibration: $\mathbf{v} = \mathbf{v}_0 + B\mathbf{x}$, $J = \partial \mathbf{q} / \partial \mathbf{x}$, $\mathbf{x} = K J^T W^2 \mathbf{q}^{\text{in}}$ (20.20).

Exercise 20.8 ((not from the course)).

- Show that the total number of cells of a triangular grid with n quarters on a side is $n(n+1)/2$ and compute the memory (in GB, doubles) of the square matrix of the full problem for $n = 240$.
- With a tensor-product cubic B-spline basis of m functions per direction, how many coefficients, and what is the size of the matrix K in (20.20)?
- Explain why $J^T W^2 J$ has rank at most the number of instruments, and why K nevertheless exists.

Chapter 21

Commodity Models

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2013 L13	A. Eydeland (Morgan Stanley)	What quants do in commodity markets: contango and the Trafigura trade; storage as a portfolio of calendar-spread options and the storage-optimisation problem; the power plant as a strip of spark-spread options; fat tails and spikes of power prices; why spot models fail; the “stack” (supply and demand) model of power prices.
Files. Slides MIT18_S096F13_lecnote13.pdf (47 pages: 44 of content, then references and disclaimers). Problem sets. None; all exercises (each after the section it practises) are <i>not from the course</i>. Year. Part II of these notes collects the 2013-only topics; there is no 2024 counterpart. The guest is Alexander Eydeland (the slides cite his book with K. Wolyniec, <i>Energy and Power Risk Management</i>, Wiley 2002, and a chapter with H. Geman). Not in the slides. The whole discussion of the trade, of the bid for storage and of the correlation structure comes from the video; the slides give the formulas and the pictures. The sections marked (<i>not discussed in class</i>) are additions of these notes, needed to make the lecture’s claims precise (Sections 21.1, 21.7 and part of 21.6, 21.8).		

Motivation

Every other lecture of the course prices financial claims whose underlying can be bought, held and sold at negligible cost. A commodity is different: oil sits in tanks, gas in salt caverns, electricity nowhere (it cannot be stored at scale). The quantities that matter are therefore physical: how much can be stored and how fast it can be injected, how efficiently a plant turns fuel into power, what the supply curve of generators looks like. Eydeland’s message is that the mathematics of derivatives ([chapter 15](#)) applies, but the *objects* to be valued are physical assets, and each asset turns out to be a portfolio of options on a *spread*:

asset	underlying spread	option
oil or gas storage	calendar spread $F(T_2) - F(T_1)$	Section 21.2
gas-fired power plant	power – heat rate $\times$ gas (spark spread)	Section 21.4
refinery	products – crude (crack spread)	Section 21.5 , section 9.6.4

The plan of the chapter is that of the lecture: a trade in a contango market ([section 21.1](#)); the value of storage, the Margrabe spread-option formula and how to hedge it ([sections 21.2](#) and [21.3](#)); the merchant power plant ([section 21.4](#)); why power prices resist the standard

spot models (sections 21.6 and 21.7) and the alternative that models supply and demand (section 21.8). The chapter uses the forward and futures language of chapter 6 (futures are margined, forwards are not) and of proposition 15.1, the Black–Scholes machinery of chapter 15, and the Ornstein–Uhlenbeck process of chapter 18; the crack-spread example of section 9.6.4 is the same economic object as the refinery above.

21.1 Contango, backwardation and cash-and-carry

The lecture opens with a news item of March 2009: the trading house Trafigura expects its best results ever *because* oil prices are low and the market is in contango [▶ 2013 L13 @ 1:06]. A year earlier oil had reached \$159 per barrel and traders had been blamed for it; in 2009 it fell to about \$30 and the same traders made record profits [▶ 2013 L13 @ 2:07]. The apparent paradox is resolved by looking at the futures curve.

21.1.1 The futures curve

A *futures contract* lets one buy today, at a price fixed today, a barrel to be delivered at a future date; the prices $F(t, T)$ for all maturities T form the *futures* (or *forward*) *curve* [▶ 2013 L13 @ 3:09]. In commodity markets there is no “spot price” as such: what television calls the spot price is the price of the nearest futures contract [▶ 2013 L13 @ 4:10].

Definition 21.1 (Contango and backwardation). The curve $T \mapsto F(t, T)$ is in *contango* if it is increasing and in *backwardation* if it is decreasing [▶ 2013 L13 @ 4:10]. (Loosely, one says that the market is in contango or backwardation when the curve is so at the front, i.e. when $F(t, T) \geq S_t$.)

On 15 January 2009 the WTI curve was in steep contango: about \$35 for the nearest contract (February 2009), \$55 for August, \$58 for December and \$60 for February 2010 (fig. 21.1; the intermediate points are read off the slide and are approximate).

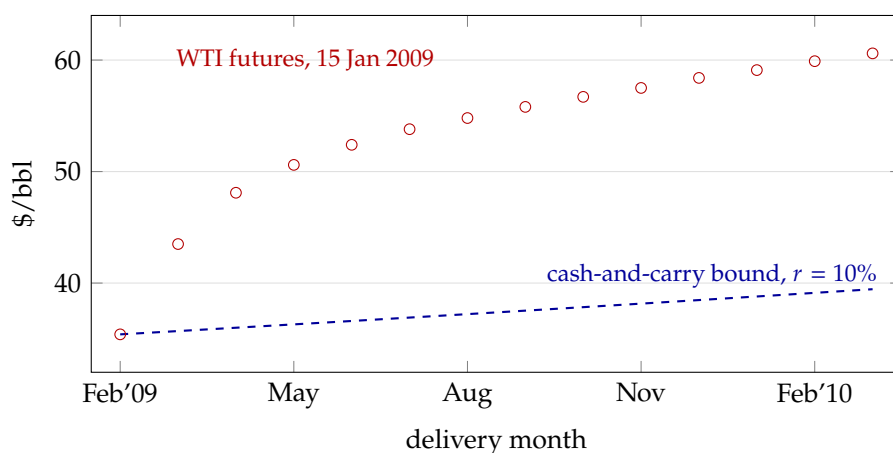


Figure 21.1: The WTI futures curve of 15 January 2009 (points read off slide 4 of the lecture) against the upper bound $S e^{rT}$ that a holder of free storage would impose with $r = 10\%$ (proposition 21.2). The gap between the two is what the market charges for storage.

Exercise 21.1 (not from the course: implied convenience yield). On a given day $S_t = \$70$, $r = 3\%$, and the one-year futures trades at \$66.

- (a) Using (21.1) with negligible storage cost, find the implied convenience yield y .
- (b) If storage actually costs $w = 2\%$ per year, what is y now?
- (c) Is the market in contango or backwardation, and is this consistent with a positive convenience yield?

21.1.2 The Trafigura trade

Suppose one owns storage. Then on 15 January 2009 one can, at no risk [► 2013 L13 @ 5:11]:

1. borrow \$35 and buy one barrel (the nearby price);
2. store it for a year;
3. sell one February-2010 futures contract, fixing the sale price at \$60.

The lock-in is $60 - 35 = \$25$ minus the interest on the loan: \$3.5 at 10% simple interest, so a profit of \$21.5 per barrel (the lecturer says “about 21”); with continuous compounding the interest is $35(e^{0.10} - 1) = 3.68$ and the profit \$21.3. With 50–60 million barrels of storage the total is enormous and risk-free [► 2013 L13 @ 6:16]. The trade needs low prices *because* the interest is proportional to the price paid: at \$125–160 a barrel the same contango would leave about half the profit [► 2013 L13 @ 7:21]. What it needs is *storage*: the strategy is “long the February 2009 contract, short the February 2010 contract”, i.e. a long *calendar spread*, plus a tank to receive the barrel [► 2013 L13 @ 7:21].

Proposition 21.2 (Cash-and-carry bounds). *Let the interest rate r and the storage cost rate u (proportional to the price, per unit time) be constant, and let S be the price of the physical commodity, which is storable. In the absence of arbitrage for an agent that can store,*

$$F(0, T) \leq S_0 e^{(r+u)T}.$$

The opposite inequality $F \geq S_0 e^{(r+u)T}$ does *not* follow, because it would require selling the commodity today and buying it back later: only somebody who holds inventory can do that, and holding inventory may itself be valuable (a refinery that runs out of crude stops). The shortfall is captured by the *convenience yield* y defined by

$$F(0, T) = S_0 e^{(r+u-y)T}, \quad y = r + u - \frac{1}{T} \ln \frac{F(0, T)}{S_0} \geq 0 \text{ for a storable commodity. (21.1)}$$

(Not discussed in class.) Backwardation corresponds to a large convenience yield, $y > r + u$: inventory is scarce and valuable, spot is expensive against later delivery.

IN FINANCE: Physical assets and *who* can arbitrage

The arbitrage of proposition 21.2 is not open to a financial investor: without storage one cannot take the physical leg. Contango therefore pays the owners of scarce infrastructure (tanks, supertankers used as floating storage, salt caverns for gas) and the shape of the curve is best read as the price of the option to store. The remainder of the lecture asks the inverse question: given the curve, *what is a tank worth?*

Proof. If $F(0, T) > S_0 e^{(r+u)T}$, borrow S_0 , buy the commodity, pay the storage cost, and sell the future. At T deliver against $F(0, T)$ and repay the loan with its interest and the accumulated storage cost $S_0 e^{(r+u)T}$: the net cash flow $F(0, T) - S_0 e^{(r+u)T} > 0$ is certain and costs nothing. This is the argument of proposition 15.1 with a storage cost. □

Example 21.3 (The 2009 contango). With $S_0 = 35$, $F = 60$ and $T = 1$ year the implied cost of carry is $\ln(60/35) = 54\%$ per year: at $r = 10\%$ the bound of proposition 21.2 would require $u \geq 44\%$ per year to remove the arbitrage, i.e. $y = r + u - 0.539 < 0$ for any reasonable u . The resolution is that the arbitrage is available only to those who own *empty storage*; storage was scarce (the market was oversupplied) and its shadow price was the difference $60 - 35e^{0.10} = 21.3$ per barrel per year. Contango of this size is the market's bid for tanks.

INTUITION: Physicist's reading

Think of the curve as a potential energy landscape along the time axis. A steep upward slope in T is a gradient that an owner of storage can “roll down” at fixed cost: buy at the bottom, sell at the top. Backwardation is the reversed slope: the tank owner is paid to empty the tank now and refill it later. The convenience yield is the price of the dissipation that keeps the slope from exceeding $r + u$ when storage is plentiful.

21.2 The value of storage: a calendar-spread option

21.2.1 The bid for a tank

The lecturer inverts the question [► 2013 L13 @ 7:21]. Your boss hands you a credit card and asks you to obtain storage from August to December in an auction, with the same curve: $F_{\text{Aug}} = 55$, $F_{\text{Dec}} = 58$. Bid too little and you lose; bid too much and you suffer the winner's curse. The maximal bid is the amount that one can be *sure to recover*, so a bidder needs, before bidding, a strategy that recovers the price paid [► 2013 L13 @ 9:23].

The trader's answer. Buy the August contract and sell the December contract on 1 January; on 1 August take delivery of the barrel and store it; in December deliver against the December contract. This locks $58 - 55 = 3$ dollars per barrel: one can pay up to \$3 (say \$2.99, to keep a penny of profit) [► 2013 L13 @ 10:25]. Interest is ignored “for simplicity” (slide 9 adds: it should not be) [► 2013 L13 @ 11:31]. With interest at rate r the locked amount, in December dollars, is $F_{\text{Dec}} - F_{\text{Aug}} e^{r(T_{\text{Dec}} - T_{\text{Aug}})}$, since the \$55 paid in August must be financed until the sale: at $r = 10\%$ and four months the financing costs about $55(e^{0.10/3} - 1) \approx 1.9$, more than half of the locked \$3, which is why the lecturer says that one must not really forget it.

The quant's answer. On 1 January sell to somebody an *August–December calendar spread option*, exercised on 31 July, with payoff

$$[F_{\text{Dec}}(T) - F_{\text{Aug}}(T)]^+, \quad T = 31 \text{ July}, \quad (21.2)$$

the futures prices being those quoted on the exercise date [► 2013 L13 @ 12:35]. There is no strike: the “strike” is the other futures price, so the option is a function of two random variables and is more difficult than an option on a single stock [► 2013 L13 @ 14:39]. Using the formula of [theorem 21.4](#) below, the lecture quotes $V = 4.4677$ dollars per barrel [► 2013 L13 @ 15:39]. So one can bid, for instance, \$4.20 [► 2013 L13 @ 15:39]: the bid beats the trader's ceiling of \$3, and the margin ($4.4677 - 4.20 = 0.27$) is much larger than a penny.

Why is the value larger than the \$3 of the trader? Because $3 = F_{\text{Dec}} - F_{\text{Aug}}$ is the *intrinsic value* of the option, its value at zero volatility, and an option is worth more than its intrinsic value when the volatility is positive; energy volatilities are far larger than in rates, currencies or equity indices [► 2013 L13 @ 16:44].

Where is the risk? Selling the option exposes one to an unbounded payment. Suppose on 31 July $F_{\text{Aug}} = 55$ and $F_{\text{Dec}} = 80$ (slide 11 uses 65 and 80): the seller owes \$25 (slide: \$15) [► 2013 L13 @ 17:47]. But he owns the tank: he buys August and sells December *at the same moment* using futures, and later moves the physical barrel, collecting exactly $F_{\text{Dec}} - F_{\text{Aug}}$ [► 2013 L13 @ 18:48]. If instead December falls below August the option pays nothing and

the tank is left unused [► 2013 L13 @ 20:55]. So the tank *is* the option: a full hedge, whatever happens.

21.2.2 The spread-option formula

The value quoted on slide 10 is the formula for an option to exchange one lognormal asset for another (Margrabe, 1978), applied to two futures prices. The natural setting is that of [section 15.8](#): futures are costless to enter and margined daily, so under the risk-neutral measure $\mathbb{Q}$ a futures price is a martingale (the futures analogue of the statement that the discounted stock is a martingale).

Theorem 21.4 (Zero-strike spread option; Margrabe). Let F_1, F_2 be $\mathbb{Q}$ -martingales with $dF_i = \sigma_i F_i dW_i$, $dW_1 dW_2 = \rho dt$, constant $\sigma_i > 0$ and $|\rho| < 1$, and let the discount factor to the expiry T be e^{-rT} . Then

$$V = e^{-rT} \mathbb{E}_{\mathbb{Q}}[(F_2(T) - F_1(T))^+] = e^{-rT} [F_2(0) \Phi(d_1) - F_1(0) \Phi(d_2)], \quad (21.3)$$

$$d_1 = \frac{\ln(F_2(0)/F_1(0)) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad \sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2,$$

and Φ is the standard normal distribution function. (Slide 10, with “DiscFactor” = e^{-rT} and $N = \Phi$.)

The proof used only that the *ratio* is lognormal; the numeraire is the asset F_1 (this change-of-numeraire trick is the standard proof of Margrabe’s formula). Three properties are immediate.

Proposition 21.5 (Properties).

- (i) $V \geq e^{-rT} (F_2(0) - F_1(0))^+$, with equality in the limit $\sigma \rightarrow 0$: the option is worth at least its intrinsic value (Jensen, since $F_2 - F_1$ is a martingale and $x \mapsto x^+$ is convex).
- (ii) V is homogeneous of degree one in (F_1, F_2) and increasing in σ , and depends on $(\sigma_1, \sigma_2, \rho)$ only through the spread volatility σ ; in particular a higher correlation ρ lowers the value.
- (iii) The derivatives are $\partial V / \partial F_2 = e^{-rT} \Phi(d_1)$ and $\partial V / \partial F_1 = -e^{-rT} \Phi(d_2)$.

The numbers of slide 10. The slide gives neither the volatilities, nor the correlation, nor the time to expiry. Taking the exercise date 31 July, $T = 211/365 = 0.578$ years from 1 January, and zero interest, one finds that $\sigma_1 = \sigma_2 = 50\%$ and $\rho = 0.95$ reproduce the printed value: $\sigma = 0.5\sqrt{2(1 - 0.95)} = 0.1581$ and

$$V = 58 \Phi(0.5019) - 55 \Phi(0.3817) = 58 \cdot 0.6921 - 55 \cdot 0.6486 = 4.4677$$

to four decimals (verified numerically; solving directly for σ gives 0.15811). This is our reconstruction, not something said in the lecture. The value splits as 3.00 of intrinsic and 1.47 of time value. The dependence on the correlation is dramatic ([table 21.1](#)): $\rho = 1$ gives exactly the intrinsic value, and a fall of ρ from 0.99 to 0.95 raises the value by 1.2 dollars.

Proof. Write $(F_2 - F_1)^+ = F_1(T)(X_T - 1)^+$ with $X = F_2/F_1$. Define a new measure $\mathbb{Q}^1$ on $\mathcal{F}_T$ by $d\mathbb{Q}^1/d\mathbb{Q} = F_1(T)/F_1(0)$, a positive random variable of mean one because F_1 is a $\mathbb{Q}$ -martingale. Then

$$\mathbb{E}_{\mathbb{Q}}[F_1(T)(X_T - 1)^+] = F_1(0) \mathbb{E}_{\mathbb{Q}^1}[(X_T - 1)^+].$$

Under $\mathbb{Q}^1$ the ratio X is a martingale: for $A \in \mathcal{F}_t$,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}^1}[X_T \mathbf{1}_A] &= \mathbb{E}_{\mathbb{Q}}[F_1(T)F_1(0)^{-1}F_2(T)F_1(T)^{-1}\mathbf{1}_A] = \mathbb{E}_{\mathbb{Q}}[F_2(T)\mathbf{1}_A] \\ &= \mathbb{E}_{\mathbb{Q}}[F_2(t)\mathbf{1}_A]/F_1(0) = \mathbb{E}_{\mathbb{Q}^1}[X_t \mathbf{1}_A]. \end{aligned}$$

By Ito’s formula $dX/X = \sigma_2 dW_2 - \sigma_1 dW_1 + (\text{drift})dt$, with quadratic variation $(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)dt = \sigma^2 dt$; the volatility of a process does not change when we change to an equivalent measure (as recalled after [eq. \(15.15\)](#)), and the drift is then forced by the martingale property. So X is a driftless lognormal, $X_T = X_0 e^{-\sigma^2 T/2 + \sigma\sqrt{T}Z}$ under $\mathbb{Q}^1$, and the Black formula with unit strike and zero rate ([theorem 15.6](#)) gives $\mathbb{E}_{\mathbb{Q}^1}[(X_T - 1)^+] = X_0 \Phi(d_1) - \Phi(d_2)$. Multiplying by $F_1(0)$ gives (21.3). $\square$

Proof. (i) is Jensen.

(ii) Degree one is visible in (21.3); the vega is $e^{-rT} F_2 \varphi(d_1) \sqrt{T} > 0$ (φ the normal density; the same computation as the vega of [chapter 15](#)).

(iii) Differentiating, $\partial_{F_2} V = e^{-rT} [\Phi(d_1) + (F_2 \varphi(d_1) - F_1 \varphi(d_2)) \partial_{F_2} d_1]$ (using $d_2 = d_1 - \sigma\sqrt{T}$), and the bracket vanishes because $\varphi(d_2)/\varphi(d_1) = e^{d_1 \sigma \sqrt{T} - \sigma^2 T/2} = F_2/F_1$. The other derivative follows by Euler’s identity $V = F_1 \partial_1 V + F_2 \partial_2 V$ for degree-one functions. $\square$

ρ	1	0.99	0.98	0.95	0.90	0.80
σ (spread vol.)	0	0.0707	0.1000	0.1581	0.2236	0.3162
V (\$/bbl)	3.000	3.259	3.615	4.468	5.512	7.037

Table 21.1: Aug–Dec spread option, $F_{\text{Aug}} = 55$, $F_{\text{Dec}} = 58$, $T = 211/365$, $r = 0$, $\sigma_1 = \sigma_2 = 50\%$. Computed for these notes.

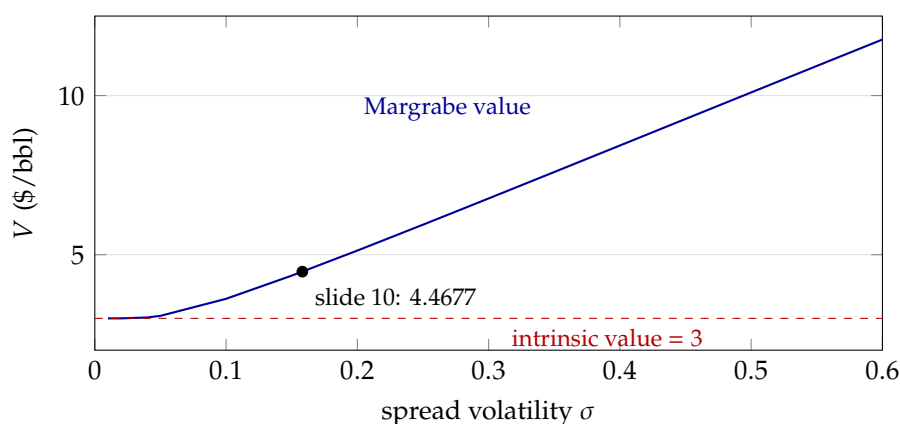


Figure 21.2: Value of the August–December spread option as a function of the spread volatility, for $F = (55, 58)$, $T = 211/365$, no interest. The dot is the number printed on slide 10.

Remark 21.6 (Why correlation between contract months is the model’s hard part). The value of a calendar-spread option depends on the joint law of two points of the forward curve, in particular on their correlation (table 21.1). A model that generates all forward prices from one random factor gives $\rho = 1$ and hence an option worth its intrinsic value: it cannot price the storage. This is the content of the slide-31 objection “cannot model the correlation structure between forward contracts” and is made precise in section 21.7.

Exercise 21.2 (not from the course: Margrabe with correlated calendar legs). Two futures $F_1 = 60$ (near), $F_2 = 63$ (far), $\sigma_1 = \sigma_2 = 45\%$, $T = 0.5$, $r = 0$.

- Compute the spread option value for $\rho = 0.3, 0.7, 0.95$ using Theorem 21.4 and comment on the monotonicity in ρ found in table 21.1.
- Recompute for $\rho = 1$: show the value reduces to the intrinsic value $(F_2 - F_1)^+$ and explain why in terms of Proposition 21.5.

21.3 Hedging the tank, and the real storage problem

21.3.1 There is no market: replicate

The lecture makes a confession [► 2013 L13 @ 37:39]: there is no market for spread options, nobody will buy the option that the quant “sold” for 4.4677. What survives is the *method* behind Black–Scholes: the value is also the cost of a dynamic strategy that reproduces the payoff [► 2013 L13 @ 38:43]. The quant therefore does not sell anything; he owns the tank,

1 COURSE NOTE: Errata and inconsistencies in this part

- Slide 11 repeats the example with $F_{\text{Aug}} = 65$, $F_{\text{Dec}} = 80$ (owe \$15) while the lecturer speaks of 55 and 80 (owe \$25); both are consistent examples.
- Slide 11 says the bid can be raised to \$4.46, the truncation of 4.4677 (the lecturer says 447 cents and bids 420).
- Slide 5 gives the profit of the Trafigura trade as \$21.5 with 10% *simple* interest, the speaker says “21”.
- Slide 9 says “interest rates are ignored for simplicity (should not be)”: formula (21.3) holds with the discount factor, but the trader’s static ceiling of \$3 needs the financing correction above.

whose payoff is that of the option, and delta-hedges it, and he tells the trader what to do every day, and the tank operator when to inject and withdraw [► 2013 L13 @ 40:49].

To make this precise (the lecture does not write it down) let $V(F_1, F_2)$ be the value (21.3). *Owning* the option (the tank) and hedging means holding $-\Phi(d_1)$ Dec futures and $+\Phi(d_2)$ Aug futures, by proposition 21.5(iii) rebalanced continuously. Futures cost nothing to hold, so the hedge earns $-\Phi(d_1) dF_2 + \Phi(d_2) dF_1$ and the terminal value of the whole position is

$$(F_2(T) - F_1(T))^+ - \int_0^T \Phi(d_1) dF_2 + \int_0^T \Phi(d_2) dF_1 = V(F_1(0), F_2(0))$$

(continuous rebalancing, $r = 0$),

the standard Black–Scholes replication statement (chapter 15): Ito’s formula applied to $V(F_1, F_2)$, which solves the two-dimensional Black equation with terminal condition (21.2), gives $dV = \partial_{F_1} V dF_1 + \partial_{F_2} V dF_2$ (time-derivative and second-order terms cancel). The hedger who wins the tank at price $P < V$ therefore locks in $V - P$. When the option ends in the money both deltas tend to one: the hedge is “long August, short December”, exactly the trader’s static position of section 21.2, which the tank then converts into physical barrels. In this sense the trader’s strategy is the limit $\sigma \rightarrow 0$ of the quant’s.

hedging dates n	1	4	16	64	211
mean of final P&L	4.462	4.466	4.472	4.469	4.468
standard deviation	2.033	1.078	0.567	0.290	0.159

Table 21.2: Discrete delta-hedging of the option of table 21.1 ($\sigma_1 = \sigma_2 = 50\%$, $\rho = 0.95$; $V = 4.4677$): terminal payoff plus hedge P&L. Computed for these notes: $2 \cdot 10^4$ paths, n equally spaced dates between 1 January and 31 July, no interest.

The simulation (table 21.2) shows the two things one expects: the average of the final position is the premium V whatever n , and the residual risk falls like $n^{-1/2}$, from 2.0 dollars per barrel with a single rebalancing to 0.16 with daily hedging.

Remark 21.7. The storage owner is *long* the option, while the lecture speaks of “selling” it and covering the short position with the tank. The two views are the same portfolio seen from the two sides: sold option plus tank is riskless with value $V - P$; the hedge of the previous paragraph reaches the same result without a counterparty. A subtlety: the option is settled on 31 July on the *futures* prices, while the barrels move later, so between the two dates the operator is committed by the futures he has entered, which is why the recipe “buy August, sell December at the same time” works.

21.3.2 A portfolio of options under physical constraints

A real tank is available for two years, not for four months, and one may sell many spread options against it: injection in any month T_i and withdrawal in any later month T_j , and also the opposite (withdraw first, refill later), which pays when the curve is in backwardation [► 2013 L13 @ 21:58]. The first problem is to choose *which portfolio of options* is the most valuable one that the tank can support, and this is an optimisation [► 2013 L13 @ 23:00]. The constraints are technical, contractual and legal: injection and withdrawal rates, tank capacity, and above all the requirement that *under no circumstances* the tank is asked to do something impossible [► 2013 L13 @ 24:02]. Slides 13–17 write the problem as follows. With dates $T_1 < \dots < T_N$ and today’s futures prices F_i (slide 13):

- $S_{i,j}$ ($i < j$) is the value of the option to inject at T_i and withdraw at T_j , with payoff $\max\{F_j - F_i - \text{Cost}, 0\}$ at exercise; $U_{i,j}$ the value of the option to withdraw at T_i and inject at the later T_j , with payoff $\max\{F_i - F_j - \text{Cost}, 0\}$ (slide 14) [► 2013 L13 @ 27:11]. The volumes sold against the tank are $x_{i,j} \geq 0$ and $v_{i,j} \geq 0$;
- $y_i \geq 0$ and $z_j \geq 0$ are the volumes committed today by *futures* for injection at T_i and withdrawal at T_j (the trader's static strategy);
- the objective is the cash collected now:

$$V = \max_{x,v,y,z} \left\{ \sum_{i < j} x_{i,j} S_{i,j} + \sum_{i < j} v_{i,j} U_{i,j} - \sum_i y_i F_i + \sum_j z_j F_j \right\} \quad (21.4)$$

[► 2013 L13 @ 25:05]

(the first two sums are premia received, the last two are the cost of the futures bought to inject and the revenue of the futures sold to withdraw);

- Boolean variables record the exercise: $\Omega_{i,j}^S = 1$ if $S_{i,j}$ expires in the money (then the tank *must* act), 0 otherwise, and the same for $\Omega_{i,j}^U$ [► 2013 L13 @ 28:16].

Define the net injection at date T_i (positive means into the tank)

$$q_i = \sum_{j > i} x_{i,j} \Omega_{i,j}^S - \sum_{\ell < i} x_{\ell,i} \Omega_{\ell,i}^S + \sum_{\ell < i} v_{\ell,i} \Omega_{\ell,i}^U - \sum_{j > i} v_{i,j} \Omega_{i,j}^U + y_i - z_i. \quad (21.5)$$

An exercised option of type S injects $x_{i,j}$ at T_i and withdraws it at T_j , so the two terms cancel if both legs fall on the same date (a withdrawal already required cancels an injection required now, [► 2013 L13 @ 30:19]); for type U the signs are reversed. The constraints of slides 16–17 are then

$$-W_i \leq q_i \leq I_i, \quad i = 1, \dots, N \quad (\text{withdrawal and injection rates}) \quad (21.6)$$

$$C_i^{\min} \leq C_0 + \sum_{k \leq i} q_k \leq C_i^{\max}, \quad i = 1, \dots, N \quad (\text{capacity}) \quad (21.7)$$

where C_0 is the initial inventory. On slide 17 the capacity constraint is written $C_0 + \sum_{k \leq i} \{ \sum_{j > i} x_{k,j} \Omega_{k,j}^S - \sum_{j > i} v_{k,j} \Omega_{k,j}^U + y_k - z_k \}$: this is the same thing, because in $\sum_{k \leq i} q_k$ every pair (k, j) with $k < j \leq i$ contributes $+x_{k,j}$ at T_k and $-x_{k,j}$ at T_j and cancels, so only the pairs still open at T_i (those with $j > i$) survive.

Remark 21.8 (Why the problem is ugly). The variables Ω depend on prices at the exercise dates: they are random. The constraints must hold in every price scenario, so (21.6)–(21.7) are in effect a *robust* problem, one system for each of the $2^{n_{\text{opt}}}$ exercise patterns. The unknowns are x, y, z, v and, unfortunately, the Boolean Ω , and the constraints are non-linear in them [► 2013 L13 @ 31:20]. For two years of monthly dates there are $24 \cdot 23/2 = 276$ pairs of each type, plus an equally large set of constraints [► 2013 L13 @ 32:24]. (The lecturer stumbles over the count; this is the count of pairs $i < j$. The reading of the constraints as a robust problem is ours.)

21.3.3 The linear core: intrinsic value

If the option premia are removed and only the futures part is kept, the problem (21.4)–(21.7) with $x = v = 0$ is a *linear program*: it gives the *intrinsic value* of the tank, what can be locked in with futures alone (the trader's strategy, made rigorous). (Not discussed in class.) Take

twelve monthly dates with the approximate curve of [fig. 21.1](#) (February 2009 to January 2010), a tank with $C_0 = 0$, $C^{\max} = 100$, injection rate $I = 30$ and withdrawal rate $W = 40$ per month, no costs. Maximising $\sum_i F_i(z_i - y_i)$ subject to $0 \leq y_i \leq I$, $0 \leq z_i \leq W$ and $0 \leq \sum_{k \leq i} (y_k - z_k) \leq C^{\max}$ gives

month	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan
F_i	35.4	43.5	48.1	50.6	52.4	53.8	54.8	55.8	56.7	57.5	58.4	59.1
inject y_i	30	30	30	10	0	0	0	0	0	0	0	0
withdraw z_i	0	0	0	0	0	0	0	0	0	20	40	40

and an intrinsic value of $5850 - 4316 = 1534$ dollars (average purchase price 43.16, average sale price 58.50). Without the rate limits one would fill the tank in February for 3540 and empty it in January for 5910 (value 2370): the injection limit alone costs 836, and the LP finds the cheapest way around it, filling in the cheap months first. (Computed for these notes with `scipy.optimize.linprog`.) The quant's value is this plus the value of the flexibility, the time value of the spread options.

Exercise 21.3 (not from the course: intrinsic vs extrinsic value of storage). A storage facility can inject/withdraw at most 10 units per month and holds at most 50 units, currently empty. Given monthly futures prices $F = (20, 24, 19, 26, 21, 18, 25, 30, 22, 19, 23, 27)$ (a stylised seasonal curve)

- find the intrinsic value (21.4) by inspection (buy at the cheapest months subject to the capacity/rate constraints, sell at the priciest)
- explain qualitatively, without solving it, why the *extrinsic* value (section 21.3.4) would be positive even if this particular forward curve were flat.

21.3.4 Methods, robustness and calibration

The lecture lists three ways to attack the full problem [[2013 L13 @ 32:24](#)]: approximate it by a linear or quadratic programme; Monte Carlo simulation of the prices; or *stochastic control*, for which the reference is Carmona and Ludkovski (2010), “Valuation of energy storage: an optimal switching approach”, *Quantitative Finance* 10(4) (slide 18): the decision inject/withdraw/hold is a control chosen as a function of the current prices and inventory, and the value is the solution of a dynamic-programming equation. None of the methods is “better”: the inputs (volatilities, correlations) are known only to a rough accuracy, and an exact method fed with parameters that are 50% off is still wrong [[2013 L13 @ 33:31](#)].

IN FINANCE: Robust, calibration-free models

Eydeland's desk, in his words, does not use the stochastic-control solution (“we do something different”) and deliberately chooses a method that is *robust*: little changes in the parameters must not change the value much, and each ingredient must be something that can be traded in the market and verified by outside controllers and regulators, “in this day and age extremely important” [[2013 L13 @ 34:35](#)]. He warns against over-parametrised models (stochastic volatility, regime switching, several jump processes), which capture the richness of price behaviour at the cost of stability [[2013 L13 @ 35:35](#)]. *Calibration*, the fitting of unobservable parameters by least squares to market data, is fragile: the objective is non-convex, the solution set can be very flat, one may stop early [[2013 L13 @ 36:35](#)]. Compare the regularisation of ill-posed calibration problems in [chapter 20](#) and the least-squares theory of [chapter 5](#).

21.4 The merchant power plant and the spark spread

21.4.1 A plant is a strip of daily options

The same reasoning applies to any physical asset: tankers, refineries, pipelines, power lines [► 2013 L13 @ 41:53]; only the constraints change (a tanker needs the routes and port limits) [► 2013 L13 @ 42:59]. The second half of the lecture works out the power plant.

A *merchant* plant decides itself, every morning, whether to run [► 2013 L13 @ 42:59]. The price of power for the day is known; to produce one megawatt-hour it burns fuel. The efficiency is measured by the *heat rate* h : the number of MMBtu of fuel per MWh (typically 7 for a modern gas plant, up to 20 for an inefficient one that runs rarely) [► 2013 L13 @ 45:05]. With the fuel price G (dollars per MMBtu) and the other variable costs K (dollars per MWh: labour, cooling, ...) the profit of the day is (slide 20, where the heat rate is written in Btu/kWh, hence the division by 1000)

$$\Pi = \max\{P - hG - K, 0\}, \quad P = \text{power price (\$/MWh)}, \quad (21.8)$$

and the plant is run when the argument is positive [► 2013 L13 @ 47:13]. With $h = 8$, $G = 4$, $K = 3$ the plant costs $8 \cdot 4 + 3 = 35$ per MWh: at $P = 50$ the profit is 15 and the plant runs; at $P = 30$ the owner “goes back to sleep” and earns zero [► 2013 L13 @ 47:13]. So operating a merchant plant is financially equivalent to owning a portfolio of *daily options on the spread* between electricity and fuel, the *spark spread options* (slide 20). The price the buyer should be willing to pay is the expectation of the sum of these payoffs, taken under the pricing measure, over the remaining life: a two-dimensional distribution of (P, G) , with its correlation [► 2013 L13 @ 50:21].

21.4.2 Pricing one daily option

For $K = 0$ the option (21.8) is a Margrabe option (exchange power for h units of gas): [theorem 21.4](#) with $F_2 = P$ and $F_1 = hG$ prices it. For $K > 0$ there is no closed form; a classic approximation (Kirk, 1995; not in the lecture) treats $hG + K$ as lognormal with volatility $\omega\sigma_G$, $\omega = hG/(hG + K)$, and prices a Margrabe option between P and $hG + K$:

$$V \approx e^{-rT} [P \Phi(d_1) - (hG + K) \Phi(d_2)], \quad \sigma_K^2 = \sigma_P^2 + \omega^2 \sigma_G^2 - 2\rho \omega \sigma_P \sigma_G, \quad (21.9)$$

with $d_{1,2}$ as in [theorem 21.4](#) with $F_2/F_1 \rightarrow P/(hG + K)$ and $\sigma \rightarrow \sigma_K$ (P, G here are the forward prices for the delivery date). Numerically (forward prices $P = 50$, $G = 4$, $h = 8$, $K = 3$, $\sigma_P = 80\%$, $\sigma_G = 50\%$, $\rho = 0.6$, three months, $r = 0$): intrinsic value 15.00, Kirk’s approximation 15.887, Monte Carlo (lognormal P, G , $2 \cdot 10^6$ paths) 15.889 ± 0.012 . The correlation matters (Kirk, same data): $\rho = 0, 0.3, 0.6, 0.9$ give 17.39, 16.65, 15.89, 15.20. This is why the joint law of power and gas is a modelling target and not a detail (slide 28: “if the model does not capture this structure it may misprice spread options (tolling contracts, power plants, etc.)”).

Remark 21.9 (What is idealised). Equation (21.8) ignores start-up costs, minimum run times, ramp rates and outages; these turn the strip of independent daily options into a path-dependent (real-option) problem, and it is for these that the Monte Carlo methods of [section 21.3.4](#) are needed. The lecture keeps to the daily-option picture.

INTUITION: Options without an underlying you can hold

Power cannot be stored, so there is no cash-and-carry argument ([proposition 21.2](#)) linking the forward price of power for a given day to the spot price today. The only no-arbitrage requirement is that forward prices be martingales under *some* pricing measure, $F(0, T) = \mathbb{E}_Q[P_T]$, and the spot process does not have to be tied to the forward curve by a carry relation. This is the sense of the slide-31 objection “no no-arbitrage argument: how to price forward contracts and options?”.

Exercise 21.4 (not from the course: heat rate and the spark spread). A gas plant has heat rate $H = 8$ (MMBtu/MWh) and variable cost $c = \$2/\text{MWh}$.

- At $F_{\text{power}} = \$45/\text{MWh}$ and $F_{\text{gas}} = \$4/\text{MMBtu}$, is the plant in the money on a forward basis?
- Using Kirk’s approximation ([21.9](#)) with $\sigma_P = 60\%$, $\sigma_G = 35\%$, $\rho = 0.5$, $T = 1$ month, price one spark-spread option (one contract = one hour of capacity).
- Explain, using [Proposition 21.13](#), why ρ between power and gas should be expected to fall when demand is very high (the top of the stack switches to oil or curtailment, decoupling power from the gas price).

21.5 Three assets, three spreads

The examples of the lecture are three instances of one construction: a physical asset gives the right, not the obligation, to transform one commodity (or the same commodity at one date) into another with a fixed conversion, so its value is that of an option on the difference of two price indices.

asset	input $\rightarrow$ output	spread	exercise decision
storage	barrel at $T_i \rightarrow$ barrel at T_j	$F(T_j) - F(T_i) - \text{cost}$	inject/withdraw
gas power plant	h MMBtu of gas $\rightarrow$ 1 MWh	$P - hG - K$	run or stay off
refinery	1 bbl of crude $\rightarrow$ products	products – crude	run at what mix

The refinery’s spread is the crack spread of [section 9.6.4](#), where the “cracks” were the differences between gasoline or heating oil (converted to dollars per barrel with 42 gallons per barrel) and crude, and where the three futures were found to be cointegrated. The link is useful in two ways. First, cointegration says that the crack spread is mean reverting, so its variance does *not* grow linearly with time (as in the lognormal spread of [theorem 21.4](#)); for long-dated options the Margrabe formula overstates the volatility of a spread that is pulled back to an equilibrium (not discussed in class). Second, the seasonal pull of heating oil in winter is a seasonal component of this spread, treated in [section 21.7.4](#) below.

21.6 What power prices look like

To value the plant one needs the distribution of P and G , and the lecture now turns to the data [[2013 L13 @ 49:21](#)]. The first idea is Brownian motion (or its exponential): the terminal distribution is then normal (lognormal for the price) [[2013 L13 @ 50:21](#)]. The S&P 500 is

often said to have fat tails; the lecturer’s point is that by commodity standards it looks normal [► 2013 L13 @ 51:25].

	annual volatility	skewness	kurtosis
Nord Pool (Scandinavian power), daily	182%	1.468	26.34
Nord Pool, 6 p.m. price	238%	2.079	76.82
DAX (equity index)	23%	0.004	3.33

Table 21.3: Distribution parameters of returns (slide 23, from A. Werner, *Risk Management in the Electricity Market*, 2003). Kurtosis is $\mathbb{E}[(X - \mu)^4]/\sigma^4$, equal to 3 for a normal law.

The kurtosis of a normal variable is exactly 3; for the DAX it is 3.33, for power 26 to 77 [► 2013 L13 @ 51:25]. And annual volatilities are in the hundreds of percent, against about 10% for the S&P at the time of the lecture [► 2013 L13 @ 53:35]: intuition built on volatilities of 10–20% fails at 100–300% [► 2013 L13 @ 53:35]. The time series of Texas (ERCOT), New England (NEPOOL) and PJM prices on slides 24–26 show the feature that jumps out: *spikes* [► 2013 L13 @ 52:27]. The slide-27 list of what a model must capture is: mean reversion, spikes, high kurtosis, regime switching, lack of data and non-stationarity [► 2013 L13 @ 54:42].

21.6.1 Jumps produce kurtosis; aggregation removes it

(Not discussed in class.) A minimal fat-tailed model is a diffusion plus compound Poisson jumps. Let a log-return over a horizon Δ be $X_\Delta = \sigma\sqrt{\Delta}Z + \sum_{i=1}^{N_\Delta} J_i$ with $Z \sim N(0, 1)$, $N_\Delta \sim \text{Poisson}(\lambda\Delta)$ and $J_i \sim N(0, s^2)$, all independent.

Proposition 21.10 (Excess kurtosis of jump-diffusion returns). *The cumulants of X_Δ are $\kappa_2 = (\sigma^2 + \lambda s^2)\Delta$ and $\kappa_4 = 3\lambda s^4\Delta$, so*

$$\frac{\mathbb{E}[(X_\Delta - \mathbb{E}X_\Delta)^4]}{\text{Var}(X_\Delta)^2} - 3 = \frac{\kappa_4}{\kappa_2^2} = \frac{3\lambda s^4}{(\sigma^2 + \lambda s^2)^2} \frac{1}{\Delta}.$$

Proof. Cumulants of independent sums add. The Gaussian part contributes only to κ_2 . For the compound Poisson sum, $\ln \mathbb{E}[e^{\theta \sum J_i}] = \lambda\Delta(\mathbb{E}[e^{\theta J}] - 1) = \lambda\Delta \sum_{n \geq 1} \theta^n \mathbb{E}[J^n]/n!$, so $\kappa_n = \lambda\Delta \mathbb{E}[J^n]$; here $\mathbb{E}[J^2] = s^2$, $\mathbb{E}[J^4] = 3s^4$. The fourth cumulant of a variable with zero mean is $\mathbb{E}[X^4] - 3(\mathbb{E}[X^2])^2$. □

For $\lambda = 20$ per year, $s = 15\%$, $\sigma = 40\%$ the excess kurtosis is 29.8 for daily returns ($\Delta = 1/365$: power trades every calendar day), 4.26 weekly, 0.99 monthly and 0.08 yearly (Monte Carlo: 29.6, 4.22, 1.00, 0.07): a kurtosis of the size of the Nord Pool figures at the daily scale, disappearing on aggregation as the central limit theorem requires (chapter 3). Fat tails of *daily* power prices therefore do not contradict a normal law for annual returns; the spikes must be modelled at the daily (or hourly) scale, and it is there that the option payoffs (21.8) live.

Exercise 21.5 (not from the course: jumps and aggregation). Using Proposition 21.10’s formula for the excess kurtosis of a jump-diffusion’s returns over an interval dt ,

- (a) show that it vanishes as $dt \rightarrow \infty$ for fixed jump intensity λ and jump size distribution, i.e. that kurtosis is a *high-frequency* phenomenon, consistent with the large *daily* kurtosis of table 21.3 and the aggregation numbers of section 21.6.1.
- (b) Explain why the stack model’s spikes (fig. 21.4) can average away much more slowly than independent jumps when high demand is persistent (a heat wave lasting weeks keeps the system at the top of the stack), and what this implies

for someone trying to fit a constant-parameter jump-diffusion to both daily and monthly power returns.

21.6.2 Mean reversion and the objections to spot models

The lecturer's step-by-step argument against the standard toolbox [► 2013 L13 @ 56:47]: geometric Brownian motion has no spikes, no correlation structure, nothing; a spike is a rise followed by a return, so add mean reversion; a rise from \$30 to \$1000 needs a *jump*; the return from \$1000 to \$30 in three or four days needs a very strong reversion speed, but a reversion so strong would freeze the price at the \$30 level in normal times; so one lets the parameters depend on the level, i.e. introduces regime switching, and one ends with 10–12 parameters “impossible to manage” [► 2013 L13 @ 58:56]. Stochastic volatility, regime switching and multiple jump processes are all used in the industry but come from other markets and are not what the lecturer recommends [► 2013 L13 @ 1:00:02].

A comment (ours): the scale of the reversion is easy to compute. For $dX = -\kappa X dt + \sigma dW$ (chapter 18) one has $\text{Corr}(X_t, X_{t+\Delta}) = e^{-\kappa\Delta}$, half-life $\ln 2/\kappa$ and stationary standard deviation $\sigma/\sqrt{2\kappa}$; a half-life of 1 day is $\kappa = 365 \ln 2 = 253$ per year (calendar days) and of 4 days $\kappa = 63$ per year, with daily autocorrelation 0.50 and 0.84. The tension is really one of *two time scales*: the spike relaxes in days, while the base level of prices, driven by fuel, moves over weeks and months. A one-factor mean-reverting model has a single scale.

21.7 Spot models and forward curves

21.7.1 The models of the slides

Slide 29 lists the spot processes. All are written for the relative change dS_t/S_t (here W is a Brownian motion and q a Poisson process of intensity λ with jump size Y_t , $k = \mathbb{E}[Y - 1]$):

$$\text{GBM: } dS_t/S_t = \mu dt + \sigma dW_t, \quad (21.10)$$

$$\text{with mean reversion: } dS_t/S_t = \kappa(\theta - \ln S_t) dt + \sigma dW_t, \quad (21.11)$$

$$\text{with jumps: } dS_t/S_t = (\mu - \lambda k) dt + \sigma dW_t + (Y_t - 1) dq_t, \quad (21.12)$$

$$\text{jumps and mean reversion: } dS_t/S_t = (\kappa(\theta - \ln S_t) - \lambda k) dt + \sigma dW_t + (Y_t - 1) dq_t. \quad (21.13)$$

The subtraction of λk compensates the jumps so that the expected relative change stays μdt (or zero from mean reversion). Equation (21.13) is written on slide 29 as $(\mu - \lambda k - \log S_t) dt$, which drops the reversion speed; we give the corrected form. The more complicated models on slide 30 (stochastic convenience yield, stochastic volatility, regime switching, multiple jump processes, term-structure models) are only listed.

21.7.2 The Schwartz one-factor model and its futures curve

(Not discussed in class; the lecture only states (21.11) and criticises it.) Write $X = \ln S$ and take the mean-reverting log price, in the form of Schwartz (1997),

$$dX_t = \kappa(\alpha - X_t) dt + \sigma dW_t, \quad \kappa, \sigma > 0, \quad (21.14)$$

which is (21.11) with $\theta = \alpha + \sigma^2/(2\kappa)$. It is solved by $X_T = \alpha + e^{-\kappa T}(X_0 - \alpha) + \sigma \int_0^T e^{-\kappa(T-s)} dW_s$, which is Gaussian with mean $\alpha + e^{-\kappa T}(X_0 - \alpha)$ and variance $\sigma^2(1 -$

2 COURSE NOTE: Erratum: slide 29, last model

The drift of the last line should be $\kappa(\theta - \ln S_t) - \lambda k$ (or $\mu - \lambda k - \kappa \ln S_t$ with $\mu = \kappa\theta$); as printed, $-\log S_t$ has no coefficient, so it cannot be a speed of reversion. Also, by Ito, $d \ln S = [\kappa(\theta - \ln S) - \sigma^2/2] dt + \sigma dW$: (21.11) is an Ornstein–Uhlenbeck process for $\ln S$ with long-run mean $\theta - \sigma^2/(2\kappa)$, not θ .

$e^{-2\kappa T})/(2\kappa)$ (multiply the equation by $e^{\kappa t}$ and integrate; the variance is the Itô isometry, chapter 17).

A futures price is the expectation of the spot price at delivery under the pricing measure. Because commodity spot prices are not traded assets, the drift under $\mathbb{Q}$ is not fixed by no-arbitrage; assume a constant market price of risk λ_r , so that $dW = dW^{\mathbb{Q}} - \lambda_r dt$ and, under $\mathbb{Q}$, $dX = \kappa(\alpha^* - X)dt + \sigma dW^{\mathbb{Q}}$ with $\alpha^* = \alpha - \sigma\lambda_r/\kappa$ (the Girsanov change of drift; chapter 18).

Theorem 21.11 (Futures prices in the one-factor mean-reverting model). Under (21.14) with risk-neutral long-run mean α^* , for $\tau = T - t$,

$$F(t, T) = \mathbb{E}_{\mathbb{Q}}[S_T | \mathcal{F}_t] = \exp\left\{e^{-\kappa\tau}X_t + (1 - e^{-\kappa\tau})\alpha^* + \frac{\sigma^2}{4\kappa}(1 - e^{-2\kappa\tau})\right\}. \quad (21.15)$$

Moreover $dF(t, T)/F(t, T) = \sigma e^{-\kappa(T-t)} dW_t^{\mathbb{Q}}$.

Proof. Under $\mathbb{Q}$, X_T given $\mathcal{F}_t$ is $N(m, v)$ with $m = \alpha^* + e^{-\kappa\tau}(X_t - \alpha^*)$ and $v = \sigma^2(1 - e^{-2\kappa\tau})/(2\kappa)$. For a Gaussian X_T , $\mathbb{E}[e^{X_T}] = e^{m+v/2}$, which is (21.15). Since $F(t, T)$ is a $\mathbb{Q}$ -martingale in t with $F = \exp(\text{deterministic} + e^{-\kappa(T-t)}X_t)$, its differential has no drift and the diffusion coefficient $e^{-\kappa(T-t)}\sigma F$ (only X_t is random, and dX_t has diffusion σ). $\square$

Three consequences:

- the curve tends, as $T \rightarrow \infty$, to $\exp(\alpha^* + \sigma^2/(4\kappa))$;
- contango or backwardation depends on where the spot sits;
- the volatility of a futures contract is $\sigma e^{-\kappa\tau}$, decreasing with maturity (the “Samuelson effect”): the front of the curve moves violently, the back hardly at all.

For contango versus backwardation differentiate (21.15) in T at $t = 0$:

$$\frac{\partial \ln F(0, T)}{\partial T} = \kappa e^{-\kappa T}(\alpha^* - X_0) + \frac{\sigma^2}{2}e^{-2\kappa T},$$

so the curve slopes upward at maturity T iff $X_0 < \alpha^* + \sigma^2 e^{-\kappa T}/(2\kappa)$. Compare with the convenience yield of (21.1): as $T \rightarrow 0$, $y - (r + u) = -\partial_T \ln F$ equals $\kappa(X_0 - \alpha^*) - \sigma^2/2$, which is large when the spot is high relative to its long-run level, the “inventory is scarce” behaviour. Mean reversion plays the role of a convenience yield that depends on the price level.

T (years)	0.25	0.5	1	2	∞
$F(0, T), S_0 = 35$	39.68	43.13	47.36	50.44	51.35
$F(0, T), S_0 = 70$	63.90	59.84	55.28	52.22	51.35
$\sigma_F(T)$ (volatility of the future)	0.27	0.19	0.09	0.02	0

Table 21.4: One-factor model with $\kappa = 1.5$, $\alpha^* = \ln 50$, $\sigma = 0.4$ (so $\sigma_{\text{spot}} = 40\%$): futures curves from a low spot (contango) and a high spot (backwardation). Formula (21.15) checked against a Monte Carlo of the exact Gaussian transition at $T = 1$ (47.358 against 47.360). Computed for these notes.

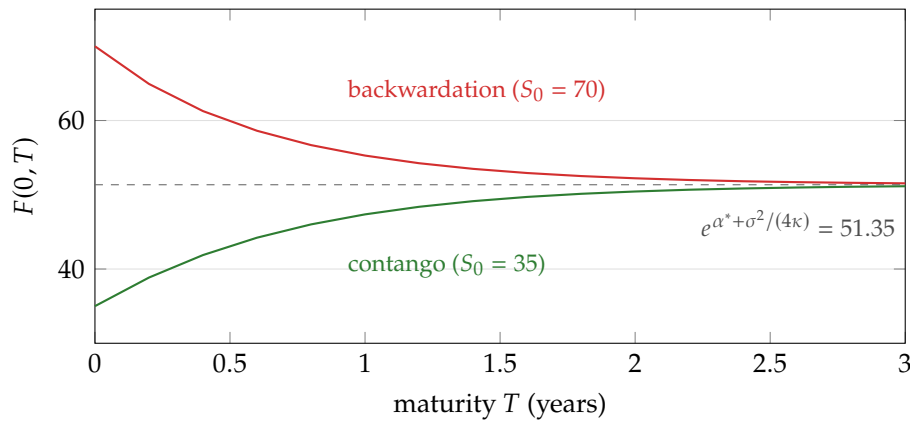


Figure 21.3: Futures curves of the one-factor mean-reverting model (theorem 21.11, parameters of table 21.4): the same model gives contango or backwardation depending on the spot.

Why one factor cannot price spread options. Since $dF(t, T_i)/F(t, T_i) = \sigma e^{-\kappa(T_i-t)} dW^\mathbb{Q}$ involves the *same* Brownian motion for every maturity, all futures returns are perfectly correlated, $\rho = 1$, and the spread volatility of theorem 21.4 is $\sigma_{\text{spread}} = |\sigma_1 - \sigma_2| = \sigma|e^{-\kappa\tau_1} - e^{-\kappa\tau_2}|$: with the parameters above, two contracts with $\tau = 0.6$ and 0.9 give 0.059 and hence an option practically at its intrinsic value, against the 0.158 that reproduces the number of slide 10. The slide-31 list of “cons” is now transparent: a spot model with one factor *cannot model the correlation structure between forward contracts* and cannot produce a complex volatility structure ($\sigma e^{-\kappa\tau}$ is monotone).

21.7.3 Stochastic convenience yield: the two-factor model

Slide 30 lists “models with stochastic convenience yield”. The prototype is Gibson and Schwartz (1990): the spot follows $dS/S = (\mu - \delta)dt + \sigma_1 dW_1$ with a *stochastic* convenience yield δ , itself mean reverting, $d\delta = \kappa(\alpha - \delta)dt + \sigma_2 dW_2$, $dW_1 dW_2 = \rho dt$. Under $\mathbb{Q}$ the spot grows at $r - \delta$ and δ has risk-adjusted level $\hat{\alpha}$. Define $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$.

Proposition 21.12 (Futures in the two-factor model). *With constant r and, under $\mathbb{Q}$, $d \ln S = (r - \delta - \sigma_1^2/2)dt + \sigma_1 dW_1$, $d\delta = \kappa(\hat{\alpha} - \delta)dt + \sigma_2 dW_2$,*

$$\begin{aligned} \ln F(0, T) = & \ln S_0 - \delta_0 B + \left(r - \hat{\alpha} + \frac{\sigma_2^2}{2\kappa^2} - \frac{\rho\sigma_1\sigma_2}{\kappa} \right) T \\ & + \left(\hat{\alpha} - \frac{\sigma_2^2}{\kappa^2} + \frac{\rho\sigma_1\sigma_2}{\kappa} \right) B + \frac{\sigma_2^2}{4\kappa^3} (1 - e^{-2\kappa T}), \quad B = B(T), \end{aligned} \quad (21.16)$$

and $dF(t, T)/F(t, T) = \sigma_1 dW_1 - \sigma_2 B(T - t) dW_2$.

Proof. Integrate: $\delta_s = \delta_0 e^{-\kappa s} + \hat{\alpha}(1 - e^{-\kappa s}) + \sigma_2 \int_0^s e^{-\kappa(s-u)} dW_2(u)$, so $\int_0^T \delta_s ds = \delta_0 B + \hat{\alpha}(T - B) + \sigma_2 \int_0^T B(T - u) dW_2(u)$ (Fubini). Hence $\ln S_T$ is Gaussian with mean $m = \ln S_0 + (r - \sigma_1^2/2)T - \delta_0 B - \hat{\alpha}(T - B)$ and variance $v = \sigma_1^2 T + \sigma_2^2 \int_0^T B^2 - 2\rho\sigma_1\sigma_2 \int_0^T B$, where $\int_0^T B(\tau) d\tau = (T - B)/\kappa$ and $\int_0^T B^2 = \kappa^{-2} [T - 2B + (1 - e^{-2\kappa T})/(2\kappa)]$. Then $F = \mathbb{E}_\mathbb{Q}[S_T] = e^{m+v/2}$; collecting the coefficients of T , of B and the exponential term gives (21.16). The volatility statement follows from $F(t, T) = S_t \exp(-\delta_t B(T - t) + \text{deterministic})$. $\square$

Checks (ours): for $\sigma_2 = 0$ and constant δ one recovers $F = S_0 e^{(r-\delta)T}$, the cost-of-carry formula (21.1) with $u = 0$; and a Monte Carlo of the two SDEs for $S_0 = 50$, $\delta_0 = 0.08$, $r = 0.03$, $\kappa = 1.2$, $\hat{\alpha} = 0.05$, $\sigma_1 = 0.35$, $\sigma_2 = 0.5$, $\rho = 0.6$, $T = 1$ gives 47.302 against 47.304 from (21.16) ($4 \cdot 10^5$ paths, 400 Euler steps); the curve is $49.31, 48.58, 47.30, 45.54, 42.44$ for $T = 0.25, 0.5, 1, 2, 5$.

The two Brownian motions decorrelate the front and the back of the curve: from the volatility formula, $\sigma_F(\tau)^2 = \sigma_1^2 + \sigma_2^2 B^2 - 2\rho\sigma_1\sigma_2 B$ and the correlation between contracts at

τ_1, τ_2 is

$$\text{Corr} = \frac{\sigma_1^2 + \sigma_2^2 B(\tau_1)B(\tau_2) - \rho\sigma_1\sigma_2(B(\tau_1) + B(\tau_2))}{\sigma_F(\tau_1)\sigma_F(\tau_2)};$$

with the parameters above this is 0.977 for $(\tau_1, \tau_2) = (0.6, 0.9)$, 0.966 for $(1, 2)$ and 0.77 for $(0.5, 3)$. So calendar spreads acquire a volatility of their own, of the order of the values needed in [table 21.1](#).

21.7.4 Seasonality

(Not discussed in class beyond the observation of heating oil in winter, [section 9.6.4](#).) Energy prices have a deterministic seasonal pattern: gas and heating oil dearer in winter, power in summer and winter peaks. The simplest device is $\ln S_t = f(t) + X_t$, f a deterministic periodic function (for example a truncated Fourier series in t with period one year) and X a mean-reverting process as in [\(21.14\)](#). Then $F(t, T) = e^{f(T)} \cdot \mathbb{E}_{\mathbb{Q}}[e^{X_T} | \mathcal{F}_t]$, so the seasonal shape is multiplied onto the non-seasonal curve of [\(21.15\)](#), and f can be chosen so that the model reproduces *exactly* the observed forward curve today. This is the mechanism by which the lecture's constants $\alpha_1, \alpha_2, \alpha_3$ (next section) “are chosen to match market data”: shape parameters absorb what the model does not explain and leave the *dynamics* to the model.

21.8 Why spot models fail for power, and the stack alternative

The one- and two-factor models of [section 21.7](#) are the standard toolkit for oil and gas, but the lecture is explicit that for *power* they are the wrong starting point [[2013 L13 @ 54:12](#)]. Electricity cannot be stored (at grid scale): there is no cash-and-carry argument linking today's price to tomorrow's, no arbitrage forces the spot process to look like anything in particular, and the empirical price series shows huge, short-lived spikes ([table 21.3](#)) that no diffusion, and not even the simple jump-diffusion of [\(21.12\)](#), reproduces convincingly: a jump-diffusion adds symmetric or one-sided jumps to a smooth base, but real spikes are triggered by a specific, identifiable mechanism – the market running out of cheap generation – and revert almost immediately once it eases [[2013 L13 @ 56:30](#)].

INTUITION: a market with no inventory

Every other underlying in this course can be held: a bond, a stock, a barrel of oil in a tank. Holding is what forces $F(t, T) \leq S_t e^{(r+u)(T-t)}$ ([eq. \(21.1\)](#)): if the forward is too rich you buy spot, store, and sell forward. Power has no “store”: you cannot buy megawatt-hours at 3 a.m. and sell them at 6 p.m. instead. The only link between today's price and the value of a future delivery is *physical* (what plants will be running then), not financial. This is why the lecture turns from diffusions to an engineering model of the market itself.

21.8.1 The merit-order stack

The alternative is to model the market micro-structure directly [[2013 L13 @ 57:40](#)]. At each instant a set of generating units is available, each with a capacity $q_i > 0$ (MW) and a marginal cost c_i (\$/MWh, the fuel cost divided by the plant's efficiency, i.e. its *heat rate* h_i times the fuel price, plus a small operating cost); a system operator dispatches units in increasing order of c_i (cheapest first: nuclear and hydro, then coal, then gas turbines of

increasing heat rate, then oil peakers) until cumulative capacity meets demand D . The market-clearing price is the cost of the *last*, marginal unit dispatched:

$$P(D) = c_{(k)}, \quad k = \min \left\{ j : \sum_{i=1}^j q_{(i)} \geq D \right\}, \quad (21.17)$$

where $c_{(1)} \leq c_{(2)} \leq \dots$ is the cost-sorted (“stacked”) list of units. Plotting cumulative capacity on the horizontal axis and $c_{(i)}$ on the vertical gives the *supply stack*, a non-decreasing step function; (21.17) says the price is where the vertical line at D crosses the stack (not discussed in class beyond the qualitative picture; the step-function formalisation is ours).

Justification (not discussed in class). Immediate from (21.17): between two consecutive kink points $D \in [Q_{k-1}, Q_k]$ the marginal unit is fixed at k , so $P = c_{(k)}(G)$, linear (indeed usually constant, or linear if unit k itself burns gas); at $D = Q_k$ the marginal unit switches to $k+1$ and P jumps (it is flat in D while a single unit is marginal, in a discrete stack) by $c_{(k+1)} - c_{(k)}$; if $D > \sum_i q_{(i)}$ (total capacity), (21.17) has no solution and the price is set by the market’s scarcity mechanism (an administrative cap, e.g. \$1,000/MWh or higher in some US markets, or unserved load). $\square$

Proposition 21.13 (Two sources of randomness, one nonlinear map). Let D_t (demand, driven mainly by temperature and the business cycle) and G_t (the marginal fuel price, typically natural gas) be the two state variables. If a gas unit of heat rate h is marginal, $c = hG_t + m$ (m a small fixed cost); (21.17) makes $P_t = \Phi(D_t, G_t)$ for a function Φ that is piecewise linear in G_t (for fixed D_t , as long as the same unit stays marginal) but is a non-decreasing step function of D_t , jumping up wherever dispatch moves to the next unit on the stack, and an essentially vertical segment once the stack is exhausted (price spike: any further demand can only be met by very expensive last-resort generation, demand-side curtailment, or a shortage price cap).

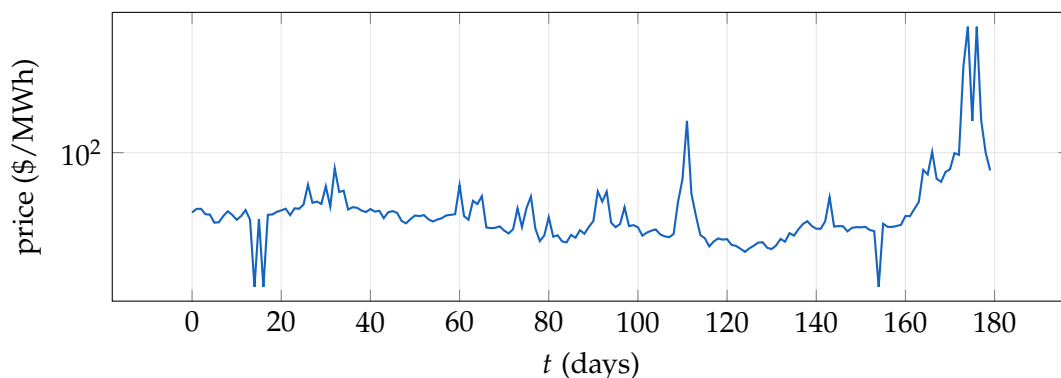


Figure 21.4: 180-day window of the simulated stack price of Example 21.14 (log scale). Sharp up-spikes occur when demand exceeds the stack (capped at \$500) or dips to a low-cost segment (the \$18 troughs, an artifact of the toy grid); between spikes the series reverts to the \$30–\$50 range in a few periods, exactly the pattern of Table 21.3. Simulated for these notes.

IN FINANCE: trading the stack, not the spot

Because Φ is known (approximately) to a utility that owns the plants and knows their costs, a merchant generator does not need a spot *model* of power at all to value

Example 21.14 (A numerical stack). Take a caricature grid: 400 MW nuclear at \$8, 300 MW coal at \$18, 400 MW of gas units with heat rates 7, 7.5, 8, 8.5 (100 MW each, cost $hG + 3$), 200 MW of peaking gas with heat rates 11, 13, 15, 18 (50 MW each), and two 40 MW oil peakers at \$150 and \$300 (a stand-in for a price cap). With demand $D_t = 950 + 130X_t$ (X_t an AR(1) with $\phi = 0.8$, roughly a daily load shape) and $\ln G_t$ an OU process ($\phi = 0.995$ per period, $\sigma = 0.03$, so a slowly wandering gas price around $G_0 = 4$), simulating 2×10^4 periods gives a price series with mean \$46.5, median \$39.7, excess kurtosis 47 (against 3 for a Gaussian) and skewness 4.3: the distribution is exactly of the shape of table 21.3, with a heavy right tail generated *purely* by the nonlinearity of Φ , not by any exogenous jump process imposed on the price. Conditional on the marginal unit, price and gas price are strongly correlated ($\text{Corr} \approx 0.94$ when a gas unit is marginal in the mid-stack); once demand exceeds the whole stack the price is capped at \$500 regardless of G_t : the correlation with gas *breaks down* exactly in the regime that matters most for a hedger, another symptom of Proposition 21.13. (Simulation for these notes; the numbers are illustrative, not calibrated to a real grid.)

physical assets: [section 21.4](#)'s spark-spread option is priced from F_{power} and F_{gas} , i.e. market-observed *forward* prices, and the stack only enters as an internal simulation tool (to size hedges, to stress-test the effect of an outage, or to price options that depend on the *path* of spikes, which forward prices alone do not pin down) [► 2013 L13 @ 58:50]. This is the practical resolution of the lecture's warning that spot models fail: one does not need the spot model to be "right" everywhere, only to be a useful simulator of the risk that the forward market does not already hedge away.

KEY FORMULAS

Cash-and-carry:	$F(t, T) = S_t e^{(r+w-y)(T-t)}$ (contango if $r + w > y$)
Margrabe (zero-strike spread):	$V = e^{-rT} [F_2 \Phi(d_1) - F_1 \Phi(d_2)]$, $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
Storage value: (rate/capacity limits)	$\max_{\{q_i\}} \sum_i q_i F(t, T_i) - \text{costs}$ s.t. (21.6)–(21.7)
Spark spread:	$\Pi = (P - H \cdot G - c)^+$, V Kirk-approximated, (21.9)
One-factor Schwartz:	$F(t, T) = S_t \exp[\text{deterministic function of } X_t, \tau]$, (21.15)
Two-factor (stoch. conv. yield):	$F(t, T) = S_t \exp[-\delta_t B(\tau) + A(\tau)]$, (21.16)
Stack price:	$P(D) = c_{(k)}$, (21.17) (non-decreasing step function of D)

3 COURSE NOTE: what the slides do and do not formalise

The slides show the stack picture and assert its consequences (fat tails, spikes, the breakdown of price/fuel correlation at the top of the stack) without deriving them; the formal statement of [Proposition 21.13](#), [Example 21.14](#) and [fig. 21.4](#) are additions of these notes built to make the qualitative claims precise and checkable. No specific numerical grid was given in the lecture.

Exercise 21.6 (not from the course: three-asset crack spread and the stack). Consider a refinery (crack spread, [section 21.5](#)) sitting downstream of a gas-fired power plant that competes for the same natural-gas supply.

- Using the cointegration result of [section 9.6.4](#), argue that a shock to the crude price is transmitted to the crack spread only through its *stationary* component.
- Using [Proposition 21.13](#), argue instead that a shock to the gas price is transmitted to the power price directly whenever a gas unit is marginal, but not at all once demand pushes past the gas units on the stack. Contrast the two propagation mechanisms in one sentence.

Chapter 22

Option Prices and Probability Duality

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2013 L20	S. Blythe (Harvard Management Company)	Guest lecture, transcript only (no slides, no notes on OCW). Three payoffs (call, zero-coupon bond, digital); the call spread as a discrete first strike-derivative and its limit, the digital; FTAP and the risk-neutral density as the second strike-derivative of the call price; the call butterfly as a traded proxy for the density; the “other direction” (density $\rightarrow$ price via FTAP/LOTUS); calls as a spanning set for European payoffs, first by piecewise-linear replication and then by an exact second-order Taylor identity; a remark that the terminal distribution for every maturity does not pin down the stock’s stochastic process (Dupire–Derman), with the practical consequence for exotic-option pricing.
Files. None: OCW lists no slides or lecture notes for L20, only the video transcript. ¹ Problem sets. None. The exercises (each after the section it practises) are all “(not from the course)”. Reconstruction. Blythe states the Breeden–Litzenberger relation and the spanning formula but proves neither in full generality (the spanning proof is done only for piecewise-linear g, by inspection). section 22.2 is a two-line recap of the proof already given in corollary 3.10. section 22.2.2 adds the triangle/rectangle area argument only sketched in class. section 22.4 gives the exact Taylor-with-remainder computation in full (the lecture writes it down and calls it “exact”, without proof, and skips the discretisation $g \rightarrow$ piecewise-linear step). All numerics (table 22.1, fig. 22.1, and example 22.6) are computed for these notes, with the Black–Scholes parameters of example 15.7 ($S_0 = K = 80$, $T = 2$, $r = 5\%$, $\sigma = 20\%$) so that the digital and call-spread numbers reproduce eq. (15.22)–(15.23).		

1 COURSE NOTE: 2013 L22 has no material at all

The following lecture, 2013 L22, “Calculus of Variations and its Application in FX Execution” (E. Pan, Morgan Stanley), has neither a transcript nor slides on OCW and is not covered anywhere in these notes.

Motivation

Chapter 15 already used one instance of a strike derivative: the digital option is $-\partial C/\partial K$ ([section 15.11](#)), and [corollary 3.10](#) proved that a call price viewed as a function of strike encodes the whole distribution of the underlying, $C''(K) = f(K)$. This chapter is about what that fact *means*. Blythe’s title for it, “option-probability duality”, is the claim that a call option is not a separate, exotic object requiring a pricing model: it *is* a piece of probability information, tradeable and observable, and the map between “the set of all call prices” and “the risk-neutral distribution of the underlying” goes both ways. Read one way, market call

prices reveal the market's implied probabilities (this is genuinely done by trading desks, via the *call butterfly* of [section 22.2.2](#)). Read the other way, the Fundamental Theorem of Asset Pricing ([theorem 2.9](#), generalised here to a continuum of states in [section 22.1](#)) turns any candidate distribution into a full price system for every European payoff. The two directions meet in the *spanning* result of [section 22.4](#): knowing the call price at every strike is exactly as much information as knowing the whole payoff-to-price map, because every payoff is a specific, explicit portfolio of zero-coupon bond, stock and calls. The chapter closes in [section 22.5](#) with what this does *not* give: the terminal distribution at every maturity is not enough to pin down the process the stock follows in between, which is why option markets alone cannot price every exotic derivative and a model ([chapter 18](#)) is still needed for path-dependent payoffs.

22.1 Three payoffs and one assumption

Fix a maturity T and let S_T be the price of the underlying at T . Following the lecture [[2013 L20 @ 13:00](#)], consider three contracts, all defined purely by their payout at T :

- a *call* with strike K : payout $(S_T - K)^+$;
- a *zero-coupon bond*: payout 1;
- a *digital* with strike K : payout $\mathbf{1}\{S_T > K\}$.

Write $C_K(t, T)$, $Z(t, T)$, $D_K(t, T)$ for their prices at $t \leq T$; $Z(T, T) = 1$. As in [definition 2.7](#), a payoff that is a deterministic function of S_T alone is called a *European claim*, written $g(S_T)$ for a payout function g : $g(x) = (x - K)^+$ for the call, $g \equiv 1$ for the bond, $g(x) = \mathbf{1}\{x > K\}$ for the digital [[2013 L20 @ 14:46](#)].

IN FINANCE: Why call it $C_K(t, T)$

Blythe is explicit that the notation separates the two roles of time: K and T are contract terms fixed once and for all, while t is the calendar date at which the price is observed [[2013 L20 @ 15:44](#)]. Determining $C_K(t, T)$ for $t < T$ from a model is exactly the Black–Scholes problem of [chapter 15](#); this chapter instead asks what can be said *without* a model, from the family $\{C_K(t, T)\}_K$ taken as given market data.

The one assumption of the chapter is the Fundamental Theorem of Asset Pricing, already proved for a finite state space in [theorem 2.9](#) (positive state prices $\Leftrightarrow$ no arbitrage, price $= \psi^T C_T = \beta \mathbb{E}_Q[C_T]$). Its continuum version is:

Theorem 22.1 (FTAP, continuum of states; not proved here). *Let $D(t, T)$ be the price at t of a European claim with payout $g(S_T)$ at T . Under no arbitrage (and suitable regularity, e.g. a complete diffusion market) there is a probability measure $\mathbb{Q}$, the risk-neutral measure, such that $D(t, T)/Z(t, T)$ is a $\mathbb{Q}$ -martingale with respect to the filtration generated by S ; in particular, conditionally on S_t ,*

$$D(t, T) = Z(t, T) \mathbb{E}_Q[g(S_T) | S_t]. \quad (22.1)$$

(22.1) is exactly [theorem 2.9\(iii\)](#), $C_0 = \psi^T C_T = \beta \mathbb{E}_Q[C_T]$, with the finite sum $\sum_i \psi_i(\cdot)$ replaced by the integral $\int(\cdot) d\mathbb{Q}$ and the state-price vector ψ replaced by its continuum analogue: the density $\psi(x) = Z(t, T) f_Q(x)$, where f_Q is the $\mathbb{Q}$ -density of S_T given S_t , plays the role of the Arrow–Debreu price of state x ([section 2.4](#) discusses the discrete Arrow–Debreu security e_i paying \$1 in state i ; here the “state” is a point $x \in \mathbb{R}_+$ and the security paying

Proof. Not given here. [theorem 2.9](#) is the exact finite-dimensional statement and its complete proof (Stiemke’s lemma plus a separating hyperplane); the continuum, continuous-time version is due to Harrison and Kreps (1979) and, as Blythe remarks, “however many times you look at it, you’re only probably going to get through two or three pages before thinking, OK, that’s hard” [[2013 L20 @ 51:57](#)]. [sections 15.4](#) and [15.5](#) prove it in the one-step and binomial-tree cases, which already contain the essential idea: replicate the claim, then no-arbitrage forces its price to equal the replicating portfolio’s price, [theorem 2.9\(iii\)](#). $\square$

\$1 in a neighbourhood dx of x costs $\psi(x) dx$). Applying (22.1) to the digital recovers the intuitive statement Blythe leads with: the price of \$1-if-the-coin-is-heads is the probability of heads, discounted [► 2013 L20 @ 23:11]–[► 2013 L20 @ 24:48],

$$D_K(t, T) = Z(t, T) \mathbb{Q}(S_T > K \mid S_t). \quad (22.2)$$

22.2 From call prices to the density

22.2.1 The digital, twice

The digital is priced two different ways: by (22.2) above, and as the limit of a *call spread*, exactly the computation of section 15.11. Long λ calls struck at K and short λ calls struck at $K + 1/\lambda$ has payout 0 below K , slope λ on $[K, K + 1/\lambda]$, and 1 above $K + 1/\lambda$ [► 2013 L20 @ 17:07]–[► 2013 L20 @ 19:19]; letting $\lambda \rightarrow \infty$ this payout tends pointwise to $1\{S_T > K\}$, and its price tends to

$$\lim_{\lambda \rightarrow \infty} \lambda [C_K(t, T) - C_{K+1/\lambda}(t, T)] = -\frac{\partial C_K}{\partial K}(t, T). \quad (22.3)$$

This is (15.22) with $\varepsilon = 1/(2\lambda)$ and a one-sided spread instead of a centred one; the limit is the same. Equating (22.3) with (22.2) gives

$$\mathbb{Q}(S_T > K \mid S_t) = -\frac{1}{Z(t, T)} \frac{\partial C_K}{\partial K}(t, T), \quad F_{\mathbb{Q}}(K) := \mathbb{Q}(S_T \leq K \mid S_t) = 1 + \frac{1}{Z(t, T)} \frac{\partial C_K}{\partial K}(t, T), \quad (22.4)$$

and one more strike derivative gives the density, restating corollary 3.10 with the discount factor made explicit:

$$f_{\mathbb{Q}}(K) = \frac{1}{Z(t, T)} \frac{\partial^2 C_K}{\partial K^2}(t, T). \quad (22.5)$$

(22.5) is the *Breeden–Litzenberger relation*, already proved in corollary 3.10 (there for $Z \equiv 1$, i.e. undiscounted expectations) and used in section 15.11 for the digital alone; nothing here is new mathematics, only the observation, which is the point of the lecture, that *iterating the same one-strike-derivative trick that gives the digital, one more time, gives the entire density, not just one tail probability*.

INTUITION: One relation, no model

Nothing about $\mathbb{Q}$ or S entered the derivation beyond FTAP itself: (22.5) holds for *whatever process* generates the risk-neutral distribution, log-normal (Black–Scholes) or not [► 2013 L20 @ 39:03]–[► 2013 L20 @ 39:31]. Plugging the Black–Scholes formula (15.18) into $\partial_K^2 C$ recovers the log-normal density of eq. (15.6), but (22.5) is agnostic to that choice: it converts *observed* call prices into a distribution without ever positing a model for S .

22.2.2 The call butterfly: trading the density

A single strike derivative (the call spread) has an obvious trading analogue; Blythe’s second construction shows that the *second* strike derivative does too [► 2013 L20 @ 30:56]. Consider the portfolio, long λ calls at $K - 1/\lambda$, short 2λ calls at K , long λ calls at $K + 1/\lambda$ (a *call butterfly*, traded in practice, unlike the idealised limit of section 22.2.1 [► 2013 L20 @ 32:22]–[► 2013 L20 @ 32:37]). Its price is

$$B_K(\lambda; t, T) := \lambda \left[C_{K-1/\lambda}(t, T) - 2C_K(t, T) + C_{K+1/\lambda}(t, T) \right], \quad (22.6)$$

a centred second difference of the call price, scaled by λ (one more power of λ than the call spread, matching the second derivative being approximated by $C(K-\varepsilon)-2C(K)+C(K+\varepsilon) \approx \varepsilon^2 C''(K)$ with $\varepsilon = 1/\lambda$):

$$\lim_{\lambda \rightarrow \infty} \lambda B_K(\lambda; t, T) = \frac{\partial^2 C_K}{\partial K^2}(t, T) = Z(t, T) f_Q(K). \quad (22.7)$$

Proposition 22.2 (The butterfly payoff is a triangle). *The payout of the portfolio in (22.6) at maturity is the triangular tent function of base $[K - 1/\lambda, K + 1/\lambda]$ and height 1 at K ,*

$$h_\lambda(x) = \lambda \left[(x - K + \frac{1}{\lambda})^+ - 2(x - K)^+ + (x - K - \frac{1}{\lambda})^+ \right] = (1 - \lambda|x - K|)^+.$$

This makes (22.7) the exact analogue of the elementary fact $\mathbb{Q}(K - \delta < S_T < K + \delta) \approx 2\delta f_Q(K)$ for small δ : the triangle of proposition 22.2 has base $2/\lambda$ and height 1, hence area $1/\lambda$, and by FTAP its discounted expected payout is

$$B_K(\lambda; t, T) = Z(t, T) \mathbb{E}_Q[h_\lambda(S_T) | S_t] \approx Z(t, T) \cdot \frac{1}{\lambda} f_Q(K) \quad \text{for large } \lambda, \quad (22.8)$$

which is (22.7) divided by λ on both sides [► 2013 L20 @ 35:39]–[► 2013 L20 @ 38:16]. Unlike the call spread, the butterfly is genuinely traded (interest-rate desks quote butterflies 10–20 basis points wide, as noted in class [► 2013 L20 @ 33:24]–[► 2013 L20 @ 33:37]), so (22.8) is not a theoretical limit alone: a market maker who believes $\mathbb{Q}(S_T \approx K)$ is larger than the price of the corresponding butterfly implies will buy it [► 2013 L20 @ 38:23]–[► 2013 L20 @ 38:48].

Table 22.1: Butterfly price $\lambda B_K(\lambda; 0, T)$ converging to $C''(K) = e^{-rT} f_Q(K)$, at $K = 80$, Black–Scholes with $S_0 = 80$, $r = 5\%$, $\sigma = 20\%$, $T = 2$ (parameters of example 15.7). Computed for these notes.

λ	$\varepsilon = 1/\lambda$	$\lambda B_{80}(\lambda)$
0.05	20	0.014656
0.1	10	0.015354
0.2	5	0.015536
0.5	2	0.015588
1	1	0.015596
2	0.5	0.015598
5	0.2	0.015598
exact $e^{-rT} f_Q(80)$		0.015598

Proof. Direct case check on the three pieces of the piecewise-linear function; each summand is a hockey stick with a kink at $K \mp 1/\lambda$ or K , and the three slopes $(\lambda, -2\lambda, \lambda)$ sum to 0 outside $[K - 1/\lambda, K + 1/\lambda]$ (so the payout is 0 there) and change by $\mp\lambda$ at each kink, giving slope $+\lambda$ on $[K - 1/\lambda, K]$ and $-\lambda$ on $[K, K + 1/\lambda]$, height $1 = \lambda \cdot \frac{1}{\lambda}$ at K . This is exactly the sketch in class [► 2013 L20 @ 36:17]–[► 2013 L20 @ 36:41], made explicit. □

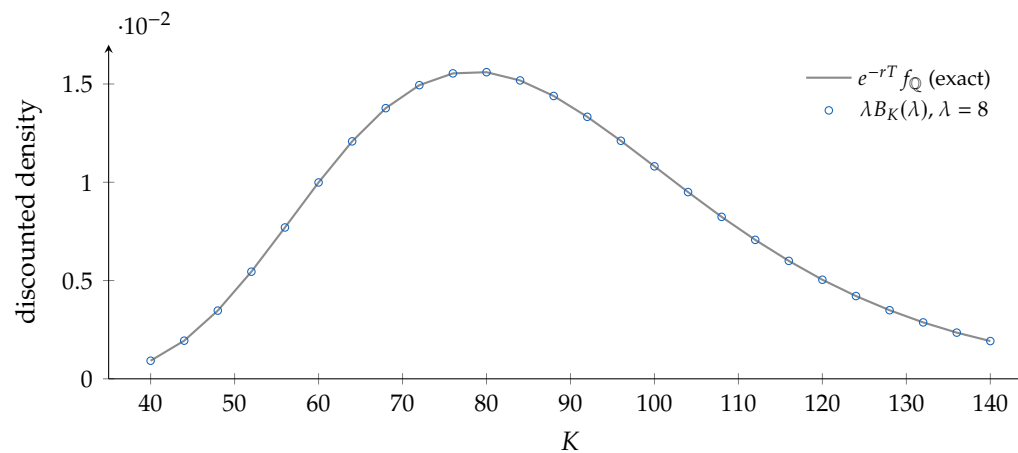


Figure 22.1: Discounted risk-neutral density recovered from the call-butterfly price (22.6) at $\lambda = 8$ (width $2/\lambda = 0.25$), against the exact Black–Scholes log-normal density (same parameters as table 22.1); markers overlap the line to plotting accuracy. Computed for these notes.

Exercise 22.1 (Butterfly width and bias). For the Black–Scholes parameters of table 22.1 ($S_0 = 80$, $r = 5\%$, $\sigma = 20\%$, $T = 2$), let $\lambda = 4$ (butterfly of half-width $1/\lambda = 0.25$ around each wing, full base 0.5). Compute $B_{80}(4)$ from the closed-form Black–Scholes call price (15.18) and compare $\lambda B_{80}(4)$ to the exact value in table 22.1. Express the leading-order error as a function of λ using a fourth-order Taylor expansion of C around $K = 80$, and check it is consistent with the pattern in the table.

Exercise 22.2 (Digital from a butterfly, not a call spread). Show that a digital option cannot be replicated as a *limit of call butterflies* the way it was replicated as a limit of call spreads in section 22.2.1: compute $\lim_{\lambda \rightarrow \infty} B_K(\lambda; t, T)$ (not $\lambda B_K(\lambda; t, T)$) and interpret the result in terms of the payoff triangle of proposition 22.2 shrinking to a point.

22.3 From the density to any price: FTAP the other way

section 22.2 went from call prices to a distribution. The Fundamental Theorem (22.1) already goes the other way, and the resulting statement is worth writing out explicitly [► 2013 L20 @ 49:41]–[► 2013 L20 @ 54:09]: if $f_{\mathbb{Q}}(\cdot | S_t)$ is known, the price of *any* European claim with payout g is

$$D(t, T) = Z(t, T) \int_0^\infty g(x) f_{\mathbb{Q}}(x | S_t) dx, \quad (22.9)$$

an application of the law of the unconscious statistician (LOTUS): $\mathbb{E}_{\mathbb{Q}}[g(S_T)]$ is computed by integrating g against the density of S_T , without first working out the (generally intractable) density of the transformed variable $g(S_T)$ itself. Taking $g(x) = (x - K)^+$ in (22.9) recovers theorem 3.9 (there stated for a general random variable X under the physical measure; here $X = S_T$ under $\mathbb{Q}$, with the discount factor $Z(t, T)$ attached). So the two directions of the duality are: call prices $\rightarrow$ density via two strike derivatives (section 22.2), density $\rightarrow$ any European price via one integral ((22.9)), and both use nothing beyond FTAP.

22.4 Calls span every European payoff

The two directions combine into a genuinely new statement, not reducible to Breeden–Litzenberger alone: knowing $C_K(t, T)$ for every K determines the price of *every* European claim, and moreover gives an *explicit static replicating portfolio* built only from the zero-coupon bond, the stock, and calls [► 2013 L20 @ 55:44]–[► 2013 L20 @ 56:09].

22.4.1 Piecewise-linear payouts

If g is piecewise linear with kinks at $K_1 < K_2 < \dots < K_n$, its graph is matched exactly, for $x \geq K_1$ say, by a portfolio holding θ_i calls of strike K_i , where θ_i is the change in slope of g at K_i (a call of strike K_i contributes slope 1 for $x > K_i$ and 0 below, so a linear combination reproduces any sequence of slope changes) [► 2013 L20 @ 56:47]–[► 2013 L20 @ 58:13]. Since g and this call portfolio have *identical* payoff at T for every value of S_T , no-arbitrage (theorem 2.9(iii): a replicable claim has a unique price, the price of any replicating portfolio) forces their prices to agree at every $t \leq T$,

$$D(t, T) = \sum_i \theta_i C_{K_i}(t, T), \quad \theta_i = \Delta g'(K_i). \quad (22.10)$$

A continuous g is a uniform limit of piecewise-linear functions, so (22.10) suggests (but does not by itself prove, because of the limit) that *every* continuous payout is spanned by calls. section 22.4.2 makes this precise and quantitative for $g \in C^2$, which is the practically relevant case and the one the lecture treats.

22.4.2 The exact Carr–Madan identity

Proposition 22.3 (Static replication of a smooth payoff). Let $g : \mathbb{R}_+ \rightarrow \mathbb{R}$ be twice continuously differentiable. For every $x \geq 0$,

$$g(x) = g(0) + g'(0)x + \int_0^\infty g''(K)(x - K)^+ dK. \quad (22.11)$$

Remark 22.4. The identity holds pointwise, for every $x \geq 0$, with no approximation and no limit: this is the sense in which it is “exact”. It is the content of section 22.4.1 made rigorous: $g''(K)dK$ replaces the discrete slope changes $\theta_i = \Delta g'(K_i)$ of a piecewise-linear function by a continuum of infinitesimal slope changes, each priced by $(x - K)^+$, the call payoff. Where the lecture stops after the piecewise-linear case and asserts the smooth case by continuity [► 2013 L20 @ 1:02:52]–[► 2013 L20 @ 1:03:03], proposition 22.3 needs no limiting argument at all: it is an identity between functions, true for every fixed x , and the piecewise-linear intuition of section 22.4.1 is recovered by taking g''

2 COURSE NOTE: A digression worth keeping

A student asks whether replication at T really forces equality of prices at $t < T$ for maturities decades away, given that no one can actually hold the position without funding or margin risk in the meantime

[► 2013 L20 @ 59:44]–[► 2013 L20 @ 1:02:13].

Blythe’s answer: mathematically yes, under no-arbitrage; empirically, for very long maturities the two prices *have* been observed to drift apart, because “no arbitrage” assumes unlimited risk capital willing to hold the position for the full horizon, which breaks down since 2008. This is a caveat on the economic content of theorem 2.9(iii), not on its mathematics, and it is not pursued further here.

Proof. This is the lecture’s “exact Taylor series to second order” [► 2013 L20 @ 1:03:33]–[► 2013 L20 @ 1:04:20], here proved rather than only asserted. Start from the fundamental theorem of calculus twice,

$$g(x) = g(0) + \int_0^x g'(K) dK, \quad g'(K) = g'(0) + \int_0^K g''(c) dc$$

substitute and swap the order of integration in the double integral $\int_0^x \int_0^K g''(c) dc dK$ over the triangle $0 \leq c \leq K \leq x$:

$$\begin{aligned} \int_0^x \int_0^K g''(c) dc dK &= \int_0^x g''(c) \int_c^x dK dc = \int_0^x g''(c)(x - c) dc \\ &= \int_0^\infty g''(c)(x - c)^+ dc, \end{aligned}$$

where the last equality extends the integral to (x, ∞) for free because $(x - c)^+ = 0$

to be a sum of Dirac masses (formally, outside the C^2 hypothesis, but the calls-with-strikes-at-the-kinks picture is exactly this limit).

Set $x = S_T$ in (22.11) (a pointwise identity between random variables, valid on every sample path) and apply the discounted-expectation operator $Z(t, T) \mathbb{E}_Q[\cdot | S_t]$ to both sides. The left side becomes $D(t, T)$, the price of the claim with payout g . On the right, linearity of expectation gives three terms: $g(0)$ times the price of $g(0)$ zero-coupon bonds is $Z(t, T)g(0)$; the term $g'(0) \mathbb{E}_Q[S_T | S_t]$, discounted, is $g'(0)$ times the price of *one unit of stock* (the stock, viewed as its own European claim with payout S_T and replicating portfolio “hold the stock”, so $Z(t, T) \mathbb{E}_Q[S_T | S_t] = S_t$ by [theorem 22.1](#) applied to $g(x) = x$ [[2013 L20 @ 1:07:25](#)]–[\[2013 L20 @ 1:07:45\]](#)); and the integral term becomes, exchanging expectation and the K -integral (justified when $g'' \geq 0$ by Tonelli, or under an integrability bound on g'' in general),

$$\begin{aligned} Z(t, T) \mathbb{E}_Q \left[\int_0^\infty g''(K) (S_T - K)^+ dK \mid S_t \right] &= \int_0^\infty g''(K) Z(t, T) \mathbb{E}_Q [(S_T - K)^+ | S_t] dK \\ &= \int_0^\infty g''(K) C_K(t, T) dK. \end{aligned}$$

Collecting the three terms:

Theorem 22.5 (Calls span all European payoffs; Carr–Madan). For $g \in C^2(\mathbb{R}_+)$ with the payoff at maturity $g(S_T)$,

$$D(t, T) = g(0) Z(t, T) + g'(0) S_t + \int_0^\infty g''(K) C_K(t, T) dK. \quad (22.12)$$

Equivalently, the claim is statically replicated at t by $g(0)$ zero-coupon bonds, $g'(0)$ units of stock, and $g''(K) dK$ calls of strike K for every $K > 0$.

KEY FORMULAS

Call prices $\rightarrow$ distribution: $\mathbb{Q}(S_T > K | S_t) = -\frac{1}{Z} \partial_K C_K$, $f_Q(K) = \frac{1}{Z} \partial_K^2 C_K$ (22.4)–(22.5) (Breeden–Litzenberger, [corollary 3.10](#)).

Distribution $\rightarrow$ any price: $D(t, T) = Z(t, T) \int g(x) f_Q(x | S_t) dx$ (22.9) (FTAP/LOTUS).

Static replication (Carr–Madan): $D(t, T) = g(0) Z(t, T) + g'(0) S_t + \int_0^\infty g''(K) C_K(t, T) dK$ (22.12).

Traded proxy for the density: call butterfly $\lambda B_K(\lambda) \rightarrow Z(t, T) f_Q(K)$ (22.7).

Exercise 22.3 (Straddle via Carr–Madan). A straddle pays $g(S_T) = |S_T - K_0|$ for a fixed $K_0 > 0$. This g is not C^2 at K_0 (it has a kink), so (22.11) must be applied with g'' interpreted as a Dirac mass $2\delta_{K_0}$ (the jump in g' at K_0 is 2). Using (22.10) directly instead (the straddle is piecewise linear with one kink), find the static replicating portfolio of zero-coupon bonds, stock and calls of a single strike, and check it agrees with formally substituting $g'' = 2\delta_{K_0}$ into (22.12).

there. Renaming the dummy variable $c \rightarrow K$ gives (22.11). (Equivalently, integrate the right-hand side of (22.11) by parts twice, as the lecture suggests [[2013 L20 @ 1:04:20](#)]–[\[2013 L20 @ 1:04:31\]](#), which reverses this computation.) $\square$

Example 22.6 (Checking the identity on $g(x) = x^2$). Take $g(x) = x^2$, so $g(0) = 0$, $g'(0) = 0$, $g''(K) = 2$ for all K . (22.11) claims $x^2 = \int_0^\infty 2(x - K)^+ dK$, and indeed $\int_0^x 2(x - K) dK = 2[xK - K^2/2]_0^x = x^2$: a direct check of [proposition 22.3](#) with an elementary integral. Numerically (trapezoidal rule, grid $[0, 800]$, 4×10^5 points, computed for these notes) the same integral gives 6400.0000 against the exact $80^2 = 6400$. A second check with a bounded, non-polynomial $g(x) = e^{-0.01x}$ ($g''(K) = 10^{-4} e^{-0.01K}$) gives, at $x = 80$, exact $g(80) = 0.449329$ against the numerical reconstruction 0.449329: (22.11) is confirmed to 6 digits both for a payoff whose second

22.5 What the terminal distribution does not determine

Everything above concerns a single fixed maturity T : knowing all call prices $C_K(t, T)$ for that one T determines the distribution of S_T given S_t , and hence the price of every payoff that depends on S_T alone. The natural next question is whether knowing this for *every* maturity T (a full implied-volatility surface, in market language) determines the entire stochastic process $(S_u)_{u \geq t}$ [► 2013 L20 @ 1:12:11]–[► 2013 L20 @ 1:13:07].

Proposition 22.7 (Marginals do not determine the process; not proved here). *Knowledge of the marginal (risk-neutral) distribution of S_T given S_t , for every $T \geq t$, does not determine the law of the process $(S_u)_{u \in [t, T]}$. If in addition S is assumed to be a one-dimensional diffusion driven by Brownian motion, $dS_u = \mu(u, S_u) du + \sigma(u, S_u) dW_u$, then the marginals do determine the process, via the local volatility function $\sigma_{\text{loc}}(u, x)$ (Dupire, Derman–Kani, early 1990s).*

IN FINANCE: Why this matters for a trading desk

A desk that has priced, and can perfectly hedge with the observed vanilla surface, every European payoff via (22.12), still cannot claim the same for a path-dependent exotic (a barrier, an Asian option, or, as Blythe’s example, a derivative paying on the number of times S crosses a level in a small interval [► 2013 L20 @ 1:19:05]–[► 2013 L20 @ 1:19:10]): its price is not pinned down by the vanilla surface alone, precisely because [proposition 22.7](#) says the surface does not pin down the process. A student’s question makes the link to market completeness explicit [► 2013 L20 @ 1:16:44]–[► 2013 L20 @ 1:18:52]: a jump/flip process for which the marginals do not determine the law corresponds to an *incomplete* market ([definition 2.7](#)) with respect to the vanilla calls, exactly as more states than independent assets leave state prices non-unique in the one-period model of [section 2.4](#). The risk-neutral density $f_Q(\cdot | S_t)$ at each fixed T is still uniquely pinned down by the calls (Blythe is emphatic on this point when pressed [► 2013 L20 @ 1:19:17]–[► 2013 L20 @ 1:19:48]); what is not determined is the joint law across different T ’s, i.e. the path.

Remark 22.8 (An alternative spanning set: Arrow–Debreu securities). A question from the audience asks whether calls are the *only* convenient spanning set [► 2013 L20 @ 1:16:16]–[► 2013 L20 @ 1:17:41]. They are not: knowing all digitals gives the CDF directly (one derivative instead of two), and knowing the price of every Arrow–Debreu security $1\{S_T \in dx\}$ gives the density directly, with no derivative at all. Blythe’s preference for calls is that the identity (22.12) is unusually clean, expressing an arbitrary smooth payoff through an explicit, liquidly-traded static hedge, whereas Arrow–Debreu securities of infinitesimal width are not traded and digitals are less liquid than calls ([section 15.11](#)). This is also exactly the representation [chapter 25](#) will use (state prices as a discrete matrix, recovered via Perron–Frobenius, [section 2.8](#)) in a setting where the underlying state space is finite rather than a continuum of strikes.

Exercise 22.4 (Two maturities, same terminal law, different paths). Let W be a Brownian motion and $Z \sim N(0, 1)$. Show that $S_t = 1 + W_t$ and $\tilde{S}_t = 1 + \sqrt{t} Z$ have the same marginal law $N(1, t)$ at every $t \in [0, 2]$, but different laws as processes: compute $\text{Cov}(S_1, S_2)$ and $\text{Cov}(\tilde{S}_1, \tilde{S}_2)$. Which of the two is a martingale? This illustrates [proposition 22.7](#) concretely: a claim paying $(S_2 - S_1)^2$ has different prices under the two processes although all the terminal laws, hence all the call prices, agree.

derivative is constant and for one that decays.

Discussion, not a proof. Blythe gives only the intuition, not the construction, and none is attempted here (it belongs to stochastic calculus proper, beyond [chapter 18](#)): take a denser and denser grid of maturities with prescribed, say symmetric, marginals that spread out smoothly in T ; the stock can still be made to jump discontinuously between two adjacent grid times while landing in a distribution consistent with the (denser and denser) marginals at both ends, because *nothing in the one-maturity-at-a-time data constrains the joint law across maturities* [► 2013 L20 @ 1:13:34]–[► 2013 L20 @ 1:14:07]. Ruling out such jumps, by assuming a diffusion, is what lets Dupire’s formula reconstruct a unique local-volatility function from the full surface of (22.5)-derived marginals. □

Chapter 23

Quanto Credit Hedging

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2013 L23	S. Andreev (Morgan Stanley)	FX spot and forwards, interest-rate parity; the FX betting-game paradox; sovereign bonds issued in two currencies (Italy) and the credit-spread puzzle; a minimal jump-on-default FX model; pricing and dynamic hedging of zero-recovery USD/EUR bonds in that model, extension to $R > 0$; the jump-diffusion model left as a final-paper topic; quanto credit default swaps and other applications.

Files. 2013 slides MIT18_S096F13_lecnote23.pdf (32 slides); accompanying written notes with the derivations (referenced repeatedly on the slides as “see class notes”) were not among the materials available for these notes, so every derivation below is reconstructed from the slides and from the blackboard work described in the lecture transcript, and re-checked numerically in a Python script written for these notes; reconstructions are flagged explicitly, and all such numerical checks are marked “computed for these notes.” **Problem sets.** None; the exercises (each after the section it practises) are all marked “not from the course”. **Not covered here.** The stochastic-hazard-rate, stochastic-interest-rate, correlated multi-factor version of the model is only sketched on the last slides (“Real Life”); [chapter 24](#) returns to term-structure modelling of credit.

Motivation

Chapter 8 showed that a change of numeraire turns a ratio of two traded assets into a martingale, and listed the FX forward $F_{FX}(t, T) = S_{FX}(t)P^F(t, T)/P^D(t, T)$ as one instance of the pattern ([proposition 8.17](#)). This chapter asks what changes when one of the two “assets” can *jump*: a sovereign issuer defaults, and its currency devalues at the same instant. Two zero-coupon bonds of the same issuer, one paying 1 USD and one paying 1 EUR, are no longer related by a simple deterministic forward rate, because the FX rate that converts one payoff into the other is itself correlated with the credit event. Pricing and hedging one bond with the other – a *quanto* problem, in the sense that a payoff is being valued in a currency other than its natural one – requires modelling the FX jump explicitly.

The chapter builds the smallest possible model that has this feature (a constant-hazard-rate default time with a deterministic percentage FX jump on default), and carries it through the three-step recipe of [section 15.4](#):

1. write the payoffs,
2. price by risk-neutral expectation and find the replicating portfolio,

3. extract the economic intuition (the sign and size of the “quanto adjustment”).

The compensator identity used along the way is the jump-process analogue of the risk-neutral drift condition of [section 15.8](#), and the pricing measure used throughout is exactly the risk-neutral measure $\mathbb{Q}$ of [chapter 15](#); [section 23.6](#) makes the connection to [proposition 8.17](#) precise.

INTUITION: Why symmetric bets are not symmetric

If I bet you \$100 that a coin lands heads and you bet me 100 EUR that it lands tails, the two bets look symmetric. They are not, if the coin is Italy defaulting and the EUR devalues whenever it does: winning USD then happens exactly when EUR is weak, and winning EUR happens exactly when EUR is strong. The currency of the payoff carries information about the payoff, and a fair price must account for it. This is the whole content of the chapter, before any formula is written down.

23.1 FX, interest-rate parity, and the currency of a payoff

Write S_t for the spot FX rate USD/EUR, i.e. the price in USD of 1 EUR. In the (no-default) Garman–Kohlhagen model, with domestic (USD) and foreign (EUR) risk-free rates r_d, r_f ,

$$\frac{dS_t}{S_t} = (r_d - r_f) dt + \sigma dW_t, \quad S_T = S_0 \exp\left[(r_d - r_f - \frac{1}{2}\sigma^2)T + \sigma W_T\right], \quad \mathbb{E}[S_T] = S_0 e^{(r_d - r_f)T}. \quad (23.1)$$

This is the same lognormal SDE as [eq. \(15.6\)](#) with $\mu = r_d - r_f$: under $\mathbb{Q}$, a foreign asset earns the domestic rate once its value is expressed in domestic currency, exactly as a stock earns r under the risk-neutral measure of [section 15.8](#). The forward FX rate is fixed by no-arbitrage (covered interest-rate parity), $F(t, T) = S_t e^{(r_d - r_f)(T - t)} = \mathbb{E}_t[S_T]$ – itself an instance of [proposition 8.17](#) with $N_1 = 1$ (cash) and $N_2 = P^D(\cdot, T)$, as recorded in the numeraire table of [chapter 8](#).

For the rest of the chapter, following the slides, interest rates are set to zero, $r_d = r_f = 0$, so that the only driver of S_t is the credit event: this isolates the “quanto credit” effect from the ordinary FX quanto adjustment of [eq. \(23.1\)](#), which is unaffected by anything in this chapter and is recovered by simply adding back $r_d - r_f$ to every drift below.

IN FINANCE: The FX betting game

Slide 7 poses two \$100-notional bets on whether USD/EUR (currently 1) is above or below 1 in one month. Bet A pays $\mp \$100$ in USD; Bet B pays ∓ 100 EUR. If the exchange rate is uncorrelated with the outcome the two bets are worth the same. But if, say, USD/EUR > 1 (euro strong, dollar weak) exactly in the scenarios where the bet is lost, then Bet A ($-\$100$ regardless of the rate) is strictly better than Bet B (-100 EUR, converted at an unfavourable rate). [Table 23.1](#) reproduces the slide’s numbers: Bet A minus Bet B is always positive because the loss in Bet B is amplified, and its gain reduced, by exactly the FX move correlated with it. The general principle: *whenever a payoff and the FX rate used to convert it are correlated, pricing the payoff in the “wrong” currency and simply converting at the current spot or forward rate is not valid* – one must price the converted payoff directly under a risk-neutral expectation, as done for CDS below.

Table 23.1: FX betting game, slide 8: two symmetric-looking \$100/EUR100 bets are worth different amounts once the payoff currency is correlated with the FX outcome.

Scenario (1 month)	Bet A (USD)	Bet B (EUR, in USD)	A – B
USD/EUR = 1.25	–100	–125	+25
USD/EUR = 0.75	+100	+75	+25

23.2 A reality check: sovereign bonds in two currencies

Slides 11–14 use Italy as a running example: a sovereign that issues bonds in EUR and in USD (and other currencies), with a *cross-default* clause (default on one currency triggers default on all). Two natural questions: which currency does Italy prefer to borrow in, and which currency do bond investors prefer? The credit spread (the extra yield over the local risk-free curve) need not be the same in the two currencies, and the slides show that Italy’s EUR and USD five-year credit spreads did in fact move apart during the 2011–2012 sovereign crisis. Comparing the two spreads directly conflates two different things: the market’s view of Italy’s creditworthiness (currency-independent, in a first approximation) and the market’s view of what happens to EUR *if Italy defaults* (which only affects the EUR-denominated bond, since a EUR bond’s payoff is naturally denominated in the currency that would devalue). Section 23.3 builds the smallest model that captures this second effect, so that the two spreads can be compared on a like-for-like basis.

23.2.1 A naive replication attempt, and why it fails

Consider two zero-coupon, zero-recovery bonds of the same issuer and maturity T : bond U pays 1 USD, bond E pays 1 EUR. With P_u, P_e their prices (in USD and EUR respectively), spot rate S_t and forward rate F_t (USD/EUR, maturity T), a naive covered-interest-rate-parity argument suggests buying 100,000 E-bonds, selling 100,000 F_t U-bonds, and entering a forward to sell 100,000 EUR at F_t (slide 15). If both bonds survive to T , the position pays exactly 0: the 100,000 EUR received is exchanged at the pre-agreed forward rate for 100,000 F_t USD, which exactly cancels the short U-bonds. But if the issuer defaults with recovery rate 25% (slide 17), the E-bonds pay 25,000 EUR and the short U-bonds owe 25,000 F_t USD, while the forward contract, which does not know that the issuer defaulted, still delivers the full 100,000 EUR notional against 100,000 F_t USD. The strategy is left with a large *unhedged* FX forward position (short 75,000 EUR against 75,000 F_t USD) exactly in the scenario – default – in which EUR is most likely to move sharply (as it did for the Argentine peso in December 2001, slide 18). The naive strategy is not an arbitrage and is not a valid hedge: the forward is the wrong instrument because it does not distinguish the default and no-default states. What is needed is a model in which the number of hedging instruments equals the number of sources of randomness (default timing and the FX jump it triggers) – a complete market in the sense of section 15.4 – so that pricing by replication is again possible.

23.3 The minimal jump-on-default FX model

23.3.1 Hazard rate and survival probability

Model the default time τ of the issuer as the arrival time of the first jump of a Poisson process of constant intensity $h > 0$ (the *hazard rate*): from the perspective of time $t < \tau$,

$$\mathbb{P}_t(\tau > T) = e^{-h(T-t)}, \quad T \geq t. \quad (23.2)$$

Definition 23.1 (Hazard rate). h is the instantaneous default intensity: $\mathbb{P}_t(\tau \in [t, t + dt) \mid \tau > t) = h \, dt$.

Proposition 23.2 (Survival probability and default density). If τ has the memoryless survival function eq. (23.2), then, conditional on $t < \tau$, the density of τ at $T > t$ is $\varphi(t, T) = he^{-h(T-t)}$, and the current ($T \rightarrow t$) default density is $\varphi(t, t) = h$.

Proof. $\mathbb{P}_t(\tau \leq T) = 1 - e^{-h(T-t)}$; differentiate in T to get $\varphi(t, T) = he^{-h(T-t)}$; set $T = t$. $\square$

This is exactly the exponential distribution: constant hazard rate $\Leftrightarrow \tau - t \sim \text{Exp}(h)$ given survival to t , the continuous-time analogue of a Poisson arrival process (section 8.7 used the same intensity idea for the CDS survival measure).

23.3.2 FX jump on default

The model's only new ingredient is that S_t (USD per EUR) jumps by a fixed multiplicative factor at τ :

$$S_\tau = S_{\tau-} e^J, \quad J \in \mathbb{R}, \quad (23.3)$$

so $J < 0$ means EUR devalues on default (the Italy/Argentina case) and $J > 0$ an appreciation. Let $N_t = \mathbf{1}_{\{t \geq \tau\}}$ be the default indicator, a jump process with a single unit jump at τ , dN_t the corresponding point-process differential ($dN_t = 1$ at $t = \tau$, 0 elsewhere). Away from the jump, $\log S_t$ is postulated to drift at some rate μ_t to be determined:

$$d \log S_t = \mu_t \, dt + J \, dN_t. \quad (23.4)$$

No-arbitrage (interest rates being 0) requires S_t to be a $\mathbb{Q}$ -martingale, in particular $\mathbb{E}[S_T] = S_0$ for every T – the jump analogue of the risk-neutral drift condition that fixes $\mu = r$ in eq. (15.6).

Proposition 23.3 (Compensator condition). If $\mu_t = h(1 - e^J)\mathbf{1}_{\{t < \tau\}}$ in eq. (23.4) (and $\mu_t = 0$ once τ has occurred, since after default there is nothing left to compensate), then $\mathbb{E}[S_T] = S_0$ for every $T \geq 0$.

The quantity $h(1 - e^J)$ is called the *compensator*: exactly as a Poisson counting process N_t has mean ht so that $N_t - ht$ is a martingale, here the drift $\mu_t \, dt$ must exactly offset, in expectation, the mean rate of change contributed by the jump term $J \, dN_t$, whose expected instantaneous contribution (while τ has not yet occurred) is $h \cdot J \, dt$ plus a jump-size correction coming from working with $\log S$ rather than S itself; this is made precise in the proof below via Itô's formula for jump processes.

Lemma 23.4 (Itô's formula for a jump process, reconstruction). Let X_t solve $dX_t = \mu_t \, dt + J \, dN_t$ for a point process N_t with a single jump at τ (no diffusive part), and let $f \in C^1$.

Proof. Between jumps X_t is absolutely continuous with $\dot{X}_t = \mu_t$, so $df(X_t) = f'(X_t)\dot{X}_t \, dt$ by the ordinary chain rule; at a jump time, $f(X_t) - f(X_{t-}) = f(X_{t-} + J) - f(X_{t-})$ by eq. (23.3)-type jumps of size J in X . Summing the con-

Then

$$df(X_t) = f'(X_t) \mu_t dt + [f(X_{t-} + J) - f(X_{t-})] dN_t. \quad (23.5)$$

INTUITION: Reading the compensator

If EUR devalues on default ($J < 0$, so $e^J < 1$), the compensator $h(1 - e^J) > 0$: between defaults the model must make EUR *appreciate* against USD, on average, at rate $h(1 - e^J)$, precisely so that the (rare, large, negative) jump at default leaves the unconditional mean unchanged. This upward drift before default is a purely technical artefact of no-arbitrage in a jump model, not a forecast: it is the exact analogue of the fact that a stock with a possible downward jump (bankruptcy) must drift upward the rest of the time to keep $\mathbb{E}[S_T]$ fixed at the forward level, a fact well known from equity jump-diffusion models.

Exercise 23.1 (Compensator with two jump sizes, not from the course). Suppose S_t can jump on default to either $S_{\tau-}e^{J_1}$ (probability p) or $S_{\tau-}e^{J_2}$ (probability $1 - p$), independently drawn at τ . Redo the proof of [proposition 23.3](#) to find the compensator drift μ_t that keeps S_t a $\mathbb{Q}$ -martingale, and check that it reduces to [proposition 23.3](#) when $p = 1$.

tinuous drift contribution and the (single) jump contribution gives [eq. \(23.5\)](#). (This is the pure-jump special case of the general jump-diffusion Itô formula; the diffusive analogue for Brownian motion is [theorem 17.9](#), with the second-derivative term there playing the role that the finite difference $f(X_{t-} + J) - f(X_{t-})$ plays here – both are “convexity” corrections beyond the naive chain rule, one from quadratic variation, the other from a discrete increment.) $\square$

Proof of [proposition 23.3](#). Write $X_t = \log S_t$, so $S_t = e^{X_t} = f(X_t)$ with $f(x) = e^x$. By [lemma 23.4](#) with μ_t as in the statement,

$$dS_t = S_{t-} h(1 - e^J) \mathbf{1}_{\{t < \tau\}} dt + S_{t-} (e^J - 1) dN_t,$$

using $f(X_{t-} + J) - f(X_{t-}) = e^{X_{t-}} (e^J - 1) = S_{t-} (e^J - 1)$. Divide by S_{t-} :

$$\frac{dS_t}{S_t} = h(1 - e^J) \mathbf{1}_{\{t < \tau\}} dt + (e^J - 1) dN_t. \quad (23.6)$$

Both terms vanish in expectation instantaneously conditional on survival: on $\{t < \tau\}$, $\mathbb{E}[dN_t | \mathcal{F}_t] = h dt$, so $\mathbb{E}[dS_t/S_t | \mathcal{F}_t, t < \tau] = h(1 - e^J) dt + (e^J - 1)h dt = 0$; after default nothing changes ($\mu_t = 0$ and N_t no longer jumps). Hence S_t is a $\mathbb{Q}$ -martingale and $\mathbb{E}[S_T] = S_0$. $\square$

23.3.3 Explicit solution for S_t

Proposition 23.5 (Path of the FX rate). Under [eq. \(23.4\)](#) with the compensator drift of [proposition 23.3](#),

$$S_T = S_0 \exp \left[h(1 - e^J)(T \wedge \tau) + J \mathbf{1}_{\{\tau \leq T\}} \right], \quad T \wedge \tau = \min(T, \tau). \quad (23.7)$$

Equivalently: if $\tau > T$, $S_T = S_0 \exp[h(1 - e^J)T]$ (pure drift); if $\tau \leq T$, $S_T = S_0 \exp[h(1 - e^J)\tau + J]$ (drift up to default, then the jump).

Proof. Integrate [eq. \(23.4\)](#): $\log S_T - \log S_0 = \int_0^T h(1 - e^J) \mathbf{1}_{\{t < \tau\}} dt + \int_0^T dN_t$. The first integral runs only while $t < \tau$, i.e. up to $T \wedge \tau$, contributing $h(1 - e^J)(T \wedge \tau)$; the second is J if a jump occurred by T ($\tau \leq T$) and 0 otherwise. Exponentiate. $\square$

23.4 Pricing and hedging zero-recovery bonds

Consider zero-coupon, zero-recovery bonds of maturity T from the same issuer: U pays 1 USD at T if $\tau > T$ (and 0 otherwise); E pays 1 EUR at T under the same condition. Both are priced, in USD, by risk-neutral expectation (interest rates 0, so the money-market numeraire is $\beta(t) \equiv 1$, as in [section 8.7](#)):

$$P_u(0, T) = \mathbb{E}[\mathbf{1}_{\{\tau > T\}}] = \mathbb{P}(\tau > T) = e^{-hT}, \quad (23.8)$$

Example 23.6 (Numerical check of the martingale property). Take $h = 0.05$, $J = -0.30$ (a 30% devaluation on default, roughly the order of magnitude of the Argentine peso move of slide 18), $T = 3$, $S_0 = 1$. Monte Carlo with 6×10^6 paths of $\tau \sim \text{Exp}(h)$ and [eq. \(23.7\)](#) gives $\mathbb{E}[S_T] = 0.99997$, matching the required value $S_0 = 1$ to four significant figures (computed for these notes;). The exact identity $\mathbb{E}[S_T] = 1$ can also be checked analytically by splitting the expectation on $\{\tau > T\}$ and $\{\tau \leq T\}$ and integrating $\varphi(0, \tau) = he^{-h\tau}$ against [eq. \(23.7\)](#); both routes agree.

by eq. (23.2), exactly as in section 15.8: the price is the expected discounted payoff. For the EUR bond, the USD-payoff is $S_T \mathbf{1}_{\{\tau > T\}}$ (converting the 1 EUR payoff at the terminal FX rate), so

$$P_e(0, T) = \mathbb{E}[S_T \mathbf{1}_{\{\tau > T\}}]. \quad (23.9)$$

Proposition 23.7 (Zero-recovery bond prices).

$$P_u(0, T) = e^{-hT}, \quad P_e(0, T) = S_0 e^{-he^J T}. \quad (23.10)$$

Formula (23.10) was checked numerically for these notes: with $h = 0.05$, $J = -0.30$, $T = 3$, $S_0 = 1$, eq. (23.10) gives $P_u = 0.86071$, $P_e = 0.89483$, against Monte Carlo estimates 0.86059 and 0.89470 from 6×10^6 paths – agreement to three decimals.

Proof. P_u is eq. (23.8). For P_e , on $\{\tau > T\}$, eq. (23.7) gives $S_T = S_0 e^{h(1-e^J)T}$ (a constant, not random, given survival), so

$$P_e(0, T) = S_0 e^{h(1-e^J)T} \mathbb{P}(\tau > T) = S_0 e^{h(1-e^J)T} e^{-hT} = S_0 e^{-he^J T} \quad \square$$

INTUITION: Reading P_e

$P_e/S_0 = e^{-he^J T}$ rather than e^{-hT} : the EUR bond is priced with an *effective hazard rate* $h_{\text{eff}} = he^J$ once expressed in USD. If EUR devalues on default ($J < 0$, $e^J < 1$), $h_{\text{eff}} < h$ and $P_e/S_0 > P_u$: in USD terms the EUR bond looks *less* risky than the USD bond of the same issuer, because its payoff is worth relatively little exactly in the states where default has happened – the market has less to lose, in USD, from that scenario. This is the quanto adjustment for credit: the raw survival probability e^{-hT} is corrected by e^J to account for the correlation between the default event and the currency of the payoff. It matches the qualitative content of the Italy example of section 23.2: everything else equal, a devaluing local-currency bond should trade with a *lower* USD-equivalent implied hazard rate than the hard-currency bond of the same issuer – exactly the pattern the slides point to (EUR spread below USD spread once basis-adjusted).

Exercise 23.2 (Sign of the quanto adjustment, not from the course). Using eq. (23.10), show that $P_e(0, T)/S_0 > P_u(0, T)$ if and only if $J < 0$ (devaluation on default), for every $h > 0, T > 0$. Interpret: does a devaluing local currency make the local-currency bond, expressed in hard currency, look safer or riskier than the equivalent hard-currency bond of the same issuer?

Exercise 23.3 (Small-jump limit, not from the course). Expand $P_e(0, T)$ of eq. (23.10) to first order in J around $J = 0$ and show $P_e(0, T) \approx S_0 e^{-hT} (1 - hTJ)$. Compare with $P_u(0, T) = e^{-hT}$ and give a one-line rule of thumb for the size of the quanto adjustment to the credit spread (defined as $-\frac{1}{T} \log P$) when J is small.

23.4.1 The replicating portfolio

Sell one U bond and buy $N_E(t) = P_u(t, T)/P_e(t, T)$ E -bonds so that the portfolio costs 0 at every rebalancing date; by eq. (23.10) (evaluated at t instead of 0, since proposition 23.7 only uses the survival probability from t to T , hence holds for any $t < \tau$),

$$N_E(t) = \frac{P_u(t, T)}{P_e(t, T)} = \frac{e^{-h(T-t)}}{S_t e^{-he^J(T-t)}} = \frac{1}{S_t} \exp[h(T-t)(e^J - 1)]. \quad (23.11)$$

Proof. $\Pi_t = -P_u(t, T) + N_E(t)P_e(t, T) = -P_u(t, T) + P_u(t, T) = 0$ by construction of $N_E(t)$, for every $t < \tau$ (not only $t = 0$), because

Proposition 23.8 (The portfolio is a hedge). The portfolio $\Pi_t = -P_u(t, T) + N_E(t)P_e(t, T) \equiv 0$ for every $t < \tau$; if default occurs, both bonds are worth 0 (zero recovery) so $\Pi_\tau = 0$ as well. Hence the portfolio replicates the (trivial) zero payoff in every state, and the hedge ratio [eq. \(23.11\)](#) is the unique arbitrage-free hedge of one bond against the other.

This was checked numerically: with the parameters of [example 23.6](#) and $T = 3$, $N_E(0) = 0.96187 (= e^{hT(e^J - 1)})$ since $S_0 = 1$); stepping $\Delta t = 0.4$ forward with no default, both P_u and $N_E(t)P_e(t, T)$ were recomputed from [eq. \(23.10\)–\(23.11\)](#) and agreed to machine precision, confirming [proposition 23.8](#) directly rather than only through the algebraic cancellation in the proof.

Remark 23.9 (Why this is harder than static replication). $N_E(t)$ depends on t (through the remaining maturity) and on S_t (through the $1/S_t$ factor): the hedge must be *continuously rebalanced*, unlike a naive static forward hedge. It also depends on h and J : how many EUR bonds are needed to hedge one USD bond depends on both the issuer's credit riskiness and the size of the FX move it would trigger, not only on the current spot or forward FX rate. This is the chapter's central point: an FX forward, which prices only the correlation-free part of the FX/credit relationship, cannot replicate a credit-quanto payoff; only a model with an explicit joint dynamics for (τ, S) can.

Exercise 23.4 (No devaluation, not from the course). Show that if $J = 0$ (no FX jump on default), [eq. \(23.10\)](#) collapses to $P_u(0, T) = P_e(0, T)/S_0 = e^{-hT}$ and the hedge ratio [\(23.11\)](#) collapses to $N_E(t) = 1/S_t$ – the ordinary FX conversion, with no quanto correction, as expected when default carries no currency information.

Exercise 23.5 (Monte Carlo verification, not from the course). Reproduce [example 23.6](#) in Python: simulate $N = 10^6$ exponential default times with hazard rate $h = 0.08$, $J = -0.15$, $T = 2$, compute S_T from [eq. \(23.7\)](#), and check numerically that $\widehat{\mathbb{E}}[S_T] \approx 1$, $\widehat{\mathbb{E}}[\mathbf{1}_{\{\tau > T\}}] \approx P_u(0, T)$ and $\widehat{\mathbb{E}}[S_T \mathbf{1}_{\{\tau > T\}}] \approx P_e(0, T)/S_0$ from [eq. \(23.10\)](#).

23.5 Non-zero recovery

If the recovery rate is $R \in (0, 1)$, the bond payoffs at T become 1 (USD, resp. EUR) if $\tau > T$ and R if $\tau \leq T$.

Proposition 23.10 (Bond prices with recovery, reconstruction).

$$P_u^R(0, T) = R + (1 - R)e^{-hT}, \quad P_e^R(0, T) = S_0 \left[R + (1 - R)e^{-he^J T} \right]. \quad (23.12)$$

The two formulas in [eq. \(23.12\)](#) are strikingly parallel: each is $R + (1 - R)$ times the corresponding zero-recovery survival probability, with h replaced by he^J in the EUR case. This was checked both by numerical quadrature of the integral above and by Monte Carlo (parameters as in [example 23.6](#), $R = 0.25$): $P_u^R = 0.89553$ (formula) vs. 0.89544 (MC), $P_e^R = 0.92112$ (formula, matching the quadrature to 6 digits) vs. 0.92102 (MC).

Remark 23.11 (Exact replication is lost). With $R > 0$ the ratio $P_u^R(t, T)/P_e^R(t, T)$ no longer has the clean multiplicative form of [eq. \(23.11\)](#) (the R and $(1 - R)e^{-h(\cdot)T}$ terms scale differently with S_t and with $T - t$), so the zero-cost ratio P_u^R/P_e^R no longer matches

[eq. \(23.10\)](#) depends on t only through the remaining maturity $T - t$ and through S_t , and $N_E(t)$ was defined by dividing them; the S_t dependence cancels exactly between $P_e(t, T)$'s explicit S_t factor and $N_E(t)$'s $1/S_t$ factor. At default both bonds default to 0 recovery, so $\Pi_\tau = 0$ trivially. What makes this a genuine hedge, and not a tautology, is that the rebalancing is *self-financing*: before default $d \log P_u(t, T) = h dt$ and $d \log P_e(t, T) = d \log S_t + he^J dt = h dt$ (by [eq. \(23.4\)](#) with the compensator drift), and both bonds jump to 0 at τ , so $dP_u/P_u = dP_e/P_e = h dt - dN_t$. Hence $-dP_u + N_E dP_e = 0$: the gains of the position are identically zero, and the $N_E(t)$ E-bonds, rebalanced at no cost, replicate one U-bond. $\square$

Proof. $P_u^R(0, T) = \mathbb{E}[\mathbf{1}_{\{\tau > T\}} + R\mathbf{1}_{\{\tau \leq T\}}] = \mathbb{P}(\tau > T) + R\mathbb{P}(\tau \leq T) = e^{-hT} + R(1 - e^{-hT})$, which rearranges to the stated form. For $P_e^R(0, T)/S_0 = \mathbb{E}[e^{X_T} \mathbf{1}_{\{\tau > T\}} + R e^{X_T} \mathbf{1}_{\{\tau \leq T\}}]$ with $X_T = \log(S_T/S_0)$ from [eq. \(23.7\)](#): the first term is $P_e(0, T)/S_0 = e^{-he^J T}$ by [proposition 23.7](#). The second term integrates over the default density $\varphi(0, \tau) = he^{-h\tau}$ of [proposition 23.2](#), using $X_T = h(1 - e^J)\tau + J$ on $\{\tau \leq T\}$:

$$R \int_0^T e^{h(1 - e^J)\tau + J} h e^{-h\tau} d\tau = R h e^J \int_0^T e^{-he^J \tau} d\tau = R h e^J \cdot \frac{1 - e^{-he^J T}}{he^J} = R(1 - e^{-he^J T}).$$

Adding the two terms, $P_e^R(0, T)/S_0 = e^{-he^J T} + R(1 - e^{-he^J T}) = R + (1 - R)e^{-he^J T}$. $\square$

the two bonds' jumps at default, and a portfolio of the two bonds alone is no longer a hedge. The model is still complete (one jump, one risky hedge instrument): holding $N_E(t) = \Delta P_u^R / \Delta P_e^R$ E -bonds, the ratio of the jumps at default ($\Delta P_u^R = R - P_u^R(t, T)$, $\Delta P_e^R = RS_t e^J - P_e^R(t, T)$), plus the USD cash needed to match the value, replicates the U -bond, because both prices are $\mathbb{Q}$ -martingales and their drifts are fixed by their jumps (reconstruction, not in the slides). The slides stop here ("even in such a simple model, hedging is not trivial"); [section 23.8](#) sketches the further step (adding a diffusive component) proposed as a final-project topic.

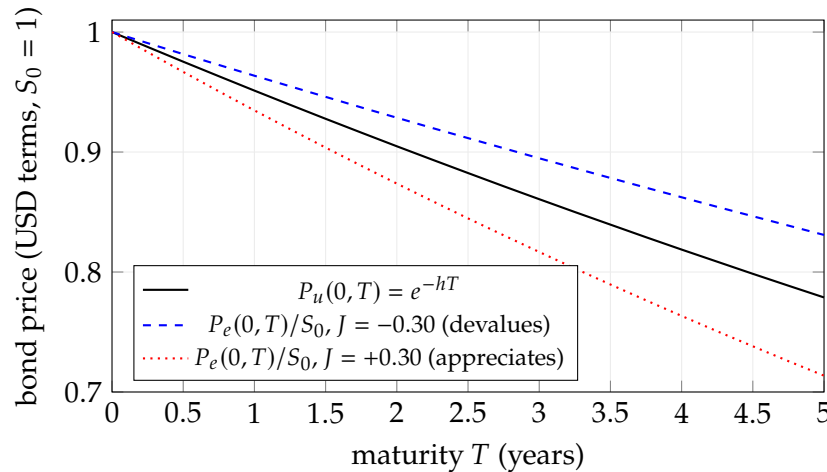


Figure 23.1: Zero-recovery bond prices (23.10), $h = 0.05$. A currency that devalues on default ($J < 0$) makes the local-currency bond look *less* risky in hard-currency terms ($P_e > P_u$); a currency that appreciates on default ($J > 0$, unusual but not impossible for a safe-haven funding currency) makes it look *more* risky ($P_e < P_u$). Coordinates computed for these notes in Python.

Exercise 23.6 (Recovery limits, not from the course). Check that [eq. \(23.12\)](#) reduces to [eq. \(23.10\)](#) when $R = 0$, and to $P_u^R = P_e^R/S_0 = 1$ for every T when $R = 1$ (a bond that always pays its notional regardless of default is riskless). Explain why the second limit no longer depends on h or J .

23.6 Change of numeraire view

The pricing formula (23.9), $P_e(0, T)/S_0 = \mathbb{E}[e^{X_T} \mathbf{1}_{\{\tau > T\}}]$, can be read as a change of numeraire in the sense of [proposition 8.17](#). Take $N_1(t) = S_t$ (EUR converted to USD, a traded USD asset since interest rates are 0) and $N_2(t) = 1$ (USD cash). By [proposition 8.17](#),

$$\mathbb{E}_0^{\mathbb{Q}}[Y_T] = \frac{N_1(0)}{N_2(0)} \mathbb{E}_0^{\mathbb{Q}_S} \left[Y_T \frac{N_2(T)}{N_1(T)} \right] \implies \mathbb{E}_0^{\mathbb{Q}}[S_T Y_T] = S_0 \mathbb{E}_0^{\mathbb{Q}_S}[Y_T],$$

where $\mathbb{Q}_S$ is the measure associated with numeraire S (the “EUR risk-neutral measure expressed in USD units,” by the Radon–Nikodým derivative $d\mathbb{Q}_S/d\mathbb{Q}|_{\mathcal{F}_T} = S_T/S_0$, since $S_t/1$ is itself the $\mathbb{Q}$ -martingale that must be used in [proposition 8.17](#)). With $Y_T = \mathbf{1}_{\{\tau > T\}}$,

$$P_e(0, T) = S_0 \mathbb{P}_0^{\mathbb{Q}}(\tau > T) \Big|_{\text{naive}} \neq P_e(0, T) = S_0 \mathbb{Q}_S(\tau > T), \quad (23.13)$$

and [proposition 23.7](#) says precisely that $\mathbb{Q}_S(\tau > T) = e^{-he^J T} \neq e^{-hT} = \mathbb{Q}(\tau > T)$: default is *less* likely under the EUR-numeraire measure $\mathbb{Q}_S$ than under $\mathbb{Q}$ when $J < 0$ (EUR weakens on default) – because $\mathbb{Q}_S$ tilts probability towards paths on which S_T is large, i.e. towards the surviving, no-jump scenario. This is exactly the same mechanism as the FX-forward row of the numeraire table in [section 8.7](#), $F_{FX}(t, T) = S_{FX}(t)P^F(t, T)/P^D(t, T)$, except that here the “foreign bond” P^F is itself default-risky and its numeraire property breaks down exactly at τ (it is only a valid numeraire, i.e. strictly positive, on $\{t < \tau\}$) – the same obstruction [section 8.7](#) met when trying to use the risky annuity $\tilde{A}_j$ as a numeraire for the CDS survival measure, and resolved there the same way: work with $\mathbf{1}_{\{t < \tau\}}$ times the relevant traded quantity, which *is* a genuine $\mathbb{Q}$ -martingale even though the numeraire itself is not always positive.

INTUITION: Quanto adjustment = change of measure

Every “quanto adjustment” in this course – FX quanto options, quanto CDS here – is a change of numeraire from the natural currency’s risk-neutral measure to the payoff currency’s risk-neutral measure. When the underlying (here, the default indicator) is independent of the FX rate, the two measures agree and quanto pricing reduces to discounting by the forward FX rate. When they are correlated (through the jump J), the Radon–Nikodým derivative S_T/S_0 tilts the default probability, and no amount of forward-FX-based reasoning like [section 23.2.1](#) can substitute for computing the expectation under the correct measure.

23.7 Quanto credit default swaps and other applications

A credit default swap (CDS) is a bilateral insurance contract on the issuer’s default: the protection buyer pays a periodic spread (the premium leg) until default or maturity, and receives $1 - R$ per unit notional from the protection seller if default occurs (the protection leg). A *quanto CDS* is a CDS whose notional and premium are denominated in a currency different from the reference obligation’s natural currency – e.g. a CDS on Brazil denominated in Brazilian reais rather than in USD. Exactly the mechanism of this chapter applies: if the reference entity’s local currency devalues on default (as is typical for a sovereign or an issuer whose revenues are tied to its home currency), then a protection payment fixed in local currency is worth less, in hard-currency terms, precisely when it is needed – the quanto CDS spread must be adjusted relative to the hard-currency CDS spread by a factor analogous to e^J in [eq. \(23.10\)](#), and the correct hedge ratio between the two currencies’ CDS again requires the jump-on-default FX model rather than a simple FX forward. Historically, quanto CDS were often priced by simply converting the hard-currency CDS spread at the current or forward FX rate, ignoring the correlation; this reproduces neither the observed market prices nor a valid hedge ratio, which is exactly why a model with the jump feature became necessary in practice.

CASE STUDY: Sovereign quanto CDS baskets

During the European sovereign crisis (2010–2012), USD- and EUR-denominated CDS on peripheral euro-area sovereigns (Italy, Spain, Portugal) diverged, giving rise to an actively quoted “quanto” basis. Market participants used models of the type developed here (often with stochastic hazard rates and a jump-diffusion FX process, as sketched in [section 23.8](#)) to convert between the USD and EUR curves and to size cross-currency hedges, essentially betting on how far the euro would fall if a member state defaulted

or exited the currency union.

23.8 Towards a more realistic model

The pure-jump FX process of [eq. \(23.6\)](#) has no day-to-day volatility except at τ : a natural next step (posed on the slides as a final-project topic) is to add a diffusive component,

$$\frac{dS_t}{S_t} = h(1 - e^J) \mathbf{1}_{\{t < \tau\}} dt + (e^J - 1) dN_t + \sigma dW_t, \quad (23.14)$$

a genuine jump-diffusion in the sense of [section 17.5](#) plus [section 23.3.2](#) combined: away from τ , S_t behaves like the ordinary geometric Brownian motion of [eq. \(23.1\)](#); at τ , it also jumps by e^J . Pricing bonds in this model requires Itô's formula with *both* a diffusive and a jump term (the sum of the Brownian second-derivative correction of [theorem 17.10](#) and the finite-difference jump correction of [lemma 23.4](#)), and hedging now needs (at least) two instruments – an FX forward for the diffusive risk and a position sensitive to the jump (e.g. a short-dated deep out-of-the-money FX option, or the bonds themselves) – since the model has two independent sources of randomness (W and N), matching the “number of hedging instruments = number of stochastic sources” completeness criterion of [section 15.4](#). Once a closed-form or PDE solution is obtained, it can (and should) be checked against a Monte Carlo simulation that simulates τ , W , and hence S_T , jointly, exactly as done at smaller scale in [example 23.6](#). In the more realistic desk models this course only sketches, both the hazard rate and interest rates are themselves stochastic and correlated with FX (“Real Life”, slide 29); [chapter 24](#) develops the term-structure machinery needed to make hazard rates stochastic in a no-arbitrage way, and correlating them with FX and with rates is a natural (unaddressed here) extension of the multi-factor models built there. Because defaults are rare, naive Monte Carlo is inefficient (most paths never see a jump); practitioners typically integrate the payoff analytically over the default-time density $\varphi(0, \tau)$ (as done by hand in [proposition 23.10](#)) and simulate only the diffusive factors (FX, rates, hazard-rate level) by Monte Carlo, mixing the two techniques.

23.9 Girsanov and the diffusive quanto adjustment

It is worth relating [eq. \(23.14\)](#) back to the purely diffusive quanto adjustment familiar from FX options (not covered in the 2013 lecture, but a standard consequence of [section 17.9](#)). If a foreign asset V_t (say, a EUR equity index) follows a geometric Brownian motion under the foreign risk-neutral measure $\mathbb{Q}^F$ and the FX rate S_t one under the domestic measure $\mathbb{Q}^D$,

$$\frac{dV_t}{V_t} = r_f dt + \sigma_V dW_t^F, \quad \frac{dS_t}{S_t} = (r_d - r_f) dt + \sigma_S d\tilde{W}_t^D, \quad dW^F d\tilde{W}^D = \rho dt,$$

then pricing a USD-denominated (*quantoed*) payoff on V_T requires switching from $\mathbb{Q}^F$ to the domestic measure $\mathbb{Q}^D$. Girsanov's theorem ([section 17.9](#)) says this changes the drift of V by $-\rho\sigma_V\sigma_S$: under $\mathbb{Q}^D$, V grows at $r_f - \rho\sigma_V\sigma_S$ rather than at r_f . This is the textbook “quanto drift adjustment,” and it is the diffusive sibling of the jump compensator [proposition 23.3](#): in both cases, expressing a foreign payoff in domestic currency under the correlated FX process shifts the effective drift of the payoff by an amount proportional to the FX/payoff correlation – $-\rho\sigma_V\sigma_S$ in the Brownian case, $h(e^J - 1)$ (via the jump-driven change of measure of [section 23.6](#)) in the jump case. [eq. \(23.14\)](#) simply has both effects at once.

KEY FORMULAS

Survival:	$\mathbb{P}_t(\tau > T) = e^{-h(T-t)} \quad (\text{eq. (23.2)})$
FX dynamics:	$dS_t/S_t = h(1 - e^J)\mathbf{1}_{\{t < \tau\}} dt + (e^J - 1) dN_t \quad (\text{eq. (23.6)})$
Zero-recovery bonds:	$P_u(0, T) = e^{-hT}, \quad P_e(0, T) = S_0 e^{-he^J T} \quad (\text{eq. (23.10)})$
Hedge ratio:	$N_E(t) = \frac{1}{S_t} \exp[h(T - t)(e^J - 1)] \quad (\text{eq. (23.11)})$
Bonds, recovery R :	$P_u^R(0, T) = R + (1 - R)e^{-hT}$
	$P_e^R(0, T) = S_0[R + (1 - R)e^{-he^J T}] \quad (\text{eq. (23.12)})$

Chapter 24

The HJM Model for Interest Rates and Credit

SOURCES FOR THIS CHAPTER

Lecture	Speaker	Content
2013 L24	D. Gorokhov (Morgan Stanley)	Monte Carlo pricing motivated by dynamic hedging of stock options (Black–Scholes recap, Ito’s lemma “under the microscope”); why yield-curve dynamics is harder than a point process; the classic short-rate models (Ho–Lee, Hull–White, CIR) listed without derivation; the HJM forward-rate SDE, the no-arbitrage drift condition derived on the slides via the zero-coupon bond price and Ito’s lemma; a practical recipe for pricing with HJM; credit derivatives (survival probabilities, hazard rates, CDS pricing), the hazard-rate HJM analogue and its drift condition left as an exercise on the slides; structured notes as a worked motivation for simulating rates, equity and credit jointly.

Files. MIT18_S096F13_lecnote24.pdf (34 slides). **Problem sets.** None assigned on this topic in either year. The exercises (each after the section it practises) are marked (*not from the course*). **Additions of these notes** (marked as such where they occur): the full derivation of the HJM drift condition from Ito’s lemma with every step spelled out (the slides state the result and sketch the computation in compressed notation); the explicit recovery of Ho–Lee, Vasicek/Hull–White and the impossibility of an exact CIR recovery as special cases of the HJM volatility structure; the real-world/risk-neutral distinction via Girsanov’s theorem ([chapter 17](#)); a symmetric full derivation of the credit (hazard-rate) drift condition, of which the slides only pose the no-correlation special case as an exercise; and all numerical checks.

Motivation

[Chapter 6](#) built today’s discount curve $Z(0, T)$ from swap quotes and priced linear products from it. [Chapter 18](#) showed how to write a stochastic differential equation for a single number, the short rate r_t , and [section 18.3.3](#) used the Ornstein–Uhlenbeck process for this purpose. But a bank does not trade one number: it trades against the whole forward curve, and pricing an option on a swap fifteen years out requires knowing how the *entire curve* can move between now and then, not just its short end. Heath, Jarrow and Morton’s 1992 answer is disarmingly simple: write down a stochastic differential equation directly for the forward-rate curve $T \mapsto f(t, T)$, one Brownian increment dB_t shared by all maturities but a maturity-dependent volatility $\sigma(t, T)$, and ask what no-arbitrage requires. The answer – the *HJM drift condition* – says that the drift of $f(t, T)$ is not free: it is pinned down by $\sigma(t, \cdot)$ alone, exactly as the drift of a stock disappeared from the Black–Scholes equation in [chapter 15](#) once the stock was used to hedge. Four questions organise the chapter:

- How do you write an SDE for a curve, and what does absence of arbitrage force on its drift? Sections 24.1 and 24.2.
- Is this new, or have we seen it before? Ho–Lee, Vasicek/Hull–White and (almost) CIR are the special cases where the whole curve dynamics collapses onto the short rate alone (section 24.3).
- Does the condition actually work? A Monte Carlo simulation of the full curve is checked against the known bond price (section 24.4).
- Does the same idea price credit risk? Replace the discount curve by the survival-probability curve and the forward rate by the forward hazard rate; the same drift argument returns, term by term (section 24.5).

For a physicist, the HJM equation is a stochastic field theory in one spatial dimension (maturity T) driven by a single noise; the no-arbitrage condition is a compatibility constraint between the drift and covariance kernels, of the same flavour as a fluctuation–dissipation relation.

24.1 The forward-rate curve as a stochastic field

24.1.1 Notation: two clocks

Section 6.6 built the discount curve $Z(0, T)$ observed *today* ($t = 0$) for every future maturity T . The instantaneous forward rate seen today is $f(0, T) = -\partial_T \ln Z(0, T)$ (proposition 6.14, continuous-time limit). The HJM model asks: as calendar time t moves forward, how does the *whole function* $T \mapsto f(t, T)$, $T \geq t$, evolve? This introduces two clocks, exactly as section 18.1 of chapter 18 introduces $\mu(t, x)$: t is the running (stochastic) time and T is a fixed label, the maturity, carried along unchanged as t increases. This is the one genuinely new feature relative to a stock price S_t , which has no second label [► 2013 L24 @ 45:22].

Definition 24.1 (Forward-rate curve, zero-coupon bond, short rate). For each $t \geq 0$, $f(t, \cdot) : [t, \infty) \rightarrow \mathbb{R}$ is the instantaneous forward curve observed at time t : $f(t, T)$ is the rate, contracted at t , for infinitesimal borrowing starting at T . The time- t price of the zero-coupon bond maturing at $T \geq t$ is

$$Z(t, T) = \exp\left(-\int_t^T f(t, s) \, ds\right) =: e^{-X_t^T}, \quad X_t^T := \int_t^T f(t, s) \, ds. \quad (24.1)$$

The short rate is the forward curve evaluated at its own running end, $r_t := f(t, t)$.

[► 2013 L24 @ 46:04] This is exactly (6.4) of chapter 6 taken to the continuous-time, instantaneous limit, now allowed to be random and re-observed at every t ; $Z(0, T)$ recovers today's discount curve, and $Z(t, t) = 1$ for every t (a bond is worth its notional at its own maturity).

24.1.2 The HJM stochastic differential equation

The starting postulate of the model, parallel to $dS_t = \mu S_t dt + \sigma S_t dB_t$ for a stock, is that for every fixed T the process $t \mapsto f(t, T)$ ($t \leq T$) is an Ito process

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dB_t, \quad f(0, T) = f_0(T) \text{ given}, \quad (24.2)$$

► 2013 L24 @ 45:22] where B is a single standard Brownian motion (one-factor case; [remark 24.4](#) extends this to several factors) and $\alpha(t, \cdot), \sigma(t, \cdot)$ are, for each t , deterministic or path-dependent functions of $T \geq t$, adapted in t and smooth enough in T to differentiate and integrate freely (technical conditions as in [theorem 18.1](#), not discussed on the slides). One equation (24.2) for every T is really a continuum of coupled SDEs, all driven by the *same* B_t : a single shock moves the whole curve, with an amount and a sign that can differ by maturity through $\sigma(t, T)$.

INTUITION: A field theory with one noise

Think of $f(t, \cdot)$ as a field on the half-line $T \geq t$ and (24.2) as its (stochastic) equation of motion. Because a single B_t drives every T , the instantaneous correlation between any two points of the curve is ± 1 (they are driven by the same shock); what makes the curve twist and not just shift in parallel is that $\sigma(t, T)$ can change sign or magnitude with T . A short-lived, front-loaded $\sigma(t, T)$ (large near $T = t$, decaying in $T - t$) gives a curve that mostly moves at the short end, exactly the "level, slope, curvature" factor decomposition of [chapter 7](#).

24.1.3 From the curve SDE to the bond-price SDE

Everything the model needs to price a derivative – the growth rate of tradable zero-coupon bonds – comes from differentiating $Z(t, T) = e^{-X_t^T}$. This requires the stochastic differential of $X_t^T = \int_t^T f(t, s) ds$, an integral over s of a process that is itself stochastic in t ; this is the one calculation the slides perform in compressed notation ► 2013 L24 @ 1:02:41], spelled out in full here.

Proof. Write $X_t^T = \int_0^T f(t, s) ds - \int_0^t f(t, s) ds$ is not quite right because the lower limit is also t ; instead fix t and differentiate $X_{t+h}^T - X_t^T$ directly. Split

$$X_{t+h}^T - X_t^T = \underbrace{\int_{t+h}^T [f(t+h, s) - f(t, s)] ds}_{(I)} - \underbrace{\int_t^{t+h} f(t, s) ds}_{(II)}$$

For (II), $f(t, \cdot)$ is continuous in s near $s = t$, so (II) = $f(t, t) h + o(h) = r_t h + o(h)$. For (I), insert (24.2) pointwise in s and interchange $\int_{t+h}^T (\cdot) ds$ with the (finite, over $s \in [t+h, T]$) sum of infinitesimal increments in t :

$$(I) = \int_t^T \alpha(t, s) ds h + \int_t^T \sigma(t, s) ds \Delta B_t + o(h) + O(h) \cdot h,$$

where the upper limit was extended from $t+h$ to t at a cost $O(h^2)$, and $\Delta B_t = B_{t+h} - B_t$ is the same shock for every s (the interchange of the ds -integral and the stochastic increment is the stochastic Fubini theorem: valid here because $\sigma(t, s)$ is bounded and jointly measurable on the compact range $s \in [t, T]$). Collecting the $O(h)$ and $O(\sqrt{h})$ terms and passing to the Ito differential gives (24.3). $\square$

Lemma 24.2 (Differential of the integrated forward curve). Under (24.2) and Leibniz's rule for differentiating under the integral sign (applied formally to the Ito differential, justified

by Fubini for the drift and by stochastic Fubini for the martingale part – both quoted, not proved in class),

$$dX_t^T = -f(t, t) dt + \left(\int_t^T \alpha(t, s) ds \right) dt + \left(\int_t^T \sigma(t, s) ds \right) dB_t. \quad (24.3)$$

Applying Ito's formula (section 17.5) to $Z_t^T = e^{-X_t^T}$, a smooth function of the single Ito process X^T ,

$$\begin{aligned} dZ_t^T &= -e^{-X_t^T} dX_t^T + \frac{1}{2} e^{-X_t^T} (dX_t^T)^2 \\ &= Z_t^T \left[-dX_t^T + \frac{1}{2} (dX_t^T)^2 \right] \\ &= Z_t^T \left[r_t dt - \left(\int_t^T \alpha(t, s) ds \right) dt - \left(\int_t^T \sigma(t, s) ds \right) dB_t + \frac{1}{2} \left(\int_t^T \sigma(t, s) ds \right)^2 dt \right], \end{aligned} \quad (24.4)$$

using $(dX_t^T)^2 = \left(\int_t^T \sigma(t, s) ds \right)^2 dt$ (the dt and cross terms of (24.3) do not contribute to the quadratic variation, as in section 17.5). This is exactly the slide's computation [► 2013 L24 @ 1:03:52], reproduced here with every substitution shown.

24.2 The HJM no-arbitrage drift condition

24.2.1 Statement and proof

A bond is a tradable asset with no dividend. By the fundamental theorem of asset pricing (section 15.8, proposition 8.17), under the risk-neutral measure $\mathbb{Q}$ every such asset, discounted by the money-market account $\exp(\int_0^t r_s ds)$, is a martingale; equivalently, its instantaneous drift under $\mathbb{Q}$ must equal r_t times its price – exactly the statement that made the stock's drift μ disappear from the Black–Scholes equation in chapter 15. Reading off the dt -coefficient of (24.4) and setting it equal to $r_t Z_t^T$ gives the condition.

Theorem 24.3 (HJM drift condition). *Under the risk-neutral measure $\mathbb{Q}$, for the bond price (24.1) to grow on average at the risk-free rate – $dZ_t^T = r_t Z_t^T dt + (\dots) dB_t$, no dt -drift beyond r_t – the drift and volatility of the forward curve (24.2) must satisfy, for every $t \leq T$,*

$$\alpha^{\mathbb{Q}}(t, T) = \sigma(t, T) \int_t^T \sigma(t, s) ds. \quad (24.5)$$

This matches the slide's result exactly [► 2013 L24 @ 1:04:40]: “ $\alpha_{tT}^{\mathbb{Q}} = \sigma_{tT} \int_t^T \sigma_{ts} ds$ ”. The logic is identical to Black–Scholes: the model started from an *arbitrary* drift α for the forward curve (there was no reason to guess it), and the requirement that hedged, tradable instruments earn the risk-free rate on average removed that freedom entirely. Only $\sigma(t, \cdot)$, the part that can in principle be read off derivative prices (remark 24.7), survives as a free input.

1 COURSE NOTE: Reading the slide's compressed derivation

The 2013 slide (which writes the drift as μ_{tT} instead of $\alpha(t, T)$) states $f(t, t) = r_t$ and writes $dX_t = -f_{tT} dt - \int_t^T df_{ts} ds$, then substitutes (24.2) under the integral sign in one line [► 2013 L24 @ 1:02:41]. The boundary term $-f(t, t) dt$ is right (the integration window $[t, T]$ loses its left end as t advances), but the sign in front of the integral is a typo: it should be $+\int_t^T df(t, s) ds$, as in (24.3). The slide's next line, the bond-price SDE, has the correct signs and agrees with (24.4), so the typo does not propagate.

Proof. Setting the dt -coefficient in (24.4) equal to r_t (after dividing by $Z_t^T \neq 0$) gives

$$r_t - \int_t^T \alpha^{\mathbb{Q}}(t, s) ds + \frac{1}{2} \left(\int_t^T \sigma(t, s) ds \right)^2 = r_t \iff \int_t^T$$

Both sides vanish at $T = t$ (empty integral, and the square of an empty integral), so differentiating both sides in T using the fundamental theorem of calculus on the left and the chain rule on the right,

$$\alpha^{\mathbb{Q}}(t, T) = 2 \cdot \frac{1}{2} \left(\int_t^T \sigma(t, s) ds \right) \cdot \partial_T \left(\int_t^T \sigma(t, s) ds \right) = \sigma(t, T),$$

which is (24.5). $\square$

INTUITION: Why the drift is an integral of the volatility

(24.5) says the risk-neutral drift at T equals the covariance between the shock at T and the shock to *every rate between t and T* , summed up: $\alpha^{\mathbb{Q}}(t, T) = \text{Cov}(\mathrm{d}f(t, T), \mathrm{d}X_t^T) / \mathrm{d}t$ up to the sign convention. A bond price is a (negative) exponential of the accumulated curve, so its convexity (the $\frac{1}{2}(\cdot)^2$ term from Ito's lemma, exactly the second-order term of [example 18.2's](#) microscope) must be compensated by a corresponding upward push in the curve's own drift for the bond's growth rate to stay pinned at r_t . This is the same mechanism as the $-\frac{1}{2}\sigma^2$ correction between the drift of $\ln S_t$ and the drift of S_t in [example 18.3](#).

Exercise 24.1 (not from the course). Verify (24.5) by direct substitution for the Ho–Lee case $\sigma(t, T) \equiv \sigma_0$: show $\int_t^T \alpha^{\mathbb{Q}}(t, s) \mathrm{d}s = \frac{1}{2}\sigma_0^2(T-t)^2$ two ways, first from $\alpha^{\mathbb{Q}}(t, s) = \sigma_0^2(s-t)$ integrated in s , then from $\frac{1}{2}(\int_t^T \sigma_0 \mathrm{d}s)^2$, and confirm they agree.

24.2.2 The real-world drift and Girsanov's theorem

[Theorem 24.3](#) pins down the drift only under $\mathbb{Q}$. Under the physical measure $\mathbb{P}$, Girsanov's theorem ([theorem 17.23](#)) allows an arbitrary (adapted, technically admissible) market price of risk λ_t , with $\mathrm{d}B_t = \mathrm{d}B_t^{\mathbb{Q}} + \lambda_t \mathrm{d}t$, so that

$$\alpha^{\mathbb{P}}(t, T) = \alpha^{\mathbb{Q}}(t, T) - \sigma(t, T) \lambda_t = \sigma(t, T) \int_t^T \sigma(t, s) \mathrm{d}s - \sigma(t, T) \lambda_t, \quad (24.6)$$

exactly as [chapter 21](#) shifts the mean-reversion level of a commodity spot process by a constant market price of risk, and as [section 18.3.3](#) shifts θ in Vasicek's model. Since λ_t is unobservable from bond prices alone (it cancels out of every $\mathbb{Q}$ -priced payoff), pricing derivatives never needs $\alpha^{\mathbb{P}}$: the whole point of risk-neutral valuation, restated once more, is that only $\sigma(t, \cdot)$ and today's curve $f(0, \cdot)$ enter a price.

IN FINANCE: What this buys a trading desk

The slide's practical summary ▶ 2013 L24 @ 1:06:30 is:

1. bootstrap $Z(0, \cdot)$, hence $f(0, \cdot)$, from swap quotes ([section 6.6](#));
2. specify (or calibrate, [remark 24.7](#)) the volatility structure $\sigma(t, T)$;
3. compute the unique arbitrage-free drift from (24.5);
4. simulate (24.2) forward under $\mathbb{Q}$ on a grid of (t, T) pairs, exactly as [chapter 15](#) simulated S_t ;
5. for a derivative paying g at τ , price it as $\mathbb{E}_{\mathbb{Q}}[e^{-\int_0^\tau r_s \mathrm{d}s} g]$, averaging over paths and discounting.

This is the same five-step Monte Carlo recipe of [chapter 15](#), with “simulate the stock” replaced by “simulate the curve”.

Remark 24.4 (Several factors). With n independent Brownian motions $B^{(1)}, \dots, B^{(n)}$ and volatilities $\sigma_1(t, T), \dots, \sigma_n(t, T)$, $\mathrm{d}f(t, T) = \alpha(t, T) \mathrm{d}t + \sum_{i=1}^n \sigma_i(t, T) \mathrm{d}B_t^{(i)}$, the same computation (each σ_i contributing its own square, by independence, as in [eq. \(18.16\)](#) of [chapter 18](#)) gives $\alpha^{\mathbb{Q}}(t, T) = \sum_{i=1}^n \sigma_i(t, T) \int_t^T \sigma_i(t, s) \mathrm{d}s$. Multiple factors (often chosen

as the level/slope/curvature loadings of [chapter 7](#)) let the curve twist in ways a single factor cannot; this is not discussed further in the lecture and not needed for the one-factor numerics of [section 24.4](#).

24.3 Short-rate models as special cases

The 2013 lecture lists the classic short-rate SDEs *before* deriving the HJM condition, without connecting the two [▶ 2013 L24 @ 1:02:13](#):

$$\begin{aligned} dr_t &= \theta(t) dt + \sigma dB_t \text{ (Ho–Lee)}, & dr_t &= \kappa(\theta(t) - r_t) dt + \sigma dB_t \text{ (Hull–White/Vasicek)}, \\ & & dr_t &= \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dB_t \text{ (CIR)}. \end{aligned} \quad (24.7)$$

The connection is the content of this section (an addition of these notes): a short-rate model is *equivalent* to the HJM framework precisely when the chosen $\sigma(t, T)$ makes the whole forward curve a deterministic function of r_t alone, so that simulating r_t (a single number) is enough to reconstruct $f(t, T)$ for every T – the entire motivation, in [section 24.1](#), for needing a curve model.

Proposition 24.5 (Ho–Lee is HJM with constant volatility). Take $\sigma(t, T) \equiv \sigma_0$ (constant, independent of t and T) in (24.2). Then (24.5) gives $\alpha^{\mathbb{Q}}(t, T) = \sigma_0^2(T - t)$, and

$$f(t, T) = f(0, T) + \sigma_0^2 t \left(T - \frac{t}{2}\right) + \sigma_0 B_t, \quad r_t = f(t, t) = f(0, t) + \frac{1}{2}\sigma_0^2 t^2 + \sigma_0 B_t, \quad (24.8)$$

which satisfies the Ho–Lee SDE $dr_t = \theta(t) dt + \sigma_0 dB_t$ with $\theta(t) = \partial_t f(0, t) + \sigma_0^2 t$ (the curve’s own slope plus a convexity correction).

Ho–Lee is the simplest possible HJM model and the one the slide’s zero-coupon bond computation implicitly matches when σ is taken constant [▶ 2013 L24 @ 1:03:00](#) (the algebra there is written for general $\sigma(t, T)$, but the arithmetic simplifies to Ho–Lee’s classic r_t -normal, curve-fitting form when σ is constant).

Proof. Integrate (24.2) in t at fixed T : $f(t, T) = f(0, T) + \int_0^t \sigma_0^2(T - u) du + \sigma_0 B_t = f(0, T) + \sigma_0^2 t(T - t/2) + \sigma_0 B_t$, which is (24.8); set $T = t$ for r_t . Differentiating r_t in t (Ito, with B contributing dB_t and the deterministic part contributing its derivative) gives exactly $dr_t = [\partial_t f(0, t) + \sigma_0^2 t] dt + \sigma_0 dB_t$. $\square$

Proof. The volatility factorises as $\sigma(t, T) = e^{-a(T-t)}\sigma_0$, of the separable form $\sigma(t, T) = g(T)/g(t) \cdot \sigma_0$ with $g(u) = e^{au}$: this is exactly the structure that made the Ornstein–Uhlenbeck integrating-factor trick of [section 18.3](#) work, and it is why exponential HJM volatility is the one choice that keeps the short rate Markovian (reducible to r_t alone) among mean-reverting models. Writing $f(t, T) = f(0, T) + \int_0^t \alpha^{\mathbb{Q}}(u, T) du + \sigma_0 \int_0^t e^{-a(T-u)} dB_u$ and specialising to $T = t$ gives, after differentiating in t (details: differentiate $r_t = f(t, t)$ exactly as in the proof of [lemma 24.2](#), tracking both the moving argument and the stochastic integral’s upper limit), the stated SDE; the flat-curve case matches [section 18.3.3](#) with $\kappa = a$ after evaluating the $t \rightarrow \infty$ limit of the bracket, $a\theta_\infty = \partial_T f(0, \cdot)|_{\text{flat}} + ar_0 + \sigma_0^2/(2a) = ar_0 + \sigma_0^2/(2a)$. $\square$

Proposition 24.6 (Vasicek/Hull–White is HJM with exponential volatility). Take $\sigma(t, T) = \sigma_0 e^{-a(T-t)}$ for constants $\sigma_0, a > 0$ (mean-reversion speed a). Then $\int_t^T \sigma(t, s) ds = \frac{\sigma_0}{a}(1 - e^{-a(T-t)})$ and (24.5) gives $\alpha^{\mathbb{Q}}(t, T) = \sigma(t, T) \cdot \frac{\sigma_0}{a}(1 - e^{-a(T-t)})$. The short rate $r_t = f(t, t)$

then solves the Hull–White SDE

$$dr_t = \left[\partial_T f(0, t) + a(f(0, t) - r_t) + \frac{\sigma_0^2}{2a}(1 - e^{-2at}) \right] dt + \sigma_0 dB_t, \quad (24.9)$$

which is (18.8) (section 18.3) with a time-dependent, curve-fitting mean level; for a flat initial curve $f(0, T) \equiv r_0$ it approaches, for $t \gg 1/a$, the stationary Vasicek SDE $dr_t = a(\theta_\infty - r_t) dt + \sigma_0 dB_t$ with $\theta_\infty = r_0 + \sigma_0^2/(2a^2)$.

Remark 24.7 (Calibrating $\sigma(t, T)$). As in the equity case (chapter 15), the volatility σ cannot be chosen freely and then forgotten: it is implied from the market prices of liquid options, here swaptions and caps/floors [► 2013 L24 @ 1:06:56]. An exponential $\sigma(t, T) = \sigma_0 e^{-a(T-t)}$ has only two free parameters and typically cannot fit the whole swaption grid (all expiries and tenors) simultaneously; richer parametrisations, or several factors (remark 24.4), trade this off against tractability, exactly the local-versus-global tension of section 6.6’s interpolation choice.

Exercise 24.2 (not from the course). Complete the differentiation step skipped in the proof of proposition 24.5: starting from (24.8), apply Ito’s formula to $r_t = f(0, t) + \frac{1}{2}\sigma_0^2 t^2 + \sigma_0 B_t$ directly (there is no hidden randomness in the deterministic part) and confirm the stated Ho–Lee SDE, including the exact form of $\theta(t)$ in terms of the initial curve.

Exercise 24.3 (not from the course). Fill in the differentiation step of proposition 24.6: write $r_t = f(0, t) + \int_0^t \alpha^Q(u, t) du + \sigma_0 \int_0^t e^{-a(t-u)} dB_u$ (the $T = t$ specialisation, tracking the moving upper limit of both integrals as in lemma 24.2), differentiate in t , and verify (24.9).

2 COURSE NOTE: CIR is not an exact HJM special case

The slides list CIR alongside Ho–Lee and Hull–White with no further comment [► 2013 L24 @ 1:02:13], but CIR does *not* embed into one-factor HJM the way the other two do: HJM’s volatility $\sigma(t, T)$ in (24.2) is a function of (t, T) only, never of the (random) curve level, whereas CIR’s $\sigma\sqrt{r_t}$ (section 18.8) is level-dependent. A level-dependent-volatility HJM model exists (it is the starting point of “Markov-functional” and “quadratic Gaussian” short-rate models, not discussed here) but the clean closed-form drift (24.5), which relied only on T -integrals of a curve known at time t , no longer separates from the state in the same way. CIR is best read as a short-rate model in its own right (section 18.8), historically motivated independently of HJM by the wish for a nonnegative rate, rather than as an HJM special case.

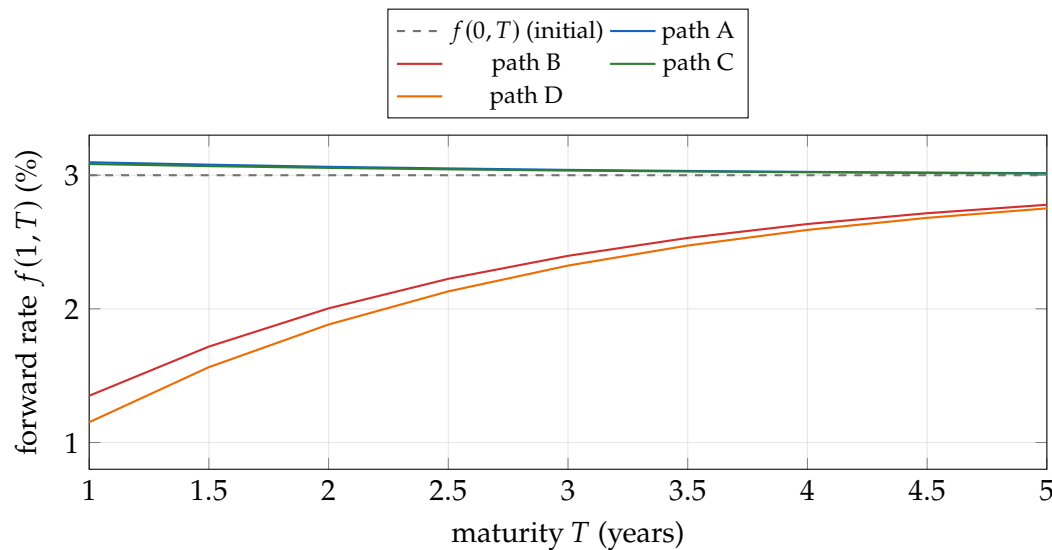
24.4 Numerical check of the drift condition

The cleanest test of theorem 24.3 is direct: simulate the full curve under the computed drift, and check that the resulting bond price matches the one implied by today’s curve – *without ever solving a bond-pricing PDE*, purely by discounting simulated paths, exactly the logic of the Monte Carlo check of the Black–Scholes formula suggested on the slides [► 2013 L24 @ 38:48](own computation; not in the lecture).

Example 24.8 (Simulating the HJM curve and pricing a bond by Monte Carlo). Take the exponential-volatility model of proposition 24.6 with $\sigma_0 = 1.2\%$, $a = 0.5$, a flat initial curve $f(0, T) \equiv 3\%$, and simulate $f(t, T)$ on a daily grid ($dt = 1/252$) up to $t = T_h = 2$ years, using the closed-form drift $\alpha^Q(t, T) = \sigma(t, T) \cdot \frac{\sigma_0}{a}(1 - e^{-a(T-t)})$ from proposition 24.6 and an Euler–Maruyama step (section 18.7) shared by every remaining maturity on the grid at each time step. Over 20,000 paths, the Monte Carlo bond price $\hat{Z}(0, T_h) = \mathbb{E}_Q[e^{-\int_0^{T_h} r_s ds}]$ and its analytic value $Z(0, T_h) = e^{-\int_0^{T_h} f(0, T_h) ds}$ (flat curve) agree to within Monte Carlo error:

$$Z(0, 2)_{\text{theory}} = e^{-0.03 \times 2} = 0.941765, \quad \hat{Z}(0, 2)_{\text{MC}} = 0.9417$$

As a second, independent check, proposition 24.6 predicts that the short rate at $t = 1$ is Gaussian



with variance $\frac{\sigma_0^2}{2a}(1 - e^{-2a}) = 9.10 \times 10^{-5}$; the simulated cross-sectional variance of r_1 across the same 20,000 paths is 9.07×10^{-5} .

Figure 24.1: Four simulated forward curves $T \mapsto f(1, T)$ one year into the simulation of [example 24.8](#) ($\sigma_0 = 1.2\%$, $a = 0.5$, flat initial curve at 3%), against the initial curve $f(0, T)$. All curves are driven by the same Brownian path within a curve but differ across paths; the exponential vol makes the short end move most (as $\sigma(t, T) \rightarrow \sigma_0$ when $T \rightarrow t$) while the long end is comparatively anchored (as $\sigma(t, T) \rightarrow 0$ when $T - t \rightarrow \infty$), the familiar level/slope pattern of [chapter 7](#).

Exercise 24.4 (not from the course). Reproduce the Monte Carlo check of [example 24.8](#) in your own code: simulate the exponential-volatility HJM curve on a daily grid, compute $\hat{Z}(0, T_h)$ for $T_h = 1, 2, 5$ years, and compare with $e^{-f_0 T_h}$. Then break the drift condition on purpose (set $\alpha \equiv 0$, “drift-free” forward rates) and show that the simulated bond price no longer matches $Z(0, T_h)$: absence of arbitrage is not automatic, it is exactly what [\(24.5\)](#) enforces.

24.5 Extension to credit: hazard rates

24.5.1 Survival probabilities as a second discount curve

[Proposition 8.17](#) priced default-free cash flows with $Z(t, T)$. A defaultable bond needs a second curve: the market-implied *survival probability* $p(t, T)$, the risk-neutral probability, seen at t , that a reference entity has not defaulted by T [[► 2013 L24 @ 1:12:45](#)]. Exactly as $Z(t, T)$ was parametrised by instantaneous forward rates, $p(t, T)$ is parametrised by instantaneous *forward hazard rates* $h(t, T)$:

$$p(t, T) = \exp\left(-\int_t^T h(t, s) ds\right), \quad p(t, t) = 1, \quad (24.10)$$

the exact analogue of [\(24.1\)](#). A zero-recovery defaultable zero-coupon bond – paying \$1 at T if and only if the reference entity has not defaulted – is priced (assuming, as the slides do, independence between interest rates and default for this formula) by

$$\tilde{Z}(t, T) = Z(t, T) p(t, T) = \exp\left(-\int_t^T [f(t, s) + h(t, s)] ds\right), \quad (24.11)$$

► 2013 L24 @ 1:18:01] shown on the slide as the running example: $\tilde{Z}_t^T = \mathbf{1}_{\{\tau > t\}} \exp(-\int_t^T [f(t, s) + h(t, s)] ds)$, with τ the (random) default time and the indicator recording that the bond is worthless once default has occurred ► 2013 L24 @ 1:18:33].

24.5.2 The HJM SDE and drift condition for hazard rates

Postulate, exactly as in (24.2), an Ito process for the hazard curve,

$$dh(t, T) = \alpha^h(t, T) dt + \sigma^h(t, T) dB_t^h, \quad (24.12)$$

► 2013 L24 @ 1:19:01] where B^h may be correlated with the rates driver B of (24.2), $dB_t dB_t^h = \rho dt$ for a constant correlation ρ (the general case; the slide's exercise asks only for $\rho = 0$ ► 2013 L24 @ 1:19:26]). The slide poses, but does not answer, the derivation of the drift condition in this case ► 2013 L24 @ 1:19:26]: "Exercise 3: Derive a drift expression for the credit HJM model." The derivation is done in full here, for general ρ , and specialises to the slide's hinted answer when $\rho = 0$.

Theorem 24.9 (HJM drift condition with credit). Under $\mathbb{Q}$, requiring that the pre-default, zero-recovery defaultable bond price $\tilde{Z}_t^T = \mathbf{1}_{\{\tau > t\}} e^{-Y_t^T}$, $Y_t^T = \int_t^T [f(t, s) + h(t, s)] ds$, grows on average at the risk-free rate r_t plus the hazard rate ($\mathbf{1}_{\{\tau > t\}}$ jumps to 0 at rate $h_t = h(t, t)$, so its compensated growth rate on the event $\{\tau > t\}$ must be $r_t + h_t$ for the discounted, default-compensated price to be a $\mathbb{Q}$ -martingale) forces

$$\begin{aligned} \alpha^h(t, T) = & \sigma^h(t, T) \int_t^T \sigma^h(t, s) ds \\ & + \rho \left[\sigma^h(t, T) \int_t^T \sigma(t, s) ds + \sigma(t, T) \int_t^T \sigma^h(t, s) ds \right]. \end{aligned} \quad (24.13)$$

When $\rho = 0$ this is $\alpha^h(t, T) = \sigma^h(t, T) \int_t^T \sigma^h(t, s) ds$, matching the slide's hint ► 2013 L24 @ 1:19:46]: "If there are no correlations between IR and hazard rates, the drift is given by the equation similar to the IR case."

INTUITION: Why a cross-term appears

When rates and hazard rates are correlated, a shock that raises $\sigma(t, s)$ -driven forward rates on $[t, T]$ also moves (through ρ) the hazard curve on the same interval; the defaultable bond price is convex in the sum $f + h$, so its quadratic-variation compensation, (24.13)'s cross-term, mixes the two volatility kernels exactly as a two-asset Ito's lemma mixes $\sigma_1 \sigma_2 \rho$ into the drift of a product (section 17.8). Setting $\rho = 0$, as the slide does, isolates the credit-only piece and reproduces the interest-rate result (24.5) verbatim with h in place of f .

Exercise 24.5 (not from the course). Specialise (24.13) to $\rho = 0$ and $\sigma^h(t, T) \equiv \sigma_0^h$ constant (a "Ho-Lee" hazard-rate model). Derive the resulting SDE for the short hazard rate $h_t = h(t, t)$ and compare its form with proposition 24.5.

Exercise 24.6 (not from the course). Redo example 24.8 with a second, correlated curve $h(t, T) = \sigma_0^h e^{-b(T-t)}$ -volatility hazard process ($\rho = 0.5$ correlation with the rates driver) and the drift (24.13). Check numerically that $\mathbb{E}_{\mathbb{Q}}[\tilde{Z}(0, T_h)]$ from simulation matches

Proof. Repeat lemma 24.2 verbatim for Y_t^T with $f + h$ in place of f and $(\sigma, \alpha) + (\sigma^h, \alpha^h)$ jointly driving it (the two curves add, so their dt - and dB -coefficients add):

$$\begin{aligned} dY_t^T = & -(r_t + h_t) dt + \left(\int_t^T [\alpha(t, s) + \alpha^h(t, s)] ds \right) dt \\ & + \left(\int_t^T \sigma(t, s) ds \right) dB_t + \left(\int_t^T \sigma^h(t, s) ds \right) dB_t^h. \end{aligned}$$

On $\{\tau > t\}$, $\tilde{Z}_t^T = e^{-Y_t^T}$ follows Ito's formula for a function of two correlated Brownian drivers (section 17.8):

$$\begin{aligned} d(e^{-Y_t^T}) = & e^{-Y_t^T} \left[-dY_t^T + \frac{1}{2} d\langle Y^T \rangle_t \right], \\ d\langle Y^T \rangle_t = & \left(\int_t^T \sigma ds \right)^2 dt + \left(\int_t^T \sigma^h ds \right)^2 dt + 2\rho \left(\int_t^T \sigma ds \right) \left(\int_t^T \sigma^h ds \right) dt \end{aligned}$$

using $d\langle B, B^h \rangle_t = \rho dt$. The jump-to-default piece contributes an extra $-\tilde{Z}_t^T dN_t$ at the default event (with dN_t the default indicator's jump, compensated intensity $h_t dt$ under $\mathbb{Q}$); requiring the compensated process to be a $\mathbb{Q}$ -martingale removes the compensator $h_t dt$ exactly against the $+h_t dt$ term already present in $-dY_t^T$ (a standard credit-modelling fact, quoted, consistent with pricing formula (24.11)). What remains, on $\{\tau > t\}$ and after this cancellation, is the same dt -matching argument as in theorem 24.3 applied to $\int_t^T [\alpha + \alpha^h]$ jointly: since the interest-rate piece already satisfies $\int_t^T \alpha = \frac{1}{2} (\int_t^T \sigma)^2$ on its own (rates alone must also be arbitrage-free), what is left is $\int_t^T \alpha^h(t, s) ds = \frac{1}{2} (\int_t^T \sigma^h)^2 + \rho (\int_t^T \sigma) (\int_t^T \sigma^h)$, and differentiating in T (product rule on the cross term, which gives two pieces) yields (24.13). This is the result of Schönbucher (1998). $\square$

$Z(0, T_h) p(0, T_h)$ from the input curves, as in (24.11).

24.5.3 Why this matters: callable and structured credit

The slide motivates the credit HJM model with corporate *callable bonds* [► 2013 L24 @ 1:14:51]: an issuer who can refinance at a lower all-in rate (risk-free plus credit spread) saves money by calling old, expensive debt, and pricing that option requires simulating *both* $f(t, \cdot)$ and $h(t, \cdot)$ jointly, exactly the setting (24.12)–(24.13) was built for.

CASE STUDY: Structured notes: coupon enhancement via joint simulation

The slide's other running example [► 2013 L24 @ 1:20:05] is a 10-year, \$100M structured note paying a 10% coupon only while *both* the 30-year swap rate exceeds the 2-year swap rate *and* the S&P 500 stays above 880: a joint condition on the rates curve and an equity index (credit enters through the issuer's own default risk on the note, priced with $h(t, \cdot)$ as above). Historically the 30y–2y spread was negative on only a few days in 2005–2006, and the S&P 500 was above 880 on 94% of trading days over the prior decade, yet the market-implied (risk-neutral) probabilities were only about 80% and 75% respectively [► 2013 L24 @ 1:28:28]: investors are compensated (a higher coupon) for bearing tail risk that is more richly priced than its historical frequency, and the bank prices and hedges the note with exactly the Monte Carlo machinery of section 24.2.2, extended to simulate equity, rates and credit together.

Example 24.10 (Refinancing savings; slide numbers [► 2013 L24 @ 1:14:51]). A corporation issues a 10-year \$100M bond at an effective (risk-free + credit) rate of $2.00\% + 3.00\% = 5.00\%$. Three years later, market rates have fallen to $1.50\% + 1.50\% = 3.00\%$ with 7 years remaining. Calling and reissuing saves $(5.00\% - 3.00\%) \times \$100M \times 7 \text{ years} = \$14M$ (undiscounted; a real valuation discounts each year's saving and nets the reissuance cost, not shown on the slide).

KEY FORMULAS

Curve and bond. $Z(t, T) = \exp(-\int_t^T f(t, s) ds)$, short rate $r_t = f(t, t)$.

HJM forward-rate SDE. $df(t, T) = \alpha(t, T) dt + \sigma(t, T) dB_t$.

HJM drift condition (no-arbitrage, one factor, under $\mathbb{Q}$). $\alpha^{\mathbb{Q}}(t, T) = \sigma(t, T) \int_t^T \sigma(t, s) ds$; with n factors, sum this over $i = 1, \dots, n$.

Real-world drift. $\alpha^{\mathbb{P}}(t, T) = \alpha^{\mathbb{Q}}(t, T) - \sigma(t, T)\lambda_t$ (Girsanov, market price of risk λ_t).

Short-rate special cases. $\sigma(t, T) \equiv \sigma_0 \Rightarrow$ Ho–Lee; $\sigma(t, T) = \sigma_0 e^{-a(T-t)} \Rightarrow$ Hull–White/Vasicek; CIR's $\sigma\sqrt{r_t}$ is level-dependent and does not embed in one-factor HJM.

Credit. $p(t, T) = \exp(-\int_t^T h(t, s) ds)$, $dh(t, T) = \alpha^h(t, T) dt + \sigma^h(t, T) dB_t^h$, $\alpha^h(t, T) = \sigma^h(t, T) \int_t^T \sigma^h(t, s) ds + \rho \left[\sigma^h(t, T) \int_t^T \sigma(t, s) ds + \sigma(t, T) \int_t^T \sigma^h(t, s) ds \right]$ ($\rho = \text{Corr}(B, B^h)$); $\rho = 0$ recovers the rates-only form with h in place of f). Defaultable zero-recovery bond: $\tilde{Z}(t, T) = Z(t, T)p(t, T)$.

Chapter 25

The Ross Recovery Theorem

SOURCES FOR THIS CHAPTER		
Lecture	Speaker	Content
2013 L25	P. Carr (Morgan Stanley), with J. Yu	Guest lecture “Can We Recover?”: Arrow–Debreu security prices and market beliefs; Ross’s Recovery Theorem for finite-state Markov chains via the Perron–Frobenius theorem; a preference-free re-derivation using a change of numeraire (Long’s numeraire portfolio) for a time-homogeneous diffusion on a bounded state space; successes and failures for unbounded state spaces (Black–Scholes fails, Cox–Ingersoll–Ross succeeds).
Files. lecnote25 (50 slides, S. Ross’s collaborators P. Carr and J. Yu, Morgan Stanley). There is no 2024 counterpart: the topic is not in the 2024 syllabus. Problem sets. None (guest lecture, no associated problem-set questions in either year). Not covered / not published. The slides’ Part I (conditioning and path-dependent Arrow–Debreu prices, multi-period Markovian consistency) is background that the lecture itself skipped over quickly (“I’m going to skip these slides”); it is summarised briefly below for completeness but not derived in full. Ross’s original paper (<i>The Recovery Theorem</i>, published in the <i>Journal of Finance</i> in 2015, then only forthcoming) is not itself a course file; statements attributed to it are as quoted or paraphrased on the slides and in the lecture.		

Motivation

Every derivative price already encodes a probability measure: the *risk-neutral* measure $\mathbb{Q}$ of [section 2.4](#), under which discounted prices are martingales. But $\mathbb{Q}$ is famously not the market’s actual view of the future — it is contaminated by risk aversion, so that, say, the risk-neutral chance of a market crash is typically far larger than anyone’s honest assessment of its physical probability. For decades the going wisdom was that untangling the two required an extra ingredient that no one could agree on: a representative investor’s risk-aversion parameter, supplied from outside the data. Ross’s 2011 insight was that a single structural assumption on how risk aversion enters — *transition independence* of the pricing kernel — is enough to pin down the real-world probabilities uniquely, with no free parameter to calibrate, provided the underlying is a finite-state, time-homogeneous Markov chain and the matrix of state prices is positive. The mechanism is one this course has already built: [section 2.8](#)’s Perron–Frobenius theorem, applied to the matrix of Arrow–Debreu prices instead of a Markov transition matrix. This chapter states the assumptions precisely, proves the recovery theorem in full using [theorem 2.14](#), verifies the construction on an explicit numerical example, and closes Part II of these notes with a look

at what happens (and what is still open) beyond the finite-state setting.

25.1 Three probability measures: P , $\mathbb{Q}$, R

Fix a single source of uncertainty X (an index level, a short rate, an exchange rate) and consider Arrow–Debreu (AD) securities written on it: a claim paying \$1 if X lands in a given state and nothing otherwise, i.e. exactly the state prices ψ_i of [definition 2.7](#), now indexed by both the *current* state and the terminal state [[▶ 2013 L25 @ 12:49](#)]. Three probability measures matter here [[▶ 2013 L25 @ 5:22](#)]:

- P , the *physical* (objective, “God’s-eye”) probability measure: the true frequency with which X visits each state. No one observes P directly.
- $\mathbb{Q}$, the *risk-neutral* measure of [theorem 2.9](#): it exists under no arbitrage and satisfies $A_0 = \beta \mathbb{E}_{\mathbb{Q}}[C_T]$ for every replicable claim, but it is not a measure of frequency — it is a repackaging of AD prices, normalised to integrate to one, and under it every asset grows at the risk-free rate [[▶ 2013 L25 @ 9:37](#)].
- R , the *recovered* (or *natural*) measure: what one gets by trying to strip $\mathbb{Q}$ of its risk-aversion and time-value-of-money content and recover a measure of actual frequencies. Ross calls this the market’s “natural probabilities”. R need not equal P — the market can be wrong (2005 mortgage-backed pricing being Carr’s stock example) — so R is honestly only the market’s *belief* about P ; a believer in market efficiency is free to set $R = P$ [[▶ 2013 L25 @ 10:40](#)].

INTUITION: Why $\mathbb{Q}$ is not a frequency

A risk-neutral probability of 40% for “the index is up” means only that a security paying \$1 in that event costs \$0.40 today, i.e. it is a *price*, not a count of outcomes. Two effects separate this price from the true chance of the event: discounting (a sure \$1 tomorrow is worth less than \$1 today, even with no uncertainty at all) and risk aversion (a payoff arriving exactly when the rest of one’s wealth is doing badly is worth more per dollar than one arriving in good times, so its price is inflated relative to its frequency). Recovering R from $\mathbb{Q}$ means undoing both effects [[▶ 2013 L25 @ 42:38](#)].

25.2 The pricing matrix and the pricing kernel

Let X be a Markov chain on a finite state space $\{1, \dots, n\}$, and write $A(x, y) > 0$ for the price, at state x today, of an AD security paying \$1 if X is at state y one period later [[▶ 2013 L25 @ 40:28](#)]. Collecting these into a matrix $A \in \mathbb{R}^{n \times n}$, no-arbitrage consistency across horizons forces X ’s one-step transition structure under A to be genuinely Markovian and, if desired, time-homogeneous, i.e. A does not depend on calendar time (*Appendix, Part I of the slides*, sketched only: with three dates 0, 1, 2, path-dependent AD prices $A_{jk|i}$ factor as $A_{jk|i} = A_{ji} \cdot A_{k|ji}$, and Markov means the one-step-ahead conditional price $A_{k|ji} = A_{k|j}$ does not depend on the earlier state i [[▶ 2013 L25 @ 39:23](#)]).

Write $p(x, y)$ for the (unknown) *natural* one-step transition probabilities under R : $p(x, y) \geq 0$, $\sum_y p(x, y) = 1$ for every x . The *pricing kernel* is the entrywise ratio

$$\varphi(x, y) = \frac{A(x, y)}{p(x, y)}, \quad (25.1)$$

capturing exactly the two contaminating effects: time value of money and risk aversion

1 COURSE NOTE: Notation clash with [chapter 2](#)

The letter A here denotes the $n \times n$ matrix of one-step Arrow–Debreu prices between states of a single Markov chain, following Ross’s and Carr’s notation; it is unrelated to the $m \times n$ payoff matrix A of [definition 2.7](#) (assets in columns, states in rows). Both are conventional in their respective

[► 2013 L25 @ 47:03]. If interest rates were zero and investors risk-neutral, $\varphi \equiv 1$ and $A = p$; in general φ is an unknown positive function of two variables, and the n^2 equations (25.1) involve $2n^2 - n$ unknowns (the n^2 values of φ and the $n^2 - n$ free entries of p , whose rows sum to one), far too many to pin down p — this is why decontamination was thought hopeless before Ross [► 2013 L25 @ 47:03].

sources and are kept as in the lecture material.

Definition 25.1 (Transition independence, Ross’s assumption). The pricing kernel is *transition-independent* if it factors as

$$\varphi(x, y) = \delta \frac{z(y)}{z(x)}, \quad x, y = 1, \dots, n, \quad (25.2)$$

for some scalar $\delta > 0$ and some function $z : \{1, \dots, n\} \rightarrow (0, \infty)$.

In a world with a representative agent and additively time-separable utility, $z(x) = U'(c(x))$ (marginal utility of consumption at state x) and δ is the subjective discount factor, so that $\varphi(x, y) = \delta U'(c(y))/U'(c(x))$: risk aversion enters only through how marginal utility differs *between* the two states, not through the path or timing otherwise [► 2013 L25 @ 49:10]. Carr’s restatement, avoiding the representative-agent language, is purely dimension-counting (remark 25.2); either reading is the same assumption (25.2).

Remark 25.2 (Degrees of freedom). Without (25.2), φ has n^2 free values. Assumption (25.2) reduces this to $n + 1$: the n values $z(1), \dots, z(n)$ (only their ratios matter, so really $n - 1$ free numbers) plus δ . For $n = 10$ this is a drop from 100 to 11 unknowns, which is what makes identification from A alone possible [► 2013 L25 @ 59:42].

25.3 The Recovery Theorem

Theorem 25.3 (Ross’s Recovery Theorem). Let $A \in \mathbb{R}^{n \times n}$ be a matrix of one-step AD prices for a time-homogeneous Markov chain, with all entries $A(x, y) > 0$. Under transition independence (25.2), there is a unique triple (δ, z, p) with $\delta > 0$, $z \in \mathbb{R}^n$, $z > 0$ (up to an overall positive scale factor, which cancels out of p and φ), and p a row-stochastic matrix with $p(x, y) > 0$, consistent with (25.1)–(25.2) and the given A . Explicitly, δ is the Perron–Frobenius eigenvalue of A , $w(x) = 1/z(x)$ is (up to scale) its positive eigenvector, and

$$p(x, y) = \frac{1}{\delta} A(x, y) \frac{w(y)}{w(x)}, \quad x, y = 1, \dots, n. \quad (25.3)$$

Proof. Write $D_z = \text{diag}(z(1), \dots, z(n))$. Assumption (25.2) in (25.1) reads $A(x, y) = \delta p(x, y) z(y)/z(x)$, i.e. $z(x)A(x, y) = \delta p(x, y)z(y)$, or in matrix form

$$D_z A = \delta p D_z. \quad (25.4)$$

Let $w = D_z^{-1} \mathbf{1} = (1/z(1), \dots, 1/z(n))^T$, where $\mathbf{1} = (1, \dots, 1)^T$. Multiply (25.4) on the right by $w = D_z^{-1} \mathbf{1}$:

$$D_z A w = \delta p D_z D_z^{-1} \mathbf{1} = \delta p \mathbf{1} = \delta \mathbf{1},$$

using that p is row-stochastic ($p \mathbf{1} = \mathbf{1}$). Multiplying through by D_z^{-1} ,

$$A w = \delta D_z^{-1} \mathbf{1} = \delta w. \quad (25.5)$$

IN FINANCE: Reading the recovered kernel

Once p is known, (25.2) gives back the pricing kernel itself, $\varphi(x, y) = \delta z(y)/z(x) = \delta w(x)/w(y)$, fully decontaminated: δ is the (state-independent) discount factor and $w(x)/w(y)$ isolates the risk-aversion premium of moving from x to y . If the recovered marginal utility $z = 1/w$ is decreasing in the state index (i.e. w is increasing) when the index orders states from “bad” to “good”, a transition into a worse state costs relatively more than a fair bet on its recovered probability alone would suggest — exactly the insurance-value story of section 25.1. Unlike the one-period model of chapter 2, where *any* positive ψ with $A_0 = \psi^\top A$ is an admissible state-price vector when the market is incomplete, here positivity of the whole matrix A plus (25.2) is what removes all the remaining freedom in one step, without ever specifying a utility function numerically.

Exercise 25.1 (Irreducible but not positive (not from the course)). Let $A = \begin{pmatrix} 0 & 0.5 & 0.3 \\ 0.4 & 0 & 0.6 \\ 0.2 & 0.4 & 0 \end{pmatrix}$ (zero diagonal, but irreducible: every state reachable from every other in finitely many steps).

- Check numerically that A still has a positive Perron eigenvector, consistent with the irreducible case of theorem 2.14 mentioned in chapter 2.
- Recover p via (25.3) and check it is a valid (row-stochastic, non-negative) transition matrix with some zero entries.
- Explain financially what $A(x, x) = 0$ means (no one-period AD security paying off in the same state you start in) and why this does not break recovery.

So w is an eigenvector of A with eigenvalue δ , and $w > 0$ because $z > 0$.

Existence and uniqueness of (δ, w) . A has all entries strictly positive, so theorem 2.14 (Perron–Frobenius, proved in chapter 2) applies directly: A has a real eigenvalue $\lambda_0 > 0$ strictly larger in modulus than every other eigenvalue, with a positive eigenvector unique up to positive scaling, and no other eigenvalue has a positive eigenvector (every other eigenvector has entries of mixed sign or is complex). Equation (25.5) exhibits $w > 0$ as a positive eigenvector of A , so by theorem 2.14(iii) it must be the Perron eigenvector, unique up to scale, and δ must be the Perron eigenvalue λ_0 : no other pair (δ', w') with $w' > 0$ solving (25.5) can exist. This proves δ and $z = 1/w$ (up to the same overall scale) are unique.

Recovering p . Solve (25.4) for p : $p = \delta^{-1} D_z A D_z^{-1}$, i.e. entrywise $p(x, y) = \delta^{-1} z(x) A(x, y) / z(y) = \delta^{-1} A(x, y) w(y) / w(x)$, which is (25.3). Positivity $p(x, y) > 0$ is immediate from $A, w, \delta > 0$. Row-stochasticity is automatic and does not need to be imposed separately: by (25.5),

$$\begin{aligned} \sum_{y=1}^n p(x, y) &= \frac{1}{\delta w(x)} \sum_{y=1}^n A(x, y) w(y) = \frac{(Aw)(x)}{\delta w(x)} \\ &= \frac{\delta w(x)}{\delta w(x)} = 1, \end{aligned}$$

for every x . Since (δ, w) are the unique Perron data of A , p in (25.3) is the unique matrix consistent with (25.2) and the given A . $\square$

2 COURSE NOTE: What the lecture actually showed

Carr states Ross’s Theorem 1 essentially verbatim (“if the pricing matrix A is positive or irreducible, then there exists a unique solution... via the Perron–Frobenius theorem”) but says explicitly that he has no slide on *how* the $n + 1$ unknowns are actually computed, referring the audience to Ross’s paper [► 2013 L25 @ 1:01:49]. The derivation above (via (25.4)–(25.5)) is a reconstruction, not shown on the slides or in the transcript, built from the definitions Carr does give (the pricing kernel (25.1), transition independence (25.2), and the explicit statement that Perron–Frobenius is the tool used). The theorem also holds, with $|\lambda| \leq \delta$ rather than strict inequality, for *irreducible non-negative* A (some $A(x, y)$ may vanish): this is the version of theorem 2.14 noted in chapter 2 as the one needed for Markov chains, and it is the version Ross actually states.

25.4 Worked example and numerical verification

CASE STUDY: Numerical check, Python

The following reconstructs [example 25.4](#): build A from (p, δ, z) , forget (p, δ, z) , and recover them from A alone via `numpy.linalg.eig`.

```
import numpy as np

P = np.array([[0.6, 0.3, 0.1],
              [0.2, 0.5, 0.3],
              [0.1, 0.4, 0.5]])
delta, z = 0.97, np.array([1.0, 1.6, 2.5])
A = delta * P * (z[None, :] / z[:, None]) # build the AD price matrix

eigvals, eigvecs = np.linalg.eig(A) # forget P, delta, z; recover them
i = np.argmax(eigvals.real) # Perron eigenvalue = largest one
lam0, w = eigvals[i].real, np.abs(eigvecs[:, i].real)
w /= w[0] # fix the free positive scale

P_recovered = (1 / lam0) * A * (w[None, :] / w[:, None])
print(lam0, w) # -> 0.97, [1.0, 0.625, 0.4] = 1/z (up to scale)
print(np.max(np.abs(P_recovered - P))) # -> ~4e-16
```

The recovered eigenvalue is $\lambda_0 = 0.97 = \delta$ to machine precision, the eigenvector is $w = (1, 0.625, 0.4) = 1/z$ up to the chosen scale, and $P_{\text{recovered}}$ matches the true p to within 4×10^{-16} (floating-point roundoff): [eq. \(25.3\)](#) reproduces the natural transition matrix exactly from the price matrix alone.

INTUITION: Why positivity of A is doing all the work

Nothing in the recovery step uses probabilistic reasoning about p directly; it is pure linear algebra on A . The only place probability enters is in guaranteeing, via no-arbitrage, that A is a matrix of strictly positive prices in the first place, and in the modelling assumption [\(25.2\)](#) that restricts what A can look like. Once both hold, the Perron eigenvector is forced to be positive and simple by [theorem 2.14](#), and positivity of the reconstructed p and its automatic row-sum-one property follow for free — there is no need to separately check that the answer is a sensible probability matrix.

Exercise 25.2 (Sensitivity of the recovered kernel (not from the course)). Using the Python code of [example 25.4](#), perturb one entry of A (e.g. multiply $A(1, 3)$ by 1.05) while leaving the others fixed, keeping A positive.

- Recompute δ , w and p ; by how much does each change relative to the unperturbed values?
- Which entries of p are most sensitive to a change in $A(1, 3)$, and does this match the intuition that Perron–Frobenius data depend on the *whole* matrix, not just one entry?
- Repeat with a 5×5 random positive matrix and comment on whether the recovered p remains close to row-stochastic under floating-point rounding alone.

Example 25.4 (Recovering a 3-state chain). Take three states (say, three ranges of a 1-year index return: down, flat, up) with the “true” natural transition matrix

$$p = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{pmatrix},$$

a discount factor $\delta = 0.97$, and marginal utilities $z = (1.0, 1.6, 2.5)$ (higher in the “up” state, i.e. increasing marginal utility of consumption in the better state — an unusual but not excluded case, matching Ross’s remark that the recovered kernel could be increasing in y [\[► 2013 L25 @ 55:28\]](#)). By [\(25.2\)–\(25.1\)](#), the (only observable, in this thought experiment) AD price matrix is

$$A(x, y) = \delta p(x, y) \frac{z(y)}{z(x)} = \begin{pmatrix} 0.5820 & 0.4656 & 0.2425 \\ 0.1213 & 0.4850 & 0.4547 \\ 0.0388 & 0.2483 & 0.4850 \end{pmatrix}.$$

Given only A , [theorem 25.3](#) recovers δ as its Perron eigenvalue and $w = 1/z$ as its Perron eigenvector (fixing the scale by, say, $w(1) = 1$); [\(25.3\)](#) then reconstructs p exactly, with no knowledge of z , δ , or p used in the recovery step itself.

25.5 Beyond finite-state chains: change of numeraire

Ross's own proof works in the finite-state, discrete-time setting above. Carr and Yu's contribution is a preference-free re-derivation, avoiding the representative-agent language, that extends the same conclusion to a continuous-state diffusion. The key substitute for the discrete pricing kernel is a change of numeraire.

Definition 25.5 (Numeraire, numeraire portfolio). A *numeraire* is a self-financing portfolio whose value is strictly positive at all times (a swap can be zero or negative, hence disqualified; a stock or a money market account qualifies, as long as it never hits zero) [► 2013 L25 @ 25:20]. Long (1990) showed that, given no arbitrage among a money market account and n risky assets, there is always a self-financing portfolio of them, of value $L_t > 0$, the *numeraire portfolio*, such that every relative price S_t^i/L_t is an R -martingale — i.e. expressing P&L in units of L turns the (unobservable) natural measure R into the measure under which discounted prices behave like ordinary risk-neutral martingales [► 2013 L25 @ 1:16:14]. The same object is also called the *growth-optimal portfolio* (Kelly, 1956: over an infinite horizon it has the largest mean growth rate among all portfolios, at the price of being quite risky along the way) or the *natural numeraire* [► 2013 L25 @ 19:04].

Assume X is a univariate, time-homogeneous diffusion under $\mathbb{Q}$,

$$dX_t = b^{\mathbb{Q}}(X_t) dt + a(X_t) dW_t,$$

on a *bounded* state space (ℓ, u) , driving the short rate $r_t = r(X_t)$ and the value of n risky assets, all with known $\mathbb{Q}$ -coefficients; and that the numeraire portfolio's value is a function $L_t = L(X_t, t)$ solving $dL_t/L_t = r(X_t) dt + \sigma_L(X_t) dW_t$ for an unknown volatility function σ_L [► 2013 L25 @ 1:17:17]. Self-financing forces L to solve the pricing PDE $\partial_t L + \frac{1}{2}a^2 \partial_x^2 L + b^{\mathbb{Q}} \partial_x L = rL$; writing $\sigma_L(x) = a(x) \partial_x \ln L(x, t)$ (from Itô's formula) and separating variables $L(x, t) = \pi(x)e^{\lambda t}$ turns this PDE into a *regular Sturm–Liouville problem*

$$\frac{a^2(x)}{2} \pi''(x) + b^{\mathbb{Q}}(x) \pi'(x) - r(x) \pi(x) = -\lambda \pi(x), \quad x \in (\ell, u), \quad (25.6)$$

with separated boundary conditions at ℓ, u [► 2013 L25 @ 1:25:39].

IN FINANCE: Sturm–Liouville as the continuum Perron–Frobenius

Equation (25.6) is the exact continuous-state analogue of the eigenvalue problem (25.5): Sturm–Liouville theory guarantees a smallest eigenvalue λ_0 , strictly below every other eigenvalue, with an associated eigenfunction $\pi_0(x) > 0$ everywhere and unique up to positive scaling, while every other eigenfunction changes sign at least once [► 2013 L25 @ 1:26:40]—precisely the conclusions (i)–(iii) of [theorem 2.14](#) carried over from finitely many states to a continuum of them. Once π_0 and λ_0 are known (numerically, in general), $L(x, t) = \pi_0(x)e^{\lambda_0 t}$ is known up to scale, hence so is $\sigma_L(x) = a(x) \partial_x \ln \pi_0(x)$, and Girsanov's theorem then gives the R -drift of X as $b^{\mathbb{Q}}(x) + a(x)\sigma_L(x) = b^{\mathbb{Q}}(x) + a^2(x) \partial_x \ln \pi_0(x)$ and the R -dynamics of every asset in the set [► 2013 L25 @ 1:19:23]. This is the “11 unknowns” of [remark 25.2](#) replaced by one unknown function π_0 and one unknown scalar λ_0 .

KEY FORMULAS

Pricing kernel:	$\varphi(x, y) = A(x, y)/p(x, y)$
Transition independence:	$\varphi(x, y) = \delta z(y)/z(x) \iff D_z A = \delta p D_z$
Recovery ($w = 1/z$):	$Aw = \delta w$ (Perron eigenpair of A , theorem 2.14)
Recovered transitions:	$p(x, y) = \delta^{-1} A(x, y) w(y)/w(x), \quad \sum_y p(x, y) = 1$
Diffusion analogue:	$\frac{1}{2} a^2 \pi'' + b^Q \pi' - r \pi = -\lambda \pi$ (Sturm–Liouville) $\sigma_L(x) = a(x) \partial_x \ln \pi_0(x), \quad b^R(x) = b^Q(x) + a(x) \sigma_L(x)$

Exercise 25.3 (From the Sturm–Liouville problem to the discrete one (not from the course)). Discretise (25.6) on a grid $x_1 < \dots < x_n$ in (ℓ, u) using centred differences for $\pi''(x_i) \approx (\pi_{i+1} - 2\pi_i + \pi_{i-1})/h^2$ and $\pi'(x_i) \approx (\pi_{i+1} - \pi_{i-1})/(2h)$, with absorbing (Dirichlet) boundary conditions $\pi_0 = \pi_{n+1} = 0$.

- Write the resulting eigenvalue problem as $M\pi = -\lambda\pi$ for an explicit tridiagonal M in terms of $a(x_i)$, $b^Q(x_i)$, $r(x_i)$, h .
- For constant a, b^Q, r , compute M 's eigenvalues in closed form and identify the smallest one and its eigenvector's sign pattern; check it matches [theorem 2.14](#)'s conclusion that only the extremal eigenvalue has an everywhere-positive eigenvector.
- Implement this in Python for $a = 0.2$, $b^Q(x) = 0.1(0.05 - x)$, $r(x) = 0.03$, on $(\ell, u) = (-0.1, 0.2)$ with $n = 50$ grid points, and plot π_0 ; is it unimodal?

Remark 25.6 (End of Part II). This closes Part II of these notes. The recovery theorem was a fitting last stop: it leans on nearly every tool built up along the way — state prices and no-arbitrage ([chapter 2](#)), the risk-neutral measure ([chapter 15](#)), Markov chains and Perron–Frobenius ([chapter 2](#) again), and, in its diffusion form, stochastic calculus and PDEs ([chapters 17](#) and [18](#)). The appendices that follow are practical rather than mathematical: how to run the course's R material, and the logistics of both years of the course.

3 COURSE NOTE: Boundedness is essential, and still an open problem beyond it

The construction needs X 's state space to be an interval (ℓ, u) with ℓ, u finite (Carr is candid that some economists object to bounding, say, the S&P 500 above, and that he personally has no problem with a bound of \$20 trillion [[► 2013 L25 @ 27:29](#)]). For an *unbounded* state space the argument needs a Hilbert-space (square-integrability) version of Sturm–Liouville theory, and it does not always go through: it *fails* for the standard geometric Brownian motion of Black–Scholes (so the natural measure of a stock price cannot be recovered this way from stock option prices under these assumptions), but it *succeeds* for the Cox–Ingersoll–Ross short-rate model, recovering an R-dynamics that is again CIR but with a higher mean-reversion speed. More generally it succeeds for affine diffusions (Linetsky and a student, cited but not derived in the lecture). At the time of the 2013 lecture, necessary and sufficient conditions for recovery on an unbounded state space were an open research question [[► 2013 L25 @ 1:27:41](#)].

Exercises

Standalone exercises follow the section they practise. The two exercises below are a connected series (the second starts from the matrix and the recovered p of the first), so they stay together here.

Exercise 25.4 (Recovering a 2-state chain by hand (not from the course)). Let $A = \begin{pmatrix} 0.45 & 0.30 \\ 0.20 & 0.40 \end{pmatrix}$.

- Find the Perron eigenvalue δ and a positive eigenvector w of A by solving the 2×2 characteristic equation directly (no software).
- Use (25.3) to recover p , and check it is row-stochastic.
- Recover $z = 1/w$ up to scale and the pricing kernel $\varphi(x, y) = \delta z(y)/z(x)$; is φ larger for a transition into state 1 or state 2? Which state carries the “insurance” premium?

Exercise 25.5 (Recovery is not just normalising rows (not from the course)). For the A of [Exercise 25.4](#), compute the naive row-normalisation $\hat{p}(x, y) = A(x, y) / \sum_z A(x, z)$.

- (a) Verify $\hat{p} \neq p$ from [Exercise 25.4\(b\)](#).
- (b) Explain, in terms of [\(25.4\)](#), why row-normalising A recovers the natural probabilities only in the special case $z \equiv \text{const}$ (no risk aversion), and identify what $\hat{p}$ actually equals in general (a $\mathbb{Q}$ -like object, not R).

Appendix A

Map of the Course Materials

Every file downloaded from the two OCW sites, with the chapter where it is used. Paths are relative to the course folder; static_resources/ is abbreviated sr/. Status: **done** = already integrated in a written chapter, *partly* = some of the problems or chapters that use it are written, *planned* = assigned to a chapter still in preparation.

18.642, Fall 2024 (18.642-fall-2024/)

File	Used in	Status
sr/mit18_642_f24_lec01_1.pdf	chapter 1	done
sr/mit18_642_f24_lec01_2.pdf	chapter 1 , chapter B	done
sr/mit18_642_f24_lec02_1.pdf	chapter 2	done
sr/mit18_642_f24_lec02_2.pdf	chapter 2	done
sr/mit18_642_f24_lec02_3.pdf	chapter 2 , chapter B	done
sr/mit18_642_f24_lec04_1.pdf, lec04_2.pdf	chapter 3	done
sr/mit18_642_f24_lec05.pdf	chapter 4	done
sr/mit18_642_f24_lec06_1.pdf – lec06_4.pdf	chapter 5	done
sr/mit18_642_f24_lec07.pdf	chapter 6	done
sr/mit18_642_f24_lec08.pdf	chapter 5	done
sr/mit18_642_f24_lec09.pdf	chapter 7	done
sr/mit18_642_f24_lec10.pdf	chapter 8	done
sr/mit18_642_f24_lec11-12.pdf	chapter 9	done
sr/mit18_642_f24_lec13.pdf	chapter 10	done
sr/mit18_642_f24_lec14_1.pdf – lec14_9.pdf	chapter 11	done
sr/mit18_642_f24_lec17_1.pdf, lec17_2.pdf	chapter 13 (2024 L19)	done
sr/mit18_642_f24_lec19_1.pdf, lec19_2.pdf	chapter 17 (2024 L24)	done
sr/mit18_642_f24_lec21.pdf	chapter 15	done
sr/mit18_642_f24_lec23.pdf	chapter 16	done
sr/mit18_642_f24_lec24.pdf	chapter 18 (2024 L25)	done
sr/mit18_642_f24_pset1.pdf	chapter 2 (all problems)	done
sr/mit18_642_f24_pset2.pdf	chapters 3 and 4 (P1–2), chapter 7 (P3), chapter C (P4)	done
sr/mit18_642_f24_pset3.pdf	chapter 3 (Stein’s lemma), chapter 10 , chapter 7	done

File	Used in	Status
sr/mit18_642_f24_pset4.pdf	chapter 11 (Brownian motion), chapter 9 (AR, ARMA)	done
sr/mit18_642_f24_pset5.pdf	chapters 4 and 17 , chapter 13	done
sr/mit18_642_f24_pset_analysis.pdf, pset_rcode.pdf	chapter 7 (PCA of Treasury yields)	done
sr/mit18_642_f24_rstudio.zip (FM_Intro1.Rmd)	chapter 1	done
sr/mit18_642_f24_equal_weighted_portfolios.zip	chapter 2	done
sr/mit18_642_f24_etf_casestudy.zip	chapter 5	done
sr/mit18_642_f24_rstudio_pset5.zip	chapter 13	done
sr/fedbondyielddata_20230911.csv, sr/feddataanalysis.ipynb	chapter 6 , chapter 7	done
sr/*_transcript.pdf (23 files)	PDF copies of the .txt transcripts; the .txt files are used instead	–
sr/21m383-s23-lecture-28-apr-28-v3_transcript.pdf	to be identified	<i>planned</i>
pages/syllabus, calendar, assignments, problem-sets	chapter C	done
pages/investment-game	chapter 1 (Exercise 1.1), chapter C	done
pages/group-project, pages/final-paper	chapter C	done
pages/week-1 – week-15	mapping lectures ↔ files (this appendix)	done
static_shared/, *.html, *.js, *.css, fonts, images	website infrastructure, no content	–

18.S096, Fall 2013 (18.s096-fall-2013/)

File names in sr/ start with a hash, omitted here.

File	Used in	Status
MIT18_S096F13_lecnote1.pdf	chapter 1	done
lecnote2.pdf	chapter 2	done
lecnote3.pdf	chapter 3	done
lecnote5.pdf	chapter 4	done
lecnote6.pdf	chapter 5	done
lecnote7.pdf	chapter 19	done
lecnote8.pdf, lecnote11.pdf, lecnote12.pdf	chapter 9	done
lecnote9.pdf	chapter 13	done
lecnote10.pdf	chapter 20	done
lecnote13.pdf	chapter 21	done
lecnote14.pdf, lecnote16.pdf	chapter 10	done
lecnote15.pdf	chapter 7	done
lecnote17.pdf	chapter 11	done
lecnote18.pdf	chapter 17	done
lecnote19.pdf	chapter 15	done
lecnote21.pdf	chapter 18	done
lecnote23.pdf	chapter 23	done
lecnote24.pdf	chapter 24	done

File	Used in	Status
lecnote25.pdf	chapter 25	done
lecnote26.pdf	chapter 8	done
CaseStudy1.pdf, CaseStudy2.pdf, fm_casestudy_1_rcode.r, fm_casestudy_1_0.r, fm_casestudy_2_rcode.r, fm_casestudy_fx_1.r	chapter 5	done
CaseStudy3.pdf, CaseStudy5.pdf	chapter 9	done
CaseStudy4.pdf	chapter 13	done
CaseStudy6.pdf, Smltn_TwoAst.pdf, ETF_pridA_15/30.pdf, ETF_pridB_15/30.pdf	chapter 10	done
CaseStudy7.pdf	chapter 7	done
fm_casestudy_0_InstallOrLoadLibraries*.r, R_Rstudio_Instructions_CaseStudy1/2.txt	chapter B	done
pset1.pdf	chapter 2	done
pset2.pdf	chapters 3 and 4	done
pset3.pdf	chapter 5	done
pset4.pdf	chapter 9	done
pset5.pdf	chapter 13	done
pset6.pdf	chapter 9 (VAR) , chapter 10	done
pset7.pdf	chapter 7	done
pset8.pdf	chapter 17	done
pset9.pdf	chapter 18	done
<youtube-id>.pdf (24 files)	PDF copies of the .txt transcripts	–
pages/syllabus, calendar, lecture-notes, case-studies, instructor-insights, projects, related-resources	chapter C	done

Transcripts

Lectures 2024 transcript/ (22 files) and Lectures 2013 transcript/ (24 files): used in every chapter for explanations given orally and for the video timestamps [▶ 2024 L1 @ m:ss](#). Lectures with no recording: 2024 L3, L15–L17 (student presentations), L22, L26 (presentations); 2013 L4, L22.

Appendix B

R and RStudio Setup

Several case studies in both courses are distributed as runnable R code rather than only as PDF write-ups: the 2013 case studies ([chapter 5](#)) ship as plain `.r` scripts, and the 2024 case studies ([chapters 1, 2 and 13](#)) ship as RStudio projects with R Markdown (`.Rmd`) files. This appendix collects, from the actual install scripts and instruction files found in the course materials, what is needed to run them and which chapter each file belongs to. It is a practical reference, not a tutorial on R itself.

B.1 2013 case studies: plain R scripts

The 18.S096 (2013) case studies used in [chapter 5](#) come with a small family of files whose names are hash-prefixed in `static\_resources/` (the hash is omitted below, as in [chapter A](#)): a library-install script, one or two data/plotting scripts per case study, and a plain-text instructions file.

The two `R\_Rstudio\_Instructions\_CaseStudy1/2.txt` files give the following steps (paraphrased; both files agree except for the file lists in step 2):

1. Install R and RStudio (<http://www.r-project.org/> and <http://www.rstudio.com/ide/download/desktop>).
2. Create a project directory (e.g. `fm\_casestudy1` or `fm\_casestudy2`) and copy into it the case study's `.r` script files (for Case Study 2, also the `fred\_fxrates.txt` data file).
3. In RStudio: Project → Create Project → Create Project from Existing Directory, and select that directory.
4. Open the script files, listed in the lower-right Files pane, in the order given by the instructions file.
5. Source (or run with `Ctrl+Enter`) the install/load script first, with `ind.install0 <- TRUE` on the very first run; then source the data-download/plotting script; then the case study's `_rcode.r` script, which reproduces the R commands shown in the case study's PDF.
6. Restart RStudio between steps if prompted (e.g. after installing packages), answering `n` to "save workspace" – the scripts reload the data explicitly and do not rely on a saved `.RData` workspace.

1 COURSE NOTE: Install script pattern

`fm\_casestudy\_0\_InstallOrLoadLibraries`
(used for Case Study 1) and
`fm\_casestudy\_0\_InstallOrLoadLibraries`
`Case\_Study\_2.r` (Case Study 2) both
follow the same pattern: a boolean flag
`ind.install0` guards a block of
`install.packages(...)` calls, followed by
`library(...)` calls that load the same packages.
Set `ind.install0 <- TRUE` the first time the
script runs on a machine (or to update the
packages), then switch it back to `FALSE` on later
runs so the script only loads what is already
installed. The packages required are:

`quantmod, tseries, vars, fxregime, moments,`
`rugarch, xts`

(`xts` appears only in the Case Study 1 variant of
the install script, and only in its
`install.packages` block, not among the
`library` calls; the Case Study 2 variant installs
and loads the other six.)

Case study	Files
Case Study 1 (regression, FX & rates)	fm_casestudy_0_InstallOrLoadLibraries.r, fm_casestudy_1_0.r, fm_casestudy_1_rcode.r
Case Study 2 (FX regime)	fm_casestudy_0_InstallOrLoadLibraries Case_Study_2.r, fm_casestudy_fx_1.r, fm_casestudy_2_rcode.r

Both case studies are used in [chapter 5](#); see [chapter A](#) for the exact correspondence with the PDF write-ups (CaseStudy1.pdf, CaseStudy2.pdf).

B.2 2024 case studies: RStudio projects

The 18.642 (2024) case studies distribute a full RStudio project per case study (a .Rproj file, one or more .Rmd files, and any data the case study needs as an .RData or .csv file), rather than loose scripts. Opening the .Rproj file in RStudio sets the working directory and loads the project automatically; there is no separate install-libraries script in these archives, so the required packages must be installed with `install.packages(...)` once, before knitting the .Rmd file, if not already present.

Archive / project	Used in	R packages (<code>library(...)</code>)
mit18_642_f24_rstudio.zip (FM_Intro1.Rmd)	chapter 1	tidyquant, tidyverse, ggplot2
mit18_642_f24_equal_weighted_ portfolios.zip	chapter 2	tidyquant, tidyverse, ggplot2
mit18_642_f24_ etf_casestudy.zip	chapter 5	tidyquant, tidyverse, car, broom
mit18_642_f24_ rstudio_pset5.zip (ARKK/S&P 500 volatility)	chapter 13	tidyquant, tidyverse, fpp2, ggfortify, glmnet

(This is the union of `library(...)` calls actually found across the .Rmd files in these four archives; a given .Rmd may only use a subset of the packages listed for its row. No combined install script for the 2024 material was found in the static resources – unlike the 2013 scripts, package installation here is left implicit and must be done once per package with `install.packages("<name>")` before the first knit.)

B.3 Try it yourself

- Run FM_Intro1.Rmd ([chapter 1](#)) end to end in RStudio with "Knit" once the packages above are installed, and compare the output to FM_Intro1.pdf in the same archive.
- Change `ind.install0` back and forth in `fm_casestudy_0_InstallOrLoadLibraries.r` and confirm that with it set to `FALSE` the script still runs (using the already-installed packages) but skips the `install.packages` calls.

2 COURSE NOTE

Two of these are financial-data packages worth calling out: `tidyquant` and `quantmod` both pull historical price/rate series from the internet (typically Yahoo Finance or FRED), so an internet connection is needed the first time a script or .Rmd runs, even though the 2013 scripts also cache the result in an .RData workspace for later offline reruns.

Appendix C

Course Logistics

This appendix collects the practical, non-mathematical information about how the two courses were run: syllabus and grading, calendar structure, problem-set and assignment logistics, the 2024 investment game, the 2024 group project and final paper, and the 2013 instructor interview. It is assembled from the `pages/` subdirectories of the two OCW course sites (`syllabus`, `calendar`, `assignments`, `problem-sets`, `investment-game`, `group-project`, `final-paper` for 2024; `syllabus`, `instructor-insights`, `projects`, `related-resources` for 2013), reporting only what those pages state.

C.1 18.642, Fall 2024

C.1.1 Syllabus and grading

Two 1.5-hour lectures per week. Prerequisites: 18.03 (Differential Equations), 18.05 or 18.600 (Probability), and 18.06 (Linear Algebra); prior finance knowledge is not required. Mathematics lectures were given by MIT mathematicians (mainly Peter Kempthorne), applications lectures by industry professionals (a mix of investment banks, FinTech and hedge funds).

The class had five problem sets, an optional investment game, a mid-semester group project (a written lecture note plus an in-class presentation), and, in place of a final exam, a final paper on a math finance topic of the student’s choice. The grading page states the weights explicitly:

Component	Weight
Homework (problem sets)	40%
Group project presentation	20%
Participation	10%
Final paper	30%

C.1.2 Calendar structure

The calendar page lists 26 lecture slots, alternating math lectures (mostly Kempthorne, covering the topics of [part I](#)) with applications lectures by outside speakers (e.g. Jeff Shen of BlackRock on quantitative equity investing, L3; Andrew Gunstensen of Mizuho on rates, L7; Stefan Andreev of Two Sigma on PCA, L9; Andrew W. Lo on AI in biomedical

portfolios, L18; John Hull on machine learning, L23). Problem sets were due at lectures 3, 7, 9, 18 and 22. Lectures 15–17 were student presentations of the group project, and lecture 26 was the final paper presentations.

C.1.3 Problem sets

Five problem sets, each covering the math lecture topics up to that point (see [chapter A](#) for the file-to-chapter mapping). The problem-sets page lists two accompanying readings: Gron, “Optimal Portfolio Choice and Stochastic Volatility” (*Applied Stochastic Models in Business and Industry*, 2012), referenced by Problem 2 of Problem Set 2, and, for the PCA problem, a note on principal component analysis of interest rates plus sample R code (both used in [chapter 7](#)); Problem Set 5 references Parkinson, “The Extreme Value Method for Estimating the Variance of the Rate of Return” (*The Journal of Business*, 1980) and ships a companion R Markdown volatility case study.

C.1.4 Investment game

The investment game (ungraded) is stated in full detail in [Exercise 1.1](#) of [chapter 1](#); it is not repeated here. Two points from the course page are purely administrative and not part of the exercise: students had to submit their choice of instrument to the instructors before 14 September 2024, and their optional switch before 12 October 2024, and at the end of the term the instructors announced the top three performers by sharing the tracking spreadsheets.

C.1.5 Group project

Groups worked on an introductory, written lecture note aimed at fellow students with no prior knowledge of the topic; each group member submitted their own 5–7 page draft on their section. The first submission (“Draft 1”) was graded only on evidence of individual, focused effort (5+ pages of draft text), and closed with a reflection paragraph asking for specific feedback. It was followed by a peer review of another group’s draft, revision, and a 15–20 minute oral presentation in class (lectures 15–17 on the calendar).

The topics offered in 2024 were: How Human Fallacies Lead to Market Anomalies; Regression Modeling of UK Property Prices; The Use of Machine Learning in Predicting Future Interest Rates; Alternative Data; Jump-Diffusion Model; Maximizing Utility While Minimizing Risk; Fourier Analysis and Its Applications in the Stock Market; Black Scholes Merton Model; Implied Volatility; Partial Differential Equation Models for Option Pricing; Manipulating the Sports Betting Market; Time Series Modeling; Analyzing Volatility Curves in Option Pricing; The Black-Scholes Model; and Delta-Hedging.

C.1.6 Final paper

Every student wrote their own final paper, individually graded even when it continued a group project topic; sections of a joint paper had to be clearly attributed to their author. Papers were due on the last day of lectures, typically 10–15 pages, graded in part on mathematical content, writing quality, and organization. As with the group project, a Draft 1 was required first (3–4 polished pages plus 1–2 pages of motivation, or 5–7 pages of developing notes, with a closing reflection section), assessed only on evidence of focused effort, followed by peer review the next week.

The final-paper page lists 33 topics proposed for 2024, spanning market microstructure (credit expansion and land prices, election effects on volatility), derivatives pricing (SABR, Heston, jump-diffusion Monte Carlo, comparative Black-Scholes/Monte Carlo studies), portfolio and risk methods (Black-Litterman, PCA, tail-risk hedging), machine learning applications (LSTM volatility forecasting, reinforcement learning for Greeks hedging, AI-driven trading), and a few lighter entries (sports-betting binary options, “The Wolf of Wall Street” and stock-market cinematography, DEI-focused ETFs). The full list is not reproduced here; consult the source page for all 33 titles.

C.2 18.S096, Fall 2013

C.2.1 Syllabus, goals and grading

Same meeting pattern (two 1.5-hour lectures/week) and a longer prerequisite list: 18.01, 18.02, 18.03, 18.05 or 18.440, and 18.06. The course had no exams: problem sets were due two weeks after each math lecture, and the final grade was 75% homework, 25% final paper. The syllabus page lists ten explicit course goals, later repeated verbatim on the instructor-insights page, e.g. deriving the price-yield relationship and convexity, bootstrapping a yield curve, computing standard VaR, estimating option volatility, deriving Black-Scholes via risk-neutral arguments, PCA as matrix decomposition, limiting theorems, Itô’s lemma, and Girsanov’s theorem and change of measure – i.e. essentially the syllabus of [parts I and II](#) combined. The course also offered an optional field trip to Morgan Stanley’s New York offices.

C.2.2 Instructor insights (2013)

The course grew out of Jake Xia’s and Vasily Strela’s wish, as MIT alumni working in industry, to give back to the department; it was first taught in Fall 2012 as invited industry lectures only, and the pure-math component was added the following year to better integrate theory and practice. Enrollment was about 50 students, mostly undergraduates with a few graduate students, mainly mathematics majors with some from EECS, economics, and aero/astro; about half had no prior finance background. Students were expected to spend about 12 hours/week on the course, split between the two lecture types and, outside class, problem sets (due two weeks after each lecture) and the final paper.

C.2.3 Final paper topics (2013)

Twenty-five percent of the grade was the final paper (see above); the projects page gives sample topics – portfolio risk management, regime-shift modeling, low-volatility investing, modeling financial bubbles (citing Didier Sornette’s work), the Black-Scholes/heat-equation correspondence, hybrid FX/rate products, HJM versus short-rate models, and Ross recovery – plus a list of topics actually chosen by students the previous year, mostly variations on the same themes (Black-Scholes to heat-equation derivations, HJM/Ho-Lee models, FX jump-diffusion bond pricing, Asian options, minimal-entropy martingale measures, PCA of energy/currency swap curves around the 2008 crisis, and Monte Carlo/Heston option pricing).

1 COURSE NOTE

The instructors’ own reflection: designing problem sets for a mixed-background cohort was an acknowledged difficulty (“Future offerings will distinguish core versus advanced problems”), which is one motivation, together with the addition of the applications lectures, for how the course evolved from 2013 to the 2024 edition mapped in [chapter A](#).

C.2.4 Related resources (2013)

The related-resources page lists six textbooks (Hull's *Options, Futures, and Other Derivatives*; Baxter and Rennie's *Financial Calculus*; Wilmott, Howison and Dewynne's *The Mathematics of Financial Derivatives*; Grinold and Kahn's *Active Portfolio Management*; Fabozzi, Focardi and Kolm's *Financial Modeling of the Equity Market*; and Tsay's *Analysis of Financial Time Series*) together with a financial glossary of about 40 terms (alpha, beta, the Greeks, VaR, swaptions, zero-coupon bonds, and so on). The glossary substantially overlaps with the terminology already introduced across [parts I and II](#); [Exercise 1.6](#) in [chapter 1](#) is the course's own suggested exercise for building such a glossary, so it is not reproduced again here.

C.3 Sources not covered

The 2024 pages/week-1 through week-15 pages are a lecture-by-lecture index used to build [chapter A](#) and are not repeated here. No case-studies or lecture-notes index page beyond what is already listed in [chapter A](#) added further logistics content for the 2013 course.